

COMPUTATIONAL METHODS FOR LARGE SYSTEMS

COMPUTATIONAL METHODS FOR LARGE SYSTEMS

Electronic Structure Approaches for Biotechnology and Nanotechnology

Edited by

Jeffrey R. Reimers

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To Noel Hush
who showed me the importance of doing things to understand the critical
experiments of the day and the need for simple models of complex phenomena,
and
to George Bacskay
who taught me the importance of getting the right answer for the right reason.

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Preface: Choosing the Right Method for Your Problem

Computational methods have now advanced to the point where there is *choice* available for almost any problem in nanotechnology and biotechnology. In this book, the various methods available are presented and applications developed.

Given the difficulty in solving (relativistic) quantum mechanical equations for systems containing thousands of atoms, this situation is truly amazing and demonstrates the results of dedicated work by many researchers over a long period of time. Once demeaned by researchers as being useless for everything practical, computational methods have come into their own, providing fresh insight and predictive design power for wide-ranging problems: from superconductivity to semiconductivity to giant magnetoresistance to molecular electronics to spintronics to natural and synthetic polymer composition and properties to color design to nonlinear optics to energy flow to electron transport to catalysis to protein function to drug design.

Although much modern software is to be commended for its accessibility and ease of use, this advantage can be a luring trap. Electronic structure calculations on systems of any size are never simple. Many things can go wrong, and just because a method has always done the job in the past doesn't mean that it will continue to do so for a new problem that may appear very similar but which in fact embodies an additional unexpected effect. Proper understanding of the methods, including their strengths and weaknesses, is always essential. This book sets out to provide the background required for a range of approaches, containing extensive literature references to many of the subtle features that can arise. Practical examples of how this knowledge should be applied are then given.

Amazing as progress has been, many significant problems in physics, chemistry, biology, and engineering will forever remain outside the reach of direct quantum mechanical electronic structure calculations. By no means does this mean that the technologies now available cannot be usefully employed to tackle these problems, however, and a significant part of this book is devoted to multiscale-linking methods. For example, the surfaces of most heterogeneous catalysts are extremely complex, and hundreds of chemical reactions may be involved. Applications of this type of problem include the combustion of fossil fuels, atmospheric pollution modeling, and many industrial chemical reactions and smelting processes. Natural and synthetic polymers present similar challenges. What existing electronic structure methods offer is the data to go into more complex, perhaps multiscale models of the phenomena. Other

examples in quite different areas include protein folding, biological processes on the microsecond-to-second time scale, including the origin of intelligence, and long-range strong electron correlations in superconductors and other materials.

The fortunate position that we are in today is owed primarily to the development of density functional theory (DFT). This is the basic workhorse for electronic structure computations on large systems, being appropriate for biological, chemical, and physical problems. Part A of the book is devoted to the fundamentals of DFT, stressing the basics in Chapter 1 and then its two most common implementations strategies, atomic basis sets in Chapter 2 and plane-wave basis sets in Chapter 3. In the early days, atomic basis sets were designed to solve the burning issues at the time, such as the nature of the hydrogen molecule and the water molecule, while plane-wave basis sets could tackle problems of similar difficulty, such as the structure of simple metals. Today, both types of methods can be applied to almost any problem, each with its own advantages and disadvantages. An important feature of Chapter 1 is that it describes not only traditional DFT for the ground state of molecules and materials but also modern time-dependent approaches designed for excited states and nonequilibrium transport environments.

Deliberately missing from this book is an extensive discussion of which density functional to use. This may seem a terrible oversight in a book that is really intended as a practical tool for a new science. DFT gives the exact answer if the exact density functional is used, but alas this is unknown and perhaps even unknowable. So what we now have is a situation in which computational programs can let the user select between hundreds of proposed approximate functionals, or even make a new one. However, from a practical perspective, the situation is not that bad. Only a handful of density functionals are in common use, with just 14 mentioned in this book (B3LYP, B97D, BLYP, BOP, BP86, CAM-B3LYP, LC-BOP, LDA, LDA+DMFT, LDA+U, PBE, PBE0, PW91, and SOAP), with the most commonly used functionals being B3LYP, LDA, PBE, and PW91. B3LYP is the most commonly used functional for chemical problems, owing to its inclusion of more physical effects, whereas PW91 and PBE are the most commonly used functionals in the physics community, as they are typically good enough in these applications and are much faster to implement.

A density functional is not a single unit but usually comes as a combination of various parts, each intended to include some physical effect. Choosing a functional that includes all of the physical effects relevant to a particular application is thus essential. In this book the applications chapters provide significant discussion as to which functionals are appropriate for common applications. Many specialized functionals exist that are not discussed, so although the book describes what is good for most, experienced users should be aware that other attractive options do exist.

The most common physical effects included in modern density functionals are short-range correlation, short-range exchange, long-range correlation, long-range exchange, asymptotic correction, and strong correlation. All density functionals include short-range correlation and short-range exchange, with LDA including

only these contributions and thus being one of the simplest and most computationally expedient functionals available. LDA gives the exact answer for the free-electron gas, a problem to which many simple metals can realistically be compared. When the nature of the atomic nuclei become important, this functional takes the wrong qualitative form, however. Nevertheless, it provides a useful point even in the worst-case scenarios and hence forms a simple and useful approach. It does not provide results of sufficient accuracy to address any chemical question, however, so its realistic use is confined to a few problems involving simple metals. The next simplest functionals improve on LDA by adding a derivative correction to the local correlation description and are generically termed generalized-gradient (GGA) approximations, with classic functionals of this type including BP86, PW91, and PBE. In general, GGAs provide descriptions that attain chemical accuracy and hence can be widely applied. Sometimes LDA provides results in better agreement with experiment than common GGAs, however, and researchers are thus tempted simply to use LDA. This is a very bad practice, as GGAs always contain more of the essential physics than does LDA, and what is required instead is to move to a more complex functional that includes even more interactions. Get the right answer for the right reason.

In widespread use for chemical properties are hybrid functionals such as B3LYP and PBE0, which include long-range exchange contributions in the density functional. This improves magnetic properties, long-range interactions, excited- and transition-state energetics, and so on. Such methods are intrinsically much more expensive than GGAs, however. Recent advances of great relevance to biological simulations include the development of density functionals containing long-range exchange, such as B97D, as is required to model dispersive van der Waals intermolecular interactions. As the exchange and correlation parts of the density functionals are obtained independently, physical constraints concerning their balance are not usually met, leading to errors in their properties at long range that become important for charge separation processes, extended conjugation, band alignment at metal–molecule interfaces, and so on. Modern functionals such as CAM-B3LYP and LC-BOP contain corrections that reestablish the proper balance, improving properties computed. Finally, approaches such as LDA+U provide explicit empirical corrections for the extremely short range, strong electron correlation effects that dominate the chemistry of the rare earth elements, for example, and are often relevant for metal-to-insulator transitions and superconductivity.

Over the next decade, the future for density functional theory looks bright. There is much current interest not only in developing corrections to account for the shortcomings of standard GGA-type functionals, but there is also keen interest in developing new classes of functionals that contain intrinsically the correct asymptotic properties for electrons in molecules. This should dramatically simplify functional design and implementation, making the use of DFT much easier for users.

Certainly the most significant issue with current implementations of DFT is that no systematic process exists for improving functionals toward the illusive

exact functional. This is where alternative computational strategies of an *ab initio* nature can be very useful. Part B of the book looks at methods that can be used when modern DFT just doesn't work. Historically, the most common *ab initio* method for electronic structure calculation has been Hartree–Fock configuration-interaction theory. This involves use of a simplistic approximation, that proposed by Hartree and Fock, followed by expansions that converge to or even explicitly determine the exact answer (within the basis set used). The Hartree–Fock approximation itself is about as accurate as LDA and is not suitable for studying chemical problems, but like LDA can provide good insight into the operation of more realistic approaches. Although codes exist that can in principle give the exact solution to any problem, in practice this can only be achieved for the smallest systems, certainly nothing of relevance to this book. As a result, some empirically determined level of truncation of the *ab initio* expansion is necessary (coupled to a choice of basis set, of course), making their practical use rather similar to that of DFT—always find out what works for your problem using model systems for which the correct answer is known. The coupled-cluster method provides the “gold standard” for chemical problems, often producing results to an order-of-magnitude higher accuracy than can be achieved by DFT, but at much greater computational expense. Nevertheless, how such methods can be applied to large systems of nanotechnological and biotechnological relevance is shown in Chapter 5. These methods fail for metals, however, and so are less popular in solid-state physics applications. They handle strong electron correlations properly and easily, of course, and how they may be combined with DFT to solve such key problems as those relevant to metal–insulator transitions and superconductivity, the combination allowing the strengths of each method to be exploited while circumventing the weaknesses, is described in Chapter 6.

Hartree–Fock-based approaches will always scale extremely poorly as the system size increases, and an alternative *ab initio* method exists that scales much better while being applicable to molecules and metals alike: quantum Monte Carlo. The problem with this method has always been its startup cost, as even the simplest systems require enormous computational effort. But the time has now come where algorithms and computers are fast enough to solve many chemical and physical problems to a specifiable degree of accuracy. The method has come of age, and these advances are reviewed in Chapter 4. Because of the excellent scaling properties of this method, applications to larger and larger systems can now be expected to appear at a rapid rate.

But no matter how far computational methods such as DFT, configuration interaction, or quantum Monte Carlo methods advance, the researcher will hunger for the ability to treat larger systems, even if at a more approximate level. Part C of this book addresses these needs. Chapter 7 covers approximate but accurate schemes for implementing DFT and other methods that allow complex systems to be broken down into discrete fragments, achieving considerable computational savings while allowing chemical intuition to be used to ensure accuracy. Chapter 8 describes semiempirical Hartree–Fock-based approaches in which most of the interactions are neglected and the remainder parameterized,

leaving a priori computation schemes that at times achieve chemical accuracy and are available for all atoms except the rare earths. A similar approach, but this time modeled after DFT, is described in Chapter 9. The DFT approach widely applicable to both biological systems and materials science but requires parameters to be determined for every pair of atoms in the periodic table, providing increased accuracy at the expense of severe implementational complexity. It is now sufficiently parameterized to meet wide-ranging needs in biotechnology and nanotechnology.

Even so, some problems, such as superconductivity and the Kondo effect, require the study of electron correlations on length scales well beyond the reach of semiempirical electronic structure calculations. In Chapter 10 we look at a range of basic chemical models that describe the essential features of such systems empirically, leaving out all nonessential aspects of the phenomena in question. These methods follow from the analytical models used to put together the basics of chemical bonding and band structure theories in the 1930s–1960s, with the semiempirical methods described in Chapters 8 and 9 also originating from these sources. Accurate electronic structure calculations remain important, but in Chapter 10 we see that they only need to be applied to model systems to generate the empirical parameters that go in the electronic structure problem of the full system.

So, no matter what the size of the system, electronic structure methods are now in a position to contribute to the modeling of real-world problems in nanotechnology and biotechnology. Choosing whether to use empirical models parameterized by high-level calculations, use the DFT workhorse, or use methods that allow systematic improvement toward the exact answer is now a pleasant problem for researchers to ponder. Just because a certain type of problem has been solved historically by one type of approach does not mean that this is the best thing to do *now*. I hope that this book will allow informed choices to be made and set new directions for the future.

Part D presents applications of electronic structure methods to nanoparticle and graphene structure (Chapter 11), photobiology (Chapter 12), control of polymerization processes (Chapter 13), nonlinear optics (Chapter 14), nanoparticle optics (Chapter 15), heterogeneous catalysis (Chapters 16 and 17), spintronics (Chapter 18), and molecular electronics (Chapter 19).

This book has its origins in the Computational Methods for Large Systems satellite meeting at the very successful WATOC-2008 conference organized by Leo Radom in Sydney, Australia. I hope the book captures some of the excitement of that meeting and the overwhelming feeling that we are now at the tip of an enormous expansion of electronic structure computation into everyday research in newly emerging technologies and sciences.

I have had a go at most things described in this book at some stage of my career, and can vouch for a lot of it. As for the rest, well, they are things that I always wanted to do! I hope that you enjoy reading the book as much as I have enjoyed editing it.

Color Figures

Color versions of selected figures can be found online at

ftp://ftp.wiley.com/public/sci_tech_med/computational_methods

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PART A

DFT: The Basic Workhorse

1 Principles of Density Functional Theory: Equilibrium and Nonequilibrium Applications

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Arguably, the most important method for electronic structure calculations in intermediate- and large-scale atomic or molecular systems is density functional theory (DFT). In this introductory chapter we discuss fundamental theoretical aspects underlying this framework. Our aim is twofold. First, we briefly explain our view on several aspects of DFTs as we understand them. Second, we discuss the fundamentals underlying applications of DFT to transport problems. Here, we offer a derivation of the salient equations which is based on single-particle scattering theory; the more standard approach relies on the nonequilibrium Green's function (or Keldysh) technique.

More practical aspects of applying DFT to large systems such as nanoparticles, liquids, large molecules, and proteins are described in Chapter 2 (using atomic basis sets) and Chapter 3 (using plane-wave basis sets). Other recent reviews of basic application procedures by Kümmel and Kronik¹ and Neese² are also available. Chapters 11 to 19 focus on applications, introducing extensions of the basis methods when required.

1.1 EQUILIBRIUM THEORIES

The interacting N -electron problem is a formidable challenge for the theoretical disciplines of physics and chemistry. It is formulated in terms of a Hamiltonian, $\hat{H}$, which has the general structure

$$\hat{H} = \sum_i [\varepsilon(\hat{\mathbf{p}}_i) + v_{\text{ex}}(\hat{\mathbf{r}}_i)] + \frac{1}{2} \sum_{ij} u(\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j) \quad (1.1)$$

Here we have introduced the following notation: v_{ex} describes the system-specific time-independent external potential, which is generated, for example, due to the atomic nuclei. $\varepsilon(\mathbf{p})$ denotes the *dispersion* of the free particle, establishing the relation between the momentum of the particle and its energy in *free space* (i.e., in the absence of v_{ex} and the third term in u). For example, a single free particle with mass m has a dispersion $\varepsilon(\mathbf{p}) = \mathbf{p}^2/2m$. The third term introduces the two-particle interactions [e.g., $u(\mathbf{r}) = e^2/|\mathbf{r}|$ for the Coulomb case]. (We indicate an operator by $\hat{O}$ to distinguish it from its eigen- or expectation values.)

Density functional theory in its simplest incarnation serves to calculate several ground-state (GS) properties of this interacting many-body system. For example, one obtains the GS particle density, $n(\mathbf{r})$, the GS energy, E_0 , or the workfunction (ionization energy), W . DFT owes its attractiveness to the fact that all of this can be obtained, in principle, by solving an optimization problem for the GS density alone without going through the often impractical explicit calculation of the GS wavefunction, Ψ_0 , of the Hamiltonian (1.1). The actual task is to find a density profile, $n(\mathbf{r})$, so that the functional inside the brackets,

$$E_0 = \min_{\tilde{n}(\mathbf{r})} \left[F[\tilde{n}] + \int d\mathbf{r} v_{\text{ex}}(\mathbf{r}) \tilde{n}(\mathbf{r}) \right] \quad (1.2)$$

is invariant under small variations, $\delta\tilde{n}(\mathbf{r})$. Here F is a certain functional of the test density $\tilde{n}(\mathbf{r})$ that depends on the free dispersion, $\varepsilon(\mathbf{p})$, and the type of two-particle interactions, but not on the (static) environment, $v_{\text{ex}}(\mathbf{r})$. [The explicit definition of F is given in Eq. (1.10)]. The optimizing density coincides with the GS density and the optimum value of the functionals inside brackets delivers the GS energy.

1.1.1 Density as the Basic Variable

At first sight, the very existence of a formalism that allows us to obtain the GS properties mentioned without evaluating Ψ_0 itself may perhaps be surprising. After all, the particle density appears to involve a lot fewer degrees of freedom than Ψ_0 , which is the canonical starting point for calculation of the expectation values of the observables. Indeed, $\Psi_0(\mathbf{r}_1, \dots, \mathbf{r}_N)$ is a complex field that depends on the individual coordinates of each of the N particles. By contrast, the density is an expectation value of the density operator:

$$\hat{n}(\mathbf{r}) = \sum_{i=1}^N \delta(\mathbf{r} - \hat{\mathbf{r}}_i) \quad (1.3)$$

which may be obtained by integrating out most of the coordinates (“details”) of Ψ_0 :

$$n(\mathbf{r}) = \sum_i \int d\mathbf{r}_1 \cdots d\mathbf{r}_N \delta(\mathbf{r} - \mathbf{r}_i) |\Psi_0(\mathbf{r}_1, \dots, \mathbf{r}_N)|^2 \quad (1.4)$$

$n(\mathbf{r})$ is a real field depending on a single coordinate only.

At a second glance, however, the essential concepts underlying DFT are quite naturally understood. From a certain perspective, most of the information content of the ground state Ψ_0 is redundant.

To see why this is a case, we discuss an example. Consider all thermodynamic properties of a system described by the Hamiltonian (1.1). Each property corresponds to calculating some ratio of expectation values:

$$O = \frac{\text{Tr}[\hat{O}e^{-\beta\hat{H}}]}{\text{Tr}[e^{-\beta\hat{H}}]} \quad (1.5)$$

with an inverse temperature, $\beta = 1/kT$, and $\hat{O}$ denoting the operator corresponding to the observable of interest. The important thing to notice is that the system characteristics enter the average only via $\hat{H}$. Therefore, *within a given set of systems with members sharing the same kinetic energy and two-body interaction* (“universality class”), *all system specifics (i.e., observables) are determined uniquely by specifying the external potential*, so O is a functional of v_{ex} : $O[v_{\text{ex}}]$. This simple observation already implies that within such a universality class, the system behavior can be reconstructed from knowledge of a scalar field [here $v_{\text{ex}}(\mathbf{r})$], and in this sense most of the information content of Ψ_0 is redundant.

In the Schrödinger theory, the classifying scalar field is the external potential. DFT amounts to a change of variables that replaces $v_{\text{ex}}(\mathbf{r}) \rightarrow n(\mathbf{r})$. Such a transformation is feasible because the density operator and the external potential v_{ex} enter $\hat{H}$ as a product, $\sum_{i=1}^N v_{\text{ex}}(\hat{\mathbf{r}}_i) = \int d\mathbf{r} v_{\text{ex}}(\mathbf{r}) \hat{n}(\mathbf{r})$. Therefore, the average density and v_{ex} are *conjugate variables* and a relation

$$n(\mathbf{r}) = \frac{\partial E_0[v_{\text{ex}}]}{\partial v_{\text{ex}}(\mathbf{r})} \quad (1.6)$$

holds true. Under the assumption that Eq. (1.6) can be inverted (at least “piecewise”), we can employ a Legendre transformation to facilitate the change in variables from v_{ex} to n :

$$F[n] = E_0[v_{\text{ex}}] - \int d\mathbf{r} n(\mathbf{r}) v_{\text{ex}}(\mathbf{r}) \quad (1.7)$$

where the external potential is now the dependent variable given by

$$v_{\text{ex}}(\mathbf{r}) = \frac{-\partial F[n]}{\partial n(\mathbf{r})} \quad (1.8)$$

Thus, it is suggested that the density n can also be considered the fundamental variable, so that observables are functionals thereof. The ground-state energy is an example of this.

Summarizing: Underlying DFT is the insight that within a given universality class, each physical system can be identified uniquely either by the belonging “environment,” $v_{\text{ex}}(\mathbf{r})$, or by its GS density, $n(\mathbf{r})$. Therefore, in principle, knowing just the ground-state density is enough information to determine any observable (equilibrium) quantity of a many-body system.

Remarks

- A formal proof that the density can act as the fundamental variable was presented by Hohenberg and Kohn³; see Section 1.1.1.
- A generalization of DFT to spin or current DFT may be indicated for systems with degeneracies. Then additional fields such as magnetization and current density are needed to distinguish among the system states.

1.1.2 Variational Principle and Levy's Proof

Just the mere statement that equilibrium expectation values of observables can be calculated from some functionals once the GS density, n , is known, is not very helpful. For DFT to be self-consistent, also needed is a procedure to obtain this GS density by not referring to anything other than the functionals of n itself. This is where the variational principle kicks in, which says that the GS has a unique property in that it minimizes the system's total energy. This implies, in particular, that the GS has a density that minimizes (for a fixed environment v_{ex}) the functional $E_0[n]$. Hence, we can find n by solving the optimization problem (1.2), involving only variations of the density.

A particularly instructive derivation of Eq. (1.2) has been given by Levy.⁴ We summarize the essential logical steps, to remind ourselves that the connection between the variational principle and DFT is actually deep and not related only to practical matters. In fact, Levy's proof starts with the variational principle for the GS. It implies that there is a configuration space, C , of totally anti-symmetric functions, $\tilde{\Psi}$, with the normalization property $N = \int d\mathbf{r} \langle \tilde{\Psi} | \hat{n}(\mathbf{r}) | \tilde{\Psi} \rangle$, together with a functional $E[\tilde{\Psi}] = \langle \tilde{\Psi} | \hat{H} | \tilde{\Psi} \rangle$ defined on this space, which is optimized by the GS, Ψ_0 , with the GS energy, E_0 , being the optimum value; explicitly,

$$E[\tilde{\Psi}] = \langle \tilde{\Psi} | \hat{T} + \hat{U} | \tilde{\Psi} \rangle + \int d\mathbf{r} v_{\text{ex}}(\mathbf{r}) \tilde{n}(\mathbf{r}) \quad (1.9)$$

where $\hat{T}$ abbreviates the kinetic energy and $\hat{U}$ the interaction energy appearing in Eq. (1.1), and $\tilde{n}$ is the particle density associated with $\tilde{\Psi}$. The trick in Levy's argument is to organize the minimum search in two steps. In the first step the total configuration space, C , is subdivided into subspaces such that all wavefunctions inside a given subspace have identical density profiles $\tilde{n} = \langle \tilde{\Psi} | \hat{n}(\mathbf{r}) | \tilde{\Psi} \rangle$. Next, within each subspace a search is launched for the elements that minimize E . Thus, a submanifold, M_{preopt} , is identified which contains a set of "preoptimized" elements. By construction, each element $\Psi_{\tilde{n}}$ of M_{preopt} is uniquely labeled by the associated density profile $\tilde{n}$ (see Fig. 1.1). In the second step, the minimum search is continued, but it can now be restricted to finding the one element, Ψ_0 , of M_{preopt} that minimizes E .

The motivation behind this particular way of organizing the search is the following: The division procedure in step 1 has been constructed such that the second term in Eq. (1.9) does not contribute to preoptimizing; within a given

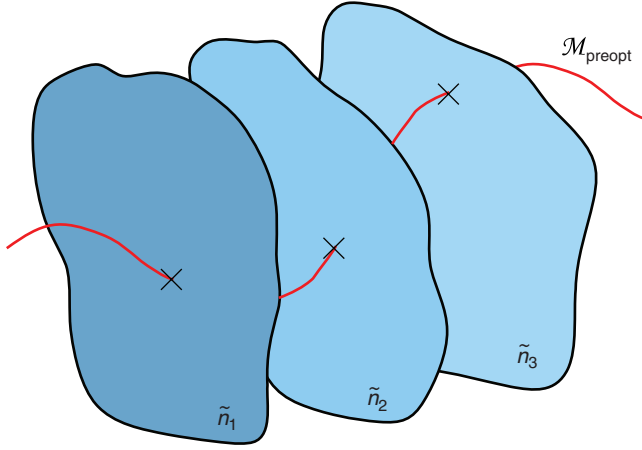


Fig. 1.1 (color online) Schematic AI representation of the constraint search strategy in C space. One sorts the space of all possible (i.e., antisymmetrized, normalizable) wavefunctions into submanifolds. By definition, wavefunctions belonging to the same submanifold generate the same density profile, $\tilde{n}(\mathbf{r})$. Each submanifold has a wavefunction $\Psi[\tilde{n}(\mathbf{r})]$ (at fixed external potential v_{ex}), which has the lowest energy. These wavefunctions sit on a *hypersurface* (a “line”) in the configuration space which is parameterized by $\tilde{n}(\mathbf{r})$. The surface is continuously connected if the evolution of $\Psi[\tilde{n}(\mathbf{r})]$ with the density profile is smooth (i.e., if degenerate shells with more than one optimum state Ψ do not exist). (We identify with each other states that differ only by a spatially homogeneous phase.) Typically, for every external potential, v_{ex} , there is exactly one such surface. The ground-state energy is found by going over the surface and searching for the global energy minimum.

subspace it is just a constant. In this step, only the first term is minimized, with an extremal value,

$$F[\tilde{n}] \equiv \langle \Psi_{\tilde{n}} | \hat{T} + \hat{U} | \Psi_{\tilde{n}} \rangle \quad (1.10)$$

The important observation is that by construction the functional $F[\tilde{n}]$ is universal (i.e., *independent of external conditions*, v_{ex}). (This statement is contained in the Hohenberg–Kohn theorem.³) Therefore, F is found by preoptimizing once and for all.

After F has been identified, the calculation of system-specific properties (depending on v_{ex}), which was described in Eq. (1.2), requires only a restricted search within the submanifold M_{preopt} . The benefit is tremendous, since the volume to be searched, M_{preopt} , is tiny compared to the original wavefunction space C .

Remarks

- $F[n]$ has the exact property

$$\left. \frac{\partial F[\tilde{n}]}{\partial \tilde{n}(\mathbf{r})} \right|_{\tilde{n}=n} + v_{\text{ex}}(\mathbf{r}) = \mu$$

Proof: The ground-state density, n , is an extremal point by construction under the constraint $N = \int d\mathbf{r} \tilde{n}(\mathbf{r})$. Introducing a Lagrange parameter, μ , we can release the constraint and perform an unrestricted search minimizing $F[\tilde{n}] + \mu N + \int d\mathbf{r} [v_{\text{ex}}(\mathbf{r}) - \mu] \tilde{n}(\mathbf{r})$. The claim follows after functional differentiation.

- The minimum search in Eq. (1.2) is in a space of scalar functions $\tilde{n}$, which have the property that they are “ Ψ -representable”: For a given $\tilde{n}(\mathbf{r})$ there is at least one element of C with the property $\tilde{n}(\mathbf{r}) = \langle \tilde{\Psi} | \hat{n}(\mathbf{r}) | \tilde{\Psi} \rangle$. This implies, for example, positivity: $\tilde{n} \geq 0$.
- We presented Levy’s argument for ground-state DFT. It is obvious, however, that the restriction to GS and the collective mode “density” was not crucial. Only the variational principle and a linear coupling of an environmental field to some collective mode (e.g., density, spin density, current density) should be kept. Therefore, generalizations of ground-state DFT to many other cases have been devised: for example, (equilibrium) thermodynamic DFT at nonzero temperature, magnetic properties (spin DFT and current DFT), and relativistic DFTs.

Moreover, it has been shown that certain excited states can also be calculated exactly with a ground-state (spin) DFT. This happens when the Hamiltonian, $\hat{H}$, exhibits symmetries, such as spin rotational invariance. Then the Hilbert space decomposes into invariant subspaces each carrying its own quantum number(s), q : for example, a spin multiplicity. The minimum search may then proceed in every subspace, separately, giving a separate functional F_q for each of them. The local q -minima thus obtained are valid eigenstates of the full Hamiltonian (Gunnarsson–Lundqvist theorem⁵).

1.2 LOCAL APPROXIMATIONS

The precise analytical dependency of the energy functional $F[n]$ on the density $n(\mathbf{r})$ is not known, of course. Available approximations employ knowledge, analytical and computational, about *homogeneous* interacting Fermi gases (i.e., the case $v_{\text{ex}} = \text{const}$). Indeed, it turns out that the homogeneous system also provides a very useful starting point to build up a zeroth-order description in the inhomogeneous environments that are relevant for describing atoms and molecules.

1.2.1 Homogeneous Electron Gas

Homogeneous gases are relatively simple. The particle density, n , is just a parameter and all functionals, which in general involve multiple spatial integrals over expressions involving $n(\mathbf{r})$ at different positions in space (nonlocality property), turn into functions of n . Analytical expressions for them can usually be derived from perturbative treatments of $E_0(n)$, which are justified in two limiting cases: where a control parameter, r_s , is either very large or very small.

For the homogeneous electron gas, r_s can easily be identified: It is the ratio of two energies. The first energy is the typical strength of the interaction that two

particles feel in the electron gas in three-dimensional space: $(e^2/\epsilon_0)n^{1/3}$. To see whether or not this energy is actually sizable, one should compare it to another energy. The correct energy scale to consider will be a measure of the kinetic energy of the particles.

The average kinetic energy of a fermion depends on the gas density, n . To derive an explicit expression, we recall that due to the Pauli principle, all particles that share the same spin state must be in different momentum states, $|\mathbf{p}\rangle$. Therefore, when filling up the volume, higher and higher momentum states, up to a maximum momentum value, p_F , will be occupied. The kinetic energy of the particles occupying the highest-energy (*Fermi energy*) states, $\epsilon_F(n) \equiv \epsilon(p_F)$, will be a good measure for the typical kinetic energy of a gas particle.

The situation is best visualized recalling the familiar quantum mechanical textbook problem of “a particle in a box” with box size L . The energy levels of the box can be ordered according to the number of nodes exhibited by the corresponding wavefunctions. The spatial distance between two nodes gives half the wavelength, $\lambda/2$, with an associated wavenumber $k = 2\pi/\lambda$ and momentum $p = \hbar k$. The maximum wavelength reached by N particles (with spin $\frac{1}{2}$) filling the box is $\lambda_F/2 = L/(N/2) = 2/n$, giving rise to a maximum wavenumber, the *Fermi wavenumber* $k_F = \pi n/2$, and a maximum momentum $p_F = \hbar k_F$. In three dimensions, similar considerations yield $\pi k_F^3/3 = (2\pi)^3(n/2)$.

Employing these results, our dimensionless parameter can now be specified as $r_s \sim e^2 n^{1/3}/\epsilon_0 \epsilon_F(n)$, which conventionally is cast into the form

$$\frac{4\pi}{3} r_s^3 = \frac{1}{n a_0^3}$$

stipulating a parabolic dispersion $\epsilon(p) = p^2/2m$ (ϵ_0 : effective dielectric constant; $a_0 = 4\pi\epsilon_0\hbar^2/me^2 \approx 0.529 \text{ \AA}$: Bohr’s radius). Analytical expansions of $E_0(n)$ are available in the limiting cases $1/r_s \ll 1$ or $r_s \ll 1$. Typically, in particular with molecular systems, one has the marginal case $r_s \simeq 1$. Here, computational methods such as quantum Monte Carlo calculations (see Chapter 4) help to interpolate the gap.

Motivated from the weakly interacting limit ($r_s \ll 1$), conventionally we consider the following splitting of the GS energy per unit volume[†]:

$$\check{\epsilon}_0(n) = 2 \sum_{|\mathbf{k}| \leq k_F(n)} \epsilon(\mathbf{k}) + v_{XC}(n) \quad (1.11)$$

[†]For homogeneous densities, the Hartree term reads $n \int d\mathbf{r}' u(\mathbf{r} - \mathbf{r}')$. Since the spatial summation over the Coulomb potential, $\sim 1/r$, does not converge, the integral makes a contribution to the energy balance which is formally infinite. This divergency is an artifact of modeling the interacting electron gas without taking into consideration the (positive) charge of those atomic nuclei (“counter charges”) that provide the source of the electrons to begin with. The physical system is always (close to) charge neutral, so that (on average) $n_{\text{nuclei}} = -n_{\text{electrons}}$. This implies that the nuclei provide a “background” potential, $n_{\text{nuclei}} \int d\mathbf{r}' u(\mathbf{r} - \mathbf{r}')$, that leads to an exact cancellation of the divergent contribution in the Hartree term. Therefore, this particular term should be ignored when dealing with the homogeneous electron system (the Jellium model).

where the factor of 2 accounts for the electron spin. The first term comprises the kinetic energy of the free gas. Its dependency on the density is regulated via the Fermi wavenumber, $k_F(n)$. The second term includes the remaining correlation effects and therefore has a weak coupling expansion. For the Coulomb case, the leading term is $\sim 1/r_s$ with subleading corrections,⁶

$$v_{XC}(n) = -n \frac{0.9163}{r_s} + n[-0.094 + 0.0622 \ln r_s + 0.018 r_s \ln r_s + O(r_s)] \quad (1.12)$$

in Rydberg units ($E_{\text{Ry}} = E_{\text{Hartree}}/2 \approx 13.6$ eV).

1.2.2 Local Density Functional

The information taken from homogeneous systems for constructing functionals describing inhomogeneous systems is the dependency of the GS energy per volume on the particle density, $\tilde{\epsilon}_0(n)$. A leading-order approximation for the general F -functional is obtained by

$$F[n] = \int d\mathbf{r} \tilde{\epsilon}_0(n(\mathbf{r})) \quad (1.13)$$

This approximation is valid if the inhomogeneous system is *real-space separable*, meaning that it can be decomposed into a large number of subsystems that (1) still contain sufficient particles to allow for treatment as an electron gas with a finite density, (2) are already small enough to be nearly homogeneous in density, and (3) have negligible interaction with each other. Systems exhibiting a relative change of density, which is large even on the shortest length scale available, the Fermi wavelength λ_F , do not satisfy (1) and (2) simultaneously. So a minimal condition for the applicability of Eq. (1.13) is

$$\frac{\lambda_F \nabla n}{n} \ll 1 \quad (1.14)$$

Remarks

- Condition (3) implies that the interaction is short range, ideally $u(\mathbf{r} - \mathbf{r}') \sim \delta(\mathbf{r} - \mathbf{r}')$. For the Coulomb case, we separate from the $1/|\mathbf{r} - \mathbf{r}'|$ -interaction a long-range term, which is then treated by introducing an extra term, the Hartree potential.
- Since the Fermi wavelength itself depends on the density, $\lambda_F \sim n^{-1/d}$, relation (1.14) is satisfied typically only in the large n -limit. There, the main contribution to the energy (1.13) stems from the kinetic term in Eq. (1.11). Therefore, the leading error in the local functional (1.14) usually comes from the fact that the *Thomas–Fermi approximation* [$k_F(\mathbf{r}) \equiv k_F(n(\mathbf{r}))$]

$$\langle \hat{T} \rangle \approx 2 \int d\mathbf{r} \sum_{|\mathbf{k}| \leq k_F(r)} \epsilon(\mathbf{k}) \quad (1.15)$$

gives only a very poor estimate of the kinetic energy of an inhomogeneous electron gas, even for noninteracting particles.

- The failure of the Thomas–Fermi approximation is the main reason that orbital-free DFT has a predictive power too limited for most practical demands. The search for more accurate representations of the kinetic energy in terms of n -functionals is at present an active field of research.^{7,8}

1.3 KOHN–SHAM FORMULATION

Better estimates for the kinetic energy can be obtained within the *Kohn–Sham formalism*.⁹ One addresses the optimization problem (1.2) by reintroducing an orbital representation of the density with single-particle states,

$$n(\mathbf{r}) = \sum_{\ell=1}^{\tilde{N}} |\phi_{\ell}(\mathbf{r})|^2 \quad (1.16)$$

called the *Kohn–Sham* or *molecular orbitals*. The orbitals ϕ_{ℓ} are sought to be ortho-normalized; the parameter $\tilde{N}$ is free, in principle. However, with an eye on approximating the kinetic energy of the interacting system by the energy of the free gas, $\tilde{N}$ is usually chosen to be equal to the number of particles, $\tilde{N} = N$. With this choice, the optimization problem formally reads

$$\frac{1}{2} \frac{\partial}{\partial \phi_{\ell}^*(\mathbf{r})} [E_0[n(\mathbf{r})] - \varepsilon_{\ell}(\langle \phi_{\ell} | \phi_{\ell} \rangle - 1)] = 0 \quad (1.17)$$

featuring the *Kohn–Sham energies* (or *molecular orbital energies*), ε_{ℓ} , which play the role of Lagrange multipliers ensuring normalization. Equation (1.17) can be cast conveniently into a form reminiscent of a Schrödinger equation of N single particles:

$$[\varepsilon(\mathbf{p}) + v_s(\mathbf{r})]\phi_{\ell}(\mathbf{r}) = \varepsilon_{\ell}\phi_{\ell}(\mathbf{r}) \quad (1.18)$$

where we have employed a substitution ($\mathbf{p} = -i\hbar\partial_x$),

$$\frac{1}{2} \frac{\partial}{\partial \phi_{\ell}^*(\mathbf{r})} E_0[n(\mathbf{r})] = [\varepsilon(\mathbf{p}) + v_s(\mathbf{r})]\phi_{\ell}(\mathbf{r}) \quad (1.19)$$

which is merely a definition of an auxiliary quantity, the effective potential $v_s(\mathbf{r})$. The set of N equations given by Eq. (1.18) constitutes the *Kohn–Sham equations*.

Remarks

- The Kohn–Sham (KS) formalism should give a much improved description of the kinetic energy, because by construction it reproduces exactly the kinetic energy of the inhomogeneous, noninteracting gas.

- The fictitious KS particles live in an effective potential which modulates their environment such that their density and all related properties coincide with those of a true many-body system.
- The potential term has a decomposition

$$v_s(\mathbf{r}) = v_{\text{ex}}(\mathbf{r}) + v_{\text{H}}(\mathbf{r}) + v_{\text{XC}}(\mathbf{r}) \quad (1.20)$$

where the second term includes the Hartree interaction, which for a specific two-body interaction potential $u(\mathbf{r} - \mathbf{r}')$ reads $v_{\text{H}}(\mathbf{r}) = \int d\mathbf{r}' u(\mathbf{r} - \mathbf{r}')n(\mathbf{r}')$. The third term, the *exchange–correlation potential*, incorporates all the remaining, more complicated many-body contributions. In particular, we have also lumped the difference between the free and interacting kinetic energies into this term.

- Solving the KS equations requires diagonalization of a *KS-Hamiltonian*:

$$\hat{H}_{\text{KS}} = \varepsilon(\hat{\mathbf{p}}) + v_s(\hat{\mathbf{r}}) \quad (1.21)$$

The dimension of the corresponding Hilbert space, N_ϕ , usually exceeds the particle number substantially: $N_\phi \gg N$. Therefore, occupied (real) eigenstates that finally enter the construction of the density [Eq. (1.16)] need to be distinguished from unoccupied (virtual) ones. The selection process follows the variational principle.

- Similar to the Hartree theory and in pronounced contrast to the Schrödinger equation for a single particle, the KS equations pose a self-consistency problem: The potential $v_s(\mathbf{r})$ is a functional of $n(\mathbf{r})$, so it needs to be determined “on the fly.” We emphasize that even though the functional $v_s[n](\mathbf{r})$ may exhibit a very complicated—in particular, *nonlocal*—dependency on the ground-state particle density, the effective potential that finally is felt by the KS particles is perfectly local in space. It provides an effective environment for the KS particles, so that the many-body density can be reproduced.
- The self-consistent field (SCF) problem in DFT is much easier to solve than the Hartree–Fock (HF) equations, which are nonlocal in space and, what is much worse, even orbital dependent. As a consequence of the orbital dependency of the Fock operator, a real HF orbital interacts with $N - 1$ other real orbitals, whereas a virtual orbital interacts with N real orbitals. The situation in DFT is much simpler in the sense that occupied and unoccupied orbitals all feel the same effective potential $v_s[n](\mathbf{r})$. Notice, however, that this computational advantage comes at the expense of the *derivative discontinuity*, an unphysical feature of exact exchange correlation functionals (see Section 1.5.3) that is very difficult to implement in efficient approximation schemes.
- Our derivation of the Kohn–Sham equations was tacitly assuming the following: The density of *any* electron system, including the interacting systems, can be represented in the manner of Eq. (1.16), where the orbitals

ϕ_ℓ are normalizable solutions of a (single-particle) Schrödinger equation. Is this really true? The answer is: Not always. That is, systems with degenerate ground states may exhibit a particle density that can only be represented as a sum of independent contributions coming from a number g of single Slater determinants. A general statement that is valid for all practical purposes is that any fermionic density may be represented uniquely as a weighted average of g degenerate ground-state densities of some effective single-particle Schrödinger problem [Eq. (1.18)].^{10,11}

1.3.1 Is the Choice of the KS–Hamiltonian Unique?

For an interacting many-body system, splitting between kinetic and potential energy as suggested in Eqs. (1.19) and (1.20) is not as unique as it may appear at first sight. To give a straight argument, recall that the dispersion relation of the free particles, $\varepsilon(\mathbf{p})$, can be altered substantially by interaction effects. For example, the mass of the electron describes how the particle’s energy depends on its momentum. In the presence of interactions, an electron always moves together with its own screening cloud, brought about by the presence of other electrons. Although this does not change the wavelength (i.e., the momentum) of the electron, it does change its velocity. It tends to make it slower, so that the “effective” mass increases. Such interaction effects on parameters such as the mass, the thermodynamic density of states, and the magnetic susceptibility are called *Fermi-liquid renormalizations*.

Having this in mind, one could easily imagine another splitting featuring a *renormalized* kinetic energy, $\varepsilon^*(\mathbf{p})$, which would describe a more adapted description of the dispersion of charged excitations (e.g., the propagation of screened electrons) in the interacting quantum liquid.¹² A remaining, *residual interaction*, V_{XC}^{res} , would appear to be designed so that the ground-state density produced by this effective system would also coincide with the true density. Such a renormalized splitting is rarely employed in practice, perhaps because a good approximation for the residual functionals is not available.

For the effective single-particle problem that yields the exact ground-state density, we conclude that various choices are possible, the choices differing from one another in the dispersion $\varepsilon(\mathbf{p})$ that enters the kinetic part of the KS–Hamiltonian. Very few restrictions on the possible functional forms of $\varepsilon(\mathbf{p})$ exist; the parabolic shape and the trivial form $\varepsilon \equiv 0$ (with proper readjustments of v_{XC}) are just two choices out of many.

1.4 WHY DFT IS SO SUCCESSFUL

The precise dependency of the exchange–correlation potential v_{XC} on the density $n(\mathbf{r})$ is not known. In the simplest approximation, the *local density approximation* (LDA), one takes for v_{XC} the result obtained from the homogeneous electron gas [Eq. (1.12)], but replacing the homogeneous density with $n(\mathbf{r})$ (see Section 1.2.2).

Remarks

- The universal success of DFT in chemistry and condensed matter physics came with the empirical finding that the combination of KS theory with LDA (and its close relatives) works in a sufficiently quantitative way to make it possible to calculate ground-state energies (and hence to determine molecular and crystal structure) even *outside* the naive regime of the validity of LDA as given by relation (1.14). This is due to a cancellation of errors in the kinetic and exchange correlation part of the KS-Hamiltonian (1.21).¹³
- In analogy with Hartree–Fock theory, a fictitious “KS–ground state” wavefunction, Φ , is often considered. It is constructed by building a Slater determinant from the real KS orbitals. In contrast to HF, this state is *not* optimal in an energetic sense. It does, however, reproduce the exact particle density. In the same spirit, KS energies are often interpreted as single-particle energies, even though from a dogmatic point of view there is no (close) connection between the Lagrange multipliers and the true many-body excitations; indeed, to the best of our knowledge, a precise justification of this practice has never been given. Still, the pragmatic approach has established itself widely, since it often gives semiquantitative estimates for Fermi-liquid renormalizations, which are important, for example, in band structure calculations.
- The implementation of efficient codes is much easier in DFT than in HF theory, due to the fact that functionals are only density and not orbital dependent. For this reason, many powerful codes are readily available in the marketplace.
- At present, because of the virtues noted above, DFT is by far the most widely used tool in electronic structure theory (lattice structures, band structures) and quantum chemistry (molecular configurations), with further applications in many other fields, such as nuclear physics, strongly correlated systems, and material science.

1.5 EXACT PROPERTIES OF DFTs

Since there is no analytic solution of the general interacting many-body problem, it is not surprising that exact statements about exchange correlation functionals are scarce. Precise information is, however, available in the presence of an interface to the vacuum.

Imagine a situation in which a molecule or a piece of material is embedded in a vacuum. The material is associated with an attractive KS potential “well,” v_s , which binds N electrons to the nuclei (or atomic ion cores). Outside the material, the binding potential and the particle density rapidly approach their asymptotic zero values. Exact information is available about how the asymptotic value is approached.

1.5.1 Asymptotic Behavior of v_{XC}

Consider the Hartree term

$$v_{\text{H}}(\mathbf{r}) = \sum_{\ell'=1}^{\text{occ}} \int d\mathbf{r}' u(\mathbf{r} - \mathbf{r}') |\phi_{\ell'}(\mathbf{r}')|^2 \quad (1.22)$$

in the KS equations

$$[\varepsilon(\hat{\mathbf{p}}) + v_{\text{ex}}(\mathbf{r}) + v_{\text{H}}(\mathbf{r}) + v_{\text{XC}}(\mathbf{r})] \phi_{\ell}(\mathbf{r}) = \varepsilon_{\ell} \phi_{\ell}(\mathbf{r}) \quad (1.23)$$

It contains at $\ell' = \ell$ a piece $u(\mathbf{r} - \mathbf{r}') |\phi_{\ell}(\mathbf{r}')|^2$, which incorporates an interaction of a particle in the occupied orbital ϕ_{ℓ} with its own density. This spurious, nonphysical interaction is known as a *self-interaction error*. In principle, it should be eliminated by a counterpiece contained in the exchange part of v_{XC} .[†] The construction and application of empirical corrections for this effect are the subject of Chapter 14.

The Hartree term is known exactly in the asymptotic region. This is the reason that it is possible to draw a rigorous conclusion about v_{XC} . To be specific, we consider the case of Coulomb interactions. In the asymptotic regime a distance r away from the materials center, where the particle density is totally negligible, all spurious contributions made by an occupied orbital add up to e^2/r . To cancel this piece we must have

$$v_{\text{XC}}(r) \xrightarrow{r \rightarrow \infty} -\frac{e^2}{r} + \frac{-\alpha_{N-1}}{2r^4} + \dots \quad (1.24)$$

whenever the particle density vanishes. The correction term, which we have also given here, describes the polarizability, α_{N-1} , of the many-body system (with $N - 1$ particles). This term incorporates the interactions with the fluctuating charge density of the mother system that particles feel when they explore the asymptotic region.

[†]This cancellation may be seen explicitly within the Hartree–Fock approximation. That is, the interaction term reads

$$\sum_{\sigma'=\uparrow,\downarrow} \sum_{\ell'} \int d\mathbf{r}' u(\mathbf{r} - \mathbf{r}') \phi_{\ell'\sigma'}^*(\mathbf{r}') [\phi_{\ell'\sigma'}(\mathbf{r}') \phi_{\ell\sigma}(\mathbf{r}) - \delta_{\sigma\sigma'} \phi_{\ell'\sigma'}(\mathbf{r}') \phi_{\ell\sigma}(\mathbf{r}')]]$$

so that the piece with $\ell' = \ell$, $\sigma' = \sigma$ in the first (Hartree) term is eliminated by a corresponding piece in the second (Fock) term.

Remarks

- A more intuitive way to rationalize the leading asymptotics of v_{XC} is to recall that an electron that makes a virtual excursion from its host material into vacuum still interacts with the hole that it leaves behind. The first term in Eq. (1.24) describes the interaction with this virtual hole.
- Both terms appearing in Eq. (1.24) are not recovered in local approximation schemes, such as LDAs and generalized-gradient approximations (GGAs), which stipulate the form $v_{\text{XC}}(\mathbf{r}) \approx v_{\text{XC}}(n(\mathbf{r}), \nabla n(\mathbf{r}), \dots)$. The statement is obvious, because the density is exponentially small in the asymptotic region (see Section 1.5.2), whereas the potential (1.24) is not. This defect has very serious consequences, since the van der Waals dispersion interactions, $v_{\text{XC}} \sim -\alpha_{N-1}/r^4$, ignored in LDAs and GGAs, provide the dominating intermolecular forces that prevail, for example, in biochemical environments. To address this problem, Grimme¹⁴ has proposed an ad hoc empirical procedure that adds a long-range term to standard energy functionals. The functional contains specific parameters, essentially modeling the local polarizability of single atoms or molecular groups chosen so that a rough description of the van der Waals interaction is retained.

1.5.2 Workfunction

Now, consider the KS potential well in its ground state with N occupied bound orbitals ϕ . Generically, every such orbital contributes to the particle density $n(\mathbf{r})$ at a point $\mathbf{r}$ unless it happens that ϕ has a node there: $\phi(\mathbf{r}) = 0$. This is also true in the asymptotic region far away from the well's center. However, in this region the state ϕ_{HOMO} with the largest KS energy [highest occupied molecular (or material) orbital (HOMO)] gives the dominating contribution almost everywhere (i.e., at all points where $|\phi_{\text{HOMO}}(\mathbf{r})|^2 > 0$). It is easy to see why this is. In the asymptotic region $v_s(\mathbf{r})$ decays in a power-law manner with the distance r from the well's center (Fig. 1.2). Therefore, the KS equations read

$$-\frac{\hbar^2}{2m}\partial_r^2(r\phi_\ell) = \varepsilon_\ell(r\phi_\ell) \quad (1.25)$$

where $\varepsilon_\ell < 0$ denotes the ionization energy of a bound KS state. The solution is

$$\phi_\ell \sim \frac{1}{r}e^{-\sqrt{2m|\varepsilon_\ell|/\hbar^2}r} \quad (1.26)$$

so that generically the HOMO orbital has the smallest KS energy by modulus, $|\varepsilon_{\text{HOMO}}|$. At large enough distances, it will give the only relevant contribution. [Exceptions to the rule occur only in the case of a vanishing prefactor not written in Eq. (1.26).] For this reason, the KS energy of the highest occupied molecular level is actually a physical observable; it gives the ionization energy or *workfunction* (Janak's theorem^{15,16}).

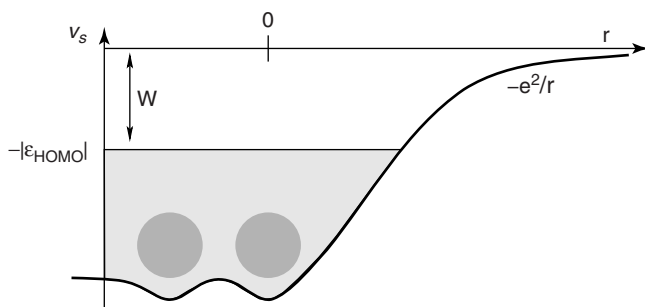


Fig. 1.2 Effective potential (solid line) near a surface of a simple metal. Surface atoms (dark balls) and the electron liquid (light background) are also indicated.

1.5.3 Derivative Discontinuity

The derivative discontinuity^{17,18} (DD) is perhaps one of the less intuitive properties that an exact XC potential must exhibit. We discuss it here in some detail, since the fact that local approximations are not capable of capturing it even qualitatively often leads to very important artifacts in the KS spectra which are not a genuine feature of DFT itself but, rather, of the LDA.

We will see that the DD is related intimately to the fact that the N (real) particles in a many-body system interact with only $N - 1$ partners, while an infinitesimal test (virtual) charge in such a system would interact with N (i.e., *all* the other particles). Since $v_{\text{XC}}[n]$ has access to the total density only, it cannot easily distinguish real and virtual orbitals with their different interacting environments (as HF does). It turns out that the way DFT implements such behavior is via a very sharp (i.e., nonanalytic) behavior of $v_{\text{XC}}[n]$ on the particle density $n(\mathbf{r})$.

1.5.3.1 Isolated System Consider an isolated quantum dot, such as a single atom or a molecule, with N electrons. The corresponding KS system exhibits a number of N KS particles that occupy the N lowest-lying KS states. It is important to recall that each KS particle interacts with the *total* charge density, $v_{\text{XC}}[n_N]$, only, including the density contribution that comes from itself. In this respect, KS particles are fundamentally different from physical particles, which do not interact with themselves, of course.

Next, add one additional particle, the *excess charge*, $\delta N = 1$; to be specific, put it into the lowest unoccupied molecular orbital (LUMO ^{N}). The new XC functional of the “anion” will be $v_{\text{XC}}[n_{N+1}]$. What are the consequences of charging for the KS energies? Due to the change $n_N \rightarrow n_{N+1}$, every original particle interacts with one more charge, δN , the excess particle in the LUMO ^{N} . Therefore, the energy of every one of the first N orbitals shifts by the amount U , which measures the interaction with the excess particle (see Fig. 1.3). Notice also that the energy of the LUMO ^{N} (now, better, HOMO ^{$N+1$}) has shifted by U after it was occupied. This is because in KS theory, all orbitals, occupied and unoccupied, are calculated in *the same* effective potential.

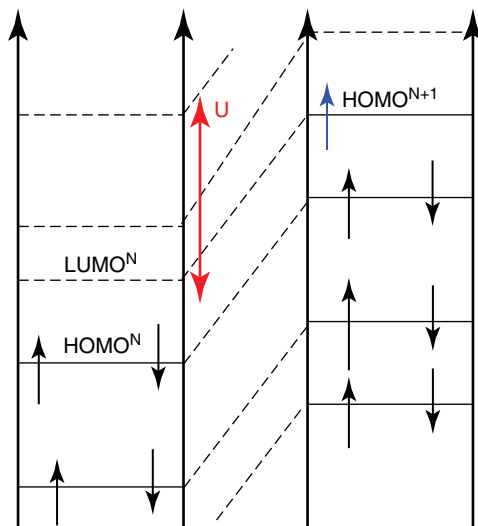


Fig. 1.3 Evolution of the energy of KS-frontier orbitals with increasing electron number from N (left) to $N + 1$ (right). The KS-LUMO N jumps upon occupation by an amount U . By contrast, in Hartree–Fock (HF) energy the HF-LUMO N is already calculated anticipating an interaction with one more particle (as compared to HF-HOMO N). Therefore, such a jump does not occur in HF theory.

So far, no peculiarities have appeared. To see that there is indeed something looming on the horizon, now add a fractional excess charge, say an infinitesimally small one, $\delta N \lll 1$, rather than an integer charge. Then the original KS orbitals should remain invariant by definition, since the perturbation is infinitesimally small so that the charge density is not disturbed. But, what are the energy and shape of the newly occupied orbital? The salient point is that a real particle does not interact with itself. Therefore, the energy of a physical orbital should not be sensitive to its occupation. Hence, the workfunction of an atom with a fractionally occupied HOMO is the same as that of one with an integer occupation.

We conclude that the fractionally occupied orbital must have the energy HOMO $^{N+1}$, which exceeds the energy of the empty orbital LUMO N by the amount U . So evolution of the energy of HOMO $^{N+\delta N}$ with δN is *not* smooth; an arbitrarily small change in the density, δN , must result in a finite reaction of $v_{\text{XC}}[n]$ if the particle number, N , is near integer values:

$$\Delta_{\text{XC}}(\mathbf{r}) = \left. \frac{\delta E_{\text{XC}}[n]}{\delta n(\mathbf{r})} \right|_{N+\delta N} - \left. \frac{\delta E_{\text{XC}}[n]}{\delta n(\mathbf{r})} \right|_{N-\delta N} \quad (1.27)$$

This is the (in)famous *derivative discontinuity* (DD).

1.5.3.2 Coupled Subsystems (Partial Charge Transfer) To illustrate the importance of the DD, we now give a typical example where fractional charge occurs.

Consider two subsystems, which are partially decoupled in the sense that electronic wavefunctions interact only weakly. Such could be, for example, two functional groups in the same molecule or two neighboring molecules in a biological environment. To be specific, we imagine here the atom from Section 1.5.3.1 and a second many-body system, a metal surface.

Each system has its own workfunction: for example, $W_A^{N+1} > W_S$. Let us bring the atom into the vicinity of the surface, but keeping their distance d extremely large. Since only the total particle number $N = N_A + N_S$ is conserved, there will be a net exchange of charge, δN , between $S \rightarrow A$. This implies that the atomic orbitals acquire a finite broadening, Γ , which however is small, $\Gamma \ll |W_A^{N+1} - W_S|$, since d is large. In this situation and in the absence of ionization, the net particle flow from $S \rightarrow A$ is exponentially small. As a consequence, the HOMO_A^{N+1} fills up, but only with a very small fraction of an electron.

To describe correctly how the HOMO_A^{N+1} fills upon approach of the two subsystems, it is crucial that the piece of the XC functional describing A indeed reacts to the flow, so that the LUMO_A^N of the coupled atom is shifted upward against the uncoupled atom by U . If U is on the order of the mean level spacing or even bigger—as it tends to be for nanoscopic systems such as atoms and small molecules—this shift is important for understanding charge transfer in DFT. On a qualitative level, the DD suppresses charge fluctuations between weakly coupled subsystems.

Remarks

- The spatial modulation of v_{XC} induced by the DD reflects the differences in the workfunction seen in different charge states of the isolated subsystems before they have been coupled. Therefore, quantitative estimates about the size of the DD-induced modulations can be obtained by calculating workfunctions of the constituting subsystems and their anions/cations.
- The DD enters in a crucial way the DFT-based description of the gate dependence of the charge inside a quantum dot. Without DD, the width of the Coulomb oscillations is U rather than $\max(\Gamma, T)$ and therefore qualitatively wrong.¹⁹
- In LDA-type approximations the DD is missing, since by construction the potentials evolve smoothly when an infinitesimal probing charge is added. Currently, attempts are under way to design orbital-dependent functionals which can take the DD into account (in a spirit similar to HF theory). Kümmel and Kronik¹ have compiled a review about the most recent developments in this direction.

1.6 TIME-DEPENDENT DFT

Since the 1980s, attempts have been made to generalize equilibrium theory into time dependent phenomena. A detailed account of its foundations may be found in recent monographs.^{20,21} We discuss only those most basic aspects which are

important to shed some light on the connection between TDDFT and transport calculations.

Consider the time-dependent Schrödinger equation

$$i\hbar\partial_t\Psi(t) = \left[\hat{T} + \hat{U} + \hat{V}_{\text{ex}} + \int d\mathbf{r} \phi_{\text{ex}}(\mathbf{r}t)\hat{n}(\mathbf{r}) \right] \Psi(t) \quad (1.28)$$

where $\hat{T}$ and $\hat{U}$ abbreviate the kinetic and interaction energies given explicitly in Eq. (1.2) and, again,

$$\hat{V}_{\text{ex}} = \int d\mathbf{r} v_{\text{ex}}(\mathbf{r})\hat{n}(\mathbf{r})$$

describes the electrostatic environment. The time evolution of all observables is fixed by (1) the time-dependent external potential $\phi_{\text{ex}}(\mathbf{r}t)$ and (2) the initial conditions (i.e., the wavefunction Ψ_i at the initial time $t = 0$). This suggests that the response of all those systems, which have been prepared in an identical way and therefore share the same initial state, is dictated by a single scalar field $v_{\text{ex}}(t)$. In this respect, the situation is very reminiscent of the equilibrium case.

To prove also that for time-dependent phenomena the density may serve as the fundamental variable, one should demonstrate that an invertible relation analog to Eq. (1.6) exists, at least in principle, which allows reconstruction of the probing potential $\phi_{\text{ex}}(t)$ from knowledge of $n(t)$ (and Ψ_i) at all times $t \geq 0$. A proof that this indeed is the case for a wide class of potentials $\phi_{\text{ex}}(t)$ was constructed first by Runge and Gross²² and corroborated by many later authors, in particular by van Leeuwen.²³

1.6.1 Runge–Gross Theorem

The Runge–Gross theorem emphasizes that the time evolution of the density $n(t)$ is a *unique* characteristic of the probing potential $\phi_{\text{ex}}(t)$: Two probing fields, which differ by more than a homogeneous shift in space, invoke two different density evolutions. This insight is then later used to argue that a density profile, $n(\mathbf{r}t)$, that is driven in one system with interaction $\hat{U}$ by $\phi_{\text{ex}}(t)$ can also be seen in another system with a different interaction $\hat{U}'$ after $\phi_{\text{ex}}(t)$ has been replaced by the appropriate modulation $\phi'_{\text{ex}}(t)$. In particular, $\hat{U}'$ can also be zero, which is the foundation of the time-dependent DFT.

We offer a proof of these statements which relies on the familiar fact that a solution of a partial differential equation (here in time) is unique once the initial situation and the evolution law have been specified.

Proof The strategy is to relate the probing field ϕ_{ex} to the second time derivatives $\ddot{n}$. For the first time derivative, Heisenberg's equation of motion tells us that

$$\dot{n}(\mathbf{r}t) = \frac{1}{i\hbar} \langle \Psi(t) | [\hat{n}(\mathbf{r}), \hat{T}] | \Psi(t) \rangle \quad (1.29)$$

because all other terms in $\hat{U}$, $\hat{V}_{\text{ex}}$, and ϕ_{ex} commute with the density operator $\hat{n}(\mathbf{r})$. By comparing with the continuity equation,

$$\dot{n}(\mathbf{r}t) + \partial_{\mathbf{r}} \langle \Psi(t) | \hat{\mathbf{j}}(\mathbf{r}) | \Psi(t) \rangle = 0 \quad (1.30)$$

one may identify the proper definition of a current density operator, $\hat{\mathbf{j}}(\mathbf{r})$. The procedure is familiar from elementary textbooks on quantum mechanics. The second derivative reads

$$\ddot{n}(\mathbf{r}t) = \left(\frac{1}{i\hbar} \right)^2 \langle \Psi(t) | \left[[\hat{n}(\mathbf{r}), \hat{T}], \hat{H}(t) \right] | \Psi(t) \rangle \quad (1.31)$$

where $\hat{H}(t)$ is the Hamiltonian driving the time evolution in Eq. (1.28). This equation is readily recast into the shape

$$\delta \ddot{n}(\mathbf{r}t) = - \int d\mathbf{r}' \Xi(\mathbf{r}t, \mathbf{r}'t) \partial_{\mathbf{r}'} \phi_{\text{ex}}(\mathbf{r}'t) \quad (1.32)$$

where we have introduced a correlator,

$$\Xi(\mathbf{r}t, \mathbf{r}'t) = \frac{i}{\hbar} \langle \Psi(t) | \left[\hat{\mathbf{j}}(\mathbf{r}'), \hat{n}(\mathbf{r}) \right] | \Psi(t) \rangle \quad (1.33)$$

and the abbreviation

$$\delta \ddot{n}(\mathbf{r}t) = \ddot{n}(\mathbf{r}t) + \frac{1}{i\hbar} \partial_{\mathbf{r}} \langle \Psi(t) | \left[\hat{\mathbf{j}}(\mathbf{r}), \hat{T} + \hat{U} + \hat{V}_{\text{ex}} \right] | \Psi(t) \rangle \quad (1.34)$$

The second term appearing in this expression describes the internal relaxation of the electron system (“gas” or “liquid”; e.g., due to viscoelastic forces).

The equal-time commutator in Eq. (1.33) is closely related to the density matrix; in terms of fermionic field operators, one has

$$n(\mathbf{r}t, \mathbf{r}'t) = \frac{1}{2} \langle \Psi(t) | \hat{\psi}^\dagger(\mathbf{r}) \hat{\psi}(\mathbf{r}') + \hat{\psi}^\dagger(\mathbf{r}') \hat{\psi}(\mathbf{r}) | \Psi(t) \rangle$$

so that

$$\Xi(\mathbf{r}t, \mathbf{r}'t) = \frac{1}{m} [n(\mathbf{r}t, \mathbf{r}'t) \partial_{\mathbf{r}'} \delta(\mathbf{r} - \mathbf{r}') - \delta(\mathbf{r} - \mathbf{r}') \partial_{\mathbf{r}} n(\mathbf{r}t, \mathbf{r}'t)] \quad (1.35)$$

Feeding this expression back into Eq. (1.32) and recalling that $n(\mathbf{r}t, \mathbf{r}t) \equiv n(\mathbf{r}t)$, we recover Newton’s third law,

$$\delta \ddot{n}(\mathbf{r}t) = \frac{1}{m} \partial_{\mathbf{r}} n(\mathbf{r}t) \partial_{\mathbf{r}} \phi_{\text{ex}}(\mathbf{r}t) \quad (1.36)$$

as we should. Clearly, a spatially homogeneous part of the probing potentials can never be recovered from the density evolution, since such potentials do not

exert a force. By contrast, the inhomogeneous piece can be reconstructed from its accelerating effect on the density.[†] Technically speaking, Eq. (1.36) represents a linear, first-order (in space) differential equation for the probing field $\phi_{\text{ex}}(t)$. Combining with the Schrödinger equation (1.28),

$$i\hbar\partial_t\Psi(t) = \hat{H}(t)\Psi(t)$$

one obtains a system of two linear equations, which are local in time and readily integrated starting from the initial time $t = 0$.

This is how, in principle, the probing field may be reconstructed (up to a homogeneous constant), if only $n(\mathbf{r}t)$ is known: $n(\mathbf{r}t) \rightarrow \phi_{\text{ex}}(\mathbf{r}t)$. Since the other direction, $\phi_{\text{ex}}(\mathbf{r}t) \rightarrow n(\mathbf{r}t)$, is provided trivially by the Schrödinger equation, we readily conclude that

$$\phi_{\text{ex}}(\mathbf{r}t) \leftrightarrow n(\mathbf{r}t)$$

Extension So far we have shown how the probing potential $\phi_{\text{ex}}(\mathbf{r}t)$ can be calculated if the density evolution and the initial state are given. It is also tacitly understood here that the Hamiltonian (i.e., the dispersion, $\hat{T}$, the electrostatic environment, $\hat{V}_{\text{ex}}$, and the interaction, $\hat{U}$) are known. Their structure cannot be reconstructed with $n(\mathbf{r}t)$.

In conjunction with Eq. (1.36), this last observation has an important implication. Consider, for example, two systems with two different interactions, $\hat{U}$ and $\hat{U}'$, and two different initial states, Ψ_i and Ψ'_i , that both satisfy the condition that their initial density $n(\mathbf{r}t_i)$, together with the time derivative $\dot{n}(\mathbf{r}t_i)$, coincide. Under this condition, for both systems an equation of the type (1.36) holds true, since the derivation made no special assumption about the structure of $\hat{U}$. Therefore, for any (reasonable) interaction $\hat{U}$ we can find a time-dependent single-particle potential such that the density of the many-body system follows a predefined time evolution $n(\mathbf{r}t)$.

We can even go a step further. In fact, we have shown how to calculate $\hat{U}$ -depending single-particle potentials, v_s , such that systems with different interactions can exhibit the same time-dependent density. This means, in particular, that we can model the time evolution $n(\mathbf{r}t)$ of interacting systems driven by $\phi_{\text{ex}}(\mathbf{r}t)$ by studying a reference system of noninteracting particles that experience a particular driving field $v_s(\mathbf{r}t)$. This field can be constructed from the (invertible) mapping

$$\phi_{\text{ex}}(\mathbf{r}t) \underset{\text{Eq. (1.28)}}{\overset{\hat{U}}{\longleftrightarrow}} n(\mathbf{r}t) \underset{\text{Eq. (1.36)}}{\overset{\hat{U}'=0}{\longleftrightarrow}} v_s(\mathbf{r}t) \quad (1.37)$$

at least in principle. Some of the conclusions, which we have arrived at here, were presented earlier by van Leeuwen²⁴ based on the same equations but with somewhat different arguments.[‡]

[†]This statement is true in those spatial regions where the particle density is nonvanishing $n(\mathbf{r}) \geq 0$.

[‡]We thank G. Stefanucci for bringing Ref. 24 to our attention and for a related discussion.

Remarks

- By including in addition to the scalar probing potential $\phi_{\text{ex}}(t)$ a vector probing potential, $A_{\text{ex}}(t)$, and keeping the current density explicit as a second collective field, one can generalize the argument presented above to derive a time-dependent current DFT. A proof in the spirit of van Leeuwen²⁴ has been given by Vignale.²⁵
- Exactly the same arguments that have been presented for the case of a single wavefunction $\Psi(t)$ also apply to an ensemble of wavefunctions characterized by a statistical operator $\hat{\rho}$ with only minor modifications: (1) quantum mechanical expectation values turn into ensemble averages, and (2) the Schrödinger equation is replaced by the von Neumann equation

$$\dot{\hat{\rho}} = \frac{i}{\hbar} [\hat{\rho}, \hat{H}(t)] \quad (1.38)$$

This prompts a generalization of TDDFT to finite temperatures.

- In principle, one can in this way also consider systems with a coupling to a heat bath (e.g., bosons). The only essential modification occurs in Newton's law, which now needs to account, for example, for a change in the effective dispersion $1/m$ due to the electron–boson coupling. First attempts to develop a TDDFT for a system coupled to reservoirs have been reported.^{26–28}
- Notice that the appearance of the gradients in Eq. (1.36) is due to particle number conservation. The reason is that symmetric correlators of the type

$$\langle \Psi(t) | [[\hat{n}(\mathbf{r}), \hat{O}], \hat{n}(\mathbf{r}')] | \Psi(t) \rangle$$

vanish after integration over one of the spatial coordinates if $\hat{O}$ commutes with the total particle number operator, $[\hat{O}, \hat{N}] = 0$. Indeed, in Eq. (1.31) this is the case, because any term in the Hamiltonian commutes with the total particle number operator $\hat{N}$. Hence, such correlators have vanishing (real space) Fourier components at zero wavenumbers, $\mathbf{q} = 0$. Assuming analyticity, we can say that the correlator is proportionate to the product of two wavenumbers, $\mathbf{q}$ and $\mathbf{q}'$, and for this reason two gradients appear in Eq. (1.36).

- The validity of time-dependent DFT is based on three elementary observations all of which relate to the fact that (quantum) mechanics is governed by linear differential equations in time:
 1. The total force can be deduced from its action on the particle density.
 2. This force can be split into an external and internal component; the internal component acting at time t can be calculated knowing just $\Psi(t)$.
 3. To calculate $\Psi(t)$, only forces acting prior to t and the initial conditions have to be known.

1.6.2 Dynamical Kohn–Sham Theory

The Runge–Gross theorem and its extensions teach us that there is a reference system of noninteracting particles living in a potential $v_s(\mathbf{r}t)$ [Eq. (1.37)], so that at $t > 0$ its density evolves in time in exactly the same way that it does for many-body system. The dynamics of this reference system are governed by an effective Schrödinger-type equation, the dynamic Kohn–Sham equations. With the decomposition $v_s = v_{\text{ex}} + v_{\text{H}} + v_{\text{XC}} + \phi_{\text{ex}}$, they read

$$\frac{i}{\hbar} \partial_t \phi_\ell(\mathbf{r}) = [\varepsilon(\hat{\mathbf{p}}) + v_{\text{ex}}(\mathbf{r}) + \phi_{\text{ex}}(\mathbf{r}t) + v_{\text{H}}(\mathbf{r}t) + v_{\text{XC}}(\mathbf{r}t)] \phi_\ell(\mathbf{r}) \quad (1.39)$$

where $\phi_{\text{ex}}(\mathbf{r}t)$ is the time-dependent probing field and

$$n(\mathbf{r}, t) = \sum_{\ell=1}^N |\phi_\ell(\mathbf{r}t)|^2$$

$$v_{\text{H}}[n](\mathbf{r}t) = \int d\mathbf{r}' u(\mathbf{r} - \mathbf{r}') n(\mathbf{r}'t) \quad (1.40)$$

The functional $v_{\text{XC}}[n](\mathbf{r}t)$ is the piece of $v_s[n](\mathbf{r}t)$ that accommodates the interactions beyond the mean field (Hartree) type. It depends on the time-dependent particle density, including its history. Moreover, as a first-order differential equation, Eq. (1.39) needs to be complemented with an initial condition. Part of this is, of course, that $n(\mathbf{r}, t = 0)$ coincides with the density of the many-body system at $t = 0$. However, in addition, the functional v_{XC} will in general also depend on the many-body wavefunction of the initial state, $\Psi_1 \equiv \Psi(t = 0)$, which may—but does not have to be—an equilibrium state.

1.6.3 Linear Density Response

Consider a situation where the many-body system is in thermal equilibrium at times $t < 0$ before the probing field $\phi_{\text{ex}}(\mathbf{r}t)$ is switched on. Moreover, assume that the perturbation is going to be very weak, so that the requirements for the application of the linear response theory are met. Under this condition, an explicit expression for the XC-functional v_{XC} is readily written down.

Indeed, there is a matrix $\chi(\mathbf{r}t, \mathbf{r}'t')$, the density susceptibility, which relates the probing field to the (linear) system response, $\Delta n = n - n^{\text{eq}}$:

$$\Delta n(\mathbf{r}t) = \int dt' \int d\mathbf{r}' \chi(\mathbf{r}t, \mathbf{r}'t') \phi_{\text{ex}}(\mathbf{r}'t') \quad (1.41)$$

The matrix $\chi(t, t')$ is an equilibrium correlation function of the system, and it therefore depends only on the time differences $t - t'$. We can use its inverse, χ^{-1} , to define an operator kernel f_{XC} via the decomposition

$$\chi^{-1} = \chi_{\text{KS}}^{-1} - f_{\text{H}} - f_{\text{XC}} \quad (1.42)$$

The operator χ_{KS} describes the density response of the equilibrium KS system, ignoring the feedback of $\phi_{\text{ex}}(t)$ into v_{H} and v_{XC} [Eq. (1.39)]; explicitly,

$$\chi_{\text{KS}}(\mathbf{r}\mathbf{r}'z) = \frac{1}{\hbar} \sum_{\ell, \ell'} \langle \ell | \hat{n}(\mathbf{r}) | \ell' \rangle \langle \ell' | \hat{n}(\mathbf{r}') | \ell \rangle \frac{f(\varepsilon_{\ell}) - f(\varepsilon_{\ell'})}{\varepsilon_{\ell'} - \varepsilon_{\ell} - z}$$

where $|\ell\rangle, |\ell'\rangle$ and $\varepsilon_{\ell, \ell'}$ denote the unperturbed ($\phi_{\text{ex}} \equiv 0$) KS orbitals and KS energies and $z = \omega + i\eta$ lies in the complex plane. The feedback is then taken into account by $f_{\text{H}} = u(\mathbf{r} - \mathbf{r}')$ for the Hartree term v_{H} and by f_{XC} for the exchange correlation potential, v_{XC} , in Eq. (1.39).

From this point of view it is obvious how to construct the dynamic correction of the XC functional to the equilibrium functional:

$$v_{\text{XC}}[n](\mathbf{r}t) = v_{\text{XC}}^{\text{eq}}[n_{\text{eq}}](\mathbf{r}) + \int dt' \int d\mathbf{r}' f_{\text{XC}}(\mathbf{r}, \mathbf{r}'; t - t') \Delta n(\mathbf{r}'t') \quad (1.43)$$

Remarks

- We have just constructed a single-particle theory, which has the property that it gives the correct linear dynamical response of the many-body system. The procedure relies on the familiar notions of linear response theory only and does not make reference to the underpinnings of the time-dependent DFT. It is emphasized here that the genuine statements of time-dependent DFT, when applied to systems that are in equilibrium at $t < 0$, reside in the claim that an effective single-particle description exists even outside the linear regime.
- Much of the recent improvement²⁹ in quantitative calculations of optical spectra of single molecules is due to including the terms f_{H} and in particular f_{XC} into the analysis (in addition to χ_{KS}), which have often been ignored before. In this way the single-particle spectrum of the bare Kohn–Sham system is dressed so as to produce the correct many-body excitations. Often, the success of this procedure is attributed to the time-dependent DFT. This is misleading, however, since it is merely the consequence of a proper application of the standard theory of linear responses.
- The best used approximation on f_{XC} is the adiabatic LDA (ALDA). It comprises two steps. First is the adiabatic approximation,

$$f_{\text{XC}}^{\text{ad}}(\mathbf{r}t, \mathbf{r}'t') = \left. \frac{\partial v_{\text{XC}}^{\text{eq}}[n](\mathbf{r}')}{\partial n(\mathbf{r})} \right|_{n(\mathbf{r}t)} \delta(t - t') \quad (1.44)$$

This step, by definition, erases all memory effects, so a δ -function in time appears. The complete absence of memory suggests one more approximation, which also eliminates nonlocal correlations in space. This is necessary, because signal propagation occurs with a finite velocity and therefore always

has a retardation time. Therefore, density fluctuations in different spatial regions cannot be correlated instantaneously. This aspect is built into

$$f_{\text{XC}}^{\text{ALDA}}(\mathbf{r}t, \mathbf{r}'t') = \left. \frac{dv_{\text{XC}}^{\text{eq}}(n)}{dn} \right|_{n(\mathbf{r}t)} \delta(\mathbf{r} - \mathbf{r}') \delta(t - t') \quad (1.45)$$

automatically, where in Eq. (1.44) $v_{\text{XC}}^{\text{eq}}$ has been replaced by its LDA approximant.

1.6.4 Time-Dependent Current DFT

The frequency structure of f_{XC} has been worked out in the hydrodynamic regime of small wavenumbers and frequencies by Kohn, Vignale, and co-workers.^{30,31} It is seen explicitly there that severe memory effects indeed exist due to general conservation laws, which express themselves as singular behavior in correlation functions with respect to wavenumber and frequency.

As usual, singularities may be partly eliminated by reformulating in terms of correlation functions of the (generalized) velocities. In the case of the particle density, one introduces the longitudinal current density,

$$j^\ell(\mathbf{q}\omega) = \frac{-i\omega}{qn(\mathbf{q}\omega)} \quad (1.46)$$

In this way one absorbs factors q^{-1} , thus removing nonlocal behavior in the density kernels, which indicates, for example, the slow density relaxation due to particle number conservation. In this spirit the time-dependent current DFT (TDCDFT) was developed.^{30,31}

Apart from the fact that it works with current-density kernels, which are more local than those in TDDFT, TDCDFT offers yet another attraction. In addition to the density [or j^ℓ , Eq. (1.48)] it also features a second independent collective field, the transverse currents $j^\perp$. Therefore, TDCDFT can in principle also describe the orbital response to probing vector potentials (i.e., magnetic fields).

1.6.5 Appendix: Variational Principle

Unlike the case with equilibrium theory, a variational principle is not required in order to derive the dynamical Kohn–Sham equations. Still, it is desirable to have a formulation of TDDFT available in terms of an action, for example, because one may hope to be able to calculate v_s by performing a functional derivative. In this section we investigate the “naive” trial action

$$\begin{aligned} A[\tilde{\Psi}] &= \int_0^\infty dt \langle \tilde{\Psi}(t) | i\hbar \partial_t - \hat{H}(t) | \tilde{\Psi}(t) \rangle \\ &= \int_0^\infty dt \langle \tilde{\Psi}(t) | i\hbar \partial_t - \hat{T} - \hat{U} - \hat{V}_{\text{ex}} | \tilde{\Psi}(t) \rangle - \int_0^\infty dt \int d\mathbf{r} \phi_{\text{ex}}(\mathbf{r}t) \tilde{n}(\mathbf{r}t) \end{aligned} \quad (1.47)$$

which is defined over the space C_{Ψ_I} of complex fields $\tilde{\Psi}(t)$ with constraints given by (1) the antisymmetry requirement in all N coordinates $\mathbf{r}_1 \cdots \mathbf{r}_N$, and (2) the initial condition $\tilde{\Psi}(0) = \Psi_I$. The solution of the Schrödinger equation for a given external field $\phi_{\text{ex}}(\mathbf{r}t)$ is the one element $\Psi(t)$ of C_{Ψ_I} that optimizes $A[\tilde{\Psi}]$.

In full analogy to the equilibrium case, the functional equation (1.47) can be used as a basis to find an action functional of the density alone by preoptimizing. We first perform a decomposition of C_{Ψ_I} into subsets; the elements of each subset have the same evolution $\tilde{n}(\mathbf{r}t)$. Second, we find within each one of these subsets those states $\Psi_{\tilde{n}(\mathbf{r}t)}$ that are optimal with respect to $A[\tilde{\Psi}]$. These states form the ensemble M_{preopt} of preoptimized fields.[†]

In this way we arrive at an action functional, which is defined on M_{preopt} :

$$S_{\Psi_I}[\tilde{n}] = \int_0^\infty dt \langle \Psi_{\tilde{n}}(t) | i\hbar \partial_t - \hat{T} - \hat{U} | \Psi_{\tilde{n}}(t) \rangle \quad (1.48)$$

$S_{\Psi_{\tilde{n}}}$ is the dynamical analog of F [Eq. (1.37)]. The Schrödinger time evolution of the density, $n(\mathbf{r}t)$, is the single one that optimizes the full action,

$$A_{\Psi_I}[v_{\text{ex}}, \tilde{n}] = S_{\Psi_I}[\tilde{n}] - \int_0^\infty dt [v_{\text{ex}}(\mathbf{r}) + \phi_{\text{ex}}(\mathbf{r}t)] \tilde{n}(\mathbf{r}t) \quad (1.49)$$

The variational space associated with this action is spanned by all those $\tilde{n}(\mathbf{r}t)$ which are $\tilde{\Psi}$ -representable: There is at least one element $\tilde{\Psi}(t)$ of C_{Ψ_I} such that $\tilde{n}(\mathbf{r}t) = \langle \tilde{\Psi}(t) | \hat{n}(\mathbf{r}) | \tilde{\Psi}(t) \rangle$.

Remarks

- Preoptimizing is a constrained minimum search in the subspace of possible wavefunctions that satisfy the initial condition (2). Therefore, each initial condition carries its own functional: $S_{\Psi_I}[n]$.
- By construction, the search over Ψ -representable densities leads to a variational equation,

$$\left. \frac{\delta S_{\Psi_I}[\tilde{n}]}{\delta \tilde{n}(\mathbf{r}t)} \right|_{\tilde{n}(\mathbf{r}t)=n(\mathbf{r}t)} = \phi_{\text{ex}}(\mathbf{r}t) + v_{\text{ex}}(\mathbf{r}) \quad (1.50)$$

Its solution, $n(\mathbf{r}t)$, defines the Schrödinger dynamics for the density corresponding to a given probing field $\phi_{\text{ex}}(\mathbf{r}t)$. A more explicit expression for the left-hand side may be obtained by taking the time derivative and comparing with Eq. (1.36).

[†]With every optimum $\Psi_{\tilde{n}}(t)$, the related function $e^{i\varphi(t)}\Psi_{\tilde{n}}(t)$ with $\varphi(0) = 0$ is an optimum, which differs by a time-dependent, spatially homogeneous phase shift. The shift merely reflects the necessity to fix the zero of energy. We identify all those states with one another that differ only by a spatially homogeneous phase $\varphi(t)$.

- Consider to generate all possible solutions of Eq. (1.50) by scanning through the space of all allowed (i.e., sufficiently smooth) probing fields $\phi_{\text{ex}}(\mathbf{r}t)$. This subset of the Ψ -representable variational space is called *v-representable*. An arbitrary element of the variational space $\tilde{n}(\mathbf{r}t)$ is certainly Ψ -representable but may not be *v-representable*.
- The Schrödinger dynamics is unitary: $N = \int d\mathbf{r} n(\mathbf{r}t)$ is an invariant of motion. *v-representable* states obey unitarity, but Ψ -representable states may not.
- By taking a functional derivative,

$$\left. \frac{\partial}{\partial n(\mathbf{r}t)} \frac{\delta S_{\Psi_I}[\tilde{n}]}{\delta \tilde{n}(\mathbf{r}t')} \right|_{\tilde{n}=n} = \frac{\partial \phi_{\text{ex}}(\mathbf{r}'t')}{\partial n(\mathbf{r}t)} = \chi^{-1}(\mathbf{r}'\mathbf{r}, (t' - t)) \quad (1.51)$$

a relation to the reciprocal of the density correlation function is derived. Note that the ∂ derivative relates to density differences within the set of all $n(\mathbf{r}t)$ that are *v-representable*. Our notation emphasizes this difference with the earlier δ derivative [Eq. (1.50)].

- The right-hand side of Eq. (1.51) is subject to causality; the density $n(\mathbf{r}t)$ indicates changes in the probing potential $\phi_{\text{ex}}(\mathbf{r}'t')$ only at later times, $t > t'$. Equation (1.51) pays respect to this asymmetry, since the ∂ and δ derivatives must not be interchanged.
- The causality issue noted above makes it very obvious that an action principle should not be based solely on the variational space of *v-representable* histories $n(\mathbf{r}t)$. This issue has been discussed in detail by van Leeuwen.^{23,32} In response, this author derives an action S employing the Keldysh formalism. The procedure by itself does appear to lead to fundamentally new insights. However, it has the charming feature against the naive starting point [Eq. (1.47)] that only one (enlarged) variational space for $n(\mathbf{r}t)$ appears. In addition, there is an important conceptual advantage, since—in principle—within this approach it is clear how one can calculate v_{XC} in a systematical perturbation theory.

1.7 TDDFT AND TRANSPORT CALCULATIONS

In this section we discuss the application of TDDFT in the context of charge transport. The focus will be on the dc limit. There are various ways how to formulate the transport problem; we shall elaborate on the consequences from linear response and scattering approaches. We concentrate on the presentation of those elementary facts that are specific of a treatment of transport within the framework of TDDFT. An attempt is being made to be as self-contained as possible.

1.7.1 Linear Current Response

One way to establish a current flow in a system, which initially is in a thermodynamic equilibrium, is to switch on an electric field $\mathbf{E}_{\text{ex}}(\mathbf{r}t)$. This field is not

the one that an electron feels when it accelerates. The accelerating (local) field, $\mathbf{E}$, also contains an induced component,

$$\mathbf{E} = \mathbf{E}_{\text{ex}} + \mathbf{E}_{\text{ind}} \quad (1.52)$$

We restrict ourselves to initial situations that respect time-reversal invariance. Then the induced field is generated by a shift of charges, $e \Delta n$, under the influence of $\mathbf{E}_{\text{ex}}$; we have

$$\mathbf{E}_{\text{ind}}(\mathbf{r}t) = -\partial_{\mathbf{r}} \int d\mathbf{r}' u(\mathbf{r} - \mathbf{r}') \Delta n(\mathbf{r}'t) \quad (1.53)$$

By definition, the conductivity matrix, σ_{ij} , relates only the total field, $\mathbf{E}$, to the linear response of the current density by

$$j_i(\mathbf{r}\omega) = \int d\mathbf{r}' \sigma_{ij}(\mathbf{r}, \mathbf{r}', \omega) E_j(\mathbf{r}'\omega) \quad (1.54)$$

To make contact to TDDFT, we decompose $\mathbf{j}$ into a longitudinal (curl free) piece, $\mathbf{j}^\ell$, and a transverse (source free) field, $\mathbf{j}^t$.

1.7.1.1 Magnetization (Transverse) Currents By construction, $\mathbf{j}^t$ incorporates the orbital ring currents that may be understood as a local magnetization density defined via $\mathbf{j}^t(\mathbf{r}t) = c\partial_{\mathbf{r}} \times \mathbf{m}(\mathbf{r}t)$, where c denotes the velocity of light. Non-vanishing magnetizations occur in equilibrium systems only in the presence of (spontaneously) broken time-reversal invariance. In these cases, the current DFT (CDFT) has to be employed, where the magnetization is explicitly kept as a second collective field in addition to the particle density. We consider here only systems that are invariant under time reversal. Then, ring currents vanish in the initial state, $\mathbf{j}^t = 0$. In such systems transverse currents can emerge in the presence of external driving fields.[†] Since they are not accompanied by density fluctuations, TDDFT does not monitor them. This implies, in particular, that the transverse currents of the time-dependent KS system do not, in general, coincide with the physical magnetization currents.

1.7.1.2 Longitudinal Currents The continuity equation connects $\mathbf{j}^\ell$ with the time dependency of the particle density. Therefore, the physical longitudinal current density and the longitudinal KS currents coincide. Hence, it makes sense to introduce a conductivity of the KS particles via

$$j_i(\mathbf{r}, \omega) = \int d\mathbf{r}' \sigma_{\text{KS},ij}(\mathbf{r}, \mathbf{r}', \omega) [E_{\text{ex}} + E_{\text{ind}} + E_{\text{XC}}]_j(\mathbf{r}', \omega) \quad (1.55)$$

[†]As an example we mention a ring current flowing in a perfectly conducting cylinder that closes around a time-dependent magnetic flux.

Just like physical particles, KS particles do not react to the external field but, rather, to the local field. This field contains the same Hartree-type term that originates from v_H in Eq. (1.39) and that was already present for the physical particles [Eq. (1.53)]. However, for KS particles not only v_H but also v_{XC} acquires a correction with a change in the density since

$$f_{XC}(\mathbf{r}, \mathbf{r}', t - t') = \frac{\partial v_{XC}[n](\mathbf{r}t)}{\partial n(\mathbf{r}'t')} \quad (1.56)$$

does not vanish [see Eq. (1.43)]. The resulting excess force $\mathbf{E}_{XC}$ from this contribution reads

$$\mathbf{E}_{XC}(r\omega) = -\partial_{\mathbf{r}} \int d\mathbf{r}' f_{XC}(\mathbf{r}, \mathbf{r}', \omega) \Delta n(\mathbf{r}', \omega) \quad (1.57)$$

in full analogy with Eq. (1.53).

Remark

- The exchange–correlation field $\mathbf{E}_{XC}$ comprises a piece that originates from the adiabatic term given in Eq. (1.44). On the level of the ALDA, we have

$$\mathbf{E}_{XC}^{\text{ALDA}}(\mathbf{r}\omega) = -\partial_{\mathbf{r}} \left. \frac{dv_{XC}^{\text{eq}}(n)}{dn} \right|_{n^{\text{eq}}(\mathbf{r})} \Delta n(\mathbf{r}, \omega) \quad (1.58)$$

In addition, $\mathbf{E}_{XC}$ also comprises a second piece, which brings in the viscoelastic properties of the correlated electron liquid. This piece is usually ignored in TDDFT, because it is very difficult to formulate in a purely density-based language. This is not surprising, because the viscosity is intimately related to shear forces within the liquid that derive from mixed terms $\partial j_x / \partial y$ typical of transverse current patterns. Such forces are more naturally described within time-dependent current DFT.^{30,31}

1.7.1.3 Quasi-One-Dimensional Wire We consider as an illustrative example the dc response of a quasi-one-dimensional wire of length L to an electric field in longitudinal direction, $\mathbf{E}(\mathbf{r}) = e_z E(z)$. The dc current, I , is given by

$$I = \int_0^L dz' g_{KS}(z, z') [E_{\text{ex}} + E_{\text{ind}} + E_{XC}](z') \quad (1.59)$$

$$g_{KS}(z, z') = \iint d\mathbf{r}'_{\perp} d\mathbf{r}_{\perp} \sigma_{KS}(\mathbf{r}, \mathbf{r}') \quad (1.60)$$

where it was assumed that the longitudinal field components have negligible variation in the perpendicular wire direction $\mathbf{r}_{\perp}$. Since any configuration of driving fields has as an associated dc current I that is the same for all observation points

z , we conclude that the kernel (1.60) is independent of its arguments and define a KS conductance: $G_{\text{KS}} = g_{\text{KS}}(z, z')$.

$$I = G_{\text{KS}} \int_0^L dz' [E_{\text{ex}} + E_{\text{ind}} + E_{\text{XC}}](z') \quad (1.61)$$

The first two terms in the integral add up to the physical voltage drop, V , along the wire. The appearance of the third term indicates that the KS particles experience another voltage, which differs by the amount

$$V_{\text{XC}} = \int_0^L dz' E_{\text{XC}}(z') \quad (1.62)$$

Remarks

- The ALDA contribution to the effective driving field is conservative, so it may be written as a gradient of a potential,

$$\int_0^L dz' E_{\text{XC}}^{\text{ALDA}}(z') = -v_{\text{XC}}^{\text{eq}}(n(z)) \Big|_{n(0)}^{n(L)}$$

As long as observation times are considered such that the effect of the charge transfer on the local charge density is still negligibly small (long wire limit), we can take $n(L) = n(0)$, so that the ALDA contribution vanishes (for macroscopically homogeneous wires). Nonzero contributions to V_{XC} come from the viscous term. The viscosity tends to reduce the response of the electron liquid to external forces. Density functional theories take this behavior into account by “renormalizing” the true forces with $\mathbf{E}_{\text{XC}}$.

- On a very qualitative level, the viscous forces tend to hinder the current flow through narrow constrictions with “sticky” walls. For this reason, their effect has been investigated in the context of current flows through single molecules.³³ However, as pointed out previously¹⁹ (and what underlies the debate^{34,35}), borrowing concepts from hydrodynamics to apply them on the molecular scale is not straightforward—for example, the viscosity: This describes how much momentum is transferred per time from a fast-moving stream to a neighboring one that flows into the same direction but with a lower speed. On a microscopic level, momentum exchange is mediated via collisions between the flowing particles. Therefore, it is clear that a description in terms of the macroscopic parameter “viscosity” can be valid only on length and time scales that substantially exceed the interparticle scattering length and time. Both scales become very large in fermion systems at low temperature, and in particular can easily exceed the dimensions of those atomistic or molecular systems that one would like to treat. Applications in mesoscopic semiconductors enjoy a much better justification.

1.7.2 Scattering Theory

The linear response theory is a framework for calculating the dynamical reaction to linear order in the probing field of any many-body system. Its advantage is that it is completely generally applicable. For the same reason, situations are easily identified, where alternative formalisms are better adapted and therefore allow a simpler and more transparent analysis.

In this section we consider an example thereof—the dc transport through a quantum dot (e.g., a molecule) which has been wired to a left and a right reservoir (see Fig. 1.4). We consider quasi-one-dimensional well-screened wires, so that particles inside the wire do not interact with each other. The traveling waves along the wire are categorized by *scattering states*. Each such state is equipped with a continuous longitudinal degree of freedom associated with a wavenumber, k , a discrete transverse degree, the *channel index* n [which should not be confused with the particle density $n(\mathbf{r})$], and a dispersion relation $E_n(k)$. In this language the current flowing through the wire is described by a superposition of scattering states.

How the particles that enter the wire from a reservoir distribute over the available scattering states is dictated by distribution functions, $f_{L,R}(E)$, which are properties solely of the left and right reservoirs. The specifics of the quantum dot enter the construction of the scattering states in terms of the reflection and transmission coefficients, $\tilde{r}_{nn'}(E, E')$ and $\tilde{t}_{nn'}(E, E')$. They describe the probability amplitude for a particle that approaches the quantum dot with energy E in channel n to be either reflected or transmitted into the channel n' with energy E' .

1.7.2.1 Landauer Theory The scattering description is particularly convenient if scattering is elastic, so in each single scattering process the state of the quantum dot is preserved; in particular, each scattering event conserves the energy of the incoming particle, $E = E'$. Under this specific condition, the current is simply given by the Landauer formula,

$$I = \int dE T(E) [f_L(E) - f_R(E)] \quad (1.63)$$

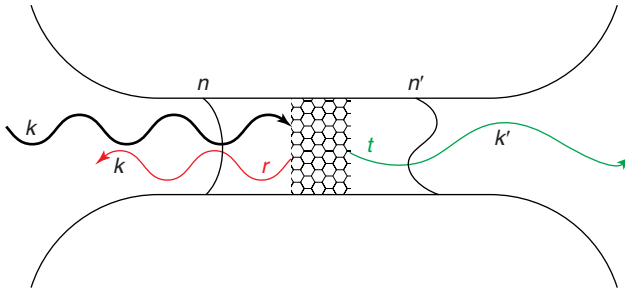


Fig. 1.4 (color online) Wiring a molecule to source and drain reservoirs: scattering states description with longitudinal (k) and transverse (n) quantum numbers.

with a transmission function

$$T(E) = \sum_{nn'} |t_{n'n}(E)|^2 \equiv \text{Tr } t t^\dagger \quad (1.64)$$

where $t_{n'n} = \tilde{t}_{v'v} (v_v/v_{v'})^{1/2}$, with $v_v = \partial \varepsilon_v(k)/\partial k$ being the group velocity of particles traveling in channel n with energy E . Here we follow the common convention that each reservoir acts as a thermal bath characterized by a temperature and an electrochemical potential, $\mu_{L,R}$. Then the distributions $f_{L,R}$ are simply Fermi functions with bath parameters.

1.7.2.2 Scattering Theory and TDDFT: Relaxation Problem Scattering theory describes a nonequilibrium situation that is (quasi-)stationary in time. Even though a current flows, expectation values of local (intensive) operators, in particular of $\hat{j}(\mathbf{r})$ and $\hat{n}(\mathbf{r})$, are time independent.[†] By contrast, TDDFT has been developed to describe the time evolution of the density, $n(\mathbf{r}t)$, under the action of a time-dependent potential, $\phi_{\text{ex}}(t)$, away from some initial condition. Both approaches may apply simultaneously if in the course of time evolution a quasistationary nonequilibrium situation develops.^{36–38}

This can happen if the superposition of $\phi_{\text{ex}}(t)$ and the induced field, $v_{\text{ind}}(t)$, shifts the electrochemical potentials of the two reservoirs against each other:

$$[v_{\text{ex}}(\mathbf{r}t) + v_{\text{ind}}(\mathbf{r}t)]_L^R \xrightarrow{t \gg \tau_{\text{trans}}} \mu_R - \mu_L \quad (1.65)$$

Then, after waiting a time τ_{trans} in which transient dynamic phenomena have died out due to internal relaxation processes, a flow may establish that indeed it is quasistationary. The current will be monitored properly by TDDFT, since it equals the flux of particles out of one of the reservoirs: $I = \dot{N}_L = -\dot{N}_R$.

In this quasistationary regime, by definition the particle and current densities are time independent. One might then suspect that the KS potentials should also have become stationary. This point is perhaps not quite as obvious as it might look. Namely, the fact that the density is time independent by itself does not always imply that the Hamiltonian is stationary. For example, homogeneous ring systems that close around time-dependent fluxes can exhibit time-dependent ring currents that leave the density completely invariant. To exclude such artifacts, one can operate with probing fields $\phi_{\text{ex}}(t)$ that couple to the density itself and that become time independent after switching on. Then, at least in the linear response regime, functionals are guaranteed to become time independent, since they derive from linear-response kernels [Eq. (1.43)] (see the remark below).

Once we accept that potentials become stationary, we may define scattering states. However, whether this concept is useful or not depends on whether one

[†]We are assuming here that the reservoirs are ideal. They remain in thermodynamic equilibrium with fixed temperature, chemical potential, and so on, even in the presence of a current flow. In reality, this condition requires a separation of scales: macroscopic reservoirs and microscopic currents.

can identify the rules pertaining to how the physical current should be constructed from them. Whether or not the same rules apply for the KS scattering states of TDDFT that work for the truly noninteracting case is not a priori clear, however.

Indeed, after switching on the bias voltage, V , the workfunction of each reservoir shifts against the vacuum level. Apart from this effect, each reservoir stays in complete thermal equilibrium due to their macroscopic size each all the time. According to the general principles of the DFT outlines in earlier sections, the distribution function of KS particles inside each reservoir should still be given by $f_{L,R}$ with the appropriate chemical potentials $\mu_{L,R}$ and $eV = \mu_L - \mu_R$, as usual. This was the point of view that has been adopted elsewhere.³⁶

However, this conclusion is not fully consistent with a result that we derived above. Namely, as we have seen in the linear response theory, the KS voltage does *not* in general coincide with the difference of the reservoir workfunctions. This effect has been incorporated^{37,38} using Fermi functions with chemical potentials that do not coincide with physical values. Here it remains an open question as to how this finding could be reconciled with the requirement that each reservoir must stay in its own equilibrium. This apparent inconsistency of DFT-based scattering theory at the moment is seemingly unresolved.

Remarks

- The precise conditions under which a nonequilibrium current flows in a quasistationary manner are very difficult to state. That flow at small enough currents is always quasistationary is supported by linear response analysis. It suggests (1) that linear responses to a sufficiently weak field never mix frequencies (i.e., they simply follow the external stimulus in time). Furthermore, (2) slow-enough driving fields, $\omega\tau_{\text{trans}} \ll 1$, signalize the dc behavior. So, combining (1) and (2), one concludes that the linear regime should always be quasistationary.
- A breakdown of the quasistationary regime at sufficiently large currents is suggested by analogy to hydrodynamics as described by the Navier–Stokes equations. Here it is known that a laminar (i.e., quasistationary) regime should be separated from turbulence that develops at larger currents. Since at least on a qualitative level, the micro- or nanoscopic flow of the electron liquid is also a hydrodynamic phenomenon, a “turbulent” regime could exist here as well. This is also supported from the observation that the TDDFT equations are nonlinear in the density and therefore should host chaotic regimes.

1.8 MODELING RESERVOIRS IN AND OUT OF EQUILIBRIUM

1.8.1 External and Internal Hilbert Spaces

Scattering theory operates in a basis of scattering states; that is, it uses those quantum numbers that reflect the behavior of wavefunctions in the asymptotic (i.e., free of scattering potential) region of space (the *external Hilbert space*).

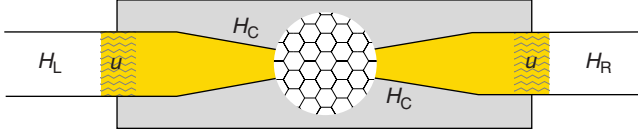


Fig. 1.5 (color online) Partitioning of the scattering zone near a molecule or quantum dot underlying the Hamiltonian equation (1.66).

For some applications, this representation is suboptimal. From a computational perspective, this can happen if the Hilbert space of states in the vicinity of the scatterer (the *internal* or *microscopic Hilbert space*) is very large or complicated, so that computations do not allow us to keep explicit track of additional degrees of freedom. For example, if one is to describe the current flow through a molecule (molecular electronics) or a quantum dot, one can keep molecular states that incorporate the molecule itself plus the states of a few lead atoms. The entire contact, which encompasses 10^{23} atoms, can certainly not be dealt with in a computer.

In more technical terms, we consider a partitioning of the system into left and right asymptotic regions, which are connected by a center region as given in Fig. 1.5 and detailed in the Hamiltonian

$$H = \begin{pmatrix} H_L & u^\dagger & 0 \\ u & H_C & v \\ 0 & v^\dagger & H_R \end{pmatrix} \quad (1.66)$$

The matrices $H_{L,R}$ comprise all the leads and are macroscopic, whereas H_C describes only the scattering region and therefore should have a microscopic size. If H_C is still very complicated, a formulation is desired that does not refer explicitly to the external, macroscopic Hilbert space (leads and reservoirs) but just focuses on the internal space. Roughly speaking, one would like to convert the trace over the external, channel degrees of freedom [Eq. 1.64] into another trace, which is only over the internal space of the molecule or quantum dot.

A formal way to derive such a representation employs the *Keldysh technique*, also referred to as the *nonequilibrium Green's function method*.³⁹ For noninteracting particles it yields predictions for physical observables which are identical to the scattering theory. Similar to earlier authors,⁴⁰ we employ the latter method here to derive the key formulas that underlie a great many applications of ab initio transport calculations for nanostructures.

1.8.2 Born Approximation, $\hat{T}$ -Matrix, and Transmission Function

Consider the situation where the left and the right leads are decoupled, $u = v = 0$ at $t = 0$. As before, we denote their eigenstates by a pair of indices $|nk\rangle$ (left) and $|n'k'\rangle$ (right). When contact is established at $t = 0$, an initial state $|nk\rangle$ becomes unstable. It can decay into the state $|n'k'\rangle$. The rate for this process is given

to lowest order by the Born approximation, which is equivalent to the familiar “golden rule” when applied to the scattering problem:

$$\hbar\tau_{n'n}^{-1}(E_n(k)) = 2\pi\delta(E_n(k) - E_{n'}(k'))|\langle n'k'|\hat{T}(E_n(k))|nk\rangle|^2 \quad (1.67)$$

Here, we have already refined the bare expression by introducing the $\hat{T}$ -matrix, which makes it formally exact. How to relate $\hat{T}$ to the original Hamiltonian, (1.66), will be shown in Section 1.8.3.

The right-going current injected in this way from a left-hand-side wire state $|nk\rangle$ into the right lead is just

$$\sum_{n'} \int dk' \tau_{n'n}^{-1}(E_n(k)) f_L(E_n(k)) (1 - f_R(E_{n'}(k')))$$

where $f_L(E_n(k))$ is the occupation of the initial state and $1 - f_R(E_{n'}(k'))$ is a measure of the available space in the final state. The total current is the difference between all right- and left-flowing components:

$$I = e \sum_{n'n} \int dk dk' \tau_{n'n}^{-1}(E_n(k)) [f_L(E_n(k)) - f_R(E_{n'}(k'))] \quad (1.68)$$

Comparing this expression with the Landauer formula, Eq. (1.63), we conclude that

$$T(E) = \sum_{n'n} \int dk dk' \delta(E - E_n(k)) \tau_{n'n}^{-1}(E) \quad (1.69)$$

$$= (2\pi)^2 \sum_{n'n} \int dk dk' \delta(E - E_n(k)) \delta(E - E_{n'}(k')) |\langle n'k'|\hat{T}(E)|nk\rangle|^2 \quad (1.70)$$

$$= \sum_{n'n} \frac{(2\pi)^2}{|v_n v_{n'}|} |\langle n'k'|\hat{T}(E)|nk\rangle|^2 \quad (1.71)$$

where the last line should be complemented with $E = E_n(k) = E_{n'}(k')$. Keeping Eq. (1.64) in mind, we have the identification (up to a phase factor)

$$t_{n'n} = \frac{2\pi}{\sqrt{|v_n v_{n'}|}} \langle n'k'|\hat{T}(E)|nk\rangle \quad (1.72)$$

Equation (1.70) has a compact notation if one introduces separate traces $\text{Tr}_{L,R,C}$ over the Hilbert spaces of $H_{L,R,C}$:

$$T(E) = (2\pi)^2 \text{Tr}_R [\delta(E - H_R) \hat{T}(E) \delta(E - H_L) \hat{T}^\dagger(E)] \quad (1.73)$$

1.8.3 $\hat{T}$ -Matrix and Resolvent Operator

We now specify how to relate $\hat{T}$ to the original Hamiltonian, H , detailed in Eq. (1.66). Our derivation starts with the observation that all information about transport across the center region is encoded in the *resolvent operator*,

$$G(z) = \frac{1}{z - H} \quad (1.74)$$

Retarded (advanced) operators are defined via $G^{\text{ret}}(E) = G(E + i\eta)[G^{\text{av}}(E) = G(E - i\eta)]$; the matrix elements $\langle \mathbf{x} | G^{\text{ret,av}}(E) | \mathbf{x}' \rangle$ define the Green's functions.[†]

Actually, we care only for transfer processes, so only those matrix elements $\langle n'k' | G(z) | nk \rangle$ are of interest that connect states in the left and right leads. The corresponding off-diagonal sector of the full resolvent matrix may be obtained from an elementary matrix inversion. Its matrix elements have the property

$$\langle n'k' | G(z) | nk \rangle = \langle n'k' | g_{\text{R}}(z) [v^\dagger G_{\text{C}}(z) u] g_{\text{L}}(z) | nk \rangle \quad (1.75)$$

The matrix product that appears here inside $\langle \cdot \cdot \cdot \rangle$ has the form familiar from the Dyson equation in T -matrix notation⁴¹:

$$G = G_0 + G_0 \hat{T} G_0 \quad (1.76)$$

where $G_0^{-1} = z - H_0$ is the bare Green's function in the absence of an interlead coupling, $u, v = 0$. In Eq. (1.75) the first term in the Dyson equation is missing, since the off-diagonal matrix elements that connect different leads vanish if there is no transmission.

Thus it is clear that the desired relation is just

$$\hat{T}(z) = v^\dagger G_{\text{C}}(z) u \quad (1.77)$$

with the resolvent operators of the central region and the leads

$$G_{\text{C}}(z) = \frac{1}{z - H_{\text{C}} - \Sigma_{\text{R}} - \Sigma_{\text{L}}} \quad (1.78)$$

$$g_{\text{R,L}}(z) = \frac{1}{z - H_{\text{R,L}}} \quad (1.79)$$

and self-energies

$$\Sigma_{\text{L}}(z) = u g_{\text{L}}(z) u^\dagger \quad \Sigma_{\text{R}}(z) = v g_{\text{R}}(z) v^\dagger \quad (1.80)$$

[†]The infinitesimal parameter η in Eq. (1.74) shifts the poles of G into the complex plane. In this way it is ensured that the density of states, $-(1/\pi)\Im G(E + i\eta)$, becomes a smooth function of energy. Otherwise, the Hamiltonian (1.66) could not model metallic reservoirs, which by definition have a smooth, nonvanishing density of states near the Fermi energy.

Notice that G and $\Sigma_{R,L}$ act on the Hilbert space of H_C only, whereas $g_{R,L}$ acts on the spaces of $H_{R,L}$.

With this result, we can rewrite Eq. (1.73),

$$T(E) = \text{Tr}_C[\Gamma_L G_C^{\text{ret}}(E) \Gamma_R G_C^{\text{av}}(E)] \quad (1.81)$$

where we have introduced

$$\Gamma_L = 2\pi u \delta(E - H_L) u^\dagger \quad \Gamma_R = 2\pi v \delta(E - H_R) v^\dagger \quad (1.82)$$

so that $\Gamma_{R,L} = -2\Im \Sigma_{R,L}^{\text{ret}}{}^\dagger$. Equation (1.81) is the desired relation. The leads appear only implicitly in the self-energies, $\Sigma_{L,R}$; they have been “integrated out.”

Remarks

- Formula (1.81) is most useful whenever (1) one can give recursive algorithms, so Σ can be calculated without having to deal with the full Hilbert space at a time, or (2) one can design approximations for Σ so that it is not necessary to deal with the Hilbert space of the leads at all. One can argue that simple but highly accurate approximations can indeed be given if H_C is “large enough”, (i.e., comprises a sufficiently large part of the leads).
- Almost all scientific works that perform a *channel decomposition* begin by rewriting Eq. (1.81), which employs the matrix

$$\tau = \Gamma_L^{1/2} G_C \Gamma_R^{1/2} \quad (1.83)$$

so that by construction, $T(E) = \text{Tr}_C \tau \tau^\dagger$. Authors interpret τ as a transmission matrix and hence identify the eigenvectors of $\tau \tau^\dagger$ as the transmission channels. We wish to point out here that this widespread practice has to be taken with a grain of salt.

1. The trace in Eq. (1.81) is over the states of the central region and *not* over the (transverse) Hilbert space of the leads. Ironically, this is why we have derived it in the first place. Therefore, the matrix product in $\text{Tr}_C[\cdots]$ acts on a Hilbert space that is disconnected from the transverse lead space, where the product $tt^\dagger$ that appears in the Landauer formula, Eq. (1.63), lives. Hence, the channels of the leads and the eigenvectors of $\tau \tau^\dagger$ have nothing to do with each other.
2. In particular, τ should not be confused with the true transfer matrix t , given in Eq. (1.72).
3. One of the irritating artifacts that an un contemplated adoption of this practice may prompt is related to the fact that the size of the central Hilbert space is a matter of convention. For this reason, the common channel analysis produces results that cannot be, in general, model

[†]We have used $\delta(E) = (i/2\pi)[G(E + i\eta) - G(E - i\eta)]$.

independent. For example, the number of transmitting states (evanescent and propagating ones) may increase with the Hilbert space size. A more detailed discussion of this and related issues can be found elsewhere.^{42,43}

1.8.4 Nonequilibrium Density Matrix

So far, we have used scattering theory to describe the current flow through a nanojunction or molecule. A very similar analysis allows us to derive even a slightly more general object, the density matrix, $\rho(\mathbf{x}, \mathbf{x}')$, in the presence of nonequilibrium. It is a matrix representation of the operator

$$\hat{\rho} = \sum_n \int dk |nk\rangle^r \langle nk| f_L(E_n(k)) + \sum_{n'} \int dk' |n'k'\rangle^l \langle n'k'| f_R(E_{n'}(k')) \quad (1.84)$$

where $|nk\rangle^r$ ($|n'k'\rangle^l$) denote the right (left)-going states emerging from the left (right) electrodes. The diagonal elements are of particular importance, since they give the particle density, $n(\mathbf{x}) = \rho(\mathbf{x}, \mathbf{x})$, at any position $\mathbf{x}$:

$$\begin{aligned} \rho(\mathbf{x}, \mathbf{x}) &= \sum_n \int dk |\langle \mathbf{x} | nk \rangle^r|^2 f_L(E_n(k)) \\ &\quad + \sum_{n'} \int dk' |\langle \mathbf{x} | n'k' \rangle^l|^2 f_R(E_{n'}(k')) \end{aligned} \quad (1.85)$$

In this section we repeat what we did in the previous section for the Landauer formula, but now for the density matrix. We derive an expression that relates those elements of $\hat{\rho}$ from the central Hilbert space only, in terms of G_C and $\Sigma_{L,R}$ alone.

Indeed, consider the expression for the equilibrium density per spin inside the central region:

$$n^{\text{eq}}(\mathbf{x}) = \int dE \langle \mathbf{x} | \delta(E - H) | \mathbf{x} \rangle f^{\text{eq}}(E) \quad (1.86)$$

Employing a series of standard transformations, which rely upon nothing but the definitions given in the preceding section, we may cast it into a form that is already similar to Eq. (1.85):

$$n^{\text{eq}}(\mathbf{x}) = -\frac{1}{2i\pi} \int dE \langle \mathbf{x} | G_C^{\text{ret}}(E) - G_C^{\text{av}}(E) | \mathbf{x} \rangle f^{\text{eq}}(E) \quad (1.87)$$

$$= -\frac{1}{\pi} \int dE \langle \mathbf{x} | G_C^{\text{ret}}(E) [\Im \Sigma_L^{\text{ret}} + \Im \Sigma_R^{\text{ret}}] G_C^{\text{av}}(E) | \mathbf{x} \rangle f^{\text{eq}}(E) \quad (1.88)$$

$$= \frac{1}{2\pi} \int dE \langle \mathbf{x} | G_C^{\text{ret}}(E) [\Gamma_L + \Gamma_R] G_C^{\text{av}}(E) | \mathbf{x} \rangle f^{\text{eq}}(E) \quad (1.89)$$

$$\begin{aligned}
&= \sum_n \int dk |\langle \mathbf{x} | G_C^{\text{ret}}(E_n(k)) u | nk \rangle|^2 f^{\text{eq}}(E_n(k)) \\
&\quad + \sum_{n'} \int dk' |\langle \mathbf{x} | G_C^{\text{ret}}(E_{n'}(k')) v | n'k' \rangle|^2 f^{\text{eq}}(E_{n'}(k')) \quad (1.90)
\end{aligned}$$

The states $|nk\rangle$ ($|n'k'\rangle$) denote the eigenstates of the left (right) lead in the absence of a coupling, $u, v = 0$. Comparing Eq. (1.90) with the equilibrium limit of Eq. (1.85), $f^{\text{eq}} = f_L = f_R$, suggests the identification

$$\langle \mathbf{x} | nk \rangle^r = \langle \mathbf{x} | G_C^{\text{ret}}(E_n(k)) u | nk \rangle \quad (1.91)$$

$$\langle \mathbf{x} | n'k' \rangle^l = \langle \mathbf{x} | G_C^{\text{ret}}(E_{n'}(k')) v | n'k' \rangle \quad (1.92)$$

for point $\mathbf{x}$ inside the central region. The educated reader may recognize the relations above as an incarnation of the well-known Lippmann–Schwinger equation. Thus equipped, we rephrase the original expression for the density operator in the following way:

$$\hat{\rho} = \int \frac{dE}{2\pi} [G_C^{\text{ret}} \Gamma_L G_C^{\text{av}} f_L(E) + G_C^{\text{ret}} \Gamma_R G_C^{\text{av}} f_R(E)] \quad (1.93)$$

which is valid inside the central region (matrix notation suppresses the argument energy, E). This equation is the main result of the present section. Needless to say, by differentiating off-diagonal elements of $\hat{\rho}$, the current density and therefore also the Landauer formula may be rederived.

1.8.5 Comment on Applications

By far the largest fraction of the vast body of DFT-based transport literature employs scattering theory in the formulation of the preceding section. The logic is that one solves the KS equations (1.39) with a particle density, $n(\mathbf{x})$, which is calculated from the nonequilibrium density operator (1.93), which also takes the reservoirs into account. The KS-Hamiltonian is then used, in turn, to construct the central Green's function and finally, also, the transmission function, (1.81), and the current, (1.63). In this final section we comment briefly on several general aspects of this research. Also, practical aspects of applications in spintronics and molecular electronics are highlighted in Chapters 18 and 19, respectively.

Transmission functions, $T(E)$, are of interest mostly near the Fermi energy, E_F , since one has for the zero-bias conductance, $G = T(E_F)$. In this region, $T(E)$ usually is dominated by the resonances originating from just two (transport) frontier orbitals. Calculations should yield the positions $E_{\text{Ho, Lu}}$ and the broadenings $\Gamma_{\text{Ho, Lu}}$ of the resonances.

In the case of resonances that do not interfere with others (*isolated resonances*), these parameters may be extracted by simply fitting a Breit–Wigner (Lorentzian) lineshape to $T(E)$. Sometimes more complicated situations exist,

where electrons can flow through the molecule via different paths that interfere with each other.⁴⁴ In this case the lineshape is not just a Lorentzian, but may, for example, be of the Fano type. Also, this structure is characterized by very few parameters only, which may be extracted from a suitable fit.

The numerical accuracy of both types of parameters, resonance positions and line widths, that one can get from the DFT-transport calculation depends on the approximations made in the underlying exchange correlation (XC) functional, of course. In transport calculations additional complications arise due to the presence of the electrodes (or reservoirs), which make it necessary to find a good approximation for the self-energies $\Sigma_{R,L}$.

1.8.5.1 Self-Energies $\Sigma_{R,L}$ The self-energies are crucial for the calculation of the resonance width. This is obvious, since without them, $\Sigma_{R,L} = 0$, there would be no level broadening at all: Each transport resonance would be arbitrarily sharp. Therefore, care is needed with the construction of these objects.

However, quite in contrast to a widespread perception in the scientific community, it is *not* necessary—and in practice not even always helpful—to perform an exact construction of $\Sigma_{R,L}$ along the lines of Eq. (1.80). This point has been made earlier^{19,45,46} and we rephrase it here.

Consider the KS equation of the central region in the presence of a coupling to the electrodes:

$$[E - H_C - \Sigma_L(E) - \Sigma_R(E)]\Psi = 0 \quad (1.94)$$

The Hermitian sector of Σ adds to the Hamiltonian H_C and therefore shifts the bare eigenvalues of H_C . The anti-Hermitian sector, $\Gamma_{L,R}$, leads to a violation of the continuity equation; it shifts eigenvalues away from the real axis into the complex plane, thus providing a finite lifetime.

The physics that is incorporated in this way is transparent: Any traveling wave that moves toward the interface between the central region and the left and right electrodes will just penetrate it without being backscattered. From the viewpoint of the central system, the interface is absorbing. It is well known since the early days of nuclear physics that proper modeling of absorbing boundaries is via optical (i.e., non-Hermitian) potentials. This is exactly what the self-energy does.

With this picture in mind, it is obvious that an interface modeling of $\Gamma_{L,R}$ with the property that incident waves are fully absorbed will give the same values for positions and lifetimes of transport resonances. Therefore, as long as the boundary of the central region does not itself hinder the current flow, a modeling of Γ in terms of an optical potential will give accurate results. All the material specifics that are contained in the exact $\Gamma_{L,R}$ matrices can readily be ignored. To meet the condition for simple modeling, in practical terms the central region should comprise pieces of the electrodes that are large enough. Then complete absorption may be achieved with a leakage rate per interface site η that is still sufficiently small, to prevent feedback into the resonance energies.

1.8.5.2 System-Size Dependency: Separation of Scales To the best of our knowledge, all prominent DFT-based transport codes work with approximated self-energies. Unfortunately, a systematic check of quantitative results on the approximation scheme used is still not a standard procedure. If optical potentials with strength η are employed, the transmission resonances, Γ , that we would ultimately like to calculate should be invariant under a change of η by a factor of 10 or more.

The existence of such an invariance is a consequence of a separation of scales. The transport resonances reflect the lifetime of a state located in that subregion (“bottleneck”) of the central region, which determines the resistance (see Fig. 1.5). If the particle has escaped this region, it vanishes into the leads once and for all—in reality. To catch this aspect, the modeling parameter η has just to be big enough to prevent the model particle from returning to the bottleneck. If the size of the central region is taken sufficiently large, much larger than the bottleneck, one can allow for $\eta \ll \Gamma$, and a separation of scales has been achieved.

Remark

- Self-energies, Σ , offer a rich toolbox for including effects of reservoirs with precision without keeping a large number of degrees of freedom explicit in the calculations. Recent applications of the principle describe systems with an inhomogeneous magnetization.⁴⁷ Also in this context, working with model self-energies rather than (formally) exact expressions proves reasonably accurate and highly useful.⁴⁸

Acknowledgments

In this chapter I give a pedagogical introduction to the field, which has grown partly out of several lectures given at Karlsruhe University in recent years. This explicit style is at the expense of accounting for a great many interesting developments pursued by many of my colleagues. Therefore, the chapter cannot serve as—and certainly has not been meant to be—a fair and proper review of the field. Finally, it is a pleasure to thank numerous colleagues for generously sharing their insights with me. Most notably, I am indebted to Alexei Bagrets, Kieron Burke, Peter Schmitteckert, and Gianluca Stefanucci for useful discussions that took place over recent years. Also, I am grateful to Alexei Bagrets and Soumya Bera for critical proofreading of the manuscript.

REFERENCES

1. Kümmel, A.; Kronik, L. *Rev. Mod. Phys.* **2008**, *80*, 3.
2. Neese, F. *Coord. Chem. Rev.* **2009**, *253*, 526–563.
3. Hohenberg, P.; Kohn, W. *Phys. Rev.* **1964**, *136*, 864.
4. Levy, M. *Proc. Natl. Acad. Sci. USA* **1979**, *76*, 6062.
5. Gunnarsson, O.; Lundqvist, B. I. *Phys. Rev. B* **1976**, *13*, 4274; *ibid.*, **1977**, *15*, 6006.

6. Mahan, G. D. *Many Particle Physics*, Plenum Press, New York, 2000.
7. Parr, R.; Yang, W. *Density Functional Theory of Atoms and Molecules*, Oxford University Press, New York, 1989.
8. Igor, V.; Ovchinnikov,.; Neuhauser, D. *J. Chem. Phys.* **2006**, *124*, 024105.
9. Kohn, W.; Sham, L. J. *Phys. Rev.* **1965**, *140*, 1133.
10. Ullrich, C. A.; Kohn, W. *Phys. Rev. Lett.* **2002**, *89*, 156401–1.
11. Chayes, J. T.; Chayes, L.; Ruskai, M. B. *J. Stat. Phys.* **1985**, *38*, 497.
12. Ho, K. M.; Schmalian, J.; Wang, C. Z. *Phys. Rev. B* **2008**, *77*, 073101.
13. Burke, K. The ABC of DFT, chem.ps.uci.edu, 2007.
14. Grimme, S. *J. Comput. Chem.* **2004**, *15*, 1463.
15. Janak, J. F. *Phys. Rev. B* **1978**, *18*, 7165–7168.
16. Almbladh, C.-O.; von Barth, U. *Phys. Rev. B* **1985**, *31*, 3231.
17. Perdew, J. P.; Parr, R. G.; Levy, M.; Balduz, J. L. *Phys. Rev. Lett.* **1982**, *49*, 1691.
18. Perdew, J. P.; Levy, M. *Phys. Rev. Lett.* **1983**, *51*, 1884.
19. Koentopp, M.; Burke, K.; Evers, F. *Phys. Rev. B* **2006**, *73*, 121403.
20. Dreizler, R. M.; Gross, E. K. U. *Density Functional Theory*, Springer-Verlag, Berlin, 1990.
21. Marques, M. A. L.; Ullrich, C. A.; Nogueira, F.; Rubio, A.; Burke, K.; Gross, E. K. U., Eds. *Time-Dependent Density-Functional Theory*, Springer Lecture Notes in Physics, Vol. 706. Springer-Verlag, Berlin, 2006.
22. Runge, E.; Gross, E. K. U. *Phys. Rev. Lett.* **1984**, *52*, 997.
23. van Leeuwen, R. *Phys. Rev. Lett.* **1998**, *80*, 1280.
24. van Leeuwen, R. *Phys. Rev. Lett.* **1999**, *82*, 3863.
25. Vignale, G. *Phys. Rev. B* **2004**, *70*, 201102.
26. Burke, K.; Car, R.; Gebauer, R. *Phys. Rev. Lett.* **2005**, *94*, 146803.
27. D'Agosta, R.; Di Ventura, M. *Phys. Rev. B* **2008**, *78*, 165105.
28. Hyldgaard, P. *Phys. Rev. B* **2008**, *78*, 165109.
29. Onida, G.; Reining, L.; Rubio, A. *Rev. Mod. Phys.* **2002**, *74*, 601–659.
30. Vignale, G.; Kohn, W. *Phys. Rev. Lett.* **1996**, *77*, 2037–2040.
31. Vignale, G.; Ullrich, C. A.; Conti, S. *Phys. Rev. Lett.* **1997**, *79*, 4878.
32. van Leeuwen, R. *Int. J. Mod. Phys. B* **2001**, *15*, 1969.
33. Sai, N.; Zwolak, M.; Vignale, G.; Di Ventura, M. *Phys. Rev. Lett.* **2005**, *94*, 186810.
34. Sai, N.; Zwolak, M.; Vignale, G.; Di Ventura, M. *Phys. Rev. Lett.* **2007**, *98*, 259702.
35. Jung, J.; Bokes, P.; Godby, R. W. *Phys. Rev. Lett.* **2007**, *98*, 259701.
36. Evers, F.; Weigend, F.; Koentopp, M. *Phys. Rev. B* **2004**, *69*, 235411.
37. Stefanucci, G.; Almbladh, C.-O. *Europhys. Lett.* **2004**, *67*, 14.
38. Stefanucci, G.; Almbladh, C.-O. *Phys. Rev. B* **2004**, *69*, 195318.
39. Meir, Y.; Wingreen, N. S. *Phys. Rev. Lett.* **1992**, *68*, 2512.
40. Khomyakov, P. A.; Brocks, G.; Karpan, V.; Zwierzycki, M.; Kelly, P. J. *Phys. Rev. B* **2005**, *72*, 035450.
41. Ferry, D. K.; Goodnick, S. M. *Transport in Nanostructures*, Cambridge Studies in Semiconductor Physics and Microelectronic Engineering, Cambridge University Press, New York, 1997.

- 42. Bagrets, A.; Papanikolaou, N.; Mertig, I. *Phys. Rev. B* **2007**, *75*, 235448.
- 43. Solomon, G. C.; Gagliardi, A.; Pecchia, A.; Frauenheim, T.; Di Carlo, A.; Reimers, J. R.; Hush, N. S. *Nano Lett.* **2006**, *6*, 2431–2437.
- 44. Cardamone, D. M.; Stafford, C. A.; Mazumdar, S. *Nano Lett.* **2006**, *6*, 2422.
- 45. Evers, F.; Arnold, A. Molecular conductance from *ab initio* calculations: self energies from absorbing boundary conditions, arXiv:cond-mat/0611401, Lecture Notes, Summerschool on Nano-Electronics, Bad Herrenalb, Germany, 2005.
- 46. Arnold, A.; Weigend, F.; Evers, F. *J. Chem. Phys.* **2007**, *126*, 174101.
- 47. Jacob, D.; Rossier, J. F.; Palacios, J. J. *Phys. Rev. B* **2005**, *71*, 220403.
- 48. Bagrets, A. Unpublished, 2009.

2 SIESTA: A Linear-Scaling Method for Density Functional Calculations

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This chapter provides a practical overview of the basic theory required to perform density functional calculations on nanoparticles, materials, and large biological systems using the SIESTA program. This program uses discrete atomic basis sets to enable rapid interpretation of results in terms of chemical models, a feature key to many applications, including an understanding of transport properties of materials. It achieves linear scaling (the computer resources required scale linearly with system size for very large systems) using basis set confinement techniques. Many examples of the use of SIESTA are provided in Chapter 11.

2.1 INTRODUCTION

The past two decades have seen the rise of density functional theory (DFT) from a technique largely confined to solid-state physics to arguably the most popular quantum mechanical technique, embraced by chemists, geologists, and most scientific disciplines concerned with the atomic structure of nature. This popularity has arisen largely from its ability to provide a reasonable quality description of properties at a relative modest computational cost in comparison to traditional wavefunction theory-based approaches. Whereas DFT in its purest sense is an exact theory,¹ the practical realization through modern functionals is recognized as having several limitations, including the lack of a pathway for continuous improvement of the answers in the manner possible within post-Hartree–Fock techniques. Despite such caveats, there are many systems for which density functional theory is a valuable and worthwhile approach.

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In this chapter we do not set out to critique the use of DFT, but assume that the reader has already studied Chapter 1, which covers this approach to electronic structure theory, and determined that it represents an appropriate choice to solve the problem of interest. Instead, we focus on another aspect of DFT that has led to its widespread use: the plurality of numerical implementations of the method and the availability of efficient software. Because of the focus on the density for the exchange and correlation potentials, which typically represent the most complex contributions to calculate within electronic structure theory, Kohn–Sham DFT has lent itself to a far greater diversity of practical calculation schemes. While wavefunction theory (WFT) has been dominated by the use of Gaussian basis sets to expand the eigenstates (see, e.g., Chapter 5), DFT has seen a plethora of choices, including plane waves (see Chapter 3), Slater orbitals (see Chapter 15), Gaussians, grids, finite elements, and wavelets, to name but a few.

Nanoscience has pushed experiments to the lower limits of the length scale for the fabrication of materials. Conversely, for computational methods it has led to push toward calculations with a greater number of atoms than ever before. Given that many nanoscale phenomena are related to the effects of quantum confinement on electronic properties, this has, in particular, driven the desire to perform large-scale theoretical studies based on electronic structure techniques rather than force-field approaches. Although simplified quantum mechanical approaches, such as tight binding (see Chapter 10) or semiempirical (see Chapter 8) methods, have a valuable role to play in this realm, ideally it would be possible to use first-principles methods to ensure the reliability of results.

In light of the above and the fact that there are many different numerical schemes for density functional theory, it is possible to reconsider the choice of algorithms and ask what represents the optimal implementation for large systems? Although there will never be an unambiguous answer to this question, we can define the key characteristics of any such method. First, the method must scale with the lowest power possible of the size of the system, typically related to the number of basis functions required, N , or number of atoms. Second, the cost per basis function, which represents the prefactor, or slope of the cost versus system size, must be as low as possible.

If we consider Hartree–Fock or Kohn–Sham theory specifically, there are two main steps in a calculation: the construction of the Hamiltonian matrix and determination of the eigenstates at a self-consistent field. For small system sizes in a localized basis set, such as Gaussians, the first step is the dominant expense and scales formally as N^4 , since the Hartree energy depends on the interaction of two density contributions and therefore up to four different basis functions.² However, this can be reduced to N^3 for Kohn–Sham theory via density fitting.² In practice, for large systems, the scaling is typically reduced through neglect of terms against a threshold. As system size increases, the solution for the eigenstates becomes the major cost since they must be orthogonalized with respect to each other, which leads to a scaling of N^3 .

The key to achieving improved scaling is locality, which is usually considered to be in real space. For example, if an atom were only to interact with other

particles out to a given radius, then once the dimensions of the system exceed this cutoff value, the number of interactions per atom remains constant regardless of increasing size. In other words, the total cost will scale linearly with the dimensions of the system. This will be equally true regardless of whether the system is a finite nanoparticle or a periodic solid. This raises the question of whether it is likely to be feasible in electronic structure theory to confine interactions to within a finite range. Given the central role of the long-range Coulomb potential in the Hamiltonian, at first sight it might be thought that this would not be possible. However, through screening, it turns out that even such interactions lead to quite short-ranged behavior in real space, leading to the *near-sightedness principle*.³ For example, in an insulating or semiconducting material it is known that states decay exponentially with distance, where the rate of decay depends on the bandgap of the substance. Even metals, where there is no gap, exhibit power-law decay behavior.

Provided that it is possible to reformulate density functional theory in a way which ensures that both the generation and solution of the Kohn–Sham equations exploits the inherent locality that exists in many systems, it should be possible to achieve linear scaling of the computational expense for large enough problems. The challenge then becomes to lower the prefactor (i.e., cost per atom) sufficiently that the crossover point at which such algorithms become more efficient than traditional ones is as low as possible. Linear-scaling methods will only be of value if this occurs for numbers of atoms that are currently accessible and of interest for scientific study. Although the specific crossover point can vary strongly according to the details of the method, linear-scaling methods typically become competitive with established algorithms for a few hundred atoms in density functional theory.

Having set the scene, the objective of this chapter is to present an overview of one approach for achieving linear-scaling density functional theory, known as the SIESTA methodology⁴ and embodied in the code of the same name. This is just one of several possible methods, and a list of some of the other most widely used candidates is given in Table 2.1. It would take too long to review the relative strengths and weaknesses of each particular implementation. However, the main differences between methods usually involve a compromise between the ability to have a systematically improvable basis set (similar to the manner that is possible with plane waves) and the lowering of the prefactor of the linear scaling, which requires the most compact basis set representation. To place the SIESTA approach in context, it targets the lowest prefactor by using physically motivated basis functions while sacrificing the arbitrary convergence with respect to the size of the basis.

The aim of this chapter is to provide a conceptual and practical guide to the use of SIESTA that will be useful to those encountering the program that implements the methodology for the first time. For full mathematical details of the SIESTA methodology we refer the reader to the original manuscripts where this can be found.⁴ Although the focus will be specifically on SIESTA, it is hoped that an understanding of the motivation and background will also be valuable to those wishing to engage in linear-scaling electronic structure theory, regardless of the particular implementation.

TABLE 2.1 Various Methodologies for Linear-Scaling Density Functional Theory, Classified According to the Nature of the Basis Functions^a

Basis Set	Implementation	Availability	Ref.
Gaussian atomic orbitals	FreeOn (MONDO set)	GPL	5
	GAUSSIAN	Commercial ^c	
	Q-CHEM ^b	Commercial	6
Gaussians/plane waves	QUICKSTEP	GPL	7
Numerical atomic orbitals	SIESTA	Free to academics	8
	PLATO	Contact authors	9
	OpenMX	GPL	10
Blips	CONQUEST	Contact authors (GPL proposed)	11
Periodic sync functions	ONETEP	Commercial	12

^aNote that this tabulation aims to highlight the most widely known implementations rather than being exhaustively comprehensive. It is also subject to constant change due to developments in the field.

^bThe construction of the Fock matrix can be linear scaling, but diagonalization is used to solve the SCF.

^cFeatures required for a fully linear-scaling calculation may not be available in the distributed version.

2.2 METHODOLOGY

2.2.1 Density Functional Theory

The fundamentals of density functional theory were outlined in Chapter 1, so only a concise statement of the relevant aspects is made here. For the purposes of the present discussion, we focus solely on the Kohn–Sham formulation of DFT, where a set of orthogonal wavefunction-like one-electron states are introduced to facilitate calculation of the kinetic energy, and the exchange-correlation potential is formulated as a local functional of the density and, where appropriate, its curvature. Thus, we will consider the linear-scaling implementation of the local density approximation (LDA) and the generalized-gradient approximation (GGA) formulations of DFT.¹³ Extension to other forms of approximation, such as meta-GGAs,¹⁴ hybrid functionals,¹⁵ or LDA + U¹⁶ is possible, but beyond the scope of the present chapter.

2.2.2 Pseudopotentials

When solving for the electronic structure of a system, in principle, all electrons must be included since they contribute to the potential experienced by other particles and determine the nodal structure of the eigensolutions. In practice, it is intuitive that the core electrons of an atom are weakly perturbed by chemical changes to the geometry and bonding arrangements, in comparison to the valence

electrons, and therefore, several approximate methods have evolved to treat these core states in order to reduce computational expense. At the simplest level, the frozen-core approximation can be made in which the occupancy of the core states is fixed to remove them from the self-consistent procedure. Alternatively, the core electrons and nucleus, which have opposite sign charges and therefore partially cancel each other, can be replaced by a combined effective potential, known as a *pseudopotential*.

In brief, the concept of a pseudopotential is that it replaces the exact potential due to nucleus and core electrons, within a given radius of the atomic center, by an effective potential. Within this distance, known as the *core radius*, the potential is smoothed and tends to a finite value at the nucleus while matching the true potential at the boundary. Due to the smoothing of the potential, the radial nodes of the valence states are lost in the core region since there is no longer a requirement to maintain orthogonality to the core states. In nearly all cases, a nonlocal pseudopotential is used, which implies that there is a different potential for each angular momentum channel, with a separate core radius, r_{core}^l , appropriate to that channel. Outside the core radii, all channels, regardless of angular momentum, experience exactly the same potential, known as the *local component*. Thus, the nonlocal contribution to the pseudopotential acts only within a small spherical region close to the nucleus.

Nonlocal pseudopotentials are most commonly formulated according to the prescription of Kleinmann and Bylander.¹⁷ While in many implementations the local component of the pseudopotential is chosen to be one of the angular momentum channels, there is no requirement to do so. Indeed, SIESTA exploits the freedom to select the local component independently and chooses the potential that results from the smooth electron density:

$$\rho_{\text{local}}(r) \propto \exp \left[- \left(\frac{\sinh(1.82r/r_{\text{core}})}{\sinh(1)} \right)^2 \right] \quad (2.1)$$

The construction of a pseudopotential generally involves satisfying at least four criteria:

1. *Boundary matching.* Beyond the core radius, the all-electron and pseudo-wavefunctions must match for each angular momentum channel.
2. *Smoothness.* Within the core radius, the pseudovalence wavefunction should have no radial nodes.
3. *Eigenvalue matching.* The eigenvalues for the pseudopotential problem must match the all-electron values for the atomic reference state chosen.
4. *Norm conservation.* The integral of the valence electron density from the nucleus to the core radius must be equal in the pseudopotential and all-electron cases.

Other conditions may also be imposed; for example, the logarithmic derivative and their first energy derivative may also be required to match outside the core region.¹⁸ An all-electron and a pseudo-wavefunction are compared in Fig. 2.1.

Although the conditions noted above are necessary for most pseudopotentials, this does not lead to a unique definition of what form the potential should take, so numerous schemes for the generation of pseudopotentials have arisen. In the case of SIESTA, pseudopotentials are usually generated through the use of a separate program known as *ATOM*, which presently supports three types of pseudopotential; improved Troullier–Martins (TM2),¹⁹ Hamann–Schlüter–Chiang (HSC),¹⁸ and Kerker.²⁰ Of these, the Troullier–Martins scheme has become the standard choice for use with SIESTA.

In the plane-wave community, the use of pseudopotentials is almost mandatory for practical calculations since the effective potential is smoothed out and the nuclear cusp removed, thereby drastically reducing the number of basis functions required to construct the Fourier expansion of the eigenstates. Even when working with localized orbitals there are some benefits to the use of pseudopotentials, aside from the reduction of the number of electrons and orbitals. The core electrons are much more strongly bound than the valence electrons and therefore dominate the total energy. Because electronic structure calculations often rely on computing small energy differences between large total energies, the inclusion of the core electrons can decrease the level of numerical precision in such quantities. Furthermore, as the atomic number of an element increases, it becomes important to correct the calculation for relativistic effects, which most strongly affect the core electrons. Through the use of a pseudopotential it is possible to

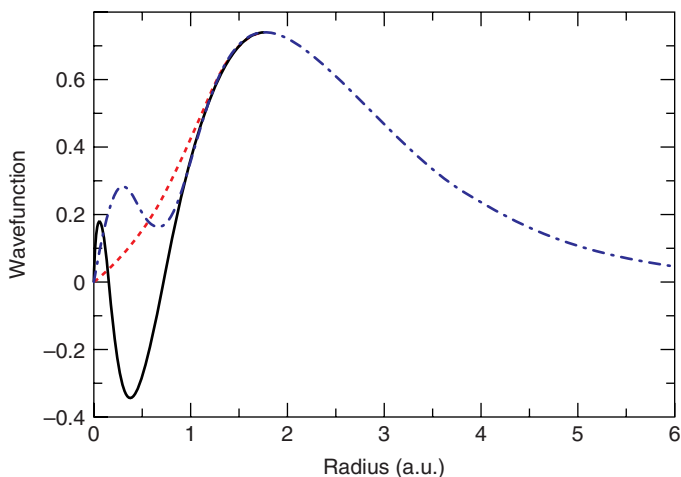


Fig. 2.1 All-electron (—) versus pseudovalence state (---) for the silicon 3s orbital. The core radius for the 3s state is 1.9 a.u. For comparison, a poorly constructed pseudo-3s state (— · —) is included for the case when the core radius is too small (1.1 a.u.), leading to an inner maximum.

subsume the majority of the relativistic effects into the effective potential, such that a full relativistic calculation is required for the isolated atom only during generation of the pseudopotential, rather than for the entire problem. Of course, it is important to note that some relativistic effects must be taken into account explicitly when necessary, such as spin-orbit coupling.

Recent years have seen a number of developments in the area of pseudopotentials with the advent of the ultrasoft pseudopotential (USP)²¹ and projector augmented wave (PAW)²² methods. For USPs, the requirement of norm conservation is relaxed and this is compensated for by the addition of an augmentation charge density. The PAW approach focuses on the augmentation of the wavefunction, rather than the density, and thus makes it possible to recover all-electron properties in the frozen core limit. Both methods lead to a dramatic reduction in the reciprocal space cutoff associated with the pseudopotential, which greatly accelerates the computation. In the case of SIESTA, which as we shall see works with real space-localized basis functions, there is likely to be little benefit associated with a switch to either of these more contemporary pseudopotential types, while the complexity of implementation is greatly increased. Consequently, SIESTA continues to employ norm-conserving pseudopotentials, which are generally more robust and easier to construct (see, e.g., an article by Bilić and Gale²³).

Although it is impossible to give a comprehensive guide to the generation of pseudopotentials, some important general guidelines can be given.

2.2.2.1 Choice of Electronic Configuration When generating a norm-conserving pseudopotential it is necessary to specify an atomic configuration whose eigenvalues and wavefunctions will be reproduced outside the core region. Usually, this is chosen to be the ground state for the isolated atom. However, for the study of ionic materials there may be merit in using a positively ionized state if this is closer to the real oxidation state of the cation. Although, in principle, a pseudopotential is supposed to be transferable across a range of charge states, it will be more accurately closer to the state for which it is generated. In the case of anions in ionic materials (e.g., the oxide ion), it is not generally a good idea to use the negatively charged state since this will be very diffuse and may be unbound (as is the case for O^{2-}).

2.2.2.2 Choice of Functional It is important to use the same density functional for generation of the pseudopotential as you intend to employ in the explicit valence calculation. Although the use of an LDA pseudopotential in a GGA calculation can often lead to fortuitously good results with respect to experimental data, it is important to remember that the objective is to reproduce the all-electron limit for a single given functional.

2.2.2.3 Choice of Core Radius The general guiding principle in the choice of the core radius is that a larger radius leads to a softer (and for plane waves, therefore more efficient) pseudopotential, whereas a smaller radius should ensure

greater transferability and reliability. Beyond this broad statement, there are a number of limitations on the upper and lower bounds to the core radius. If the radius becomes too large, there is a risk that the core regions of two adjacent atoms might overlap and this would invalidate the calculation. On the lower bound, the core radius must lie farther from the nucleus than the last radial node of the all-electron wavefunction; otherwise, the removal of nodal structure will not be possible. In practice, making the core radius too small can lead to spurious features in the pseudo-wavefunction, such as inner maxima, due to enforcement of the norm-conservation condition (see Fig. 2.1 for an example of what happens as the core radius becomes too small). The optimal choice for the core radius usually will lie close to the outer maximum in the all-electron wavefunction. With the Troullier–Martins construction scheme, the core radius can lie outside the maximum, and the wavefunction will still be well reproduced beyond the turning point.

2.2.2.4 Choice of Core–Valence Split For many elements, especially those toward the right-hand side of the periodic table, there is no ambiguity as to the valence electrons of an atom. However, for quite a large number of elements there may be cause for careful consideration, depending on the material to be studied. For example, aluminum has the electron configuration $[1s^2 2s^2 2p^6] 3s^2 3p^1$, where the brackets delimit the conventional core electrons. If one were to perform a study of aluminum nanoparticles, for example, only including the 3s and 3p electrons in the valence would be a reasonable choice, since the atom is close to the charge neutral state. However, if one were instead to study the material Al_2O_3 , where the nominal oxidation state is Al(III), the 3s and 3p electrons have been largely ionized. Here the 2p electrons then become the highest occupied state of aluminum, and the conventional choice of valence would lead to a poor pseudopotential description. For elements toward the beginning of a new block of the periodic table, it is therefore necessary to modify the pseudopotential choice to allow for these semicore states.

2.2.2.5 Evaluating Pseudopotential Accuracy A good indicator as to whether semicore states need to be included is whether there is any significant overlap between the electron density of the valence and core electrons (see Fig. 2.2, which shows the case of Fe where there is significant overlap between the 4s/3d states and the underlying 3s/3p). There are two common methods for handling semicore states; either the electrons can be explicitly included in the calculation, or partial core corrections can be applied.²⁴ *Partial core corrections*, also known as *nonlinear core corrections*, aim to correct for the fact that exchange–correlation potential depends on the *total* electron density and is therefore not readily separable into core and valence contributions if there is any overlap of the density between regions. To handle this, partial core corrections operate by including a smooth piece of frozen electron density that matches the exact core density down to a given radius and then tends smoothly to zero at the nucleus. This density is then added back during calculation of the exchange–correlation potential to capture the nonlinearity in the region of density overlap. Note that this extra density

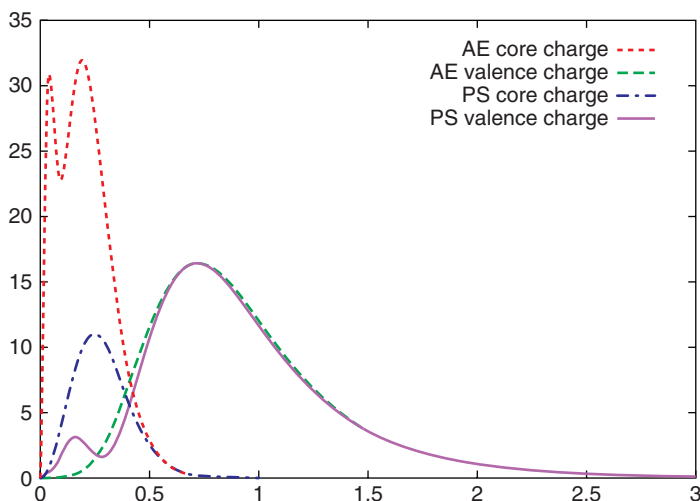


Fig. 2.2 Electron density for an iron atom, showing the all-electron curve (core contribution in - - - and valence in - · -), the valence-only contribution from the pseudopotential-generated orbitals (——), and the partial core correction density (— · —) as a function of radius (in a.u.). Note the overlap between the core and valence densities in the region between 0.2 and 0.7 a.u. that leads to the need for partial core correction.

is not included in the norm-conservation requirement of the pseudopotential. The choice of the radius for the partial core corrections is a compromise between being small enough to describe sufficient core electron density and large enough to minimize the computational work associated with evaluating accurately the exchange-correlation potential for the combined density. While for plane-wave methods the use of partial core corrections is often the preferable approach to semicore states since it reduces the size of the basis set significantly, for the SIESTA method the two approaches are similar in cost, and therefore the use of explicit semicore states may be favored.

Having generated a new pseudopotential and inspected its properties visually to check that there are no untoward characteristics, the next important step is to test it by comparing the energies for changes in atomic state between the all-electron- and the pseudopotential-based calculation. Configurations for testing might usually include ionization from the various valence orbitals, as well as promoting electrons from one angular momentum to another. If the pseudopotential passes this examination, it is ready for validation in a full calculation of a molecule or solid.

2.2.3 Basis Sets

Numerical solution of the Kohn–Sham equations is performed by expanding the orbitals or bands in terms of a computationally convenient mathematical function: the basis set. The coefficients that determine how much these functions contribute

are found by applying the variational principle. As mentioned in the introduction, there are many possible choices that could be made for the basis set, although Gaussians²⁵ have dominated the molecular community while plane waves have been the de facto standard for solid-state physics.

In choosing the optimal basis set for large linear-scaling calculations, we are guided by the need for locality in real space and the requirement to minimize the number of basis functions needed to obtain reasonable numerical precision. Clearly, a physically motivated basis set that takes into account the shape of atomic orbitals will best satisfy the latter criterion. If pseudopotentials of the form described in the preceding section are employed, then neither existing Slater, or Gaussian, basis sets will be of the correct form, due to the modification of shape in the nuclear region.

Taking the discussion above into account, it can be seen that the optimal compact basis set is to work with exact solutions to the pseudopotential form of the atomic problem, provided that they can be represented. Following the approach taken by other researchers, such as Becke and Dickson²⁶ in the NUMOL code and Delley²⁷ in DMol, the basis set can conveniently be represented by a numerical tabulation rather than a specific, but approximate, analytical form. In the SIESTA methodology, the standard choice of basis set is pseudoatomic orbitals (PAOs), φ_{nlm}^A for atom A, which are tabulated on a logarithmic radial grid for each angular momentum and then multiplied by the appropriate spherical harmonics:

$$\varphi_{nlm}^A(r, \theta, \varphi) = R_{nl}^A(r) Y_{lm}(\theta, \varphi) \quad (2.2)$$

These PAOs can be determined conveniently during generation of the pseudopotential and represent a “perfect” basis set for describing the isolated atom.

While the PAOs above decay rapidly with distance, as do other atomic-centered basis functions, they only tend asymptotically to zero at infinite radius. To achieve linear scaling it is necessary to impose on the Hamiltonian strict locality in real space. The most common approach to achieving this is to introduce a drop tolerance in some form and to neglect integrals when they fall below a certain magnitude. However, this is fundamentally unappealing since it corresponds to modifying the Hamiltonian being solved, although this may be a philosophical point rather than a practical difficulty. In the SIESTA methodology, an alternative approach is taken in which the basis functions are localized rather than modifying the Hamiltonian. Following the fireball concept of Sankey and Niklewski,²⁸ the eigenfunctions of the pseudoatomic problem are found within the confines of a spherical boundary at which the potential becomes infinite. In this way, the tails of the PAOs are modified such that they go rigorously to zero at a given radius, as shown in Fig. 2.3. This radius, r_c , can be selected to be different for each angular momentum.

Radial confinement is clearly an approximation, but it allows a choice to be made readily between higher precision, corresponding to large r_c , or greater computational efficiency as the radius decreases. Although there is the flexibility

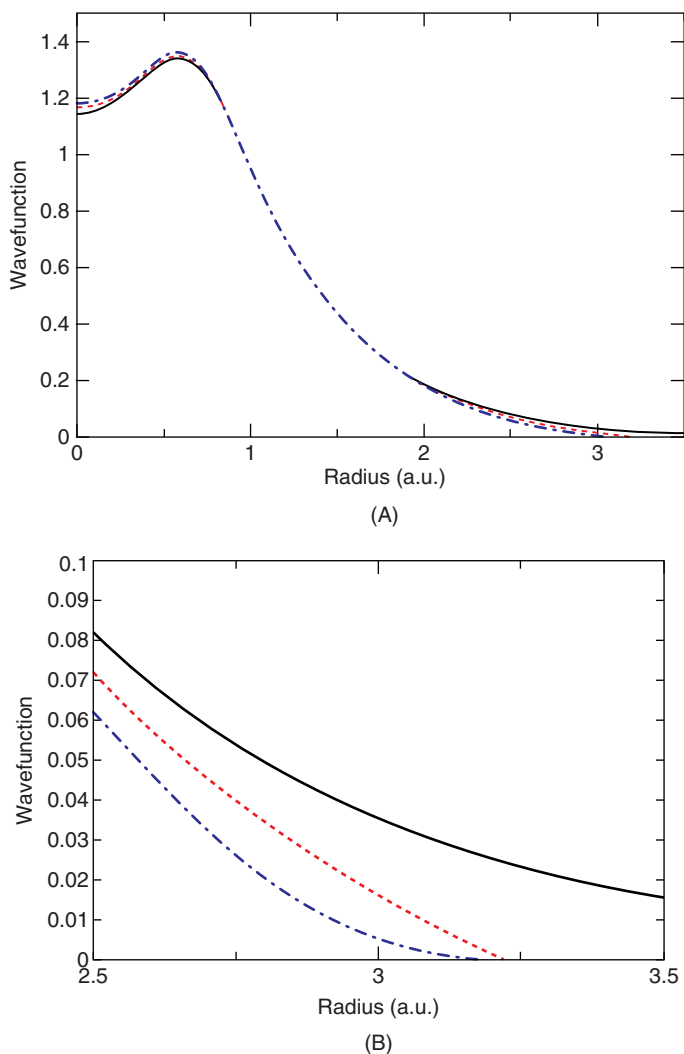


Fig. 2.3 (A) Pseudoatomic orbitals (PAOs) for oxygen 2s, illustrating the shape for the unconfined orbital (—), hard confinement with an energy shift of 0.02 Ry (---), and soft confinement with an energy shift of 0.02 Ry, a potential V_0 of 50 Ry, and a radius of soft confinement commencement of 0.8 times the hard confinement radius (— · —). (B) Close-up of the region where the confined orbitals approach the cutoff radius of 3.2 a.u.

to choose an individual radius for each orbital in the valence of every atom, it is preferable to have a more systematic method for selecting radii. Choosing a single fixed radius of confinement for all atoms is obviously not a sensible approach, since atoms with different atomic radii will be affected to varying extents. Hence, the calculation would be biased toward the precise description

of light atoms. When an orbital is radially confined, its energy increases with respect to the free atom. Therefore, a natural concept to aid in the selection of appropriate radii is the *energy shift*. Here, a single energy value is specified for all atoms and the radius of confinement found that raises the energy of each orbital by this amount. Typically, energy shifts in the range 0.001 to 0.02 Ry (1 Ry = 0.5 Ha $\sim$ 13.6 eV) are useful depending on whether precision or speed is being sought, respectively. As with all approximations, it is important to test the consequences of a given choice for the specific property of interest before proceeding.

Although the default energy shift-based scheme provides a good first estimate of the radii in many cases, there are alternative approaches to refining the truncation of the orbitals.

2.2.3.1 Soft Confinement In the default confinement scheme the orbital goes to zero at the cutoff radius. However, there is a discontinuity in the derivatives of the orbital, which can lead to difficulties during structural optimization and more acutely during phonon calculations. The solution to this problem is to use a potential that tends asymptotically to infinity in a smooth manner rather than applying a discontinuous hard-wall potential.²⁹ The form of the potential currently used is

$$V_{\text{soft}}(r) = V_0 \frac{e^{-(r_c-r)(r-r_s)}}{r_c - r} \quad (2.3)$$

This introduces two new parameters that determine the shape of the basis set tail by determining the radius at which the potential begins, r_s , and the magnitude, V_0 .

2.2.3.2 Basis Set Enthalpy In a further alternative scheme, an external pressure, P_{ext} , can be applied to the atomic orbitals. This leads to determination of the radii through the associated enthalpy by adding a $P_{\text{ext}}V$ term to the intrinsic energy, where V represents the volume of the confinement sphere.³⁰ Under this scheme, the confinement radii now correspond to equal hardness among the basis functions, rather than energy perturbation.

Occasionally, it may be beneficial to intervene manually in the choice of radii. For example, in the case of negatively charged species such as the oxide ion, which is nominally O^{2-} , the radii determined by typical energy shift values as being appropriate for a neutral oxygen atom may be too confined to allow a good description of the anion in an ionic crystal.

Although the formulation of PAOs above provides a good starting point for a basis set, it is well known that increased variational freedom is required to allow the system to respond to the changes associated with chemical bonding, external fields, or other perturbations to the electronic structure. In the Gaussian community this is achieved through the use of multiple-zeta basis sets, where one or more Gaussians (usually, the outermost function) is decontracted from the

Slater-type orbital to allow the effective atom size to respond to its environment. When working with a numerical representation of the valence orbitals on a radial grid, there is no equivalent means of creating distinct “zetas.” Indeed, there is the flexibility to choose any arbitrary partitioning of the valence orbital into multiple components.

From experience it is known that the objective is to allow the outermost part of the radial function to vary independent of the inner part while maintaining the smoothness of the basis functions. In the current SIESTA methodology, the division of the radial function into multiple components is achieved using the *split-norm concept*. Here a second, or higher, radial function is designed to possess the same tail as the full valence orbital, $\varphi_l^{1\zeta}$, outside a split radius, r_s , while inside this value it decays according to a polynomial to be zero at the nucleus:

$$\varphi_l^{2\zeta}(r) = \begin{cases} r(a_1 - b_1 r^2) & r < r_s \\ \varphi_l^{1\zeta}(r) & r \geq r_s \end{cases} \quad (2.4)$$

The polynomial coefficients are determined by matching the function and its derivative at the split radius. If this new function is subtracted from the original valence orbital, the result is a contracted basis function that goes to zero at the split radius. Motivated by similar arguments to the use of the energy shift, the split radius is usually chosen indirectly by specifying the norm of the valence state to be included in the outer function. Typically, an outer zeta should contain on the order of 15% of the total norm. For hydrogen, in a double-zeta basis set, a value closer to 50% can prove more effective, given that the variation in effective size between a neutral hydrogen atom and a proton-like state can be particularly extreme. Conversely, very small values for the split norm can represent a poor choice since their effect is negligible and can lead to linear-dependence issues in the basis set.

There are several things to note regarding the choice of the split-norm approach to increasing the radial variational freedom of the basis set. As already pointed out, this is just one possible choice and there are many other possible approaches. In the all-electron numerical methodology of Delley,²⁷ an alternative strategy is employed in which the basis functions for charged atomic states are used for the additional radial functions to describe more contracted environments. Alternatively, one could use extra Gaussian functions to mimic a standard multiple-zeta basis set from conventional molecular quantum mechanics.³¹ A strength of the split-norm approach is that the operation can be applied as many times as desired to create a basis set of arbitrary size in a systematic fashion. Usually, a double- or triple-zeta basis is sufficient unless trying to achieve plane-wave levels of numerical convergence. We should note that the use of terms *double zeta* (DZ) and *triple zeta* (TZ) is a matter of conforming to the nomenclature that has arisen in the Gaussian community, although strictly speaking it is incorrect since there are no “zetas” (i.e., Gaussian exponents) in the present approach. In the terminology of Delley, the basis sets are referred to more correctly as double numeric (DN), triple numeric (TN), and so on.

It may be questioned whether an approach that allows atoms to adopt a smaller effective radius, but not a larger one, is always sufficient. The answer is usually in the affirmative. If the minimal basis set is constructed for the neutral atom, when an atom is placed in a crystal, or even in a molecule, the rate of decay of the valence states will usually be increased by Pauli repulsion due to the neighboring atoms. Hence, a shorter-range basis set is generally appropriate, although with some exceptions.

Although the split-norm approach provides increased radial variational freedom, there is also the need to consider angular augmentation of the basis set. For example, a minimal basis set for hydrogen would only include the 1s orbital, but the moment an external field is applied, or the hydrogen forms a covalent bond to another atom, there is a need to describe asymmetric contributions to the electronic structure about the hydrogen nucleus. Therefore, it is necessary to include basis functions of higher angular momentum than those from the occupied valence states alone, and these are known as *polarization functions*. Typically, functions with a value of the angular momentum quantum number, l , one higher than that of the highest occupied state, are needed as a minimum requirement for a reliable description of the electronic structure (i.e., 2p for H, 3d for C, 4f for Fe, etc). Hence, the default basis set, and minimum recommended quality, for SIESTA would be double-zeta polarized (DZP). Although some special cases, such as bulk silicon, are relatively well described with a minimal basis set, these are the exceptions rather than the rule.

The key question with polarization functions is how to obtain the radial form of these basis functions. Unfortunately, the excited states of the pseudopotential atomic problem tend to be either rather extended in space, or even unbound, and therefore taking the hard confined unoccupied orbitals, as basis functions can often be unsatisfactory. In an attempt to circumvent this problem, the default method for the generation of polarization functions uses perturbation theory. By applying an electric field to the atomic problem, states of higher angular momentum are created, and these are taken as the polarization functions.

The choice of good polarization functions is the most difficult part of the basis set creation and is often responsible for lower-quality results, as can be demonstrated in an example. If we consider the comparison of results for the molecule SO_2 , as obtained using the default DZP basis set in SIESTA and from the use of the same density functional with a range of standard Gaussian basis sets, it can be seen that there is some discrepancy (Table 2.2). If instead of using the default polarization functions, the shape of the radial part of this basis set is tuned by using a soft-confinement potential to lower the energy of the system variationally, a significant improvement is achieved. Indeed, the results for the DZP basis set are now very close to those for the equivalent Gaussian basis set.

While default basis sets can be generated within the SIESTA methodology, according to the energy shift, split-norm, and perturbative polarization function schemes described above, there is also a possibility for the user to control the

TABLE 2.2 Comparison of Optimized Structural Parameters for the Molecule SO₂ with the PBE Functional as a Functional of Basis Set Quality

Basis Set	$r(\text{S-O})$ (Å)	$\angle(\text{O-S-O})$ (deg)
STO-6G	1.628	107.40
6-31G	1.634	114.67
6-311G	1.630	114.66
6-31G*	1.483	119.34
6-311G*	1.477	119.04
DZP (standard/0.01 Ry)	1.509	118.71
DZP (optimized polarization)	1.482	119.34

basis fully. Accordingly, there are methods to tune to the basis set performance in a number of ways.

2.2.3.3 Charge State By default the basis set is generated for the reference state used in pseudopotential generation. However, a charge on a species can also be specified during basis set creation. Here a positive charge will lead to more contracted basis functions, while a small negative charge will result in more diffuse PAOs. Note that a large negative charge would not be sensible since species become formally unbound.

2.2.3.4 Variational Optimization The experience of other communities that have adapted molecular basis sets to the solid state shows that optimization of the basis set parameters with respect to the total energy of a target material can improve the results substantially.³² Although compromising the transferability, it allows the best results to be obtained for a particular problem while maintaining a low prefactor for the computational cost.

As with all numerical approximations, it is important to test the influence of basis set quality before embarking on any scientific study. While DZP should be adequate to obtain at least qualitatively correct results for most problems, this should not be assumed a priori for a new class of problem. It is also important to consider the consequences of radial confinement for the study to be undertaken. For example, if considering the decay properties of the electronic states of a surface into vacuum, by construction the answer will be in error unless steps are taken to rectify this.³³ The present method will also share much of the cautionary advice common to all localized, atomic-centered basis sets, including basis set superposition error (BSSE) and the need for floating functions when describing states that involve electron density in a region away from atomic centers (e.g., a defect such as an F-center). BSSE can be a particular issue, since the overlap of basis functions from different atoms allows the radial confinement to be released, thereby artificially inflating the binding energy even more than usual. Therefore, when considering molecular adsorption, particularly if it is weak, it is essential to work with a low value for the energy shift and to apply a counterpoise correction³⁴ to the final result in order to extract a meaningful binding energy.

2.2.4 Construction of the Kohn–Sham Equations

Once the basis set is defined, it is then possible to define the Kohn–Sham equations for the system of interest (see Section 1.3). Note that because the basis set is nonorthogonal, the overlap matrix must also be computed, in addition to the Hamiltonian. Although the average user of the SIESTA methodology need not understand all the details of how the elements of the Hamiltonian and overlap matrices are computed, it is essential to possess some appreciation of the underlying concepts and the numerical approximations that influence calculation quality.

In considering the construction of the Kohn–Sham equations, it is possible to break the problem down into several components:

- Overlap matrix elements between basis functions
- Kinetic energy of basis functions
- Nonlocal contribution of the pseudopotential (confined to core region)
- Local contribution of the pseudopotential (long-range)
- Hartree potential (mean-field Coulomb interaction of electrons)
- Exchange-correlation contribution; either LDA or GGA

As emphasized previously, the key is to evaluate the terms in a manner that is linear scaling and efficient. The components naturally break down into two different classes of integral to be evaluated: those that depend on the basis functions only, and those that depend on the electron density or are potentially long-range.

Considering first the overlap matrix elements, kinetic energy matrix elements, and the nonlocal contribution of the pseudopotential, these are all strictly local in real space, due to the finite range of the basis set. The first two terms depend on pairs of overlapping orbitals, and therefore the range is at most twice the largest orbital cutoff radius for any species. In the case of the nonlocal pseudopotential projectors, these give rise to matrix elements between the atomic center associated with the pseudopotential and the basis functions of up to two neighboring atoms. Hence, the range is slightly greater, spanning twice the largest orbital cutoff radius, plus twice the largest core radius for any pseudopotential. However, the range of interaction is still readily predefined. Evaluation of these two- or three-center integrals can be performed readily by use of a Fourier expansion (see the original papers for full details^{8,28}). The key point is that these integrals are performed with a default reciprocal space cutoff of 2000 Ry, which is sufficient to ensure that they are numerically well converged in all but the most extreme circumstances. Furthermore, the cost of these matrix elements is usually a minor part of the total computing time of any calculation. Therefore, the user need not be particularly concerned with the evaluation of these contributions to the Hamiltonian and overlap matrix.

The remaining contributions to the potential and energy are more complex than the terms above since they involve the *electron density* rather acting directly on the basis functions. The electron density is, of course, expanded in terms of

the basis functions:

$$\rho(r) = \sum_{\mu\nu} \rho_{\mu\nu} \phi_\nu^*(r) \phi_\mu(r) \quad (2.5)$$

$$\rho_{\mu\nu} = \sum_i \int_{\text{BZ}} c_{\mu i}(k) o_i(k) c_{i\nu}(k) e^{ik(r_\nu - r_\mu)} dk \quad (2.6)$$

where the coefficients are stored as the density matrix elements, $\rho_{\mu\nu}$. Here integration over the Brillouin zone is explicitly included and $o_i(k)$ represents the occupancy of eigenstate i at a given point in reciprocal space. If evaluated simplistically, this would make the Coulomb interaction between two points of electron density a long-range interaction that scales as the fourth power of the number of basis functions. Fortunately, this is less problematic than it appears for two reasons. First, the contribution due to the local part of the pseudopotential is of opposite sign to the interaction with the electron density. For a charge-neutral system, these two contributions cancel in the long-range limit, so the Coulomb interaction is ultimately screened. Second, the use of an auxiliary basis set to represent the electron density is well known to reduce the scaling problem and improve computational efficiency.²

Many different choices could be made to converge Coulomb sums efficiently, such as fast multipole methods,³⁵ and to represent the electron density in an auxiliary basis set. In the SIESTA methodology, the choice was made to represent the electron density on a uniform Cartesian grid of points in real space. This decision can be justified for a number of reasons. First, unlikely in some localized basis sets, there is no natural representation to choose for the density expansion; although the basis functions themselves have some of the correct properties, it is difficult to extend the minimal set to ensure an accurate representation of the density at all points. A Cartesian grid is systematic and basis set shape independent; as the fineness of the grid increases, the aliasing error should decrease, as all Fourier components become representable. Second, the construction of the electron density is rigorously linear scaling. As shown in Fig. 2.4, only basis functions within the maximum cutoff radius can contribute to the electron density at a given grid point, and therefore the cost per point does not depend on the overall system size. Third, calculation of the exchange-correlation contribution for both LDA and GGA becomes a trivial summation over grid points. In the case of GGAs, calculation of the gradient of the density is facilitated by the use of a finite difference expansion³⁶ over the neighboring grid points (and equally important, the additional contribution to the potential from the GGA is straightforward to determine in the same way).

Once the total electron density on the grid points is known, it is possible to begin computation of the electrostatic potential, consisting of the electron–electron interaction (Hartree potential) and the electron–local component of the pseudopotential interaction. We note that the Hartree term is based on the interaction between the electron density at all points to give a single orbital-independent potential and therefore contains the self-interaction of an

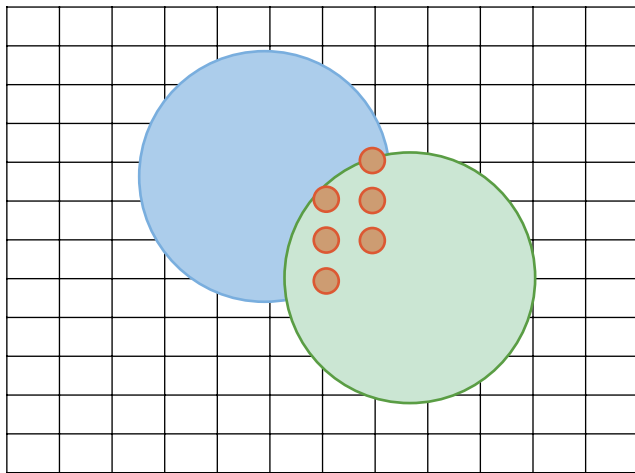


Fig. 2.4 Calculation of the density based on two orbitals (large circles) on an underlying Cartesian mesh. Here the density contribution would only be nonzero at the mesh points (small circles).

electron with its own density, as is the norm within standard Kohn–Sham theory. Rather than working directly with the total electron density, it is advantageous to divide the electrostatic contributions into two parts: the neutral contribution and the deformation density. The electron density of the neutral atoms can readily be computed on the grid and subtracted from the total electron density to leave the deformation density. The neutral atom density can then be added to the local part of the pseudopotential to yield a potential that goes strictly to zero at the outermost core radius. Being local, the electrostatic contribution of the neutral atoms is readily computed.

Having determined the deformation density on a uniform grid, $\delta\rho$, the calculation of electrostatic potential due to this quantity, δV_H , can be made through solution of Poisson’s equation:

$$\delta\rho(r) = \rho_{\text{tot}}(r) - \rho_{\text{NA}}(r) = -\frac{1}{4\pi}\nabla^2\delta V_H(r) \quad (2.7)$$

At present, SIESTA solves for the potential through the use of a fast Fourier transform (FFT), as many efficient libraries are available to perform this task. Although this approach is not actually linearly scaling ($N \ln N$), the relative low scaling, combined with the efficiency of the method, ensures that the contribution to the computational cost is negligible and therefore the deviation from linear scaling due to this contribution has yet to be observed.

Arguably a more significant drawback of the use of FFTs, with practical consequences for the user, is the requirement that all systems must have three-dimensional periodic boundary conditions. In the implementation of the SIESTA method, all systems are automatically enclosed within a periodic cell, regardless

of whether it is a molecule, a polymer, a surface, or a solid. For cases where there is no natural periodicity, the fictitious cell parameter(s) is chosen so as to ensure that there is no overlap between the basis functions of images. Although this guarantees that there are no direct matrix elements between periodic repeats, there is a potential for interaction via electrostatic terms. Consequently, for systems with a strong dipole or higher-order moment, it is recommended that the explicit convergence with respect to cell size be tested. Unlike plane-wave methods, the cost of including a large region of vacuum is generally small since there is no change in the basis set associated with this, and the only computational cost lies in the Fourier transform step to compute the potential. Hence, it is usually straightforward and inexpensive to ensure that the interaction between periodic images is negligible.

An alternative to the use of fast Fourier transforms is to employ multigrid methods to solve the problem.^{37,38} This has the advantage of being linear scaling and can be adapted to any set of boundary conditions that are required. Although it has been explored in conjunction with the SIESTA method,³⁹ the absolute performance remains slower than the use of FFTs, so it has not yet been adopted within the distributed implementation. Once the potential due to the deformation density is determined, by either FFTs or multigrid, the contribution to the energy from this term can be calculated by summing the product of this potential with the total electron density across the mesh.

Having discussed the background to the evaluation of the electron density-oriented contributions to the Hamiltonian, it remains to consider the practical consequences for the use of the methodology. The most significant point is that there will always be a numerical error in the integral of quantities involving the electron density. While the description of the electron density at the grid points is correct, the integration between adjacent points is approximate. As the grid spacing is reduced, the numerical integration becomes more precise. Rather than specifying the grid spacing directly, the fineness is controlled by a kinetic energy value, known as the *mesh cutoff*, for the highest-energy Fourier component that can be represented. For periodic systems, the grid spacings allowed are constrained by the requirement to be commensurate with the unit cell, so the nearest mesh cutoff above the target specified is chosen. Typical mesh cutoffs are between 80 and 400 Ry, although higher values may be required for very precise calculations. Ultimately, the value required will depend on the pseudopotentials present or basis set shape and must be tested for convergence behavior. Note that the use of partial core corrections often necessitates the use of higher mesh cutoffs, due to the larger total electron density to be integrated.

The practical consequence of the numerical integration error above is that there will be a small breaking of translational invariance (i.e., the energy of a system will change slightly according to its absolute Cartesian position relative to the underlying mesh). This is referred to as *space rippling* or the “egg-box” effect. In addition to affecting the energy, this will also lead to numerical deviations in the

forces. As a result, there can be slight symmetry breaking of structures or convergence slowdown during geometry optimization if the mesh cutoff is too low. It should be noted that this issue is common to most methods that use non-atom-centered basis (or auxiliary basis) sets, although it can be hidden through explicit symmetry constraints, or reduced through the use of softer pseudopotentials/basis function shapes.

A number of practical schemes to reduce the influence of the “egg box” have evolved. Obviously, increasing the mesh cutoff is one, but since the mesh dominates the computational expense for small to moderately sized systems, this is not the ideal solution. A more efficient technique is referred to as *grid-cell sampling*. Imagine an isolated atom being displaced relative to the underlying grid. The energy of the system will vary with the periodicity of the grid and may exhibit a behavior that to first order resembles either a simple sine or cosine wave (see Fig. 2.5). If this were the case, the energy and forces could be evaluated for two positions displaced by half of a grid spacing relative to each other and then averaged. The result would then be invariant to absolute position. While the situation for molecules and solids is more complex, with many Fourier components, the averaging over several displacements with respect to the grid points can lead to a reduction of the numerical error in the forces. This is the grid-cell sampling technique. On the face of it, this may not appear to represent a computational saving over increasing the mesh cutoff, since multiple energy/force evaluations appear to be required. However, it transpires that the breaking of translational invariance is much more significant for the forces than for the potential. Consequently, the self-consistent field procedure (see Section 2.2.5) can be performed for a single mesh position and then only the force evaluation need be conducted

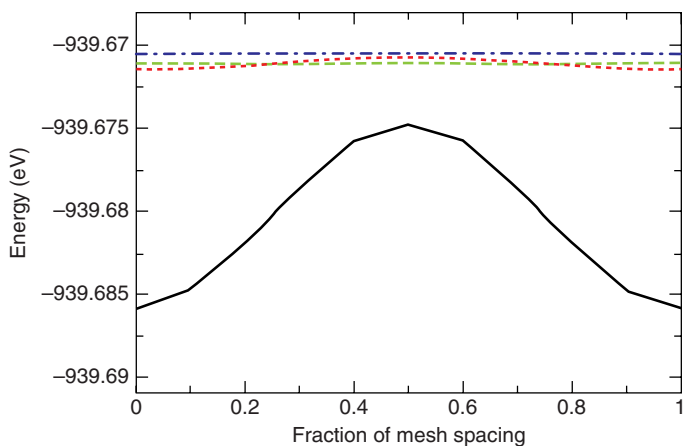


Fig. 2.5 Egg-box effect for a Ne atom with a DZP basis set and an energy shift of 0.01 Ry. The total energy is plotted as a function of atom position relative to the underlying mesh in fractions of the mesh spacing. The curves shown are for a cutoff of 150 Ry (—), 250 Ry (---), 450 Ry (-.-), and 250 Ry with a two-point grid cell sampling (....).

for multiple grid positions, thereby representing a considerable efficiency gain. The validity of this approximation can be seen in Fig. 2.5, where the grid-cell sampling correction largely removes the oscillation for a single atom.

There are several further methodologies for the reduction of space-rippling effects. For example, the basis functions and pseudopotentials can be explicitly Fourier filtered to reduce the components beyond the mesh cutoff.⁴⁰ Although this guarantees almost no invariance breaking for an isolated atom, it is difficult to limit the Fourier components that arise from combinations of basis functions from different atoms when they overlap. Ultimately, the only way to ensure that translational invariance is obeyed exactly is to use atom-centered integration grids, such as the radial grid techniques that have been employed for numerical basis sets.⁴¹ In such cases it is necessary to include the derivatives associated with the movement of the integration grid and the change of weights; terms that are often neglected for simplicity in some implementations, although there can also be numerical benefits to considering the grid to be fixed in some cases.

So far we have focused on the requirements to achieve linear scaling in the CPU time cost of a calculation. However, for a scheme to be useful it is also necessary for the memory usage of an algorithm to increase linearly while being small in absolute size; otherwise, this will become the bottleneck that prevents large-scale calculations from being performed. The memory usage of a SIESTA calculation can be dominated by one of two things. First, there is the storage of the matrices used in the construction of the Kohn–Sham equations and subsequent quantities, which consists of the Hamiltonian, overlap, density, and energy-density matrices. Second, storage of the nonzero orbital values at the mesh points can represent a large amount of data, especially for high mesh cutoffs, and is often the dominant memory use. Other mesh-related quantities are typically much smaller since there can be several tens of orbitals that contribute to each mesh point in a dense solid, whereas other arrays involve just one number per grid point. In cases where the storage of the orbitals on the grid becomes a limiting factor, there is a *direct-phi algorithm* in which orbital values are recomputed on the fly (analogous to the direct SCF concept in Gaussian methods, but for different quantities). This approach greatly reduces memory usage at the expense of additional computational cost.

The key to reducing the memory usage to linear scaling is to recognize that the Hamiltonian and overlap matrices are both sparse, due to the finite basis set range. Indeed, the number of nonzero elements per row or column remains fixed as the system size increases once the dimensions of the problem exceed the maximum interaction range. To exploit this, all matrices are stored in compressed row storage format, which is a standard technique for storing just the nonzero elements of a sparse array, at the cost of storing two extra integer pointer arrays to allow mapping of the stored elements to the dense matrix representation. To reduce this overhead, the overlap matrix is presently treated as possessing the same sparsity pattern as the Hamiltonian, even though it actually has a greater number of null elements. Along similar lines, the approximation is made that the density matrix obeys the same sparsity pattern as the Hamiltonian. Although the

density matrix is not physically constrained to be zero where the Hamiltonian is, the matrix elements that match the nonzero terms in the Hamiltonian capture the contributions that are important for the total energy.

2.2.5 Solving the Kohn–Sham Equations

Once the Hamiltonian and overlap matrices have been constructed, the next key step in any calculation is to solve for the new density matrix and then to iterate to self-consistency. The traditional approach to this problem has been to use matrix diagonalization to determine the Kohn–Sham eigenstates and then to use the coefficients of the basis functions to construct the next density matrix in the iterative sequence. This approach has the benefit of being able to determine both the occupied and unoccupied Kohn–Sham eigenstates, making it possible to compute properties such as the bandgap and densities of states. We note, of course, that these quantities should be interpreted with care since the Kohn–Sham wavefunctions do not represent true one-electron eigenstates as a result of self-interaction error.

For periodic systems it is necessary to integrate all observables across the Brillouin zone. This is usually approximated by a sum over discrete points in reciprocal space, and most commonly a uniform grid of k -points is chosen according to the scheme of Monkhorst and Pack.⁴² In the case of small unit cells it is necessary to take the same approach within the SIESTA methodology. One specific feature of the actual implementation is the standard method of choosing the grid size. Here a quantity called the *K-grid cutoff* can be chosen as a single value with units of distance. This methodology, due to Moreno and Soler,⁴³ exploits the relationship between reciprocal space sampling on a grid of k -points and the equivalent sampling through the use of supercells (e.g., a $2 \times 2 \times 2$ grid of k -points allows the same phase factors to be sampled as creating a $2 \times 2 \times 2$ supercell in real space). By specifying the real space supercell length that is desired, the equivalent reciprocal space sampling for a single cell can be determined. Through the use of a single control value it is possible to try to achieve consistent convergence across a range of different systems, provided that the bandgap and dispersion are similar. Of course, to be certain, the user must always check the convergence for each system.

The SIESTA methodology is designed to target large systems containing several hundreds to thousands of atoms. Thus, by the time such dimensions are reached, it is often a good approximation to consider only the Brillouin zone center (gamma point) for sampling purposes. This greatly simplifies the calculation and leads to a dramatic increase in computational speed since the Hamiltonian and overlap matrices become real rather than complex. Hence, from this point onward the assumption will be made that the integration over the Brillouin zone can be dropped and the system will be treated at the gamma point only.

Since there are many efficient machine-optimized libraries for dense matrix diagonalization, usually based on the LAPACK and BLAS routines, this approach can be highly competitive up to relatively large system sizes. However, the problem of cubic scaling and the need to work typically with dense matrices

ultimately dominates the computational cost. As a result, there has been considerable research over the last two decades into alternative techniques to determine the density matrix during self-consistency.^{44,45} Although improvements can be made to the diagonalization approach, such as solving for only the occupied states and iterative techniques for sparse matrices,⁴⁶ there is a need for more radical alternatives to achieve linear scaling.

The major difficulty when working with a localized atomic orbital basis set is the need to solve the generalized eigenvalue problem:

$$H\Psi = \varepsilon S\Psi \quad (2.8)$$

which involves first transforming the problem to a standard eigenvalue equation:

$$H'\Psi' = \varepsilon\Psi' \quad (2.9)$$

To do this implies the multiplication of the Hamiltonian by the effective inverse of the overlap matrix, which is often achieved indirectly through the use of Cholesky decomposition. Although both the Hamiltonian and overlap matrices may be very sparse, the difficulty is that the inverse of the overlap matrix is potentially much less sparse or even dense. While reordering techniques can reduce the degree of potential fill-in that occurs,⁴⁷ and other factorization schemes⁴⁸ may improve the level of sparsity of an effective inverted overlap matrix, the main challenge remains how to handle the nonorthogonality of the basis set while achieving linear scaling.

One of the first linear-scaling methods to be proposed was the *divide-and-conquer method* of Yang.⁴⁹ The principle of the approach is to reduce the total set of Kohn–Sham equations into a series of smaller overlapping subproblems from which the overall electron density could be constructed. For example, a partition could be created centered on each atom of the system whereby all Hamiltonian and overlap matrix elements within a cutoff distance are collected and solved using diagonalization. Provided that the cutoff radius is much smaller than the total system size, the cost of each separate diagonalization is much less than that for solving for the whole system together, and will be independent of the number of atoms for the entire problem. Hence, linear scaling is achieved while retaining the use of efficient matrix diagonalization for small problems. The remaining issue is how to reconstruct the total density from the sum of the subproblems, since the same contribution will appear in many different partitions. While first formulated in terms of the electron density itself, the divide-and-conquer scheme was later also cast in terms of the coefficients of a density matrix,⁵⁰ which is more appropriate here. Accordingly, the overlapping contributions can be partitioned as follows:

$$\rho_{\mu\nu} = \sum_{\alpha} P_{\mu\nu}^{\alpha} P_{\mu\nu}^{\alpha} \quad (2.10)$$

$$P_{\mu\nu}^{\alpha} = \begin{cases} 1 & \mu \in \alpha, \nu \in \alpha \\ \frac{1}{2} & \mu \in \alpha, \nu \notin \alpha \text{ or } \mu \notin \alpha, \nu \in \alpha \\ 0 & \mu \notin \alpha, \nu \notin \alpha \end{cases} \quad (2.11)$$

where α represents a partition label. The density matrix divide-and-conquer approach above has recently been implemented in SIESTA and shown to be an effective linear-scaling solution.⁵¹

Divide and conquer, as described above, is a simple and appealing approach to achieving linear scaling and has found considerable favor in some communities.⁵² However, it is important to recognize the limitations. First, for reasons of simplicity, the division of the Hamiltonian into submatrices is usually made based on a distance cutoff. However, decay lengths for matrix elements and the density matrix in different systems can vary substantially according to the nature of the bandgap, atoms involved, and so on. Therefore, truncation methods that are more adaptable to the physical problem are arguably superior. Second, the prefactor for the divide-and-conquer method is relatively high because a large amount of duplicate work is being performed (i.e., the same density matrix element is being computed many times over as a result of partition overlap). Third, all the subsystems are connected by the requirement that the Fermi energy must be globally the same; otherwise, electron density would flow from one partition to another until the chemical potential was equalized. Hence, once the submatrices have been diagonalized to obtain the local eigenspectrum, the population of the states cannot be determined without knowledge of the eigenvalues for all partitions simultaneously. Consequently, either the eigenvalues and eigenvectors for all subsystems must be stored, which represents a large amount of memory, or multiple diagonalizations must be performed for each partition, thus further raising the prefactor.

Because of the issues described above relating to divide and conquer, especially the second factor, there has been a search for more efficient algorithms that act on a single sparse density matrix. All methods involve dropping negligible contributions to the density matrix in one way or another, and are generally applicable to materials with a HOMO/LUMO or bandgap. Within this there are two general classes of method: those that impose truncation on the density matrix and those that invoke localization of the wavefunction, similar to divide and conquer. Considering first the former class of methods, they recognize that the density matrix can be used directly without recourse to the Kohn–Sham wavefunctions. However, in doing so, the conditions of N -representability must be observed (i.e., the density matrix must be derivable from an underlying antisymmetric N -particle wavefunction).⁵³ For an orthonormal basis set, the density matrix must therefore obey the following conditions:

- *Symmetry*. $D = D^T$, where D is the density matrix and D^T is its transpose.
- *Trace*. $\text{Tr}(D) = N_e$, where Tr represents the trace of a matrix and N_e is the number of electrons.
- *Idempotency*. $D^2 = D$, since eigenvalues are either 0 or 1.

Given these constraints, a trial density matrix can be converged to an approximation to the true density matrix by one invoking one of two broad classes of approach. In the first class, purification formulas are used to iteratively transform an approximate density matrix, D , into one that is more nearly idempotent, $\tilde{D}$. The most widely known purification transformation is that due to McWeeny⁵⁴:

$$\tilde{D} = 3D^2 - 2D^3 \quad (2.12)$$

although this has recently been generalized to higher orders by Niklasson.⁵⁵ The second class of density matrix-based methods involve minimization of an energy functional of the trial matrix, subject to the constraints above, based on the Hamiltonian. One of the best known examples is the method of Li et al.,⁵⁶ with further refinements by other groups.^{57,58}

All of the techniques above are valuable approaches to linear-scaling generation of the density matrix. However, they perform optimally for a basis set that is orthonormal. For a localized atom-centered basis there is the extra complexity of transforming the Hamiltonian or carrying the effective inverse of the overlap matrix through the formulas. For this reason, the SIESTA methodology currently employs a different class of method that focuses on the localization of the wavefunction. It is possible to perform a unitary transformation of a set of extended wavefunctions into a localized set of states known as *Wannier functions*. Although this is a nonunique transformation, there are well-developed approaches for this process, such as maximally localized Wannier functions.⁵⁹ It should also be noted that when discussing the locality of these Wannier functions, this usually implies an exponential decay rather than strict confinement.

The culmination of several developments led to the Kim–Mauri–Galli (KMG)⁶⁰ order- N functional for linear-scaling construction of the Wannier functions, and thereby the density matrix. This represents the default approach for achieving true linear scaling within SIESTA. Here the Wannier functions are forced to be strictly local through the use of a cutoff radius, so the approach has much in common with the philosophy of the density matrix divide-and-conquer method, but avoids the duplicate generation of matrix elements. Each atomic center carries a number of localized Wannier functions (LWFs), such that the total number of localized states exceeds the number of occupied states. The number assigned to a given atom is specified by $(N_e + 2)/2$ for KMG.

Within the KMG method, the orbital coefficients within the localized states are determined by minimization of a functional that depends on the Hamiltonian and overlap matrix, as well as the chemical potential, μ , of the electrons:

$$U_{\text{KMG}} = 2 \sum_{ij} (2\delta_{ij} - S_{ij})(H_{ij} - \mu S_{ij}) \quad (2.13)$$

Here the use of the distinct subscripts i and j indicate that the Hamiltonian and overlap matrices have been transformed to the basis of the localized Wannier functions according to the coefficients of the orbital basis set within the LWFs.

The conceptual key to achieving linear scaling is that this expression avoids the need for explicit orthogonalization, but instead, imposes an energy penalty for the deviation from orthogonality [the first term in parentheses in expression (2.13) represents a truncated polynomial expansion of the inverse of the overlap matrix]. During the minimization the localized states therefore gradually become orthonormal until this condition is met at convergence. It is important to note that this minimization is an extra iterative step that lies within each self-consistent field (SCF) cycle.

The greatest challenge within the KMG approach is the determination of the chemical potential, which represents the Fermi energy of the system. Because there is no determination of eigenstates in this method, the Fermi energy is not computed directly, although techniques exist to evaluate subsets of the eigenvalue spectrum of a matrix at a considerably lower cost than full diagonalization. However, this extra calculation is generally undesirable and would have to be repeated at every step of the self-consistent field procedure, since the Fermi energy changes as a function of the density matrix.

In the KMG method, the chemical potential need not be exactly equal to the true Fermi energy; it must just lie above the top of the valence band/HOMO and below the conduction band/LUMO. For an insulator, or even many semiconducting materials, the bandgap is sufficiently large and the Fermi energy is known to be in the vicinity of zero, such that it is possible to “guess” a value of the chemical potential that satisfies this requirement. Alternatively, a trial-and-error approach can be used. If the chemical potential is set too low, the number of electrons in the system will lie below the actual number, while if it is set too high, the converse will be true. Should the value lie within a band, the minimization procedure can diverge, again providing an indication that the value chosen is not suitable.

Where it can be afforded, a practical scheme that avoids the difficulty in setting the chemical potential correctly is the following. First, a small number of iterations of diagonalization are performed to obtain a good approximation to the density matrix, and the Fermi energy can be determined, as well as being seen to be stable. Having written out the unconverged density matrix, the calculation can then be restarted to use the KMG scheme, taking the Fermi energy from this calculation. Although the first step may represent a considerable initial overhead for the initial geometry, the cost rapidly becomes insignificant if an extensive geometry optimization or molecular dynamics simulation is subsequently to be run.

Let us now consider the convergence behavior of the minimization of the KMG functional, assuming that the chemical potential has been chosen correctly to lie within the correct energy window. In Table 2.3 the number of iterations required to achieve minimization of the KMG functional is quoted for the simple case of bulk silicon. There are several trends to note in the behavior. First, the initial minimization of the orbital coefficients within the LWFs is very slow to converge and can take over 1000 iterations. This is because the initial guess for the localized states involves the use of random coefficients to avoid artificially biasing the symmetry of the solutions. Minimization uses conjugate gradients, and therefore

TABLE 2.3 Number of Iterations Required to Converge the Localized Wannier Functions at Each of the First Five SCF Iterations and the Total Number of SCF Iterations Required for Convergence for Bulk Si^a

RcLWF (bohr)	Iter. 1	Iter. 2	Iter. 3	Iter. 4	Iter. 5	No. of SCF Cycles
6	502	16	6	6	6	7
8	902	171	30	18	10	12
10	1202	302	302	100	6	10
12	902	302	5	5	7	13
14	1502	302	7	7	5	9
16	902	302	1	1	3	8

^aThe basis set and parameters are as in Fig. 2.6.

convergence is naturally slow. Attempts at using more sophisticated minimization algorithms have, however, generally proved no more effective. Second, subsequent SCF cycles require progressively fewer minimization steps since the LWFs from the previous cycle are reused and the number of iterations drops rapidly to less than 10. Third, the number of iterations required can decrease as the radius of confinement for the LWF (RcLWF) increases, especially for the later SCF cycles. Consequently, a more accurate calculation can actually be as fast overall, so the use of very small radii to confine the LWFs is not advisable.

The variation of calculation quality as a function of the radius used for the localized states is illustrated in Fig. 2.6 for the case of bulk silicon. As can be

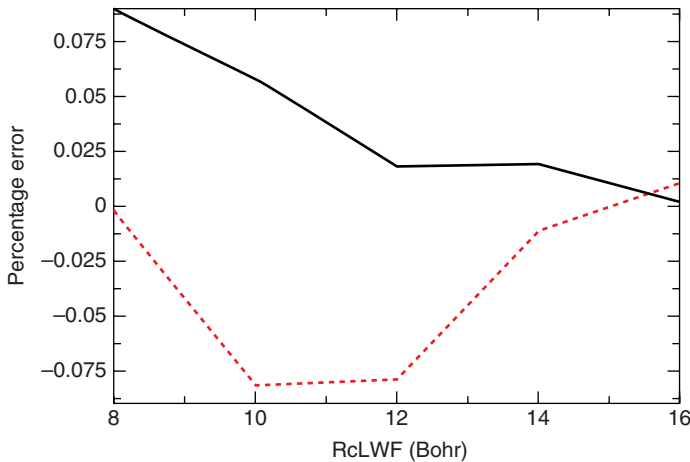


Fig. 2.6 Percentage error in the total energy (—) and optimized lattice parameter (----) as a function of the localization radius for bulk silicon. Calculations are based on a $3 \times 3 \times 3$ supercell containing 216 atoms for a SZ basis set and an energy shift of 0.01 Ry. The mesh cutoff is 250 Ry and the converged reference is for diagonalization using the gamma point only. The converged values for the total energy per atom and single-cell lattice parameter are -106.98172 eV and 5.541 Å, respectively.

seen, sensitivity to the localization radius varies according to the property being studied. While the energy converges to within an acceptable error (i.e., less than ambient thermal energy) relatively quickly, the error in lattice parameter is slightly larger, and the curvature-related properties, such as bulk modulus, greater still. Of course, the rate of convergence is also dependent on the bandgap, which influences the decay of the states, and therefore testing the influence of this approximation is important for each material of interest.

Before concluding the topic of solving the Kohn–Sham equations, it is worth briefly mentioning two topics that are common to all numerical implementations: spin and SCF convergence acceleration.

For the case where diagonalization is used to achieve self-consistency, the SIESTA code allows the user to include spin polarization where either the total spin may be fixed or the electrons allowed to flow between spin states to attain a common Fermi energy. In addition, there is the option to use noncollinear spin to describe spiral magnetic states.⁶¹ If using a linear-scaling solver, in particular the KMG form, the options for treatment of spin are more limited. Spin polarization is still allowed, but control of the spin state is achieved via the specification of two separate values of the chemical potential for alpha and beta spin.

Turning to the second topic, there are a number of methods for assisting the convergence of the self-consistent field procedure that might otherwise diverge or require a larger number of iterations. The simplest technique is static mixing, which may be applied to either the Hamiltonian or the density matrix, but is applied more conventionally to the latter. Here the density matrix for a new iteration is taken to be a combination of the old density matrix with the undamped result of the current solution step, i (either diagonalization or order N), in a proportion controlled by the mixing parameter, α :

$$D_{\text{in}}^{i+1} = \alpha D_{\text{out}}^i + (1 - \alpha) D_{\text{in}}^i \quad (2.14)$$

Typically, values of the mixing parameter in the range 0.05 to 0.35 are used, where a small value is used for a poorly convergent system, while the larger value is appropriate for a wide-gap material. If too large a value is used, there is a risk that the SCF procedure may start to oscillate. Even in cases that are intrinsically convergent, the iterative process may take numerous cycles to converge as a result of the damped mixing, so there are acceleration techniques to deal with this. SIESTA has the option to use either Pulay mixing⁶² or the Broyden–Vanderbilt–Louie–Johnson scheme,⁶³ both of which store information from previous iterations, such as the density matrix, and then extrapolate forward. These methods can reduce the number of iterations considerably, though as a caution it should be noted that they could also prevent convergence in some problematic cases. Although there are numerous other convergence techniques, such as level shifting,⁶⁴ dynamic mixing, and exponential transformation,⁶⁵ these have yet to be combined with the SIESTA implementation but may be available in the future.

2.3 FUTURE PERSPECTIVES

This chapter has sought to present a perspective on the key background aspects of the SIESTA methodology that will be of value to a new user of the technique. A complementary chapter in this volume (Chapter 11) highlights some applications of the SIESTA approach that are possible, with a focus on the area of nanoscience.

Unlike other mature computational methods, the SIESTA methodology could be considered an evolving approach that may develop further in the future as we learn about the optimal methods for creating numerical basis sets in particular. In addition, implementation in the SIESTA code will develop in response to new trends and advances in the field of density functional theory, where this is compatible with linear scaling. For example, there is no reason why the method cannot be extended to encompass Hartree–Fock exchange, hybrid functionals, and localized post-HF correlation methods, as has been the case for other solid-state codes.

Acknowledgments

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REFERENCES

1. Hohenberg, P.; Kohn, W. *Phys. Rev.* **1964**, *136*, B864.
2. Dunlap, B. I.; Connolly, J. W. D.; Sabin, J. R. *J. Chem. Phys.* **1979**, *71*, 3396.
3. Kohn, W. *Phys. Rev. Lett.* **1996**, *76*, 3168.
4. Artacho, E.; Sánchez-Portal, D.; Ordejoń, P.; García, A.; Soler, J. M. *Phys. Status Solidi (b)* **1999**, *215*, 809.
5. Bock, N.; Challacombe, M.; Chee-Kwan, G.; Henkleman, G.; Nemeth, K.; Niklasson, A.-M.-N.; Odell, A.; Schwegler, E.; Tymczak, C.-J.; Weber, V. Los Alamos National Laboratory (LA-CC 01-2. LA-CC-04-086).
6. Shao, Y. et al., *PCCP* **2006**, *8*, 3172.
7. VandeVondele, J.; Krack, M.; Mohamed, F.; Parrinello, M.; Chassaing, T.; Hutter, J. *Comput. Phys. Commun.* **2005**, *167*, 103.
8. Soler, J. M.; Artacho, E.; Gale, J. D.; García, A.; Junquera, J.; Ordejon, P.; Sanchez-Portal, D. *J. Phys. Condens. Matter* **2002**, *14*, 2745.
9. Kenny, S. D.; Horsfield, A. P.; Fujitani, H. *Phys. Rev. B* **2000**, *62*, 4899.
10. Ozaki, T. *Phys. Rev. B* **2003**, *67*, 155108.
11. Bowler, D. R.; Choudhury, R.; Gillan, M. J.; Miyazaki, T. *Phys. Status Solidi (b)* **2006**, *243*, 989.

12. Skylaris, C. K.; Haynes, P. D.; Mostofi, A. A.; Payne, M. C. *J. Phys. Condens. Matter* **2008**, *20*, 064209.
13. Perdew, J. P. *Physica B* **1991**, *172*, 1.
14. Perdew, J. P.; Kurth, S.; Zupan, A.; Blaha, P. *Phys. Rev. Lett.* **1999**, *82*, 2544.
15. Becke, A. D. *J. Chem. Phys.* **1993**, *98*, 5648.
16. Anisimov, V. I.; Zaanen, J.; Andersen, O. K. *Phys. Rev. B* **1991**, *44*, 943.
17. Kleinman, L.; Bylander, D. M. *Phys. Rev. Lett.* **1982**, *48*, 1425.
18. Hamann, D. R.; Schlüter, M.; Chiang, C. *Phys. Rev. Lett.* **1979**, *43*, 1494.
19. Troullier, N.; Martins, J. L. *Phys. Rev. B* **1991**, *43*, 1993.
20. Kerker, G. P. *J. Phys. C* **1980**, *13*, L189.
21. Vanderbilt, D. *Phys. Rev. B* **1990**, *41*, 7892.
22. Blöchl, P. E. *Phys. Rev. B* **1994**, *50*, 17953.
23. Bilić, A.; Gale, J. D. *Phys. Rev. B* **2009**, *79*, 174107.
24. Louie, S. G.; Froyen, S.; Cohen, M. L. *Phys. Rev. B* **1982**, *26*, 1738.
25. Ahlrichs, R.; Taylor, P. R. *J. Chim. Phys. Phys. Chim. Biol.* **1981**, *78*, 315.
26. Becke, A. D.; Dickson, R. M. *J. Chem. Phys.* **1990**, *92*, 3610.
27. Delley, B. *J. Chem. Phys.* **1990**, *92*, 508.
28. Sankey, O. F.; Niklewski, D. J. *Phys. Rev. B* **1989**, *40*, 3979.
29. Junquera, J.; Paz, O.; Sanchez-Portal, D.; Artacho, E. *Phys. Rev. B* **2001**, *64*.
30. Anglada, E.; Soler, J. M.; Junquera, J.; Artacho, E. *Phys. Rev. B* **2002**, *66*, 205101.
31. Sanchez-Portal, D.; Ordejon, P.; Artacho, E.; Soler, J. M. *Int. J. Quantum Chem.* **1997**, *65*, 453.
32. Causa, M.; Dovesi, R.; Pisani, C.; Roetti, C. *Phys. Rev. B* **1986**, *33*, 1308.
33. García-Gil, S.; García, A.; Lorente, N.; Ordejon, P. *Phys. Rev. B* **2009**, *79*, 075441.
34. Boys, S. B.; Bernardi, F. *Mol. Phys.* **1970**, *19*, 553.
35. Greengard, L.; Rokhlin, V. *J. Comput. Phys.* **1987**, *73*, 325.
36. Chelikowsky, J. R.; Troullier, N.; Wu, K.; Saad, Y. *Phys. Rev. B* **1994**, *50*, 11355.
37. Brandt, A. *Math. Comput.* **1977**, *31*, 333.
38. Briggs, E. L.; Sullivan, D. J.; Bernholc, J. *Phys. Rev. B* **1995**, *52*, R5471.
39. Artacho, E.; Anglada, E.; Dieguez, O.; Gale, J. D.; García, A.; Junquera, J.; Martin, R. M.; Ordejon, P.; Pruneda, J. M.; Sanchez-Portal, D.; Soler, J. M. *J. Phys. Condens. Matter* **2008**, *20*, 064208.
40. Anglada, E.; Soler, J. M. *Phys. Rev. B* **2006**, *73*, 115122.
41. Becke, A. D. *J. Chem. Phys.* **1988**, *88*, 2547.
42. Monkhorst, H. J.; Pack, J. D. *Phys. Rev. B* **1976**, *13*, 5188.
43. Moreno, J.; Soler, J. M. *Phys. Rev. B* **1992**, *45*, 13891.
44. Goedecker, S. *Rev. Mod. Phys.* **1999**, *71*, 1085.
45. Bowler, D. R.; Fattetbert, J. L.; Gillan, M. J.; Haynes, P. D.; Skylaris, C. K. *J. Phys. Condens. Matter* **2008**, *20*, 290301.
46. Lehoucq, R. B.; Sorensen, D. C.; Yang, C. *ARPACK Users' Guide: Solution of Large-Scale Eigenvalue Problems with Implicitly Restarted Arnoldi Methods*, Society for Industrial and Applied Mathematics, Philadelphia, 1998.
47. Karypis, G.; Kumar, V. *SIAM J. Sci. Comput.* **1999**, *20*, 359.

48. Benzi, M.; Meyer, C. D.; Tuma, M. *SIAM J. Sci. Comput.* **1996**, *17*.
49. Yang, W. *Phys. Rev. Lett.* **1991**, *66*, 1438.
50. Yang, W.; Lee, T.-S. *J. Chem. Phys.* **1995**, *103*, 5674.
51. Cankurtaran, B. O.; Gale, J. D.; Ford, M. J. *J. Phys. Condens. Matter* **2008**, *20*, 294208.
52. van der Vaart, A.; Gogonea, V.; Dixon, S. L.; Merz, Jr, K. M. *J. Comput. Chem.* **2000**, *21*, 1494.
53. Coleman, A. J. *Rev. Mod. Phys.* **1963**, *35*, 668.
54. McWeeny, R. *Rev. Mod. Phys.* **1960**, *32*, 335.
55. Niklasson, A. M. N. *Phys. Rev. B* **2002**, *66*, 155115.
56. Li, X. P.; Nunes, R. W.; Vanderbilt, D. *Phys. Rev. B* **1993**, *47*, 10891.
57. Millam, J. M.; Scuseria, G. E. *J. Chem. Phys.* **1997**, *106*, 5569.
58. Challacombe, M. *J. Chem. Phys.* **1999**, *110*, 2332.
59. Mazari, N.; Vanderbilt, D. *Phys. Rev. B* **1997**, *56*, 12847.
60. Kim, J.; Mauri, F.; Galli, G. *Phys. Rev. B* **1995**, *52*, 1640.
61. García-Suárez, V. M.; Newman, C. M.; Lambert, C. J.; Pruneda, J. M.; Ferrer, J. J. *Phys. Condens. Matter* **2004**, *16*, 5453.
62. Pulay, P. *Chem. Phys. Lett.* **1980**, *73*, 393.
63. Johnson, D. D. *Phys. Rev. B* **1988**, *38*, 12807.
64. Saunders, V. R.; Hillier, I. H. *Int. J. Quantum Chem.* **1973**, *7*, 699.
65. Douady, J.; Ellinger, Y.; Subra, R.; Levy, B. *J. Chem. Phys.* **1980**, *72*, 1452.

3 Large-Scale Plane-Wave-Based Density Functional Theory: Formalism, Parallelization, and Applications

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The basic density functional formalism presented in Chapter 1 is applied to the simulation of large materials, solutions, and molecules using plane-wave basis sets. This parallels the applications developed in Chapter 2 for similar systems using atomic basis sets. Much attention is focused on the pseudopotentials that describe the interaction of the atomic nuclei and their inner-shell electrons (“ions”) with the valence electrons. Methods for simulating charged systems are described, as well as the use of hybrid density functionals in simulations of chemical properties. Advances in numerical methods and software (contained in the NWChem package) are described that allow for both geometry optimization and multi-picosecond time scale Car–Parinello molecular dynamic simulations of very large systems. Sample applications including the structure of hematite and the aqueous solvation of cations are described.

3.1 INTRODUCTION

The development of fast and efficient ways to calculate density functional theory (DFT) using plane-wave basis sets^{1–8} combined with parallel supercomputers^{7,9–16} has opened the door to new classes of large-scale first-principles simulations. It is now routine at this level of theory to perform simulations containing hundreds of atoms,¹⁷ and simulations containing over 1000 atoms are feasible on today's parallel supercomputers,²⁰ making realistic descriptions of a variety of systems possible. Several techniques are responsible for the efficiency of plane-wave DFT programs. The central feature is the representation of the electronic orbitals in terms of a plane-wave basis set. In this representation, one can take advantage of fast fourier transform (FFT) algorithms²¹ for fast calculations of total energies and forces. Periodic boundary conditions (PBCs) are also incorporated automatically as a result. However, the plane-wave basis sets do have an important shortcoming: their inefficient description of the electronic wavefunction in the vicinity of the atomic nucleus or core region. Valence wavefunctions vary rapidly in this region and much more slowly in the interstitial regions (or bonding regions) (see Fig. 3.1). Accurate description of the rapid variation of the wavefunction inside the atomic or core region would require very large plane-wave basis sets.

The pseudopotential plane-wave (PSPW) method can be used to resolve this problem.^{22–25} In this approach the fast-varying core regions of the atomic potentials and the core electrons are removed or pseudized and replaced by smoothly varying pseudopotentials. The pseudopotentials are constructed such that the scattering properties of the resulting pseudoatoms are the same as those of the original atoms.^{26,27} The rationale behind the pseudopotential approach is that changes in the electronic wavefunctions during bond formation occur only in the valence region, and therefore proper removal of the core from the problem should not affect the prediction of bonding properties of the system. The projector augmented plane-wave (PAW) method developed by Blöchl is a further enhancement of the pseudopotential in that it addresses some of the shortcomings encountered in a traditional PSPW approach. Since the main computational algorithms are essentially the same in the two approaches, we will not specifically discuss the PAW approach and refer the reader to comprehensive reviews.^{8,15,28–31}

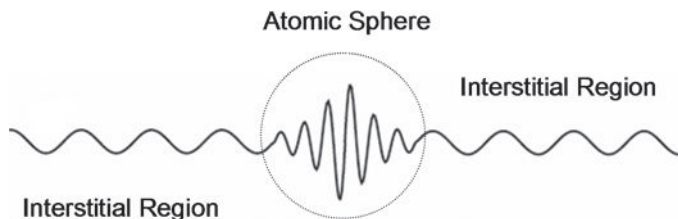


Fig. 3.1 Valence wavefunction.

3.2 PLANE-WAVE BASIS SET

Plane waves are natural for solid-state applications, since crystals are readily represented using periodic boundary conditions where the system is enclosed in a unit cell defined by the primitive lattice vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$, as shown in Fig. 3.2. However, periodic plane-wave basis sets can also be used for molecular simulations as long as the unit cell is large enough to minimize the image interactions between cells. In terms of plane waves, the molecular orbitals are represented as

$$\psi_i(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} \sum_{\{\mathbf{G}\}} \psi_i(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}} \quad (3.1)$$

where Ω is the volume of the primitive cell ($\Omega = [\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3] = \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)$). Since the system is periodic, the plane-wave expansion must consist of only the plane waves $e^{i\mathbf{G} \cdot \mathbf{r}}$ that have the periodicity of the lattice, which can be determined using the constraint

$$e^{i\mathbf{G} \cdot (\mathbf{r} + \mathbf{L})} = e^{i\mathbf{G} \cdot \mathbf{r}} \quad (3.2)$$

where $\mathbf{L}$ is the Bravais lattice vector ($\mathbf{L} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3$, with $n_1, n_2, n_3 = \text{integers}$) and $\mathbf{G}$ represents the wave vectors, which can be defined in terms of the reciprocal lattice vectors:

$$\mathbf{G}_{i_1 i_2 i_3} = \left(i_1 - \frac{N_1}{2}\right) \mathbf{b}_1 + \left(i_2 - \frac{N_2}{2}\right) \mathbf{b}_2 + \left(i_3 - \frac{N_3}{2}\right) \mathbf{b}_3 \quad (3.3)$$

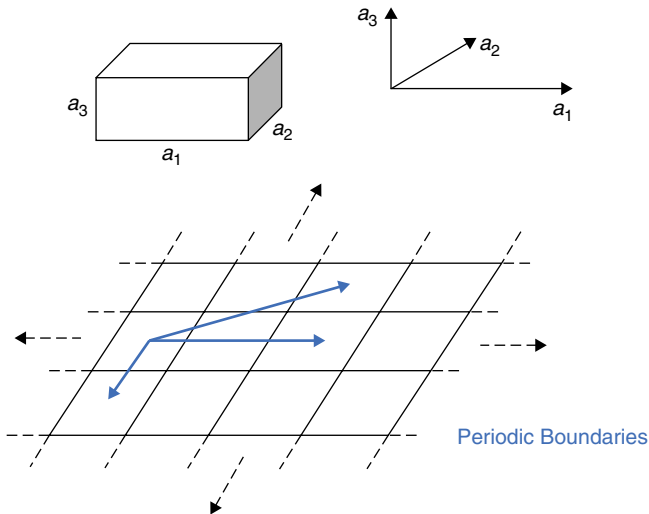


Fig. 3.2 Unit cell in periodic boundary conditions. The solid arrows represent the Bravais lattice vectors.

where N_1 , N_2 , and N_3 are chosen sizes of the lattice vector grid, which can range from 1 to ∞ ; i_1 , i_2 , and i_3 are integers defined in the ranges of $1 \cdots N_1$, $1 \cdots N_2$, and $1 \cdots N_3$, respectively, and

$$\mathbf{b}_1 = 2\pi \frac{\mathbf{a}_2 \times \mathbf{a}_3}{\Omega} \quad \mathbf{b}_2 = 2\pi \frac{\mathbf{a}_3 \times \mathbf{a}_1}{\Omega} \quad \mathbf{b}_3 = 2\pi \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\Omega} \quad (3.4)$$

are the primitive reciprocal lattice vectors. A real space grid that is dual to the reciprocal lattice grid can be defined and is given by

$$\mathbf{r}_{i_1 i_2 i_3} = \left(\frac{i_1}{N_1} - \frac{1}{2} \right) \mathbf{a}_1 + \left(\frac{i_2}{N_2} - \frac{1}{2} \right) \mathbf{a}_2 + \left(\frac{i_3}{N_3} - \frac{1}{2} \right) \mathbf{a}_3 \quad (3.5)$$

The transformation between the reciprocal and real space representations is achieved via the discrete Fourier transform:

$$\begin{aligned} f(\mathbf{r}_{i_1 i_2 i_3}) &= \frac{1}{\sqrt{\Omega}} \sum_{j_1=1}^{N_1} \sum_{j_2=1}^{N_2} \sum_{j_3=1}^{N_3} F(\mathbf{G}_{j_1 j_2 j_3}) e^{i\mathbf{G}_{j_1 j_2 j_3} \cdot \mathbf{r}_{i_1 i_2 i_3}} \\ F(\mathbf{G}_{i_1 i_2 i_3}) &= \frac{\sqrt{\Omega}}{N_1 N_2 N_3} \sum_{j_1=1}^{N_1} \sum_{j_2=1}^{N_2} \sum_{j_3=1}^{N_3} f(\mathbf{r}_{j_1 j_2 j_3}) e^{-i\mathbf{G}_{j_1 j_2 j_3} \cdot \mathbf{r}_{i_1 i_2 i_3}} \end{aligned} \quad (3.6)$$

These transformations can be calculated efficiently via fast Fourier transform (FFT) algorithms.²¹

In typical plane-wave calculations, the plane-wave expansion is truncated in that only the reciprocal lattice vectors whose kinetic energy is lower than a predefined maximum cutoff energy,

$$\frac{1}{2} |\mathbf{G}|^2 < E_{\text{cut}} \quad (3.7)$$

are kept in the expansion, while the rest of the coefficients are set to zero. The density is also expanded using plane waves,

$$\rho(\mathbf{r}) = \sum_i \psi_i^*(\mathbf{r}) \psi_i(\mathbf{r}) = \sum_{\mathbf{G}} \rho(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}} \quad (3.8)$$

Since the density is the square of the wavefunctions, it can vary twice as rapidly. Hence, for translational symmetry to be formally maintained, the density should contain eight times more plane waves than the corresponding wavefunction expansion. Often, the density cutoff energy is chosen to be the same as the wavefunction cutoff energy; this approximation is known as *dualing*.

An added complication arises in the calculation of crystalline systems. In these systems the orbitals may have long-wavelength contributions that span over a large number of primitive unit cells. To account for the infinite number of electrons in the periodic system, an infinite number of k -points are required.

The Bloch theorem, however, helps restate this problem of calculating an infinite number of wavefunctions to one of calculating a finite number of wavefunctions at an infinite number of k -points or BZ points:

$$\psi_i(\mathbf{r}) = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{\Omega}} \sum_{\mathbf{G}} \psi_i(\mathbf{G}) e^{i\mathbf{G}\cdot\mathbf{r}} \quad (3.9)$$

Since the occupied states at each k -point contribute to the electronic potential, an infinite number of calculations are required in principle. However, experience tells us that wavefunctions at k -points that are nearby are almost identical. As a result, one can redefine the k -point summations or integrals in the DFT expressions to those that just span only a small set of special k -points in the Brillouin zone. There are a number of prescriptions to generate these special points. Since a detailed discussion of the various prescriptions is beyond the scope of this chapter, we refer the reader to more comprehensive papers and reviews.^{1,32–34} Obviously, for molecular systems there is no need for k -point sampling. Systems with large unit cells (disordered systems) and large bandgap systems also do not require or require a limited k -point sampling because the long-wavelength components are typically contained within the unit cell as in the former, or the electronic states are localized as in the latter. In this work we restrict ourselves to the Γ -point ($\mathbf{k} = 0$), since we are interested in isolated systems and systems with large unit cells.

3.3 PSEUDOPOTENTIAL PLANE-WAVE METHOD

The pseudopotential plane-wave method (PSPW) has its roots in the work on orthogonalized plane waves³⁵ and core state projector methods,²³ and empirical pseudopotentials have been used for some time in plane-wave calculations.^{25,36–38} However, this method was not considered entirely reliable until the development of norm-conserving pseudopotentials.^{26,39–41} It is currently a very popular method for solving DFT equations. In particular, PSPW can perform ab initio molecular dynamics very efficiently,³ and treat unit cells up to a couple of thousand atoms.^{4,6,7,17} Another advantage of PSPW methods is their transferability across a wide range of systems. In this section we describe implementation of the norm-conserving PSPW method. Formulas for the total energy, wavefunction gradient, and nuclear gradients are given in terms of a plane-wave basis set at the Γ -point.

3.3.1 Pseudopotentials

Pseudopotentials (effective core potentials) are based on two observations. First, in almost any system one can identify a set of *core orbitals* which change little from their atomic counterparts. Second, the remainder, or *valence orbitals*, acquire their oscillating behavior as a result of their orthogonality to the core orbitals. This also keeps valence electrons away from the core. In the

pseudopotential approximation the original atoms that constitute a given chemical system are modified by removing core states and replacing their effect via a repulsive pseudopotentials. This removes the rapid oscillations from the atomic valence orbitals and allows efficient application of plane-wave basis set expansion. The resulting pseudoatoms will in general acquire a nonlocal potential term.

There have been many ways to define pseudopotentials.^{1,23,24,27,40–58} The original procedure of Phillips and Kleinman formed pseudopotentials from pseudo wavefunctions in which atomic core wavefunctions were added to the valence wavefunctions.²³ Unfortunately, this procedure and related later developments^{44–46} resulted in “hard-core” potentials that contained singularities. These pseudopotentials were not useful in plane-wave calculations, since the nonregularized singularities could not be expanded using a reasonable number of plane waves. At about the same time, “soft-core” empirical pseudopotentials were developed.^{24,25,36–38} These potentials, which were made up of smooth functionals with a few parameters, were fitted to reproduce one-electron eigenvalues and orbital shapes. Such soft-core pseudopotentials were readily expanded using plane waves. However, pseudopotentials generated in this way were not transferable, yielding pseudowavefunctions that were different from the true valence wavefunctions by a few percent outside the core. Later it was realized that soft-core pseudopotentials needed to maintain norm conservation for them to be transferable.^{26,39–41} The principle of norm conservation states that if the charge of the real valence densities and the pseudovalence densities are identical inside the core region, the real valence wavefunction and pseudowavefunction will be identical outside the core region. This procedure was refined over the years and now most soft-core pseudopotentials are designed to have the following properties⁵⁴:

- The valence pseudowavefunction generated from the pseudopotentials should not contain nodes.
- The pseudowavefunctions near zero approach $\tilde{\varphi}_l(\mathbf{r}) \rightarrow \mathbf{r}^{l+1}$. This criterion removes the singularities from the pseudopotential.
- Real and pseudovalence eigenvalues agree for a chosen “prototype” atomic configuration ($\epsilon_l^{\text{AE}} = \epsilon_l^{\text{PP}}$).
- Real and pseudoatomic valence wavefunctions agree beyond a chosen core radius $\mathbf{r}_c$.
- Real and pseudovalence charge densities agree for $\mathbf{r} > \mathbf{r}_c$.
- Logarithmic derivatives and the first energy derivatives agree for $\mathbf{r} > \mathbf{r}_c$.

These types of pseudopotentials are called *norm-conserving pseudopotentials*.

Here we review briefly the construction of pseudopotentials suggested by Troullier and Martins.⁵⁴ The first step is to solve the radial Kohn–Sham equation self-consistently for a given atom:

$$\left[-\frac{1}{2} \frac{d^2}{dr^2} + \frac{l(l+1)}{2r^2} + V_{\text{AE}}(r) \right] \varphi_{nl}(r) = \epsilon_{nl} \varphi_{nl}(r) \quad (3.10)$$

to obtain a set of radial atomic orbitals, $\{\varphi_{nl}\}$. The self-consistent potential $V_{\text{AE}}(r)$ is given by

$$V_{\text{AE}}(r) = -\frac{Z}{r} + \int \frac{\rho(r')}{|r - r'|} dr' + V_{\text{xc}}(\rho(r)) \quad (3.11)$$

where the density, $\rho(r)$, is given by the sum of the occupied orbital densities,

$$\rho(r) = \sum_{nl} f_{nl} \left| \frac{\varphi_{nl}(r)}{r} \right|^2 \quad (3.12)$$

and $V_{\text{xc}}(\rho(r))$ is the exchange–correlation potential. In Eq. (3.12), f_{nl} is the occupancy of the nl state. Pseudopotential construction starts by introducing a smooth pseudovalence wavefunction, $\tilde{\varphi}_l(r)$, such that it and at least one derivative continuously approaches the all-electron valence wavefunction, $\varphi_l^{\text{AE}}(r)$, beyond a chosen cutoff radius r_{cl} . In addition, to avoid a hard-core pseudopotential (i.e., a singularity in the pseudopotential), the pseudowavefunctions near zero have to approach $\tilde{\varphi}_l(r) \rightarrow r^{l+1}$. The actual functional form of $\tilde{\varphi}_l(r)$ could be chosen in many different ways. Troullier and Martins suggested the following form for the pseudowavefunctions:

$$\tilde{\varphi}_l(r) = \begin{cases} \varphi_l^{\text{AE}}(r) & \text{if } r \geq r_{cl} \\ r^{l+1} e^{p(r)} & \text{if } r < r_{cl} \end{cases} \quad (3.13)$$

where $p(r)$ is a polynomial of order 12:

$$p(r) = \sum_{n=0}^6 c_n r^{2n} \quad (3.14)$$

The seven coefficients are then determined using the following constraints:

- Norm conservation with the core
- Continuity of the pseudowavefunction and its first four derivatives at r_{cl}
- The curvature of the screened pseudopotential at the origin defined to be zero

An explicit procedure to do this can be found in the paper by Troullier and Martins.⁵⁴

The next step is to generate the screened pseudopotentials, which are easily obtained by inverting the radial Schrödinger equation:

$$\begin{aligned} V_l^{\text{scr}}(r) &= \varepsilon_l - \frac{l(l+1)}{2r^2} + \frac{1}{2\tilde{\varphi}_l(r)} \frac{d^2}{dr^2} \tilde{\varphi}_l(r) \\ &= \begin{cases} V_{\text{AE}}(r) & \text{if } r \geq r_{cl} \\ \varepsilon_l + \frac{l+1}{r} \frac{p'(r)}{2} + \frac{p''(r) + [p'(r)]^2}{2} & \text{if } r < r_{cl} \end{cases} \end{aligned} \quad (3.15)$$

Three important properties of pseudopotentials result from Eq. (3.15). First, the pseudopotential will not be continuous if the pseudowavefunction does not have at least two continuous derivatives. Second, a hard-core singularity will be present in Eq. (3.15) if $\tilde{\varphi}_l(r) \neq r^{l+1}$ at zero. Third, the pseudopotentials may contain discontinuities if the pseudowavefunctions have nodes.

For rare gases, where all the electrons are in the core, these are the correct pseudopotentials to use. However, in cases where one wants to include valence electrons in a calculation, the screened potentials must be unscreened to remove the effects of the valence electrons from the pseudopotential, thus generating an *ionic* pseudopotential. This is done by subtracting off the Hartree and exchange–correlation potentials that are calculated from the valence pseudowavefunctions from the screened pseudopotential:

$$V_l^{\text{ion}}(r) = V_l^{\text{scr}}(r) - \frac{4\pi}{r} \int_0^r \tilde{\rho}(r') r'^2 dr' - 4\pi \int_r^\infty \tilde{\rho}(r') r' dr' - V_{\text{xc}}(\tilde{\rho}(r)) \quad (3.16)$$

where

$$\tilde{\rho}(r) = \sum_l f_l \left| \frac{\tilde{\varphi}_l(r)}{r} \right|^2 \quad (3.17)$$

In Section 3.4.8, f_l is the occupancy of the valence state l .

Based on these atomic pseudopotentials, the pseudopotential for the entire system takes the form

$$V_{\text{psp}}(\mathbf{r}, \mathbf{r}') = \sum_{l=0}^{l_{\text{max}}} \sum_{m=-l}^l Y_{lm}(\hat{\mathbf{r}}) (V_l^{\text{ion}}(|\mathbf{r}|) \delta(|\mathbf{r}| - |\mathbf{r}'|)) Y_{lm}^*(\hat{\mathbf{r}}') \quad (3.18)$$

where $Y_{lm}(\hat{\mathbf{r}})$ are spherical harmonic functions. Because of the explicit angular dependence of the pseudopotentials, the formula for applying ionic pseudopotentials of Eq. (3.18) to nonspherical systems is fairly difficult. In this semilocal form, the pseudopotential is computationally difficult to calculate with a plane-wave basis set, since the kernel integration is not separable in $\mathbf{r}$ and $\mathbf{r}'$. This form of the pseudopotential is usually simplified by rewriting the potential kernel into a separable form suggested by Kleinman and Bylander,⁵⁹ which was later shown by Blöchl⁶⁰ to be the first term of a complete series expansion using atomic pseudowavefunctions. Equation (3.18) rewritten within the Kleinman–Bylander form is

$$V_{\text{psp}}^{\text{KB}}(\mathbf{r}, \mathbf{r}') = V_{\text{local}}(\mathbf{r}) + \sum_{l=0}^{l_{\text{max}}} \sum_{m=-l}^l P_{lm}(\mathbf{r}) h_l P_{lm}^*(\mathbf{r}') \quad (3.19)$$

where the atom-centered projectors $P_{lm}(\mathbf{r})$ are of the form

$$P_{lm}(\mathbf{r}) = [V_l^{\text{ion}}(|\mathbf{r}|) - V_{\text{local}}(|\mathbf{r}|)] \tilde{\varphi}_l(|\mathbf{r}|) Y_{lm}(\hat{\mathbf{r}}) \quad (3.20)$$

and the coefficient

$$h_l = \left\{ 4\pi \int_0^\infty [V_l^{\text{ion}}(r) - V_{\text{local}}(r)] \tilde{\varphi}_l(r) r^2 dr \right\}^{-1} \quad (3.21)$$

where $\tilde{\varphi}_l(r)$ are the zero radial node pseudowavefunctions corresponding to $V_l^{\text{ion}}(r)$. The choice of the local potential, $V_{\text{local}}(r)$, is somewhat arbitrary but is usually chosen to be the highest angular momentum pseudopotential.^{27,54} When a larger series expansion atomic wavefunction is used,^{49,60} it is easy to show that Eq. (3.19) will have the general form

$$V_{\text{psp}}(\mathbf{r}, \mathbf{r}') = V_{\text{local}}(\mathbf{r}) + \sum_{l=0}^{l_{\text{max}}} \sum_{m=-l}^l \sum_{n=1}^{n_{\text{max}}} \sum_{n'=1}^{n_{\text{max}}} P_{nlm}(\mathbf{r}) h_l^{n,n'} P_{n'lm}^*(\mathbf{r}') \quad (3.22)$$

It is known that the norm-conservation condition results in harder pseudopotentials for some elements. For example, the p states in the first-row elements (oxygen, 2p) and the d states in the second-row transition elements (copper, 3d) do not have core counterparts of the same angular momentum. As a result, these states are compact and close to the core compared to the other valence states, resulting in higher plane-wave cutoffs. The ultrasoft pseudopotentials developed by Vanderbilt^{52,61} relax the norm-conservation condition by generalizing the norm-conservation sum rule. This results in pseudopotentials that are smoother and consequently require a lower plane-wave cutoff. We do not discuss the details of these pseudopotentials in this chapter and refer the reader to more comprehensive reviews.^{7,8,28,31,62}

3.3.2 Total Energy

The total energy in the pseudopotential plane-wave method can be written as a sum of kinetic, external (i.e., pseudopotential), electrostatic, and exchange and correlation energies:

$$E_{\text{total}} = E_{\text{kinetic}} + E_{\text{pseudopotential}} + E_{\text{electrostatic}} + E_{\text{xc}} \quad (3.23)$$

The kinetic energy can be written

$$E_{\text{kinetic}} = \frac{1}{2} \sum_{i,\mathbf{G}} f_i G^2 |\psi_i(\mathbf{G})|^2 \quad (3.24)$$

where f_i are the occupation numbers. To simplify our presentation here we restricted ourselves to spin-unpolarized systems, with $f_i = 2$. The extension to spin-polarized systems is straightforward and will not be discussed here. The pseudopotential energy $E_{\text{pseudopotential}}$ is given as a sum of local and nonlocal contributions:

$$E_{\text{pseudopotential}} = E_{\text{pseudopotential}}^{\text{local}} + E_{\text{pseudopotential}}^{\text{nonlocal}} \quad (3.25)$$

The local portion of the pseudopotential energy can be evaluated as

$$E_{\text{pseudopotential}}^{\text{local}} = \sum_I \int_{\Omega} \hat{V}_{\text{local}}^I(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r} = \sum_{I, \mathbf{G}} \hat{V}_{\text{local}}^I(\mathbf{G}) \rho^*(\mathbf{G}) \quad (3.26)$$

The valence electron density in reciprocal space $\rho(\mathbf{G})$ is obtained from its real space representation, $\rho(\mathbf{r}) = \sum f_n |\psi_n(\mathbf{r})|^2$, using a fast Fourier transform. The local potential is defined to be periodic and is represented as a sum of piecewise functions on the Bravais lattice by

$$\hat{V}_{\text{local}}^I(\mathbf{r}) = \sum_{\mathbf{L}} V_{\text{local}}^I(|\mathbf{r} - \mathbf{R}_I - \mathbf{L}|) \quad (3.27)$$

where $\mathbf{R}_I$ is the location of atom I , $\mathbf{L}$ is a Bravais lattice vector, and $V_{\text{local}}^I(r)$ is the radial local potential for the ion defined in Section 3.3.1. The local pseudopotential in reciprocal space is found by a spherical Bessel transform

$$\hat{V}_{\text{local}}^I(\mathbf{G}) = \frac{4\pi}{\sqrt{\Omega}} e^{i\mathbf{G} \cdot \mathbf{R}_I} \int_0^\infty V_{\text{local}}^I(r) j_0(r) r^2 dr \quad (3.28)$$

where $j_0(r) = \frac{\sin(r)}{r}$ is the spherical Bessel function.

The nonlocal part of the pseudopotential energy is given by

$$E_{\text{pseudopotential}}^{\text{nonlocal}} = \sum_i \sum_I \sum_{\mathbf{G}, \mathbf{G}'} f_i \psi_i^*(\mathbf{G}) \hat{V}_{\text{NL}}^I(\mathbf{G}, \mathbf{G}') \psi_i(\mathbf{G}') \quad (3.29)$$

where

$$\hat{V}_{\text{NL}}^I(\mathbf{G}, \mathbf{G}') = \sum_{lm} P_{lm}^I(\mathbf{G}) h_l^I P_{lm}^{I*}(\mathbf{G}') \quad (3.30)$$

and $P_{lm}^I(\mathbf{G})$ is the reciprocal space representation of the nonlocal projector [e.g., Eq. (3.20)], which can be obtained using the spherical Bessel transform

$$P_{lm}^I(\mathbf{G}) = \frac{4\pi}{\sqrt{\Omega}} e^{-i\mathbf{G} \cdot \mathbf{R}_I} Y_{lm}(\hat{\mathbf{G}}) \int_0^\infty P_{lm}^I(r) j_l(r) r^2 dr \quad (3.31)$$

The electron–electron repulsion energy can be written as

$$E_{\text{electrostatic}}^{\text{e-e}} = \frac{1}{2} \int_{\Omega} V_{\text{H}}(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r} = \frac{1}{2} \sum_{\mathbf{G}} \rho(\mathbf{G}) V_{\text{H}}^*(\mathbf{G}) \quad (3.32)$$

where the Hartree potential, $V_{\text{H}}(\mathbf{r})$, is defined as

$$V_{\text{H}}(\mathbf{r}) = \sum_L \int_{\Omega} \frac{\rho(\mathbf{r}' - \mathbf{L})}{|\mathbf{r} - \mathbf{r}' + \mathbf{L}|} d\mathbf{r}' \quad (3.33)$$

and in reciprocal space it is calculated as

$$V_H(\mathbf{G}) = \begin{cases} \frac{4\pi}{G^2} \rho(\mathbf{G}) & G \neq 0 \\ 0 & G = 0 \end{cases} \quad (3.34)$$

The ion–ion electrostatic energy for a periodic system can be facilitated using the Ewald decomposition⁶³:

$$\begin{aligned} E_{\text{electrostatic}}^{\text{ion-ion}} &= \frac{1}{2\Omega} \sum_{G \neq 0} \frac{4\pi}{|G|^2} \exp\left(-i \frac{|G|^2}{4\epsilon}\right) \\ &\quad \left[\sum_{I,J} Z_I \exp(i\mathbf{G} \cdot \mathbf{R}_I) Z_J \exp(-i\mathbf{G} \cdot \mathbf{R}_J) \right] \\ &\quad + \frac{1}{2} \sum_{\mathbf{L}} \sum_{I,J \in |\mathbf{R}_I - \mathbf{R}_J + \mathbf{L}| \neq 0} Z_I Z_J \frac{\text{erf}(\epsilon|\mathbf{R}_I - \mathbf{R}_J + \mathbf{L}|)}{|\mathbf{R}_I - \mathbf{R}_J + \mathbf{L}|} \\ &\quad - \frac{\epsilon}{\sqrt{\pi}} \sum_I Z_I^2 - \frac{\pi}{2\epsilon^2\Omega} \left(\sum_I Z_I \right)^2 \end{aligned} \quad (3.35)$$

where ϵ is a constant (typically on the order of 1) and $\mathbf{L}$ is a lattice vector.

The exchange–correlation energy E_{xc} with LDA or GGA approximation is given by

$$\begin{aligned} E_{\text{xc}} &= \int_{\Omega} f_{\text{xc}}(\rho(\mathbf{r}), |\nabla\rho(\mathbf{r})|) d\mathbf{r} \\ &\approx \frac{\Omega}{N} \sum_{\mathbf{r}_{i1i2i3}} f_{\text{xc}}(\rho(\mathbf{r}_{i1i2i3}), |\nabla\rho(\mathbf{r}_{i1i2i3})|) \end{aligned} \quad (3.36)$$

where f_{xc} is the exchange–correlation energy density, Ω is the volume of the unit cell, and N is the number of real-space grid points in the FFT grid $\mathbf{r}_{i1i2i3}$.

3.3.3 Electronic Gradient

During the course of total energy minimization or Car–Parrinello molecular dynamics simulation it is required to calculate the electron gradient, defined as

$$S_i = \frac{\delta E_{\text{total}}}{\delta \psi_i^*} \quad (3.37)$$

Part of the electron gradient is evaluated in reciprocal space and the other in real space:

$$S_i = S_i^G + S_i^r \quad (3.38)$$

The reciprocal-space portion contains contributions from the kinetic and nonlocal pseudopotential energy terms:

$$\begin{aligned} S_i^G(\mathbf{G}) &= \frac{\partial E_{\text{kinetic}}}{\partial \psi_i^*(\mathbf{G})} + \frac{\partial E_{\text{pseudopotential}}^{\text{nonlocal}}}{\partial \psi_i^*(\mathbf{G})} \\ &= \frac{1}{2} G^2 \psi_i(\mathbf{G}) + \sum_I \sum_{\mathbf{G}, \mathbf{G}'} \hat{V}_{\text{NL}}^I(\mathbf{G}, \mathbf{G}') \psi_i(\mathbf{G}') \end{aligned} \quad (3.39)$$

The real-space portion is given by

$$\begin{aligned} S_i^r(\mathbf{r}) &= \frac{\partial}{\partial \psi_i^*(\mathbf{r})} \left(E_{\text{pseudopotential}}^{\text{local}} + E_{\text{electrostatic}}^{\text{e-e}} + E_{\text{xc}} \right) \\ &= \left[V_{\text{H}}(\mathbf{r}) + \sum_I \hat{V}_{\text{local}}^I(\mathbf{r}) + V_{\text{xc}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) \end{aligned} \quad (3.40)$$

where $V_{\text{H}}(\mathbf{r})$ and $\hat{V}_{\text{local}}^I(\mathbf{r})$ are the Hartree potential and the local pseudopotential, respectively. The exchange–correlation potential is given by⁶⁴

$$\begin{aligned} V_{\text{xc}}(\mathbf{r}_{i1i2i3}) &= \frac{\delta E_{\text{xc}}}{\delta \rho(\mathbf{r}_{i1i2i3})} \\ &= \frac{\partial f_{\text{xc}}}{\partial \rho(\mathbf{r}_{i1i2i3})} - \frac{1}{N} \sum_{\mathbf{G}, \mathbf{r}'} i\mathbf{G} \cdot \frac{\nabla \rho(\mathbf{r}')}{|\nabla \rho(\mathbf{r}')|} \frac{\partial f_{\text{xc}}}{\partial \nabla \rho(\mathbf{r}')} e^{i\mathbf{G} \cdot (\mathbf{r}_{i1i2i3} - \mathbf{r}')} \end{aligned} \quad (3.41)$$

Equivalently, all the real-space expressions above can be derived from a completely reciprocal space representation using the convolution theorem. The real-space forms above are, however, considerably more efficient to compute.

3.3.4 Atomic Forces

The force acting on the atoms in the system is defined as

$$F_I = - \frac{\partial E_{\text{total}}}{\partial \mathbf{R}_I} \quad (3.42)$$

Only the pseudopotential and ion–ion electrostatic energies contribute to the force:

$$F_I = F_{\text{pseudopotential}}^I + F_{\text{ion-ion}}^I \quad (3.43)$$

The force due to the pseudopotential is give by

$$F_{\text{pseudopotential}}^I = - \frac{\partial E_{\text{pseudopotential}}^{\text{local}}}{\partial \mathbf{R}_I} - \frac{\partial E_{\text{pseudopotential}}^{\text{nonlocal}}}{\partial \mathbf{R}_I}$$

$$\begin{aligned}
&= i \sum_{\mathbf{G}} \mathbf{G} \rho^*(\mathbf{G}) \hat{V}_{\text{local}}^I(\mathbf{G}) \\
&\quad - 2 \operatorname{Re} \sum_i \sum_I \sum_{lm} \left[\sum_{\mathbf{G}} \psi_i^*(\mathbf{G}) P_{lm}^I(\mathbf{G}) \right] h_I \nabla_{R_I} \\
&\quad \left[\sum_{\mathbf{G}'} P_{lm}^{I*}(\mathbf{G}') \psi_i(\mathbf{G}') \right]
\end{aligned} \tag{3.44}$$

where $\nabla_{R_I} [\sum_{\mathbf{G}'} P_{lm}^{I*}(\mathbf{G}') \psi_i(\mathbf{G}')] = i \sum_{\mathbf{G}'} \mathbf{G}' P_{lm}^{I*}(\mathbf{G}') \psi_i(\mathbf{G}')$.

The force due to the ion-ion interaction is given by

$$\begin{aligned}
F_{\text{ion-ion}}^I &= - \frac{\partial E_{\text{electrostatic}}^{\text{ion-ion}}}{\partial \mathbf{R}_I} \\
&= - \sum_L \sum_{J \in |\mathbf{R}_I - \mathbf{R}_J + \mathbf{L}| \neq 0} Z_I Z_J (\mathbf{R}_I - \mathbf{R}_J + \mathbf{L}) \\
&\quad \times \left[\frac{\operatorname{erf}(\varepsilon |\mathbf{R}_I - \mathbf{R}_J + \mathbf{L}|)}{|\mathbf{R}_I - \mathbf{R}_J + \mathbf{L}|^3} + \frac{2}{\sqrt{\pi} \varepsilon} \frac{\exp(-\varepsilon^2 |\mathbf{R}_I - \mathbf{R}_J + \mathbf{L}|^2)}{|\mathbf{R}_I - \mathbf{R}_J + \mathbf{L}|^2} \right] \\
&\quad + \frac{1}{\Omega} \sum_{\mathbf{G} \neq 0} \mathbf{G} \frac{4\pi}{|\mathbf{G}|^2} \exp\left(-\frac{|\mathbf{G}|^2}{4\varepsilon}\right) Z_I \\
&\quad \times \operatorname{Im} \left[\exp(i \mathbf{G} \cdot \mathbf{R}_I) \sum_J Z_J \exp(-i \mathbf{G} \cdot \mathbf{R}_J) \right]
\end{aligned} \tag{3.45}$$

3.4 CHARGED SYSTEMS

As we have discussed so far, plane waves are ideal to describe systems that are intrinsically periodic. However, periodic and aperiodic systems are very different within a periodic boundary condition (PBC) framework and this is compounded further if the system is charged (e.g., charged defects, charged ions). The electrostatic energy in these systems is, in principle, divergent. A standard approach to dealing with this issue is to impose a charge-neutrality condition via a uniform charge background. This implicitly introduces a jellium background. Makov and Payne⁶⁶ have shown that this procedure results in errors which go as L^{-1} for charged systems and L^{-3} for isolated neutral systems in three dimensions, where L is size of a cubic unit cell. One approach to minimizing these errors is to use the scheme developed by Leslie and Gillan⁶⁵ and improved by Makov and Payne.⁶⁶ They derived an analytic expression for the electrostatic correction between charged unit cells as follows:

$$E_{\text{Makov-Payne}} = E_{\text{total}} - \frac{q^2 \alpha}{2L} - \frac{2\pi q Q}{3L^3} + O\left(\frac{1}{L^5}\right) \tag{3.46}$$

where E_{total} is the calculated energy of the charged cell, α is the Madelung constant for the lattice, q is the total charge of the cell, and Q is the quadrupole moment of the cell, given by

$$Q = \int_{\Omega} r^2 \rho(\mathbf{r}) d\mathbf{r} \quad (3.47)$$

Another approach for charged systems is via free-space boundary conditions. Provided that the density has decayed to zero at the edge of the supercell, free-space boundary conditions can be implemented by restricting the integration to just one isolated supercell, Ω ,

$$\begin{aligned} E_{\text{Coulomb}} &= \frac{1}{2} \int_{\Omega} \int_{\Omega} \rho(\mathbf{r}) g(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}') d\mathbf{r} d\mathbf{r}' \\ V_{\text{H}}(\vec{\mathbf{r}}) &= \int_{\Omega} g(\mathbf{r}, \mathbf{r}') \rho(\mathbf{r}') d\mathbf{r}' \end{aligned} \quad (3.48)$$

This essentially defines a modified Coulomb interaction

$$g(\mathbf{r}, \mathbf{r}') = \begin{cases} \frac{1}{|\mathbf{r} - \mathbf{r}'|} & \text{for } \mathbf{r}, \mathbf{r}' \in \Omega \\ 0 & \text{otherwise} \end{cases} \quad (3.49)$$

Hockney and Eastwood showed that an interaction of the form of Eq. (3.49) could still be used in conjunction with the fast Fourier transform convolution theorem.^{67,68} In their algorithm, the interaction between neighboring supercells is removed by padding the density with an external region of zero density, or in the specific case of a density defined in cubic supercell of length L , the density is extended to a cubic supercell of length $2L$, where the original density is defined as before on the $[0, L]^3$ domain and the remainder of the $[0, 2L]^3$ domain is set to zero. The grid is eight times larger than the conventional grid. The Coulomb potential is calculated by convoluting the density with the Green's function kernel on the extended grid. The density on the extended grid is defined by expanding the conventional grid to the extended grid and putting zeros where the conventional grid is not defined. After the aperiodic convolution, the free-space potential is obtained by restricting the extended grid to the conventional grid.

In his original work, Hockney suggested that the cutoff Coulomb kernel could be defined by

$$g(\mathbf{r}_{i,j,k}) = \begin{cases} \frac{\text{constant}}{h} & \text{for } |\mathbf{r}_{i,j,k}| = 0 \\ \frac{1}{|\mathbf{r}_{i,j,k}|} & \text{otherwise} \end{cases} \quad (3.50)$$

where h^3 is the constant volume of subintervals, defined by the unit cell divided by the number of conventional FFT grid points.⁶⁷ Hockney suggested a constant

at $|\mathbf{r}| = 0$ to be between 1 and 3. Barnett and Landman in their implementation defined the constant to be⁶⁹

$$\frac{1}{h^2} \int_{h^3} \frac{1}{r} d\mathbf{r} \approx \begin{cases} 2.380077 & \text{for SC lattice} \\ 0.910123 & \text{for FCC lattice} \\ 1.447944 & \text{for BCC lattice} \end{cases} \quad (3.51)$$

Regardless of the choice of the constant, the singular nature of $g(\mathbf{r})$ in real space can lead to significant numerical error. James addressed this problem somewhat by expanding the Coulomb kernel to higher orders in real space.⁷⁰

The convolution theorem suggests that defining $g(\mathbf{r})$ in reciprocal space will lead to much higher accuracy. A straightforward definition in reciprocal space is

$$g(\mathbf{r}) = \sum_{\mathbf{G}} g_{\text{uniform}}(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}}$$

$$g_{\text{uniform}}(\mathbf{G}) = \frac{1}{h^3} \int_{\Omega'} \frac{e^{-i(\mathbf{G} \cdot \mathbf{r}/2)}}{r} d\mathbf{r} \quad (3.52)$$

where Ω' is the volume of the extended unit cell and h^3 is the volume of the unit cell divided by the number of conventional FFT grid points. The reciprocal space definition gains accuracy because the singularity at $\mathbf{r} = \mathbf{r}'$ in Eq. (3.48) is integrated out analytically. Even when Eq. (3.52) is used to define the kernel, a slight inexactness in the calculated electron–electron Coulomb energy will always be present, due to the discontinuity introduced in the definition of the extended density where the extended density is forced to be zero in the extended region outside Ω . However, this discontinuity is small, since the densities we are interested in decay to zero within Ω , thus making the finite Fourier expansion of the extended densities extremely close to zero in the extended region outside Ω .

Equation (3.52) could be calculated numerically; however, we have found that alternative definitions can be used with little loss of numerical accuracy. In an earlier work^{71,72} we suggested that the cutoff Coulomb kernel could be defined as

$$g(\mathbf{r}) = \begin{cases} \sum_{\mathbf{G}} g_a(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}} & \text{for } |\mathbf{r}| \leq R_{\text{max}} - \delta \\ \frac{1}{|\mathbf{r}|} & \text{otherwise} \end{cases}$$

$$g_a(\mathbf{G}) = \begin{cases} \frac{2\pi(R_{\text{max}})^2}{h^3} & \text{for } G = 0 \\ \frac{4\pi}{h^3 G^2} [1 - \cos(G^2 R_{\text{max}})] & \text{otherwise} \end{cases}$$

$$R_{\max} = \begin{cases} L & \text{(simple cubic)} \\ \frac{\sqrt{2}}{2}L & \text{(face-centered cubic)} \\ \frac{\sqrt{3}}{2}L & \text{(body-centered cubic)} \end{cases}$$

$$\delta = \text{small constant} \quad (3.53)$$

Other forms have been suggested and could also be used.^{7,73–75} The Fourier-represented kernels improve the integration accuracy by removing the singularity at $|\mathbf{r} - \mathbf{r}'|$ in a trapezoidal integration. A disadvantage of the kernel defined by Eq. (3.53) is that only regular-shaped cells can be used. To extend this method to irregular-shaped cells, short- and long-range decomposition can be used¹⁵:

$$g(\mathbf{r}) = g_{\text{shortrange}}(\mathbf{r}) + g_{\text{longrange}}(\mathbf{r})$$

$$g_{\text{shortrange}}(\mathbf{r}) = \sum \mathbf{g}_{\text{shortrange}}(\mathbf{G}) e^{i\mathbf{G} \cdot \mathbf{r}}$$

$$g_{\text{shortrange}}(\mathbf{G}) = \begin{cases} \frac{4\pi}{h^3 G^2} (1 - e^{-(|\mathbf{G}|^2/4\epsilon^2)}) & \text{for } G \neq 0 \\ \frac{\pi}{h^3 \epsilon^2} & \text{for } G = 0 \end{cases}$$

$$g_{\text{longrange}}(\mathbf{r}) = \begin{cases} \frac{\text{erf}(\epsilon r)}{r} & \text{for } r \neq 0 \\ \frac{2\epsilon}{\sqrt{\pi}} & \text{for } r = 0 \end{cases} \quad (3.54)$$

We have found this kernel to give very high accuracy, even for highly non-cubic supercells. Marx and Hutter recently proposed the use of this kernel as well.⁷ Other kernel definitions are possible (e.g., using short- and long-range decomposition based on a Lorentzian).⁷⁴

Other schemes involve the use of countercharges, represented by Gaussian densities, whose potential can be derived analytically. Since a detailed discussion of the various approaches to this problem is beyond the scope of this chapter, we refer the reader to various papers on the subject.^{65,66,76–78}

3.5 EXACT EXCHANGE

A number of failures are known to exist in DFT (see Chapter 1), such as underestimating bandgaps, the inability to localize excess spin density, and underestimating chemical reaction barriers. These problems are a consequence of having to rely on computationally efficient approximations to the exact exchange–correlation functional (e.g., LDA and GGA) used by plane-wave DFT programs—that is an accuracy–performance trade-off. It is generally agreed

that the largest error in these approximations is their failure to completely cancel out the orbital self-interaction energies, or in plain terms that electrons partially “see” themselves.^{79,80} In the Hartree–Fock approximation, the exchange energy is calculated exactly and no self-interaction is present; however, by definition all electron correlation effects are missing from it. In all practical implementations of DFT the exchange energy is calculated approximately, and cancellation of the self-interaction is incomplete.

Experience has shown that many of the failures associated with the erroneous self-interaction term can be corrected by approaches in which DFT exchange–correlation functionals are improved by inclusion of the nonlocal exchange term (hybrid-DFT, e.g., B3LYP and PBE0⁸¹),⁸²

$$E_{x\text{-exact}} = -\frac{1}{2} \sum_{\sigma=\uparrow,\downarrow} \sum_n \sum_m \iint \frac{\rho_{ij}^\sigma(\mathbf{r}) \rho_{ij}^\sigma(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \quad (3.55)$$

were the overlap densities are given by

$$\rho_{ij}^\sigma(\mathbf{r}) = \psi_i^{\sigma*}(\mathbf{r}) \psi_j^\sigma(\mathbf{r}) \quad (3.56)$$

Using the expanded Bloch states⁸³ representation

$$\psi_{i\mathbf{k}}^\sigma(\mathbf{r}) = \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{\sqrt{\Omega}} \sum_{\mathbf{G}} \psi_{i\mathbf{k}}^\sigma(\mathbf{G}) e^{i\mathbf{G}\cdot\mathbf{r}} \quad (3.57)$$

the exchange term takes the form

$$E_{x\text{-exact}} = \frac{-1}{2\Omega} \left(\frac{\Omega}{8\pi^3} \right)^2 \sum_{\sigma=\uparrow,\downarrow} \int_{\text{BZ}} d\mathbf{k} \int_{\text{BZ}} d\mathbf{l} \left[\sum_n \sum_m \sum_G \frac{4\pi}{|\mathbf{G} - \mathbf{k} + \mathbf{l}|^2} \rho_{j\mathbf{l};i\mathbf{k}}^\sigma(-\mathbf{G}) \rho_{i\mathbf{k};j\mathbf{l}}^\sigma(\mathbf{G}) \right] \quad (3.58)$$

where

$$\rho_{i\mathbf{k};j\mathbf{l}}^\sigma(\mathbf{G}) = \sum_{\mathbf{G}'} \psi_{i\mathbf{k}}^{\sigma*}(\mathbf{G}') \psi_{j\mathbf{l}}^\sigma(\mathbf{G}' + \mathbf{G}) \quad (3.59)$$

As pointed out by Gygi and Baldereschi^{84–86} and others,^{87–91} this expression must be evaluated with some care, especially for small Brillouin zone samplings and small unit cell size, because of the singularity at $\mathbf{G} - \mathbf{k} + \mathbf{l} = 0$.

A better alternative for the evaluation of $E_{x\text{-exact}}$ for Γ -point ($\mathbf{k} = 0$) calculations with large unit cells can be found in terms of localized Wannier orbitals.^{92,93}

The standard approach for the generation of Wannier orbitals using unitary transformation over $\mathbf{k}$,

$$w_i^\sigma(\mathbf{r} - \mathbf{L}) = \frac{\Omega}{8\pi^3} \int_{\text{BZ}} e^{-i\mathbf{k}\cdot\mathbf{L}} \psi_{i\mathbf{k}}^\sigma(\mathbf{r}) d\mathbf{k} \quad (3.60)$$

is not applicable for the Γ -point case. Instead, one can follow a Marzari–Vanderbilt localization procedure (which is the counterpart of the Foster–Boys transformation for finite systems)^{92–94} forming linear combinations of $\psi_{i\mathbf{k}=0}^\sigma(\mathbf{r})$ over different n to produce a new set of Γ -point Bloch functions, $\hat{w}_{i\mathbf{k}=0}^\sigma(\mathbf{r})$. These new periodic orbitals are extremely localized within each cell for nonmetallic systems with sufficiently large unit cells⁹³ (see Fig. 3.3). In that case $\hat{w}_{i\mathbf{k}=0}^\sigma(\mathbf{r})$ can be represented as a sum of piecewise localized functions, $w_i^\sigma(\mathbf{r} - \mathbf{L})$, on the Bravais lattice

$$\hat{w}_{i\mathbf{k}=0}^\sigma(\mathbf{r}) = \sum_{\mathbf{L}} w_i^\sigma(\mathbf{r} - \mathbf{L}) \quad (3.61)$$

with the exchange term per unit cell written as

$$E_{x\text{-exact}} = -\frac{1}{2} \sum_i \sum_j \iint \frac{w_i^*(\mathbf{r}) w_j(\mathbf{r}) w_j^*(\mathbf{r}') w_i(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}' \quad (3.62)$$

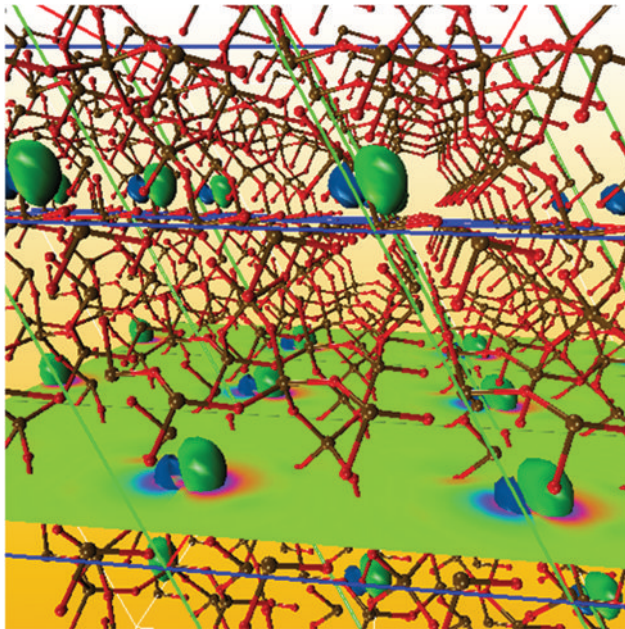


Fig. 3.3 (color online) Periodic localized function $\hat{w}_{i\mathbf{k}=0}(\mathbf{r})$ for a 72-atom unit cell of a SiO_2 crystal.

Evaluation of this integral in a plane-wave basis set requires some care, since representing overlap densities $[w_i^*(\mathbf{r})w_j(\mathbf{r})]$ with a plane-wave expansion [i.e., $\hat{w}_i^*(\mathbf{r})\hat{w}_j(\mathbf{r})$] will result in the inclusion of redundant periodic images. Interactions between such images can be eliminated^{95,96} by replacing the standard Coulomb kernel, $1/r$, in Eq. (3.13) by the following cutoff Coulomb kernel:

$$f_{\text{cutoff}}(r) = \frac{1 - [1 - e^{-(r/R_c)^{N_c+2}}]^{N_c}}{r} \quad (3.63)$$

where N_c and R_c are adjustable parameters. This kernel decays rapidly to zero at distances larger than R_c . Hence, Eq. (3.62) can be transformed to

$$E_{x\text{-exact}} = -\frac{1}{2} \sum_{\sigma=\uparrow,\downarrow} \sum_i \sum_j \iint \hat{w}_i^{\sigma*}(\mathbf{r}) \hat{w}_j^{\sigma}(\mathbf{r}) f_{\text{cutoff}}(|\mathbf{r} - \mathbf{r}'|) \hat{w}_j^{\sigma*}(\mathbf{r}') \hat{w}_i^{\sigma}(\mathbf{r}') d\mathbf{r} d\mathbf{r}' \quad (3.64)$$

That is, replacing $w_i(\mathbf{r})$ with $\hat{w}_i(\mathbf{r})$, combined with using Eq. (3.14), in Eq. (3.13) will give the same energy, since the cutoff Coulomb interaction is nearly $1/r$ with itself and zero with its periodic images. The parameter R_c must be chosen carefully. It has to exceed the size of each Wannier orbital to include all of the orbital in the integration, while concurrently having $2R_c$ be smaller than the shortest linear dimension of the unit cell to exclude periodic interactions. Finally, we note that when one uses the cutoff Coulomb kernel, localized orbitals are not needed to calculate the exchange term since Eq. (3.62) can be unitary transformed, resulting in

$$E_{x\text{-exact}} = -\frac{1}{2} \sum_{\sigma=\uparrow,\downarrow} \sum_i \sum_j \iint \psi_i^{\sigma*}(\mathbf{r}) \psi_j^{\sigma}(\mathbf{r}) f_{\text{cutoff}}(|\mathbf{r} - \mathbf{r}'|) \psi_j^{\sigma*}(\mathbf{r}') \psi_i^{\sigma}(\mathbf{r}') d\mathbf{r} d\mathbf{r}' \quad (3.65)$$

and

$$\frac{\delta E_{x\text{-exact}}}{\delta \psi_i^{\sigma*}(\mathbf{r})} = - \sum_j \psi_j^{\sigma}(\mathbf{r}) \int f_{\text{cutoff}}(|\mathbf{r} - \mathbf{r}'|) \psi_j^{\sigma*}(\mathbf{r}') \psi_i^{\sigma}(\mathbf{r}') d\mathbf{r}' \quad (3.66)$$

We note that while using the localized functions here is not required in this formulation, one should still evaluate the set of maximally localized Wannier functions in order to estimate their extent and, consequently, the minimal size of the unit cell.

3.6 WAVEFUNCTION OPTIMIZATION FOR PLANE-WAVE METHODS

In DFT calculations it is necessary to determine the set of orthonormal one-electron wavefunctions $\{\psi_i\}$ that minimize the Kohn–Sham energy functional. There are two classes of methods available for optimizing the Kohn–Sham energy functional: the self-consistent field approach and the direct minimization approach.

3.6.1 Self-Consistent Field Method

The steps involved in the self-consistent field procedure are as follows:

1. Set the iteration number $m = 0$ and choose an initial set of trial molecular orbitals $\{\psi_n\}$ and input charge $\rho(\mathbf{r})$; for example,

$$\rho^{(0)}(\mathbf{r}) = \sum_{i=1}^{\text{occ}} |\psi_i(\mathbf{r})|^2$$

2. Use the input charge density to construct an effective potential which is a sum of the Hartree and exchange–correlation potentials, respectively:

$$V_{\text{eff}}(\mathbf{r}) = V_{\text{H}}[\rho^{(m)}, \mathbf{r}] + V_{\text{xc}}[\rho^{(m)}, \mathbf{r}]$$

3. Generate a new set of molecular orbitals by solving the linearized Kohn–Sham equations via an iterative scheme:

$$\left[-\frac{1}{2}\nabla^2 + \sum_I (\hat{V}_{\text{local}}^I(\mathbf{r}) + \hat{V}_{\text{NL}}^I) + V_{\text{eff}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \varepsilon_i \psi_i(\mathbf{r})$$

4. Use the new set of molecular orbitals to construct an output density:

$$\rho_{\text{out}}^{(m)}(\mathbf{r}) = \sum_{n=1}^{\text{occ}} |\tilde{\psi}_n(\mathbf{r})|^2$$

5. Generate a new input density by mixing the output density with the previous input density:

$$\rho^{(m+1)}(\mathbf{r}) \Leftarrow \rho^{(m)}, \rho_{\text{out}}^{(m)}$$

6. If self-consistency is not achieved, $m = m + 1$; go to step 2.

In this scheme, self-consistency is achieved when the distance between the input and output charge densities is zero:

$$D[\rho_{\text{out}}, \rho] = \sqrt{\langle \rho_{\text{out}} - \rho | \rho_{\text{out}} - \rho \rangle} \quad (3.67)$$

For plane-wave methods, where the molecular orbitals are expanded using $\sim 10,000$ to several million basis functions, an efficient iterative method for diagonalizing the Kohn–Sham Hamiltonian is needed. Many iterative methods have been developed^{4,6,97–101} and several good reviews on the subject are available in the literature. Two of the more popular algorithms used for plane-wave methods include the conjugate gradient algorithm applied to plane-wave calculations proposed by Teter et al.⁹⁹ and the residual minimization method direct inversion

in the iterative subspace (RMM-DIIS) proposed by Pulay.^{97,98} A preconditioning scheme is generally used with these methods.^{4,6,7,99}

An important step in the self-consistent field procedure is the generation of a new trial density, $\rho^{(m+1)}$, from prior input, $\rho^{(m)}$, and output, $\rho_{\text{out}}^{(m)}$, densities. A simpleminded iteration,

$$\rho^{(m+1)} = \rho_{\text{out}}^{(m)} \quad (3.68)$$

in which the input density is replaced by the output density will usually result in the development of charge oscillations which cause the algorithm to diverge. The simplest way to control these oscillations is to dampen them during the iteration process by a simple mixing algorithm,

$$\rho^{(m+1)} = (1 - \alpha)\rho^{(m)} + \alpha\rho_{\text{out}}^{(m)} \quad (3.69)$$

where α is a parameter between [0,1]. In many cases convergence can be achieved by using a suitable choice of α (e.g., $0.1 \leq \alpha \leq 0.5$). Several other iteration schemes have been developed besides simple mixing.^{6,97,102–113}

3.6.2 Direct Methods

An alternative approach is to treat the DFT energy functional as an optimization problem and minimize it directly.^{4,7,114–116} Interest in this method began with the introduction of the Car–Parrinello algorithm.³ These methods stand out in that they rarely, if ever, fail to achieve self-consistency. The simplest of this class of methods is the fixed-step steepest descent algorithm, which is effectively the Car–Parrinello algorithm (see Section 3.7) with the velocity set to zero at every step in the iteration. Orthonormality constraints are handled by Lagrange multipliers. A significantly more powerful approach is the conjugate gradient method on the Grassmann manifold developed by Edelman et al.¹¹⁷ This method is very fast and has been shown to demonstrate superlinear speedup near the minimum. In this algorithm, the set of wavefunctions ψ_i are written in terms of a tall and skinny $N_{\text{basis}} \times N_e$ matrix:

$$Y = \begin{bmatrix} \psi_1(\phi_1) & \psi_2(\phi_1) & \cdots & \psi_{N_e}(\phi_1) \\ \psi_1(\phi_2) & \psi_2(\phi_2) & \cdots & \psi_{N_e}(\phi_2) \\ \psi_1(\phi_3) & \psi_2(\phi_3) & \cdots & \psi_{N_e}(\phi_3) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_1(\phi_{N_{\text{basis}}}) & \psi_2(\phi_{N_{\text{basis}}}) & \cdots & \psi_{N_e}(\phi_{N_{\text{basis}}}) \end{bmatrix} \quad (3.70)$$

where the matrix is written in terms of the orthonormal basis $\phi_j(\mathbf{r})$ (or $e^{i\mathbf{G}_j \cdot \mathbf{r}}$ for a plane-wave basis) by

$$\psi_i(\vec{r}) = \sum_{j=1}^{N_{\text{basis}}} \psi_i(\phi_j) \phi_j(\vec{r}) \quad (3.71)$$

and obeys the orthogonality constraint $Y^t Y = I$. The following steps illustrate this algorithm:

1. Given an initial Y_0 such that $Y_0^t Y_0 = I$, calculate the tangent residual:

$$G_0 = (1 - Y_0 Y_0^t) \left[\frac{\delta E}{\delta Y^t} \right]_{Y=Y_0}$$

2. Set $H_0 = -G_0$ and $E_{\text{new}} = E_{\text{total}}(Y_0)$.
3. Find the compact singular value decomposition of H_0 :

$$H_0 \rightarrow U \Sigma V^t$$

4. Minimize $E_{\text{total}}(Y(\theta))$ in the following geodesic line parameterized by θ :

$$Y(\theta) = YV \cos(\Sigma\theta) V^t + U \sin(\Sigma\theta) V^t$$

5. Set $Y_1 = Y(\theta)$, $E_{\text{old}} = E_{\text{new}}$, and $E_{\text{new}} = E_{\text{total}}(Y_1)$.
6. Calculate the tangent residual:

$$G_1 = (1 - Y_1 Y_1^t) \left[\frac{\delta E}{\delta Y^t} \right]_{Y=Y_1}$$

7. Parallel-transport the previous search direction along the geodesic:

$$T_0 = [-Y_0 V \sin(\Sigma\theta) + U \cos(\Sigma\theta)] \Sigma V^t$$

8. Compute the new search direction,

$$H_1 = -G_1 + \frac{\text{Tr}[G_1, G_1]}{\text{Tr}[G_0, G_0]} T_0$$

9. Set $Y_0 = Y_1$, $G_0 = G_1$, and $H_0 = H_1$.
10. If $E_{\text{old}} - E_{\text{new}} > \text{tolerance}$, go to step 3.

3.7 CAR-PARRINELLO MOLECULAR DYNAMICS

The development of fast and efficient ab initio molecular dynamics methods (AIMD), such as Car-Parrinello molecular dynamics,³ has opened the door to the study of strongly interacting many-body systems by direct dynamics simulation without the introduction of empirical interactions. In AIMD simulations the electronic degrees of freedom are continuously updated at each step in the simulation and all the changes in the electronic structure are properly accounted for. The forces are calculated as derivatives of the total energy calculated with respect to the atomic positions. Hence, the dynamical simulation automatically includes

all many-body interactions and effects, such as changes in coordination, bond saturation, and polarization. Applications for this first-principles method include the calculation of free energies, search for global minima, explicit simulation of solvated molecules, and so on.

This important generalization of molecular dynamics methods to include the essential physics of the interactions of complex systems comes at a considerable price. However, with present-day algorithms and parallel supercomputers, simulations of hundreds of atoms for a time scale of several picoseconds are feasible. Although this is far less, both in numbers of particles and in time, than is possible with conventional MD, AIMD simulations might be the only option for systems with complex chemistry where even qualitative interpretation requires proper description of interatomic interactions.

In the Car-Parrinello version of AIMD the electronic and ionic degrees of freedom are updated simultaneously. This is accomplished by introducing a fictitious electronic kinetic energy functional

$$\text{KE}(\{\psi_i\}) = \frac{1}{2} \sum_i \mu \int \dot{\psi}_i^*(\mathbf{r}) \dot{\psi}_i(\mathbf{r}) d\mathbf{r} \quad (3.72)$$

where μ is a fictitious mass assigned to electron degrees of freedom. The equations of motion for the ion, $\mathbf{R}_I$, and the Kohn-Sham orbitals, ψ_i , are found by taking the first variation of the auxiliary Lagrangian:

$$\begin{aligned} L(\{\psi_i\}, \{\mathbf{R}_I\}) = & \frac{1}{2} \sum_i \mu \int \dot{\psi}_i^*(\mathbf{r}) \dot{\psi}_i(\mathbf{r}) d\mathbf{r} + \frac{1}{2} \sum_I M_I |\dot{\mathbf{R}}_I|^2 \\ & - E_{\text{total}}(\{\psi_i\}, \{\mathbf{R}_I\}) + \sum_{i,j} \left[\int \psi_i^*(\mathbf{r}) \psi_j(\mathbf{r}) d\mathbf{r} - \delta_{i,j} \right] \Lambda_{j,i} \end{aligned} \quad (3.73)$$

The resulting equations of motion are

$$\mu \ddot{\psi}_i(\mathbf{r}) = -H \psi_i(\mathbf{r}) + \sum_j \psi_j(\mathbf{r}) \Lambda_{j,i} \quad (3.74)$$

$$M_I \ddot{\mathbf{R}}_I = \mathbf{F}_I \quad (3.75)$$

where

$$\frac{\delta E_{\text{total}}}{\delta \psi_i^*(\mathbf{r})} = H \psi_i(\mathbf{r}) \quad (3.76)$$

Given the equations of motion (Sections 3.3.3 and 3.3.4), the electronic and ionic degrees of freedom can be integrated using the Verlet algorithm:

$$\psi_i^{t+\Delta t}(\mathbf{r}) = 2\psi_i^t(\mathbf{r}) - \psi_i^{t-\Delta t}(\mathbf{r}) + \frac{(\Delta t)^2}{\mu} \left[-H \psi_i^t(\mathbf{r}) + \sum_j \psi_j^t(\mathbf{r}) \Lambda_{j,i}^{t+\Delta t} \right] \quad (3.77)$$

$$\mathbf{R}_I^{t+\Delta t} = 2\mathbf{R}_I^t - \mathbf{R}_I^{t-\Delta t} + \frac{(\Delta t)^2}{M_I} \mathbf{F}_I \quad (3.78)$$

The matrix $\Lambda_{j,i}^{t+\Delta t}$ is determined by the orthogonality constraint

$$\int \psi_i^{*t+\Delta t}(\mathbf{r}) \psi_i^{t+\Delta t}(\mathbf{r}) d\mathbf{r} = \delta_{i,j} \quad (3.79)$$

This constraint yields the matrix Riccati equation [to simplify the following equations the following symbols are used: $\bar{\psi}_i^t(\mathbf{r}) = 2\psi_i^t(\mathbf{r}) - \psi_i^{t-\Delta t}(\mathbf{r}) + (t^2/\mu)H\psi_i^t(\mathbf{r})$, $\alpha = \Delta t^2/\mu$]:

$$\begin{aligned} I &= \int \psi_i^{*t+\Delta t}(\mathbf{r}) \psi_j^{t+\Delta t}(\mathbf{r}) d\mathbf{r} \\ &= \int \left\{ \psi_i^{*t}(\mathbf{r}) - \alpha \left[\bar{\psi}_i^{*t}(\mathbf{r}) + \sum_k \Lambda_{i,k} \psi_k^{*t}(\mathbf{r}) \right] \right\} \\ &\quad \left\{ \psi_j^t(\mathbf{r}) - \alpha \left[\bar{\psi}_j^t(\mathbf{r}) + \sum_l \psi_l^t(\mathbf{r}) \Lambda_{l,j} \right] \right\} d\mathbf{r} \\ &= A + X^t B + B^t X + X^t C X \end{aligned} \quad (3.80)$$

where $X_{ij} = \alpha \Lambda_{ij}$ and the matrices $A_{i,j}$, $B_{i,j}$, and $C_{i,j}$ are given by

$$A_{ij} = \int \{ \psi_i^{*t}(\mathbf{r}) - \alpha [\bar{\psi}_i^{*t}(\mathbf{r})] \} \{ \psi_j^t(\mathbf{r}) - \alpha [\bar{\psi}_j^t(\mathbf{r})] \} d\mathbf{r} \quad (3.81)$$

$$B_{ij} = \int [\psi_i^{*t}(\mathbf{r})] \{ \psi_j^t(\mathbf{r}) - \alpha [\bar{\psi}_j^t(\mathbf{r})] \} d\mathbf{r} \quad (3.82)$$

$$C_{ij} = \int \psi_i^{*t}(\mathbf{r}) \psi_j^t(\mathbf{r}) d\mathbf{r} \quad (3.83)$$

Blöchl²⁸ suggested the following iteration for solving this matrix Riccati equation:

$$A^{(0)} = A \quad (3.84)$$

$$A^{(n)} = A^{(n-1)} + X^{(n-1)t} B + B X^{(n-1)} + X^{(n-1)t} C X^{(n-1)} \quad (3.85)$$

$$X_{rs}^{(n+1)} = \sum_{i,j,k,l} \frac{U_{ri}^t U_{ij} (A_{jk}^{(n)} - \delta_{jk}) U_{kl}^t U_{ls}}{b_i + b_l} \quad (3.86)$$

where the eigenvalues b and the unitary matrix U are obtained from diagonalizing $B_{ij} = \sum_l U_{il}^t b_l U_{lj}$.

3.8 PARALLELIZATION

During the course of a total energy minimization or molecular dynamics simulation the electron gradient $\delta E_{\text{total}}/\delta \psi_i^*$ [Eq. (3.37)] needs to be calculated as efficiently as possible. For a pseudopotential plane-wave calculation the main parameters that determine the cost of a calculation are N_g , N_e , N_a , and N_{proj} , where N_g is the size of the three-dimensional FFT grid, N_e is the number of occupied orbitals, N_a is the number of atoms, and N_{proj} is the number of projectors per atom. In most plane-wave DFT programs the solution of eigenvalue equations is typically approached by means of a conjugate gradient algorithm or, for dynamics, a Car–Parrinello algorithm that requires many evaluations of the electron gradient. The operation counts for each part of the electron gradient are shown in Fig. 3.4. The three (or four) major computational pieces of the gradient are:

1. The Hartree potential V_H , including the local exchange and correlation potentials $V_x + V_c$. The main computational kernel in these computations is the calculation of N_e three-dimensional FFTs.
2. The nonlocal pseudopotential, $\hat{V}_{\text{NL}}$. The major computational kernel in this computation can be expressed by the following matrix multiplications: $\mathbf{W} = \mathbf{P}^t \cdot \mathbf{Y}$, and $\mathbf{Y}_2 = \mathbf{P} \cdot \mathbf{W}$, where $\mathbf{P}$ is an $N_g \times (N_{\text{proj}} \cdot N_a)$ matrix, $\mathbf{Y}$ and $\mathbf{Y}_2$ are $N_g \times N_e$ matrices, and $\mathbf{W}$ is an $(N_{\text{proj}} \cdot N_a) \times N_e$ matrix. We note that for most pseudopotential plane-wave calculations, $N_{\text{proj}} \cdot N_a \approx N_e$.
3. Enforcing orthogonality. The major computational kernels in this computation are following matrix multiplications: $\mathbf{S} = \mathbf{Y}^t \cdot \mathbf{Y}$ and $\mathbf{Y}_2 = \mathbf{Y} \cdot \mathbf{S}$, where $\mathbf{Y}$ and $\mathbf{Y}_2$ are $N_g \times N_e$ matrices, and $\mathbf{S}$ is an $N_e \times N_e$ matrix.
4. When exact exchange is included, the exact exchange operator $\Sigma K_{ij} \psi_j$. The major computational kernel in this computation involves the calculation of $(N_e+1) \cdot N_e$ three-dimensional FFTs.

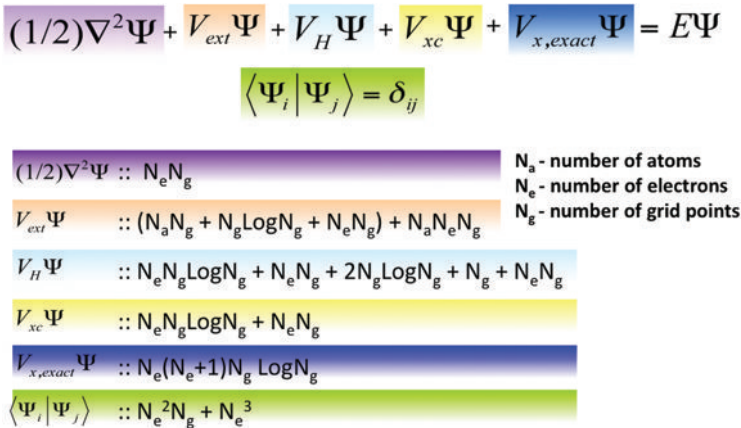


Fig. 3.4 (color online) Operation count of $H\psi$ in a plane-wave DFT simulation.

There are several ways to parallelize a plane-wave Hartree–Fock and DFT program.^{7,9,11,12,15,18} For many solid-state calculations the computation can be distributed over the Brillouin zone sampling space.¹¹ This approach cannot be used for Γ -point ($\mathbf{k} = 0$) calculations with large unit cells. Another approach is to distribute the one-electron orbitals across processors.¹² The drawback of this method is that orthogonality parts of the computation will involve a lot of message passing. Furthermore, this method will not work for simulations with very large cutoff energy requirements (i.e., using large numbers of plane waves to describe the one-electron orbitals) on parallel computers that have nodes with a small amount of memory, because a complete one-electron must be stored on each node. Hence this approach is not practical for Car–Parrinello simulations with large unit cells; however, this approach can work well for simulations with modest-size unit cells and with small cutoff energies, when used in combination with minimization algorithms that perform orthogonalization sparingly (e.g., RMM-DIIS).

Another straightforward way is to do a spatial decomposition of the one-electron orbitals.^{7,9,15} This approach is versatile, easily implemented, and is well suited for performing Car–Parrinello simulations with large unit cells and cutoff energies. However, a parallel three-dimensional fast Fourier transform (FFT) must be used, which is known not to scale beyond $\sim N_g^{1/3}$ processors (or processor groups), where N_g is the number of FFT grid points. In Fig. 3.5, an example of timings versus the number of CPUs for this type of parallelization is shown. These simulations were taken from a Car–Parrinello simulation of the hydrated uranyl cation $\text{UO}_2^{2+} + 122\text{H}_2\text{O}$ using the plane-wave DFT module (PSPW) in NWChem.¹¹⁸ These calculations were performed on all four cores on the quad-core Cray-XT4 system (NERSC Franklin), composed of a 2.3-GHz single-socket quad-core AMD Opteron processor (Budapest). The NWChem program was compiled using a Portland Group FORTRAN 90 compiler, version 7.2.4, linked with the Cray MPICH2 library, version 3.02, for message passing. The performance of the program is reasonable with an overall parallel efficiency of 84% on 128 CPUs, dropping to 26% on 1024 CPUs. However, not every part of the program scales in exactly the same way. For illustrative purposes, the timings of the FFTs, nonlocal pseudopotential, and orthogonality are also shown. The efficiency of the FFTs are by far the biggest bottleneck in this implementation. At smaller processor sizes the inefficiency of the FFTs are damped out, due to the fact that these parts of the code make up less than 5% of the overall computation, and the largest part of the calculation is the nonlocal pseudopotential evaluation. Ultimately, however, the lack of scalability of the three-dimensional FFT algorithm beyond the $\sim N_g^{1/3}$ processor prevails, causing the simulation not to speed up.

Recently, Gygi et al. have come up with an approach that can be used to improve the overall efficiency of a plane-wave DFT program.¹⁸ In this approach, both the spatial and the orbitals are distributed in a two-dimensional processor geometry, as shown in Fig. 3.6. Using simple scaling arguments, it can be shown that with this decomposition the algorithms will require only $O(\log(p_1)) + O(\log(p_2))$ communications per CPU as opposed to $O(\log(P))$ communications per CPU for algorithms in which only the spatial or orbital dimensions are

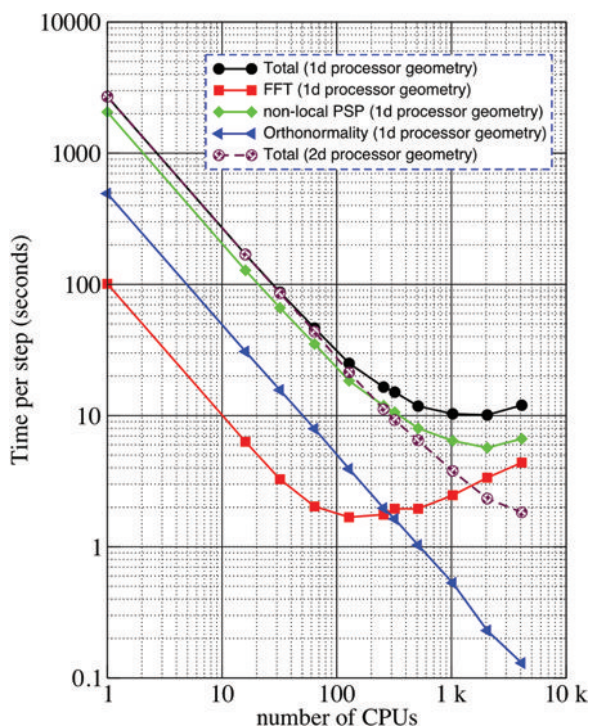


Fig. 3.5 (color online) Overall and component timings and component from AIMD simulations of $\text{UO}_2^{2+} + 122\text{H}_2\text{O}$ using one-dimensional processor geometry. Overall best timings are also shown for a two-dimensional processor grid. Timings are from calculations on the Franklin Cray-XT4 computer system at NERSC.

1d and 2d Processor Geometries

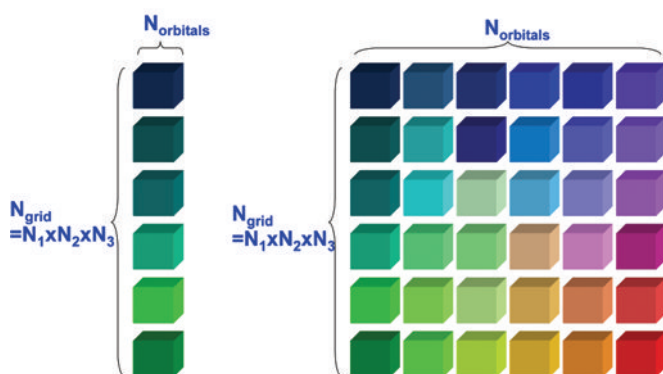


Fig. 3.6 (color online) Parallel distribution (shown on the left), implemented in most plane-wave DFT software. Each of the one-electron orbitals is identically spatially decomposed. The two-dimensional parallel distribution suggested by Gygi et al.¹⁸ is shown on the right.

distributed (a processor for where the total number of processors, P , can be written as $P = p_1 p_2$). The overall performance of our plane-wave DFT simulations were found to improve considerably using this new approach. Using the optimal processor geometries, the running time per step took 2699 s (45 min) for 1 CPU down to 3.7 s with a 70% parallel efficiency on 1024 CPUs. The fastest running time found was 1.8 s with 36% parallel efficiency on 4096 CPUs. As shown in Fig. 3.7, these timings were found to be very sensitive to the layout of the two-dimensional processor geometry. For 256, 512, 1024, and 2048 CPUs, the optimal processor geometries were 64×4 , 64×8 , 128×8 , and 128×16 processor grids, respectively. The timings of the FFTs, nonlocal pseudopotential, and orthogonality are also shown in Fig. 3.7. Not every part of the program scaled perfectly. The parallel efficiency of several other key operations depends strongly on the shape of the processor geometry. It was found that distributing the processors over the orbitals significantly improved the efficiency of the FFTs and the nonlocal pseudopotential, while distributing the processors over the spatial dimensions favored the orthogonality computations.

The two-dimensional processor geometry method can also be used to parallelize the computation of the exact exchange operator. This operator has a cost of $O(N_e^2 \cdot N_g \cdot \log(N_g))$, and when it is included in a plane-wave DFT calculation it is by far the most demanding term. The exchange term is well suited for this method. Whereas if only the spatial or orbital dimensions are distributed, the exchange term does not scale well. When only the spatial dimensions are distributed, each of the $N_e(N_e + 1)$ FFT are computed one at a time, using the entire machine for each evaluation. The drawback of this approach is that we are underutilizing the resources; parallel efficiency is effectively bounded to $\sim N_g^{1/3}$ processors. When only the orbital dimensions are distributed, the parallelization is realized by multicasting the $O(N_e)$ orbitals to set up the $O(N_e^2)$ wavefunction pairs. This multicast is followed by a multireduction which reverses the pattern. We note that with this type of

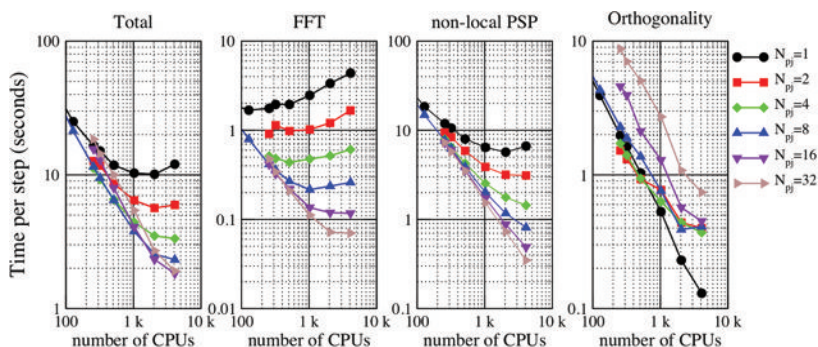


Fig. 3.7 (color online) Overall and component timings in seconds for $\text{UO}_2^{2+} + 122\text{H}_2\text{O}$ plane-wave DFT simulations at various processor sizes (N_p) and processor grids (n_j , $n_i = N_p/n_j$). Timings are from calculations on the Franklin Cray-XT4 computer system at NERSC.

algorithm one could readily fill a very large parallel machine by assigning each a few FFTs to each processor. However, to obtain reasonable performance from this algorithm it is vital to mask latency, since the interconnects between the processors will be flooded with $O(N_e)$ streams, each on long messages comprising N_g floating-point words of data. When both the spatial and orbital dimension are distributed, only the parallel three-dimension FFTs along the processor grid columns need to be computed. Compared with a multicast across all processors the benefit of this approach is to reduce latency costs, since broadcasting is done across the rows of the two-dimensional processor grid only.

The overall best timings for hybrid-DFT calculations of an 80-atom supercell of hematite (Fe_2O_3) with an FFT grid of $N_g = 72^3$ ($N_e^{\text{up}} = 272$, $N_e^{\text{down}} = 272$), and a 160-atom supercell of hematite (Fe_2O_3) with an FFT grid of $N_g = 144 \times 72 \times 72$ ($N_e^{\text{up}} = 544$ and $N_e^{\text{down}} = 544$) (wavefunction cutoff energy = 100 Ry and density cutoff energy = 200 Ry) and orbital occupations of $N_e^{\text{up}} = 272$ and $N_e^{\text{down}} = 272$ are shown in Fig. 3.8. The overall best timing per step found for the 80-atom supercell was 3.6 s on 9792 CPUs, and for the 160-atom supercell

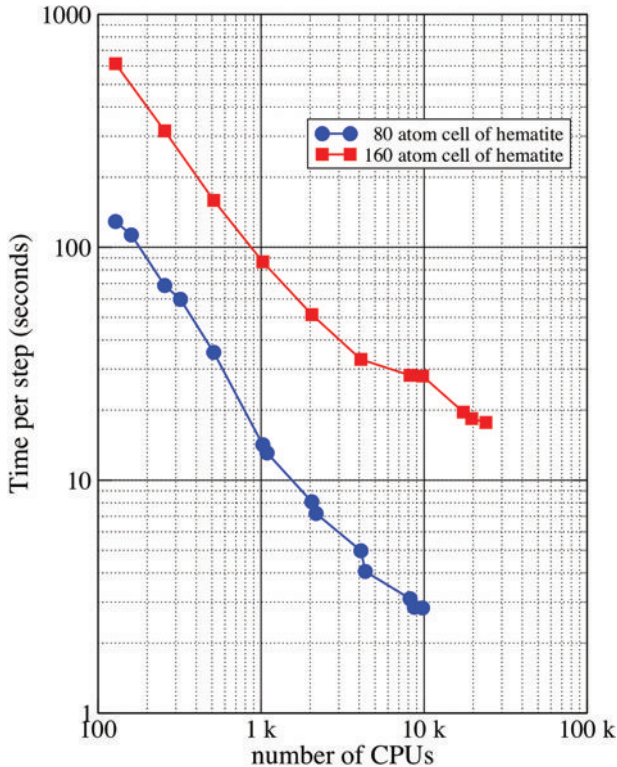


Fig. 3.8 (color online) Overall fastest timings taken for an 80- and 160-atom Fe_2O_3 hybrid-DFT energy calculations. Timings are from calculations on the Franklin Cray-XT4 computer system at NERSC.

of hematite, was 17.7 s on 23,936 CPUs. The timings results are somewhat uneven, since limited numbers of processor grids were tried at each processor size. However, even with this limited amount of sampling, these calculations were found to have speedups to at least 25,000 CPUs. We expect that further improvements will be obtained by trying more processor geometry layouts.

3.9 AIMD SIMULATIONS OF HIGHLY CHARGED IONS IN SOLUTION

An understanding of the structure and dynamics of the water molecules in the hydration shells surrounding ions is essential to the interpretation of many chemical processes in aqueous solutions. X-ray and neutron scattering results have been reported which provide direct results about shell structure for many ionic species.^{119,120} Information about the dynamics of water molecules in this region has also been obtained from other probes, such as NMR, infrared spectroscopy, and inelastic neutron scattering.^{119,120} For singly charged ions (Na^+ , Li^+), a structured first hydration shell can be identified. The residence time in this shell is short (e.g. <1 ps for Na^+) and the change in the solution structure due to the presence of the ion does not extend far beyond first-nearest-neighbor water molecules.^{121,122} However, for highly charged metal ions such as Al^{3+} , which play a role in many important chemical processes, experimental information providing the structure in the region near the ion is much more difficult to obtain. From the interpretation of EXAFS, NMR, and neutron scattering data there is evidence of a well-structured first solvation shell with six (Al^{3+} , Cr^{3+})^{123–128} to eight (Dy^{3+} , Yb^{3+})^{129–131} strongly bound waters with very long lifetimes. The residence times of the waters in the first hydration shell of triply charged ions vary over many orders of magnitude (Ti^{3+} , $\approx 10^{-5}$ s; Ir^{3+} , $\approx 10^{10}$ s).

The bonding pattern and the dynamics of the molecules in the second shell are critical to the interpretation of the ion association properties and reaction chemistry of ions in solution. Unfortunately, determining the structure of the second shell and the existence of an ion-related structure in the third shell are very difficult to obtain from x-ray or neutron data. In order to relate the scattering intensities in terms of a shell structure, a geometric model of the molecular arrangement must be invoked. The validity of the model assumed must be demonstrated by the agreement of the scattering prediction from the model with observations.¹²⁴ Although this type of fitting can produce excellent agreement between the experimental data and the results of model calculations, small but important changes in structure are difficult to resolve. As a consequence, information about the structure of the solution beyond the second shell and the behavior of the solution as it returns to the bulk structure¹³² is not available for most highly charged metal ions.

Since these are strongly interacting systems, the most reliable path to theoretical interpretation is direct simulation at the molecular level. For singly charged ions, both first-principles [ab initio molecular dynamics^{3,4,7,8} (AIMD)] and classical molecular dynamics (MD; based on empirical intermolecular potentials) methods are in reasonable agreement with each other and with the experimental

data. However, even for these simple systems, important differences in solvent structure are seen between AIMD and MD simulations.^{121,122,133,134} For highly charged species such as Al^{3+} , conventional MD methods are even more difficult to apply, because the strong polarization of the surrounding water molecules from the $+3$ ion makes it difficult to develop an accurate representation of the solvent–ion interaction.^{135,136} AIMD methods, on the other hand, calculate the interaction between species directly from the electronic Schrödinger equation as the simulation proceeds. They do not need, therefore, to introduce an intermolecular potential. In Fig. 3.9, snapshots from various AIMD (Car–Parrinello)

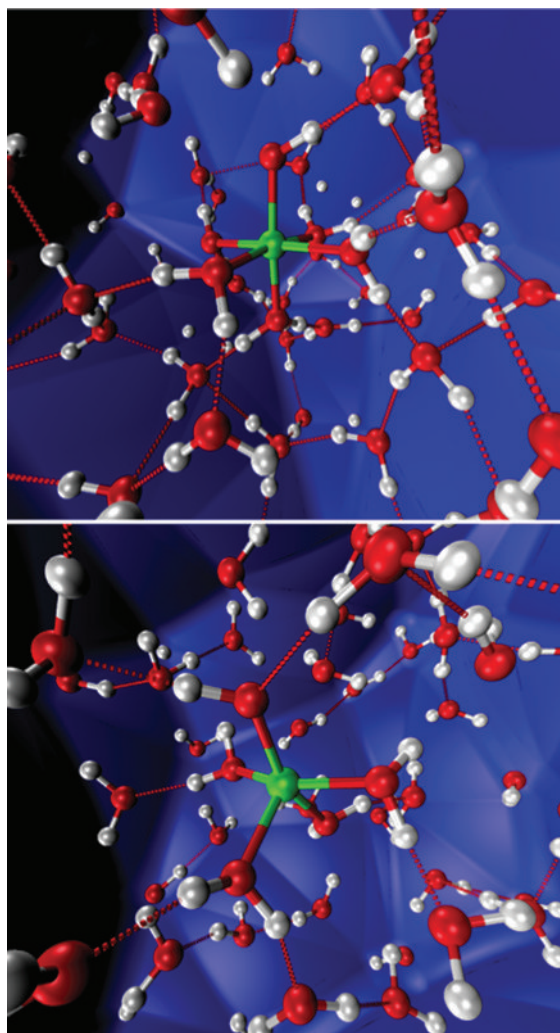


Fig. 3.9 (color online) (Top) $\text{Al}^{3+}(\text{H}_2\text{O})_6$ AIMD simulation. (Bottom) Five-coordinate AlOH_2^+ + ion that forms after hydrolysis. Dotted lines represent hydrogen bonds.

plane-wave DFT simulations of the Al^{3+} cation in aqueous solution are shown. This method was found to agree remarkably well with the measured octahedral structure of the first solvation shell of Al^{3+} . The average second-shell radius was also in excellent agreement with the value measured. Less is known experimentally about the structure of the second shell. These simulations suggest that this shell contains ~ 12 water molecules, which are trigonally coordinated to the first-shell waters. This exact structure cannot be measured directly. However, the number of water molecules in the second shell predicted by the simulation is consistent with experimental estimates. Tetrahedral bulk water coordination reappears in the third shell. Although the time scale of our simulation is not long enough to observe transfers of waters from the first to the second shell, we do see transfers occurring on a picosecond time scale between the second and third shells via an associative mechanism. This is faster than, but consistent with, the results of measurements on the more tightly bound Cr^{3+} system. It was also found that directly removing a proton from the hexaaqua Al^{3+} ion leads to a much more labile solvation shell and to a five-coordinated Al^{3+} ion. This is consistent with very recent rate measurements of ligand exchange,¹³⁷ but it is in contrast with other trivalent metal aqua ions, for which there is no evidence for stable pentacoordinate hydrolysis products.

Another example of the use of AIMD simulations is the study of hydrated radionuclides under extreme conditions. A major obstacle to the development of nuclear power is the ability to safely store highly toxic waste materials containing uranium and other radionuclides. Most current storage strategies are designed to store waste in saturated and unsaturated geological formations. Stored in this way, the most likely means for uranium to migrate into the biosphere is through groundwater contact with containment canisters, resulting in a solvated UO_2^{2+} cation (or complexes). To reliably predict the behavior of the radionuclide (e.g., uranium, thorium, plutonium) waste products of nuclear power production over the range of conditions encountered in a storage facility requires a theory based at the most fundamental level on the electronic Schrödinger equation. For example, a detailed understanding of chemical processes occurring in aqueous solutions, as well as a better understanding of the surrounding hydration shells, for this cation is needed. In particular, the bond pattern and dynamics of the second hydration shell is critical to the interpretation of the ion-association properties and reaction chemistry of the UO_2^{2+} cation in solution.

In Fig. 3.10, results from an AIMD simulation of the UO_2^{2+} cation in aqueous solution are shown.¹³⁸ The UO_2^{2+} cation was found to have a first solvation shell composed of five water molecules bonded to the uranium atom along its equatorial plane, and a second solvation shell of 10 water molecules hydrogen bonded to the first-shell water molecules. In addition, about five or six water molecules on average were found to hydrogen bond sporadically to the oxygen atoms of the UO_2^{2+} cation in the apical region. The second and apical solvent regions were found to be very dynamic, with many water transfers into and out of the equatorial and apical second solvation shells occurring on a picosecond time scale via dissociative mechanisms. Beyond these shells,

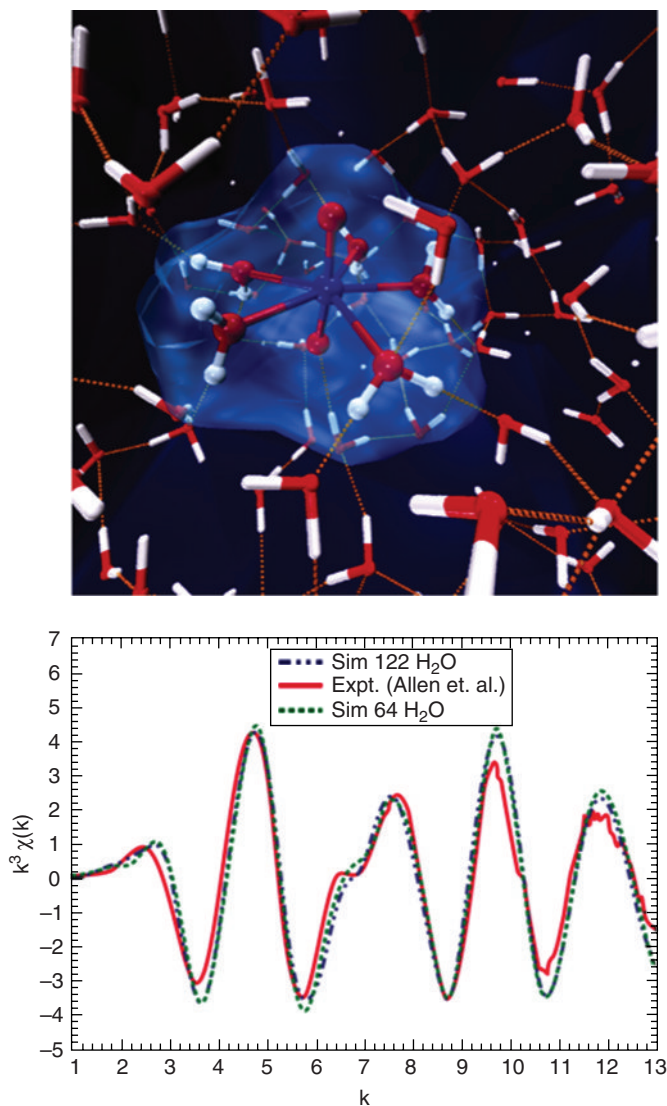


Fig. 3.10 (color online) (Top) Solvent structure of the UO_2^{2+} ion from AIMD simulations. (Bottom) EXAFS spectra calculated for the $\text{UO}_2^{2+} + 64\text{H}_2\text{O}$ and $\text{UO}_2^{2+} + 122\text{H}_2\text{O}$ simulations compared to the experimental data of Allen¹⁹ for free uranyl cation in solution.

the bonding pattern returned substantially to the tetrahedral structure of bulk water.

Even though the UO_2^{2+} cation has been studied extensively over the years using a variety of static ab initio (e.g., static coupled cluster calculations) and classical molecular dynamics methods, these simulations have been either incomplete or inaccurate. Static ab initio simulations have been incomplete because they

were not able to take into account the motion of the water molecules in the second and apical solvent shells, and classical molecular dynamic simulations have been plagued by inaccurate force fields of the complex interactions between the UO_2^{2+} cation and water. AIMD simulations, which are able to take into account complex interactions and dynamics, are able overcome the well-known deficiencies of other molecular simulation methods. In fact, this AIMD simulation was the first molecular simulation able to reproduce the measured extended x-ray absorption fine structure (EXAFS) spectrum from experiments.¹³⁸

3.10 CONCLUSIONS

In this chapter we have provided a thorough description of plane-wave-based density functional theory, including detailed formulas for energies and gradient, an overview of algorithms for minimization and Car–Parrinello molecular dynamics, and strategies for implementing exact exchange and large-scale parallelization. This method is well suited for the implementation of first-principles molecular dynamics simulations since the use of a plane-wave basis set allows a rapid update of electron degrees of freedom and ionic forces, leading to fast update times. However, to obtain convergence within the plane-wave basis representation, something has to be done to modify the rapid variation of the electron atom potential near the atomic centers. The common approach to this problem is the replacement of the atomic potentials by slowly varying norm-conserving pseudopotentials, which conserve the scattering properties of the original atomic species. The pseudopotential method can provide accurate calculations for a wide range of chemical systems.

To illustrate the capability of plane-wave DFT methods, we showed results from the AIMD simulations to the solvation of Al^{3+} and UO_2^{2+} aqua ions. These systems are difficult to simulate with conventional molecular dynamics methods, because the strong polarization of the surrounding water molecules from a highly charged ion makes it difficult to develop an accurate representation of the solvent–ion interaction.^{135,136} AIMD methods, on the other hand, calculate the interaction between species directly from the electronic Schrödinger equation as the simulation proceeds. They therefore do not need to introduce an intermolecular potential.

It is becoming clear that the levels of approximation of the electronic structure problem currently used in standard plane-wave DFT simulations will not reliably predict the properties for many of the materials encountered in applications. Examples include charge localization in elements with tightly bound *d* electrons, the prediction of barriers to proton transfer, and the relative stability of intermediate states. Experience has shown that many of these failures can be corrected by approaches in which DFT exchange–correlation functionals are improved by inclusion of the nonlocal exchange term. Very recent developments of parallel algorithms for incorporating exact exchange in a plane-wave DFT calculation have shown great promise. Although still computationally challenging, we have shown that even on modest-sized parallel machines, exact exchange

can be readily incorporated into plane-wave DFT methods, and simulations containing 300 orbitals at a cutoff energy 100 Ry can be computed in roughly 4 s per step, making Car–Parrinello simulations of 20+ ps quite feasible. With the advent of new petascale machines along with additional algorithmic refinements and code optimizations, we expect that similar-length Car–Parrinello simulations will be possible for systems containing 500+ orbitals in the very near future, significantly advancing the state of the art in predictive modeling of materials. A more difficult problem with the use of the DFT method is its poor representation of long-range van der Waals interactions. In this case the inclusion of exchange is not sufficient to improve the accuracy. This problem must be treated by using higher-level approximations to the electronic structure problem that are more reliable (e.g., MP2). In fact, Kresse and co-workers have recently developed MP2 within the plane-wave framework.¹³⁹

Significant progress has been made in terms of accuracy, efficiency, and scalability of plane-wave DFT methods in recent years. However the algorithms and implementations of these methods need to be upgraded constantly to capture the performance of emerging massively parallel computers. The projected size of the next-generation supercomputers are very large (10^6 to 10^7 threads of computation), suggesting that current limitations in simulation times, particle sizes, and levels of theory will be overcome in the coming years by brute-force increases in computer size.

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REFERENCES

1. Pickett, W. *Comput. Phys. Rep.* **1989**, 9, 115.
2. Ihm, J.; Zunger, A.; Cohen, M. L. *J. Phys. C* **1979**, 12, 4409.
3. Car, R.; Parrinello, M. *Phys. Rev. Lett.* **1985**, 55, 2471.

4. Payne, M. C.; Teter, M. P.; Allan, D. C.; Arias, T. A.; Joannopoulos, J. D. *Rev. Mod. Phys.* **1992**, *64*, 1045.
5. Remler, D. K.; Madden, P. A. *Mol. Phys.* **1990**, *70*, 921.
6. Kresse, G.; Furthmüller, J. *Comp. Mater. Sci.* **1996**, *6*, 15.
7. Marx, D.; Hutter, J. *Ab initio* molecular dynamics: theory and implementation. In *Modern Methods and Algorithms of Quantum Chemistry*, Vol. 1, Grotendorst, J., Ed., Forschungszentrum, Jülich, Germany, 2000, p. 301.
8. Valiev, M.; Bylaska, E. J.; Gramada, A.; Weare, J. H. *Ab initio* molecular dynamics simulations using density-functional theory. In *Reviews in Modern Quantum Chemistry: A Celebration of the Contributions of R. G. Parr*, Sen, K. D., Ed., World Scientific, Singapore, 2002.
9. Nelson, J. S.; Plimpton, S. J.; Sears, M. P. *Phys. Rev. B* **1993**, *47*, 1765.
10. Brommer, K. D.; Larson, B. E.; Needels, M.; Joannopoulos, J. D. *J. Comput. Phys.* **1993**, *7*, 350.
11. Clarke, L. J.; Stich, I.; Payne, M. C. *Comput. Phys. Commun.* **1992**, *72*, 14.
12. Wiggs, J.; Jonsson, H. *Comput. Phys. Commun.* **1994**, *81*, 1.
13. Wiggs, J.; Jonsson, H. *Comput. Phys. Commun.* **1995**, *87*, 319.
14. Cavazzoni, C. Large scale first-principles simulations of water and ammonia at high pressure and temperature, Scuola Internazionale Superiore di Studi Avanzati (SISSA), Trieste, Italy, 1998.
15. Bylaska, E. J.; Valiev, M.; Kawai, R.; Weare, J. H. *Comput. Phys. Commun.* **2002**, *143*, 11.
16. Kendell, R. A.; Apra, E.; Bernholdt, E. J.; Bylaska, E. J.; Dupuis, J.; Fann, G. I.; Harrison, R. J.; Ju, J.; Nichols, J. A.; Nieplocha, J.; Straatsma, T. P.; Windus, T. L.; Wong, A. T. *Comput. Phys. Commun.* **2000**, *128*, 260.
17. Apra, E.; Bylaska, E. J.; Dean, D. J.; Fortunelli, A.; Gao, F.; Krstic, P. S.; Wells, J. C.; Windus, T. L. *Comp. Mater. Sci.* **2003**, *28*, 209.
18. Gygi, F.; Draeger, E. W.; Schulz, M.; de Supinski, B. R.; Gunnels, J. A.; Austel, V.; Sexton, J. C.; Franchetti, F.; Kral, S.; Ueberhuber, C. W.; Lorenz, J. Large-scale electronic structure calculations of high-Z metals on the BlueGene/L platform. In *SC '06: Proceedings of the 2006 ACM/IEEE Conference on Supercomputing*, 2006.
19. Allen, P. G.; Bucher, J. J.; Shuh, D. K.; Edelstein, N. M.; Reich, T. *Inorg. Chem.* **1997**, *36*, 4676.
20. Schwegler, E.; Galli, G.; Gygi, F.; Hood, R. Q. *Phys. Rev. Lett.* **2001**, *87*, 265501.
21. Brigham, E. O. *The Fast Fourier Transform*, Prentice-Hall, Englewood Cliffs, NJ, 1974.
22. Phillips, J. C. *Phys. Rev.* **1958**, *112*, 685.
23. Phillips, J. C.; Kleinman, L. *Phys. Rev.* **1959**, *116*, 287.
24. Austin, B. J.; Heine, V.; Sham, L. J. *Phys. Rev.* **1962**, *127*, 276.
25. Yin, M. T.; Cohen, M. L. *Phys. Rev. B* **1982**, *25*, 7403.
26. Bachelet, G. B.; Hamann, D. R.; Schlüter, M. *Phys. Rev. B* **1982**, *26*, 4199.
27. Hamann, D. R. *Phys. Rev. B* **1989**, *40*, 2980.
28. Blöchl, P. E. *Phys. Rev. B* **1994**, *50*, 17953.
29. Holzwarth, N. A. W.; Mathews, G. E.; Tackett, A. R.; Dunning, R. B. *Phys. Rev. B* **1997**, *57*, 11827.

30. Valiev, M.; Weare, J. H. *J. Phys. Chem. B* **1999**, *103*, 10588.
31. Kresse, G.; Joubert, D. *Phys. Rev. B* **1999**, *59*, 1758.
32. Chadi, D. J.; Cohen, M. L. *Phys. Rev. B* **1973**, *8*, 5747.
33. Monkhorst, H. J.; Pack, J. D. *Phys. Rev. B* **1976**, *13*, 5188.
34. Pack, J. D.; Monkhorst, H. J. *Phys. Rev. B* **1977**, *16*, 1748.
35. Herring, C. *Phys. Rev.* **1940**, *57*, 1169.
36. Heine, V.; Weaire, D. *Phys. Rev.* **1966**, *152*, 603.
37. Appelbaum, J. A.; Hamann, D. R. *Phys. Rev. B* **1973**, *8*, 1777.
38. Schluter, M.; Chelikowsky, J. R.; Louie, S. G.; Cohen, M. L. *Phys. Rev. B* **1975**, *12*, 4200.
39. Topp, W. C.; Hopfield, J. J. *Phys. Rev. B* **1973**, *7*, 1295.
40. Starkloff, T.; Joannopoulos, J. D. *Phys. Rev. B* **1977**, *16*, 5212.
41. Hamann, D. R.; Schluter, M.; Chiang, C. *Phys. Rev. Lett.* **1979**, *43*, 1494.
42. Weeks, J. D.; Hazi, A.; Rice, S. A. *Adv. Can. Chem. Phys.* **1969**, *16*, 283.
43. Kahn, L. R.; Goddard, W. A. *J. Chem. Phys.* **1972**, *16*, 2685.
44. Topiol, S.; Zunger, A.; Ratner, M. A. *Chem. Phys. Lett.* **1977**, *49*, 367.
45. Zunger, A.; Cohen, M. L. *Phys. Rev. B* **1978**, *18*, 5449.
46. Lam, P. K.; Cohen, M. L.; Zunger, A. *Phys. Rev. B* **1980**, *22*, 1698.
47. Kerker, G. P. *J. Phys. C* **1980**, *13*, L189.
48. Bachelet, G. B.; Schluter, M. *Phys. Rev. B* **1982**, *25*, 2103.
49. Vanderbilt, D. *Phys. Rev. B* **1985**, *32*, 8412.
50. Bachelet, G. B.; Hamann, D. R.; Schluter, M. *Phys. Rev. B* **1988**, *37*, 4798.
51. Shirley, E. L.; Allan, D. C.; Martin, R. M.; Joannopoulos, J. D. *Phys. Rev. B* **1989**, *40*, 3652.
52. Vanderbilt, D. *Phys. Rev. B* **1990**, *41*, 7892.
53. Rappe, A. M.; Rabe, K. M.; Kaxiras, E.; Joannopoulos, J. D. *Phys. Rev. B* **1990**, *41*, 1227.
54. Troullier, N.; Martins, J. L. *Phys. Rev. B* **1991**, *43*, 1993.
55. Lin, J. S.; Qteish, A.; Payne, M. C.; Heine, V. *Phys. Rev. B* **1993**, *47*, 4174.
56. Goedecker, S.; Teter, M.; Hutter, J. *Phys. Rev. B* **1996**, *54*, 1703.
57. Hartwigsen, C.; Goedecker, S.; Hutter, J. *Phys. Rev. B* **1998**, *58*, 3641.
58. Fuchs, M.; Scheffler, M. *Comput. Phys. Commun.* **1999**, *119*, 67.
59. Kleinman, L.; Bylander, D. M. *Phys. Rev. Lett.* **1982**, *48*, 1425.
60. Blöchl, P. E. *Phys. Rev. B* **1990**, *41*, 5414.
61. Laasonen, K.; Car, R.; Lee, C.; Vanderbilt, D. *Phys. Rev. B* **1991**, *43*, 6796.
62. Holzwarth, N. A. W.; Tackett, A. R.; Matthews, G. E. *Comput. Phys. Commun.* **2001**, *135*, 329.
63. Ewald, P. P. Z. *Kristallogr.* **1921**, *56*, 129.
64. White, J. A.; Bird, D. M. *Phys. Rev. B* **1994**, *50*, 4954.
65. Leslie, M.; Gillan, M. J. *J. Phys. C* **1985**, *18*, 973.
66. Makov, G.; Payne, M. C. *Phys. Rev. B* **1995**, *51*, 4014.
67. Hockney, R. W. *Methods Comput. Phys.* **1970**, *9*, 135.

68. Hockney, R.; Eastwood, J. W. *Computer Simulations Using Particles*, McGraw-Hill, New York, 1981.
69. Barnett, R. N.; Landman, U. *Phys. Rev. B* **1993**, 48, 2081.
70. James, R. A. *J. Comput. Phys.* **1977**, 35, 71.
71. Lubin, M. I.; Bylaska, E. J.; Weare, J. H. *Chem. Phys. Lett.* **2000**, 322, 447.
72. Bylaska, E. J.; Taylor, P. R.; Kawai, R.; Weare, J. H. *J. Phys. Chem.* **1996**, 100, 6966.
73. Jarvis, M. R.; White, I. D.; Godby, R. W.; Payne, M. C. *Phys. Rev. B* **1997**, 56, 14972.
74. Kawai, R. Private communication.
75. Martna, G. J.; Tuckerman, M. E. *J. Chem. Phys.* **1999**, 110, 2810.
76. Kantorovich, L. N. *Phys. Rev. B* **1999**, 60, 15476.
77. Blöchl, P. E. *J. Chem. Phys.* **1995**, 103, 7422.
78. Dabo, I.; Kozinsky, B.; Singh-Miller, N. E.; Marzari, N. *Phys. Rev. B* **2008**, 77.
79. Parr, R. G. *Density-Functional Theory of Atoms and Molecules*, Vol. 16, Oxford University Press, New York, 1989.
80. Dreizler, R. M.; Gross, E. K. U. *Density Functional Theory: An Approach to the Quantum Many-Body Problem*, Springer-Verlag, Berlin, 1990.
81. Perdew, J. P.; Burke, K.; Wang, Y. *Phys. Rev. B* **1996**, 54, 16533.
82. Szabo, A.; Ostlund, N. S. *Modern Quantum Chemistry: Introduction to Advanced Electronic Structure Theory*, Dover, Mineola, NY, 1996.
83. Ashcroft, N. W.; Mermin, N. D. *Solid State Physics*, Holt, Rinehart and Winston, New York, 1976.
84. Gygi, F.; Baldereschi, A. *Phys. Rev. Lett.* **1989**, 62, 2160.
85. Gygi, F.; Baldereschi, A. *Helv. Phys. Acta* **1986**, 59, 972.
86. Gygi, F.; Baldereschi, A. *Helv. Phys. Acta* **1985**, 58, 928.
87. Chawla, S.; Voth, G. A. *J. Chem. Phys.* **1998**, 108, 4697.
88. Sorouri, A.; Foulkes, W. M. C.; Hine, N. D. M. *J. Chem. Phys.* **2006**, 124.
89. Marsman, M.; Paier, J.; Stroppa, A.; Kresse, G. *J. Phys. Condens. Matter* **2008**, 20.
90. Gorling, A. *Phys. Rev. B* **1996**, 53, 7024.
91. Carrier, P.; Rohra, S.; Gorling, A. *Phys. Rev. B* **2007**, 75.
92. Marzari, N.; Vanderbilt, D. *Phys. Rev. B* **1997**, 56, 12847.
93. Silvestrelli, P. L. *Phys. Rev. B* **1999**, 59, 9703.
94. Foster, J. M.; Boys, S. F. *Rev. Mod. Phys.* **1960**, 32, 300.
95. Bylaska, E. J.; Tsemekhman, K.; Gao, F. *Phys. Scr.* **2006**, T124, 86.
96. Du, J. C.; Corrales, L. R.; Tsemekhman, K.; Bylaska, E. J. *Nucl. Instrum. Methods Phys. Res. B* **2007**, 255, 188.
97. Pulay, P. *Chem. Phys. Lett.* **1980**, 73, 393.
98. Wood, D. M.; Zunger, A. *J. Phys. A* **1985**, 18, 1343.
99. Teter, M. P.; Payne, M. C.; Allan, D. C. *Phys. Rev. B* **1989**, 40, 12255.
100. Bylander, D. M.; Kleinman, L.; Lee, S. *Phys. Rev. B* **1990**, 42, 1394.
101. Pollard, W. T.; Friesner, R. A. *J. Chem. Phys.* **1993**, 99, 6742.
102. Anderson, D. G. *J. ACM* **1965**, 12, 547.

103. Kerker, G. P. *Phys. Rev. B* **1981**, 23, 3082.
104. Purvis, G. D.; Bartlett, R. J. *J. Chem. Phys.* **1981**, 75, 1284.
105. Pulay, P. *J. Comput. Chem.* **1982**, 3, 556.
106. Ho, K. M.; Ihm, J.; Joannopoulos, J. D. *Phys. Rev. B* **1982**, 25, 4260.
107. Bendt, P.; Zunger, A. *Phys. Rev. B* **1982**, 26, 3114.
108. Vanderbilt, D.; Louie, S. G. *Phys. Rev. B* **1984**, 30, 6118.
109. Dennis, J. E.; Schnabel, R. B. *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, Society for Industrial and Applied Mathematics, Philadelphia, 1996.
110. Johnson, D. D. *Phys. Rev. B* **1988**, 38, 12807.
111. Brown, P. N.; Saad, Y. *SIAM J. Sci. Stat. Comput.* **1990**, 11, 450.
112. Fokkema, D. R.; Sleijpen, G. L. G.; van der Vorst, H. A. *SIAM J. Sci. Comput.* **1998**, 19, 657.
113. Harrison, R. J. *J. Comput. Chem.* **2004**, 25, 328.
114. Stich, I.; Car, R.; Parrinello, M.; Baroni, S. *Phys. Rev. B* **1989**, 39, 4997.
115. Hutter, J.; Parrinello, M.; Vogel, S. *J. Chem. Phys.* **1994**, 101, 3862.
116. Pfrommer, B. G.; Demmel, J.; Simon, H. *J. Comput. Phys.* **1999**, 150, 287.
117. Edelman, E.; Arias, T.; Smith, S. T. *SIAM J. Matrix Anal. Appl.* **1998**, 20, 303.
118. Bylaska, E. J.; de Jong, W. A.; Govind, N.; Kowalski, K.; Straatsma, T. P.; Valiev, M.; Wang, D.; Apra, E.; Windus, T. L.; Hammond, J.; Nichols, P.; Hirata, S.; Hackler, M. T.; Zhao, Y.; Fan, P.-D.; Harrison, R. J.; Dupuis, M.; Smith, D. M. A.; Nieplocha, J.; Tipparaju, V.; Krishnan, M.; Wu, Q.; Voorhis, T. V.; Auer, A. A.; Nooijen, M.; Brown, E.; Cisneros, G.; Fann, G. I.; Fruchtl, H.; Garza, J.; Hirao, K.; Kendall, R.; Nichols, J. A.; Tsemekhman, K.; Wolinski, K.; Anchell, J.; Bernholdt, D.; Borowski, P.; Clark, T.; Clerc, D.; Dachsel, H.; Deegan, M.; Dylla, K.; Elwood, D.; Glendening, E.; Gutowski, M.; Hess, A.; Jaffe, J.; Johnson, B.; Ju, J.; Kobayashi, R.; Kutteh, R.; Lin, Z.; Littlefield, R.; Long, X.; Meng, B.; Nakajima, T.; Niu, S.; Pollack, L.; Rosing, M.; Sandrone, G.; Stave, M.; Taylor, H.; Thomas, G.; van Lenthe, J.; Wong, A.; Zhang, Z. *NWChem: A Computational Chemistry Package for Parallel Computers, Version 5.1.1*, Pacific Northwest National Laboratory, Richland, WA, 2009.
119. Ohtaki, H.; Radnai, T. *Chem. Rev.* **1993**, 93, 1157.
120. Richens, D. T. *The Chemistry of Aqua Ions: Synthesis, Structure, and Reactivity*, Wiley, New York, 1997.
121. White, J. A.; Schwegler, E.; Galli, G.; Gygi, F. *J. Chem. Phys.* **2000**, 113, 4668.
122. Lyubartsev, A. P.; Laasonen, K.; Laaksonen, A. *J. Chem. Phys.* **2001**, 114, 3120.
123. Bol, W.; Welzen, T. *Chem. Phys. Lett.* **1977**, 49, 189.
124. Caminiti, R.; Licheri, G.; Piccaluga, G.; Pinna, G. *J. Chem. Phys.* **1978**, 69, 1.
125. Caminiti, R.; Licheri, G.; Piccaluga, G.; Pinna, G.; Radnai, T. *J. Chem. Phys.* **1979**, 71, 2473.
126. Neilson, G. W.; Ansell, S.; Wilson, J. Z. *Naturforsch. A* **1995**, 50, 247.
127. Herdman, G. J.; Salmon, P. S. *J. Am. Chem. Soc.* **1991**, 113, 2930.
128. Salmon, P. S.; Herdman, G. J.; Lindgren, J.; Read, M. C.; Sandstrom, M. *J. Phys. Condens. Matter* **1989**, 1, 3459.

129. Mayanovic, R. A.; Jayanetti, S.; Anderson, A. J.; Bassett, W. A.; Chou, I. M. *J. Phys. Chem. A* **2002**, *106*, 6591.
130. Mayanovic, R. A.; Jayanetti, S.; Anderson, A. J.; Bassett, W. A.; Chou, I. M. *J. Chem. Phys.* **2003**, *118*, 719.
131. Cossy, C.; Barnes, A. C.; Enderby, J. E.; Merbach, A. E. *J. Chem. Phys.* **1989**, *90*, 3254.
132. Bockris, J. O. M.; Reddy, A. K. N. *Modern Electrochemistry*, 2nd ed., Plenum Press, New York, 1998.
133. Lightstone, F. C.; Schwegler, E.; Allesch, M.; Gygi, F.; Galli, G. *ChemPhysChem* **2005**, *6*, 1745.
134. Lightstone, F. C.; Schwegler, E.; Hood, R. Q.; Gygi, F.; Galli, G. *Chem. Phys. Lett.* **2001**, *343*, 549.
135. Wasserman, E.; Rustad, J. R.; Xantheas, S. S. *J. Chem. Phys.* **1997**, *106*, 9769.
136. Spangberg, D.; Hermansson, K. *J. Chem. Phys.* **2004**, *120*, 4829.
137. Swaddle, T. W.; Rosenqvist, J.; Yu, P.; Bylaska, E.; Phillips, B. L.; Casey, W. H. *Science* **2005**, *308*, 1450.
138. Nichols, P.; Bylaska, E. J.; Schenter, G. K.; de Jong, W. *J. Chem. Phys.* **2008**, *128*, 124507.
139. Marsman, M.; Gruneis, A.; Paier, J.; Kresse, G. *J. Chem. Phys.* **2009**, *130*, 184103.

PART B

Higher-Accuracy Methods

4 Quantum Monte Carlo, Or, Solving the Many-Particle Schrödinger Equation Accurately While Retaining Favorable Scaling with System Size

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I introduce and discuss the quantum Monte Carlo method, a state-of-the-art computer simulation technique capable of solving the equations of quantum mechanics with extremely high accuracy, while remaining tractable for systems with relatively large numbers of constituent particles. The CASINO code, developed in our group in Cambridge over many years, is reviewed briefly and results obtained with it are used to illuminate the discussion.

4.1 INTRODUCTION

Of course we all know how to solve the Schrödinger equation for a given quantum mechanical system as accurately as we like: just do a full configuration interaction calculation¹ with an overwhelmingly large basis set. Simple. The only problem with such an approach is that for more than a few tens of electrons, the calculation will take *forever* and this requires more patience than most people have.

The main problem in computational electronic structure theory, therefore, amounts to getting the required computer time to scale nicely as a function of system size without a significant accompanying loss of accuracy. In other words, how can we approximately solve the Schrödinger equation using a highly accurate method that increases in computational cost as a low-order polynomial with the size of the system? Currently the *only* known way to do this, using a

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method that remains tractable for systems containing up to at least a few thousand electrons, is the *continuum quantum Monte Carlo (QMC) method*. With some relatively simple tricks this can be made to scale as the square of the system size. A quantum chemistry “gold-standard” technique with the requisite accuracy, such as the coupled-cluster method CCSD(T), scales in most circumstances as the seventh power of the system size. By restricting one’s attention to atoms and very small molecules it is possible to pretend that this doesn’t matter, but it does, as anyone who wishes to study solids, surfaces, realistic industrial chemical reactions, or biochemistry will readily appreciate. If you double the size of the system, the cost of a QMC calculation goes up by a factor of 4 or maybe 8; that of a CCSD(T) calculation by a factor of 128. No contest.

Given this overwhelmingly favorable scaling, one might assume that QMC would be in widespread use throughout the quantum chemistry community. However, it is not, and one might wish to speculate as to why this is the case. One obvious reason is that despite the favorable scaling, the method has a large prefactor; that is, it is just an expensive technique. In the worst cases the computational cost might be, for example, around a thousand times more than the cost of a low-accuracy technique with comparable scaling, such as density functional theory (DFT).² This might sound disastrous, but it turns out that QMC is ideally suited to parallel computing and can exploit modern multiprocessor machines with significantly more efficiency than can most other methods. Use a 100-processor machine and it is only 10 times slower than DFT. Clearly, this is beginning to sound reasonable. The widespread lack of availability of parallel machines in the past is one reason for the limited takeup of QMC, but we are surely not so far off the time when it will be routine to have such a thing under one’s desk, and as we move into and then beyond the petascale era, it pays to be prepared.

From the usual quantum chemistry perspective one may derive a quantum of solace from the fact that the mainstream accurate techniques fall primarily into a relatively well-defined hierarchy with all levels systematically derivable from simple Hartree–Fock theory and the fact that all these methods are implemented in standard widely available software packages such as the GAUSSIAN-09 code.³ In general, in this field, analytically integrable basis sets—usually Gaussians—are preferred when representing the wavefunction, with an implied distaste for the messiness of numerical integration. Unfortunately, QMC is not directly related to the usual hierarchy of techniques, nor is it implemented in standard commercial codes. Furthermore, the way it works is very far from the quantum chemistry comfort zone; it is effectively a clever but brute-force way of carrying out numerical integration of high-dimensional wavefunctions, usually wedded to optimization techniques for improving the shape of the integrand. “Quantum Monte Carlo” is actually a catch-all term for various methods having in common only the use of *random sampling*, which turns out to be by far the most efficient way to do many-dimensional numerical integration. Exaggerating only slightly, a stochastic technique such as QMC has the disconcerting property of being *accurate* but not *precise* (it gives numbers essentially in agreement with

experiment but possessing a finite error bar), in contrast to the more usual techniques still tractable for large systems which are *precise* but not accurate (they give the wrong answer to 16 decimal places). Nevertheless, we all derive comfort from precision, and for good reason. QMC calculations may have to be run for a considerable time to get the error bar down to an acceptable value, particularly in cases where one has to compute *differences* between two similar but imprecise numbers (the error bar in the numerical derivative computed by a two-point finite difference might be around 100 times the error bar in the individual energies).

Where the method is used at all, two particular varieties of QMC are commonly employed: variational Monte Carlo (VMC) and diffusion Monte Carlo (DMC).^{4,5} In this chapter I give a short introduction to both. As will become apparent, VMC is a conceptually very simple technique; it can calculate arbitrary expectation values by sampling a given fixed many-electron wavefunction using the standard Monte Carlo integration algorithm. As the method turns out to be variational, it is also possible, to some extent, to *optimize* suitably parameterized wavefunctions using standard techniques. Not insisting that the wavefunction be expanded in analytically integrable functions confers the advantage that such wavefunctions can be *explicitly correlated*; that is, they can have a direct dependence on interparticle distances and thus a vastly more compact representation. This is not the final answer, of course, as no one knows any practical way to parameterize arbitrarily accurate wavefunctions; the problem here is clearly one of *complexity*. However, there exists a class of methods, collectively called *projector Monte Carlo*, which in principle solve quantum problems exactly. They attempt the much more difficult job of simultaneously creating and sampling the unknown exact ground-state wavefunction. Diffusion Monte Carlo is one such method. In principle, such techniques depend on guessed properties of the many-body wavefunction only in their computational efficiency, if at all. In particular, since the wavefunction is not represented in terms of a basis set but by the average distribution of a time-varying ensemble of particles, for most practical purposes the basis set problem is greatly minimized in DMC. Although a basis set is generally used, its sole purpose is to represent a *guiding function* required for importance sampling—that is, grouping the sampling points in regions where they are most required—and the final DMC energy depends only weakly on the *nodal surface* of this guiding function (i.e., the set of points in configuration space at which the function is zero). It should be noted for completeness that the use of wavefunctions implies zero temperature; for finite temperature simulations one is required to use density matrix formalisms, which are somewhat underdeveloped in this field.

People not familiar with this technique might wonder what sorts of quantum mechanical systems it can be or has been applied to. Specialists in density functional theory should certainly have heard of it, as its most famous early application was in Ceperley and Alder's essentially exact QMC calculations⁶ of the correlation energy of the homogeneous electron gas in 1980. The first popular local density approximations (LDAs) to the correlation energy density interpolated the accurate QMC values obtained at various densities while reproducing the exactly known limiting behavior at high and low densities. Various different

analytic approaches led to a number of widely used LDAs for the correlation functional such as Vosko–Wilk–Nusair (VWN)⁷ and Perdew–Zunger (PZ81).⁸ One could thus argue that QMC was largely responsible for the huge growth of the DFT industry during the 1980s and beyond. In this chapter, however, I shift the focus away from QMC calculations of model systems toward real continuum problems involving real atoms (as opposed to “pretend” atoms without separate nuclei and electrons, which are sometimes used). It took some time before such applications became routine. Significant calculations of “chemical” systems began to appear in the late 1980s and early 1990s (although completely general and publically available software had to wait for the twenty-first century). The accuracy of the technique was amply demonstrated; in molecules containing light atoms such as H and He, total energies accurate to greater than 0.01 kcal mol⁻¹ ($\approx 1.5 \times 10^{-5}$ Ha!) were produced, for example, on tens of thousands of points on the $\text{H} + \text{H}_2 \rightarrow \text{H}_2 + \text{H}$ potential energy surface.⁹ A list of interesting references to systems and physical problems that have been treated with QMC since then might include equations of state in solids,^{10–13} solid-state structural phase transitions,¹⁴ binding of molecules and their excitation energies,^{15–19} studies of exchange and correlation,^{20–23} optical bandgaps of nanocrystals,^{24,25} defects in semiconductors,^{26–28} transition metal oxide chemistry,^{29–31} band structures of insulators,^{32–34} quantum dots,³⁵ reconstruction of crystalline surfaces,³⁶ molecules on surfaces,^{37,38} and pairing in ultracold atomic gases.^{39–41} DFT, of course, continues to be the preeminent technique for treating problems in computational electronic structure theory. However, it has now been demonstrated convincingly that systems and problems for which an accurate determination of the total energy actually matters, and for which DFT is not sufficiently accurate, are more numerous than is generally believed. QMC calculations are particularly useful in trying to understand when this is the case, and at the time of writing, it is now believed to be the most accurate available technique that remains applicable to medium-sized and large systems.

This increase in accuracy can be traced to our insistence on retaining the $3N$ -dimensional many-electron wavefunction as our basic quantity despite the emphasis normally placed on reducing the dimensionality of the quantum problem. For example, the density used in DFT depends, of course, on only *three* independent variables, a somewhat clear improvement on 6×10^{23} . The fundamental point in favor of QMC is that the many-electron wavefunction satisfies the rather well-known and essentially simple fundamental differential equation⁴² discovered by Professor Schrödinger in 1926: namely, $\hat{H}\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) = E\Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$. If we wish to reformulate the problem in terms of the density, then it is an unfortunate fact that the exact equation satisfied by the ground-state density is *completely unknown*. In DFT, the complicated many-body problem is effectively subsumed in the definition of the exchange-correlation functional, whose correct mathematical expression is unlikely ever to be known exactly. The inevitable approximations to this quantity, from the various LDAs referred to above up to the best modern functionals, substantially reduce the attainable accuracy and the predictability of the method.

QMC has an additional advantage over conventional high-accuracy quantum chemistry techniques in that it may be trivially adapted to work in condensed-matter systems such as crystalline solids rather than just in atoms and molecules. All but the most modern standard solid-state texts in fact routinely deny the possibility of solving the many-particle Schrödinger equation in any meaningful way for large crystalline systems. For example, in Ashcroft and Mermin's well-known *Solid State Physics*⁴³ it is explicitly stated—referring to the many-electron Schrödinger equation—that “one has no hope of solving an equation such as this” and one must reformulate the problem in such a way as “to make the one-electron equations least unreasonable.” This ultimately turns out to mean Hartree–Fock theory; that is, we assume that the many-electron wavefunction can be represented as a single determinant of one-electron orbitals, and the solutions to the Hartree–Fock equations then give the optimum form of these orbitals. This presentation is slightly misleading in that the simplifying physical idea that allows one to use QMC, for example, in cases such as these is not the use of one-electron orbitals but rather the imposition of periodic boundary conditions. There can then be any form of many-electron wavefunction you like (including the possibility of having explicit dependence on the interparticle separations) representing a box of particles embedded in an infinite number of copies of itself. The “particles” sampling the many-body wavefunction can be visualized as a periodic array of electrons moving in tandem with each other rather than as individual electrons. It is clear that for this to have any chance of being an accurate approximation, the width of the exchange-correlation hole surrounding each electron must be substantially less than the lattice constant, and the box must be large enough so that the forces on the particles within it are very close to those in the bulk. If this is not the case, the calculation may have significant “finite-size errors” (these must anyway be monitored and corrected for).

How accurate must the total energies be in QMC calculations in order to get chemically useful information? We note that the main goal is often, even usually, the calculation of differences in energy between two arrangements of a set of atoms, giving us the energy barrier to some process, the energy required to create a defect, or whatever. Under most circumstances, high-quality QMC calculations will give the lowest energy (i.e., they will recover more of the “correlation energy” missing in Hartree–Fock theory) compared to other variational techniques. This does not guarantee that the *difference* in energy of the two systems, which may be very different from each other, will be similarly accurate. This is, of course, a standard problem for all electronic structure methods, which in large systems must rely on a cancellation of errors in energy differences. It is well known that, for example, the ordering of binding energies of isomeric atomic clusters in DFT often come out essentially randomly, depending on which exchange-correlation functional is used, meaning that it is not a reliable technique for answering questions such as: What is the smallest stable fullerene?⁴⁴ For high-quality error cancellation it is required that the error in the energy per atom be proportional to the number of atoms; otherwise, for example, a cohesive energy would not have a well-defined limit for large

systems. In general, the error cancellation in QMC has been found to be reliable. It has been demonstrated repeatedly that VMC calculations lead to errors that are proportional to the number of atoms, recovering something like 70 to 90% of the correlation energy independent of system size. In DMC the error is also proportional to the number of atoms, but the method can recover up to 100% of the correlation energy in favorable cases. For tractable QMC algorithms we also require that the number of wavefunction parameters must not increase too rapidly with system size (fortunately, this number generally increases only linearly, or at worst quadratically, and the function can be evaluated in a time period that rises as a low power of the system size). The wavefunction must also be easily computable at a given point in the configuration space.

Over the last decade a decent number of software packages have become available for performing QMC calculations of the type described in this chapter (a reasonably up-to-date list of such codes is maintained online⁴⁵). Our own efforts in this direction—due to various members of our group in Cambridge University’s Cavendish Laboratory—began in 1999 with publication of the first version of our general-purpose QMC computer program, CASINO.^{46,47} This code, which continues to be actively developed, is designed to perform variational and diffusion Monte Carlo calculations on essentially any conceivable system (subject to the obvious memory and time constraints). As well as the usual model systems, it can treat realistic finite objects such as atoms and molecules, or one may impose periodic boundary conditions in one, two, or three dimensions (corresponding to polymers, slabs, and solids) in order to model crystalline systems. We have coded up a large number of different types of interactions, particles, and many-body wavefunctions (the latter usually generated through interfaces to appropriate third-party codes). It can also calculate a great many properties in addition to accurate total energies. To complement the CASINO code and provide instruction in its use, and to counter the widespread lack of knowledge of these methods, we have for the last six years also run a pedagogical program of annual QMC summer schools in Tuscany, Italy.⁴⁸

4.2 VARIATIONAL MONTE CARLO

VMC is a relatively straightforward stochastic numerical integration method. It is in principle capable of computing quantum mechanical expectation values for any many-electron wavefunction whose value can be evaluated at arbitrary points in its configuration space. Given some trial wavefunction Ψ_T satisfying the appropriate boundary conditions, one may, for example, simply calculate the total energy as the expectation value of the many-body Hamiltonian operator $\hat{H}$,

$$\frac{\int \Psi_T^*(\mathbf{R}, \{\alpha\}) \hat{H} \Psi_T(\mathbf{R}, \{\alpha\}) d\mathbf{R}}{\int \Psi_T^*(\mathbf{R}, \{\alpha\}) \Psi_T(\mathbf{R}, \{\alpha\}) d\mathbf{R}} = E(\{\alpha\}) \geq E_0 \quad (4.1)$$

where $\mathbf{R}$ is a $3N$ -dimensional vector giving the configuration coordinates $(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$ of the N particles in the system (we ignore spin for the moment). The numerical integration is performed by sampling the configuration space of the wavefunction at random points. Now for a wavefunction with appropriate properties, the usual variational theorem tells us that by evaluating this integral we obtain an energy that is an upper bound to the exact ground-state energy; that is, energies for approximate wavefunctions are always higher than that of the true ground state. Such wavefunctions generally depend on some parameters—here collectively denoted by $\{\alpha\}$ —and these parameters may thus be varied to minimize an objective function such as the energy; in so doing, the “shape” of the wavefunction can be optimized.

Why do we sample the wavefunction at *random* points? Most people are familiar with ordinary fixed-grid quadrature methods for, say, integrating a regular one-dimensional function $f(x)$ over some interval. The simplest such method would be to define a regular grid of M points along the x -axis, evaluate the function at each of these points, and then use these numbers to calculate the mean value $\langle f \rangle$ of the function in that interval. The integral—the area under the function—is then just that mean value times the length of the interval. One can define more complicated and efficient schemes, all amounting simply to changing the positions of the points and/or assigning weights to them. The principal difference between such schemes and a Monte Carlo algorithm, then, is that in the latter the sampling points are chosen at random. This allows us to assume that the well-known *central limit theorem* is valid, and then for large enough M the estimate of the mean value will be normally distributed. The statistical uncertainty in the mean value $\langle f \rangle$ is then given by $\sigma = \sigma_{\text{sample}}/\sqrt{M}$ with $\sigma_{\text{sample}} = \sqrt{\langle f^2 \rangle - \langle f \rangle^2}$; that is, the error in the integral decreases as the square root of the number M of sampling points *irrespective of the dimensionality d of the integral*. For a standard grid method such as the trapezoidal rule, the error decreases as $O(M^{-2/d})$. Monte Carlo wins in more than four dimensions. To see the importance of this, let us take the particular example of a 100-dimensional integral. To make the estimate of this integral 10 times more accurate requires 100 times more work with Monte Carlo integration. With the trapezium rule it would require 10 to the power 50 times more work. Now it is the case that when evaluating quantum mechanical expectation values for an N -particle system we must do $3N$ -dimensional integrals; thus, for high-dimensional numerical integrations such as these, there is effectively *no alternative* to Monte Carlo methods. We conclude this point by noting that although good enough for many purposes, the assumption of the validity of the central limit theorem in this context is only approximate; a good technical discussion of the subtleties involved may be found in two recent papers by Trail.^{49,50}

Now in practice we do not wish to sample the random points from a uniform probability distribution—as effectively implied above—but, rather, to group the points in regions where the integrand is finite, and to do this in such a way as to minimize the sample variance. Such *importance sampling* requires us to generate points distributed according to some nonuniform probability distribution

$p(\mathbf{R})$, following which the calculation of the mean proceeds as usual. This sampling can be accomplished using a random walk moved according to the rules of the *Metropolis algorithm*.⁵¹ We propose random moves taken from some standard distribution (usually, Gaussians of appropriate width centered on the current points), always accepting moves to points of higher probability, and occasionally rejecting moves to regions of lower probability according to a particular formula obeying a detailed balance condition. Assuming *ergodicity*—that is, any point in the configuration space can be reached in a finite number of moves—the distribution of the moving points will converge to the desired $p(\mathbf{R})$ after some appropriate period of *equilibration*.

To apply this procedure to evaluate an integral such as Eq. (4.1), we need to rewrite the expression slightly so that the integrand looks like a probability distribution times some appropriate function to be evaluated at each point (that can later be averaged). What is the best probability distribution to use; that is, what distribution of points will minimize the sample variance? It can be shown that it is $p_{\text{best}}(\mathbf{R}) = |f(\mathbf{R})|/\int |f(\mathbf{R}')| d\mathbf{R}'$ (i.e., we concentrate sampling points in regions where the absolute value of the integrand is large). Note that one cannot do this exactly because in general we do not know the normalization integral in the denominator; the best one can do is to make $p(\mathbf{R})$ look as much like this as possible. It is readily seen that in our case $p(\mathbf{R}) = |\Psi_T(\mathbf{R})|^2$ —and although this is not the best sampling distribution to use according to the criterion above, it is close to it for an approximate eigenstate. We may therefore rewrite the expectation value of the Hamiltonian $\hat{H}$ with respect to the trial wavefunction Ψ_T as

$$\langle \hat{H} \rangle = \frac{\int |\Psi_T(\mathbf{R})|^2 E_L(\mathbf{R}) d\mathbf{R}}{\int |\Psi_T(\mathbf{R})|^2 d\mathbf{R}} \quad (4.2)$$

where the function to be evaluated, $E_L(\mathbf{R}) = \hat{H}(\mathbf{R})\Psi_T(\mathbf{R})/\Psi_T(\mathbf{R})$, is known as the *local energy*. If Ψ_T were in fact the exact ground-state wavefunction note that, according to Schrödinger's equation, the local energy should have a constant value E_0 over the entire configuration space. For an *approximate* wavefunction this is no longer the case; a plot of the local energy for 1000 Metropolis-sampled points in the configuration space of an approximate trial function for a hydrogen atom ("approximate" since it is expanded in a finite Gaussian basis set) might look like the wiggly line in Fig. 4.1. So having used the Metropolis algorithm to generate a sequence of configurations $\mathbf{R}$ distributed according to $\Psi_T^2(R)$, we may then compute the desired expectation value by averaging the set of local energies:

$$\langle \hat{H} \rangle = \frac{1}{M} \sum_{i=1}^M E_L(\mathbf{R}_i) = \frac{1}{M} \sum_{i=1}^M \frac{\hat{H}\Psi_T(\mathbf{R}_i)}{\Psi_T(\mathbf{R}_i)} \quad (4.3)$$

It should be clear from the figure that for hydrogen the energy thus obtained should (correctly) be -0.5 Ha plus or minus some small error bar. The error bar

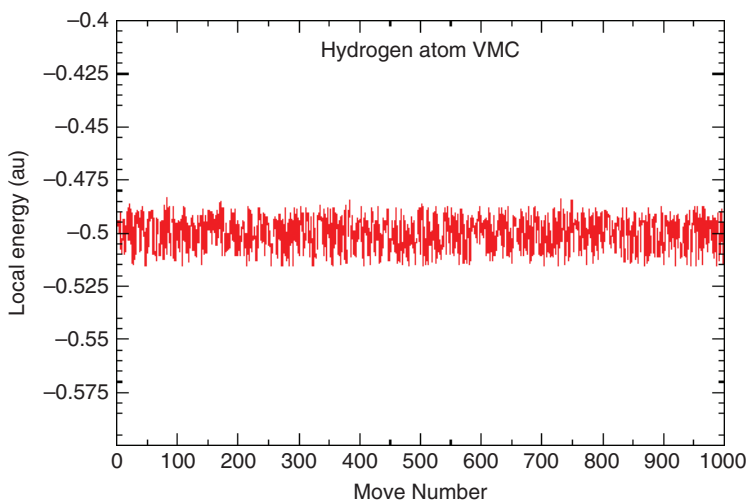


Fig. 4.1 Local energies of points in the random walk for a VMC run with an approximate wavefunction. All values are clustered around the true value of -0.5 Ha.

may need to be refined somewhat by particular statistical techniques to account for serial correlation of the points along the walk. Clearly, expectation values other than the energy could be calculated in a similar way.

4.3 WAVEFUNCTIONS AND THEIR OPTIMIZATION

As implied previously, VMC is to a large extent a basis set method. You get out what you put in. It is often difficult to prepare wavefunctions of equivalent quality for two systems whose energy we may wish to compare, and hence it is difficult to get an efficient cancellation of errors in the difference. If we had some way of writing down a formula representing the exact wavefunction for a given system, then great—that would be the end of the story. Unfortunately, that can essentially never be done exactly, as we know, and in practice it turns out that the main use of VMC is to *prepare the input* for DMC calculations, which are much more computationally efficient if supplied with a good “starting guess” for the many-body wavefunction. The better the guess, the more efficient the DMC, so we need now to consider how to represent wavefunctions and how to improve or optimize them within the limitations of VMC in order to supply that better guess.

In all optimization schemes, one needs a choice of objective function, that is, a function depending on Ψ_T , whose value is to be optimized. We have seen that the energy is one possibility; the variational theorem tells us that the answer will approach the true energy from above as we use better and better wavefunctions. Another approach is suggested by the *zero-variance principle*: Better wavefunctions will flatten out the wiggly line in Fig. 4.1. That is, as Ψ_T approaches

an exact eigenstate, the fluctuations will become smaller and the local energy $\hat{H}\Psi/\Psi$ approaches a constant, E_0 , everywhere in configuration space. Hence, the sample variance approaches zero. Through its direct influence on the variance of the energy, the accuracy of the trial wavefunction thus determines the amount of computation required to achieve a specified accuracy. When optimizing wavefunctions, one can therefore choose to use energy, variance, or possibly a combination of both as the objective function to be minimized.

Now before we consider the process of optimization in more detail, it is necessary to discuss how the many-electron wavefunction is to be represented in our calculations. It is an unfortunate fact that when learning about density functional theory, students are often presented with slightly hysterical motivation (with a generous sprinkling of exclamation marks) for avoiding playing with the wavefunction at all. For example, we might learn that the wavefunction

is a high dimensional object, dependent on $3N$ electron Cartesian coordinates, N electron spin coordinates, and $3N_n$ nuclear coordinates. For a molecule such as benzene, with 12 nuclei and 42 electrons, this object exists in a $162 - 6 = 156$ -dimensional Cartesian space! The determinant has $42! = 1.4 \times 10^{51}$ terms! We can't even make a picture of it. If we were to store it as a numerical object, with a resolution of 0.1 au out to 10 au, we would need 10^{312} numbers to store, at single precision (4 bytes/number), at 2 Gbytes/cm² (counting electrical connections access), we would need more than 10^{293} km² of surface to store that information. The Earth has a surface area of less than 10^9 km². The promised knowledge hidden in the Schrödinger equation is not quite easily accessible! We must make do with much much less. How much less can we do with?⁵⁵

Faced with the awful prospect of finding 10^{284} planets in order to store the wavefunction for a single benzene molecule (and forgetting that the Kohn–Sham single-determinant wavefunction for the noninteracting system that he'll be forced to use in practical DFT calculations has just as much information), the horrified student gives up the game, and grabs at the density with a sigh of relief.

But, of course, you don't need to store every possible value of the wavefunction. From a VMC perspective, you just need to be able to *sample* it. We evaluate the formula for Ψ_T at a small subset of sampling points in the configuration space; that is, if the electrons and nuclei have certain fixed positions in space, what is the value of the wavefunction at these points? In DMC, where we don't have a formula for the wavefunction, we avoid having to do that directly by appropriately *weighting* the calculated values of a trial importance-sampling function (usually, just the optimized VMC wavefunction).

So what formula shall we use for Ψ_T ? We begin by noting that the simplest basic techniques in quantum chemistry—Hartree–Fock theory, say—give in general approximately the right density but the wrong pair-correlation function. Knowing that with Monte Carlo methods we can simply multiply in appropriate pair-correlation functions, this gives us confidence that a good starting point would be the usual 1920s-style legacy of appropriate fictions. First the fiction that each electron has its own personal wavefunction (an orbital). The simplest

way of making a many-electron wavefunction out of a set of orbitals is to multiply all the orbitals together (the Hartree product), but the electrons are then all completely uncorrelated. We then note that if we use one-electron functions as building blocks, we require that the many-electron function built from them satisfies the Pauli principle. The simplest way of doing this and thereby making the entire wavefunction correctly antisymmetric is to say that the many-electron wavefunction is a *single determinant* of orbitals, that is, an antisymmetrized linear combination of however many Hartree products are necessary to obtain all possible permutations of the electron labels among the orbitals.

It is important to note how we are dealing with spin. Strictly speaking, of course, the Schrödinger equation is for *spinless* particles only. We must therefore *invent* a two-valued property of particles called *spin* that is tacked onto the Schrödinger theory by modifying the form of the wavefunction. We say that there are two types of electron, *spin-up* and *spin-down*, and we define two corresponding orthonormal spin functions (formally, we define *spin orbitals* by multiplying the one-electron spatial wavefunction by a simple *spin function* with the appropriate properties). The orthogonality imposed introduces a type of correlation (confusingly referred to as *exchange*), which results in same-spin electrons tending to avoid each other. At a point in the configuration space where two electrons with up-spin labels (say) are very close to each other, the value of the entire many-electron wavefunction is correspondingly small; each electron is said to be surrounded by a *Fermi hole* or *exchange hole*. In the normal way of doing this we declare that there is a global quantization axis with respect to which the spins are measured, thus allowing us to deal with simple spin orbitals rather than more complicated two-component spinor orbitals; the *spin density* in this sense then becomes a scalar quantity rather than a vector magnetization density.

Solving the usual Hartree–Fock equations gives the optimum form of the spin orbitals—under the usually incorrect assumption that the true many-electron wavefunction can be accurately represented by a single determinant. To go beyond the Hartree–Fock level, quantum chemists normally expand the difference between this single-determinant wavefunction and the exact one using a basis set of appropriate many-electron functions: namely, all possible excited determinants obtained by replacing orbitals occupied in the ground-state determinant by unoccupied ones. The fact that literally *billions* of terms are normally required for this expansion to converge tells us that, in fact, the basis set looks nothing like the function we are trying to expand. This would ordinarily be understood to be a bad thing and, indeed, the approach becomes completely intractable for more than a few tens of electrons. Note also that formally speaking, the method (known as *full configuration interaction*) converges to the right answer only if the one-electron basis set used to expand the orbitals is infinitely large. We shall adopt a different approach.

An aside: It is instructive to reflect on the nature of spin using an interpretation of quantum mechanics with a realistic ontology such as the de Broglie–Bohm pilot-wave theory.⁵² In this interpretation—in apparent agreement with much experimental evidence, if you think about it—*wave–particle duality* is taken

literally by assuming that particles *and* a wave field (represented mathematically by the wavefunction) both have a permanent objective existence. An analysis of the particle probability current then shows us that the field apparently exerts a force on the particles, and that particles guided by a wave undergoing Schrödinger evolution eventually become distributed as the square of the wave over time, even if they were not so distributed initially—hence the Born rule.¹⁶³ Because the potential energy function of the wave field turns out to depend on the spin, it is then evident that spin is not a property of particles at all, but a property of the wave field; in fact, it is the *polarization-dependent part of the wave field's angular momentum*. In a relativistic theory the wave field would be a vector field with states of polarization, and this may be represented by the scalar Schrödinger field with tacked-on spinors. The effects of the Pauli exclusion principle turn out to be due to the influence of the “quantum force” on the electron trajectories (the electrons have trajectories!) and their consequent inability to pass through nodal surfaces in the wave field. Interesting!

We have now defined our starting point. How, then, do we get to properly correlated wavefunctions which depend explicitly on the distances between particles? We simply multiply what we have already by an appropriate correlation function. The resulting functional form is known as the *Slater–Jastrow wavefunction*, and this is the most common choice for quantum Monte Carlo calculations. It consists of a single Slater determinant (or sometimes a linear combination of a small number of them) multiplied by a positive-definite Jastrow correlation function,^{53,54} which is symmetric in the electron coordinates and depends on the interparticle distances. The Jastrow factor allows efficient inclusion of both long- and short-range correlation effects. As we shall see, however, the final DMC answer depends only on the nodal surface of the wavefunction, and this cannot be affected by the nodeless Jastrow. In DMC it serves mainly to decrease the amount of computer time required to achieve a given statistical error bar and to improve the stability of the algorithm.

So the basic functional form of the Slater–Jastrow function is

$$\Psi_T(\mathbf{X}) = e^{J(\mathbf{X})} \sum_n c_n D_n(\mathbf{X}) \quad (4.4)$$

where $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N)$ and $\mathbf{x}_i = \{\mathbf{r}_i, \sigma_i\}$ denotes the space-spin coordinates of electron i , $e^{J(\mathbf{X})}$ is the Jastrow factor, the c_n are coefficients, and the $D_n(\mathbf{X})$ are Slater determinants of single-particle orbitals,

$$D(\mathbf{X}) = \begin{vmatrix} \psi_1(\mathbf{x}_1) & \psi_1(\mathbf{x}_2) & \cdots & \psi_1(\mathbf{x}_N) \\ \psi_2(\mathbf{x}_1) & \psi_2(\mathbf{x}_2) & \cdots & \psi_2(\mathbf{x}_N) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_N(\mathbf{x}_1) & \psi_N(\mathbf{x}_2) & \cdots & \psi_N(\mathbf{x}_N) \end{vmatrix} \quad (4.5)$$

The orbitals in the determinants are usually obtained from self-consistent DFT or Hartree–Fock calculations, and are assumed to be products of spatial and spin

factors,

$$\psi_\alpha(\mathbf{x}) = \psi_\alpha(\mathbf{r})\delta_{\sigma,\sigma_\alpha} \quad (4.6)$$

Here $\delta_{\sigma,\sigma_\alpha} = 1$ if $\sigma = \sigma_\alpha$ and zero otherwise. If the determinant contains $N_\uparrow$ orbitals with $\sigma_\alpha = \uparrow$ and $N_\downarrow = N - N_\uparrow$ with $\sigma_\alpha = \downarrow$, it is an eigenfunction of $\hat{S}_z$ with eigenvalue $(N_\uparrow - N_\downarrow)/2$. To avoid having to sum over spin variables in QMC calculations, one generally replaces the determinants D_n by products of separate up- and down-spin determinants,

$$\Psi_T(\mathbf{R}) = e^{J(\mathbf{R})} \sum_n c_n D_n^\uparrow(\mathbf{r}_1, \dots, \mathbf{r}_{N_\uparrow}) D_n^\downarrow(\mathbf{r}_{N_\uparrow+1}, \dots, \mathbf{r}_N) \quad (4.7)$$

where $\mathbf{R} = (\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N)$ denotes the spatial coordinates of all the electrons. This function is not antisymmetric under exchange of electrons with opposite spins, but it can be shown that it gives the same expectation value as $\Psi_T(\mathbf{X})$ for any spin-independent operator.

Now what about the Jastrow factor $e^{J(\mathbf{R})}$? Through its explicit dependence on interparticle distances, this introduces correlation directly in such a way that the overall wavefunction is smaller when particles with repulsive interactions—two electrons, for example—are close together and larger when particles with attractive interactions (such as an electron and a positron) approach each other. The Jastrow is also useful for ensuring that the system obeys the appropriate Kato cusp conditions⁵⁶ for two-particle coalescences, a known property of the exact many-body wavefunction. A correctly shaped cusp gives rise to a divergence in the local kinetic energy equal and opposite to the divergence in the interparticle Coulomb interaction as two particles move toward each other, which is helpful given that their sum—the local energy—is supposed to be constant. This is particularly important in QMC, where large local-energy divergences can lead to inaccurate statistics or, in DMC, severe numerical instabilities. Note that with regular quantum chemistry methods one does not in general worry about either electron–electron or electron–nuclear cusps, as the wavefunction forms generally employed do not have the correct structure to represent them. The use of Gaussian basis sets, for example, which have zero gradient at the nucleus, forgoes in principle any possibility of representing a gradient discontinuity there. One can get away with this since the usual methods are effectively *analytically integrating* the wavefunction, rather than sampling quantities at individual points in its configuration space as in QMC; the small differences in the “area under the graph” due to having an incorrect cusp can be seen to be relatively unimportant.

The full Jastrow function that we have typically used in CASINO contain one- and two-electron terms and may be inhomogeneous (i.e., depend on the distances of the electrons from the nuclei). The exact functional form is quite complicated and is presented in detail elsewhere.⁵⁴ Essentially, our Jastrow consists of separate electron–electron (u), electron–nucleus (χ), and electron–electron–nucleus (f) terms, which are expanded in appropriate polynomials with optimizable coefficients and are forced to go to zero at some optimizable cutoff radii (as they must

do in periodic systems). In systems with no nuclei, such as the homogeneous electron gas, we might use a much simpler one-parameter Jastrow function, such as

$$J(\mathbf{R}) = - \sum_{i>j}^N u_{\sigma_i, \sigma_j}(r_{ij}) \quad \text{and} \quad u_{\sigma_i, \sigma_j}(r_{ij}) = \frac{A}{r_{ij}} \left(1 - e^{-r_{ij}/F_{\sigma_i, \sigma_j}} \right) \quad (4.8)$$

Here r_{ij} is the distance between the N electrons i and j , the σ subscript is the spin label, and the parameter F is chosen so that the electron–electron cusp conditions are obeyed (i.e., $F_{\uparrow\uparrow} = \sqrt{2A}$ and $F_{\uparrow\downarrow} = \sqrt{A}$). The value of A could be optimized using variance minimization or whatever. For systems with both electrons and nuclei present, one can write a standard Jastrow with all three terms (ignoring the spin dependence for clarity) as follows:

$$J(\mathbf{R}, \{\mathbf{r}_I\}) = \sum_{i>j}^N u(r_{ij}) + \sum_{i=1}^N \sum_{I=1}^{N_I} \chi_I(r_{iI}) + \sum_{i>j}^N \sum_{I=1}^{N_I} f_I(r_{ij}, r_{iI}, r_{jI}) \quad (4.9)$$

Other terms, such as an extra plane-wave expansion in the electron–electron separation for periodic systems or an additional three-body term, are part of our standard Jastrow⁵⁴ and can be useful in certain circumstances but are not usually necessary. For particles with attractive interactions one finds that the usual Slater–Jastrow form is not appropriate, and in order to get a better description of exciton formation one might use a determinant of “pairing orbitals” instead.⁵⁷ A further recent advance by members of our group has been the development of a completely general functional form for the Jastrow factor which allows the inclusion of arbitrary higher-order terms (depending on, for example, the separation of four or more particles); this has now been implemented in our code.⁵⁸

To convince yourself that the Slater–Jastrow function is doing what it should, consider Fig. 4.2. These are the results of simple VMC calculations of the spin-dependent pair correlation function (PCF) in a silicon crystal with an electron fixed at a bond center.²¹ The figure on the left is for parallel spins and corresponds to the Fermi or *exchange hole*. The figure on the right is for antiparallel spins and corresponds to the *correlation hole*; note that the former is much wider and deeper than the latter. We have here then a pretty illustration of the different levels of theory that we use. In Hartree theory (where we use a Hartree product of all the orbitals as a wavefunction, and which thus corresponds to entirely uncorrelated electrons), both PCFs would have a value of 1 everywhere. In Hartree–Fock theory, the left-hand plot would look very similar, but the antiparallel PCF on the right would be 1 everywhere. The energy lowering over Hartree theory caused by the fact that parallel spin electrons tend to avoid each other is essentially the *exchange energy*, which correctly has a negative sign. It is slightly sobering to note that the entire apparatus of quantum chemistry (an expansion in billions of determinants) is devoted to modeling the little hole on the right and thereby evaluating the correlation energy. In QMC our quantum of solace comes from

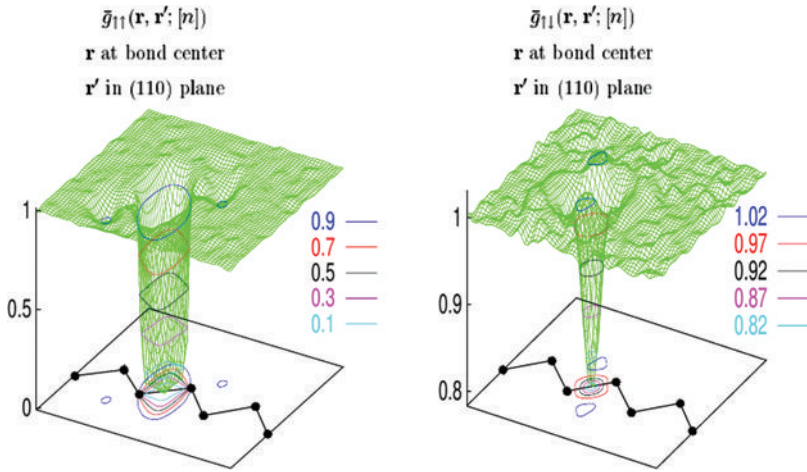


Fig. 4.2 (color online) VMC plots of the pair correlation function for (on the left) parallel spins and (on the right) antiparallel spins using a Slater–Jastrow wavefunction. The data are shown for crystalline silicon in the (110) plane passing through the atoms and shows the pair correlation functions around a single electron fixed at a bond center. The atoms and bonds in the (110) plane are represented schematically. (From Ref. 20, with permission. Copyright © 1997 by The American Physical Society.)

our *compact representation*; with a Slater–Jastrow function we can do the same thing in VMC using a simple polynomial expansion involving a few tens of parameters, and if this is not accurate enough we can make the necessary minor corrections to it using the DMC algorithm.

However, we do not know a priori what the shape of the hole is, and we must therefore optimize the various parameters in the Slater–Jastrow function in order to find out. The usual procedure is to leave the Slater determinant part alone and optimize the Jastrow factor. With a full inhomogeneous Jastrow such as that of Eq. (4.9), we generally optimize the coefficients of the various polynomial expansions (which appear linearly in the Jastrow factor) and the cutoff radii of the various terms (which are nonlinear). The linearity or otherwise of these terms clearly has a bearing on their ease of optimization. There is, of course, no absolute prohibition on optimizing the Slater part and one might also envisage, for example, optimization of the coefficients of the determinants of a multideterminant wavefunction, or even the orbitals in the Slater determinants themselves (although the latter is quite difficult to do in general, and often pointless). A higher-order technique called *backflow*, to be explained in a subsequent section, also involves functions with optimizable parameters. We thus turn our attention to the technicalities of the optimization procedure.

Now, optimization of the wavefunction is clearly a critical step; it is also a numerically difficult one. It is apparent that the parameters appear in many different contexts, they need to be optimized in the presence of noise, and there can be a great many of them. As has already been stated, there are two basic

approaches. Until recently, the most widely used was the optimization of the variance of the energy,

$$\sigma_E^2(\alpha) = \frac{\int [\Psi_T^\alpha(\mathbf{R})]^2 [E_L^\alpha(\mathbf{R}) - E_V^\alpha]^2 d\mathbf{R}}{\int [\Psi_T^\alpha(\mathbf{R})]^2 d\mathbf{R}} \quad (4.10)$$

where E_V is the variational energy, with respect to the set of parameters $\{\alpha\}$. Now, of course, there is no reason that one may not optimize the energy directly, and because wavefunctions corresponding to the minimum energy turn out to have more desirable properties, this has become the preferred approach in the last few years. Historically, variance minimization was much more widely used^{60,61}—not just for trivial reasons such as the variance having a known lower bound of zero—but most important because of the difficulties encountered in designing a robust, numerically stable algorithm to minimize the energy, particularly in the case of large systems.

First, I briefly summarize how a simple variance minimization is done. Beginning with an initial set of parameters α_0 (generated, for example, simply by zeroing the Jastrow polynomial coefficients), we proceed to minimize $\sigma_E^2(\alpha)$ with respect to them. A correlated-sampling approach turns out to be most efficient. First, a set of some thousands of *configurations* distributed according to $|\Psi_T^{\alpha_0}|^2$ is generated. Practically speaking, a configuration in this sense is just a snapshot of the system taken at intervals during a preliminary VMC run and consists of the current particle positions and the associated interaction energies written on a line of a file. We then calculate the variance in the energies for the fully sampled set of configurations. This is the objective function to be minimized. Now, unfortunately, every time we modify the parameters slightly, the wavefunction changes and our configurations are no longer distributed according to the square of the current Ψ_T^α , but to the square of the initial wavefunction $\Psi_T^{\alpha_0}$. In principle, therefore, we should regenerate the configurations, a relatively expensive procedure. The correlated sampling is what allows us to avoid this; we reuse the initial set of configurations simply by including appropriate weights w in the formula for the variance:

$$\sigma_E^2(\alpha) = \frac{\int [\Psi_T^{\alpha_0}(\mathbf{R})]^2 w_\alpha^{\alpha_0} [E_L^\alpha(\mathbf{R}) - E_V(\alpha)]^2 d\mathbf{R}}{\int [\Psi_T^{\alpha_0}(\mathbf{R})]^2 w_\alpha^{\alpha_0} d\mathbf{R}} \quad (4.11)$$

where

$$E_V(\alpha) = \frac{\int [\Psi_T^{\alpha_0}(\mathbf{R})]^2 w_\alpha^{\alpha_0} E_L^\alpha(\mathbf{R}) d\mathbf{R}}{\int [\Psi_T^{\alpha_0}(\mathbf{R})]^2 w_\alpha^{\alpha_0} d\mathbf{R}} \quad (4.12)$$

and the weight factors $w_{\alpha}^{\alpha_0}$ are given simply by

$$w_{\alpha}^{\alpha_0} = \frac{[\Psi_T^{\alpha}(\mathbf{R})]^2}{[\Psi_T^{\alpha_0}(\mathbf{R})]^2} \quad (4.13)$$

The parameters α are then adjusted until $\sigma_E^2(\alpha)$ is minimized. This may be done using standard algorithms which perform an unconstrained minimization of a sum of m squares of functions that contain n variables (where $m \geq n$) without requiring the derivatives of the objective function (see, e.g., Ref. 59). Although in principle we do not need to regenerate the configurations at all, one finds in practice that it usually pays to recalculate them occasionally when the wavefunction strays very far from its initial value. Generally, this needs to be done only a couple of times before we obtain complete convergence within the statistical noise.

There is a problem, however. Thus far we have described the optimization of what is known as the *reweighted variance*. In the limit of perfect sampling, the reweighted variance is equal to the actual variance, and is therefore independent of the initial parameters and the configuration distribution, so that the optimized parameters would not change over successive cycles. The problem arises from the fact that the weights may vary rapidly as the parameters change, especially for large systems. This can lead to severe numerical instabilities. For example, one or a few configurations acquire an exceedingly large weight, incorrectly reducing the estimate of the variance almost to zero. Somewhat surprisingly, perhaps, it usually turns out that the best solution to this is to do without the weights at all; that is, we minimize the *unreweighted variance*. We can do this because the minimum value of the variance (zero) is obtained only if the local energy is constant throughout configuration space, and this is possible only for eigenfunctions of the Hamiltonian. This procedure turns out to have a number of advantages beyond improving the numerical stability. The self-consistent minimum in the unreweighted variance almost always turns out to give lower energies than the minimum in the reweighted variance. (For some examples of this for model systems, see Ref. 62.)

It was recognized only relatively recently⁶² that one can obtain a huge speed-up in the optimization procedure for parameters that occur linearly in the Jastrow, that is, for Jastrows expressible as $\sum_{\alpha} \alpha_n f_n(\mathbf{R})$. These are the most important optimizable parameters in almost all wavefunctions that we use. The reason this can be done is that the unreweighted variance can be written analytically as a quartic function of the linear parameters. This function usually has a single minimum in the parameter space, and as the minima of multidimensional quartic functions may be found very rapidly, the optimization is extraordinarily efficient compared to the regular algorithm, in particular because we no longer need to generate large numbers of configurations to evaluate the variance. The main *non-linear* parameters in the Jastrow factor are the cutoff lengths where the function is constrained to go to zero. These are important variational parameters, and some attempt to optimize them should always be made. We normally recommend that

a (relatively cheap) calculation using the standard variance minimization method should be carried out in order to optimize the cutoff lengths, followed by an accurate optimization of the linear parameters using the fast minimization method. For some systems, good values of the cutoff lengths can be supplied immediately (e.g., in periodic systems at high density with small simulation cells, the cutoff length L_u should be set equal to the Wigner–Seitz radius of the simulation cell).

Let us now move on to outlining the theory of *energy minimization*. We know that except in certain trivial cases the usual trial wavefunction forms cannot in general provide an exact representation of energy eigenstates. The minima in the energy and variance therefore do not coincide. Energy minimization should thus produce lower VMC energies, and although it does not necessarily follow that it produces lower DMC energies, experience indicates that more often than not, it does. It is also normally stated that the variance of the DMC energy is more or less proportional to the difference between the VMC and DMC energies,^{63,64} so one might suppose that energy-optimized wavefunctions may be more efficient in DMC calculations.

For a long time, efficient energy minimization with QMC was extremely problematic. The methods that have now been developed are based on a well-known technique for finding approximations to the eigenstates of a Hamiltonian. One expands the wavefunction in some set of basis states, $\Psi_T(\mathbf{R}) = \sum_{i=1}^N a_i \phi_i(\mathbf{R})$. Following calculation of the Hamiltonian and overlap matrix elements, $H_{ij} = \langle \phi_i | \hat{H} | \phi_j \rangle$ and $S_{ij} = \langle \phi_i | \phi_j \rangle$, the two-sided eigenproblem $\sum_j H_{ij} a_j = E \sum_j S_{ij} a_j$ may be solved through standard diagonalization techniques. People have tried to do this in QMC directly,⁶⁵ but it is apparent that the number of configurations used to evaluate the integrals converges slowly because of statistical noise in the matrix elements. As shown in Ref. 66, however, far fewer configurations are required if the diagonalization is first reformulated as a least-squares fit. Let us assume that the result of operating with $\hat{H}$ on any basis state ϕ_i is just some linear combination of all the functions ϕ_i (technically speaking, the set $\{\phi_i\}$ is then said to span an *invariant subspace* of $\hat{H}$). We may thus write (for all i)

$$\hat{H}\phi_i(\mathbf{R}) = \sum_{j=1}^N A_{ij}\phi_j(\mathbf{R}) \quad (4.14)$$

To compute the required eigenstates and associated eigenvalues of $\hat{H}$, we then simply diagonalize the A_{ij} matrix. Within a Monte Carlo approach we could evaluate the $\phi_i(\mathbf{R})$ and $\hat{H}\phi_i(\mathbf{R})$ for N uncorrelated configurations generated by a VMC calculation and solve the resulting set of linear equations for the A_{ij} . For problems of interest, however, the assumption that the set $\{\phi_i\}$ spans an invariant subspace of $\hat{H}$ does not hold, and there exists no set of A_{ij} that solves Eq. (4.14). If we took N configurations and solved the set of N linear equations, the values of A_{ij} would depend on which configurations had been chosen. To overcome this problem, a number of configurations $M \gg N$ is sampled to obtain

an overdetermined set of equations which can be solved in a least-squares sense using the singular value decomposition technique. In Ref. 66 it is recommended that Eq. (4.14) be divided by $\Psi_T(\mathbf{R})$ so that in the limit of perfect sampling the scheme corresponds precisely to standard diagonalization.

The method of Ref. 66 is pretty good for linear parameters. How might we generalize it for nonlinear parameters? The obvious way is to consider the basis of the initial trial wavefunction ($\phi_0 = \Psi_T$) and its derivatives with respect to the variable parameters, $\phi_i = \partial \Psi_T / \partial a_i|_{a_i^0}$. The simplest such algorithm is, in fact, unstable, and this turns out to be because the implied first-order approximation is often not good enough. To overcome this problem, Umrigar et al. introduced a stabilized method^{67,68} that works well and is quite robust (the details need not concern us here). The VMC energies given by this method are usually slightly lower than those obtained from variance minimization.

David Ceperley once asked: “How many graduate students’ lives have been lost optimizing wavefunctions?”⁶⁹ That was in 1996. To give a more twenty-first century feeling for the time scale involved in optimizing wavefunctions, I can tell you about the weekend a few years back when I added the entire G2-1 set^{70,71} to the examples included with the CASINO distribution. This is a standard set of 55 molecules with various experimentally well-characterized properties intended for benchmarking of different quantum chemistry methods (see, e.g., Ref. 72). Grossman has published the results of DMC calculations of these molecules using pseudopotentials,¹⁶ and we have now done the same with all-electron calculations.^{73,74} It took a little over three days using only a few single-processor workstations to create all 55 sets of example files from scratch, including optimizing the Jastrow factors for each molecule. Although if one concentrated very hard on each individual case, one might be able to pull a little more energy out of a VMC simulation, the optimized Jastrow factors were all good enough to be used as input to DMC simulations. The entire procedure of variance minimization can be, and in CASINO is, thoroughly automated, and provided that a systematic approach is adopted, optimizing VMC wavefunctions is not the complicated time-consuming business that it once was. This is certainly the case if one requires the optimized wavefunction only for input into a DMC calculation, in which case one need not be overly concerned with lowering the VMC energy as much as possible. I suggest that the process is sufficiently automated these days that graduate students are better employed elsewhere; certainly we have not suffered any fatalities here in Cambridge.

4.4 DIFFUSION MONTE CARLO

Let us imagine that we are ignorant, or have simply not been paying attention in our quantum mechanics class, and that we believe that the wavefunction of the hydrogen atom has the shape of a big cube centered on the nucleus. If we tried to calculate the expectation value of the Hamiltonian using VMC, we would obtain an energy that was substantially in error. What DMC does, in essence, is to automatically correct the shape of the guessed square box wavefunction so

that it looks like the correct exponentially decaying one before calculating the expectation value. In principle it can do this even though our formula for the VMC wavefunction that we have spent so long justifying turns out not to have enough variational freedom to represent the true wavefunction. This is clearly a nice trick, particularly when—as is more usual—we have very little practical idea of what the *exact* many-electron wavefunction looks like.

As one might expect, the DMC algorithm is necessarily rather more involved than that for VMC. I think that an approachable way of understanding it is to focus on the properties of quantum mechanical propagators, so we begin by reminding ourselves about these. Let's say that we wish to integrate the time-dependent Schrödinger equation,

$$i\hbar \frac{\partial \Psi(\mathbf{R}, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi(\mathbf{R}, t) + V(\mathbf{R}, t) \Psi(\mathbf{R}, t) = \hat{H} \Psi(\mathbf{R}, t) \quad (4.15)$$

where $\mathbf{R} = \{\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N\}$, V is the potential energy operator, and $\nabla = (\nabla_1, \nabla_2, \dots, \nabla_N)$ is the $3N$ -dimensional gradient operator. Integrating this is equivalent to wanting a formula for Ψ , and to find this, we must invert this differential equation. The result is an integral equation involving the *propagator* K :

$$\Psi(\mathbf{R}, t) = \int K(\mathbf{R}, t; \mathbf{R}', t') \Psi(\mathbf{R}', t') d\mathbf{R}' \quad (4.16)$$

The propagator is interpreted as the probability amplitude for a particle to travel from one place to another (in this case, from $\mathbf{R}'$ to $\mathbf{R}$) in a given time $t - t'$. It is a *Green's function* for the Schrödinger equation. We see that the probability amplitude for a particle to be at $\mathbf{R}$ sometime in the future is given by the probability amplitude of it traveling there from $\mathbf{R}'$ —which is just $K(\mathbf{R}, t; \mathbf{R}', t')$ —weighted by the probability amplitude of it actually starting at $\mathbf{R}'$ in the first place—which is $\Psi(\mathbf{R}', t')$ —summed over all possible starting points $\mathbf{R}'$. This is a straightforward concept.

How might we calculate the propagator? A typical way might be to use the *Feynman path-integral* method. For given start and end points $\mathbf{R}'$ and $\mathbf{R}$, one gets the overall amplitude by summing the contributions of the infinite number of *all* possible “histories” or paths that include those points. It doesn't matter why for the moment (look it up!), but the amplitude contributed by a particular history is proportional to $e^{iS_{\text{cl}}/\hbar}$ where S_{cl} is the classical *action* of that history (i.e., the time integral of the classical Lagrangian $\frac{1}{2}mv^2 - V$ along the corresponding phase-space path of the system). The full expression for the propagator in Feynman's method may then be written as

$$K^F(\mathbf{R}, t; \mathbf{R}', t') = N \sum_{\text{all paths}} \exp \left[\frac{i}{\hbar} \int_{t'}^t L_{\text{cl}}(t'') dt'' \right] \quad (4.17)$$

An alternative way to calculate the propagator is to use the de Broglie–Bohm pilot-wave interpretation of quantum mechanics,⁵² where the electrons both objectively exist and have the obvious definite trajectories derived from a straightforward analysis of the streamlines of the quantum mechanical probability current. From this perspective we find that we can achieve precisely the same result as that obtained using the Feynman method, by integrating the *quantum* Lagrangian $L_q(t) = \frac{1}{2}mv^2 - (V + Q)$ along precisely *one* path—the path that the electron actually follows—as opposed to linearly superposing amplitudes obtained from the *classical* Lagrangian associated with the infinite number of all possible paths. Here Q is the *quantum potential*, which is the potential energy function of the quantum force (the force that the wave field exerts on the electrons). It is easy to show that the equivalent pilot-wave propagator is

$$K^B(\mathbf{R}, t; \mathbf{R}', t') = \frac{1}{J(t)^{1/2}} \exp \left[\frac{i}{\hbar} \int_{t'}^t L_q(t'') dt'' \right] \quad (4.18)$$

where J is a simple Jacobian factor. This formula should be contrasted with Eq. (4.17). One should also note that because de Broglie–Bohm trajectories do not cross, one need not sum over all possible starting points $\mathbf{R}'$ to compute $\Psi(\mathbf{R}, t)$ —one simply uses the $\mathbf{R}'$ that the unique trajectory passes through.

What is the connection of all this with the diffusion Monte Carlo method? Well, in DMC an arbitrary starting wavefunction is evolved using a (Green’s function) propagator just like the ones we have been discussing. The main difference is that the propagation occurs in *imaginary time* $\tau = it$ as opposed to real time t . For reasons that will shortly become apparent, this has the effect of “improving” the wavefunction (i.e., making it look more like the ground state as imaginary time passes). For technical reasons, it also turns out that the propagation has to take place in a sequence of very short hops in imaginary time, so our evolution equation now looks like this:

$$\Psi(\mathbf{R}, \tau + \delta\tau) = \int K^{\text{DMC}}(\mathbf{R}, \mathbf{R}', \delta\tau) \Psi(\mathbf{R}', \tau) d\mathbf{R}' \quad (4.19)$$

The evolving wavefunction is not represented in terms of a basis set of known analytic functions but by the distribution in space and time of randomly diffusing electron positions over an ensemble of copies of the system (“configurations”). In other words, the DMC method is a *stochastic projector method* whose purpose is to evolve or project out the solution to the *imaginary-time Schrödinger equation* from an arbitrary starting state. We shall write this equation—which is simply what you get by taking the regular time-dependent equation and substituting τ for the time variable it —in atomic units as

$$-\frac{\partial \Psi^{\text{DMC}}(\mathbf{R}, \tau)}{\partial \tau} = -\frac{1}{2} \nabla^2 \Psi(\mathbf{R}, \tau) + (V(\mathbf{R}) - E_T) \Psi(\mathbf{R}, \tau) \quad (4.20)$$

Here the real variable τ measures the progress in imaginary time, and for purposes to be revealed presently, I have included a constant E_T , an energy offset to the zero of the potential which affects only the wavefunction normalization.

How, then, does propagating our trial function in imaginary time “improve” it? For eigenstates, the general solution to the usual time-dependent Schrödinger equation is clearly $\phi(\mathbf{R}, t) = \phi(\mathbf{R}, 0)e^{-i(\hat{H}-E_T)t}$. By definition, we may expand an arbitrary “guessed” $\Psi(\mathbf{R}, t)$ in terms of a complete set of these eigenfunctions of the Hamiltonian $\hat{H}$:

$$\Psi(\mathbf{R}, t) = \sum_{n=0}^{\infty} c_n \phi_n(\mathbf{R}) e^{-i(E_n - E_T)t} \quad (4.21)$$

On substituting it with imaginary time τ the oscillatory time dependence of the complex exponential phase factors becomes an exponential decay:

$$\Psi(\mathbf{R}, \tau) = \sum_{n=0}^{\infty} c_n \phi_n(\mathbf{R}) e^{-(E_n - E_T)\tau} \quad (4.22)$$

Let us assume that our initial guess for the wavefunction is not orthogonal to the ground state (i.e., $c_0 \neq 0$). Then if we magically choose the constant E_T to be the ground-state eigenvalue E_0 (or, in practice, keep very tight control of it through some type of feedback procedure), it is clear we should eventually get *imaginary-time independence* of the probability distribution, in the sense that as $\tau \rightarrow \infty$, our initial $\Psi(\mathbf{R}, 0)$ comes to look more and more like the stationary ground state $\phi_0(\mathbf{R})$ as the contribution of the excited-state eigenfunctions dies away:

$$\Psi(\mathbf{R}, \tau) = c_0 \phi_0 + \sum_{n=1}^{\infty} c_n \phi_n(\mathbf{R}) e^{-(E_n - E_0)\tau} \quad (4.23)$$

So now we know *why* we do this propagation. How, in practice, do we find an expression for the propagator K ? Consider now the imaginary-time Schrödinger equation in two parts:

$$\frac{\partial \Psi(\mathbf{R}, \tau)}{\partial \tau} = \frac{1}{2} \nabla^2 \Psi(\mathbf{R}, \tau) \quad (4.24)$$

$$\frac{\partial \Psi(\mathbf{R}, \tau)}{\partial \tau} = -(V(\mathbf{R}) - E_T) \Psi(\mathbf{R}, \tau) \quad (4.25)$$

These two formulas have the form of the usual diffusion equation and of a rate equation with a position-dependent rate constant, respectively. The appropriate propagator for the diffusion equation is well known; it is a $3N$ -dimensional Gaussian with variance $\delta\tau$ in each dimension. The propagator for the rate equation is also known; it gives a *branching factor* which can be interpreted as a position-dependent weight or stochastic survival probability for a member of an ensemble.

Multiplying the two together to get the following propagator for the imaginary-time Schrödinger equation is an approximation, the *short-time approximation*, valid only in the limit of small $\delta\tau$ (which is why we need to do the evolution as a sequence of short hops):

$$K^{\text{DMC}}(\mathbf{R}, \mathbf{R}', \delta\tau) = \frac{1}{(2\pi\delta\tau)^{3N/2}} \exp\left(-\frac{|\mathbf{R} - \mathbf{R}'|^2}{2\delta\tau}\right) \exp\left[-\delta\tau \frac{V(\mathbf{R}) + V(\mathbf{R}') - 2E_T}{2}\right] \quad (4.26)$$

Let us then summarize with a simple example how the DMC algorithm works. If we interpret Ψ as a *probability density*, the diffusion equation $\partial\Psi/\partial\tau = \frac{1}{2}\nabla^2\Psi$ represents the movement of N diffusing particles. If we turn this around, we may decide to *represent* $\Psi(x, \tau)$ by an *ensemble* of such sets of particles. Each member of such an ensemble will be called a *configuration*. We interpret the full propagator $K^{\text{DMC}}(\mathbf{R}, \mathbf{R}', \delta\tau)$ as the probability of a configuration moving from $\mathbf{R}'$ to $\mathbf{R}$ in a time $\delta\tau$. The branching factor in the propagator will generally be interpreted as a stochastic survival probability for a given configuration rather than as a simple weight, as the latter is prone to numerical instabilities. This means that the configuration population becomes dynamically variable; configurations that stray into regions of high V have a good chance of being killed (removed from the calculation); in low- V regions, configurations have a high probability of multiplying (i.e., they create copies of themselves, which then propagate independently). It is solely this branching or reweighting that “changes the shape of the wavefunction” as it evolves. So, as we have seen, after a sufficiently long period of imaginary-time evolution, all the excited states will decay away, leaving only the ground-state wavefunction, at which point the propagation may be continued to accumulate averages of interesting observables.

As a simple example, consider Fig. 4.3. Here we make a deliberately bad guess that the ground-state wavefunction for a single electron in a harmonic potential well is a constant in the vicinity of the well and zero everywhere else. We begin with seven copies of the system or configurations in our ensemble; the electrons in this ensemble are initially randomly distributed according to the uniform probability distribution in the region where the trial function is finite. The particle distribution is then evolved in imaginary time according to the scheme developed above. The electrons are subsequently seen to become distributed according to the proper Gaussian shape of the exact ground-state wavefunction. It is evident from the figure that the change in shape is produced by the branching factor occasionally eliminating configurations in high- V regions and duplicating them in low- V regions.

This “pure DMC” algorithm works very well in a single-particle system with a nicely behaved potential, as in the example. Unfortunately, it suffers from two very serious drawbacks which become evident in multiparticle systems with divergent Coulomb potentials.

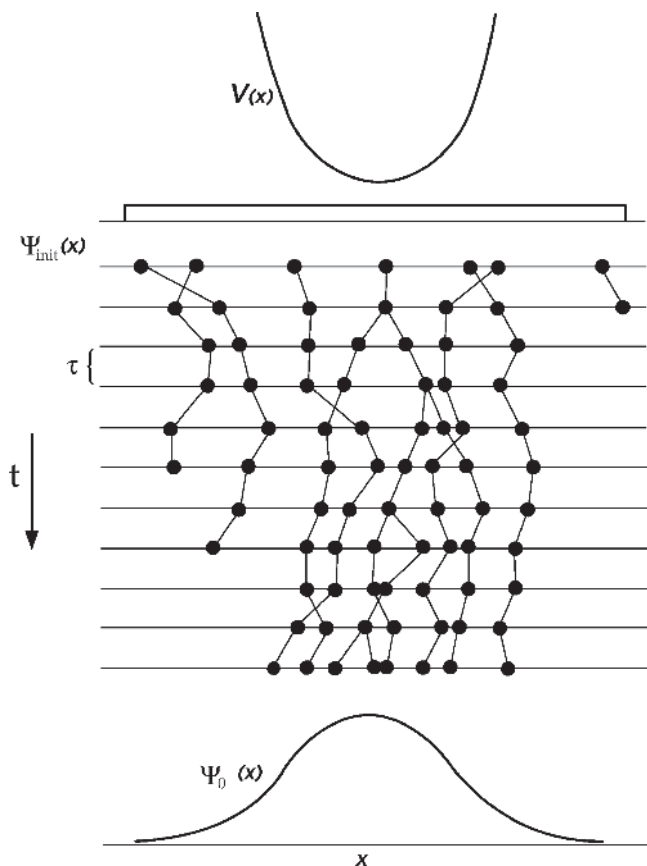


Fig. 4.3 Figure 4.3: Schematic illustration of the DMC algorithm for a single electron in a harmonic potential well, showing the evolution of the shape of the wavefunction due to propagation in imaginary time. (From Ref. 5, with permission. Copyright © 2001 by The American Physical Society.)

The first problem arises due to our assumption that Ψ is a probability distribution—necessarily positive everywhere—even though the antisymmetric nature of multiparticle fermionic wavefunctions means that it must have both positive and negative parts separated by a *nodal surface*, that is, a $(3N - 1)$ -dimensional hypersurface on which it has the value zero. One might think that two separate populations of configurations with attached positive and negative weights might get around this problem (essentially, the well-known *fermion sign problem*), but in practice there is a severe signal-to-noise issue. It is possible to construct formally exact algorithms of this nature which overcome some of the worst practical problems,⁷⁵ but to date all seem highly inefficient, with poor system-size scaling.

The second problem is less fundamental but in practice very severe. The required rate of removing or duplicating configurations diverges when the

potential energy diverges (which occurs whenever two particles are coincident) due to the presence of V in the branching factor of Eq. (4.26). This leads to stability problems and poor statistical behavior.

These problems may be dealt with at the cost of introducing the most important approximation in the DMC algorithm: the *fixed-node approximation*.⁷⁶ We say, in effect, that particles may not cross the nodal surface of the *trial* wavefunction Ψ_T ; that is, there is an infinite repulsive potential barrier on the nodes. This forces the DMC wavefunction Ψ to be zero on that hypersurface. If the nodes of the trial function coincide with the exact nodes, such an algorithm will give the exact ground-state energy (it is, of course, well known that the exact de Broglie–Bohm particle trajectories cannot pass through the nodal surface). If the trial function nodes do not coincide with the exact nodes, the DMC energy will be higher than the ground-state energy (but less than or equal to the VMC energy). The variational principle thus applies.

To make such an algorithm efficient we must introduce importance sampling, and this is done in the following way. We require that the imaginary-time evolution produces the *mixed distribution* $f = \Psi_T \Psi$ rather than the pure distribution. Substituting this into the imaginary-time Schrödinger equation, Eq. (4.20), we obtain

$$-\frac{\partial f(\mathbf{R}, \tau)}{\partial \tau} = -\frac{1}{2}\nabla^2 f(\mathbf{R}, \tau) + \nabla \cdot [v_D(\mathbf{R})f(\mathbf{R}, \tau)] + (E_L(\mathbf{R}) - E_T)f(\mathbf{R}, \tau) \quad (4.27)$$

where $v_D(\mathbf{R})$ is the $3N$ -dimensional *drift velocity vector*, defined by

$$v_D(\mathbf{R}) = \nabla \ln |\Psi_T(\mathbf{R})| = \frac{\nabla \Psi_T(\mathbf{R})}{\Psi_T(\mathbf{R})} \quad (4.28)$$

and

$$E_L(\mathbf{R}) = \Psi_T^{-1} \left(-\frac{1}{2}\nabla^2 + V(\mathbf{R}) \right) \Psi_T \quad (4.29)$$

is the usual local energy. The propagator from $\mathbf{R}'$ to $\mathbf{R}$ for the importance sampled algorithm now looks like this:

$$\begin{aligned} K^{\text{DMC}}(\mathbf{R}, \mathbf{R}', \delta\tau) &= \frac{1}{(2\pi\delta\tau)^{3N/2}} \exp \left[-\frac{(\mathbf{R} - \mathbf{R}' - \delta\tau F(\mathbf{R}'))^2}{2\delta\tau} \right] \\ &\quad \exp \left[-\frac{\delta\tau}{2}(E_L(\mathbf{R}) + E_L(\mathbf{R}') - 2E_T) \right] \end{aligned} \quad (4.30)$$

Because the nodal surface of Ψ is constrained to be that of Ψ_T , their product f is positive everywhere and can now be properly interpreted as a probability distribution. The time evolution generates the distribution $f = \Psi_T \Psi$, where Ψ is now the lowest-energy wavefunction with the same nodes as Ψ_T . This solves

the first of our two problems. The second problem of the poor statistical behavior due to the divergences in the potential energy is also solved because the term $V(\mathbf{R}) - E_T$ in Eq. (4.20) has been replaced by $E_L(\mathbf{R}) - E_T$ in Eq. (4.27), which is much smoother. Indeed, if Ψ_T was an exact eigenstate, $E_L(\mathbf{R}) - E_T$ would be independent of position in configuration space. Although we cannot in practice find the exact Ψ_T , it is possible to eliminate the local energy divergences due to coincident particles by choosing a trial function that has the correct cusplike behavior at the relevant points in the configuration space.⁵⁶ Note that this is all reflected in the branching factor of the new propagator of Eq. (4.30).

The nodal surface partitions the configuration space into regions that we call *nodal pockets*. The fixed-node approximation implies that we are restricted to sampling only those nodal pockets that are occupied by the initial set of configurations, and this appears to introduce some kind of ergodicity concern, since at first sight it seems that we ought to sample every nodal pocket. This would be an impossible task in large systems. However, the *tiling theorem* for exact fermion ground states^{77,78} asserts that all nodal pockets are in fact equivalent and related by permutation symmetry; one need therefore only sample one of them. This theorem is intimately connected with the existence of a variational principle for the DMC ground-state energy.⁷⁸ Other interesting investigations of properties of nodal surfaces have been published.^{79–81}

A practical importance-sampled DMC simulation proceeds as follows. First we pick an ensemble of a few hundred configurations chosen from the distribution $|\Psi_T|^2$ using VMC and the standard Metropolis algorithm. This ensemble is then evolved according to the short-time approximation to the Green's function of the importance-sampled imaginary-time Schrödinger equation [Eq. (4.27)], which involves repeated steps of biased diffusion followed by the deletion and/or duplication of configurations. The bias in the diffusion is caused by the drift vector arising out of the importance sampling, which directs the sampling toward parts of configuration space where $|\Psi_T|$ is large (i.e., it plays the role of an Einsteinian *osmotic velocity*). This drift step is always directed away from the node, and $\nabla\Psi_T$ is in fact a normal vector of the nodal hypersurface. After a period of equilibration the excited-state contributions will have largely died out and the configurations start to trace out the probability distribution $f(\mathbf{R})/\int f(\mathbf{R}) d\mathbf{R}$. We can then start to accumulate averages, in particular the DMC energy. Note that throughout this process the reference energy E_T is varied to keep the configuration population under control through a specific feedback mechanism. The initial stages of a DMC simulation—for solid antiferromagnetic NiO crystal with 128 atoms per cell using unrestricted Hartree–Fock trial functions of the type discussed in Refs. 82 and 83—are shown in Fig. 4.4.

The DMC energy is given by

$$E_{\text{DMC}} = \frac{\int f(\mathbf{R}) E_L(\mathbf{R}) d\mathbf{R}}{\int f(\mathbf{R}) d\mathbf{R}} \approx \sum_i E_L(\mathbf{R}_i) \quad (4.31)$$

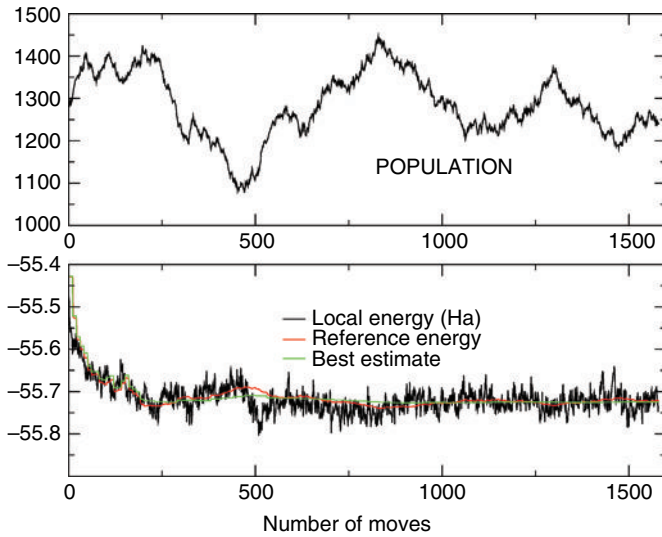


Fig. 4.4 (color online) DMC simulation of solid antiferromagnetic NiO. In the lower panel, the noisy black line is the local energy after each move, the smoother green line is the current best estimate of the DMC energy, and the red line is E_T in Eq. (4.27), which is varied to control the population of configurations through a feedback mechanism. As the simulation equilibrates, the best estimate of the energy, initially equal to the VMC energy, decreases significantly, then approaches a constant, which is the final DMC energy. The upper panel shows the variation in the population of the ensemble during the simulation as walkers are created or destroyed.

This energy expression would be exact if the nodal surface of Ψ_T were exact, and the fixed-node error is second order in the error in the Ψ_T nodal surface (when a variational theorem exists⁷⁸). The accuracy of the fixed-node approximation can be tested on small systems and normally leads to very satisfactory results. The trial wavefunction thus limits the final accuracy that can be obtained and it also controls the statistical efficiency of the algorithm. Like VMC, the DMC algorithm satisfies a zero-variance principle (i.e., the variance of the energy goes to zero as the trial wavefunction goes to an exact eigenstate). For other expectation values of operators that do not commute with the Hamiltonian, the DMC mixed estimator is biased and other techniques are required in order to sample the pure distribution.^{84–86}

A final point: The necessity of using the fixed-node approximation suggests that the best way of optimizing wavefunctions would be to do it in DMC directly. The nodal surface could then in principle be optimized to the shape that minimizes the DMC energy. The backflow technique discussed in Section 4.5.1 has some bearing on the problem, but the usual procedure involving optimization of the energy or variance in VMC will not usually lead to the optimal nodes in the sense that the fixed-node DMC energy is minimal. The large number of parameters—up to a few hundred—in your typical Slater–Jastrow(–backflow)

wavefunction means that direct variation of the parameters in DMC is too expensive (although this has been done, see, e.g., Refs. 87 and 88). Furthermore, we note that optimizing the energy in DMC is tricky for the nodal surface, as the contribution of the region near the nodes to the energy is small. More exotic ways of optimizing the nodes are still being actively developed.^{89,90}

4.5 BITS AND PIECES

4.5.1 More About Wavefunctions, Orbitals, and Basis Sets

Single-determinant Slater–Jastrow wavefunctions often work very well in QMC calculations since the orbital part alone provides a pretty good description of the system. In the ground state of the carbon pseudoatom, for example, a single Hartree–Fock determinant retrieves about 98.2% of the total energy. The remaining 1.8%, which at the VMC level must be recovered by the Jastrow factor, is the correlation energy, and in this case it amounts to 2.7 eV—clearly important for an accurate description of chemical bonding. By definition a determinant of Hartree–Fock orbitals gives the lowest energy of all single-determinant wavefunctions, and DFT orbitals are often very similar to them. These orbitals are not optimal when a Jastrow factor is included, but it turns out that the Jastrow factor does not change the detailed structure of the optimal orbitals very much, and the changes are well described by a fairly smooth change to the orbitals. This can conveniently be included in the Jastrow factor itself.

How, though, might we improve on the Hartree–Fock/DFT orbitals in the presence of the Jastrow factor? One might naturally consider optimizing the orbitals themselves. This has been done, for example, with the atomic orbitals of a neon atom by Drummond et al.,⁹¹ optimizing a parameterized function that is added to the self-consistent orbitals. This was found to be useful only in certain cases. In atoms one often sees an improvement in the VMC energy but not in DMC, indicating that the Hartree–Fock nodal surface is close to optimal even in the presence of a correlation function. Unfortunately, direct optimization of both the orbitals and the Jastrow factor cannot easily be done for large polyatomic systems because of the computational cost of optimizing large numbers of parameters, so it is difficult to know how far this observation extends to more complex systems. One technique that has been tried^{92,93} is to optimize the potential that generates the orbitals rather than the orbitals themselves. It was also suggested by Grossman and Mitas⁹⁴ that another way to improve the orbitals over the Hartree–Fock form is to use a determinant of the *natural orbitals*, which diagonalize the one-electron density matrix. While the motivation here is that the convergence of configuration interaction expansions is improved by using natural orbitals instead of Hartree–Fock orbitals, it is not clear why this would work in QMC. The calculation of reasonably accurate natural orbitals costs a lot, and such an approach is therefore less attractive for large systems.

It should be noted that all such techniques which move the nodal surface of the trial function (and hence potentially improve the DMC energy) make

wavefunction optimization with fixed configurations more difficult. The nodal surface deforms continuously as the parameters are changed, and in the course of this deformation the fixed set of electron positions of one of the configurations may end up being on the nodal surface. As the local energy $\hat{H}\Psi/\Psi$ diverges on the nodal surface, the unreweighted variance of the local energy of a fixed set of configurations also diverges, making it difficult to locate the global minimum of the variance. A discussion of what one might do about this can be found elsewhere.⁶² In some cases it is necessary to use multideterminant wavefunctions to preserve important symmetries of the true wavefunction. In other cases a single determinant may give the correct symmetry, but a significantly better wavefunction can be obtained by using a linear combination of a few determinants. Multideterminant wavefunctions have been used successfully in QMC studies of small molecules and even in periodic calculations such as the study of the neutral vacancy in diamond due to Hood et al.²⁷ However, other studies have shown that although using multideterminant functions improves VMC, this sometimes does not extend to DMC, indicating that the nodal surface has not been improved.⁹¹ Of course, there is very little point in using methods that employ expansions of large numbers of determinants to generate QMC trial functions, not only because the use of methods that scale so badly as a *preliminary calculation* completely defeats the entire point of QMC, but because the medium- and short-range correlation which these expansions describe^{95,96} is dealt with directly and vastly more efficiently by the Jastrow factor.

By far the most useful way to go beyond the Slater–Jastrow form is the *backflow technique*, to which we have already alluded. Backflow correlations were originally derived from a current conservation argument by Feynman⁹⁷ and by Feynman and Cohen⁹⁸ to provide a picture of the excitations in liquid ^4He and the effective mass of a ^3He impurity in ^4He . In a modern context they can also be derived from an imaginary-time evolution argument.^{99,100} In the simplest form of backflow trial function the electron coordinates $\mathbf{r}_i$ appearing in the Slater determinants of Eq. (4.7) are replaced by *quasiparticle coordinates*,

$$\mathbf{r}_i = \bar{\mathbf{r}}_i + \sum_{j \neq i}^N \eta(r_{ij})(\mathbf{r}_i - \mathbf{r}_j) \quad (4.32)$$

where $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$. This is supposed to represent the characteristic flow pattern where the quantum fluid is “pushed out of the way” in front of a moving particle and fills in the space behind it. The optimal function $\eta(r_{ij})$ may be determined variationally, and in so doing the nodal surface is shifted. Backflow thus represents another practical possibility for relaxing the constraints of the fixed-node approximation in DMC. Kwon et al.^{99,101} found that the introduction of backflow significantly lowered the VMC and DMC energies of the two- and three-dimensional uniform electron gas at high densities. The use of backflow has also been investigated for metallic hydrogen.¹⁰² For real polyatomic systems, a much more complicated inhomogeneous backflow function is required; the one

developed in our group and implemented in the CASINO program by López Ríos¹⁰³ has the following functional form:

$$\Psi_{\text{BF}}(\mathbf{R}) = e^{J(\mathbf{R})} \det[\psi_i(\mathbf{r}_i^\uparrow + \xi_i(\mathbf{R}))] \det[\psi_i(\mathbf{r}_i^\downarrow + \xi_j(\mathbf{R}))] \quad (4.33)$$

with the backflow displacement for electron i in a system of N electrons and N_n nuclei given by

$$\xi_i = \sum_{j \neq i}^N \eta_{ij} r_{ij} + \sum_I^{N_{\text{ion}}} \mu_{iI} r_{iI} + \sum_{j \neq i}^N \sum_I^{N_{\text{ion}}} (\Phi_i^{jI} r_{ij} + \Theta_i^{jI} r_{iI}) \quad (4.34)$$

Here $\eta_{ij} = \eta(r_{ij})$ is a function of electron–electron separation, $\mu_{iI} = \mu(r_{iI})$ is a function of electron–ion separation, and $\Phi_i^{jI} = \Phi(r_{iI}, r_{jI}, r_{ij})$ and $\Theta_i^{jI} = \Theta(r_{iI}, r_{jI}, r_{ij})$. The functions η , μ , Φ , and Θ are parameterized using power expansions with optimizable coefficients.¹⁰³

Now, of course, the use of backflow wavefunctions can significantly increase the cost of a QMC calculation. This is largely because every element of the Slater determinant has to be recomputed each time an electron is moved, whereas only a single column of the Slater determinant has to be updated after each move when the basic Slater–Jastrow wavefunction is used. The basic scaling of the algorithm with backflow (assuming localized orbitals and basis set) is thus N^3 rather than N^2 . Backflow functions also introduce more parameters into the trial wavefunction, making the optimization procedure more difficult and costly. However, the reduction in the variance normally observed with backflow greatly improves the statistical efficiency of QMC calculations in the sense that the number of moves required to obtain a fixed error in the energy is smaller. In our Ne-atom calculations,⁹¹ for example, it was observed that the computational cost per move in VMC and DMC increased by a factor of between 4 and 7, but overall the time taken to complete the calculation to a fixed error bar increased only by a factor of between 2 and 3. One interesting thing that we found is that energies obtained from VMC with backflow approached those of DMC without backflow. VMC with backflow may thus represent a useful level of theory since it is significantly less expensive than DMC (although the problem with obtaining accurate energy *differences* in VMC presumably remains). Finally, it should be noted that backflow is expected to improve the QMC estimates of all expectation values, not just the energy. We like it.

We now move on to consider the issue of basis sets. The importance of using good-quality single-particle orbitals in building up the Slater determinants in the trial wavefunction is clear. The determinant part accounts for by far the most significant fraction of the variational energy. However, the evaluation of single-particle orbitals and their first and second derivatives can sometimes take up more than half of the total computer time, and consideration must therefore be given to obtaining accurate orbitals that can be evaluated rapidly at arbitrary points in space. It is not difficult to see that the most critical thing is to expand

the single-particle orbitals in a basis set of *localized* functions. This ensures that beyond a certain system size, only a fixed number of the localized functions will give a significant contribution to a particular orbital at a particular point. The cost of evaluating the orbitals does not then increase rapidly with the size of the system. Note that *localized basis functions* can (1) be strictly zero beyond a certain radius, or (2) can decrease monotonically and be prescreened before the calculation starts, so that only those functions that could be significant in a particular region are considered for evaluation.

An alternative procedure is to tabulate the orbitals and their derivatives on a grid, and this is feasible for small systems such as atoms, but for periodic solids or larger molecules the storage requirements quickly become enormous. This is an important consideration when using parallel computers, as it is much more efficient to store the single-particle orbitals on every node. Historically, a very large proportion of condensed matter electronic structure theorists have used plane-wave basis sets in their DFT calculations. However, in QMC, plane-wave expansions are normally extremely inefficient because they are not localized in real space; every basis function contributes at every point, and the number of functions required increases linearly with system size. Only if there is a short repeat length in the problem are plane waves not totally unreasonable. Note that this does not mean that all plane-wave DFT codes (such as CASTEP,¹⁰⁴ ABINIT,¹⁰⁵ and PWSCF¹⁰⁶) are useless for generating trial wavefunctions for CASINO; a postprocessing utility can be used to reexpand a function expanded in plane waves in another localized basis before the wavefunction is read into CASINO. The usual thing here is to use some form of localized spline functions on a grid such as “blip” functions.^{107,108}

Another reasonable way to do this is to expand the orbitals in a basis of Gaussian-type functions. These are localized, relatively quick to evaluate, and are available from a wide range of sophisticated software packages. Such a large expertise has been built up within the quantum chemistry community with Gaussians that there is significant resistance to using any other type of basis. A great many Gaussian-based packages have been developed by quantum chemists for treating molecules. The best known of these are probably the various versions of the GAUSSIAN software.³ In addition to the regular single-determinant methods, these codes implement various techniques involving multideterminant correlated wavefunctions and are flexible tools for developing accurate molecular trial wavefunctions. For systems with periodic boundary conditions, the Gaussian basis set program CRYSTAL¹⁰⁹ turns out to be very useful; it can perform all-electron or pseudopotential Hartree–Fock and DFT calculations both for molecules and for systems periodic in one, two, or three dimensions. For some systems, Slater basis sets may be useful in QMC (since they provide a more compact representation than Gaussians, and hence more rapidly calculable orbitals).⁷⁴ To this end, we have implemented an interface to the program ADF.¹¹⁰

There is one more issue we must consider that is relevant to all basis sets but is particular to the case of Gaussian-type functions. This has to do with *cusp conditions*. At a nucleus the exact wavefunction has a cusp so that the divergence

in the potential energy is canceled by an equal and opposite divergence in the kinetic energy. Therefore, if this cusp is represented accurately in the QMC trial wavefunction, the fluctuations in the local energy will be greatly reduced. It is relatively easy to produce an accurate representation of this cusp when using a grid-based numerical representation of the orbitals. However, as we have already remarked, such representations cannot really be used for large polyatomic systems because of the excessive storage requirements, and we would prefer to use a Gaussian basis set. But then there can be no cusp in the wavefunction since Gaussians have zero gradient at $r = 0$. The local energy thus diverges at the nucleus. In practice, one finds that the local energy has wild oscillations close to the nucleus, which can lead to numerical instabilities in DMC calculations. To solve this problem we can make small corrections to the single-particle orbitals close to the nuclei, which impose the correct cusp behavior; these need to be applied at each nucleus for every orbital which is larger than a given tolerance at that nucleus. The scheme we developed to correct for this is outlined elsewhere.⁷³ Generalizations of this method have been developed for other basis set types.

To see the cusp corrections in action, let us first look at a hydrogen atom where the basis set has been made to model the cusp very closely by using very sharp Gaussians with high exponents. Visually (top left in Fig. 4.5), the fact that the orbital does not obey the cusp condition is not immediately apparent. If we zoom in on the region close to the nucleus (top right), we see the problem; the black line is the orbital expanded in Gaussians and the red line is the cusp-corrected orbital. The effect on the gradient and local energy is clearly significant. This scheme has been implemented within the CASINO code for both finite and periodic systems, and produces a significant reduction in the computer time required to achieve a specified error bar, as one can appreciate from looking at the bottom two panels in Fig. 4.5, which show the local energy as a function of move number for a carbon monoxide molecule with and without cusp corrections.

The problem with electron–nucleus cusps is clearly more significant for atoms of higher atomic number. To understand how this helps to do all-electron DMC calculations for heavier atoms, and to understand how the necessary computer time scales with atomic number, we performed calculations for various noble gas atoms.⁶⁴ By ensuring that the electron–nucleus cusps were represented accurately, it proved perfectly possible to produce converged DMC energies with acceptably small error bars for atoms up to xenon ($Z = 54$).

4.5.2 Pseudopotentials

Well, “perfectly possible,” I said. Possible, maybe, but definitely somewhat tiresome. On trying to do all-electron calculations for heavier atoms than xenon, we were quickly forced to stop when smoke was observed coming out of the side of the computer.¹¹¹ Might it therefore be better to do heavy atoms using pseudopotentials, as is commonly done with other methods, such as DFT? In electronic structure calculations pseudopotentials or effective core potentials are used to remove the inert core electrons from the problem and to improve the computational efficiency. Although QMC scales very favorably with system size

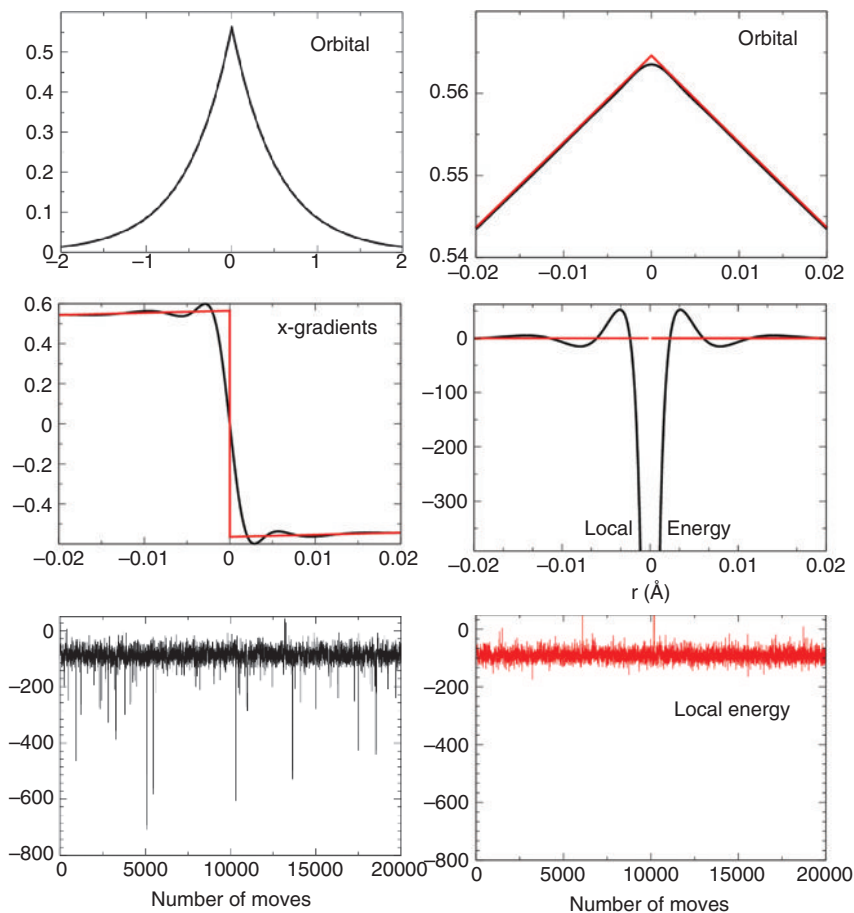


Fig. 4.5 (color online) The top two rows show the effect of Gaussian basis set cusp corrections in the hydrogen atom (red straight-line segments corrected; black lines not corrected). The bottom row shows local energy as a function of move number in a VMC calculation for a carbon monoxide molecule with a standard reasonably good Gaussian basis set. The cusp corrections are imposed only in the figure on the right. The reduction in the local energy fluctuations with the new scheme is clearly apparent.

in general, it has been estimated⁶³ that the scaling of all-electron calculations with the atomic number Z is approximately $Z^{5.5}$, which in the relatively recent past was generally considered to rule out applications to atoms with Z greater than about 10. Our paper⁶⁴ pushing all-electron QMC calculations to $Z = 54$ was therefore a significant step. The use of a pseudopotential then serves to reduce the effective value of Z and to improve the scaling to $Z^{3.5}$. Although errors are inevitably introduced, the gain in computational efficiency is easily sufficient to make pseudopotentials preferable in heavier atoms. They also offer a simple way to incorporate approximate relativistic corrections.

Accurate pseudopotentials for single-particle theories such as DFT or Hartree–Fock theory are well developed, but pseudopotentials for correlated wavefunction techniques such as QMC present additional challenges. The presence of core electrons causes two related problems. The first is that the shorter length-scale variations in the wavefunction near a nucleus of large Z require the use of a small time step. In VMC this problem can, at least in principle, be somewhat reduced by the use of acceleration schemes.^{112,113} The second problem is that the fluctuations in the local energy tend to be large near the nucleus because both the kinetic and potential energies are large.

The central idea of pseudopotential theory is to create an effective potential that reproduces the effects of both the nucleus and the core electrons on the valence electrons. This is done separately for each of the different angular momentum states, so the pseudopotential contains angular momentum projectors and is therefore a nonlocal operator. It is convenient to divide the pseudopotential for each atom into a local part $V_{\text{loc}}^{\text{ps}}(r)$ common to all angular momenta and a correction, $V_{\text{nl},l}^{\text{ps}}(r)$, for each angular momentum l . The electron–ion potential energy term in the full many-electron Hamiltonian of the atom then takes the form

$$V_{\text{loc}} + \hat{V}_{\text{nl}} = \sum_i V_{\text{loc}}^{\text{ps}}(r_i) + \sum_i \hat{V}_{\text{nl},i}^{\text{ps}} \quad (4.35)$$

where $\hat{V}_{\text{nl},i}^{\text{ps}}$ is a nonlocal operator that acts on an arbitrary function $g(\mathbf{r}_i)$ as follows:

$$\hat{V}_{\text{nl},i}^{\text{ps}} g(\mathbf{r}_i) = \sum_l V_{\text{nl},l}^{\text{ps}}(r_i) \sum_{m=-l}^l Y_{lm}(\Omega_{r_i}) \int Y_{lm}^*(\Omega_{r'_i}) g(\mathbf{r}'_i) d\Omega'_i \quad (4.36)$$

where the angular integration is over the sphere passing through the $\mathbf{r}_i$. This expression can be simplified by choosing the z -axis along $\mathbf{r}_i$, noting that $Y_{lm}(0,0) = 0$ for $m \neq 0$, and using the definition of the spherical harmonics to give

$$\hat{V}_{\text{nl},i}^{\text{ps}} g(\mathbf{r}_i) = \sum_l V_{\text{nl},l}^{\text{ps}}(r_i) \frac{2l+1}{4\pi} \int P_l[\cos(\theta'_i)] g(\mathbf{r}'_i) d\Omega'_i \quad (4.37)$$

where P_l denotes a Legendre polynomial.

While the use of nonlocal pseudopotentials is relatively straightforward in a VMC calculation,^{115,116} there is an issue with DMC. The fixed-node boundary condition turns out not to be compatible with the nonlocality. This forces us to introduce an additional approximation (the *locality approximation*¹¹⁷) whereby the nonlocal pseudopotential operator $\hat{V}_{\text{nl}}$ acts on the trial function rather than the DMC wavefunction; that is, we replace $\hat{V}_{\text{nl}}$ by $\Psi_T^{-1} \hat{V}_{\text{nl}} \Psi_T$. The leading-order error term is proportional to $(\Psi_T - \Psi_0)^2$, where Ψ_0 is the exact fixed-node ground-state wavefunction.¹¹⁷ Unfortunately, this error may be positive or negative, so the method is no longer strictly variational. An alternative to this approximation

is the *semilocalization* scheme for DMC nonlocal pseudopotentials introduced by Casula et al. in 2005^{118,119}; as well as restoring the variational property, this method appears to have better numerical stability than the older scheme.

It is not currently possible to construct pseudopotentials for heavy atoms entirely within a QMC framework, although progress in this direction was made by Acioli and Ceperley.¹¹⁴ It is therefore currently necessary to use pseudopotentials generated within some other framework. Possible schemes include Hartree–Fock theory and local DFT, where there is a great deal of experience in generating accurate pseudopotentials. There is evidence to show that Hartree–Fock pseudopotentials give better results within QMC calculations than DFT pseudopotentials,¹²⁰ although the latter work quite well in many cases. The problem with DFT pseudopotentials appears to be that they already include a (local) description of correlation which is quite different from the QMC description. Hartree–Fock theory, on the other hand, does not contain any effects of correlation. The QMC calculation puts back the valence–valence correlations but neglects core–core correlations (which have only an indirect and small effect on the valence electrons) and core–valence correlations. Core–valence correlations are significant when the core is highly polarizable, such as in alkali-metal atoms. The core–valence correlations may be approximately included by using a *core polarization potential* (CPP), which represents the polarization of the core due to the instantaneous positions of the surrounding electrons and ions. Another issue is that relativistic effects are important for heavy elements. It is still, however, possible to use a QMC method for solving the Schrödinger equation with the scalar relativistic effects obtained within the Dirac formalism incorporated within the pseudopotentials. The combination of Dirac–Hartree–Fock pseudopotentials and CPPs appears to work well in many QMC calculations. CPPs have been generated for a wide range of elements (see, e.g., Ref. 121).

Many Hartree–Fock pseudopotentials are available in the literature, mostly in the form of sets of parameters for fits to Gaussian basis sets. Unfortunately, many of them diverge at the origin and it is well known that this can lead to significant time step errors in DMC calculations.¹²⁰ It was thus apparent a few years ago that none of the available sets were ideal for QMC calculations, and it was decided that it would be helpful if we generated an online periodic table of smooth nondivergent Hartree–Fock pseudopotentials (with relativistic corrections) developed specifically for QMC. This project has now been completed and has been described in detail by Trail and Needs.^{122,123} The resulting pseudopotentials are available online¹²⁴; the repository includes both Dirac–Fock and Hartree–Fock potentials, and a choice of small or large core potentials (the latter being more amenable to plane-wave calculations). Burkatzki et al. have since developed another set of pseudopotentials, also intended for use in QMC calculations.¹²⁵ Although data are limited, tests^{126,127} appear to show that the Trail–Needs pseudopotentials give essentially the same results as the Burkatzki pseudopotentials, although the smaller core radii of the former appear to lead to a slight increase in efficiency.

4.5.3 Periodic Systems

As with other methods, QMC calculations for extended systems may be performed using finite clusters or infinitely large crystals with periodic boundary conditions. The latter are generally preferred because they approximate the desired large-size limit (i.e., the infinite system size *without* periodic boundary conditions) more closely. One can also use the standard supercell approach for *aperiodic* systems such as point defects. For such cases, cells containing a point defect and a small part of the host crystal are repeated periodically throughout space; the supercell must clearly be made large enough so the interactions between defects in different cells are negligible.

In periodic DFT calculations the charge density and potentials are taken to have the periodicity of a suitably chosen lattice. The single-particle orbitals can then be made to obey Bloch's theorem, and the results for the infinite system are obtained by summing quantities obtained from the different Bloch wave vectors within the first Brillouin zone. The situation with many-particle wavefunctions is rather different, since it is not possible to reduce the problem to solving within a primitive unit cell. Such a reduction is allowed in single-particle methods because the Hamiltonian is invariant under the translation of a single electronic coordinate by a translation vector of the primitive lattice, but this is not a symmetry of the many-body Hamiltonian.^{129,128} Consequently, QMC calculations must be performed at a single k -point. This normally gives a poor approximation to the result for the infinite system, unless one chooses a pretty large nonprimitive simulation cell. One may also average over the results of QMC calculations done at different single k -points.¹³⁰

There are also a number of problems associated with the long-range Coulomb interaction in many-body techniques such as QMC. It is well known that simply summing the $1/r$ interaction out over cells on the surface of an ever-expanding cluster never settles down because of the contribution from shape-dependent arrangements of surface charge. The usual solution to this problem is to employ the Ewald method.¹³¹ The Ewald interaction contains an effective *depolarization field* intended to cancel the field produced by the surface charges (and is thus equivalent to what you get if you put the large cluster in a medium of infinite dielectric constant). Long-range interactions also induce long-range exchange-correlation interactions, and if the simulation cell is not large enough, these effects are described incorrectly. Such effects are absent in local DFT calculations because the interaction energy is written in terms of the electronic charge density, but Hartree–Fock calculations show very strong effects of this kind, and various ways to accelerate the convergence have been developed. The finite-size effects arising from the long-range interaction can be divided into potential and kinetic energy contributions.^{132,133} The potential energy component can be removed from the calculations by replacing the Ewald interaction by the *model periodic Coulomb* (MPC) interaction.^{134–136} Recent work has added substantially to our understanding of finite-size effects, and theoretical expressions have been derived for them,^{132,133} but at the moment it seems that they cannot entirely

replace extrapolation procedures. An alternative approach to estimating finite-size errors in QMC calculations has been developed recently.¹³⁷ DMC results for the three-dimensional homogeneous electron gas are used to obtain a system-size-dependent local-density approximation functional. The correction to the total energy is given by the difference between the DFT energies for finite-sized and infinite systems. This approach is interesting, although it does rely on the LDA giving a reasonable description of the system. As will be shown later, DMC calculations using periodic boundary conditions with thousands of atoms per cell have now been done, and the technology is clearly approaching maturity.

4.5.4 Differences, Derivatives, and Forces

Calculations in computational electronic structure theory almost always involve the evaluation of *differences* in energy, and all methods that work in complex systems rely for their accuracy on the cancellation of errors in such energy differences. Apart from the statistical errors, all known errors in DMC have the same sign and partially cancel out in the subtraction because the method is variational. That said, incomplete cancellation of nodal errors is the most important source of error in DMC results, even though DMC often retrieves 95% or more of the correlation energy. *Correlated sampling*¹³⁸ is one way of improving computation of the energy difference between two similar systems with a smaller statistical error than those obtained for the individual energies. This is relatively straightforward in VMC, and a version of it was described briefly in Section 4.3 when discussing variance minimization.

As well as simple differences, we would quite often like to calculate *derivatives*. Many quantities of physical interest can be formulated as an energy derivative, and thus an ability to calculate them accurately in QMC considerably enhances the scope of the method. Normally, of course, this sort of thing would be encountered in the calculation of forces on atoms, but if we expand the energy in a Taylor series in a perturbation such as the strength of an applied electric field, for example, the coefficients of the first- and second-order terms, respectively, give the *dipole moment* and the various elements of the *dipole polarizability tensor*:

$$E(F_i) = E(0) + \underbrace{F_i \left[\frac{\partial E}{\partial F_i} \right]_{F_i=0}}_{\text{dipole moment}} + \frac{1}{2} \sum_{j=1}^3 \underbrace{F_i F_j \left[\frac{\partial^2 E}{\partial F_i \partial F_j} \right]_{F_i=0, F_j=0}}_{\text{dipole polarizability tensor}} + \cdots \quad (4.38)$$

One may also calculate the dipole moment (no surprise) by evaluating the expectation value of the dipole-moment operator. However, since the operator doesn't commute with the Hamiltonian, there will be a significant error using the mixed distribution in DMC—you need to use the pure distribution using future walking^{84,85} or whatever. This is a significant extra complication, and by formulating the thing as a derivative, you avoid having to do that. As well as the electric field, the perturbation could be the displacement of nuclear positions

(giving forces, etc.) or a combination of both (e.g., the intensity of peaks in infrared spectra depends on changes in the dipole moment corresponding to changes in geometry). Such energy derivatives can, of course, be computed numerically (by finite differencing) or analytically (by differentiating the appropriate energy expressions), the latter being clearly preferable in this case.

First, we focus on atomic forces. These are generally used in three main areas of computational electronic structure theory: structural optimization, the computation of vibrational properties, and in explicit molecular dynamics simulations of atomic behavior.¹³⁹ Unfortunately, methods for calculating accurate forces in QMC in a reasonable amount of computer time have proved elusive, at least until relatively recently, due to the lack of readily calculable expressions with reasonable statistical properties.

As usual, we begin with a discussion of the Hellmann–Feynman theorem (HFT), which in this context is the statement that the force is the expectation value of the gradient of the Hamiltonian $\hat{H}$:

$$F = -\nabla E = -\frac{\int \Psi \nabla \hat{H} \Psi d\mathbf{R}}{\int \Psi \Psi d\mathbf{R}} \quad (4.39)$$

The other terms in the expression for the gradient of the expectation value of the energy (the ones involving derivatives of the wavefunction itself) have disappeared only because we are assuming that the wavefunction is an exact eigenstate. Inevitably, then, the use of the HFT is an approximation in QMC because we have only an inexact trial function. The correct QMC expressions for the forces must contain additional (“Pulay”) terms, which depend on wavefunction derivatives. There is also an additional term which accounts for the action of the gradient operator on parameters which couple only indirectly with the nuclear positions (e.g., orbital coefficients), but this can be greatly reduced by optimizing the wavefunction through minimization of the *energy* rather than the variance. There is another type of Pulay term which arises in DMC. The HFT is expected to be valid for the exact DMC algorithm since it solves for the ground state of the fixed-node Hamiltonian exactly. However, this Hamiltonian differs from the physical one due to the presence of the infinite potential barrier on the trial nodal surface, which constrains the DMC wavefunction ϕ_0 to go to zero there. As we vary the nuclear position(s), the nodal surface moves, and hence the infinite potential barrier moves, giving a contribution to $\nabla \hat{H}$ that depends on both Ψ_T and its first derivative.^{140–142}

To calculate the Pulay terms arising from the derivative of the mixed estimator of Eq. (4.31), we need in principle to calculate a derivative of the DMC wavefunction ϕ_0 . Because we don’t have any kind of formula for ϕ_0 , this derivative cannot be readily evaluated, and what has been done in the past is to use the expression for the derivative of the trial function Ψ_T in its place.^{142–150} The resulting errors are of first order in $(\Psi_T - \phi_0)$ and $(\Psi'_T - \phi'_0)$; therefore, its accuracy depends sensitively on the quality of the trial function and its derivative.

In practice the results obtained from this procedure are not generally accurate enough.

Instead of using the usual mixed DMC energy expression, one may calculate forces from the “pure DMC” energy given by $E_D = \int \phi_0 \hat{H} \phi_0 d\mathbf{R} / \int \phi_0 \phi_0 d\mathbf{R}$, which, by construction, is equal to the mixed DMC energy. It is more expensive to do things this way, but the benefits are now clear. Despite the fact that the derivative E'_D contains the derivative of the DMC wavefunction, ϕ'_0 , Badinski et al.¹⁴² were able to show that ϕ'_0 can be eliminated from the pure DMC formula to give the following exact expression (where $d\mathbf{S}$ is a nodal surface element):

$$E'_D = \frac{\int \phi_0 \phi_0 \phi_0^{-1} \hat{H}' \phi_0 d\mathbf{R}}{\int \phi_0 \phi_0 d\mathbf{R}} - \frac{1}{2} \frac{\int \phi_0 \phi_0 \Psi_T^{-2} |\nabla_R \Psi_T| \Psi'_T d\mathbf{S}}{\int \phi_0 \phi_0 d\mathbf{R}} \quad (4.40)$$

Of course it is not easy to compute integrals over the nodal surface, and luckily, the expression can be converted into a regular volume integral with no ϕ'_0 . The error in the required approximation is then of order $(\Psi_T - \phi_0)^2$, giving

$$E'_D = \frac{\int \phi_0 \phi_0 [\phi_0^{-1} \hat{H}' \phi_0 + \Psi_T^{-1} (\hat{H} - E_D) \Psi'_T] d\mathbf{R}}{\int \phi_0 \phi_0 d\mathbf{R}} + \frac{\int \Psi_T \Psi_T (E_L - E_D) \Psi_T^{-1} \Psi'_T d\mathbf{R}}{\int \Psi_T \Psi_T d\mathbf{R}} + O[(\Psi_T - \phi_0)^2] \quad (4.41)$$

One may readily evaluate this expression by generating configurations distributed according to the pure (ϕ_0^2) and variational (Ψ_T^2) distributions. The approximation is in the Pulay terms, which are smaller in pure than in mixed DMC, and in addition, the approximation in equation (4.41) is second order, in contrast to the first-order error obtained by simply substituting Ψ'_T for ϕ'_0 . This equation satisfies the zero-variance condition; if Ψ_T and Ψ'_T are exact, the variance of the force obtained from this formula is zero (the variance of the Hellman–Feynman estimator is, strictly speaking, infinite!). Although it remains true that not many papers have been published with actual applications of these methods (some calculations of very accurate forces in small molecules can be found, e.g., in Refs. 150 and 151), one can certainly say that reasonable approximations for the difficult expressions have been found and that the outlook for QMC forces is very promising.

4.6 APPLICATIONS

Time and space preclude me from presenting a long list of applications. Here are two: (1) a somewhat unfair comparison of the worst DFT functional with VMC

and DMC for some cohesive energies of tetrahedrally bonded semiconductors, and (2) the equations of state of diamond and iron. Many other applications can be found, for example, in Ref. 5.

4.6.1 Cohesive Energies

A number of VMC and DMC studies have been performed on the cohesive energies of solids. This quantity is given by the difference between the summed energies of the appropriate isolated atoms and the energies of the same atoms in the bulk crystal. This is generally reckoned to be a severe test of QMC methods because the trial wavefunctions used in the two cases must be closely matched in quality to maximize the effective cancellation of errors. Data for Si, Ge, C, and BN have been collected in Table 4.1. The local spin density approximation (LSDA) density functional theory data shows the standard overestimation of the cohesive energy, while the QMC data is in good agreement with experiment. Studies such as these have been important in establishing DMC as an accurate method for calculating the energies of crystalline solids.

4.6.2 Equations of State of Diamond and Iron

The equation of state is the equilibrium relationship between the pressure, volume, and temperature. Computed equations of state are of particular interest in regions where experimental data are difficult to obtain. Diamond anvil cells are

TABLE 4.1 Cohesive Energies of Tetrahedrally Bonded Semiconductors Calculated Within the LSDA, VMC, and DMC Methods and Compared with Experimental Values^a

Method	Si	Ge	C	BN
LSDA	5.28 ^b	4.59 ^b	8.61 ^b	15.07 ^c
VMC	4.38(4) ^d	3.80(2) ^e	7.27(7) ^f	12.85(9) ^c
	4.82(7) ^f	—	7.36(1) ^g	
	4.48(1) ^h			
DMC	4.63(2) ^h	3.85(2) ^e	7.346(6) ^g	
Expt.	4.62(8) ^b	3.85 ^b	7.37 ^b	12.9 ⁱ

^aThe energies for Si, Ge, and C are quoted in eV per atom, while those for BN are in eV per two atoms.

^bFrom Ref. 152 and references therein.

^cFrom Ref. 153.

^dFrom Ref. 162.

^eFrom Ref. 128.

^fFrom Ref. 115. Zero-point energy corrections of 0.18 eV for C and 0.06 eV for Si have been added to the published values for consistency with the other data in the table.

^gFrom Ref. 27.

^hFrom Ref. 26.

ⁱFrom Ref. 154, estimated from experimental results on hexagonal BN.

widely used in high-pressure research, and one of the important problems is the measurement of the pressure inside the cell. The most common approach is to place a small grain of ruby in the sample chamber and measure the frequency of a strong laser-stimulated fluorescence line. The resolution is, however, poor at pressures above about 100 GPa, and alternative methods are being investigated. One possibility is to measure the Raman frequency of diamond itself, assuming that the highest frequency derives from the diamond faces adjacent to the sample chamber. Calibrating such a scale requires an accurate equation of state and the corresponding pressure dependence of the Raman frequency.

Maezono et al. performed VMC, DMC, and DFT calculations of the equation of state of diamond.¹² The DMC and DFT data are shown in Fig. 4.6, along with equations of state derived from experimental data.^{155,156} The experimentally derived equations of state differ significantly at high pressures. It is now believed that the pressure calibration in the more modern experiment of Occelli et al.¹⁵⁶ is inaccurate, and our DMC data support this view. As can be seen in Fig. 4.6, the equations of state calculated within DFT depend on the choice of exchange-correlation functional, undermining confidence in the DFT method. A recent QMC study of the equation of state and Raman frequency of cubic boron nitride has produced data that could be used to calibrate pressure measurements in diamond anvil cells.¹⁵⁷

Another example of a DMC equation of state was produced by Sola et al.,¹⁵⁸ who calculated the equation of state of hexagonal close-packed (hcp) iron under Earth's core conditions. With up to 150 atoms or 2400 electrons per

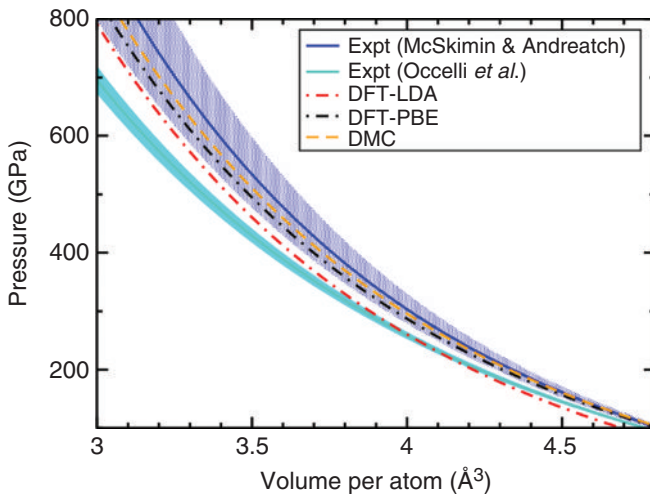


Fig. 4.6 (color online) Equation of state of diamond at high pressures from measurements by McSkimin and Andreatch¹⁵⁵ and Occelli et al.,¹⁵⁶ and as calculated using DFT with two different functionals and DMC.¹² The shaded areas indicate the uncertainty in the experimental equations of state. The zero-point phonon pressure calculated using DFT with the PBE functional is included in the theoretical curves.

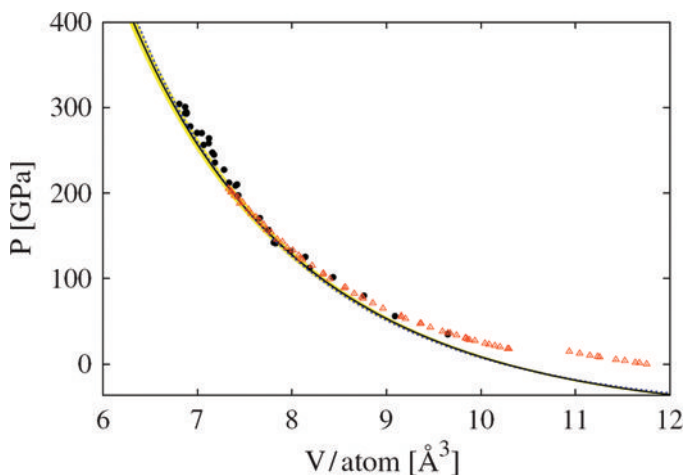


Fig. 4.7 (color online) Pressure–volume curve in iron obtained from DMC calculations (solid line¹⁵⁸). The small yellow error band above the DMC curve is due to the errors in the parameters of a fit to the Birch–Murnaghan equation of state. DFT-PW91 results (dotted line¹⁶⁰) and experimental data (circles¹⁶¹ and open triangles¹⁵⁹) are reported for comparison.

cell, these represent some of the largest systems studied with DMC to date and demonstrate the ability of QMC to treat heavier transition metal atoms. Figure 4.7 shows the calculated equation of state, which agrees closely with experiments and with previous DFT calculations. (DFT is expected to work well in this system and the DMC calculations appear to confirm this.) Notice the discontinuity due to the hcp–bcc (body-centered cubic) phase transition in the experimental values reported by Dewaele et al.¹⁵⁹ At low pressures, the calculations and experiments differ because of the magnetism, which is not taken into account in these particular calculations (although it could be in principle).

4.7 CONCLUSIONS

Quite a lot of progress has been made in the theory and practical implementation of quantum Monte Carlo over the past few years, but certainly many interesting problems remain to be solved. For its most important purpose of calculating highly accurate total energies, the method works well and currently has no serious competitors for medium-sized and large systems. Our group has developed the software package CASINO,^{46–48} which has been designed to allow researchers to explore the potential of QMC in arbitrary molecules, polymers, slabs, and crystalline solids and in various model systems, including standard electron and electron–hole phases such as the homogeneous electron gas and Wigner crystals. Many young people also seem to believe that QMC is way cooler than boring old density functional theory, and they’re probably right. So that’s all right, then.

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REFERENCES

1. Cramer, C. J. *Essentials of Computational Chemistry*, Wiley, Hoboken, NJ, 2002, pp. 191–232.
2. Parr, R. G.; Yang, W. *Density Functional Theory of Atoms and Molecules*, Oxford University Press, New York, 1994.
3. Frisch, M. J.; et al. *Gaussian 09*, Gaussian Inc., Wallingford, CT, 2009.
4. Hammond, B. L.; Lester, W. A., Jr.; Reynolds, P. J. *Monte Carlo Methods in Ab Initio Quantum Chemistry*, World Scientific, Singapore, 1994.
5. Foulkes, W. M. C.; Mitas, L.; Needs, R. J.; Rajagopal, G. *Rev. Mod. Phys.* **2001**, *73*, 33.
6. Ceperley, D. M.; Alder, B. J. *Phys. Rev. Lett.* **1980**, *45*, 566.
7. Vosko, S. H.; Wilk, L.; Nusair, M. *Can. J. Phys.* **1980**, *58*, 1200.
8. Perdew, J. P.; Zunger, A. *Phys. Rev. B* **1981**, *23*, 5048.
9. Wu, Y. S. M.; Kuppermann, A.; Anderson, J. B. *Phys. Chem. Chem. Phys.* **1999**, *1*, 929.
10. Natoli, V.; Martin, R. M.; Ceperley, D. M. *Phys. Rev. Lett.* **1993**, *70*, 1952.
11. Delaney, K. T.; Pierleoni, C.; Ceperley, D. M. *Phys. Rev. Lett.* **2006**, *97*, 235702.
12. Maezono, R.; Ma, A.; Towler, M. D.; Needs, R. J. *Phys. Rev. Lett.* **2007**, *98*, 025701.
13. Pozzo, M.; Alfè, D. *Phys. Rev. B* **2008**, *77*, 104103.
14. Alfè, D.; Alfredsson, M.; Brodholt, J.; Gillan, M. J.; Towler, M. D.; Needs, R. J. *Phys. Rev. B* **2005**, *72*, 014114.
15. Manten, S.; Lüchow, A. *J. Chem. Phys.* **2001**, *115*, 5362.
16. Grossman, J. C. *J. Chem. Phys.* **2002**, *117*, 1434.
17. Aspuru-Guzik, A.; El Akramine, O.; Grossman, J. C.; Lester, W. A., Jr. *J. Chem. Phys.* **2004**, *120*, 3049.
18. Gurtubay, I. G.; Drummond, N. D.; Towler, M. D.; Needs, R. J. *J. Chem. Phys.* **2006**, *124*, 024318.
19. Gurtubay, I. G.; Needs, R. J. *J. Chem. Phys.* **2007**, *127*, 124306.
20. Hood, R. Q.; Chou, M.-Y.; Williamson, A. J.; Rajagopal, G.; Needs, R. J.; Foulkes, W. M. C. *Phys. Rev. Lett.* **1997**, *78*, 3350.
21. Hood, R. Q.; Chou, M.-Y.; Williamson, A. J.; Rajagopal, G.; Needs, R. J. *Phys. Rev. B* **1998**, *57*, 8972.
22. Nekovee, M.; Foulkes, W. M. C.; Needs, R. J. *Phys. Rev. Lett.* **2001**, *87*, 036401.
23. Nekovee, M.; Foulkes, W. M. C.; Needs, R. J. *Phys. Rev. B* **2003**, *68*, 235108.

24. Williamson, A. J.; Grossman, J. C.; Hood, R. Q.; Puzder, A.; Galli, G. *Phys. Rev. Lett.* **2002**, *89*, 196803.
25. Drummond, N. D.; Williamson, A. J.; Needs, R. J.; Galli, G. *Phys. Rev. Lett.* **2005**, *95*, 096801.
26. Leung, W.-K.; Needs, R. J.; Rajagopal, G.; Itoh, S.; Ihara, S. *Phys. Rev. Lett.* **1999**, *83*, 2351.
27. Hood, R. Q.; Kent, P. R. C.; Needs, R. J.; Briddon, P. R. *Phys. Rev. Lett.* **2003**, *91*, 076403.
28. Alfè, D.; Gillan, M. J. *Phys. Rev. B* **2005**, *71*, 220101.
29. Towler, M. D.; Needs, R. J. *Int. J. Mod. Phys. B* **2003**, *17*, 5425.
30. Wagner, L. K.; Mitas, L. *Chem. Phys. Lett.* **2003**, *370*, 412.
31. Wagner, L. K.; Mitas, L. *J. Chem. Phys.* **2007**, *126*, 034105.
32. Mitas, L.; Martin, R. M. *Phys. Rev. Lett.* **1994**, *72*, 2438.
33. Williamson, A. J.; Hood, R. Q.; Needs, R. J.; Rajagopal, G. *Phys. Rev. B* **1998**, *57*, 12140.
34. Towler, M. D.; Hood, R. Q.; Needs, R. J. *Phys. Rev. B* **2000**, *62*, 2330.
35. Ghosal, A.; Guclu, A. D.; Umrigar, C. J.; Ullmo, D.; Baranger, H. *Nature Phys.* **2006**, *2*, 336.
36. Healy, S. B.; Filippi, C.; Kratzer, P.; Penev, E.; Scheffler, M. *Phys. Rev. Lett.* **2001**, *87*, 016105.
37. Filippi, C.; Healy, S. B.; Kratzer, P.; Pehlke, E.; Scheffler, M. *Phys. Rev. Lett.* **2002**, *89*, 166102.
38. Kim, Y.-H.; Zhao, Y.; Williamson, A.; Heben, M. J.; Zhang, S. *Phys. Rev. Lett.* **2006**, *96*, 016102.
39. Carlson, J.; Chang, S.-Y.; Pandharipande, V. R.; Schmidt, K. E. *Phys. Rev. Lett.* **2003**, *91*, 050401.
40. Astrakharchik, G. E.; Boronat, J.; Casulleras, J.; Giorgini, S. *Phys. Rev. Lett.* **2004**, *93*, 200404.
41. Carlson, J.; Reddy, S. *Phys. Rev. Lett.* **2008**, *100*, 150403.
42. Schrödinger, E. *Ann. Phys.* **1926**, *79*, 361.
43. Ashcroft, N. W.; Mermin, N. D. *Solid State Physics*, W. B. Saunders, Philadelphia, 1976, p. 330.
44. Kent, P. R. C.; Towler, M. D.; Needs, R. J.; Rajagopal, G. *Phys. Rev. B* **2000**, *62*, 15394.
45. http://www.qmcwiki.org/index.php/Research_resources.
46. Needs, R. J.; Towler, M. D.; Drummond, N. D.; López Ríos, P. *CASINO Version 2.5 User Manual*, Cambridge University, Cambridge, UK, 2009.
47. CASINO Web site: <http://www.tcm.phy.cam.ac.uk/~mdt26/casino2.html>.
48. <http://www.vallico.net/tti/tti.html>. Click on "PUBLIC EVENTS."
49. Trail, J. R. *Phys. Rev. E* **2008**, *77*, 016703.
50. Trail, J. R. *Phys. Rev. E* **2008**, *77*, 016704.
51. Metropolis, N.; Rosenbluth, A. W.; Rosenbluth, M. N.; Teller, A. M.; Teller, E. *J. Chem. Phys.* **1953**, *21*, 1087.

52. Towler, M. D. De Broglie-Bohm pilot-wave theory and the foundations of quantum mechanics. Graduate lecture course, available at http://www.tcm.phy.cam.ac.uk/~mdt26/pilot_waves.html, 2009.
53. Jastrow, R. J. *Phys. Rev.* **1955**, 98, 1479.
54. Drummond, N. D.; Towler, M. D.; Needs, R. J. *Phys. Rev. B* **2004**, 70, 235119.
55. Aragon, S. *Density Functional Theory: A Primer*, San Francisco State University teaching material, available at www.wag.caltech.edu/PASI/lectures/SFSU-ElectronicStructure-Lect-6.doc.
56. Kato, T. *Commun. Pure Appl. Math.* **1957**, 10, 151.
57. de Palo, S.; Rapisarda, F.; Senatore, G. *Phys. Rev. Lett.* **2002**, 88, 206401.
58. López Ríos, P.; Needs, R. J. Unpublished.
59. Dennis, J. E.; Gay, D. M.; Welsch, R. E. *ACM Trans. Math. Software* **1981**, 7, 369.
60. Umrigar, C. J.; Wilson, K. G.; Wilkins, J. W. *Phys. Rev. Lett.* **1988**, 60, 1719.
61. Kent, P. R. C.; Needs, R. J.; Rajagopal, G. *Phys. Rev. B* **1999**, 59, 12344.
62. Drummond, N. D.; Needs, R. J. *Phys. Rev. B* **2005**, 72, 085124.
63. Ceperley, D. M. *J. Stat. Phys.* **1986**, 43, 815.
64. Ma, A.; Drummond, N. D.; Towler, M. D.; Needs, R. J. *Phys. Rev. E* **2005**, 71, 066704.
65. Riley, K. E.; Anderson, J. B. *Mol. Phys.* **2003**, 101, 3129.
66. Nightingale, M. P.; Melik-Alaverdian, V. *Phys. Rev. Lett.* **2001**, 87, 043401.
67. Umrigar, C. J.; Toulouse, J.; Filippi, C.; Sorella, S.; Hennig, R. G. *Phys. Rev. Lett.* **2007**, 98, 110201.
68. Toulouse, J.; Umrigar, C. J. *J. Chem. Phys.* **2007**, 126, 084102.
69. Ceperley, D. M. Top-ten reasons why no-one uses quantum Monte Carlo, Ceperley group Web site, 1996; since removed.
70. Pople, J. A.; Head-Gordon, M.; Fox, D. J.; Raghavachari, K.; Curtiss, L. A. *J. Chem. Phys.* **1989**, 90, 5622.
71. Curtiss, L. A.; Jones, C.; Trucks, G. W.; Raghavachari, K.; Pople, J. A. *J. Chem. Phys.* **1990**, 93, 2537.
72. Curtiss, L. A.; Raghavachari, K.; Redfern, P. C.; Pople, J. A. *J. Chem. Phys.* **1997**, 106, 1063.
73. Ma, A.; Drummond, N. D.; Towler, M. D.; Needs, R. J. *J. Chem. Phys.* **2005**, 122, 224322.
74. Nemec, N.; Towler, M. D.; Needs, R. J. *J. Chem. Phys.* **2010**, 132, 034111.
75. Kalos, M. H.; Colletti, L.; Pederiva, F. *J. Low Temp. Phys.* **2005**, 138, 747.
76. Anderson, J. B. *J. Chem. Phys.* **1975**, 63, 1499; *Ibid.*, **1976**, 65, 4121.
77. Ceperley, D. M. *J. Stat. Phys.* **1991**, 63, 1237.
78. Foulkes, W. M. C.; Hood, R. Q.; Needs, R. J. *Phys. Rev. B* **1999**, 60, 4558.
79. Glauser, W.; Brown, W.; Lester, W.; Bressanini, D.; Hammond, B. *J. Chem. Phys.* **1992**, 97, 9200.
80. Bressanini, B.; Reynolds, P. J. *Phys. Rev. Lett.* **2005**, 95, 110201.
81. Bajdich, M.; Mitás, L.; Drobný, G.; Wagner, L. K. *Phys. Rev. B* **1999**, 60, 4558.
82. Towler, M. D.; Allan, N. L.; Harrison, N. M.; Saunders, V. R.; Mackrodt, W. C.; Aprà, E. *Phys. Rev. B* **1994**, 50, 5041.

83. Needs, R. J.; Towler, M. D. *Int. J. Mod. Phys. B* **2003**, *17*, 5425.
84. Liu, S. K.; Kalos, M. H.; Chester, G. V. *Phys. Rev. A* **1974**, *10*, 303.
85. Barnett, R. N.; Reynolds, P. J.; Lester, W. A., Jr. *J. Comput. Phys.* **1991**, *96*, 258.
86. Baroni, S.; Moroni, S. *Phys. Rev. Lett.* **1999**, *82*, 4745.
87. Drummond, N. D.; Radnai, Z.; Trail, J. R.; Towler, M. D.; Needs, R. J. *Phys. Rev. B* **2004**, *69*, 085116.
88. Drummond, N. D.; Needs, R. J. *Phys. Rev. Lett.* **2009**, *102*, 126402.
89. Lüchow, A.; Petz, R.; Scott, T. C. *J. Chem. Phys.* **2007**, *126*, 144110.
90. Reboredo, F. A.; Hood, R. Q.; Kent, P. R. C. *Phys. Rev. B* **2009**, *79*, 195117.
91. Drummond, N. D.; López Ríos, P.; Ma, A.; Trail, J. R.; Spink, G.; Towler, M. D.; Needs, R. J. *J. Chem. Phys.* **2006**, *124*, 224104.
92. Fahy, S. In *Quantum Monte Carlo Methods in Physics and Chemistry*, Nato Science Series C: Mathematical and Physical Sciences, Vol. 525, Nightingale, P., Umrigar, C. J., Eds., Kluwer Academic, Dordrecht, The Netherlands, 1999, p. 101.
93. Filippi, C.; Fahy, S. *J. Chem. Phys.* **2000**, *112*, 3523.
94. Grossman, J. C.; Mitas, L. *Phys. Rev. Lett.* **1995**, *74*, 1323.
95. Kutzlning, W.; Morgan, J. D., III. *J. Phys. Chem.* **1992**, *96*, 4484.
96. Prendergast, D.; Nolan, M.; Filippi, C.; Fahy, S.; Greer, J. C. *J. Chem. Phys.* **2001**, *115*, 1626.
97. Feynman, R. P. *Phys. Rev.* **1954**, *94*, 262.
98. Feynman, R. P.; Cohen, M. *Phys. Rev.* **1956**, *102*, 1189.
99. Kwon, Y.; Ceperley, D. M.; Martin, R. M. *Phys. Rev. B* **1993**, *48*, 12037.
100. Holzmann, M.; Ceperley, D. M.; Pierleoni, C.; Esler, K. *Phys. Rev. E* **2003**, *68*, 046707.
101. Kwon, Y.; Ceperley, D. M.; Martin, R. M. *Phys. Rev. B* **1998**, *58*, 6800.
102. Pierleoni, C.; Ceperley, D. M.; Holzmann, M. *Phys. Rev. Lett.* **2004**, *93*, 146402.
103. López Ríos, P.; Ma, A.; Drummond, N. D.; Towler, M. D.; Needs, R. J. *Phys. Rev. E* **2006**, *74*, 066701.
104. Segall, M. D.; Lindan, P. L. D.; Probert, M. J.; Pickard, C. J.; Hasnip, P. J.; Clark, S. J.; Payne, M. C. *J. Phys. Condens. Matter* **2002**, *14*, 2717.
105. Gonze, X.; Beuken, J.-M.; Caracas, R.; Detraux, F.; Fuchs, M.; Rignanese, G.-M.; Sindic, L.; Verstraete, M.; Zerah, G.; Jollet, F.; Torrent, M.; Roy, A.; Mikami, M.; Ghosez, Ph.; Raty, J.-Y.; Allan, D. C. *Comput. Mater. Sci.* **2002**, *25*, 478.
106. Baroni, S.; Dal Corso, A.; de Gironcoli, S.; Giannozzi, P. <http://www.pwscf.org>.
107. Hernandez, E.; Gillan, M. J.; Goringe, C. M. *Phys. Rev. B* **1997**, *55*, 13485.
108. Alfè, D.; Gillan, M. J. *Phys. Rev. B* **2004**, *70*, 161101.
109. Dovesi, R.; Saunders, V. R.; Roetti, C.; Orlando, R.; Zicovich-Wilson, C. M.; Pascale, F.; Civalieri, B.; Doll, K.; Harrison, N. M.; Bush, I. J.; D'Arco, Ph.; Llunell, M. *CRYSTAL06 User's Manual*, University of Torino, Torino, Italy, 2006.
110. te Velde, G.; Bickelhaupt, F. M.; Baerends, E. J.; Guerra, C. F.; van Gisbergen, S. J. A.; Snijders, J. G.; Ziegler, T. *J. Comput. Chem.* **2001**, *22*, 931.
111. This practice has recently been outlawed in our department by new university anti-smoking legislation. My thanks to an anonymous referee for supplying me with this joke.

112. Umrigar, C. J. *Phys. Rev. Lett.* **1993**, *71*, 408.
113. Stedman, M. L.; Foulkes, W. M. C.; Nekovee, M. J. *Chem. Phys.* **1998**, *109*, 2630.
114. Acioli, P. H.; Ceperley, D. M. *J. Chem. Phys.* **1994**, *100*, 8169.
115. Fahy, S.; Wang, X. W.; Louie, S. G. *Phys. Rev. B* **1990**, *42*, 3503.
116. Fahy, S.; Wang, X. W.; Louie, S. G. *Phys. Rev. Lett.* **1998**, *61*, 1631.
117. Mitas, L.; Shirley, E. L.; Ceperley, D. M. *J. Chem. Phys.* **1991**, *95*, 3467.
118. Casula, M.; Filippi, C.; Sorella, S. *Phys. Rev. Lett.* **2005**, *95*, 100201.
119. Casula, M. *Phys. Rev. B* **2006**, *74*, 161102.
120. Greeff, C. W.; Lester, W. A., Jr. *J. Chem. Phys.* **1998**, *109*, 1607.
121. Shirley, E. L.; Martin, R. M. *Phys. Rev. B* **1993**, *47*, 15413.
122. Trail, J. R.; Needs, R. J. *J. Chem. Phys.* **2005**, *122*, 174109.
123. Trail, J. R.; Needs, R. J. *J. Chem. Phys.* **2005**, *122*, 014112.
124. http://www.tcm.phy.cam.ac.uk/~mdt26/casino2_pseudopotentials.html.
125. Burkatzki, M.; Filippi, C.; Dolg, M. *J. Chem. Phys.* **2007**, *126*, 234105; *ibid.*, **2008**, *129*, 164115.
126. Trail, J. R.; Needs, R. J. *J. Chem. Phys.* **2008**, *128*, 204103.
127. Santra, B.; Michaelides, A.; Fuchs, M.; Tkatchenko, A.; Filippi, C.; Scheffler, M. *J. Chem. Phys.* **2008**, *129*, 194111.
128. Rajagopal, G.; Needs, R. J.; James, A. J.; Kenny, S. D.; Foulkes, W. M. C. *Phys. Rev. B* **1995**, *51*, 10591.
129. Rajagopal, G.; Needs, R. J.; Kenny, S. D.; Foulkes, W. M. C.; James, A. J. *Phys. Rev. Lett.* **1994**, *73*, 1959.
130. Lin, C.; Zong, F. H.; Ceperley, D. M. *Phys. Rev. E* **2001**, *64*, 016702.
131. Ewald, P. P. *Ann. Phys.* **1921**, *64*, 25.
132. Chiesa, S.; Ceperley, D. M.; Martin, R. M.; Holzmann, M. *Phys. Rev. Lett.* **2006**, *97*, 076404.
133. Drummond, N. D.; Needs, R. J.; Sorouri, A.; Foulkes, W. M. C. *Phys. Rev. B* **2008**, *78*, 125106.
134. Fraser, L. M.; Foulkes, W. M. C.; Rajagopal, G.; Needs, R. J.; Kenny, S. D.; Williamson, A. J. *Phys. Rev. B* **1996**, *53*, 1814.
135. Williamson, A. J.; Rajagopal, G.; Needs, R. J.; Fraser, L. M.; Foulkes, W. M. C.; Wang, Y.; Chou, M.-Y. *Phys. Rev. B* **1997**, *55*, R4851.
136. Kent, P. R. C.; Hood, R. Q.; Williamson, A. J.; Needs, R. J.; Foulkes, W. M. C.; Rajagopal, G. *Phys. Rev. B* **1999**, *59*, 1917.
137. Kwee, H.; Zhang, S.; Krakauer, H. *Phys. Rev. Lett.* **2008**, *100*, 126404.
138. Dewing, M.; Ceperley, D. M. Methods for coupled electronic–ionic Monte Carlo. In *Recent Advances in Quantum Monte Carlo Methods*, Part II, Lester, W. A., Rothstein, S. M., and Tanaka, S., Eds., World Scientific, Singapore, 2002.
139. Grossman, J. C.; Mitas, L. *Phys. Rev. Lett.* **2005**, *94*, 056403.
140. Huang, K. C.; Needs, R. J.; Rajagopal, G. *J. Chem. Phys.* **2000**, *112*, 4419.
141. Schautz, F.; Flad, H.-J. *J. Chem. Phys.* **2000**, *112*, 4421.
142. Badinski, A.; Haynes, P. D.; Needs, R. J. *Phys. Rev. B* **2008**, *77*, 085111.
143. Reynolds, P. J.; Barnett, R. N.; Hammond, B. L.; Grimes, R. M.; Lester, W. A., Jr. *Int. J. Quantum Chem.* **1986**, *29*, 589.

- 144. Assaraf, R.; Caffarel, M. *Phys. Rev. Lett.* **1999**, *83*, 4682.
- 145. Casalegno, M.; Mella, M.; Rappe, A. M. *J. Chem. Phys.* **2003**, *118*, 7193.
- 146. Assaraf, R.; Caffarel, M. *J. Chem. Phys.* **2003**, *119*, 10536.
- 147. Lee, M. W.; Mella, M.; Rappe, A. M. *J. Chem. Phys.* **2005**, *122*, 244103.
- 148. Badinski, A.; Needs, R. J. *Phys. Rev. E* **2007**, *76*, 036707.
- 149. Badinski, A.; Needs, R. J. *Phys. Rev. B* **2008**, *78*, 035134.
- 150. Badinski, A.; Trail, J. R.; Needs, R. J. *J. Chem. Phys.* **2008**, *129*, 224101.
- 151. Badinski, A.; Haynes, P. D.; Trail, J. R.; Needs, R. J. *J. Phys. Condens. Matter* **2010**, *22*, 074202.
- 152. Farid, B.; Needs, R. J. *Phys. Rev. B* **1992**, *45*, 1067.
- 153. Malatesta, A.; Fahy, S.; Bachelet, G. B. *Phys. Rev. B* **1997**, *56*, 12201.
- 154. Knittle, E.; Wentzcovitch, R.; Jeanloz, R.; Cohen, M. L. *Nature* **1989**, *337*, 349.
- 155. McSkimin, H. J.; Andreatch, P. *J. Appl. Phys.* **1972**, *43*, 2944.
- 156. Occelli, F.; Loubeyre, P.; LeToullec, R. *Nature Mater.* **2003**, *2*, 151.
- 157. Esler, K. P.; Cohen, R. E.; Militzer, B.; Kim, J.; Needs, R. J.; Towler, M. D. *Phys. Rev. Lett.* **2010**, *104*, 185702.
- 158. Sola, E.; Brodholt, J. P.; Alfè, D. *Phys. Rev. B* **2009**, *79*, 024107.
- 159. Dewaele, A.; Loubeyre, P.; Occelli, F.; Mezouar, M.; Dorogokupets, P. I.; Torrent, M. *Phys. Rev. Lett.* **2006**, *97*, 215504.
- 160. Söderlind, P.; Moriarty, J. A.; Wills, J. M. *Phys. Rev. B* **1996**, *53*, 14063.
- 161. Mao, K.; Wu, Y.; Chen, L. C.; Shu, J. F. *J. Geophys. Res.* **1990**, *95*, 21737.
- 162. Li, X.-P.; Ceperley, D. M.; Martin, R. M. *Phys. Rev. B* **1991**, *44*, 10929.
- 163. Towler, M.D.; Russell, N.J.; Valentini, A. *arXiv* **2011**, 1103.1589v1 [quant-ph].

5 Coupled-Cluster Calculations for Large Molecular and Extended Systems

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The ever-increasing power of modern computer systems is advancing many areas of computational chemistry and allowing one to study significantly larger systems with extremely accurate quantum chemistry methods. This has been made possible, in part, by the developments of highly scalable implementations of core quantum chemistry methodologies. In particular, there has been significant progress in the parallel implementations of coupled-cluster (CC) methods, which has become a method of choice for studying complex chemical processes that require accurate treatment of the electron correlation. In this chapter we outline the various CC formalisms available in NWChem and discuss the parallel implementation of these methods in our code. Performance issues, system-size limitations, and the accuracies that can be achieved with these calculations are also discussed. Representative examples from two key domains of CC theory (excited-state formalism and linear response studies) are reviewed and the possibilities of coupling CC methods with different multiscale approaches are highlighted.

5.1 INTRODUCTION

Many aspects of computational chemistry require accuracies that can only be achieved by highly accurate computational methods that account appropriately for the instantaneous interactions or correlations between electrons in molecules.¹ Including these electronic correlation effects is necessary to be able to compare theory and experiment in a precise manner. Even though these correlation effects contribute less than 1% of the total energy, they are fundamental to an understanding of the electronic structure of various systems and in the development of predictive models. For this reason these methods have become an integral part of many computational chemistry packages.

Among the many methods that describe correlation effects systematically, the coupled-cluster (CC) formalism^{2,3} has evolved into a widely used and very accurate method for solving the electronic Schrödinger equation. Compared with other formalisms, such as perturbative methods or approaches based on the linear expansion of the wavefunction (e.g., configuration interaction methods), the main advantage of CC methods lies in the fact that the correlation effects are elegantly captured in the exponential form of the wavefunction. A simple consequence of this ansatz is the size extensivity of the resulting energies or, equivalently, proper scaling of the energy with the number of electrons. Although the CC method was initially proposed in nuclear physics,^{4,5} it was quickly adopted by quantum chemists, and since the late 1960s there has been steady development that has spawned a variety of CC methodologies. In the last decade this formalism has been “rediscovered” by the nuclear physics community.^{6–8} This clearly demonstrates the universal applicability of the method across a wide energy scale.

Despite these successes, the inherent numerical cost of CC methods, which grows rapidly with system size, significantly hampers the wide applicability of this formalism. This difficulty may be overcome through the use of massively parallel computer systems and highly scalable CC implementations. The parallel implementations available in quantum chemistry programs such as ACES II MAB,⁹ ACES III,^{10,11} PQS,^{12–15} MOLPRO,¹⁶ GAMESS(US),^{17–19} and NWChem implementations^{20–24} are excellent examples of recent developments.

In this chapter we demonstrate the capabilities and review the parallel CC implementation in NWChem. We refer the reader to other papers listed above for discussions on other implementations. The rest of this chapter is organized as follows. An overview of CC theory for ground or excited states and CC linear response theory is given in Section 5.2. The details of our parallel CC implementation are described in Section 5.3. In Section 5.4 we present various ground- and excited-state examples and studies involving coupling CC methodologies with multiphysics approaches.

5.2 THEORY

The details of the CC formalism have been discussed in many review articles.^{1,25–27} For the purpose of this chapter we present only the most

important approaches within the single reference formulation, where the CC ground-state wavefunction $|\Psi_0\rangle$ is represented in the form of the exponential Ansatz,

$$|\Psi_0\rangle = e^T |\Phi\rangle \quad (5.1)$$

where the reference function $|\Phi\rangle$ is usually chosen as a Hartree–Fock (HF) determinant and the cluster operator T is represented as

$$T = \sum_{i=1}^N T_i \quad (5.2)$$

where N refers to the total number of correlated electrons. Each component T_n takes the form

$$T_n = \sum_{i_1 < \dots < i_n; a_1 < \dots < a_n} t_{a_1 \dots a_n}^{i_1 \dots i_n} X_{a_1}^+ \dots X_{a_n}^+ X_{i_n} \dots X_{i_1} \quad (5.3)$$

and produces n -tuply excited configurations when acting on the reference function $|\Phi\rangle$. X_p^+ (X_p) is the creation (annihilation) operator for the p th spin-orbital. In the notation above we adopt the convention that $i_1, \dots, i_n$ ($a_1, \dots, a_n$) spin-orbitals refer to spin-orbitals occupied (unoccupied) in the reference function $|\Phi\rangle$. The cluster amplitudes ($t_{a_1 \dots a_n}^{i_1 \dots i_n}$) and CC energies are obtained by solving the electronic Schrödinger equation defined by Hamiltonian H ,

$$He^T |\Phi\rangle = Ee^T |\Phi\rangle \quad (5.4)$$

Premultiplying Eq. (5.4) from the left by e^{-T} leads to the separation of the equations for cluster amplitudes and energy:

$$Qe^{-T} He^T |\Phi\rangle = 0 \quad (5.5)$$

$$E = \langle \Phi | e^{-T} H e^T | \Phi \rangle \quad (5.6)$$

where the Q operator is the projection operator onto the subspace orthogonal to the reference function $|\Phi\rangle$. Using the well-known Hausdorff formula, one can show that the similarity-transformed Hamiltonian $\overline{H} = e^{-T} H e^T$ contains only connected diagrams, which assures the additive separability of the CC energies in the dissociation limit (assuming that the reference function dissociates properly). The connected form of the Schrödinger equation is used as a theoretical foundation for almost all CC implementations.

The approximate CC methods are usually defined by restricting the excitation level included in the cluster operator. In this way we can define a standard hierarchy of approximations:

$T \approx T_2$	CCD (CC with doubles) ²
$T \approx T_1 + T_2$	CCSD (CC with singles and doubles) ^{28,29}
$T \approx T_1 + T_2 + T_3$	CCSDT (CC with singles, doubles, and triples) ^{30,31}
$T \approx T_1 + T_2 + T_3 + T_4$	CCSDTQ (CC with singles, doubles, triples, and
$\vdots$	quadruples) ^{32,33}

For each approximation, the equations for the cluster amplitudes are obtained by projecting the connected form of the Schrödinger equation onto the excited configurations that correspond to the excitations included in cluster operator. This assures that the CC equations are well defined; that is, the number of unknowns (cluster amplitudes) is equal to the number of equations. For example, the CCSD equations for cluster amplitudes can be expressed as

$$\langle \Phi_{i_1}^{a_1} | [H(1 + T_1 + T_2 + \frac{1}{2}T_1^2 + T_1T_2 + \frac{1}{6}T_1^3)]_C | \Phi \rangle = 0 \quad \text{for all } i_1, a_1 \quad (5.7)$$

$$\langle \Phi_{i_1i_2}^{a_1a_2} | [H(1 + T_1 + T_2 + \frac{1}{2}T_1^2 + T_1T_2 + \frac{1}{6}T_1^3 + \frac{1}{2}T_2^2 + \frac{1}{2}T_1^2T_2 + \frac{1}{24}T_1^4)]_C | \Phi \rangle = 0 \quad \text{for all } i_1, i_2, a_1, a_2 \quad (5.8)$$

where the subscript C designates a connected part of a given operator expression (see Refs. 28 and 29 for details). These nonlinear equations are solved iteratively. The most efficient solvers are based on the DIIS formalism.³⁴ The CCD, CCSD, CCSDT, and CCSDTQ approximations scale with the system size (S) as S^6 (CCD), S^6 (CCSD), S^8 (CCSDT), and S^{10} (CCSDTQ), respectively. Unfortunately, this high numerical overhead significantly limits the application range of the CC methodologies, making high-level methods such as the CCSDT and CCSDTQ approaches prohibitively expensive even for small systems. For example, a single CCSD calculation for a water dimer is $2^6 = 64$ times more expensive than a single calculation with the same basis set for a water monomer. The analogous increase in numerical cost for the CCSDT and CCSDTQ methods amounts to 256 and 1024, respectively. However, in many areas of computational chemistry related to calculations of the transition states, activation barriers, intermolecular interactions, CCSDT accuracies are required. Noniterative approaches accounting for the effect of triply excited clusters, such as the CCSD(T) approach,³⁵ are ideally suited for this purpose. At the same time they constitute a widely accepted compromise between numerical cost and accuracy. The S^7 scaling of CCSD(T) and related approaches is between the costs of the CCSD and CCSDT formalisms. Another advantage of noniterative methods for the approximate treatment of triple excitations is their algorithmic structure. Unlike CCSDT, which is iterative and requires storage of the triple amplitudes, the (T) energy correction can be evaluated by generating needed triply excited amplitudes on the fly and parallelizes extremely well because very little communication (scattering of the single and double amplitudes followed by an accumulation of the scalar energy) is required and because very good load balancing can be achieved.

Given the tremendous interest in the comprehensive understanding of photo processes such as solar energy conversion and interaction of biological systems with radiation, it is evident that in years to come, special attention will be given to the development of accurate CC methods capable of characterizing complex excited states. The excited-state extension of the CC method based on the equation-of-motion CC formalism EOMCC^{36–39} is ideally suited for this purpose. The EOMCC wavefunction $|\Psi_K\rangle$ for the K th state is obtained by acting with the excitation operator R_K onto the already correlated ground-state wavefunction in the CC parameterization; that is,

$$|\Psi_K\rangle = R_K e^T |\Phi\rangle \quad (5.9)$$

The excitation operator R_K can be expressed as a sum of its many-body components $R_{K,i}$:

$$R_K = \sum_{i=0}^N R_{K,i} \quad (5.10)$$

where $R_{K,0}$ is a scalar operator. The equations for excited-state energies E_K (or excitation energies ω_K) and the excitation operator are obtained by substituting the wavefunction (5.9) in the Schrödinger equation,

$$H R_K e^T |\Phi\rangle = E_K R_K e^T |\Phi\rangle \quad (5.11)$$

which after premultiplying from the left by the e^{-T} (and taking into account that the R_K and T operators commute, i.e., $[R_K, T] = 0$) takes the form of a non-Hermitian (e^T is not a unitary operator) eigenvalue problem:

$$\overline{H} R_K |\Phi\rangle = E_K R_K |\Phi\rangle \quad (5.12)$$

which, once the exact theory is invoked, is diagonalized in a full configurational space. In practice, we have to resort to various approximations (some of which are listed below) that share the same cost as their CC counterparts:

$R_K \approx R_{K,0} + R_{K,1} + R_{K,2}$	EOMCCSD (EOMCC with singles and doubles) ^{36,37}
$R_K \approx R_{K,0} + R_{K,1} + R_{K,2} + R_{K,3}$	EOMCCSDT (EOMCC with singles, doubles, and triples) ^{40–42}
$R_K \approx R_{K,0} + R_{K,1} + R_{K,2} + R_{K,3} + R_{K,4}$	EOMCCSDTQ (EOMCC with singles, doubles, triples, and quadruples) ⁴³
$\vdots$	

The diagonalization procedure for all approximate methods is carried out in the space of excited configurations included in the definition of the R_K operator. The characteristic feature of standard EOMCC approaches is the size-intensive

character of resulting excited-state energies,⁴⁴ which means that the EOMCC energy is additively separable if the system dissociates into two constituent parts, one in its ground state and the other in an excited state. The EOMCC accuracies depend heavily on the level of theory applied and the character of excited state to be described. It is broadly accepted that the EOMCCSD approach is capable of providing a satisfactory description of singly excited states, while more challenging states, which involve the excitation of two electrons, require at least the EOMCCSDT level of theory. For this reason, in close analogy to the ground-state problem, many noniterative approaches accounting for the effect of triply excited configurations were proposed over the last decade to improve the quality of the EOMCCSD results. One should mention the genuine EOM-CCSD(T) and EOMCCSD($\tilde{T}$) approaches by Watts and Bartlett,^{45,46} which were the first attempts to estimate the effects due to triples in the EOMCC formalism. Other noniterative formulations, such as the CCSDR(3), CCSDR(T), CCSDR(1a), and CCSDR(1b)⁴⁷ methods, which have their basis in linear response CC theory; perturbative approaches based on the partitioning of similarity transformed Hamiltonian—EOM-CC(m)PT(n), EOM-CCSD(2)_T, EOM-CCSD(3)_Q, or EOM-CCSD(2)_{TQ} by Hirata et al.^{48,49}; or the most recent EOM-SF-CCSD(fT) and EOM-SF-CCSD(dT) by Manohar and Krylov,⁵⁰ provide further refinements in the accuracies attainable with noniterative methods.

The excited-state extension of the method of moments of coupled-cluster equations (MMCC)⁵¹ provides a very convenient way of hierarchical characterization of the excited-state correlation effects in terms of the moments of EOMCC equations. In the most rudimentary variant, the EOMCCSD total energies are corrected by adding corrections expressed through triply excited moments:

$$\delta_K(T) = \frac{\langle \Psi_K | M_{K,3}^{\text{EOMCCSD}} | \Phi \rangle}{\langle \Psi_K | R_K^{\text{EOMCCSD}} e^{T_{\text{CCSD}}} | \Phi \rangle} \quad (5.13)$$

where the $M_{K,3}^{\text{EOMCCSD}}$ is the triply excited moment operator of the EOMCCSD equations and $|\Psi_K\rangle$ represents the trial wavefunction.⁵¹ The ensuing completely renormalized EOMCCSD(T) [CR-EOMCCSD(T)] methods⁵² offer significant improvements in the quality of the excited-state PESs. In calculating excitation energies, the best results are obtained when the CR-EOMCCSD(T) corrections are added directly to the EOMCCSD excitation energies ($\omega_K^{\text{EOMCCSD}}$):

$$\omega_K^{\text{CR-EOMCCSD(T)}} = \omega_K^{\text{EOMCCSD}} + \delta_K^{\text{CR-EOMCCSD(T)}} \quad (5.14)$$

This fact is a reflection of the problems the class of noniterative approaches correcting *total energies* are facing when balancing ground- and excited-state correlation effects.^{49,53} The recently introduced nested EOMCCSD(T) formalism⁵⁴ seems to provide an efficient way of balancing these effects. In contrast to the CR-EOMCCSD(T) and EOM-CCSD(2)_T index EOM-CCSD(2)T formalisms, no unphysical assumptions are needed in the calculations of excitation energies. In analogy to the CCSD(T) approach, the numerical scaling of all noniterative

methods discussed here is proportional to S^7 per excited state. For systems composed of 40 to 80 atoms, this scaling makes the noniterative EOMCC approaches a workable alternative to various multireference perturbative approaches.

The last theoretical component discussed in this section is linear response CC theory, which enables one to describe the behavior of molecular systems in an external electric field. Of special importance are the calculations of molecular properties, such as the static and frequency-dependent polarizabilities. The linear-response CC (LR-CC) theory was first developed by Monkhorst and Dalggaard^{55,56} and later by Koch and Jørgensen.⁵⁷ The general formulation of response theory using time-averaged quasi-energy Lagrangians was reviewed by Christiansen et al.,⁵⁸ who present a detailed discussion of the LR-CC equations. For other LR-CC developments, see Refs. 59 to 64.

The LR-CC theory is closely related to the time-dependent Schrödinger equation

$$i \frac{\partial}{\partial t} |\Psi(t)\rangle = H |\Psi(t)\rangle \quad (5.15)$$

where the Hamiltonian H can be decomposed into a time-independent part representing the electronic Hamiltonian (H_0) and a time-dependent part $V(t)$, which represents time-dependent perturbation. The CC parameterization for the time-dependent wavefunction $|\Psi(t)\rangle$ takes the form

$$|\Psi(t)\rangle = e^{i\delta(t)} e^{T(t)} |\Phi\rangle \quad (5.16)$$

where $\delta(t)$ stands for the time-dependent phase factor and $T(t)$ represents the time-dependent cluster operator. The cluster operator $T(t)$ can be expanded in order of perturbation, that is,

$$T(t) = T^{(0)} + T^{(1)}(t) + T^{(2)}(t) + \dots \quad (5.17)$$

This perturbative approach leads to the following expression for the frequency-dependent dipole polarizabilities:

$$\alpha_{ij}(\omega) = - \ll \mu_i; \mu_j \gg_\omega \quad (5.18)$$

where the linear response function $\ll A; B \gg_\omega$ is defined as

$$\begin{aligned} \ll A; B \gg_\omega = & \frac{1}{2} \hat{C}^{\pm\omega} \hat{P}_{A;B} \{ \langle \Phi | (1 + \Lambda) [\bar{A}, T_B^{(1)}(\omega)] | \Phi \rangle \\ & + \langle \Phi | (1 + \Lambda) [[\bar{H}, T_A^{(1)}(\omega)], T_B^{(1)}(-\omega)] | \Phi \rangle \} \end{aligned} \quad (5.19)$$

where the Λ operator and Fourier transform components $T_A^{(1)}(\omega)$ of the first-order cluster operator $T^{(1)}$ were obtained from the equations

$$\langle \Phi | (1 + \Lambda) \bar{H} | \Phi_{i_1 \dots i_n}^{a_1 \dots a_n} \rangle = 0 \quad (5.20)$$

$$\langle \Phi_{i_1 \dots i_n}^{a_1 \dots a_n} | [\bar{H}, T_\gamma^{(1)}(\omega)] - \omega T_\gamma^{(1)}(\omega) | \Phi \rangle + \langle \Phi_{i_1 \dots i_n}^{a_1 \dots a_n} | \bar{\mu}_\gamma | \Phi \rangle = 0 \quad \gamma = x, y, z \quad (5.21)$$

The equations above are defined by projections onto the excited configurations $|\Phi_{i_1 \dots i_n}^{a_1 \dots a_n}\rangle$ used to define the T , Λ , and $T^{(1)}$ operators. Also, the notation convention $\bar{A} = e^{-T} A e^T$ is invoked. As in the case of the CC and EOMCC theories, the level of approximations applied to the T , Λ , and $T^{(1)}$ operators plays a key role in achieving high accuracies. Over the last decade, a significant amount of work has been invested in developing theories, including the effect of triply excited configurations. The CC3 implementation of Christiansen et al.^{65,66} and the arbitrary order response of Kállay and Gauss^{67,68} are also worth noting. The recent parallel implementation of the LR-CCSDT approach by Hammond et al.⁶⁹ allows highly precise estimates of static and dynamic polarizabilities of small diatomic molecules with augmented (doubly augmented) cc-pVNZ basis sets ($N = D, T, Q, 5$). From a practical viewpoint, calculating dipole polarizabilities is a very laborious process consisting of several iterative steps: solving ground-state CC equations, finding the Λ operator, and calculating three components ($\gamma = x, y, z$) of the static/frequency-dependent $T^{(1)}$ operator. Nevertheless, accurate predictions for molecular properties can take advantage of highly scalable LR-CC implementations. In Section 5.3 we discuss state-of-the-art property calculations with parallel LR-CC codes implemented in NWChem.

5.3 GENERAL STRUCTURE OF PARALLEL COUPLED-CLUSTER CODES

The parallel CC codes discussed in this chapter draw heavily on the tensor contraction engine (TCE) implementation by Hirata. We refer the reader to the original TCE papers^{70,71} for more details of the TCE. Here we discuss only the basic tenets of TCE and recent modifications that strive to improve both the speed and scalability of the TCE-generated codes. The key factor determining the parallel structure of the TCE CC codes is the *tile structure* of all tensors used in CC equations. The tiling scheme corresponds to a grouping in the spin-orbital domain into smaller subsets containing the spin-orbitals of the same spin and spatial symmetries (the *tiles*). This partitioning of the spin-orbital domain entails the blocking of all tensors corresponding to one- and two-electron integrals, cluster amplitudes, and all recursive intermediates, into smaller blocks of the size defined by the size of the tile (or *tile size* for short). Since the parallel scheme used in all TCE-generated codes is deeply rooted in dynamic load-balancing techniques, the tile structure defines the granularity of the work to be distributed.

Each CC/EOMCC calculation consists of several steps:

1. Solving the appropriate independent-particle model, providing an appropriate reference function for single-reference CC theories. This usually refers to solving Hartree–Fock (HF) equations, which provides the optimized form of a single-determinant wavefunction. Several variants of the HF theory can be used as a reference for CC calculations. For closed-shell systems, the restricted HF (RHF) approach is the most obvious choice; for open-shell problems, restricted

open-shell or unrestricted HF determinants can be used (ROHF and UHF). We stress the spin-orbital character of the TCE CC implementations which allow users to employ all these references. The performance of the TCE CC implementation for the closed-shell case is additionally enhanced by eliminating the redundant cluster amplitudes.

2. Having obtained the HF solution, the molecular one- and two-electron integrals are generated in the next step. This part, usually referred to as the *four-index transformation*, is an integral part of the TCE CC code. Over the last few years several efficient algorithms have been developed. Of particular interest are the four-index transformations for RHF and ROHF cases, where the orbital form of the transformation can significantly reduce the time required to generate molecular integrals. A very efficient transformation for the RHF and ROHF cases will be available in the next release of NWChem. There are number of features that make this algorithm appealing from the point of view of large-scale calculations. First, this fully fused algorithm scales as S^5 and belongs to the class of “in-core” algorithms (i.e., algorithms that do not use disk). Second, memory management can be tuned to the parameters of a given computer architecture. This is achieved using a twofold strategy based on the use of atomic tiles and the possibility of generating the atomic two-electron integrals in several passes, which can significantly reduce the global memory requirements. The largest CC calculations discussed in the next chapter were performed using this type of the algorithm. Our preliminary tests (see Fig. 5.1) clearly show that our implementation of four-index transformation is capable of scaling across many thousands of CPUs. In the example discussed in Fig. 5.1 we find almost perfect scaling across 3072 CPUs and some flattening of the scalability curve for a larger number of processors. This can be attributed to the fact that the calculations are performed using two passes for generating atomic integrals (if one pass was used, we would observe good scaling up to 6144 processors). In future developments we are planning to significantly revise the management of two-electron integrals, which in its present form is a very memory-consuming process. Feasible alternatives include direct algorithms (atomic integrals are computed “on-the-fly,” used immediately, and then discarded) or by low-rank Cholesky decomposition of the two-electron integral matrix.⁷²

3. The iterative CC/EOMCC calculations, which follow the four-index transformation, are especially challenging, for several reasons. Large amounts of communications are required due to the algebraic structure of the CC equations, and large memory requirements are associated with storing cluster amplitudes and recursive intermediates. These problems are described exhaustively in the original TCE implementation paper.⁷⁰ Here we review the most important approaches in the current TCE CC implementation in NWChem.

The use of recursive intermediates is absolutely essential for reducing the inherent cost of any CC formulation. This “pulling-out-of-bracket” technique can easily be explained using a simple tensor expression:

$$R_j^i = A_k^i B_l^k C_j^l \quad (5.22)$$

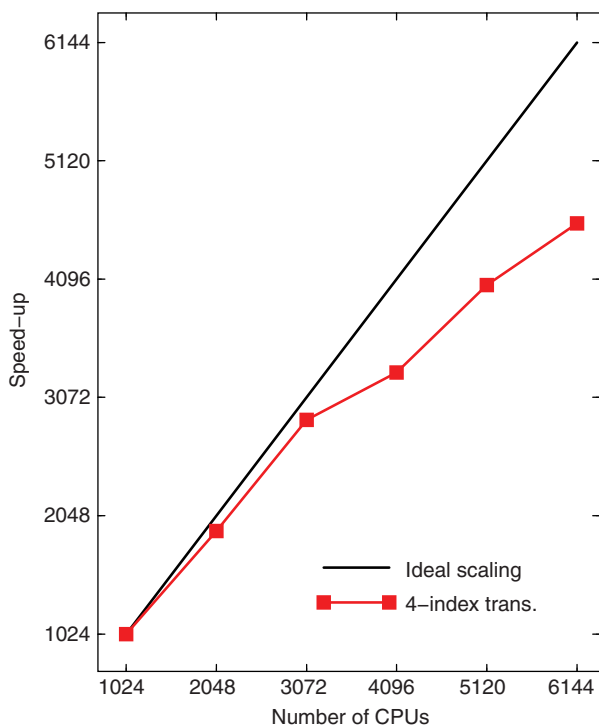


Fig. 5.1 (color online) Scalability of the four-index transformation for the P_2TA system described by a basis set composed of 578 basis set functions.

where the Einstein summation convention over repeated indices is assumed. If we assume that each index runs from 1 to L , the cost of calculating all components of the R -tensor by direct summation of all products on the right-hand side amounts to L^4 . However, if we calculate the auxiliary tensor (or intermediate), defined as

$$I_j^k = B_l^k C_j^l \quad (5.23)$$

first, and then

$$R_j^i = A_k^i I_j^k \quad (5.24)$$

the overall cost is reduced to L^3 , which is achieved at the expense of storing the I -tensor in memory. Similar techniques are used extensively throughout all efficient CC implementations. In our TCE CCSD and EOMCCSD codes we have significantly reduced the maximum size for recursive intermediates down to $n_o^2 n_u^2$, where n_o and n_u refer to the total number of occupied and unoccupied spin-orbitals, respectively. This was achieved by fusing certain types of diagrams that utilize the most expensive class of two-electron integrals, the four-particle (or four-virtual) integrals. Another important improvement in performance was achieved for the

RHF and ROHF (high-spin) references. As stated earlier, the TCE-CC codes are rooted in spin-orbital formalism. However, storing two-electron integrals in spin-orbital form for RHF and ROHF cases is rather ineffective. Instead, for the RHF and ROHF references, the required antisymmetrized two-electron integrals $v_{r\sigma_r s\sigma_s}^{p\sigma_p q\sigma_q}$ are generated on the fly using the formula

$$v_{r\sigma_r s\sigma_s}^{p\sigma_p q\sigma_q} = (pr|qs)\delta_{\sigma_p\sigma_r}\delta_{\sigma_q\sigma_s} - (ps|qr)\delta_{\sigma_p\sigma_s}\delta_{\sigma_q\sigma_r} \quad (5.25)$$

where σ_p corresponds to a spin function of p th orbitals and

$$(pq|rs) = \iint \varphi_p^*(1)\varphi_q(1)\frac{1}{r_{12}}\varphi_r^*(2)\varphi_s(2) dr_1 dr_2$$

is a spin-free two-electron integral for orbitals p, q, r , and s . Further improvements in the parallel performance of CCSD and EOMCCSD were recently achieved by localizing (replicating) certain classes of low-rank tensors and with more efficient memory-lookup procedures for high-rank tensors.

4. To achieve chemical accuracy in many applications, one must account for the effect of triply excited configurations, preferably in a noniterative way. Several related methodologies, such as the CCSD(T) and CR-EOMCCSD(T) approaches, were discussed earlier. Due to its S^7 scalability, this noniterative part can dominate the entire calculation, especially those for large molecular systems. For this reason, an efficient parallel implementation of these approaches is absolutely pivotal for reducing the time to solution. Fortunately, due to their noniterative character, all (T)-type methods are ideally suited for highly efficient implementation (see Fig. 5.2). Generally speaking, all corrections can be represented as the following tensor contraction:

$$\delta(T) = \alpha * \sum_{i,j,k;a,b,c} A_{abc}^{ijk} B_{ijk}^{abc} \quad (5.26)$$

where the A and B tensors are functions of lower-order tensors. This expression can be effectively implemented using a simple (schematic) parallel scheme:

```
do [i] (occupied tiles)
do [j] (occupied tiles)
do [k] (occupied tiles)
do [a] (unoccupied tiles)
do [b] (unoccupied tiles)
do [c] (unoccupied tiles)
  ■ dynamic load balancing starts here;
  workload for CPU(n):
    - forming  $A_{[a][b][c]}^{[i][j][k]}$  and  $B_{[i][j][k]}^{[a][b][c]}$ 
    - accumulating:
```

```


$$\delta(T)(n) = \delta(T)(n) + \alpha * \beta(n) * A_{[a][b][c]}^{[i][j][k]} B_{[i][j][k]}^{[a][b][c]}$$

    enddo
  enddo
enddo
enddo
enddo
enddo
gathering:  $\delta(T) = \sum_n \delta(T)(n)$ 

```

where the α and $\beta(n)$ are the method/expression specific constants and $\delta(T)(n)$ represents the contribution to $\delta(T)$ accumulated on the n th CPU. This strategy, shown in Fig. 5.2, can effectively harness the computational power offered by several thousands of CPUs. The calculations above, representing CR-EOMCCSD(T) calculations for a P₂TA system, were performed on the EMSL HP Linux cluster with the Infiniband Network. Obviously, the noniterative procedure can easily be extended to include higher-order effects, such as quadruples and pentuples.

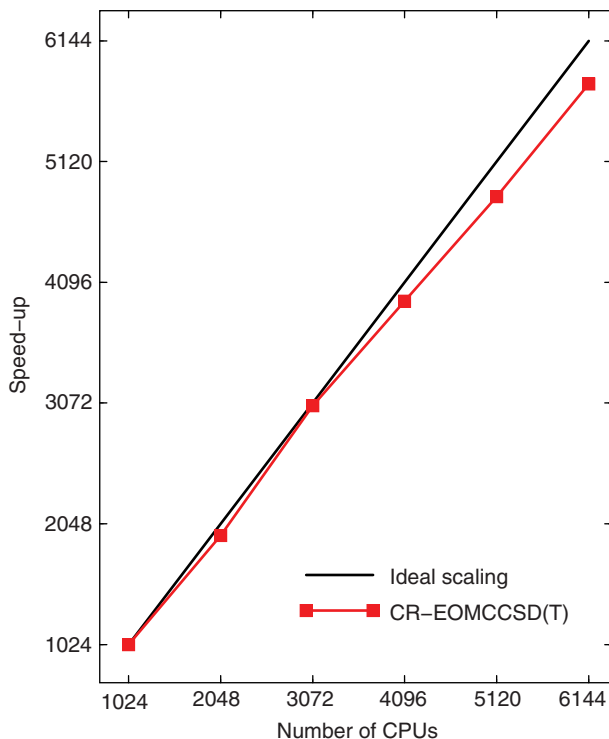


Fig. 5.2 (color online) Scalability of the CR-EOMCCSD(T) method (triples part) for a P₂TA system described by a basis set composed of 578 basis set functions.

The modular character of NWChem makes the integration of each implementation discussed in this section with other modules of NWChem a relatively easy task. Of particular importance are the multiscale modules such as QM/MM and the embedded cluster approach, which offer a unique opportunity for high-level methods to be used in different arenas beyond pure gas-phase calculations. Of particular interest are multiscale approaches that allow one to calculate ground- and excited-state properties of molecular systems in realistic settings at finite temperatures and pressures. This issue is addressed later in the chapter.

5.4 LARGE-SCALE COUPLED-CLUSTER CALCULATIONS

The area of large-scale calculations with the CC methodology is evolving at a very quick pace, and large-scale CC calculations performed on massively parallel machines are reported in the literature practically on a monthly basis. This is possible thanks to the development of several parallel CC codes in packages such as ACES(MAB), ACES III, GAMESS(US), MOLPRO, NWChem, and PQS. Most of the well-documented applications are related to ground-state problems with the CCSD and CCSD(T) methods used as drivers. However, over the last couple of years a few large-scale CC excited-state and property calculations have been carried out. In this section we focus entirely on the recent implementation of the EOMCC and LR-CC methodologies in NWChem. We also discuss various multiscale schemes that can take advantage of these parallel implementations. Part of the material discussed in this section is related to studies of EOMCC formalism,²² LR-CC formalism,^{24,69} and combined multiscale approaches.^{73–75} In all our calculations we have used various Pople^{76–80} and Dunning^{81–83} basis sets.

5.4.1 Equation-of-Motion Coupled-Cluster Approaches to Large Molecular Systems

The excited-state calculations are in many aspects different from the ground-state calculations. For example, when calculating excited-state potential energy surfaces, the configurational structures of the corresponding excited-state wavefunctions can undergo significant changes, which makes the failure of the lower-rank approaches more evident than in the case of ground-state calculations. An excellent illustration of this fact is provided by the EOMCC calculations for the two PESs corresponding to the two lowest excited states of A' symmetry ($2^1A'$ and $3^1A'$) of water molecule as functions of asymmetric O—H stretch. While the EOMCCSD and EOMCCSDT curves for $R_{OH} < 1.2$ Å are relatively close, for larger distances the EOMCCSD curve barely resembles the EOMCCSDT curves (see Fig. 5.3). This is a simple consequence of the increasing role played by the doubly excited component for both states for larger O—H distances. Relatively good agreement of the EOMCCSD and EOMCCSDT results close to the equilibrium geometry ($R_{OH} = 0.9572$ Å; $\alpha_{HOH} = 104.52^\circ$) can be attributed to the singly excited character of these states in this region. Although the water

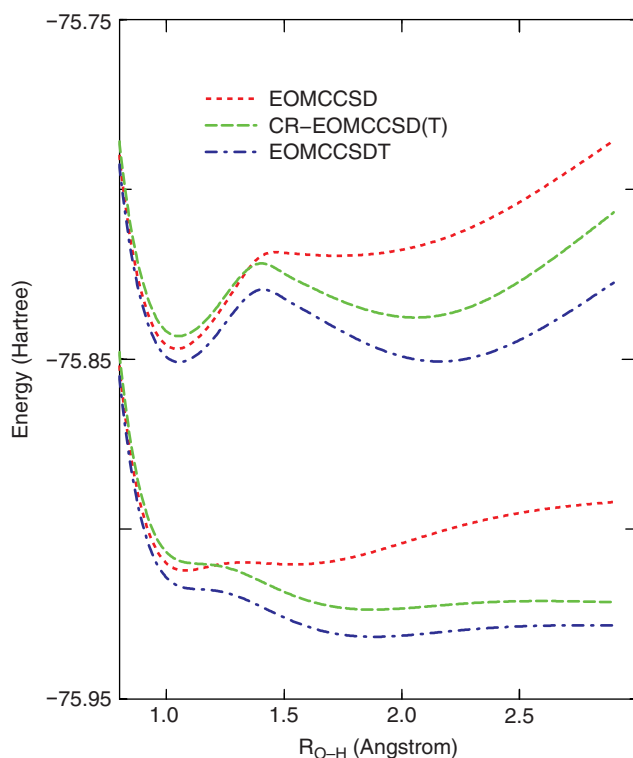


Fig. 5.3 $2^1A'$ and $3^1A'$ PESs of a water molecule as functions of a asymmetric O—H stretch. The calculations were performed using an aug-cc-pVDZ basis set.

molecule is a rather small system, similar effects can be expected for much larger systems where the effective inclusion of the effect of triply excited configurations may be essential for accurate results. Using the noniterative CR-EOMCCSD(T) approach, which mediates between the iterative EOMCCSD and EOMCCSDT approaches in accuracy and computational cost, one can effectively correct the quality of the EOMCCSD curves. This pertains to the location of minima, barrier heights, and asymptotic behavior in the noninteracting limit.

As representative examples of large-scale EOMCC calculations, we have chosen two systems: zinc-porphyrin (ZnP) and oligoporphyrin dimer (P_2TA),⁸⁴ which due to the roles they play in studies of light-harvesting systems, design of molecular switches, and molecular wires, have been and continue to be the focus of high-level excited-state studies. The ZnP, P_2TA , and related systems, such as the free-base porphyrin/zinc porphyrin dimers have been the subject of intensive studies with wavefunction-based approaches, including the SAC-CI (equivalent of the EOMCC approach),^{85–88} STEOMCC,^{89–92} and CASPT2^{84,93,94} methods. Due to the size of ZnP dimers and the P_2TA molecule, severe restrictions have been imposed either on the number of correlated orbitals used in the

SAC-CI calculations [for the zinc porphyrin dimers only molecular orbitals with corresponding orbital energies falling into the $(-1.2, 1.3)$ a.u. interval were correlated in SAC-CI calculations⁸⁷] or on the number of active orbitals included in the CASPT2 calculations.⁸⁴ For example, a total number of 373 orbitals was used in the largest SAC-CI calculations for the ZnP dimer (model C) of Ref. 87.

We begin our discussion with the results for the ZnP molecule (see Table 5.1). The calculations were performed using a combined QM/MM approach with TD-DFT (based on the B3LYP^{95,96} and CAM-B3LYP^{97,98} functionals; for NWChem CAM-B3LYP implementation, see Refs. 99 and 100), EOMCCSD, and CR-EOMCCSD(T) methods for the quantum region. The DFT(B3LYP)/MM optimization of the entire system for the ZnP monomer embedded in a 30-Å cubic box with 869 water molecules leads to the equilibrium ground-state geometry that exhibits small out-of-plane bend. For this reason all calculations were performed using C_1 symmetry. The pairs of 2^1A and 3^1A , 4^1A and 5^1A , and 6^1A and 7^1A states should be attributed to the Q-, B-, and N-bands of ZnP in D_{4h} symmetry. Comparison of the results in Table 5.1 shows clearly that the time-dependent DFT(B3LYP) [TD-DFT(B3LYP)] results, in contrast to the results obtained with the TD-DFT(CAM-B3LYP) and all EOMCC methodologies, interchange the ordering of the B- and N-bands. This is a consequence of the CT character of corresponding excited states, which can be handled effectively by the TD-DFT(CAM-B3LYP) approach. It is interesting to analyze the effect of triple corrections added to the EOMCCSD excitation energies. While the EOMCCSD excitation energies are off by approximately 0.2 eV from the experimental values for the Q-band, the CR-EOMCCSD(T) excitation energies for this case are in perfect agreement with the experimental data (2.25 eV given by CR-EOMCCSD(T) for the 2^1A state versus 2.28 inferred from the experiment). For the B-band the triple correction improves the EOMCCSD results by 0.25 eV, but there is a significant gap (~ 0.75 eV) between the CR-EOMCCSD(T) and experimental results. It is interesting to observe that the TD-DFT(CAM-B3LYP) results for the Q- and

TABLE 5.1 EOMCCSD, CR-EOMCCSD(T), and TD-DFT (B3LYP and CAM-B3LYP) Excitation Energies (in eV) for ZnP System in Aqueous Solution^a

Method	2^1A	3^1A	4^1A	5^1A	6^1A	7^1A
B3LYP ^b	2.40	2.41	3.51	3.51	3.36	3.38
CAM-B3LYP ^c	2.35	2.35	3.65	3.66	3.89	3.92
EOMCCSD ^b	2.49	2.50	4.01	4.02	4.29	4.32
CR-EOMCCSD(T) ^b	2.25	2.26	3.74	3.75	4.18	4.20
Expt. ^d	2.28	—	3.00			

^aFor EOMCC methods the r_1 -B₁ basis set²² was used and all core electrons were kept frozen. The values for the α , β , and γ parameters used by the CAM-B3LYP approach are 0.19, 0.46, and 0.33, respectively.

^bFrom Ref. 22.

^cFrom Refs. 99 and 100.

^dFrom Ref. 101.

B-bands are in a very good agreement with the CR-EOMCCSD(T) ones. For the N-band the TD-DFT(CAM-B3LYP) results are below the CR-EOMCCSD(T) excitation energies and observed discrepancies are around 0.3 eV.

Our next example is the P₂TA molecule (see Fig. 5.4), which was studied previously with the CASPT2 method using a 6-31G basis set.⁸⁴ The high degree of symmetry of this molecule makes EOMCC calculations feasible. Even after freezing 116 core electrons, 270 electrons need to be correlated. In our calculations we used the cc-pVDZ basis set, which translates into 942 basis set functions. Only the lowest excited states corresponding to the 1^1B_{3u} and 1^1B_{1g} states were the targets of our calculations. We begin our discussion of the results with the CASPT2 calculations, which are about 0.2 eV below the experimental values. This may be attributed to the problems with the definition of the active space used in calculations,⁸⁴ small basis sets, and the well-documented tendency of the CASPT2 method to underestimate excitation energies (see, e.g., Ref. 102). The EOMCCSD as well as the CC2 energies are above the experimental value. For the 1^1B_{3u} excitation energy the difference between EOMCCSD and experimental values amounts to 0.29 eV. By accounting for the effects of the triples, we can significantly improve the quality of the EOMCCSD results. Our estimates of the effect of triples are based on the CR-EOMCCSD(T) correction obtained for the 1^1B_{3u} state in calculations that correlated all orbitals (excluding the frozen core) with corresponding energies below 1.5 a.u. This correction was subsequently added to the cc-pVDZ EOMCCSD excitation energy (reported in Table 5.2), which led to 1.97 eV. This should be viewed as a significant improvement over EOMCCSD. Further improvements should be expected when larger numbers of virtual orbitals are included in the construction of the CR-EOMCCSD(T) correction. It is also interesting to notice that the TD-DFT(CAM-B3LYP) approach yields excitation energies of EOMCCSD quality.

In the last paragraph of this section we discuss briefly the timings per EOM-CCSD iterations for various systems: di-8-ANEPPS fluorescent probe¹⁰³ and P₂TA. (The aliphatic tail of di-8-ANEPPS was removed for all calculations,

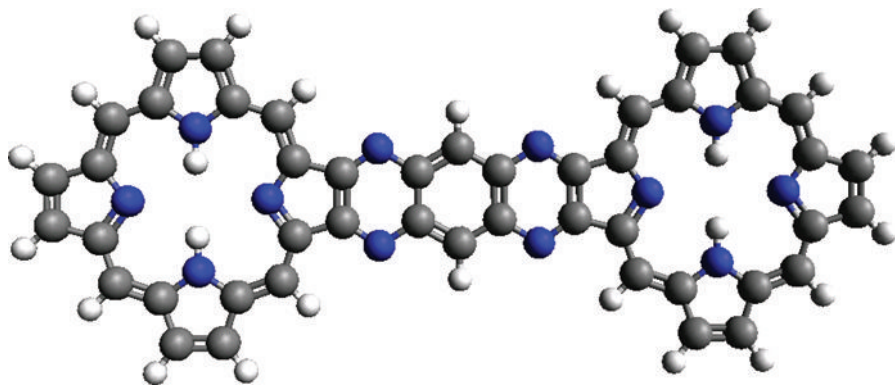


Fig. 5.4 (color online) The geometrical structure of the P₂TA system.

TABLE 5.2 EOMCC, CASPT2, and CAM-B3LYP Excitation Energies (eV) for 1^1B_{3u} and 1^1B_{1g} Transitions (Assigned to the Q_x -Band) of the P_2TA System^a

State	CC2	EOMCCSD	CR-EOMCCSD(T) ^b	CASPT2 ^b	CAM-B3LYP	Expt. ^c
1^1B_{3u}	2.17	2.13	1.97	1.66	2.12	1.84
1^1B_{1g}	—	2.14	—	1.67	2.14	

^aThe CC2, EOMCCSD, CR-EOMCCSD(T), and CAM-B3LYP results were obtained with a cc-pVDZ basis set (942 orbitals), while the CASPT2 results were obtained with a 6-31G basis set. In all CC calculations, core electrons were kept frozen.

^bThe CR-EOMCCSD(T) correction was calculated in the space of all orbitals with corresponding orbital energies being less than 1.5 a.u. Subsequently, this correction was added to the EOMCCSD results performed for a cc-pVDZ basis set (excluding the core electrons, which were kept frozen).

^cFrom Ref. 84.

TABLE 5.3 Timings of the EOMCCSD Iterations for Two Benchmark Systems: di-8-ANEPPS and P_2TA ^a

System	Time per EOMCCSD Iteration (1024 CPUs) (1)
di-8-ANEPPS	
6-31 + G* (570 bsf ^b)	1680 (no symm.)
cc-pVDZ (516 bsf)	1500 (no symm.)
P_2TA	
cc-pVDZ (942 bsf)	1120 (D2h symm.)

^aSee the text for details.

^bBasis set functions.

which is not expected to affect the overall results.) As seen from Table 5.3, time per single EOMCCSD iteration for di-8-ANEPPS ($C_{22}H_{24}N_2O_3S$) in a 6-31 + G* basis set [total 570 basis set functions (bsf)] takes about 28 min on 1024 CPUs. The analogous time for the P_2TA system ($C_{46}H_{26}N_{12}$) is about 19 min (with symmetry). The timings discussed show clearly that the use of large numbers of CPUs can effectively reduce the time required to get a single EOMCCSD root to within a few hours.

5.4.2 Property Calculations Using the Coupled-Cluster Method

In this part we describe the results for static and dynamic dipole polarizabilities obtained with the recent LR-CC implementation in NWChem. In particular, we focus on two representative examples that not only provide an excellent example of large-scale calculations that can be done on massively parallel machines and provide an independent estimate of accuracies obtained using lower-order methods. Our benchmark systems for this purpose will be the C_{60} system and polyaromatic hydrocarbons. We also assess the equally important impact of higher-rank clusters included in linear response CC theory. This part is illustrated by the

LR-CCSD and LR-CCSDT calculations for relatively small systems with large basis sets.

The properties of C_{60} have been intensively studied over the last two decades. Two cornerstone dipole polarizability measurements have provided the experimental values for the static and dynamic ($\lambda = 1064$ nm) case. Antoine et al.¹⁰⁴ used beam deflection techniques to assess the static polarizability of the isolated C_{60} molecule. The value of $76.5 \pm 8 \text{ \AA}^3$ obtained is characterized by significant uncertainty. A later measurement by Ballard et al.¹⁰⁵ led to $79 \pm 4 \text{ \AA}^3$ for the optical polarizability at a wavelength of 1064 nm. For theory the calculation of electric properties of the C_{60} molecule constitutes a significant challenge, which is best illustrated by the range of different theoretical estimates of its dipole polarizability, which vary from 36 to $154 \text{ \AA}^3$ (see Ref. 23 and references therein). Increased computing power combined with increasingly accurate ab initio methods should be able to provide accurate estimates for the (dynamic) polarizability.

Owing to the delocalized character of the electronic structure of C_{60} , proper choice of the basis set and accurate inclusion of the correlation effects are necessary for obtaining reliable results for dipole polarizabilities. Ruud et al.¹⁰⁶ generated results of 75.1 and $76.4 \text{ \AA}^3$, in good agreement with the experimental values 76.5 ± 8 and $79 \pm 4 \text{ \AA}^3$ for $\lambda = \infty$ ($\omega = 0.0$) and $\lambda = 1064$ nm ($\omega = 0.04282270$). However, the authors employed a rather small 6-31++G basis set and did not account for correlation effects. A later study by Pedersen et al.¹⁰⁷ clearly showed the strong basis set/correlation effect dependence of the polarizability. For example, for static polarizability the difference between the 6-31++G and aug-cc-pVDZ¹⁰⁷ is as large as $5.5 \text{ \AA}^3$ (37 a.u.). For dynamic polarizabilities the differences are even larger. Due to the large size of the molecular system, very few coupled-cluster (CC) calculations have ever been performed for C_{60} . Pedersen et al.¹⁰⁷ used the CC2 method¹⁰⁸ with an aug-cc-pVDZ basis set and the Cholesky algorithm to approximate two-electron integrals, to obtain polarizabilities equal to $92.33 \text{ \AA}^3$ (623.70 a.u.) and $94.77 \text{ \AA}^3$ (640.15 a.u.) for $\lambda = \infty$ and $\lambda = 1064$ nm, respectively, which are too high compared to experiment. These discrepancies may be caused to some extent by the incomplete treatment of correlation effects due to singles and doubles offered by the CC2 formalism (the CC2 methods include only a small part of all diagrams included in the full LR-CCSD approach). Moreover, the results by Pedersen et al. appear to indicate that the CC2 method combined with sufficiently flexible basis sets may overestimate the electron correlation contribution to the dipole polarizability of C_{60} . To estimate the effect of missing correlation effects, we performed the calculations for the static and dynamic polarizabilities using our LR-CCSD implementation in NWChem. The best choice of the basis set is provided by properly tailored basis sets, which are suitable for calculations of electric dipole polarizabilities, yet small enough to make these calculations feasible. The present calculations use the recently developed reduced-size polarized basis sets^{109,110} denoted hereafter by ZPolX, where X is the element symbol. The ZPolC basis set used in this study corresponds to the Zm3PolC basis set of Refs. 109 and

110. The ZPolX basis sets originate from earlier ideas¹¹¹ concerning the form of the electric field dependence of basis set functions and have lead to the development of PolX sets (or Sadlej's pVTZ).¹¹² In comparison with PolX sets, the ZPolX basis sets offer significant reduction (by about one-third) of the basis set size with negligible (1 to 2%) loss of accuracy of the polarizabilities calculated. They are presumably the smallest basis sets that lead to reliable theoretical data for dipole polarizabilities. In the case of C₆₀, their use corresponds to the total basis set size of 1080 functions, compared to 1440 functions with the PolC basis set. The smaller dimension of the basis set has an immediate bearing on the numerical overhead associated with use of the rather expensive LR-CCSD approach, characterized by S^6 scaling with the system size. For the C₆₀ molecule it translates into a fivefold speed-up in CCSD, Λ -CCSD, and LR-CCSD calculations, which are needed in order to calculate the tensor of dipole polarizability. The results of our calculations are shown in Table 5.4. In all calculations the D_{2h} symmetry was assumed and all core electrons were frozen. Due to symmetry, $\alpha_{XX}(\omega) = \alpha_{YY}(\omega) = \alpha_{ZZ}(\omega) = \alpha(\omega)$. All CCSD iterations were converged to 10^{-4} . To make direct comparisons with earlier studies we have used the same geometry of C₆₀ as in Refs. 106 and 107. The earlier studies show that the ZPolC basis set performs similarly to the aug-cc-pVDZ and aug-cc-pVTZ sets. Hence, by comparing our results with the CC2 data of Pedersen et al. we can estimate the importance of terms that are not included in the CC2 formalism. For the ground-state formulation, these terms correspond to CCSD diagrams for doubly excited configurations, which involve a T_2 operator; only the most rudimentary term, $\langle \Phi_{ij}^{ab} | [F, T_2] | \Phi \rangle$, where F represents the Fock operator, is retained in the CC2 approximation. The CC2 value of $\alpha(0)$ for the aug-cc-pVDZ basis set is equal to 623.70 a.u. (92.33 Å³) and increases to 640.15 a.u. (94.77 Å³) for $\lambda = 1064$ nm. Hence, for static and dynamic cases the differences between the CC2 data and the present LR-CCSD values of α are equal to 10.13 Å³ and 11.15 Å³, respectively. This indicates that T_2 -dependent terms play an important role in the evaluation of linear response properties in C₆₀. This conclusion is in line with the hierarchical structure of CC methods (CCS, CC2, CCSD, CC3, CCSDT, etc.) discussed by Christiansen et al.¹⁰⁸ Finally, it is pleasing to note that the calculated LR-CCSD static polarizability of C₆₀ is well within error bars of the experimental value. The LR-CCSD dynamic polarizability $\omega = 0.04282270$ ($\lambda = 1064$ nm) is only marginally higher (by 0.62 Å³) than the upper limit for the experimental estimate (see Table 5.4).

Since 240 electrons were correlated in the C₆₀ calculations, it gives us a unique opportunity to study the scalability characteristic of our codes in the regime of large numbers of correlated electrons described by a large number of basis set functions. The observed speed-up going from 256 to 1024 CPUs is about 3.75 (see Fig. 5.5). Individual timings per CCSD iteration amount to 2924 s and 776 s for 256 and 1024 CPUs, respectively. This good scalability is a direct consequence of equal load balancing, which is much easier to achieve in the case of large numbers of correlated particles. Our scalability figures are in line with the earlier performance analysis for the zinc-porphyrin molecule.²²

TABLE 5.4 Static and Frequency-Dependent Polarizabilities of C₆₀ in Å³ (a.u.)^a

Wavelength (nm)	CCSD ^b	Experiment
∞	82.20 (555.27 a.u.)	76.5 ± 8 ^c
1064	83.62 (564.85 a.u.)	79 ± 4 ^d
532	88.62 (598.64 a.u.)	

^aThe ZPolC basis set was used and all 1s electrons were frozen (total number of basis set functions is equal to 1080). All calculations were performed with D_{2h} geometry used in Ref. 106.

^bFrom Ref. 23.

^cFrom Ref. 104.

^dFrom Ref. 105.

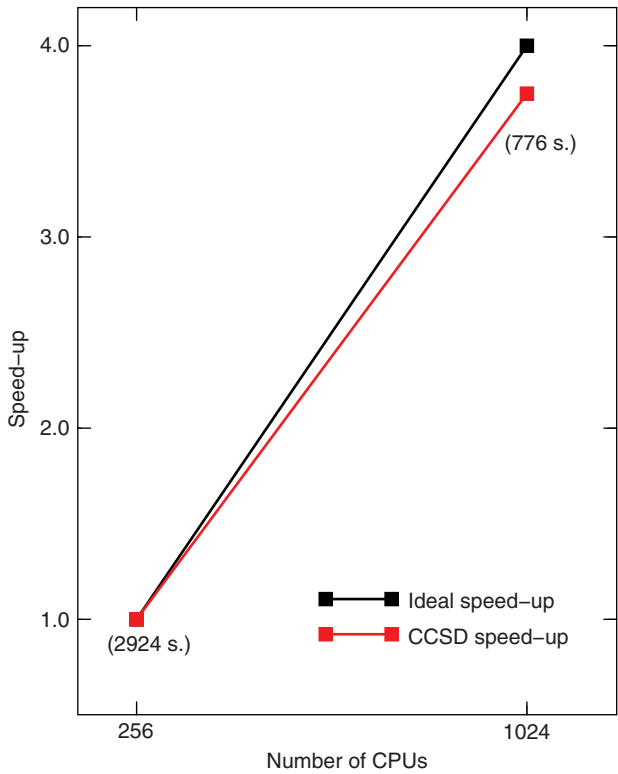


Fig. 5.5 (color online) Scalability of the CCSD code: C₆₀ molecule.

It is well known that the DFT theory experiences serious difficulties in the prediction of polarizabilities and hyperpolarizabilities of conjugated systems.^{113–116} The availability of highly scalable CC property codes are invaluable in quantifying the accuracies of various DFT approaches. The existing comparisons of the DFT and CC formalism are based on the estimates obtained with finite

field techniques suitable only for static properties for systems described using relatively small basis sets. Again, as in the case of the C_{60} calculations, the choice of the basis set is an important issue, which was discussed extensively by Hammond et al.²⁴ for the benzene molecule. It was found that the Dunning basis sets outperform the Pople basis sets. For example, the aug-cc-pVDZ basis, with only 192 functions, produces equally accurate results to the 6-311++G(3df,3pd) Pople basis set, which has 342 basis set functions. It is also worth noticing that the Pople basis set results converge systematically to the results obtained with a large aug-cc-pVQZ basis set (756 basis set functions) from below, while the Dunning basis sets appear to converge from above (if we neglect the smallest case, the aug-cc-pVDZ basis set). The Sadlej pVTZ basis set with a smaller number of functions yields aug-cc-pVTZ/d-aug-cc-pVTZ quality. For this reason a Sadlej pVTZ is ideally suited for large-scale property calculations.

The representative CCSD calculations for static- and frequency-dependent polarizabilities of linear oligoacenes are collected in Tables 5.5 and 5.6. In Table 5.5 static CCSD polarizabilities obtained with a Sadlej basis set are

TABLE 5.5 Static Dipole Polarizabilities of Linear Oligoacenes for $n = 1$ to 6^a

Ring	CCSD ^b			B3LYP ^c		
	α_{LL}	α_{MM}	α_{NN}	α_{LL}	α_{MM}	α_{NN}
1	80.57	80.57	44.66	79.38	79.38	42.97
2	166.61	123.39	66.43	168.59	121.70	63.58
3	281.60	166.00	87.58	291.56	164.63	83.83
4	423.83	209.77	108.61	447.60	209.00	103.97
5	589.97	254.92	129.58	634.65	255.01	124.00
6	776.83	301.26	150.55	849.55	302.79	143.95

^aPolarizabilities are given in atomic units. The CCSD calculations were performed with a Sadlej pVTZ basis set and the B3LYP calculations with a 6-311G(2d,1p) basis set. The ranks of the atomic basis set used in CCSD calculations are equal to 198, 312, 426, 540, 654, and 768 basis set functions for benzene, naphthalene, anthracene, tetracene, pentacene, and hexacene, respectively.

^bFrom Ref. 24.

^cFrom Ref. 117.

TABLE 5.6 Static and Dynamic Polarizabilities of Pentacene Obtained with the PBE and CCSD Approaches Using a Sadlej pVTZ Basis Set^a

Method	$\omega = 0.0$			$\omega = 0.072$		
	α_{LL}	α_{MM}	α_{NN}	α_{LL}	α_{MM}	α_{NN}
PBE ^b	642.22	259.61	120.20	773.43	(175.85)	122.59
CCSD ^b	589.97	254.92	129.58	672.07	284.54	132.06

^aAll polarizabilities and frequencies are given in atomic units. The α_{MM} component of the dynamic polarizability (in parentheses) is erroneous for PBE since the frequency is greater than the excitation energy of corresponding dipole-allowed transition.

^bFrom Ref. 24.

compared with B3LYP polarizabilities obtained with a 6-311G(2d,1p) basis set. These results can be used to determine the saturation rate with respect to the ring size. One can see that the difference between the CCSD and B3LYP results is approximately 10% for $n = 6$ (hexacene). We should expect this difference to be larger if the Sadlej pVTZ basis set was used. The results for dynamic polarizabilities of pentacene are shown in Table 5.6, where we compare CCSD and PBE results obtained with a Sadlej pVTZ basis set. In DFT calculations, the α_{MM} component of the dynamic polarizability is erroneous due to the proximity of a pole or pole crossing. Additionally, the error in PBE increases substantially for finite frequency.

Although the numerical cost makes the LR-CCSDT approach prohibitively expensive for systems tractable by the cheaper LR-CCSD method, it is very important to characterize qualitatively the role of triply excited clusters in retaining correlation effects for molecular properties in various basis sets. The role of the triples in describing accurate dynamic polarizabilities and hyperpolarizabilities has been discussed based on the CC3 approach.^{65,66} Recently reported results by Kállay and Gauss⁶⁷ and O'Neill et al.⁶⁸ for the CCSDT polarizabilities and hyperpolarizabilities were obtained with double-zeta basis sets. Also,⁶⁹ using the NWChem implementation of LR-CCSDT theory, studies have been extended to much larger basis sets, including basis sets of TZ/QZ/5Z quality. From a computational point of view the LR-CCSDT calculations with these basis sets, even for small systems, constitute a significant challenge. As a specific example, let us consider the CN molecule in various basis sets. The results for the parallel components of the CCSD and CCSDT dynamic polarizabilities are shown in Fig. 5.6. The parallel component reveals a strong dependence on the basis set employed and the excitation included in the cluster operator. In agreement with previous studies by Kállay and Gauss,⁶⁷ the inclusion of triply excited effects is important for obtaining accurate results, especially at larger frequencies. In addition, it was found that the choice of reference—ROHF versus UHF—is very important for accurate polarizabilities, due to the large amount of spin contamination present in the UHF reference for CN. The spin-orbital nature of the TCE CC codes allows all references to be treated using the same framework.

5.4.3 Multiscale Coupled-Cluster Formulations

Many important chemical processes such as the energy storage or energy conversion in biological or light-harvesting systems are very sensitive to the surrounding environment. In many cases gas-phase calculations provide an inadequate description of such systems, as the interaction with the environment is not properly accounted for.¹¹⁸ There is also a pressing need to develop multiscale formalisms capable of describing finite-temperature ground- and excited-state properties of molecular systems relevant to structural biology, hydrogen economy, and environmental and material science problems. To address this challenging task, several multiscale approaches that combine the accuracy and computational complexity of coupled-cluster (CC) methods with the efficiency of classical molecular

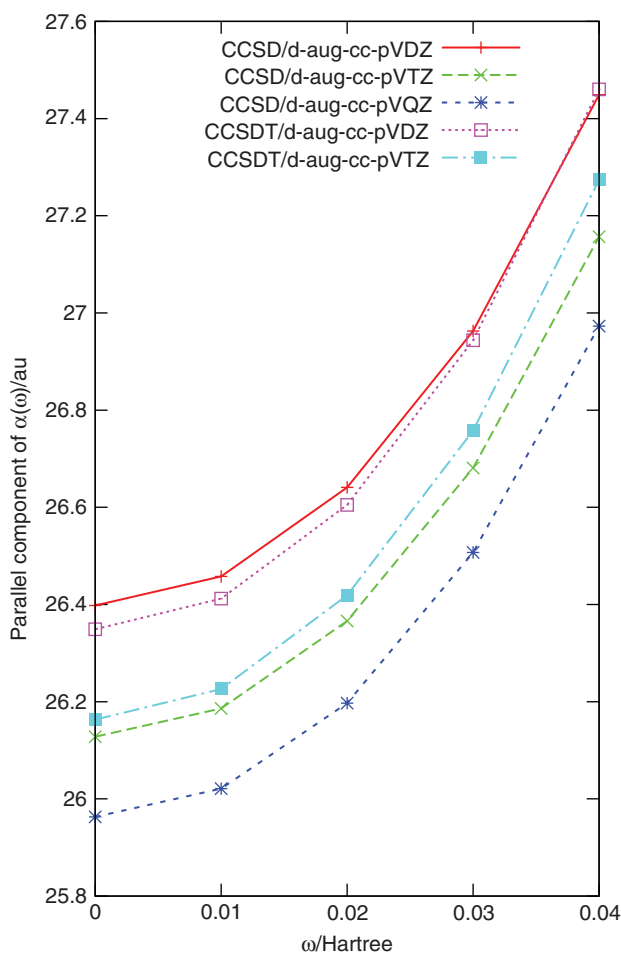


Fig. 5.6 (color online) Basis set dependence of the parallel component of dynamic polarizability of the CN molecule in various basis sets.

dynamics simulations have been implemented.^{73,103,119–122} Here we describe briefly two multiscale approaches that are applicable in different arenas: combined QM/MM formalism and the embedded cluster approach.

5.4.3.1 Combined QM/MM Formalism The main motivation behind the development of the multiscale QM/MM methodologies is to provide an accurate and efficient approach to describing chemical transformations in complex environments such as biological systems. The approach is defined by a hybrid Hamiltonian that consists of three components:

$$H = H_{\text{QM}} + H_{\text{QM/MM}} + H_{\text{MM}} \quad (5.27)$$

Here H_{QM} is a standard many-electron Hamiltonian which describes the internal energy of the quantum (QM) region, $H_{\text{QM/MM}}$ is the interaction between the QM region and its surroundings (MM), and H_{MM} describes the internal energy of the MM part. The H_{QM} and $H_{\text{QM/MM}}$ components can be represented as

$$H_{\text{QM}} = E_{\text{QM}}^{(0)} + \sum_{\mu, \nu} f_{\nu}^{\mu} X_{\nu}^{+} X_{\mu} + \frac{1}{4} \sum_{\mu, \nu, \lambda, \kappa} v_{\lambda \kappa}^{\mu \nu} X_{\lambda}^{+} X_{\kappa}^{+} X_{\nu} X_{\mu} \quad (5.28)$$

and

$$H_{\text{QM/MM}} = \sum_{i, \mu, \nu} \left\langle \mu \left| \frac{Q_i^m}{|\vec{R}_i^m - \vec{r}|} \right| \nu \right\rangle X_{\mu}^{+} X_{\nu} + V(\{R^m\}, \{R\}) \quad (5.29)$$

where the indices ν, μ, λ , and κ designate single-particle states, the elements f_{ν}^{μ} and $v_{\lambda \kappa}^{\mu \nu}$ represent one- and two-electron integrals, respectively, and the X_{λ}^{+} (X_{λ}) operators are the usual creation (annihilation) operators (see Section 5.2). In formula (5.29), Q_i^m and R_i^m denote charges and coordinates of the MM region. The $V(\{R^m\}, \{R\})$ term represents the interaction between nuclei in the MM and QM regions ($\{R\}$ represents symbolically the set of nuclear coordinates of the QM region). The Hamiltonian $\tilde{H}$, given by the expression

$$\tilde{H} = H_{\text{QM}} + H_{\text{QM/MM}} \quad (5.30)$$

effectively includes the interaction of the environment with the QM region and therefore was used throughout our QM/MM calculations. The modular structure of NWChem enables seamless integration between the generic QM/MM interface and all CC codes implemented with TCE.

One of the applications of the coupled-cluster base QM/MM approach has involved the mechanism of ultrafast conversion observed in DNA bases, with several scenarios of this process discussed in the literature.^{123–127} The proper inclusion of correlation effects and the native environment are very important in the comprehensive understanding of the remarkable photostability of DNA. This approach was first calibrated on the water molecule in aqueous solution. The 0.63 eV value calculated for the blue shift was in good agreement with original calculations by Kongsted et al.¹¹⁹ The blue shift was obtained as a result of thermal averaging of EOMCCSD excitation energies ω using a non-Boltzmann sampling scheme:

$$\langle \omega \rangle = \frac{\langle \omega e^{-\beta(E-E_0)} \rangle}{\langle e^{-\beta(E-E_0)} \rangle} \quad (5.31)$$

where dynamics of the entire system was driven by a lower level of theory (E_0) and where the resulting trajectories were then processed with higher-level theory (E) (see Refs. 128 and 129 for details).

TABLE 5.7 EOMCCSD/MM and CR-EOMCCSD(T)/MM Calculations for the Cytosine and Guanine Molecules^a

	EOMCCSD		CR-EOMCCSD(T)	
	$^1\pi\pi^*$	$^1n\pi^*$	$^1\pi\pi^*$	$^1n\pi^*$
Cytosine				
ω_{gas}	5.02	5.44	4.76	5.24
ω_{DNA} (t.a.)	5.27	6.01	5.01	5.79
Guanine				
ω_{gas}	5.31	5.61	5.03	5.47
ω_{DNA}	5.33	6.00	5.05	5.70
ω_{water}	5.38	5.99	5.10	5.84

^a All excitation energies given in eV. The time-averaged (t.a.) values were calculated only for cytosine molecule. Otherwise, the excitation energies were obtained for optimized ground-state geometries (see Refs. 73 and 75).

The formalism above was used to calculate the thermal average of the $\pi\pi^*$ and $n\pi^*$ transitions of cytosine (see Table 5.7). Our model consisted of the 12-mer fragment of B-DNA (3'-TCGCGTTGCGCT-5') solvated in a rectangular box ($51 \times 51 \times 69$ Å) of SPC/E water. To neutralize the charge, 22 sodium ions (or counterions) were also added to the system, resulting in a total of 18,060 atoms. After initial optimization the system was brought to equilibrium by warming up in stages (50-K increments) over the course of 60 ps of classical molecular dynamics simulation. The excitation energies of quantum representation of cytosine base capped with a hydrogen link atom in the field of the entire DNA–water complex (18,048 point charges) were calculated every 0.5 ps in the context of combined CR-EOMCCSD(T)/MM approach. In the CR-EOMCCSD(T) calculations we used the cc-pVDZ basis set, and all core electrons were kept frozen. Our results demonstrate the impact of the fluctuating environment. After thermal averaging based on formula (5.31) we obtained excitation energies of 5.01 and 5.79 eV for $^1n\pi^*$ and $^1\pi\pi^*$ transitions, respectively. These values should be contrasted with the gas-phase (carried out for the ground-state optimized geometry on the B3LYP level) calculations for these states, which are equal to 4.76 and 5.24 eV, respectively. Therefore, a significant “blue shift” of 0.25 and 0.54 eV was detected for these states in our simulations. Effectively, one can observe that the environment “stabilizes” the lowest excited states ($^1\pi\pi^*$ transition) by increasing the gap between excited states corresponding to the $^1\pi\pi^*$ and $n\pi^*$ transitions. The increasing role of doubly excited configurations in thermal probing of the $n\pi^*$ transition is also worth mentioning. For this transition, use of a high-level method seems to play a key role in obtaining reliable results. Although the results obtained for guanine were performed without thermal averaging, the main conclusions obtained for cytosine still holds. In particular, in DNA settings (ω_{DNA}) as well as in aqueous solution (ω_{water}), we observe a significant widening of the gap between $^1\pi\pi^*$ and $^1n\pi^*$ transitions.

5.4.3.2 Embedded Cluster Model Surface excitons (or electron–hole pairs) are transitions corresponding to the lowest photon energy necessary for efficient surface electronic excitation. Compared with the bulk, electronic spectra displaying surface exciton band positions are rarely available because of experimental difficulties. The recently developed *surface electronic spectroscopy detected by atomic desorption* (SESDAD) *technique* seems to be the only surface-sensitive technique capable of detecting the signature of the surface exciton unambiguously.^{130,131} This method is unique in that it combines a theoretical model of desorption with experimental observations to interpret the results, thus making it an excellent playing field for theory and experiment. Despite substantial progress in the understanding of self-trapped excitons, the early-stage dynamics still poses a challenge for electronic structure methods. The primary difficulty lies in the fact that the relevant quasiparticle excitation undergoes a transition between a delocalized state and a localized one. In fact, this forms the underlying basis for the SESDAD technique. In brief, the surface excitation results in the formation of one-center self-trapped excitons. Unlike in the bulk, where the one-center self-trapped excitons are strongly localized and produce fast luminescence, the self-trapping process at the surface is accompanied by atom desorption. It has recently been demonstrated that photoexcitation of alkali halide surfaces with photons with specially tuned energies is accompanied by halide atom desorption with hyperthermal velocities. From a theory point of view, this self-trapping process can be explained within the adiabatic approximation in terms of a potential energy surface (PES) connecting the delocalized and localized states. Accurate calculations of the excited-state PES can yield valuable information about desorption dynamics. Embedding models are very well suited for this purpose since they mediate between the extended and cluster representation of the system.¹³² The embedding approach we have used involves embedding a quantum cluster of atoms in an environment of appropriate potentials and point charges at the lattice sites of the host environment. The atoms closest to the quantum cluster are treated using a shell-model representation, while the far field is represented using point charges or a rigid-ion representation. Since all embedding schemes are based on a systematic partitioning of the total system, the total energy of the system has to be clearly defined with respect to various subsystems: quantum (qm), environment (env), and their interactions (int). Formally, this can be written as

$$E_{\text{total}}[\rho](Z, R, q, \tau) = E_{\text{qm}}[\rho](Z, R) + E_{\text{env}}(q, \tau) + E_{\text{int}}[\rho](Z, R, q, \tau) \quad (5.32)$$

where ρ is the electronic charge density, R and Z are the coordinates and charges of the nuclei in the quantum region, and τ and q are the coordinates and charges of the particles in the host system.¹³² The first application of combined embedded approach and high-level CC theories was related to the surface excitations of the crystalline potassium bromide (KBr) system.⁷⁴ Our model for the KBr system consists of a finite nanocube of $24 \times 24 \times 24$ ions which is divided into regions, as shown in Fig. 5.7. The quantum cluster of atoms is embedded in an

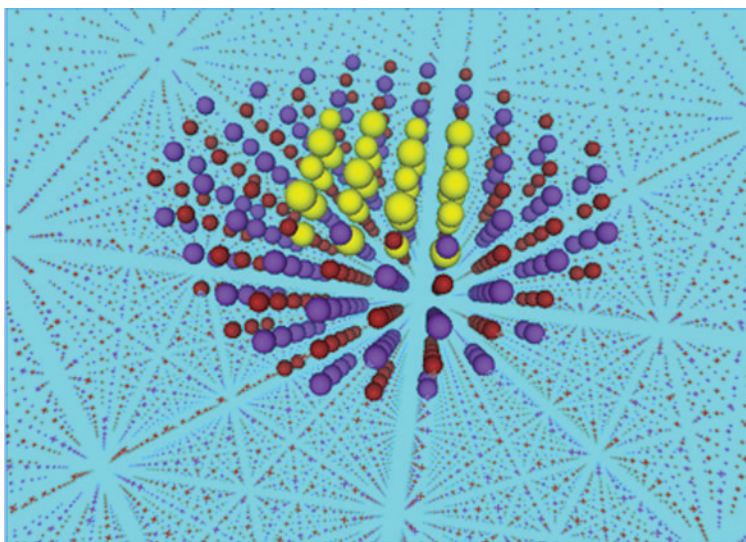


Fig. 5.7 (color online) Schematic representation of the surface KnBr_n cluster.

array of polarizable ions which is further subdivided into two regions: 1 and 2, respectively. The atoms in region 1 are represented using the shell model and are allowed to relax and respond to changes in the electronic density within the quantum cluster. The point charges in region 2 are nonpolarizable and are kept fixed at the lattice sites of the surrounding environment. Since we have a finite system, the crystalline potential using this approach is determined up to a constant and does not affect relative energies. To ensure convergence we have modeled the (001) surface of the KBr system using a series of embedded quantum clusters of increasing size (K_4Br_4 , K_9Br_9 , $\text{K}_{16}\text{Br}_{16}$). The two-layered surface models were found to be sufficient to capture the lowest surface excitations.

The representative calculations for the K_nBr_n ($n = 4, 9, 16$) are shown in Table 5.8. Due to the large number of correlated electrons for $\text{K}_{16}\text{Br}_{16}$ (256 electrons), the calculations were performed for smaller number of orbitals, including

TABLE 5.8 EOMCC Excitation Energies (in eV) for selected KnBr_n Surface Clusters^a

Cluster	EOMCCSD	CR-EOMCCSD(T)	EOMCCSDt
K_4Br_4	7.05	7.10	7.05 ^b
K_9Br_9	6.68	6.73	
$\text{K}_{16}\text{Br}_{16}$	6.55 ^c		

^aSee Ref. 74. The LANL2DZ ECP basis¹³⁴ was used in the calculations.

^bThe active space used in the EOMCCSDt calculations contained two HOMO and two LUMO orbitals.

^cOnly orbitals with corresponding orbital energies below 1.2 Ha were correlated in calculations.

only those with corresponding energies below 1.2 Ha. To characterize excited-state correlation effects corresponding to different levels of approximation, several methods were used, including EOMCCSD, CR-EOMCCSD(T), and the active-space approach EOMCCSDt (version II of Ref. 133). It is interesting to observe that in all cases when the inclusion of triply excited clusters in either noniterative or iterative fashion is possible, the discrepancies between EOMCCSD and CR-EOMCCSD(T)/EOMCCSDt excitation energies are negligible. For example, the difference between EOMCCSD and CR-EOMCCSD(T) excitation energies is as small as 0.05 eV, which is strong evidence in favor of the high quality of EOMCCSD results. This is also a consequence of the singly excited character of the lowest excited state, which is dominated almost entirely by HOMO–LUMO excitation, where the HOMO orbital is composed of p-orbitals localized on Br atoms, whereas the LUMO orbital corresponds to a mixture of s- and p-orbitals on K atoms. The EOMCCSD result of 6.55 eV for $\text{K}_{16}\text{Br}_{16}$ is in good agreement with the experimentally inferred value of 6.35 ± 0.1 eV (an even more accurate estimate can be obtained when the EOMCCSD results are extrapolated to infinite size of the cluster using the formula $\omega = ae^{-bn} + c$, which leads to 6.38 eV). In future work we aim to extend this study to other excited-state aspects of ionic systems and other insulating materials. It is conceivable that we can also study the small-system analog of the collective plasmon excitations typically seen in metal nanoparticles.

5.5 CONCLUSIONS

The work described here gives a general outline of current capabilities of TCE-based CC implementation in NWChem. Our main aim was to show that certain classes of applications, which require accuracies provided by the CC methodology, can greatly capitalize on highly scalable codes capable of harnessing the power of existing computer architectures. It is interesting to stress that the use of a thousand or several thousand CPUs will become routine in the coming few years. Several examples, such as the CR-EOMCCSD(T) calculations performed for the P_2TA system on 6144 CPUs, provide an indication that we will be able to meet this challenge. We have clearly showed that the LR-CCSD or EOMCCSD models based on the manifold of single and double excitations can tackle the problems characterized by molecular sizes of nearly 100 atoms (84 atoms for the P_2TA system). This may significantly enhance our understanding of many processes related to systems being on the crossroad between nano and molecular science. Noniterative approaches are by their nature highly scalable and, as shown on the example of P_2TA , can provide very accurate estimates for experimentally inferred excitation energies. We plan to pursue new theoretical strategies that provide a better balance between particular correlation effects in excited states. Additionally, studies can provide the results necessary to calibrate lower-order methods. In particular, we seek to establish strong ties between the EOMCC methodology and various DFT approaches broadly used in excited-state and property calculations. Another important issue concerns use of the CC

formalism in a much broader context defined by multiple components, such as the QM/MM and embedded cluster methodologies. When coupled with these hybrid approaches, the CC formalism can be applied to a wide range of problems in the presence of an environment.

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REFERENCES

1. Bartlett, R. J.; Musial, M. *Rev. Mod. Phys.* **2007**, *79*, 291–352.
2. Čížek, J. *J. Chem. Phys.* **1966**, *45*, 4256–4266.
3. Paldus, J.; Shavitt, I.; Čížek, J. *Phys. Rev. A* **1972**, *5*, 50–67.
4. Coester, F. *Nucl. Phys.* **1958**, *7*, 421–424.
5. Coester, F.; Kummel, H. *Nucl. Phys.* **1960**, *17*, 477–485.
6. Heisenberg, J. H.; Mihaila, B. *Phys. Rev. C* **1999**, *59*, 1440–1448.
7. Dean, D. J.; Hjorth-Jensen, M. *Phys. Rev. C* **2004**, *69*, 14.
8. Kowalski, K.; Dean, D. J.; Hjorth-Jensen, M.; Papenbrock, T.; Piecuch, P. *Phys. Rev. Lett.* **2004**, *92*, 4.
9. Stanton, J. F.; J. G., Watts, J. D.; Szalay, P. G.; Bartlett, R. J.; with contributions from Auer, A. A.; Bernholdt, D. E.; Christiansen, O.; Harding, M. E.; Heckert, M.; Heun, O.; Huber, C.; Jonsson, D.; Jusélius, J.; Lauderdale, W. J.; Metzroth, T.; Michauk, C.; O'Neill, D. P.; Price, D. R.; Ruud, K.; Schiffmann, F.; Varner, M. E.; Vázquez, and the integral packages *MOLECULE*: Almlöf, J.; Taylor, P. R.; *PROPS*: Taylor, P. R.; and *ABACUS*: Helgaker, T.; Jensen, H. J.; Jørgensen, P.; Olsen, J. For the current version, see <http://www.aces2.de/>.
10. Lotrich, V.; Flocke, N.; Ponton, M.; Yau, A. D.; Perera, A.; Deumens, E.; Bartlett, R. J. *J. Chem. Phys.* **2008**, *128*, 194104.
11. Kus, T.; Lotrich, V. F.; Bartlett, R. J. *J. Chem. Phys.* **2009**, *130*, 124122.
12. Janowski, T.; Pulay, P. *Chem. Phys. Lett.* **2007**, *447*, 27–32.

13. Baker, J.; Wolinski, K.; Malagoli, M.; Kinghorn, D.; Wolinski, P.; Magyarfalvi, G.; Saebo, S.; Janowski, T.; Pulay, P. *J. Comput. Chem.* **2009**, *30*, 317–335.
14. Janowski, T.; Ford, A. R.; Pulay, P. *J. Chem. Theory Comput.* **2007**, *3*, 1368–1377.
15. Janowski, T.; Pulay, P. *J. Chem. Theory Comput.* **2008**, *4*, 1585–1592.
16. Werner, H.-J.; Knowles, P. J.; Lindh, R.; Manby, F. R.; Schütz, M.; Celani, P.; Korona, T.; Mitrushenkov, A.; Rauhut, G.; Adler, T. B.; Amos, R. D.; Bernhardsson, A.; Berning, A.; Cooper, D. L.; Deegan, M. J. O.; Dobbyn, A. J.; Eckert, F.; Goll, E.; Hampel, C.; Hetzer, G.; Hrenar, T.; Knizia, G.; Köppl, C.; Liu, Y.; Lloyd, A. W.; Mata, R. A.; May, A. J.; McNicholas, S. J.; Meyer, W.; Mura, M. E.; Nicklaß, A.; Palmieri, P.; Pflüger, K.; Pitzer, R.; Reiher, M.; Schumann, U.; Stoll, H.; Stone, A. J.; Tarroni, R.; Thorsteinsson, T.; Wang, M.; Wolf, A. MOLPRO, Version 2010.1, a package of ab initio programs.
17. Schmidt, M. W.; Baldridge, K. K.; Boatz, J. A.; Elbert, S. T.; Gordon, M. S.; Jensen, J. H.; Koseki, S.; Matsunaga, N.; Nguyen, K. A.; Su, S. J.; Windus, T. L.; Dupuis, M.; Montgomery, J. A. *J. Comput. Chem.* **1993**, *14*, 1347–1363.
18. Olson, R. M.; Bentz, J. L.; Kendall, R. A.; Schmidt, M. W.; Gordon, M. S. *J. Chem. Theory Comput.* **2007**, *3*, 1312–1328.
19. Bentz, J. L.; Olson, R. M.; Gordon, M. S.; Schmidt, M. W.; Kendall, R. A. *Comput. Phys. Commun.* **2007**, *176*, 589–600.
20. Bylaska, E. J.; de Jong, W. A.; Govind, N.; Kowalski, K.; Straatsma, T. P.; Valiev, M.; Wang, D.; Apra, E.; Windus, T. L.; Hammond, J.; Nichols, P.; Hirata, S.; Hackler, M. T.; Zhao, Y.; Fan, P.-D.; Harrison, R. J.; Dupuis, M.; Smith, D. M. A.; Nieplocha, J.; Tipparaju, V.; Krishnan, M.; Wu, Q.; Van Voorhis, T.; Auer, A. A.; Nooijen, M.; Brown, E.; Cisneros, G.; Fann, G. I.; Fruchtl, H.; Garza, J.; Hirao, K.; Kendall, R.; Nichols, J. A.; Tsemekhman, K.; Wolinski, K.; Anchell, J.; Bernholdt, D.; Borowski, P.; Clark, T.; Clerc, D.; Dachsel, H.; Deegan, M.; Dyall, K.; Elwood, D.; Glendening, E.; Gutowski, M.; Hess, A.; Jaffe, J.; Johnson, B.; Ju, J.; Kobayashi, R.; Kutteh, R.; Lin, Z.; Littlefield, R.; Long, X.; Meng, B.; Nakajima, T.; Niu, S.; Pollack, L.; Rosing, M.; Sandrone, G.; Stave, M.; Taylor, H.; Thomas, G.; van Lenthe, J.; Wong, A.; Zhang, Z. Pacific Northwest National Laboratory, Richland, WA, 2007.
21. Pollack, L.; Windus, T. L.; de Jong, W. A.; Dixon, D. A. *J. Phys. Chem. A* **2005**, *109*, 6934–6938.
22. Fan, P. D.; Valiev, M.; Kowalski, K. *Chem. Phys. Lett.* **2008**, *458*, 205–209.
23. Kowalski, K.; Hammond, J. R.; de Jong, W. A.; Sadlej, A. J. *J. Chem. Phys.* **2008**, *129*, 226101.
24. Hammond, J. R.; Kowalski, K.; de Jong, W. A. *J. Chem. Phys.* **2007**, *127*, 144105.
25. Paldus, J.; Li, X. Z. In *Advances in Chemical Physics*, Vol. 110, Wiley, New York, 1999, pp. 1–175.
26. Crawford, T. D.; Schaefer, H. F. In *Reviews in Computational Chemistry*, Vol. 14, Wiley-VCH, New York, 2000, pp. 33–136.
27. Piecuch, P.; Kowalski, K.; Pimienta, I. S. O.; McGuire, M. J. *Int. Rev. Phys. Chem.* **2002**, *21*, 527–655.
28. Purvis, G. D.; Bartlett, R. J. *J. Chem. Phys.* **1982**, *76*, 1910–1918.
29. Cullen, J. M.; Zerner, M. C. *J. Chem. Phys.* **1982**, *77*, 4088–4109.
30. Noga, J.; Bartlett, R. J. *J. Chem. Phys.* **1987**, *86*, 7041–7050.

31. Scuseria, G. E.; Schaefer, H. F. *Chem. Phys. Lett.* **1988**, *152*, 382–386.
32. Oliphant, N.; Adamowicz, L. *J. Chem. Phys.* **1991**, *95*, 6645–6651.
33. Kucharski, S. A.; Bartlett, R. J. *J. Chem. Phys.* **1992**, *97*, 4282–4288.
34. Pulay, P. *Chem. Phys. Lett.* **1980**, *73*, 393–398.
35. Raghavachari, K.; Trucks, G. W.; Pople, J. A.; Head-Gordon, M. *Chem. Phys. Lett.* **1989**, *157*, 479–483.
36. Geertsens, J.; Rittby, M.; Bartlett, R. J. *Chem. Phys. Lett.* **1989**, *164*, 57–62.
37. Comeau, D. C.; Bartlett, R. J. *Chem. Phys. Lett.* **1993**, *207*, 414–423.
38. Stanton, J. F.; Bartlett, R. J. *J. Chem. Phys.* **1993**, *98*, 7029–7039.
39. Sekino, H.; Bartlett, R. J. *Int. J. Quantum Chem.* **1984**, 255–265.
40. Kowalski, K.; Piecuch, P. *J. Chem. Phys.* **2001**, *115*, 643–651.
41. Kowalski, K.; Piecuch, P. *Chem. Phys. Lett.* **2001**, *347*, 237–246.
42. Kucharski, S. A.; Wloch, M.; Musial, M.; Bartlett, R. J. *J. Chem. Phys.* **2001**, *115*, 8263–8266.
43. Hirata, S. *J. Chem. Phys.* **2004**, *121*, 51–59.
44. Koch, H.; Jensen, H. J. A.; Jørgensen, P.; Helgaker, T. *J. Chem. Phys.* **1990**, *93*, 3345–3350.
45. Watts, J. D.; Bartlett, R. J. *Chem. Phys. Lett.* **1995**, *233*, 81–87.
46. Watts, J. D.; Bartlett, R. J. *Chem. Phys. Lett.* **1996**, *258*, 581–588.
47. Christiansen, O.; Koch, H.; Jørgensen, F. *J. Chem. Phys.* **1996**, *105*, 1451–1459.
48. Hirata, S.; Nooijen, M.; Grabowski, I.; Bartlett, R. J. *J. Chem. Phys.* **2001**, *114*, 3919–3928.
49. Shiozaki, T.; Hirao, K.; Hirata, S. *J. Chem. Phys.* **2007**, *126*, 224106.
50. Manohar, P. U.; Krylov, A. I. *J. Chem. Phys.* **2008**, *129*, 194105.
51. Kowalski, K.; Piecuch, P. *J. Chem. Phys.* **2001**, *115*, 2966–2978.
52. Kowalski, K.; Piecuch, P. *J. Chem. Phys.* **2004**, *120*, 1715–1738.
53. Kowalski, K.; Valiev, M. *Int. J. Quantum Chem.* **2008**, *108*, 2178–2190.
54. Kowalski, K. *J. Chem. Phys.* **2009**, *130*, 194110.
55. Monkhorst, H. J. *Int. J. Quantum Chem.* **1977**, 421–432.
56. Dalgaard, E.; Monkhorst, H. J. *Phys. Rev. A* **1983**, *28*, 1217–1222.
57. Koch, H.; Jørgensen, P. *J. Chem. Phys.* **1990**, *93*, 3333–3344.
58. Christiansen, O.; Jørgensen, P.; Hattig, C. *Int. J. Quantum Chem.* **1998**, *68*, 1–52.
59. Sekino, H.; Bartlett, R. J. In *Advances in Quantum Chemistry*, Vol. 35, Academic Press, San Diego, CA, 1999, pp. 149–173.
60. Rozyczko, P. B.; Perera, S. A.; Nooijen, M.; Bartlett, R. J. *J. Chem. Phys.* **1997**, *107*, 6736–6747.
61. Russ, N. J.; Crawford, T. D. *Chem. Phys. Lett.* **2004**, *400*, 104–111.
62. Kondo, A. E.; Piecuch, P.; Paldus, J. *J. Chem. Phys.* **1995**, *102*, 6511–6524.
63. Piecuch, P.; Paldus, J. *J. Math. Chem.* **1997**, *21*, 51–70.
64. Kondo, A. E.; Piecuch, P.; Paldus, J. *J. Chem. Phys.* **1996**, *104*, 8566–8585.
65. Christiansen, O.; Gauss, J.; Stanton, J. F. *Chem. Phys. Lett.* **1998**, *292*, 437–446.
66. Gauss, J.; Christiansen, O.; Stanton, J. F. *Chem. Phys. Lett.* **1998**, *296*, 117–124.
67. Kállay, M.; Gauss, J. *J. Mol. Struct. (Theochem)* **2006**, *768*, 71–77.

68. O'Neill, D. P.; Kállay, M.; Gauss, J. *J. Chem. Phys.* **2007**, *127*, 134109.
69. Hammond, J. R.; de Jong, W. A.; Kowalski, K. *J. Chem. Phys.* **2008**, *128*, 224102.
70. Hirata, S. *J. Phys. Chem. A* **2003**, *107*, 9887–9897.
71. Hirata, S. *Theor. Chem. Acc.* **2006**, *116*, 2–17.
72. Koch, H.; de Meras, A. S.; Pedersen, T. B. *J. Chem. Phys.* **2003**, *118*, 9481–9484.
73. Valiev, M.; Kowalski, K. *J. Chem. Phys.* **2006**, *125*, 211101.
74. Govind, N.; Sushko, P. V.; Hess, W. P.; Valiev, M.; Kowalski, K. *Chem. Phys. Lett.* **2009**, *470*, 353–357.
75. Kowalski, K.; Valiev, M. *Res. Lett. Phys. Chem.* **2007**, ID85978.
76. Hariharan P. C.; Pople, J. A. *Theor. Chim. Acta* **1973**, *28*, 213–222.
77. Krishnan, R.; Binkley, J. S.; Seeger, R.; Pople, J. A. *J. Chem. Phys.* **1980**, *72*, 650–654.
78. Francl, M. M.; Pietro, W. J.; Hehre, W. J.; Binkley, J. S.; Gordon, M. S.; Defrees, D. J.; Pople, J. A. *J. Chem. Phys.* **1982**, *77*, 3654–3665.
79. Clark, T.; Chandrasekhar, J.; Spitznagel, G. W.; Schleyer, P. V. *J. Comput. Chem.* **1983**, *4*, 294–301.
80. Gill, P. M. W.; Johnson, B. G.; Pople, J. A.; Frisch, M. J. *Chem. Phys. Lett.* **1992**, *197*, 499–505.
81. Dunning, T. H. *J. Chem. Phys.* **1989**, *90*, 1007–1023.
82. Kendall, R. A.; Dunning, T. H.; Harrison, R. J. *J. Chem. Phys.* **1992**, *96*, 6796–6806.
83. Woon, D. E.; Dunning, T. H. *J. Chem. Phys.* **1994**, *100*, 2975–2988.
84. Sendt, K.; Johnston, L. A.; Hough, W. A.; Crossley, M. J.; Hush, N. S.; Reimers, J. R. *J. Am. Chem. Soc.* **2002**, *124*, 9299–9309.
85. Nakatsuji, H. *Chem. Phys. Lett.* **1978**, *59*, 362–364.
86. Nakatsuji, H. *Chem. Phys. Lett.* **1979**, *67*, 329–333.
87. Miyahara, T.; Nakatsuji, H.; Hasegawa, J.; Osuka, A.; Aratani, N.; Tsuda, A. *J. Chem. Phys.* **2002**, *117*, 11196–11207.
88. Tokita, Y.; Hasegawa, J.; Nakatsuji, H. *J. Phys. Chem. A* **1998**, *102*, 1843–1849.
89. Nooijen, M. *J. Chem. Phys.* **1996**, *104*, 2638–2651.
90. Nooijen, M.; Bartlett, R. J. *J. Chem. Phys.* **1997**, *106*, 6441–6448.
91. Nooijen, M.; Bartlett, R. J. *J. Chem. Phys.* **1997**, *106*, 6449–6455.
92. Gwaltney, S. R.; Bartlett, R. J. *J. Chem. Phys.* **1998**, *108*, 6790–6798.
93. Andersson, K.; Malmqvist, P. A.; Roos, B. O.; Sadlej, A. J.; Wolinski, K. *J. Phys. Chem.* **1990**, *94*, 5483–5488.
94. Serrano-Andres, L.; Merchan, M.; Rubio, M.; Roos, B. O. *Chem. Phys. Lett.* **1998**, *295*, 195–203.
95. Becke, A. D. *J. Chem. Phys.* **1993**, *98*, 5648–5652.
96. Lee, C. T.; Yang, W. T.; Parr, R. G. *Phys. Rev. B* **1988**, *37*, 785–789.
97. Yanai, T.; Tew, D. P.; Handy, N. C. *Chem. Phys. Lett.* **2004**, *393*, 51–57.
98. Peach, M. J. G.; Helgaker, T.; Salek, P.; Keal, T. W.; Lutnaes, O. B.; Tozer, D. J.; Handy, N. C. *Phys. Chem. Chem. Phys.* **2006**, *8*, 558–562.
99. Jensen, L.; Govind, N. In preparation, *J. Phys. Chem. A* **2009**, *113*, 9761–9765.

100. Govind, N.; Valiev, M.; Jensen, L.; Kowalski, K. *J. Phys. Chem. A* **2009**, *113*, 6041–6043.
101. Aratani, N.; Osuka, A.; Kim, Y. H.; Jeong, D. H.; Kim, D. *Angew. Chem. Int. Ed.* **2000**, *39*, 1458.
102. Schreiber, M.; Silva, M. R.; Sauer, S. P. A.; Thiel, W. *J. Chem. Phys.* **2008**, *128*, 134110.
103. Rusu, C. F.; Lanig, H.; Othersen, O. G.; Krysch, C.; Clark, T. *J. Phys. Chem. B* **2008**, *112*, 2445–2455.
104. Antoine, R.; Dugourd, P.; Rayane, D.; Benichou, E.; Broyer, M.; Chandezon, F.; Guet, C. *J. Chem. Phys.* **1999**, *110*, 9771–9772.
105. Ballard, A.; Bonin, K.; Louderback, J. *J. Chem. Phys.* **2000**, *113*, 5732–5735.
106. Ruud, K.; Jonsson, D.; Taylor, P. R. *J. Chem. Phys.* **2001**, *114*, 4331–4332.
107. Pedersen, T. B.; de Meras, A.; Koch, H. *J. Chem. Phys.* **2004**, *120*, 8887–8897.
108. Christiansen, O.; Koch, H.; Jørgensen, P. *Chem. Phys. Lett.* **1995**, *243*, 409–418.
109. Benkova, Z.; Sadlej, A. J.; Oakes, R. E.; Bell, S. E. *J. Theor. Chem. Acc.* **2005**, *113*, 238–247.
110. Baranowska, A.; Siedlecka, M.; Sadlej, A. J. *Theor. Chem. Acc.* **2007**, *118*, 959–972.
111. Baranowska, A.; Sadlej, A. J. *Chem. Phys. Lett.* **2004**, *398*, 270–275.
112. Sadlej, A. J. *Collect. Czech. Chem. Commun.* **1988**, *53*, 1995–2016.
113. Champagne, B.; Perpète, É. A.; van Gisbergen, S. J. A.; Baerends, E. J.; Snijders, J. G.; Soubra-Ghaoui, C.; Robins, K. A.; Kirtman, B. *J. Chem. Phys.* **1998**, *109*, 10489–10498.
114. Champagne, B.; Perpète, É. A.; van Gisbergen, S. J. A.; Baerends, E. J.; Snijders, J. G.; Soubra-Ghaoui, C.; Robins, K. A.; Kirtman, B. *J. Chem. Phys.* **1999**, *110*, 11664–11664.
115. van Gisbergen, S. J. A.; Schipper, P. R. T.; Gritsenko, O. V.; Baerends, E. J.; Snijders, J. G.; Champagne, B.; Kirtman, B. *Phys. Rev. Lett.* **1999**, *83*, 694–697.
116. Champagne, B.; Perpète, É. A.; Jacquemin, D.; van Gisbergen, S. J. A.; Baerends, E. J.; Soubra-Ghaoui, C.; Robins, K. A.; Kirtman, B. *J. Phys. Chem. A* **2000**, *104*, 4755–4763.
117. Hinchcliffe, A.; Machado, H. J. S. *Int. J. Mol. Sci.* **2000**, *1*.
118. Sinicropi, A.; Andruniów, T.; De Vico, L.; Ferré, N.; Olivucci, M. *Int. Union Pure Appl. Chem.* **2005**, *77*, 977–993.
119. Kongsted, J.; Osted, A.; Mikkelsen, K. V.; Christiansen, O. *J. Chem. Phys.* **2003**, *118*, 1620–1633.
120. Kongsted, J.; Osted, A.; Mikkelsen, K. V.; Christiansen, O. *J. Chem. Phys.* **2003**, *119*, 10519–10535.
121. Osted, A.; Kongsted, J.; Mikkelsen, K. V.; Astrand, P. O.; Christiansen, O. *J. Chem. Phys.* **2006**, *124*, 16.
122. Xu, Z. R.; Matsika, S. *J. Phys. Chem. A* **2006**, *110*, 12035–12043.
123. Sobolewski, A. L.; Domcke, W. *Phys. Chem. Chem. Phys.* **2004**, *6*, 2763–2771.
124. Ismail, N.; Blancafort, L.; Olivucci, M.; Kohler, B.; Robb, M. A. *J. Am. Chem. Soc.* **2002**, *124*, 6818–6819.

125. Zgierski, M. Z.; Patchkovskii, S.; Fujiwara, T.; Lim, E. C. *J. Phys. Chem. A* **2005**, *109*, 9384–9387.
126. Merchan, M.; Serrano-Andrés, L. *J. Am. Chem. Soc.* **2003**, *125*, 8108–8109.
127. Matsika, S. *J. Phys. Chem. A* **2004**, *108*, 7584–7590.
128. Torrie, G. M.; Valleau, J. P. *Chem. Phys. Lett.* **1974**, *28*, 578–581.
129. Wood, R. H.; Liu, W. B.; Doren, D. J. *J. Phys. Chem. A* **2002**, *106*, 6689–6693.
130. Hess, W. P.; Joly, A. G.; Gerrity, D. P.; Beck, K. M.; Sushko, P. V.; Shluger, A. L. *J. Chem. Phys.* **2001**, *115*, 9463–9472.
131. Hess, W. P.; Joly, A. G.; Beck, K. M.; Henyk, M.; Sushko, P. V.; Trevisanutto, P. E.; Shluger, A. L. *J. Phys. Chem. B* **2005**, *109*, 19563–19578.
132. Huang, P.; Carter, E. A. *Annu. Rev. Phys. Chem.* **2008**, *59*, 261–290.
133. Kowalski, K.; Hirata, S.; Wloch, M.; Piecuch, P.; Windus, T. L. *J. Chem. Phys.* **2005**, *123*, 074319.
134. Hay, P. J.; Wadt, W. R. *J. Chem. Phys.* **1985**, *82*, 299–310.

6 Strongly Correlated Electrons: Renormalized Band Structure Theory and Quantum Chemical Methods

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This chapter provides means of dealing with materials in which the electrons are strongly correlated, affecting and controlling properties such as conductivity, magnetism, and high-temperature superconductivity. Quantitative measures are described to allow systems with strong electron correlations to be identified and hence treated properly. Conceptual approaches, powered by renormalized band structure theory (e.g., LDA plus phase-shift corrections) or wavefunction-based electronic structure theory (e.g., CASSCF and MRCI embedded clusters), are developed and shown to reproduce key system observables.

6.1 INTRODUCTION

The physics of strongly correlated electrons has become one of the major fields in solid-state science. To a large extent this is due to the upsurge the field has obtained after the discovery of high-temperature superconductivity by Bednorz and Müller.¹ In the superconducting cuprate perovskites, electron correlations are strong to the extent that the parent compounds are correlation-induced insulators (i.e., Mott–Hubbard systems).^{2,3} But even before the discovery of high-temperature superconductivity, metals with heavy quasiparticles (heavy-fermion systems, as they are often called) had caught the attention of condensed matter

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physicists.^{4,5} This raises the question: When do we speak of strongly correlated systems, and when do we consider the electronic system to be weakly correlated? For example, is graphene or are the Fe pnictides, which are presently much discussed due to their superconducting properties,⁶ weakly or strongly correlated electronic materials?

One way of answering this question is by comparing the size of the Coulomb repulsion between electrons at a given site with the hybridization energy (i.e., with the energy gain of an electron by delocalizing). Surely those two energies work against each other. Delocalization of electrons due to hybridization matrix elements increases the fluctuation of the electron number n_i at a given site i (i.e., $\langle (n_i - \langle n_i \rangle)^2 \rangle$), since electrons hop on and off from it (charge fluctuations). Here the expectation value is stated with respect to the ground state of the system. An increase of the fluctuations leads to an increase of the Coulomb repulsion since the latter is minimized when the electrons are distributed over the lattice as uniformly as possible.

Thus, in the Hubbard model when we characterize the on-site Coulomb repulsion of two electrons by an energy U and the hybridization (interatomic interaction or resonance energy) by a “hopping” matrix element to a nearest-neighbor (NN) site t , the ratio $U/|t|$ enables us to distinguish between strong and weak correlations. When $U/|t| \gg 1$, the Coulomb repulsion and hence the tendency to minimize charge fluctuations are dominant, whereas for $U/|t| \ll 1$ the electrons move as freely as independent particles. In an alkali metal such as Na, the s-wave orbitals are relatively extended, implying that U is not particularly strong and the NN overlap is large. Therefore, $U/|t|$ is small and the electrons are weakly correlated only. At the other end of the scale are electrons in the 4f shell, which are close to the nuclei. The NN overlap is in this case small, and $U/|t| \gg 1$. Thus, these electrons are strongly correlated.

The reduction of charge fluctuations relative to those one would obtain in the Hartree–Fock self-consistent field (SCF) approximation may be used to define a measure for the strength of electron correlations in a system. Accordingly, one may classify, for example, different bonds in molecules or solids with respect to their correlation strength. Electrons in incompletely filled 4f shells of rare-earth ions are the strongest correlated valence electrons in solids. Except for Ce^{3+} with a $4f^1$ configuration and Yb^{3+} with $4f^{13}$, fluctuations in the f -electron number at a site are practically zero, implying that the 4f electrons remain highly localized. Note that we leave aside here valence-fluctuating ions such as Sm and Eu. Then the 4f shell is characterized by its total angular momentum J , which can be calculated by applying Hund’s rules. The $(2J + 1)$ -fold degeneracy of the ground-state J multiplet is split in the crystalline electric field set up by the neighboring ions. Typical splitting energies are of the order of few meV and therefore much less than a typical Fermi energy. They define a low-energy scale caused by correlations. This scale will prevail even when the 4f electrons become somewhat delocalized, as in the case of Ce^{3+} or Yb^{3+} . New low-energy scales are an earmark of strong correlation effects. They lead to a large density of low-energy excitations that show up, for example, in a large linear specific-heat coefficient. The latter

constitute another earmark of strong correlations. The enhanced specific heat and a correspondingly enhanced spin susceptibility let these systems appear like ordinary Fermi liquids but with strongly renormalized quasiparticle masses. Since at low temperatures de Haas–van Alphen experiments can be carried out, the Fermi surface and the strongly anisotropic mass enhancement can be determined. The theory has to provide a reliable framework to calculate these quantities and to predict trends for different compounds. (For a recent review, see, e.g., Ref. 7.)

Currently, the common approach to the computation of electronic structures is density functional theory (DFT) and its various approximations: for example, the local density approximation (LDA). DFT and the LDA have led to unprecedented progress in computational solid-state physics and to surprisingly good results for numerous compounds. However, it is also known and well documented^{8–12} that DFT has problems with describing strongly correlated electron systems which usually involve 3d or 4f elements. As mentioned above, the fundamental point underlying the physics of these compounds is the interplay between electron localization effects as a result of strong repulsive interactions and itinerancy as a result of translational symmetry and intersite orbital overlap. Standard band theories based on the DFT model mainly emphasize the latter aspect. To cope with the former, the LDA results were imported into many-body schemes based on dynamic mean-field theory (DMFT).⁸ In the LDA + DMFT approach, the local correlations missed in LDA are described by effective interactions such as the Hubbard U , interorbital Coulomb repulsion U' , and on-site exchange J . Even so, the correlation treatment is limited to on-site effects. To deal with intersite correlations, the DMFT scheme has been generalized to cluster DMFT.^{9,10} Equivalent projection operator methods¹³ based on coherent potential approximations have also been devised. However, computation of the Coulomb repulsion parameters U , U' (and intersite V) is particularly difficult. Use of constrained LDA to estimate these couplings entails a large degree of uncertainty, while their self-consistent estimation within LDA + DMFT is highly problematic.^{11,12}

A semiempirical approach, yet one with high predictive power for Fermi surfaces and strongly anisotropic heavy-quasiparticle masses, is the renormalized band structure theory approach, which is described in the next section. It has been exploited by Zwicky and co-workers,^{14–16} who could build on earlier formulations.^{17,18} Renormalized band structure calculations are based on a properly modified density functional approach. The strongly correlated f or d electrons are described by parameterized energy-dependent phase shifts, which usually contain one adjustable parameter, while for the remaining weakly correlated electrons the LDA is used. A rather different approach to strongly correlated electron systems is based on wavefunction methods as they are often applied to molecular compounds. Wavefunction-based quantum chemistry is known to provide a rigorous theoretical framework¹⁹ for addressing the correlation problem. With the quantum chemical approach, many-body wavefunctions can be explicitly constructed at levels of increasing sophistication and accuracy. The standard quantum chemical methods¹⁹ thus offer a systematic

path to converged results. Nevertheless, rigorous schemes for *periodic systems* have been developed only recently. For weakly correlated electron systems they rely on second-order perturbation theory,²⁰ extensions to Green's function techniques,²¹ or coupled-cluster theory²² and are mostly restricted to closed-shell systems and single-reference wavefunctions. Our discussion, however, is concerned primarily with open-shell transition metal solid-state compounds for which a multiconfiguration (and multireference) treatment¹⁹ is mandatory.

The present contribution is structured as follows. First we discuss a measure of the strength of electron correlations. After that we discuss in detail two methods which are particularly suitable for strongly correlated electrons: the renormalized band structure approach and the quantum chemical wavefunction-based approach. Chapter 10 provides a thorough review of empirical approaches such as the Hubbard model, emphasizing the basic nature of key properties such as metal–insulator transitions and superconductivity. An alternative approach to the *ab initio* study of strongly correlated electrons, the quantum Monte Carlo method, is described in Chapter 4.

6.2 MEASURE OF THE STRENGTH OF ELECTRON CORRELATIONS

The simplest case for studying the strength of electron correlations between different sites (i.e., interatomic correlations) is a H_2 molecule when the distance between the two sites is considered as a parameter. In order to exclude correlations between electrons when they are occupying the same site (i.e., intraatomic correlations), let us allow for one fixed orbital per site only. Then for small distances the molecular orbital (MO) or Hückel theory is a good approximation. The corresponding wavefunction $|\Psi_{MO}(\mathbf{r}_1, \mathbf{r}_2)\rangle$, where $\mathbf{r}_1$ and $\mathbf{r}_2$ denote the positions of the two electrons, describes the electrons as independent or uncorrelated. Consequently, charge fluctuations at a proton site are large (i.e., the ionic configurations have 50% weight).

On the other hand, when the distance between the atoms is large, the Heitler–London wavefunction $|\Psi_{HL}(\mathbf{r}_1, \mathbf{r}_2)\rangle$ is a good approximation to the ground state. In that limit the ionic configurations vanish, and therefore electronic charge fluctuations between the sites are completely suppressed. The wavefunction corresponds to the strong correlation limit. The energy gain due to electron hopping between the two sites decreases with increasing distance, while the on-site Coulomb repulsion remains finite in the atomic limit.

These two limiting cases suggest immediately introducing the reduction of charge fluctuations compared to the uncorrelated case, as a measure of the strength of interatomic correlations.²³ Let $|\Psi_0\rangle$ denote the exact ground state of the electrons and $|\Phi_{SCF}\rangle$ the state of uncorrelated electrons. The reduction of the normalized mean-square deviation of the electron number n_i at site i , with $i = 1, 2$, can be quantified by computing

$$\Sigma(i) = \frac{\langle \Phi_{SCF} | \delta n_i^2 | \Phi_{SCF} \rangle - \langle \Psi_0 | \delta n_i^2 | \Psi_0 \rangle}{\langle \Phi_{SCF} | \delta n_i^2 | \Phi_{SCF} \rangle} \quad (6.1)$$

where $\delta n_i^2 = n_i^2 - \bar{n}_i^2$. Thus, $\Sigma(i) = 0$ implies a MO-like ground state of independent electrons for $|\Psi_0\rangle$, while $\Sigma(i) = 1$ describes the Heitler–London limit of strong interatomic correlations.

One can also define a correlation strength for different bonds instead of atoms. In that case the denominator is modified when heteropolar bonds are considered. Then we must subtract from $\langle \Phi_{\text{SCF}} | \delta n_i^2 | \Phi_{\text{SCF}} \rangle$ a term δn_{pc}^2 . It takes into account the fact that some number fluctuations are required even when the electrons are perfectly correlated to ensure a heteropolar charge distribution within the bond. Let α_p denote the bond polarity, defined as the difference in the average occupation numbers of the two hybrid functions (half-bonds) 1 and 2, which form the heteropolar bond [i.e., $\bar{n}_{1(2)} = (1 \pm \alpha_p)$]. In that case, $\delta n_{\text{pc}}^2 = \alpha_p(1 - \alpha_p)$. These hybrid functions are, for example, of the form sp^3 , sp^2 , or other, and the considerations apply to a solid as well as to a molecule. How strong interatomic correlations in chemical bonds depends considerably on the bond length and only to a small extent on the chemical environment. The bond length of a C–C σ -bond, for example, is nearly the same in different molecules and in diamond. Therefore, we can determine $\Sigma(i)$ values for different bonds by looking at corresponding molecules. (See Ref. 23 for a detailed study of various molecular systems, including effects of the chemical environment; the original literature is also cited therein.) To give a few examples: For a C–C or N–N π -bond, one finds $\Sigma \approx 0.5$, while for C–C and N–N σ -bonds, Σ is 0.18 and 0.19, respectively. Thus, we can immediately conclude that the π -electrons in graphene are fairly correlated (i.e., they are just between the MO and Heitler–London limits).

It is of particular interest to know how strongly valence electrons are correlated in the CuO_2 planes of the high-temperature superconducting cuprates. Here one has to distinguish between a partial suppression of charge fluctuations on the Cu and O sites. By applying the considerations noted above, one finds that $\Sigma(\text{Cu}) \approx 0.8$ and $\Sigma(\text{O}) \approx 0.7$.²⁴ So, indeed, correlations are quite strong in cuprates. Nevertheless, they are still smaller than those of 4f electrons in a system such as CeAl_3 . There the 4f electron of Ce^{3+} is nearly localized such that $\Sigma(4f) \approx 1$.

When correlations are so strong that the electrons remain localized, like the 4f electrons in most rare-earth systems, we observe a separation of the spin and charge degrees of freedom. Spin degrees may lead to excitations in the form of magnons or crystal-field excitations, while the charge degrees of freedom are seen in photoemission experiments. In cuprates, the values of Σ are too small to expect spin-charge separation, except in the crudest approximation.

In addition to interatomic correlations, we have to consider the intraatomic correlations. A measure of the strength of the latter is more difficult to define. One way is to find out to what extent Hund's rule correlations are operative at a given site i . To this end one can define the degree of spin alignment,

$$S_i^2 = \langle \Psi_0 | \mathbf{S}^2(i) | \Psi_0 \rangle \quad (6.2)$$

where $\mathbf{S}(i) = \sum_v \mathbf{s}_v(i)$ and $\mathbf{s}_v(i)$ is the spin operator for orbital v . The quantity S_i^2 should be compared with the value corresponding to the case when the SCF ground-state wavefunction $|\Phi_{\text{SCF}}\rangle$ is used and with the situation in which the ground-state wavefunction $|\Psi_{\text{loc}}\rangle$ is taken in the limit of complete suppression of interatomic charge fluctuations (i.e., large interatomic distances). Therefore, we may define

$$\delta S_i^2 = \frac{\langle \Psi_0 | \mathbf{S}^2(i) | \Psi_0 \rangle - \langle \Phi_{\text{SCF}} | \mathbf{S}^2(i) | \Phi_{\text{SCF}} \rangle}{\langle \Psi_{\text{loc}} | \mathbf{S}^2(i) | \Psi_{\text{loc}} \rangle - \langle \Phi_{\text{SCF}} | \mathbf{S}^2(i) | \Phi_{\text{SCF}} \rangle} \quad (6.3)$$

for a measure of the strength of intraatomic correlations, with $0 \leq \delta S_i^2 \leq 1$. For elemental transition metals such as Fe, Co, and Ni, δS_i^2 is approximately 0.5. These estimates are based on five-band model Hamiltonians with parameter values used in the literature.²⁵ Corresponding numbers for charge fluctuations are given in Ref. 26. This shows that the much discussed transition metals are just in the middle between the limits of uncorrelated and strongly correlated electrons. Hund's rule correlations are important, but the relatively large overlap of the atomic wavefunctions on neighboring sites makes them less effective.

6.3 RENORMALIZED BAND STRUCTURE THEORY

Systems with heavy-quasiparticle excitations have strongly anisotropic masses which can be determined together with the Fermi surface in de Haas–van Alphen experiments. The energy dispersion near the Fermi surface is given by

$$E(\mathbf{k}) = \mathbf{v}_F(\hat{k})(\mathbf{k} - \mathbf{k}_F) \quad (6.4)$$

where $\mathbf{k}_F$ describes the Fermi surface and $\mathbf{v}_F(\hat{k})$ is the Fermi velocity in direction $\hat{k}$ of the Fermi surface. The effective potential in which the quasiparticles move can be described completely by a set of energy-dependent phase shifts $\eta_l^A(\epsilon)$, where A denotes the atoms within a unit cell and l is the orbital momentum quantum number at site A . The phase shifts contain all necessary information about the effective potential. The partial electronic densities are given by

$$n_l^A = \frac{2}{\pi} (2l + 1) \eta_l^A(\epsilon_F) \quad (6.5)$$

Knowing all n_l^A is equivalent to having the Fermi surface. The Fermi velocity follows from the derivatives

$$\dot{\eta}_l^A(\epsilon_F) = \left(\frac{d\eta_l^A(\epsilon)}{d\epsilon} \right)_{\epsilon=\epsilon_F} \quad (6.6)$$

which also yield the effective quasiparticle masses. Luttinger's theorem requires that the volume enclosed by the Fermi surface must equal half of the number of

valence electrons. Thus Luttinger's theorem reduces the number of independent $\eta_l^A(\epsilon_F)$ by one. When we deal with heavy quasiparticles involving 4f electrons such as in the Ce intermetallic compounds, the f electrons must be included in the counting. The main idea of renormalized band theory is to use for the $\eta_l^A(\epsilon_F)$ the electrons obtained from LDA band structure calculations except for the phase shifts of the strongly correlated electrons. For Ce compounds, which we consider next, these are the phase shifts of the 4f electrons. Those phase shifts would come out incorrectly in the LDA treatment because of the shortcomings of this approximation for strongly correlated electrons. Proper f-electron phase shifts are determined as follows. Ce^{3+} has a $4f^1$ configuration, and the lowest J multiplet is, according to Hund's rules, $J = \frac{5}{2}$, with J denoting the total angular momentum. This multiplet is split by the crystalline electric field (CEF) of the surrounding of the Ce ions. Usually, the ground state is a Kramers doublet characterized by a pseudospin $\tau = 1, 2$. Thus, of the phase shifts $\eta_{l=3}^{\text{Ce}}(\epsilon)$, only $\eta_{\tau}^{\text{Ce}}(\epsilon)$ contributes. We make for it the two-parameter ansatz

$$\eta_{\tau}^{\text{Ce}}(\epsilon) = \eta_{\tau}^{\text{Ce}}(\epsilon_F) + \frac{1}{k_B T^*}(\epsilon - \epsilon_F) \quad (6.7)$$

where $k_B T^*$ is a characteristic low energy usually called *coherent Kondo temperature*. Instead of Eq. (6.7), we can make as well the ansatz

$$\eta_{\tau}^{\text{Ce}}(\epsilon) = \arctan \frac{\tilde{\Gamma}}{\tilde{\epsilon} - \epsilon} \quad (6.8)$$

which again has two parameters in the form of a resonance (Kondo or Abrikosov–Suhl resonance). One of the two parameters [i.e., $\eta_{\tau}^{\text{Ce}}(\epsilon_F)$] is fixed by the requirement of Luttinger's theorem. The remaining parameter is, for example, adjusted to the low-temperature γ coefficient in the specific heat $C(T) = \gamma T$. (See, e.g., Ref. 15 for more details.) The results obtained have generally been very good. As an example we present in Table 6.1 a comparison

TABLE 6.1 Comparison of de Haas–van Alphen Data for $\text{CeRu}_2\text{Si}_2^{27}$ with Theoretical Results^a

CeRu_2Si_2		Experiment		LDA		RB	
Orbit	Field	Area (MG)	Mass ratio	Area (MG)	Mass ratio	Area (MG)	Mass ratio
α	(110)	4.7	12.3	≈ 10	≈ 1.5	≈ 10	≥ 10
ϵ	(110)	25.0	19.7	23	1.2	20	≈ 20
δ	(001)	12.2	4.0	24	1.5	26	2.1
ψ	(100)	53.6	120.0	70		≈ 62	≈ 100

Source: Ref. 14.

^aShown are some of the extremal areas of the Fermi surface (megagauss) and the effective mass ratio m^*/m_0 . Unlike the LDA, the renormalized band theory (RB) reproduces well the large experimental mass anisotropies.

between experiments²⁷ and theory¹⁵ for CeRu_2Si_2 . Similar good results were obtained for CeCu_2Si_2 , UPt_3 , and UPd_2Al_3 , to name a few. The considerations noted above hold for sufficiently low temperatures. When T becomes of order T^* , Eq. (6.7) no longer holds. The small hybridization of the 4f electrons leading to a large coefficient $(k_B T^*)^{-1}$ then becomes ineffective because of temperature fluctuations, and the 4f electrons must be treated as being localized. The volume enclosed by the Fermi surface must shrink correspondingly since the 4f electrons no longer take part in the valence electron counting. Thus, we expect a transition from a large to a small Fermi surface with increasing temperature.^{28,29} This has indeed been observed for CeRu_2Si_2 . While de Haas–van Alphen measurements performed around 4 K give a large Fermi surface, photoemission experiments at $T = 25$ K yield band structure results which resemble those for LaRu_2Si_2 (i.e., without the signature of the 4f electrons).³⁰ The characteristic Kondo temperature is 15 K in this case.

6.4 QUANTUM CHEMICAL METHODS

Two issues must be addressed carefully in the theoretical investigation of 3d- and 4f-metal compounds: (1) a rigorous treatment of the ubiquitous strong short-range correlations, and (2) a realistic representation of the crystalline environment. Real-space wavefunction-based quantum chemistry provides the right framework to cope with the former issue. The momentum ($\mathbf{k}$)-dependent quasiparticle bands, for example, can be constructed at a later stage after accounting for the short-range correlations. The strategy is to use a finite atomic cluster $\mathcal{C}$, cut out from the infinite solid and large enough for describing the crucial short-range correlations. The presence of partially filled d-electron shells requires a multiconfiguration representation of the many-electron wavefunction. This is achieved in quantum chemistry with the complete-active-space (CAS) self-consistent field (i.e., CASSCF) method, which we discuss below.

Regarding the second point mentioned above, the modeling of the crystalline environment, there are few different ways of describing the surroundings of the region where the many-body correlation treatment is carried out. The simplest approximation relies on the fully ionic model. In such a picture, the remaining part of the crystal is represented as an array of point charges at the lattice positions, with formal valence states for each species. In La_2CuO_4 , for example, one of the parent compounds of the cuprate superconductors, the formal valence states for La, Cu, and O are 3+, 2+, and 2−, respectively. In more advanced schemes, the crystalline environment is modeled as an effective potential obtained on the basis of prior periodic Hartree–Fock (HF) or DFT calculations. The HF-based approach is reviewed in the following section, while DFT embeddings are described elsewhere.^{31–34}

6.4.1 Cluster-in-Solid Approach and Ab Initio Embedding

The main point when applying quantum chemical methods to correlation calculations is to make use of the local character of the correlation hole surrounding

an electron. The latter optimizes the Coulomb repulsion between electrons, and its accurate description is the essence of the correlation treatment. To a good approximation, in the ground state of a system, the correlation hole extends over a few lattice sites. A finite cluster is therefore sufficient for its description. The cluster must, however, be properly embedded to avoid spurious charge polarization effects. For electron-removal (oxidation) and electron-addition (reduction) states the correlation hole is no longer short-ranged, due to the long-range polarization cloud of the added electron or hole. When comparing energies of different electron-removal (or electron-addition) states, however, the long-range polarization cloud drops out. Thus, it is again the short-range part that is responsible for these energy differences. If the dielectric constant of the system is known, it is also possible to treat the long-range polarization contribution to the correlation energy by a continuum approximation.²³ This allows for calculating ionization energies, for example, although the accuracy for such quantities does not match the accuracy for ground-state properties.

An important step in the *ab initio* computation of correlated electronic structures in solids was recently achieved at the Max-Planck-Institute for the Physics of Complex Systems in Dresden. In this scheme,^{35–37} the crystalline environment is modeled as an effective one-electron potential which is extracted from prior HF calculations for the periodic lattice. The HF calculations are carried out with the program package CRYSTAL,³⁸ and the embedding potential is obtained on the basis of the CRYSTAL output data with computational tools developed in our laboratory. The methodology has been applied to insulating materials with closed-shell ground states such as MgO³⁶ and BN.³⁷ Excellent results were obtained for the fundamental gaps in these compounds and the widths of the quasiparticle dispersive bands. To extend the applicability of this embedding technique to open-shell systems, two different paths at different levels of approximation can be followed. In a simplified approach, the transition metal ions in the surroundings of the finite “quantum” cluster can be modeled as closed-shell ions. If the cluster for which the many-body calculations are carried out is large enough, a closed-shell HF representation of the remaining part of the crystal is acceptable for understanding the ubiquitous strong short-range correlations. Embeddings constructed on the basis of periodic closed-shell HF calculations were used in previous studies on LaCoO₃³⁹ and LiFeAs.⁴⁰ The best approach, however, is to represent the crystalline environment at the restricted open-shell HF (ROHF) level. Work to extend the cluster-in-solid embedding technique³⁵ to the ROHF case is under way in Dresden. In the following paragraphs, we illustrate the cluster-in-solid embedding scheme by the example of the cobalt oxide perovskite LaCoO₃.

LaCoO₃ has attracted considerable attention due to a number of puzzling phase transitions induced by changes in temperature,^{41,42} doping,^{42,43} and/or strain.^{44,45} Up to now, most of the experimental work has been aimed at understanding the nature of the phase transitions in the undoped compound, from a nonmagnetic insulator at low temperatures to a paramagnetic semiconductor above 90 K and to a metal for $T > 500$ K. The low-temperature ground state

was assigned to a closed-shell $t_{2g}^6 e_g^0$ configuration [low-spin (LS), $S = 0$] of the Co ions.⁴¹ For temperatures in the vicinity of 90 K, however, the available experimental results are rather contradictory. While recent x-ray absorption spectroscopy⁴⁶ and inelastic neutron scattering⁴⁷ measurements indicate with increasing T a gradual transition into an $S = 2$ ($t_{2g}^4 e_g^2$) high-spin (HS) configuration of the Co 3d electrons, the observation of Co–O bond-length alternation⁴⁸ suggests the formation of a $S = 1$ ($t_{2g}^5 e_g^1$) intermediate-spin (IS) state for $T > 90$ K, susceptible to Jahn–Teller distortions. On the theoretical side, the transition to an HS state around 90 K is favored by many-body model-Hamiltonian calculations.⁴⁶ The IS electron configuration is supported by LDA + U calculations.^{49,50}

The periodic HF calculations yield for LaCoO_3 a t_{2g}^6 electron configuration at the Co sites and a finite gap at the Fermi level,³⁹ in agreement with the interpretation of the low-temperature experimental data.^{41,42} An interesting finding is that the Co t_{2g} levels are located below the O 2p bands (i.e., the HF gap is defined by the O 2p and Co e_g bands). This indicates that ligand-to-metal charge-transfer effects are important. A detailed analysis requires, however, quantum chemical methods beyond the HF approximation.

With the cluster-in-solid embedding technique,³⁵ the cluster $\mathcal{C}$ where the correlated quantum chemical calculations are carried out is divided into an “active” region $\mathcal{C}_A$ and a “buffer” domain $\mathcal{C}_B$ (i.e., $\mathcal{C} = \mathcal{C}_A + \mathcal{C}_B$). Orbitals centered at sites within $\mathcal{C}_A$ are explicitly correlated in the post-HF calculations, while the orbitals centered at atomic sites in $\mathcal{C}_B$ are held frozen. The role of the atomic sites in $\mathcal{C}_B$ is to ensure an accurate description of the “tails” of the orbitals centered within the active region $\mathcal{C}_A$.

The orbital set associated with the finite cluster $\mathcal{C}$ is generated from the HF crystal Wannier orbitals (WOs). The latter are obtained via a Wannier–Boys localization scheme⁵¹ for both the occupied bands, up to Co t_{2g} and O 2p, and the lowest conduction-band HF states (i.e., Co e_g). The original WOs $|w_n(\mathbf{r})\rangle$ are next projected onto the set of Gaussian atomic basis functions¹⁹ attached to the sites in $\mathcal{C}$:

$$|w'_n(\mathbf{r})\rangle = \sum_{\beta, \alpha \in \mathcal{C}} |\beta\rangle S_{\beta\alpha}^{-1} \langle\alpha|w_n(\mathbf{r})\rangle \quad (6.9)$$

where α and β are Gaussian basis functions centered within the region $\mathcal{C}$, $S_{\beta\alpha}^{-1}$ represents the inverse overlap matrix for the basis set attached to $\mathcal{C}$, and the WO $|w_n(\mathbf{r})\rangle$ is centered in the unit cell corresponding to the position vector $\mathbf{r}$. The projection procedure is confined to HF WOs centered within $\mathcal{C}$. The projected Wannier orbitals $|w'_n(\mathbf{r})\rangle$ are neither normalized nor orthogonal since the small, longer-range tails involving atomic sites outside the fragment $\mathcal{C}$ were cut off. Therefore, they are groupwise orthonormalized in the following order: active core, active valence, active low-lying conduction-band orbitals, buffer core, buffer valence, and buffer low-lying conduction-band orbitals. This set of orthonormal orbitals is denoted as $|w''_n(\mathbf{r})\rangle$.

For sufficiently large active and buffer regions, the original WOs at the active sites of $\mathcal{C}$ remain practically unaffected by the projection. In LaCoO_3 , for example, with a buffer region $\mathcal{C}_B$ consisting of the nearest-neighbor CoO_6 octahedra and nearest-neighbor La sites around $\mathcal{C}_A$, the norms of the projected Co 3d and O 2p orbitals are not smaller than 98.9% of the original crystal WOs. Whereas a high-quality description can easily be achieved for the orbitals centered in the active region, the representation of the WOs in the buffer zone is less accurate. The impact of this deficiency on the correlation calculations, however, is fully compensated by an appropriate choice of the embedding potential. The embedding potential V_{emb} is constructed as

$$V_{\text{emb}} = F_{\text{crys}}^0 - F_{\mathcal{C}}^0 = F_{\text{crys}}^0 - (F_A^0 + F_B^0) \quad (6.10)$$

where F_{crys}^0 is the self-consistent Fock operator from the periodic HF calculation and $F_{\mathcal{C}}^0 = F_A^0 + F_B^0$ is the Fock operator associated with the density operator

$$P^{\mathcal{C}} = 2 \sum_{\nu}^{\mathcal{C}, \text{occ}} |w_{\nu}''\rangle \langle w_{\nu}''| \quad (6.11)$$

arising from all occupied orbitals $|w_{\nu}''\rangle$ in $\mathcal{C}_A$ and $\mathcal{C}_B$. For the post-HF cluster calculations, we can define an effective Fock operator $F_{\text{corr}} = F_{\mathcal{C}} + V_{\text{emb}}$, where V_{emb} is given above and $F_{\mathcal{C}} = F_A + F_B$. We can then write

$$F_{\text{corr}} = F_A + F_B + F_{\text{crys}}^0 - F_A^0 - F_B^0 \quad (6.12)$$

Since within $\mathcal{C}_B$ all electrons are still described at the HF level and all orbitals are kept frozen, $F_B = F_B^0$ and

$$F_{\text{corr}} = F_A + F_{\text{crys}}^0 - F_A^0 \quad (6.13)$$

The expression of F_A for the case of CASSCF calculations is discussed in Section 6.4.2. It is now clear from Eq. (6.13) that the correlation calculations are effectively performed in an *infinite* frozen HF environment.

For the construction of the full variational space to be used in subsequent correlation calculations, one can apply the prescription of Saebø and Pulay.⁵² Virtual projected atomic orbitals (PAOs) are generated in this scheme from the Gaussian basis functions associated with the cluster $\mathcal{C}$ by projecting out the occupied and low-lying conduction-band orbitals $|w_n''(\mathbf{r})\rangle$. Thereafter, the PAOs are Löwdin orthonormalized among themselves (see also the discussion elsewhere).^{35–37}

6.4.2 CASSCF: Multireference Configuration interaction

The electronic wavefunction must often comprise several electron configurations in order to be even qualitatively correct. The natural extension of the single-configuration HF approximation to systems where strong configurational mixing

occurs is the multiconfiguration SCF (MCSCF) method. The most common approach is currently the CASSCF scheme, where the user selects a set of chemically important “active” orbitals and the multiconfiguration wavefunction is constructed as a full configuration interaction (CI) within that active space.¹⁹ The MCSCF wavefunction can be written as a linear combination of Slater determinants $|i\rangle$ or configuration state functions (CSFs) $|m\rangle$,

$$|\Psi\rangle = \sum_i C_i |i\rangle = \sum_m C_m |m\rangle \quad (6.14)$$

The CSFs $|m\rangle$ are spin (and symmetry)-adapted combinations of Slater determinants, that is, eigenfunctions of the operators for the projected and total spins. In turn, the Slater determinants are constructed from a set of real and orthonormal spin orbitals $\{\phi_p(\mathbf{r}, \sigma)\}$, where $\mathbf{r}$ and σ are the spatial and spin coordinates, respectively:

$$|m\rangle = \det |\phi_1(\mathbf{r}_1, \sigma_1) \cdots \phi_N(\mathbf{r}_N, \sigma_N)| \quad (6.15)$$

In determining the wavefunction $|\Psi\rangle$, the orbitals are variationally optimized simultaneously with the coefficients of the CSFs, which makes the MCSCF method quite flexible. A detailed discussion of the practical aspects of the optimization of the MCSCF wavefunction can be found in Ref. 19. With respect to the form of the CASSCF “Fock operator” in Eq. (6.13), this can be written¹⁹

$$F_A = \sum_{pq} f_{pq} E_{pq} = \frac{1}{2} \sum_{pq} \sum_{\sigma} \langle \Psi | \left[a_{q\sigma}^{\dagger}, \left[a_{p\sigma}, \hat{H} \right] \right]_+ | \Psi \rangle E_{pq} \quad (6.16)$$

Here $\hat{H}$ is the Hamiltonian operator, $(a_{p\alpha}^{\dagger}, a_{q\alpha})$ are pairs of spin-orbital creation and annihilation operators, and $E_{pq} = a_{p\alpha}^{\dagger} a_{q\alpha} + a_{p\beta}^{\dagger} a_{q\beta}$ is a singlet excitation operator.

CASSCF calculations for the Co oxide compound LaCoO_3 provide new insight into the correlated multiorbital physics underlying the spin-state transition at about 90 K. To account for both on-site and metal–ligand intersite correlation effects, the active region $\mathcal{C}_A$ should consist of one CoO_6 octahedron. For an accurate description of the tails of the Co 3d and O 2p Wannier orbitals in $\mathcal{C}_A$, the buffer $\mathcal{C}_B$ incorporates the six NN CoO_6 octahedra and the eight NN La ions. The basis sets employed for these calculations are described in Ref. 39.

In a first step, a minimal active space can be constructed with the five 3d orbitals at the “central” Co^{3+} site and six electrons. Intriguingly, for this minimal active space, which we refer to as CAS-5, CASSCF predicts a HS $t_{2g}^4 e_g^2$ (${}^5T_{2g}$) ground-state configuration. The LS state, $t_{2g}^6 e_g^0$ (1A_g), is 1.26 eV higher in energy. The IS state, $t_{2g}^5 e_g^1$ (${}^3T_{1g}$), is 0.21 eV above the LS state (see Fig. 6.1). Clearly, the minimal active space is insufficient for this system. Inclusion of O 2p functions in the multiconfiguration treatment appears to be essential.

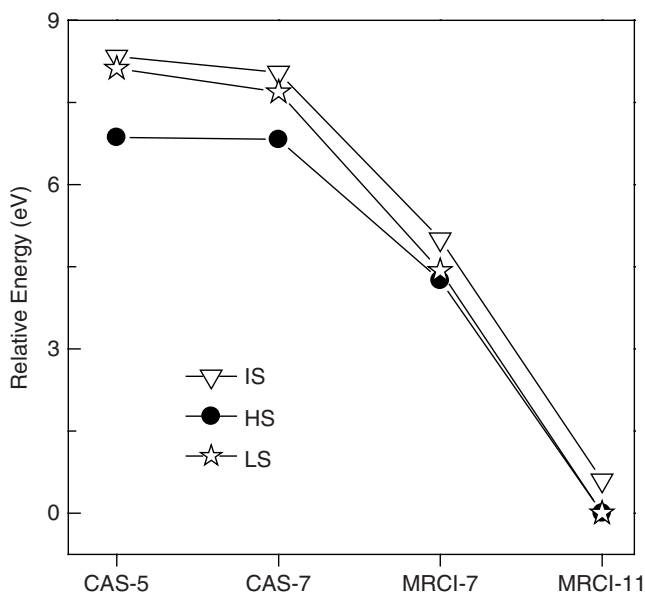


Fig. 6.1 Energy diagram for the lowest N -particle states in LaCoO_3 . The MRCI treatment is based on a CAS-7 reference (see the text). Only the Co 3d and O 2p- e_g electrons are correlated in the MRCI-7 calculations. For MRCI-11, the Co 3s and 3p electrons are also included. The reference is the energy of the MRCI-11 LS state.

The two σ -like e_g combinations of 2p orbitals at the NN O sites have the largest overlap with the Co 3d functions. When these two combinations of e_g symmetry are added to the active space, which we denote as the CAS-7 active orbital space, the CASSCF HS-LS splitting is reduced from 1.26 eV to 0.87 eV. This is related to the strong interaction between the O 2p- e_g to Co 3d- e_g charge-transfer (CT) configurations and the “non-CT” $t_{2g}^6 e_g^0$ CSFs in the LS CASSCF wavefunction. However, the HS state is still the lowest (see Fig. 6.1). Other types of charge fluctuations are taken into account by multireference CI (MRCI) calculations¹⁹; we account first for all single and double excitations from the Co 3d and O 2p- e_g levels into the virtual orbitals. The reference is the CAS-7 wavefunction. The HS-LS splitting is now reduced by an additional 0.70 to 0.17 eV, with the HS state remaining lower in energy. A switch in the energy order of these two states is only obtained when accounting for correlation effects (single and double excitations in the MRCI treatment) that involve the semicore Co 3s and 3p electrons. These correlations are related mainly to the coupling between the O 2p- e_g to Co 3d- e_g CT, on-site Co 3s, 3p $\rightarrow$ 3d, 4s, 4p, and Co 3d $\rightarrow$ 4s, 4p excitations. For such MRCI wavefunctions, which are referred to as MRCI-11, the LS state is found indeed to be the lowest, and the first N -particle excited state is the HS state, with an excitation energy of 6 meV (≈ 70 K, not visible in Fig. 6.1).

It is clear now that the quantum chemical investigation provides unique information about the importance of different types of correlations in this compound. It unravels features beyond models^{49,50} where various physical effects are wrapped up into generic U parameters. In particular, the data discussed above demonstrate that inclusion of the O 2p orbitals in the correlation treatment is crucial. In many-body techniques such as DMFT, the ligand p functions are usually excluded from the correlation treatment.

Although the quantum chemical electronic structure calculations were performed at zero temperature, the energy splittings computed between the different N -particle states allow us to draw a more complete picture for the spin-state transition at about 90 K. In particular, the MRCI results show that for an undistorted lattice the HS state is lower in energy as compared to the IS state. In the MRCI-11 calculation this energy difference is 0.60 eV, in contrast to LDA + U calculations^{49,50} which predict that the IS arrangement is lower in energy even when Jahn–Teller-type couplings are neglected. Whether this HS–IS splitting is sufficiently low to be compensated by the energy gain associated with a Jahn–Teller ordered configuration of e_g orbitals in the IS state, as suggested in Ref. 48, remains to be clarified in future work. An alternative scenario, which can reconcile the recent x-ray absorption⁴⁶ and crystallographic⁴⁸ measurements, is that Jahn–Teller effects also occur for the HS $t_{2g}^4 e_g^2$ configuration, with a long-range ordering of doubly occupied t_{2g} orbitals above 90 K.

6.4.3 Quasiparticle Bands in Layered Copper Oxides

The two-dimensional CuO_2 layers constitute the most important structural element of the cuprate high-temperature superconducting pseudoperovskites. Apical oxygen ions may supplement these planes by being either on one (e.g., as in $\text{YBa}_2\text{Cu}_3\text{O}_6$ and $\text{YBa}_2\text{Cu}_3\text{O}_7$) or on both sides (as in La_2CuO_4) of the copper–oxygen sheets. The planes themselves consist of corner-sharing CuO_4 plaquettes. The in-plane Cu–O bond lengths along the x and y axes are shorter than the apical bond lengths. That lifts the degeneracy between the Cu $3d_{3z^2-r^2}$ and $3d_{x^2-y^2}$ orbitals. The layers of CuO_6 octahedra or CuO_5 pyramids are separated from each other by additional layers of alkaline and/or rare-earth ions.

Electron correlations in the CuO_2 planes are strong, and without doping, as in the La_2CuO_4 compound, they are in an antiferromagnetic (AF) Mott–Hubbard insulating state. In that case there is one hole per CuO_4 unit and the hole is essentially in the Cu $3d_{x^2-y^2}$ orbital. The formal valence electron configuration is $\text{O}^{2-}2p^6$ and $\text{Cu}^{2+}t_{2g}^6d_{3z^2-r^2}^2d_{x^2-y^2}^1$.

Here we want to compute the energy dispersion of doped holes and of doped electrons as well. We consider two cases. In one case one hole is moving in the presence of AF long-range order. In the other case the hole is moving in a paramagnet with short-range AF correlations. At the beginning we want to make some general remarks. Let us assume that the hole (electron) doping is sufficiently large that the antiferromagnetism has been destroyed and the system is paramagnetic. Nonetheless, short-range AF correlations remain strong. This

implies that on a short spatial scale the system still looks antiferromagnetic, although long-range AF order is absent. In calculating the energy dispersion of, for example, doped holes, a quasiparticle approximation is made; that is, the doped hole, together with its modified surrounding (correlation hole), is moving in the form of a Bloch wave coherently through the CuO_2 plane. This poses a problem in some respects, for the following reasons. From quantum Monte Carlo studies of the one-band Hubbard model, which is considered to be the most simplified model for the description of holes in the CuO_2 plane, it is known that for a given momentum $\mathbf{k}$, several excitations are present. They include incoherent excitations due to internal degrees of freedom of the correlation hole as well as a coherent excitation.⁵³ For no or low doping, the spectral weight of the dominant low-energy excitations undergoes drastic changes along the symmetry axis $(0,0) - (\pi, \pi)$. Although it is large from $(0,0)$ to $(\pi/2, \pi/2)$, it nearly disappears for $\mathbf{k}$'s between $(\pi/2, \pi/2)$ and (π, π) . In the latter interval a pole at much higher energy (i.e., on the order of the on-site Coulomb repulsion U) dominates (the upper Hubbard band). In other words, the quasiparticle excitations are restricted approximately to a reduced Brillouin zone, as in an antiferromagnet. This goes hand in hand with the finding of spin-wave-like excitations when the spin response is calculated for that model. The spin response changes dramatically at higher doping. Here the quasiparticle dispersion extends throughout the paramagnetic Brillouin zone and the spin-wave-like excitations are gone. The coherence length of the AF spin fluctuations is on the order of the Cu–Cu distance, or smaller in this case.⁵³

Returning to our problem, we note that a vanishing weight of the quasiparticle pole in parts of the paramagnetic Brillouin zone, seen in Green's function or exact diagonalization approaches, is outside the range of the quasiparticle approximation. In the quasiparticle approximation the weight of the quasiparticle pole is taken to be unity.

In our study the quasiparticle approximation is combined with a CASSCF approach. Since the O 2p levels are formally filled, a minimal orbital active space that consists of one in-plane $d_{x^2-y^2}$ orbital at each Cu site would be sufficient for undoped cuprates. Such CAS wavefunctions are actually quite similar to the variational wavefunctions used in numerical studies of the two-dimensional one-band Hubbard model.⁵⁴ The main difference is that *all* integrals are computed in a totally ab initio way in quantum chemistry.

If extra holes are created, the active space must be enlarged with orbitals from the inactive orbital group. Each additional doped hole requires one orbital to be transferred from the inactive to the active space. For the lowest electron-removal state, for example, the orbital added to the active space turns into a Zhang–Rice (ZR) type of p-d composite⁵⁵ in the variational calculation,^{56,57} localized on a given CuO_4 plaquette and involving the four σ -like O 2p_x and 2p_y functions on that plaquette. However, in distinction from the original ZR model,⁵⁵ the quantum chemical CASSCF calculations⁵⁷ show that the Cu–O intersite charge fluctuations are important: the dominant contribution to the lowest electron-removal state is a superposition of three $S = 0$ configurations: $|d_{x^2-y^2}^1 p_\sigma^1\rangle$, $|d_{x^2-y^2}^2 p_\sigma^0\rangle$, and

$|d_{x^2-y^2}^0 p_\sigma^2\rangle$ (hole notation), where p_σ denotes the in-plane ZR combination of O $2p_x/2p_y$ orbitals on one CuO_4 plaquette. The weights of these configurations in the CASSCF wavefunction are 0.70, 0.14, and 0.11, respectively. In the Heitler–London limit (i.e., for infinitely large Coulomb integrals at the copper and oxygen sites), the weights of the $|d_{x^2-y^2}^2 p_\sigma^0\rangle$ and $|d_{x^2-y^2}^0 p_\sigma^2\rangle$ configurations would vanish since double hole occupation of a given site is excluded in that limit.

Moreover, there are contributions of a few percent to the multiconfiguration wavefunction which are related to inter-plaquette d–p–d excitations and responsible for the occurrence of short-range ferromagnetic (FM) d–d correlations around the O 2p hole. In a simplified three-site picture, Cu–O–Cu, these higher-energy CSFs responsible for stabilization of the d–d “FM configuration” $|d_{x^2-y^2}^\uparrow p_{x/y}^\downarrow d_{x^2-y^2}^\uparrow\rangle$ can be written as $|d_{x^2-y^2}^2 p_{x/y}^0 d_{x^2-y^2}^\uparrow\rangle$, $|d_{x^2-y^2}^\uparrow p_{x/y}^0 d_{x^2-y^2}^2\rangle$, $|d_{x^2-y^2}^0 p_{x/y}^2 d_{x^2-y^2}^\uparrow\rangle$, and $|d_{x^2-y^2}^\uparrow p_{x/y}^2 d_{x^2-y^2}^0\rangle$. When moving through the AF background, the O 2p hole must drag along this FM spin polarization cloud at NN Cu sites, which produces a strong renormalization of the effective hoppings. The 2p hole together with the spin polarization cloud at NN Cu sites is referred to as an FM *spin polaron* (see Fig. 6.2). With regard to the lowest electron-addition conduction-band states, these turn out to have Cu $3d^{10}$ character. In a first approximation, a CAS including only the $3d_{x^2-y^2}$ orbitals is in this case sufficient.

The next step is the computation of the effective hopping integrals. An effective “site” consists here of one CuO_4 plaquette. We distinguish between NN hopping t , second-NN hopping t' , and third-NN hopping t'' . These

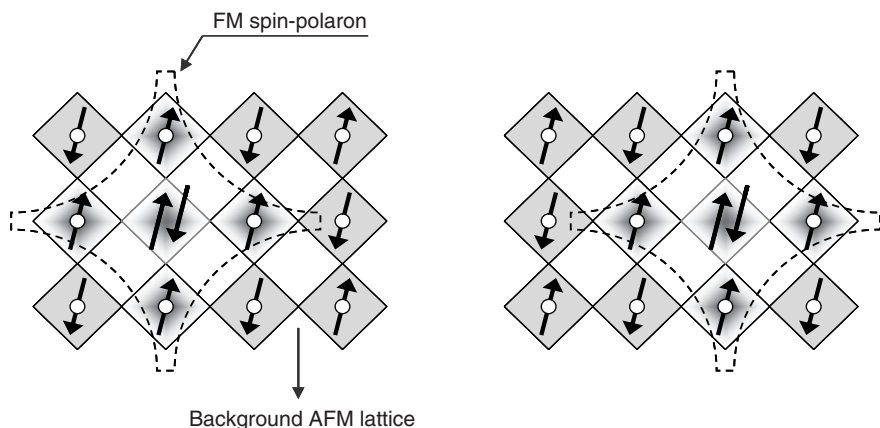


Fig. 6.2 FM correlations among Cu sites around a doped O hole as inferred from CASSCF, see text. The Zhang-Rice plaquette is drawn with thick gray lines. The uniform-gray plaquettes illustrate the antiferromagnetic background. The motion of the O hole is coherent when the FM spin polarization “cloud” at adjacent Cu sites moves solidarily with the hole.

quantities can be calculated by using the overlap, S_{ij} , and Hamiltonian, H_{ij} , matrix elements (MEs) between $(N \mp 1)$ -particle wavefunctions having the additional particle (hole or electron) located on different plaquettes ($i, j, \dots$) of a given cluster. To account for both charge *and* spin polarization and relaxation effects, each of these $(N \mp 1)$ wavefunctions, $|\Psi_i^{N \mp 1}\rangle$, is obtained by *separate* CASSCF optimizations. For degenerate (i.e., $H_{ii} = H_{jj}$) $(N \mp 1)$ states, $t \equiv (\epsilon_j - \epsilon_i)/2 = (H_{ij} - S_{ij}H_{ii})/(1 - S_{ij}^2)$, where ϵ_i and ϵ_j are the eigenvalues of the 2×2 secular problem. For nondegenerate ($H_{ii} \neq H_{jj}$) states, $t \equiv 1/2[(\epsilon_j - \epsilon_i)^2 - (H_{jj} - H_{ii})^2/(1 - S_{ij}^2)]^{1/2}$. The S_{ij} and H_{ij} terms are computed using the state-interaction (SI) method.⁵⁸ The basis sets employed for these quantum chemical calculations are described in Ref. 57. To localize the ZR-like hole on a given plaquette, prior CASSCF calculations are performed having the in-plane Cu–O bonds on that particular plaquette shortened by about 6%. The orbitals associated with this distorted geometry are used next as a starting guess for a CASSCF calculation with no distortions. In this new CASSCF calculation, the hole remains localized on the same plaquette for which the Cu–O bonds were shortened initially by 6%. A similar procedure is followed for the electron-addition, Cu d^{10} states; the Cu–O bonds on a given plaquette are first elongated by a few percent and the CASSCF orbitals associated with that distorted $(N + 1)$ configuration are used as starting guess for a new CASSCF calculation with no distortions. In the latter CASSCF calculation the extra electron remains localized on the same plaquette. We note that for the calculations discussed in this section, simple point-charge embeddings were employed (see the Section 6.2 and Ref. 57).

When the hole is moving in an antiferromagnet with long-range order, the NN hopping matrix element is renormalized to zero because of the mismatch of the spin configuration in the AF surroundings (see Fig. 6.2). For the case of a paramagnet, the short-range magnetic correlations can be accounted for by constructing clusters where few extra CuO_4 units are added around those plaquettes directly involved in the hopping process (see Fig. 6.3). The size of the cluster determines the AF correlation length. The NN hopping, in particular, depends on the size of the cluster because the mismatch of a hole with its AF surroundings in the initial and final states increases with correlation length (i.e., cluster size).

“Unrenormalized” (or bare) hoppings can also be computed, by imposing a FM arrangement of spins at the nearby Cu sites (i.e., a FM lattice). The bare hoppings are substantially smaller for the $(N + 1)$ Cu d^{10} states (see Table 6.2), because the Cu 3d functions are more compact than the O 2p orbitals. However, the renormalization effects are stronger for the O 2p hole carriers, due to the FM spin polaron physics, which is absent for the doped electrons (d^{10} quasiparticles).

One can divide the CuO_2 plane into sublattices A and B. The hopping matrix elements t' and t'' take us from a given plaquette of a sublattice to another one of the same sublattice. The overall AF spin background is not modified when the doped particle, hole or electron, hops and the renormalized hopping matrix elements t' and t'' computed on clusters such as those sketched in Fig. 6.3 are reliable even in the presence of long-range AF correlations. They enter the

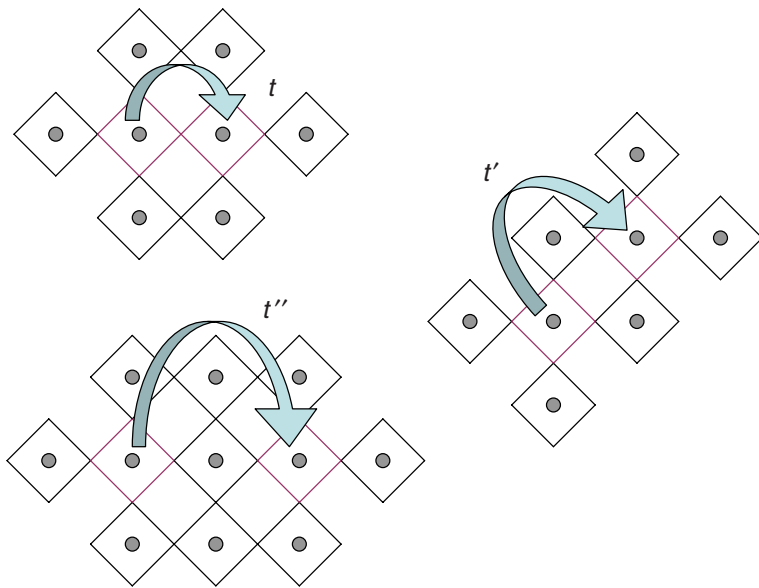


Fig. 6.3 (color online) Finite clusters employed for calculation of the effective valence- and conduction-band hoppings. Each square with a dot in the center is a CuO_4 plaquette.

dispersion of (x^2-y^2) -like states on a square lattice in the form

$$\varepsilon_{23}(\mathbf{k}) = 4t' \cos k_x a \cos k_y a - 2t''(\cos 2k_x a + \cos 2k_y a) \quad (6.17)$$

The situation is different with respect to the NN hopping t . As pointed out above, if the AF correlations are significant over large distances, as in the presence of long-range order, the effective hopping t is zero because of the “mismatch” of the correlation clouds when the hole is at site i and at a NN site j . The effective hopping between NN sites calculated for the cluster in Fig. 6.3 corresponds to an AF correlation length on the order of one lattice constant. In that case a contribution

$$\varepsilon_1(\mathbf{k}) = -2t(\cos k_x a + \cos k_y a) \quad (6.18)$$

has to be added to the dispersion in Eq. (6.17). In the general case we have

$$\begin{aligned} \varepsilon(\mathbf{k}) = & -2t(\cos k_x a + \cos k_y a) + 4t' \cos k_x a \cos k_y a \\ & - 2t''(\cos 2k_x a + \cos 2k_y a) \end{aligned} \quad (6.19)$$

In effective one-band models,⁵⁵ the ZR p-d state is regarded as a vacant, or unoccupied, d-like site. Consequently, there is no renormalization of the second-NN and third-NN hoppings t' and t'' because these connect sites of the same magnetic sublattice. In contrast, the interplay between short-range FM correlations and

TABLE 6.2 Hopping Matrix Elements for Zhang–Rice-like States in the p -Type Cuprate La_2CuO_4 and for Electron-Addition d^{10} States in n -Type SrCuO_2 ^a

Hopping MEs (eV)	Bare	Renormalized
ZR state		
t	0.540	0.135
t'	0.305	0.015
t''	0.115	0.075
d^{10} state		
t	0.290	0.115
t'	0.130	0.130
t''	0.045	0.015

^aThe renormalized value of the NN hopping t corresponds to an AF correlation length of about one lattice constant (see the text and Fig. 6.3). The bare t 's were computed by imposing HS couplings among nearby Cu spins. Crystal structures as measured by Cava et al.⁵⁹ for La_2CuO_4 and by Smith et al.⁶⁰ for SrCuO_2 were employed. For LDA MEs, see the text below.

longer-range AF couplings (see Fig. 6.2) produces large renormalization effects for *all* hopping MEs with the quantum chemical approach. For t' and t'' , in particular, the hopping of the O 2p hole implies “flips” of the Cu d spins on the two plaquettes involved directly in the hopping process. Nevertheless, spin correlations decay rapidly with distance, so the renormalization effects are less drastic for the third-NN ME, t'' . We thus find $t'' \simeq t/2$ and $t' \ll t$ (see Table 6.2), an unexpected result.

For a description of the electron-addition d^{10} states, an effective one-band model is seen to be justified. As shown in Table 6.2, in this case only t is substantially affected by nonlocal spin correlations. Again, t is zero in the presence of AF long-range order and it depends on the coherence length of AF correlations in a paramagnetic state. For AF correlations over distances on the order of one lattice constant, the renormalized hoppings for the lowest $(N + 1)$ state satisfy $t' \simeq t$, again an unexpected result. The particle–hole asymmetry (see Fig. 6.4) is now readily understood from the very different t'/t and t''/t values for the $(N - 1)$ ZR-like and $(N + 1)$ Cu d^{10} bands.

Another important finding in the quantum chemical study is the substantial mixing between the ZR state and the next-higher-lying electron-removal state, a state for which the additional hole occupies the Cu $3d_{3z^2-r^2}$ orbital. In contrast to the ZR configuration, the $d_{3z^2-r^2}$ hole is coupled to a triplet with the hole in the $d_{x^2-y^2}$ orbital at the same site. For this reason, the ZR- $d_{3z^2-r^2}$ interband mixing involves only the NN ZR- $d_{3z^2-r^2}$ hopping t_m . The CASSCF and SI calculations yield $t_m = 0.20$ eV.⁵⁷

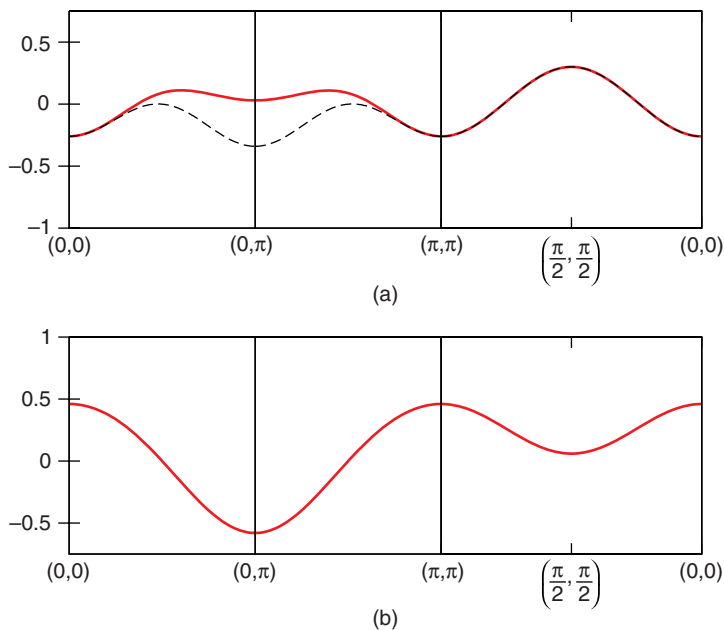


Fig. 6.4 (color online) (a) ZR-like electron-removal band for CuO_2 planes in La_2CuO_4 in the presence of long-range AF order ($t = 0$; see the text), without including the interaction with the $d_{3z^2-r^2}$ hole state (dashed line) and after including this interaction (thick continuous line). For clarity, the $d_{3z^2-r^2}$ band is not shown in the figure. The zero of energy is the value of the on-site Hamiltonian ME of the ZR-like state, H_{ii}^{ZR} . (b) Quasi-particle dispersion for the electron-addition Cu d^{10} state in SrCuO_2 in the presence of AF order. The reference energy is the value of the on-site Hamiltonian ME, $H_{ii}^{d^{10}}$. Units of eV are used in both panels.

In $\mathbf{k}$ -space, the hybridization ME between the ZR-like and $d_{3z^2-r^2}$ bands reads $\gamma_m(\mathbf{k}) = t_m(\cos k_x a - \cos k_y a)$. Further, on a given plaquette, the ZR and $d_{3z^2-r^2}(N-1)$ states are separated by an energy $\Delta\varepsilon$. It turns out that the corrections to the CASSCF energy separation due to dynamic correlation effects are substantial. These were obtained by second-order perturbation theory calculations performed on top of the CASSCF data, which is referred to as the CASPT2 method (for a monograph, see Ref. 19). Within CASPT2, we find that $\Delta\varepsilon = 1.70$ eV. It is now straightforward to diagonalize the $\mathbf{k}$ -dependent 2×2 matrix,

$$\begin{pmatrix} \varepsilon_{\text{ZR}}(\mathbf{k}) & \gamma_m(\mathbf{k}) \\ \gamma_m(\mathbf{k}) & \varepsilon_{z^2}(\mathbf{k}) + \Delta\varepsilon \end{pmatrix}$$

to yield the renormalized bands. This constitutes a nontrivial extension of the three-band Hubbard model, where the additional renormalization from the “apical” link is not considered.

Plotted in Fig. 6.4a is the resulting dispersion of the “generalized” ZR singlet generated by a doped hole in the presence of long-range AF order ($t = 0$). It shows good agreement with the dispersion of the lowest angle-resolved photoemission band reported for La_2CuO_4 in Ref. 61. In particular, the flat dispersion around $(\pm\pi, 0)$ and $(0, \pm\pi)$ and a renormalized bandwidth of less than 1 eV are well reproduced by the theoretical results. The importance of the ZR- $d_{3z^2-r^2}$ mixing is apparent: it produces pronounced flattening in the antinodal regions of the Brillouin zone.

As discussed above, the quasiparticle bands in Fig. 6.4 should best describe a doping regime, where the AF correlations extend over many lattice sites up to long-range order. For a paramagnetic state with AF correlations over distances of about one lattice constant (see Fig. 6.3), the renormalized NN hopping is finite, as shown in Table 6.2.

For the Cu d^{10} states (see Fig. 6.4b), a lack of detailed data for the photoemission lineshapes in n -type cuprates precludes a direct comparison between theory and experiment. One distinctive feature observed in the photoemission experiments is that for low electron doping, the “Fermi surface” is defined by small electron pockets centered around the $(\pm\pi, 0)$ and $(0, \pm\pi)$ points.^{62,63} This feature is reproduced in our plot in Fig. 6.4b, where the minimum of the electron-addition quasiparticle dispersion is found as well at the $(\pm\pi, 0)$ and $(0, \pm\pi)$ points. With respect to the energy difference between the electron-addition [$E(N + 1)$] and electron-removal [$E(N - 1)$] states relative to the N -electron state, we found that $\Delta E = E(N + 1) + E(N - 1) - 2E(N)$ is positive and a few eV in magnitude.⁵⁷ Thus, the ground state of the undoped N -electron system is found to be a Mott–Hubbard type of insulating state,² in agreement with standard knowledge for the parent cuprate compounds.

We note that the Zhang–Rice physics and the strong renormalization of the effective hoppings are *not* accessible by standard DFT/LDA calculations. The set of effective hopping MEs associated with the LDA valence bands⁶⁴ are not very different from the bare terms obtained by CASSCF. The LDA NN hopping t , for example, is 0.4 to 0.5 eV (see Ref. 64), somewhere in between the unrenormalized t values obtained for ZR holes and those obtained for electron-addition d^{10} states (see Table 6.2). The ratio between the NN and second-NN LDA hoppings t'/t is about 0.17 for La_2CuO_4 and 0.33 for $\text{Tl}_2\text{Ba}_2\text{CuO}_6$.⁶⁴

6.5 CONCLUSIONS

We have described two different approaches that enable us to deal with problems which strongly correlated electrons pose in periodic solids. The first case, the renormalized band structure approach, applies to Ce- and Yb-based intermetallic compounds with heavy quasiparticles. The strong correlations of the 4f electrons are treated by introducing a one-parameter phase shift for the appropriate $l = 3$ subchannel. It has proven in the past to have predictive power in particular for Fermi surfaces and the strongly anisotropic effective masses on them which are measured by using the de Haas–van Alphen method.

The second approach is based on the *ab initio* computation of many-body wavefunctions. Here it has been demonstrated that it is now possible to calculate both ground- and excited-state wavefunctions for strongly correlated electronic systems. The strong correlations are treated by a CASSCF approximation, while the remaining correlation effects are handled by either multireference techniques or second-order perturbation theory. CASSCF calculations can be done on embedded clusters only. When quasiparticle bands are computed and the short-range antiferromagnetic correlations are strong, quantum chemical calculations on finite clusters are facing problems that were pointed out and described for doped CuO₂ layers. Those are related to the limitations of the quasiparticle picture for strongly correlated systems. Therefore, one would like to extend the CASSCF calculations to Green's functions. The poles yield in addition to the quasiparticle pole the incoherent excitations resulting from the internal degrees of freedom of the correlation hole. This remains an important project for the future. Nevertheless, despite the difficulties mentioned above, the use of CASSCF calculations for the description of strongly correlated electrons marks significant progress.

REFERENCES

1. Bednorz, J. G.; Müller, K. A. *Z. Phys. B* **1986**, *64*, 189.
2. More precisely, we are dealing here with a charge-transfer gap (see Ref. 3).
3. Zaanen, J.; Sawatzky, G. A.; Allen, J. W. *Phys. Rev. Lett.* **1985**, *55*, 418.
4. Andres, K.; Graebner, J. E.; Ott, H. R. *Phys. Rev. Lett.* **1975**, *35*, 1779.
5. Stewart, G. R. *Rev. Mod. Phys.* **1984**, *56*, 755.
6. Kamihara, Y.; Watanabe, T.; Hirano, M.; Hosono, H. *J. Am. Chem. Soc.* **2008**, *130*, 3296.
7. Fulde, P.; Thalmeier, P.; Zwicky, G. In *Solid State Physics*, Vol. 60, Ehrenreich, H., and Spaepen, F.; Eds., Elsevier, Amsterdam, 2006.
8. Georges, A.; Kotliar, G.; Krauth, W.; Rozenberg, M. J. *Rev. Mod. Phys.* **1996**, *68*, 13.
9. Maier, T.; Jarrell, M.; Pruschke, T.; Hettler, M. H. *Rev. Mod. Phys.* **2005**, *77*, 1027.
10. Kotliar, G.; Savrasov, S. Y.; Haule, K.; Oudovenko, V. S.; Parcollet, O.; Marianetti, C. A. *Rev. Mod. Phys.* **2006**, *78*, 865.
11. Aryasetiawan, F.; Imada, M.; Georges, A.; Kotliar, G.; Biermann, S.; Lichtenstein, A. I. *Phys. Rev. B* **2004**, *70*, 195104.
12. Solov'yev, I. V.; Imada, M. *Phys. Rev. B* **2005**, *71*, 045103.
13. Kakehashi, Y.; Fulde, P. *J. Phys. Soc. Jpn.* **2007**, *76*, 074702.
14. Zwicky, G.; Runge, E.; Christensen, N. E. *Physica B* **1990**, *163*, 97.
15. Zwicky, G. *Adv. Phys.* **1992**, *41*, 203.
16. Zwicky, G.; Pulst, U. *Physica B* **1993**, *186*, 895.
17. Razafimandimby, H.; Fulde, P.; Keller, J. Z. *Phys. B* **1984**, *54*, 111.
18. d'Ambrumenil, N.; Fulde, P. *J. Magn. Magn. Mater.* **1985**, *47–48*, 1.
19. Helgaker, T.; Jørgensen, P.; Olsen, J. *Molecular Electronic-Structure Theory*, Wiley, New York, 2000.

20. See, e.g., Pisani, C.; Busso, M.; Capecchi, G.; Casassa, S.; Dovesi, R.; Maschio, L.; Zicovich-Wilson, C.; Schütz, M. *J. Chem. Phys.* **2005**, *122*, 094113, and references therein.
21. See, e.g., Buth, C.; Birkenheuer, U.; Albrecht, M.; Fulde, P. *Phys. Rev. B* **2005**, *72*, 195107, and references therein.
22. See, e.g., Podeszwa, R.; Tobita, M.; Bartlett, R. J. *J. Chem. Phys.* **2004**, *120*, 2581, and references therein.
23. Fulde, P. *Electron Correlations in Molecules and Solids*, 3rd ed., Springer Series in Solid-State Sciences, Vol. 100, Springer-Verlag, Berlin, 1995.
24. Oleś, A.; Zaanen, J.; Fulde, P. *Physica B* **1987**, *148*, 260.
25. Fulde, P.; Kakehashi, Y.; Stollhoff, G. In *Metallic Magnetism*, Capellmann, H., Ed., Topics in Current Physics, Vol. 42, Springer-Verlag, Berlin, 1987.
26. Stollhoff, G.; Thalmeier, P. *Z. Phys. B* **1981**, *43*, 13.
27. Lonzarich, G. G. *J. Magn. Magn. Mater.* **1988**, *76–77*, 1.
28. Fulde, P. In *Narrow Band Phenomena: Influence of Electrons with Both Band and Localized Character*, Fuggle, J. C., Sawatzky, G. A., and Allen, J., Eds., Plenum Press, New York, 1988.
29. Zwicky, G. *Phys. Scr. T* **1993**, *49*, 34.
30. Denlinger, J. D.; et al. *J. Electron Spectrosc. Relat. Phenom.* **2001**, *117–118*, 347.
31. Govind, N.; Wang, Y. A.; Carter, E. A. *J. Chem. Phys.* **1999**, *110*, 7677.
32. Huang, P.; Carter, E. A. *J. Chem. Phys.* **2006**, *125*, 084102.
33. Pereira Gomes, A. S.; Jacob, C. R.; Visscher, L. *Phys. Chem. Chem. Phys.* **2008**, *10*, 5353.
34. Burrow, A. M.; Sierka, M.; Döbler, J.; Sauer, J. *J. Chem. Phys.* **2009**, *130*, 174710.
35. Birkenheuer, U.; Fulde, P.; Stoll, H. *Theor. Chem. Acc.* **2006**, *116*, 398.
36. Hozoi, L.; Birkenheuer, U.; Fulde, P.; Mitrushchenkov, A.; Stoll, H. *Phys. Rev. B* **2007**, *76*, 085109.
37. Stoyanova, A.; Hozoi, L.; Fulde, P.; Stoll, H. *J. Chem. Phys.* **2009**, *131*, 044119.
38. Crystal 2000, University of Torino, Torino, Italy.
39. Hozoi, L.; Birkenheuer, U.; Stoll, H.; Fulde, P. *New J. Phys.* **2009**, *11*, 023023.
40. Hozoi, L.; Fulde, P. *Phys. Rev. Lett.* **2009**, *102*, 136405.
41. Goodenough, J. B. In *Progress in Solid State Chemistry*, Vol. 5, Reiss, H., Ed., Pergamon Press, London, 1971.
42. Imada, M.; Fujimori, A.; Tokura, Y. *Rev. Mod. Phys.* **1998**, *70*, 1039; Sec. IV.G.4.
43. Itoh, M.; Natori, I.; Kubota, S.; Motoya, K. *J. Phys. Soc. Jpn.* **1994**, *63*, 1486.
44. Rata, A. D.; Herklotz, A.; Nenkov, K.; Schultz, L.; Dörr, K. *Phys. Rev. Lett.* **2008**, *100*, 076401.
45. Fuchs, D.; Arac, E.; Pinta, C.; Schuppler, S.; Schneider, R.; von Löhneysen, H. *Phys. Rev. B* **2008**, *77*, 014434.
46. Haverkort, M. W.; et al. *Phys. Rev. Lett.* **2006**, *97*, 176405.
47. Podlesnyak, A.; et al. *Phys. Rev. Lett.* **2006**, *97*, 247208.
48. Maris, G.; Ren, Y.; Volotchaev, V.; Zobel, C.; Lorenz, T.; Palstra, T. T. M. *Phys. Rev. B* **2003**, *67*, 224423.
49. Korotin, M. A.; Ezhov, S. Yu.; Solovyev, I. V.; Anisimov, V. I.; Khomskii, D. I.; Sawatzky, G. A. *Phys. Rev. B* **1996**, *54*, 5309.

50. Rondinelli, J. A.; Spaldin, N. A. *Phys. Rev. B* **2009**, 79, 054409.
51. Zicovich-Wilson, C. M.; Dovesi, R.; Saunders, V. R. *J. Chem. Phys.* **2001**, 115, 9708.
52. Saebø, S.; Pulay, P. *Chem. Phys. Lett.* **1985**, 113, 13.
53. Preuss, R.; Hanke, W.; Gröber, C.; Evertz, H. G. *Phys. Rev. Lett.* **1997**, 79, 1122.
54. Capello, M.; Becca, F.; Fabrizio, M.; Sorella, S.; Tosatti, E. *Phys. Rev. Lett.* **2005**, 94, 026406.
55. Zhang, F. C.; Rice, T. M. *Phys. Rev. B* **1998**, 37, 3759.
56. Hozoi, L.; Laad, M. S. *Phys. Rev. Lett.* **2007**, 99, 256404.
57. Hozoi, L.; Laad, M. S.; Fulde, P. *Phys. Rev. B* **2008**, 78, 165107.
58. Malmqvist, P.-Å. *Int. J. Quantum Chem.* **1986**, 30, 479.
59. Cava, R. J.; Santoro, A.; Johnson, D. W.; Rhodes, W. W. *Phys. Rev. B* **1987**, 35, 6716.
60. Smith, M. G.; Manthiram, A.; Zhou, J.; Goodenough, J. B.; Markert, J. T. *Nature* **1991**, 351, 549.
61. Ino, A.; et al. *Phys. Rev. B* **2000**, 62, 4137.
62. Damascelli, A.; Hussain, Z.; Shen, Z.-X. *Rev. Mod. Phys.* **2003**, 75, 473.
63. Armitage, N. P.; et al. *Phys. Rev. Lett.* **2002**, 88, 257001.
64. Pavarini, E.; Dasgupta, I.; Saha-Dasgupta, T.; Jepsen, O.; Andersen, O. K. *Phys. Rev. Lett.* **2001**, 87, 047003.

PART C

More-Economical Methods

7 The Energy-Based Fragmentation Approach for Ab Initio Calculations of Large Systems

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The basic principles of the energy-based fragmentation (EBF) approach and its generalized version (GEBF) for approximately computing energies and molecular properties of large molecules at the full quantum mechanical level are presented. The GEBF (with EBF as a special case) approach is easily implemented, applicable at various ab initio theoretical levels, and computationally scales linearly with the system size. The GEBF approach is applied to investigate relative energies of different conformers, optimized structures, vibrational frequencies and intensities (as well as thermochemistry quantities), and some electric properties (dipole moments and polarizabilities) for a number of medium-sized or large molecules, including small proteins, carbon nanotubes, peptides, and water clusters. It is shown that total energies, structures, and properties of these systems predicted with the GEBF approach are very consistent with those from the corresponding conventional calculations (whenever the conventional results are available). The GEBF approach thus provides a fast and reliable theoretical tool for evaluating energies, structures, and properties of large systems with hundreds of atoms at various ab initio levels (with tens of computer nodes).

7.1 INTRODUCTION

The development of fast and effective algorithms for various electronic structure methods has greatly expanded the applications of computational quantum

chemistry. Now *ab initio* quality calculations can be performed for very large biological systems and various nanoparticles (or nanotubes) to study their structures and physical properties.^{1–6} It has long been recognized that the steep computational scaling of traditional algorithms for electronic structure calculations with the system size is due to the use of delocalized molecular orbitals in solving various electronic structure equations. By exploiting the locality of the density matrix or the localized molecular orbitals, many groups have developed various linear scaling algorithms,^{2,7–39} for Hartree–Fock (HF), density functional theory (DFT), and post-HF electron correlation calculations. These linear scaling methods are based on solid physical approximations and can be applied to various large and complex systems. However, the crossover between traditional algorithms and linear scaling algorithms for various electronic structure calculations usually occurs at systems with hundreds of atoms (especially for those with complicated three-dimensional structures). Thus, these methods have not become widely used in practical applications on ordinary PC workstations, but it can be expected that their applications will be greatly enhanced with the development of computer hardware in the future.

On the other hand, many groups have pursued simpler fragment-based approaches for *ab initio* quality calculations of very large systems, in which a large system is broken into many small subsystems and each subsystem is treated using traditional electronic structure methods. This type of method is much rooted in the chemistry community, as many quantities, such as bond energies and local molecular orbitals, are well known to be approximately transferable in structurally similar compounds. Many variants of fragment-based approaches,^{40–67} based on different local quantities, have been developed. Some approaches^{40–45} are aimed to evaluate the total density matrices from the density matrices of various subsystems. These density matrix–based approaches are suitable for single-point energy or one-electron properties calculations at the HF or DFT level. A popular approach is the fragment molecular orbital approach (FMO), which has been developed extensively for *ab initio* energy, energy gradient, and dipole moment calculations at a variety of theoretical levels.^{46–49} In this approach, a large molecule is first divided into fragments, and these fragments are capped with necessary local molecular orbitals to form monomers. Then monomers, dimers, or trimers are put in the Coulomb field of all other monomers, and traditional electronic structure calculations are performed for these subsystems to obtain monomer, dimer, or trimer energies, which are summed up to give the total energy of the target molecule.^{46,47} The FMO approach has been applied to a wide range of large and complex systems for obtaining *ab initio* quality energies and structures.^{48,49}

Besides density matrix–based approaches and the FMO approach, another type of fragment-based approach may be called the energy-based fragmentation (EBF) approach,^{50–64} in which the total energy and energy derivatives of a large system are obtained directly as the linear combination of the corresponding quantities from small subsystems. For this type of approach, each subsystem is constructed from a given fragment capped with its local environmental groups in

the parent system (hydrogen atoms are used as capping atoms when necessary). For covalently bonded molecules, the EBF approach was first proposed by us⁵⁴ and Collins's group⁵⁵ independently. The approach we advocated is formally derived from the fact that intra- and interfragment energy components are approximately constant in structurally analogous compounds, whereas the systematic fragmentation approach developed by Deev and Collins^{55,56} is based on the transferability of bond energies in the parent molecule and its constituent fragments. It should be noted that both approaches were inspired by the "molecular fractionation with conjugate caps" approach proposed by Zhang and Zhang,^{69,70} which was originally developed to evaluate the intermolecular interaction energy for large molecules. Before our work and Collins's work, the cardinality guided molecular tailoring approach developed by Gadre and his co-workers⁷¹ has also been employed to compute the one-electron properties for various large systems. Then it was recognized that this approach could also be employed for energy and energy derivative calculations.⁶¹ In fact, on the basis of the transferability of fragment energies, various fragmentation schemes can be implemented, in which different ways are suggested to assemble fragment energies into the total energy of a target molecule.^{54,55,57,61} However, energy-based fragmentation approaches described above are applicable only for neutral systems without many polar groups. For charged systems like those of many biological molecules or systems with polar groups, a subsystem (or fragment) is affected significantly by remote fragments within the target molecule because the electrostatic interaction between two polar or charged fragments is long range. Thus, different subsystems should be somewhat mutually dependent in charged or polar systems. To refine the EBF approach, Jiang et al.⁵⁹ put point charges on the charge centers and place each subsystem in the background point charges. Afterward, we suggested that each fragment (neutral or charged) might be modeled as an array of point charges, and each subsystem should be embedded in the presence of background point charges generated by all other fragments. This refined approach is called a generalized energy-based fragmentation (GEBF) approach, for brevity.^{65–68} As these "embedded" subsystems can also be treated readily by conventional electronic structure methods, the GEBF approach, like the original EBF approach, is still simple and cost-effective. The accuracy of the GEBF approach is very satisfactory for a variety of systems with (or without) charged or polar groups. It is worthwhile pointing out that the EBF or GEBF approaches mentioned above are also applicable for weakly bound molecular clusters. On the other hand, several EBF approaches^{50–52,63,64} specifically designed for weakly bound molecular clusters have also emerged. These methods include the integrated multicenter molecular orbital (IMiCMO) method developed by Sakai and Morita,^{50,51} a simplified FMO approach suggested by Hirata et al.,⁵² and the electrostatically embedded many-body expansion method developed by Dahlke and Truhlar.^{63,64}

In comparison with other fragment-based approaches, the energy-based approaches can achieve similar accuracy but with significantly less computational effort. This is mainly because calculations on various subsystems are

almost independent of each other in the energy-based approaches. A very important advantage of various EBF approaches is their simplicity, since no extra programming efforts are required as long as subsystems can be treated with existing quantum chemistry packages. Thus, energy-based approaches can be applied to obtain energy and molecular properties (which are derived from energy derivatives with respect to an external electric or magnetic field) at various theoretical levels (HF, DFT, MP2, etc). For small or medium-sized systems, efficient methods^{72,73} have been established for the analytic evaluation of first and second (or higher) energy derivatives at various *ab initio* theoretical levels. With these energy derivatives, computational chemists have been able to calculate the optimized molecular structures and many important molecular properties, including harmonic vibrational frequencies, infrared (IR) and Raman intensities, as well as dipole moments, polarizabilities, and so on. However, the analytic calculations of energy derivatives for large systems still require large memory and disk space, and substantial computational cost. Nevertheless, within the EBF or GEBF approach, the energy derivatives of a large system can be obtained approximately from energy derivative calculations on small subsystems. As a result, the EBF or GEBF approach provides a fast and reliable way for evaluating *ab initio* quality molecular structures and properties of very large systems. A number of test applications have shown that the EBF or GEBF approach can give very satisfactory predictions for energies,^{50–68} optimized structures,^{54,61,62,66} vibrational frequencies^{51,55,62,66} and corresponding IR or Raman intensities,⁶⁶ dipole moments,^{65,67} polarizabilities,^{65,67} and nuclear magnetic shielding tensors,⁵⁸ which reproduce well the corresponding results from conventional electronic structure calculations.

In this review we first give a brief description of the basic principles and practical procedures of the EBF and GEBF approaches, and discuss how to evaluate molecular properties from energy derivatives within the EBF or GEBF approach in Section 7.2. The purpose of Section 7.3 is to show the applicability of the EBF and GEBF approaches in predicting relative energies (Section 7.3.1), optimized structures (Section 7.3.2), vibrational spectra and thermochemical data (Section 7.3.3), and some molecular properties (Section 7.3.4) for a variety of large systems, including macromolecules and molecular clusters. Since the physical foundations behind the EBF or GEBF approach and some illustrative applications were included in our previous review,⁶⁷ here we put more emphasis on applications not described before. In Section 7.4 we give a brief summary and future prospect on the GEBF approach.

7.2 THE ENERGY-BASED FRAGMENTATION APPROACH AND ITS GENERALIZED VERSION

7.2.1 EBF Approach

The main idea of various EBF approaches is to evaluate the total energy (and its derivatives) of a large system as the linear combination of the energies (or energy

derivatives) of many small subsystems. The basic procedure of energy-based fragmentation approaches includes the following steps:

1. Divide a target molecule into many fragments of about equal size, each of which usually contains at least three nonhydrogen atoms.
2. Construct subsystems from fragments according to some rules, and determine the expansion coefficients of all subsystems.
3. Perform conventional quantum chemistry calculations on these subsystems.
4. Obtain the total energy of the target molecule from the energies of all subsystems with the formula

$$E_{\text{tot}} \approx \sum_k^M C_k E_k \quad (7.1)$$

Here E_k represents the total energy (including the electronic energy and nuclear repulsion energy) of the k th subsystem, C_k is the expansion coefficient of this subsystem, which can be determined unambiguously by following some rules, and M is the total number of constructed subsystems. Various EBF schemes differ from each other in different ways to construct subsystems.^{50–68}

We have suggested a fast and effective way^{65–68} to generate various subsystems from all fragments assigned manually. Our strategy consists of the following steps:

1. For a given fragment (called the *central fragment*), add its neighboring fragments to construct a primitive closed-shell subsystem. The maximum number of fragments (η) is a parameter used to control the size of each subsystem. All other fragments are first ranked according to their distances from this central fragment, and then fragments are added sequentially according to their distances [those beyond a distance threshold (6 Å) are ignored], but the maximum number of fragments in this subsystem cannot exceed η . It should be mentioned that hydrogen atoms are always used as capping atoms whenever necessary.⁷⁴
2. Compare all primitive subsystems, and remove those subsystems whose fragments are all embedded in larger subsystems. The coefficients of all retained primitive subsystems are set to be unity.
3. Construct derivative subsystems with $\eta - 1$ fragments first, then subsystems with $\eta - 2$ fragments, and finally, subsystems with one fragment.

Here we show how to determine subsystems with $\eta - 1$ fragments and their coefficients. We first expand the total energy of all primitive subsystems with η fragments into the sum of m -fragment ($m = 1, 2, \dots, \eta$) energy terms. For example, the energy of a subsystem with fragments I, J, K , and L (abbreviated

as $IJKL$ hereafter) is expanded as

$$E(IJKL) = [E(I) + \cdots + E(L)] + [E(I, J) + \cdots + E(K, L)] \\ + [E(I, J, K) + \cdots + E(J, K, L)] + E(I, J, K, L) \quad (7.2)$$

Here $E(I)$, $E(I, J)$ (and so on) are the intrafragment and two-fragment energy components. If one sums up the $(\eta - 1)$ -body energy terms in the energy expansions of all subsystems with η fragments, one can derive the coefficients of all $(\eta - 1)$ -body energy terms. If a specific $(\eta - 1)$ -body term has a coefficient d , whose value is different from unity, we should add a $(\eta - 1)$ -fragment derivative subsystem k corresponding to this energy term, with $C_k = 1 - d$. By doing so, the coefficient of this specific $(\eta - 1)$ -body term in the total energy expression will be exactly unity, as it should be. For example, if we already have two primitive subsystems (1234) and (1235), we should construct a derivative subsystem (123), whose coefficient is required to be -1 . In a similar way, one can build all m -fragment ($m = \eta - 2, \dots, 1$) derivative subsystems and their coefficients. It should be pointed out that in the derivation described above, a given m -fragment energy term ($m \neq 1$) is implicitly assumed to be constant in different subsystems containing this term. This was demonstrated to be a good approximation in our previous work.⁶⁵⁻⁶⁸ However, a one-fragment term like $E(I)$ is strongly dependent on the subsystem. For example, $E(I)$ in a subsystem with the fragment I capped with a hydrogen atom is quite different from that in another subsystem in which the fragment I is bonded to other fragments. But according to our construction rules, those one-fragment energy terms for fragments with capping hydrogen atoms will be canceled out so that they will not appear in the final energy expression. To conclude from discussions above, we can find that if the energies of the target molecule and all constructed subsystems (primitive and derivative) are expanded up to the η -body energy terms, the linear combination of all subsystems is just equal to the total energy of the target molecule, as indicated by Eq. (7.1). For a pentapeptide with each residue a fragment, all its subsystems and their coefficients required for the EBF calculation with the parameter $\eta = 3$ [called the EBF(3) for short] are displayed in Fig. 7.1.

7.2.2 The GEBF Approach

Within various EBF schemes, for a given fragment only its spatially close fragments within a distance threshold (usually 6 Å) are included in constructing the corresponding subsystem. So the influence of distant fragments on each subsystem is totally ignored. This turns out to be a very good approximation for neutral nonpolar systems. However, for systems with polar or charged groups, the neglect of the interaction between distant two fragments will bring significant errors, as the electrostatic interaction between charge centers or between a charge center and a dipole (even between two dipoles) decays slowly with their distance. Clearly, the long-range interaction between any two fragments should be taken into account.

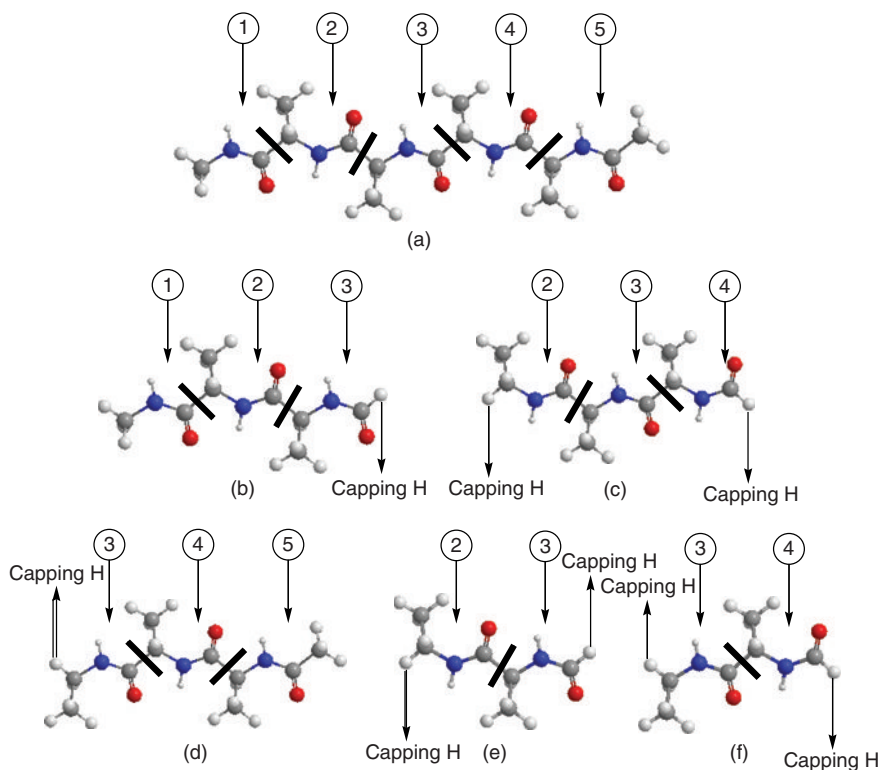


Fig. 7.1 (color online) Fragmentation scheme and all constructed subsystems required for the EBF(3) approach for a pentapeptide: (a) target system; (b) subsystem 1 ($C_1 = 1$); (c) subsystem 2 ($C_2 = 1$); (d) subsystem 3 ($C_3 = 1$); (e) subsystem 4 ($C_4 = -1$); (f) subsystem 5 ($C_5 = -1$).

In the GEBF approach,^{65–68} a simple but effective way is employed to include such long-range interfragment interactions. For a given subsystem, all atoms outside this subsystem are modeled as an array of point charges on the nuclear centers, and this subsystem is then placed in the field of background point charges generated by all atoms outside this subsystem. By this way, the polarization and electrostatic interaction between fragments in a given subsystem and all other fragments is modeled approximately. Thus, the GEBF approach differs from the EBF approach only in that each subsystem is embedded in the presence of an array of background point charges. It is well known that there is no unique way of assigning point charges for atoms in a molecule. We have tested several different charges and found that charges from natural population analysis (or NPA charges)^{75,76} usually give better performance. It is also worthwhile mentioning how to obtain these point charges. First, we may do HF calculations for all primitive subsystems without point charges, and then we extract charges on central fragments from natural population analysis on each primitive subsystem. Then we

perform HF calculations again for all primitive subsystems, which are embedded in the point charges obtained in the first run. NPA charges from calculations on embedded subsystems are then used in subsequent GEBF calculation, in which all subsystems (primitive and derivative) are embedded in the point charges. Our experience shows that the NPA charges from HF calculations could be employed for GEBF calculations at other theoretical levels (such as MP2).

The total energy of a large system in the GEBF approach can be computed approximately from the formula

$$E_{\text{tot}} \approx \sum_k^M C_k \tilde{E}_k - \left(\sum_k^M C_k - 1 \right) \sum_A \sum_{B>A} \frac{Q_A Q_B}{R_{AB}} \quad (7.3)$$

where $\tilde{E}_k$ represents the total energy (including the self-energy of point charges) of the k th “embedded” subsystem, and Q_A is the point charge on atom A . The second term in Eq. (7.3) is included so that the interactions between fragments are counted once. For more details on the derivation of this equation, refer to our original paper.⁶⁵ Clearly, if the point charges on all atoms are zero, Eq. (7.3) will reduce to Eq. (7.1). Thus, the EBF approach can be considered as a special case of the GEBF approach.

To help readers understand the GEBF procedure, in the appendix we give an illustrative example of how to estimate the GEBF-HF energy for dotriacontane.

7.2.3 Evaluation of Energy Derivatives and Molecular Properties within the EBF or GEBF Approach

Molecular property can be considered as the response of the molecular system to an external electric (or magnetic) field perturbation. Thus, all molecular properties can be formulated in terms of energy derivatives with respect to an external electric or magnetic field (electric or magnetic field derivatives). For example, in the presence of an external electric field $\mathbf{F}$, the total energy of a molecular system can be expressed

$$\begin{aligned} E(F) = E(0) &+ \sum_i \left[\frac{\partial E(F)}{\partial F_i} \right]_{F \rightarrow 0} F_i + \frac{1}{2!} \sum_{ij} \left[\frac{\partial E(F)}{\partial F_i \partial F_j} \right]_{F \rightarrow 0} F_i F_j \\ &+ \frac{1}{3!} \sum_{ijk} \left[\frac{\partial E(F)}{\partial F_i \partial F_j \partial F_k} \right]_{F \rightarrow 0} F_i F_j F_k + \dots \end{aligned} \quad (7.4)$$

Here $E(0)$ represents the total energy without the external electric field, and F_i is the component of the electric field $\mathbf{F}$ along the i axis that corresponds to one of the x , y , and z Cartesian coordinates. From Eq. (7.4), the components of the dipole moment (μ_i), polarizability (α_{ij}), first hyperpolarizability (β_{ijk}) can be extracted,

$$\mu_i = \left[\frac{\partial E(F)}{\partial F_i} \right]_{F \rightarrow 0} \quad (7.5)$$

$$\alpha_{ij} = \left[\frac{\partial E(F)}{\partial F_i \partial F_j} \right]_{F \rightarrow 0} \quad (7.6)$$

$$\beta_{ijk} = \left[\frac{\partial E(F)}{\partial F_i \partial F_j \partial F_k} \right]_{F \rightarrow 0} \quad (7.7)$$

On the other hand, the derivatives of the energy E with respect to displacements of nuclear centers (geometrical derivatives) can be derived in a similar manner, by taking the change in the electronic Hamiltonian generating from displacements of nuclear centers as a perturbation.

Two approaches have been developed to compute molecular properties, in which one is the analytic approach⁷⁷ and another is the finite-field (FF) approach.⁷⁸ The analytic approach is usually preferred and thus has been developed and implemented for various ab initio methods. The finite-field approach is usually employed to test newly developed electronic structure methods, due to its easy implementation. In the FF approach, for example, the dipole moment and polarizability of a molecular system can be determined numerically as below by calculating its total energies in an electric field with several different field strengths⁷⁸:

$$\begin{aligned} \mu_i = & -\{45[E(F_i) - E(-F_i)] - 9[E(2F_i) - E(-2F_i)] \\ & + [E(3F_i) - E(-3F_i)]\}/60F_i \end{aligned} \quad (7.8)$$

$$\begin{aligned} \alpha_{ii} = & \{490E(0) - 270[E(F_i) + E(-F_i)] + 27[E(2F_i) + E(-2F_i)] \\ & - 2[E(3F_i) + E(-3F_i)]\}/180F_i^2 \end{aligned} \quad (7.9)$$

$$\begin{aligned} \alpha_{ij} = & [E(2F_i, 2F_j) - E(2F_i, -2F_j) - E(-2F_i, 2F_j) + E(-2F_i, -2F_j)] \\ & - 16[E(F_i, F_j) - E(F_i, -F_j) - E(-F_i, F_j) + E(-F_i, -F_j)]/48F_i F_j \end{aligned} \quad (7.10)$$

where $E(kF_i)$ and $E(kF_i, kF_j)$ ($k = \pm 1, \pm 2, \pm 3$) represent the total energies in the presence of the electric field along the i axis or along both the i and j axes ($i, j = x, y, z$). In practical calculations, F_i and F_j are usually set as 0.005 a.u. These equations, which were first given by Kamada et al.,⁷⁸ correspond to the numerical differentiation of the energy $E(F)$ [Eq. (7.4)] with triple finite electric fields.

7.2.3.1 Geometrical Derivatives, Geometry Optimizations, and Vibrational Frequencies The first energy derivative with respect to displacements of nuclear centers is called the *energy gradient* g . From Eq. (7.3) one can see that the gradient g of a large molecule can be evaluated approximately using the formula⁶⁶

$$g_i = \frac{\partial E_{\text{tot}}}{\partial q_{It}} \approx \sum_{I \in k} C_k \frac{\partial E_k}{\partial q_{It}} - \sum_k C_k \sum_{A \notin k} f_{At,It}$$

$$t = x, y, z; i = 1x, 1y, 1z, 2x, \dots \quad (7.11)$$

where

$$E_k = \tilde{E}_k - \sum_{\substack{A < B \\ A, B \notin k}} \frac{Q_A Q_B}{R_{AB}} \quad (7.12)$$

$$f_{At,It} \equiv Q_A Q_I \frac{q_{At} - q_{It}}{R_{AI}^3} \quad (7.13)$$

$$R_{AI} \equiv [(q_{Ax} - q_{Ix})^2 + (q_{Ay} - q_{Iy})^2 + (q_{Az} - q_{Iz})^2]^{1/2} \quad (7.14)$$

In the equations above, q_{It} represents a certain Cartesian coordinate t ($t = x, y, z$) of the atom I , $f_{At,It}$ denotes the t component of the Coulomb force between charge on A and charge on I separated by a distance R_{AI} , and E_k is the total energy of the k th subsystem without the self-energy of background point charges. It should be mentioned that the first summation of Eq. (7.11) only runs over those subsystems containing the atom I as a real atom. As the gradient for each “embedded” subsystem ($\partial E_k / \partial q_{It}$) can be obtained with existing quantum chemistry packages, the energy gradient g of a large system can readily be evaluated.

Following the similar idea, the second derivatives of the total energy with respect to the displacements of nuclear centers, which form the Hessian matrix H , can be expressed approximately as

$$H_{ij} = \frac{\partial^2 E_{\text{tot}}}{\partial q_{It} \partial q_{Js}} \approx \sum_{I, J \in k} C_k \frac{\partial^2 E_k}{\partial q_{It} \partial q_{Js}} - \sum_k C_k \left(\sum_{A \notin k} G_{A,It,Js} \right) \quad t, s = x, y, z \quad (7.15)$$

with

$$G_{A,It,Js} \equiv Q_A Q_I \left[\frac{(-3)(q_{At} - q_{It})(q_{As} - q_{Is})}{R_{AI}^2} + \delta_{Is} \right] R_{AI}^{-3} (\delta_{AJ} - \delta_{IJ}) \quad (7.16)$$

Thus, the Hessian matrix H of a large system can also be obtained from the Hessian matrices $H^{(k)}$ of various embedded subsystems [the first term in Eq. (7.15)] subtracted by a term from the electrostatic charge–charge interaction.

With the gradient g and the Hessian matrix H of the target system, we can perform geometry optimization from an initial structure to locate local or global minimum structures and transition states connecting different minimum

structures. Any existing optimization codes can be employed. If an optimized structure is obtained with the GEBF approach, we can proceed to construct the mass-weighted Hessian matrix H' (as shown below) with the Hessian matrix H , and then diagonalize H' to obtain vibrational frequencies ω_k and their corresponding normal modes Q_k :

$$H'_{ij} = H_{ij}(m_i m_j)^{-1/2} \quad (7.17)$$

$$H' Q_k = \omega_k^2 Q_k \quad (7.18)$$

Here m_i denotes the atomic mass of a certain atom. If the background point charges are absent, the second term in Eqs. (7.11) and (7.15) does not appear. The resulting gradient and Hessian matrix have been employed for geometry optimizations⁵⁴ and vibrational frequency calculations.⁶²

7.2.3.2 Electric Field Derivatives, Electric Properties, and Vibrational Intensities If a small external electric field $\mathbf{F}$ is applied to a large system, one would expect that an approximate relationship such as Eq. (7.3) may still hold true for the total energy of this system: that is,

$$E_{\text{tot}}(F) \approx \sum_k^M C_k \tilde{E}_k(F) - \left(\sum_k^M C_k - 1 \right) \sum_A \sum_{B>A} \frac{Q_A Q_B}{R_{AB}} \quad F \text{ is small} \quad (7.19)$$

Here $\tilde{E}_k(F)$ stands for the total energy of the k th subsystem in the presence of both background point charges and the external electric field $\mathbf{F}$. In terms of the definitions of electric properties given in Eqs. (7.5) to (7.7), we have suggested the following formula for evaluating various electric properties⁶⁵ (assume that point charges are not dependent on the external electric field):

$$\Omega \approx \sum_k^M C_k \Omega_k \quad \Omega = \mu_i, \alpha_{ij}, \dots \quad (7.20)$$

Here Ω may stand for the dipole moment, polarizability, and so on. Thus, the electric properties of a large system can be estimated from the corresponding quantities of small “embedded” subsystems, which are readily available with the analytic approach. On the other hand, we may use Eq. (7.19) directly to obtain the total energies in the presence of different external electric fields and derive the dipole moment and polarizability (and so on) from the finite-field equations (7.8) to (7.10). Both approaches will be tested in the next section for evaluating the dipole moments and static polarizabilities of some medium-size systems, in which results from conventional full system calculations are available for comparison. It should be pointed out that the derivatives with respect to the external magnetic field should be obtained in the same way within the GEBF approach, which will be reported in our future work. Lee and Bettens have demonstrated that the energy-based fragmentation approach (without background point charges)

can provide accurate predictions on the nuclear magnetic shielding tensor for a number of neutral molecular systems.⁵⁸

It is well known that the infrared (IR) intensity I_k and Raman intensity R_k corresponding to a given normal mode Q_k can be calculated with the expressions below,

$$I_k = \left| \frac{\partial \vec{\mu}}{\partial Q_k} \right|^2 \quad (7.21)$$

$$R_k = (45\bar{\alpha}_k^2 + 7\gamma_k^2) \quad (7.22)$$

where

$$\bar{\alpha}_k = \frac{1}{3} \sum_{a=x,y,z} \frac{\partial \alpha_{aa}}{\partial Q_k} \quad (7.23)$$

$$\gamma_k^2 = \frac{1}{2} \left[3 \sum_{a,b=x,y,z} \left(\frac{\partial \alpha_{ab}}{\partial Q_k} \right)^2 - 9\bar{\alpha}_k^2 \right] \quad (7.24)$$

According to the equations above, the derivatives of the dipole moment and polarizability with respect to the normal mode Q_k are required to be determined. Since the normal mode Q_k is a linear combination of displacements of all atoms, the required derivatives can be obtained through the chain rule with the corresponding derivatives with respect to the Cartesian coordinates of all atoms q_{It} ($t = x, y, z$):

$$\frac{\partial \Omega}{\partial Q_k} = \sum_{a=1}^{3N} \frac{\partial \Omega}{\partial \xi_a} \frac{\partial \xi_a}{\partial Q_k} \quad k = 1, 2, \dots, N_f \quad (7.25)$$

$$\xi_1 = \sqrt{m_1}q_{1x}, \xi_2 = \sqrt{m_1}q_{1y}, \xi_3 = \sqrt{m_1}q_{1z}, \xi_4 = \sqrt{m_2}q_{2x}, \dots, \quad (7.26)$$

$$\frac{\partial \Omega}{\partial q_{It}} \approx \sum_{I \in k}^M C_k \frac{\partial \Omega_k}{\partial q_{It}} \quad \Omega = \mu_i, \alpha_{ij}, \dots; t = x, y, z \quad (7.27)$$

Here N_f is the number of vibrational degrees of freedom, ξ_a is the mass-weighted Cartesian coordinate, and Eq. (7.27) is derived by differentiating Eq. (7.20) with respect to the Cartesian coordinate of the atom I , q_{It} ($t = x, y, z$).

7.3 RESULTS AND DISCUSSION

7.3.1 Ground-State Energies

In this subsection we apply the GEBF approach to compute the ground-state energies for two systems: the epidermal growth factor subdomain (PDB⁷⁹ id: 1FGD) protein (see Fig. 7.2) and a water cluster, $(\text{H}_2\text{O})_{16}$. For each system,

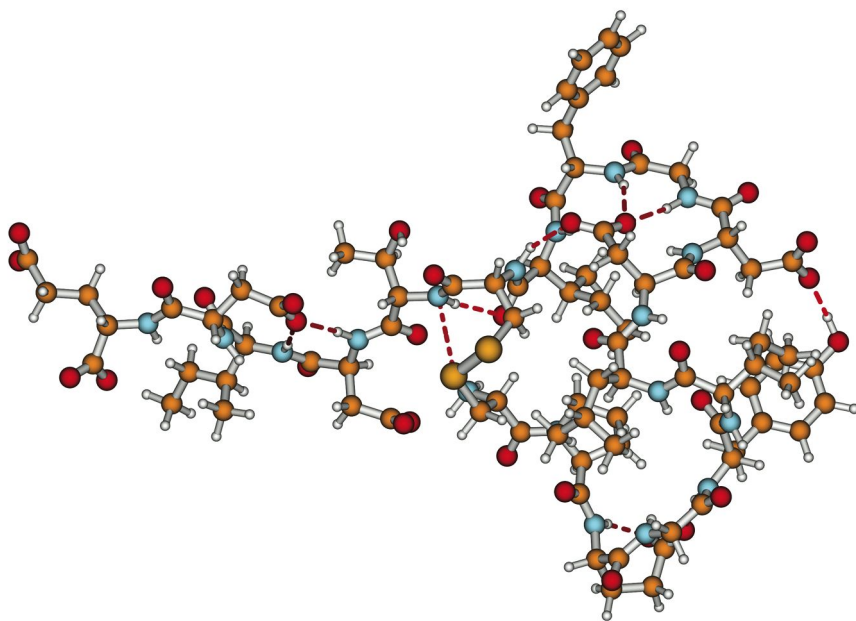


Fig. 7.2 (color online) Structure of a conformer of the epidermal growth factor subdomain (PDB id: 1FGD).

its five conformers will be studied, and their relative energies calculated with the GEBF approach will be compared with those from conventional calculations. For 1FGD, we have performed AMBER99⁸⁰ force field calculations with the TINKER⁸¹ package to locate the five lowest-energy conformers. For $(\text{H}_2\text{O})_{16}$, its five lowest-energy structures (see Fig. 7.3) determined with the TIP4P⁸² force field are taken from an online database.⁸³ HF and coupled-cluster singles and doubles (CCSD) calculations with the conventional and GEBF approaches will be performed for 1FGD and $(\text{H}_2\text{O})_{16}$, respectively. The LSQC⁸⁴ program is designed for performing all GEBF calculations reported in this chapter, which include the construction of subsystems, the geometry optimization, vibrational frequencies and vibrational intensity calculations, and total energy and molecular properties calculations. The GAUSSIAN03⁸⁵ package is linked to the LSQC program for performing subsystem (or full system) calculations. For $(\text{H}_2\text{O})_{16}$, the GAMESS⁸⁶ package is employed for conventional CCSD calculations.⁸⁷ In GEBF calculations for both systems, the maximum number of fragments for any subsystem is set to be 6 (i.e., $\eta = 6$). For 1FGD, the fragmentation scheme is chosen to be the same as described for proteins in our previous work⁶⁵; that is, the cut bonds are the C—C bond between α -carbon and the carbonyl group in the central residues, the S—S bond between two residues, and the C—C bond between β - and γ -carbons in five residues with large side chains (Arg, Lys, Phe, Trp, Tyr). For $(\text{H}_2\text{O})_{16}$, each water molecule is selected as a fragment. The 6-31G(d) basis set with 6d Cartesian polarization functions for heavy atoms is employed in all calculations.

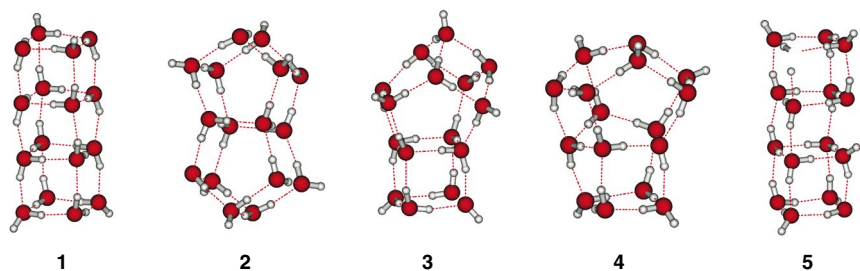


Fig. 7.3 (color online) Five lowest-energy conformers of $(\text{H}_2\text{O})_{16}$ obtained with the TIP4P force field. (From Ref. 83.)

TABLE 7.1 Conventional HF and GEBF-HF Energies for Five Conformers of 1FGD with the 6-31G(d) Basis Set

Conformer	Basis Functions ^a	HF Energy	
		Conventional (a.u.)	GEBF(6) (mH) ^b
1	2348/910	−7609.98126	2.08
2	2348/910	−7610.02424	2.14
3	2348/910	−7610.02323	2.35
4	2348/938	−7609.98940	2.81
5	2348/870	−7609.97223	3.49

^aThe number of basis functions for the entire system and the largest subsystem are listed before and after the slash, respectively.

^bThe relative energies with respect to the conventional HF energies.

The conventional HF and GEBF-HF energies for five conformers of 1FGD are compared in Table 7.1. For all five conformers, the GEBF-HF energies deviate from the conventional HF energies by less than 3.5 millihartree (mHa), although these conformers have 260 atoms with eight charged centers (1^+7^-). For each conformer, the largest subsystem has a basis function of 938, about 40% of the total number of basis functions (2348) in the entire system (as shown in the second column of Table 7.1). Furthermore, for these five conformers, we compare their relative energies calculated with the GEBF-HF and conventional HF approaches in Fig. 7.4. For better comparison, the relative energies with respect to the energy of the first conformer are defined for both conventional HF and GEBF-HF results. It can be seen from Fig. 7.4 that the energy curve obtained from GEBF-HF calculations is remarkably close to that from conventional HF calculations. The largest difference between the HF relative energies and the GEBF-HF relative energies is only 1.41 mHa ($0.88 \text{ kcal mol}^{-1}$). Thus, the relative energies of different conformers for 1FGD can be predicted accurately using the GEBF approach.

The energies from conventional CCSD and GEBF-CCSD calculations for five lowest-energy conformers of $(\text{H}_2\text{O})_{16}$ are listed in Table 7.2. In our previous work⁶⁸ we have investigated the performance of the GEBF approach in describing

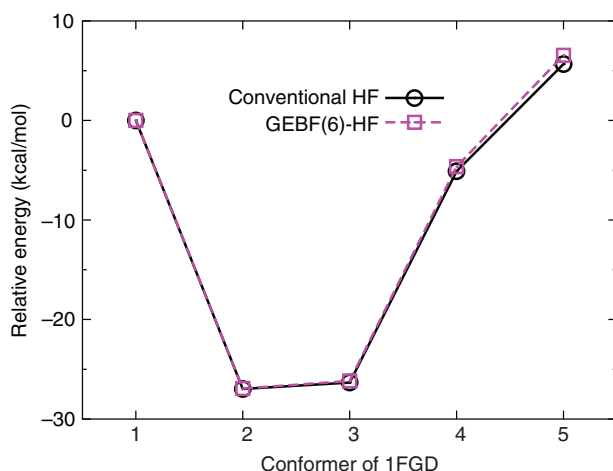


Fig. 7.4 (color online) Relative energies obtained by conventional HF and GEBF-HF calculations for five conformers of 1FGD with the 6-31G(d) basis set. Relative energies are defined with respect to the energy of the first conformer for both HF and GEBF-HF results.

TABLE 7.2 Conventional CCSD and GEBF-CCSD Energies for Five Isomers of $(\text{H}_2\text{O})_{16}$ with the 6-31G(d) Basis Set

Structure	Basis Functions ^a	CCSD Energy	
		Conventional (a.u.)	GEBF(6) (mH) ^b
1	304/114	−1219.61449	1.08
2	304/114	−1219.60632	1.28
3	304/114	−1219.60505	0.89
4	304/114	−1219.60186	0.89
5	304/114	−1219.60838	0.63

^aThe number of basis functions for the entire system and the largest subsystems are listed before and after the slash, respectively.

^bThe relative energies with respect to the conventional CCSD energies.

the relative energies of $(\text{H}_2\text{O})_{20}$ at the HF, MP2, and DFT levels. So it may be interesting to know the performance of the GEBF approach at the CCSD level. The results collected in Table 7.2 show that the largest difference between the GEBF-CCSD and conventional CCSD results is only 1.28 mHa, although the largest subsystems in GEBF calculations only involve six water molecules. The relative energies for five structures of $(\text{H}_2\text{O})_{16}$ with GEBF and conventional calculations are compared in Fig. 7.5. When the energy of the first structure is set to zero, the GEBF-CCSD relative energies deviate from the conventional CCSD values by less than $0.28 \text{ kcal mol}^{-1}$. Hence, the results described previously⁶⁸ and here indicate that the GEBF approach can provide satisfactory descriptions for relative energies of water clusters at various theoretical levels (HF, DFT, MP2, and CCSD).

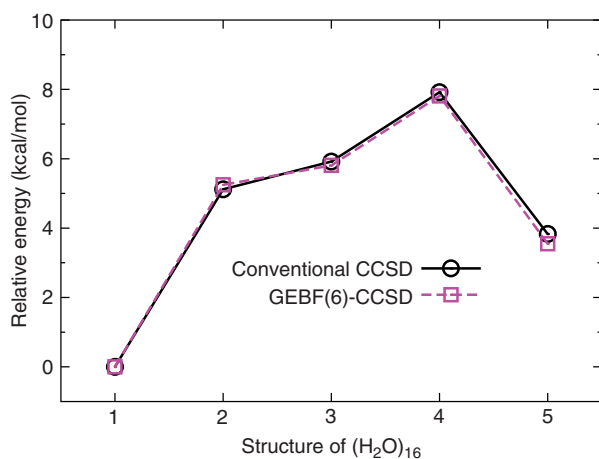


Fig. 7.5 (color online) Relative energies obtained by conventional CCSD and GEBF-CCSD calculations for five conformers of $(\text{H}_2\text{O})_{16}$ with the 6-31G(d) basis set. Relative energies are defined with respect to the energy of the first conformer for both CCSD and GEBF-CCSD results.

7.3.2 Geometry Optimizations

The EBF and GEBF approaches have been employed previously^{54,66} to optimize the structures of several systems, including BN nanotubes, short peptides, and water clusters. Here, the results of $(\text{H}_2\text{O})_{20}$ (denoted as W20), a 36-layer Armchair(2,2) single-walled carbon nanotube (SWCNT-36), and a 21-amino acid tricyclic peptide (PDB id: 1RPB),⁸⁸ will be shown to demonstrate the applicability of the GEBF (or EBF) approach. SWCNT-36 contains 296 atoms, being the smallest diameter among all the carbon nanotubes. Hydrogen atoms are used to cap the terminal carbons. 1RPB has 280 atoms (with one negative charge center). The latter two molecules are representatives of nanotubes and biological systems, and their geometry optimizations with conventional methods would be very time consuming. For these three systems, we compare the GEBF-optimized structures with those from the conventional approach for W20, and compare EBF-optimized structures obtained at two fragmentation levels for SWCNT-36 and 1RPB.

We first describe some details for GEBF geometry optimizations. For W20, each water is chosen as a fragment, and its geometry optimization is carried out at the GEBF(5)-B3LYP/cc-pVTZ level, starting from a TIP4P-optimized minimum structure. For SWCNT-36, we choose both three layers (3L) and four layers (4L) as a fragment, respectively, and perform geometry optimizations at the EBF(3)-B3LYP/6-31G level. For simplicity, we denote the results with different fragmentations schemes as EBF(3)-3L and EBF(3)-4L, respectively. As the EBF energy for this system is very close to the GEBF energy, the EBF approach is employed here (which corresponds to GEBF without background point charges). In the case of 1RPB, we use the same fragmentation scheme as described above for 1FGD, and GEBF(5) and GEBF(6) optimizations are carried

out at the HF/6-31G level. For all three systems, geometry optimizations are done with the quasi-Newton algorithm,^{89–91} and the BFGS procedure is used for updating the Hessian matrix.⁹² For W20 and SWCNT-36, the redundant internal coordinates are used, and the convergence criteria for the energy change and the maximum gradient are set to 0.01 mHa and 1.0 mHa/bohr, respectively. While for 1RPB, we use the Cartesian coordinates and the convergence criteria for the maximum gradient and maximum displacement are set to be 1.125 mHa/bohr and 0.0045 bohr, respectively.

The GEBF-optimized structures for three systems are shown in Fig. 7.6, and their Cartesian coordinates are available online.⁹³ In Table 7.3 we list the total energies (at the optimized geometries) and root-mean-square distance (RMSD) between the B3LYP and GEBF-B3LYP geometries for W20, between the EBF(3)-3L and EBF(3)-4L geometries for SWCNT-36, and between GEBF(5) and GEBF(6) geometries for 1RPB. The superposition picture of each molecule is also shown in Fig. 7.6. In all the GEBF (or EBF) calculations shown here, the largest subsystem is less than one-third of the entire system, but the GEBF-optimized geometry agrees well with that from the conventional calculations for W20, with the RMSD being only 0.073. For SWCNT-36 and 1RPB, two GEBF (or EBF)-optimized structures with two different fragmentation levels are in good agreement with each other, with the RMSDs less than 0.34 Å. Although the optimized structures with the conventional DFT or HF calculations are not available for these two systems, we have performed instead a single-point full-system calculation at the GEBF (or EBF)-optimized geometries. The GEBF (or EBF) energies deviate no more than 1.0 mHa from the conventional values, indicating the high accuracy of the GEBF (or EBF) approach. One may notice that the RMSD between two GEBF-optimized structures for 1RPB is significantly larger than that for SWCNT-36, this is because 1RPB is much more flexible than SWCNT-36, and two GEBF optimizations with different fragmentation schemes lead to two slightly different local minima (they differ primarily in the location of side chains such as Tyr15 and Trp21, as shown in Fig. 7.6). We have done all the calculations with the message-passing interface (MPI)⁹⁴ parallel technique on a platform of 10 nodes, each of which contains eight Intel Xeon 2.66-GHz processors. It is worthwhile mentioning the computational cost for GEBF calculations. Let us take 1RPB as an example. If each subsystem is calculated with two processors via shared memory, it takes only 13 and 18 min per step for GEBF(5) and GEBF(6) optimization, respectively, whereas for the conventional full-system calculation, one optimization step will need 42 min on one node (eight processors) via shared memory.

From the discussions above, one can see that the optimized structures from the GEBF approach at the HF or DFT level are quite reliable, either consistent with those from the standard approach, or nearly convergent with increasing the size of subsystems. In general, when the GEBF approach is employed to treat large systems because conventional *ab initio* calculations are not available, the first step is to select the appropriate parameter (η) for systems under study. If the system

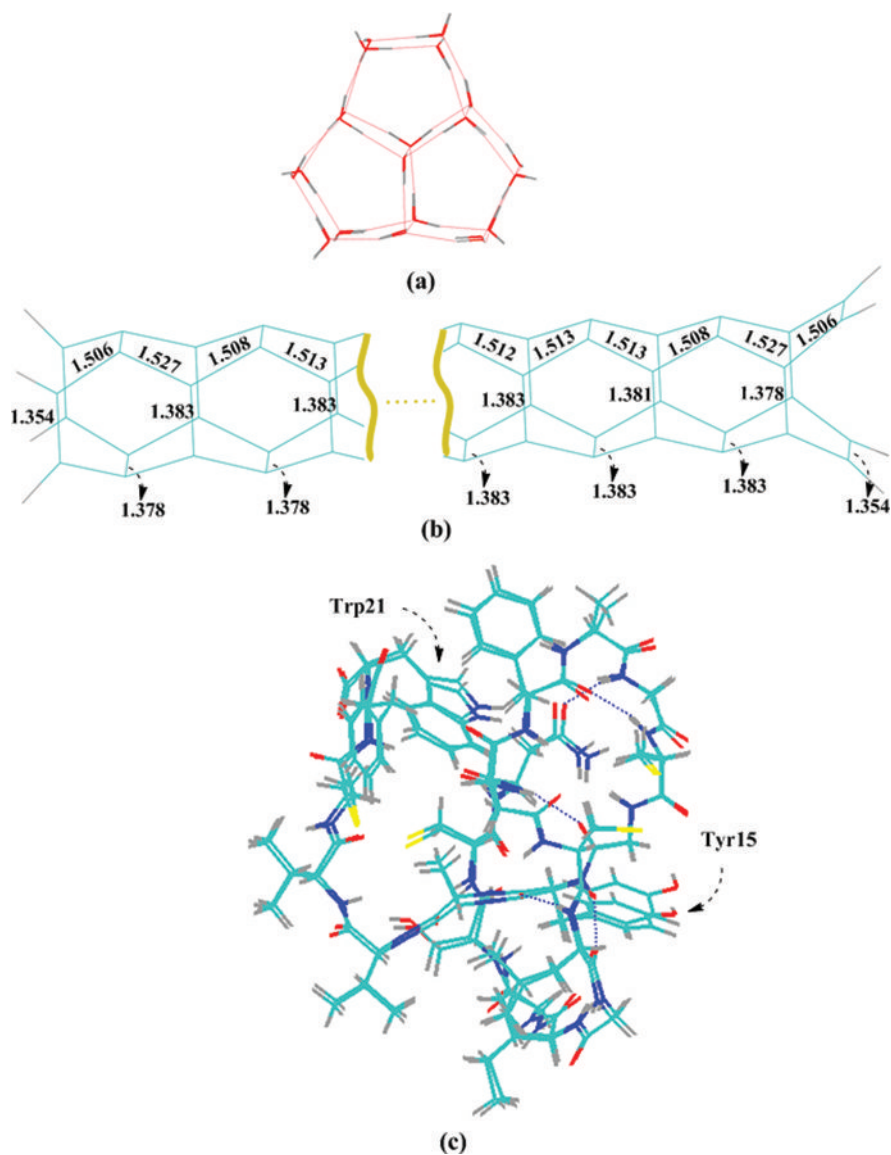


Fig. 7.6 (color online) Superposition of the optimized geometries obtained at various levels: (a) between conventional and GEBF(5) for W20; (b) between EBF(3)-3L and EBF(3)-4L for SWCNT-36 (length = 10.3 nm); (c) between GEBF(5) and GEBF(6) for 1RPB. Some selected bond lengths of SWCNT-36 are also shown in (b). The maximum bond length difference between EBF(3)-3L and EBF(3)-4L is less than 0.001 Å.

TABLE 7.3 Total Energies (at the Optimized Geometries) and RMSDs Between Conventional and GEBF-Optimized Structures or Between GEBF-Optimized Structures Obtained at Two Different Fragmentation Levels

System	Basis Set ^a	Method (Basis Set)	Basis Function ^b	E(optimized) ^c (a.u.)	RMSD (Å)
W20	cc-pVTZ (1160)	HF	1160	−1529.588 10	0.073
		GEBF(5)-HF	290	−1529.588 85 (−1529.586 39)	
SWCNT- 36	6-31G (2608)	EBF(3)- B3LYP(3L) ^b	664	−10967.589 21 (−10967.589 13)	0.0011
		EBF(3)- B3LYP(4L) ^b	880	−10967.588 83 (−10967.588 68)	
1RPB	6-31G (1626)	GEBF(5)-HF	391	−8535.135 67 (−8535.136 46)	0.34
		GEBF(6)-HF	455	−8535.123 33 (−8535.123 67)	

^aThe number of basis functions for the entire system.^bThe number of basis functions for the largest subsystem in GEBF or EBF calculations.^cSingle-point energies at each (G)EBF-optimized geometry calculated with the conventional approach is included in parentheses for comparison.^dTwo different fragmentation schemes are described in the text.

are quasi-one-dimensional, $\eta = 3$ or 4 can be chosen to be the default value. For systems with two- or three-dimensional structures, $\eta = 5$ or 6 can usually give reasonably accurate results. A more reliable method is to compare the difference between the GEBF(η) and GEBF($\eta + 1$) energies for a given geometry. If the difference is within a few millihartrees the GEBF(η) approach may be considered to be suitable for the system studied. This strategy has been employed here in the GEBF-based geometry optimizations of SWCNT-36 and 1RPB. Due to its linear scaling computational cost, the GEBF-based geometry optimizations at the HF or DFT levels are expected to possess wide applications in exploring the structures of large biological systems and nanosized tubes or particles.

7.3.3 Vibrational Frequencies, IR and Raman Intensities, and Thermochemistry Data

The applicability of the GEBF approach for computing vibrational frequencies and intensities as well as thermochemistry data has been validated previously for two medium-sized systems, (Gly)₁₂ and (H₂O)₂₈,⁶⁶ in which conventional results are available for comparison. Recently, the cardinality-guided molecular tailoring approach has also been applied to optimize the structures and do frequency calculations for several quite large systems, such as cholesterol, capreomycin, and a (H₃BO₃)₄₀ nanotube.⁶² Here we choose SWCNT-36 and 1RPB to show the performance of the GEBF (or EBF) approach in computing the vibrational spectra

of large molecules. As vibration frequencies from the standard approach are not available, our analysis is based on a comparison of GEBF or EBF results obtained with two fragmentation levels. For simplicity, only the highest 30 vibrational frequencies of each system are listed in Table 7.4 for comparison. For SWCNT-36, one can see that the RMSD and maximum difference for the highest 30 frequencies are 0.14 and 0.21 cm^{-1} , respectively. For 1RPB, the RMSD is 14.41 cm^{-1} , but the maximum difference is 44.62 cm^{-1} (the corresponding relative error is still less than 1.12%). As discussed above, this relatively larger difference for 1RPB arises from two slightly different local minima, which is understandable for a biological molecule with many local minima. In summary, the reasonable

TABLE 7.4 Highest 30 Vibrational Frequencies (cm^{-1}) at Different Fragmentation Levels for SWCNT-36 (at the B3LYP/6-31G Level) and 1RPB (at the HF/6-31G Level)

Normal Mode	SWCNT-36 (B3LYP/6-31G)		1RPB (HF/6-31G)	
	EBF(3)-3L	EBF(3)-4L	GEBF(5)	GEBF(6)
1	1523.76	1523.64	3392.29	3389.52
2	1527.36	1527.24	3400.42	3399.72
3	1531.02	1530.91	3406.60	3410.63
4	1534.78	1534.66	3439.00	3435.57
5	1538.60	1538.48	3693.05	3726.01
6	1542.41	1542.29	3747.02	3740.82
7	1546.14	1546.02	3762.99	3765.70
8	1549.77	1549.64	3775.04	3776.55
9	1553.24	1553.11	3794.89	3783.16
10	1556.51	1556.37	3798.12	3794.61
11	1559.56	1559.43	3802.37	3832.25
12	1562.34	1562.21	3816.59	3838.04
13	1564.84	1564.72	3825.73	3838.51
14	1567.04	1566.91	3839.80	3845.53
15	1568.88	1568.76	3848.43	3857.36
16	1570.37	1570.23	3850.94	3857.92
17	1571.44	1571.32	3851.86	3862.36
18	1572.09	1571.97	3854.75	3866.86
19	1588.62	1588.65	3856.89	3873.55
20	1588.62	1588.66	3879.05	3879.75
21	1601.92	1601.92	3885.49	3895.45
22	1601.92	1601.92	3896.76	3896.38
23	3205.54	3205.74	3899.45	3897.42
24	3205.55	3205.76	3908.51	3908.85
25	3206.34	3206.54	3909.60	3909.57
26	3206.35	3206.56	3917.69	3917.30
27	3228.68	3228.87	3922.29	3940.19
28	3228.68	3228.89	3925.55	3970.18
29	3230.20	3230.39	4044.61	4031.30
30	3230.21	3230.41	4055.61	4045.06

match between vibrational frequencies from two GEBF (or EBF) calculations with different fragmentation schemes shows that the GEBF (or EBF) approach is a very useful tool for computing ab initio quality vibrational frequencies of large systems. The GEBF vibrational frequency calculations for systems with several hundreds of atoms can be routinely done with a PC cluster with tens of computer nodes. For example, for 1RPB, the wall time of GEBF frequency calculations on the same platform as described above is 351 and 643 minutes for GEBF(5) and GEBF(6), respectively.

With the obtained vibrational frequencies, the calculated thermochemistry data for these two molecules are listed in Table 7.5. For zero-point vibrational energies, the differences from two GEBF (or EBF) calculations are less than 0.5 mHa for both systems. While for the thermal capacity and the entropy, the relative deviations are also less than 1.4%. For SWCNT-36, other thermochemistry quantities calculated at the room temperature and 1.0 atm are also very close to each other. For 1RPB, these thermochemistry quantities from two GEBF calculations have slightly larger differences, due mainly to their relative larger electronic energy difference (since two slightly different local minimum structures are obtained, as mentioned earlier). To conclude, the thermochemistry quantities from GEBF (or EBF) calculations are also encouraging.

The IR/Raman spectra calculated for SWCNT-36 and 1RPB are shown in Figs. 7.7 and 7.8, respectively. The bar spectra is Lorentzian convoluted using half-width at half-maximum (HWHM) 30 cm^{-1} (see Figs. 7.7c and 7.8c) and all the main peaks in the range are labeled. It is clear that the spectra from two fragmentation schemes match quite well with each other. For SWCNT-36, the positions of main peaks differ by less than 1 and 2 cm^{-1} in the IR and Raman spectra, respectively; and for 1RPB, the differences are no more than 6 cm^{-1} in both IR and Raman spectra. With the results with larger subsystems [EBF(3)-4L or GEBF(6)] as the reference, for SWCNT-36 the mean (or maximum) relative deviation between IR intensities of main peaks is less than 1% (or 1%), and the mean (or maximum) relative deviation between Raman intensities of main

TABLE 7.5 Thermochemistry Data Obtained from (G)EBF Calculations^a

Quantity	SWCNT-36 (B3LYP/6-31G)		1RPB (HF/6-31G)	
	EBF(3)-3L	EBF(3)-4L	GEBF(5)	GEBF(6)
ZPVE (a.u.)	1.804 46	1.803 76	2.493 61	2.493 16
E_0 (a.u.)	-10,965.785 29	-10,965.785 15	-8,532.642 06	-8,532.630 18
U (a.u.)	-10,965.665 68	-10,965.665 30	-8,532.500 60	-8,532.488 44
H (a.u.)	-10,965.664 74	-10,965.664 36	-8,532.499 66	-8,532.487 50
G (a.u.)	-10,965.919 72	-10,965.923 05	-8,532.848 86	-8,532.838 11
C_v (cal · mol ⁻¹ K ⁻¹)	613.873	614.253	526.144	526.900
S (cal · mol ⁻¹ K ⁻¹)	536.646	544.468	734.957	737.931

^a E_0 represents the sum of electronic and zero-point vibrational energies (ZPVE). U , H , G , C_v , and S denote the internal energy, enthalpy, Gibbs free energy, thermal capacity, and entropy, respectively. The enthalpies and Gibbs free energies are calculated at the room temperature and 1 atm.

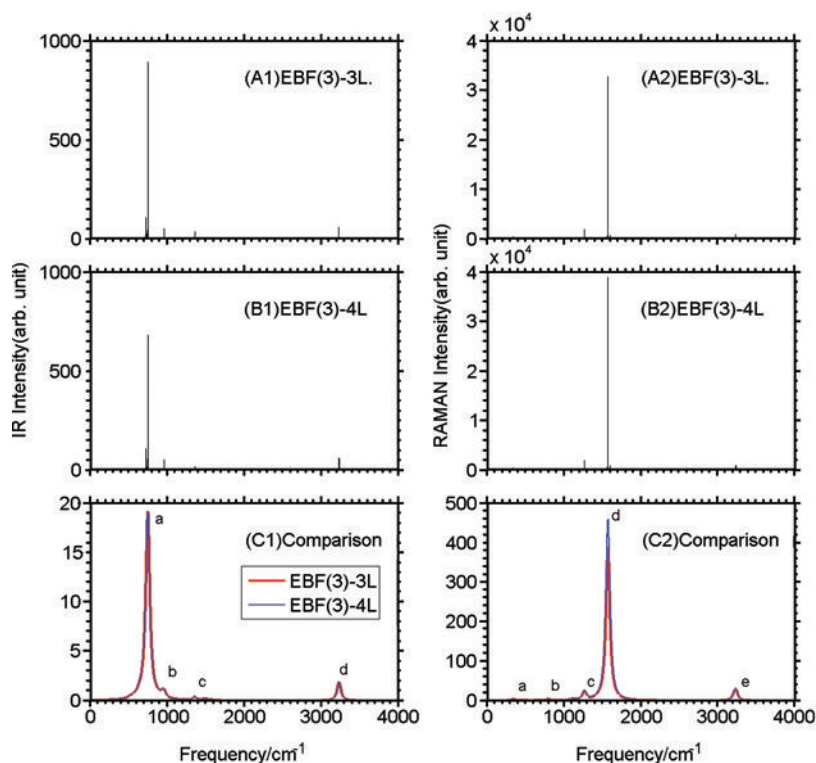


Fig. 7.7 (color online) IR (left panel) and Raman (right panel) spectra of SWCNT-36 at the B3LYP/6-31G level calculated with different fragmentation schemes: (a) EBF(3)-3L; (b) EBF(3)-4L; (c) superposition of both results using Lorentz broadening with HWHM = 30 cm^{-1} .

peaks is about 5% (or 15%). While in the 1RPB case, the mean (or maximum) relative deviation is 5% (or 17%) for the IR intensities and 5% (or 18%) for the Raman intensities. Relatively larger deviations for 1RPB are understandable because the location of some side chains is somewhat different in two minimum structures [from GEBF(5) to GEBF(6)]. From the results here and those reported earlier,⁶⁶ one can conclude that the IR or Raman intensities from the GEBF (or EBF) approach are also fairly reliable for large systems. Such information should be very useful for experimentalists to interpret and assign their vibration spectra measured for large nanosized clusters as well as large biomolecules.

7.3.4 Electric Properties

The GEBF-HF approach is applied for computing electric properties (the dipole moments and static polarizabilities) of four systems, which include dotriacontane, β -strand acetyl(ala)₁₀NH₂, α -conotoxin imi (PDB⁷⁹ id: 1CNL), and (H₂O)₃₂ (shown in Fig. 7.9). In GEBF calculations, $\eta = 5$ is set for the

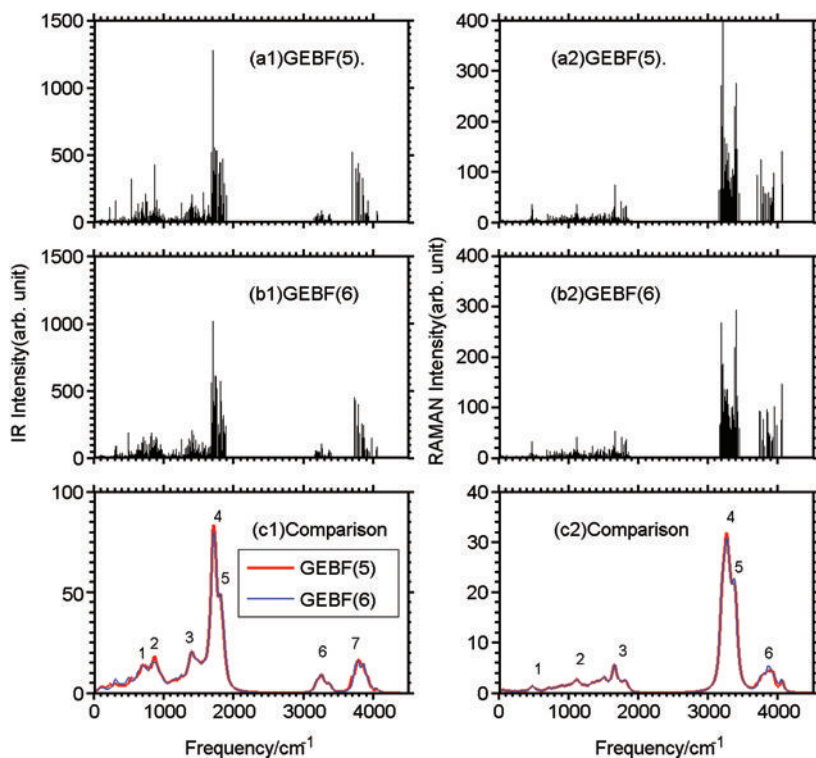


Fig. 7.8 (color online) IR (left panel) and Raman (right panel) spectra of 1RPB at the HF/6-31G level calculated with different fragmentation schemes: (a) GEBF(5); (b) GEBF(6); (c) superposition of both results using Lorentz broadening with HWHM = 30 cm⁻¹.

first three covalently bonded systems and $\eta = 6$ for (H₂O)₃₂, in which one water molecule is chosen as a fragment. For dotriacontane, each fragment includes two neighboring carbon atoms with their hydrogen atoms. For β -strand acetyl(ala)₁₀NH₂ and 1CNL, the fragmentation scheme is the same as that described in Section 7.3.1 for 1FGD. For all systems studied, both analytic and finite-field approaches are employed in the conventional and GEBF-HF calculations. The 6-311G(d) basis set is used for dotriacontane and β -strand acetyl(ala)₁₀NH₂, and the 6-31G(d,p) and 6-311G(d,p) basis sets are used for 1CNL and (H₂O)₃₂, respectively. In our previous work⁶⁵ we compared the performance of the GEBF-HF approach against the standard HF approach in computing the dipole moments and static polarizabilities with the analytic approach for some systems with small basis sets. Here the performance of the GEBF approach with some medium-sized basis sets is investigated, and results obtained with both the analytic and finite-field approaches are provided. It should be mentioned that for static polarizabilities only the averaged (or isotropic) polarizabilities, $\alpha = (\alpha_{xx} + \alpha_{yy} + \alpha_{zz})/3$ (called *polarizabilities* for convenience), are reported here.

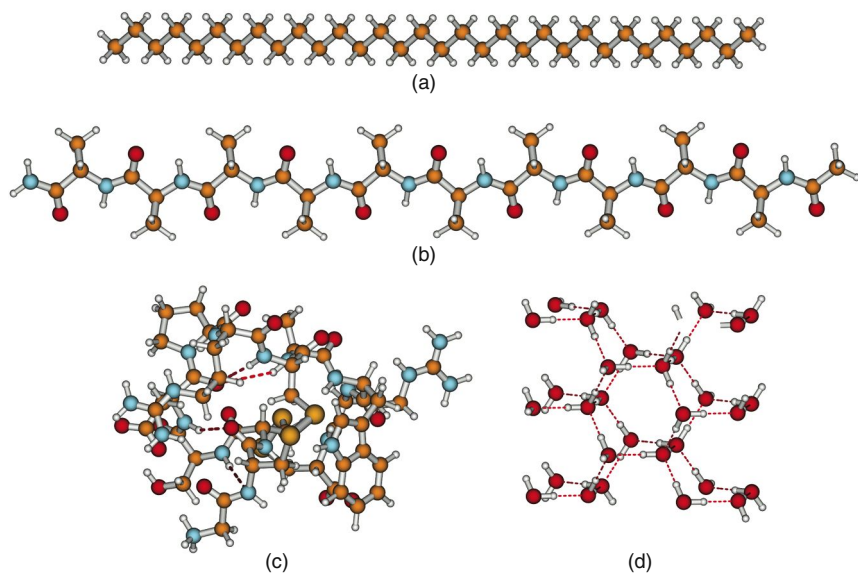


Fig. 7.9 (color online) Systems selected for electric property calculations: (a) dotriacontane; (b) β -strand acetyl(ala)₁₀NH₂; (c) α -conotoxin imi (1CNL); (d) (H₂O)₃₂.

The dipole moments and polarizabilities computed by conventional HF and GEBF-HF calculations are compared in Table 7.6. It can be seen that the results obtained with the analytic approach are almost identical to those with the finite-field approach [using Eqs. (7.8), (7.9), and (7.10)], and the GEBF results from both approaches agree well with those from conventional calculations. For simplicity we focus on a comparison of conventional and GEBF results obtained using the analytical approach. From Table 7.6 one can see that for the dipole moments the largest difference between the GEBF and conventional results is only 0.17 debye, and the relative deviations of the GEBF values with respect to the conventional values are less than 1.8% for the four systems under study. For example, for dotriacontane, the GEBF approach predicts a nearly zero value for its dipole moment, as expected (due to the C_{2h} symmetry of this molecule). For the hydrogen-bonded water cluster, (H₂O)₃₂, with the largest subsystem being only about one-fifth of the entire system, the dipole moment computed by the GEBF approach deviates from the conventional result only by 0.1 debye, while for static polarizabilities, the GEBF approach can achieve similar accuracy. The relative deviations of the GEBF values with respect to the conventional values are less than 0.6% for all four molecules. For example, the polarizability of 1CNL from the GEBF-HF calculation is 683.83 a.u., which is very close to 682.45 a.u. from the conventional HF calculation. Thus, the results presented here and reported previously⁶⁵ demonstrate that the GEBF approach is capable of providing highly accurate dipole moments and static polarizabilities for a wide range of molecules. The GEBF approach is expected to be applicable for computing other

TABLE 7.6 Dipole Moments (μ) and Average Polarizabilities (α) of Selected Systems from Conventional HF and GEBF-HF Calculations^a

Molecule	Basis Sets ^b	μ (Debye) ^c		α (a.u.) ^c	
		Conven.	GEBF	Conven.	GEBF
Dotriacontane	6-311G(d)	0.00	0.00	351.82	349.59
	(774/246)	(0.00)	(0.00)	(351.82)	(342.73)
β -Strand	6-311G(d)	22.62	22.48	431.16	429.86
Acetyl(ala) ₁₀ NH ₂	(1137/588)	(22.62)	(22.49)	(431.16)	(428.65)
α -Conotoxin imi (1CNL) (H ₂ O) ₃₂	6-31G(d,p)	9.27	9.10	682.45	683.83
	(1771/702)	(9.27)	(9.30)	(682.45)	(674.76)
	6-311G(d,p)	19.26	19.18	215.54	216.25
	(960/180)	(19.26)	(19.22)	(215.54)	(213.57)

^aResults from the analytic and finite-field approaches are listed.

^bThe number of basis functions for the entire system and the largest subsystems are included in parentheses (listed before and after the slash, respectively).

^cThe values in parentheses are from the finite-field method.

electric properties or magnetic properties of large molecules. More applications will be reported in the near future.

7.4 CONCLUSIONS

In this chapter we have reviewed the recent development and applications of the energy-based fragmentation approach and its generalized version, which are used to approximate the energies and molecular properties of large molecules or molecular clusters at the full quantum mechanical level. The basic principles and implementation details of this approach for evaluating total energies, energy derivatives, and molecular properties are described in some detail. The most favorable feature of this approach is that the GEBF (with EBF as a special case) approach can be combined directly with existing electronic structure packages (with little additional effort) to perform full quantum mechanical calculations for large systems at various theoretical levels (HF, DFT, CCSD, etc.). By applying the GEBF approach to investigate relative energies of different conformers, optimized structures, vibrational frequencies and intensities (as well as thermochemistry quantities), and some electric properties (dipole moments and polarizabilities) for a number of medium-sized and large molecules, we demonstrate that the GEBF approach provides a cost-effective and reliable way to evaluate structures and properties of large systems at the ab initio level. With the MPI parallel technique, the GEBF approach can be employed to study systems with hundreds of atoms at the full quantum mechanical level on an ordinary PC workstation with tens of computer nodes. Thus, many large and complicated systems that were studied with force-field methods or semiempirical quantum chemistry methods can now be studied with the more accurate GEBF approach at various ab initio levels.

It should be mentioned that the GEBF approach has its inherent limitations. This approach (or other fragmentation approaches) is applicable only for closed-shell systems with localized electronic structures or for open-shell systems with localized unpaired electrons. For example, although the GEBF approach is able to treat systems containing small conjugated components (e.g., benzene), it cannot be used to treat strongly delocalized conjugated systems. In our present implementation of the GEBF approach, fragments are always formed by cutting the single bonds, which should be defined manually at the very beginning, and fragments remain unchanged during the geometry optimizations. An automatic fragmentation procedure is possible and will be implemented in the near future. On the other hand, we notice that there are still many applications to be explored for the GEBF approach. For example, the GEBF approach can be employed to compute more electric properties corresponding to third (or higher)-energy derivatives or magnetic properties. Another interesting direction is to develop GEBF-based *ab initio* molecular dynamics or Monte Carlo simulations for complex systems.

Supporting Information Available

Cartesian coordinates of all compounds studied in this work, and fragmentation schemes of selected systems, can be found on the book Web site.

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7.5 APPENDIX: ILLUSTRATIVE EXAMPLE OF THE GEBF PROCEDURE

1. *Define fragments.* Here we choose the dotriacontane molecule, $C_{32}H_{66}$, as the target molecule. In Fig. 7.A1a, the structure of this molecule, as well as its fragmentation scheme, is shown (the Cartesian coordinates are given in the supporting information). We define every two carbon units (together with the hydrogens bonded to them) as a fragment; thus, there is a total of 16 fragments. The numbering of heavy atoms for all fragments is given explicitly in Table 7.A1.

2. *Construct subsystems.* The principle of constructing subsystems from information on fragments is discussed in the text. For this system, the maximum

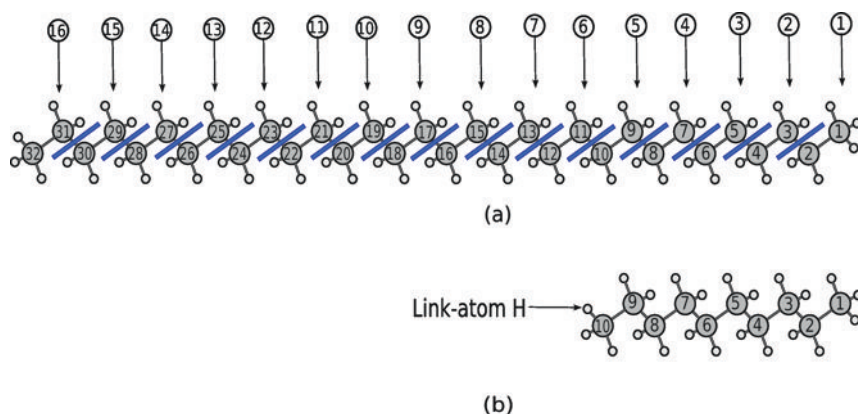


Fig. 7.A1 (color online) Fragmentation scheme for dotriacontane: (a) all fragments; (b) one subsystem ($k = 1$).

TABLE 7.A1 Numbering of the Heavy Atoms

Fragment	Heavy Atoms	Fragment	Heavy Atoms
1	1–2	9	17–18
2	3–4	10	19–20
3	5–6	11	21–22
4	7–8	12	23–24
5	9–10	13	25–26
6	11–12	14	27–28
7	13–14	15	29–30
8	15–16	16	31–32

number of fragments (η) is set to 5. For each subsystem, hydrogens are added as link atoms for saturating dangling bonds whenever necessary, the positions of these added hydrogens are determined following the rules given in our previous work.⁶⁵ For the target system we can construct 12 primitive subsystems ($k = 1, 2, \dots, 12$ in Table 7.A2) and 11 derivative subsystems ($k = 13, 14, \dots, 23$ in Table 7.A2). The components and coefficient of each subsystem are also given in the table. The first subsystem ($k = 1$) is also shown schematically in Fig. 7.A1b.

3. Perform subsystem calculations. We perform HF/6-311G(d) calculations for all subsystems and the entire system using GAUSSIAN03 package. Generally, the GEBF calculation includes two steps. First, we determine the NPA charge of each atom in an iterative way (usually, two iterations) by carrying out calculations on primitive subsystems only. The details on how to determine NPA charges are discussed in the text. The NPA charges obtained are

TABLE 7.A2 Components (Only Heavy Atoms) and Total Energies of All Subsystems, and the Total Energies Obtained with the GEBF-HF and Conventional HF Methods at the 6-311G(d) Level

k	C_k	Fragments	Heavy Atoms	$\tilde{E}_k$ (a.u.)
1	1	1–5	1–10	–392.455 04
2	1	2–6	3–12	–392.626 16
3	1	3–7	5–14	–392.627 09
4	1	4–8	7–16	–392.626 92
5	1	5–9	9–18	–392.626 86
6	1	6–10	11–20	–392.626 86
7	1	7–11	13–22	–392.626 86
8	1	8–12	15–24	–392.626 86
9	1	9–13	17–26	–392.626 92
10	1	10–14	19–28	–392.627 09
11	1	11–15	21–30	–392.626 16
12	1	12–16	23–32	–392.455 04
13	–1	2–5	3–10	–314.607 85
14	–1	3–6	5–12	–314.608 78
15	–1	4–7	7–14	–314.608 61
16	–1	5–8	9–16	–314.608 56
17	–1	6–9	11–18	–314.608 55
18	–1	7–10	13–20	–314.608 55
19	–1	8–11	15–22	–314.608 55
20	–1	9–12	17–24	–314.608 56
21	–1	10–13	19–26	–314.608 61
22	–1	11–14	21–28	–314.608 78
23	–1	12–15	23–30	–314.607 85
GEBF-HF				–1250.484 64
Conventional HF				–1250.483 74

listed in Table 7.A3. Second, after NPA charges are available, each subsystem is calculated in the presence of background charges generated by all atoms outside this subsystem. The energies of all subsystems are also listed in Table 7.A2.

4. *Evaluate the total energy in the GEBF approach.* Within the GEBF approach, the total energy is calculated by

$$E_{\text{tot}} \approx \sum_k^M C_k \tilde{E}_k - \left(\sum_k^M C_k - 1 \right) \sum_A \sum_{B>A} \frac{Q_A Q_B}{R_{AB}}$$

One can find that the GEBF-HF energy differs from the conventional HF energy by less than 1 mHa (see Table 7.A2). It should be mentioned that other properties can be calculated similarly as a linear combination of corresponding properties of all subsystems.

TABLE 7.A3 NPA Charges of All Atoms Used in the GEBF Approach

Atom	Element	Charge	Atom	Element	Charge	Atom	Element	Charge
1	C	-0.478580	34	H	0.152910	67	H	0.151950
2	C	-0.306510	35	H	0.151950	68	H	0.151950
3	C	-0.312180	36	H	0.151950	69	H	0.151950
4	C	-0.307940	37	H	0.165180	70	H	0.151950
5	C	-0.306800	38	H	0.159400	71	H	0.151950
6	C	-0.306840	39	H	0.159400	72	H	0.151950
7	C	-0.306660	40	H	0.154180	73	H	0.151950
8	C	-0.306660	41	H	0.154180	74	H	0.151950
9	C	-0.306660	42	H	0.152640	75	H	0.151950
10	C	-0.306660	43	H	0.152640	76	H	0.151940
11	C	-0.306670	44	H	0.152960	77	H	0.151940
12	C	-0.306670	45	H	0.152960	78	H	0.151940
13	C	-0.306670	46	H	0.152910	79	H	0.151940
14	C	-0.306670	47	H	0.152910	80	H	0.151920
15	C	-0.306670	48	H	0.152550	81	H	0.151920
16	C	-0.306670	49	H	0.152550	82	H	0.151920
17	C	-0.306670	50	H	0.151880	83	H	0.151920
18	C	-0.306670	51	H	0.151880	84	H	0.151910
19	C	-0.306670	52	H	0.151910	85	H	0.151910
20	C	-0.306670	53	H	0.151910	86	H	0.151880
21	C	-0.306670	54	H	0.151920	87	H	0.151880
22	C	-0.306670	55	H	0.151920	88	H	0.152550
23	C	-0.306660	56	H	0.151920	89	H	0.152550
24	C	-0.306660	57	H	0.151920	90	H	0.152910
25	C	-0.306660	58	H	0.151940	91	H	0.152960
26	C	-0.306660	59	H	0.151940	92	H	0.152960
27	C	-0.306840	60	H	0.151940	93	H	0.152640
28	C	-0.306800	61	H	0.151940	94	H	0.152640
29	C	-0.307940	62	H	0.151950	95	H	0.154180
30	C	-0.312180	63	H	0.151950	96	H	0.154180
31	C	-0.306510	64	H	0.151950	97	H	0.165180
32	C	-0.478580	65	H	0.151950	98	H	0.159400
33	H	0.159400	66	H	0.151950			

REFERENCES

1. Alsenoy, C. V.; Yu, C.-H.; Peeters, A.; Martin, J. M. L.; Schäfer, L. *J. Phys. Chem. A* **1998**, *102*, 2246.
2. Scuseria, G. E. *J. Phys. Chem. A* **1999**, *103*, 4782.
3. Inaba, T.; Tahara, S.; Nisikawa, N.; Kashiwagi, H.; Sato, F. *J. Comput. Chem.* **2005**, *26*, 987.
4. Xu, H.; Ma, J.; Chen, X.; Hu, Z.; Huo, K.; Chen, Y. *J. Phys. Chem. B* **2004**, *108*, 4024.

5. Gao, B.; Jiang, J.; Liu, K.; Wu, Z.; Lu, W.; Luo, Y. *J. Comput. Chem.* **2008**, *29*, 434.
6. Brothers, E. N.; Izmaylov, A. F.; Scuseria, G. E. *J. Phys. Chem. C* **2008**, *112*, 1396.
7. Strout, D. L.; Scuseria, G. E. *J. Chem. Phys.* **1995**, *102*, 8448.
8. Strain, M. C.; Scuseria, G. E.; Frisch, M. J. *Science* **1996**, *271*, 51.
9. White, C. A.; Head-Gordon, M. *J. Chem. Phys.* **1994**, *101*, 6593.
10. Schwegler, E.; Challacombe, M. *J. Chem. Phys.* **1996**, *105*, 2726.
11. Ochsenfeld, C.; White, C. A.; Head-Gordon, M. *J. Chem. Phys.* **1998**, *109*, 1663.
12. Burant, J. C.; Strain, M. C.; Scuseria, G. E.; Frisch, M. J. *J. Chem. Phys. Lett.* **1996**, *248*, 43.
13. Kudin, K. N.; Scuseria, G. E. *Phys. Rev. B* **2000**, *61*, 16440.
14. Stratmann, R. E.; Scuseria, G. E.; Frisch, M. J. *J. Chem. Phys. Lett.* **1996**, *257*, 213.
15. Millam, J. M.; Scuseria, G. E. *J. Chem. Phys.* **1997**, *106*, 5569.
16. Li, X.; Millam, J. M.; Scuseria, G. E.; Frisch, M. J.; Schlegel, H. B. *J. Chem. Phys.* **2003**, *119*, 7651.
17. Leczyszynski, J. *Computational Chemistry: Review of Current Trends*, World Scientific, Singapore, 2002.
18. Pulay, P. *J. Chem. Phys. Lett.* **1983**, *100*, 151.
19. Saebø, S.; Pulay, P. *Annu. Rev. Phys. Chem.* **1993**, *44*, 213.
20. Hampel, C.; Werner, H.-J. *J. Chem. Phys.* **1996**, *104*, 6286.
21. Schütz, M.; Hetzer, G.; Werner, H.-J. *J. Chem. Phys.* **1999**, *111*, 5691.
22. Schütz, M.; Werner, H.-J. *J. Chem. Phys.* **2001**, *114*, 661.
23. Werner, H.-J.; Manby, F. R.; Knowles, P. J. *J. Chem. Phys.* **2003**, *118*, 8149.
24. Ayala, P. Y.; Scuseria, G. E. *J. Chem. Phys.* **1999**, *110*, 3660.
25. Scuseria, G. E.; Ayala, P. Y. *J. Chem. Phys.* **1999**, *111*, 8330.
26. Ayala, P. Y.; Kudin, K. N.; Scuseria, G. E. *J. Chem. Phys.* **2001**, *115*, 9698.
27. Almlöf, J. *J. Chem. Phys. Lett.* **1991**, *181*, 319.
28. Head-Gordon, M.; Maslen, P. E.; White, C. A. *J. Chem. Phys.* **1998**, *108*, 616.
29. Nakao, Y.; Hirao, K. *J. Chem. Phys.* **2004**, *120*, 6375.
30. Christiansen, O.; Manninen, P.; Jørgensen, P.; Olsen, J. *J. Chem. Phys.* **2006**, *124*, 084103.
31. Förner, W.; Ladik, J.; Otto, P.; Eížek, J. *J. Chem. Phys.* **1985**, *97*, 251.
32. Li, S.; Ma, J.; Jiang, Y. *J. Comput. Chem.* **2002**, *23*, 237.
33. Li, S.; Shen, J.; Li, W.; Jiang, Y. *J. Chem. Phys.* **2006**, *125*, 074109.
34. Saebø, S.; Baker, J.; Wolinski, K.; Pulay, P. *J. Chem. Phys.* **2004**, *120*, 11423.
35. Azhary, A. E.; Rauhut, G.; Pulay, P.; Werner, H.-J. *J. Chem. Phys.* **1998**, *108*, 5185.
36. Rauhut, G.; Werner, H.-J. *J. Phys. Chem. Chem. Phys.* **2001**, *3*, 4853.
37. Schütz, M.; Werner, H.-J.; Lindh, R.; Manby, F. R. *J. Chem. Phys.* **2004**, *121*, 737.
38. Yang, W. *Phys. Rev. Lett.* **1991**, *66*, 1438.
39. Yang, W.; Lee, T.-S. *J. Chem. Phys.* **1995**, *103*, 5674.
40. Exner, T. E.; Mezey, P. G. *J. Phys. Chem. A* **2004**, *108*, 4301.
41. He, X.; Zhang, J. Z. H. *J. Chem. Phys.* **2005**, *122*, 031103.
42. Chen, X.; Zhang, Y.; Zhang, J. Z. H. *J. Chem. Phys.* **2005**, *122*, 184105.

43. Chen, X.; Zhang, J. Z. H. *J. Chem. Phys.* **2006**, *125*, 044903.
44. Li, W.; Li, S. *J. Chem. Phys.* **2005**, *122*, 194109.
45. Gu, F. L.; Aoki, Y.; Korchowiec, J.; Imamura, A.; Kirtman, B. *J. Chem. Phys.* **2004**, *121*, 10385.
46. Kitaura, K.; Ikeo, E.; Asada, T.; Nakano, T.; Uebayasi, M. *Chem. Phys. Lett.* **1999**, *313*, 701.
47. Fedorov, D. G.; Kitaura, K. *J. Chem. Phys.* **2004**, *120*, 6832.
48. Fedorov, D. G.; Ishida, T.; Uebayasi, M.; Kitaura, K. *J. Phys. Chem. A* **2007**, *111*, 2722.
49. Fedorov, D. G.; Kitaura, K. *J. Phys. Chem. A* **2007**, *111*, 6904.
50. Morita, S.; Sakai, S. *J. Comput. Chem.* **2001**, *22*, 1107.
51. Sakai, S.; Morita, S. *J. Phys. Chem. A* **2005**, *109*, 8424.
52. Hirata, S.; Valiev, M.; Dupuis, M.; Xantheas, S. S.; Sugiki, S.; Sekino, H. *Mol. Phys.* **2005**, *103*, 2255.
53. Li, W.; Li, S. *J. Chem. Phys.* **2004**, *121*, 6649.
54. Li, S.; Li, W.; Fang, T. *J. Am. Chem. Soc.* **2005**, *127*, 7215.
55. Deev, V.; Collins, M. A. *J. Chem. Phys.* **2005**, *122*, 154102.
56. Collins, M. A.; Deev, V. A. *J. Chem. Phys.* **2006**, *125*, 104104.
57. Bettens, R. P. A.; Lee, A. M. *J. Phys. Chem. A* **2006**, *110*, 8777.
58. Lee, A. M.; Bettens, R. P. A. *J. Phys. Chem. A* **2007**, *111*, 5111.
59. Jiang, N.; Ma, J.; Jiang, Y. *J. Chem. Phys.* **2006**, *124*, 114112.
60. Li, W.; Fang, T.; Li, S. *J. Chem. Phys.* **2006**, *124*, 154102.
61. Ganesh, V.; Dongare, R. K.; Balanarayan, P.; Gadre, S. R. *J. Chem. Phys.* **2006**, *125*, 104109.
62. Rahalkar, A. P.; Ganesh, V.; Gadre, S. R. *J. Chem. Phys.* **2008**, *129*, 234101.
63. Dahlke, E. E.; Truhlar, D. G. *J. Chem. Theory Comput.* **2007**, *3*, 46.
64. Dahlke, E. E.; Truhlar, D. G. *J. Chem. Theory Comput.* **2007**, *3*, 1342.
65. Li, W.; Li, S.; Jiang, Y. *J. Phys. Chem. A* **2007**, *111*, 2193.
66. Hua, W.; Fang, T.; Li, W.; Yu, J.-G.; Li, S. *J. Phys. Chem. A* **2008**, *112*, 10864.
67. Li, S.; Li, W. *Annu. Rep. Prog. Chem. Sect. C* **2008**, *104*, 256.
68. Li, W.; Dong, H.; Li, S. *Progress in Theoretical Chemistry Physics*, Vol. 18, *Frontiers in Quantum Systems in Chemistry Physics*, Wilson, S., Grout, P. J., Maruani, J., Delgado-Barrio, G., and Piecuch, P., Eds., Springer-Verlag, Berlin, 2008, pp. 289–299.
69. Zhang, D. W.; Zhang, J. Z. H. *J. Chem. Phys.* **2003**, *119*, 3599.
70. Zhang, D. W.; Xiang, Y.; Zhang, J. Z. H. *J. Phys. Chem. B* **2003**, *107*, 12039.
71. Gadre, S. R.; Shirsat, R. N.; Limaye, A. C. *J. Phys. Chem.* **1994**, *98*, 9165.
72. Pulay, P. *Adv. Chem. Phys.* **1987**, *69*, 241.
73. Amos, R. D.; Rice, J. E. *Comput. Phys. Rep.* **1989**, *10*, 147.
74. The criterion for hydrogen bonds $X-H\cdots Y$ in our calculations is $r_{H\cdots Y} \leq 2.9\text{\AA}$, $r_{X\cdots Y} \leq 3.5\text{\AA}$, $\angle X-H\cdots Y \geq 120^\circ$.
75. Foster, J. P.; Weinhold, F. *J. Am. Chem. Soc.* **1980**, *102*, 7211.
76. Reed, A. E.; Weinstock, R. B.; Weinhold, F. *J. Chem. Phys.* **1985**, *83*, 735.

77. Hurst, J. B.; Dupuis, M.; Clementi, E. *J. Chem. Phys.* **1989**, *89*, 385.
78. Kamada, K.; Ueda, M.; Nagao, H.; Tawa, K.; Sugino, T.; Shmizu, Y.; Ohta, K. *J. Phys. Chem. A* **2000**, *104*, 4723.
79. Berman, H. M.; Westbrook, J.; Feng, Z.; Gilliland, G.; Bhat, T. N.; Weissig, H.; Shindyalov, I. N.; Bourne, P. E. *Nucleic Acids Res.* **2000**, *28*, 235.
80. Case, D. A.; Cheatham, T. E., III; Darden, T.; Gohlke, H.; Luo, R.; Merz, K. M., Jr.; Onufriev, A.; Simmerling, C.; Wang, B.; Woods, R. J. *J. Comput. Chem.* **2005**, *26*, 1668.
81. Ponder, J. W. *Tinker Software Tools for Molecular Design*, 4.2 ed., <http://dasher.wustl.edu/tinker>, 2004.
82. Jørgensen, W. L.; Chandrasekhar, J.; Madura, J. D.; Impey, R. W.; Klein, M. L. *J. Chem. Phys.* **1983**, *79*, 926.
83. <http://www.pci.tu-bs.de/agbauerecker/Sigurd/WaterClusterDatabase/>.
84. Li, S.; Li, W.; Fang, T.; Ma, J.; Jiang, Y. *LSQC Program, version 1.1*, Nanjing University, Nanjing, China, 2006.
85. Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; et al. In *Gaussian 03, Revision D.01*, Gaussian, Inc., Wallingford, CT, 2004.
86. Schmidt, M. W.; Baldridge, K. K.; Boatz, J. A.; Elbert, S. T.; Gordon, M. S.; Jensen, J. H.; Koseki, S.; Matsunaga, N.; Nguyen, K. A.; Su, S. J.; Windus, T. L.; Dupuis, M.; Montgomery, J. A. *J. Comput. Chem.* **1993**, *14*, 1347.
87. Li, W.; Piecuch, P.; Gour, J. R.; Li, S. *J. Chem. Phys.* **2009**, *131*, 114109.
88. Frechet, D.; Guitton, J. D.; Herman, F.; Faucher, D.; Helynck, G.; du Sorbier, B. M.; Ridoux, J. P.; James-Surcouf, E.; Vuilhorgne, M. *Biochemistry* **1994**, *33*, 42.
89. Farkas, O.; Schlegel, H. B. *J. Chem. Phys.* **1999**, *111*, 10806.
90. Schlegel, H. B. *J. Comput. Chem.* **1982**, *3*, 214.
91. Pulay, P.; Fogarasi, G. *J. Chem. Phys.* **1992**, *96*, 2856.
92. Leach, A. R. *Molecular Modelling: Principles and Applications*, Addison Wesley Longman, London, 1996.
93. Structures available at <http://itcc.nju.edu.cn/itcc/shuhua/Mol/>.
94. <http://www-unix.mcs.anl.gov/mpi/>.

8 MNDO-like Semiempirical Molecular Orbital Theory and Its Application to Large Systems

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In this chapter we describe modern MNDO-like semiempirical theory and its application either to very large molecules or to a very large number of smaller ones. We use the term *MNDO-like* to describe methods that use variations of the original MNDO¹ and MNDO/d²⁻⁶ techniques. This covers essentially all commonly used techniques, which all use the original multipole formulation for the two-electron integrals, and many of the original MNDO approximations. We first outline the theory of LCAO-SCF methods in general, followed by a more detailed discussion of the neglect of diatomic differential overlap (NDDO) approximation and the MNDO technique. We discuss individual Hamiltonians and their parameterization and describe the strengths of these remarkably powerful methods and their application to large systems.

8.1 BASIC THEORY

8.1.1 LCAO-SCF Theory

The two approximations *linear combination of atomic orbitals* (LCAO) and *self-consistent field* (SCF) form the core of modern (MNDO-like) semiempirical molecular orbital theory. They have been described in many standard textbooks

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but are important for understanding MNDO-like techniques and so are outlined briefly here.

We can write the Hamiltonian for a molecule that consists of M nuclei and N electrons as

$$H = \sum_{i=1}^N \frac{1}{2} \nabla_i^2 + \sum_{A=1}^M \frac{1}{2M_A} \nabla_A^2 - \sum_{i=1}^N \sum_{A=1}^M \frac{Z_A}{R_{Ai}} + \sum_{i=1}^N \sum_{j>i}^N \frac{1}{r_{ij}} + \sum_{A=1}^M \sum_{B>A}^M \frac{Z_A Z_B}{R_{AB}} \quad (8.1)$$

where the indices i and j run over the electrons and A and B over the nuclei. The individual terms that make up the Hamiltonian are defined in Table 8.1.

We make use of the Born–Oppenheimer approximation,⁷ which in turn uses the fact that the nuclei move so much more slowly than the electrons that the former can, in effect, be regarded as being stationary. This reduces the kinetic energy of the nuclei to zero and makes the nucleus–nucleus repulsion term a constant, so that they can be neglected in the electronic Hamiltonian:

$$H = H_{\text{nuclear}} + H_{\text{electronic}} = H_{\text{nuclear}} + \sum_{i=1}^N \frac{1}{2} \nabla_i^2 - \sum_{i=1}^N \sum_{A=1}^M \frac{Z_A}{R_{Ai}} + \sum_{i=1}^N \sum_{j>i}^N \frac{1}{r_{ij}} \quad (8.2)$$

TABLE 8.1 Definitions of the Individual Terms in Eq. (8.1)

Term	Definition	Variables
$\sum_{i=1}^N \frac{1}{2} \nabla_i^2$	Kinetic energy of the electrons	∇_i = the first derivative of the position of electron i with respect to time (its velocity)
$\sum_{A=1}^M \frac{1}{2M_A} \nabla_A^2$	Kinetic energy of the nuclei (zero within the Born–Oppenheimer approximation)	∇_A = the first derivative of the position of nucleus A with respect to time (its velocity)
$\sum_{i=1}^N \sum_{A=1}^M \frac{Z_A}{R_{Ai}}$	Nucleus–electron attraction	Z_A is the nuclear charge of atom A and R_{Ai} is the distance between atom A and electron i
$\sum_{i=1}^N \sum_{j>i}^N \frac{1}{r_{ij}}$	Electron–electron repulsion	r_{ij} is the distance between electrons i and j
$\sum_{A=1}^M \sum_{B>A}^M \frac{Z_A Z_B}{R_{AB}}$	Nucleus–nucleus repulsion (constant within the Born–Oppenheimer approximation)	R_{AB} is the distance between atoms A and B

where the total Hamiltonian H has now been separated into nuclear and electronic components.

This allows us to write the total energy as the sum of the nuclear repulsion energy and the electronic energy defined by the Hamiltonian $H_{\text{electronic}}$:

$$E_{\text{total}} = E_{\text{electronic}} + \sum_{A=1}^M \sum_{B>A}^M \frac{Z_A Z_B}{R_{AB}} \quad (8.3)$$

Thus, we “only” need to calculate the electronic energy, which according to the Schrödinger equation⁸ is obtained from the electronic wavefunction. The electronic wavefunction $\Phi_{\text{electronic}}$ in turn is a function of the positions and spins of the N electrons of the system:

$$\Phi_{\text{electronic}} = \Phi(x_1, x_2, x_3, \dots, x_N) \quad \text{where } x_i = \{\mathbf{r}_i, \omega_i\} \quad (8.4)$$

Here $\mathbf{r}_i$ denotes the (vector) position of electron i and ω_i its spin. Thus, the wavefunction is a function of $4N$ variables (the three coordinates and the spin per electron).

To cut a long story short, we can only solve Schrödinger’s equation for systems with only one electron, so we are forced to introduce approximations. The first of these is the SCF (also known as *mean-field* or *Hartree–Fock*) approximation.^{9,10} Basically, rather than solving the Schrödinger equation for many particles, we approximate the many-particle solution in terms of many one-electron wavefunctions, which are solvable. This means that we make the approximation that

$$H_{\text{electronic}} \approx \sum_{i=1}^N h_i \quad (8.5)$$

where h_i is the one-electron Hamiltonian for electron i . This leads to the Hartree product, Φ^{HP} , which is an approximation for a many-electron wavefunction, $\Phi_{\text{electronic}}$:

$$\Phi^{\text{HP}}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N) = \chi_1(\mathbf{x}_1)\chi_2(\mathbf{x}_2) \cdots \chi_N(\mathbf{x}_N) \quad (8.6)$$

In Eq. (8.6), χ_i are the spin orbitals, which are one-electron wavefunctions. The Schrödinger equation based on the Hartree approximation becomes

$$H\Phi^{\text{HP}} = E\Phi^{\text{HP}}, \quad (8.7)$$

so that the eigenvalues ε_i of the one-electron wavefunctions χ_i can be summed to give the electronic energy:

$$E_{\text{electronic}} = \sum_{i=1}^N \varepsilon_i \quad (8.8)$$

This would all be fine except for one significant complication. Because electrons are fermions (i.e., they have spin), they must obey the Pauli exclusion principle,¹¹ which can be formulated as the antisymmetry principle, which states that the wavefunction must be antisymmetric with respect to the exchange of any two electrons. Fock's contribution was to point out that the Hartree product does not obey the antisymmetry principle. Slater¹² later pointed out that the wavefunction suggested by Fock can be expressed as a determinant now known as a Slater determinant, Φ^{Slater} :

$$\Phi^{\text{Slater}} = \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_1(\mathbf{x}_1), \chi_2(\mathbf{x}_1), \dots, \chi_N(\mathbf{x}_1) \\ \chi_1(\mathbf{x}_2), \chi_2(\mathbf{x}_2), \dots, \chi_N(\mathbf{x}_2) \\ \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\ \chi_1(\mathbf{x}_N), \chi_2(\mathbf{x}_N), \dots, \chi_N(\mathbf{x}_N) \end{vmatrix} \quad (8.9)$$

The prefactor $1/\sqrt{N!}$ is simply a normalization constant.

This is the Hartree–Fock (or SCF) wavefunction, but the question remains as to how we define the spin orbitals χ_i . This is where the almost universal LCAO approximation, introduced by Erich Hückel,¹³ comes into play. Hückel's idea was that molecular orbitals (in our case the χ_i introduced above) can be represented as a linear combination of atomic orbitals appropriate for the constituent atoms. For a system constituted of N atomic orbitals (AOs),

$$\chi_i = \sum_{j=1}^{N_{\text{AOs}}} c_j^i \varphi_j \quad (8.10)$$

where c_j^i is the coefficient of atomic orbital φ_j in molecular orbital χ_i , and the coefficients are normalized so that $\sum_{j=1}^{N_{\text{AOs}}} (c_j^i)^2 = 1$.

We still cannot solve for the wavefunction directly, even using the SCF and LCAO approximations. This is where the variational principle, which says that there are no solutions with a lower energy than the correct wavefunction, comes into play. Solutions are generally found by starting with a set of guessed molecular orbitals χ_i and iterating until the energy converges to its minimum value and the electron density does not vary. We discuss this algorithm in more detail below.

8.1.2 Implications of LCAO-SCF Theory

LCAO-SCF theory is remarkably successful but has two limitations that we need to discuss in order to understand MNDO-like theories better. The first is a consequence of the SCF approximation and is known as *electron correlation*.

Physically, the introduction of the Hartree product [Eq. (8.6)] means that the electrons do not feel each other individually. Instead, each electron feels the electron density (but not the instantaneous positions) of the others. This means that the individual electrons are not given the opportunity to avoid each other

instantaneously, which they would obviously do because they are negatively charged. Thus, the SCF approximation means that the electron–electron repulsion is overestimated. This effect, which is purely a consequence of the SCF approximation, is known as *dynamic correlation*.¹⁴ A second type of correlation (nondynamic or static correlation) has also been defined. It is a consequence of using only a single Slater determinant to describe the wavefunction. Although most “normal” molecules can be described very well using a single Slater determinant, some (such as diradicals) cannot. This is essentially because the wavefunction cannot be described adequately by a single scheme in which a single set of molecular orbitals is occupied by zero, one, or two electrons. This second type of correlation is very different from the first and not as easily treated. However, the implicit treatment of dynamic correlation in MNDO-like theories is poorly appreciated and will be discussed below.

The second implication of the LCAO-SCF approximations concerns the limitations placed on the wavefunction by the atomic orbitals used to form the MOs. Although the LCAO approximation is very instinctive and actually forms the basis of our qualitative understanding of bonding effects,¹⁵ it nevertheless has no physical basis. It is very convenient for calculations, but we can also describe MOs as combinations of non-atom-centered functions or simply as numerical grids. The LCAO approach, however, does bring some limitations. We can only describe wavefunctions that are linear combinations of the atomic orbitals [which are usually called the basis set in *ab initio* and density functional theory (DFT) calculations]. Current MNDO-like semiempirical techniques use single-valence basis sets. This means that each atomic orbital in the valence shell is represented by only one basis function. This, in turn, means that the size of the orbital is fixed, although in reality some valence orbitals are more or less diffuse than others. This is a serious limitation in *ab initio* and DFT calculations, but appears to be less serious in MNDO-like techniques. The one possible exception is hydrogen, for which a single valence 1s orbital is not ideal in some bonding situations.¹⁶

8.1.3 Neglect of Diatomic Differential Overlap

The NDDO approximation is perhaps the key simplification made in MNDO-like semiempirical MO theories. Interestingly, although some adverse effects of other approximations have been identified (see below), the NDDO approximation appears to be extremely robust and does not lead to identifiable systematic errors. In full (*ab initio*) Hartree–Fock theory, calculating the electron–electron repulsion requires that all integrals of the type $(\mu\nu|\lambda\sigma)$ (i.e., all integrals in which the indices μ , ν , λ , and σ vary from 1 to N_{AOs} , the number of atomic orbitals) be calculated. This means that a very large number of integrals (formally $N_{\text{AOs}}^4/4$, if we ignore symmetry) must be calculated and processed in every iteration of the SCF procedure. The NDDO approximation sets all integrals $(\mu\nu|\lambda\sigma)$ to zero in which either atomic orbitals μ and ν or λ and σ are on different atoms. The combinations $\mu\nu$ and $\lambda\sigma$ are known as *charge distributions*, so that the NDDO approximation can also be expressed as meaning that we only consider integrals between charge distributions $\mu\nu$ and $\lambda\sigma$ situated on single, but not necessarily

the same, atoms. Thus, the NDDO approximation reduces the problem of calculating and using the two-electron integrals (i.e., those needed for calculating the electron–electron repulsion) from one of four centers to one of only two; we calculate only one- and two-center two-electron integrals and ignore three- and four-center two-electron integrals.

Having reduced the number of integrals to be calculated, we need an efficient technique to calculate them. *Ab initio* and DFT calculations often use basis sets based on Gaussian functions because these are particularly suitable for calculating the integrals. Gaussian orbitals have the form

$$\varphi_l^m(\mathbf{r}) = Y_l^m e^{-\zeta r^2} \quad (8.11)$$

where Y_l^m is the angular part (a spherical harmonic function) of the orbital with principal quantum number l and angular momentum quantum number m . The expression $e^{-\zeta r^2}$ describes the radial behavior of the wavefunction, where ζ is the exponent that governs how fast the wavefunction falls off with increasing distance r from the nucleus. Despite their almost universal use as atom-centered basis sets in *ab initio* and DFT techniques, Gaussian functions are far from ideal. Because the distance from the nucleus is squared in the exponent, the wavefunction falls off far faster than it should do and also does not describe the wavefunction at the nucleus correctly. A far better choice would be Slater orbitals, which have the form

$$\varphi_l^m(\mathbf{r}) = Y_l^m e^{-\zeta|r|} \quad (8.12)$$

However, the two-electron integrals are very expensive to calculate for Slater orbitals, so that they are not used as often as Gaussians, despite their inherent advantages.

MNDO-like techniques use Slater-type orbitals, but must therefore resort to a fast, approximate method for calculating the two-electron integrals. This is the multipole approach introduced with MNDO¹ and extended to d-orbitals for MNDO/d.² In this approximation, the interactions between Slater orbitals are approximated as interactions between electrostatic monopoles, dipoles, and quadrupoles, which allows the integrals to be calculated very effectively and with reasonable accuracy. The multipole model has been used to calculate the molecular electrostatic potential for MNDO-like wavefunctions, and the definitions for all the multipoles for the 45 charge distributions that arise with an s-, p-, d-basis set have been listed.¹⁷

An important approximation in standard MNDO-like theories is that the basis set (the atomic orbitals) is assumed to be orthogonal (i.e., the orbitals have zero overlap with each other). This saves an initial orthogonalization step in the SCF calculation, which would slow semiempirical calculations considerably. Jorgensen et al.¹⁸ reintroduced this orthogonalization into MNDO and found that the resulting method (NO-MNDO) performed as well as later, more highly parameterized, methods and gave improvements in two problem areas: the rotational

barriers about C—C single bonds and the relative stabilities of branched and unbranched hydrocarbons. NO-MNDO require about twice the CPU time needed for a standard MNDO calculation. A better known solution to the orthogonalization problem is to add an orthogonalization correction that mimics the effects of the orthogonalization step at less cost in CPU time. This is the basis of the OM n ($n = 1$ to 3) methods introduced by Thiel and co-workers.^{19–22} These methods are probably the most sophisticated MNDO-like techniques available.

One of the most difficult areas in MNDO-like theories is the treatment of the nucleus–nucleus repulsion. What appears initially in Eq. (8.1) and Table 8.1 to be a very simple Coulomb repulsion is, in fact, a fairly complex entity in MNDO-like theories. The problem arises from the fact that the Coulomb interactions in MNDO-like theories are not all treated equally well. Whereas we treat the nucleus–nucleus repulsion exactly in Eq. (8.1), introducing the NDDO approximation leads to some neglect of Coulomb terms involving the electrons. Specifically, the long-range behavior of the electron–electron and nucleus–electron integrals is not correct, so that the simple, physically correct nucleus–nucleus repulsion term in Eq. (8.1) would lead to a net repulsion between neutral atoms or molecules at distances outside their van der Waals radii. Thus, an artificial screening effect must be introduced. In MNDO, the nucleus–nucleus repulsion term E_{AB} becomes

$$E_{AB}^{\text{MNDO}} = Z_A Z_B (s_A s_A | s_B s_B) (1 + e^{-\alpha_A R_{AB}} + e^{-\alpha_B R_{AB}}) \quad (8.13)$$

where the integral is treated in the same way as the electron–electron integrals and the two constants α_A and α_B are parameters specific to the elements A and B. However, MNDO is not able to reproduce hydrogen bonds, an effect that was,²³ probably erroneously,¹⁶ attributed to the nucleus–nucleus repulsion being too strong. Therefore, this term was modified by the addition of up to four Gaussian terms in MNDO/H.²³ These Gaussian terms were later adopted for other methods (see below), but lead to some artifacts. The corresponding expression for E_{AB} becomes

$$E_{AB} = E_{AB}^{\text{MNDO}} + \frac{Z_A Z_B}{R_{AB}} \left(\sum_i a_{A,i} e^{-b_{A,i} (R_{AB} - c_{A,i})^2} + \sum_j a_{B,j} e^{-b_{B,j} (R_{AB} - c_{B,j})^2} \right) \quad (8.14)$$

where there are i Gaussian functions for atom A and j for atom B. The variables a , b , and c are parameterized for each element [A and B in Eq. (8.14)] and each individual Gaussian function [$1 - i$ and $1 - j$ in Eq. (8.14)].

Use of these Gaussian functions is not without hazard because they can lead to spurious minima²⁴ and is generally undesirable because the function introduce a large number of additional parameters for each element. A solution that has been found more practical and yields very good results is to introduce two-center terms in to the nucleus–nucleus repulsion, as suggested originally for AM1(d)

by Voityuk and Rösch.²⁵ The nucleus–nucleus repulsion term then becomes

$$E_{AB} = E_{AB}^{\text{MNDO}}(1 + \delta_{AB}e^{-\alpha_{AB}R_{AB}}) \quad (8.15)$$

where δ_{AB} and α_{AB} are parameters specific to the pair of elements AB. In addition, it is common to use distance-dependent expressions for metal–hydrogen nucleus–nucleus interactions.

The problem with all these corrections is that they essentially represent fixes to a fundamental deficiency of current MNDO-like theories. In addition, they all represent modifications to a two-center potential and can adversely affect the parameterization of other such interactions because the effects of the two potentials are not independent of each other.

8.1.4 SCF Iterations and Pseudodiagonalization

Figure 8.1 is a standard flow diagram for a semiempirical MO SCF iteration algorithm. Given a set of Cartesian coordinates, the number of electrons, and the spin multiplicity, the program first assigns atomic orbitals (the basis set) to the atoms and calculates the one-electron matrix, which contains all the interactions except the electron–electron term. In order to proceed, an initial guess density matrix is required. In standard semiempirical MO programs, this initial guess consists of simply dividing the electrons evenly over the available atomic orbitals. More sophisticated initial guesses, such as extended Hückel MOs, could be envisaged but would involve an extra diagonalization. The two-electron contribution is then added to the one-electron matrix to give the Fock matrix. This two-electron contribution depends on the density matrix and the two-electron integrals, which are generally precalculated and stored in memory. The Fock matrix is then diagonalized to give a new set of MOs, from which a new density matrix can be generated. The total energy and the density matrix are then tested

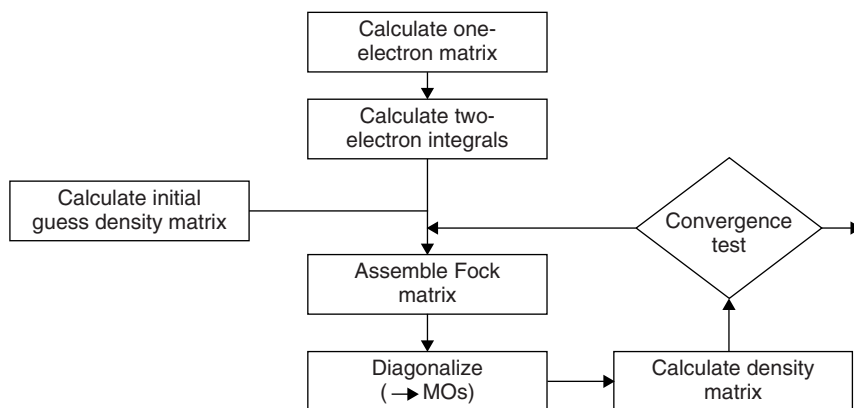


Fig. 8.1 Standard semiempirical MO SCF flow diagram.

for convergence by comparison with the last cycle, and if they have not yet converged, another SCF cycle is started using the new density matrix. The energy improves from cycle to cycle and the density converges steadily until they are both static within predefined thresholds, after which the program exits the SCF cycles. In practice, additional features, such as interpolation schemes, damping, or level shifting, are often included to improve convergence, but Fig. 8.1 gives the basics of the algorithm.

However, because the other steps of the calculation are so fast, the diagonalization of the Fock matrix typically takes up approximately 50% of the CPU time for an implementation such as that shown in Fig. 8.1. This is often not appreciated because the diagonalization is a relatively minor component of the calculation for *ab initio* or DFT calculations. Modern semiempirical programs therefore do not perform full diagonalizations in every SCF cycle but, rather, switch to *pseudodiagonalization*²⁶ as soon as the SCF converges far enough. This is shown in Fig. 8.2.

The pseudodiagonalization procedure is key to the remaining discussion and therefore is described in detail. The principle of pseudodiagonalization is that the MO eigenvectors are updated but not their eigenvalues. However, as the differences between eigenvalues are needed for the pseudodiagonalization procedure, full diagonalizations must be performed until the eigenvalues have settled to more or less constant values. This is shown in Fig. 8.2. Full diagonalizations are performed until a given threshold (usually, convergence on the density matrix, although convergence of the eigenvalues would be more relevant), after which the pseudodiagonalization can be used until the SCF criteria are met. A final full diagonalization must be performed after convergence to obtain the final eigenvalues and eigenvectors. Using the pseudodiagonalization procedure rather than full diagonalizations at every cycle does not slow convergence and speeds up the calculation by approximately a factor of 2. Just as important, the

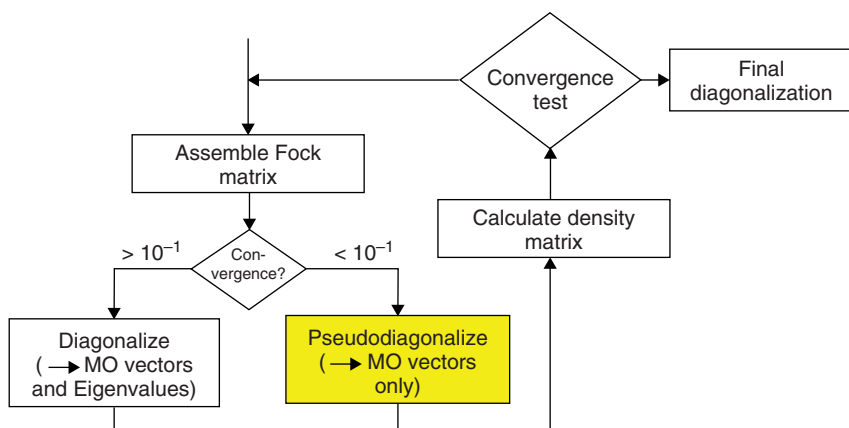


Fig. 8.2 Cyclic section of the SCF iteration algorithm with pseudodiagonalization.

pseudodiagonalization procedure has properties that can be exploited for alternative SCF iteration schemes, as outlined below. Note that separate calculation of the eigenvalues and pseudodiagonalization can be used to replace the full diagonalizations in Fig. 8.2. Alternatively, if the initial guess is close enough to the final solution, no initial full diagonalizations are needed.

The principle behind pseudodiagonalization is that improvements in the eigenvectors for the occupied MOs must come from mixing with virtual MOs. Essentially, there is nothing to win by mixing two occupied MOs. Therefore, the first step is to calculate the occupied-virtual block of the Fock matrix, $\mathfrak{F}$, in the current MO basis:

$$\mathfrak{F} = c_o^+ F c_v \quad (8.16)$$

where the subscripts o and v denote the occupied and virtual blocks, respectively, c are the current eigenvector coefficients, and F is the Fock matrix. Large elements of $\mathfrak{F}$ indicate strong interactions between occupied and virtual MOs, which must be removed by mixing the two. The mixing is achieved by a Givens rotation.

For an updated occupied eigenvector $\tilde{c}_o$,

$$\tilde{c}_o = x_{ov}c_o - \sqrt{(1 - x_{ov}^2)}c_v \quad (8.17)$$

where c_o and c_v are the coefficients of the relevant occupied and virtual eigenvectors, respectively, and x_{ov} is the rotation angle between the two eigenvectors. The expression for the corresponding updated virtual eigenvector is

$$\tilde{c}_v = x_{ov}c_v + \sqrt{(1 - x_{ov}^2)}c_o \quad (8.18)$$

Thus, the Givens rotations simply mix an occupied MO with a virtual MO with which it interacts strongly. However, the rotation angle x_{ov} must be determined before the rotation can be carried out. This is achieved using what is essentially a first-order perturbation theory expression:

$$x_{ov} = \frac{\mathfrak{F}_{ov}}{\epsilon_o - \epsilon_v} \quad (8.19)$$

where $\mathfrak{F}_{ov}$ is the element of $\mathfrak{F}$ that connects the occupied and virtual orbitals o and v , and ϵ_o and ϵ_v are the eigenvalues of these two orbitals, respectively. This expression explains the need for relatively constant eigenvalues (or eigenvalues calculated explicitly from the eigenvectors) before using the pseudodiagonalization, as these determine the rotation angles.

The importance of the pseudodiagonalization procedure is that it allows us to select which orbitals to mix in a very transparent way. This feature is used, for example, in the MOZYME algorithm (see below). For normal-sized molecules, one possible implementation is to calculate $\mathfrak{F}$ and to select a certain proportion

of the largest elements (the details of this step vary from implementation to implementation) in order to carry out the rotations between the orbitals connected by these elements. After testing for convergence and calculating the new density and Fock matrices, $\mathfrak{S}$ is calculated for the new Fock matrix and the process is repeated until convergence.

8.1.5 Dispersion

MNDO-like semiempirical MO techniques exhibit the weakness also found for ab initio Hartree–Fock and DFT: that weak (van der Waals) interactions (dispersion) are not reproduced. This problem is more severe than might seem at first sight because, in addition to the obvious intermolecular interaction energies, the intramolecular dispersion energies, which become very significant for large molecules such as those now treated routinely by MNDO-like methods, are also affected. The solution that was introduced for ab initio Hartree–Fock²⁷ and has also been used for DFT^{28–30} has been to add a classical two-center potential with a damping function for short distances to the DFT Hamiltonian. A similar correction has been added to SCC-DFTB calculations (see Chapter 9).³¹ Such corrections are very successful, but suffer from the inherent problem for MNDO-like methods that they represent an additional two-center potential that can lead to linear dependencies with the nucleus–nucleus potential function. This is not a problem if the dispersion term is added after parameterization, as in OMn-D,³² although some methods have been reported in which a dispersion potential was parameterized together with the remaining parameters.³³ A more consistent way to treat this problem is to modify the existing two-center potential (the nucleus–nucleus repulsion potential) to include the effects of dispersion. This is the approach used by PM6,³⁴ for which the core–core term is given by

$$E_{AB}^{\text{PM6}} = E_{AB}^{\text{MNDO}}(1 + \delta_{AB}e^{-\alpha_{AB}(R_{AB}+0.0003R_{AB}^6)}) \quad (8.20)$$

This modification of Voityuk and Rösch’s formula [Eq. (8.15)] behaves very similarly at short distances, but at distances of 3 Å and larger gives a noticeably smaller repulsion. This, together with an additional correction to take account of the nonvalence electrons (which are neglected in MNDO-like methods), leads to better performance and behavior similar to that expected from a method that includes dispersion.

Each of these modifications assumes that the dispersion interaction attributable to a given atom is isotropic. Even if we accept the hypothesis that dispersion interactions can be assigned on an atom–atom basis, this is probably not a good approximation, for example, for sp^2 -hybridized carbon atoms or atoms with lone pairs. One Ansatz that takes this effect into account also has the advantage that the dispersion term can be separated from other two-center potentials because it is based on (and parameterized for) the polarizability. In the early 1970s, Rinaldi and Rivail introduced a variational treatment for calculating molecular electronic polarizabilities using MNDO-like methods.³⁵ This approach leads to very fast calculations but is not very accurate. However, Schürer et al.³⁶ were

able to show that parameterizing the atomic multipole integrals (three per non-hydrogen element), rather than using the analytical values, gave very accurate molecular electronic polarizabilities. Furthermore, this technique lends itself to (arbitrarily) partitioning the molecular electronic polarizability into atomic, or even atomic-orbital, contributions.³⁷ The “atomic polarizability tensors” thus obtained can be used in conjunction with the London equation³⁸ and a damping function at short distances to provide a dispersion correction to MNDO.³⁹

8.1.6 Need for Linear-Scaling Methods

Using current, readily available computers, conventional semiempirical SCF methods are limited to systems of only a few hundred atoms; above that, the computational effort becomes prohibitive. This limit is a direct consequence of the use of matrix algebra for solving the SCF equations, for which several operations, such as inversion and diagonalization, scale as the third power of the size of the system. By using special methods, such as pseudodiagonalization, this effort can be minimized, but elimination of the N^3 dependency is impossible when matrix algebra is used. Before larger systems could be studied, alternatives to matrix algebra methods had to be developed; two of the more successful are the divide-and-conquer linear-scaling method, and the localized molecular orbital method MOZYME.

8.1.7 Divide-and-Conquer Linear Scaling

Given that the N^3 dependency cannot be eliminated, the computational effort required to solve the SCF for a large system can be reduced by splitting the system into smaller ones, which can then be solved separately. Thus, if a system of N atoms is split into m equal parts, each of the m parts will require a computational effort approximately proportional to $(N/m)^3$. That is, the total effort is reduced by a factor of m^2 . This is the basis for the divide-and-conquer (D&C) method.⁴⁰ Once special care is taken to ensure that the joins between the various parts are handled correctly, the results are almost indistinguishable from those obtained using exact matrix algebra methods.⁴¹ The computational effort involved in the D&C method scales linearly with the size of system, which makes it suitable for modeling phenomena in very large species, including protein-protein interactions.⁴²

8.1.8 Localized Orbital SCF

For a self-consistent field to exist, it is a necessary and sufficient condition that all Fock integrals involving occupied and virtual molecular orbitals be zero. On the assumption that a rough approximation to the electronic structure of a molecule is provided by its Lewis structure, the conditions necessary for an SCF provide a guide for moving from the simple Lewis structure to the optimized electronic structure. This is the premise for MOZYME⁴³: Starting with a Lewis structure represented by localized molecular orbitals (LMOs) on one or at most two atoms,

in order to generate an NSCF it is sufficient to eliminate the Fock terms between these LMOs and the nearby virtual LMOs. For each pair of LMOs, this operation is very fast and can be performed using a 2×2 Givens rotation. The operation is carried out on every occupied LMO and every nearby virtual LMO.

A result of this operation is to move the system in the direction of the SCF. However, because each Givens rotation modifies the occupied and virtual LMOs, the result of one annihilation rotation is to cause some matrix elements that had been eliminated by earlier Givens rotations now to become nonzero. This means that the process of annihilating occupied-virtual LMO interactions must be repeated. Over the first few complete sweeps of Givens rotations, the size of the LMOs, represented by the number of atoms on which the LMO has significant intensity, increases rapidly, and then tapers off as the system converges toward self-consistency. To the degree that each complete set of annihilation steps results in the system moving closer to the energy minimum, the MOZYME method is similar to the conventional matrix algebra procedure. Indeed, when an SCF is achieved, MOZYME and conventional matrix algebra give rise to identical electron density distributions.

Surprisingly, the MOZYME method is intrinsically more arithmetically stable than the conventional method. Using conventional methods, an SCF sometimes fails to form—the charge distribution simply oscillates from iteration to iteration. This propensity increases as the HOMO-LUMO energy gap decreases. When the gap is very small, the polarizabilities of the HOMO and LUMO become very large, and autoregenerative charge fluctuations effectively prevent an SCF from forming. In conventional methods the MOs are eigenvectors; therefore, the HOMO-LUMO gap is irreducibly small. By contrast, when LMOs are used, the HOMO-LUMO gap is at or near its maximum possible value, and the polarizability of the HOMO is correspondingly small. One practical consequence is that, in general, the MOZYME procedure requires fewer iterations to achieve an SCF.

Using the MOZYME technique, the computational effort scales approximately as $N^{1.4}$, and much larger systems can be studied, with the upper limit now being on the order of 15,000 atoms.⁴⁴ Because having a starting Lewis structure is a prerequisite, the MOZYME method is limited to systems for which a Lewis structure can be defined. At present, only closed-shell systems are allowed, so while ferrocene, $\text{Fe}^{\text{II}}(\text{Cp})_2$, and crystalline potassium chromate, $\text{K}_2\text{Cr}^{\text{VI}}\text{O}_4$, can be modeled, no open-shell system (e.g., $[\text{Cr}^{\text{III}}(\text{H}_2\text{O})_6]^{3+}$) can be run. Similarly, systems with extended π -conjugation cannot be treated using the MOZYME or D&C techniques because individual orbitals are delocalized across the boundaries between subsystems or cannot be localized.

8.2 PARAMETERIZATION

Many of the equations used in semiempirical methods contain adjustable parameters. Within the broad family of NDDO^{45,46} methods, the main difference between the various methods lies in the values of these parameters. Provided that the set

of approximations is sufficiently flexible and physically realistic, the accuracy of a semiempirical method depends on precisely two quantities: the accuracy and range of the reference data used in determining the values of the parameters and the thoroughness of optimization of the parameters.

8.2.1 Data

The set of reference data used in parameterization must satisfy several criteria: It obviously must be as accurate as possible, it must represent a wide range of chemical systems and properties, and it must be manipulated easily by the parameter optimization program. Several useful collections of reference data are available, such as the NIST databases of atomic energy levels,⁴⁷ reference heats of formation,⁴⁸ and atomic⁴⁷ and molecular ionization potentials,⁴⁹ and the Cambridge Structural Database⁵⁰ for molecular geometries. Despite the large amount of available experimental reference data, important gaps or deficiencies exist. For the organic elements C, H, N, and O, this is not a problem, but for less popular elements, particularly transition metals, such as Sc and Tc, there is a paucity of reliable reference data. Where data are missing or are incomplete, the few data that do exist can be augmented by using reference data generated from the results of high-level (i.e., highly accurate) theoretical calculations. Of course, since the objective of a semiempirical method is to model the real world, great care must be taken to maximize confidence in the accuracy of all calculated reference data. In the most recent parameterization, the training set consisted of over 10,000 individual data representing over 9000 separate species.

8.2.2 Parameterization Techniques

Although parameterization might initially appear to be a complicated process, in principle it is really very simple⁵¹: Given a set of reference data, x^{ref} , and a set of adjustable parameters, P_i , the values of the parameters are modified so as to minimize the root-mean-square difference between the data predicted and the reference data. That is, given

$$S = \sum_i (x_i - x_i^{\text{ref}})^2 \quad (8.21)$$

parameters are considered optimized when $\partial S / \partial P_i = 0$ and $\partial^2 S / \partial P_i^2 > 0$ for all parameters.

The first step is to take all the various reference data (dipole moments, bond lengths, heats of formation, etc.) and render them dimensionless, so that they can be manipulated using standard mathematical tools. Default weighting factors for this operation are shown in Table 8.2.

In the early days of parameter optimization, making decisions regarding the initial values for the various parameters for the different elements was difficult⁵²; in that groundbreaking work, there was no precedent to refer to. A real risk at that time was that an incorrect choice could result in the parameters converging

TABLE 8.2 Weighting Factors for Reference Data

Reference Data	Weight
ΔH_f^0	1.0 mol · kcal ⁻¹
Bond length	0.7 Å mol · kcal ⁻¹
Angle	0.7 Å mol · kcal ⁻¹
Dipole	20 debye ⁻¹
Ionization potential	10 V ⁻¹

on a false minimum. This risk was not hypothetical; computers available in the 1970s were much less powerful than now and only a small number of reference data could be used in a parameter optimization. This increased the probability that spurious minima might be encountered. Over time, and by dint of hard work, these issues were resolved, and now, more than 30 years later, there is a wealth of knowledge of suitable starting values for parameter optimization.

8.2.3 Methods and Hamiltonians

In *ab initio* work, different methods (e.g., Hartree–Fock and density functional) can be defined using quantum mechanical terms such as the one- and two-electron operators and instantaneous correlation. These terms are a natural consequence of the underlying quantum theory. Within a given method, a balance can be struck between computational effort and accuracy. In part, this is achieved by the choice of basis set—a small set would give rise to a faster but less accurate method, and vice versa. *Ab initio* methods are thus defined by two quantities: the method and the basis set.

The NDDO-based semiempirical methods, on the other hand, use similar sets of approximations and are best distinguished by the values of the parameters. Minor differences do exist in the approximations, with most of these having to do with the core–core terms. Thus, the oldest NDDO method, MNDO,¹ had the simplest core–core term; AM1,⁵³ PM3,^{54,55} and RM1⁵⁶ had terms added to mimic the van der Waals attraction; and in PM6³⁴ diatomic parameters were used. These changes were the results of attempts to make the set of approximations more realistic.

That the main difference between the methods lies in the values of the parameters can be readily shown. If the original MNDO set of approximations were used and the parameters for H, C, N, and O were reoptimized using modern reference data and modern optimization techniques, the accuracy of the resulting method would be significantly higher than that of the original MNDO method. This is not to disparage the quality of parameterization in MNDO (when it was first developed, it represented a large improvement over even older methods); rather, it demonstrates how the accuracy of methods can be increased as the quality of parameter optimization improves.

NDDO methods are best defined by the set of approximations and the set of parameters. This definition is easily seen to be necessary: If the set of parameters

is not specified, the three methods AM1, PM3, and RM1, methods of very different accuracies, would become indistinguishable.

8.2.3.1 MNDO First published in 1977, MNDO^{1,52} is the oldest of the NDDO methods. At that time it represented a large increase in accuracy over the then-popular MINDO/3.⁵⁷ There were two reasons for this increase in accuracy: For the first time, a semiempirical method could represent the lone-pair/lone-pair interaction of the type found in hydrazine and in hydrogen peroxide (hitherto, such interactions had simply been ignored) and also for the first time reference data based on experimental results for molecular systems were used in the parameter optimization. Parameters for H, C, N, and O were optimized using data on 34 compounds. The much-increased accuracy of MNDO resulted in its becoming instantly popular. But as it was applied to more and more species, various systematic errors became apparent, the most serious of these being the almost complete absence of a hydrogen bond.

8.2.3.2 AM1 Hydrogen bonds are much weaker than covalent bonds and can best be represented by three terms: an electrostatic, a covalent, and a third term variously called the instantaneous correlation, dispersion, or van der Waals interaction. MNDO included the electrostatic and covalent terms, but not the VDW term. To mimic the effect of the VDW term, during the development of AM1 the core–core interaction in MNDO was modified by the addition of simple Gaussian functions to provide a weak attractive force. This extra stabilization allowed hydrogen bonds to form. Parameters for H, C, N, and O were again optimized, now using a larger set of reference data, and the resulting AM1 method was published in 1985.⁵³ Over the following few years, parameters were optimized for many more main-group elements. Each new element was parameterized without changing the parameters for the original AM1 elements. This resulted in a piecemeal method—the values of the parameters depended on the sequence in which the parameterizations were done.

At the time the parameters in the AM1 method were being optimized, two different philosophical approaches were explored. One, advocated by Michael Dewar, was to guide the progress of the optimization by using chemical knowledge. At the same time, by carefully selecting the reference data used in the parameterization, the size of the training data set could be kept to a minimum. The quality of such a method could then be determined by its accuracy and predictive power; that is, the ability of the method to predict the properties of systems not used in the training set. As Dewar had an encyclopedic knowledge in this field, this approach had obvious merit. The other approach, advocated by one of us (J.S.), was to provide the parameter optimization procedure with a wide range of reference data, in the hope that if enough data were provided, the rules of chemistry would be implicitly provided to the parameter optimization. In the development of AM1, the first of these two approaches was used.

8.2.3.3 PM3 In contrast to the approach used in AM1, a large amount of reference data was used in the training set for the development of PM3.^{54,55} In

the initial parameter optimization, parameters for 12 elements, H, C, N, O, F, Al, Si, P, S, Cl, Br, and I, were optimized simultaneously. Also, in contrast to the development of AM1, no external constraints based on chemical experience were applied. When PM3 was completed, it was found that the average errors for common properties such as heats of formation were lower than those in AM1, but the troubling question of predictive power of PM3 versus AM1 became more difficult to answer. Possibly because of this, although PM3 was widely used, it was never as widely used as was AM1. PM3 was soon extended to include most,⁵⁸ and ultimately all,⁵⁹ of the main group. As with AM1, the later parameterizations were carried out using fixed values for the elements that had previously been parameterized.

In the initial PM3 work, parameters for all 12 elements were optimized simultaneously, this eliminating any error due to undesired restrictions on the values of the parameters. At the same time, the training set increased in both size and quality. Each entry in it was checked for consistency with the other data. Errors due to incomplete parameterization and inconsistent reference data were minimized. Despite all this, the average unsigned error in the heat of formation remained stubbornly and unacceptably large.

8.2.3.4 PM6 In 2000, in an attempt to improve the accuracy of a method for modeling systems containing molybdenum, Voityuk and Rösch²⁵ proposed using diatomic core–core parameters. This modification was tested using various pairs of elements in the first PM3 set. In every case, the average error decreased. The next step was obvious: to replace the original MNDO core–core term with a simple function that used diatomic parameters. A few other minor modifications were made to the core–core term, mainly to cater for highly specific interactions such as the acetylenic triple bond. Parameters for the whole of the main group, plus Zn, Cd, and Hg (three elements that behave like main-group elements), 42 elements in all, were then optimized simultaneously. This was followed by the remaining 27 transition metals of periods 4, 5, and 6, and the fourteenth lanthanide, Lu. Two other approaches had been considered, but these were not completed (PM4) or not published (PM5), so the new method was named PM6.

A reasonable question to ask is: How does the accuracy of PM6, the most recent semiempirical method, compare with standard *ab initio* methods? This can best be answered by comparing standard quantities. In PM6, the accuracy of prediction of heats of formation of common organic compounds is somewhat better than those predicted by B3LYP DFT calculations using the 6-31G(d) basis set,⁶⁰ which in turn is significantly better than Hartree–Fock, using the same basis set. Unfortunately, ΔH_f^0 is the only property for which PM6 is superior to B3LYP, for geometries it is somewhat worse, and for ionization potentials and dipole moments—purely electronic properties—it is significantly worse.

There is a reason for this initially surprising high accuracy relative to standard *ab initio* HF and DFT methods, methods that require considerably more computational effort than PM6. Semiempirical methods are parameterized to reproduce experimental reference data, which by definition take into account

all possible phenomena. Many of these phenomena (e.g., instantaneous correlation) are extremely difficult to calculate *ab initio*, but in semiempirical methods their effects are simply absorbed into the values of the parameters, and, in turn, when the methods are used in modeling chemical systems, the effects are reproduced. This benefit comes at a price: In semiempirical methods, each atomic basis set is normally referred to by using the standard principal quantum number (PQN), but because the associated parameters are optimized using experimental data, the basis set cannot strictly be identified with a specific PQN. Instead, it represents the blend of atomic functions that most precisely reproduces the phenomena observed. A result of this is that the theoretical underpinnings of semiempirical methods cannot, and should not, be compared with those of *ab initio* methods.

8.2.3.5 AM1* AM1*^{61–66} provides an interesting contrast to PM6. In AM1*, d-orbitals were added to various elements that had previously been parameterized at the AM1 level, but the original AM1 parameterization was retained for the elements H, C, N, O, and F. Using the original AM1 parameters for these elements obviously limits its ultimate accuracy. Unlike other methods, where the objective was to increase accuracy, the motivation for the development of AM1* was an exploration of the role of the training data and development of a strategy for increasing the robustness or predictive power. To this end, training data calculated using DFT or *ab initio* techniques were used extensively to supplement the experimental data available. Also in contrast to PM6, the “chemical intuition” approach was used to provide a “reasonable” parameterization. The resulting method performs very similarly to PM6 in terms of its overall statistics. AM1* is usually statistically better than PM6 for its own training data, but usually not for the PM6 training data set. This is expected for local parameterizations, especially so for cases in which it is impossible to use an independent validation data set because of the lack of experimental data. Together, PM6 and AM1* provide an opportunity to validate results by comparing the results of the two methods, which are essentially identical quantum mechanically but were parameterized using different data and philosophies.

8.2.3.6 Methods with Orthogonalization Corrections The desirability of either explicit orthogonalization of the atomic orbitals¹⁸ or a more computationally efficient orthogonalization correction was discussed above. The latter technique has been used by Thiel and co-workers in the OM*n* methods. The first such method, OM1,¹⁹ introduced orthogonalization corrections to the one-electron terms within the NDDO approximation. This work was extended to include two-center corrections and the use of effective core potentials in place of the frozen-core approximation in OM2.²⁰ The faster OM3 method²² neglects some of the expensive, but less important, terms included in OM2. The benefits of orthogonalization corrections lie predominantly in improved performance in reproducing relative conformational energies in, for example, peptides.²¹ OM2 combined with a multireference configuration-interaction technique performs extremely well for excited states (see below).⁶⁷

8.2.3.7 Other Hamiltonians Over the past 30 years, several avenues for improving semiempirical methods have been explored. In each instance there were good reasons to believe that the proposed change would be beneficial. Sometimes this was true; other times the proposed benefit did not materialize or there were competing factors that militated against the change being adopted. Some of the more important ideas that were examined will now be described.

MNDOC An increase in accuracy should occur if correlation effects were included in semiempirical methods such as MNDO. This principle was examined by Thiel⁶⁸ in 1982, when parameters for H, C, N, and O were optimized using a modification of MNDO in which a perturbational correction for electron correlation was included explicitly. Whereas the results obtained using the new method, MNDOC, were better than for stand-alone MNDO, the computational effort was significantly larger, and MNDOC was not widely used.

MNDO/d In its original form, MNDO was limited to an *sp*-basis set. This obviously constrained its use to modeling normal-valent systems; the study of hypervalent species such as $\text{H}_2\text{S}^{\text{VI}}\text{O}_4$ and $\text{P}^{\text{V}}\text{Cl}_5$, which occur frequently in normal chemistry, was precluded. During chemical reactions, many main-group elements expand their valency temporarily to form extra bonds with ligands; such phenomena could not be modeled using MNDO. In 1992, Thiel and Voityuk² added *d*-orbitals to some elements, and in 1996 demonstrated⁶ that this resulted in a significant increase in accuracy, particularly in reducing the average unsigned errors (AUE) in ΔH_f^0 . The new method involved optimizing parameters for several elements that could be hypervalent, but did not involve reoptimizing those for the other MNDO elements. As such, it was a piecemeal approach. Nevertheless, the demonstration was convincing, and all subsequent methods employed used Thiel and Voityuk's multipole formalism for the integrals involving *d*-orbitals.

SAM1 While modifications to the core–core repulsion function have resulted in large improvements in accuracy, another function, the electron repulsion integral (ER), should also be regarded as a candidate for examination. Various forms of the ER were examined, and parameters for H, C, N, and O were optimized. When it was published in 1993, the new method, SAM1,⁶⁹ was shown to be more accurate than the then-current methods AM1 and PM3.

It is unfortunate that no further work has been reported on this topic: If the improvements resulting from modifying the ER approximation are real, and there is no reason to doubt that, there is a high probability that further work on modifying the ER term would result in significant improvements over current methods.

PDDG As just mentioned, a computationally inexpensive way to reduce error in NDDO-type methods is by modification of the core–core term. In MNDO itself, the analytic expression $Z_A Z_B / R_{AB}$ had been replaced by an approximation that took into account the long-range electron–nuclear attraction and

electron–electron repulsion terms. The core–core term had been further modified in AM1 and PM3, and in PDDG, Jorgensen et al., explored the effects of using a pairwise distance-directed Gaussian modification.⁷⁰ At the heart of the PDDG method is a modification of the core repulsion function, the modification being the addition of the following term:

$$\text{PDDG(A,B)} = \sum_A \sum_{B \neq A} \frac{1}{n_A + n_B} \left\{ \sum_{i=1}^2 \sum_{j=1}^2 (n_A P_{A_i} + n_B P_{B_j}) \exp[-10(R_{AB} - D_{A_i} - D_{B_j})^2] \right\} \quad (8.22)$$

where n_A is the number of valence electrons on atom A, and P_{A_i} and D_{A_i} are parameters.

As with SAM1, the PDDG method resulted in an increase in accuracy over AM1 and PM3.

RM1 A convincing demonstration of the importance of training set and parameterization is provided by RM1.⁵⁶ Starting with the AM1 method, and without making any change to the formalism, parameters for H, C, N, P, S, F, Cl, Br, and I were reoptimized. The AUE for heats of formation dropped to about half of that for AM1, and for dipole moments the accuracy exceeded that of PM6.

8.3 NATURAL HISTORY OR EVOLUTION OF MNDO-LIKE METHODS

The evolution of NDDO methods has followed a completely logical course. When it first appeared, MNDO represented a large improvement over the earlier purely atom-based method, MINDO/3. This improvement was due to the more sophisticated set of approximations and to the use of molecular reference data. Only after it had been used for awhile did severe errors in MNDO become apparent, the most important of these being the almost complete lack of a hydrogen bond. This deficiency contained within it an indication of the direction for further improvement—to add a term to represent the hydrogen bond. Still using a small set of reference data, parameters for H, C, N, and O were reoptimized; this resulted in AM1. A consequence of piecemeal parameterization of AM1, in which the first elements parameterized were not reoptimized when more elements were added, was that the final set of parameters were by no means optimal. An obvious next step to correct this was to investigate the consequences of optimizing many elements simultaneously using large amounts of reference data. This gave rise to PM3.

No further reduction in accuracy could be achieved by better parameterization or better reference data, so the focus turned to the third and last possible cause of error: the set of approximations used. The core–core terms were modified

to include diatomic parameters, and a reparameterization involving the entire main group resulted in a dramatic drop in AUE for heats of formation. The new method was named PM6. Each modification addressed a definite fault in the earlier method and resulted in a significant improvement in accuracy. This sequence of incremental improvement is both clear and simple and the overall effect is a natural evolution in the direction of increased accuracy. As the accuracy improves, various faults in any given method that were hidden by much more severe errors in earlier methods become apparent, and these could then be addressed. There is every indication that this sequence will continue far into the future. As just mentioned, the most recent method, PM6, represents a large improvement over PM3. Nevertheless, soon after it was released, errors that were masked by the relatively large errors in PM3 became apparent, the most important of these being a bias in favor of zwitterions instead of neutral biochemical species. It is likely that such errors had existed in earlier methods, but they only became obvious in PM6. In principle, correcting such an error is straightforward—simply adding appropriate reference data to the training set and rerunning the parameterization. In practice, such operations are time consuming, as checks have to be run to ensure that none of the previous gains made are compromised.

8.3.1 Strengths of MNDO-like Methods

The most recent methods developed from the MNDO line, PM6 and AM1*, are particularly useful, that is, accurate, in modeling the structural and thermochemical properties of a broad swath of ordinary chemistry, particularly biochemical systems. However, like the earlier methods, their accuracy is much reduced when they are used for modeling exotic systems, such as transition states, electronic excited states, high-energy systems such as radicals, and solids with low or zero bandgaps, such as metals. For such systems, *ab initio* methods still reign supreme. In part, this reflects the emphasis or bias imposed on the parameterization: Since one of the objectives of the development of PM6 was to focus on systems of biochemical interest, it is not surprising that it is particularly suitable for modeling such systems. This accuracy comes at a price: A direct consequence of the increased emphasis on ordinary chemistry is the inability to model exotic systems accurately. AM1* provides some contrast because of the conscious attempt to represent “more chemistry” in its parameterization. Once again, the dominant effect of the training data on determining the range of applicability of a semiempirical molecular orbital method cannot be overemphasized. Nevertheless, MNDO-like methods as a general class have important strengths that have tended to be forgotten since the rise of DFT techniques. We outline some of these below.

8.3.1.1 Correlation in MNDO-like Methods As outlined in Section 8.1, MNDO-like methods are based on the LCAO-SCF approximations. They do not, therefore, explicitly include electron correlation. However, in an analogy

to DFT that is often overlooked, dynamic correlation is included implicitly in MNDO-like techniques. This is achieved through parameterization (experimental results clearly include correlation) and through scaling of the two-electron integrals so that they are correct at the one-center limit (i.e., at $R_{AB} = 0$). Perhaps the best known pre-MNDO scaling scheme is that of Klopman–Ohno.^{71,72} In MNDO¹ this scaling is achieved by constructing the multipoles used to calculate the two-electron integrals so that they give the correct values at $R_{AB} = \infty$ and at the one-center limit. The values at the one-center limit are determined by fitting to atomic spectra using Oleari's method.^{73,74} This restriction was relaxed when PM3 was introduced⁵⁴ and the one-center two-electron repulsion integrals were treated as variable parameters.

The result of this integral scaling is similar to that of treating electron correlation using a functional of the density in DFT. Dynamic correlation can be treated quite effectively in this fashion and the implicit consideration of dynamic correlation in MNDO-like methods has important consequences for configuration-interaction (CI) calculations on excited states, as discussed below.

8.3.2 One-Electron Properties

One-electron properties,⁷⁵ in this case primarily the molecular electrostatic potential and field and electrostatic and transition moments, are generally reproduced very well by MNDO-like methods, almost independent of the particular Hamiltonian being used. As an example, we can think of the molecular electrostatic potential (MEP), which has been shown to be a dominant factor in determining intermolecular interactions.⁷⁶ The MNDO formalism offers a convenient model for representing the electrostatics of molecules because we can derive an atom-centered multipole model⁷⁷ (up to quadrupoles) directly from the MNDO multipole model for the two-electron integrals.¹ Using the AM1* Hamiltonian,^{61–66} for a small test set of diverse molecules, standard deviations between AM1* multipole MEPs at points on the isodensity surfaces of the molecules and those calculated at the same points using MP2/6-31G(d) or B3LYP/6-31G(d) was only on the order of 2 kcal mol⁻¹ if a simple linear scaling factor was used. This observation has significant consequences for many branches of chemistry. It means, for example, that we can happily use MNDO-like methods to calculate solvation energies using polarizable continuum methods because the electrostatics of the molecules are correct. Further examples are given below for the use of transition moments in ensemble models.

8.3.3 Excited States

Semiempirical molecular orbital techniques were used very early to investigate excited states and to predict spectra. The early π -only Pople–Pariser–Parr technique⁷⁸ was quite successful in predicting ultraviolet/visible spectra.⁷⁹ Later, the development of the specially parameterized INDO/S technique,⁸⁰ which used CI calculations limited to single excitations, became the method of choice for calculating spectra of organic and inorganic molecules.⁸¹ In the late 1990s, INDO/S

allowed calculation of the excited states of systems as large as a bacteriochlorophyll hexadecamer with 704 atoms, more than 2000 electrons, and a CI expansion of 4096 symmetry-selected configurations.⁸² Semiempirical CI calculations are not limited to INDO/S. Even “general purpose” methods such as AM1 give surprisingly good results for predicting absorption and fluorescence spectra and nonlinear optical (NLO) properties.^{83,84} It is probably fair to say that semiempirical CI calculations can give similar agreement with experimental excitation energies as current standard time-dependent DFT (TDDFT) methods, although the latter clearly have considerable potential for improvement. Multireference semiempirical techniques can provide remarkably accurate results when used with an orthogonalization correction and are eminently suitable for geometry optimizations on excited states.⁶⁷ One major advantage of semiempirical CI calculations is that they are computationally very efficient, so that we can afford to perform tens of thousands of calculations on snapshots from classical molecular-dynamics simulations. This is the basis of the ensemble model, which has been used to simulate fluorescence resonant energy transfer (FRET) in proteins⁸⁵ and field-dependent second-harmonic generation by a dye embedded in a biological membrane.⁸⁶ Such applications demonstrate the real potential and one of the most promising areas of application for MNDO-like methods.

8.4 LARGE SYSTEMS

By *large systems* we mean both very large molecules and large databases of smaller molecules. Semiempirical molecular orbital methods are useful for the former because of their potential linear scaling. Their inherent speed makes them the ideal choice for both applications.

8.4.1 Databases

Because of their ability to deliver accurate geometries, energies, and one-electron properties, semiempirical MO methods are ideally suited for providing extra information about, for example, druglike molecules.⁸⁷ It is important to emphasize that the all-important⁷⁶ molecular electrostatic potential (MEP) is reproduced very poorly by the atomic monopoles commonly used in force fields. The MEP calculated from an atomic-monopole model may even be so much in error as to preclude important intermolecular bonding effects, such as halogen bonding.⁸⁸ The MEP generated from common semiempirical methods is, however, in very good agreement with that calculated by DFT or *ab initio* methods.⁷⁷ Furthermore, semiempirical MO techniques can be used to calculate an array of local properties that describe intermolecular interactions.⁸⁹ It is therefore not surprising that a complete database of 53,000 compounds was treated (the geometries of all molecules optimized) with AM1 as early as 1998⁹⁰ and to process the entire NCI database (250,000 compounds) in 2005.⁹¹ Several in-house databases of companies in the pharmaceutical industry (1 to 2 million compounds) have been treated similarly.

8.4.2 Ensemble Models

Large databases are not the only area in which very many calculations are required. The two major challenges that face computational chemistry are to represent the potential energy hypersurface of the system correctly (the Hamiltonian) and, for large flexible systems, to sample the conformational space adequately to be able to calculate thermodynamic or spectral properties of the real system (sampling). Clearly, we cannot calculate Avogadro's number of molecules in order to simulate a mole of substance. We can, however, use the ergodic hypothesis,^{92,93} which basically proposes that if we sample long enough, we will obtain a distribution of conformations for a single molecule that corresponds to that of an ensemble of very many molecules. This leads to the ensemble models⁹⁴ for simulating macroscopic systems. In these models, very many snapshots (instantaneous geometries of the system) are taken from a single (or several) molecular-dynamics simulations, their properties calculated by a suitable method (in the examples below semiempirical CI) and the properties of the real system calculated as the average of those of the individual snapshots. Such models have been very successful in calculating the details of FRET in the tetracycline repressor protein⁸⁵ and simulating the effects of an applied potential on an NLO dye embedded in a cell membrane.⁸⁶ Semiempirical CI calculations are the only techniques that can provide the necessary accuracy and throughput for such applications.

8.4.3 Proteins

Linear scaling techniques have made the calculation of protein properties—structure, energetics, interactions—possible with quantum mechanical techniques. In part, this was due to the fact that the computational effort required in solving the SCF equations had limited the size of the systems to just a few hundred atoms; this meant that only the smaller proteins, such as crambin, could be studied. More important, weak interatomic interactions such as those found in hydrogen bonds and $\pi - \pi$ stacking, were poorly represented by the “fast” quantum mechanical techniques (semiempirical and DFT). As interactions of this type are important in proteins, this fault cast doubt on any predicted results. But now, with the development of linear scaling methods, the properties of proteins containing up to 15,000 atoms can be modeled; less than 13% of all entries in the Protein Data Bank⁹⁵ are larger than that, and with the advent of PM6, weak interactions of the type found in proteins can also be reproduced with unprecedented accuracy using semiempirical MO theory. These developments have resulted in the ability to model protein chemistry with relative ease; using PM6 and the linear scaling function MOZYME, the properties of over 40 proteins were modeled using a simple desktop computer.⁹⁶ Among these properties are structure (albeit starting from the PDB geometry), heat of formation, transition states for enzyme-catalyzed reaction, and elastic modulus for structural proteins. The more general problem of de novo predicting protein structure is still unsolved.

D&C methods were the first to be used for calculations on moderately sized proteins, both with⁹⁷ and without⁹⁸ solvent effects simulated using the Poisson–Boltzmann equation. Both AM1 and PM3 have proven to be useful in distinguishing between native and misfolded protein structures.⁹⁹

The more recent PM6 technique in combination with the LMO linear scaling approach has proven to be very useful for studying proteins.⁹⁶ Many phenomena in proteins can be modeled with good accuracy using PM6, but significant limitations remain. The long-standing fault of semiempirical methods—that predicted barrier heights for covalent reactions are of low accuracy—still exists in PM6. Another fault is that despite the improvements in modeling weak interactions, intermolecular interactions of the type that occurs when a substrate binds to a protein are also poorly reproduced. Very recent work suggests that by making simple modifications to the core–core interactions, to include¹⁰⁰ an explicit correction for hydrogen bonds involving oxygen or nitrogen, and adding in a correlation term,²⁹ the accuracy of prediction of intermolecular interactions can be increased significantly. Thus, for the S22 data set,¹⁰¹ intermolecular interactions were reproduced with chemical accuracy (average unsigned error = 0.8 kcal mol^{−1}), considerably less than the 3.4 kcal mol^{−1} found when PM6 was used.

REFERENCES

1. Dewar, M. J. S.; Thiel, W. *J. Am. Chem. Soc.* **1977**, *99*, 4899.
2. Thiel, W.; Voityuk, A. A. *Theor. Chim. Acta* **1992**, *81*, 391.
3. Thiel, W.; Voityuk, A. A. *Theor. Chim. Acta* **1996**, *93*, 315.
4. Thiel, W.; Voityuk, A. A. *Int. J. Quantum Chem.* **1994**, *44*, 807.
5. Thiel, W.; Voityuk, A. A. *J. Mol. Struct.* **1994**, *313*, 141.
6. Thiel, W.; Voityuk, A. A. *J. Phys. Chem.* **1996**, *100*, 616.
7. Born, M.; Oppenheimer, J. R. *Ann. Phys. (Leipzig)* **1927**, *84*, 457.
8. Schrödinger, E. *Phys. Rev.* **1926**, *28*, 1049.
9. Hartree, D. R. *Proc. Cambridge Phil. Soc.* **1928**, *24*, 89, 111, 426.
10. Fock, V. *Z. Phys.* **1930**, *61*, 126.
11. Pauli, W. *Z. Phys.* **1925**, *31*, 765.
12. Slater, J. C. *Phys. Rev.* **1929**, *34*, 1293; **1930**, *35*, 509.
13. Hückel, E. *Z. Phys.* **1931**, *70*, 204; **1931**, *72*, 310; **1932**, *76*, 628; **1933**, *83*, 632.
14. Sinanoglu, O.; Fu-Tai Tan, D. *Chem. Phys.* **1963**, *38*, 1740.
15. Clark, T.; Koch, R. *The Chemist's Electronic Book of Orbitals*, Springer-Verlag, Berlin, 1999.
16. Winget, P.; Selçuki, C.; Horn, A. H. C.; Martin, B.; Clark, T. *Theor. Chem. Acc.* **2003**, *110*, 254.
17. Horn, A. H. C.; Lin, J.-H.; Clark, T. *Theor. Chem. Acc.* **2005**, *114*, 159–168; erratum: *Theor. Chem. Acc.* **2007**, *117*, 461–465.
18. Sattelmeyer, K. W.; Tubert-Brohmann, I.; Jørgensen, W. L. *J. Chem. Theor. Comput.* **2006**, *2*, 413.

19. Kolb, M.; Thiel, W. *J. Comput. Chem.* **1993**, *14*, 775.
20. Weber, W.; Thiel, W. *Theor. Chem. Acc.* **2000**, *103*, 495.
21. Mohle, K.; Hofmann, H.-J.; Thiel, W. *J. Comput. Chem.* **2001**, *22*, 509.
22. Scholten, M. Ph.D. dissertation, Heinrich-Heine-Universität, Düsseldorf, Germany, 2003.
23. Burstein, K. Y.; Isaev, A. N. *Theor. Chim. Acta* **1984**, *64*, 397.
24. Csonka, G. I.; Ángyán, J. G. *J. Mol. Struct. (Theochem)* **1997**, *393*, 31.
25. Voityuk, A. A.; Rösch, N. *J. Phys. Chem. A* **2000**, *104*, 4089.
26. Stewart, J. J. P.; Császár, P.; Pulay, P. *J. Comput. Chem.* **1982**, *3*, 227.
27. Ahlrichs, R.; Penco, R.; Scoles, G. *Chem. Phys.* **1977**, *19*, 119.
28. Grimme, S. *J. Comput. Chem.* **2004**, *25*, 1463.
29. Jurecka, J.; Cerny, J.; Hobza, P.; Salahub, D. *J. Comput. Chem.* **2007**, *28*, 555.
30. Cerny, J.; Jurecka, J.; Hobza, P.; Valdes, H. *J. Phys. Chem. A* **2007**, *111*, 1146.
31. Elstner, M.; Hobza, P.; Frauenheim, T.; Suhai, S.; Kaxiras, E. *J. Chem. Phys.* **2001**, *114*, 5149.
32. Tuttle, T.; Thiel, W. *Phys. Chem. Chem. Phys.* **2008**, *10*, 2159.
33. McNamara, J. P.; Hillier, I. H. *Phys. Chem. Chem. Phys.* **2007**, *9*, 2362.
34. Stewart, J. J. P. *J. Mol. Model.* **2007**, *13*, 1173.
35. Rinaldi, D.; Rivail, J.-L. *Theor. Chim. Acta* **1973**, *32*, 57; **1974**, *32*, 243.
36. Schürer, G.; Gedeck, P.; Gottschalk, M.; Clark, T. *Int. J. Quantum Chem.* **1999**, *75*, 17.
37. Martin, B.; Gedeck, P.; Clark, T. *Int. J. Quantum Chem.* **2000**, *77*, 473.
38. Eischenschitz, R.; London, F. Z. *Phys.* **1930**, *60*, 491.
39. Martin, B.; Clark, T. *Int. J. Quantum Chem.* **2006**, *106*, 1208.
40. Yang, W. *Phys. Rev. Lett.* **1991**, *66*, 1438.
41. Dixon, S. L.; Merz, K. M., Jr. *J. Chem. Phys.* **1997**, *107*, 879.
42. Ababoua, A.; van der Vaart, A.; Gogonea, V.; Merz, K. M., Jr. *Biophys. Chem.* **2007**, *125*, 221.
43. Stewart, J. J. P. *Int. J. Quantum Chem.* **1996**, *58*, 133.
44. Stewart, J. J. P. *J. Mol. Model.* **2009**, *15*, 765.
45. Pople, J. A.; Santry, D. P.; Segal, G. A. *J. Chem. Phys.* **1965**, *43*, S129.
46. Pople, J. A.; Beveridge, D. L.; Dobosh, P. A. *J. Chem. Phys.* **1967**, *47*, 2026.
47. Kramida, A. E.; Martin, W. C.; Musgrove, A.; Olsen, K.; Reader, J.; Saloman, E. B. http://physicsnistgov/cgi-bin/ASBib1/Elevbib/search_formcgi, 2009.
48. Afeefy, H. Y.; Liebman, J. F.; Stein, S. E. Neutral thermochemical data. In *NIST Chemistry WebBook*, Linstrom, P. J., and Mallard, W. G., Eds., NIST Standard Reference 69, National Institute of Standards and Technology, Gaithersburg, MD, 2003. Available at <http://webbooknistgov/chemistry>.
49. Levin, R. D.; Lias, S. G. *Ionization Potentials and Appearance Potential Measurements*, National Standards Reference Data Series, Vol. 71, National Bureau of Standards, Washington, DC, 1982.
50. Allen, F. H. *Acta Crystallogr. B* **2007**, *58*, 380.

51. Stewart, J. J. P. Parameterization of semiempirical M.O. methods. In *Encyclopedia of Computational Chemistry*, Vol. 3, Schleyer, P. v. R., Allinger, N. L., Clark, T., Gasteiger, J., Kollman, P. A., Schaefer, H. F. S., III, and Schreiner, P. R., Eds., Wiley, Chichester, UK, 2000.
52. Dewar, M. J. S.; Thiel, W. *J. Am. Chem. Soc.* **1977**, *99*, 4907.
53. Dewar, M. J. S.; Zoebisch, E. G.; Healy, E. F.; Stewart, J. J. P. *J. Am. Chem. Soc.* **1985**, *107*, 3902.
54. Stewart, J. J. P. *J. Comput. Chem.* **1989**, *10*, 209.
55. Stewart, J. J. P. *J. Comput. Chem.* **1989**, *10*, 221.
56. Rocha, G. B.; Freire, R. O.; Simas, A. M.; Stewart, J. J. P. *J. Comput. Chem.* **2006**, *27*, 1101.
57. Bingham, R. C.; Dewar, M. J. S.; Lo, D. H. *J. Am. Chem. Soc.* **1975**, *97*, 1285.
58. Stewart, J. J. P. *J. Comput. Chem.* **1991**, *12*, 320.
59. Stewart, J. J. P. *J. Mol. Model.* **2004**, *10*, 155.
60. (a) Ditchfield, R.; Hehre, W. J.; Pople, J. A. *J. Chem. Phys.* **1971**, *54*, 724. (b) Hehre, W. J.; Ditchfield, R.; Pople, J. A. *J. Chem. Phys.* **1972**, *56*, 2257. (c) Hariharan, P. C.; Pople, J. A. *Mol. Phys.* **1974**, *27*, 209. (d) Gordon, M. S. *Chem. Phys. Lett.* **1980**, *76*, 163. (e) Hariharan, P. C.; Pople, J. A. *Theor. Chim. Acta* **1973**, *28*, 213. (f) Blaudeau, J. -P.; McGrath, M. P.; Curtiss, L. A.; Radom, L. *J. Chem. Phys.* **1997**, *107*, 5016. (g) Francl, M. M.; Pietro, W. J.; Hehre, W. J.; Binkley, J. S.; DeFrees, D. J.; Pople, J. A.; Gordon, M. S. *J. Chem. Phys.* **1982**, *77*, 3654. (h) Binning, R. C., Jr.; Curtiss, L. A. *J. Comput. Chem.* **1990**, *11*, 1206. (i) Rassolov, V. A.; Pople, J. A.; Ratner, M. A.; Windus, T. L. *J. Chem. Phys.* **1998**, *109*, 1223. (j) Rassolov, V. A.; Ratner, M. A.; Pople, J. A.; Redfern, P. C.; Curtiss, L. A. *J. Comput. Chem.* **2001**, *22*, 976. (k) Frisch, M. J.; Pople, J. A.; Binkley, J. S. *J. Chem. Phys.* **1984**, *80*, 3265.
61. Winget, P.; Horn, A. H. C.; Selçuki, C.; Martin, B.; Clark, T. *J. Mol. Model.* **2004**, *9*, 408.
62. Winget, P.; Clark, T. *J. Mol. Model.* **2005**, *11*, 439.
63. Kayi, H.; Clark, T. *J. Mol. Model.* **2007**, *13*, 965.
64. Kayi, H.; Clark, T. *J. Mol. Model.* **2009**, *15*, 295.
65. Kayi, H.; Clark, T. *J. Mol. Model.* **2009**, *15*, 1253.
66. Kayi, H.; Clark, T. *J. Mol. Model.* **2010**, *16*, 29.
67. Koslowski, A.; Beck, M. E.; Thiel, W. *J. Comput. Chem.* **2003**, *24*, 714–726.
68. Thiel, W. *Quantum Chemistry Program Exchange*, QCPE 438, University of Indiana, Bloomington, IN, 1982.
69. Dewar, M. J. S.; Jie, C.; Yu, J. *Tetrahedron* **1993**, *49*, 5003.
70. Repasky, M. P.; Chandrasekhar, J.; Jørgensen, W. L. *J. Comput. Chem.* **2002**, *23*, 1601.
71. Klopman, G. *J. Am. Chem. Soc.* **1964**, *86*, 4550.
72. Ohno, K. *Theor. Chim. Acta* **1964**, *3*, 219.
73. Oleari, L.; DiSipio, L.; DeMichelis, G. *Mol. Phys.* **1966**, *10*, 97.
74. Dewar, M. J. S.; Lo, D. H. *J. Am. Chem. Soc.* **1972**, *94*, 5296.
75. See Karplus, M.; Kuppermann, A.; Isaacson, L. M. *J. Chem. Phys.* **1958**, *29*, 1240.

76. Murray, J. S.; Politzer, P. *J. Mol. Struct. (Theochem)* **1998**, 425, 107; Murray, J. S.; Lane, P.; Brinck, T.; Paulsen, K.; Grince, M. E.; Politzer, P. *J. Phys. Chem.* **1993**, 97, 9369.
77. Horn, A. H. C.; Lin, J.-H.; Clark, T. *Theor. Chem. Acc.* **2005**, 114, 159; erratum: *Theor. Chem. Acc.* **2007**, 117, 461.
78. Pariser, R.; Parr, R. G. *J. Chem. Phys.* **1963**, 21, 466.
79. See, e.g., Griffiths, J. *Dyes Pigments* **1982**, 3, 211.
80. Ridley, J.; Zerner, M. C. *Theor. Chim. Acta*, **1973**, 32, 111.
81. Zerner, M. C. In *Reviews of Computational Chemistry*, Vol. 2, Lipkowitz, K. B., Ed., VCH, New York, 1991, p. 313.
82. Cory, M. G.; Zerner, M. C.; Hu X.; Schulten, K. *J. Phys. Chem. B* **1998**, 102, 7640.
83. Clark, T.; Chandrasekhar, J. *Israel J. Chem.* **1993**, 33, 435.
84. Göller, A.; Grummt, U. W. *Int. J. Quantum Chem.* **2000**, 77, 727.
85. Beierlein, F. R.; Othersen, O. G.; Lanig, H.; Schneider, S.; Clark, T. *J. Am. Chem. Soc.* **2006**, 128, 5142.
86. Rusu, C.; Lanig, H.; Clark, T.; Krysch, C. *J. Phys. Chem. B* **2008**, 112, 2445.
87. Clark, T. In *Molecular Informatics: Confronting Complexity*, Hicks M. G., and Kettner C., Eds., Logos Verlag, Berlin, 2003, p. 193.
88. Politzer, P.; Murray, J. S.; Concha, M. *J. Mol. Model.* **2008**, 14, 659.
89. Clark, T.; Byler, K. G.; de Groot M. J. In *Molecular Interactions: Bringing Chemistry to Life*, Hicks M. G., and Kettner C., Eds., Logos Verlag, Berlin, 2008, p. 129.
90. Beck, B.; Horn, A. H. C.; Carpenter, J. E.; Clark, T. *J. Chem. Inf. Comput. Sci.* **1998**, 38, 1214.
91. Murray-Rust, P.; Rzepa, H. S.; Stewart J. J. P.; Zhang, Y. *J. Mol. Model.* **2005**, 11, 532.
92. Boltzmann, L. *Einige allgemeine Sätze über das Wärmegleichgewicht*, Vienna, Austria, 1871.
93. Boltzmann, L. *Creeles J.* **1884**, 98, 68.
94. Lee, M.; Tang, J.; Hochstrasser, R. M. *Chem. Phys. Lett.* **2001**, 344, 501.
95. <http://www.pdb.org/>, Research Collaboratory for Structural Bioinformatics, The San Diego Supercomputer Center, San Diego, CA, 2007.
96. Stewart, J. J. P. *J. Mol. Model.* **2008**, 15, 765.
97. Gogonea, V.; Merz, K. M., Jr. *J. Phys. Chem. A* **1999**, 103, 5171.
98. For a review, see van der Vaart, A.; Gogonea, V.; Dixon, S. L.; Merz, K. M., Jr. *J. Comput. Chem.* **2000**, 21, 1494.
99. Wollacott, A. M.; Merz, K. M., Jr. *J. Chem. Theor. Comput.* **2007**, 3, 1609.
100. Rezac, J.; Fanfrlik, J.; Salahub, D.; Hobza, P. *J. Chem. Theor. Comput.* **2009**, 5, 1749.
101. Jurecka, P.; Spöner, J.; Cerny, J.; Hobza, P. *Phys. Chem. Chem. Phys.* **2006**, 8, 1985.

9 Self-Consistent-Charge Density Functional Tight-Binding Method: An Efficient Approximation of Density Functional Theory

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In this chapter we describe the derivation of the approximate DFT method SCC-DFTB from DFT. The basic formalism of SCC-DFTB results from a second-order expansion of the DFT total energy, followed by appropriate approximations. The formal basis of SCC-DFTB is the non-self-consistent Harris functional. We discuss the performance of SCC-DFTB as well as recent extensions such as the inclusion of third-order terms and van der Waals corrections.

9.1 INTRODUCTION

Most semiempirical (SE) methods are derived either from Hartree–Fock (HF) or density functional theory (DFT) applying two types of approximations: first, they are based primarily on a minimal atomic orbital-like basis set; second, the numerous integrals, which have to be evaluated in HF and DFT, are partially neglected and the remaining ones can be calculated either using further approximations or can be substituted by parameters, which in turn are fitted to reproduce experimental data. As a result, no integrals have to be evaluated during the runtime of the program, and the dominant computational cost is given by the diagonalization of the Fock (Hamilton) matrix. Since this matrix is represented in a minimal atomic basis set, solution of the eigenvalue problem is much less expensive than for full DFT and HF methods, which usually

apply more extended basis sets. Typically, SE methods are about three orders of magnitude faster than HF/DFT methods using double-zeta basis sets. They exhibit an $O(N^3)$ scaling behavior, that means that the computing time increases cubically with the system size (which is roughly proportional to the number of atoms, or, more correctly, proportional to the number of electrons N). Since DFT is also $O(N^3)$ scaling, the factor of 1000 gained in computational speed with respect to DFT means that about 10-fold larger systems can be treated. For example, today about 100 atoms can be handled by DFT on standard desktop PCs, while roughly 1000 atoms can be treated using SE methods. The bottleneck here is the diagonalization of the Fock–Hamilton matrix, and methods that avoid this step, such as $O(N)$ scaling algorithms,¹ help to increase the system size dramatically, as discussed in Chapters 2 and 8. However, in many cases the system size is not the limiting issue. Chemistry often occurs in localized regions and the “active site” of interest often contains only several 10 to 100 atoms [i.e., a quantum mechanical (QM) treatment is needed only for this small subsystem (this applies often in biological systems)]. The remainder of the system can be treated by empirical potentials [molecular mechanics (MM)]. A combination of QM methods with MM force fields in QM/MM methods can now be applied routinely (for recent comprehensive reviews, see, e.g., Refs. 2 and 3). A major issue however, is the time scale that can be reached using molecular dynamics (MD) simulations. HF and DFT make it possible to follow the system dynamics (for several tens of atoms) in the picosecond regime. In this case, the factor 1000 gained in computational speed by SE methods allows for 1000-fold longer MD simulations (i.e., the nanosecond time scale is easily accessible). In many applications, this helps to follow the relevant conformational changes or, much more important, to compute free-energy changes along reaction pathways.⁴ This is probably the main reason why SE methods have been used increasingly in the past years, although they sacrifice accuracy compared to DFT in many cases (note that this can be reversed for specific applications).

In quantum chemistry, the classical route to deriving SE methods is to start from HF theory and fit the remaining parameters (integrals) to experimental data. This approach leads to a family of SE methods, with MNDO, AM1, and PM3 being the best known. The latest and most accurate members of this family are discussed by Clark and Stewart in Chapter 8.

In solid-state physics, tight-binding (TB) approaches have been used extensively to study the properties of solids and clusters,^{5,6} directly paralleling the development of the Hückel model in chemistry; these methods are reviewed in Chapter 10. Standard tight-binding methods are usually based on the Harris functional approach⁷ (i.e., they diagonalize a suitable Hamiltonian once and use this non-self-consistent solution to derive further properties, such as forces and second derivatives). The relation of DFT and TB methods has been discussed in detail by Foulkes and Haydock.⁸ TB methods can be understood as a stationary approximation to DFT and tend to work well when the “guess” density, which is incorporated into the predetermined Hamilton matrix, is a good approximation to the DFT ground-state density.

SCC-DFTB is an approximate quantum chemical method that is derived from DFT by a second-order expansion of the DFT total energy with respect to density fluctuations around a suitable reference density.⁹ On the other hand, SCC-DFTB can be viewed as an extension of a tight-binding method, which includes charge self-consistency and is parameterized using DFT. Energy in tight-binding methods consists of two parts: electronic and repulsive. The electronic part is described by a Hamiltonian, which is usually represented in a minimal basis of atom-centered basis functions. In DFTB, this Hamilton matrix is derived from DFT using as a reference density the superposition of neutral atomic densities and a minimal basis of atomic wavefunctions, which is calculated explicitly.^{10–14} The repulsive energy, which consists of the DFT double-counting contributions and the core–core repulsion, can be approximated as a sum of atomic pair repulsion functions.

SCC-DFTB is parameterized using the generalized-gradient approximation (GGA). In the actual version the electronic parameters are calculated using the PBE functional.¹⁵ This means, however, that the well-known DFT-GGA deficiencies are inherited by SCC-DFTB. Of particular relevance is the DFT-GGA tendency to overpolarize extended π -conjugate systems,¹⁶ the problems of ionic and charge-transfer excited states,¹⁷ and the missing dispersion interactions, which have been included by augmenting SCC-DFTB using an empirical extension.¹⁸ The performance and deficiencies of SCC-DFTB with respect to biological applications have been reviewed recently,^{19,20} and methodological developments have been described elsewhere.²¹

9.2 THEORY

The derivation of SCC-DFTB starts from the DFT total energy. In a first step, we discuss the Harris functional approximation as the basis for non-self-consistent TB methods. In a second step, second-order corrections to Harris functional theory are introduced, leading after further approximations to the SCC-DFTB formalism. In a next step, the remaining approximations, the performance and possible extensions of this methodology, are discussed.

9.2.1 DFT and the Harris Functional

The DFT total energy reads

$$E[\rho] = T[\rho] + \int v^{\text{ext}}(r)\rho(r) dr + \frac{1}{2} \int \frac{\rho(r)\rho(r')}{|r - r'|} dr dr' + E^{\text{xc}}[\rho] + \frac{1}{2} \sum_{\alpha\beta} \frac{Z_\alpha Z_\beta}{R_{\alpha\beta}} \quad (9.1)$$

where $\rho(r)$ is the electron density, $T[\rho]$ the kinetic energy of the electrons, v^{ext} the external potential arising from the nuclei with charge Z , and $E^{\text{xc}}[\rho]$ is the

exchange-correlation energy. Application of the variational principle leads to the well-known Kohn–Sham (KS) equations,

$$\left[-\frac{1}{2}\nabla^2 + v^{\text{eff}}[\rho]\right]\phi_i = \varepsilon_i \phi_i \quad (9.2)$$

with $v^{\text{eff}}[\rho]$ being the KS effective potential, which determines the KS eigenvalues (molecular orbital energies) ε_i and KS (molecular) orbitals ϕ_i . Since $v^{\text{eff}}[\rho]$ already contains the electron density, which is calculated as

$$\rho = \sum_i |\phi_i|^2 \quad (9.3)$$

these equations have to be solved iteratively until self-consistency is achieved. Using the Kohn–Sham energies ε_i , the total energy can be written²²

$$\begin{aligned} E[\rho] = & \sum_i^{\text{occ}} \varepsilon_i - \frac{1}{2} \int \frac{\rho(r)\rho(r')}{|r-r'|} dr dr' + E^{\text{xc}}[\rho] \\ & - \int v^{\text{xc}}(r)\rho(r) dr + \frac{1}{2} \sum_{\alpha\beta} \frac{Z_\alpha Z_\beta}{R_{\alpha\beta}} \end{aligned} \quad (9.4)$$

In the Harris-functional approach,⁷ an initial density ρ^0 is constructed as a superposition of fragment densities ρ_α^0 ,

$$\rho^0 = \sum_\alpha \rho_\alpha^0 \quad (9.5)$$

and it can be shown that the total energy can be approximated in first order as

$$\begin{aligned} E[\rho] = & \sum_i^{\text{occ}} \varepsilon_i^{\text{H}} - \frac{1}{2} \int \frac{\rho^0(r)\rho^0(r')}{|r-r'|} dr dr' + E^{\text{xc}}[\rho^0] \\ & - \int v^{\text{xc}}(r)\rho^0(r) dr + \frac{1}{2} \sum_{\alpha\beta} \frac{Z_\alpha Z_\beta}{R_{\alpha\beta}} \end{aligned} \quad (9.6)$$

where the ε_i^{H} are determined from Eq. (9.2) using ρ^0 instead of the true density ρ , which would have to be determined self-consistently by iterating Eqs. (9.2) and (9.3). Any DFT method has to be initialized by choosing a proper initial density ρ^0 , which is usually taken as a superposition of atomic densities.

As pointed out by Harris,⁷ the KS equations (9.2) do not have to be solved iteratively if the starting density ρ^0 is close to the ground-state density ρ^{G} (introducing an error of second order in the difference density $\delta\rho = \rho - \rho^0$). This non-self-consistent solution of the KS equations is the basis of the Harris functional approach, and proper implementation boils down to the question of how to find a good starting density ρ^0 , which has been elaborated in particular in TB theory.

9.2.2 Non-Self-Consistent TB Methods

To get started, consider a case where one already knows the ground-state density ρ^0 to sufficient accuracy. In this case, one can omit the self-consistent solution of the KS equations and get the orbitals immediately through

$$\left[-\frac{1}{2}\nabla^2 + v^{\text{eff}}[\rho^0]\right]\phi_i = \varepsilon_i\phi_i \quad (9.7)$$

(ρ^0 stands for a properly chosen input density in the following). This saves a factor of 5 to 10 already; however, it is the starting point for further approximations.

Consider a minimal basis set consisting of atomic orbitals: that is, $\eta_\mu = 2s, 2p_x, 2p_y$, and $2p_z$ for first-row elements (core orbitals are usually omitted) and $\eta_\mu = 1s$ for H. With the basis set expansion

$$\phi_i = \sum_{\mu} c_{\mu i} \eta_{\mu}$$

and the Hamiltonian

$$\hat{H}[\rho^0] = \hat{T} + v^{\text{eff}}[\rho^0]$$

we find that

$$\sum_{\mu} c_{\mu i} \hat{H}[\rho^0]|\eta_{\mu}\rangle = \varepsilon_i \sum_{\mu} c_{\mu i} |\eta_{\mu}\rangle \quad (9.8)$$

Multiplication with $\langle \eta_{\nu}|$ leads to

$$\sum_{\mu} c_{\mu i} \langle \eta_{\nu}|\hat{H}[\rho^0]|\eta_{\mu}\rangle = \varepsilon_i \sum_{\mu} c_{\mu i} \langle \eta_{\nu}|\eta_{\mu}\rangle \quad (9.9)$$

or equivalently, in matrix notation,

$$H^0 C = S C \varepsilon \quad (9.10)$$

This means that we just have to solve the *eigenvalue equation* once; that is, we have to diagonalize the Hamilton matrix $H_{\mu\nu}^0 = \langle \eta_{\nu}|\hat{H}[\rho^0]|\eta_{\mu}\rangle$. The superscript zero indicates that the matrix elements are evaluated using the reference density ρ^0 . Diagonalization leads to the one-particle energies ε_i , that is, to the *electronic energy*:

$$E^{\text{elec}} = \sum_i \varepsilon_i \quad (9.11)$$

Note that the basis set is nonorthogonal; that is, the overlap matrix $S_{\mu\nu} = \langle \eta_{\nu}|\eta_{\mu}\rangle$ appears in the eigenvalue equations. In such a scheme, the Hamilton and overlap matrix elements have to be determined. Effectively, the Hamilton matrix

elements can be fitted to reproduce properties of well-chosen benchmark systems. Goringe et al.⁵ and Colombo⁶ discuss several examples. Since the general form of the Hamilton operator is always known, fitting determines implicitly a proper starting density, as pointed out by Foulkes and Haydock.⁸ The overlap matrix, however, is difficult to achieve if matrix elements are not computed from first principles but are fitted to experimental data. Therefore, orthogonal TB methods are usually employed.

9.2.2.1 Orthogonal Empirical Tight Binding (ETB) or Hückel Theory In empirical schemes, the basis functions are taken to be orthogonal (i.e., $S_{\mu\nu} = \delta_{\mu\nu}$). The background is the *Löwdin orthogonalization*, where we get orthonormal orbitals through

$$\eta' = S^{1/2}\eta$$

Introducing orthonormal orbitals means multiplying with $S^{-1/2}$ and inserting a “1”:

$$S^{-1/2}HS^{-1/2}S^{1/2}C = S^{-1/2}S^{1/2}S^{1/2}C\varepsilon$$

to get the orthonormal equations ($C' = S^{1/2}C$, $H' = S^{-1/2}HS^{-1/2}$):

$$H'C' = C'\varepsilon$$

Introducing orthonormal orbitals means effectively changing the Hamiltonian. This is convenient, since in empirical schemes the Hamilton matrix is completely fitted to empirical data: for example, for carbon to the solid-state band structures of several crystal structures (e.g., diamond, graphite, body-centered cubic) or, in Hückel theory, to properties of hydrocarbons.^{5,6}

9.2.2.2 Density Functional Tight Binding (DFTB) The derivation of parameters via fitting is a quite involved process. If one could derive the parameters from DFT calculations, one would gain much more flexibility and a simplified parameterization scheme.

In a first step, one has to choose a basis set. In TB theory, basis functions are atomic orbitals η_μ , and these can be calculated from the atomic KS equations:

$$\left[-\frac{1}{2}\nabla^2 + v^{\text{eff}}[\rho^{\text{atom}}]\right]\eta_\mu = \varepsilon_\mu\eta_\mu \quad (9.12)$$

The choice of a basis is to a large degree arbitrary, and several functional forms have been applied in quantum chemistry. Atomic orbitals have the disadvantage that they are very diffuse compared to the bonding situation in solids, molecules, or clusters, where atomiclike orbitals would be “compressed” due to interaction with the neighbors. Therefore, it would be wise to use orbitals, which anticipate this interaction/compression to some degree. One way to enforce this is to add

an additional (harmonic) potential to the atomic Kohn–Sham equations, which leads to compressed atomic orbitals or optimized atomic orbitals (O-LCAO), as introduced by Eschrig²³:

$$\left[-\frac{1}{2}\nabla^2 + v^{\text{eff}}[\rho^{\text{atom}}] + \left(\frac{r}{r_0}\right)^2 \right] \eta_\mu = \varepsilon_\mu \eta_\mu \quad (9.13)$$

A measure of the distance between neighbors is given by the covalent radius r^0 and is determined for all atoms empirically. This parameter enters the evaluation of the matrix elements and is, of course, of empirical nature. As a result of the atomic calculations, we get the orbitals η_μ , the electron density at (the charge neutral) atom α ,

$$\rho_\alpha^0 = \sum |\eta_\mu|^2 \quad (9.14)$$

and the overlap matrix $S_{\mu\nu} = \langle \eta_\nu | \eta_\mu \rangle$.

To solve the eigenvalue problem in Eq. (9.9) or (9.10), we only need the Hamiltonian matrix. This leads to further approximations, since although we have the complete input density $\rho^0 = \sum \rho_\alpha^0$, the Hamiltonian evaluation would be very complicated:

$$H_{\mu\nu} = \langle \eta_\nu | \hat{H}[\rho^0] | \eta_\mu \rangle = \langle \eta_\nu | \hat{H} \left[\sum \rho_\alpha^0 \right] | \eta_\mu \rangle$$

We therefore usually make the *two-center approximation* for $\mu \neq \nu$:

$$H_{\mu\nu} = \langle \eta_\nu | \hat{H}[\rho^0] | \eta_\mu \rangle = \langle \eta_\nu | \hat{H}[\rho_\alpha^0 + \rho_\beta^0] | \eta_\mu \rangle \quad (9.15)$$

where the orbital ν is located on atom α and the orbital μ is located on atom β . The diagonal Hamiltonian elements $H_{\mu\mu} = \varepsilon_\mu$ are taken from Eq. (9.13).

The two-center approximation neglects two types of integrals which contain contributions of density ρ_γ . The terms that would enter the diagonal $H_{\mu\mu}$ are *crystal field terms*, while the terms missing on the off-diagonal terms $H_{\mu\nu}$ are *three-center terms*. These approximations are discussed in detail elsewhere.^{24,25} As can be shown, the neglect of crystal field terms becomes more severe for short interatomic distances, which, however, may be compensated for by a properly chosen repulsive potential.²⁵ The missing crystal field terms may also be responsible for errors in the cohesive energies for highly coordinated systems, as has been described for some bulk silicon systems.²⁶ In the context of semiempirical MO theory, the neglect of three-center terms has been discussed as being responsible for an underestimation of rotational barriers. In DFTB, this may have a similar consequence. Rotational barriers are slightly underestimated, which manifests itself in an underestimation of vibrational frequencies of the low-lying vibrational modes.

In DFTB,^{10–13} $H_{\mu\nu}$ and $S_{\mu\nu}$ are tabulated for various distances between atom pairs up to 10 a.u., where they vanish (also due to compression!). For any molecular geometry, these matrix elements are read in based on the distance between two atoms and then oriented in space using the Slater–Koster sin/cos

combination rules²⁷ (see, e.g., Ref. 6). Then the generalized eigenvalue problem (9.10) is solved and the electronic part of the energy, E^{elec} , from Eq. (9.11) can be calculated. It should be emphasized that this is a nonorthogonal TB scheme, which is more *transferable*, due to the appearance of the overlap matrix.

9.2.3 Repulsive Energy E^{rep}

Up to now, we have only discussed the first part of the total energy in DFT in Eq. (9.6), the sum over the Kohn–Sham energies ε_i^{H} as calculated in Eq. (9.11):

$$E[\rho] = \sum_i^{\text{occ}} \varepsilon_i^{\text{H}} - \frac{1}{2} \int \frac{\rho^0(r)\rho^0(r')}{|r-r'|} dr dr' + E^{\text{xc}}[\rho^0] - \int v^{\text{xc}}(r)\rho^0(r) dr + \frac{1}{2} \sum_{\alpha\beta} \frac{Z_\alpha Z_\beta}{R_{\alpha\beta}} \quad (9.16)$$

In TB theory, the remaining terms, the DFT *double-counting* and core–core repulsion terms are put together into an energy term called repulsive energy, E^{rep} , that the TB total energy reads:

$$E^{\text{TB}}[\rho] = \sum_i^{\text{occ}} \varepsilon_i^{\text{H}} + E^{\text{rep}}[\rho] \quad (9.17)$$

First, it is interesting to note that the double-counting terms depend on the input/reference density ρ^0 only. If we introduce the atomic density decomposition, $\rho^0 = \sum_\alpha \rho_\alpha^0$, where the atomic densities are computed according to Eq. (9.14), the Coulomb contributions

$$\frac{1}{2} \sum_{\alpha\beta} \left[\frac{Z_\alpha Z_\beta}{R_{\alpha\beta}} - \int \frac{\rho_\alpha^0(r)\rho_\beta^0(r')}{|r-r'|} dr dr' \right]$$

decay exponentially with distance $R_{\alpha\beta}$, since the overlap of the atomic densities decays exponentially. The Coulomb terms therefore can be regarded as a sum of two-body interactions, which is not the case for the exchange–correlation part in Eq. (9.4). Foulkes and Haydock⁸ suggested applying a cluster expansion,

$$E^{\text{xc}}[\rho^0] = \sum_\alpha E^{\text{xc}}[\rho_\alpha^0] + \frac{1}{2} \sum_{\alpha\beta} (E^{\text{xc}}[\rho_\alpha^0 + \rho_\beta^0] - E^{\text{xc}}[\rho_\alpha^0] - E^{\text{xc}}[\rho_\beta^0]) + \dots \quad (9.18)$$

The three-center terms are assumed to be small and are neglected. Therefore, the repulsive potential E^{rep} is approximated as the sum of a set of pairwise atom–atom potentials. Because ρ_α^0 corresponds to the charge density of a neutral atom, the electron–electron and nucleic–nucleic repulsions cancel for

large interatomic distances. Therefore, E^{rep} can be assumed to be short-ranged. However, due to the first term on right-hand side of Eq. (9.18), the repulsive potential does not approach zero for large interatomic distances R .²⁸ Because in DFTB E^{rep} is assumed to be short-range anyhow, an additive constant has to be taken into account for some applications (e.g., when computing proton affinities).

Early ETB models had the form

$$E^{\text{tot}} = \sum_i \varepsilon_i + \frac{1}{2} \sum_{\alpha\beta} U_{\alpha\beta}$$

with the two-body terms $U_{\alpha\beta}$ being exponentials fitted to reproduce, for example, geometries, vibrational frequencies, and reaction energies of suitable systems.

There are various approaches in the literature to treating this repulsive part, including attempts to account for the many-body nature of E^{rep} . In DFTB,

$$E^{\text{rep}}[\rho^0] = \frac{1}{2} \sum_{\alpha\beta} U_{\alpha\beta}$$

is calculated pointwise as follows: To get the repulsive potential for carbon, for example, one could take the carbon dimer C_2 , stretch its bond, and for each distance calculate the total energy with DFT and the electronic TB part $\sum_i \varepsilon_i$. $U_{\text{CC}}(R_{\text{C}-\text{C}})$ is given pointwise for every $R_{\text{C}-\text{C}}$ by

$$U_{\text{CC}}(R_{\text{C}-\text{C}}) = E_{\text{tot}}^{\text{DFT}}(R_{\text{C}-\text{C}}) - \sum_i \varepsilon_i \quad (9.19)$$

Since for the varying $R_{\text{C}-\text{C}}$ in the carbon dimer a lot of state crossings appear in DFT calculations, this example becomes more complex. Another possibility is to include information of a C—C single, double, and triple bond.²⁰ Here for various carbon–carbon distances, $R_{\text{C}-\text{C}}$ of the molecules ethyne, ethene, and ethane DFT calculations are performed and the resulting curves connected. This example is illustrated in Fig. 9.1.

The repulsive potential is shifted so that it goes to zero at the cutoff distance. This shift makes the construction of repulsive potentials the most time-consuming part in a new parameterization. The shift affects the atomization energy and, consequently, the heat of formation of a molecule. More important, reaction energies are controlled by the relative shifts of two potentials. Additionally, no arbitrary shift of a potential is possible, due to restrictions at the cutoff radius. Further restrictions apply for the slope and the curvature of a potential which is directly connected to the description of bond lengths and harmonic vibrational frequencies.

With this conventional approach, every repulsive potential was individually hand-constructed. For illustration, we take the example of the C—H bond. Practically, one C—H bond of methane is stretched and compressed, and the DFT total energy and DFTB electronic energy are recorded pointwise for a sufficient number of geometries. Then the difference in the energies according to Eq. (9.19)

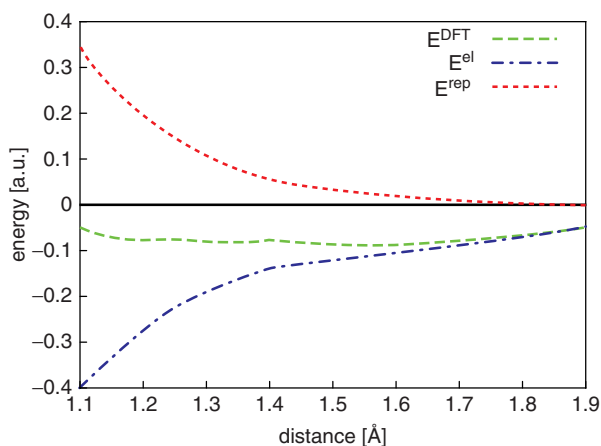


Fig. 9.1 E^{DFT} shows the (shifted) total energy versus C—C distance for $\text{HC}\equiv\text{CH}$, $\text{H}_2\text{C}=\text{CH}_2$, and $\text{H}_3\text{C}-\text{CH}_3$, E^{el} represents $\sum_i \varepsilon_i$ + shift for the same structures [the second term on right-hand side of Eq. (9.19)], and E^{rep} is the difference of these two curves.

is fitted to a polynomial (or a spline), yielding the desired repulsive potential. At the end, the repulsive potential is shifted in order to match the atomization energy of methane. Practically, the potentials could not be shifted upward sufficiently; therefore, the potentials were constructed to yield a consistent overbinding for every bond type, as noted recently.²⁹

Recent work has been carried out to find an automated approach. Knaup et al. use a genetic algorithm to reproduce reference forces and reaction barriers.³⁰ Gaus et al. solve a linear equation system containing parameters for the repulsive potentials as unknowns in order to fit them to reference geometries, atomization energies, reaction energies, and vibrational frequencies.³¹

The resulting DFTB method works very well for homonuclear systems, where charge transfer between the atoms in the system does not occur or is very small. As soon as charge is flowing between atoms because of an electronegativity difference, the resulting density is no longer well approximated by the superposition of the atomic densities $\rho^0 = \sum \rho_\alpha$. As examples of the breakdown of the standard non-self-consistent method, the molecules CO_2 and formamide have been discussed.⁹ However, the formalism works very well when the charge flow is small; therefore, an extension will try to start from the non-self-consistent scheme and augment the Hamiltonian with appropriate additional terms.

9.2.4 Second-Order Approximation of the DFT Total Energy: Self-Consistent-Charge Density Functional Tight-Binding Method

The problem with the charge transfer is that the effective Kohn–Sham potentials contain only the neutral reference density ρ^0 , which does not account for charge

transfer between atoms. Let's try a Taylor series expansion (functional expansion) of the potential with the ground-state density ρ around the reference density ρ^0 :

$$v^{\text{eff}}[\rho] = v^{\text{eff}}[\rho^0] + \int \frac{\delta v^{\text{eff}}[\rho]}{\delta \rho} \delta \rho dr \quad (9.20)$$

This potential could be inserted into Eqs. (9.9) and (9.10). The first term on the right-hand side of Eq. (9.20) would lead to the zero-order terms in Eqs. (9.9) and (9.10), $H_{\mu\nu}[\rho^0]$, depending on the reference density, while the second term on the right-hand side of Eq. (9.20) would lead to corrections for charge transfer. In a second step, one would have to find approximations for the functional derivatives.

Since we need the total energy and not only the KS equations, it is better to start the functional expansion with the DFT total energy. The SCC-DFTB method is derived from density functional theory (DFT) by a second-order expansion of the DFT total energy functional with respect to the charge-density fluctuations $\delta \rho$ around a given reference density ρ^0 [$\rho^0 = \rho^0(\vec{r}')$, $\int' = \int d\vec{r}'$]:

$$\begin{aligned} E = \sum_i^{\text{occ}} < \Psi_i | \hat{H}^0 | \Psi_i > + \frac{1}{2} \iint' \left(\frac{1}{|\vec{r} - \vec{r}'|} + \left. \frac{\delta^2 E^{\text{xc}}}{\delta \rho \delta \rho'} \right|_{\rho^0} \right) \Delta \rho \Delta \rho' \\ - \frac{1}{2} \iint' \frac{\rho^0 \rho^0}{|\vec{r} - \vec{r}'|} + E^{\text{xc}}[\rho^0] - \int V^{\text{xc}}[\rho^0] \rho^0 + E^{\text{cc}} \end{aligned} \quad (9.21)$$

After introducing an LCAO ansatz $\Psi_i = \sum c_{\mu i} \eta_{\mu}$, the first term becomes

$$< \Psi_i | \hat{H}^0 | \Psi_i > = \sum c_{\mu i} c_{\nu i} H_{\mu\nu}^0$$

and can be evaluated as discussed above. The last four terms in Eq. (9.21) depend only on the reference density ρ^0 and represent the repulsive energy contribution E^{rep} , as discussed above. Therefore, we only have to deal with the second-order terms.

Going from DFTB to SCC-DFTB, the second-order term $E^{2\text{nd}}$ in the charge density fluctuations $\Delta \rho$ [second term in Eq. (9.21)] is approximated by writing $\Delta \rho$ as a superposition of atomic contributions:

$$\Delta \rho = \sum_{\alpha} \Delta \rho_{\alpha}$$

To further simplify $E^{2\text{nd}}$, we apply a monopole approximation

$$\Delta \rho_{\alpha} \approx \Delta q_{\alpha} F_{\alpha}^{00} Y^{00} \quad (9.22)$$

Basically, $\Delta \rho_{\alpha}$ is assumed to look like an 1s orbital. F_{α}^{00} denotes the normalized radial dependence of the density fluctuation on atom α , which is constrained (approximated) to be spherical (Y^{00}) (i.e., the angular deformation of the charge

density change in second order is neglected):

$$E^{2\text{nd}} \approx \frac{1}{2} \sum_{\alpha\beta} \Delta q_\alpha \Delta q_\beta \int \int' \left(\frac{1}{|\vec{r} - \vec{r}'|} + \frac{\delta^2 E^{\text{xc}}}{\delta \rho \delta \rho'} \bigg|_{n_0} \right) F_\alpha^{00} F_\beta^{00} (Y^{00})^2 dr dr' \quad (9.23)$$

This formula looks complicated but has a quite simple curve shape:

- For large distances, $R_{\alpha\beta} = |\vec{r} - \vec{r}'| \rightarrow \infty$, the XC terms vanish and the integral describes the Coulomb interaction of two spherical normalized charge densities, which reduces basically to $1/R_{\alpha\beta}$; that is, we get

$$E^{2\text{nd}} \approx \frac{1}{2} \sum_{\alpha\beta} \frac{\Delta q_\alpha \Delta q_\beta}{R_{\alpha\beta}}$$

- For vanishing interatomic distance, $R_{\alpha\beta} = |\vec{r} - \vec{r}'| \rightarrow 0$, the integral describes the electron–electron interaction on atom α . We can approximate the integral as

$$E^{2\text{nd}} \approx \frac{1}{2} \Delta q_\alpha^2 \frac{\partial^2 E_\alpha}{\partial^2 q_\alpha} = \frac{1}{2} \Delta q_\alpha^2 U_\alpha$$

U_α , known as the *Hubbard parameter* (which is twice the chemical hardness), describes how much the energy of a system changes upon adding or removing electrons.

Now we need a formula γ to interpolate between these two cases. A very similar situation appears in semiempirical quantum chemical methods such as MNDO, AM1, or PM3, where γ has a simple form, as given, for example, by the Klopman–Ohno approximation,

$$\gamma_{\alpha\beta} = \frac{1}{\sqrt{R_{\alpha\beta}^2 + 0.25(1/U_\alpha + 1/U_\beta)^2}} \quad (9.24)$$

To derive an expression analytically, we approximate the charge density fluctuations with spherical charge densities. Slater-like distributions

$$F_\alpha^{00} = \frac{\tau_\alpha}{8\pi} \exp(-\tau_\alpha |r - R_\alpha|) \quad (9.25)$$

located at R_α allow for an analytical evaluation of the Hartree contribution of two spherical charge distributions. This leads to a function of $\gamma_{\alpha\beta}$, which depends on the parameters τ_α and τ_β , determining the extension of the charge densities of atoms α and β . This function has a $1/R_{\alpha\beta}$ dependence for large $R_{\alpha\beta}$ and

approaches a finite value for $R_{\alpha\beta} \rightarrow 0$. For zero interatomic distances (i.e., $\alpha = \beta$) one finds that

$$\tau_{\alpha} = \frac{16}{5} \gamma_{\alpha\alpha} \quad (9.26)$$

The function $\gamma_{\alpha\beta}$ is shown schematically in Fig. 9.2.

After integration, $E^{2\text{nd}}$ becomes a simple two-body expression depending on atomic-like charges:

$$E^{2\text{nd}} = \frac{1}{2} \sum_{\alpha\beta} \Delta q_{\alpha} \Delta q_{\beta} \gamma_{\alpha\beta} \quad (9.27)$$

The diagonal terms $\gamma_{\alpha\alpha}$ model the dependence of the total energy on charge density fluctuations (decomposed into atomic contributions) in second order. The monopole approximation restricts the change of the electron density considered and no spatial deformations are included; only the change of energy with respect to change of charge on atom α is considered. By neglecting the effect of the chemical environment on atom α , the diagonal part of γ can be approximated by the chemical hardness η of the atom,

$$\gamma_{\alpha\alpha} = 2\eta_{\alpha} = U_{\alpha} = \frac{\partial^2 E_{\alpha}}{\partial^2 q_{\alpha}} \quad (9.28)$$

where E_{α} is the energy of the isolated atom α . U_{α} , the Hubbard parameter, is twice the chemical hardness of atom α , which can be estimated from the difference in

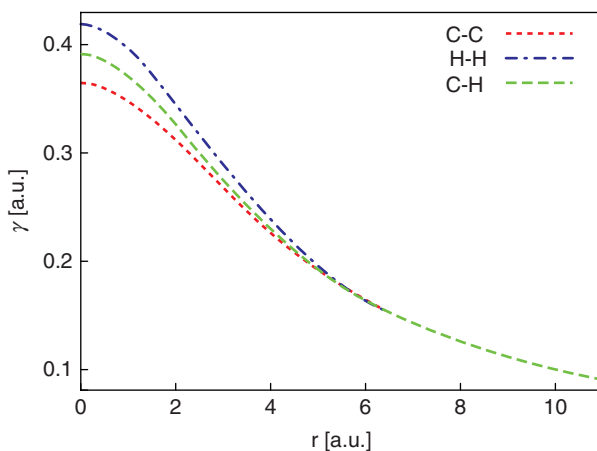


Fig. 9.2 Function γ_{CC} for two carbon atoms with the Hubbard parameter $U_C = 0.3647$ a.u. and γ_{HH} for two hydrogen atoms with $U_H = 0.4195$ a.u. over the interatomic distance. The function γ_{CH} differs from γ_{CC} and γ_{HH} for short interatomic distances. Clearly, the case $R_{C-H} = 0$ a.u. will not appear in a calculation.

the ionization potential and the electron affinity of atom α . For SCC-DFTB, it is calculated using Janak's theorem, by taking the first derivative of the energy of the highest-occupied molecular orbital with respect to the occupation number. Therefore, Eq. (9.26) implies that the extension of the charge distribution is inversely proportional to the chemical hardness of the respective atom (i.e., the size of an atom is inversely related to its chemical hardness). This is an important finding which is discussed in more detail below. The total SCC-DFTB finally reads

$$E^{\text{SCC-DFTB}} = \sum_{i\mu\nu} c_{\mu i} c_{\nu i} H_{\mu\nu}^0 + E^{2\text{nd}} + E^{\text{rep}} \quad (9.29)$$

9.3 PERFORMANCE OF STANDARD SCC-DFTB

9.3.1 Timings

The substantial advantage of using SCC-DFTB is its time/performance efficiency. Before showing the performance of several properties in the following subsections, Table 9.1 shows benchmark calculations for the CPU time of a single-point energy calculation on C_{60} , polyaniline, and some water clusters. All calculations were carried out on a single processor of a standard desktop PC. For SCC-DFTB the DFTB+ code³² was used. The DFT values were obtained using the TURBOMOLE program package.³³ For the PBE functional calculations the resolution of the identity (RI) integral evaluation has been used.³⁴ As a basis set for the DFT methods we chose 6-31G(d), which is a rather small basis set for practical use.

Table 9.1 shows that SCC-DFTB is at least 250 times faster than RI-PBE and more than 1000 times faster than B3LYP. This acceleration is due primarily to two issues: (1) the use of a minimal basis set within SCC-DFTB, and (2) the tabulation and neglect of integrals. For the water cluster $(\text{H}_2\text{O})_{48}$, for example, $N = 288$ basis functions are needed for a minimal basis set and $N = 864$ basis functions for the 6-31G(d) basis set. The time-limiting step for obtaining the

TABLE 9.1 Calculation Time (s) for Various Molecules with DFT and SCC-DFTB

Molecule	n^a	SCC-DFTB	RI-PBE ^b	B3LYP ^{b,c}
C_{60}^d	60	1	1,112	9,398
$(\text{Ala})_{10}^e$	112	4	966	6,655
$(\text{Ala})_{20}^e$	212	12	3,418	27,605
$(\text{H}_2\text{O})_{48}^f$	144	3	769	3,466
$(\text{H}_2\text{O})_{123}^f$	369	15	5,488	30,822

^aNumber of atoms.

^bBasis set 6-31G(d).

^cB3LYP_Gaussian keyword in TURBOMOLE.

^dBuckminsterfullerene C_{60} .

^ePolyaniline in α -helical form and including capping groups.

^fWater cluster.

energy with all methods discussed here is a matrix diagonalization, which scales with N^3 . Thus, an acceleration just from using the minimal basis of the factor 27 is achieved. The remaining factor is due to the tabulation and neglect of integrals; in this example this factor is roughly 10 and 40, for comparison with RI-PBE and B3LYP, respectively.

9.3.2 Small Organic Molecules

SCC-DFTB has been tested for various properties of small organic molecules, such as heats of formations, geometries, vibrational frequencies, and dipole moments, as documented in several recent publications. It should be noted that all these test sets contain a large number of molecules, representative of many chemical bonding situations.

In general, SCC-DFTB is excellent in reproducing geometries. Also, reaction energies are reproduced reasonably well on average,^{9,35} while heats of formation are overestimated, owing to the overbinding tendency of SCC-DFTB. Recently, the SCC-DFTB heats of formation have been tested systematically. It turned out that reparametrization of atomic contributions can improve the performance for heats of formation significantly; however, refined NDDO methods such as OM2³⁶ or PDDG/PM3³⁷ are still superior to SCC-DFTB in this respect.^{29,38} For a set of 622 neutral molecules containing the elements C, H, N, and O, Sattelmeyer et al. found a mean absolute error (MAE) in heats of formation for PDDG/PM3 of 3.2 and 5.8 kcal mol⁻¹ for SCC-DFTB.³⁸ Similarly, for a set of 140 CHNO-containing molecules, the respective mean absolute errors for OM2 and SCC-DFTB are 3.1 and 7.7 kcal mol⁻¹.²⁹

The performance of SCC-DFTB for vibrational frequencies, although reasonable on average, is less satisfactory than for geometries. However, vibrational frequencies could also be improved significantly after reparametrization.³⁹ The MAE for harmonic vibrational frequencies of 14 hydrocarbons drops from 59 cm⁻¹ for the standard parameterization to 33 cm⁻¹ for the reparameterized version. The MAE for the GGA-functional BLYP with the Dunning-type basis set cc-pVTZ is 25 cm⁻¹. Currently, parameters are available for O, N, C, H,⁹ S,⁴⁰ Zn,²⁸ Mg,⁴¹ and many transition metals.⁴²

9.3.3 Peptides

A good performance for small molecules does not guarantee a good description for larger molecules. A good example are the structures and relative energies of peptides, which pose significant problems for semiempirical models such as AM1⁴³ and PM3⁴⁴ but are well described at the SCC-DFTB level,^{45,46} or more elaborate NDDO methods such as OM1⁴⁷ OM2.^{36,48} Therefore, the performance for small organic molecules does not necessarily tell much about the performance for larger complexes, and SE methods should be benchmarked carefully before applying them to new classes of molecules.

9.3.4 Hydrogen-Bonded Systems

Standard SCC-DFTB slightly underestimates the dipole moments of polar molecules, as discussed, for example, for peptides.^{45,46,49} This leads to a slight underestimation of binding energies of weak hydrogen-bonded complexes^{18,49} by 1 to 2 kcal mol⁻¹ (e.g., the binding of the water dimer is found to be 3.3 kcal mol⁻¹, in contrast to 5 kcal mol⁻¹ at a high computational level). Also, relative energies of peptide conformations are underestimated due to this error. It should be noted that this underestimation is quite systematic (i.e., the relative stability of different conformers is preserved).

9.4 EXTENSIONS OF STANDARD SCC-DFTB

9.4.1 Inclusion of Dispersion Forces

SCC-DFTB is derived from DFT and therefore inherits the well-known failures of the gradient-corrected (GGA) DFT functionals. This concerns the problem of overpolarizability,¹⁶ the problem of charge transfer and ionic excited states,⁵⁰ and deficiencies in describing van der Waals interactions. These problems have been reviewed briefly by Elstner.²⁰

Dispersion interactions become important for larger molecules, since they stabilize more complex structures. Therefore, we proposed to include them empirically on top of DFT and implemented this for SCC-DFTB.¹⁸ This approach was adopted to DFT later^{51,52} and has become increasingly available in many DFT codes.

We have shown that DFT would fail to describe the stacking interaction between DNA bases without proper inclusion of dispersion interactions.¹⁸ DNA would not be stable. Surprisingly, dispersion interactions are also vital for stable peptide and protein structures. Neglecting dispersion forces, many peptide and protein conformations would not be stable; that is, standard DFT and SCC-DFTB are not able to describe the structure and dynamics of complex biological matter (and other materials, where dispersion forces are important).

To include dispersion forces, simple two-body potentials with $1/R^6$ dependence are added to the DFTB total energy. However, they have to be damped using a properly chosen damping function $f(R_{\alpha\beta})$ for short distances¹⁸:

$$E^{\text{SCC-DFTB-D}} = E^{\text{SCC-DFTB}} - \sum_{\alpha \neq \beta} f(R_{\alpha\beta}) \frac{C_{\alpha\beta}^6}{R_{\alpha\beta}^6} \quad (9.30)$$

with $C_{\alpha\beta}^6$ being properly chosen van der Waals parameters. Note that including such an extension to DFT leads to very different results, depending on the DFT functional used for exchange and dispersion.⁵¹ Only a properly chosen scaling function leads to quantitatively satisfying results.⁵² More details may be found elsewhere.²⁰

9.4.2 Beyond Standard Second-Order DFTB

The approximation of the second derivatives of the total DFT energy by the γ function in order to model charge-transfer effects contains several approximations. As we have discussed in detail, the use of the γ function implicitly assumes that the size of an atom is represented by the inverse of the Hubbard (chemical hardness) parameter U_α , which enters the γ function.^{20,53} This relation holds quite well for many main-group elements but is completely wrong for the hydrogen atom.⁵³ Therefore, the function γ has been modified to account for this irregularity. This leads to a significant improvement in hydrogen-bonding energies. The large error of 1 to 2 kcal mol⁻¹ per hydrogen bond in the standard SCC-DFTB scheme can be reduced to about 0.5 kcal mol⁻¹ using the modified γ function.

Whereas for the description of hydrogen bonds a second-order expansion of total energy seems to be adequate, the calculation of proton affinities have been shown to be largely in error. This property is crucial, however, for an appropriate description of proton transfer reactions, and semiempirical methods in general have problems predicting this value accurately.⁵⁴ The second-order approximation of DFTB works well for many systems, including charged systems, where the charge is delocalized over extended molecular fragments. For charged molecules, however, where the charge is localized, this approximation breaks down. It has been shown that for these cases the total energy [Eq. (9.21)] has to be expanded up to third order in the density fluctuations.^{20,53,55} This is crucial in particular for the calculation of deprotonation energies, where the inclusion of third-order terms leads to significant improvement. For example, the deprotonation energy of water is in error by nearly 30 kcal/mol in standard SCC-DFTB, whereas it has an error of a few kcal mol⁻¹ in the third-order formulation.

Formally, the expansion of the DFT total energy is carried out up to third order, and similar approximations are made as in the second-order case.⁵³ In third order, the Hubbard parameter U_α becomes charge dependent. Since $1/U_\alpha$ reflects the atom size, the charge dependence of U_α can account for the larger size of anions compared to neutral atoms or cations. In third-order DFTB, a new parameter occurs, the derivative of the Hubbard parameter, which can be calculated from DFT⁵³ or fitted to minimize the error in the deprotonation energies of a suitably chosen reference set of molecules.⁵⁵

9.4.3 Excited States via Time-Dependent DFT

The core of SCC-DFTB is an efficient approximation of the second derivatives of the total energy by the function $\gamma_{\alpha\beta}$. Such a second derivative also appears in the TD-DFT linear response formalism, which makes it possible to compute excited-state energies within the DFT framework. We have implemented this formalism for SCC-DFTB,⁴⁰ finding surprisingly good results for singlet excitations at very low computational cost, while the problems of TD-DFT for

higher excitation, charge transfer, and ionic excited states are retained.⁵⁰ More details are available in a recent review by Niehaus.⁵⁶

9.4.4 QM/MM Methods

To effectively represent coupling between the environment and the quantum region, quantum mechanical methods have been coupled to empirical force-field methods in the QM/MM methods. Although introduced as early as in 1976,⁵⁷ it was not until the early 1990s that QM/MM methods became widely used in the study of biological systems (a recent comprehensive review can be found in Ref. 2). Several QM/MM implementations with SCC-DFTB as the QM part have been realized up to now, incorporating it into various empirical force-field packages.^{58–62} But even for QM/MM approaches using SE methods as QM, the collective reorganization in the environment can become a computational bottleneck. Therefore, much effort is invested in developing *multiscale methods*, which combine QM/MM with continuum electrostatic methods (CM) for an integrated treatment of large systems. DFTB QM/MM coupling to CHARMM has been combined with a continuum approach,^{63,64} the generalized solvent-boundary potential developed originally by Roux and co-workers⁶⁵ for classical simulations. The SCC-DFTB/MM methodology^{19,20} as well as the SCC-DFTB/MM/CM methodology^{63,66} has recently been reviewed.

9.5 CONCLUSIONS

SCC-DFTB is a semiempirical method derived from DFT-GGA. This means that all deficiencies of DFT-GGA are inherited directly. Note that SCC-DFTB applies pure GGA functionals (PBE) (i.e., no hybrid variant is available), which can ameliorate these failures to some degree. On the other hand, SCC-DFTB also inherits the merits of DFT, its conceptual simplicity in incorporating correlation effects, and its good performance for many molecular properties of interest. As a result, SCC-DFTB predicts molecular geometries surprisingly well; vibrational frequencies are also satisfactory. Reproduction of heats of formation for small organic molecules is comparable to the performance of modern semiempirical methods, although new variants such as PDDG-PM3 or OM2 are still slightly superior in this respect. It should be noted that approximate methods should be carefully benchmarked for classes of molecules and not applied blindly.[†]

REFERENCES

1. Bowler, D. R.; Aoki, M.; Goringe, C. M.; Horsfield, A. P.; Pettifor, D. G. *Model. Simul. Mater. Sci. Eng.* **1997**, 5, 199.

[†]This also applies to DFT methods (although to a lesser degree), since their approximate nature leads to a variety of problems and failures.

2. Senn, H. M.; Thiel, W. *Curr. Opin. Chem. Biol.* **2007**, *11*, 182.
3. Senn, H. M.; Thiel, W. *Angew. Chem. Int. Ed.* **2009**, *48*, 1198.
4. Elstner, M.; Cui, Q. *Multi-scale Methods for the Description of Chemical Events in Biological Systems*, Multiscale Simulation Methods in Molecular Sciences, NIC-Serie, Publikationsreihe des John von Neumann-Instituts für Computing, Jülich, Germany, 2009.
5. Goringe, C. M.; Bowler, D. R.; Hernandez, E. *Rep. Prog. Phys.* **1997**, *60*, 1447.
6. Colombo, L. *Riv. Nuovo Cimento Soc. Ital. Fisi.* **2005**, *28*, 1.
7. Harris, J. *Phys. Rev. B* **1985**, *31*, 1770.
8. Foulkes, W. M. C.; Haydock, R. *Phys. Rev. B* **1989**, *39*, 12520.
9. Elstner, M.; Porezag, D.; Jungnickel, G.; Elstner, J.; Haugk, M.; Frauenheim, T.; Suhai, S.; Seifert, G. *Phys. Rev. B* **1998**, *58*, 7260.
10. Porezag, D.; Frauenheim, T.; Köhler, T.; Seifert, G.; Kaschner, R. *Phys. Rev. B* **1995**, *51*, 12947.
11. Seifert, G.; Eschrig, H.; Bieger, W. *Z. Phys. Chem. (Leipzig)* **1986**, *267*, 529.
12. Widany, J.; Frauenheim, T.; Köhler, T.; Sternberg, M.; Porezag, D.; Jungnickel, G.; Seifert, G. *Phys. Rev. B* **1996**, *53*, 4443.
13. Seifert, G. *J. Phys. Chem. A* **2007**, *111*, 5609.
14. Witek, H. A.; Köhler, C.; Frauenheim, T.; Morokuma, K.; Elstner, M. *J. Phys. Chem. A* **2007**, *111*, 5712.
15. Perdew, J. P.; Burke, K.; Ernzerhof, M. *Phys. Rev. Lett.* **1996**, *77*, 3865.
16. Wanko, M.; Hoffmann, M.; Frauenheim, T.; Elstner, M. *J. Comput. Aided Mol. Des.* **2006**, *20*, 511.
17. Wanko, M.; Hoffmann, M.; Strodel, P.; Koslowski, A.; Thiel, W.; Neese, F.; Frauenheim, T.; Elstner, M. *J. Phys. Chem. B* **2005**, *109*, 3606.
18. Elstner, M.; Hobza, P.; Frauenheim, T.; Suhai, S.; Kaxiras, E. *J. Chem. Phys.* **2001**, *114*, 5149.
19. Elstner, M.; Frauenheim, T.; Suhai, S. *J. Mol. Struct. (Theochem)* **2003**, *632*, 29.
20. Elstner, M. *Theor. Chem. Acc.* **2006**, *116*, 316.
21. Frauenheim, T.; Seifert, G.; Elstner, M.; Niehaus, T.; Köhler, C.; Amkreutz, M.; Sternberg, M.; Hajnal, Z.; Di Carlo, A.; Suhai, S. *J. Phys. Condens. Matter* **2002**, *14*, 3015.
22. Parr, R. G.; Yang, W. *Density-Functional Theory of Atoms and Molecules*; Oxford University Press, New York, 1989.
23. Eschrig, H. *Optimized LCAO Method and Electronic Structure of Extended Systems*, Springer-Verlag, Berlin, 1989.
24. Seifert, G. *J. Phys. Chem. A* **2007**, *111*, 5609.
25. Seifert, G.; Porezag, D.; Frauenheim, T. *Int. J. Quantum Chem.* **1996**, *58*, 185.
26. Frauenheim, T.; Weich, F.; Köhler, T.; Uhlmann, S.; Porezag, D.; Seifert, G. *Phys. Rev. B* **1995**, *52*, 11492.
27. Slater, J. C.; Koster, G. F. *Phys. Rev.* **1954**, *94*, 1498.
28. Elstner, M.; Cui, Q.; Munih, P.; Kaxiras, E.; Frauenheim, T.; Karplus, M. *J. Comput. Chem.* **2003**, *24*, 565.
29. Otte, N.; Scholten, M.; Thiel, W. *J. Phys. Chem. A* **2007**, *111*, 5751.

30. Knaup, J. M.; Hourahine, B.; Frauenheim, T. *J. Phys. Chem. A* **2007**, *111*, 5637.
31. Gaus, M.; Chou, C.; Witek, H.; Elstner, M. *J. Phys. Chem. A* **2009**, *113*, 11866.
32. DFTB+, a development of Bremen Center of Computational Material Science (Prof. Frauenheim), available at <http://www.dftb.org>.
33. TURBOMOLE V6.1 2009, a development of University of Karlsruhe and Forschungszentrum Karlsruhe GmbH, 1989–2007, TURBOMOLE GmbH, since 2007; available at <http://www.turbomole.com>.
34. Ahlrichs, R. *Phys. Chem. Chem. Phys.* **2004**, *6*, 5119.
35. Krüger, T.; Elstner, M.; Schiffels, P.; Frauenheim, T. *J. Chem. Phys.* **2005**, *122*, 114110.
36. Weber, W.; Thiel, W. *Theor. Chem. Acc.* **2000**, *103*, 495.
37. Repasky, M. P.; Chandrasekhar, J.; Jørgensen, W. L. *J. Comput. Chem.* **2002**, *23*, 1601.
38. Sattelmeyer, K. W.; Tirado-Rives, J.; Jørgensen, W. L. *J. Phys. Chem. A* **2006**, *110*, 13551.
39. Małolepsza, E.; Witek, H. A.; Morokuma, K. *Chem. Phys. Lett.* **2005**, *412*, 237.
40. Niehaus, T. A.; Suhai, S.; Della Sala, F.; Lugli, P.; Elstner, M.; Seifert, G.; Frauenheim, T. *Phys. Rev. B* **2001**, *6308*, 085108.
41. Cai, Z.; Lopez, P.; Reimers, J. R.; Cui, Q.; Elstner, M. *J. Phys. Chem. A* **2007**, *111*, 5743.
42. Zheng, G.; Witek, H. A.; Bobadova-Parvanova, P.; Irle, S.; Musaev, D. G.; Prabhakar, R.; Morokuma, K.; Lundberg, M.; Elstner, M.; Khler, C.; Frauenheim, T. *J. Chem. Theory Comput.* **2007**, *3*, 1349.
43. Dewar, M. J. S.; Ziebis, E. G.; Healy, E. F.; Stewart, J. J. P. *J. Am. Chem. Soc.* **1985**, *107*, 3902.
44. Stewart, J. J. P. *J. Comput. Chem.* **1989**, *10*, 209.
45. Elstner, M.; Jalkanen, K.; Knapp-Mohammady, M.; Frauenheim, T.; Suhai, S. *Chem. Phys.* **2000**, *256*, 15.
46. Elstner, M.; Jalkanen, K.; Knapp-Mohammadi, M.; Frauenheim, T.; Suhai, S. *Chem. Phys.* **2001**, *263*, 203.
47. Kolb, M.; Thiel, W. *J. Comput. Chem.* **1993**, *14*, 775.
48. Möhle, K.; Hofmann, H.-J.; Thiel, W. *J. Comput. Chem.* **2001**, *22*, 509.
49. Elstner, M.; Frauenheim, T.; Kaxiras, E.; Seifert, G.; Suhai, S. *Phys. Status Solidi B* **2000**, *217*, 357.
50. Wanko, M.; Garavelli, M.; Bernardi, F.; Niehaus, T. A.; Frauenheim, T.; Elstner, M. *J. Chem. Phys.* **2004**, *120*, 1674.
51. Wu, Q.; Yang, W. *J. Chem. Phys.* **2002**, *116*, 515.
52. Grimme, S. *J. Comput. Chem.* **2004**, *25*, 1463.
53. Elstner, M. *J. Phys. Chem. A* **2007**, *111*, 5614.
54. Range, K.; Riccardi, D.; Elstner, M.; Cui, Q.; York, D. *Phys. Chem. Chem. Phys.* **2005**, *7*, 3070.
55. Yang, Y.; Yu, H.; York, D.; Cui, Q.; Elstner, M. *J. Phys. Chem. A* **2007**, *111*, 10861.
56. Niehaus, T. A. *J. Mol. Struct. (Theochem)* **2009**, *914*, 38.
57. Warshel, A.; Levitt, M. *J. Mol. Biol.* **1976**, *103*, 227.

58. Han, W.; Elstner, M.; Jalkanen, K. J.; Frauenheim, T.; Suhai, S. *Int. J. Quantum Chem.* **2000**, *78*, 459.
59. Cui, Q.; Elstner, M.; Kaxiras, E.; Frauenheim, T.; Karplus, M. *J. Phys. Chem. B* **2001**, *105*, 569.
60. Seabra, G. D. M.; Walker, R. C.; Elstner, M.; Case, D. A.; Roitberg, A. E. *J. Phys. Chem. A* **2007**, *111*, 5655.
61. Hu, H.; Elstner, M.; Hermans, J. *Proteins Struct. Funct. Genet.* **2003**, *50*, 451.
62. Liu, H.; Elstner, M.; Kaxiras, E.; Frauenheim, T.; Hermans, J.; Yang, W. *Proteins Struct. Funct. Genet.* **2001**, *44*, 484.
63. Riccardi, D.; Schaefer, P.; Yang, Y.; Yu, H.; Ghosh, N.; Prat-Resina, X.; König, P.; Li, G.; Xu, D.; Guo, H.; Elstner, M.; Cui, Q. *J. Phys. Chem. B* **2006**, *110*, 6458.
64. König, P. H.; Ghosh, N.; Hoffmann, M.; Elstner, M.; Tajkhorshid, E.; Frauenheim, T.; Cui, Q. *J. Phys. Chem. A* **2006**, *110*, 548.
65. Im, W.; Berneche, S.; Roux, B. *J. Chem. Phys.* **2001**, *114*, 2924.
66. Cui, Q. *Theor. Chem. Acc.* **2006**, *116*, 51.

10 Introduction to Effective Low-Energy Hamiltonians in Condensed Matter Physics and Chemistry

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In this chapter I discuss some simple effective Hamiltonians that have widespread applications to solid-state and molecular systems. Although meant to be an introduction to a beginning graduate student, I hope that it may also help to break down the divide between the physics and chemistry literatures.

After a brief introduction to second quantization notation (Section 10.1), which is used extensively, I focus on the “four H’s”: the Hückel (or tight binding; Section 10.2), Hubbard (Section 10.3), Heisenberg (Section 10.4), and Holstein (Section 10.6) models. These models play central roles in our understanding of condensed matter physics, particularly for materials where electronic correlations are important but are less well known to the chemistry community. Some related models, such as the Pariser–Parr–Pople model, the extended Hubbard model, multiorbital models, and the ionic Hubbard model, are also discussed in Section 10.6. As well as their practical applications, these models allow us to investigate electronic correlations systematically by “turning on” various interactions in the Hamiltonian one at a time. Finally, in Section 10.7, I discuss the epistemological basis of effective Hamiltonians and compare and contrast this approach with *ab initio* methods before discussing the problem of the parameterization of effective Hamiltonians.

As this chapter is intended to be introductory, I do not attempt to make frequent comparisons to the latest research problems; rather, I compare the predictions of model Hamiltonians with simple systems chosen for pedagogical reasons. Similarly, references have been chosen for their pedagogical and historical value rather than on the basis of scientific priority.

Given the similarity in the problems addressed by theoretical chemistry and theoretical condensed matter physics, surprisingly few advanced texts discuss the interface of two subjects. Unfortunately, this leads to many cultural differences between the fields. Nevertheless, some textbooks do try to bridge the gap, and the reader in search of more than the introductory material presented here is referred to a book by Fulde¹ and several other chapters in this book: Chapter 6 describes the state of the art in using density functional theory and *ab initio* Hartree–Fock-based approaches to the *a priori* evaluation of properties of systems involving strongly correlated electrons, and Chapter 4 describes *ab initio* approaches based on quantum Monte Carlo.

10.1 BRIEF INTRODUCTION TO SECOND QUANTIZATION NOTATION

The models discussed in this chapter are easiest to understand if one employs the *second quantization formalism*. In this section we introduce its basic formalism briefly and informally. More details may be found in many textbooks (e.g., Schatz and Ratner² or Mahan³). Readers already familiar with this notation may wish to skip this section, although the last two paragraphs do define some nomenclature that is used throughout the chapter.

10.1.1 Simple Harmonic Oscillator

Let us begin by considering a particle of mass m moving in a one-dimensional harmonic potential:

$$V(x) = \frac{1}{2}kx^2 \quad (10.1)$$

This may be familiar as the potential experienced by an ideal spring displaced from its equilibrium position by a distance x , in which context k is known as the *spring constant*.⁴ Equation (10.1) is also the potential felt by an atom as it is displaced (by a small amount) from its equilibrium position in a molecule.⁵ Classically, this problem is straightforward to solve,⁴ and as well as the trivial solution, one finds that the particle may oscillate with a resonant frequency $\omega = \sqrt{k/m}$. The time-independent Schrödinger equation for a simple harmonic oscillator is therefore

$$\hat{H}_{\text{sho}}\psi_n \equiv \left(\frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \right) \psi_n = E_n\psi_n \quad (10.2)$$

where $\hat{p} = (\hbar/i)(\partial/\partial x)$ is the particle's momentum and ψ_n is the n th wavefunction or eigenfunction, which has energy, or eigenvalue, E_n .

This problem is solved in many introductory texts on quantum mechanics⁶ using the standard methods of “first quantized” quantum mechanics. However, a more elegant way to solve this problem is to introduce the *ladder operator*:

$$\hat{a} \equiv \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + i \frac{\hat{p}}{\sqrt{2m\hbar\omega}} \quad (10.3)$$

and its hermitian conjugate:

$$\hat{a}^\dagger \equiv \sqrt{\frac{m\omega}{2\hbar}} \hat{x} - i \frac{\hat{p}}{\sqrt{2m\hbar\omega}} \quad (10.4)$$

One of the most important features of quantum mechanics is that momentum and position do not commute⁶ (i.e., $[\hat{p}, \hat{x}] \equiv \hat{p}\hat{x} - \hat{x}\hat{p} = -i\hbar$). From this commutation relation it is straightforward to show that

$$\hat{H}_{\text{sho}} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \quad (10.5)$$

and

$$[\hat{a}, \hat{a}^\dagger] \equiv \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} = 1 \quad (10.6)$$

One can also show that $[\hat{H}_{\text{sho}}, \hat{a}] = [\hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2}), \hat{a}] = \hbar\omega[\hat{a}^\dagger, \hat{a}]\hat{a} = -\hbar\omega\hat{a}$, in a similar manner. Therefore, $[\hat{H}_{\text{sho}}, \hat{a}]\psi_n = -\hbar\omega\hat{a}\psi_n$, and hence

$$\hat{H}_{\text{sho}}\hat{a}\psi_n = (E_n - \hbar\omega)\hat{a}\psi_n \quad (10.7)$$

Equation (10.7) tells us that $\hat{a}\psi_n$ is an eigenstate of $\hat{H}_{\text{sho}}$ with energy $E_n - \hbar\omega$, provided that $\hat{a}\psi_n \neq 0$. That is, the operator $\hat{a}$ moves the system from one eigenstate to another whose energy is lower by $\hbar\omega$; thus, $\hat{a}$ is known as the lowering or destruction operator.

Note that for any wavefunction ϕ , $\langle \phi | \hat{p}^2 | \phi \rangle \geq 0$ and $\langle \phi | \hat{x}^2 | \phi \rangle \geq 0$. Therefore, it follows from Eq. (10.2) that $E_n \geq 0$ for all n . Hence, there is a lowest energy state, or ground state, which we will denote as ψ_0 . Therefore, there is a limit to how often we can keep lowering the energy of the state, (i.e., $\hat{a}\psi_0 = 0$). We can now calculate the ground-state energy of the harmonic oscillator,

$$\hat{H}_{\text{sho}}\psi_0 = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \psi_0 = \frac{1}{2}\hbar\omega \quad (10.8)$$

In the same way as we derived Eq. (10.7), one can easily show that $\hat{H}_{\text{sho}}\hat{a}^\dagger\psi_n = (E_n + \hbar\omega)\hat{a}^\dagger\psi_n$. Therefore, $\hat{a}^\dagger$ moves us up the ladder of states that $\hat{a}$ moved us down. Hence $\hat{a}^\dagger$ is known as a raising or creation operator. Thus, we have

$$\hat{a}^\dagger\psi_n = \sqrt{n+1}\psi_{n+1} \quad (10.9)$$

and

$$\hat{a}\psi_n = \sqrt{n}\psi_{n-1} \quad (10.10)$$

where the terms inside the radicals are required for the correct normalization of the wavefunctions.⁷ Therefore, $\psi_n = (1/\sqrt{n!})(\hat{a}^\dagger)^n\psi_0$ and

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right) \quad (10.11)$$

Notice that above we solved the simple harmonic oscillator (i.e., calculated the energies of all of the eigenstates) without needing to find explicit expressions for any of the first quantized eigenfunctions, ψ_n . This general feature of the second quantized approach is extremely advantageous when we are dealing with the complex many-body wavefunctions typical in condensed matter physics and chemistry.

10.1.2 Second Quantization for Light and Matter

We can extend the second quantization formalism to light and matter. Let us first consider bosons, which are not subject to the Pauli exclusion principle (e.g., phonons, photons, deuterium nuclei, ^4He atoms). We define the bosonic field operator $\hat{b}^\dagger(\mathbf{r})$ as creating a boson at position $\mathbf{r}$; similarly, $\hat{b}(\mathbf{r})$ annihilates a boson at position $\mathbf{r}$. The bosonic field operators obey the commutation relations $[\hat{b}(\mathbf{r}), \hat{b}(\mathbf{r}')] = 0$, $[\hat{b}^\dagger(\mathbf{r}), \hat{b}^\dagger(\mathbf{r}')] = 0$, and

$$[\hat{b}(\mathbf{r}), \hat{b}^\dagger(\mathbf{r}')] = \delta(\mathbf{r} - \mathbf{r}') \quad (10.12)$$

This is just the generalization of Eq. (10.6) for the field operators. We can create any state by acting products, or sums of products, of the $\hat{b}^\dagger(\mathbf{r})$ on the vacuum state (i.e., the state that does not contain any bosons), which is usually denoted as $|0\rangle$.

Many body wavefunctions for fermions (e.g., electrons, protons, neutrons, ^3He atoms) are complicated by the need for the antisymmetrization of the wavefunction (i.e., the wavefunction must change sign under the exchange of any two fermions). Therefore, if we introduce the fermionic field operators $\hat{\psi}^\dagger(\mathbf{r})$ and $\hat{\psi}(\mathbf{r})$, which, respectively, create and annihilate fermions at position $\mathbf{r}$, we must make sure that any wavefunction that we can make by acting on some set of these operators on the vacuum state is properly antisymmetrized. This is ensured⁸ if one insists that the field operators anticommute, that is, if

$$\{\hat{\psi}(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}')\} \equiv \hat{\psi}(\mathbf{r})\hat{\psi}^\dagger(\mathbf{r}') + \hat{\psi}^\dagger(\mathbf{r}')\hat{\psi}(\mathbf{r}) = \delta(\mathbf{r} - \mathbf{r}') \quad (10.13)$$

$$\{\hat{\psi}(\mathbf{r}), \hat{\psi}(\mathbf{r}')\} = 0 \quad (10.14)$$

$$\{\hat{\psi}^\dagger(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}')\} = 0 \quad (10.15)$$

This guarantee of an antisymmetrized wavefunction is one of the most obvious advantages of the second quantization formalism, as it is much easier than having to deal with the Slater determinants that are typically used to ensure the antisymmetrization of the many-body wavefunction in the first quantized formalism.²

For any practical calculation one needs to work with a particular basis set, $\{\phi_i(\mathbf{r})\}$. The field operators can be expanded in an arbitrary basis set as

$$\hat{\Psi}(\mathbf{r}) = \sum_i \hat{c}_i \phi_i(\mathbf{r}) \quad (10.16)$$

$$\hat{\Psi}^\dagger(\mathbf{r}) = \sum_i \hat{c}_i^\dagger \phi_i^*(\mathbf{r}) \quad (10.17)$$

Thus, $\hat{c}_i^{(\dagger)}$ annihilates (creates) a fermion in the state $\phi_i(\mathbf{r})$. These operators also obey fermionic anticommutation relations,

$$\{\hat{c}_i, \hat{c}_j^\dagger\} = \delta_{ij} \quad (10.18)$$

$$\{\hat{c}_i, \hat{c}_j\} = 0 \quad (10.19)$$

$$\{\hat{c}_i^\dagger, \hat{c}_j^\dagger\} = 0 \quad (10.20)$$

As fermions obey the Pauli exclusion principle, there can be at most one fermion in a given state. We denote a state in which the i th basis function contains zero (one) particles by $|0_i\rangle$ ($|1_i\rangle$). Therefore,

$$\begin{aligned} \hat{c}_i |0_i\rangle &= 0 & \hat{c}_i |1_i\rangle &= |0_i\rangle \\ \hat{c}_i^\dagger |0_i\rangle &= |1_i\rangle & \hat{c}_i^\dagger |1_i\rangle &= 0 \end{aligned} \quad (10.21)$$

It is important to realize that the number 0 is very different from the state $|0_i\rangle$.

Any operator acting on a system of fermions can be expressed in terms of the $\hat{c}$ operators. A particularly important example is the number operator, $\hat{n}_i \equiv \hat{c}_i^\dagger \hat{c}_i$, which simply counts the number of particles in the state i , as can be confirmed by explicit calculation from Eqs. (10.21). The total number of particles in the system is therefore simply the expectation value of the operator $\hat{N} = \sum_i \hat{n}_i = \sum_i \hat{c}_i^\dagger \hat{c}_i$. Importantly, because we can write any operator in terms of the $\hat{c}$ operators, we can calculate any observable from the expectation value of some set of $\hat{c}$ operators. Thus we have access to a complete description of the system from the second quantization formalism. Further, we can always write the wavefunction in terms of the $\hat{c}$ operators if an explicit description of the wavefunction is required. For example, the sum of Slater determinants,

$$\Psi(r_1, r_2) = \alpha \begin{vmatrix} \phi_1(r_1) & \phi_2(r_1) \\ \phi_1(r_2) & \phi_2(r_2) \end{vmatrix} + \beta \begin{vmatrix} \phi_3(r_1) & \phi_4(r_1) \\ \phi_3(r_2) & \phi_4(r_2) \end{vmatrix} \quad (10.22)$$

describes the same state as

$$|\Psi\rangle = (\alpha \hat{c}_1 \hat{c}_2 + \beta \hat{c}_3 \hat{c}_4) |0\rangle \quad (10.23)$$

where $|0\rangle = |0_1, 0_2, 0_3, 0_4, \dots\rangle$ is the vacuum state, as $\Psi(r_1, r_2) = \langle r_1, r_2 | \Psi \rangle$ (cf., e.g., Ref. 7).

Often, in order to describe solid-state and chemical systems, one needs to describe a set of N electrons whose behavior is governed by a Hamiltonian of the form

$$H = \sum_{n=1}^N \left[-\frac{\hbar^2 \nabla_n^2}{2m} + U(\mathbf{r}_n) + \frac{1}{2} \sum_{m \neq n} V(\mathbf{r}_n - \mathbf{r}_m) \right] \quad (10.24)$$

where $V(\mathbf{r}_n - \mathbf{r}_m)$ is the potential describing the interactions between electrons and $U(\mathbf{r}_i)$ is an external potential (including interactions with ions or nuclei, which may often be considered to be stationary on the time scales relevant to electronic processes, although we discuss effects due to the displacement of the nuclei in Section 10.6). In terms of our second quantization operators, this Hamiltonian may be written

$$\hat{H} = - \sum_{ij} t_{ij} \hat{c}_i^\dagger \hat{c}_j + \frac{1}{2} \sum_{ijkl} V_{ijkl} \hat{c}_i^\dagger \hat{c}_k^\dagger \hat{c}_l \hat{c}_j \quad (10.25)$$

where

$$t_{ij} = - \int d^3\mathbf{r} \phi_i^*(\mathbf{r}) \left[-\frac{\hbar^2 \nabla^2}{2m} + U(\mathbf{r}) \right] \phi_j(\mathbf{r}) \quad (10.26)$$

$$V_{ijkl} = \int d^3\mathbf{r}_1 \int d^3\mathbf{r}_2 \phi_i^*(\mathbf{r}_1) \phi_j(\mathbf{r}_1) V(\mathbf{r}_1 - \mathbf{r}_2) \phi_k^*(\mathbf{r}_2) \phi_l(\mathbf{r}_2) \quad (10.27)$$

and the labels i, j, k , and l are taken to define the spin as well as the basis function. This is exact, provided that we have an infinite complete basis. But practical calculations require the use of finite basis sets and often use incomplete basis sets. The simplest approach is simply to ignore this problem and calculate t_{ij} and V_{ijkl} directly from the finite basis set. However, this is often not the best approach. We delay until Section 10.7 a detailed discussion of why this is and of the deep philosophical issues that it raises. We also delay until Section 10.7 discussion of how to calculate these parameters. Until then we simply assume that t_{ij} , V_{ijkl} , and similar parameters required are known and focus instead on how to perform practical calculations using models of the form of Eq. (10.25) and closely related Hamiltonians.

In what follows we assume that the states created by the $\hat{c}_i^\dagger$ operators form an orthonormal basis. This greatly simplifies the mathematics but differs from the approach usually taken in introductory chemistry textbooks, as most quantum chemical calculations are performed in nonorthogonal bases for reasons of computational expedience.

10.2 HÜCKEL OR TIGHT-BINDING MODEL

The simplest model with the form of Eq. (10.25) is usually called the *Hückel model* in the context of molecular systems⁹ and the *tight-binding model* in the context of crystals.¹⁰ In these models one makes the approximation that $V_{ijkl} = 0$ for all i, j, k , and l . Therefore, these models explicitly neglect interactions between

electrons. The models are identical, but slightly different notation is standard in the different traditions. We assume that our basis set consists of orbitals centered on particular sites, as we will for all of the models considered in this chapter. These sites might, for example, be atoms in a molecule or solid, chemical groups within a molecule, p-d hybrid states in a transition metal oxide, entire molecules in a molecular crystal, or even larger structures. We will often use a nomenclature motivated by the case where the sites are atoms below; however, this does not mean that the mathematics is only applicable to that case. In the simplest case of only one orbital per spin state on each site

$$\hat{H}_{\text{tb}} = - \sum_{ij\sigma} t_{ij} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \quad (10.28)$$

where $\hat{c}_{i\sigma}^\dagger$ annihilates (creates) an electron with spin σ in an orbital centered on site i .

10.2.1 Molecules (the Hückel Model)

The standard notation in this context is $t_{ii} = -\alpha_i$, $t_{ij} = -\beta_{ij}$ if sites i and j are connected by a chemical bond, and $t_{ij} = 0$ otherwise. Note that the subscripts on α and β are also often dropped, but they are usually implicit; if the molecule contains more than one species of atom, the α 's will clearly be different on the different species and the β 's will depend on the species of each of the atoms between which the electron is hopping. Therefore,

$$\hat{H}_{\text{Hückel}} = \sum_{i\sigma} \alpha_i \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} + \sum_{\langle ij \rangle} \beta_{ij} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \quad (10.29)$$

where $\langle ij \rangle$ serves to remind us that the sum is only over those pairs of atoms joined by a chemical bond. Note that β_{ij} is typically negative.

10.2.1.1 Molecular Hydrogen Clearly, in H_2 there is only a single atomic species. In this case one can set $\alpha_i = \alpha$ for all i without loss of generality. Further, as there is also only a single bond, we may choose $\beta_{ij} = \beta$, giving

$$\hat{H}_{\text{Hückel}} = \alpha \sum_{\sigma} (\hat{n}_{1\sigma} + \hat{n}_{2\sigma}) + \beta \sum_{\sigma} (\hat{c}_{1\sigma}^\dagger \hat{c}_{2\sigma} + \hat{c}_{2\sigma}^\dagger \hat{c}_{1\sigma}) \quad (10.30)$$

where we have labeled the two atomic sites 1 and 2. This Hamiltonian has two eigenstates: one is known as the *bonding orbital*,

$$|\psi_{b\sigma}\rangle = \frac{1}{\sqrt{2}} (\hat{c}_{1\sigma}^\dagger + \hat{c}_{2\sigma}^\dagger) |0\rangle \quad (10.31)$$

and the other is known as the *antibonding orbital*,

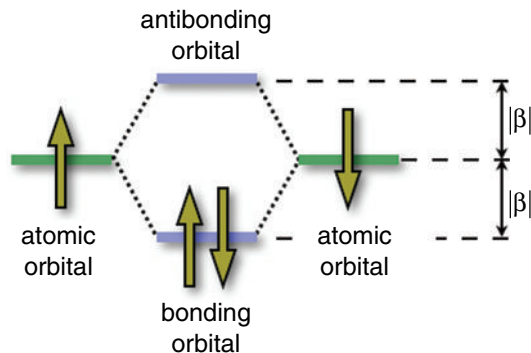


Fig. 10.1 (color online) Energy levels of the atomic and molecular orbitals in the Hückel description of H_2 . The bonding orbital is $|\beta|$ lower in energy than the atomic orbital, whereas the antibonding orbital is $|\beta|$ higher in energy than the atomic orbital. Therefore, neutral H_2 is stabilized by $2|\beta|$ relative to 2H .

$$|\psi_{a\sigma}\rangle = \frac{1}{\sqrt{2}}(\hat{c}_{1\sigma}^\dagger - \hat{c}_{2\sigma}^\dagger)|0\rangle \quad (10.32)$$

The bonding orbital has energy $\alpha + \beta$, whereas the antibonding orbital has energy $\alpha - \beta$. Recall that $\beta < 0$; therefore, every electron in the bonding state stabilizes the molecule by an amount $|\beta|$, whereas electrons in the antibonding state destabilize the molecule by an amount $|\beta|$, hence the nomenclature.[†] This is sketched in Fig. 10.1.

Because $V_{ijkl} = 0$, the electrons are noninteracting, so the molecular orbitals are not dependent on the occupation of other orbitals. Therefore to calculate the total energy of the ground state of the molecule, one simply fills up the states, starting with the lowest-energy states and respecting the Pauli exclusion principle. If the two protons are infinitely separated, $\beta = 0$ and the system has total energy $N\alpha$, where N is the total number of electrons. H_2^+ has only one electron, which, in the ground state, will occupy the bonding orbital, so H_2^+ has a binding energy of β . H_2 has two electrons; in the ground state these electrons have opposite spin and therefore can both occupy the bonding orbital. Thus, H_2 has a binding energy of 2β . H_2^- has three electrons, so while two can occupy the bonding state, one must be in the antibonding state; therefore, the binding energy is only β . Finally, H_2^{2-} has four electrons, so one finds two in the each molecular orbital. Therefore, the bonding energy is zero: the molecule is predicted to be unstable.

Thus, the Hückel model makes several predictions: neutral H_2 is predicted to be significantly more stable than any of the ionic states; the two singly ionic species are predicted to be equally stable; and the doubly cationic species is predicted to be unstable. Further, the lowest optical absorption is expected to correspond to the transition between the bonding orbital and the antibonding

[†]Note that in a nonorthogonal basis, the antibonding orbital may be destabilized by a greater amount than the bonding orbital is stabilized.

orbital. The energy gap for this transition is $2|\beta|$. Therefore, the lowest optical absorption is predicted to be the same in the neutral species as in the singly cationic species. Further, this absorption is predicted to occur at a frequency with the same energy as the heat of formation for the neutral species. Although these predictions do capture qualitatively what is observed experimentally, they are certainly not within chemical accuracy (i.e., within $k_B T \sim 1 \text{ kcal mol}^{-1} \sim 0.03 \text{ eV}$ for $T = 300 \text{ K}$). For example, the experimentally determined binding energies⁹ are 2.27 eV for H_2^+ , 4.74 eV for H_2 , and 1.7 eV for H_2^- , while H_2^{2-} is indeed unstable.

10.2.1.2 π -Hückel Theory of Benzene For many organic molecules a model known as π -Hückel theory is very useful. In π -Hückel theory one considers only the π -electrons. A simple example is a benzene molecule. The hydrogen atoms have no π -electrons and therefore are not represented in the model. This leaves only the carbon atoms, so again we can set $\alpha_i = \alpha$ and $\beta_{ij} = \beta$. Because of the ring geometry of benzene (and assuming that the molecule is planar), the Hamiltonian becomes

$$\hat{H}_{\text{Hückel}} = \alpha \sum_{i\sigma} \hat{n}_{i\sigma} + \beta \sum_{i\sigma} (\hat{c}_{i\sigma}^\dagger \hat{c}_{i+1\sigma} + \hat{c}_{i+1\sigma}^\dagger \hat{c}_{i\sigma}) \quad (10.33)$$

where the addition in the site index is defined modulo six (i.e., site seven is site one). For benzene we have six solutions per spin state:

$$\begin{aligned} |\psi_{A_{2u}}\rangle &= \frac{1}{\sqrt{6}} (\hat{c}_{1\sigma}^\dagger + \hat{c}_{2\sigma}^\dagger + \hat{c}_{3\sigma}^\dagger + \hat{c}_{4\sigma}^\dagger + \hat{c}_{5\sigma}^\dagger + \hat{c}_{6\sigma}^\dagger) |0\rangle \\ |\psi_{E_{1g}}\rangle &= \frac{1}{\sqrt{6}} (\hat{c}_{1\sigma}^\dagger + \varepsilon \hat{c}_{2\sigma}^\dagger + \varepsilon^2 \hat{c}_{3\sigma}^\dagger - \hat{c}_{4\sigma}^\dagger - \varepsilon \hat{c}_{5\sigma}^\dagger - \varepsilon^2 \hat{c}_{6\sigma}^\dagger) |0\rangle \\ |\psi_{E'_{1g}}\rangle &= \frac{1}{\sqrt{6}} (\hat{c}_{1\sigma}^\dagger - \varepsilon^2 \hat{c}_{2\sigma}^\dagger - \varepsilon \hat{c}_{3\sigma}^\dagger - \hat{c}_{4\sigma}^\dagger + \varepsilon^2 \hat{c}_{5\sigma}^\dagger + \varepsilon \hat{c}_{6\sigma}^\dagger) |0\rangle \\ |\psi_{E_{2u}}\rangle &= \frac{1}{\sqrt{6}} (\hat{c}_{1\sigma}^\dagger + \varepsilon^2 \hat{c}_{2\sigma}^\dagger - \varepsilon \hat{c}_{3\sigma}^\dagger + \hat{c}_{4\sigma}^\dagger + \varepsilon^2 \hat{c}_{5\sigma}^\dagger - \varepsilon \hat{c}_{6\sigma}^\dagger) |0\rangle \\ |\psi_{E'_{2u}}\rangle &= \frac{1}{\sqrt{6}} (\hat{c}_{1\sigma}^\dagger - \varepsilon \hat{c}_{2\sigma}^\dagger + \varepsilon^2 \hat{c}_{3\sigma}^\dagger + \hat{c}_{4\sigma}^\dagger - \varepsilon \hat{c}_{5\sigma}^\dagger + \varepsilon^2 \hat{c}_{6\sigma}^\dagger) |0\rangle \end{aligned}$$

and

$$|\psi_{B_{2g}}\rangle = \frac{1}{\sqrt{6}} (\hat{c}_{1\sigma}^\dagger - \hat{c}_{2\sigma}^\dagger + \hat{c}_{3\sigma}^\dagger - \hat{c}_{4\sigma}^\dagger + \hat{c}_{5\sigma}^\dagger - \hat{c}_{6\sigma}^\dagger) |0\rangle$$

where $\varepsilon = e^{i\pi/3}$. These wavefunctions are sketched in Fig. 10.2. The energies of these states are $E_{A_{2u}} = \alpha - 2|\beta|$, $E_{E_{1g}} = E_{E'_{1g}} = \alpha - |\beta|$, $E_{E_{2u}} = E_{E'_{2u}} = \alpha + |\beta|$, and $E_{B_{2g}} = \alpha + 2|\beta|$. The subscripts are symmetry labels^{11,12} for the group D_{6h} ; one should recall that because we are dealing with π -orbitals, all of the orbitals sketched here are antisymmetric under reflection through the plane of

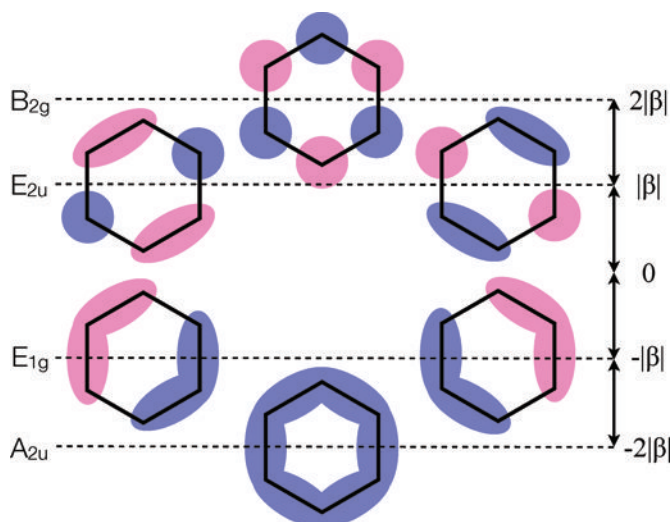


Fig. 10.2 (color online) Molecular orbitals for benzene from π -Hückel theory. Different colors indicate a change in sign of the wavefunction. In the neutral molecule the A_{2u} and both E_{1g} states are occupied, while the B_{2g} and E_{2u} states are virtual. Note that we have taken real superpositions⁹ of the twofold degenerate states in these plots.

the page. The degenerate (E_{1g} and E_{2u}) orbitals are typically written or drawn rather differently (see Lowe and Peterson⁹). However, any linear combination of degenerate eigenstates is also an eigenstate; this representation was chosen as it highlights the symmetry of the problem. For a more detailed discussion of this problem, see *Coulson's Valence*.¹³

10.2.1.3 Electronic Interactions and Parameterization of the Hückel Model

As noted above, the Hückel model does not explicitly include interactions between electrons. This leads to serious qualitative and quantitative failures of the model, some of which we have seen above and discuss further below. However, given the (mathematical and conceptual) simplicity and the computational economy of the method, one would like to improve the method as far as possible. So far we have treated the theory as parameter free. However, if we treat the model as a semiempirical method instead, we can include some of the effects due to electron–electron interactions without greatly increasing the computational cost of the method. For example, one can make α dependent on the charge on the atom. This is reasonable, as the more electrons we put on an atom, the more difficult it is to add another, due to the additional Coulomb repulsion from the extra electrons. The simplest way to account for this is by use of the ω technique,⁹ where one replaces

$$\alpha_i \rightarrow \alpha_i' = \alpha_i + \omega(q_0 - q_i)\beta \quad (10.34)$$

where q_i is the charge on atom i , q_0 is a (fixed) reference charge, and ω is a parameter. The ω technique suppresses the unphysical fluctuations of the electron density, which are often predicted by the Hückel model (cf. the discussion of H_2 above). Similar techniques can also be applied to β . These parameterizations only slightly complicate the model and do not lead to a major inflation of the computational cost, but can significantly improve the accuracy of the predictions of the Hückel model.¹⁴

10.2.2 Crystals (the Tight-Binding Model)

For infinite systems it is necessary to work with a fixed chemical potential rather than a fixed particle number. Therefore, before we discuss the tight-binding model, we briefly review the chemical potential (see also the discussion by Aktins and de Paula⁵ of the chemical potential in a chemical context).

10.2.2.1 Chemical Potential When one is dealing with a large system, keeping track of the number of particles can become difficult. This is particularly true in the thermodynamic limit, where the number of electrons $N_e \equiv \langle \hat{N} \rangle \rightarrow \infty$ and the volume of the system $V \rightarrow \infty$ in such a way as to ensure that the electronic density, $n_e = N_e/V$, remains constant. Lagrange multipliers¹⁵ are a powerful and general method for imposing constraints on differential equations (such as the Schrödinger equation) without requiring the solution of integrodifferential equations. Briefly, consider a function, $f(x, y, z, \dots)$ that we wish to extremize (minimize or maximize) subject to a constraint which means that $x, y, z, \dots$ are no longer independent. In general, we may write the constraint in the form $\phi(x, y, z, \dots) = 0$. This allows us to define the function $g(x, y, z, \dots, \lambda) \equiv f(x, y, z, \dots) + \lambda\phi(x, y, z, \dots)$, where λ is known as a Lagrange multiplier. One may show¹⁵ that the extremum of $g(x, y, z, \dots, \lambda)$ with respect to $x, y, z, \dots$ and λ is the extremum of $f(x, y, z, \dots)$ with respect to $x, y, z, \dots$ subject to the constraint that $\phi(x, y, z, \dots) = 0$.

Typically, the problem we wish to solve in chemistry and condensed matter physics is to minimize the free energy, F (which reduces to the energy, E , at $T = 0$) subject to the constraint of having a fixed number of electrons (determined by the chemistry of the material in question). This suggests that one should simply introduce a Lagrange multiplier to resolve the difficulty of constraining the number of electrons in the thermodynamic limit. A suitable constraint could be introduced by adding the term $\lambda(N_0 - \hat{N})$ to the Hamiltonian, where N_0 is the chemically required number of electrons, and requiring that the free energy is an extremum with respect to λ . However, one can also impose the same constraint and achieve additional physical insight by subtracting the term $\mu\hat{N}$ from the Hamiltonian and requiring that

$$N_0 = -\frac{\partial F}{\partial \mu} \quad (10.35)$$

The chemical potential (for electrons), μ , is then given by

$$\mu = -\frac{\partial F}{\partial N_e} \quad (10.36)$$

Therefore, specifying a system's chemical potential is equivalent to specifying the number of electrons, but provides a far more powerful approach for bulk systems.

Physically, this approach is equivalent to thinking of the system as being attached to an infinite bath of electrons (i.e., one is working in the grand canonical ensemble).¹⁶ Thus, the Fermi distribution for the system is given by

$$f(E, T) = \frac{1}{1 + e^{(E-\mu)/k_B T}} \quad (10.37)$$

Therefore, at $T = 0$ all of the states with energies lower than the chemical potential are occupied, and all of the states with energies greater than the chemical potential are unoccupied. Therefore, the Fermi energy, $E_F = \mu(T = 0)$. Note that as F is temperature dependent, Eq. (10.36) shows that, in general, μ will also be temperature dependent.[†] Nevertheless, Eq. (10.37) gives a clear interpretation of the chemical potential at any nonzero temperature: $\mu(T)$ is the energy of a state with a 50% probability of occupation at temperature T .

10.2.2.2 Tight-Binding Model For periodic systems (crystals) one usually refers to the Hückel model as the tight-binding model. Often, one only considers models with nearest-neighbor terms; that is, one takes $t_{ii} = -\epsilon_i$, $t_{ij} = t$ if i and j are at nearest-neighbor sites, and $t_{ij} = 0$ otherwise. Thus, for nearest-neighbor hopping only,

$$\hat{H}_{\text{tb}} - \mu \hat{N} = -t \sum_{\langle ij \rangle \sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \sum_{i\sigma} (\epsilon_i - \mu) \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \quad (10.38)$$

where μ is the chemical potential and $\langle ij \rangle$ indicates that the sum is over nearest neighbors only. Further, if we consider materials with only a single atomic species, we can set $\epsilon_i = 0$, yielding

$$\hat{H}_{\text{tb}} - \mu \hat{N} = -t \sum_{\langle ij \rangle \sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} - \mu \sum_{i\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \quad (10.39)$$

10.2.2.3 One-Dimensional Chain The simplest infinite system is a chain with nearest-neighbor hopping only. As we are on a chain, the sites have a natural ordering and the Hamiltonian may be written as

[†]In contrast, as E_F is only defined at $T = 0$, it is not temperature dependent.

$$\hat{H}_{\text{tb}} - \mu \hat{N} = -t \sum_{i\sigma} (\hat{c}_{i\sigma}^\dagger \hat{c}_{i+1\sigma} + \hat{c}_{i+1\sigma}^\dagger \hat{c}_{i\sigma}) - \mu \sum_{i\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \quad (10.40)$$

We can solve this model exactly by performing a lattice Fourier transform. We begin by introducing the reciprocal space creation and annihilation operators:

$$\hat{c}_{i\sigma} = \frac{1}{\sqrt{N}} \sum_k \hat{c}_{k\sigma} e^{ikR_i} \quad (10.41)$$

and

$$\hat{c}_{i\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_k \hat{c}_{k\sigma}^\dagger e^{-ikR_i} \quad (10.42)$$

where k is the lattice wavenumber or crystal momentum and R_i is the position of the i th lattice site. Therefore,

$$\hat{H}_{\text{tb}} - \mu \hat{N} = \frac{1}{N} \sum_{ikk'\sigma} \hat{c}_{k\sigma}^\dagger \hat{c}_{k'\sigma} e^{i(k'-k)R_i} [-t(e^{ik'a} + e^{-ika}) - \mu] \quad (10.43)$$

where a is the lattice constant (i.e., the distance between neighboring sites R_i and R_{i+1}). As $(1/N) \sum_i e^{i(k'-k)R_i} = \delta(k' - k)$,¹⁷ therefore,

$$\begin{aligned} \hat{H}_{\text{tb}} - \mu \hat{N} &= \sum_{k\sigma} [-2t \cos(ka) \hat{c}_{k\sigma}^\dagger \hat{c}_{k\sigma} - \mu \hat{c}_{k\sigma}^\dagger \hat{c}_{k\sigma}] \\ &= \sum_{k\sigma} (\varepsilon_k - \mu) \hat{c}_{k\sigma}^\dagger \hat{c}_{k\sigma} \end{aligned} \quad (10.44)$$

where $\varepsilon_k = -2t \cos ka$ is known as the *dispersion relation*. Notice that Eq. (10.44) is diagonal (i.e., it depends only on number operator terms, $n_{k\sigma} = \hat{c}_{k\sigma}^\dagger \hat{c}_{k\sigma}$). Therefore, the energy is just the sum of ε_k for the states $k\sigma$ that are occupied, and we have solved the problem. We plot the dispersion relation in Fig. 10.3a. For a tight-binding model, calculating the dispersion relation is equivalent to solving the problem.

The chemical potential, μ , must be chosen to ensure that there are the physically required number of electrons. Changing the chemical potential has the effect of moving the Fermi energy up or down the band and hence changing the number of electrons in the system. For example (cf. Fig. 10.3b to d), in the problem above, the half-filled band corresponds to $\mu = 0$, the quarter-filled band corresponds to $\mu = -t$, and the three-quarter-filled band corresponds to $\mu = t$.

10.2.2.4 Square, Cubic, and Hypercubic Lattices In more than one dimension the notation becomes slightly more complicated, but the mathematics does not necessarily become any more difficult. The simplest generalization of the chain we have solved above is the two-dimensional square lattice, where

$$\hat{H}_{\text{tb}} - \mu \hat{N} = -t \sum_{\langle ij \rangle \sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} - \mu \sum_{i\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \quad (10.45)$$

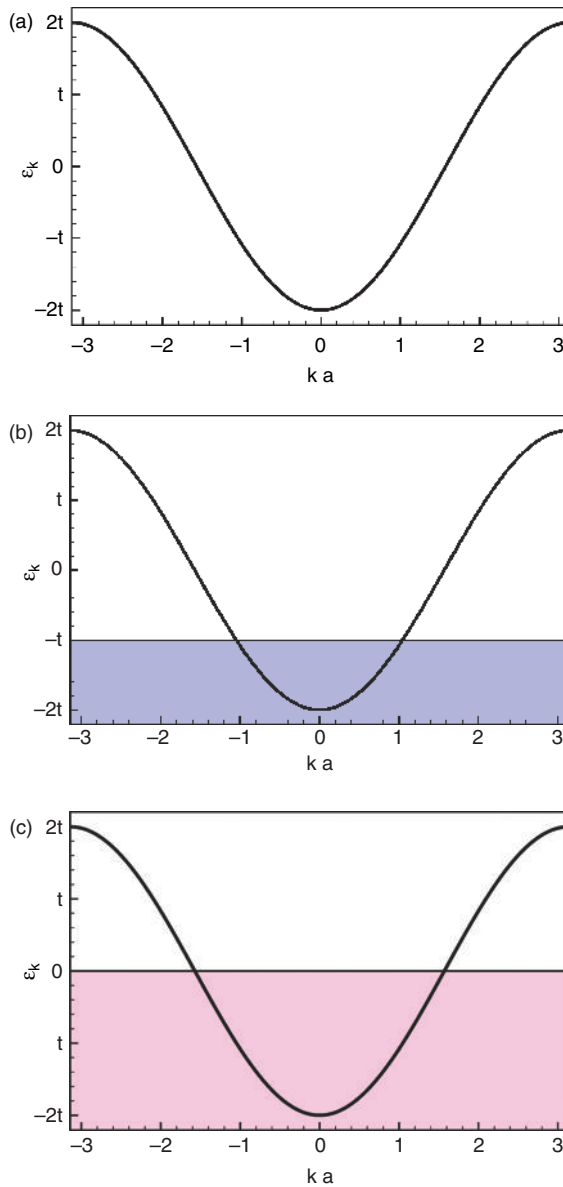


Fig. 10.3 (color online) (a) The dispersion relation, $\varepsilon_k = -2t \cos(ka)$, of the one-dimensional tight-binding chain with nearest neighbour hopping only. (b) Shaded area shows the filled states for $\mu = t$. (c) Shaded area shows the filled states for $\mu = t$. (d) Shaded area shows the filled states for $\mu = t$.

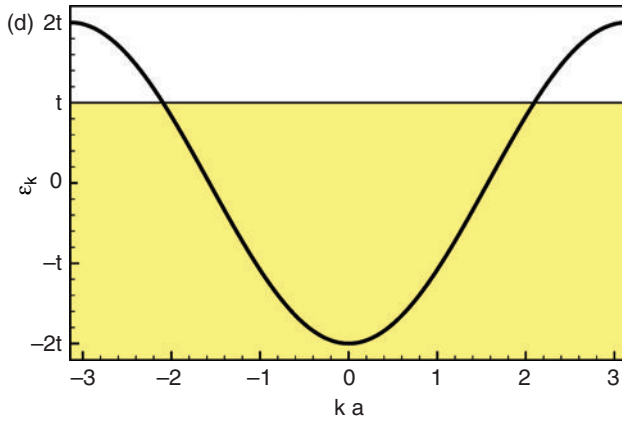


Fig. 10.3 (color online) (continued)

Recall that $\langle ij \rangle$ indicates that the sum is over nearest neighbors only. To solve this problem we simply generalize our reciprocal lattice operators to

$$\hat{c}_{i\sigma} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} \hat{c}_{\mathbf{k}\sigma} e^{i\mathbf{k} \cdot \mathbf{R}_i} \quad (10.46)$$

$$\hat{c}_{i\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} \hat{c}_{\mathbf{k}\sigma}^\dagger e^{-i\mathbf{k} \cdot \mathbf{R}_i} \quad (10.47)$$

where $\mathbf{k} = (k_x, k_y)$ is the lattice wavevector or crystal momentum and $\mathbf{R}_i = (x_i, y_i)$ is the position of the i th lattice site. We then simply repeat the process we used to solve the one-dimensional chain. As the lattice only contains bonds in perpendicular directions, the calculations for the x and y directions go through independently and one finds that

$$\hat{H}_{\text{tb}} - \mu \hat{N} = \sum_{\mathbf{k}\sigma} (\epsilon_{\mathbf{k}} - \mu) \hat{c}_{\mathbf{k}\sigma}^\dagger \hat{c}_{\mathbf{k}\sigma} \quad (10.48)$$

where the dispersion relation is now $\epsilon_{\mathbf{k}} = -2t(\cos k_x a_x + \cos k_y a_y)$ and a_v represents the lattice constants in the v direction.

A three-dimensional cubic lattice is not any more difficult. In this case, $\mathbf{k} = (k_x, k_y, k_z)$ and the solution is of the form of Eq. (10.48) but with $\epsilon_{\mathbf{k}} = -2t(\cos k_x a_x + \cos k_y a_y + \cos k_z a_z)$. Indeed, as long as we keep all the bonds mutually perpendicular, we can keep generalizing this solution to higher dimensions. This may sound somewhat academic, as no materials live in more than three dimensions, but the infinite-dimensional hypercubic lattice has become important in recent years because many models that include interactions can be solved exactly in infinite dimensions, as we discuss in Section 10.3.4.2.

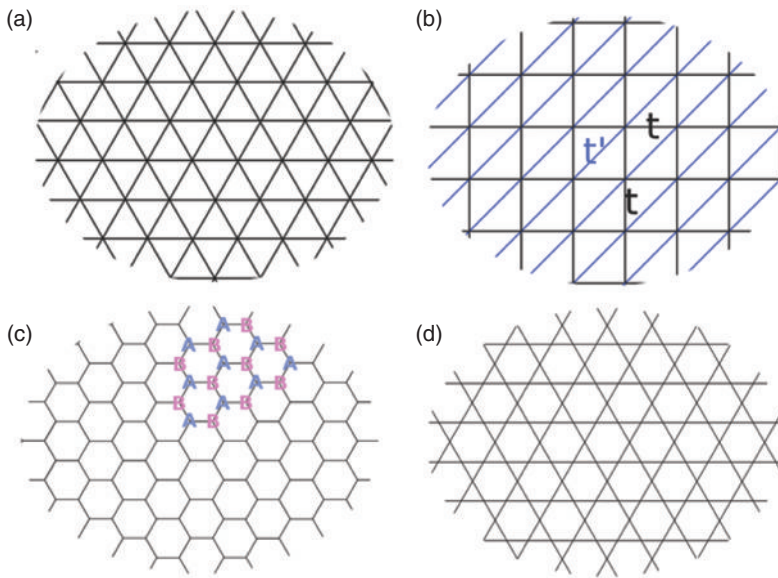


Fig. 10.4 (color online) (a) Hexagonal (triangular), (b) anisotropic triangular, (c) honeycomb, and (d) kagome lattices. The hexagonal lattice contains two inequivalent types of lattice site, some of which are labeled A and B. The sets of equivalent sites are referred to as sublattices.

10.2.2.5 Hexagonal and Honeycomb Lattices Even if the bonds are not all mutually perpendicular the solution to the tight-binding model can still be found by Fourier-transforming the Hamiltonian. Three important examples of such lattices are the hexagonal lattice (which is often referred to as the triangular lattice, although this is formally incorrect), the anisotropic triangular lattice, and the honeycomb lattice, which are sketched in Fig. 10.4. For each lattice the solution is of the form of Eq. (10.48). For the hexagonal lattice,

$$\varepsilon_{\mathbf{k}} = -2t \cos k_x a_x - 4t \cos \left(\frac{\sqrt{3}}{2} k_y a_y \right) \cos \frac{k_x a_x}{2} \quad (10.49)$$

For the anisotropic triangular lattice,

$$\varepsilon_{\mathbf{k}} = -2t (\cos k_x a_x + \cos k_y a_y) - 2t' \cos(k_x a_x + k_y a_y) \quad (10.50)$$

The honeycomb lattice has an important additional subtlety: that there are two inequivalent types of lattice site (see Fig. 10.4c), which it is worthwhile to work through. We begin by introducing new operators, $\hat{c}_{i\nu\sigma}$, which annihilate an electron with spin σ on the ν th sublattice in the i th unit cell, where $\nu = A$ or B .

Therefore, we can rewrite Eq. (10.45) as

$$\begin{aligned}
 \hat{H}_{\text{tb}} &= -t \sum_{\langle ij \rangle \sigma} \hat{c}_{iA\sigma}^\dagger \hat{c}_{jB\sigma} + \hat{c}_{jB\sigma}^\dagger \hat{c}_{iA\sigma} \\
 &= -t \sum_{\langle ij \rangle \sigma} \begin{pmatrix} \hat{c}_{iA\sigma} \\ \hat{c}_{iB\sigma} \end{pmatrix}^\dagger \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \hat{c}_{iA\sigma} \\ \hat{c}_{iB\sigma} \end{pmatrix} \\
 &= -t \sum_{k\sigma} \begin{pmatrix} \hat{c}_{kA\sigma} \\ \hat{c}_{kB\sigma} \end{pmatrix}^\dagger \begin{pmatrix} 0 & h_k \\ h_k^* & 0 \end{pmatrix} \begin{pmatrix} \hat{c}_{kA\sigma} \\ \hat{c}_{kB\sigma} \end{pmatrix} \quad (10.51)
 \end{aligned}$$

where $h_k = e^{ik_x a} + e^{-i(k_x + \sqrt{3}k_y)a/2} + e^{-i(k_x - \sqrt{3}k_y)a/2}$. Therefore,

$$\begin{aligned}
 \varepsilon_{\mathbf{k}} &= \pm t |h_{\mathbf{k}}| \\
 &= \pm t \sqrt{3 + 2 \cos \sqrt{3} k_y a + 4 \cos \frac{\sqrt{3}}{2} k_y a \cos \frac{3k_x a}{2}} \quad (10.52)
 \end{aligned}$$

We plot this dispersion relation in Fig. 10.5.

The most interesting features of this band structure are the *Dirac points*. The Dirac points are located at $\mathbf{k} = n\mathbf{K} + m\mathbf{K}'$, where n and m are integers, $\mathbf{K} = (2\pi/3a, 2\pi/3\sqrt{3}a)$, and $\mathbf{K}' = (2\pi/3a, -2\pi/3\sqrt{3}a)$. To see why these points are interesting, consider a point $\mathbf{K} + \mathbf{q}$ in the neighborhood of $\mathbf{K}$. Recalling that $\cos(K + q) = \cos K - q \sin K + \frac{1}{2}q^2 \cos K + \dots$, one finds that for small $|\mathbf{q}|$,

$$\varepsilon_{\mathbf{K}+\mathbf{q}} = \hbar v_F |\mathbf{q}| + \dots \quad (10.53)$$

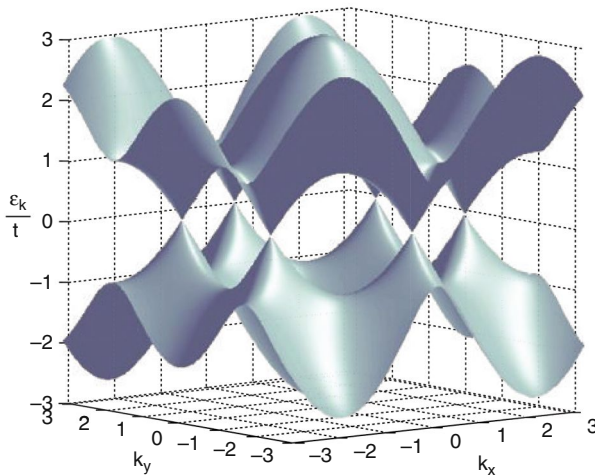


Fig. 10.5 Dirac dispersion of the honeycomb lattice.

where $v_F = 3ta/2\hbar$ is known as the Fermi velocity.

This result should be compared with the relativistic result

$$E_{\mathbf{k}}^2 = m^2 c^4 + \hbar^2 c^2 |\mathbf{k}|^2 \quad (10.54)$$

where m is a particle's rest mass and c is the speed of light. This reduces to the famous $E = mc^2$ for $k = 0$, but for massless particles such as photons, one finds that $E_{\mathbf{k}} = \hbar c |\mathbf{k}|$. Thus, the low-energy electronic excitations on a honeycomb lattice behave as if they are massless relativistic particles, with the Fermi velocity playing the role of the speed of light in the theory. Therefore, much excitement¹⁸ has been caused by the recent synthesis of atomically thick sheets of graphene,¹⁹ in which carbon atoms form a honeycomb lattice. In graphene $v_F \simeq 1 \times 10^6$ m s⁻¹, two orders smaller than the speed of light in the vacuum. This has opened the possibility of exploring and controlling “relativistic” effects in a solid-state system.¹⁸

10.3 HUBBARD MODEL

So far we have neglected electron–electron interactions. In real materials the electrons repel each other, due to the Coulomb interaction between them. The most obvious extension to the tight-binding model that describes some of the electron–electron interactions is to allow only on-site interactions (i.e., if $V_{ijkl} \neq 0$ if and only if i, j, k , and l all refer to the same orbital). For one orbital per site we then have the Hubbard model,

$$\hat{H}_{\text{Hubbard}} = -t \sum_{(ij)\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + U \sum_i \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} \quad (10.55)$$

where we have assumed nearest-neighbor hopping only. It follows from Eq. (10.27) that $U > 0$ (i.e., electrons repel one another).

10.3.1 Two-Site Hubbard Model: Molecular Hydrogen H₂

The two-site Hubbard model is a nice context in which to consider some of the basic properties of the chemical bond. The two-body term in the Hubbard model greatly complicates the problem relative to the tight-binding model. Therefore, the Hubbard model also presents a nice context in which to introduce one of the most important tools in theoretical physics and chemistry: mean-field theory.

10.3.1.1 Mean-Field Theory, the Hartree–Fock Approximation, and Molecular Orbital Theory To construct a mean-field theory of any two as-yet-unspecified physical quantities, $m = \bar{m} + \delta m$ and $n = \bar{n} + \delta n$, where $\bar{n}(\bar{m})$ is the mean value of n (m) and δn (δm) are the fluctuations about the mean, which are assumed to be small, one notes that

$$\begin{aligned}
mn &= (\bar{m} + \delta m)(\bar{n} + \delta n) \\
&= \bar{m}\bar{n} + \bar{m}\delta n + \delta m\bar{n} + \delta m\delta n \\
&\approx \bar{m}\bar{n} + \bar{m}\delta n + \delta m\bar{n}
\end{aligned} \tag{10.56}$$

Thus, mean-field approximations neglect terms that are quadratic in the fluctuations.

Hartree theory is a mean field in the electron density; that is,

$$\begin{aligned}
\hat{c}_\alpha^\dagger \hat{c}_\beta \hat{c}_\gamma^\dagger \hat{c}_\delta &= [(\hat{c}_\alpha^\dagger \hat{c}_\beta) + (\hat{c}_\alpha^\dagger \hat{c}_\beta - \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle)] [(\hat{c}_\gamma^\dagger \hat{c}_\delta) + (\hat{c}_\gamma^\dagger \hat{c}_\delta - \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle)] \\
&\approx \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \hat{c}_\gamma^\dagger \hat{c}_\delta + \hat{c}_\alpha^\dagger \hat{c}_\beta \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle - \langle \hat{c}_\alpha^\dagger \hat{c}_\beta \rangle \langle \hat{c}_\gamma^\dagger \hat{c}_\delta \rangle
\end{aligned} \tag{10.57}$$

However, it was quickly realized that this does not allow for electron exchange; that is, one should also include averages such as $\langle \hat{c}_\alpha^\dagger \hat{c}_\delta \rangle$. Therefore, a better mean-field theory is Hartree–Fock theory, which includes these terms. However, because of the limited interactions included in the Hubbard model, Hartree theory is identical to Hartree–Fock theory if one assumes that spin-flip terms are negligible (i.e., that $\langle \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow} \rangle = 0$), which we will.

The Hartree–Fock approximation to the Hubbard Hamiltonian is therefore

$$\begin{aligned}
\hat{H}_{\text{HF}} &= -t \sum_{(ij)\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + U \sum_i \left[\langle \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \rangle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} + \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \langle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} \rangle \right. \\
&\quad \left. - \langle \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \rangle \langle \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} \rangle \right] \\
&= -t \sum_{(ij)\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + U \sum_i \left[n_{i\uparrow} \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} + n_{i\downarrow} \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} - n_{i\uparrow} n_{i\downarrow} \right]
\end{aligned} \tag{10.58}$$

where $n_{i\sigma} = \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \rangle$. Thus, we have a Hamiltonian for a single electron moving in the mean field of the other electrons. Note that this Hamiltonian is equivalent to the ω -method parameterization of the Hückel model [see Section 10.2.1.3, particularly Eq. (10.34)] if we set $\omega = U/\beta$. Thus, the ω method is just a parameterization of the Hubbard model solved in the Hartree–Fock approximation.

The Hubbard model with two sites and two electrons can be taken as a model for molecular hydrogen. In the Hartree–Fock ground state, $|\Psi_{\text{HF}}^0\rangle$, the two electrons have opposite spin and each occupies the bonding state, which we found to be the ground state of the Hückel model in Section 10.2.1.1:

$$|\Psi_{\text{HF}}^0\rangle = |\Psi_{b\downarrow}\rangle \otimes |\Psi_{b\uparrow}\rangle = \frac{1}{2}(\hat{c}_{1\uparrow}^\dagger + \hat{c}_{2\uparrow}^\dagger)(\hat{c}_{1\downarrow}^\dagger + \hat{c}_{2\downarrow}^\dagger)|0\rangle \tag{10.59}$$

$$= \frac{1}{2}(\hat{c}_{1\uparrow}^\dagger \hat{c}_{1\downarrow}^\dagger + \hat{c}_{1\uparrow}^\dagger \hat{c}_{2\downarrow}^\dagger - \hat{c}_{1\downarrow}^\dagger \hat{c}_{2\uparrow}^\dagger + \hat{c}_{2\uparrow}^\dagger \hat{c}_{2\downarrow}^\dagger)|0\rangle \tag{10.60}$$

Notice that $|\Psi_{\text{HF}}^0\rangle$ is just a product of two single-particle wavefunctions [one for the spin-up electron and another for the spin-down electron; cf. Eq. (10.59)].

Thus, we say that the wavefunction is *uncorrelated* and that the two electrons are unentangled.

An important prediction of the Hartree–Fock theory is that if we pull the protons apart, we are equally likely to get two hydrogen atoms ($H + H$) or two hydrogen ions ($H^+ + H^-$). This is not what is observed experimentally. In reality the former is far more likely.

10.3.1.2 Heitler–London Wavefunction and Valence-Bond Theory Just a year after the appearance of Schrödinger’s wave equation,²⁰ Heitler and London²¹ proposed a theory of the chemical bond based on the new quantum mechanics. Explaining the nature of the chemical bond remains one of the greatest achievements of quantum mechanics. Heitler and London’s theory led to the valence-bond theory of the chemical bond.²² The two-site Hubbard model of H_2 is the simplest context in which to study this theory.

The Heitler–London wavefunction is

$$|\Psi_{HL}^0\rangle = \frac{1}{\sqrt{2}}(\hat{c}_{1\uparrow}^\dagger \hat{c}_{2\downarrow}^\dagger - \hat{c}_{1\downarrow}^\dagger \hat{c}_{2\uparrow}^\dagger)|0\rangle \quad (10.61)$$

Notice that the wavefunction is *correlated*, as it cannot be written as a product of a wavefunction for each of the particles. Equivalently, one can say that the two electrons are entangled. The Heitler–London wavefunction overcorrects the physical errors in the Hartree–Fock molecular orbital wavefunction, as it predicts zero probability of H_2 dissociating to an ionic state but is, nevertheless, a significant improvement on molecular orbital theory.

10.3.1.3 Exact Solution of the Two-Site Hubbard Model The Hilbert space of the two-site, two-electron Hubbard model is sufficiently small that we can solve it analytically; nevertheless, this problem can be greatly simplified by using the symmetry properties of the Hamiltonian. First, note that the total spin operator commutes with the Hamiltonian equation (10.55), as none of the terms in the Hamiltonian cause spin flips. Therefore, the energy eigenstates must also be spin eigenstates. For two electrons this means that all of the eigenstates will be either singlets ($S = 0$) or triplets ($S = 1$).

Let us begin with the triplet states, $|\Psi_1^m\rangle$. Consider a state with two spin-up electrons, $|\Psi_1^1\rangle$. Because there is only one orbital per site, the Pauli exclusion principle ensures that there will be exactly one electron per site (i.e., $|\Psi_1^1\rangle = \hat{c}_{1\uparrow}^\dagger \hat{c}_{2\uparrow}^\dagger|0\rangle$). The electrons cannot hop between sites, as the presence of the other electron and the Pauli principle forbid it. Therefore, $\langle\Psi_1^1|(-t\hat{c}_{1\sigma}^\dagger \hat{c}_{2\sigma})|\Psi_1^1\rangle = \langle\Psi_1^1|(-t\hat{c}_{2\sigma}^\dagger \hat{c}_{1\sigma})|\Psi_1^1\rangle = 0$ for $\sigma = \uparrow$ or $\downarrow$. There is exactly one electron on each site, so $\langle\Psi_1^1|U \sum_i \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow}|\Psi_1^1\rangle = 0$. Thus, the total energy of this state is $E_1^1 = 0$.

The same chain of reasoning shows that $|\Psi_1^{-1}\rangle = \hat{c}_{1\downarrow}^\dagger \hat{c}_{2\downarrow}^\dagger|0\rangle$ and $E_1^{-1} = 0$. It then follows from spin rotation symmetry that $|\Psi_1^0\rangle = (1/\sqrt{2})(\hat{c}_{1\uparrow}^\dagger \hat{c}_{2\downarrow}^\dagger + \hat{c}_{1\downarrow}^\dagger \hat{c}_{2\uparrow}^\dagger)|0\rangle$ and $E_1^0 = 0$.

As the Hilbert space contains six states, this leaves three singlet states. A convenient basis for these is formed by the Heitler–London state and the two charge-transfer states: $|\Psi_{\text{HL}}\rangle = (1/\sqrt{2})(\hat{c}_{1\uparrow}^\dagger \hat{c}_{2\downarrow}^\dagger - \hat{c}_{1\downarrow}^\dagger \hat{c}_{2\uparrow}^\dagger)|0\rangle$, $|\Psi_{\text{ct}+}\rangle = (1/\sqrt{2})(\hat{c}_{1\uparrow}^\dagger \hat{c}_{1\downarrow}^\dagger + \hat{c}_{2\uparrow}^\dagger \hat{c}_{2\downarrow}^\dagger)|0\rangle$, and $|\Psi_{\text{ct}-}\rangle = (1/\sqrt{2})(\hat{c}_{1\uparrow}^\dagger \hat{c}_{1\downarrow}^\dagger - \hat{c}_{2\uparrow}^\dagger \hat{c}_{2\downarrow}^\dagger)|0\rangle$. Note that $|\Psi_{\text{HL}}\rangle$ and $|\Psi_{\text{ct}+}\rangle$ are even under “inversion” symmetry,[†] which swaps the site labels $1 \leftrightarrow 2$, whereas $|\Psi_{\text{ct}-}\rangle$ is odd under inversion symmetry. As the Hamiltonian is symmetric under inversion the eigenstates will have a definite parity, so $|\Psi_{\text{ct}-}\rangle$ is an eigenstate, with energy $E_{\text{ct}-} = U$. The other two singlet states are not distinguished by any symmetry of the Hamiltonian, so they do couple, yielding the Hamiltonian matrix

$$\begin{aligned} H &= \begin{pmatrix} \langle \Psi_{\text{HL}} | \hat{H}_{\text{Hubbard}} | \Psi_{\text{HL}} \rangle & \langle \Psi_{\text{HL}} | \hat{H}_{\text{Hubbard}} | \Psi_{\text{ct}+} \rangle \\ \langle \Psi_{\text{ct}+} | \hat{H}_{\text{Hubbard}} | \Psi_{\text{HL}} \rangle & \langle \Psi_{\text{ct}+} | \hat{H}_{\text{Hubbard}} | \Psi_{\text{ct}+} \rangle \end{pmatrix} \\ &= \begin{pmatrix} 0 & -2t \\ -2t & U \end{pmatrix} \end{aligned} \quad (10.62)$$

This has eigenvalues, $E_{\text{CF}} = \frac{1}{2}(U - \sqrt{U^2 + 16t^2})$ and $E_{\text{S}^2} = \frac{1}{2}(U + \sqrt{U^2 + 16t^2})$. The corresponding eigenstates are

$$\begin{aligned} |\Psi_{\text{CF}}\rangle &= \cos \theta |\Psi_{\text{HL}}\rangle + \sin \theta |\Psi_{\text{ct}+}\rangle \\ &= \left[\frac{\cos \theta}{\sqrt{2}} (\hat{c}_{1\uparrow}^\dagger \hat{c}_{2\downarrow}^\dagger - \hat{c}_{1\downarrow}^\dagger \hat{c}_{2\uparrow}^\dagger) \right. \\ &\quad \left. + \frac{\sin \theta}{\sqrt{2}} (\hat{c}_{1\downarrow}^\dagger \hat{c}_{1\uparrow}^\dagger + \hat{c}_{2\downarrow}^\dagger \hat{c}_{2\uparrow}^\dagger) \right] |0\rangle \end{aligned} \quad (10.63)$$

and

$$\begin{aligned} |\Psi_{\text{S}^2}\rangle &= \sin \theta |\Psi_{\text{HL}}\rangle + \cos \theta |\Psi_{\text{ct}+}\rangle \\ &= \left[\frac{\sin \theta}{\sqrt{2}} (\hat{c}_{1\uparrow}^\dagger \hat{c}_{2\downarrow}^\dagger - \hat{c}_{1\downarrow}^\dagger \hat{c}_{2\uparrow}^\dagger) \right. \\ &\quad \left. + \frac{\cos \theta}{\sqrt{2}} (\hat{c}_{1\downarrow}^\dagger \hat{c}_{1\uparrow}^\dagger + \hat{c}_{2\downarrow}^\dagger \hat{c}_{2\uparrow}^\dagger) \right] |0\rangle \end{aligned} \quad (10.64)$$

where $\tan \theta = (U - \sqrt{U^2 + 16t^2})/4t$. For $U > 0$, as is required physically, the state $|\Psi_{\text{CF}}\rangle$ is the ground state for all values of U/t . $|\Psi_{\text{CF}}\rangle$ is often called the *Coulson–Fischer* wavefunction.

Inspection of Eq. (10.63) reveals that for $U/t \rightarrow \infty$, the Coulson–Fischer state tends to the Heitler–London wavefunction, while for $U/t \rightarrow 0$ we regain the molecular orbital picture (Hartree–Fock wavefunction).

[†]It may not be immediately obvious that $|\Psi_{\text{HL}}\rangle$ is even under inversion symmetry, but this is easily confirmed as $\hat{I}|\Psi_{\text{HL}}\rangle = (1/\sqrt{2})(\hat{c}_{2\uparrow}^\dagger \hat{c}_{1\downarrow}^\dagger - \hat{c}_{2\downarrow}^\dagger \hat{c}_{1\uparrow}^\dagger)|0\rangle = (1/\sqrt{2})(-\hat{c}_{1\downarrow}^\dagger \hat{c}_{2\uparrow}^\dagger + \hat{c}_{1\uparrow}^\dagger \hat{c}_{2\downarrow}^\dagger)|0\rangle = |\Psi_{\text{HL}}\rangle$, where $\hat{I}$ is the inversion operator, which swaps the labels 1 and 2.

10.3.2 Mott Insulators and the Mott–Hubbard Metal–Insulator Transition

In 1949, Mott²³ asked an apparently simple question with a profound and surprising answer. As we have seen above, for the two-site Hubbard model both the molecular orbital (Hartree–Fock) and valence-bond (Heitler–London) wavefunctions are just approximations of the exact (Coulson–Fischer) wavefunction. Mott asked whether the equivalent statement is true in an infinite solid and, surprisingly, found that the answer is no. Further, Mott showed that the Hartree–Fock and Heitler–London wavefunctions predict very different properties for crystals.

One of the most important properties of a crystal is its conductivity. In a metal the conductivity is high and increases as the temperature is lowered, whereas in a semiconductor or an insulator the conductivity is low and decreases as the temperature is lowered. These behaviors arise because of fundamental differences between the electronic structures of metals and semiconductors/insulators.¹⁰ In metals there are excited states at arbitrarily low energies above the Fermi energy. This means that even at the lowest temperatures, electrons can move in response to an applied electric field. In semiconductors and insulators there is an energy gap between the highest occupied electronic state and the lowest unoccupied electronic state at zero temperature. This means that a thermal activation energy must be provided if electrons are to move in response to an applied field. The difference between semiconductors and insulators is simply the size of the gap; therefore, we will not distinguish between the two below and will refer to any material with a gap as an insulator.

Consider a Hubbard model at half-filling, that is, with the same number of electrons as lattice sites. For a macroscopic current to flow, an electron must move from one lattice site (leaving an empty site with a net positive charge) to a *distant* site (creating a doubly occupied site with a net negative charge). The net charges may move through the collective motions of the electrons. One could keep track of this by describing the movement of all the electrons, but it is easier to introduce an equivalent description where we treat the net charges as particles moving in a neutral background. Therefore, we refer to the positive charge as a *holon* and the negative charge as a *doublon*. In the ground state of valence-bond theory, all of the sites are neutral and there are no holons or doublons [cf. Eq. (10.61)]. However, it is reasonable to postulate that there are low-lying charge-transfer excited states and hence thermal states that contain a few doublons and holons. These doublons and holons interact via the Coulomb potential, $V(r) = -e^2/\kappa r$, where κ is the dielectric constant of the crystal. We know from the theory of the hydrogen atom (or, better, positronium; see Gasiorowicz⁷) that this potential gives rise to bound states. Therefore, one expects that in valence-bond theory, holons and doublons are bound and that separating holon–doublon pairs costs a significant amount of energy. Thus, one expects the number of distant holon–doublon pairs to decrease as the temperature is lowered. Therefore, valence-bond theory predicts that a half-filled Hubbard model is an insulator.

In contrast, molecular orbital theory has large numbers of holons and doublons [cf. Eq. (10.60), which suggests that for an N -site model there will be $N/2$ neutral sites, $N/4$ empty sites, and $N/4$ doubly occupied sites]. Mott reasoned

that if there are many holon–doublon pairs “it no longer follows that work must necessarily be done to form some more.” This is because the holon and doublon now interact via a screened potential, $V(r) = -(e^2/\kappa r) \exp(-qr)$, where q is the Thomas–Fermi wavevector (see Ashcroft and Mermin¹⁰). For sufficiently large q there will be no bound states, hence molecular orbital theory predicts that the half-filled Hubbard model is metallic.

Thus, Mott argued that there are two (local) minima of the free energy in a crystal (see Fig. 10.6). One of the minima corresponds to a state with no holon–doublon pairs that is well approximated by a valence-bond wavefunction and is now known as the *Mott insulating state*. The second minimum corresponds to a state with many doublon–holon pairs that is well approximated by a molecular orbital wavefunction and is metallic. As we saw above, valence-bond theory works well for $U \gg t$ and molecular orbital theory works well for $U \ll t$. Therefore, in the half-filled Hubbard model we expect a Mott insulator for large U/t and a metal for small U/t . Further the “double-well” structure of the energy predicted by Mott’s argument (Fig. 10.6) suggests that there is a first-order metal–insulator phase transition, known as the *Mott transition*. Mott predicted that this metal–insulator transition can be driven by applying pressure to a Mott insulator. This has now been observed in a number of systems; perhaps the purest examples are the organic charge-transfer salts (BEDT-TTF)₂X.²⁴

It is interesting to note that this infusion of chemical ideas into condensed matter physics has remained important in studies of the Mott transition. Of particular note is Anderson’s resonating valence-bond theory of superconductivity in high-temperature superconductors,^{26,27} which describes superconductivity in a doped Mott insulator in terms of a generalization of the valence-bond theory discussed above. This theory can also be modified to describe superconductivity on the metallic side of the Mott transition for a half-filled lattice. This theory then

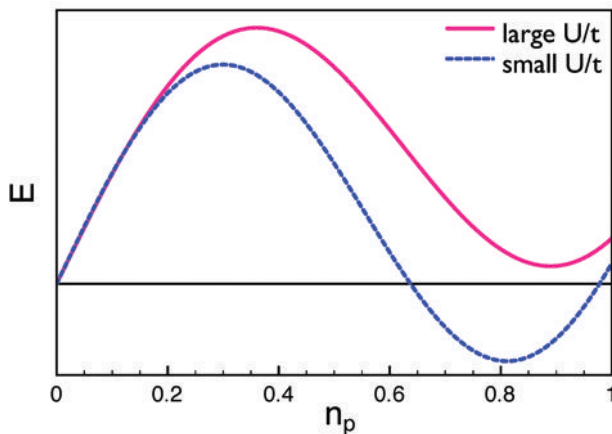


Fig. 10.6 (color online) Mott’s proposal for the energy of the Hubbard model as a function of the number of holon–doublon pairs, n_p , at low (zero) temperature(s) for large and small U/t .

provides a good description of the superconductivity observed in (BEDT-TTF)₂X salts.²⁸

Note that theories such as Hartree–Fock and density functional²⁵ that do not include the strong electronic correlations present in the Hubbard model do not predict a Mott insulating state. Thus, weakly correlated theories make the qualitatively incorrect prediction that materials such as NiO, V₂O₃, La₂CuO₄, and κ-(BEDT-TTF)₂Cu[N(CN)₂]Cl are metals, whereas experimentally, all are insulators.

We will discuss a quantitative theory of the Mott transition in Section 10.3.3.2.

10.3.3 Mean-Field Theories for Crystals

10.3.3.1 Hartree–Fock Theory of the Hubbard Model: Stoner Ferromagnetism

In a manner similar to that in which we constructed the Hartree–Fock mean-field theory for the two-site Hubbard model in Section 10.3.1.1, we can also construct a Hartree–Fock theory of the infinite lattice Hubbard model. Again, we simply replace the number operators in the two-body term by their mean values, $n_{i\sigma} \equiv \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \rangle$, plus the fluctuations about the mean, $(\hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} - n_{i\sigma})$, and neglect terms that are quadratic in the fluctuations:

$$\begin{aligned} U \sum_i \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} &= U \sum_i [n_{i\uparrow} + (\hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} - n_{i\uparrow})][n_{i\downarrow} + (\hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} - n_{i\downarrow})] \\ &\simeq U \sum_i [n_{i\downarrow} \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} + n_{i\uparrow} \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} - n_{i\uparrow} n_{i\downarrow}] \end{aligned} \quad (10.65)$$

If we make the additional approximation that $n_{i\sigma} = n_\sigma$ for all i (i.e., that the system is homogeneous and does not spontaneously break translational symmetry), we find that the Hartree–Fock Hamiltonian for the Hubbard model is

$$\hat{H}_{\text{HF}} - \mu \hat{N} = -t \sum_{\langle ij \rangle \sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \sum_{i\sigma} (Un_{\bar{\sigma}} - \mu) \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} - UNn_\uparrow n_\downarrow \quad (10.66)$$

where N is the number of lattice sites and $\bar{\sigma}$ is the opposite spin to σ . It is convenient to write this Hamiltonian in terms of the total electron density, $n = n_\uparrow + n_\downarrow$, and the magnetization density, $m = n_\uparrow - n_\downarrow$, which gives

$$\begin{aligned} \hat{H}_{\text{HF}} - \mu \hat{N} &= -t \sum_{\langle ij \rangle \sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} - \mu \sum_{i\sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \\ &\quad + U \sum_i \left[\frac{1}{2}(n - m) \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\uparrow} + \frac{1}{2}(n + m) \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\downarrow} - \frac{1}{4}(n + m)(n - m) \right] \\ &= \sum_{\mathbf{k}\sigma} \left(\varepsilon_{\mathbf{k}}^0 + \bar{\sigma} \frac{Um}{2} \right) \hat{n}_{\mathbf{k}\sigma} - \left(\mu - \frac{Un}{2} \right) \sum_{\mathbf{k}\sigma} \hat{n}_{\mathbf{k}\sigma} - \frac{NU}{4} (n^2 - m^2) \end{aligned} \quad (10.67)$$

where $\varepsilon_{\mathbf{k}}^0$ is the dispersion relation for $U = 0$ and $\sigma = \pm 1 = \uparrow\downarrow$. The last term is just a constant and will not concern us greatly. The penultimate term is the “renormalized” chemical potential; that is, the chemical potential, μ , of the system with $U = 0$ is decreased by $Un/2$ due to the interactions. The first term is just the renormalized dispersion relation; in particular, we find that if the magnetization density is nonzero the dispersion relation for spin-up electrons is different from that for spin-down electrons (see Fig. 10.7). It is important to note that the Hartree–Fock approximation has reduced the problem to a single-particle (single-determinant) theory. Thus, we can write

$$\hat{H}_{\text{HF}} - \mu \hat{N} = \sum_{\mathbf{k}\sigma} (\varepsilon_{\mathbf{k}\sigma}^* - \mu^*) \hat{n}_{\mathbf{k}\sigma} - \frac{NU}{4} (n^2 - m^2) \quad (10.68)$$

where $\varepsilon_{\mathbf{k}\sigma}^* = \varepsilon_{\mathbf{k}}^0 - \frac{1}{2}\sigma Um$ and $\mu^* = \mu - \frac{1}{2}Un$.

We can now calculate the magnetization density (magnetic moment):

$$\begin{aligned} m &= n_{\uparrow} - n_{\downarrow} \\ &= \int_{-\infty}^0 d\varepsilon [D_{\uparrow}(\varepsilon - \mu^*) - D_{\downarrow}(\varepsilon - \mu^*)] \\ &= \int_{-\infty}^0 d\varepsilon \left[D_0\left(\varepsilon - \frac{1}{2}Um + \frac{1}{2}Un - \mu\right) - D_0\left(\varepsilon + \frac{1}{2}Um + \frac{1}{2}Un - \mu\right) \right] \\ &\equiv f(m) = D_0(0)Um + O(m^2) \end{aligned} \quad (10.69)$$

where $D_0(\varepsilon) = \partial N_0(\varepsilon)/\partial \varepsilon|_{\varepsilon}$ is the density of states (DOS; see Ashcroft and Mermin¹⁰) per spin for $U = 0$, $N_0(\varepsilon)$ is the number of electrons (per spin species)

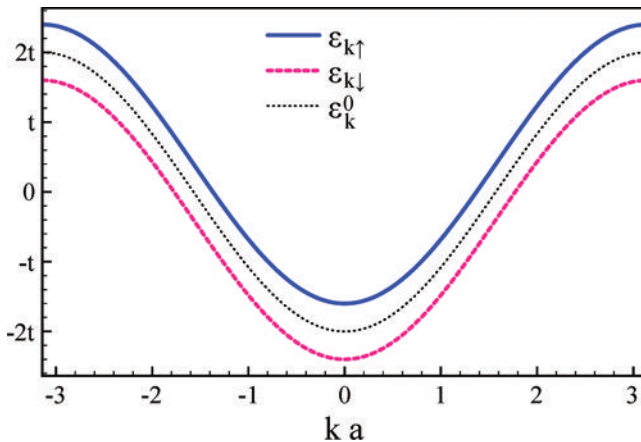


Fig. 10.7 (color online) Dispersion relations for spin-up and spin-down electrons in the Hartree–Fock theory of the Hubbard chain (Stoner model of ferromagnetism) with $m = 0.8t/U$.

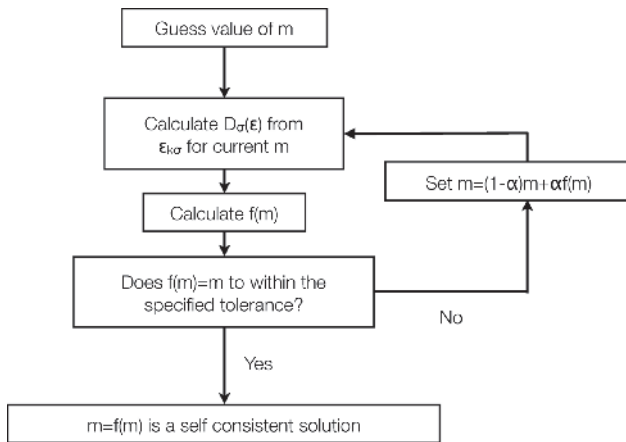


Fig. 10.8 How to find the self-consistent solution of Eq. (10.69). If the convergence works well, one can take $\alpha = 1$, but for some problems convergence can be reached more reliably with a small value of α (often a value as small as ~ 0.05 is used).

for which $\epsilon_{\mathbf{k}}^0 \leq \epsilon$ for $U = 0$, $D_\sigma(\epsilon) = \partial N_\sigma(\epsilon) / \partial \epsilon|_\epsilon$ is the full interacting DOS for spin σ electrons, and $N_\sigma(\epsilon)$ is the number of electrons with spin σ for which $\epsilon_{\mathbf{k}\sigma} \leq \epsilon$. The standard way to solve mean-field theories, known as the method of self-consistent solution, is illustrated in Fig. 10.8. The major difficulty with self-consistent solutions is that it is not possible to establish whether or not one has found all of the self-consistent solutions, and therefore it is not possible to establish whether or not one has found the global minimum. Therefore, it is prudent to try a wide range of initial guesses for m (or whatever variable the initial guess is made in).

Clearly, $m = 0$ is always a solution of Eq. (10.69), and for $U D_0(0) < 1$ this turns out to be the only solution. But for $U D_0(0) > 1$ there are additional solutions with $m \neq 0$. This is easily understood from the sketch in Fig. 10.9. Furthermore, the $m \neq 0$ solutions typically have lower energy than the $m = 0$ solution, and therefore for $U D_0(0) > 1$ the ground state is ferromagnetic. $U D_0(0) \geq 1$ is known as the *Stoner condition* for ferromagnetism. For the Stoner condition to be satisfied, a system must have narrow bands [small t , and hence large $D(0)$] and strong interactions (large U).

There are three elemental ferromagnets, Fe, Co, and Ni, each of which is also metallic. As the Hartree–Fock theory of the Hubbard model predicts metallic magnetism if the Stoner criterion is satisfied and these materials have narrow bands of strongly interacting electrons, it is natural to ask whether this is a good description of these materials. However, if one extends the treatment above to finite temperatures,²⁹ one finds that the Hartree–Fock theory of the Hubbard model does not provide a good theory of the three elemental magnets. The Curie temperatures, T_C (i.e., the temperature at which the material becomes ferromagnetic) of Fe, Co, and Ni are ~ 1000 K (see, e.g., Table 33.1 of Ashcroft and Mermin¹⁰). Hartree–Fock theory predicts

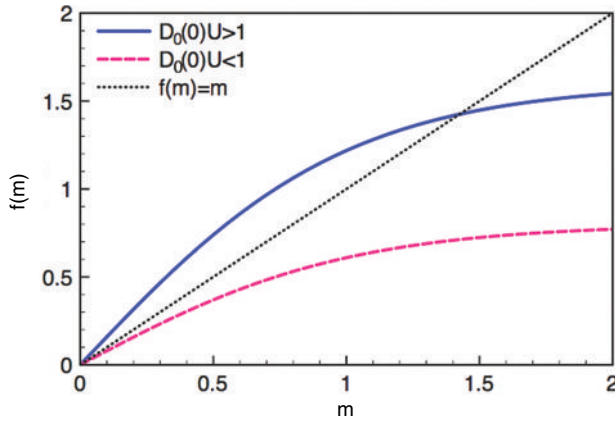


Fig. 10.9 (color online) Graphical solution of the self-consistency equation [Eq. (10.69)] for the Stoner model of ferromagnetism.

that $T_c \sim Um_0$, where m_0 is the magnetization at $T = 0$. If the parameters in the Hubbard model are chosen so that Hartree–Fock theory reproduces the observed m_0 , the predicted critical temperature is $\sim 10,000$ K. This order-of-magnitude disagreement with experiment results from the failure of the mean-field Hartree–Fock approximation to account properly for the fluctuations in the local magnetization. This is closely related to the (incorrect) prediction of the Hartree–Fock approximation that there are no local moments above T_c . (Experimentally local moments *are* observed above T_c .) However, for weak ferromagnets, such as ZrZn_2 ($T_c \sim 30$ K) the Hartree–Fock theory of the Hubbard model provides an excellent description of the behavior observed.³⁰

The effects missed by Hartree–Fock theory are referred to as *electronic correlations*. The dramatic failure of Hartree–Fock theory in Fe, Co, and Ni shows that electron correlations are very important in these materials, as do other comparisons of theory and experiment.³¹ However, it is important to note that mean-field theory is not limited to Hartree–Fock theory (although the terms are often, but imprecisely, used synonymously). Rather, Hartree–Fock theory is the mean-field theory of the electronic density. By constructing mean-field theories of other properties it is possible to construct mean-field theories that capture (some) electronic correlations. We now consider an example of a rather different mean-field theory.

10.3.3.2 Gutzwiller Approximation, Slave Bosons, and the Brinkman–Rice Metal–Insulator Transition In 1963, Gutzwiller³² proposed a variational wavefunction for the Hubbard model:

$$\begin{aligned}
 |\Psi_G\rangle &= \prod_i (1 - \alpha \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}) |\Psi_0\rangle \\
 &= \exp \left(-g \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \right) |\Psi_0\rangle
 \end{aligned} \tag{10.70}$$

where $g = -\ln(1 - \alpha)$ is a variational parameter and $|\Psi_0\rangle$ is the ground state for uncorrelated electrons. One should note that the Gutzwiller wavefunction is closely related to the coupled cluster ansatz,¹ which is widely used in both physics and chemistry. Gutzwiller used this ansatz to study the problem of itinerant ferromagnetism. This leads to an improvement over the Hartree–Fock theory discussed above. However, in 1970, Brinkman and Rice³³ showed that this wavefunction also describes a metal–insulator transition, now referred to as a Brinkman–Rice transition. Rather than studying this wavefunction in detail, we use an equivalent technique known as *slave bosons*. This has the advantage of making it clear that the Brinkman–Rice theory is just a mean-field description of the Mott transition.

The i th site in a Hubbard model has four possible states: the site can be empty, $|e_i\rangle$; can contain a single spin σ ($=\uparrow$ or $\downarrow$) electron $|\sigma_i\rangle$; or can contain two electrons, $|d_i\rangle$. The Kotliar–Ruckenstein slave boson technique introduces an overcomplete description of these states:

$$|e_i\rangle = \hat{e}_i^\dagger |0_i\rangle \quad (10.71)$$

$$|\sigma_i\rangle = \hat{p}_{i\sigma}^\dagger \hat{c}_{i\sigma}^\dagger |0_i\rangle \quad (10.72)$$

$$|d_i\rangle = \hat{d}_i^\dagger \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow}^\dagger |0_i\rangle \quad (10.73)$$

where $\hat{e}_i^\dagger$, $\hat{p}_{i\sigma}^\dagger$, and $\hat{d}_i^\dagger$ are bosonic creation operators which correspond to empty, partially filled, and doubly occupied sites. $|0_i\rangle$ is a state with no fermions and no bosons on site i ; note that this is not a physically realizable state. This transformation is not only kosher, but also exact, as long as we also introduce the constraints

$$\hat{e}_i^\dagger \hat{e}_i + \sum_{\sigma} \hat{p}_{i\sigma}^\dagger \hat{p}_{i\sigma} + \hat{d}_i^\dagger \hat{d}_i = 1 \quad (10.74)$$

which ensures that there is exactly one boson per site and therefore that each site is either empty, partially occupied, or doubly occupied, and

$$\hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} - \hat{p}_{i\sigma}^\dagger \hat{p}_{i\sigma} - \hat{d}_i^\dagger \hat{d}_i = 0 \quad (10.75)$$

which ensures that if a site contains a spin σ electron, it is either singly occupied (with spin σ) or doubly occupied.

Writing the Hubbard Hamiltonian in terms of the slave bosons yields

$$\hat{H}_{\text{Hubbard}} = -t \sum_{\langle ij \rangle \sigma} \hat{z}_{i\sigma}^\dagger \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \hat{z}_{j\sigma} + U \sum_i \hat{d}_i^\dagger \hat{d}_i \quad (10.76)$$

where $\hat{z}_{j\sigma} = \hat{e}_j^\dagger \hat{p}_{j\sigma} + \hat{p}_{j\bar{\sigma}}^\dagger \hat{d}_j$.

We now make a mean-field approximation and replace the bosonic operators by the expectation values: $\langle e_i \rangle = e$, $\langle p_{i\uparrow} \rangle = \langle p_{i\downarrow} \rangle = p$, $\langle d_i \rangle = d$. Note that we have

additionally assumed that the system is homogeneous (the expectation values do not depend on i) and paramagnetic ($\langle p_{i\uparrow} \rangle = \langle p_{i\downarrow} \rangle$). Therefore, the constraints reduce to

$$|e|^2 + 2|p|^2 + |d|^2 = 1 \quad (10.77)$$

and

$$|p|^2 + |d|^2 = \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \rangle = \frac{n}{2} \quad (10.78)$$

where n is the average number of electrons per site. This amounts only to enforcing the constraints, on average.

This theory does not reproduce the correct result for $U = 0$. However, this deficiency can be fixed if $\hat{z}_{j\sigma}$ is replaced by the “renormalized” quantity, $\tilde{z}_{j\sigma}$, defined such that

$$\langle \tilde{z}_{j\sigma}^\dagger \tilde{z}_{j\sigma} \rangle = \frac{(n/2) - |d|^2}{(1 - n/2)(n/2)} \left(d + \sqrt{1 - n + |d|^2} \right) \quad (10.79)$$

Let us specialize to a half-filled band, $n = 1$. The constraints now allow us to eliminate $|p|^2 = \frac{1}{2} - |d|^2$ and $|e|^2 = |d|^2$. Thus, we find that

$$\begin{aligned} \hat{H}_{\text{Hubbard}} &\simeq -t \sum_{\langle ij \rangle \sigma} \frac{1}{8} (|d|^2 - 2|d|^4) \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + UN_0 |d|^2 \\ &= \frac{1}{8} (|d|^2 - 2|d|^4) \sum_{k\sigma} \epsilon_k^0 \hat{n}_{k\sigma} + UN_0 |d|^2 \end{aligned} \quad (10.80)$$

where ϵ_k^0 is the dispersion for $U = 0$ and N is the number of lattice sites. Recall that $|d|^2 = \langle d_i^\dagger d_i \rangle$ (i.e., $|d|^2$ is the probability of site being doubly occupied). We construct a variational theory by ensuring that the energy is minimized with respect to $|d|$, which yields

$$\frac{\partial E}{\partial |d|} = \frac{1}{4} (|d| - 4|d|^3) \sum_{k\sigma} \epsilon_k^0 \langle \hat{n}_{k\sigma} \rangle + 2UN_0 |d| = 0 \quad (10.81)$$

Equation (10.81) allows one to solve the problem self-consistently (see Fig. 10.8). For small U this equation has more than one minimum and the lowest-energy state has $|d|^2 > 0$, which corresponds to a correlated metallic state (the details of this minimum depend on ϵ_k^0). But above some critical U the ground-state solution has $|d|^2 = 0$, which corresponds to no doubly occupied states (i.e., the Mott insulator). Thus, the dependence of the energy on the number holon-doublon pairs ($n_p = |d|^2$) calculated from the mean-field slave boson theory is exactly as Mott predicted on rather general grounds (shown in Fig. 10.6).

10.3.4 Exact Solutions of the Hubbard Model

10.3.4.1 One Dimension Lieb and Wu³⁴ famously solved the Hubbard chain at $T = 0$ using the Bethe ansatz.^{35,36} Lieb and Wu found that the half-filled Hubbard chain is a Mott insulator for any nonzero U . Nevertheless, the Bethe ansatz solution is not straightforward to understand, and weighty textbooks have been written on the subject.^{35,36}

10.3.4.2 Infinite Dimensions: Dynamical Mean-Field Theory As one increases the dimension of a lattice, the coordination number (the number of nearest neighbors for each lattice site) also increases. In infinite dimensions each lattice site has infinitely many nearest neighbors. For a classical model, mean-field theory becomes exact in infinite dimensions, as the environment (the infinite number of nearest neighbors) seen by each site is exactly the same as the mean field. However, quantum mechanically, things are complicated by the internal dynamics of the site. In the Hubbard model each site can contain zero, one, or two electrons, and a dynamic equilibrium between the different charge and spin states is maintained. However, the environment is still described by a mean field, even though the dynamics are not. Therefore, although the Hartree–Fock theory of the Hubbard model does not become exact in infinite dimensions, it is possible to construct a theory that treats the on-site dynamics exactly and the spatial correlations at the mean-field level; this theory is known as *dynamic mean-field theory* (DMFT).³⁷

The importance of DMFT is not in the somewhat academic limit of infinite dimensions. Rather, DMFT has become an important approximate theory in the finite numbers of dimensions relevant to real materials.³⁷ It has been found that DMFT captures a great deal of the physics of strongly correlated electrons. Typically, the most important correlations are on-site and therefore are described correctly by DMFT. These include the correlations that are important in metallic magnetism³⁸ and many other strongly correlated materials.^{24,37} Cluster extensions to DMFT, such as cellular dynamical mean-field theory (CDMFT) and the dynamical cluster approximation (DCA), which capture some of the non-local correlations, have led to further insights into strongly correlated materials.³⁹ Considerable success has also been achieved by combining DMFT with density functional theory.⁴⁰

10.3.4.3 Nagaoka Point The Nagaoka point in the phase diagram of the Hubbard model is the $U \rightarrow \infty$ limit when we add one hole to a half-filled system. Nagaoka rigorously proved^{41,42} that at this point the state that maximizes the total spin of the system [i.e., the state with $\langle S_z \rangle = (N - 1)/2$, for an N -site lattice] is an extremum in energy (i.e., either the ground state or the highest-lying excited state). On most bipartite lattices (cf. Fig. 10.11a) one finds that this “Nagaoka state” is indeed the ground state.⁴² However, on frustrated lattices (Fig. 10.11b) the Nagaoka state is typically only the ground state for one sign of t .⁴³

It is quite straightforward to understand why the Nagaoka state is often the ground state. As we are considering the $U \rightarrow \infty$ limit there will strictly be no

double occupation of any sites. One therefore need only consider the subspace of states with no double occupation. As none of these states contain any potential energy (i.e., terms proportional to U), the ground state will be the state that minimizes the kinetic energy (the term proportional to t). Thus, the ground state is the state that maximizes the magnitude of the kinetic energy with a negative sign. In the Nagaoka state all of the electrons align, which means that the holon can hop unimpeded by the Pauli exclusion principle, thus maximizing the magnitude of the kinetic energy. It is a simple matter to check whether this is the ground state or the highest-lying excited state, as we just compare the energy of the Nagaoka state with that of any other state satisfying the constraint of no double occupation.

Nagaoka's rigorous treatment has not been extended to doping by more than one hole and it remains an outstanding problem to further understand this interesting phenomenon, which shares important features with the magnetism observed in the elemental magnets³⁸ and many strongly correlated materials.⁴³

10.4 HEISENBERG MODEL

Like the Stoner ferromagnetism we discussed above in the context of the Hartree–Fock solution for the Hubbard model (Section 10.3.3.1) and Hund's rules (which we discuss in Section 10.5.2), the Heisenberg model is an important paradigm for understanding magnetism. The Heisenberg model does not provide a realistic description of the three elemental ferromagnets (Fe, Co, and Ni) as they are metals, whereas the Heisenberg model only describes insulators. However, as we will see in Section 10.4.3, the Heisenberg model is a good description of Mott insulators such as La_2CuO_4 (the parent compound of the high-temperature superconductors) and $\kappa\text{-(BEDT-TTF)}_2\text{Cu}[\text{N}(\text{CN})_2]\text{Cl}$ (the parent compound for the organic superconductors). The Heisenberg model also plays an important role in the valence-bond theory of the chemical bond.⁴⁴

In the Heisenberg model one assumes that there is a single (unpaired) electron localized at each site and that the charge cannot move. Therefore, the only degrees of freedom in the Heisenberg model are the spins of each site (the model can also be generalized to spin $> \frac{1}{2}$). The Hamiltonian for the Heisenberg model is

$$\hat{H}_{\text{Heisenberg}} = \sum_{ij} J_{ij} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j \quad (10.82)$$

where $\hat{\mathbf{S}}_i = (\hat{S}_i^x, \hat{S}_i^y, \hat{S}_i^z) = \frac{1}{2} \sum_{\alpha\beta} \hat{c}_{i\alpha}^\dagger \vec{\sigma}_{\alpha\beta} \hat{c}_{i\beta}$ is the spin operator on site i , $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ is the vector of Pauli matrices, and J_{ij} is the exchange energy between sites i and j .

10.4.1 Two-Site Model: Classical Solution

In the classical Heisenberg model one replaces the spin operator, $\hat{\mathbf{S}}_i$, with a classical spin (i.e., a real vector, $\mathbf{S}_i$). Thus, on two sites, with $J_{12} = J$, the energy of the model is

$$E_{\text{Heisenberg}}^{(2)} = J \mathbf{S}_1 \cdot \mathbf{S}_2 = J |\mathbf{S}_1| |\mathbf{S}_2| \cos \phi \quad (10.83)$$

where ϕ is the angle between the two spins (vectors). The classical energy is minimized by $\phi = \pi$ for $J > 0$ and $\phi = 0$ for $J < 0$. Thus, for $J > 0$ the lowest-energy solution is for the two spins to point antiparallel (i.e., in opposite directions to one another); we refer to this as the *antiferromagnetic solution*. For $J < 0$ the lowest-energy solution is for the two spins to point parallel to one another; we refer to this as the *ferromagnetic solution*. Note that the difference in energy between the antiferromagnetic solution and the ferromagnetic solution is $2J |\mathbf{S}_1| |\mathbf{S}_2|$; so for $S = |\mathbf{S}_1| = |\mathbf{S}_2| = \frac{1}{2}$, the energy difference is $J/2$.

10.4.2 Two-Site Model: Exact Quantum Mechanical Solution

To solve the quantum mechanical version of the two-site Heisenberg model, it is useful to define the spin raising and lowering operators,

$$\hat{S}_i^+ \equiv \hat{S}_i^x + i \hat{S}_i^y = \hat{c}_{i\uparrow}^\dagger \hat{c}_{i\downarrow} \quad (10.84)$$

and

$$\hat{S}_i^- \equiv \hat{S}_i^x - i \hat{S}_i^y = \hat{c}_{i\downarrow}^\dagger \hat{c}_{i\uparrow} \quad (10.85)$$

Let us denote the state with spin-up on site i as $|\uparrow_i\rangle$ and the state with spin-down on site i as $|\downarrow_i\rangle$. Therefore, $\hat{S}_i^+ |\uparrow_i\rangle = 0$, $\hat{S}_i^+ |\downarrow_i\rangle = |\uparrow_i\rangle$, $\hat{S}_i^- |\uparrow_i\rangle = |\downarrow_i\rangle$, and $\hat{S}_i^- |\downarrow_i\rangle = 0$. Further, it is straightforward to confirm that

$$\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 = \frac{1}{2} (\hat{S}_1^+ \hat{S}_2^- + \hat{S}_1^- \hat{S}_2^+) + \hat{S}_1^z \hat{S}_2^z \quad (10.86)$$

We now note that the Hilbert space of the two-site Heisenberg model is spanned by four states (the spin on each site may be up or down; in general, for an N -site Heisenberg model the Hilbert space is 2^N -dimensional). Further notice that the total spin of the model ($\hat{\mathbf{S}} = \hat{\mathbf{S}}_1 + \hat{\mathbf{S}}_2$) commutes with the Hamiltonian; therefore, the eigenstates will also be eigenstates of the total spin. Thus, the four eigenstates must be a singlet,

$$|\Psi_s\rangle = \frac{1}{\sqrt{2}} (|\uparrow_1\rangle |\downarrow_2\rangle - |\downarrow_1\rangle |\uparrow_2\rangle) \equiv \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (10.87)$$

and a triplet,

$$|\Psi_t^+\rangle = |\uparrow_1\rangle |\uparrow_2\rangle \equiv |\uparrow\uparrow\rangle \quad (10.88)$$

$$|\Psi_t^0\rangle = \frac{1}{\sqrt{2}} (|\uparrow_1\rangle |\downarrow_2\rangle + |\downarrow_1\rangle |\uparrow_2\rangle) \equiv \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) \quad (10.89)$$

$$|\Psi_t^-\rangle = |\downarrow_1\rangle |\downarrow_2\rangle \equiv |\downarrow\downarrow\rangle \quad (10.90)$$

It is now straightforward to calculate the total energy of the model for these states:

$$\begin{aligned}
 E_s &= J \langle \Psi_s | \hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 | \Psi_s \rangle \\
 &= \frac{J}{2} (\langle \uparrow\downarrow | - \langle \downarrow\uparrow |) \left[\frac{1}{2} (\hat{S}_i^+ \hat{S}_j^- + \hat{S}_i^- \hat{S}_j^+) + \hat{S}_i^z \hat{S}_j^z \right] (| \uparrow\downarrow \rangle - | \downarrow\uparrow \rangle) \\
 &= -\frac{3J}{4}
 \end{aligned} \tag{10.91}$$

and

$$\begin{aligned}
 E_t &= J \langle \Psi_t^+ | \hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 | \Psi_t^+ \rangle \\
 &= J \langle \Psi_t^0 | \hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 | \Psi_t^0 \rangle = J \langle \Psi_t^- | \hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 | \Psi_t^- \rangle \\
 &= J \langle \downarrow\downarrow | [\frac{1}{2} (\hat{S}_i^+ \hat{S}_j^- + \hat{S}_i^- \hat{S}_j^+) + \hat{S}_i^z \hat{S}_j^z] | \downarrow\downarrow \rangle \\
 &= +\frac{J}{4}
 \end{aligned} \tag{10.92}$$

Thus, we find that the singlet–triplet splitting for the quantum mechanical two-site Heisenberg model is J .

10.4.3 Heisenberg Model as an Effective Low-Energy Theory of the Hubbard Model

Consider a two-site Hubbard model with two electrons and $U \gg t$, which is known as the atomic limit. $U \gg t$ implies that $\theta \rightarrow 0$ in Eq. (10.63) and that the ground state is the Heitler–London state, which is a singlet. The two other singlet eigenstates are the charge-transfer states, which have energy $\sim U$ and so will not participate in any low-energy processes (i.e., will not be involved in the interesting physics or chemistry). Therefore, we can “integrate out” the charge-transfer states and derive a simpler model with a smaller Hilbert space. A model derived in this manner is known as an *effective low-energy Hamiltonian* (see Section 10.7). In this case we use second-order perturbation theory to derive our effective low-energy Hamiltonian.

We start by writing the two-site Hubbard model as

$$\hat{H}_{\text{Hubbard}} = U(\hat{H}_0 + \frac{t}{U} \hat{H}_1) \tag{10.93}$$

where $\hat{H}_0 = \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$ and $\hat{H}_1 = \sum_{\langle ij \rangle \sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma}$. Thus, it is clear that the small parameter for our perturbation theory is t/U . For $t = 0$ the ground state is fourfold degenerate; the four states involved being the Heitler–London state and the triplet states. Formally, one should therefore use degenerate perturbation theory. But the perturbation, $\hat{H}_1$, does not connect any of the four ground states; therefore, to

second order (which is all we consider), nondegenerate perturbation theory will yield the same results. As it simplifies the discussion, we frame our discussion in terms of nondegenerate perturbation theory. It is left as an exercise to the reader to show that on adding the appropriate projection operators (see, e.g., Ref. 45) to perform degenerate perturbation theory, the result is unchanged.

Consider the related Hamiltonian with $\hat{H}_0 \rightarrow \hat{H}_0 + \eta |\Psi_{\text{HL}}\rangle \langle \Psi_{\text{HL}}|$ in the limit $\eta \rightarrow 0^-$ (i.e., as η tends to 0 from below). This has the same properties as Eq. (10.93), except that $|\Psi_{\text{HL}}\rangle$ is the true ground state and we may use non-degenerate perturbation theory. $\langle \Psi_{\text{HL}} | \hat{H}_1 | \Psi_{\text{HL}} \rangle = 0$, so there is no correction to the ground-state energy to first order in t/U . The second-order change in the ground-state energy, $\Delta E^{(2)}$, is given by

$$\Delta E^{(2)} = - \sum_{s \neq \text{HL}} \frac{|\langle \Psi_{\text{HL}} | \hat{H}_1 | \Psi_s \rangle|^2}{E_s - E_{\text{HL}}} \quad (10.94)$$

where the sum over s runs over all states except the ground state. Note that, as is true in general, the second-order contribution to the ground-state energy is negative. Evaluating the matrix elements for the superexchange interactions (see Fig. 10.10), one finds that

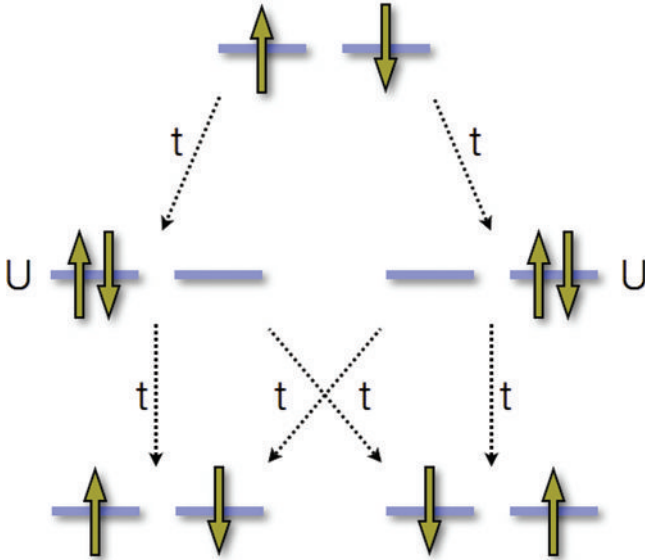


Fig. 10.10 (color online) Superexchange processes that lead to the effective antiferromagnetic Heisenberg coupling between nearest neighbors in the large U/t limit of the half-filled Hubbard model. These processes lower the energy of the singlet state by $4t^2/U$ as there are four paths, the matrix element between the ground state and the intermediate states is $-t$, and the intermediate states are higher in energy by U . The energy of the triplet states is unchanged by perturbations to second order in t/U , as the Pauli exclusion principle prevents two electrons with the same spin from occupying the same site.

$$\Delta E^{(2)} = -\frac{4|t|^2}{U} \quad (10.95)$$

In contrast, if we add an infinitesimal term to make one of the triplet states the true ground state (e.g., $0^-|\Psi_1^1\rangle\langle\Psi_1^1|$), we find that the Pauli exclusion principle ensures that $\langle\Psi_1^1|\hat{H}_1|\Psi_s\rangle = 0$ for all s . Thus, there is no change in the energy of the triplet state to second order in t/U .

Therefore, it is clear that for $U/t \rightarrow \infty$, the half-filled Hubbard model reduces to the Heisenberg model; that is, the eigenstates and energies are the same if we set $J = 4|t|^2/U$. This result is not a special property of the two-site model and is true to second order for an arbitrary lattice,⁴⁶ as second-order perturbation theory couples sites i and j only if the hopping integral between them is nonzero. For an arbitrary lattice to second order, $J_{ij} = 4|t_{ij}|^2/U$.

As the Heisenberg model is the large U/t limit of the half-filled Hubbard model, electronic correlations are vitally important in the physics of materials described by the Heisenberg model. Therefore, weakly correlated theories, such as Hartree–Fock and density functional theory, will give qualitatively incorrect results.

10.4.4 Frustration: Solution of the Three- and Four-Site Classical Heisenberg Models

Before considering the classical three-site model, let us spend a moment discussing the classical four-site model. We assume that the four sites are situated on the vertices of a square and there is an exchange interaction J between nearest neighbors (i.e., along the sides of the square), but no interactions between next nearest neighbors (i.e., along diagonals of the square). The energy of the model is

$$\begin{aligned} E_{\text{Heisenberg}}^{(4)} &= J \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j = J \sum_{\langle ij \rangle} |\mathbf{S}_i| |\mathbf{S}_j| \cos \theta_{ij} \\ &= \frac{J}{4} \sum_{\langle ij \rangle} \cos \theta_{ij} \end{aligned} \quad (10.96)$$

where θ_{ij} is the angle between the spins on sites i and j , $\mathbf{S}_i$ is the spin on the i th lattice site, and in the last equality we have specialized to the case $|\mathbf{S}_i| = \frac{1}{2}$ for all i . Notice that the Hamiltonian is just a sum over the “bonds” (sides of the square). As for the two-site model (Section 10.4.1), the solution depends on the sign of J . For $J < 0$ the lowest-energy state is ferromagnetic (all of the spins align parallel to one another) and has energy $E_{\text{Heisenberg}}^{(4)} = J$. For $J > 0$ the lowest-energy state is antiferromagnetic (each spin aligns antiparallel to its nearest neighbor; see Fig. 10.11a). Thus, the four-site cluster is split into two “sublattices” with all of the spins parallel to one another within the same sublattice and antiparallel to spins on the other sublattice. The antiferromagnetic arrangement of spins therefore has energy $E_{\text{Heisenberg}}^{(4)} = -J$. Thus, we find that the energy difference between the ferromagnetic and antiferromagnetic arrangements is $2J$.

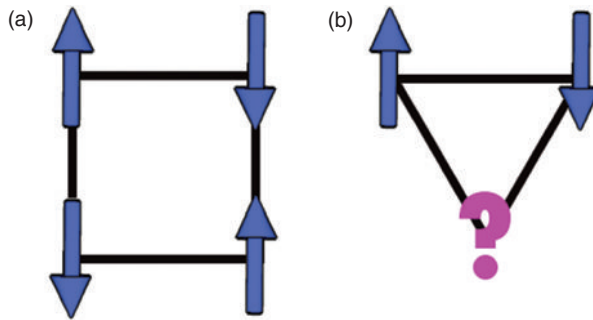


Fig. 10.11 (color online) Examples of classical spins on (a) a bipartite cluster and (b) a frustrated cluster. On the square cluster (a) one can arrange all of the spins antiferromagnetically (i.e., so that each spin is antiparallel to all of its nearest neighbors). The same is true for the square lattice. This cannot be accomplished on either the triangular cluster (b) or the triangular lattice. In each panel there is an exchange interaction, J , between any two spins joined by a line. (Modified from Ref. 81.)

A lattice that can be split, as described above, into two sublattices such that all nearest neighbors are on different sublattices is referred to as *bipartite*. Both the four-site (square) and two-site lattices are bipartite. For bipartite lattices the energy difference between the ferromagnetic and antiferromagnetic arrangements is $JNz/4$, where z is the coordination number of the lattice and N is the number of lattice sites. This is because the energy of each bond can be optimized regardless of what happens to other bonds for either sign of J . It can be seen from Fig. 10.11 that the triangular lattice is not bipartite; this leads to significant differences in its physics.

Before analyzing this model mathematically, let us consider some of those differences. Clearly, for $J < 0$ it is straightforward to arrange the spins ferromagnetically. Further, as the energy of each of the three bonds will be optimized in this arrangement, we expect the total energy of this state to be $E_{\text{Heisenberg}}^{(3)} = 3J/4$ for $S = \frac{1}{2}$. But as shown in Fig. 10.11b, one cannot arrange three spins antiferromagnetically on a triangular lattice. Thus, for $J > 0$, we cannot optimize the energy of each bond individually. When this is the case one says that the lattice is *frustrated*. For a frustrated lattice with $S = \frac{1}{2}$, we expect the solution for $J > 0$ to have energy $E_{\text{Heisenberg}}^{(3)} > -3J/4$, and thus one expects the difference in energy between this state and the ferromagnetic state to be $< JNz/4$. The concept of frustration can also be generalized to itinerant systems where a similar reduction in the bandwidth of the itinerant electrons is found.⁴³

Having outlined our expectations, let us now consider the three-site Heisenberg model more carefully. The energy is given by

$$E_{\text{Heisenberg}}^{(3)} = J \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j \quad (10.97)$$

Without loss of generality we can choose $\mathbf{S}_1 = S_1(1, 0, 0)$, $\mathbf{S}_2 = S_2(\cos \phi_2, \sin \phi_2, 0)$, and $\mathbf{S}_3 = S_3(\cos \theta_3 \cos \phi_3, \cos \theta_3 \sin \phi_3, \sin \theta_3)$. Thus, for $S_1 = S_2 = S_3 = \frac{1}{2}$,

$$E_{\text{Heisenberg}}^{(3)} = \frac{J}{4} [\cos \phi_2 + \cos \theta_3 \cos(\phi_2 - \phi_3) + \cos \theta_3 \cos \phi_3] \quad (10.98)$$

Physically, we seek the minimum energy, which yields the conditions

$$\begin{aligned} \frac{\partial E_{\text{Heisenberg}}^{(3)}}{\partial \theta_3} &= \frac{J}{4} \sin \theta_3 [\cos(\phi_2 - \phi_3) + \cos \phi_3] = 0 \\ \frac{\partial E_{\text{Heisenberg}}^{(3)}}{\partial \phi_3} &= \frac{J}{4} \cos \theta_3 [\sin(\phi_2 - \phi_3) - \sin \phi_3] = 0 \\ \frac{\partial E_{\text{Heisenberg}}^{(3)}}{\partial \phi_2} &= -\frac{J}{4} [\cos \theta_3 \sin(\phi_2 - \phi_3) + \sin \phi_2] = 0 \end{aligned}$$

For $J > 0$ the global minimum is, unsurprisingly, $\theta_3 = \phi_2 = \phi_3 = 0$ (i.e., ferromagnetism). The energy of the ferromagnetic state is $3J/4$. For $J < 0$ there are several degenerate minima, which all show the same physics. For simplicity we will just consider the minimum $\theta_3 = 0$, $\phi_2 = 2\pi/3$, and $\phi_3 = 4\pi/3$. In this solution each of the spins points 120° away from each of the other spins; hence, this is known as the 120° state. It is left as an exercise to the reader to identify the other solutions, to show that there are none with lower energy than those discussed above, and to show that all of the degenerate solutions are physically equivalent. The energy of the 120° state is $-3J/8$ and hence the energy difference between the ferromagnetic state and the 120° state is just $9J/8$, less than we would expect ($JNz/4 = 3J/2$ for $N = 3$, $z = 2$) for a bipartite lattice.

10.4.5 Three-Site Model: Exact Quantum Mechanical Solution

Group theory, the mathematics of symmetry, allows one to solve the quantum spin- $\frac{1}{2}$ three-site Heisenberg model straightforwardly. Unfortunately, space does not permit an introduction to the relevant group theory. Therefore, the reader who is not familiar with the mathematics is advised either to refer to one of the many excellent textbooks on the subject (e.g., Tinkham¹¹ or Lax¹²) or, failing that, simply to check that the wavefunctions derived by the group-theoretic arguments below are indeed eigenstates.

The Hamiltonian is

$$\begin{aligned} \hat{H}_{\text{Heisenberg}}^{(3)} &= J \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j \\ &= J \sum_{\langle ij \rangle} \left[\frac{1}{2} (\hat{S}_i^+ \hat{S}_j^- + \hat{S}_i^- \hat{S}_j^+) + \hat{S}_i^z \hat{S}_j^z \right] \end{aligned} \quad (10.99)$$

We begin by noting that $2 \otimes 2 \otimes 2 = 2 \oplus 2 \oplus 4^\dagger$; that is, a system formed from three spin- $\frac{1}{2}$ particles will have two doublets (with twofold-degenerate spin- $\frac{1}{2}$ eigenstates) and one quadruplet (with fourfold-degenerate spin- $\frac{3}{2}$ eigenstates).

There are only four possible quadruplet states consistent with C_3 point-group symmetry[‡] of the model. Each of these belongs to the A irreducible representation of C_3 . They are

$$\begin{aligned} |\Psi_{3/2}^{3/2}\rangle &= |\uparrow\uparrow\uparrow\rangle \\ |\Psi_{3/2}^{1/2}\rangle &= \frac{1}{\sqrt{3}}(|\downarrow\uparrow\uparrow\rangle + |\uparrow\downarrow\uparrow\rangle + |\uparrow\uparrow\downarrow\rangle) \\ |\Psi_{3/2}^{-1/2}\rangle &= \frac{1}{\sqrt{3}}(|\uparrow\downarrow\downarrow\rangle + |\downarrow\uparrow\downarrow\rangle + |\downarrow\downarrow\uparrow\rangle) \\ |\Psi_{3/2}^{-3/2}\rangle &= |\downarrow\downarrow\downarrow\rangle \end{aligned}$$

where $|\alpha\beta\gamma\rangle = |S_1^z, S_2^z, S_3^z\rangle$ with α, β , and $\gamma = \uparrow$ or $\downarrow$. Each of these states has energy $E = 3J/4$, and they are the (degenerate) ground states for $J < 0$.

We are left with the four doublet states. These belong to the two-dimensional E irreducible representation of C_3 , and as the Hamiltonian is time-reversal symmetric, all four doublet states are degenerate. Explicitly the states are

$$\begin{aligned} |\Psi_{1/2}^{1/2}\rangle &= \frac{1}{\sqrt{3}}(|\downarrow\uparrow\uparrow\rangle + e^{i2\pi/3}|\uparrow\downarrow\uparrow\rangle + e^{-i2\pi/3}|\uparrow\uparrow\downarrow\rangle) \\ |\Psi_{1/2}^{-1/2}\rangle &= \frac{1}{\sqrt{3}}(|\uparrow\downarrow\downarrow\rangle + e^{i2\pi/3}|\downarrow\uparrow\downarrow\rangle + e^{-i2\pi/3}|\downarrow\downarrow\uparrow\rangle) \\ |\tilde{\Psi}_{1/2}^{1/2}\rangle &= \frac{1}{\sqrt{3}}(|\downarrow\uparrow\uparrow\rangle + e^{-i2\pi/3}|\uparrow\downarrow\uparrow\rangle + e^{i2\pi/3}|\uparrow\uparrow\downarrow\rangle) \\ |\tilde{\Psi}_{1/2}^{-1/2}\rangle &= \frac{1}{\sqrt{3}}(|\uparrow\downarrow\downarrow\rangle + e^{-i2\pi/3}|\downarrow\uparrow\downarrow\rangle + e^{i2\pi/3}|\downarrow\downarrow\uparrow\rangle) \end{aligned}$$

Each of these states has energy $E = -5J/4$ and they are the (degenerate) ground states for $J > 0$. Thus, the energy difference between the highest spin state and the lowest spin state is $2J$. From the solution to the two-site model (Section 10.4.2), we expected each of the three bonds to yield an energy difference of J between the lowest and highest spin states. Thus, the frustration has a similar effect on both the quantum and classical models (i.e., frustration lowers the energy difference between the highest spin and lowest spin states).

[†]In this notation the integers are the degeneracy of the state.

[‡]One might, reasonably, take the view that the model has either D_{3h} or C_{3v} . In fact, the arguments in this section go through almost identically for either of these symmetries (with appropriate changes in notation), due to the homomorphisms from these groups to C_3 . We use C_3 notation for simplicity.

10.4.6 Heisenberg Model on Infinite Lattices

The Heisenberg model can be solved exactly in one dimension, and we discuss this further below, but not in any other finite dimension. However, in more than one dimension, physics of the Heisenberg model is typically very different from that in one dimension, so we will begin by discussing, qualitatively, the semiclassical spin-wave approximation for the Heisenberg model, which captures many important aspects of magnetism. A quantitative formulation of this theory can be found in many textbooks (e.g., Ashcroft and Mermin¹⁰ or Rössler²⁹).

In inelastic neutron scattering experiments a neutron may have its spin flipped by its interaction with the magnet; this causes a spin 1 excitation in the material. The conceptually simplest spin 1 excitation would be to flip one (spin- $\frac{1}{2}$) spin; in a one-dimensional ferromagnetic Heisenberg model, this state has energy $2|J|$ greater than the ground state. However, a much lower energy excitation is a “spin wave,” where each spin is rotated a small amount from its nearest neighbors (see Fig. 10.12). In a one-dimensional ferromagnetic Heisenberg model, spin waves have excitation energies of $\hbar\omega_k = 2|J|(1 - \cos ka)$, where a is the lattice constant.²⁹ Note, in particular, that the excitation energy vanishes for long-wavelength (small- k) spin waves. This spin-wave spectrum can indeed be observed directly in neutron-scattering experiments from suitable materials,⁴⁷ and the spectrum is found to be in good agreement with the predictions of the semiclassical theory in many materials. One can also quantize the semiclassical theory by making a *Holstein–Primakoff transformation*.²⁹ This yields a description of the low-energy physics of the Heisenberg model in terms of noninteracting bosons, known as *magnons*, which have the same dispersion relation as the classical spin waves. Similar spin-wave and magnon descriptions can be constructed straightforwardly for the antiferromagnetic Heisenberg model.²⁹

The effective low-energy physics of the one-dimensional Heisenberg model is, as noted above, rather different from the semiclassical approximation. To understand this, it is helpful to think of the Heisenberg model as a special case of the *XXZ model*:

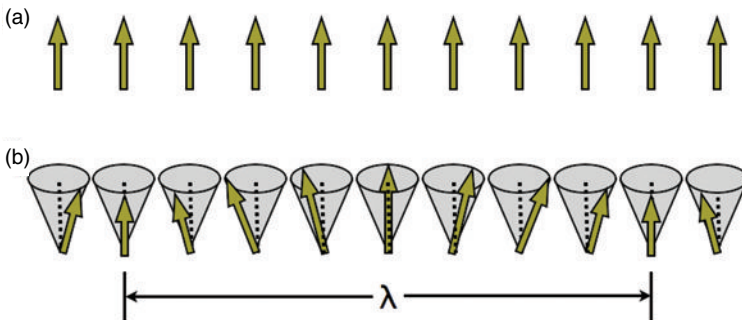


Fig. 10.12 (color online) (a) Classical ground state of a ferromagnetic Heisenberg chain; (b) spin-wave excitation with wavelength $\lambda = 1/k$ in the same model.

$$H_{\text{XXZ}} = J_{xy} \sum_i (S_i^x S_{i+1}^x + S_i^y S_{i+1}^y) + J_z \sum_i S_i^z S_{i+1}^z \quad (10.100)$$

which reduces to the Heisenberg model for $J_{xy} = J_z = J$. For $J_z < J_{xy} < 0$, the model displays an exotic quantum phase known as a *Luttinger liquid*. (At $J_{xy} = J_z$ the model undergoes a quantum phase transition from the Luttinger liquid to an ordered phase.⁴⁸)

On the energy scales relevant to chemistry, one does not need to worry about the fact that protons and neutrons are made up of smaller particles (quarks). This is because the quarks are confined within the proton or neutron.⁴⁹ Similarly, in a normal magnet it does not matter that the material is made up of spin- $\frac{1}{2}$ particles (electrons). As described above, on the energy scales relevant to magnets, the spins are confined into spin-1 particles, magnons. However, magnons can be described in terms of two spin- $\frac{1}{2}$ spinons, which are confined inside the magnon. In the Luttinger liquid the spinons are deconfined; that is, the spinons can move independent of one another (see Fig. 10.13). As the magnon is a composite particle made from two spinons, this is often referred to as *fractionalization*. A key prediction of this theory is that the spinons display a continuum of excitations in neutron-scattering experiments (as opposed to the sharp dispersion predicted for magnons). The two-spinon continuum has indeed been observed in a number of quasi-one-dimensional materials.⁵⁰

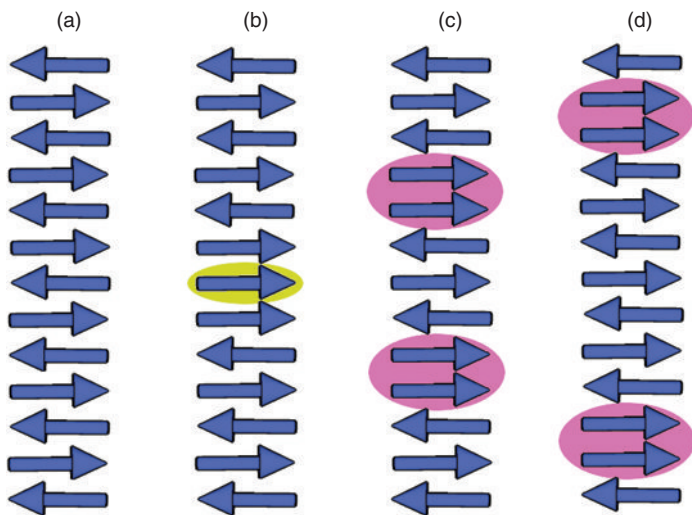


Fig. 10.13 (color online) Spinons in a one-dimensional spin chain. (a) Local antiferromagnetic correlations. (b) A neutron scattering off the chain causes one spin (circled) to flip. (c,d) Spontaneous flips of adjacent pairs of spins due to quantum fluctuations allow the spinons (circled) to propagate independently. A key open question is: Can this free propagation occur in two-dimensions, or do interactions confine the spinons? (Modified from Ref. 81.)

An open research question is: Does fractionalization occur in higher dimensions? Because of the success of spin-wave theory (implying confined spinons) in describing magnetically ordered materials, one does not expect fractionalization in materials with magnetic order. Therefore, one would like to investigate quasi-two- or three-dimensional materials whose low-energy physics is described by spin Hamiltonians (such as the Heisenberg model) but that do not order magnetically even at the lowest temperatures. Such materials are collectively referred to as *spin liquids*. There is a long history of theoretical contemplation of spin liquids, which suggests that frustrated magnets and insulating systems near to the Mott transition are strong candidates to display spin-liquid physics. However, evidence for real materials with spin-liquid ground states has been scarce until very recently,⁵¹ but there is now evidence for spin liquids in the triangular lattice compound κ -(BEDT-TTF)₂Cu(CN)₃,^{24,52} the kagome lattice (see Fig. 10.4) compound $\text{ZnCu}_3(\text{OH})_6\text{Cl}_2$,⁵³ and the hyperkagome lattice compound $\text{Na}_4\text{Ir}_3\text{O}_8$.⁵⁴ It remains to be seen whether any of these materials support fractionalized excitations.

10.5 OTHER EFFECTIVE LOW-ENERGY HAMILTONIANS FOR CORRELATED ELECTRONS

10.5.1 Complete Neglect of Differential Overlap, the Pariser–Parr–Pople Model, and Extended Hubbard Models

We now consider another model for which the quantum chemistry and condensed matter physics communities have different names. These models belong to class of models known as complete neglect of differential overlap (CNDO). For a pair of orthogonal states, $\phi(x)$ and $\psi(x)$, the integral over all space of the overlap of the two wavefunctions vanishes [i.e., $\int_{-\infty}^{\infty} \phi(x)\psi(x)dx = 0$]. If the differential overlap vanishes, the overlap of the two wavefunctions vanishes at every point in space [i.e., $\lim_{\delta \rightarrow 0} \int_{x_0}^{x_0+\delta} \phi(x)\psi(x)dx = 0$ for all x_0]. The CNDO approximation is simply to assume that the differential overlap between all basis states is negligible. Thus CNDO implies that $V_{ijkl} = V_{iikk}\delta_{ij}\delta_{kl}$ (cf. Section 10.1.2) and the general CNDO Hamiltonian is

$$\hat{H}_{\text{CNDO}} = - \sum_{ij\sigma} t_{ij} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \sum_{ij\sigma\sigma'} V_{ij} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \quad (10.101)$$

where $V_{ij} \equiv V_{iijj}$ and the number operator $\hat{n}_{i\sigma} \equiv \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma}$. The Pariser–Parr–Pople (PPP) model is the CNDO approximation in a basis that includes only the π -electrons. Often, a Hückel-like notation is used with $V_{ij} = \gamma_{ij}$; thus,

$$\hat{H}_{\text{PPP}} = \sum_{i\sigma} \alpha_i \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} + \sum_{ij\sigma} \beta_{ij} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \sum_{ij\sigma\sigma'} \gamma_{ij} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \quad (10.102)$$

The extended Hubbard model, as with the plain Hubbard model, is typically studied in a basis with one orbital per site. Further, one often makes the approximation that $V_{ii} = U$, $V_{ij} = V$, if i and j are nearest neighbors and $V_{ij} = 0$ otherwise. This yields

$$\hat{H}_{\text{eH}} = - \sum_{\langle ij \rangle \sigma} t_{ij} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} + V \sum_{\langle ij \rangle \sigma \sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \quad (10.103)$$

One can, of course, go beyond CNDO. The most general possible model for two identical sites with a single orbital per site is

$$\begin{aligned} \hat{H}_{\text{eH2}} = & - \sum_{\sigma} [t - X(\hat{n}_{1\bar{\sigma}} + \hat{n}_{2\bar{\sigma}})] (\hat{c}_{1\sigma}^\dagger \hat{c}_{2\sigma} + \hat{c}_{2\sigma}^\dagger \hat{c}_{1\sigma}) \\ & + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} + V \hat{n}_1 \hat{n}_2 + J \mathbf{S}_1 \cdot \mathbf{S}_2 \\ & + P (\hat{c}_{1\uparrow}^\dagger \hat{c}_{1\downarrow}^\dagger \hat{c}_{2\uparrow} \hat{c}_{2\downarrow} + \hat{c}_{2\uparrow}^\dagger \hat{c}_{2\downarrow}^\dagger \hat{c}_{1\uparrow} \hat{c}_{1\downarrow}) \end{aligned} \quad (10.104)$$

where $\hat{n}_i = \sum_{\sigma} \hat{n}_{i\sigma}$, $\hat{\mathbf{S}}_i = \sum_{\alpha\beta} \hat{c}_{i\alpha}^\dagger \vec{\sigma}_{\alpha\beta} \hat{c}_{i\beta}$, $\vec{\sigma}_{\alpha\beta}$ is the vector of Pauli matrices, J is the direct exchange interaction, X is the correlated hopping amplitude, and P is the pair hopping amplitude.

10.5.2 Larger Basis Sets and Hund's Rules

Thus far we have focused mainly on models with one orbital per site. Often, this is not appropriate: for example, if one were interested in chemical bonding or materials containing transition metals. Many of the models discussed in this chapter can be extended straightforwardly to include more than one orbital per site. However, while writing down models with more than one orbital per site is not difficult, these models do contain significant additional physics. Some of the most important effects are known as *Hund's rules*.¹ These rules have important experimental consequences, from atomic physics to biology. To examine Hund's rules, let us consider the atomic limit ($t = 0$) of an extended Hubbard model with two electrons in two orbitals per site:

$$\hat{H}_{\text{eH1s2o}} = U \sum_{\mu} \hat{n}_{\mu\uparrow} \hat{n}_{\mu\downarrow} + V_0 \hat{n}_1 \hat{n}_2 + J_{\text{H}} \hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 \quad (10.105)$$

where $\mu = 1$ or 2 labels the orbitals, $\hat{n}_{\mu\sigma} = \hat{c}_{\mu\sigma}^\dagger \hat{c}_{\mu\sigma}$, $\hat{n}_{\mu} = \sum_{\sigma} \hat{n}_{\mu\sigma}$, $\hat{\mathbf{S}}_{\mu} = \sum_{\alpha\beta} \hat{c}_{\mu\alpha}^\dagger \vec{\sigma}_{\alpha\beta} \hat{c}_{\mu\beta}$, U is the Coulomb repulsion between two electrons in the same orbital, V_0 is the Coulomb repulsion between two electrons in different orbitals, and J_{H} is the Hund's rule coupling between electrons in different orbitals. Notice that the Hund's rule coupling is an exchange interaction between orbitals.

Further, if we compare the Hamiltonian with the definition given in Eq. (10.28), we find that

$$\begin{aligned}
 -J_H &= \int d^3\mathbf{r}_1 \int d^3\mathbf{r}_2 \phi_1^*(\mathbf{r}_1) \phi_2(\mathbf{r}_1) V(\mathbf{r}_1 - \mathbf{r}_2) \phi_2^*(\mathbf{r}_2) \phi_1(\mathbf{r}_2) \\
 &\sim \int d^3\mathbf{r}_1 \int d^3\mathbf{r}_2 |\phi_1(\mathbf{r}_1)|^2 V(\mathbf{r}_1 - \mathbf{r}_2) |\phi_2(\mathbf{r}_2)|^2 \\
 &\geq 0
 \end{aligned} \tag{10.106}$$

as $V(\mathbf{r}_1 - \mathbf{r}_2)$ is positive semidefinite. Therefore, typically, $J_H < 0$; that is, the Hund's rule coupling favors the parallel alignment of the spins in a half-filled system.

U is the largest energy scale in the problem, so, for simplicity, let us consider the case $U \rightarrow \infty$. For $J_H = 0$ there are four degenerate ground states: a singlet, $(1/\sqrt{2})(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ (where the first arrow refers to the spin of the electron in orbital 1 and the second arrow refers to the spin in orbital 2), and a triplet: $|\uparrow\uparrow\rangle$, $|\downarrow\downarrow\rangle$, and $(1/\sqrt{2})(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$. But for $J > 0$ the energy of the triplet states is J_H lower than that of the singlet state. Indeed, even if we relax the condition $U \rightarrow \infty$, the triplet state remains lower in energy than the singlet state, as physically we require that $U > J_H$. One can repeat this argument for any number of electrons in any number of orbitals, and one always finds that the highest spin state has the lowest energy. However, if one studies models with more than one site and moves away from the atomic limit ($t = 0$), one finds that there is a subtle competition between the kinetic (hopping) term and the Hund's rule coupling which means that the high spin state is not always the lowest-energy state. Many such interesting effects can be understood on the basis of a two-site generalization of this two-orbital model.⁵⁵

10.5.3 Ionic Hubbard Model

Thus far we have assumed that all sites are identical. Of course, this is not always true in real materials. In a compound, more than one species of atom may contribute to the low-energy physics,⁵⁶ or different atoms of the same species may be found at crystallographic distinct sites.^{43,57} A simple model that describes this situation is the ionic Hubbard model:

$$\hat{H}_{\text{IH}} = -t \sum_{\langle ij \rangle \sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} + \sum_{i\sigma} \varepsilon_i \hat{n}_{i\sigma} \tag{10.107}$$

where $\varepsilon_i = t_{ii}$ is the site energy, which will be taken to be different on different sites. Note that in the standard form of the ionic Hubbard model, all sites are assumed to have the same U .

An important application of the ionic Hubbard model is in describing transition metal oxides.⁵⁶ Typically, ε_i is larger on the transition metal site than on the oxygen site; therefore, the oxygen orbitals are nearly filled. This means that there

is a low hole density in the oxygen orbitals and hence that electronic correlations are less important for the electrons in the oxygen orbitals than for electrons in transition metal orbitals. If the difference between ε_i on the oxygen sites and ε_i on the transition metal sites is large enough, the oxygen orbitals are completely filled in all low-energy states and therefore need not feature in the low-energy description of the material. However, just because the oxygen orbitals do not appear explicitly in the effective low-energy Hamiltonian of the material does not mean that the oxygen does not have a profound effect on low-energy physics.

To see this, consider a toy model with two metal sites (labeled 1 and 2) and one oxygen site (labeled O), whose Hamiltonian is

$$\hat{H}_{\text{IH3}} = -t \sum_{\sigma} (\hat{c}_{1\sigma}^{\dagger} \hat{c}_{O\sigma} + \hat{c}_{O\sigma}^{\dagger} \hat{c}_{1\sigma} + \hat{c}_{2\sigma}^{\dagger} \hat{c}_{O\sigma} + \hat{c}_{O\sigma}^{\dagger} \hat{c}_{2\sigma}) + \sum_{i\sigma} \frac{\Delta}{2} (\hat{n}_{1\sigma} + \hat{n}_{2\sigma} - \hat{n}_{O\sigma}) \quad (10.108)$$

as sketched in Fig. 10.14, which is just the ionic Hubbard model with $U = 0$ and $\Delta = \varepsilon_1 - \varepsilon_O = \varepsilon_2 - \varepsilon_O > 0$. With three electrons in the system and $t = 0$, the ground state is fourfold degenerate, the ground states have two electrons on the O atom and the other electron on one of the metal atoms. If we now consider finite, but small $t \ll \Delta$, we can construct a perturbation theory in t/Δ . One finds that there is a splitting between the bonding, $(1/\sqrt{2})(\hat{c}_{1\sigma}^{\dagger} + \hat{c}_{2\sigma}^{\dagger})\hat{c}_{O\uparrow}^{\dagger}\hat{c}_{O\downarrow}^{\dagger}|0\rangle$ and antibonding, $(1/\sqrt{2})(\hat{c}_{1\sigma}^{\dagger} - \hat{c}_{2\sigma}^{\dagger})\hat{c}_{O\uparrow}^{\dagger}\hat{c}_{O\downarrow}^{\dagger}|0\rangle$, states. The processes that lead to this splitting are sketched in Fig. 10.15. Therefore, our effective low-energy Hamiltonian is a tight-binding model involving just the metal atoms:

$$\hat{H}_{\text{eff}} = -t^* \sum_{\sigma} (\hat{c}_{1\sigma}^{\dagger} \hat{c}_{2\sigma} + \hat{c}_{2\sigma}^{\dagger} \hat{c}_{1\sigma}) \quad (10.109)$$

where, to second order in t/Δ , the effective metal-to-metal hopping integral is given by

$$t^* = -\frac{t^2}{\Delta} \quad (10.110)$$

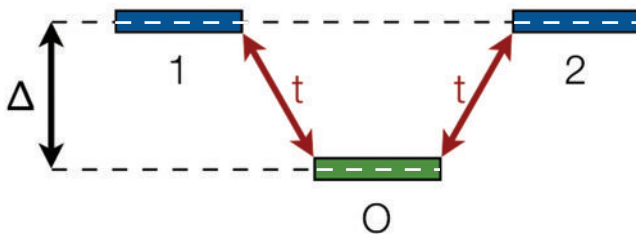


Fig. 10.14 (color online) Toy model for a transition metal oxide, Hamiltonian equation (10.108), with two transition metal sites (1 and 2) and a single oxygen site (O).

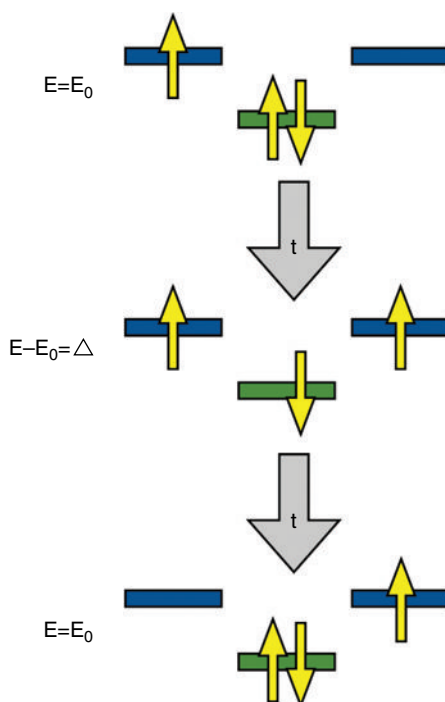


Fig. 10.15 (color online) Processes described by Hamiltonian equation (10.108) that give rise to the effective hopping integral between the two transition metal atom sites.

Note that even though t is positive, $t^* < 0$ (or, equivalently, $\beta^* > 0$), in contrast to our naive expectation that hopping integrals are positive ($\beta < 0$; cf. Section 10.2).

10.6 HOLSTEIN MODEL

So far we have assumed that the nuclei or ions form a passive background through which the electrons move. However, in many situations this is not the case. Atoms move and these lattice/molecular vibrations interact with the electrons via the electron–phonon/vibronic interaction. One of the simplest models of such effects is the Holstein model, which we discuss below. Electron–vibration interactions play important roles across science. In physics, electron–phonon interactions can give rise to superconductivity,⁵⁸ spin and charge density waves,⁵⁹ polaron formation,⁶⁰ and piezoelectricity.⁵⁸ In chemistry, vibronic interactions affect electron-transfer processes,⁶¹ Jahn–Teller effects, spectroscopy, stereochemistry, activation of chemical reactions, and catalysis.⁶² In biology the vibronic interactions play important roles in photoprotection,⁶³ photosynthesis,⁶⁴ and vision.⁶⁵ It is therefore clear that one of the central tasks for condensed matter theory and theoretical chemistry is to describe electron–vibration interactions.

In general, one may write the Hamiltonian of a system of electrons and nuclei as

$$\hat{H} = \hat{H}_e + \hat{H}_n + \hat{H}_{en} \quad (10.111)$$

where $\hat{H}_e$ contains those terms that affect only the electrons, $\hat{H}_n$ contains those terms that affect only the nuclei, and $\hat{H}_{en}$ describes the interactions between the electrons and the nuclei. $\hat{H}_e$ might be any of the Hamiltonians we have discussed above. However, for the Holstein model one assumes a tight-binding form for $\hat{H}_e$. In the normal-mode approximation,⁶² which we will make, one treats molecular and lattice vibrations as harmonic oscillators (cf. Section 10.1.1). As the ions carry a charge, any displacement of the ions from their equilibrium positions will change the potential felt by the electrons. The Holstein model assumes that each vibrational mode is localized on a single site. For this to be the case, the site must have some internal structure (i.e., the site cannot correspond to a single atom). Therefore, the Holstein model is more appropriate for molecular solids than for simple crystals. For small displacements, $x_{i\mu}$, of the μ th mode of the i th lattice site, we can perform a Taylor expansion in the dimensionless normal coordinate of the vibration, $Q_{i\mu} = x_{i\mu} \sqrt{m_{i\mu} \omega_{i\mu} / \hbar}$, where $m_{i\mu}$ and $\omega_{i\mu}$ are, respectively, the mass and the frequency of the μ th mode on the i th site, and we find that

$$\hat{H}_{en} = \sum_{ij\sigma\mu} \frac{\partial t_{ij}}{\partial Q_{i\mu}} Q_{i\mu} (\hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \hat{c}_{j\sigma}^\dagger \hat{c}_{i\sigma}) + \dots \quad (10.112)$$

In the Holstein model one assumes that the derivative vanishes for $i \neq j$. We may quantize the vibrations in the usual way (cf. Section 10.1.1), which yields

$$\hat{H}_{en} = \sum_{i\sigma\mu} g_{i\mu} (\hat{a}_{i\mu}^\dagger + \hat{a}_{i\mu}) \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \quad (10.113)$$

where $\hat{a}_{i\mu}^{(\dagger)}$ destroys (creates) a quantized vibration in the μ th mode on the i th site, $g_{i\mu} = 2^{-1/2} \partial t_{ii} / \partial Q_{i\mu}$, and $\hat{H}_n = \sum_{i\mu} \hbar \omega_{i\mu} \hat{a}_{i\mu}^\dagger \hat{a}_{i\mu}$. Thus,

$$\hat{H}_{\text{Holstein}} = -t \sum_{\langle ij \rangle \sigma} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} + \sum_{i\mu} \hbar \omega_{i\mu} \hat{a}_{i\mu}^\dagger \hat{a}_{i\mu} + \sum_{i\sigma\mu} g_{i\mu} (\hat{a}_{i\mu}^\dagger + \hat{a}_{i\mu}) \hat{c}_{i\sigma}^\dagger \hat{c}_{i\sigma} \quad (10.114)$$

10.6.1 Two-Site Holstein Model

If we assume that there is only one electron and one mode per site, the Holstein model simplifies to

$$\hat{H}_{\text{Holstein}} = -t \sum_{\sigma} (\hat{c}_{1\sigma}^\dagger \hat{c}_{2\sigma} + \hat{c}_{2\sigma}^\dagger \hat{c}_{1\sigma}) + \hbar \omega \sum_i \hat{a}_i^\dagger \hat{a}_i + g \sum_i (\hat{a}_i^\dagger + \hat{a}_i) \hat{n}_i \quad (10.115)$$

on two symmetric sites, where $\hat{n}_i = \sum_{\sigma} \hat{n}_{i\sigma} = \sum_{\sigma} \hat{c}_{i\sigma}^{\dagger} \hat{c}_{i\sigma}$. It is useful to change the basis in which we consider the phonons to that of in-phase (symmetric), $\hat{s} = (\hat{a}_1 + \hat{a}_2)/\sqrt{2}$, and out-of-phase (antisymmetric), $\hat{b} = (\hat{a}_1 - \hat{a}_2)/\sqrt{2}$, vibrations. In this basis one finds that

$$\hat{H}_{\text{Holstein}} = \hat{H}_s + \hat{H}_{be} \quad (10.116)$$

where

$$\hat{H}_s = \hbar\omega \hat{s}^{\dagger} \hat{s} + \frac{g}{\sqrt{2}} (\hat{s}^{\dagger} + \hat{s}) (\hat{n}_1 + \hat{n}_2) \quad (10.117)$$

and

$$\hat{H}_{be} = -t \sum_{\sigma} (\hat{c}_{1\sigma}^{\dagger} \hat{c}_{2\sigma} + \hat{c}_{2\sigma}^{\dagger} \hat{c}_{1\sigma}) + \hbar\omega \hat{b}^{\dagger} \hat{b} + \frac{g}{\sqrt{2}} (\hat{b}^{\dagger} + \hat{b}) (\hat{n}_1 - \hat{n}_2) \quad (10.118)$$

Note that $\hat{n}_1 + \hat{n}_2 = N$, the total number of electrons in the problem. As N is a constant of the motion, the dynamics of the electrons cannot affect the symmetric vibrations, and vice versa. Hence all of the interesting effects are contained in $\hat{H}_{be}$ and we need only study this Hamiltonian below.

10.6.1.1 Diabatic Limit, $\hbar\omega \gg t$ In the diabatic limit the vibrational modes are assumed to adapt themselves instantaneously to the particle's position. Thus,

$$\hbar\omega \hat{b}^{\dagger} \hat{b} + \frac{g}{\sqrt{2}} (\hat{b}^{\dagger} + \hat{b}) (\hat{n}_1 - \hat{n}_2) = \hbar\omega \hat{b}^{\dagger} \hat{b} \pm \frac{g}{\sqrt{2}} (\hat{b}^{\dagger} + \hat{b}) \quad (10.119)$$

The plus sign is relevant when the electron is located on site 1 and the minus sign is relevant when the electron is on site 2. We now introduce the *displaced oscillator transformation*,

$$\hat{b}_{\pm}^{\dagger} = \hat{b}^{\dagger} \pm \frac{1}{\sqrt{2}} \frac{g}{\hbar\omega} \quad (10.120)$$

Therefore, we find that

$$\hat{H}_{be} = -t \sum_{\sigma} (\hat{c}_{1\sigma}^{\dagger} \hat{c}_{2\sigma} + \hat{c}_{2\sigma}^{\dagger} \hat{c}_{1\sigma}) + \hbar\omega (\hat{b}_+^{\dagger} \hat{b}_+ + \hat{b}_-^{\dagger} \hat{b}_-) - \frac{g^2}{\hbar^2 \omega^2} \quad (10.121)$$

It is important to note that the operators $\hat{b}_+$ and $\hat{b}_-$ satisfy the same commutation relations as the $\hat{b}$ operator; therefore, they describe bosonic excitations. We define the ground states of the displaced oscillators by $\hat{b}_-|0_-\rangle = 0$ and $\hat{b}_+|0_+\rangle = 0$. Therefore,

$$\hat{b}|0_+\rangle = -\frac{1}{\sqrt{2}} \frac{g}{\hbar\omega} |0_+\rangle \quad (10.122)$$

and hence

$$\hat{b}_-|0_+\rangle = -\frac{\sqrt{2}g}{\hbar\omega}|0_+\rangle \quad (10.123)$$

Similarly,

$$\hat{b}_+|0_-\rangle = \frac{\sqrt{2}g}{\hbar\omega}|0_-\rangle \quad (10.124)$$

that is, $|0_\pm\rangle$ is an eigenstate of $\hat{b}_\mp$ with eigenvalue $\mp\sqrt{2}g/\hbar\omega$. The eigenstates of bosonic annihilation operators are known as *coherent states*.⁶⁶ Equations (10.122) to (10.124) therefore show that the ground state of one of the $\hat{b}_\pm$ operators may be written as a coherent state of the other operator⁶⁷:

$$|0_\pm\rangle = \exp\left[-\frac{\sqrt{2}g}{\hbar\omega}\left(\frac{1}{2} \pm \hat{b}_\mp^\dagger\right)\right]|0_\mp\rangle \quad (10.125)$$

Therefore,

$$\langle 0_+|0_-\rangle = \exp\left(-\frac{g^2}{\hbar^2\omega^2}\right) \quad (10.126)$$

which is known as the *Franck–Condon factor*.

The Franck–Condon factor describes the fact that in the diabatic limit, the bosons cause a “drag” on the electronic hopping. That is, we can describe the solution of the diabatic limit in terms of an effective two-site tight-binding model if we replace t by

$$t^* = t\langle 0_+|0_-\rangle = t \exp\left(-\frac{g^2}{\hbar^2\omega^2}\right) \quad (10.127)$$

Thus, the hopping integral is renormalized by the interactions of the electron with the vibrational modes (cf. Section 10.7). This renormalization is also found in the solution for an electron moving on a lattice in the diabatic limit. In this context the exponential factor is known as *polaronic band narrowing*.⁶⁰ The exponential factor results from the small overlap of the two displaced operators and may be thought of as an increase in the effective mass of the electron.

10.6.1.2 Adiabatic Limit, $\hbar\omega \ll t$ We begin by noting that as there is only one electron, the spin of the electron only leads to a trivial twofold degeneracy and therefore can be neglected without loss of generality. A useful notational change is to introduce a pseudospin notation where we define $\hat{\sigma}_z = \hat{c}_{1\sigma}^\dagger \hat{c}_{1\sigma} - \hat{c}_{2\sigma}^\dagger \hat{c}_{2\sigma}$

and $\hat{\sigma}_x = \hat{c}_{1\sigma}^\dagger \hat{c}_{2\sigma} + \hat{c}_{2\sigma}^\dagger \hat{c}_{1\sigma}$. Therefore, the one-electron two-site Holstein model Hamiltonian becomes

$$\hat{H}_{sb} = -t\hat{\sigma}_x + \hbar\omega\hat{b}^\dagger\hat{b} + \frac{g}{\sqrt{2}}(\hat{b}^\dagger + \hat{b})\hat{\sigma}_z \quad (10.128)$$

which is often referred to as the *spin-boson model*.

Let us now replace the bosonic operators by position and momentum operators for the harmonic oscillator defined as

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{b}^\dagger + \hat{b}) \quad (10.129)$$

and

$$\hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}(\hat{b}^\dagger - \hat{b}) \quad (10.130)$$

Therefore,

$$\hat{H}_{sb} = -t\hat{\sigma}_x + \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega\hat{x}^2 + g\sqrt{\frac{m\omega}{\hbar}}\hat{x}\hat{\sigma}_z \quad (10.131)$$

The adiabatic limit is characterized by a sluggish bosonic bath that responds only very slowly to the motion of the electron (i.e., $\hat{p}^2/2m \rightarrow 0$), which it is often helpful to think of as the $m \rightarrow \infty$ limit. Further, in the adiabatic limit the Born–Oppenheimer approximation^{2,67} holds, which implies that the total wavefunction of the system, $|\Psi\rangle$, is a product of a electronic (pseudospin) wavefunction, $|\phi_e\rangle$, and a vibrational (bosonic) wavefunction, $|\psi_v\rangle$ (i.e., $|\Psi\rangle = |\phi_e\rangle \otimes |\psi_v\rangle$). Therefore, the harmonic oscillator will be in a position eigenstate and we may replace the position operator, $\hat{x}$, by a classical position x , yielding

$$\hat{H}_{sb} = -t\hat{\sigma}_x + g\sqrt{\frac{m\omega}{\hbar}}x\hat{\sigma}_z + \frac{1}{2}m\omega x^2 \quad (10.132)$$

$$= \begin{pmatrix} g\sqrt{\frac{m\omega}{\hbar}}x & -t \\ -t & -g\sqrt{\frac{m\omega}{\hbar}}x \end{pmatrix} + \frac{1}{2}m\omega x^2 \quad (10.133)$$

where in the second line we have simply switched to the matrix representation of the Pauli matrices. This is easily solved and one finds that the eigenvalues are

$$E_{\pm} = \frac{1}{2}m\omega x^2 \pm \sqrt{t^2 + \frac{m\omega}{\hbar}g^2x^2} \quad (10.134)$$

$$\approx \frac{1}{2}m\omega x^2 \pm \frac{m\omega g^2x^2}{2\hbar t} \pm t \quad (10.135)$$

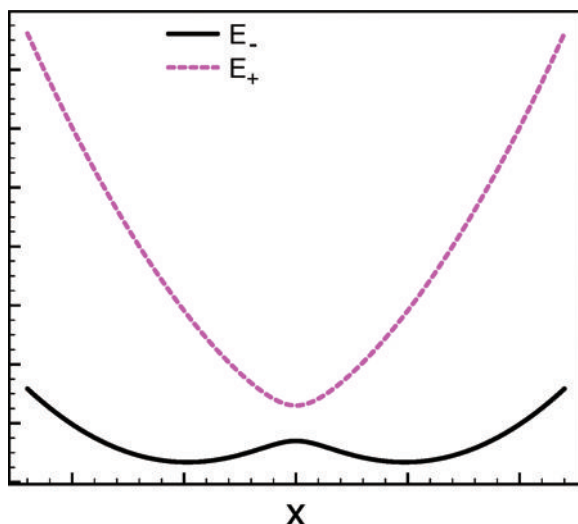


Fig. 10.16 (color online) Energies of the ground and excited states for a single electron in the two-site Holstein model in the adiabatic weak coupling limit ($t \gg g \gg \hbar\omega$), calculated from Eq. (10.134). x is the position of the harmonic oscillator describing out-of-phase vibrations.

where Eq. (10.135) holds in the weak-coupling limit, $gx \ll t$. We plot the variation of these eigenvalues with x in this limit in Fig. 10.16. Notice that for the electronic ground state, E_- , the lowest-energy states have $x \neq 0$. This is an example of spontaneous symmetry breaking,⁶⁸ as the ground state of a system has a lower symmetry than the Hamiltonian of the system. Thus, the system must “choose” either the left well or the right well (but not both) in order to minimize its energy.

10.7 EFFECTIVE HAMILTONIAN OR SEMIEMPIRICAL MODEL?

The models discussed in this chapter are generally known as either empirical or semiempirical models in a chemical context and as effective Hamiltonians in the physics community. Here the difference is not just nomenclature but is also indicative of an important difference in the epistemological status awarded to these models by the two communities. In this section I describe two different attitudes toward semiempirical models and effective Hamiltonians and discuss the epistemological views embodied in the work of two of the greatest physicists of the twentieth century.

10.7.1 Diracian Worldview

Paul Dirac famously wrote⁶⁹ that “the fundamental laws necessary for the mathematical treatment of a large part of physics and the whole of chemistry are thus

completely known, and the difficulty lies only in the fact that application of these laws leads to equations that are too complex to be solved.” There is clearly a great deal of truth in the statement. In solid-state physics and chemistry we know that the Schrödinger equation provides an extraordinarily accurate description of the phenomena observed. Gravity, the weak and strong nuclear forces, and relativistic corrections are typically unimportant; thus, all of the interactions boil down to nonrelativistic electromagnetic effects.

Dirac’s world view is realized in the *ab initio* approach to electronic structure, wherein one starts from the Hartree–Fock solution to the full Schrödinger equation in some small basis set. One then adds in correlations via increasingly complex approximation schemes and increases the size of the basis set, in the hope that with a sufficiently large computer one will find an answer that is “sufficiently close” to the exact solution (full CI in an infinite complete basis set).

In the last few decades rapid progress has been made in *ab initio* methods due to an exponential improvement in computing technology, methodological progress, and the widespread availability of implementations of these methods.⁷⁰ However, this progress is unsustainable: The complexity recognized by Dirac eventually limits the accuracy possible from *ab initio* calculations. Indeed, solving the Hamiltonian given in Eq. (10.24) is known to be computationally difficult. Feynman proposed building a computer that uses the full power of quantum mechanics to carry out quantum simulations.⁷¹ Indeed, the simplest of all quantum chemical problems, the H_2 molecule in a minimal basis set, has been solved on a prototype quantum computer.⁷² But while even a rather small scale quantum computer (containing just a few hundred qubits⁷²) would provide a speed-up over classical computation, it is believed that the solution of Hamiltonian (10.24) remains difficult even on a quantum computer [i.e., it is believed that even a quantum computer could not solve Hamiltonian equation (10.24) in a time that grows only polynomially with the size of the system⁷³]. Further, simple extensions of these arguments provide strong reasons to believe that there is no efficiently computable approximation to the exact functional in density functional theory.⁷³ Therefore, it appears that the equations will always remain “too complex to be solved” directly. This suggests that semiempirical models will always be required for large systems.

10.7.2 Wilsonian Project

Typically, one is only interested in a few low-energy states of a system, perhaps the ground state and the first few excited states. Therefore, as long as our model gives the correct energies for these low-energy states, we should regard it as successful. This apparently simple realization, particularly as embodied by Wilson’s renormalization group,⁷⁴ has had profound implications throughout modern physics from high-energy particle physics to condensed matter physics.

The basic idea of renormalization is remarkably simple. Imagine starting with some system that has a large number of degrees of freedom. As we have noted,

for practical purposes we care only about the lowest-energy states. Therefore, one might be tempted to simplify the description of the system by discarding the highest-energy states. However, simply discarding such states will cause a shift in the low-energy spectrum. Therefore, one must remove the high-energy states that complicate the description and render the problem computationally intractable in such a way as to preserve the low-energy spectrum. This is often referred to as “integrating out” the high-energy degrees of freedom (because of the way this process is carried out in the path-integral formulation of quantum mechanics⁷⁵). Typically, integrating out the high-energy degrees of freedom causes the parameters of the Hamiltonian to “flow” or “run” (i.e., change their values). When this happens, one says that the parameters are renormalized.

A simple example is the Coulomb interaction between the two electrons in a neutral helium atom. For simplicity, let’s imagine trying to calculate just the ground-state energy. We begin by analyzing the problem in the absence of a Coulomb interaction between the two electrons. In the ground state both electrons occupy the $1s$ orbital. We would like to work in as small a basis set as possible. The simplest approach is just to work in the minimal basis set, which in this case is just the two $1s$ spin-orbitals, $\phi_{1s\sigma}(r)$. The total energy of a He atom neglecting the interelectron Coulomb interaction is -108.8 eV (relative to the completely ionized state). Now we restore the Coulomb repulsion between electrons. A simple question is: How much does this change the total energy of the He atom? In the minimal basis set the solution seems straightforward:

$$\begin{aligned}\langle 1s^2 | V | 1s^2 \rangle &= \int_{-\infty}^{\infty} d^3r_1 \int_{-\infty}^{\infty} d^3r_2 \frac{e^2 |\phi_{1s\uparrow}|^2 |\phi_{1s\downarrow}|^2}{4\pi\epsilon_0 |r_1 - r_2|} \\ &\simeq 34.0 \text{ eV}\end{aligned}\tag{10.136}$$

Therefore, it is tempting to conclude that we can model the He atom by a one-site Hubbard model with $U = \langle 1s^2 | V | 1s^2 \rangle$. However, this yields a total energy for the He atom of -74.8 eV, which is not particularly close to the experimental value of -78.975 eV.⁷

Let us then continue to consider the problem in the basis set of the hydrogenic atom, which is complete due to the spherical symmetry of the Hamiltonian. One can now straightforwardly carry out a perturbation theory around the noninteracting electron solution, where we take

$$H_0 = \sum_{i=1}^2 \left(-\frac{\hbar^2 \nabla_i^2}{2m} - \frac{e^2}{\pi\epsilon_0 |r_i|} \right)\tag{10.137}$$

and

$$H_1 = \frac{e^2}{4\pi\epsilon_0 |r_1 - r_2|}\tag{10.138}$$

A detailed description of this perturbation theory is given in Chapter 18 of Gasiorowicz.⁷ However, for our discussion, the key point is that in this perturbation theory, the term $\langle 1s^2 | V | 1s^2 \rangle$ is simply the first-order correction to the ground-state energy. It is therefore clear why the minimal basis set gives such a poor result: It ignores all the higher-order corrections to the total energy.

The failure of the simple minimal basis set calculation does not, however, mean that the effective Hamiltonian approach also fails, despite the fact that the effective Hamiltonian is also in an extremely small basis set. Rather, one must realize that as well as the first-order contributions, U also contains contributions from higher orders in perturbation theory. It is therefore possible, although extremely computationally demanding, to calculate the parameters for effective Hamiltonians from this type of perturbation theory.⁷⁶

A more promising approach, which has been applied to a number of molecular crystals,^{77,78} is to use atomistic calculations to parameterize an effective Hamiltonian. For example, density functional theory gives quite reasonable values for the total energy of the ground state of many molecules. Therefore, one approach to calculating the Hubbard U is to calculate the ionization energy, $I = E_0(N - 1) - E_0(N)$, and the electron affinity, $A = E_0(N) - E_0(N + 1)$, of the molecule, where $E_0(n)$ is the ground-state energy of the molecule when it contains n electrons and N is the filling corresponding to a half-filled band. One finds that $U = I - A = E_0(N + 1) + E_0(N - 1) - 2E_0(N)$. A simple way to see this is that if we assume the molecule is neutral when it contains N electrons, then U corresponds to the energy difference in the charge disproportionation reaction $2M \rightleftharpoons M^+ + M^-$ for two well-separated molecules, M . A more extensive discussion of this approach is given by Scriven et al.⁷⁷

It is worth noting that we have actually carried out this program of parameterizing effective Hamiltonians three times in the discussion above. In Section 10.4.3 we showed that the Heisenberg model is an effective low-energy model for the half-filled Hubbard model in the limit $t/U \rightarrow 0$. In Section 10.5.3 we derived an effective tight-binding model that involved only the metal sites from an ionic Hubbard model of a transition metal oxide. Finally, in Section 10.6.1.1 we showed that vibronic interactions lead to an effective tight binding model describing the low-energy physics of the Holstein model in the diabatic limit, and that in this model the quasiparticles (electron-like excitations) are polarons, a bound state of electrons and vibrational excitations with a mass enhanced over that of the bare electron.

However, to date, the most important method for parameterizing effective Hamiltonians has been to fit the parameters to a range of experimental data—whence the name *semiempirical*. Of course, experimental data contain all corrections to all orders; therefore, this is indeed an extremely sensible thing to do. But it is important to understand that empiricism is not a dirty word. Indeed, empiricism is what distinguishes science from other belief systems. Further, this empirical approach is exactly the approach that the mathematics tells one to take. It is also important to know that no quantum chemical or solid-state calculation is truly *ab initio*—the nuclear and electronic masses and the charge

on the electron are all measured rather than calculated. Indeed, the modern view of the “standard model” of particle physics is that it, too, is an effective low-energy model.⁴⁹ For example, in quantum electrodynamics (QED), the quantum field theory of light and matter, the bare charge on the electron is, for all practical purposes, infinite. But the charge is renormalized to the value seen experimentally in a manner analogous to the renormalization of the Hubbard U of He discussed above. Therefore, as we do not at the time of writing know the correct mathematical description of processes at higher energies, all of theoretical science should, perhaps, be viewed as the study of semiempirical effective low-energy Hamiltonians.⁷⁹

Finally, the most important point about effective Hamiltonians is that they promote understanding. Ultimately, the point of science is to understand the phenomena we observe in the world around us. Although the ability to perform accurate numerical calculations is important, we should not allow this to become our main goal. The models discussed above provide important insights into the chemical bond, magnetism, polarons, the Mott transition, electronic correlations, the failure of mean-field theories, and so on. All of these effects are much more difficult to understand simply on the basis of atomistic calculations. Further, many important effects seen in crystals, such as the Mott insulator phase, are not found methods such as density functional theory or Hartree–Fock theory, while post-Hartree–Fock methods are not practical in infinite systems. Thus effective Hamiltonians have a vital role to play in developing the new concepts that are required if we are to understand the emergent phenomena found in molecules and solids.⁸⁰

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REFERENCES

1. Fulde, P. *Electron Correlations in Molecules and Solids*, Springer-Verlag, Berlin, 1995.
2. Schatz, G. C.; Ratner, M. A. *Quantum Mechanics in Chemistry*, Prentice Hall, Englewood Cliffs, NJ, 1993.

3. Mahan, G. D.; *Many-Particle Physics*, Kluwer Academic, New York, 2000.
4. Goldstein, H.; Poole, C.; Safko, J. *Classical Mechanics*, Addison-Wesley, Reading, MA, 2002.
5. Atkins, P.; de Paula, J. *Atkins' Physical Chemistry*, Oxford University Press, Oxford, UK, 2006.
6. See, e.g., Rae, A. I. M. *Quantum Mechanics*, Institute of Physics Publishing, Bristol, UK, 1996.
7. See, e.g., Gasiorowicz, S. *Quantum Physics*, Wiley, Hoboken, NJ, 2003.
8. Jordan, P.; Wigner, E. *Z. Phys.* **1928**, *47*, 631–651.
9. Lowe, J. P.; Peterson, K. A. *Quantum Chemistry*, Elsevier, Amsterdam, 2006.
10. Ashcroft, N. W.; Mermin, N. D. *Solid State Physics*, Holt, Rinehart and Winston, New York, 1976.
11. Tinkham, M. *Group Theory and Quantum Mechanics*, McGraw-Hill, New York, 1964.
12. Lax, M. *Symmetry Principles in Solid State and Molecular Physics*, Wiley, New York, 1974.
13. McWeeny, R. *Coulson's Valence*, Oxford University Press, Oxford, UK, 1979.
14. Brogli, F.; Heilbronner, E. *Theor. Chim. Acta* **1972**, *26*, 289–299.
15. See, e.g., Arfken, G. *Mathematical Methods for Physicists*, 3rd ed., Academic Press, Orlando, FL, 1985.
16. Mandl, F. *Statistical Physics*, Wiley, Chichester, UK, 1998.
17. See pp. 799–800 in Ref. 15.
18. (a) Castro Neto, A. H.; Guinea, F.; Peres, N. M. R.; Novoselov, K. S.; Geim, A. K. *Rev. Mod. Phys.* **2009**, *81*, 109–162. (b) Castro Neto, A. H.; Guinea, F.; Peres, N. M. R. *Phys. World* **2006**, *19*, 33–37.
19. (a) Novoselov, K. S.; Geim, A. K.; Morozov, S. V.; Jiang, D.; Zhang, Y.; Dubonos, S. V.; Gregorieva, I. V.; Firsov, A. A. *Science* **2004**, *306*, 666–669. (b) Choucair, M.; Thordarson, P.; Stride, J. A. *Nature Nanotechnol.* **2009**, *4*, 30–33.
20. Schrödinger, E. *Ann. Phys.* **1926**, *79*, 361–428.
21. Heitler, W.; London, F. *Z. Phys.* **1927**, *44*, 455–472.
22. Pauling, L. *The Nature of the Chemical Bond and the Structure of Molecules and Crystals*, Cornell University Press, Ithaca, NY, 1960.
23. Mott, N. F. *Proc. R. Soc. A* **1949**, *62*, 416–422.
24. Powell, B. J.; McKenzie, R. H. *J. Phys. Condens. Matter* **2006**, *18*, R827–R865.
25. Cohen, A. J.; Mori-Sanchez, P.; Yang, W. T. *Science* **2008**, *321*, 792–794.
26. (a) Anderson, P. W. *Science* **1987**, *235*, 1196–1198. (b) Zhang, F. C.; Gross, C.; Rice, T. M.; Shiba, H. *Supercond. Sci. Technol.* **1988**, *1*, 36–46.
27. Anderson, P. W. *Phys. Today* **2008**, *61* (4), 8–9.
28. Powell, B. J.; McKenzie, R. H. *Phys. Rev. Lett.* **2005**, *94*, 047004; Gan, J. Y.; Chen, Y.; Su, Z. B.; Zhang, F. C. *Phys. Rev. Lett.* **2005**, *94*, 067005; Liu, J.; Schmalian, J.; Trivedi, N. *Phys. Rev. Lett.* **2005**, *94*, 127003.
29. Rössler, U. *Solid State Theory*, Springer-Verlag, Berlin, 2004.
30. Mohn, P.; Wohlfarth, E. P. *J. Magn. Magn. Mater.* **1987**, *68*, L283–L285.
31. Jacko, A. C.; Fjærestad, J. O.; Powell, B. J. *Nature Phys.* **2009**, *5*, 422–425.

32. Gutzwiller, M. C. *Phys. Rev. Lett.* **1963**, *10*, 159–162.
33. Brinkmann, W. F.; Rice, T. M. *Phys. Rev. B* **1970**, *2*, 4302–4304.
34. Lieb, E. H.; Wu, F. Y. *Phys. Rev. Lett.* **1968**, *20*, 1445–1448.
35. Essler, F. H. L.; Frahm, H.; Göhmann, F.; Klümper, A.; Korepin, V. E. *The One-Dimensional Hubbard Model*, Cambridge University Press, Cambridge, UK, 2005.
36. Tsvelik, A. M. *Quantum Field Theory in Condensed Matter Physics*, Cambridge University Press, Cambridge, UK, 1996.
37. Kotliar, G.; Vollhardt, D. *Phys. Today* **2004**, *57*(3), 53–59.
38. Kollar, M.; Strack, R.; Vollhardt, D. *Phys. Rev. B* **1996**, *53*, 9225–9231.
39. Maier, T.; Jarrell, M.; Pruschke, T.; Hettler, M. H. *Rev. Mod. Phys.* **2005**, *77*, 1027–1080.
40. Kotliar, G.; Savrasov, S. Y.; Haule, K.; Oudovenko, V. S.; Parcollet, O.; Marianetti, C. A. *Rev. Mod. Phys.* **2006**, *78*, 865–951.
41. Nagaoka, Y. *Phys. Rev.* **1966**, *145*, 392–405.
42. Tian, G. *J. Phys. A* **1990**, *23*, 2231–2236.
43. Merino, J.; Powell, B. J.; McKenzie, R. H. *Phys. Rev. B* **2006**, *73*, 235107.
44. Shaik, S.; Hiberty, P. C. Valence bond theory: its history, fundamentals, and applications—a primer. In *Reviews in Computational Chemistry*, Lipkowitz, K. B., Larter, R., and Cundari, T. R., Eds., Wiley-VCH, Hoboken, NJ, 2004, pp. 1–100.
45. Sakurai, J. J. *Modern Quantum Mechanics*, Addison-Wesley, Reading, MA, 1994.
46. Chao, K. A.; Spałek, J.; Oleś, A. M. *J. Phys. C* **1977**, *10*, L271–L276.
47. Brockhouse, B. N. Slow neutron spectroscopy and the grand atlas of the physical world. In *Nobel Lectures in Physics, 1991–1995*, Ekspång, G., Ed.; World Scientific, Singapore, 1997. Also available at http://nobelprize.org/nobel_prizes/physics/laureates/1994/brockhouse-lecture.html.
48. Zaliznyak, I. A. *Nature Mater.* **2005**, *4*, 273–275.
49. Griffiths, D. *Introduction to Elementary Particles*, Wiley-VCH, Weinheim, Germany, 2008.
50. (a) Coldea, R.; Tennant, D. A.; Tylczynski, Z. *Phys. Rev. B* **2003**, *68*, 134424. (b) Lake, B.; Tennant, D. A.; Frost, C. D.; Nagler, S. E. *Nature Mater.* **2005**, *4*, 329–334.
51. Lee, P. A. *Science* **2008**, *321*, 1306–1307.
52. Shimizu, Y.; et al. *Phys. Rev. Lett.* **2003**, *91*, 107001.
53. Helton, J.; et al. *Phys. Rev. Lett.* **2007**, *98*, 107204.
54. Okamoto, Y.; et al. *Phys. Rev. Lett.* **2007**, *99*, 137207.
55. Raczkowski, M.; Frésard, R.; Oleś, A. M. *J. Phys. Condens. Matter* **2006**, *18*, 7449–7469.
56. Sarma, D. D. *J. Solid State Chem.* **1990**, *88*, 45–52.
57. (a) Merino, J.; Powell, B. J.; McKenzie, R. H. *Phys. Rev. B* **2009**, *79*, 161103(R). (b) Merino, J.; McKenzie, R. H.; Powell, B. J. *Phys. Rev. B* **2009**, *80*, 045116. (c) Powell, B. J.; Merino, J.; McKenzie, R. H. *Phys. Rev. B* **2009**, *80*, 085113.
58. See, e.g., Ziman, J. M. *Electrons and Phonons*, Oxford University Press, Oxford, UK, 1960.
59. For a review, see Grüner, G. *Density Waves in Solids*, Perseus Publishing, Cambridge, UK, 1994.

60. See, e.g., Alexandrov, A. S.; Mott, N. F. *Polarons and Bipolarons*, World Scientific, Singapore, 1995.
61. For a review, see Marcus, R. A. *Rev. Mod. Phys.* **1993**, *65*, 599–610.
62. See, e.g., Bersuker, I. B. *The Jahn–Teller Effect and Vibronic Interactions in Modern Chemistry*, Plenum Press, New York, 1984.
63. (a) Olsen, S.; Riesz, J.; Mahadevan, I.; Coutts, A.; Bothma, J. P.; Powell, B. J.; McKenzie, R. H.; Smith, S. C.; Meredith, P. *J. Am. Chem. Soc.* **2007**, *129*, 6672–6673. (b) Meredith, P.; Powell, B. J.; Riesz, J.; Nighswander-Rempel, S.; Pederson, M. R.; Moore, E. *Soft Matter* **2006**, *2*, 37–44.
64. Reimers, J. R.; Hush, N. S. *J. Am. Chem. Soc.* **2004**, *126*, 4132–4144.
65. Hahn, S.; Stock, G. *J. Phys. Chem. B* **2000**, *104*, 1146–1149.
66. Walls, D. F.; Milburn, G. J. *Quantum Optics*, Springer-Verlag, Berlin, 2006.
67. Weiss, U. *Quantum Dissipative Systems*, World Scientific, Singapore, 2008.
68. For an introductory discussion of broken symmetry, see, e.g., Blundell, S. J. *Magnetism in Condensed Matter*, Oxford University Press, Oxford, UK, 2001. For a more advanced discussion, see, e.g., Anderson, P. W. *Basic Notions of Condensed Matter Physics*, Benjamin-Cummings, Menlo Park, CA, 1984.
69. Dirac, P. *Proc. R. Soc. A* **1929**, *123*, 714–733.
70. (a) Pople, J. A. *Rev. Mod. Phys.* **1999**, *71*, 1267–1274. (b) Truhlar, D. G. *J. Am. Chem. Soc.* **2008**, *130*, 16824–16827.
71. Feynman, R. P. *Int. J. Theor. Phys.* **1982**, *21*, 467–488.
72. Lanyon, B. P.; Whitfield, J. D.; Gillet, G. G.; Goggin, M. E.; Almeida, M. P.; Kassal, I.; Biamonte, J. D.; Mohseni, M.; Powell, B. J.; Barbieri, M.; Aspuru-Guzik, A.; White, A. G. *Nature Chem.* **2010**, *2*, 106–111.
73. Schuch, N.; Verstraete, F. *Nature Phys.* **2009**, *5*, 732–735.
74. Goldenfeld, N. D. *Lectures on Phase Transitions and the Renormalisation Group*, Addison-Wesley, Reading, MA, 1992.
75. See, e.g., Wen, X.-G. *Quantum Field Theory of Many-Body Systems*, Oxford University Press, Oxford, UK, 2004.
76. (a) Freed, K. F. *Acc. Chem. Res.* **1983**, *16*, 137–144. (b) Gunnarsson, O. *Phys. Rev. B* **1990**, *41*, 514–518. (c) Iwata, S.; Freed, K. F. *J. Chem. Phys.* **1976**, *65*, 1071–1088. (d) Graham, R. L.; Freed, K. F. *J. Chem. Phys.* **1992**, *96*, 1304–1316. (e) Martin, C. M.; Freed, K. F. *J. Chem. Phys.* **1994**, *100*, 7454–7470. (f) Stevens, J. E.; Freed, K. F.; Arendt, F.; Graham, R. L. *J. Chem. Phys.* **1994**, *101*, 4832–4841. (g) Finley, J. P.; Freed, K. F. *J. Chem. Phys.* **1995**, *102*, 1306–1333. (h) Stevens, J. E.; Chaudhuri, R. K.; Freed, K. F. *J. Chem. Phys.* **1996**, *105*, 8754–8768. (i) Chaudhuri, R. K.; Freed, K. F. *J. Chem. Phys.* **2003**, *119*, 5995–6002. (j) Chaudhuri, R. K.; Freed, K. F. *J. Chem. Phys.* **2005**, *122*, 204111.
77. (a) Scriven, E.; Powell, B. J. *J. Chem. Phys.* **2009**, *130*, 104508. (b) *Phys. Rev. B* **2009**, *80*, 205107.
78. (a) Martin, R. L.; Ritchie, J. P. *Phys. Rev. B* **1993**, *48*, 4845–4849. (b) Antropov, V. P.; Gunnarsson, O.; Jepsen, O. *Phys. Rev. B* **1992**, *46*, 13647–13650. (c) Pederson, M. R.; Quong, A. A. *Phys. Rev. B* **1992**, *46*, 13584–13591. (d) Brocks, G.; van den Brink, J.; Morpurgo, A. F. *Phys. Rev. Lett.* **2004**, *93*, 146405. (e) Cano-Cortés, L.; Dolfen, A.; Merino, J.; Behler, J.; Delley, B.; Reuter, K.; Koch, E. *Eur. Phys. J. B* **2007**, *56*, 173–176.

79. For an accessible and highly outspoken discussion of these ideas, see Laughlin, R. B.; Pines, D. *Proc. Natl. Acad. Sci. USA* **2000**, 97, 28–31; Laughlin, R. B. *A Different Universe*, Basic Books, New York, 2005.
80. Anderson, P. W. *Science* **1972**, 177, 393–396.
81. Powell, B. J. *Chem. Aust.* **2009**, 76, 18–21.

PART D

Advanced Applications

11 SIESTA: Properties and Applications

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SIESTA provides access to the usual set of properties common to most DFT implementations:

- Total energy, charge densities, and potentials
- Atomic forces and unit cell stresses
- Geometry specification in Cartesian and/or internal z -matrix coordinates
- Geometry optimization using the conjugate gradient, modified Broyden and Fire algorithms, and simulated annealing
- Total and partial densities of states
- Band dispersions
- Constant energy, temperature, or pressure molecular dynamics
- Simulation of scanning tunneling microscope images according to the Tersoff–Hamann approximation
- Electron transport properties using the nonequilibrium Green's function approach
- Optical properties and the frequency-dependent dielectric function within the random phase approximation and using first-order time-dependent perturbation theory
- Phonon spectrum and vibrational frequencies
- Mulliken population analysis
- Born charges

In this chapter a number of these properties are discussed through examples relevant to nanoscience and technology. The SIESTA methodology is described

in detail in Chapter 2; the present chapter is intended as an accompaniment. The first three examples illustrate the general capabilities of the SIESTA code for problems containing relatively small numbers of atoms and that are amenable to standard diagonalization to solve the self-consistent problem. The last example illustrates the divide-and-conquer linear-scaling capabilities to tackle problems containing large numbers of atoms.

11.1 ETHYNYLBENZENE ADSORPTION ON AU(111)

There has been considerable interest for some time in self-assembled monolayers (SAMs) in nanotechnology. They are relatively easy to prepare on a variety of surfaces, gold being the most common, with a wide range of molecules forming ordered molecular layers.^{1–3} They are a useful platform for controlling surface properties and providing functionality with applications in, for example, molecular electronics.^{4,5}

The alkynyl group as method of anchoring SAMs to gold surfaces is a promising candidate to study. It should provide an unbroken conjugated pathway to the gold surface, unlike thiol linkers, and a wide range of terminal alkynes can be synthesized.⁶ Ethynylbenzene is a simple representative example of this class of molecule; there is some experimental evidence that it binds to gold surfaces and nanoparticles, although these studies are inconclusive about the nature of the bond.^{7,8} The calculations described below attempt to answer the question of whether this molecule is likely to form SAMs and the likely adsorption geometries and energetics.^{9,10}

The computational conditions first have to be established and an appropriate representation of the semi-infinite surface in terms of a multilayer slab needs to be determined. The slab needs to contain sufficient layers that the center of the slab is relatively bulklike, or in this particular case so that a molecule adsorbed on one side of the slab is not influenced by the other surface. Conversely, the slab should not be too big, such that the calculations are prohibitively large. Figure 11.1 shows the convergence of surface charge density above the slab layer and convergence of the workfunction for an Au(111) slab as a function of the number of layers. Convergence of the workfunction with two computational parameters, reciprocal space grid (k -grid), and orbital confinement (energy shift) are also shown in Fig. 11.1A. The workfunction is calculated as the difference between the electrostatic potential in vacuum (i.e., at a position in the unit cell far above the surface) and the Fermi level.

The charge density and density difference are extracted from the density matrix (saved to file at each SCF step) using the DENCHAR utility at the points of a user-specified plane, or volume. Charge densities can then be visualized using standard plotting packages. Alternatively, the charge densities and potentials evaluated over the real space grid used to represent the density matrix can be written to file directly from SIESTA by setting the appropriate input flags. These are written unformatted and need to be processed for plotting. The GRID2CUBE utility

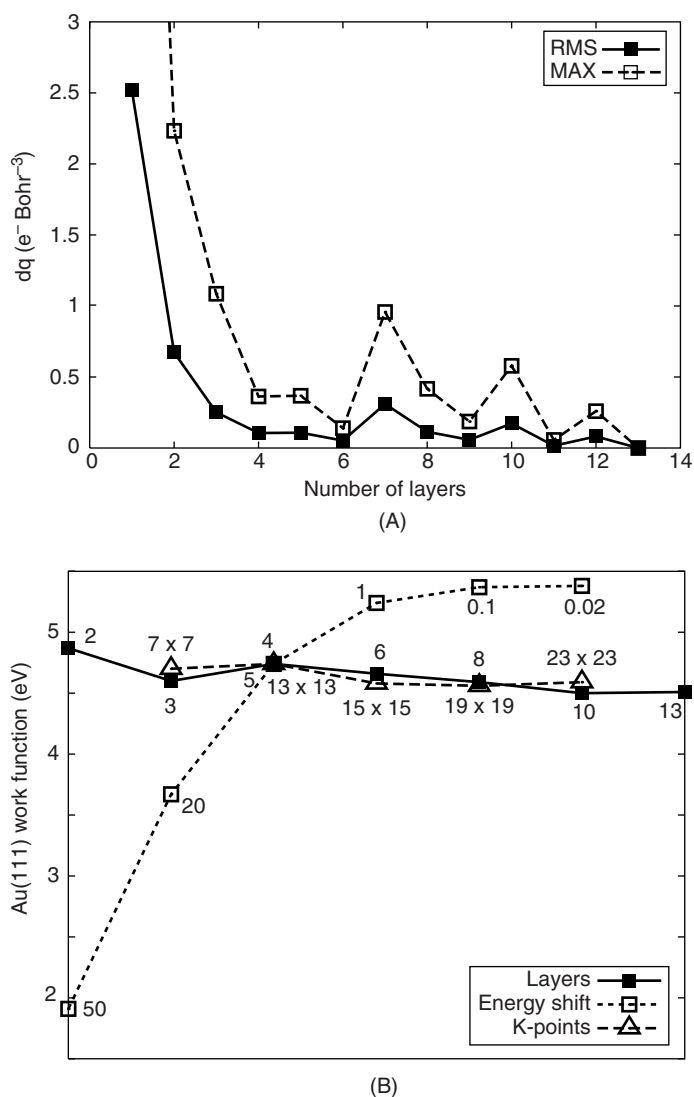


Fig. 11.1 (A) Charge density 1 Å above the Au(111) slab surface. Values are maximum and the RMS difference is with respect to a 13-layer slab. (B) Convergence of workfunction with number of slab layers, energy shift parameter (mRy), and k -point grid. [From Ref. 13 and R. C. Hoft, N. Armstrong, M. J. Ford, and M. B. Cortie, *J. Phys. Condens. Matter*, **19** 215206 (2007), with permission. Copyright © IOP Publishing.]

will generate formatted output from these files in the format of a GAUSSIAN cube file.

The calculations in Fig. 11.1 are for a 1×1 unit cell in the plane of the surface, that is, one atom per layer. The equivalent of a double-zeta plus polarization

(DZP) basis set is used. A generalized-gradient approximation to the exchange-correlation functional according to Perdew–Burke–Ernzerhof (GGA-PBE)¹¹ and a real-space integration grid with a 300-Ry cutoff are employed (1 Ry = 0.5 atomic unit of energy = ca.13.6 eV). It is often advisable to use a fine real-space grid to avoid numerical errors; the time penalty for such a grid is not generally a limiting factor. A cutoff of 300 Ry is well converged. A Troullier–Martins pseudopotential¹² with scalar relativistic corrections is used to represent the core Au electrons, with a valence of 5d¹⁰6s. Cutoff radii for each of the angular momentum channels of the pseudopotential are 2.32 Å for s and p, 1.48 Å for d and f. The quality of these pseudopotentials has been checked in the usual way by comparing against all electron calculations for the atom; they reproduce well the bulk properties of gold (lattice parameter, cohesive energy, and bulk modulus).¹³ It is interesting to note that values for the total and cohesive energies of bulk gold do not vary much between a single-zeta plus polarization (SZP) and a DZP basis set, while DZ is considerably worse. Where computational cost is a limiting factor, an SZP basis may be acceptable, although for adsorption energies DZP is probably necessary.

The Au(111) surface is unusual in that it reconstructs to form a $\sqrt{3} \times 22$ structure with a period of about 63 Å,¹⁴ although there is evidence that this reconstruction is lifted in the presence of adsorbed molecules.^{14,15} More recently, experimental measurements and calculations suggest that thiolate adsorption drives an alternative gold adatom structure and that these adatoms are an integral part of the adsorption motif.^{16–18} A detailed analysis of these points is beyond the scope of the present chapter, where we are more interested in demonstrating the utility of the SIESTA methodology. Accordingly, a bulk terminated Au(111) surface is assumed.

Temperature smearing of the electron occupation is employed in these calculations to assist convergence of the SCF steps. Both the standard Fermi–Dirac function and the function proposed by Methfessel and Paxton¹⁹ are implemented in SIESTA. In this case it is the free energy $F(T)$ that is minimized during self-consistency. The total energy in the athermal limit is then approximated by the expression

$$E_{\text{tot}}(T = 0) = \frac{1}{2}[E_{\text{tot}}(T) + F(T)] \quad (11.1)$$

The degree of smearing is determined by specifying a fictitious temperature to the electron distribution; in this case, a temperature corresponding to 25 meV is used.

Charge density close to the slab surface has converged by four layers and thereafter oscillates slightly. The charge density should be a reasonable indicator of how the adsorption properties will converge. The workfunction is less sensitive to the number of slab layers and the k -grid. Again four layers and a 15×15 k -grid are reasonably converged. Only one k -point is required perpendicular to the surface because there is no periodicity in this direction. The workfunction is very sensitive to the energy shift, with values as small as 0.1 mRy required for good

convergence. This level is impractical for realistic surface adsorption calculations, as it is extremely time intensive. It is worth noting that the converged value of the workfunction calculated here is 5.13 eV, compared with an experimental value of 5.31 eV.²⁰

The conclusion from the data in Fig. 11.1 is that a four-layer slab is the minimum for obtaining reasonably converged results. Calculations of ethynylbenzene adsorption support this conclusion, the binding energy is converged to within about 0.05 eV for four layers and is essentially fully converged at seven layers. Two additional factors need to be considered when assessing adsorption calculations: basis set superposition errors (BSSE) and dipole corrections. BSSE is inherent in the use of atom-centered basis sets. The binding energy, E_B , is determined from calculations of the total energies of slab + adsorbate, E_T , slab alone, E_S , and adsorbate alone, E_A , according to

$$E_B = E_T - (E_S + E_A) \quad (11.2)$$

The numbers of basis functions used to describe the two fragments, slab and adsorbate, are smaller than for the total system, leading to fewer variational degrees of freedom and hence overestimates of the total energies. Although this error is small for the total energies, it can amount to about 10% of the binding energy calculated from the difference of total energies according to Eq. (11.2). Here the established method of counterpoise correction is used to remove this effect.²¹ The same set of basis functions are used in the two fragment calculations, with zero charge assigned to those basis functions associated with the missing atoms, a procedure commonly referred to as *ghosting*. This is implemented in SIESTA by assigning the corresponding negative atomic number to ghosted atoms. The efficacy of counterpoise corrections has been debated in the literature and demonstrated to “correct” the binding energy in the wrong direction in certain circumstances²²; it is however, a well-established and widely used technique. Dipole corrections are an artifact of periodic boundary conditions and arise in situations where an asymmetric geometry is used.²³ Periodicity perpendicular to the slab surface imposes the condition that the potential must be identical at the cell boundary above and below the slab. However, if the slab is asymmetric, as is the case where adsorption occurs on only one slab surface, physically the potential is not identical and approaches different asymptotic values above and below. This leads to the presence of an additional unphysical potential that can distort optimized geometries and binding energies. One solution to this problem is to introduce a fictitious dipole charge layer in the vacuum portion of the unit cell parallel to the slab surface that can be included in the self-consistent field. This is not implemented in SIESTA. The problem can obviously be avoided by always using symmetric geometries, at the expense of requiring more atoms. In the present application this dipole layer is neglected, having little effect on optimized geometries and contributing less than 1% to binding energies. For more polar bonds between surface and adsorbate, one might expect the situation to be considerably worse.

Figure 11.2 shows the convergence of binding energy for ethynylbenzene on Au(111) against the number of k -points and energy shift. An energy shift of 5 mRy and 15 k -points gives well-converged values with binding energies reliable to better than 0.05 eV. The number of k -points corresponds to a 5×5 grid giving 15 symmetry unique points. SIESTA uses inversion symmetry in the reciprocal

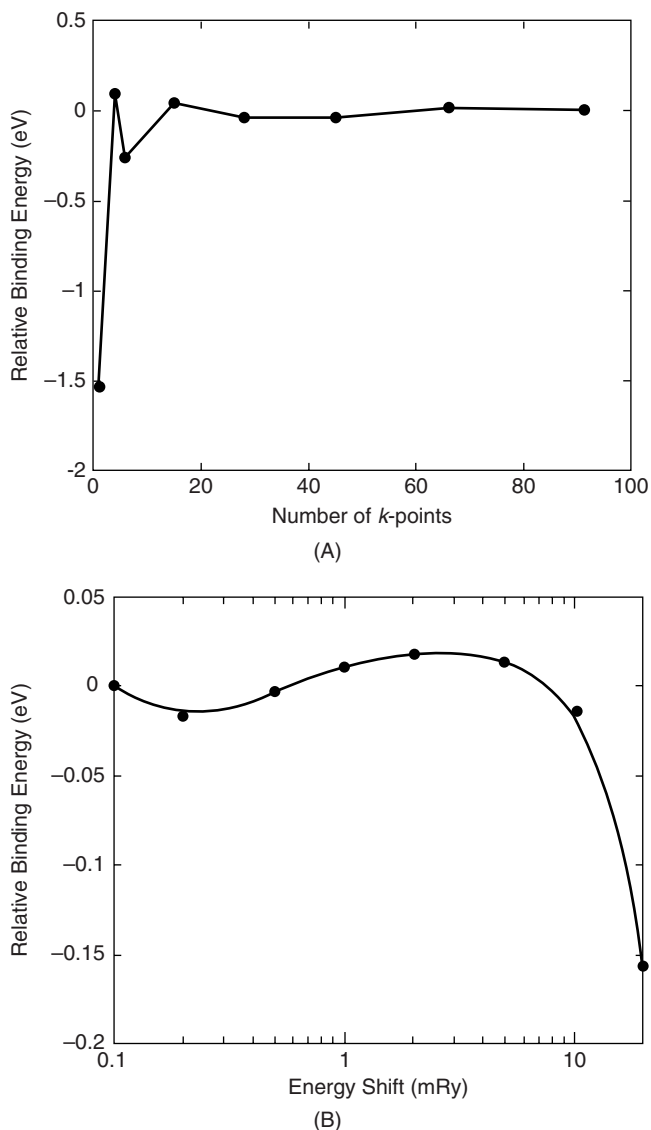


Fig. 11.2 Convergence of binding energy with (A) the number of k -points and (B) the energy shift. Binding energies are relative to value at the largest k -point grid and smallest energy shift.

cell to generate the k -grid. Fewer k -points (by a factor of 3) are needed here compared with the previous analysis because the unit cell is now a 3×3 supercell in order to accommodate the adsorbate and reduce interactions between periodic images. The use of strictly localized orbitals is an advantage in this regard because multipole interactions between periodic images of the molecule tend to zero quite rapidly with increasing unit cell size. The interaction here is essentially zero.

The likely adsorption motifs for ethynylbenzene on the gold surface are shown in Fig. 11.3. For the ethynylbenzene radical (Fig. 11.3A) the terminal C—H bond has been cleaved and the H atom removed. One might expect this to be the

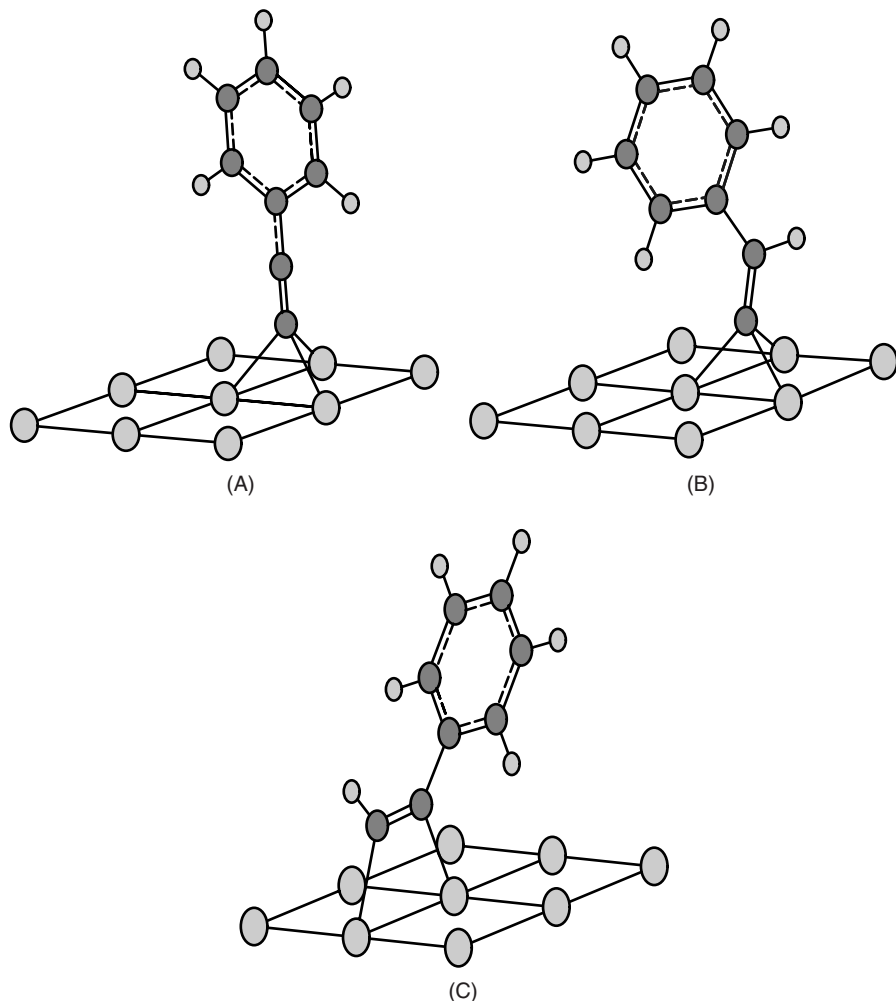


Fig. 11.3 Potential configurations of surface-bound ethynylbenzene molecule: (A) ethynylbenzene radical with terminal H atom removed; (B) vinylidene; (C) flat configuration. (From Ref. 10.)

most promising candidate for SAM formation. Two additional configurations are also possible, one where a 1,2 hydrogen shift has occurred to give vinylidene (Fig. 11.3B) and a second where the C—C triple bond opens up to give the flat configuration (Fig. 11.3C). The latter two configurations are potential intermediates to the final state of the strongly bound radical by removal of the hydrogen atom. Reactions of metals with ethynylbenzene are known to proceed via a 1,2 hydrogen shift to form metal vinylidenes.²⁴

The likely absorption sites are first identified by scanning the adsorbate across the surface with the adsorbate geometry held rigid. This involves a large number of single-point energy calculations and is therefore carried out at a low computational level. Once the potential energy surface has been mapped out roughly in this way, full geometry optimizations are carried out at a higher level using a four-layer slab, a $3 \times 3 \times 1$ k -grid, and a 5-mRy energy shift. Both adsorbate and the first layer of Au surface atoms are optimized to 0.04 eV/Å. Although this is a relatively weak force tolerance, binding energies do not change appreciably when the tolerance is improved to 0.01 eV/Å. Final binding energies are calculated using optimum geometries from the previous step, calculated at a higher level (seven slab layers, $5 \times 5 \times 1$ k -grid) and are converged to better than 0.05 eV. Further relaxation at the final step is not necessary, as it does not affect the binding energies or geometries appreciably. Table 11.1 gives the final binding energies and adsorption sites for the three motifs in Fig. 11.3.

All three motifs form strong covalent bonds to the surface, in contrast to thiol molecules where the interaction is weaker if the terminal hydrogen is not removed. Mulliken overlap populations give an indication of the character of the bond, and for both the ethynylbenzene radical and vinylidene there is considerable overlap (greater than 0.12) between three of the surface Au atoms and the nearest C atom. Adsorption heights, optimum adsorption sites, and binding energies are also nearly the same for these two motifs, suggesting they both interact with the surface in a similar manner. The flat geometry is bound through two C atoms, each forming a single bond with a surface Au atom. Again, Mulliken overlap populations suggest a covalent bond.

Overall energies in going from the gas-phase molecule in its relaxed geometry to the surface-bound species are exothermic for vinylidene and energy neutral for the flat geometry. The latter value is below the reliability of the calculations.

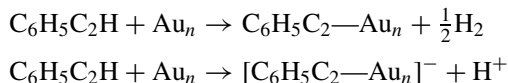
TABLE 11.1 Binding Energies and Adsorption Sites

	Site	Energy (eV)	
		Binding	Overall ^a
Vinylidene	fcc	−2.45	−0.24
Flat geometry	atop	−1.84	0.03
Ethynylbenzene	fcc	−2.99	2.54

^aOverall energies are energies of the surface-bound species relative to the relaxed, isolated molecule and slab.

This is despite a relatively large geometry change upon adsorption. These two configurations are therefore likely intermediates to the formation of a SAM. Indeed, previous surface-enhanced Raman (SERS) experiments suggest the possibility that ethynylbenzene can adsorb onto a gold surface in the flat geometry.⁷ For ethynylbenzene, C—H bond cleavage is calculated for the gas-phase molecule and leads to a very endothermic overall energy upon adsorption.

Reaction energies for formation of a SAM can be estimated from the calculations described above.



As well as C—H bond cleavage (first reaction), deprotonation (second reaction) also needs to be considered. Either of the two reactions can proceed directly or through the vinylidene or flat intermediates. Thus, calculating reaction energies for all three pathways gives a check on the reliability of the estimates since they should all give the same value. The first reaction is slightly endothermic, with an energy of about 0.5 eV; the range of values for the three pathways is 0.4 eV. Using a value for the proton solvation energy of²⁵ 11.4 eV gives a more endothermic reaction in the second case, with a value of 1.7 eV, but with more consistent values for the three pathways varying only by 0.1 eV.

These calculations demonstrate that the ethynylbenzene moiety is indeed a promising alternative to thiols for formation of SAMs on Au(111). It is strongly bound to the surface, yet has a small diffusion barrier, less than 0.2 eV,⁹ between hollows, a site that will allow ordering of the molecules. This linkage scheme may be more oxidatively stable than sulfur, and preparation of monolayers with double-ended molecules should be possible without the problem of forming multilayers. The vinylidene intermediate is a candidate pathway, although from these calculations it is difficult to determine whether subsequent C—H bond cleavage or deprotonation will lead to the surface-bound radical. The latter is known to be the case in the synthesis of metal complexes of ethynylbenzene.²⁴

11.2 DIMERIZATION OF THIOLS ON Au(111)

This example serves to illustrate the advantage of internal coordinates in surface adsorption studies. Geometries can be specified in the *z*-matrix format in SIESTA,²⁶ where one atom is specified in Cartesian coordinates and the remaining molecule is specified in terms of bond lengths, bond angles, and torsion angles relative to this atom. The objective in this example is to map out the potential energy surface (PES) for adsorption of methanethiolate and benzenethiolate on the Au(111) surface in detail and to estimate the dissociation barrier of the dimer, dimethyldisulfide, on this surface.²⁷ Previous computational studies have already reported the energetics^{28,29} of dimerization, but not the dynamics. They find that

dissociation of the surface-bound disulfide is favored, although agreement with available experimental data is limited.

Even for these relatively simple molecules there are sufficient degrees of freedom that mapping out the complete PES is not trivial. Generally, PES maps have been limited to a small subset of degrees of freedom and have been created by scanning rigid molecules across the surface.^{30,31} Using internal coordinates to describe the molecule, it is possible to perform constrained optimizations at each point on the PES and hence map this surface more completely.

Figure 11.4A shows the two thiolate molecules calculated here; note that the terminal hydrogen has been removed, and as a consequence, the sulfur is strongly chemisorbed to the surface. It has been pointed out in the literature that the term *thiolate* is misleading, as it implies an ionic bond to the surface, whereas it is actually closer to a covalently bound “thiyl.”³¹ Here we use the nomenclature prevalent in the literature. Mixed coordinates are used, with a z -matrix to specify

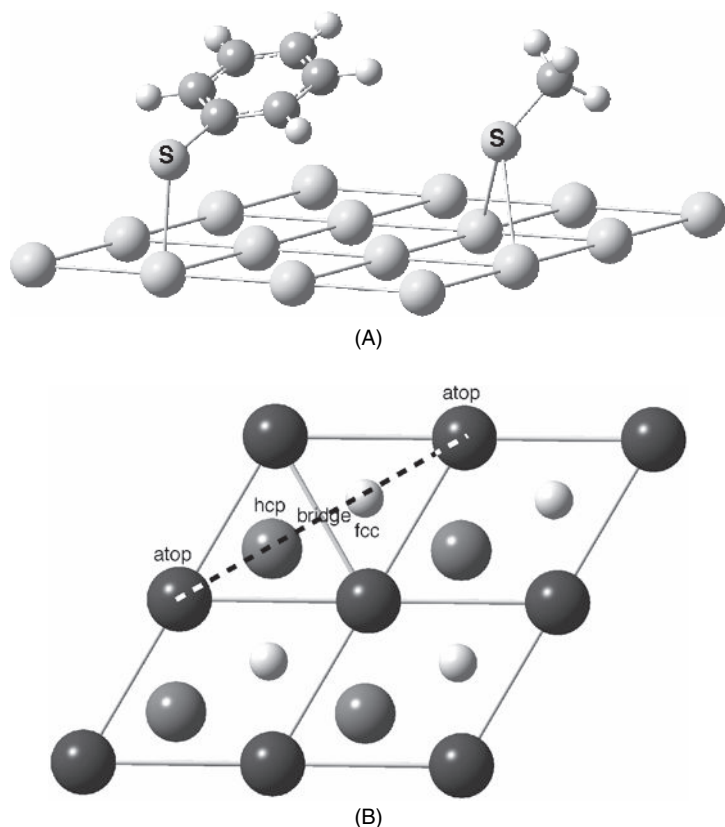


Fig. 11.4 (A) Adsorption of benzenethiolate (left) and methanethiolate onto the Au(111) surface; (B) path for the PES scan relative to surface Au atoms. Second and third layers of gold atoms are depicted by successively smaller spheres. (From Ref. 27.)

the adsorbate and Cartesian coordinates for the Au slab. For each adsorbate the PES is mapped along the atop–bridge–atop path shown in Fig. 11.4B. At each step in the PES a constrained optimization is performed with the position of the sulfur atom fixed relative to the Au surface while its height above the surface is allowed to vary. The rest of the molecule and the surface layer of Au atoms are fully relaxed. Mapping the PES in this much detail using Cartesian coordinates is not practicable. It is also possible to decouple optimization of the bond lengths and bond angles with the z -matrix approach and to specify different force tolerances for each. This is particularly advantageous where the PES is very flat in one coordinate compared to the other. This is the case for many molecular adsorption problems, where the PES is quite flat with respect to tilting of the molecular axis relative to the surface. With Cartesian coordinates it can be difficult to find the minimum of such a surface.

Provided that there is little or no coupling between coordinates, such as in cyclic molecules, internal coordinates can also lead to efficiency gains in the optimization process, as they lead to better preconditioning of the optimization algorithm. Table 11.2 compares geometry optimizations using z -matrix and Cartesian coordinates within the SIESTA code for some simple molecules.²⁶ The conjugate gradient algorithm is used in all cases, with the optimization being performed to three levels of force convergence and with different numbers of degrees of freedom. In the z -matrix optimization for N atoms, an unconstrained optimization can be achieved with $3N - 6$ variables whereas $3N - 3$ are required for Cartesian coordinates. This is because in addition to fixing the coordinates of one atom (the reference atom), in the z -matrix approach it is also possible to fix the three rotational degrees of freedom for the entire molecule.

The z -matrix approach performs better for both the simple water molecule and acyclic hexanedithiol molecule. In the latter case, fixing either three or six degrees of freedom reduces the number of CG steps for z -matrix optimization very considerably. Conversely, fixing degrees of freedom in Cartesian coordinates increases the number of steps. This is because the method used (there is no Hessian matrix) is not sensitive to the translational invariance. For the cyclic benzene molecule, Cartesian coordinates improve optimization because internal coordinates are coupled to each other. The same final geometries are obtained irrespective of the coordinates used and number of degrees of freedom in the optimization.

The computational conditions used here are essentially the same as those used for the geometry optimizations of ethynylbenzene described above. The force tolerances are set to 0.04 eV/Å for bond lengths and 0.0009 eV/deg for angles. Optimizations are performed using the conjugate gradient (CG) method. The forces are calculated by direct differentiation of the energy and are generated in the same section of code within SIESTA. The CG method is a variant of steepest descent but avoids its pitfall of successive steps being perpendicular to each other. Instead, they are constructed to be conjugate to the previous gradient and as far as possible from all previous steps. In this method it is only necessary to store information from the last CG step rather than building up the full

TABLE 11.2 Number of Conjugate Gradient Steps Required to Optimize the Geometry of Three Molecules in Z-Matrix and Cartesian Coordinates

Molecule	No. of Atoms	Coordinates	No. of Variables	No. of CG Steps ^a		
				I	II	III
Water	3	Cartesian	6	15	15	15
			9	35	37	40
		z-matrix	2	6	8	8
			3	3	6	9
			6	3	6	9
			9	4	19	21
Benzene	12	Cartesian	33	25	33	36
			36	7	9	9
		z-matrix	2	7	11	18
			11	12	14	20
			30	47	57	69
			33	45	58	63
Hexanedithiol	22	Cartesian	36	44	55	66
			63	76	108	171
		z-matrix	66	44	46	81
			60	20	33	44
			63	24	39	115
			66	32	397	

Source: Ref. 26.

^aColumns I, II, and III represent progressively stricter convergence criteria for lengths and angles: namely, I, (0.04 eV/Å, 0.0009 eV/deg); II, (0.02 eV/Å, 0.0004 eV/deg); and III, (0.01 eV/Å, 0.0002 eV/deg). For the Cartesian coordinate optimizations the angle tolerance is to be ignored.

Hessian matrix for the entire optimization. SIESTA writes the previous step to disk at every CG step, allowing for easy restarts of optimizations. In principle, for M nuclei, the CG method should converge in less than $3M$ steps. However, due to numerical errors and the fact that the potential energy surface does not necessarily have the assumed quadratic form, more steps are often required. Both Fletcher–Reeves and Polak–Ribiere CG algorithms are implemented in SIESTA, although the latter is the default and preferred option, as it reportedly performs better where the minimum is not quadratic (details of the implementations are given elsewhere³²). The modified Broyden³³ method is also available in SIESTA. In principle, the modified Broyden method, a quasi-Newton–Raphson method, would be extremely efficient if the Jacobian were known and could easily be inverted. However, this is not the case in practice; rather, the Jacobian is updated over successive steps. It is also possible to find optimum geometries using molecular dynamics (MD), and SIESTA has implemented both simulated annealing, where the temperature of the MD simulation is gradually reduced to a target temperature, and quenching, where the velocity components of the nuclei are set to zero if they are opposite the corresponding force. Although relatively easy to implement, these MD-based schemes are often not competitive compared

with the sophisticated line search–based algorithms mentioned previously. More recently, FIRE³⁴ (scheme for fast inertial relaxation engine), a new MD-based optimization method has been reported that is competitive and can be used easily for systems containing millions of degrees of freedom.

The PESs for the two monomers are shown in Fig. 11.5. It is interesting to note that with the current z -matrix constrained optimization, the hexagonal close-packed (hcp) and face-centered cubic (fcc) hollow adsorption sites are local maxima for both PESs. By contrast, a Cartesian coordinate–based scan will yield local minima at these two sites; previous studies find this result.²⁸ Bilić et al.³¹ also find the hollow sites to be saddle points for two-layer slab calculations, but minima for a four-layer calculation. There is also no barrier to diffusion at the bridge site, in contrast to some previous calculations where the PES is mapped by scanning a rigid molecule.^{28,30} The PES in this region is sensitive to the tilt angle of the molecule and also its orientation. The minimum on both sides of the bridge site is with the tail group tilted back over the bridge (i.e., as the bridge is traversed from one side to the other, the tail of the molecule swings around rather than remaining fixed in orientation). Adsorption energies of 1.85 and 1.43 eV are calculated for the optimum sites for methanethiolate and benzenethiolate, respectively. This is in good agreement with previous calculations.^{28,31}

Optimum geometries for adsorption of the dimers are shown in Fig. 11.6; here the entire dimer and surface layer are relaxed. The SIESTA implementation of z -matrix coordinates is particularly convenient for this example. Multiple z -matrix blocks can be defined, making it possible to have separate sets of internal

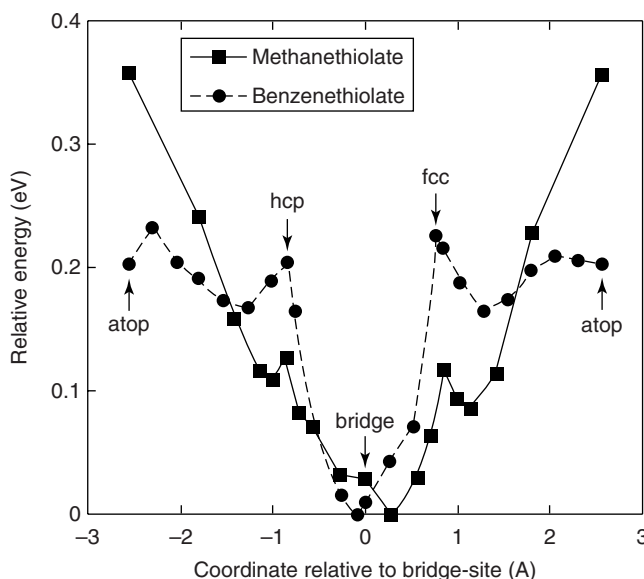


Fig. 11.5 PES for methanethiolate and benzenethiolate along the atop–bridge–atop path on the Au(111) surface.

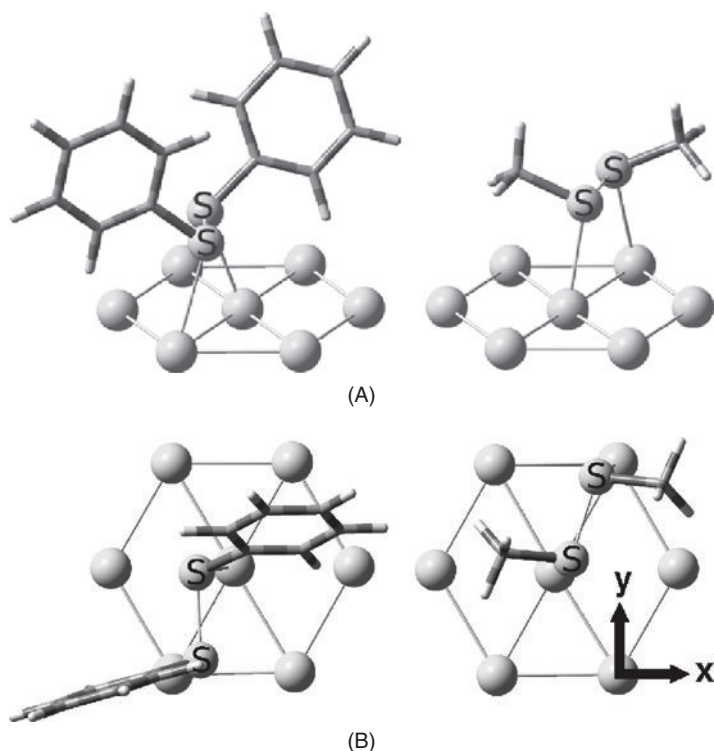


Fig. 11.6 Relaxed geometries for the two thiol dimers diphenyldisulfide (left) and dimethyldisulfide (right). Two different perspectives for each are shown in (A) and (B). (From Ref. 27.)

coordinates centered around each S atom. Adsorption occurs through the sulfur atoms, with each S atom in the dimer adsorbed near the atop site and displaced slightly toward the bridge site. The two S atoms are at similar heights above the surface. Previous studies using Cartesian coordinates find a different optimum geometry with S atoms nearer the bridge sites and at different heights above the surface.^{28,35,36} If the calculations here are repeated using Cartesian coordinates, this previously reported minimum appears to become a local minimum. This result further illustrates the robustness of internal coordinate descriptions for molecular adsorption.

Both dimers are energetically unfavorable on the surface relative to two isolated monomers, by 0.41 and 0.62 eV for dimethyldisulfide and diphenyldisulfide, respectively. This is despite the fact that geometry optimizations find a local minimum and do not dissociate the dimer. This would suggest that there is an activation barrier to dissociation. To explore this point the PES for dissociation of dimethyldisulfide was mapped and is shown in Fig. 11.7A. One S atom is fixed at its optimum site while the other is scanned over the surface with a constrained optimization of the molecule performed at each point. The PES in Fig. 11.7A

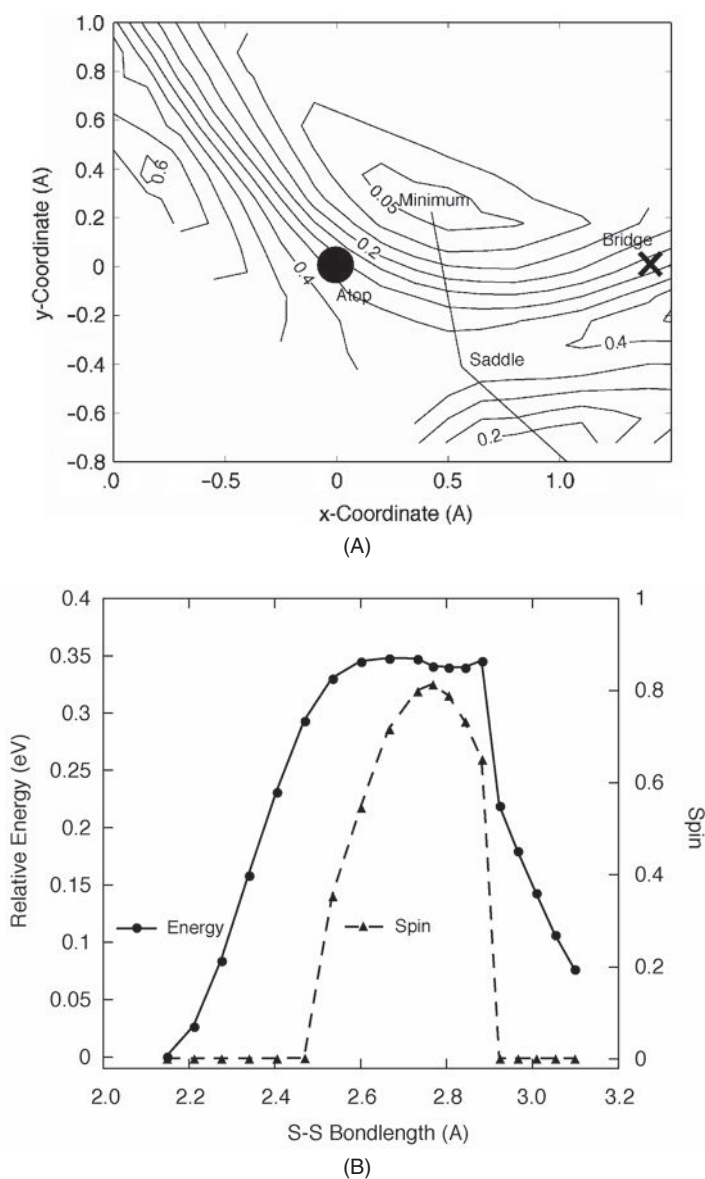


Fig. 11.7 (A) Spin-restricted PES for dissociation of dimethyldisulfide. Contours are in 0.05-eV intervals relative to energy minimum; position of surface atop and bridge sites are shown; one S atom is fixed at $x = 1.05$ and $y = 2.27$ Å. (B) Spin-unrestricted PES along the dissociation path shown in (A). Units of spin are number of electrons. (From Ref. 27.)

was mapped using spin-restricted calculations for computational efficiency. This will give a reasonable idea of the PES shape and help identify the dissociation path. A spin-unrestricted scan along this path is then performed, with the results shown in Fig. 11.7B.

As expected, DFT does not describe the region where the bond is dissociating very well; Fig. 11.7B shows that there is significant spin contamination around the saddle point. Away from this point, where the spin is zero the DFT energies are presumably quite reliable and allow us to estimate the height of the dissociation barrier to lie between 0.3 and 0.35 eV. The barrier for formation of the dimer from two surface-bound isolated monomers is estimated to lie between 0.71 and 0.76 eV.

11.3 MOLECULAR DYNAMICS OF NANOPARTICLES

So far, only ground-state properties at 0 K have been discussed. Molecular dynamics (MD) is the standard method of introducing the motion of the atomic nuclei into the problem and hence simulating various temperature-dependent properties, such as phonon spectra or melting behavior. The MD capabilities implemented in SIESTA will be illustrated in this section, where the melting behavior of the 20-atom gold cluster is examined.^{37,38} This particular size cluster is interesting because its optimum geometry is an ordered tetrahedral pyramid and is isolated by about 1 eV from its nearest-lying isomer, at least as determined in 0 K DFT calculations.^{39–42} There is experimental evidence that this structure is indeed the optimum.⁴³

The standard Verlet algorithm^{44,45} is implemented in SIESTA to propagate the MD trajectory in time. A detailed description of this algorithm and other established components of MD are given in many textbooks, for example.⁴⁶ Here the initial velocities are chosen from the Maxwell–Boltzmann distribution corresponding to a specified temperature. The total energy of the system is then kept constant throughout the trajectory: the microcanonical ensemble. Motion of the center of mass of the system is frozen out initially, although rotational motion currently is not. Nonperiodic systems such as clusters and molecules can pick up slight center-of-mass kinetic energy over a long trajectory due to numerical errors. Rotational motion is generally very small to start but can become appreciable over a long trajectory. Specifying a fine integration grid can help prevent these problems. In this example, thermal behavior in the canonical ensemble is calculated using the Nosé–Hoover^{47,48} thermostat to maintain constant temperature. Briefly, in this method the system is connected to a heat bath that can transfer energy into or out of the system to attempt to maintain constant temperature. The heat bath is realized by coupling a fictitious degree of freedom to the system. The degree of coupling is determined by the Nosé mass, which controls quite sensitively the dynamics of the simulation. Constant-pressure simulations are also implemented in SIESTA using the Parrinello–Rahman method^{49–51} where again an effective mass must be set in order to carefully thermostat the trajectory

correctly. Constant-temperature and constant-pressure methods can be combined into a single simulation.

The critical parameter to optimize is the time step; this must be small enough to capture the atomic motion but not too small that only short total times can be sampled. The MD time step is traditionally determined according to the following rule of thumb:

$$dt = \frac{1}{10} \frac{1}{c \omega_{\max}} \quad (11.3)$$

where c is the speed of light and $\omega_{\max}$ is the highest vibrational frequency. The vibrational frequencies are determined by calculating the force matrix in SIESTA and then finding the eigenvalues of this matrix using the VIBRA utility supplied with SIESTA. The energy-shift parameter needs to be set to a small value, typically better than 5 mRy, to avoid negative frequencies for the optimized structure. For the present 20-atom gold cluster the maximum frequency is 221 cm^{-1} , corresponding to a time step of 15 fs.⁵² The time step can be analyzed more rigorously by monitoring the conservation of total energy of the extended system (i.e., the 20-atom cluster plus the Nosé thermostat). In the present example time steps up to about 3 fs conserve this total energy well during the MD trajectory, but significant variations occur above this value. The time step is set to 2.5 fs for all the simulations presented here.

A large value of Nosé mass results in low coupling to the reservoir and leads to large temperature fluctuations and relatively constant total energy; thermostating is ineffective in this case. A low value, on the other hand, restrains the temperature oscillations and can lead to poor equilibration and overdamping of the dynamics. One way to assess the appropriate Nosé mass value is to observe temperature fluctuations over a number of MD steps and decide on a suitable level of temperature fluctuation. Alternatively, the statistical convergence of the trajectory can be examined where the average values of the temperature, or equivalently, kinetic energy of the ions and higher moments of these quantities are observed. While the average is a good indicator that the ensemble is converging to the correct temperature, higher moments are a more sensitive indicator of the temperature fluctuations and statistical quality.⁵³

The average kinetic energy of the ions $\langle KE_{\text{ion}} \rangle$ and second moment $\langle (KE_{\text{ion}} - \langle KE_{\text{ion}} \rangle)^2 \rangle$ are shown in Fig. 11.8 for a thermostat temperature of 900 K over 45,000 MD steps (112.5 ps) and a Nosé mass of $50 \text{ Ry} \cdot \text{fs}^2$. The energy shift is set to 20 mRy, the real-space grid is cut off to 100 Ry, and the LDA exchange-correlation function is used. Both quantities converge reasonably well over the entire trajectory but require about 10,000 steps to equilibrate. The average kinetic energy and its second moment converge to values corresponding to temperatures of 900 and 821 K, respectively.⁵³ The second moment gives slightly different ensemble average temperature because it is more sensitive to temperature fluctuations. Higher moments can be calculated to give an indication of statistical quality. These results indicate that the current number of MD steps is sufficient to provide a good statistical ensemble and that

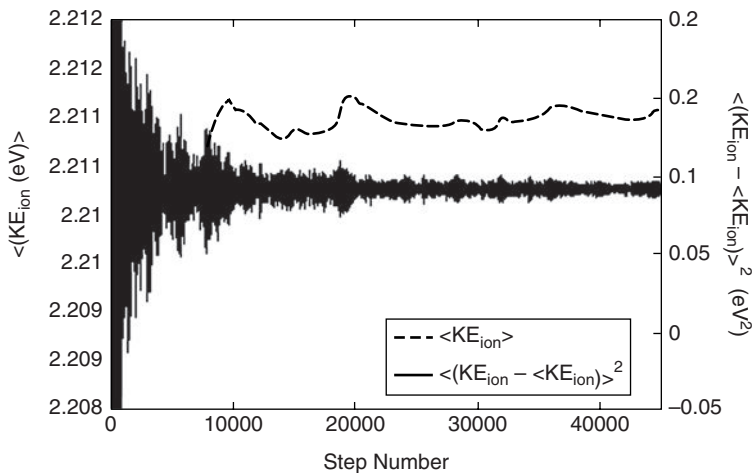


Fig. 11.8 Statistical convergence for a Nose MD simulation of a 20-atom gold cluster.

the Nosé mass gives reasonable statistical behavior. Temperature fluctuations (not shown) are also reasonable across the trajectory. Increasing or reducing the Nosé mass by a factor of 2 gives slightly worse average values.

SIESTA outputs to a formatted file the value of instantaneous temperature, Kohn–Sham energy, total energy, ion kinetic energy, and unit cell volume and pressure for each step of the MD simulation. The atomic coordinates at each step are output as a formatted xyz file for animation of the trajectory, and the coordinates and ion velocities at each step are also output to an unformatted file for subsequent postprocessing. Here, the melting behavior of the 20-atom cluster is to be investigated. The appropriate quantity to examine in this case is the root-mean-square bond-length fluctuation, defined by

$$\delta_{\text{rms}} = \frac{2}{n(n-1)} \sum_{i < j} \frac{\left(\langle r_{ij}^2 \rangle_t - \langle r_{ij} \rangle_t^2 \right)^{1/2}}{\langle r_{ij} \rangle_t} \quad (11.4)$$

where n is the number of atoms in the cluster, r_{ij} is the distance between ions i and j , and $\langle \dots \rangle_t$ denotes a time average over the trajectory. δ is calculated at each step of the trajectory, and its final value for the entire trajectory is then plotted for each temperature simulation, shown in Fig. 11.9. The abrupt changes in bond-length fluctuation are an indicator of melting and provide a value of the melting temperature. Below the melting point the atoms all vibrate about an equilibrium position, giving a small value for the bond-length fluctuation. Above the melting temperature, where the cluster is in a liquid-like state, the atoms move relative to each other and the bond-length fluctuation is large. The change from one regime to the other for the 20-atom tetrahedral cluster is quite sharp, indicating that there is a relatively well defined melting temperature.

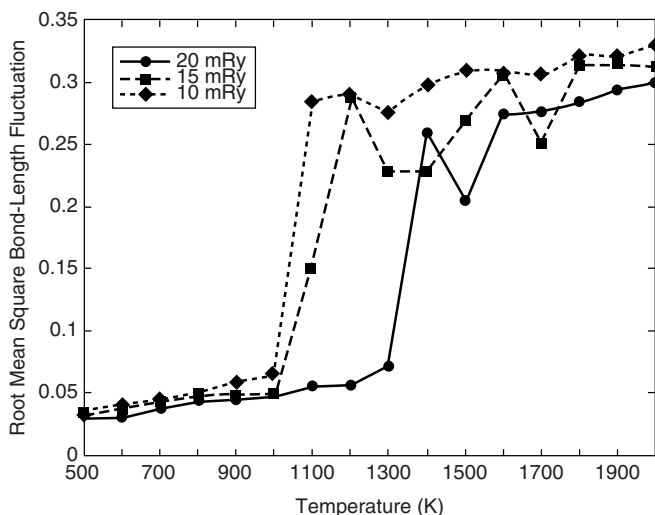


Fig. 11.9 Root-mean-square bond-length fluctuation as a function of temperature for a 20-atom Au cluster for three different value of energy shift.

The melting temperature is clearly quite dependent on the value of the energy shift, changing from about 1300 K at 20 mRy to about 1000 K at 10 mRy. This is effectively a basis set effect: Larger values of energy shift (i.e., more confined orbitals) tend to overestimate bond strengths between the Au atoms. As the energy shift is decreased and the orbitals become more extended, the bond energies converge toward the correct value. Consequently, the melting temperature is overestimated at an energy shift of 20 mRy. It appears that the melting temperature is reasonably well converged at 10 mRy given that the behavior does not change substantially between 10 and 15 mRy. It is difficult to test this directly because of the computational cost involved in these long simulations at improved values of the energy shift.

11.4 APPLICATIONS TO LARGE NUMBERS OF ATOMS

The examples discussed above all contained, at most, a few hundred atoms. In this regime standard diagonalization of the Hamiltonian can be used to find the self-consistent energy and density. As discussed in the methods section above, two linear scaling methods are implemented in SIESTA, functional minimization according to Kim, Mauri, and Galli (KMG),⁵⁴ and divide and conquer in the spirit of the original proposal by Yang.^{55,56} In this section the divide-and-conquer method is illustrated with application to large gold clusters.

11.4.1 Large Gold Clusters

The prefactor in the linear scaling of divide and conquer is critical and must be sufficiently small that the crossover with normal diagonalization occurs at a

reasonable number of atoms. Figure 11.10A compares the scaling behavior for divide and conquer versus diagonalization for gold clusters from 923 to 10,179 atoms. An energy shift of 15 mRy, real space grid cutoff of 300 Ry, and SZ basis are used, and the calculations are parallelized over 64 CPUs. The elapsed time is for the first SCF cycle. The limit for diagonalization in this case is about 5000 atoms, the calculation level is relatively low, and the scaling is reasonably well

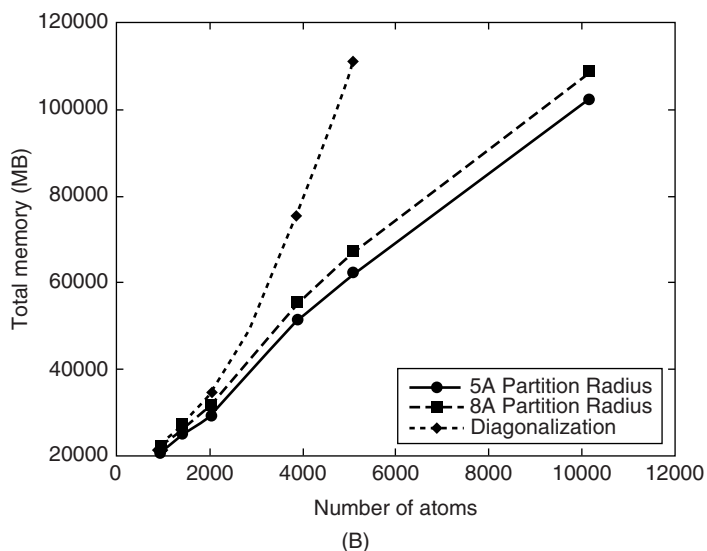
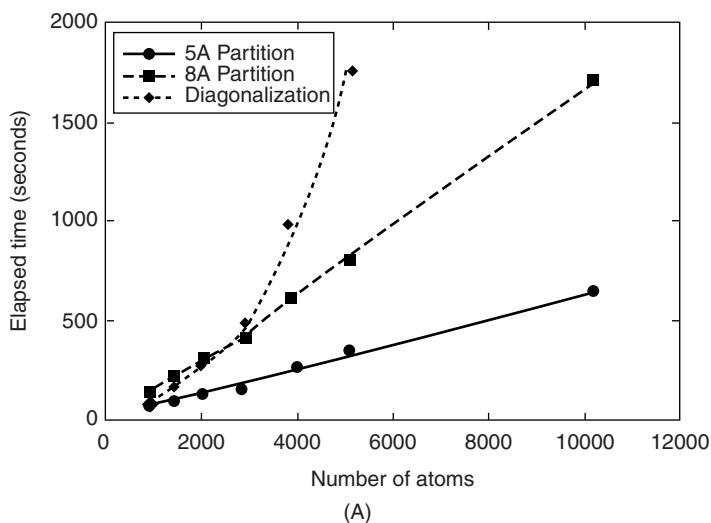


Fig. 11.10 Comparison of scaling with divide and conquer and diagonalization for (A) elapsed time for the first SCF cycle and (B) the total memory demand. Lines in (A) are linear fits to the divide-and-conquer results and a cubic fit to the diagonalization results.

approximated by a cubic curve. The divide-and-conquer calculations scale very linearly with a prefactor that depends on the partition parameter. This is the radius of the partition surrounding each core atom. A simple and universal, although not necessarily optimum, partitioning scheme is implemented in SIESTA. The partitions are spheres, with each partition containing one core atom; in other words, there is a partition for each atom in the system. The number of buffer atoms, and hence how strongly partitions are coupled to each other, will depend on the radius of this sphere. The advantage of this approach is that it can be applied to any configuration of atoms and requires specification of only a single parameter: the partition radius.

Memory scaling is also an important consideration and, rather than computational time, may be the limiting factor in many instances. This is illustrated in Fig. 11.10B, where the amount of total memory required is plotted as a function of the number of atoms. These results are the total memory requirement and include a large component for the Hamiltonian and overlap matrices. This is the dominant component for the divide-and-conquer calculations, and hence memory requirements do not change significantly as the partition radius is increased. The substantial difference between diagonalization and divide and conquer is due to the large memory overhead in the diagonalization routines.

Scaling of the calculation time with respect to the partition radius is shown in Fig. 11.11 for two cluster sizes. Each partition is diagonalized using standard linear algebra routines, and one might therefore expect the time to scale roughly to the third power in partition radius. The results in Fig. 11.11 appear to scale less favorably, however. It is clear from this example that divide and conquer is able to extend the capabilities of SIESTA to systems approaching the size of those used in experiments. A cluster of 10,179 atoms has a diameter of about 7 nm. The critical issue is how well divide-and-conquer results approximate the

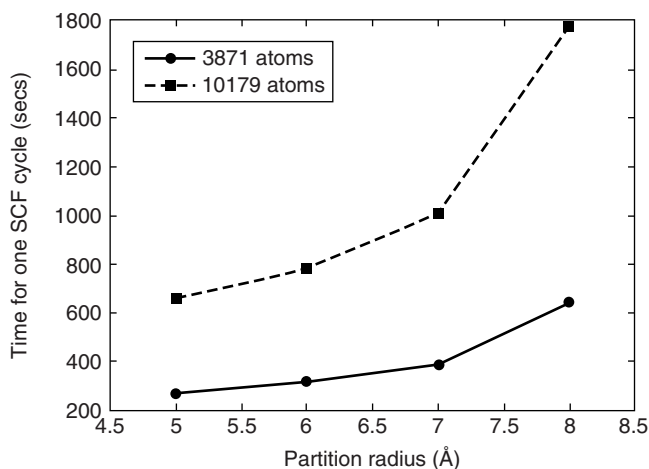


Fig. 11.11 Scaling with partition radius.

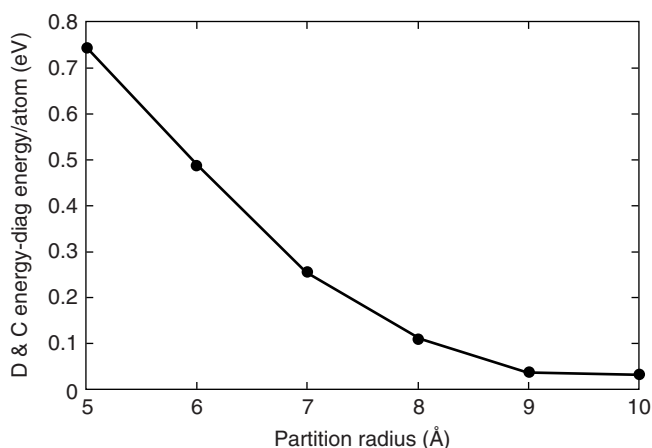


Fig. 11.12 Energy difference between divide and conquer and diagonalization for a 3871-atom Au cluster as a function of partition radius.

“correct” answer. As the partition radius increases, the energy and density should, in principle, approach those from a diagonalization calculation. The question then becomes whether a suitable value can be found that gives reliable results without increasing the calculation time unreasonably. The difference in total energy per atom between divide and conquer and diagonalization is shown in Fig. 11.12 for a 3871-atom cluster. The divide-and-conquer calculation does not converge close to the diagonalization value until the partition radius is about 10 Å; at this point the difference is 0.04 eV/atom. However, at an 8-Å radius the difference is below about 0.1 eV/atom, a value that may be sufficient in many circumstances. We have shown previously⁵⁷ that this implementation of divide and conquer performs more favorably in less extreme examples. For example, for alkane chains the total energy per methyl group converges to the diagonalization energy to better than 0.005 eV for a partition radius of only 5 Å, and is virtually the same at 10 Å. In addition, the forces in this case are within 0.05 eV/Å at a partition radius of 5 Å and within 0.00004 at 10 Å. Similar results were obtained for semiconducting bulk silicon and near-metallic carbon nanotubes, although a partition radius of 7 Å is required in the latter case to give an energy difference of less than 0.003 eV. These results are not surprising given that the central approximation in divide and conquer is that the interaction between neighboring regions should fall off rapidly with distance; for insulators and semiconductors it is close to exponential, while for metallic systems it is more like a power-law decay.

11.4.2 Graphene Nanoflakes

As a final example we consider graphene, a material that is currently attracting considerable attention due to its remarkable electronic properties. It is an

interesting case study for linear-scaling techniques because of these properties. It is a covalently bonded material but one that has a zero bandgap, at least at one point in the Brillouin zone, the K-point. Therefore, one might expect delocalized electrons and a relatively slow decay of electron states more akin to that of a metal.

The divide-and-conquer method in SIESTA is appropriate only for Γ -point calculations; however, the unusual electronic structure in periodic graphene sheets occurs at the K-point of the Brillouin zone. The primitive cell for graphene and a conventional reciprocal space cell are shown in Fig. 11.13. Using the primitive cell representation, a multiple of a 6×6 k -grid is required to capture the K-point of the Brillouin zone. Alternatively, a multiple of a 6×6 supercell will achieve the same result. In this example a 24×24 supercell is used containing 1152 atoms, so that the results of diagonalization and divide and conquer can be compared. The calculations are carried out at a high computational level using a DZP basis set, a 5-mRy energy shift, and a 350-Ry cutoff for the real-space grid. Figure 11.14 shows how the divide and conquer total energy and Fermi energy converge toward the diagonalization value. From this figure a partition radius of 7.5 Å gives a reasonable total energy with less than about 0.02 eV/atom difference. The Fermi energy oscillates considerably, but appears to be stable at 7.5 Å; this might not be such a reliable measure of convergence in any case.

The convergences of the forces and electron density toward those values calculated from diagonalization will also test how well the divide-and-conquer approach can capture the electronic structure. Figure 11.15 shows the stress-strain curve calculated for this 24×24 graphene supercell using these two methods. The lattice parameter and atomic positions are first optimized for

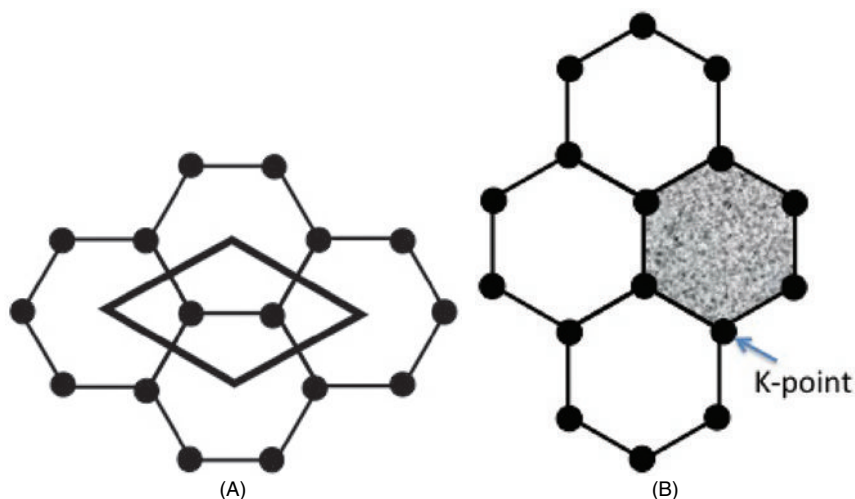


Fig. 11.13 (A) Primitive real-space cell, indicated by the heavy lines; (B) Brillouin zone (shaded) for graphene.

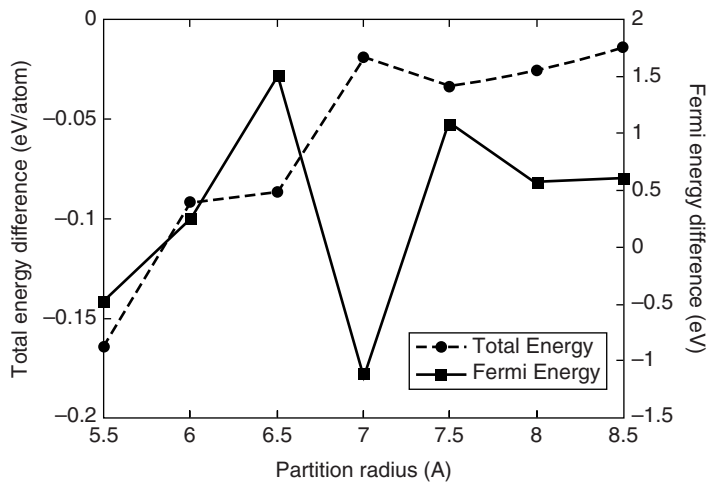


Fig. 11.14 Convergence of total energy and Fermi energy as a function of partition radius. The total energy difference per atom and Fermi energy difference are with respect to the values from a diagonalization calculation.

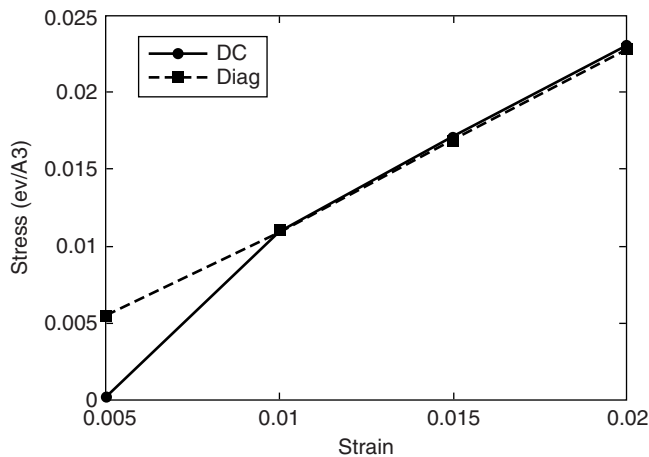


Fig. 11.15 Comparison of stress–strain curve for 24×24 graphene supercell calculated using divide and conquer or diagonalization. A isotropic strain is applied to the graphene in the plane of the sheet.

a single primitive cell using diagonalization, and the resulting relaxed structure is then replicated to produce the supercell. The stress–strain curve is generated by applying an isotropic strain to the supercell and undertaking a single-point energy calculation to generate the resulting stress tensor. No further optimization is performed. This is a relatively crude way to calculate stress–strain but is intended more to compare two methods rather than yielding a robust value

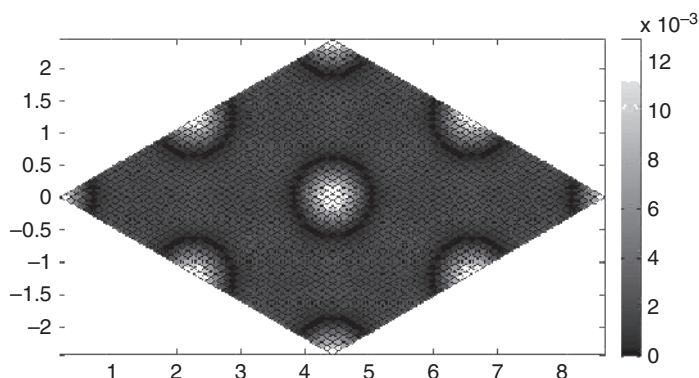


Fig. 11.16 Difference between electron density calculated from diagonalization and from divide and conquer. The fractional difference relative to the diagonalization results are plotted on a gray scale, with light colors indicating larger values. The maximum of the gray scale corresponds to a fractional difference of 0.012.

for the elastic modulus. The two curves, for diagonalization and divide and conquer are identical, indicating that the forces are also very well converged at a partition radius of 7.5 Å. Assuming an interlayer spacing of 0.335 nm for graphite, the results from Fig. 11.15 give a value for the elastic constant in the plane of the graphene sheet of about 1 TPa. This value is in good agreement with a recent experimental determination.⁵⁸

Finally we consider the difference in electron density. Figure 11.16 shows the difference in electron density between the diagonalization calculation and divide and conquer, again with a 7.5-Å partition radius. The difference is the fractional difference relative to the diagonalization calculation; the plane of the plot is parallel to the graphene surface 1.6 Å above the plane of the atoms. The maximum difference between the two calculations is about 1.2%, but this occurs in regions where the electron density is extremely small and may therefore be due to numerical errors. In any case, this shows that even for a relatively modest partition radius of 7.5 Å, there is essentially little difference between the results from divide and conquer and diagonalization.

These calculations are carried out on eight processors for divide and conquer, with a memory demand of about 14 GB, and on 16 processors for diagonalization, with about 32 GB of memory. It is possible to extend the divide-and-conquer calculations to a 48×96 supercell, approaching 10,000 atoms, on 64 processors and 132 GB of memory. This corresponds to a supercell with dimensions of about 10×20 nm. The diagonalization calculations struggle at around a quarter of this size.

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REFERENCES

1. Ulman, A. *Chem. Rev.* **1996**, 96, 1533.
2. Ulman, A. *An Introduction to Ultrathin Organic Films*, Academic Press, San Diego, CA, 1991.
3. Love, J. C.; A, E. L.; Kriebel, J. K.; Nuzzo, G.; Whitesides, G. M. *Chem. Rev.* **2005**, 105, 1103.
4. Nitzan, A.; Ratner, M. A. *Science* **2003**, 300, 1384.
5. Tour, J. M.; Jones, L., II; Pearson, D. L.; Lamba, J. J. S.; Burgin, T. P.; Whitesides, G. M.; Allara, D. L.; Parikah, A. N.; Atre, S. V. *J. Am. Chem. Soc.* **1995**, 117, 9529.
6. Long, N. J.; Williams, C. K. *Angew. Chem.* **2003**, 42, 2586.
7. Feilchenfeld, H.; Weaver, M. J. *J. Phys. Chem.* **1989**, 93, 4276.
8. Joo, S.-W.; Kim, K. *J. Raman Spectrosc.* **2004**, 35, 549.
9. Ford, M. J.; Hoft, R. C.; Gale, J. D.; McDonagh, A. A new class of self-assembled monolayers on gold using an alkynyl group as a linker. In *Proceedings of the International Conference on Nanoscience and Nanotechnology*, Brisbane, Australia, 2006.
10. Ford, M. J.; Hoft, R. C.; McDonagh, A. *J. Phys. Chem. B* **2005**, 109, 20387.
11. Perdew, J. P.; Burke, K.; Ernzerhof, M. *Phys. Rev. Lett.* **1996**, 77, 3865.
12. Troullier, N.; Martins, J. L. *Phys. Rev. B* **1991**, 43, 1993.
13. Hoft, R. C. Modelling electron tunnelling in the presence of adsorbed materials, Ph.D. dissertation, University of Technology, Sydney, Australia, 2007.
14. Kolb, D. M. *Prog. Surf. Sci.* **1996**, 51, 109.
15. Somorjai, G. A.; Van Hove, M. A. *Prog. Surf. Sci.* **1989**, 30, 201.
16. Roper, M. G.; Skegg, M. P.; Fisher, C. J.; Lee, J. J.; Dhanak, V. R.; Woodruff, D. P.; Jones, R. G. *Chem. Phys. Lett.* **2004**, 389, 87.
17. Yu, M.; Bovet, N.; Satterley, J.; Bengio, S.; Lovelock, K. R. J.; Milligan, P. K.; Jone, R. G.; Woodruff, D. P.; Dhanak, V. R. *Phys. Rev. Lett.* **2006**, 97, 166102.
18. Wang, Y.; Hush, N. S.; Reimers, J. R. *J. Am. Chem. Soc.* **2007**, 129, 14532.
19. Methfessel, M.; Paxton, A. T. *Phys. Rev. B* **1989**, 40, 3616.
20. *CRC Handbook of Chemistry and Physics*, 87th ed., CRC Press, Boca Raton, FL, 2006.
21. Boys, S. B.; Bernardi, F. *Mol. Phys.* **1970**, 19, 553.
22. Lambropoulos, N. A.; Reimers, J. R.; Hush, N. S. *J. Chem. Phys.* **2002**, 116, 10277.
23. Bengtsson, L. *Phys. Rev. B* **1999**, 59, 12301.
24. Wakatsuki, Y. *J. Organomet. Chem.* **2004**, 689, 4092.
25. Zhan, C. G.; Dixon, D. A. *J. Phys. Chem. A* **2001**, 105, 11534.

26. Hoft, R. C.; Gale, J. D.; Ford, M. J. *Mol. Simul.* **2006**, *32*, 595.
27. Ford, M. J.; Hoft, R. C.; Gale, J. D. *Mol. Simul.* **2006**, *32*, 1219.
28. Yourdshahyan, Y.; Rappe, A. M. *J. Chem. Phys.* **2002**, *117*, 825.
29. Gronbeck, H.; Curioni, A.; Andreoni, W. *J. Am. Chem. Soc.* **2000**, *122*, 3839.
30. Cometto, F. P.; Paredes-Olivera, P.; Macagno, V. A.; Patrito, E. M. *J. Phys. Chem. B* **2005**, *109*, 21737.
31. Bilić, A.; Reimers, J. R.; Hush, N. S. *J. Chem. Phys.* **2005**, *122*, 094708/1.
32. Press, W. H.; Teukolsky, S. A.; Vetterling, W. T.; Flannery, B. P. *Numerical Recipes in Fortran 90*, 2nd ed., Cambridge University Press, New York, 1996.
33. Johnson, D. D. *Phys. Rev. B* **1988**, *38*, 12807.
34. Bitzek, E.; Koskinen, P.; Gahler, F.; Moseler, M.; Gumbsch, P. *Phys. Rev. Lett.* **2006**, *97*, 170202.
35. Hayashi, T.; Morikawa, Y.; Nozoye, H. *J. Chem. Phys.* **2001**, *114*, 7615.
36. Vargas, M. C.; Giannozzi, P.; Selloni, A.; Scoles, G. *J. Phys. Chem. B* **2001**, *105*, 9509.
37. Soule de Bas, B.; Ford, M. J.; Cortie, M. B. *J. Phys. Condens. Matter* **2006**, *18*, 55.
38. Krishnamurty, S.; Shafai, G. S.; Kanhere, D. G.; Soule de Bas, B.; Ford, M. J. *J. Phys. Chem. A* **2007**, *111*, 10769.
39. Soule de Bas, B.; Ford, M. J.; Cortie, M. B. *J. Mol. Struct.* **2005**, *686*, 193.
40. Ford, M. J.; Soule de Bas, B.; Cortie, M. B. *Mater. Sci. Eng. A* **2007**, *140*, 177.
41. Wang, J.; Wang, G.; Zhai, J. *Chem. Phys. Lett.* **2003**, *380*, 716.
42. Wang, J.; Wang, G.; Zhao, J. *Phys. Rev. B* **2002**, *66*, 035418.
43. Li, J.; Li, X.; Zhai, H.-J.; Wang, L.-S. *Science* **2003**, *299*, 864.
44. Verlet, L. *Phys. Rev. B* **1967**, *159*, 98.
45. Gear, C. W. *Numerical Initial Value Problems in Ordinary Differential Equations*, Prentice-Hall, Englewood Cliffs, NJ, 1971.
46. Allen, M. P.; Tildesley, D. J. *Computer Simulation of Liquids*, Oxford University Press, New York, 1987.
47. Nosé, S. *Mol. Phys.* **1984**, *52*, 255.
48. Hoover, W. G. *Phys. Rev. A* **1985**, *31*, 1695.
49. Parrinello, M.; Rahman, A. *Phys. Rev. Lett.* **1980**, *45*, 1196.
50. Parrinello, M.; Rahman, A. *J. Appl. Phys.* **1981**, *52*, 7182.
51. Parrinello, M.; Rahman, A. *J. Chem. Phys.* **1982**, *76*, 2662.
52. Soule de Bas, B. Modelling the Properties of Gold Nanoparticles, Ph.D. dissertation, University of Technology, Sydney, Australia, 2005.
53. Cho, K.; Joannopoulos, J. D. *Phys. Rev. B* **1992**, *45*, 7089.
54. Kim, J.; Mauri, F.; Galli, G. *Phys. Rev. B* **1995**, *52*, 1640.
55. Yang, W. *Phys. Rev. Lett.* **1991**, *66*, 1438.
56. Yang, W.; Lee, T.-S. *J. Chem. Phys.* **1995**, *103*, 5674.
57. Cankurtaran, B. O.; Gale, J. D.; Ford, M. J. *J. Phys. Condens. Matter* **2008**, *20*, 294208.
58. Lee, C. L.; Wei, X.; Kysar, J. W.; Hone, J. *Science* **2008**, *321*, 385.

12 Modeling Photobiology Using Quantum Mechanics and Quantum Mechanics/Molecular Mechanics Calculations

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Photobiological systems play an essential role in living organisms. Compared to the usual enzymatic reactions occurring in the ground state, it is much more challenging to model photobiological systems involving electronic excited states using quantum mechanics (QM) and quantum mechanics/molecular mechanics (QM/MM) calculations. In this chapter, key computational methods for photobiological system are introduced and recent selected computational studies on the reaction mechanisms are discussed.

12.1 INTRODUCTION

Photobiological reactions, which start with the absorption of light as a source of energy to facilitate otherwise difficult reactions involving electronic excited state(s), are not as common as biochemical reactions in the ground state, but are vital nevertheless.¹ Examples of such processes include the conversion of solar light into electrical, then chemical, energy through photosynthesis in plants, algae, and some microorganisms; DNA repairing of photoproducts by DNA photolyase²; and light-induced signal transduction in photoreceptors³ such as rhodopsin⁴ in

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the retina for visual perception and phytochromes⁵ in plants for regulation of growth. Therefore, many extensive experimental and theoretical studies have been devoted to understanding reaction mechanisms of photophysical and/or photochemical processes. However, it is challenging to elucidate entire reaction process, partly due to difficulties in characterizing all the species involved as well as the complicated nature of the associated potential energy hypersurfaces. With the recent advancement of theoretical methods and increasing computing power, aided by new experimental findings, it is anticipated that advanced quantum mechanics (QM) and quantum mechanics/molecular mechanics (QM/MM) calculations will become more and more feasible to be applied as basic tools for understanding the reaction mechanisms involved. In this chapter we first introduce common computational strategies (methods and models) in Section 12.2, discussing recent theoretical studies of selected photobiological systems through QM and QM/MM calculations in Section 12.3. The readers should refer to excellent papers and textbooks^{6–10} related to our discussions.

Before commenting on computational strategies, some background of photochemistry that should be useful in the following discussions is briefly introduced.⁶ The schematic equations of reactions and potential energy surfaces (PESs) for some common photo-induced reactions are depicted in Eqs. (12.1) to (12.4) and Figs. 12.1 and 12.2. It should be noted that realistic photophysical or/and photochemical processes could be further complicated by involving one or more intermediates, more than one electronic excited state, different vibrational states, and/or different spin states, in addition to multidimensional energy hypersurfaces. Taking the simplest case, when a reactant R at the electronic ground state (GS) (e.g., S_0 in Fig. 12.1) absorbs the radiation of one specific wavelength ($h\nu$), R is excited to the Frank–Condon (FC) region of an electronic excited state (ES) (e.g., S_1 in Fig. 12.1). The reactant R^* in the electronic excited state can lose excess energy via radiative relaxation at the fluorescent state (FS) to give emission of light with a lower wavelength ($h\nu'$) or nonradiative relaxation [e.g., internal conversion (IC), intersystem crossing (ISC), or vibrational relaxation] and finally, yield the electronic ground-state reactant R. R^* could also go through a barrier to afford an electronic excited-state product P^* , which eventually leads to the electronic ground-state product P via either the radiative or nonradiative relaxation

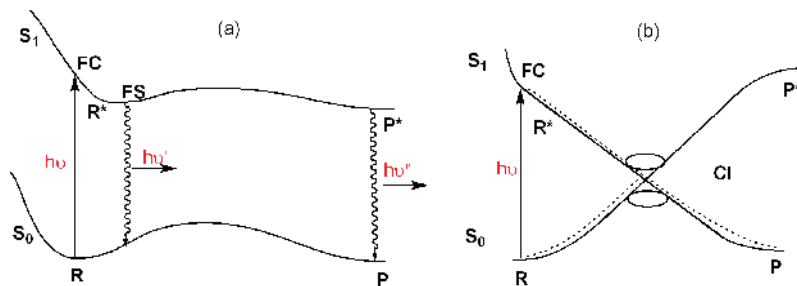


Fig. 12.1 Schematic potential energy surfaces of photo-induced reactions.

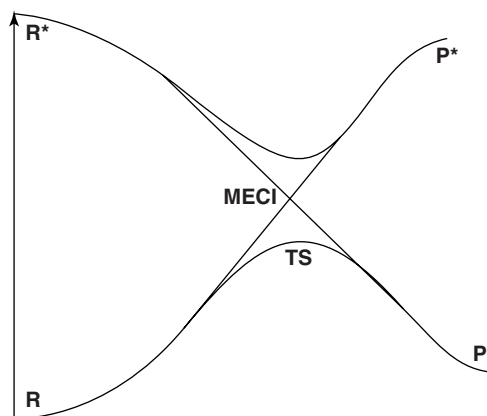


Fig. 12.2 Schematic PESs for the thermal and photo-induced reaction.

pathway (Fig. 12.1a). In addition, a branching process through a conical intersection allows R^* to generate R and P directly (Fig. 12.1b). Apart from the *photochemical reaction*,



other processes can also be initiated, such as *excitation energy transfer* (or *exciton transfer*),



electron transfer,



or its variant *hole transfer*,



Conical intersections, or seams of crossings, serve as funnels or channels connecting two electronic states^{6c,d;10f;11} and are special entities in the photophysical and photochemical processes. Conical intersections involving crossing of three electronic states have also been proposed to be involved.¹² The lowest point on the seam-of-crossing hypersurface [often abbreviated as the MSX or minimum-energy conical-intersection (MECI) point] plays a critical role in many photophysical and photochemical processes. On the other hand, conical intersections generally do not play a crucial role in thermal reactions, because the MECI is usually higher in energy than the adiabatic transition state (TS) for electronic

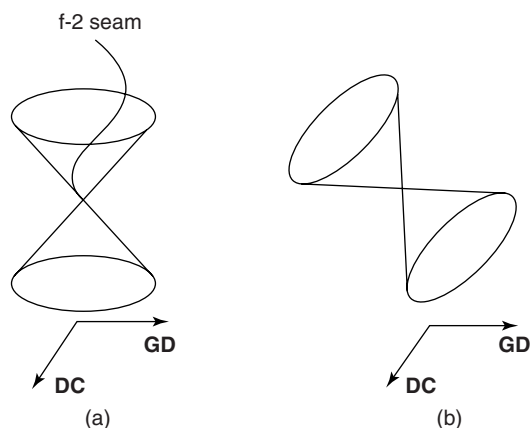


Fig. 12.3 Topology of two different conical intersections: (a) peaked; (b) sloped.

ground-state reactions (Fig. 12.2).^{7h,13} For nondiatomic molecules, when two electronic states have the same symmetry and spin, they can cross at an $(f - 2)$ -dimensional conical intersection hypersurface forming a two-dimensional funnel, where f is the number of the internal degrees of freedom ($f = 3N - 6$, where N is the number of atoms; Fig. 12.3). The two-dimensional branching space of the conical intersection consists of the gradient difference (GD) coordinate and the derivative coupling (DC) or nonadiabatic coupling coordinate. The energy degeneracy of the two electronic states is lifted when the nuclear displacement follows the GD and/or DC coordinates. Generally, conical intersection can be categorized into peaked and sloped types (Fig. 12.3).^{13,14} When the two electronic states have different symmetry or spin, these states can cross along an $(f - 1)$ -dimensional seam hypersurface and provide pathways for spin-forbidden or symmetry-forbidden processes.

12.2 COMPUTATIONAL STRATEGIES: METHODS AND MODELS

12.2.1 Quantum Mechanical Methods

Photobiological systems, involving both electronic ground- and excited-state potential energy surfaces (PESs), are challenging for quantum chemistry calculations.^{7e–g,10e,15} The nature and energetics of excitation, emission, transition states, and conical intersections are all important elements for understanding the underlying photophysical and photochemical processes.

To compute the vertical excitation energy, a reliable ground-state equilibrium geometry of the key active site (e.g., chromophore) is first needed. The density functional theory (DFT)¹⁶ method (see Chapters 1 to 3) has been used widely for ground-state geometry optimization, particularly for large systems, because it usually gives reasonable geometries, except for cases involving weak interactions,¹⁷

while requiring less computational cost than do *ab initio* wavefunction-based methods (e.g., MP2 and coupled-cluster methods). Alternatively, the multireference complete active-space self-consistent field (CASSCF)¹⁸ method is commonly employed for geometry optimization of the ground-state active site, particularly when the ground state has some multiconfigurational character (i.e., more than one electronic configuration is essentially involved); this method is also appropriate for problems involving strong electron correlations in nanotechnology (see Chapters 6 and 10). CASSCF involves an active space or orbitals in which full configuration interactions of the electrons selected is included (i.e., all possible electronic configurations in the active space are included) that has to be chosen appropriately. Due to the method's high computational cost, the choice of the active space, which is often limited to 12 orbitals, could be based on the chemical intuition and/or could be selected from the most important electrons and orbitals in some preliminary calculations [e.g., time-dependent TD-DFT, CASSCF calculations with a larger active space and very small basis sets, or a restricted active-space self-consistent field (RASSCF)¹⁹]. However, as shown in a recent rhodopsin studies,²⁰ the CASSCF does not necessarily give a better geometry than the DFT since it excludes dynamic electron correlation involving orbitals outside the active space.

According to Kasha's rule,²¹ emission of fluorescence or phosphorescence always occurs from the lowest-energy excited state of the appropriate spin manifold (e.g., S_1 or T_1). To estimate the vertical emission energy, the CASSCF method has typically been employed to give the first excited-state equilibrium geometry of the active site, largely because before this year not many QM codes provided analytic gradients for TD-DFT or CASPT2²² (CASSCF combined with multireference MP2-type perturbation theory) calculations. Also, DFT is less suited for electronic excited states because the DFT functionals are chosen or calibrated based on predicted properties of the electronic ground state, which favors an optimal "arrangement" of the electrons. In electronic excited states, multiconfigurational character is often involved. Therefore, single-reference methods (e.g., CIS, TD-DFT, or coupled-cluster-based methods) often cannot give reliable excited-state geometries. However, excitation or emission energies computed by CASSCF are not quantitatively reliable, as CASSCF does not include dynamic electron correlation. Therefore, after obtaining the optimized ground- and excited-state geometries of the active site, energetics of excitation and emission should be recalculated by more reliable excited-state methods (including single-reference methods such as TD-DFT^{7e,23} or coupled-cluster-based methods (e.g., EOM-CCSD, its equivalent SAC-CI,²⁴ CC2²⁵), as well as multireference methods such as multireference perturbation theory (e.g., CASPT2)²⁶ and multireference configuration interaction (MRCI)). Instead of the optimized geometry, snapshots from molecular dynamics simulations can also be taken to compute the excitation or emission energy. However, the computational cost is much larger and its application has been limited to less demanding excited-state methods, such as the empirical and semiempirical methods described in Chapters 8 to 10.

Different QM methods have individual advantages and weaknesses. More accurate *ab initio* QM methods are generally restricted to smaller models. On the other hand, the TD-DFT method can handle relatively large systems but is seldom very accurate or reliable, with larger errors for charge transfer (CT), Rydberg and ionic states of large π -systems, and transition metal complexes.^{7e,g} To improve excitation energy for the charge-transfer states, several long-range-corrected (LRC) DFT functionals have also been proposed and parameterized (see, e.g., Chapter 14). However, the excitation energy depends on parameters such as the range-separation parameter. Also, it is not clear yet how well these new LRC functionals describe the ground state and other excited states.²⁷ The response theory based on the coupled-cluster method, such as SAC-CI, CC2 and EOM-CC methods,^{24,25,28} can give more accurate excitation energies but requires much higher computational cost than does TD-DFT. It should be noted that single-reference TD-DFT and coupled-cluster-based methods are applicable when the ground state of the active site is determined predominantly by a single-determinant wavefunction and the electronic transition involves mainly single excitation. In comparison, the multireference methods, such as CASSCF, RASSCF, CASPT2, and MRCI, are more reliable to describe multiconfigurational electronic ground and excited states. However, because the limited active space has to be adopted for large systems, these sometimes are “irregular in behavior” and give unexpected errors (e.g., for transition metal complexes).²⁹ In addition, complicated multireference calculations involving the choice of active space have no standardized procedures and should be performed with great caution.

Recently, Thiel and co-workers performed benchmark calculations for valence excited states using TD-DFT, DFT/MRCI, CASPT2, CC2, CCSD, and CC3 methods for a set of 28 medium-sized neutral organic molecules, as representative examples of the organic chromophores.³⁰ TD-B3LYP has errors of 0.27 and 0.45 eV for the vertical singlet and triplet excitation energies, respectively. The CC3 method gives results similar to those of multistate CASPT2, while CC2 and CCSD have larger deviations.

Optimization of minima on the seam of crossing (MSXs) between two potential energy surfaces has been a very difficult task, because guessing such a structure is often beyond chemical intuition and also, for conical intersections, the energies of the two states have to be calculated from the same quantum equation. State-average approaches, in which density matrices of different states are averaged, has commonly been used in CASSCF calculations, particularly for optimization of MSXs and improving convergence difficulties due to the presence of close-lying electronic state(s).³¹ An advantage of the state-averaging approach is that the two states are obtained from the same secular equation. Therefore, they naturally represent a conical (square-root) behavior of the crossing and give a common set of orbitals for easier manipulation. However, state-averaged orbitals may not be good for either state, and the energies of either state may not be well described. Many groups developed different methods for locating MSXs.³² Our group recently developed an automatic global mapping (GRRM) search method

for MSXs,^{32c} based on a penalty function method.^{32h,k} Recently, Robb and co-workers developed a method to characterize saddle points on the intersection seam and map the minimum-energy path within the intersection space.^{32e-g}

In addition to geometry optimization, *ab initio* molecular dynamics (AIMD) simulations are important tools for studies of excited-state chemistry.^{10c,d;13a;33;34} These include both the direct dynamics methods in which PESs are calculated on the fly from QM calculations, and advanced automatic methods for incorporating such data into reliable analytic global potential energy functions for many degrees of freedom (see Chapter 7). Such treatments are often essential, as vibrationally hot excited states are often involved in the photochemical process, making the effect of internal energies significant. Trajectories with internal energies do not necessarily follow the minimum energy pathway (MEP). Additionally, the distribution of different photo-products (i.e., branching ratios) are determined by the momentum of trajectories around the conical intersections and transition states. For direct dynamics of complex systems in photobiology, classical molecular dynamics simulations are mostly employed, with the electrons treated by the QM methods to give reliable force, and then the nuclei are propagated according to Newton's equations of motion in classical mechanics. To describe nonadiabatic processes, which involve transition switching from one electronic state to another, approximate surface-hopping methods have been proposed and are widely employed in dynamics simulations.³⁵ However, the transition between two electronic states around a conical intersection or at other places where energy surfaces come very close together violates the Born–Oppenheimer approximation used to separate the electronic and nuclear motions. The quantum effects on the nuclei can sometimes be treated explicitly using more expensive quantum treatments.^{10c,d;36} In any case, the computational cost of direct dynamics simulations for excited-state calculations remains very high. Several computational tricks have been adopted to reduce the computational cost, including smaller QM models, very small active space, small basis sets, large time steps, scaled-CASSCF potential,^{34c} or even the use of TD-DFT,³⁷ semiempirical methods³⁸ (see also Chapters 8 and 9) or completely empirical methods³⁹ (see also Chapter 10).

Apart from the computed energetic profiles, the electronic nature of excitation and emission, as well as the nature of transition structures and minimum energy crossings, should be analyzed in detail by examining changes in major electronic configurations, molecular orbitals, charges, spin densities, and/or electric and magnetic dipole moments. The conical intersection can be characterized further by constructing the PES (i.e., topology) along the branching space (GD and DC, Fig. 12.3) in a simple additive manner.⁴⁰

12.2.2 Active-Site Model

Before hybrid QM/MM methods became widely available, the minimum key photo-induced fragment (e.g., the chromophore) of the protein system (the active-site model) was usually adopted and studied by gas-phase calculations. As pointed

out previously,⁴¹ the active-site model has several advantages. First, small and adequate models often give results similar to those for a very large model. Second, different feasible mechanisms can be examined carefully by efficient active-site model calculations. Third, the reliability of QM calculations for the active-site model is under much easier control than large and complex QM/MM calculations. Moreover, in many of the current QM/MM calculations, the reaction pathway is approximated by relaxed scan calculations along one or two assumed reaction coordinates, or the calculated transition state is not verified by the required but very expensive Hessian calculations. Well-characterized reaction pathways using active-site model calculations could provide the basis for QM/MM calculations. Furthermore, comparison between the active-site model calculations and QM/MM calculations can provide in-depth elucidation of the protein (e.g., geometrical and electronic) effects. To suppress excessive flexibility of the active-site model, the coordinates of some of the atoms are often kept fixed during the optimization, which yields structures closer to the model in the protein. However, this might cause some strain if the active-site changes the structure during the reaction.⁴²

12.2.3 QM/MM Methods

Although the active-site model calculation can provide some insights into photophysical and photochemical processes, the complex effects of the protein environment are totally neglected in the active-site model. Pioneered by Warshel and Levitt in 1976,⁴³ combined quantum mechanics/molecular mechanics (QM/MM) approaches were developed and have recently become a popular protocol to study the reaction mechanisms by including the entire protein as well as explicit solvent molecules.⁴⁴ Readers should refer to the excellent QM/MM reviews available.⁴⁵ In QM/MM calculations, the key and chemically important part (e.g., the chromophore), is described by a highly accurate QM method, while the rest of the protein and solvent environment is treated by very fast classical molecular mechanics (MM) force fields. There are two general approaches to evaluating the total energy (as well as gradient and Hessian) of the entire system in the QM/MM calculations: additive and extrapolation schemes. In the additive scheme, the total energy of the system is a sum of the internal energies of the QM part (E_{QM}) and MM part (E_{MM}), and the QM–MM interaction energy ($E_{\text{QM-MM}}$):

$$E_{\text{QM/MM}} = E_{\text{QM}} + E_{\text{MM}} + E_{\text{QM-MM}} \quad (12.5)$$

Alternatively, Morokuma and co-workers developed the extrapolation approach, called the *our own n-layer integrated molecular orbital + molecular mechanics* (ONIOM)⁴⁶ method (depicted in Fig. 12.4):

$$E_{\text{ONIOM(QM:MM)}} = E_{\text{QM,model}} + E_{\text{MM,real}} - E_{\text{MM,model}} \quad (12.6)$$

In both cases the QM–MM interactions include nonbonding components (van der Waals and electrostatic interactions), and bonding components (i.e., stretching, bending, and torsional terms) if a chemical bond is involved in the QM–MM

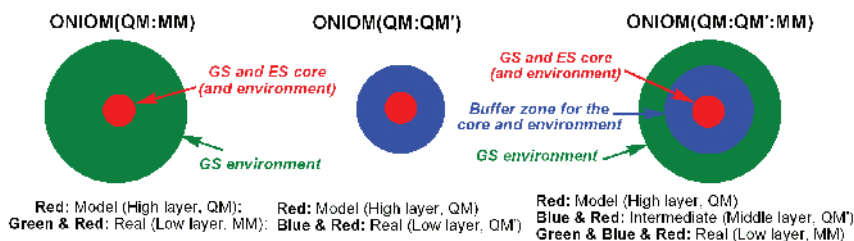


Fig. 12.4 (color online) Two- and three-layer ONIOM methods (red, center; green, outside).

boundary. In the mechanical embedding (ME) scheme, these QM–MM interactions are treated classically by the MM force field. It should be noted that almost all force fields were calibrated to describe interactions in the ground state, not in excited states. Therefore, it should be expected that the QM–MM interactions may be problematic when the QM wavefunctions for the ground state and the excited state differ considerably or change their nature during the photophysical and photochemical processes. In the electronic embedding scheme (EE),⁴⁷ the MM fixed-point charges are incorporated in the QM Hamiltonian as one-electron terms, allowing polarization of the QM wavefunction by the MM charges. The electronic embedding formalism was also developed for the ONIOM(QM:MM) and ONIOM(QM:QM') methods.^{46i,48} However, mutual polarization and charge transfer between the MM and QM parts are not accounted for in either the mechanical or the electronic embedding scheme.

Uniquely, the ONIOM extrapolation scheme can be extended to involve two different levels of QM approximation, making the ONIOM(QM:QM') and ONIOM(QM:QM':MM) schemes (also depicted in Fig. 12.4):

$$E_{\text{ONIOM(QM:QM')}} = E_{\text{QM,model}} + E_{\text{QM',real}} - E_{\text{QM',model}} \quad (12.7)$$

$$E_{\text{ONIOM(QM:QM':MM)}} = E_{\text{QM,model}} + E_{\text{QM',int}} - E_{\text{QM',model}} + E_{\text{MM,real}} - E_{\text{MM,int}} \quad (12.8)$$

Here the QM–QM' interactions, including electrostatic interactions, polarization, and charge transfer⁴⁹ (or even dispersion for some QM methods, e.g., MP2 or the B97D⁵⁰ density functional available in GAUSSIAN-09), are approximated by the lower QM method (i.e., QM'). Recently, Bearpark and co-workers employed ONIOM(QM:QM') and ONIOM(QM:MM) methods to locate conical intersection of a relatively large system, previtamin D.⁵¹ The three-layer ONIOM(QM:QM':MM) method has also been used to study the excitation energy of rhodopsin.^{46i,52} These ONIOM studies for photobiological systems significantly speed up computational time with reasonable accuracy, and should be expected to be a promising means of localized excited states of large photobiological systems.

The reliability of QM/MM methods and setups for ground-state reactions have been tested systematically by several research groups.^{45j;46f,i;53} QM/MM methods have also been applied to photobiological systems, and many QM/MM studies

have shown good agreement of the calculated absorption or emission energies with experiment. However, the QM/MM methods and setups, particularly the above-mentioned problems in the QM–MM interactions, for excited-state problems should be tested further.^{20b} In photobiological problems, one often needs to describe both the ground and several excited states at the same time. These states are often very different in electronic nature, possibly exhibiting vastly different interactions with the ground-state surroundings (the protein), which is usually treated by just the ground-state MM force field. These potential QM–MM problems can ideally be solved by expanding the QM region so that the interactions of the QM chromophore and its surroundings are treated by the QM method. Expansion of the QM region is feasible for ground-state enzymatic reactions, but it is unfortunately very difficult for photobiological problems, as these require much more expensive excited-state computational methods. Here we summarize recent approximate treatments intended to improve the description of such QM–MM interactions.

As to electrostatic interaction, in the QM/MM ME scheme, different MM charges for the QM part should be used for ground and excited states to reflect different classical QM–MM electrostatic interactions. To estimate the electrostatic interactions more accurately during the photo-induced reaction, MM charges for the QM part should be updated for both states for the classical MM calculations. The QM/MM EE scheme evaluates the QM–MM electrostatic interactions in the QM Hamiltonian, reflecting different QM densities and in principle should work for multiple electronic states in a more accurate manner, but the computational cost is increased. However, the fixed atomic MM charge approach is sometimes questioned as to whether it can realistically represent electrostatic interactions, as it may overpolarize the QM wavefunctions. The QM wavefunctions can be polarized by the MM point charges of the protein in the QM/MM EE scheme, but mutual polarization between the QM wavefunction and the MM part, and thus more realistic response of the MM part to different QM electronic states, is absent. Effects of polarization of the environment could be included through classical means⁵⁴ or quantum (via QM:QM:MM or DFI QM/MM scheme) approaches.⁵⁵

Even with QM/MM EE, charge transfer between the QM and MM parts is still excluded,⁵⁵ an effect that could be a very serious problem for some electronic states. In addition, as also discussed previously,⁵⁶ differential van der Waals interactions between the chromophore in its different electronic states and its immediate protein environment may affect the absorption energy, but the van der Waals dispersion energy evaluated by most QM/MM calculations does not depend on the QM wavefunction.

Furthermore, due to the computational cost of excited-state methods, the size of the QM part is often small, making the QM–MM boundary unavoidable close to the QM atom, significantly influencing the electronic nature of the QM part. Therefore, the QM–MM bonding interactions obtained using the common hydrogen link-atom technique might be problematic. Recently, Persico and Martínez groups adopted the connection atom approach and the effective core potential (ECP) approach, respectively, to parameterize the QM–MM boundary

individually for the ground state and excited states,⁵⁷ improving the QM–MM boundary treatment.

Although the above-mentioned “classical” means improve the QM–MM interactions, these methods are heavily depended on the parameters. To gain a better description of interactions between the QM core containing the chromophore and its immediate protein environment while minimizing computational cost and involving no parameters, the three-layer ONIOM(QM:QM':MM) method can be a feasible protocol for photobiology. Including the middle layer in the ONIOM(QM:QM':MM) method allows mutual electronic polarization and charge transfer (as well as possibly also differential dispersion) between the QM core in its different electronic states and its immediate surroundings. This layer could be treated using some lower QM' excited-state method, such as TD-DFT, TD-HF, CIS, TD-DFTB, OM2, or INDO [see Eq. 12.8 and Fig. 12.4]. Thus, the medium layer can be regarded as a QM buffer zone to combine the excited-state QM core in the high layer (e.g., by CASSCF) with the ground-state MM protein in the low layer (e.g., by AMBER).

Complexes **1** to **3**, which contain neutral phenol and anionic phenoxide moieties that represent key parts of the chromophores for green fluorescent proteins (GFP), photoactive yellow protein (PYP), or oxyluciferin, are taken as illustrative examples (Table 12.1 and Fig. 12.5). The ONIOM(QM:QM') calculations, where QM is TD-B3LYP or EOM-CCSD, reproduce the effects of the environment (H_2O in **1** and **3**; CH_3CO_2^- in **2**) on the excitation energy, and show quite a small error (at most 0.12 eV for only one case) compared to the respective full QM calculations (Table 12.1). Reliability and feasibility of ONIOM(QM:QM') and ONIOM(QM:QM':MM) methods for photobiology are being tested systematically in our laboratory and will be reported in due course. The accuracy of ONIOM(QM:QM') and ONIOM(QM:QM':MM) methods may be further enhanced by the recent QM/QM EE scheme.⁴⁸

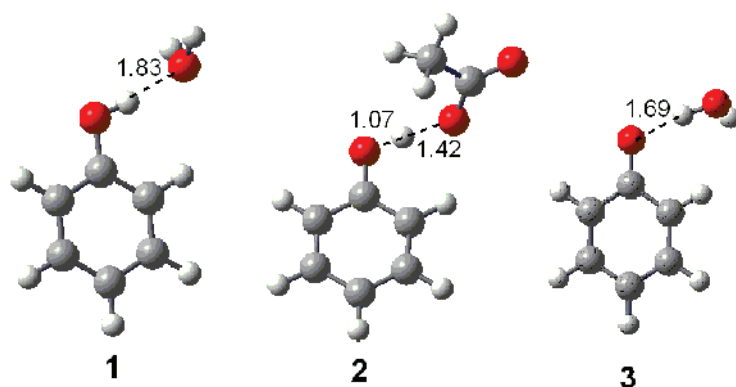


Fig. 12.5 (color online) Optimized complexes **1** to **3** by the B3LYP/6-31G* method. The phenol (**1** or **2**) or phenoxide (**3**) part was used as the QM model in the ONIOM(QM:QM') calculations.

TABLE 12.1 Vertical Excitation Energies (eV) Calculated for Three Complexes by Four Methods^{a,b}

Complex	TD-B3LYP ^c	TD-B3LYP:		EOM-CCSD ^c	EOM/CCSD:	
		TD-HF	Error		TD-HF	Error
1	5.15(−0.07)	5.18	+0.03	5.09(−0.04)	5.09	0.00
2	4.85(−0.37)	4.97	+0.12	4.91(−0.22)	4.87	−0.04
3	4.37(+0.15)	4.31	−0.06	4.49(+0.13)	4.48	−0.01

^aAll complexes were optimized by B3LYP/6-31G*. Basis set 6-31G* was also used for all excitation calculations. All calculations were carried out GAUSSIAN-03, except EOM-CCSD by MOLPRO—2006.

^bThe excitation energy of a phenol complex was calculated as 5.22 and 5.13 eV by TD-B3LYP/6-31G* and EOM-CCSD/6-31G*, respectively. The excitation energy of a phenoxide anion complex was calculated as 4.22 and 4.36 eV by TD-B3LYP/6-31G* and EOM-CCSD/6-31G*, respectively.

^cThe effects of the environment on the excitation energy are shown in parentheses.

Similarly, Elstner and co-workers developed an additive QM/QM'/MM approach with DFTB as the QM' method and studied absorption of rhodopsins, in which the Coulomb interaction between the QM and QM' parts is solved by an iterative self-consistent approach.^{55a} Fujimoto and Yang developed density–fragment interaction DFI-QM/MM and studied Mg²⁺-sensitive dye KMG-20, in which the QM layer is divided into several fragments and the interaction between the QM fragments are solved by self-consistent method (mean-field treatment).^{55b} However, charge-transfer, exchange, and dispersion interactions between the QM and QM' parts in the former approach^{55a} and those between the QM fragments in latter approach^{55b} are not included.

Modeling photobiology is one of the most challenging topics for quantum chemistry calculations. In prospect, increasing computing power and several recent methodology developments in multiscale simulations will advance our understanding of photobiology. In this regard, reliable or realistic QM/MM simulations for large systems would be achieved by combining faster or efficient QM and MM algorithms and methods, including linear-scaling methods, local correlation treatments,^{58–62} fragment-based methods (such as FMO⁶³ and Chapter 7), effective potential-based methods (e.g., effective-fragment potential (EFP)^{64–66}), semiempirical methods,³⁸ MM methods,³⁹ continuum solvent methods,⁶⁷ or even with a coarse-grain (CG) model,⁶⁸ as well as efficient sampling algorithms and methods.^{34f, 45k, 69, 70}

12.2.4 QM/MM Model and Setup

The preparation and setup for QM/MM calculations should be considered carefully, since errors caused by improper setup are difficult to estimate and correct. Various QM/MM models and setups have been addressed by different groups.^{45j; 46f,i; 53} We outline several key steps in Fig. 12.6.

1. First, the initial geometry of the photobiology system can be obtained from the Protein Data Bank, if available. Prior to the simulations, we often need

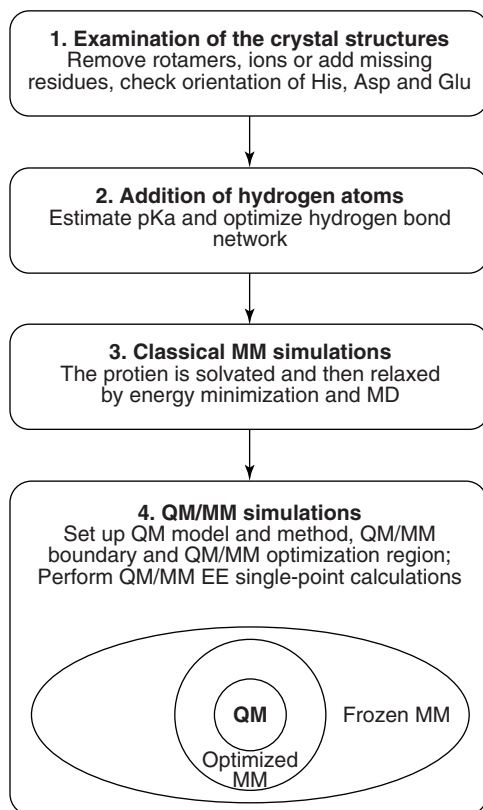


Fig. 12.6 QM/MM Setup.

to manipulate the protein structure [e.g., removing an additional rotamer and some ions for crystallization, adding missing residues (e.g., by SWISS-PDB VIEWER⁷¹) or even by homology modeling]. Since it is difficult to distinguish the electron density for carbon, nitrogen, and oxygen atoms in the modest resolution protein x-ray crystal structure, alternative orientations of histidine, asparagine, and glutamine residues should be examined by some program (e.g., WHATCHECK⁷² or MOLPROBITY⁷³) or by visual inspection in order to have better hydrogen-bonding interactions or less steric repulsion.

2. Positions of hydrogen atoms are usually missed in the x-ray crystal structures and hydrogen-bond networks could be important. The addition of hydrogen atoms and optimal hydrogen-bond networks of the proteins could be attained by some codes, (e.g., PDB2PQR, HBUILD, and WHATIF⁷⁴). Similarly, protonation states of titratable residues are also crucial, or if there is no clue from the experiment, they can be estimated by performing Poisson–Boltzmann calculations⁷⁵ or by empirical methods.⁷⁶

3. The prepared protein is then solvated by a solvent shell and relaxed by the classical MM minimization and molecular dynamics (MD) simulations. To avoid

significant changes of the protein in these simulations, the water shell could be relaxed first, followed by relaxation of all parts except the backbones, and finally, relaxation of all parts. Since common force fields for the chromophore (active site) could be missing or is not very good, the active site should be kept frozen during the MM optimization and dynamics simulations. The geometry of the active site is then refined by subsequent QM/MM optimization.

4. Based on the structures obtained from the classical simulations, the QM/MM calculations [i.e., setting QM model and method, as well as defining QM–MM boundary and QM/MM (ME or EE) method] are then performed. As far as it is affordable, it is better to have all QM–MM boundaries far away from the photo-induced active site so as to avoid discontinuity of the QM/MM PESs,⁴⁶ⁱ and key interactions with the photo-induced active site are treated by the QM method. An ideal QM–MM boundary should be an inert $C_{sp3}-C_{sp3}$ bond, if possible. These two requirements are often not met, due to the expense of excited-state calculations. The effect of expanding the QM model should always be tested. To avoid dramatic and artificial changes of the PES during QM/MM optimizations, it is highly recommended to divide the entire system into the optimized MM region surrounding the QM part and the frozen MM region in QM/MM optimization.

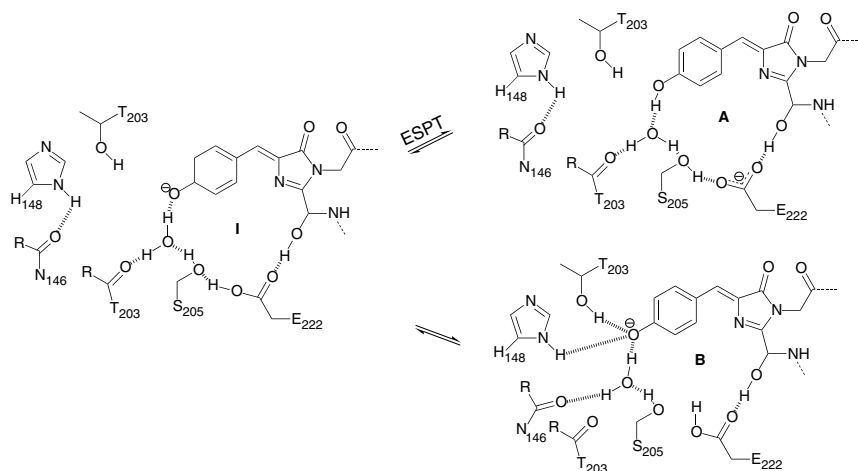
MM point charges for the QM part are often needed for classical simulations or QM/MM ME (e.g., ONIOM ME) calculations. Since procedures for computing ESP-derived charges are not implemented in some QM programs, corresponding Mulliken charges for the ground-state or excited-state QM-part can be used alternatively. Such approximated point charges led to insignificant effects in our previous studies.⁷⁷ In the ONIOM calculations, classical force field parameters of the bonding terms and dispersion for the QM part are needed and can be taken from those having atom types similar to the QM atoms.

After the QM/MM ME optimizations, energies of excitation or emission can be refined by QM/MM EE calculations. Since there could be some limitations in QM codes for QM/MM–EE calculations, for the case of single-point calculations of the excitation or emission energies, the MM contributions cancel out and only the electrostatic interactions of the MM point charges with the QM Hamiltonian needs to be evaluated. This can be accomplished by running QM calculations for the ground and excited states in the presence of the MM charges with the optimized MM positions, without using any QM/MM code. To avoid overpolarization of the QM wavefunction, the MM charges close to the QM–MM boundaries can be excluded in the QM/MM EE calculations.⁴⁶ⁱ

12.3 APPLICATIONS

12.3.1 Fluorescent Proteins

Green fluorescent protein (GFP), which was discovered from the jellyfish *Aequorea victoria*,⁷⁸ and its variants are one of the most widely studied and



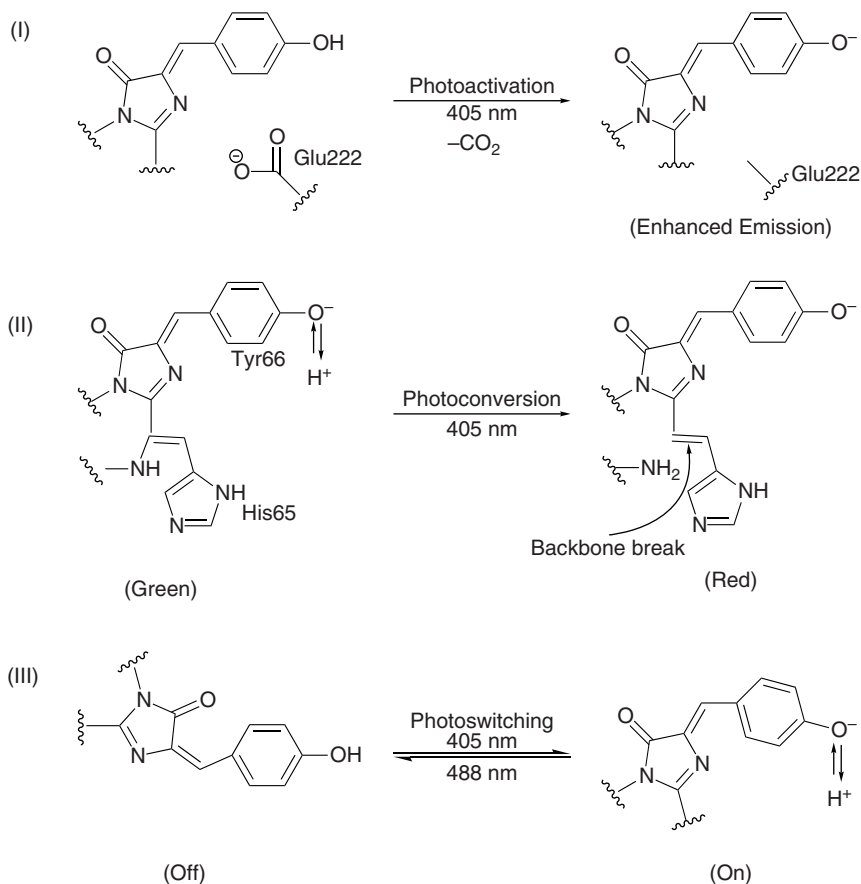
Scheme 12.1 Three-state model proposed for the photoisomerization of GFP.

exploited proteins in biochemistry and cell biology, particularly for biological imaging and analysis.^{9,79} The three-state model has been widely accepted for photoisomerization of wild-type GFP (Scheme 12.1). Upon photoexcitation of the neutral chromophore in GFP, excited-state proton transfer (ESPT) along the proton wire has been shown experimentally to operate and form the anionic state (B form), which is responsible for fluorescent emission.⁸⁰

Furthermore, a new class of fluorescent proteins, photoactivatable fluorescent proteins (PAFPs), have recently been developed, in which the photophysical and photochemical properties of the chromophore can be dramatically altered by illumination.^{81–85} Based on respective photoactivation mechanisms, PAFPs can be categorized into three types (Scheme 12.2):

1. Irreversible photoactivation (such as PA-GFP^{82e}) significantly enhances the intensity of the fluorescence by excitation with ultraviolet (UV) to violet light via decarboxylation of Glu222 to give the anionic chromophore.^{82e}
2. Irreversible photoconversion (such as Kaede, EosFP, and IrisFP): color of the fluorescence can be changed irreversibly from green to red by UV to violet light.
3. Dronpa, Padron, asFP595, IrisFP, and mTFP0.7 can undergo reversible photoswitching between fluorescent and nonfluorescent state by irradiation at appropriate wavelengths, presumably via the change of conformation and protonation state of the chromophore.

This rapid development of a new class of PAFPs has advanced fluorescent protein technologies and could be used potentially for nano applications such as molecular switches and data storage.⁸⁶



Scheme 12.2 Three types of photoactivation mechanisms.

12.3.1.1 Green Fluorescent Proteins Based on the x-ray crystal structures^{80a} and ultrafast transient infrared spectroscopy,^{80e} the neutral chromophore (called the “A form”) in GFP was proposed to be excited to undergo excited-state proton transfer (ESPT) along the hydrogen-bond wire to Glu222 via one water molecule and Ser205 (Scheme 12.1). A few theoretical studies have also been devoted to understanding the complex photoactivation process in GFP.⁸⁷ For example, recent CASPT2 calculations for the active-site model of GFP suggested that the photoactive π , π^* state is responsible for excited-state proton transfer.^{87a} rather than excited-state hydrogen transfer via the π , σ^* state with some diffuse Rydberg character in the σ^* orbital in other systems.⁸⁸ CASPT2 calculations suggested further that most energetically favorable proton-transfer pathway in the excited state initiates with a proton transfer from the serine, and finally from the chromophore to the water molecule (i.e., stepwise pathway).^{87a} A nearly concerted pathway was calculated to be only 4 kcal mol⁻¹ higher in energy than the stepwise pathway.

An essentially concerted, synchronous, and fast excited-state proton transfer and at least two dynamical regimes were later observed in a subsequent excited-state quantum dynamics simulation.^{87b}

12.3.1.2 Reversible Photoswitching Fluorescent Proteins Reversible photo-switching fluorescent proteins (RFPs) are a new class of fluorescent proteins in which fluorescent on-state and nonfluorescent off-state can be reversibly switched by irradiation at two different wavelengths (Scheme 12.2 and Fig. 12.7).^{81,84,89,90} Miyawaki and co-workers discovered one of the most promising RFPs, Dronpa, which was engineered from a coral *Pectiniidae*.^{81a} Dronpa has been employed successfully to track the protein dynamics in vivo (nucleocytoplasmic shuttling of signaling proteins).

On the basis of photophysical properties of Dronpa determined by the single-molecule spectroscopy, the reaction mechanism of reversible photoswitching of Dronpa was proposed as shown in Fig. 12.7.⁸¹ Excited-state proton transfer (ESPT) was proposed to proceed from the neutral nonfluorescent A_2 form to give an anionic nonfluorescent intermediate I, presumably in an unrelaxed protein environment, and eventually give the anionic fluorescent B form (Fig. 12.7). This reaction mechanism is analogous to the three-state photoisomerization model for wild-type GFP (Scheme 12.1).^{9b,80} The ESPT in Dronpa was supported by the kinetic deuterium isotope effect ($KIE \sim 2$).^{81c} The unknown nonfluorescent metastable D form was also proposed to account for the dynamic behavior of Dronpa.⁸¹

Recently, x-ray crystal structures of the on- and off-states of Dronpa were obtained by different groups.⁹¹ The chromophore is formed by post-translational

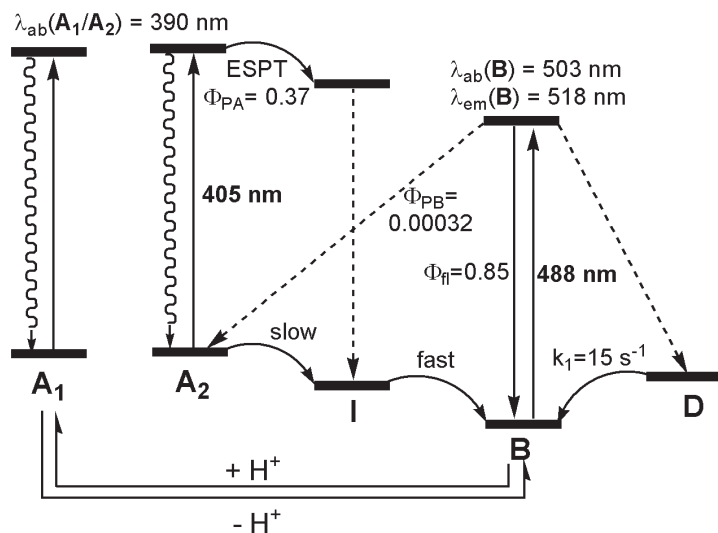


Fig. 12.7 Proposed mechanism for reversible photoswitching in Dronpa.

TABLE 12.2 Vertical Absorption Energies Calculated for Various Forms of the Chromophore in the Protein^a

Absorption (eV)	State A	State Z	State N	State C	Expt.
<i>Cis</i> (on-state)	2.36	2.21	3.09	2.42	2.46
<i>Trans</i> (off-state)	2.24	2.73	3.01	2.36	3.18

^aCalculated by the ONIOM(SAC-CI(Level2)/D95(d):AMBER)-EE method at ONIOM(B3LYP/6-31+G(d,p):AMBER)-EE optimized ground-state structures.

TABLE 12.3 Absorption and Emission Energies (eV) of *A_{cis}*, *Z_{cis}*, and *N_{trans}* in Proteins^a

Method	CASPT2	SAC-CI	Expt.
<i>Absorption</i>			
<i>A_{cis}</i>	2.71	2.42	2.46
<i>Z_{cis}</i>	3.29	2.54	
<i>N_{trans}</i>	3.24 ^{b,c}	3.18	3.18
<i>Emission</i>			
<i>A_{cis}</i>	2.42	2.08	2.39
<i>Z_{cis}</i>	1.63	0.67	
<i>N_{trans}</i>	2.84 ^{c,d}	2.12	2.76

^aCalculated by the ONIOM(CASPT2(14e,13o)/6-31G(d):AMBER)-EE and ONIOM(SAC-CI (Level2)/D95(d):AMBER)-EE methods at the ONIOM (CASSCF(14e,13o)/6-31G(d):AMBER)-ME optimized ground state (absorption) and excited state (emission) structures.

^bSA2-CASSCF(12e,12o)/6-31G(d) optimized geometry was used.

^cIt was obtained from the multistate (MS) CASPT2(12e,12o) method with including S_0 , S_1 , S_2 , and S_3 . ANO-S basis sets ([4s3p1d] for all elements except [2s1p] for the hydrogen) were used.

^dSA4-CASSCF(12e,12o)/6-31G(d) optimized geometry was used.

states were studied preliminarily by state-average (SA) CASSCF(16e,14o)/3-21G calculations to select the most important π -electrons and orbitals for our CASSCF calculations [i.e., SA-CASSCF(14e,13o)/6-31G*].

The excitation energy calculated for the anionic chromophores by the TD-B3LYP method was found to have a large error (presumably due to contributions from some character of the charge-transfer state). In addition, the TD-B3LYP method predicts similar excitation energies for different protonation states. Therefore, the TD-B3LYP method was shown to be unreliable for the determination of the protonation states in Dronpa. The conclusion based solely on the excitation energy calculated in the gas phase by SAC-CI or CASPT2 is insufficient to compare directly with experiments. Therefore, we further calculated the vertical emission in the gas phase as well as excitation and emission in the Dronpa proteins.

The ONIOM(QM:MM) EE scheme is important in evaluating reliable excitation and emission energies in the proteins, using either ONIOM-ME- or ONIOM-EE-optimized geometry. As shown in Tables 12.2 and 12.3, more elaborate ONIOM(SAC-CI:MM) calculations on the vertical absorption and emission energies in the proteins support A_{cis} and N_{trans} forms as the dominant protonation states of the chromophore in the on- and off-states, respectively. A_{cis} was supported further by the ONIOM(CASPT2:MM) calculations (Table 12.3). Both ONIOM(TD-B3LYP:MM) (not shown) and ONIOM(SAC-CI:MM) methods can reproduce absorption energy for the neutral form. However, as reported recently,⁹³ we unexpectedly found a complex situation for the neutral forms using the multireference CASSCF method. The desired excitation and emission states were found to involve S_3 , rather than S_1 , in the four-state-average SA4-CASSCF (average of S_0 , S_1 , S_2 , and S_3) calculations. The S_3 state was found to have considerable dynamic correlation and becomes lower in energy than the other two electronic excited states in the CASPT2 calculations. Thus, the SA4-CASSCF approach is required for energy evaluation and for determination of the excited-state equilibrium geometry.

The feasibility of multiconfigurational Z_{cis} , which was suggested to be involved in another RFPF, asFP595, is not supported by the excitation and emission energies calculated from multiconfiguration CASPT2 calculations. Although the excitation energy for C_{cis} is similar to the experimental value for the on-state Dronpa, it is inconsistent with the experimental pH conditions. In summary, our ONIOM calculations support the *cis*–*trans* isomerization of the chromophore along with the change in the protonation state as a feasible pathway (mechanism B in Scheme 12.3).

Although the excitation and emission energies calculated in the proteins were comparable to the experimental results, the protonation state of the chromophore in the on- and off-state Dronpa proteins was further examined by solving the linear Poisson–Boltzmann equation via the program MEAD^{75a,b} and by sampling the ensemble of protonation patterns via a Monte Carlo (MC) program, Karlsberg.⁷⁵ⁱ As shown in Table 12.4, the *trans* chromophore was calculated to be neutral (*i.e.*, N_{trans}) in the off-state Dronpa, while the *cis* chromophore was found to be essentially anionic (A_{cis}), but not zwitterionic (Z_{cis}), in the on-state Dronpa. As a result, these classical calculations further support the hypothesis that the local protein environment and *trans*–*cis* isomerization of the chromophore modulate the protonation state of the chromophore (mechanism B in Scheme 12.3).

To qualitatively estimate the effect of electrostatic interactions of nearby key residues (Glu144, Ser142, His193, and Glu211) on the stability of the anionic form in the on-state Dronpa, we further performed Poisson–Boltzmann electrostatic calculations by turning off the charges on the side chain of one of these residues. The population of A_{cis} is hardly changed by eliminating the charges of Glu144, Glu211, or His193 (ca. 93 to 100%). However, population of A_{cis} was predicted to be reduced from 100% to about 46 to 66% by excluding the charges of Ser142, which forms a hydrogen bond with the phenoxide oxygen

TABLE 12.4 Population (%) Calculated for Various Protonation States at pH 7^a

Geometry	Off-State		On-State		On-State		
	ONIOM// N _{trans} ^b	X-ray	ONIOM// A _{cis} ^c	X-ray	ONIOM// A _{cis} ^c	X-ray	ONIOM// Z _{cis} ^d
Chromophore	N: 100	N: 100	A: 100	A: 94	A: 98	A: 100	A: 100
Glu144	A: 100	A: 100	A: 100	A: 100	A: 100	A: 100	A: 100
His193	HIE: 100	HIE: 100	HIP: 99	HIP: 49	HIP: 98	HIP: 43	HIP: 92
Glu211	A: 100	A: 100	A: 99	A: 52	A: 100	A: 57	A: 88

^aThe neutral and anionic forms of chromophore are denoted by N and A, respectively. The neutral and cationic His193 are denoted by HIE and HIP, respectively. The anionic Glu144 and Glu211 are denoted by A.

^bThe ONIOM-EE optimized structure for N_{trans} in the off-state protein is used.

^cThe ONIOM-EE optimized structure for A_{cis} in the on-state protein is used.

^dThe ONIOM-EE optimized structure for Z_{cis} in the on-state protein is used.

of the chromophore. This Poisson–Boltzmann electrostatic calculation was supported qualitatively by a very recent and independent experimental mutation study, where mutation of Ser142 by Ala, Asp, Cys, or Gly in Dronpa was found to afford the neutral chromophore.^{91e} Based on systematic QM and QM/MM calculations on the excitation and emission energies as well as pK_a via the Poisson–Boltzmann electrostatic calculations, we obtained much more conclusive support of the intermediates proposed for on- and off-state Dronpa. This approach should also be useful and important in studying other photobiological systems.

The reaction mechanism for the reversibly photochromic changes in Dronpa is not clear, although it has been suggested to be affected by several factors (protonation states, conformations, nonplanarity or flexibility of the chromophore, as well as intersystem crossing). Miyawaki and co-workers first proposed that the reaction is initiated by excited-state proton transfer (ESPT) from the neutral chromophore in the off-state Dronpa and gives the anionic chromophore in the on-state Dronpa (analogous to mechanism A in Scheme 12.3).⁸¹ The mechanism involving isomerization of the chromophore leading to changes of the protonation state was calculated to be a feasible pathway in our study discussed above (mechanism B in Scheme 12.3). However, the reaction barrier of the *trans*–*cis* isomerization (the first step in the mechanism B) is still unknown. Also, this proposed pathway may not explain the observed KIE in Dronpa.^{81c} To account for the KIE observed and the conformational changes of the chromophore in the crystal structures, we proposed an alternative mechanism involving photoisomerization, followed by excited-state proton transfer in a concerted or stepwise manner, and finally by isomerization (mechanism C in Scheme 12.3). It is based on our CASSCF calculations that photoisomerization along the bond connecting to the imidazolinone ring gives a very stable twisted minimum N_{TI}. The imidazolinone and phenol rings in the excited state N_{TI} bear about -0.89 and $+0.49$ e [i.e., twisted intramolecular charge transfer

(TICT) state], respectively. Therefore, the acidity of the phenol moiety is further enhanced by the energetically favorable photoisomerization, which could promote ESPT to afford an anionic twisted intermediate (A_{TI}). Isomerization of A_{TI} eventually gives A_{cis} . Our theoretical studies on the reaction mechanism of Dronpa via photoisomerization and excited-state proton transfer are in progress.

asFP595, which was obtained from the sea anemone *Anemonia sulcata*, is another photoswitchable protein for the emission of red light.^{83b,84a} asFP595 can be switched from the nonfluorescent “off” state to the fluorescent “on” state by irradiation with green light. asFP595 has been investigated by QM and QM/MM calculations, as well as QM/MM excited-state molecular dynamics simulations.^{34f} Excited-state dynamics of the three possible protonated states (neutral, anionic, and zwitterionic) of the chromophore were explored by QM/MM(CASSCF:OPLS) molecular dynamics simulations.^{34f} Due to the substantial computational cost, a small active space and a small basis set CASSCF(6e,6o)/3-21G were used. The *trans* neutral chromophore was found to undergo *trans*-to-*cis* photoisomerization in one of the five trajectories. Upon excitation to S_1 , rotation of the imidazolinone part of the chromophore takes place to give a twisted excited-state minimum, followed by accessing conical intersection and hopping to the electronic ground state. Rotation of the phenol part of the chromophore becomes important after the decay to the electronic ground state. On the other hand, only two of the five trajectories starting from the *cis* neutral chromophore were found to decay to the electronic ground state within 10 ps, in which the *cis*-to-*trans* photoisomerization was observed in one trajectory via a pathway similar to the neutral *trans* chromophore. These simulations suggested a higher probability of the *cis*-to-*trans* isomerization than the *trans*-to-*cis* isomerization. In addition, the protein matrix was found to stabilize S_1 more than S_0 during the isomerization process, particularly at the potential surface crossing seam.

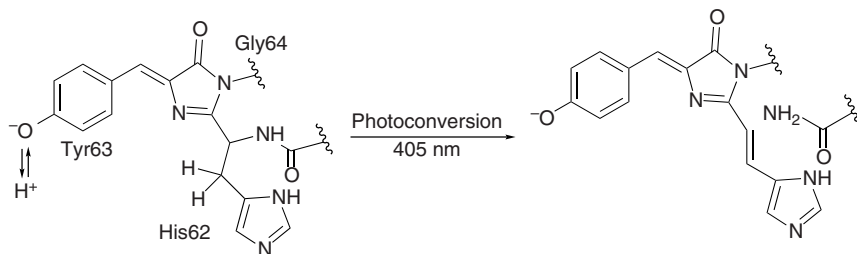
Moreover, both the *trans* and *cis* anionic chromophores were suggested to be subject to rapid radiationless decay in 20 trajectories. In addition, the lifetime of the excited-state *cis* anionic chromophore was calculated to be roughly four times longer than the *trans* anionic chromophore. Different from the neutral chromophore, the decay was driven by rotation of the phenoxide part only, and thus no isomerization was rendered. Therefore, the anionic forms were concluded to be responsible for ultrafast radiationless deactivation, especially for the non-planarity of the *trans* chromophore. Similarly, the protein environment strongly stabilized S_1 more than S_0 at the conical intersection.

The *trans* and *cis* zwitterionic chromophores were found to be stable as the planar excited-state minimum and not to undergo radiationless decay within 10 ps from 10 trajectories. This is because the minimum energy conical intersection was calculated to be higher in energy than the Frank–Condon point by 23 kJ mol⁻¹ in the gas phase. Also, the protein environment does not reduce the energy gap between S_0 and S_1 in the isomerization process. These factors were attributed to suppress radiationless relaxation for the zwitterionic chromophores. Therefore,

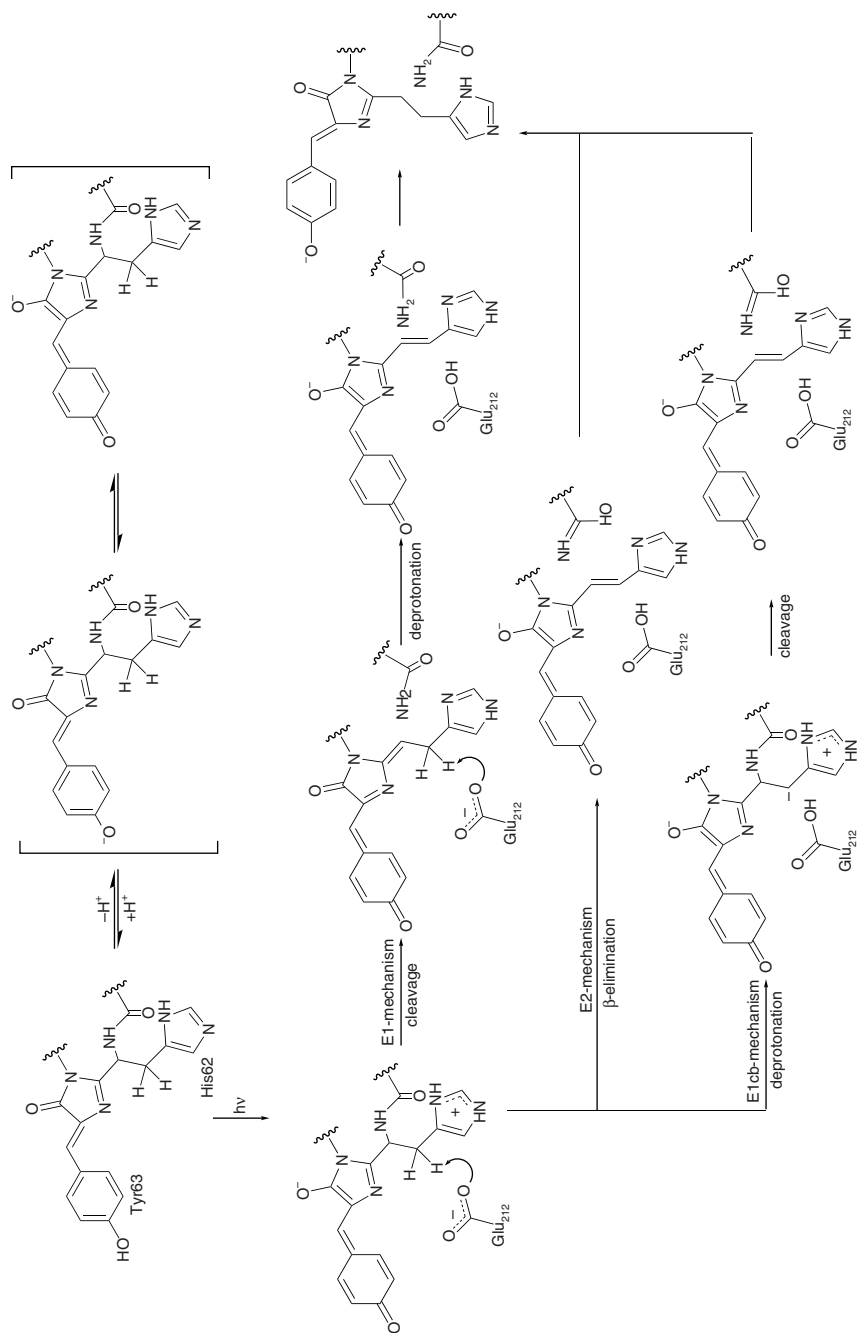
the zwitterionic forms were proposed as the putative fluorescent state for asFP595, which is different from Dronpa.

12.3.1.3 Photoconversion of Fluorescent Proteins The irreversible photoconversion fluorescent proteins, Kaede by Miyawaki's group and EosFP and IrisFP by Nienhaus's group, respectively, have been discovered from a stony coral.^{82a-c,f;84d;94} The color of fluorescence in these fluorescent proteins was found to be changed irreversibly from green to red by irradiation of UV/visible light. X-ray crystal structures of the red and green forms of Kaede, EosFP, and IrisFP have recently become available.^{82c,d;94} These three fluorescent proteins have a chromophore formed from the His62–Tyr63–Gly64 tripeptide. The UV/visible irradiation (350 to 410 nm) was shown to be important for the unusual peptide backbone breaking and extension of the π -conjugation, resulting in the change of the emission color (Scheme 12.4). The photo-induced cleavage of such peptide backbone is unique, since proteases catalyze cleavage of the more reactive peptide amide bond.^{82c,f;84d;94} The mutation studies showed that His62 and Glu212 are critical for irreversible photoconversion.^{82b,c,f;94} Accordingly, two mechanisms, the E1 and E2 mechanisms, were proposed by Miyawaki's and Nienhaus's groups, respectively (Scheme 12.5).

We performed ONIOM(B3LYP:AMBER) calculations with a very large QM model (108 QM atoms, including the chromophore, Glu212, and almost all of its nearby charged residues) to study the feasibility of the mechanisms proposed in Kaede (Scheme 12.5).⁹⁵ In the deprotonation step, electrostatic interactions of Glu212 and nearby charged residues are altered significantly. After testing several QM models in the deprotonation step, such a large QM model was found to be essential for the description of electrostatic interactions and mutual polarization. Interestingly, the stepwise E1 mechanism (i.e., the C—N bond cleavage followed by deprotonation process) was calculated to be comparable in energy to our proposed alternative E1cb mechanism (deprotonation step prior to the C—N bond cleavage) in the ONIOM(QM:MM) calculations. The reaction barriers in these two pathways are about 21 to 25 kcal mol⁻¹. However, the E2-elimination transition state (i.e., the concerted deprotonation and the C—N bond cleavage), cannot be found; instead, the calculations led to the lowest-energy E1-type transition



Scheme 12.4 UV-induced protein cleavage and green-to-red conversion of fluorescent protein Kaede or EosFP.



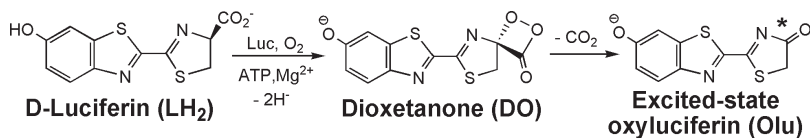
Scheme 12.5 Proposed mechanisms (E1, E2, and E1cb) of the green-to-red photoconversion reaction in Kaede and EosFP.

state. A two-dimensional PES scan showed that the proposed E2-type pathway was much higher in energy (about 34 kcal mol⁻¹).

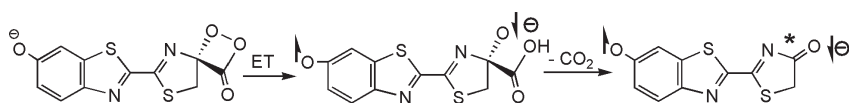
12.3.2 Luciferases

Firefly emission is a well-known efficient bioluminescence.⁶ The recently revised quantum yield (Φ_{bl}) is about 0.41.⁹⁶ The widely accepted reaction mechanism of firefly bioluminescence involves reaction of D-luciferin, ATP, Mg²⁺, and O₂ to give oxyluciferin in an electronic singlet excited state, the assumed emitter, via formation of D-luciferyl adenylate (not shown) and high-energy dioxetanone intermediate in the firefly luciferase (Luc) (Scheme 12.6). Recently, Kato and co-workers obtained x-ray crystal structures of Japanese firefly luciferase containing a high-energy intermediate analogue or oxyluciferin product,⁹⁷ which clearly showed the catalytic and emission centers of the firefly luciferase for the first time. In contrast, decomposition of simple dioxetanes or dioxetanones (without the strong electron donor) gives rise to the respective carbonyl compounds predominantly in an electronic triplet excited state rather than the electronic singlet excited state.^{6a,98} For these simple systems, a diradical mechanism or, later, “merged” mechanism initiating with homolytic O—O bond cleavage was proposed and supported by theoretical studies.⁹⁹

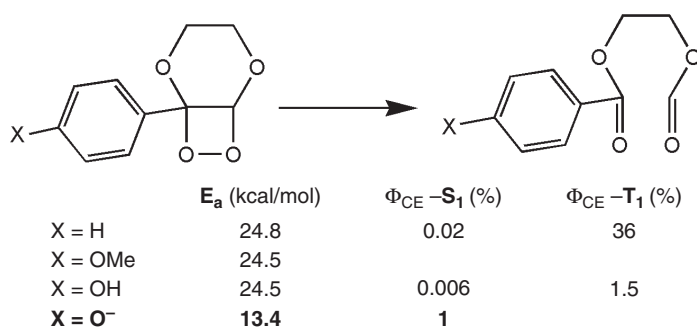
The details of efficient thermal generation of the electronic singlet excited state in firefly still remained unclear, particularly at the atomic and molecular levels. Intramolecular chemically initiated electron-exchange luminescence (CIEEL), in which one-electron transfer from the anionic phenoxide moiety to the dioxetanone part was proposed to account for chemiexcitation process in fireflies (Scheme 12.7).^{100a-c} However, the reaction mechanism via electron transfer was questioned, and a modified mechanism via charge transfer was suggested [intramolecular charge-transfer-induced chemiluminescence (CTICL)].^{98d,100d} Although a sloped conical intersection was suggested to be a key to access the electronic excited state from the ground state,^{7h,12b,14,101} the



Scheme 12.6 Firefly bioluminescence.



Scheme 12.7 Intramolecular chemically initiated electron-exchange luminescence for firefly bioluminescence.



Scheme 12.8 Substituent effect on the reaction rate of decomposition and quantum yield of the excited-state products.

nature of this peculiar channel for the efficient thermal generation of the singlet electronic excited state in fireflies was still unclear.

In addition, several model compounds were synthesized and used to study the physical and chemical properties (e.g., Scheme 12.8).^{98d,102} Similar to simple dioxetane compounds, several dioxetane compounds containing benzene or neutral phenol moieties shown in Scheme 12.8 were found to be quite stable and generate predominantly the electronic triplet excited-state product.¹⁰² However, the anionic model dioxetane compound derived from deprotonation has a considerably lower reaction barrier and with increased quantum yield for the formation of the electronic singlet excited-state product.¹⁰² In this regard, when the hydroxy group of the firefly luciferin was replaced by a methoxy group, no bioluminescence was observed.¹⁰³ These results indicate the importance of the anionic electron donor and imply the CIEEL or CTICL (electron or charge-transfer) mechanism in chemiluminescence and bioluminescence.

Recently, we performed multireference calculations (SA-CASSCF(12,12)/6-31G* and CASPT2(12,12)/6-31G*//SA-CASSCF(12,12)/6-31G*) to study bioluminescence reaction mechanism starting from the firefly dioxetanone in the active-site model.¹⁰⁴ The size of **DO** is very large, involving two important and different electronic moieties (anionic delocalized π -donor and the localized reacting four-membered ring dioxetanone). At first, we performed preliminary potential energy surface scans by B3LYP calculations for both singlet and triplet states. The DFT calculations suggested that decomposition via initial O—O bond cleavage is preferred, and this changes the ground electronic configuration of **DO** from closed-shell to an intramolecular charge-transfer (π , σ^*) state (Scheme 12.7). The intramolecular charge-transfer (π , σ^*) state formally represents the CIEEL mechanism, with one spin transferred from the π -system to the reactive dioxetanone moiety. The electronic configuration of the ground state changes from (π , σ^*) to closed shell again in the subsequent C—C cleavage process. Due to the current algorithm and computing limitations, all electrons and orbitals for the π -donor and the local excitation from the nonbonding oxygen lone pairs

from the dioxetanone moiety (n, σ^* states) cannot be included at the same time in the CASSCF calculations. The latter may contribute to the reaction, but should be less critical than the intramolecular charge transfer, according to both experimental observations and calculations.^{98d, 100a, 101, 102, 105} On the basis of the experimental observations and DFT calculations,^{102, 103} we chose (4e,4o) for the O—O and C—C bonds as well as (8e,8o) for the important π -donor as the active space of our CASSCF and CASPT2 calculations to focus on the vital intramolecular charge transfer during the reaction. In our opinion, the current state-of-art CASSCF and CASPT2 methods can provide only a qualitative, but nevertheless important picture for such complicated and large systems.

Our CASSCF and CASPT2//CASSCF calculations show that decomposition of the high-energy anionic intermediate **DO** starts with the O—O bond cleavage via an adiabatic transition state (**TS**) (Fig. 12.8). When the O—O bond is elongated to about 1.9 Å, the electronic ground- and excited-state surfaces were calculated to become similar in energy and thus the O—O bond cleavage transition state was found to be mixed with the two electronic configurations, closed-shell singlet (CSS) and π, σ^* states (Figs. 12.8 and 12.9). This suggests the occurrence of an “avoided” crossing, which creates a transition state in the ground state. Rather

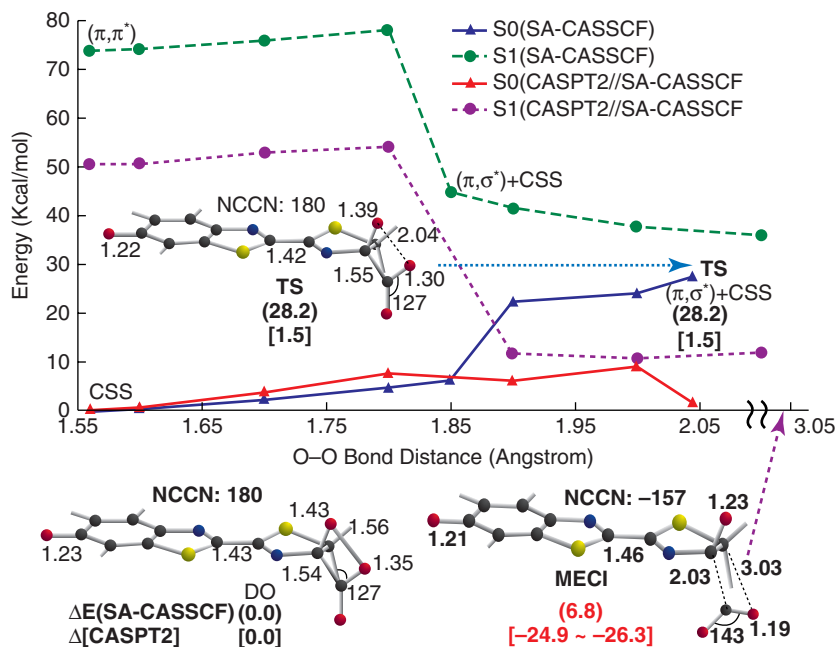


Fig. 12.8 (color online) Potential energy profile for the O—O cleavage on the S_0 and S_1 surfaces. Structures of DO, adiabatic transition state (TS), and minimum energy conical intersection (MECI) by the SA-CASSCF method are also shown. (From Ref. 102, with permission of The American Chemical Society.)

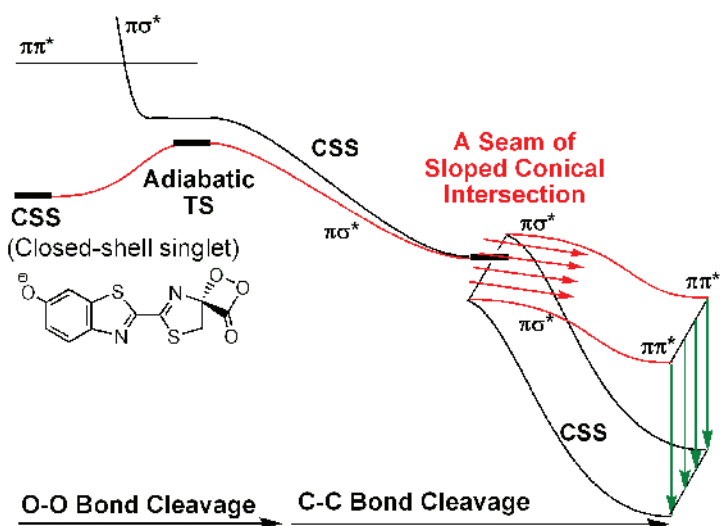


Fig. 12.9 (color online) Reaction mechanism of firefly bioluminescence based on our CASSCF calculations.

than the electron transfer proposed in CIEEL,^{100a} charge transfer was found to occur (i.e., CTICL).¹⁰⁴

When the subsequent C—C bond cleavage occurs, the two surfaces for the closed-shell and π, σ^* states can re-cross at the minimum-energy conical intersection (MECI) along MEP of the C—C bond cleavage in S_1 , a feature discovered for the first time. Interestingly, the gradient vectors on S_0 and S_1 , and the two branching space vectors (GD and DC), essentially follow the intrinsic reaction pathway of the second step (i.e., the C—C bond stretching and O—C—O bending). Since the final reaction coordinates (possible directions of the trajectories) and branching coordinates have a qualitatively similar direction, the molecule should have a high probability to encounter the MECI. In addition, the computed pathway from TS to MECI is barrierless, and a large velocity around the intersection ($\Delta E_{\text{TS}} - \Delta E_{\text{MECI}}$: 20.7 and 26.4–27.8 kcal mol⁻¹ by CASSCF and CASPT2//CASSCF, respectively) is then expected. Moreover, MECI was characterized as a sloped conical intersection, not a peaked conical intersection. Particularly, MECI resembles a $(f - 1)$ -dimensional seam [rather than the usual $(f - 2)$ -dimensional seam], in which two surfaces along one of the branching space coordinates (GD) are very close in energy (Figs. 12.3 and 12.9). Semiclassically, within the Landau–Zener model^{35a,b} and the adiabatic representation, the electronic transition probability (P_{LZ}) is equal to $\exp(-\frac{1}{4}\pi\xi)$, where the Massey parameter ξ may be estimated by $\Delta E/(\hbar v \cdot \text{DC})$.^{7d,106} The values of ΔE , v , and DC correspond to the energy difference between the two electronic states, nuclear velocity vector, and nonadiabatic coupling vector, respectively, at the crossing region. The above-mentioned features in the firefly chromophore (a small energy difference, a large kinetic energy, and a similar direction of the

final reaction coordinate and nonadiabatic coupling vector) should give a large transition probability P_{LZ} to access the electronic excited state diabatically from the electronic ground state in a widely extended channel.

In comparison, the C—C cleavage process in the ground-state surface has a much larger energy gap between the electronic ground and excited states and is also much lower in energy than MECI. Moreover, conformation change occurs, in which the oxyluciferin part becomes planar and is dominated by the closed-shell singlet state. Studies of the effects of protein and finite temperature are in progress.

12.4 CONCLUSIONS

Photobiology systems play important roles in living organisms. Thanks to recent rapid development of QM and QM/MM methods as well as increasing computational power, theoretical calculations have become an important protocol in the study of reaction mechanism in photobiology and provide new insights and understandings. Common QM and QM/MM methods to study photobiology are summarized in this chapter. Selected recent computational studies on fluorescent proteins and firefly luciferase are also discussed, which improve our understanding of photobiology systems. However, modeling reaction processes involving large QM models and different electronic states in an accurate and particularly dynamic manner is still very challenging, even for the current state-of-the-art methods, which can be regarded as one of “Holy Grails” of quantum chemistry calculations. We look forward to witnessing the development of more accurate and efficient QM and MM methodologies and algorithms so as to offer more quantitative and dynamical insights into our photobiology systems.

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REFERENCES

1. (a) Horspool, W.; Lenci, F. *CRC Handbook of Organic Photochemistry and Photobiology*, CRC Press, Boca Raton, FL, 2004. (b) Kohen, E.; Santus, R.; Hirschberg, J. G. *Photobiology*, Academic Press, San Diego, CA, 1995.
2. (a) Sancar, A. *J. Biol. Chem.* **2008**, 283, 32153. (b) Sancar, A. *Chem. Rev.* **2003**, 103, 2203.
3. (a) van der Horst, M. A.; Hellingwerf, K. J. *Acc. Chem. Res.* **2004**, 37, 13. (b) Briggs, W. R.; Huala, E. *Annu. Rev. Cell Dev. Biol.* **1999**, 15, 33. (c) Chen, M.; Chory, J.; Fankhauser, C. *Annu. Rev. Genet.* **2004**, 38, 87. (d) Lin, C.; Shalitin, D.

- Annu. Rev. Plant Biol.* **2003**, *54*, 469. (e) Hardie, R. C.; Raghu, P. *Nature* **2001**, *413*, 186. (f) Cashmore, A. R.; Jarillo, J. A.; Wu, Y.-J.; Liu, D. *Science* **1999**, *284*, 760. (g) Kennis, J. T. M.; Groot, M.-L. *Curr. Opin. Struct. Biol.* **2007**, *17*, 623. (h) Hellingwerf, K. J.; Hendriks, J.; Gensch, T. *J. Phys. Chem. A* **2003**, *107*, 1082.
4. (a) Sakmar, T. P.; Menon, S. T.; Marin, E. P.; Awad, E. S. *Annu. Rev. Biophys. Biomol. Struct.* **2002**, *31*, 443. (b) Filipek, S.; Stenkamp, R. E.; Teller, D. C.; Palczewski, K. *Annu. Rev. Physiol.* **2003**, *65*, 851. (c) Spudich, J. L.; Yang, C.-S.; Jung, K.-H.; Spudich, E. N. *Annu. Rev. Cell Dev. Biol.* **2000**, *16*, 365. (d) Hoff, W. D.; Jung, K.-H.; Spudich, J. L. *Annu. Rev. Biophys. Biomol. Struct.* **1997**, *26*, 223. (e) Rao, V. R.; Oprian, D. D. *Annu. Rev. Biophys. Biomol. Struct.* **1996**, *25*, 287. (f) Cashmore, A. R.; Jarillo, J. A.; Wu, Y.-J.; Liu, D. *Science* **1999**, *284*, 760. (g) Altun, A.; Yokoyama, S.; Morokuma, K. *Photochem. Photobiol.* **2008**, *84*, 845.
5. (a) Mroginiski, M. A.; Murgida, D. H.; Hildebrandt, P. *Acc. Chem. Res.* **2007**, *40*, 258. (b) Smith, H. *Nature* **2000**, *407*, 585.
6. (a) Turro, N. J. *Modern Molecular Photochemistry*, Benjamin-Cummings, Menlo Park, CA, 1978. (b) Adam, W.; Cilento, G. *Chemical and Biological Generation of Excited States*, Academic Press, New York, 1982. (c) Klessinger, M.; Michl, J. *Excited States and Photochemistry of Organic Molecules*, VCH, New York, 1995. (d) Domcke, W.; Yarkony, D. R.; Köppel, H. *Conical Intersections: Electronic Structure, Dynamics and Spectroscopy*, World Scientific, Singapore, 2004.
7. (a) Bernardi, F.; Olivucci, M.; Robb, M. A. *Chem. Soc. Rev.* **1996**, 321. (b) Olivucci, M. *Theoretical and Computational Chemistry*, Vol. 16, Elsevier, New York, 2005. (c) Kutateladze, A. G. *Molecular and Supramolecular Photochemistry*, Vol. 13, CRC Press, Boca Raton, FL, 2005. (d) Robb, M. A.; Garavelli, M.; Olivucci, M.; Bernardi, F. *Rev. Comput. Chem.* **2000**, *15*, 87. (e) Grimme, S. *Rev. Comput. Chem.* **2004**, *20*, 153. (f) Serrano-Andrés, L.; Merchán, M. *J. Mol. Struct. (Theochem)* **2005**, *729*, 99. (g) Dreuw, A.; Head-Gordon, M. *Chem. Rev.* **2005**, *105*, 4009. (h) Carpenter, B. K. *Chem. Soc. Rev.* **2006**, *35*, 736.
8. (a) Shimomura, O. *Bioluminescence: Chemical Principles and Methods*, World Scientific, Hackensack, NJ, 2006. (b) McCapra, F. *Methods Enzymol.* **2000**, *305*, 3. (c) Wilson, T.; Hastings, J. W. *Annu. Rev. Cell Dev. Biol.* **1998**, *14*, 197. (d) Fraga, H. *Photochem. Photobiol. Sci.* **2008**, *7*, 146.
9. (a) Conn, P. M. *Methods in Enzymology*, Vol. 302, Academic Press, San Diego, CA, 1999. (b) Zimmer, M. *Chem. Rev.* **2002**, *102*, 759. (c) Remington, S. J. *Curr. Opin. Struct. Biol.* **2006**, *16*, 714. (d) Tsien, R. Y. *Annu. Rev. Biochem.* **1998**, *67*, 509. (e) Chalfie, M.; Kain, S. R. *Green Fluorescent Protein: Properties, Applications and Protocols*, Wiley-Interscience, Hoboken, NJ, 2006. (f) Sullivan, K. F. *Methods in Cell Biology*, Vol. 85, Academic Press, London, 2008. (g) Lippincott-Schwartz, J.; Altan-Bonnet, N.; Patterson, G. H. *Nature Cell Biol.* **2003**, *5*, S7. (h) Miyawaki, A.; Sawano, A.; Kogure, T. *Nature Cell Biol.* **2003**, *5*, S1. (i) Chudakov, D. M.; Lukyanov, S.; Lukyanov, K. A. *Trends Biotechnol.* **2005**, *23*, 605. (j) Zhang, J.; Campbell, R. E.; Ting, A. Y.; Tsien, R. Y. *Nature Rev. Mol. Cell Biol.* **2002**, *3*, 906. (k) Shaner, N. C.; Patterson, G. H.; Davidson, M. W. *J. Cell Sci.* **2007**, *120*, 4247. (l) Henderson, J. N.; Remington, S. J. *Physiol.* **2006**, *21*, 162. (m) Lukyanov, K. A.; Chudakov, D. M.; Lukyanov, S.; Verkhusha, V. V. *Nature Rev. Mol. Cell Biol.* **2005**, *6*, 885. (n) Lippincott-Schwartz, J.; Patterson, G. H. *Science* **2003**, *300*, 87.
10. (a) Roos, B. O. *Acc. Chem. Res.* **1999**, *32*, 137. (b) Helms, V. *Curr. Opin. Struct. Biol.* **2002**, *12*, 169. (c) Olsen, S.; Toniolo, A.; Ko, C.; Manohar, L.; Lamothe, K.;

- Martínez, T. J. In *Theoretical and Computational Chemistry*, Vol. 16, Olivucci, M., Ed., Elsevier, New York, 2005, pp. 225–254. (d) Martínez, T. J. *Acc. Chem. Res.* **2006**, 39, 119. (e) Dreuw, A. *ChemPhysChem* **2006**, 7, 2259. (f) Levine, B. G.; Martínez, T. J. *Annu. Phys. Chem.* **2007**, 58, 613.
11. (a) Michl, J. *Mol. Photochem.* **1972**, 243. (b) Klessinger, M. *Angew. Chem. Int. Ed. Engl.* **1995**, 34, 549. (c) Yarkony, D. R. *J. Phys. Chem. A* **2001**, 105, 6277.
 12. (a) Matsika, S.; Yarkony, D. R. *J. Am. Chem. Soc.* **2003**, 125, 10672. (b) Blancafort, L.; Robb, M. A. *J. Phys. Chem. A* **2004**, 108, 10609. (c) Coe, J. D.; Martínez, T. J. *J. Am. Chem. Soc.* **2005**, 127, 4560.
 13. (a) Blancafort, L.; Ogliaro, F.; Olivucci, M.; Robb, M. A.; Bearpark, M. J.; Sinicropi, A. In *Molecular and Supramolecular Photochemistry*, Vol. 13, Kutateladze, A. G., Ed., CRC Press, Boca Raton, FL, 2005, pp. 31–110. (b) Blancafort, L.; Jolibois, F.; Olivucci, M.; Robb, M. A. *J. Am. Chem. Soc.* **2001**, 123, 722.
 14. Atchity, G. J.; Xantheas, S. S.; Ruedenberg, K. *J. Chem. Phys.* **1991**, 95, 1862.
 15. Bearpark, M. J.; Ogliaro, F.; Vreven, T.; Boggio-Pasqua, M.; Frisch, M. J.; Larkin, S. M.; Morrison, M.; Robb, M. A. *J. Photochem. Photobiol. A* **2007**, 190, 207.
 16. (a) Kohn, W.; Sham, L. J. *Phys. Rev.* **1965**, 140, A1133. (b) Hohenberg, P.; Kohn, W. *Phys. Rev.* **1964**, 136, B864. (c) Kohn, W.; Holthausen, M. C. A. *Chemist's Guide to Density Functional Theory*, Wiley-VCH, Weinheim, Germany, 2000. (d) Kohn, W.; Becke, A. D.; Parr, R. G. *J. Phys. Chem.* **1996**, 100, 12974. (e) Parr, R. G.; Yang, W. *Density Functional Theory of the Electronic Theoretical Methods for Structure of Molecules*, Oxford University Press, New York, 1989.
 17. (a) Schwabe, T.; Grimme, S. *Acc. Chem. Res.* **2008**, 41, 569. (b) Zhao, Y.; Truhlar, D. G. *Acc. Chem. Res.* **2008**, 41, 157, and references therein.
 18. (a) Roos, B. O. *Adv. Chem. Phys.* **1987**, 69, 399. (b) Cramer, C. J. *Essentials of Computational Chemistry*, Wiley, Chichester, UK, 2004, pp. 203–223.
 19. Malmqvist, P.-Å.; Rendell, A.; Roos, B. O. *J. Phys. Chem.* **1990**, 94, 5477.
 20. (a) Altun, A.; Yokoyama, S.; Morokuma, K. *J. Phys. Chem. B* **2008**, 112, 16883. (b) Wanko, M.; Hoffmann, M.; Strodel, P.; Koslowski, A.; Thiel, W.; Neese, F.; Frauenheim, T.; Elstner, M. *J. Phys. Chem. B* **2005**, 109, 3606.
 21. Kasha, M. *Discuss. Faraday Soc.* **1950**, 9, 14.
 22. (a) Roos, B. O.; Andersson, K. *Chem. Phys. Lett.* **1995**, 245, 215. (b) Andersson, K.; Malmqvist, P.-Å.; Roos, B. O. *J. Chem. Phys.* **1992**, 96, 1218.
 23. (a) Runge, E.; Gross, E. K. U. *Phys. Rev. Lett.* **1984**, 52, 997. (b) Elliot, P.; Furche, F.; Burke, K. *Rev. Comput. Chem.* **2009**, 26, 91.
 24. (a) Nakatsuji, H. *Chem. Phys. Lett.* **1978**, 59, 362. (b) Nakatsuji, H. *Chem. Phys. Lett.* **1989**, 67, 329. (c) Nakatsuji, H. *Chem. Phys. Lett.* **1989**, 67, 334.
 25. Christiansen, O.; Koch, H.; Jørgensen, P. *Chem. Phys. Lett.* **1995**, 243, 409.
 26. (a) Roos, B. O.; Andersson, K. *Chem. Phys. Lett.* **1995**, 245, 215. (b) Andersson, K.; Malmqvist, P.-Å.; Roos, B. O. *J. Chem. Phys.* **1992**, 96, 1218.
 27. (a) Yanai, T.; Tew, D. P.; Handy, N. C. *Chem. Phys. Lett.* **2004**, 393, 51. (b) Peach, M. J. G.; Benfield, P.; Helgaker, T.; Tozer, D. J. *J. Chem. Phys.* **2008**, 128, 044118. (c) Rohrdanz, M. A.; Martins, K. M.; Herbert, J. M. *J. Chem. Phys.* **2009**, 130, 054112. (d) Rohrdanz, M. A.; Herbert, J. M. *J. Chem. Phys.* **2008**, 129, 034107. (e) Jacquemin, D.; Perpète, É. A.; Scuseria, G. E.; Ciofini, I.; Adamo, C. *J. Chem. Theory Comput.* **2008**, 4, 123.

28. (a) Ehara, M.; Hasegawa, J.; Nakatsuji, H. In *Theory and Applications of Computational Chemistry: The First 40 Years*, Dykstra, C. E., Frenking, G., Kim, K. S., and Scuseria, G. E., Eds., Elsevier, Oxford, UK, 2005, pp. 1099–1141. (b) Bartlett, R. J. In *Theory and Applications of Computational Chemistry: The First 40 Years*, Dykstra, C. E., Frenking, G., Kim, K. S., and Scuseria, G. E., Eds., Elsevier, Oxford, UK, 2005, pp. 1191–1221.
29. Siegbahn, P. E. M.; Borowski, T. *Acc. Chem. Res.* **2006**, 39, 729.
30. (a) Silva-Junior, M. R.; Schreiber, M.; Sauer, S. P. A.; Thiel, W. *J. Chem. Phys.* **2008**, 129, 104103. (b) Schreiber, M.; Silva-Junior, M. R.; Sauer, S. P. A.; Thiel, W. *J. Chem. Phys.* **2008**, 129, 134110.
31. (a) Diffenderfer, R. N.; Yarkony, D. R. *J. Phys. Chem.* **1982**, 86, 5098. (b) Docken, K. K.; Hinze, J. *J. Chem. Phys.* **1972**, 57, 4928.
32. (a) Koga, N.; Morokuma, K. *Chem. Phys. Lett.* **1985**, 119, 371. (b) Cui, Q.; Morokuma, K.; Stanton, J. F. *Chem. Phys. Lett.* **1996**, 263, 46. (c) Maeda, S.; Ohno, K.; Morokuma, K. *J. Phys. Chem. A* **2009**, 113, 1704. (d) Bearpark, M. J.; Robb, M. A.; Schlegel, H. B. *Chem. Phys. Lett.* **1994**, 223, 269. (e) Sicilia, F.; Blancafort, L.; Bearpark, M. J.; Robb, M. A. *J. Chem. Theory Comput.* **2008**, 4, 257. (f) Sicilia, F.; Blancafort, L.; Bearpark, M. J.; Robb, M. A. *J. Phys. Chem. A* **2007**, 111, 2182. (g) Sicilia, F.; Bearpark, M. J.; Blancafort, L.; Robb, M. A. *Theor. Chem. Acc.* **2007**, 118, 241. (h) Levine, B. G.; Coe, J. D.; Martínez, T. J. *J. Phys. Chem. B* **2008**, 112, 405. (i) Manaa, M. R.; Yarkony, D. R. *J. Chem. Phys.* **1993**, 99, 5251. (j) Harvey, J. N.; Aschi, M.; Schwarz, H.; Koch, W. *Theor. Chem. Acc.* **1998**, 99, 95. (k) Ciminelli, C.; Granucci, G.; Persico, M. *Chem. Eur. J.* **2004**, 10, 2327. (l) Keal, T. W.; Koslowski, A.; Thiel, W. *Theor. Chem. Acc.* **2007**, 118, 837.
33. Virshup, A. M.; Punwong, C.; Pogorelov, T. V.; Lindquist, B. A.; Ko, C.; Martínez, T. J. *J. Phys. Chem. B* **2009**, 113, 3280.
34. Recent QM/MM MD simulations: (a) Hayashi, S.; Tajkhorshid, E.; Schulten, K. *Biophys. J.* **2003**, 85, 1440. (b) Hayashi, S.; Tajkhorshid, E.; Schulten, K. *Bio-phys. J.* **2009**, 96, 403. (c) Frutos, L. M.; Andruniów, T.; Santoro, F.; Ferré, N.; Olivucci, M. *Proc. Natl. Acad. Sci. USA* **2007**, 104, 7764. (d) Groenhof, G.; Bouxin-Cademartory, M.; Hess, B.; de Visser, S. P.; Berendsen, H. J. C.; Olivucci, M.; Mark, A. E.; Robb, M. A. *J. Am. Chem. Soc.* **2004**, 126, 4228. (e) Groenhof, G.; Schäfer, L. V.; Boggio-Pasqua, M.; Grubmüller, H.; Robb, M. A. *J. Am. Chem. Soc.* **2008**, 130, 3250. (f) Schäfer, L. V.; Groenhof, G.; Boggio-Pasqua, M.; Robb, M. A.; Grubmüller, H. *PLoS Comput. Biol.* **2008**, 4, e1000034. (g) Toniolo, A.; Granucci, G.; Martínez, T. J. *J. Phys. Chem. A* **2003**, 107, 3822. (h) Toniolo, A.; Olsen, S.; Manohar, L.; Martínez, T. J. *Faraday Discuss.* **2004**, 127, 149.
35. Landau–Zener model: (a) Zener, C. *Proc. R. Soc. London A* **1932**, 137, 696. (b) Landau, L. D. *Phys. Z. Sowiet.* **1932**, 2, 46. (c) Fewest switches model: Tully, J. C. *J. Chem. Phys.* **1990**, 93, 1061. (d) Hammes-Schiffer, S.; Tully, J. C. *J. Chem. Phys.* **1994**, 101, 4657.
36. Worth, G. A.; Cederbaum, L. S. *Annu. Rev. Phys. Chem.* **2004**, 55, 127.
37. (a) Tavernelli, I.; Tapavicza, E.; Rothlisberger, U. *J. Chem. Phys.* **2009**, 130, 124107. (b) Tavernelli, I.; Tapavicza, E.; Rothlisberger, U. *Phys. Rev. Lett.* **2007**, 98, 023001.
38. DFTB: (a) Niehaus, T. A.; Suhai, S.; Della Sala, F.; Lugli, P.; Elstner, M.; Seifert, G.; Frauenheim, Th. *Phys. Rev. B* **2001**, 63, 085108. OM2: (b) Weber, W.; Thiel, W. *Theor. Chem. Acc.* **2000**, 103, 495. FOMO-CI: (c) Toniolo, A.; Granucci, G.;

- Martinez, T. J. *J. Phys. Chem. A* **2003**, *107*, 3822. MNDOC-CI: (d) Klessinger, M.; Pötter, T.; van Wüllen, C. *J. Theor. Chim. Acta* **1991**, *80*, 1. FOMO: (e) Granucci, G.; Persico, M.; Toniolo, A. *J. Chem. Phys.* **2001**, *114*, 10608.
39. MMVB: (a) Bernardi, F.; Olivucci, M.; Robb, M. A. III *J. Am. Chem. Soc.* **1992**, *114*, 1606. eFF: (b) Su, J. T.; Goddard, W. A. *Phys. Rev. Lett.* **2007**, *99*, 185003.
40. An approximate second-order analysis: Paterson, M. J.; Bearpark, M. J.; Robb, M. A.; Blancafort, L. *J. Chem. Phys.* **2004**, *121*, 11562.
41. Siegbahn, P. E. M. *J. Biol. Inorg. Chem.* **2006**, *11*, 695.
42. Chen, S.-L.; Fang, W.-H.; Himo, F. *Theor. Chem. Acc.* **2008**, *120*, 515.
43. Warshel, A.; Levitt, M. *J. Mol. Biol.* **1976**, *103*, 227.
44. (a) Singh, U. C.; Kollman, P. A. *J. Comput. Chem.* **1986**, *7*, 718. (b) Field, M. J.; Bash, P. A.; Karplus, M. *J. Comput. Chem.* **1990**, *11*, 700.
45. (a) Gao, J. *Rev. Comput. Chem.* **1996**, *7*, 119. (b) Monard, G.; Merz, K. M., Jr. *Acc. Chem. Res.* **1999**, *32*, 904. (c) Gao, J.; Truhlar, D. G. *Annu. Rev. Phys. Chem.* **2002**, *53*, 467. (d) Field, M. J. *J. Comput. Chem.* **2002**, *23*, 48. (e) Garcia-Viloca, M.; Gao, J.; Karplus, M.; Truhlar, D. G. *Science*, **2004**, *303*, 186. (f) Friesner, R. A.; Guallar, V. *Annu. Rev. Phys. Chem.* **2005**, *56*, 389. (g) Mulholland, A. J. *Drug Discov. Today* **2005**, *10*, 1393. (h) Warshel, A.; Sharma, P. K.; Kato, M.; Xiang, Y.; Liu, H. B.; Olsson, M. H. M. *Chem. Rev.* **2006**, *106*, 3210. (i) Lin, H.; Truhlar, D. G. *Theor. Chem. Acc.* **2007**, *117*, 185. (j) Senn, H. M.; Thiel, W. *Top. Curr. Chem.* **2007**, *268*, 173. (k) Hu, H.; Yang, W. *Annu. Rev. Phys. Chem.* **2008**, *59*, 573. (l) Senn, H. M.; Thiel, W. *Angew. Chem., Int. Ed.* **2008**, *47*, 1198.
46. (a) Maseras, F.; Morokuma, K. *J. Comput. Chem.* **1995**, *16*, 1170. (b) Humbel, S.; Sieber, S.; Morokuma, K. *J. Chem. Phys.* **1996**, *105*, 1959. (c) Matsubara, T.; Sieber, S.; Morokuma, K. *Int. J. Quantum Chem.* **1996**, *60*, 1101. (d) Svensson, M.; Humbel, S.; Froese, R. D. J.; Matsubara, T.; Sieber, S.; Morokuma, K. *J. Phys. Chem.* **1996**, *100*, 19357. (e) Svensson, M.; Humbel, S.; Morokuma, K. *J. Chem. Phys.* **1996**, *105*, 3654. (f) Dapprich, S.; Komáromi, I.; Byun, S.; Morokuma, K.; Frisch, M. J. *J. Mol. Struct. (Theochem)* **1999**, *461*, 1. (g) Vreven, T.; Morokuma, K. *J. Comput. Chem.* **2000**, *21*, 1419. (h) Vreven, T.; Frisch, M. J.; Kudin, K. N.; Schlegel, H. B.; Morokuma, K. *Mol. Phys.* **2006**, *104*, 701. (i) Vreven, T.; Byun, K. S.; Komáromi, I.; Dapprich, S.; Montgomery, J. A. K., Jr.; Morokuma, K.; Frisch, M. J. *J. Chem. Theory Comput.* **2006**, *2*, 815. (j) Vreven, T.; Morokuma, K. *Annal. Rep. Comput. Chem.* **2006**, *2*, 35.
47. Bakowies, D.; Thiel, W. *J. Phys. Chem.* **1996**, *100*, 10580.
48. Hratchian, H. P.; Parandekar, P. V.; Raghavachari, K.; Frisch, M. J.; Vreven, T. *J. Chem. Phys.* **2008**, *128*, 34107.
49. Morokuma, K. *Acc. Chem. Res.* **1977**, *10*, 294.
50. Grimme, S. *J. Comput. Chem.*, **2006**, *27*, 1787.
51. Bearpark, M. J.; Larkin, S. M.; Vreven, T. *J. Phys. Chem. A* **2009**, *112*, 7286.
52. Hall, K. F.; Vreven, T.; Frisch, M. J.; Bearpark, M. J. *J. Mol. Biol.* **2008**, *383*, 106.
53. (a) Eurenium, K. P.; Chatfield, D. C.; Brooks, B. R.; Hodoscek, M. *Int. J. Quantum Chem.* **1996**, *60*, 1189. (b) Koenig, P.; Hoffman, M.; Frauenheim, T.; Cui, Q. *J. Phys. Chem. B* **2005**, *109*, 9082. (c) Altun, A.; Shaik, S.; Thiel, W. *J. Comput. Chem.* **2006**, *27*, 1324. (d) Zheng, J.; Altun, A.; Shaik, S.; Thiel, W. *J. Comput. Chem.* **2007**, *28*, 2147. (e) Lin, H.; Truhlar, D. G. *J. Phys. Chem. A* **2005**, *109*, 3991.

54. (a) Warshel, A.; Zhu, Z. T. *J. Phys. Chem. B* **2001**, *105*, 9857. (b) Luzhkov, V.; Warshal, A. *J. Am. Chem. Soc.* **1991**, *113*, 4491. (c) Thompson, M. A.; Schenter, G. K. *J. Phys. Chem.* **1995**, *99*, 6374. (d) Gao, J.; Alhambra, C. *J. Am. Chem. Soc.* **1997**, *119*, 2962. (e) Matsuura, A.; Sato, H.; Houjou, H.; Saito, S.; Hayashi, T.; Sakurai, M. *J. Comput. Chem.* **2006**, *27*, 1623.
55. (a) Wanko, M.; Hoffmann, M.; Frähmcke, J.; Frauenheim, T.; Elstner, M. *J. Phys. Chem. B* **2008**, *112*, 11468. (b) Fujimoto, K.; Yang, W. *J. Chem. Phys.* **2008**, *129*, 054102.
56. Wanko, M.; Hoffmann, M.; Frauenheim, T.; Elstner, M. *J. Phys. Chem. B* **2008**, *112*, 11462.
57. (a) Toniolo, A.; Ciminelli, C.; Granucci, G.; Laino, T.; Persico, M. *Theor. Chem. Acc.* **2004**, *111*, 270. (b) Slavíček, P.; Martínez, T. J. *J. Chem. Phys.* **2006**, *124*, 084107.
58. Ochsenfeld, C.; Kussmann, J.; Lambrecht, D. S. *Rev. Comput. Chem.* **2007**, *23*, 1.
59. Werner, H.-J.; Pflüger, K. *Annu. Rep. Comput. Chem.* **2006**, *2*, 53.
60. Graphics processing unit (GPU): Ufimtsev, I. S.; Martínez, T. J. *J. Chem. Theory Comput.* **2008**, *4*, 222.
61. Aquilante, F.; Malmqvist, P.-Å.; Pedersen, T. B.; Ghosh, A. B.; Roos, O. J. *Chem. Theory Comput.* **2008**, *4*, 694.
62. (a) White, S. R. *Phys. Rev. Lett.* **1992**, *69*, 2863. (b) Chan, G. K.-L.; Head-Gordon, M. *J. Chem. Phys.* **2002**, *116*, 4462. (c) Ghosh, D.; Hachmann, J.; Yanai, T.; Chan, G. K.-L. *J. Chem. Phys.* **2008**, *128*, 144117.
63. Fedorov, D. G.; Kitaura, K. *J. Phys. Chem. A* **2007**, *111*, 6904.
64. (a) Gordon, M. S.; Freitag, M. A.; Bandyopadhyay, P.; Jensen, J. H.; Kairys, V.; Stevens, W. J. *J. Phys. Chem. A* **2001**, *105*, 293. (b) Gordon, M. S.; Mullin, J. M.; Pruitt, S. R.; Roskop, L. B.; Slipchenko, L. V.; Boatz, J. A. *J. Phys. Chem. B* **2009**, *113*, 9646.
65. Ohta, Y.; Yoshioka, K.; Morokuma, K.; Kitaura, K. *Chem. Phys. Lett.* **1983**, *101*, 12.
66. Effective group potential (EGP): (a) Poteau, R.; Ortega, I.; Alary, F.; Solis, A. R.; Barthelat, J.-C.; Daudey, J.-P. *J. Phys. Chem. A* **2001**, *105*, 198. (b) Heully, J.-L.; Poteau, R.; Berasaluce, S.; Alary, F. *J. Chem. Phys.* **2002**, *116*, 4829.
67. (a) Hayik, S. A.; Liao, N.; Merz, K. M., Jr. *J. Chem. Theory Comput.* **2008**, *4*, 1200. (b) Schaefer, P.; Riccardi, D.; Cui, Q. *J. Chem. Phys.* **2005**, *123*, 14905.
68. Voth, G. A. *Coarse-Graining of Condensed Phase and Biomolecular Systems*, CRC Press, Boca Raton, FL, 2008.
69. (a) Kühne, T. D.; Krack, M.; Mohamed, F. R.; Parrinello, M. *Phys. Rev. Lett.* **2007**, *98*, 66401. (b) Laio, A.; Parrinello, M. *Proc. Natl. Acad. Sci. USA* **2002**, *99*, 12562. (c) Car, R.; Parrinello, M. *Phys. Rev. Lett.* **1985**, *55*, 2471. (d) Ong, M. T.; Leiding, J.; Tao, H.; Virshup, A. M.; Martínez, T. J. *J. Am. Chem. Soc.* **2009**, *131*, 6377.
70. QM/MM FEP: (a) Zhang, Y.; Liu, H.; Yang, W. *J. Chem. Phys.* **2000**, *112*, 3483. QM/MM MFEP: (b) Hu, H.; Lu, Z.; Yang, W. *J. Chem. Theory Comput.* **2007**, *3*, 390.
71. Guex, N.; Peitsch, M. C. *Electrophoresis* **1997**, *18*, 2714.
72. (a) Hooft, R. W. W.; Vriend, G.; Sander, C.; Abola, E. E. *Nature* **1996**, *381*, 272. (b) <http://swift.cmbi.ru.nl/gv/pdbreport/>.

73. (a) Lovell, S. C.; Davis, I. W.; Arendall, W. B., III; de Bakker, P. I. W.; Word, J. M.; Prisant, M. G.; Richardson, J. S.; Richardson, D. C. *Proteins Struct. Funct. Genet.* **2003**, *50*, 437. (b) <http://molprobit.biochem.duke.edu/>.
74. (a) Dolinsky, T. J.; Nielsen, J. E.; McCammon, J. A.; Baker, N. A. *Nucleic Acids Res.* **2004**, *32*, W665. (b) Brünger, A. T.; Karplus, M. *Proteins* **1988**, *4*, 148. (c) Hooft, R. W.; Sander, C.; Vriend, G. *Proteins* **1996**, *26*, 363.
75. (a) Bashford, D.; Karplus, M. *Biochemistry* **1990**, *29*, 10219. (b) Bashford, D.; Gerwert, K. *J. Mol. Biol.* **1992**, *224*, 473. (c) Gordon, J. C.; Myers, J. B.; Folta, T.; Shoja, V.; Heath, L. S.; Onufriev, A. *Nucleic Acids Res.* **2005**, *33*, W368. (d) <http://biophysics.cs.vt.edu/H++>. (e) <http://bioserv.rpbs.jussieu.fr/cgi-bin/PCE-pKa>. (f) Nicholls, A.; Honig, B. *J. Comput. Chem.* **1991**, *12*, 435. (g) Yang, A.-S.; Gunner, M. R.; Sampogna, R.; Sharp, K.; Honig, B. *Proteins* **1993**, *15*, 252. (h) Baker, N. A.; Sept, D.; Joseph, S.; Holst, M. J.; McCammon, J. A. *Proc. Natl. Acad. Sci. USA* **2001**, *98*, 10037. (i) Rabenstein, B. *Karlsberg Online Manual*, 1999, <http://agknapp.chemie.fu-berlin.de/karlsberg>.
76. PROPKA: (a) Li, H.; Robertson, A. D.; Jensen, J. H. *Proteins Struct. Funct. Bioinf.* **2005**, *61*, 704. (b) <http://propka.ki.ku.dk/~propka2/>.
77. (a) Vreven, T.; Morokuma, K. *Theor. Chem. Acc.* **2003**, *109*, 125. (b) Altun, A.; Yokoyama, S.; Morokuma, K. *J. Phys. Chem. B* **2008**, *112*, 6814.
78. Shimomura, O.; Johnson, F. H.; Saiga, Y. *J. Cell. Comp. Physiol.* **1962**, *59*, 223.
79. Selected theoretical works: (a) Martin, M. E.; Negri, F.; Olivucci, M. *J. Am. Chem. Soc.* **2004**, *126*, 5452. (b) Altoe, P.; Bernardi, F.; Garavelli, M.; Orlandi, G.; Negri, F. *J. Am. Chem. Soc.* **2005**, *127*, 3952. (c) Sinicropi, A.; Andruniow, T.; Ferré, N.; Basosi, R.; Olivucci, M. *J. Am. Chem. Soc.* **2005**, *127*, 11534. (d) Weber, W.; Helms, V.; McCammon, J. A.; Langhoff, P. W. *Proc. Natl. Acad. Sci. USA* **1999**, *96*, 6177. (e) Olsen, S.; Smith, S. C. *J. Am. Chem. Soc.* **2007**, *129*, 2054. (f) Olsen, S.; Smith, S. C. *J. Am. Chem. Soc.* **2008**, *130*, 8677. (g) Das, A. K.; Hasegawa, J.; Miyahara, T.; Ehara, M.; Nakatsuji, H. *J. Comput. Chem.* **2003**, *24*, 1421. (h) Nifosí, R.; Amat, P.; Tozzini, V. *J. Comput. Chem.* **2007**, *28*, 2366. (i) Bravaya, K. B.; Bochenkova, A. V.; Granovsky, A. A.; Savitsky, A. P.; Nemukhin, A. V. *J. Phys. Chem. A* **2008**, *112*, 8804. (j) Nemukhin, A. V.; Topol, I. A.; Burt, S. K. *J. Chem. Theory Comput.* **2006**, *2*, 292. (k) Voityuk, A. A.; Kummer, A. D.; Michel-Beyerle, M.-E.; Rosch, N. *Chem. Phys.* **2001**, *269*, 83.
80. (a) Brejc, K.; Sixma, T. K.; Kitts, P. A.; Kain, S. R.; Tsien, R. Y.; Ormoe, M.; Remington, S. J. *Proc. Natl. Acad. Sci. USA* **1997**, *94*, 2306. (b) Heim, R.; Prasher, D. C.; Tsien, R. Y. *Proc. Natl. Acad. Sci. USA* **1994**, *91*, 12501. (c) Palm, G. J.; Zdanov, A.; Gaitanaris, G. A.; Stauber, R.; Pavlakis, G. N.; Wlodawer, A. *Nature Struct. Biol.* **1997**, *4*, 361. (d) Chattoraj, M.; King, B. A.; Bublit, G. U.; Boxer, S. G. *Proc. Natl. Acad. Sci. USA* **1996**, *93*, 8362. (e) Stoner-Ma, D.; Jaye, A. A.; Matousek, P.; Towrie, M.; Meech, S. R.; Tonge, P. J. *J. Am. Chem. Soc.* **2005**, *127*, 2864.
81. Dronpa: (a) Ando, R.; Mizuno, H.; Miyawaki, A. *Science* **2004**, *306*, 1370. (b) Habuchi, S.; Ando, R.; Dedecker, P.; Verheijen, W.; Mizuno, H.; Miyawaki, A.; Hofkens, J. *Proc. Natl. Acad. Sci. USA* **2005**, *102*, 9511. (c) Fron, E.; Flors, C.; Schweitzer, G.; Habuchi, S.; Ando, R.; De Schryver, F. C.; Miyawaki, A. J.; Hofkens, A. *J. Am. Chem. Soc.* **2007**, *129*, 4870.
82. (a) Ando, R.; Hama, H.; Yamamoto-Hino, M.; Mizuno, H.; Miyawaki, A. *Proc. Natl. Acad. Sci. USA* **2002**, *99*, 12651. (b) Wiedenmann, J.; Ivanchenko, S.; Oswald, F.;

- Schmitt, F.; Röcker, C.; Salih, A.; Spindler, K. D.; Nienhaus, G. U. *Proc. Natl. Acad. Sci. USA* **2004**, *101*, 15905. (c) Nienhaus, K.; Nienhaus, G. U.; Wiedenmann, J.; Nar H., *Proc. Natl. Acad. Sci. USA* **2005**, *102*, 9156. (d) Gurskaya, N. G.; Verkhusha, V. V.; Shcheglov, A. S.; Staroverov, D. B.; Chepurnykh, T. V.; Fradkov, A. F.; Lukyanov, S.; Lukyanov, K. A. *Nature Biotechnol.* **2006**, *24*, 461. (e) Patterson, G. H.; Lippincott-Schwartz, J. *Science* **2002**, *297*, 1873. (f) Mizuno, H.; Mal, T. K.; Tong, K. I.; Ando, R.; Furuta, T.; Ikura, M.; Miyawaki, A. *Mol. Cell* **2003**, *12*, 1051.
83. (a) Chudakov, D. M.; Belousov, V. V.; Zeraisky, A. G.; Novoselov, V. V.; Staroverov, D. B.; Zorov, D. B.; Lukyanov, S.; Lukyanov, K. A. *Nature Biotechnol.* **2003**, *21*, 191. (b) Chudakov, D. M.; Feofanov, A. V.; Mudrik, N. N.; Lukyanov, S.; Lukyanov, K. A. *J. Biol. Chem.* **2003**, *278*, 7215. (c) Chudakov, D. M.; Verkhusha, V. V.; Staroverov, D. B.; Souslova, E. A.; Lukyanov, S.; Lukyanov, K. A. *Nature Biotechnol.* **2004**, *22*, 1435. (d) Miyawaki, A. *Nature Biotechnol.* **2004**, *22*, 1374. (e) Verkhusha, V. V.; Lukyanov, K. A. *Nature Biotechnol.* **2004**, *22*, 289.
84. (a) Andresen, M.; Wahl, M. C.; Stiel, A. C.; Gräter, F.; Schäfer, L. V.; Trowitzsch, S.; Weber, G.; Eggeling, C.; Grubmüller, H.; Hell, S. W.; Jakobs, S. *Proc. Natl. Acad. Sci. USA* **2005**, *102*, 13070. (b) Henderson, N. J.; Ai, H.-W.; Campbell, R. E.; Remington, S. J. *Proc. Natl. Acad. Sci. USA* **2007**, *104*, 6672. (c) Hofmann, M.; Eggeling, C.; Jakobs, S.; Hell, S. W. *Proc. Natl. Acad. Sci. USA* **2005**, *102*, 17565. (d) Adam, V.; Lelimousin, M.; Boehme, S.; Desfonds, G.; Nienhaus, K.; Field, M. J.; Wiedenmann, J.; McSweeney, S.; Nienhaus, G. U.; Bourgeois, D. *Proc. Natl. Acad. Sci. USA* **2008**, *105*, 18343.
85. Wilmann, P. G.; Petersen, J.; Devenish, R. J.; Prescott, M.; Rossjohn, J. *J. Biol. Chem.* **2005**, *280*, 2401.
86. Sauer, M. *Proc. Natl. Acad. Sci. USA* **2005**, *102*, 9433.
87. (a) Vendrell, O.; Gelabert, R.; Moreno, M.; Lluch, J. M. *J. Am. Chem. Soc.* **2006**, *128*, 3564. (b) Vendrell, O.; Gelabert, R.; Moreno, M.; Lluch, J. M. *J. Phys. Chem. B* **2008**, *112*, 5500. (c) Vendrell, O.; Gelabert, R.; Moreno, M.; Lluch, J. M. *J. Chem. Theory Comput.* **2008**, *4*, 1138. (d) Lill, M. A.; Helms, V. *Proc. Natl. Acad. Sci. USA* **2002**, *99*, 2778.
88. (a) Tanner, C.; Manca, C.; Leutwyler, S. *Science* **2003**, *302*, 1736. (b) Ashfold, M. N. R.; Cronin, B.; Devine, A. L.; Dixon, R. N.; Nix, M. G. D. *Science* **2006**, *312*, 1637. (c) Sobolewski, A. L.; Domcke, W. *J. Phys. Chem. A* **1999**, *103*, 4494.
89. Ando, R.; Flors, C.; Mizuno, H.; Hofkens, J.; Miyawaki, A. *Biophys. J.* **2007**, *92*, L97.
90. bsDronpa and Padron: Andresen, M.; Stiel, A. C.; Fölling, J.; Wenzel, D.; Schönle, A.; Egner, A.; Eggeling, C.; Hell, S. W.; Jakobs, S. *Nature Biotechnol.* **2008**, *26*, 1035.
91. (a) Wilmann, P. G.; Turcic, K.; Battad, J. M.; Wilce, M. C. J.; Devenish, R. J.; Prescott, M.; Rossjohn, J. *J. Mol. Biol.* **2006**, *364*, 213. (b) Stiel, A. C.; Trowitzsch, S.; Weber, G.; Andresen, M.; Eggeling, C.; Hell, S. W.; Jakobs, S.; Wahl, M. C. *Biochem. J.* **2007**, *402*, 35. (c) Nam, K.-H.; Kwon, O. Y.; Sugiyama, K.; Lee, W.-H.; Ki, Y. K.; Song, H. K.; Kim, E. E.; Park, S.-Y.; Jeon, H.; Hwang, K. S. *Biochem. Biophys. Res. Commun.* **2007**, *354*, 962. (d) Andresen, M.; Stiel, A. C.; Trowitzsch, S.; Weber, G.; Eggeling, C.; Wahl, M. C.; Hell, S. W.; Jakobs, S. *Proc. Natl. Acad. Sci. USA* **2007**, *104*, 13005. (e) Mizuno, H.; Kumar, M. T.; Wälchli,

- M.; Kikuchi, A.; Fukano, T.; Ando, R.; Jeyakanthan, J.; Taka, J.; Shiro, Y.; Ikura, M.; Miyawaki, A. *Proc. Natl. Acad. Sci. USA* **2008**, *105*, 9927.
92. Li, X.; Chung, L. W.; Mizuno, H.; Miyawaki, A.; Morokuma, K. *J. Phys. Chem. B* **2010**, *114*, 1114.
 93. Bravaya, K. B.; Bochenkova, A. V.; Granovskii, A. A.; Nemukhin, A. V. *Russ. J. Phys. Chem. B* **2008**, *2*, 671. The absorption calculated for the neutral form of the model GFP chromophore (HBIA) is 3.61, 3.71 and 3.11 eV by the MRMP2, MCQDPT2, and aug-MCQDPT2 methods, respectively.
 94. Hayashi, I.; Mizuno, H.; Tong, K. I.; Furuta, T.; Tanaka, F.; Yoshimura, M.; Miyawaki, A.; Ikura, M. *J. Mol. Biol.* **2007**, *372*, 918, and references therein.
 95. Li, X.; Chung, L. W.; Mizuno, H.; Miyawaki, A.; Morokuma, K. *J. Phys. Chem. B* **2010**, *114*, 16666.
 96. Ando, Y.; Niwa, K.; Yamada, N.; Enomoto, T.; Irie, T.; Kubota, H.; Ohmiya, Y.; Akiyama, H. *Nature Photon.* **2008**, *2*, 44.
 97. Nakatsu, T.; Ichiyama, S.; Hiratake, J.; Saldanha, A.; Kobashi, N.; Sakata, K.; Kato, H. *Nature* **2006**, *440*, 372.
 98. (a) Adam, W. In *The Chemistry of Functional Groups: Peroxides*, Patai, S., Ed., Wiley, New York, 1983, pp. 830–920. (b) Adam, W.; Baader, W. J. *J. Am. Chem. Soc.* **1985**, *107*, 410. (c) Adam, W.; Trofimov, A. V. In *The Chemistry of Peroxides*, Vol. 2, Rappoport, Z., Ed., Wiley, Hoboken, NJ, 2006, pp. 1171–1209. (d) Matsumoto, M. *J. Photochem. Photobiol. C* **2004**, *5*, 27.
 99. (a) Wilsey, S.; Bernardi, F.; Olivucci, M.; Robb, M. A.; Murphy, S.; Adam, W. *J. Phys. Chem. A* **1999**, *103*, 1669. (b) Tanaka, C.; Tanaka, J. *J. Phys. Chem. A* **2000**, *104*, 2078. (c) Rodríguez, E.; Reguero, M. *J. Phys. Chem. A* **2002**, *106*, 504. (d) De Vico, L.; Liu, Y.-J.; Krogh, J. W.; Lindh, R. *J. Phys. Chem. A* **2007**, *111*, 8013.
 100. (a) Koo, J.-Y.; Schmidt, S. P.; Schuster, G. B. *Proc. Natl. Acad. Sci. USA* **1978**, *75*, 30. (b) Zaklika, K. A.; Thayer, A. L.; Schaap, A. P. *J. Am. Chem. Soc.* **1978**, *100*, 4916. (c) Baader, W. J.; Stevani, C. V.; Bastos, E. L. In *The Chemistry of Peroxides*, Vol. 2, Rappoport, Z., Ed., Wiley, Hoboken, NJ, 2006, pp. 1211–1278. (d) Catalani, L. H.; Wilson, T. *J. Am. Chem. Soc.* **1989**, *111*, 2633.
 101. (a) Isobe, H.; Takano, Y.; Okumura, M.; Kuramitsu, S.; Yamaguchi, K. *J. Am. Chem. Soc.* **2005**, *127*, 8667. (b) Isobe, H.; Yamanaka, S.; Kuramitsu, S.; Yamaguchi, K. *J. Am. Chem. Soc.* **2008**, *130*, 132.
 102. Schaap, A. P.; Gagnon, S. D. *J. Am. Chem. Soc.* **1982**, *104*, 3504.
 103. White, E. H.; Wörther, H.; Field, G. F.; McElroy, W. D. *J. Org. Chem.* **1965**, *30*, 2344.
 104. Chung, L. W.; Hayashi, S.; Lundberg, M.; Nakatsu, T.; Kato, H.; Morokuma, K. *J. Am. Chem. Soc.* **2008**, *130*, 12880.
 105. Matsumoto, M.; Watanabe, N.; Hoshiya, N.; Ijuin, H. K. *Chem. Rec.* **2008**, *8*, 213.
 106. Desouter-Lecomte, M.; Lorquet, J. C. *J. Chem. Phys.* **1979**, *71*, 4391.

13 Computational Methods for Modeling Free-Radical Polymerization

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In this chapter electronic structure computational approaches can be used to predict the kinetics of free-radical polymerization reactions and hence the properties of complex polymers. Much effort is spent in detailing how free-energy changes for reactions and reactive transition states in condensed phases can be obtained using standard computational methods, and how these data can be used to drive complex models of reaction kinetics. Emphasis is placed on the need for calculations to attain *chemical accuracy* (better than 4 kJ mol^{-1} or 0.1 eV) and so can be used to interpret polymer properties, and various combinations of current density functional and ab initio methods are prescribed to achieve this goal.

13.1 INTRODUCTION

13.1.1 Free-Radical Polymerization

Free-radical polymerization is an important industrial process, responsible for the production of around 50% of all synthetic polymers worldwide (ca. 100 million tons per annum).¹ It is typically used to produce polymers of the form $\text{---}(\text{CR}_1\text{R}_2\text{---CR}_3\text{R}_4)_n\text{---}$ from the corresponding vinyl monomer, $\text{CR}_1\text{R}_2=\text{CR}_3\text{R}_4$. Among the most important of these monomers are ethylene and its mono- and 1,1-disubstituted derivatives, such as styrene and its substitution products, acrylonitrile, vinyl chloride, and other halogenated alkenes, and various vinyl, acrylic, and methacrylic esters, amides, and acids. Dienes can also be polymerized in this way; for example, nitrile rubber is formed via the free-radical

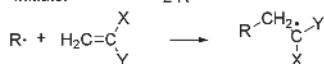
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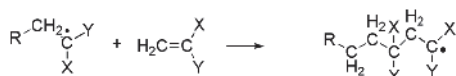
copolymerization of butadiene and acrylonitrile. More hindered monomers, such as bulky 1,2-disubstituted ethylene derivatives, can be polymerized under certain circumstances, usually as part of a copolymerization with a less hindered monomer. Polymers can also be formed from the radical ring-opening polymerization of exocyclic double bonds and the cyclopolymerization of 1,6-dienes. In a few limited cases, it is also possible to use free-radical polymerization to form polymers from monomers containing other types of π -bonds, such as SO_2 and CO , although usually, only via copolymerization with an alkene.

As in all radical chain processes, polymerization proceeds via a series of initiation, propagation, and termination reactions (see Scheme 13.1). Free radicals are typically produced in situ via thermal, photolytic, or electrochemical decomposition of an initiator. The resulting radicals then attack the π -bond of an alkene monomer via a free-radical addition reaction that results in the formation of a new σ -bond between the attacking radical and monomer, and a new

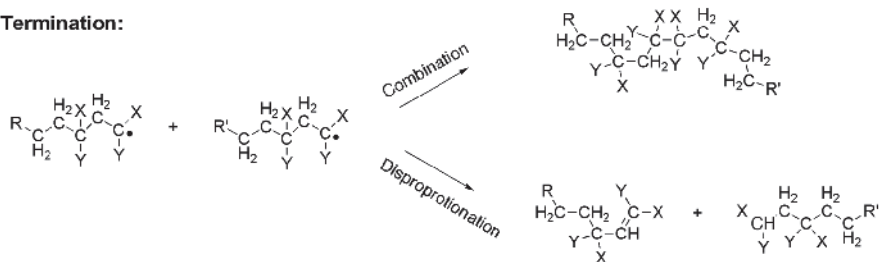
Initiation: Initiator $\longrightarrow$ $2 \text{ R}\cdot$



Propagation:



Termination:

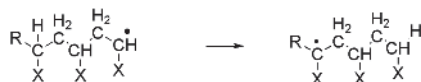


Chain Transfer:



[A·B = solvent, initiator, polymer, monomer, chain transfer agent]

Intramolecular:



Scheme 13.1 Conventional free-radical polymerization.

radical center, which can then undergo further radical addition reactions with further monomer molecules. In contrast to the ionic and transition metal-catalyzed alternatives, the propagation step in free-radical polymerization is both thermodynamically and kinetically favorable for a broad range of monomer functionalities, requires less demanding reaction conditions, and is tolerant of protic impurities such as water or alcohols. It is usually highly regioselective, with addition occurring at the least-substituted carbon center of the π -bond. As the radical center is sp^2 hybridized, attack can occur from either face, and control over the stereochemistry of the polymer resulting from *conventional* free-radical polymerization is therefore limited. However, in recent years, successful *stereocontrolled* radical polymerization has been achieved for some classes of monomer (vinyl esters, acrylamides, methacrylates) via complexation of the growing radical with Lewis acids.²

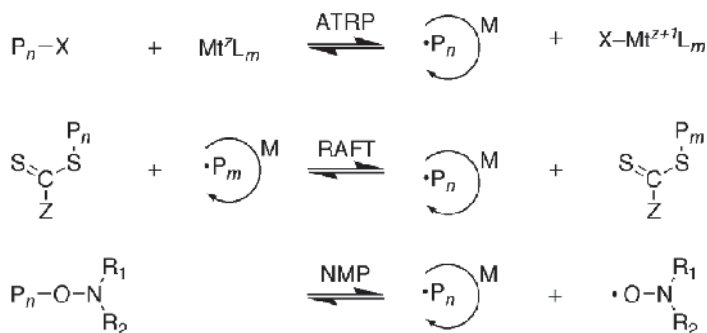
Termination of the growing radical occurs via bimolecular radical-radical termination reactions, which, depending on the system, occur via combination and/or disproportionation. These reactions tend to be inherently much faster than propagation, to the extent that they are usually diffusion limited. In conventional free-radical polymerization, bimolecular termination occurs randomly throughout the process, leading to a polymer that has a broad molecular weight distribution and end groups that are not normally suitable for further modification. The propagating radical is also highly susceptible to side reactions, such as hydrogen or halogen atom abstraction from species such as the monomer, solvent, initiator, or added chain-transfer agent. These reactions, which are typically followed by further propagation of the new radical center, can be exploited to limit the molecular weight of the resulting polymer and confer specific end groups, but are not always desirable. For example, in some cases, such as vinyl chloride polymerization, chain-transfer reactions are followed by subsequent side reactions, such as β -elimination, that result in unwanted structural defects; in other cases, such as propylene polymerization, chain transfer may lead to a new radical that is too stable to reinitiate polymerization, thereby resulting in inhibition of the polymerization. Inter- and intramolecular chain transfer to polymers, if followed by subsequent propagation, leads respectively to long- and short-chain branching of the polymer, which may or may not be desirable, depending on the application. For example, due to significant branching, free-radical polymerization of ethylene results in a low-density polymer that is suitable for many applications (such as cling film); however, when high-density polyethylene is required (as in the production of milk bottles), a nonradical polymerization process must be used.

13.1.2 Controlled Radical Polymerization

Until recently, free-radical polymerization was thought to offer only limited control over the molecular weight distribution, chain-end composition, and other aspects of the polymer microstructure. However, in the last couple of decades,

the field has been transformed by the development of controlled/living radical polymerization techniques such as reversible addition fragmentation chain transfer (RAFT) polymerization,³ atom transfer radical polymerization (ATRP),⁴ and nitroxide-mediated polymerization (NMP).⁵ The key feature of these methods is their protection of the growing polymer chains from the bimolecular-termination reactions that normally occur in conventional free-radical polymerization, through their reversible trapping as a dormant species. The chemical nature of the control equilibrium varies according to the process, with some of the leading examples provided in Scheme 13.2. Through this equilibrium, the termination rate (which depends on the square of the radical concentration) is minimized with respect to the propagation rate (which depends only on the first order of the radical concentration). Some bimolecular termination is inevitable, but provided that the process is optimized correctly, most of the chains survive throughout the polymerization and therefore have a narrow molecular weight distribution. They can then be isolated in their “dormant” form and their active end groups used for further polymerization or functionalization.

Controlled radical polymerization combines the advantages of conventional radical polymerization with the ability to produce polymers with narrow molecular weight distributions, designer end groups, and special architectures such as stars, blocks, and grafts for use in bioengineering and nanotechnology applications.^{1a} However, its development has also given rise to new technical challenges. In particular, its success is crucially dependent upon choosing control agents and reaction conditions that strike an optimal balance between the rates of several competing reactions, such that concentration of the dormant species is orders of magnitude greater than the free species, and the exchange between the two forms is rapid. It is therefore more important than ever to be able to build accurate kinetic models for these processes and to understand the effect of substituents on the rate coefficients and other parameters within these models, so as to enable the design of optimal control agents and reaction conditions for any given system.



Scheme 13.2 Main control equilibria in atom transfer radical polymerization (ATRP), reversible addition fragmentation chain transfer (RAFT) polymerization, and nitroxide-mediated polymerization (NMP).

13.1.3 Why Quantum Chemistry?

Owing to the complexity of their reaction schemes, the experimental measurement of individual rate coefficients in free-radical polymerization is not straightforward. This is because the observables of a polymerization (such as the overall reaction rate and the molecular weight distribution of the resulting polymer) are a complicated function of the individual rate coefficients, which are typically estimated by fitting an assumed kinetic model to the experimental data available. Ideally, any such assumed kinetic model should be as flexible as possible, including all conceivable side reactions and treating all reactions as potentially chain-length dependent. However, this would give rise to very complicated kinetic models, containing thousands of unknown rate coefficients, which would then need to be estimated by fitting the model to limited experimental data. As a result, in practice, various simplifying assumptions are introduced (such as omitting potential side reactions and ignoring the chain-length dependence of various rate coefficients); depending on the system, these are a potentially large source of error. In recent years, the development of elegant laser flash photolysis techniques such as pulsed-laser polymerization has made possible measurement of the individual rate coefficients for the principal reactions in a simple free-radical homopolymerization in a relatively model-free manner.⁶ However, in more complex situations, such as copolymerization or controlled radical polymerization, the inclusion of additional reactions in the kinetic scheme renders the direct (i.e., model-free) measurement of the individual rate coefficients difficult, if not impossible. This situation is exemplified by a recent controversy surrounding the causes of rate retardation in RAFT polymerization.⁷ At the center of this controversy are alternative experimental values for the fragmentation rate coefficient in cumyl dithiobenzoate-mediated polymerization of styrene at 60°C that differ by six orders of magnitude, a discrepancy that arises (at least in part) in the alternative model-based assumptions implicit in their measurement.

Quantum chemistry offers a potential solution to this problem, as it allows the individual reactions within a complex process to be studied directly, without recourse to empirical data (other than the fundamental physical constants) or model-based assumptions (other than the laws of quantum mechanics). Moreover, quantum chemistry also provides access to useful related data (such as radical stabilization energies, charge distributions, and transition-state geometries) that can assist in interpretation of the results. However, quantum chemistry is not without its own problems. As is well known, the many-electron Schrödinger equation has no analytical solution, and in order to solve it, various numerical approximations need to be made. Although accurate methods exist, these are computationally intensive, and as their computational cost scales exponentially with the size of the system, they cannot be applied practically to large polymeric molecules.

To study polymeric systems, one needs to model the polymeric propagating radical using much smaller species and, so that these models can be as realistic as possible, adopt less expensive computational methods—both of these

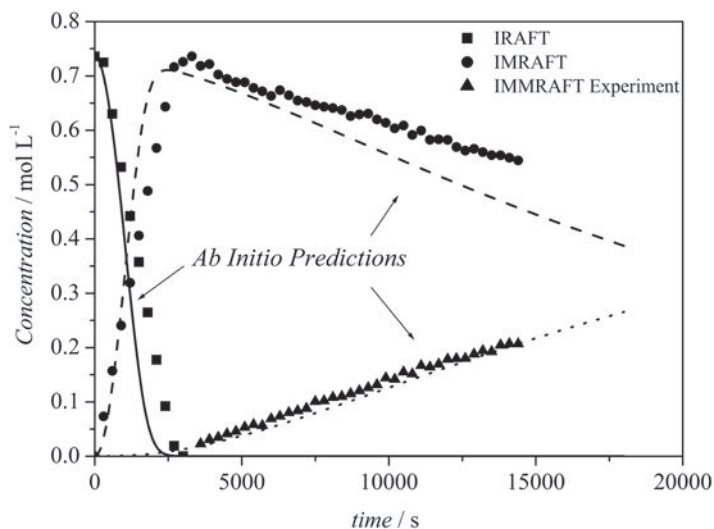


Fig. 13.1 Comparison of the experimental¹⁹ values and corresponding theoretical¹⁸ predictions for concentrations of the initial RAFT agent (IRRAFT), unimeric RAFT agent (IMRAFT), and dimeric RAFT agent (IMMRAFT), during the initialization of cyanisopropyl dithiobenzoate-mediated polymerization of styrene.

approximations are a potentially significant source of error. Nonetheless, aided by rapid and continuing advances in computer power and the development of cost-effective computational methods for larger molecules, chemical accuracy is now possible, even for solvent-sensitive systems.⁸ Indeed, computational chemistry has already helped to clarify the mechanism of various controlled^{9–11} and conventional^{8,12–14} free-radical polymerization processes, has helped to design optimal control agents,¹⁵ and has contributed to the discovery of new types of radical polymerization processes.^{16,17} Computational chemistry has even been used to build accurate kinetic models of complicated radical polymerization processes such as RAFT, and to simulate these polymerization processes successfully from first principles (see Fig. 13.1).¹⁸

13.1.4 Scope of The Chapter

In this chapter we outline the special strategies that need to be adopted for successful modeling of radical polymerization processes. We first examine how to design small model reactions that accurately mimic the behavior of their polymeric counterparts. We then identify the electronic structure procedures that deliver chemical accuracy in the most cost-effective manner: in particular, special strategies that are required when applying them to larger molecules. Finally, we describe how the electronic-structure information derived from quantum chemical calculations is used to calculate the rate and equilibrium constants of the individual reactions within a free-radical polymerization process. For examples of how

such quantum chemically derived kinetic data have been used to develop better kinetic models for free-radical polymerization processes, and/or design improved reagents and reaction conditions, the reader is referred to the original references cited above as well as to some recent review articles.^{11h,20}

13.2 MODEL REACTIONS FOR FREE-RADICAL POLYMERIZATION KINETICS

13.2.1 Background

Free-radical polymerization typically involves reactions of radicals having molecular weights on the order of 10^5 g mol^{-1} or more (i.e., thousands of nonhydrogen atoms). In contrast, even at relatively low levels of theory, quantum chemical calculations are typically practical only for systems with tens of nonhydrogen atoms. To study polymeric reactions, it is therefore necessary to model the polymeric species with related short-chain versions. For example, one might model the propagating radical in acrylonitrile polymerization as the corresponding dimer or trimer radical. Such models are feasible because provided that there is no conjugation along the polymer chain, the influence of remote substituents diminishes rapidly with their distance from the reaction center. They do, of course, contribute to the total free energy of the polymer chain, but their contribution is unperturbed by the reaction and hence cancels from the reaction barrier and enthalpy. Experimentally, this is seen in experimental studies of the chain-length dependence of propagation rate coefficients, which have been shown to converge to within a factor of 2 or 3 of the long-chain limit by the dimer or trimer stage.²¹ It is also the basis of leading models for copolymerization kinetics, which ignore all substituent effects beyond the penultimate unit of the propagating radical.²²

13.2.2 Design Considerations

When designing a small model for a polymeric reaction, one has to strike an optimal balance between the need to make the model as realistic as possible, and the need to limit its size, to facilitate the use of sufficiently high levels of theory. To this end, one must first examine the chain-length dependence of the reaction under study, in order to identify the shortest chain-length at which the chain length effects converge to within an acceptable level of accuracy. This level of accuracy will itself depend on the accuracy of the theoretical procedures to be used as well as the purpose of the computational study (e.g., whether one is interested in quantitative predictions or qualitative trends); however, convergence to within a factor of 2 or 3 is usually both desirable and possible using relatively small models.

As an example, Fig. 13.2 shows the G3(MP2)-RAD//B3LYP/6-31G(d) calculations of the propagation rate coefficients in vinyl chloride polymerization and acrylonitrile polymerization, as taken from Izgorodina and Coote.⁸ To construct

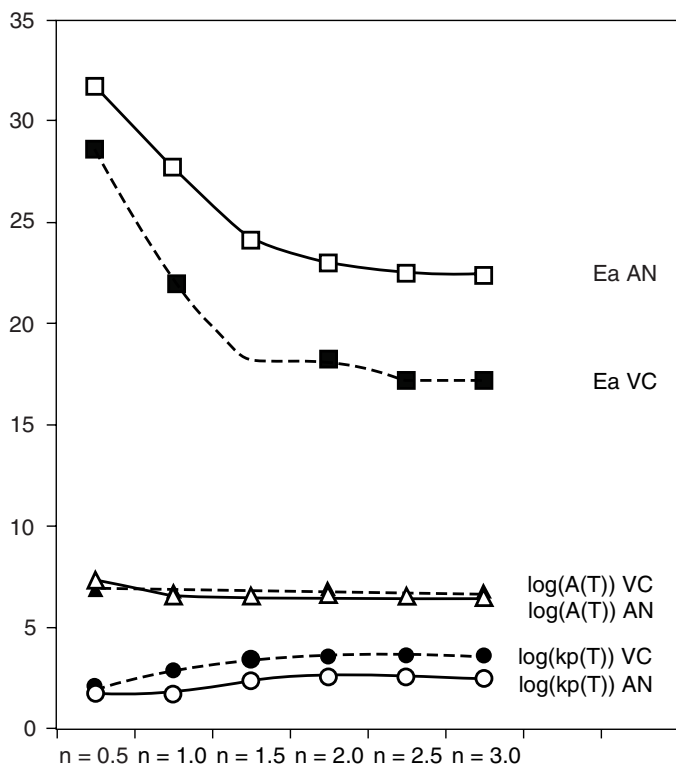
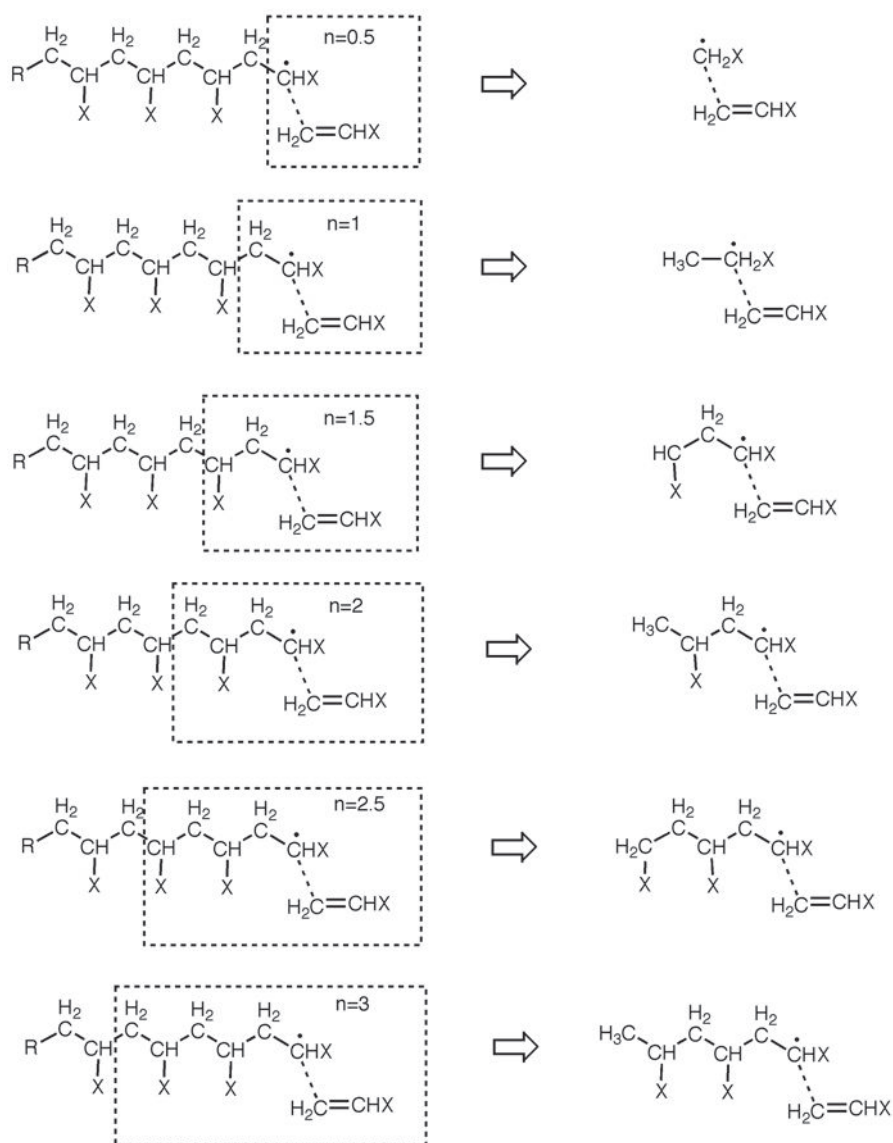


Fig. 13.2 Chain-length dependence of the propagation rate coefficient (k_p ; $\text{L mol}^{-1} \text{s}^{-1}$), Arrhenius activation energy (E_a ; kJ mol^{-1}), and frequency factor (A ; $\text{L mol}^{-1} \text{s}^{-1}$) for acrylonitrile (AN) and vinyl chloride (VC) propagation at 298.15 K. (From Ref. 8.)

this plot, the propagation rate coefficients were calculated for successively larger models of the propagating radical, as shown in Scheme 13.3. In each case, the polymer chain was truncated at increasingly larger distances from the chain end, and replaced by a hydrogen atom, so as to maintain the correct valency. From this plot it is seen that the rate coefficients converge to within less than a factor of 2 once all substituents within three backbone atoms of the chain end are included in the model ($n = 1.5$), with concurrent convergence of the barriers to within 2 kJ mol^{-1} and convergence of the frequency factors to within a factor of 1.6. Hence, this would serve as a suitable small model for the corresponding polymeric reaction, for these systems.

The exact rate of convergence of rate and equilibrium constants with respect to chain length varies according to the type of reaction under study and the nature of the substituents. Under a best practice scenario one should thus always perform a systematic study of the effects of chain length in order to identify the long-chain limit. Nonetheless, as a rough guideline, studies in the literature on a diverse range of systems, including various propagation reactions,^{8,14}



Scheme 13.3 Models for studying the chain-length dependence of the propagation step in free-radical polymerization of vinyl chloride ($\text{X} = \text{Cl}$) and acrylonitrile ($\text{X} = \text{CN}$).

radical ring-opening polymerization,^{17a} and the addition-fragmentation equilibrium of RAFT polymerization¹¹ⁱ indicate that convergence to within chemical accuracy is typically achieved once four backbone atoms are included in the model. Similar degrees of convergence with respect to chain length might also be reasonably expected when truncating larger side chains, provided conjugated functional groups are counted as a single unit.

13.3 ELECTRONIC STRUCTURE METHODS

13.3.1 Background

Having chosen a suitable small model for the polymerization process, quantum chemistry is then used to optimize the geometries of each species and calculate their frequencies and total energies. Having obtained this information, the rates, equilibrium constants, and other thermodynamic quantities of the chemical reaction can then be obtained via the equations detailed in the next section. The quantum chemical calculations can be performed using any standard software package, such as GAUSSIAN,²³ MOLPRO,²⁴ GAMESS,²⁵ Q-CHEM,²⁶ NWChem,²⁷ ADF,²⁸ and ACES II.²⁹ For details on how to use this software, the reader is referred to their respective manuals, most of which are freely accessible online. The purpose of this section is to provide guidelines for choosing appropriate levels of theory that strike the optimal balance between accuracy and computational cost for the types of reactions relevant to radical polymerization processes. We also describe a number of special theoretical procedures that have been developed for studying larger molecules and are particularly useful for studying radical polymerization processes.

13.3.2 Assessment Studies

When studying any new chemical reaction, one should always perform an assessment study (or refer to an existing one in the literature) to select appropriate theoretical procedures that balance the competing demands of accuracy and computational efficiency. Assessment studies are ideally performed using a series of small prototypical reactions, for which reliable experimental data are available and/or high-level benchmarking calculations are possible. These reactions are studied at a wide variety of levels of theory and the results are compared with the corresponding benchmark data, so as to identify the minimal requirements for chemical accuracy.

In performing an assessment study, the following points should be noted. First, the optimal theoretical procedures for performing geometry optimizations, frequency calculations, and (single-point) energy calculations are likely to be different, and hence the accuracy of theoretical procedures for each should be assessed separately. Second, in the case of geometry optimizations and frequency calculations, it is often not the accuracy of the geometries and frequencies themselves that is relevant but rather the effect of any inaccuracies on the calculated kinetics and thermodynamics. For geometries, this can be assessed by performing single-point energy calculations *at a consistent level of theory*, on the geometries optimized at the various levels of theory. One should also use the geometries obtained at the various levels of theory to calculate the rotational entropy. For frequencies, one should use frequencies calculated at the various levels of theory to obtain the zero-point vibrational energies, thermal corrections, and vibrational entropies.

Finally, it should be noted that many lower-level theoretical procedures often deliver excellent results for simple prototypical reactions, only to fail for larger

substituted systems (or vice versa). This is particularly the case for highly parameterized methods, such as popular density functional theory (DFT) methods.³⁰ For example, B3LYP calculations¹²¹ were found to reproduce the experimental³¹ ethylene propagation rate coefficients at 463 K to within an order of magnitude; however, a B3LYP study^{12h} of vinyl chloride propagation yielded results that differed from the corresponding experimental values³² by approximately three orders of magnitude at room temperature. By contrast, high-level *ab initio* calculations of the same system⁸ were successful in reproducing the experimental data to within a factor of 2. Importantly, in the latter study it was shown that errors in the reaction barriers calculated at the B3LYP level of theory for $\bullet\text{CH}_2\text{Cl}$ addition to $\text{CH}_2=\text{CHCl}$ were just 3.8 kJ mol^{-1} ; however, these had grown to 15.7 kJ mol^{-1} when the addition of the dimer vinyl chloride radical was considered. In another example, studies of radical addition to thiocarbonyl compounds (a model of the addition-fragmentation step in RAFT polymerization) found errors in the calculated B3LYP reaction enthalpies for $\text{R}\bullet + \text{S}=\text{C}(\text{CH}_3)\text{SCH}_3$ that varied from as little as 7.4 kJ mol^{-1} ($\text{R} = \text{CH}_3$) to as much as 62.1 kJ mol^{-1} [$\text{R} = \text{C}(\text{CH}_3)_2\text{Ph}$], depending on the attacking radical.^{30f} Even *ab initio* methods such as MP2 may perform well for simple radical reactions such as methyl radical addition to ethylene, but fail when the corresponding reactions of delocalized radicals such as cyanomethyl or benzyl are considered instead.^{30m}

These nonsystematic errors have important implications for the broader relevance of assessment studies based solely on small prototypical reactions. To take just one example from above, an assessment study based solely on ethylene propagation or even the shortest-chain model of vinyl chloride propagation would have concluded that B3LYP was a suitable method for this problem, despite the fact that the errors in the more realistic dimer model were three orders of magnitude larger. As a result, it is important to design the test set for an assessment study carefully, so that it tests the ability of a theoretical method to model not just the prototypical reaction but also representative types of substituent effect. For example, an assessment for a carbon-centered radical reaction might ideally include the reactions of the parent methyl radical, as well as the bulky *t*-butyl radical, and radicals bearing typical electron-withdrawing and electron-donating groups (such as cyanomethyl and hydroxymethyl, respectively). Similar variation in the substrate should also be considered as relevant.

13.3.3 Optimal Theoretical Procedures for Radical Chemistry

Since a number of the principal reactions in free-radical polymerization are of wider importance in fields as diverse as organic synthesis, combustion, atmospheric chemistry, and biochemistry, considerable effort has already been devoted to developing and benchmarking theoretical procedures for their study. Among other reactions, assessment studies for radical addition to various types of double bond and various types of chain-transfer process have been published.^{30m,33,34} In general, low levels of theory, such as DFT or HF calculations with small basis set, are suitable for geometry optimizations and frequency calculations,^{34b-d} provided that an IRCMax³⁵ approach (see Section 13.3.4) is used to correct transition

structures, and provided that frequencies are scaled by their appropriate scale factors.³⁶

However, as noted above, these lower-level theoretical procedures have been shown to fail dramatically for predicting the thermochemistry of a wide range of simple organic reactions, including a number of radical addition and abstraction processes relevant to radical polymerization.³⁰ Importantly, in many of these cases, these methods failed to model not only the correct absolute values of reaction energies but also the correct trends, thus precluding their use even in semiquantitative studies of substituent effects. Although various suggestions have been put forward as to the origin of these errors, we are not yet in a position to predict with confidence when such methods will succeed or fail.³⁰ As a result, for accurate results, it is important that improved energy calculations be performed using higher-level *ab initio* methods. Unfortunately, these are computationally expensive, particularly for larger molecules, and special strategies, such as the use of composite *ab initio* methods and ONIOM approximations, are therefore required to improve their computational efficiency. These are described in more detail in Section 13.3.4.

13.3.4 Composite Ab Initio Procedures

Assessment studies for a variety of radical (and indeed nonradical) reactions have shown that composite *ab initio* procedures can offer chemical accuracy (ca. 4 kJ mol⁻¹) at a moderate computational cost.^{8,30,34} Composite methods seek to approximate high-level correlated calculations with a large basis set using a series of lower-cost calculations in conjunction with additivity and/or extrapolation routines. Among the most prominent are the *Gn* family of methods,³⁷ which approximate CCSD(T) or QCISD(T) calculations with a large triple-zeta basis set from via a series of additivity approximations, carried out at a lower level of theory, usually MP2 or combinations of MP2 with MP3 or MP4. Several variants of *Gn* theory exist, according to the specific levels of theory employed for the component calculations. One of the lowest-cost variants is G3(MP2)-RAD,³⁸ which approximates (U)RCCSD(T) with the triple-zeta basis set G3MP2large as the sum of the corresponding (U)RCCSD(T)/6-31G(d) calculation and a basis set correction term, obtained as the difference of the corresponding calculations at the R(O)MP2/G3MP2large and R(O)MP2/6-31G(d) level of theory.

$$\begin{aligned}
 E_0[\text{G3(MP2)-RAD}] &= E_0[\text{URCCSD(T)/6-31G(d)}] \\
 &+ E_0[\text{ROMP2/G3MP2large}] - E_0[\text{ROMP2/6-31G(d)}] \\
 &+ \text{SO} + \text{HLC}
 \end{aligned} \tag{13.1}$$

As in all *Gn* theory methods, two additional corrections are included: the first-order spin-orbit correction (SO) and the higher-level correction (HLC) term. SO applies only to atoms and is taken from experiment where available and accurate

theoretical calculations in other cases. The HLC is an empirical correction, which in G3 theory and its variants is calculated as the following function of the number of alpha (n_α) and beta (n_β) valence electrons:

$$\text{HLC (molecules)} = -An_\beta - B(n_\alpha - n_\beta) \quad (13.2)$$

$$\text{HLC (atoms)} = -Cn_\beta - D(n_\alpha - n_\beta) \quad (13.3)$$

The parameters A , B , C , and D are obtained by fitting to a large test of experimental data and are specific to the particular variant of G3 theory; for G3(MP2)-RAD, the values (in mHa) are $A = 9.413$, $B = 3.969$, $C = 9.438$, and $D = 1.888$. It is important to note that since it only depends on the number of alpha and beta valence electrons, the HLC cancels entirely from most reaction energies, except when the reactions involve a mixture of atoms and molecules (as in heats of formation and bond dissociation energies) and/or when spin is not conserved (as in ionization and excitation energies). In other words, for the reactions of interest to radical polymerization, G3 methods contain no empirical correction terms and can thus be viewed as truly *ab initio* in nature.

Other important families of composite methods include the CBS methods³⁹ and the W_n methods,⁴⁰ both of which feature extrapolation to the infinite basis set limit. In the CBS procedures this is performed at the MP2 level of theory using pair natural orbital energies, and the results are then corrected to the CCSD(T) level via a series of additivity approximations, as in the G_n methods; empirical corrections are also included. In the W_n methods, extrapolation to the infinite basis set limit is carried out using coupled-cluster theory with correlation-consistent basis sets; additional corrections for relativistic effects, core correlation, and spin-orbit corrections are also included. Not surprisingly, W_n methods are generally more accurate than CBS or G_n methods (typically offering kilojoule accuracy rather than kilocalorie accuracy) but also considerably more computationally expensive and therefore limited to relatively small systems.

The only package that allows a user to run an automatic composite method is GAUSSIAN, which has automatic keywords for selected G_n , W_n , and CBS procedures. However, with the possible exception of the CBS procedures, most composite *ab initio* methods can easily be performed using any of the leading computational chemistry software packages, merely by running the component calculations (geometry optimization, frequency calculation, and various single-point energy calculations) separately and then combining them via the relevant prescribed formulas, usually by means of a spreadsheet. Indeed, even when it is possible to use direct keywords, it is usually more computationally efficient to run the component calculations separately. For exact recipes for the various composite methods, the reader is referred to their original references; fully worked examples for two key methods used in studying radical chemistry are provided in Tables 13.1 and 13.2.

TABLE 13.1 Example Calculation for the G3(MP2)-RAD Energy of $\bullet\text{CH}_3$

Protocol for G3(MP2)-RAD ^a	Energy Component	Energy (hartrees)
E(ZPVE) from OPT structure at B3LYP/6-31G(d)	—	0.02983
Scaling factor	—	0.98060
Scaled ZPVE	E(ZPVE)	0.02926
R(O)MP2/6-31G*	E(MP2S)	-39.66850
R(O)MP2/G3MP2large	E(MP2L)	-39.73046
(U)RCCSD(T)/6-31G*	E(CC)	-39.69103
Number of alpha valence electrons (n_α)	—	4
Number of beta valence electrons (n_β)	—	3
HLC = $1/1000[-An_\beta - B(n_\alpha - n_\beta)]$	E(HLC)	-0.03221
Spin-orbit splitting	E(SO)	0.00000
-E(MP2S)+E(MP2L)+ E(CC)+E(HLC)+E(SO)	G3(MP2)-RAD electronic energy	-39.78519
-E(MP2S)+E(MP2L)+E(CC)+ E(HLC)+E(SO)+E(ZPVE)	G3(MP2)-RAD energy at 0 K	-39.75594

^a $A = 9.413$ and $B = 3.969$ (in millihartrees) for the G3(MP2)-RAD method.

13.3.5 ONIOM

While composite ab initio methods extend the range of systems for which chemical accuracy is accessible, they are at best currently feasible only for systems of up to 17 nonhydrogen atoms (in C_1 symmetry), which is smaller than the typical dimer or trimer models of free-radical polymerization reactions. Moreover, for some reactions of relevance to free-radical polymerization, most notably radical addition to thiocarbonyl compounds,^{34b} even some high-level composite procedures such as G3(MP2)-RAD are not sufficiently accurate and more expensive procedures such as W1 theory are required; these are presently feasible only for systems of just six or seven nonhydrogen atoms. As a result, further approximations are needed if these methods are to be applied to radical polymerization processes.

To address this problem, we have recently developed and tested an approach based on the ONIOM (our own n -layer integrated molecular orbital + molecular mechanics) procedure of Morokuma and co-workers, in particular in its IMOMO (integrated MO/MO method) incarnation.⁴¹ In ONIOM/IMOMO methods, one first defines a “core” section of the reaction that typically includes all forming and breaking bonds and the principal substituents attached to them. In forming the core system, deleted substituents are replaced with “link atoms” (typically, hydrogens), chosen so that the core system provides a good chemical model of the reaction center. The core system is studied at both a high level of theory, and also at a lower level, while the full system is studied only at the lower level

TABLE 13.2 Example Calculation for the W1 Energy of $\bullet\text{CH}_3$

Protocol for W1 ^a	Energy Component	Energy (hartrees)
<i>E</i> (ZPVE) from OPT structure @ B3LYP/VTZ+1	—	0.02967
Scaling factor	—	0.98500
Scaled ZPVE	<i>E</i> (ZPVE)	0.02922
SCF /AVDZ+2d	E1	−39.56525
CCSD /AVDZ+2d	E2	−39.71925
CCSD(T) /AVDZ+2d	E3	−39.72239
SCF /AVTZ+2d1f	E4	−39.57790
CCSD /AVTZ+2d1f	E5	−39.75800
CCSD(T) /AVTZ+2d1f	E6	−39.76293
SCF /AVQZ+2d1f	E7	−39.58040
CCSD /AVQZ+2d1f	E8	−39.76775
CCSD(T,fc) /Mtsmall	E9	−39.76533
CCSD(T,full) /Mtsmall, int=dkh	E10	−39.83005
Spin-orbit splitting	<i>E</i> (SO)	0.00000
<i>Two pt SCF Extrapolation</i>		
(E4-E7)/((3) ^{−5} -(4) ^{−5})	E11	0.79684
E7-E11*(4) ^{−5}	<i>E</i> (SCF)	−39.58118
<i>CCSD Valence Correlation Component</i>		
E5-E4	E12	−0.18011
E8-E7	E13	−0.18735
(E12-E13)/((3 ^β) ^{−1} -(4 ^β) ^{−1})	E14	0.41247
E12-E14/3 ^β	<i>E</i> (CCSD)	−0.19210
<i>(T) Valence Correlation Component</i>		
E3-E2	E15	−0.00313
E6-E5	E16	−0.00493
(E15-E16)/((2 ^β) ^{−1} -(3 ^β) ^{−1})	E17	0.02292
E15-E17/2 ^β	<i>E</i> (T)	−0.00559
<i>Core and Scalar Relativistic Contribution</i>		
E10-E9	<i>E</i> (Core) + <i>E</i> (ScalR)	−0.06472
<i>E</i> (SCF) + <i>E</i> (CCSD) + <i>E</i> (T) + <i>E</i> (Core) + <i>E</i> (ScalR) + <i>E</i> (SO)	W1 electronic energy	−39.84359
<i>E</i> (SCF) + <i>E</i> (CCSD) + <i>E</i> (T) + <i>E</i> (Core) + <i>E</i> (ScalR) + <i>E</i> (SO) + <i>E</i> (ZPVE)	W1 energy at 0 K	−39.81437

^aβ = 3.22 for W1 method.

of theory. The high-level energy for the full system is then approximated as the sum of the high-level energy for the core system, and the substituent effect, as measured at the lower level of theory:

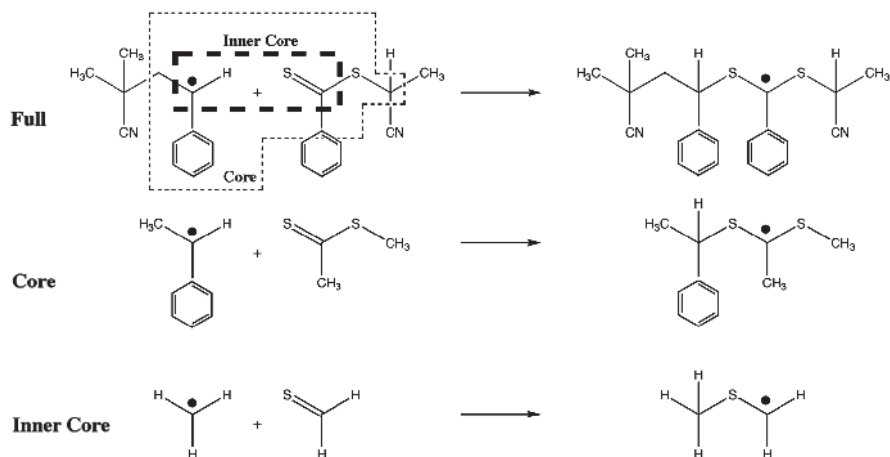
$$E_0^{\text{full}}[\text{ONIOM}] = E_0^{\text{core}}[\text{high}] - E_0^{\text{core}}[\text{low}] + E_0^{\text{full}}[\text{low}] \quad (13.4)$$

This approximation is valid provided that the low level of theory measures the substituent effect accurately; this in turn depends not only on the level of theory chosen but also the manner in which the core is defined. For example, as noted earlier, a method such as ROMP2 is not always capable of modeling the effects of substituents alpha to the radical center, as substituents such as CN or phenyl that delocalize the unpaired electron significantly give rise to larger errors than those that do not. However, if the alpha substituents are held constant across a series of calculations, this same method can provide an excellent measure of more remote substituent effects.^{30m} Thus, provided that the core contains all alpha substituents, a method such as ROMP2/6-311+G(3df,2p) is suitable for an outer ONIOM layer.

Based on our assessment studies^{30m,34e} for a broad range of radical processes, we have designed the following guidelines for calculating chemically accurate energies of the larger molecules (up to ca. 40 nonhydrogen atoms) involved in radical polymerization processes via a multilayer ab initio ONIOM process.

- The “inner core” of the reaction should be defined so as to contain all forming and breaking bonds. The deleted groups should be replaced with hydrogen link atoms. The inner core should be studied at W1 and G3(MP2)-RAD (or an equivalent G3 or CBS procedure).
- The “core” should contain all forming and breaking bonds, along with all substituents in the alpha position to these bonds and any remote substituents connected through conjugation. In certain specific cases, such as radical addition to thiocarbonyl compounds, further simplification of the core has been shown to be possible without compromising the accuracy (for an example, see Scheme 13.4 and Table 13.3)^{30f}; however, in general, this should be avoided if possible. The core should be studied at both G3(MP2)-RAD (or the same equivalent G3 or CBS procedure as above) and R(O)MP2/6-311+G(3df,2p).
- The remaining “full” system should be studied at R(O)MP2/6-311+G(3df,2p); if the full system is too big even for R(O)MP2/6-311+G(3df,2p), successive ONIOM layers using progressively lower levels of theory [such as R(O)MP2 calculations with smaller basis sets] can be used instead.
- In contrast to standard ONIOM calculations, which typically involve very large molecules for which even DFT optimizations are unfeasible, all geometries and frequencies of all species should be fully optimized at the same consistent level of theory.
- As in all ONIOM approximations, it is essential that all ONIOM layers are defined and treated consistently over the course of a chemical reaction.

Using these simple guidelines, we have been able to extend the size of systems for which chemical accuracy is feasible to those involving as many as 40 nonhydrogen atoms, which is sufficiently large to model most common radical polymerization processes to the dimer or trimer stage. For an example of



Scheme 13.4 ONIOM boundaries for the addition of a styryl radical to a RAFT agent.

TABLE 13.3 Example of ONIOM Calculation for the Addition of a Styryl Radical to a RAFT Agent

		E_0 (hartrees)			Energy
		Reactant 1	Reactant 2	Product	(kJ mol ⁻¹)
Inner core					
W1	E1	-39.84359	-438.37761	-478.27574	-143.19
G3(MP2)-RAD	E2	-39.78519	-436.97759	-476.81289	-131.56
Core					
G3(MP2)-RAD	E3	-309.80231	-913.28619	-1223.10572	-45.21
ROMP2/6-311+ G(3df,2p)	E4	-309.51471	-913.06303	-1222.60105	-61.20
Full					
ROMP2/6-311+ G(3df,2p)	E5	-519.26440	-1235.69267	-1754.99865	-109.17
ONIOM(W1)	E1-E2+E3 -E4+E5	-519.61032	-1237.31585	-1756.96617	-105.02

a successful application of this technique to an entire polymerization process, refer to the work of Coote et al.¹⁸ A worked example for RAFT-mediated polymerization of styrene is provided in Scheme 13.4 and Table 13.3.

13.3.6 Conformational Searching

Studying radical polymerization processes using dimer or trimer models typically involves the calculation of moderately sized molecules, having 20 to 40 nonhydrogen atoms. Obtaining optimized geometries and frequencies for individual molecules of this size is computationally feasible using lower-cost computational

procedures such as B3LYP/6-31G(d). While accurate energy calculations require more computationally demanding methods, we have shown above that these can also be achieved for systems of this size using carefully designed ONIOM approximations. However, to study these reactions in practical situations, a further computational bottleneck emerges: obtaining appropriate conformations for the reactants, products, and transition structures.

Conformational isomerism occurs when molecules with the same structural formula exist as different nonsuperimposable structures in three-dimensional space, separated by hindered internal rotations about one or more of their single bonds. For present purposes, rotations about double bonds (i.e., *cis-trans* isomers) are also included in this broad definition. The conformational space is the set of all possible conformers of a given molecule, and can grow very rapidly with the size of the molecule. For example, a trimer of vinyl chloride has potentially 81 different (local) minimum energy conformations, corresponding to the (nonequivalent) threefold rotations about its backbone C—C bonds; when conformations of the side chain are also relevant, this number increases further (e.g., the trimer of methyl acrylate has 5184 minimum-energy conformations due to the additional two twofold rotations on each side chain). In general, a bond between two sp^3 carbon centers should normally have at most three minima separated at approximately 120° , for a molecule with N rotatable bonds, the conformational space is thus $(360^\circ/120^\circ)^N$. In principle, the molecule will exist as an ensemble of its conformers, although in practice only the lowest-energy structures will be significantly populated, as according to Boltzmann's law. Moreover, for thermochemical purposes it is usually sufficient to locate the minimum-energy structure or a structure that is close to it in energy: For those conformations with energies similar to those of the global minimum yield similar results and hence do not affect the thermally averaged energies; those with much higher energies are not populated significantly and therefore do not contribute significantly to the thermally averaged energies. However, if one follows this approach, it is essential that the conformation chosen is the global minimum or one very close to it in energy; simply choosing a conformation at random might result in a structure that is tens or even hundreds of kJ mol^{-1} too high in energy.

Although in principle one need only consider the minimum-energy conformation of a molecule, actually locating this conformation poses a major problem. This is because the geometry optimization routines of computational chemistry packages locate the local minimum-energy structure that is nearest the starting guess. The only systematic way to ensure that the optimized geometry is a global rather than merely a local minimum is therefore to repeat the geometry optimization for starting guesses corresponding to every possible conformation. That is, one might begin with an optimized geometry for a random starting conformation and then use it to generate starting structures corresponding to all combinations of all nonequivalent threefold rotations (i.e., dihedral increments of 0° , 120° , and 240°) about every single bond (with corresponding twofold rotations for every double bond). These starting structures are then optimized as well, and the optimized energies are then compared so as to identify the

global minimum. This approach to conformational analysis, known more generally as a tree search⁴² method, is feasible for small molecules having tens of conformations, but, as seen above, rapidly becomes infeasible for larger molecules, due to the combinatorial explosion problem.

Several algorithms are available to find the low-energy conformers of a molecule without searching through the full conformational space, and these vary in their computational cost and their accuracy.⁴³ The majority of these algorithms, of which *simulated annealing*⁴⁴ would be a prominent example, are highly stochastic and have been designed for studying very large molecules (such as proteins) with full conformational spaces comprising millions or billions of conformers. For such large molecules, performing kilocalorie-accurate quantum chemical calculations on even a single conformation is in any case not practical; hence, the accuracy versus efficiency demands are very different from those required to perform accurate thermochemical calculations on smaller molecular systems. As a result, stochastic algorithms are designed to explore only a small fraction of the overall conformational space, and although they can efficiently identify relatively low-energy conformers, they do not offer the reliability needed when studying radical polymerization kinetics and thermodynamics.

To address this problem, we have recently introduced a new type of conformational searching method called *energy-directed tree search* (EDTS).⁴³ This method is designed to identify the lowest-energy conformation of a molecule (or at least a structure within 1 kcal mol⁻¹ of it) with a very high degree of reliability, without exhaustively searching all conformations. By its nature, such an algorithm remains too computationally expensive for studying very large molecules, but is suitable for the moderately large molecules (having conformational spaces in the hundreds or thousands) common to free-radical polymerization studies—molecules for which accurate single-point energy calculations are possible but for which full systematic conformational searching is impractical. EDTS is similar to a full tree search method but reduces the dimensionality of the conformational space using an approach inspired by the buildup⁴⁵ principle. In a buildup approach, one performs full systematic conformational searches on small subunits of the molecule (either as isolated fragments or in situ), and then assembles the overall conformation of the molecule from these fragments with only limited conformational searching of their relative conformations. In the EDTS algorithm, we use this general strategy but perform conformational searching on the molecule as a whole and use a built-in learning approach to identify which combinations of dihedral angles need to undergo full conformational searching rather than relying on arbitrary spatial boundaries or chemical intuition.

The basic algorithm is shown in Fig. 13.3, and two fully worked examples are provided in the supporting information section of an article by Izgorodina et al.⁴³ In essence, one begins with an optimized geometry of a randomly chosen conformation and performs an initial linear search of the conformational space. That is, one examines separately all possible conformations about each bond of the molecule in isolation; for each rotation, the rest of the molecule is optimized but no further conformational searching is undertaken. This is effectively an extreme

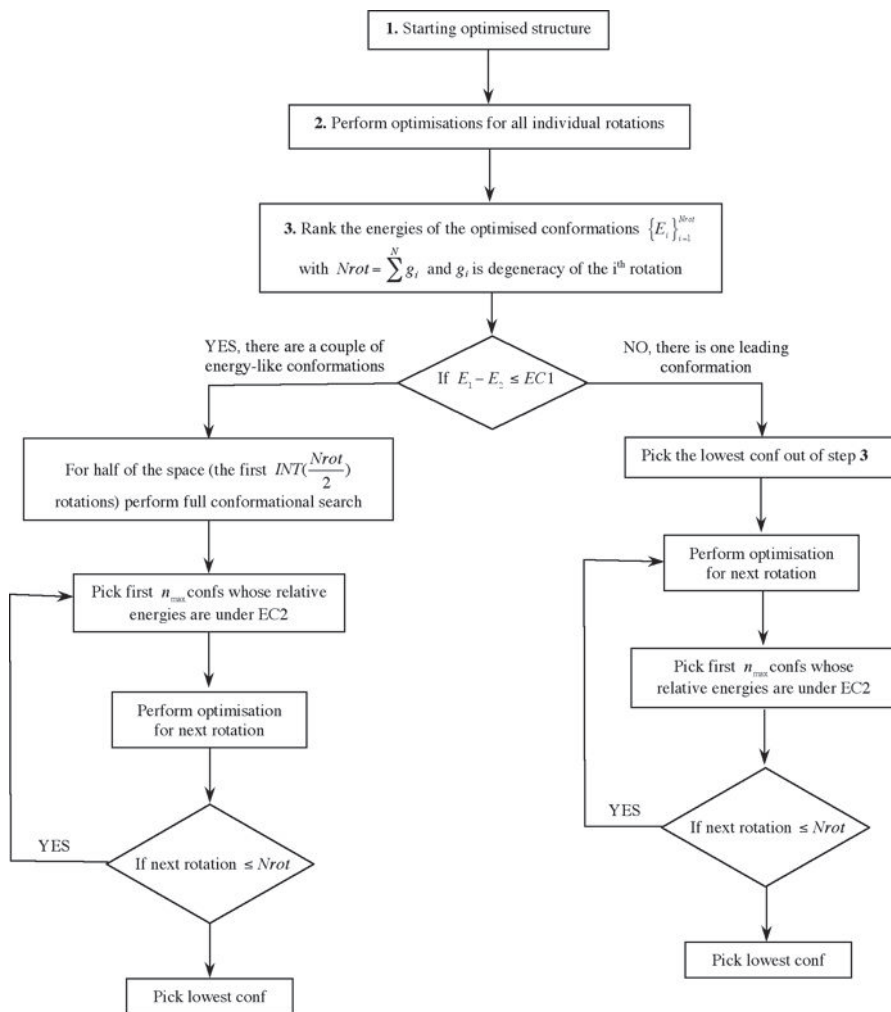


Fig. 13.3 EDTS flowchart. Optimal values of EC1, EC2, and NMAX are at 3 kJ mol⁻¹, 4 kJ mol⁻¹, and 5, respectively. (From Ref. 43.)

example of the buildup approach, whereby the conformations of all bonds of the molecule are optimized separately. Although this is extremely computationally efficient, its accuracy is crucially dependent on the starting structure and the order in which the rotations are analyzed. The key feature of EDTS is to utilize the relative energies of the conformations generated in the initial linear search to refine the starting structure and define the optimal order in which the rotations should be examined. This information is then used to perform a second and final linear search, this time with a “starting” structure that is updated to the current global minimum after each rotation. In cases where an initial linear search yields a large number of low-energy conformations with similar energies

from one to another (within the tolerance value $EC1 = 3 \text{ kJ mol}^{-1}$), an additional step in the algorithm is introduced whereby a full systematic search is performed on the lower half of the conformational space. Moreover, if at any time in the subsequent search, there are multiple conformations whose energies are similar to one another (within the tolerance value $EC2 = 4 \text{ kJ mol}^{-1}$), the linear search is continued in parallel with these multiple conformers (up to a maximum of $NMAX = 5$) as “starting” structures.

The cost of the EDTS algorithm depends on whether or not these extra precautions are required: at its best it is linear scaling; in the worst-case scenario, this method scales 2^N compared with 3^N for the corresponding full systematic search. Either way, the algorithm dramatically cuts the cost of searching conformational space for larger molecules (i.e., having 100 or more conformers) and remains sufficiently economical for most of the structures that would be encountered in a study of radical polymerization processes. Importantly, in all systems so far tested, the algorithm has successfully located either the global minimum energy structure or a species within 4 kJ mol^{-1} of it. More details regarding this method and overviews of current conformational search methods are described by Izgorodina et al.⁴³

In principle, the EDTS algorithm can be implemented “by hand” in conjunction with any computational chemistry software package. That is, one could create starting geometries for the initial linear search by hand, optimize these using any computational chemistry program, sort the resulting energies in a spreadsheet, and use the results to choose the next conformations to be searched (following the flowchart in Fig. 13.3), which could then be created by hand, optimized using the computational chemistry software, and so on. Of course, this becomes very cumbersome for large conformational searches and we have created some scripts that automate this process.

- **CONFMAKER** is a program to create a full conformational space for user-defined rotatable bonds, each of which is rotated at its own user-defined resolution. It requires a *z*-matrix input file with appropriate GAUSSIAN job specification lines where the user can specify which level of theory to use, and another file to specify which dihedral angles to rotate and the increments (e.g., 0° , 120° , 240°). This then returns a full conformational space of GAUSSIAN input files. This can easily be altered to create jobs for any other quantum chemistry package.
- **CONFSEARCH** is an automation program of the EDTS algorithm, it currently works with GAUSSIAN, but it is very easy to apply to any other quantum chemistry package. This takes the input files for the full conformational space, as generated from the CONFMAKER program, filters through any sterically hindered structures, and submits appropriate jobs to a large computer cluster automatically. Once the program is running, it will rank all the energies of the optimized conformations in the background, and, applying the algorithm, use the results to submit more structures that are likely to have low energies. Finally, it returns the lowest-energy structure according

to the EDTS algorithm and gives a report of all the energy of structures that have been done. This cannot only be used to find a global minimum, but also works with saddle points where the lowest transition-state structure can be found through exploring the entire conformation space.

Both programs are freely available at <http://rsc.anu.edu.au/~cylin/scripts.html>.

13.3.7 IRCMax

IRCMax is a method developed by Petersson and co-workers⁴⁶ for improving the accuracy of transition-structure optimizations in a computationally efficient manner; techniques based on the same principle have also been used to calculate improved imaginary frequencies and tunneling coefficients.^{47–49} It is based on the observation that stable structures can generally be optimized at relatively low levels of theory, while transition structures tend to require higher-level procedures to achieve the same level of accuracy, implying that it is probably only the stretched forming and breaking bonds of the transition state that are difficult to model. To address this problem, in the IRCMax method the minimum energy path (MEP) for a reaction is first calculated at a low level of theory and then improved via single-point energy calculations at a higher level of theory. The improved transition structure is then identified as the maximum energy structure along the MEP for the reaction. By identifying the transition structure from the high-level MEP (rather than the original low-level MEP), one effectively optimizes the reaction coordinate at the high level of theory (see Fig. 13.4). As

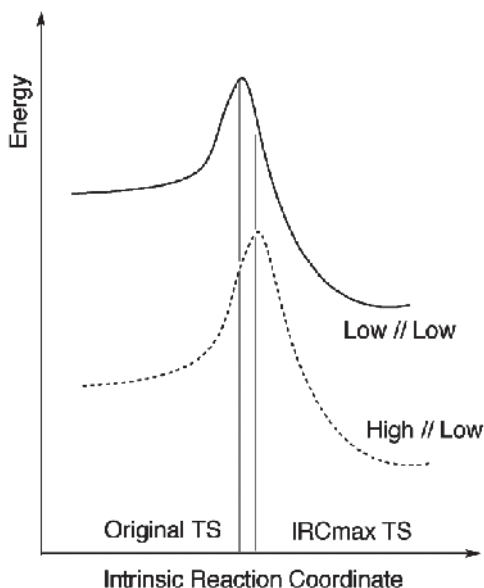


Fig. 13.4 IRCMax technique for transition-state optimization. (From Ref. 50.)

noted above, assessment studies for a wide variety of radical reactions have shown that low-level HF/6-31G(d) or B3LYP/6-31G(d) transition structures, which have been corrected to a higher level of theory, such as QCISD/6-31G(d), via the IRCMax technique, offer an excellent approximation to transition structures that have been fully optimized at the higher level of theory but at a small fraction of the computational cost.^{34b-d}

13.4 CALCULATION OF KINETICS AND THERMODYNAMICS

13.4.1 Overview

Having obtained the geometries, frequencies, and energies of the reactants and products in a chemical reaction, it is possible to use these to obtain the equilibrium constant, partition functions, and related thermochemical functions (entropies, enthalpies, free energies) using standard formulas⁵¹ derived from the statistical thermodynamics of an ideal gas under the harmonic oscillator/rigid rotor approximation. If one also treats the transition state of the reaction in the same manner, one can use transition-state theory⁵² to obtain the rate coefficient of the reaction. These calculations are based on several assumptions and approximations, and we therefore also outline some important additional corrections that are required in order to improve the accuracy of the results.

13.4.2 Basic Formulas

The equilibrium (K) and rate (k) constants for a chemical reaction can be calculated from the corresponding geometries, frequencies, and total energies as follows:

$$K(T) = (c^\circ)^{\Delta n} e^{-\Delta G/RT} = (c^\circ)^{\Delta n} \frac{\prod_{\text{products}} Q_j}{\prod_{\text{reactants}} Q_i} e^{-\Delta E/RT} \quad (13.5)$$

$$k(T) = \kappa(T) \frac{k_B T}{h} (c^\circ)^{1-m} e^{-\Delta G^\ddagger/RT} = \kappa(T) \frac{k_B T}{h} (c^\circ)^{1-m} \frac{Q_\ddagger}{\prod_{\text{reactants}} Q_i} e^{-\Delta E^\ddagger/RT} \quad (13.6)$$

In these formulas, $\kappa(T)$ is the tunneling correction factor, T is the absolute temperature; k_B is Boltzmann's constant (1.380658×10^{-23} J molec⁻¹ K⁻¹); h is Planck's constant ($6.6260755 \times 10^{-34}$ J s); c° is the standard unit of concentration (mol L⁻¹); R is the universal gas constant (8.314 J mol⁻¹ K⁻¹); m is the molecularity of the reaction; Δn is the change in moles upon reaction; $Q_\ddagger$, Q_i , and Q_j are the molecular partition functions of the transition structure, reactant i , and product j , respectively; $\Delta G^\ddagger$ is the Gibbs free energy of activation; ΔG is the Gibbs free energy of reaction; $\Delta E^\ddagger$ is the 0-K zero-point energy-corrected

energy barrier for the reaction; and ΔE is the 0-K zero-point energy-corrected energy change for the reaction. The value of c° depends on the standard-state concentration assumed in calculating the thermodynamic quantities (and translational partition function); this is discussed in more detail below. The tunneling coefficient $\kappa(T)$ corrects for quantum effects in motion along the reaction path and is close to unity in most chemical reactions, although important exceptions include hydrogen atom transfer. Details on how to calculate it are provided in Section 13.4.3.

The molecular partition functions serve as a bridge between the quantum mechanical states of a system and its thermodynamic properties, and are given by:

$$Q = \sum_i g_i \exp\left(-\frac{\varepsilon_i}{k_B T}\right) \quad (13.7)$$

The values ε_i are the energy levels of a system, each having a number of degenerate states g_i , and are obtained by solving the Schrödinger equation. In theory, this equation should be solved for all active modes but in practice the calculations can be greatly simplified by separating the partition function into the product of the translational, rotational, vibrational, and electronic terms, as follows:

$$Q = Q_{\text{trans}} \times Q_{\text{rot}} \times Q_{\text{vibr}} \times Q_{\text{elec}} \quad (13.8)$$

This is generally a reasonable assumption, provided that the reaction occurs on a single electronic surface. Finally, if we assume that reacting species are ideal gas molecules, analytical expressions for the partition functions are as follows:

$$Q_{\text{trans}} = V \left(\frac{2\pi M k_B T}{h^2} \right)^{3/2} = \frac{RT}{P} \left(\frac{2\pi M k_B T}{h^2} \right)^{3/2} \quad (13.9)$$

$$Q_{\text{rot,linear}} = \frac{1}{\sigma_r} \left(\frac{T}{\Theta_r} \right) \quad \text{where } \Theta_r = \frac{h^2}{8\pi^2 I k_B} \quad (13.10)$$

$$Q_{\text{rot,nonlinear}} = \frac{\pi^{1/2}}{\sigma_r} \frac{T^{3/2}}{(\Theta_{r,x} \Theta_{r,y} \Theta_{r,z})^{1/2}} \quad \text{where } \Theta_{r,i} = \frac{h^2}{8\pi^2 I_i k_B} \quad (13.11)$$

$$Q_{\text{vib}} = \prod_i \exp\left(-\frac{1}{2} \frac{h\nu_i}{k_B T}\right) \prod_i \frac{1}{1 - \exp(-h\nu_i/k_B T)} \quad (13.12)$$

$$Q_{\text{elec}} = \omega_0 \quad (13.13)$$

In Eqs. (13.9) to (13.13), M is the molecular mass of the species; V is the reference volume; T and P are the corresponding reference temperature and pressure; I is the principal moment of inertia of a linear molecule, while for the nonlinear case, I_x , I_y , and I_z are the principal moments of inertia about axes x , y , and z , respectively; σ_r is the symmetry number of the molecule, which counts its

number of symmetry-equivalent forms⁵³; ν_i are the vibrational frequencies of the molecule; and ω_0 is the degeneracy of the spin states. This is usually equal to the multiplicity of the molecule (i.e., $\omega_0 = 1$ for singlet species, $\omega_0 = 2$ for doublet species, etc.); however, for atoms, the degeneracy of spin states also depends on the total angular momentum. For example, for F, Cl, Br, and I, $\omega_0 = 4$ from $2J + 1$, where J is the total angular momentum and is equal to $3/2$. The information required to evaluate these partition functions is routinely accessible from quantum chemical calculations: The moments of inertia and symmetry numbers depend on the geometry of the molecule, while the vibrational frequencies are obtained from the second derivative of the energy with respect to the geometry using the harmonic approximation to generate normal coordinates and vibrational frequencies.

A number of additional comments need to be made concerning the use of Eqs. (13.9) to (13.13). First, in the calculation of the translational partition function [Eq. (13.9)], a reference volume (or equivalently, a temperature and pressure) is assumed. This is needed for the calculation of thermodynamic quantities such as enthalpy and entropy, but the assumption has no bearing on the calculated rate and equilibrium constants, as the reference volume is removed from Eqs. (13.5) and (13.6) through the parameter $c^\circ (= n/V = P/RT)$. Second, the external rotational partition function is calculated using Eq. (13.10) if the molecule is linear, and Eq. (13.11) if it is a nonlinear molecule. Third, the vibrational partition function [Eq. (13.12)] has been written as the product of two terms. The first of these corresponds to the zero-point vibrational energy of the molecule, while the latter corresponds to its additional vibrational energy at some nonzero temperature T . The zero-point vibrational energy is often included in the calculated reaction barrier E_0 . When this is the case, this first term must be removed from Eq. (13.12) so as not to count this energy twice. For atomic species both Q_{rot} and Q_{vib} are equal to unity.

The thermodynamic properties of a system (e.g., Gibbs free energy G , enthalpy, H , and entropy, S) are related to the partition functions and can also be calculated from the geometries, energies, and frequencies as follows. The Gibbs free energy of a species is related to the enthalpy and entropy via

$$G = H - T \cdot S \quad (13.14)$$

The enthalpy of a species can be written as the sum of its electronic energy (E_0), zero-point vibrational energy (ZPVE), and temperature correction ($\Delta\Delta H$), in turn calculable from the vibrational frequencies as follows:

$$H = E_0 + \text{ZPVE} + \Delta\Delta H \quad (13.15)$$

$$\text{ZPVE} = R \frac{1}{2} \sum_i \frac{h\nu_i}{k_B} \quad (13.16)$$

$$\Delta\Delta H = \frac{3}{2}RT + \frac{3}{2}RT + R \sum_i \frac{h\nu_i/k_B}{\exp(h\nu_i/k_B T) - 1} + 0 + RT \quad (13.17)$$

In Eq. (13.17), the first term is the translational contribution to the thermal correction and the second term is rotational contribution for nonlinear species in this case; for linear species, the second term becomes RT ; for atoms, the second term is zero. The third term is the vibrational contribution and the fourth term is the electronic contribution to the thermal correction, and due to no temperature-dependent term in the electronic partition function, this term is always zero. The last term, RT , is included to differentiate enthalpy from energy, so when applying ΔH for a reaction, $\Delta n RT = PV$ is included. The entropy of a species is calculated from the translational (S_{trans}), rotational (S_{rot}), vibrational (S_{vib}), and electronic (S_{elec}) contributions to the entropies of the individual species, in turn expressed as follows:

$$S = S_{\text{trans}} + S_{\text{rot}} + S_{\text{vib}} + S_{\text{elec}} \quad (13.18)$$

$$S_{\text{trans}} = R \left\{ \ln \left[\left(\frac{2\pi M k_B T}{h^2} \right)^{3/2} \frac{k_B T}{P} \right] + 1 + \frac{3}{2} \right\} \quad (13.19)$$

$$S_{\text{rot,linear}} = R \left[\ln \left(\frac{1}{\sigma_r} \frac{T}{\Theta_r} \right) + 1 \right] \quad (13.20)$$

$$S_{\text{rot,nonlinear}} = R \left[\ln \left(\frac{\pi^{1/2}}{\sigma_r} \frac{T^{3/2}}{(\Theta_{r,x} \Theta_{r,y} \Theta_{r,z})^{1/2}} \right) + \frac{3}{2} \right] \quad (13.21)$$

$$S_{\text{vib}} = R \sum_i \left(\frac{h\nu_i/k_B T}{\exp(h\nu_i/k_B T) - 1} - \ln \left[1 - \exp \left(\frac{-h\nu_i}{k_B T} \right) \right] \right) \quad (13.22)$$

$$S_{\text{elec}} = R \ln(\omega_0) \quad (13.23)$$

The parameters required to evaluate these expressions are the same as those used in evaluating the partition functions, as described above. For an atomic species, both S_{rot} and S_{vib} are zero.

Finally, by evaluating the derivative of Eq. (13.6) with respect to temperature, it is possible to derive a relationship between the thermodynamic quantities above and the empirical Arrhenius expression for reaction rate coefficients^{51d}:

$$k(T) = A e^{-E_a/RT} \quad (13.24)$$

The frequency factor (A) in this expression is related to the entropy of the system as follows:

$$A = \kappa(c^\circ)^{1-m} e^m \frac{k_B T}{h} \exp \left(\frac{\Delta S^\ddagger}{R} \right) \quad (13.25)$$

The Arrhenius activation energy (E_a) is related to the reaction barrier as follows:

$$E_a = E_0 + \text{ZPVE} + \Delta \Delta H^\ddagger + mRT \quad (13.26)$$

From these expressions it can be seen that even when tunneling can be ignored, the “temperature-independent” parameters of the Arrhenius expression are, in fact, functions of temperature, which is why the Arrhenius expression is valid only over relatively small temperature ranges. It should also be clear that the ZPVE-corrected barrier ($\Delta E^\ddagger$), the enthalpy of activation ($\Delta H^\ddagger$), and the Arrhenius activation energy (E_a) are equal to each other only at 0 K. At nonzero temperatures, these quantities are nonequivalent and thus should not be used interchangeably.

Although it is straightforward to get the energy, enthalpy and free energy of a molecule directly from a quantum chemistry package such as GAUSSIAN, it is usually more convenient to calculate these separately (e.g., to allow for scaling of the frequencies or variation of the temperature). We therefore have developed a program called T-CHEM, which reads a GAUSSIAN frequency calculation output file and gathers all the information it needs in order to generate the temperature-correction terms $\Delta\Delta H$ as in Eq. (13.17) and entropy S in Eq. (13.18); these can then be combined with the high-level zero-point vibrational energy-corrected electronic energies so as to calculate the barriers and enthalpies for the reaction and the corresponding rate and equilibrium constants. This program is freely available at <http://rsc.anu.edu.au/~cylin/scripts.html>.

13.4.3 Beyond Transition-State Theory

Transition-state theory⁵² assumes that in the space represented by the coordinates and momenta of the reacting particles, it is possible to define a dividing surface such that all reactants crossing this plane go on to form products and do not re-cross the dividing surface. The minimum-energy structure on this dividing plane is referred to as the *transition structure* or *transition state* of the reaction. Transition-state theory also assumes that there is an internal statistical equilibrium between the degrees of freedom of each type of system (reactant, product, or transition state) and that the transition state is in statistical equilibrium with the reactants. In addition, it assumes that motion through the transition state can be treated as a classical translation. These assumptions are normally appropriate for the types of reactions of relevance to free-radical polymerization; however, there are two important exceptions, as follows.

First, for low- or zero-energy barrier reactions, extra care must be taken to choose a transition state so as to make the “no re-crossing” assumption as valid as possible. In simple transition-state theory, the transition state is located as the maximum-energy structure, along the minimum-energy path connecting the reactants and products. This is generally a good approximation for reactions having barriers that are large compared to RT . However, for reactions with low or zero barriers, a more accurate approach is required. To this end, in *variational transition-state theory*, the transition state is located as the structure (on the minimum-energy path) that yields the lowest reaction rate. In thermodynamic terms, this may be thought of as the maximum in the Gibbs free energy of activation rather than the maximum internal energy of activation.

Second, the assumption that motion through the transition state can be treated as a classical translation is valid only if the reacting species are relatively large and hence their wavelengths are relatively small compared with the barrier width. Although this is the case for most chemical reactions, the assumption breaks down for electron, hydrogen, and (to a lesser extent) deuterium transfer reactions, as in these cases the molecular mass of the species being transferred is relatively small, and thus quantum effects can be very important. In such cases, it is possible for particles with energies below the reaction barrier to appear on the other side of the barrier, a phenomenon known as *tunneling*. This has the effect of increasing the reaction rate observed by as much as two orders of magnitude at room temperature compared with the value calculated via classical mechanics. It has been found that the probability of tunneling occurring relates to the height, width, and shape of the barrier and the mass of the particle; temperature also plays an important role in determining how many particles tunnel through. Corrections for quantum mechanical tunneling are incorporated into the κ coefficient of Eq. (13.6), and are known as *tunneling coefficients*.

There are an enormous variety of expressions available for calculating tunneling coefficients. The most accurate methods, such as *small curvature tunneling*,⁵⁴ *large curvature tunneling*,⁵⁵ and *microcanonical optimized multidimensional tunneling*,⁵⁶ involve solving the multidimensional Schrödinger equation describing the motion of the molecules at every position along the reaction coordinate. To calculate such tunneling coefficients, specialized software (such as POLYRATE⁵⁷) is used, and additional quantum chemical data (such as the geometries, energies, and frequencies along the entire minimum-energy path) are required. As a result, simpler (and hence less accurate) expressions are often adopted. These are derived by treating motion along the reaction coordinate as a function of one variable, the intrinsic reaction coordinate, and hence solving a one-dimensional Schrödinger equation:

$$\frac{d^2\Psi(x)}{dx^2} + \frac{2M}{\hbar^2}[W - V(x)]\Psi(x) = 0 \quad (13.27)$$

where $\Psi(x)$ is the wavefunction, W is the energy, and M is the mass of the particle of interest. When this is done using the energies calculated along the reaction path, the procedure is known as *zero-curvature tunneling*.⁵⁸ However, this procedure still entails numerical solution of the Schrödinger equation, and hence an additional simplification is also often made. Instead of using the energies calculated along this path, some assumed functional form for the potential energy is used instead. This is chosen so that the Schrödinger equation has an analytical solution, and thus a closed expression for the tunneling coefficient can be derived. The derivation of these simple tunneling coefficients is described by Bell,⁵⁹ and the most accurate of these is described below.

In the Eckart method, the change in potential energy along the minimum energy path is described by the following *Eckart function*⁶⁰:

$$V(x) = \frac{Ay}{(1+y)^2} + \frac{By}{(1+y)} \quad \text{where } y = e^{x/\Delta} \quad (13.28)$$

To ensure that the function passes through the reactants, products, and transition structures, the parameters A and B are defined as the following functions of the forward (V_f) and reverse (V_r) reaction barriers (where the reaction is taken in the exothermic direction):

$$A = (\sqrt{V_f} + \sqrt{V_r})^2 \quad \text{and} \quad B = V_f - V_r \quad (13.29)$$

The remaining parameter Δ is chosen so as to give the most appropriate fit to the minimum-energy path. If this fit is biased toward the points near the transition structure (where tunneling is most important), it can be calculated as the following function of the imaginary frequency $\nu_{\ddagger}$ (where c is the speed of light)^{47,49}:

$$\Delta = \frac{i}{2\pi c \nu_{\ddagger}} \sqrt{\frac{1}{8} \frac{(B^2 - A^2)^2}{A^3}} \quad (13.30)$$

The Δ value obtained from this expression is in mass-weighted coordinates, which enables the reduced mass to be dropped from the standard⁶⁰ Eckart formulas,⁴⁹ resulting in the following expression for the permeability of the reaction barrier $G(W)$ as a function of the energy W :

$$G(W) = 1 - \frac{\cosh(\alpha - \beta) + \cosh \delta}{\cosh(\alpha + \beta) + \cosh \delta}$$

where

$$\alpha = \frac{4\Delta\pi^2}{h} \sqrt{2W} \quad \beta = \frac{4\Delta\pi^2}{h} \sqrt{2(W-B)} \quad \delta = \frac{4\Delta\pi^2}{h} \sqrt{2A - \frac{h^2}{16\pi^2\Delta^2}} \quad (13.31)$$

The Eckart tunneling correction (κ) is then obtained by numerically integrating $G(W)$ over a Boltzmann distribution of energies, via⁵⁹

$$\kappa = \frac{\exp(V_f/k_B T)}{k_B T} \int_0^\infty G(W) \exp\left(-\frac{W}{k_B T}\right) dW \quad (13.32)$$

Although this expression requires numerical integration, it does not require sophisticated software and can be implemented easily on a spreadsheet; an Excel file for performing the calculation, given the forward and reverse barriers, the temperature, and the imaginary frequency, is freely provided at <http://rsc.anu.edu.au/~cylin/scripts.html>.

13.4.4 Beyond the Harmonic Oscillator Approximation

In most quantum chemistry packages, Eqs. (13.9) to (13.23) are used for calculating the partition functions and associated thermodynamic data, and thus assume an ideal gas under the rigid rotor/harmonic oscillator approximation. Under the harmonic oscillator approximation, the potential for each vibrational mode is treated as a parabolic well. For bond-stretching modes, this is a close approximation to the true potential energy surface; however, for a torsional mode, the potential energy surface may have multiple minima and is better described as a hindered internal rotation. For high-frequency modes ($\nu > 300 \text{ cm}^{-1}$), the contribution of these motions to the overall partition function is negligible at room temperature (i.e., $Q_{\text{vib},i} \approx 1$), and thus the error incurred in treating these modes as harmonic oscillators is not significant. However, for the low-frequency torsional modes, these errors can be significant and a more rigorous treatment is often necessary; this is especially the case for the reactions of relevance to free-radical polymerization.^{8,11,12b,c,l}

Ideally, one should solve the Schrödinger equation for the full multidimensional potential energy surface representing all active modes of a molecule, and use the resulting energy levels in Eq. (13.7) to obtain the partition functions; however, this is impractical for larger molecules. Instead, the approach that is usually adopted is to apply the harmonic oscillator approximation to all $3N - 6$ internal modes of a molecule (as in the standard formulas above), but then multiply the resulting vibrational partition function by a correction factor for each internal hindered rotor partition function. This factor is calculated as the ratio of the 1D-HR partition function to the corresponding “pure” vibrational partition function, as calculated from the second derivative of the rotational potential at the minimum-energy structure. Using approximations such as this, the 1D-HR model has been shown to provide reasonable results in situations where testing against more sophisticated treatments is possible.⁶¹

To obtain the 1D-HR partition function for any given low-frequency torsional mode, we first need to compile the full rotational potential $V(\theta)$ for the mode in question; studies have shown that a resolution of 60° is sufficient for accurate results.⁶² The potential should be compiled as a relaxed scan (i.e., at each dihedral angle, the dihedral angle is frozen but the rest of the molecule is fully optimized) and, as in ordinary geometry optimizations, low levels of theory, such as B3-LYP/6-31G(d) are usually sufficiently accurate. Having obtained the potential, this is then used to solve the one-dimensional Schrödinger equation for a rigid rotor:

$$-\frac{\hbar^2}{2I_r} \frac{d^2\Psi}{d\theta^2} + V(\theta)\Psi = \varepsilon_i\Psi \quad (13.33)$$

In this equation Ψ is the wavefunction, ε is the energy, I_r is the reduced moment of inertia, and $V(\theta)$ is the rotational potential, which for this purpose should be supplied at a high resolution. To this end, the 60° resolution potential is fitted with a Fourier series of up to 18 terms and then reevaluated at a resolution of

1.2°. The reduced moment of inertia (I_r) is assumed to be independent of θ and is calculated from the optimized geometry using the equation for $I^{(2,3)}$, as defined by East and Radom.⁶³ There is no analytical solution to this Schrödinger equation; however, it can be solved numerically for the eigenvalues, ϵ , by converting it into the Hill differential equation. Having obtained the energy levels, these are then summed in order to obtain the partition function via Eq. (13.7), in the usual manner. A program called T-CHEM for performing these calculations is freely available at <http://rsc.anu.edu.au/~cylin/scripts.html>.

Finally, in addition to the approach described above, there are a number of lower-cost methods available for calculating hindered-rotor partition functions; some of which (such as the Pitzer tables⁶⁴) are applicable only for potentials that can be described by a pure cosine function; others are approximations designed for use with any type of partition function. It is beyond the scope of this chapter to detail these here, but a description and evaluation of these methods is found in the literature.^{62,65,66}

13.4.5 Solvent Effects

The methodology described thus far is designed to reproduce chemically accurate values of the rate and equilibrium constants for gas-phase systems, and the vast majority of computational studies of radical polymerization in the literature have indeed been performed in the gas phase. In many situations, the effects of solvents on radical reactions are relatively minor and the gas-phase calculations are indicative of solution-phase behavior. For example, gas-phase calculations of the propagation rate coefficients of vinyl chloride and acrylonitrile were able to reproduce the experimental (solution-phase) rate coefficients for these monomers to within a factor of 2, and solvation effects (as calculated using simple continuum models) were minor.⁸ Gas-phase studies of the equilibrium constants in certain RAFT polymerizations have also reproduced experimental data to within chemical accuracy, for both small model reactions^{16a} and polymeric systems.^{11e,18}

Nonetheless, there are free-radical polymerizations, such as those of monomers, that are capable of undergoing hydrogen bonding or other specific interactions with the solvent, where strong solvent effects have been well documented experimentally.^{6c,67} Not unexpectedly in such cases, there can be very large differences between the gas-phase rate coefficients calculated and the corresponding solution-phase values. For example, in a recent computational study⁶⁸ of the propagation rate coefficient of ethyl- α -hydroxymethacrylate (EHMA) the gas-phase rate coefficient calculated differed from the corresponding solution-phase experimental values⁶⁹ by more than five orders of magnitude. In such cases, the correct treatment of solvent effects is therefore crucial. Unfortunately, the development of cost-effective methods for treating the solvent in chemical reactions is an ongoing area of research and there have been relatively few benchmarking studies for the specific case of radical polymerization. Nonetheless, it is worth making a few general comments on the main strategies that are available for modeling solvation effects.

The simplest and most computationally efficient methods are *continuum models*, in which each solute molecule is embedded in a cavity surrounded by a dielectric continuum of permittivity ϵ .⁷⁰ Most models, of which the ab initio conductor-like solvation model (COSMO)⁷¹ and the polarizable continuum model (PCM)⁷² are prominent examples, also include terms for the nonelectrostatic contributions of the solvent, such as dispersion, repulsion, and cavitation. Some of the more recent models also incorporate more sophisticated treatments of the solvent itself. For example, COSMO-RS⁷³ is a variant of the COSMO model that describes the interactions in a fluid as local interaction of molecular surfaces, the interaction energies being quantified by the values of the two screening charge densities that form a molecule contact. SM6⁷⁴ (Solvent Model 6) is based on a generalized Born approach, which uses a long-range dielectric continuum to treat bulk electrostatics effects combined with short-range atomic surface tensions to account for first-shell solvent effects.

Continuum solvation models can be invoked in most of the leading computational chemistry software packages, and the reader is referred to their respective manuals for specific implementation details. However, the following general points should be noted. First, continuum solvation models rely upon empirically optimized parameters, and it is important to choose radii and levels of theory that are optimized for the specific method in use. As always, the choice of solvation method for any particular system should be determined through assessment studies. Second, the specification of a particular solvent depends on several parameters in addition to the dielectric constant, including the volume, density, and solvent radius. If using a nondefault solvent model, care must be taken to set all of these parameters appropriately. Third, since the levels of theory used for solvation energy calculations, typically small basis set HF or B3LYP calculations, are not usually sufficiently accurate for gas-phase energetics, the total free energies in solution should be calculated via a simple thermodynamic cycle as follows:

$$\Delta G_{\text{soln}} = \Delta G_{\text{gas}} + \Delta G_{\text{solv}} + \Delta G^{\text{1atm} \rightarrow \text{1M}} \quad (13.34)$$

In this equation, ΔG_{gas} is the gas-phase free energy of reaction, which is calculated separately at a high level of theory, and ΔG_{solv} , the free energy of *solvation*, should not be confused with the total free energy of reaction in *solution*. In some software packages, additional keyword(s) are required for the solvation free energy (the difference of the gas- and solution-phase free energies at the same level of theory) to be calculated. In GAUSSIAN, the SCFVAC keyword is used for this purpose. The final term in Eq. (13.34), $\Delta G^{\text{1atm} \rightarrow \text{1M}}$, is required for converting from the gas-phase standard state for an ideal gas (typically, 1 atm) to 1 M in solution, and is given by

$$\Delta G^{\text{1atm} \rightarrow \text{1M}} = \Delta n RT \ln(V) = \Delta n RT \ln \frac{RT}{P} \quad (13.35)$$

where Δn is the number of moles of gas change from reactants to products. As an example, at room temperature (298.15 K) and standard pressure (1 atm), this term has a value of 7.9 kJ mol^{-1} . Finally, having made the correction for the change in state, $\Delta G^{\text{1atm} \rightarrow \text{1M}}$, the standard unit of concentration in the rate and equilibrium constant expressions [Eqs. (13.5) and (13.6)] becomes $c^\circ = 1 \text{ mol L}^{-1}$, rather than its value for an ideal gas (e.g., $0.0408 \text{ mol L}^{-1}$ at room temperature and standard pressure).

Continuum models are designed to reproduce bulk or macroscopic behavior and can fare extremely well in certain applications, not least the prediction of solvation energies of stable organic molecules.^{74,75} Continuum models have been applied to radical polymerization processes with mixed results. In an early study, Thickett and Gilbert^{12g} used a simple PCM model to study the effect of solvent on acrylic acid propagation, confirming experimental observations⁷⁶ that aqueous solvation substantially lowers the reaction barrier. However, it was noted in this work that the levels of theory used in the gas- and solution-phase calculations were not accurate enough for quantitative predictions of the reaction rate. As noted above, in our study of vinyl chloride and acrylonitrile propagation, we found that continuum models slightly improved the agreement between theory and experiment; however, in those systems the solvation effects were very small and well within the uncertainty of the experimental and theoretical data.⁸ More encouragingly, we have found that the combination of high-level *ab initio* calculations with continuum solvation models can reproduce one- and two-electron redox potentials of a wide range of open- and closed-shell systems,⁷⁷ including systems directly relevant to atom transfer radical polymerization.^{9e,f} In such systems, the solvation effects are very large, due to the presence of charged species.

Nonetheless, in other systems, the continuum solvation models have failed to redress the deviations of theory and experiment. For the problematic EHMA system described above, the use of PCM solvation energies actually increased the deviation between theory and experiment from five orders of magnitude to as much as eight orders of magnitude, depending on the solvent.⁶⁸ This is presumably because continuum models do not take into account the hydrogen-bonding interactions, expected to be important in this system. Indeed, similar failures have been noted in other (non-polymer-related) systems where hydrogen bonding is important.⁷⁸ Moreover, even where explicit solute–solvent interactions can be neglected, the use of continuum models to study polymerization *kinetics* is likely to be problematic. This is because the results obtained using continuum models are highly sensitive to the choice of cavities, and these are typically parameterized to reproduce the free energies of solvation for a set of small *stable* organic molecules. As a result, the choice of appropriate cavities for weakly bound species such as transition structures can be difficult.⁷⁵

For problematic systems where strong explicit solute–solvent interactions are important, the inclusion of explicit solvent molecules in the *ab initio* calculation is necessary. Ideally, one should include many explicit solvent molecules in the calculation and try to reproduce bulk behavior via molecular dynamics or Monte Carlo simulations, combined with the imposition of periodic boundary

conditions.⁷⁹ However, such calculations are hampered by problems such as the lack of potentials that can adequately describe both cluster and bulk behavior and the rapid increase in the conformational possibilities as the number of individual components increases. As a result, such approaches are not currently practical for polymerization systems. A less computationally demanding approach, known as a cluster-continuum model,⁸⁰ is to include a small number of explicit solvent molecules in the calculation (effectively treating them as additional reactants), while modeling the remaining solvation effects via a continuum model. However, choosing an appropriate number of explicit solvent molecules and their location, without testing all possibilities exhaustively, is always problematic, particularly for larger molecules. Further work is required to design practical guidelines for applying these methods to polymerization systems. In the meantime, it is worth noting that very promising results have recently been obtained without the need for explicit solvent molecules using COSMO-RS solvation energies in conjunction with the standard high-level gas-phase methodology.^{8b} To date, this approach has been evaluated only for the propagation kinetics of methyl acrylate and vinyl acetate, two systems where simple continuum models fail.^{8b} If its excellent performance can be maintained for other problematic systems, this methodology will further expand the scope of computational radical polymerization.

13.5 CONCLUSIONS

Computational quantum chemistry has much to offer the experimental polymer chemist. At the microscopic level, it can be used to clarify the reaction mechanism and explain the effects of substituents on the individual reactions, thereby facilitating the rational design of optimal control agents. At the macroscopic level, it can be used to build accurate kinetic models for simulating the outcome of polymerization processes as a function of the reaction conditions, for use in process optimization and control. However, the success of computational chemistry is crucially dependent on choosing realistic model reactions and applying accurate computational procedures; simultaneously satisfying these competing demands has, until recently, been difficult. Nonetheless, in recent years the development of new cost-effective computational methods, along with concurrent increases in computing power, has at last brought chemical accuracy within reach. Although the treatment of solvent effects remains problematic, even here, computational quantum chemistry has now proven itself a reliable and useful tool and an important complement to experiment.

REFERENCES

1. For more information on the chemistry and kinetics of free-radical polymerization, see, e.g., (a) Matyjaszewski, K.; Davis, T. P. *Handbook of Radical Polymerization*, Wiley, Hoboken, NJ, 2002. (b) Moad, G.; Solomon, D. H. *The Chemistry of Free-Radical Polymerization*, Pergamon Press, Oxford, UK, 1995. (c) Odian, G. *Principles of Polymerization*, Wiley-Interscience, New York, 1991.

2. Kamigaito, M.; Satoh, K. *Macromolecules* **2008**, *41*, 269–276.
3. Moad, G.; Rizzardo, E.; Thang, S. H. *Aust. J. Chem.* **2005**, *58*, 379–410.
4. Matyjaszewski, K. *Prog. Polym. Sci.* **2005**, *30*, 858–875.
5. Hawker, C. J.; Bosman, A. W.; Harth, E. *Chem. Rev.* **2001**, *101*, 3661–3688.
6. (a) Coote, M. L.; Zammit, M. D.; Davis, T. P. *Trends Polym. Sci.* **1996**, *4*, 189–196. (b) van Herk, A. M. *Macromol. Theory Simul.* **2000**, *9*, 433–441. (c) Beuermann, S.; Buback, M. *Prog. Polym. Sci.* **2002**, *27*, 191–254. (d) Barner-Kowollik, C.; Buback, M.; Egorov, M.; Fukuda, T.; Goto, A.; Olaj, O. F.; Russell, G. T.; Vana, P.; Yamada, B.; Zetterlund, P. B. *Prog. Polym. Sci.* **2005**, *30*, 605–643.
7. Barner-Kowollik, C.; Buback, M.; Charleux, B.; Coote, M. L.; Drache, M.; Fukuda, T.; Goto, A.; Klumperman, B.; Lowe, A. B.; McLeary, J. B.; Moad, G.; Monteiro, M. J.; Sanderson, R. D.; Tonge, M. P.; Vana, P. *J. Polym. Sci. A* **2006**, *44*, 5809–5831.
8. See, e.g., (a) Izgorodina, E. I.; Coote, M. L. *Chem. Phys.* **2006**, *324*, 96–110. (b) Lin, C. Y.; Izgorodina, E. I.; Coote, M. L. *Macromolecules* **2010**, *43*, 533–560.
9. (a) Gillies, M. B.; Matyjaszewski, K.; Norrby, P.-O.; Pintauer, T.; Poli, R.; Richard, P. *Macromolecules* **2003**, *36*, 8551–8559. (b) Singleton, D. A.; Nowlan, D. T., III; Jahed, N.; Matyjaszewski, K. *Macromolecules* **2003**, *36*, 8609–8616. (c) Matyjaszewski, K.; Poli, R. *Macromolecules* **2005**, *38*, 8093–8100. (d) Lin, C. Y.; Coote, M. L.; Petit, A.; Richard, P.; Poli, R.; Matyjaszewski, K. *Macromolecules* **2007**, *40*, 5985–5994. (e) Tang, W.; Kwak, Y.; Braunecker, W.; Tsarevsky, N. V.; Coote, M. L.; Matyjaszewski, K. *J. Am. Chem. Soc.* **2008**, *130*, 10702–10713. (f) Lin, C. Y.; Coote, M. L.; Gennaro, A.; Matyjaszewski, K. *J. Am. Chem. Soc.*, **2008** *130*, 12762–12774.
10. (a) Marsal, P.; Roche, M.; Tordo, P.; de Sainte Claire, P. *J. Phys. Chem. A* **1999**, *103*, 2899–2905. (b) Gimes, D.; Gaudel-Siri, A.; Marque, S. R. A.; Bertin, D.; Tordo, P.; Astolfi, P.; Greci, L.; Rizzoli, C. *Helv. Chim. Acta* **2006**, *89*, 2312–2326. (c) Kaim, A.; Megiel, E. *J. Polym. Sci. A* **2005**, *44*, 914–927. (d) Kaim, A. *J. Polym. Sci. A* **2006**, *45*, 232–241. (e) Megiel, E.; Kaim, A. *J. Polym. Sci. A* **2008**, *46*, 1165–1177.
11. (a) Farmer, S. C.; Patten, T. E. *J. Polym. Sci. A* **2002**, *A40*, 555–563. (b) Coote, M. L.; Radom, L. *J. Am. Chem. Soc.* **2003**, *125*, 1490–1491. (c) Coote, M. L.; Radom, L. *Macromolecules* **2004**, *37*, 590–596. (d) Coote, M. L. *Macromolecules* **2004**, *37*, 5023–5031. (e) Feldermann, A.; Coote, M. L.; Stenzel, M. H.; Davis, T. P.; Barner-Kowollik, C. *J. Am. Chem. Soc.* **2004**, *126*, 15915–15923. (f) Coote, M. L.; Henry, D. J. *Macromolecules* **2005**, *38*, 1415–1433. (g) Coote, M. L. *J. Phys. Chem. A* **2005**, *109*, 1230–1239. (h) Coote, M. L.; Krenske, E. H.; Izgorodina, E. I. *Macromol. Rapid Commun.* **2006**, *27*, 473–497. (i) Izgorodina, E. I.; Coote, M. L. *Macromol. Theory Simul.* **2006**, *15*, 394–403. (j) Lin, C. Y.; Coote, M. L. *Aust. J. Chem.* **2009**, *62*, 1479–1483.
12. (a) Leroy, G.; Dewispelaere, J.-P.; Benkadour, H.; Wilante, C. *Macromol. Theory Simul.* **1996**, *5*, 269–289. (b) Heuts, J. P. A.; Gilbert, R. G.; Radom, L. *J. Phys. Chem.* **1996**, *100*, 18997–19006. (c) Huang, D. M.; Monteiro, M. J.; Gilbert, R. G. *Macromolecules* **1998**, *31*, 5175–5187. (d) Toh, J. S.-S.; Huang, D. M.; Lovell, P. A.; Gilbert, R. G. *Polymer* **2001**, *42*, 1915–1920. (e) Filley, J.; McKinnon, J. T.; Wu, D. T.; Ko, G. H. *Macromolecules* **2002**, *35*, 3731–3738. (f) Zhan, C.-G.; Dixon, D. A. *J. Phys. Chem. A* **2002**, *106*, 10311–10325. (g) Thickett, S. C.; Gilbert, R. G. *Polymer* **2004**, *45*, 6993–6999. (h) Van Cauter, K.; Hemelsoet, K.; Van Speybroeck, V.; Reyniers, M. F.; Waroquier, M. *Int. J. Quantum Chem.* **2004**,

- 102, 454–460. (i) Salman, S.; Albayrak, A. Z.; Avci, D.; Aviyente, V. *J. Polym. Sci. A* **2005**, *43*, 2574–2583. (j) Günaydin, H.; Salman, S.; Tüzün, N. S.; Avci, D.; Aviyente, V. *Int. J. Quantum Chem.* **2005**, *103*, 176–189. (k) Van Cauter, K.; Van Speybroeck, V.; Vansteenkiste, P.; Reyniers, M.-F.; Waroquier, M. *ChemPhysChem* **2006**, *7*, 131–140. (l) Degirmenci, I.; Avci, D.; Aviyente, V.; Van Cauter, K.; Van Speybroeck, V.; Waroquier, M. *Macromolecules* **2007**, *40*, 9590–9602.
13. (a) Purmova, J.; Pauwels, K. F. D.; van Zoelen, W.; Vorenkamp, E. J.; Schouten, A. J.; Coote, M. L. *Macromolecules* **2005**, *38*, 6352–6366. (b) Van Cauter, K.; Van Den Bossche, B. j.; Van Speybroeck, V.; Waroquier, M. *Macromolecules* **2007**, *40*, 1321–1331. (c) Purmová, J.; Pauwels, K. F. D.; Agostini, M.; Bruinsma, M.; Vorenkamp, E. J.; Schouten, A. J.; Coote, M. L. *Macromolecules* **2008**, *41*, 5527–5539.
14. (a) Heuts, J. P. A.; Sudarko; Gilbert, R. G. *Macromol. Symp.* **1996**, *111*, 147–157. (b) Heuts, J. P. A.; Gilbert, R. G.; Maxwell, I. A. *Macromolecules* **1997**, *30*, 726–736. (c) Coote, M. L.; Davis, T. P.; Radom, L. *Theochem* **1999**, *461–462*, 91–96. (d) Coote, M. L.; Davis, T. P.; Radom, L. *Macromolecules* **1999**, *32*, 5270–5276. (e) Coote, M. L.; Davis, T. P.; Radom, L. *Macromolecules* **1999**, *32*, 2935–2940. (f) Cieplak, P.; Kaim, A. *J. Polym. Sci. A* **2004**, *42*, 1557–1565.
15. Barner-Kowollik, C. W.; Coote, M. L.; Davis, T. P.; Stenzel, M. H.; Theis, A. Polymerization agent, International Patent WO2006122344 A1, 2006. <http://v3.espacenet.com/textdoc?DB=EPODOC&IDX=WO2006122344&F=0>.
16. (a) Ah Toy, A.; Chaffey-Millar, H.; Davis, T. P.; Stenzel, M. H.; Izgorodina, E. I.; Coote, M. L.; Barner-Kowollik, C. *Chem. Commun.* **2006**, 835–837. (b) Chaffey-Millar, H.; Izgorodina, E. I.; Barner-Kowollik, C.; Coote, M. L. *J. Chem. Theory Comput.* **2006**, *2*, 1632–1645.
17. (a) Hodgson, J. L.; Coote, M. L. *Macromolecules* **2005**, *38*, 8902. (b) Coote, M. L.; Hodgson, J. L.; Krenske, E. H.; Namazian, M.; Wild, S. B. *Aust. J. Chem.* **2007**, *60*, 744–753.
18. Coote, M. L.; Izgorodina, E. I.; Krenske, E. H.; Busch, M.; Barner-Kowollik, C. *Macromol. Rapid Commun.* **2006**, *27*, 1015–1022.
19. McLeary, J. B.; Calitz, F. M.; McKenzie, J. M.; Tonge, M. P.; Sanderson, R. D.; Klumperman, B. *Macromolecules* **2004**, *37*, 2382–2394.
20. Coote, M. L. *Macromol. Theory Simul.* **2009**, *18*, 388–400.
21. See, e.g., Heuts, J. P. A.; Russell, G. T. *Eur. Polym. J.* **2006**, *42*, 3–20.
22. Coote, M. L.; Davis, T. P. *Prog. Polym. Sci.* **1999**, *24*, 1217–1251.
23. Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; Scuseria, G. E.; Robb, M. A.; Cheeseman, J. R.; Montgomery, J. A., Jr.; Vreven, T.; Kudin, K. N.; Burant, J. C.; Millam, J. M.; Iyengar, S. S.; Tomasi, J.; Barone, V.; Mennucci, B.; Cossi, M.; Scalmani, G.; Rega, N. P.; Petersson, G. A.; Nakatsuji, H.; Hada, M.; Ehara, M.; Toyota, K.; Fukuda, R.; Hasegawa, J.; Ishida, M. N.; Nakajima, T.; Honda, Y.; Kitao, O.; Nakai, H.; Klene, M.; Li, X.; Knox, J. E.; Hratchian, H. P.; Cross, J. B. A.; Adamo, C.; Jaramillo, J.; Gomperts, R.; Stratmann, R. E.; Yazyev, O.; Austin, A. J.; Cammi, R.; Pomelli, C.; Ochterski, J. W.; Ayala, P. Y.; Morokuma, K.; Voth, G. A.; Salvador, P.; Dannenberg, J. J.; Zakrzewski, V. G.; Dapprich, S.; Daniels, A. D.; Strain, M. C.; Farkas, O.; Malick, D. K.; Rabuck, A. D.; Raghavachari, K.; Foresman, J. B.; Ortiz, J. V.; Cui, Q.; Baboul, A. G.; Clifford, S.; Cioslowski, J.; Stefanov, B. B.; Liu, G.; Liashenko, A.; Piskorz, P.; Komaromi, I.; Martin, R. L.; Fox, D. J.; Keith,

- T.; Al-Laham, M. A.; Peng, C. Y.; Nanayakkara, A.; Challacombe, M.; Gill, P. M. W.; Johnson, B.; Chen, W.; Wong, M. W.; Gonzalez, C.; Pople, J. A. *Gaussian 03, Revision B.03*, Gaussian Inc., Pittsburgh, PA, 2003.
24. Werner, H.-J.; Knowles, P. J.; Lindh, R.; Manby, F. R.; Schütz, M.; Celani, P.; Korona, T.; Rauhut, G.; Amos, R. D.; Bernhardsson, A.; Berning, A.; Cooper, D. L.; Deegan, M. J. O.; Dobbyn, A. J.; Eckert, F.; Hampel, C.; Hetzer, G.; Lloyd, A. W.; McNicholas, S. J.; Meyer, W.; Mura, M. E.; Nicklass, A.; Palmieri, P.; Pitzer, R.; Schumann, U.; Stoll, H.; Stone, A. J.; Tarroni, R.; Thorsteinsson, T. *MOLPRO, Version 2006.1*, a package of ab initio programs, <http://www.molpro.net>.
 25. Schmidt, M. W.; Baldridge, K. K.; Boatz, J. A.; Elbert, S. T.; Gordon, M. S.; Jensen, J. H.; Koseki, S.; Matsunaga, N.; Nguyen, K. A.; Su, S. J.; Windus, T. L.; Dupuis, M.; Montgomery, J. A. *J. Comput. Chem.* **1993**, *14*, 1347.
 26. Shao, Y.; Molnar, L. F.; Jung, Y.; Kussmann, J.; Ochsenfeld, C.; Brown, S. T.; Gilbert, A. T. B.; Slipchenko, L. V.; Levchenko, S. V.; O'Neill, D. P.; DiStasio, R. A.; Lochan, R. C.; Wang, T.; Beran, G. J. O.; Besley, N. A.; Herbert, J. M.; Lin, C. Y.; Van Voorhis, T.; Chien, S. H.; Sodt, A.; Steele, R. P.; Rassolov, V. A.; Maslen, P. E.; Korambath, P. P.; Adamson, R. D.; Austin, B.; Baker, J.; Byrd, E. F. C.; Dachsel, H.; Doerksen, R. J.; Dreuw, A.; Dunietz, B. D.; Dutoi, A. D.; Furlani, T. R.; Gwaltney, S. R.; Heyden, A.; Hirata, S.; Hsu, C. P.; Kedziora, G.; Khalliulin, R. Z.; Klunzinger, P.; Lee, A. M.; Lee, M. S.; Liang, W.; Lotan, I.; Nair, N.; Peters, B.; Proynov, E. I.; Pieniazek, P. A.; Rhee, Y. M.; Ritchie, J.; Rosta, E.; Sherrill, C. D.; Simmonett, A. C.; Subotnik, J. E.; Woodcock, H. L.; Zhang, W.; Bell, A. T.; Chakraborty, A. K.; Chipman, D. M.; Keil, F. J.; Warshel, A.; Hehre, W. J.; Schaefer, H. F.; Kong, J.; Krylov, A. I.; Gill, P. M. W.; Head-Gordon, M. *Phys. Chem. Chem. Phys.* **2006**, *8*, 3172.
 27. Bylaska, E. J.; de Jong, W. A.; Govind, N.; Kowalski, K.; Straatsma, T. P.; Valiev, M.; Wang, D.; Apra, E.; Windus, T. L.; Hammond, J.; Nichols, P.; Hirata, S.; Hackler, M. T.; Zhao, Y.; Fan, P.-D.; Harrison, R. J.; Dupuis, M.; Smith, D. M. A.; Nieplocha, J.; Tipparaju, V.; Krishnan, M.; Wu, Q.; Voorhis, T. V.; Auer, A. A.; Nooijen, M.; Brown, E.; Cisneros, G.; Fann, G. I.; Fruchtl, H.; Garza, J.; Hirao, K.; Kendall, R.; Nichols, J. A.; Tsemekhman, K.; Wolinski, K.; Anchell, J.; Bernholdt, D.; Borowski, P.; Clark, T.; Clerc, D.; Dachsel, H.; Deegan, M.; Dyall, K.; Elwood, D.; Glendening, E.; Gutowski, M.; Hess, A.; Jaffe, J.; Johnson, B.; Ju, J.; Kobayashi, R.; Kutteh, R.; Lin, Z.; Littlefield, R.; Long, X.; Meng, B.; Nakajima, T.; Niu, S.; Pollack, L.; Rosing, M.; Sandrone, G.; Stave, M.; Taylor, H.; Thomas, G.; von Lenthe, J.; Wong, A.; Zhang, Z. *NWChem: A Computational Chemistry Package for Parallel Computers, Version 5.1*, Pacific Northwest National Laboratory, Richland, WA, 2007.
 28. Velde, G. T.; Bickelhaupt, F. M.; Baerends, E. J.; Guerra, C. F.; Van Gisbergen, S. J. A.; Snijders, J. G.; Ziegler, T. *J. Comput. Chem.* **2001**, *22*, 931.
 29. Stanton, J. F.; Gauss, J.; Perera, S. A.; Watts, J. D.; Yau, A. D.; Nooijen, M.; Oliphant, N.; Szalay, P. G.; Lauderdale, W. J.; Gwaltney, S. R.; Beck, S.; Balková, A.; Bernholdt, D. E.; Baeck, K. K.; Rozyczko, P.; Sekino, H.; Huber, C.; Pittner, J.; Cencek, W.; Taylor, D.; Bartlett, R. J. ACES II is a program product of the Quantum Theory Project, University of Florida. Integral packages included are VMOL (J. Almlöf and P. R. Taylor); VPROPS (P. Taylor); ABA-CUS (T. Helgaker, H. J. Aa. Jensen, P. Jørgensen, J. Olsen, and P. R. Taylor); HONDO/GAMESS (M. W. Schmidt, K. K. Baldridge, J. A. Boatz, S. T. Elbert, M. S. Gordon, J. J. Jensen, S. Koseki, N. Matsunaga, K. A. Nguyen, S. Su, T. L. Windus, M. Dupuis, J. A. Montgomery).

30. (a) Choi, C. C.; Kertesz, M.; Karpfen, A. *Chem. Phys. Lett.* **1997**, 276, 266. (b) Curtiss, L. A.; Raghavachari, K.; Redfern, P. C.; Pople, J. A. *J. Chem. Phys.* **2000**, 112, 7374. (c) Woodcock, H. L.; Schaefer, H. F., III; Schreiner, P. R. *J. Phys. Chem. A* **2002**, 106, 11923. (d) Izgorodina, E. I.; Coote, M. L.; Radom, L. *J. Phys. Chem. A* **2005**, 109, 7558. (e) Check, C. E.; Gilbert, T. M. *J. Org. Chem.* **2005**, 70, 9828. (f) Izgorodina, E. I.; Coote, M. L. *J. Phys. Chem. A* **2006**, 110, 2486. (g) Grimme, S. *Angew. Chem. Int. Ed.* **2006**, 45, 4460. (h) Schreiner, P. R.; Fokin, A. A.; Pascal, R. A., Jr.; de Meijere, A. *Org. Lett.* **2006**, 8, 3635. (i) Wodrich, M. D.; Corminbæf, C.; von Ragué Schleyer, P. *Org. Lett.* **2006**, 8, 3631. (j) Wodrich, M. D.; Corminbæf, C.; Schreiner, P. R.; Fokin, A. A.; von Ragué Schleyer, P. *Org. Lett.* **2007**, 9, 1851. (k) Grimme, S.; Steinmetz, M.; Korth, M. *J. Chem. Theory Comput.* **2007**, 3, 42. (l) Schreiner, P. R. *Angew. Chem. Int. Ed.* **2007**, 46, 4217. (m) Izgorodina, E. I.; Brittain, D. R. B.; Hodgson, J. L.; Krenske, E. H.; Lin, C. Y.; Namazian, M.; Coote, M. L. *J. Phys. Chem. A* **2007**, 111, 10754. (n) Brittain, D. R. B.; Lin, C. Y.; Gilbert, A. T. B.; Izgorodina, E. I.; Gill, P. M. W.; Coote, M. L. *Phys. Chem. Chem. Phys.* **2009**, 11, 1138–1142.
31. Buback, M.; Hippler, H.; Schweer, J.; Voge, H.-P. *Makromol. Chem. Rapid Commun.* **1986**, 7, 261–265.
32. (a) Kajiwar, A.; Kamachi, M. *Macromol. Chem. Phys.* **2000**, 201, 2165–2169. (b) Burnett, G. M.; Wright, W. W. *Proc. R. Soc. (Lond.) A* **1954**, 211, 41.
33. For a review of the early work in this field, see Fischer, H.; Radom, L. *Angew. Chem. Int. Ed.* **2001**, 40, 1340–1371.
34. For more recent studies, see, e.g., (a) Henry, D. J.; Parkinson, C. J.; Mayer, P. M.; Radom, L. *J. Phys. Chem. A* **2001**, 105, 6750. (b) Coote, M. L.; Wood, G. P. F.; Radom, L. *J. Phys. Chem. A* **2002**, 106, 12124–12138. (c) Coote, M. L. *J. Phys. Chem. A* **2004**, 108, 3865–3872. (d) Gómez-Balderas, R.; Coote, M. L.; Henry, D. J.; Radom, L. *J. Phys. Chem. A* **2004**, 108, 2874–2883. (e) Lin, C. Y.; Hodgson, J. L.; Namazian, M.; Coote, M. L. *J. Phys. Chem. A* **2009**, 113, 3690–3697.
35. Malick, D. K.; Petersson, G. A.; Montgomery, J. A. *J. Chem. Phys.* **1998**, 108, 5704.
36. Scott, A. P.; Radom, L. *J. Phys. Chem.* **1996**, 100, 16502.
37. (a) Pople, J. A.; Head-Gordon, M.; Fox, D. J.; Raghavachari, K.; Curtiss, L. A. *J. Chem. Phys.* **1989**, 90, 5622. (b) Curtiss, L. A.; Jones, C.; Trucks, G. W.; Raghavachari, K.; Pople, J. A. *J. Chem. Phys.* **1990**, 93, 2537. (c) Curtiss, L. A.; Raghavachari, K.; Trucks, G. W.; Pople, J. A. *J. Chem. Phys.* **1991**, 94, 7221. (d) Curtiss, L. A.; Raghavachari, K.; Redfern, P. C.; Rassolov, V.; Pople, J. A. *J. Chem. Phys.* **1998**, 109, 7764. (e) Curtiss, L. A.; Redfern, P. C.; Raghavachari, K.; Pople, J. A. *J. Chem. Phys.* **2001**, 114, 108. (f) Curtiss, L. A.; Redfern, P. C.; Raghavachari, K. *J. Chem. Phys.* **2007**, 126, 084108.
38. Henry, D. J.; Sullivan, M. B.; Radom, L. *J. Chem. Phys.* **2003**, 118, 4849.
39. Montgomery, J. A.; Frisch, M. J.; Ochterski, J. W.; Petersson, G. A. *J. Chem. Phys.* **1999**, 110, 2822.
40. Martin, J. M. L.; Parthiban, S. In *Quantum Mechanical Prediction of Thermochemical Data*, Cioslowski, J., Ed. Kluwer-Academic, Dordrecht, The Netherlands, 2001, pp. 31–65.
41. (a) Vreven, T.; Morokuma, K. *J. Chem. Phys.* **1999**, 111, 8799–8803. (b) Vreven, T.; Morokuma, K. *J. Comput. Chem.* **2000**, 21, 1419–1432.
42. Lipton, M.; Still, W. C. *J. Comput. Chem.* **1988**, 9, 343–355.

43. Izgorodina, E. I.; Lin, C. Y.; Coote, M. L. *Phys. Chem. Chem. Phys.* **2007**, *9*, 2507–2516.
44. (a) Kirkpatrick, S.; Gelatt, C. D., Jr.; Vecchi, M. P. *Science* **1983**, *220*, 671. (b) Wilson, S. R.; Cui, W.; Moskowitz, J. W.; Schmidt, K. E. *Tetrahedron Lett.* **1988**, 4343.
45. (a) Gibson, K. D.; Scheraga, H. A. *J. Comput. Chem.* **1987**, *8*, 826. (b) Pincus, M. R.; Klausner, R. D.; Scheraga, H. A. *Proc. Natl. Acad. Sci. USA* **1982**, *79*, 5107. (c) Hingerty, B. E.; Figueroa, S.; Hayden, T. L.; Broyde, S. *Biopolymers* **1989**, *28*, 1195.
46. Malick, D. K.; Petersson, G. A.; Montgomery, J. A. *J. Chem. Phys.* **1998**, *108*, 5704–5713.
47. (a) Knyazev, V. D.; Slagle, I. R. *J. Phys. Chem.* **1996**, *100*, 16899–16911. (b) Knyazev, V. D.; Bencsura, A.; Stoliarov, S. I.; Slagle, I. R. *J. Phys. Chem.* **1996**, *100*, 11346–11354.
48. Schwartz, M.; Marshall, P.; Berry, R. J.; Ehlers, C. J.; Petersson, G. A. *J. Phys. Chem. A* **1998**, *102*, 10074–10081.
49. Coote, M. L.; Collins, M. A.; Radom, L. *Mol. Phys.* **2003**, *101*, 1329–1338.
50. Coote, M. L. In *Encyclopaedia of Polymer Science and Technology*, 3rd ed., Vol. 9, Kroschwitz, J. I., Ed., Wiley, Hoboken, NJ, 2004, pp. 319–371.
51. See, e.g., (a) Benson, S. W. *Thermochemical Kinetics*, Wiley, New York, 1976. (b) McQuarrie, D. A. *Statistical Mechanics*, Harper & Row, New York, 1976. (c) Gilbert, R. G.; Smith, S. C. *Theory of Unimolecular and Recombination Reactions*, Blackwell Scientific, Oxford, UK, 1990. (d) Steinfeld, J. I.; Francisco, J. S.; Hase, W. L. *Chemical Kinetics and Dynamics*, 2nd ed., Prentice Hall, Englewood Cliffs, NJ, 1999. (e) Atkins, P. W. *Physical Chemistry*, 6th ed., W.H. Freeman, San Francisco, 2000.
52. Eyring, H. *J. Chem. Phys.* **1935**, *3*, 107.
53. For a more detailed definition of this term, see, e.g., Karas, A. J.; Gilbert, R. G.; Collins, M. A. *Chem. Phys. Lett.* **1992**, *193*, 181–184.
54. Skodje, R. T.; Truhlar, D. G.; Garrett, B. C. *J. Phys. Chem.*, **1981**, *85*, 3019.
55. Garrett, B. C.; Truhlar, D. G.; Wagner, A. F.; Dunning, T. H., Jr. *J. Chem. Phys.* **1983**, *78*, 4400.
56. Liu, Y. P.; Lu, D. H.; Gonzalez-Lafont, A.; Truhlar, D. G.; Garrett, B. C. *J. Am. Chem. Soc.* **1993**, *115*, 7806.
57. Corchado, J. C.; Chuang, Y.-Y.; Fast, P. L.; Villà, J.; Hu, W.-P.; Liu, Y.-P.; Lynch, G. C.; Nguyen, K. A.; Jackels, C. F.; Melissas, V. S.; Lynch, B. J.; Rossi, I.; Coitiño, E. L.; Fernandez-Ramos, A.; Pu, J.; Albu, T. V.; Steckler, R.; Garrett, B. C.; Isaacson, A. D.; Truhlar, D. G. *POLYRATE 9.1*, University of Minnesota, Minneapolis, MN, 2002, <http://comp.chem.umn.edu/polyrate/>.
58. (a) Kuppermann, A.; Truhlar, D. G. *J. Am. Chem. Soc.* **1971**, *93*, 1840. (b) Garrett, B. C.; Truhlar, D. G.; Grev, R. S.; Magnuson, A. W. *J. Phys. Chem.* **1980**, *84*, 1730.
59. Bell, R. P. *The Tunnel Effect in Chemistry*, Chapman & Hall, New York, 1980.
60. Eckart, C. *Phys. Rev.* **1930**, *35*, 1303.
61. See, e.g., Vansteenkiste, P.; Van Neck, D.; Van Speybroeck, V.; Waroquier, M. *J. Chem. Phys.* **2006**, *124*, 044314.
62. Lin, C. Y.; Izgorodina, E. I.; Coote, M. L. *J. Phys. Chem. A* **2008**, *112*, 1956–1964.
63. East, A. L. L.; Radom, L. *J. Chem. Phys.* **1997**, *106*, 6655.

64. (a) Pitzer, K. S.; Gwinn, W. D. *J. Chem. Phys.* **1942**, *10*, 428–440. (b) Pitzer, K. S. *J. Chem. Phys.* **1946**, *14*, 239–243. (c) Li, J. C. M.; Pitzer, K. S. *J. Phys. Chem.* **1956**, *60*, 466–474. (d) Kilpatrick, K. E.; Pitzer, K. S. *J. Chem. Phys.* **1949**, *17*, 1064–1075.
65. Ellingson, B. A.; Lynch, V. A.; Mielke, S. L.; Truhlar, D. G. *J. Chem. Phys.* **2006**, *125*, 084305.
66. Ayala, P. Y.; Schlegel, H. B. *J. Chem. Phys.* **1998**, *108*, 7560.
67. Coote, M. L.; Davis, T. P.; Klumperman, B.; Monteiro, M. J. *J. Macromol. Sci. Rev. Macromol. Chem. Phys.* **1998**, *C38*, 567–593.
68. Degirmenci, I.; Aviyente, V.; Van Speybroeck, V.; Waroquier, M. *Macromolecules* **2009**, *42*, 3033–3041.
69. Morrison, D. A.; Davis, T. P. *Macromol. Chem. Phys.* **2000**, *201*, 2128–2137.
70. Tomasi, J. *Theor. Chem. Acc.* **2004**, *112*, 184.
71. (a) Klamt, A.; Schueuermann, G. *J. Chem. Soc. Perkin Trans. 2* **1993**, 799. (b) Cossi, M.; Rega, N.; Scalmani, G.; Barone, V. *J. Comput. Chem.* **2003**, *24*, 669.
72. Miertus, S.; Scrocco, E.; Tomasi, J. *J. Chem. Phys.* **1981**, *55*, 117.
73. (a) Klamt, A. *J. Phys. Chem.* **1995**, *99*, 2224. (b) Klamt, A. *COSMO-RS: From Quantum Chemistry to Fluid Phase Thermodynamics and Drug Design*, Elsevier Science, Amsterdam, 2005. (c) Klamt, A.; Jonas, V.; Burger, T.; Lohrenz, J. C. W. *J. Phys. Chem. A* **1998**, *102*, 5074.
74. Kelly, C. P.; Cramer, C. J.; Truhlar, D. G. *J. Chem. Theory Comput.* **2005**, *1*, 1133.
75. See, e.g., Takano, Y.; Houk, K. N. *J. Chem. Theory Comput.* **2005**, *1*, 70–77.
76. Beuermann, S.; Buback, M.; Hesse, P.; Kuchta, F.-D.; Lacik, I.; Van Herk, A. M. *Pure Appl. Chem.* **2007**, *79*(8), 1463–1469.
77. See, e.g., (a) Namazian, M.; Coote, M. L. *J. Phys. Chem. A* **2007**, *111*, 7227–7232. (b) Hodgson, J. L.; Namazian, M.; Bottle, S. E.; Coote, M. L. *J. Phys. Chem. A* **2007**, *111*, 13595–13605. (c) Namazian, M.; Zare, H. R.; Coote, M. L. *Biophys. Chem.* **2008**, *132*, 64–68. (d) Namazian, M.; Siahrostami, S.; Coote, M. L. *J. Fluorine Chem.* **2008**, *129*, 222–225. (e) Blinco, J. P.; Hodgson, J. L.; Morrow, B. J.; Walker, J. R.; Will, G. D.; Coote, M. L.; Bottle, S. E. *J. Org. Chem.* **2008**, *73*, 6763–6771. (f) Zare, H.; Eslami, M.; Namazian, M.; Coote, M. L. *J. Phys. Chem. B* **2009**, *113*, 8080–8085.
78. See, e.g., Ho, J.; Coote, M. L. *J. Chem. Theory Comput.* **2009**, *5*, 295–306.
79. Levy, R. M.; Kitchen, D. B.; Blair, J. T.; Krogh-Jespersen, K. *J. Phys. Chem.* **1990**, *94*, 4470–4476.
80. Pliego, J. R., Jr.; Riveros, J. M. *J. Phys. Chem. A* **2001**, *105*, 7241–7247.

14 Evaluation of Nonlinear Optical Properties of Large Conjugated Molecular Systems by Long-Range-Corrected Density Functional Theory

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Advantages and problems of quantum chemical methods for nonlinear optical (NLO) property evaluation are discussed. Density functional theory (DFT) is the best quantum chemical tool for quantitative evaluation of the property of NLO materials that have no absorption in the response frequency region. We introduce a practical DFT method with long-range correction (LC) for the purpose. We discuss a strategy for realistic evaluation of large conjugated systems, finding sufficient the classical hypothesis that only the π -electron system needs to be considered in conjugated molecules. The errors arising from this approximation are much smaller than those caused by a deficiency in traditional DFT functionals. We examine the LC-DFT method further by comparison of the length dependence between polyyne and polyene. From a comparison with rigorous ab initio correlated methods, we conclude that the LC-DFT method can calculate NLO properties successfully without a catastrophic overestimation of the conventional DFT functionals and can provide basic information for systematic fabrication of new organic NLO materials.

14.1 INTRODUCTION

The nonlinear optical (NLO) response of materials under intense optical electromagnetic field is an important, yet challenging subject in theoretical and computational materials science that can arise through a variety of processes. However, it is evident that the nonlinear electronic response plays the most significant role, making essential rigorous quantum chemical calculations for molecular systems. While the importance of contributions from vibrational processes in determining the hyperpolarizabilities of small conjugated molecules has recently been highlighted,¹ the pure high-order electronic response is paramount, especially for the evaluation of hyperpolarizabilities of large conjugated systems.

Quantum chemical methods are the most reliable methods to use to quantitatively describe the electronic response in molecules. Although no exact analytical solution is available for the Schrödinger equation of many-electron systems, advances in quantum chemical theories and computational technologies have pushed the methods to the stage where near-equilibrium molecular electronic states can be described to chemical precision. When the intensity of the incident light field is low, the electronic response arises from states whose parity difference corresponds to that of a single photon. For high-intensity incident light fields, however, more than two photons arrive at the material within a short time and interact simultaneously with the electron. The process that describes such situations must therefore involve states whose parity differences correspond to those of multiphotons, and many higher-lying states are accessed. This nonlinear optical response involves a complex analytical formalism, and several practiced methodologies have been developed based on the energy or dipole response properties. We can further adapt these methods to consider the system as being initially in a molecular single state, typically the ground state. However, to describe the electronic response of extended materials quantitatively, we need knowledge of this initial state *in the presence* of the light field. Therefore, care must be taken to introduce extra flexibility into calculations to allow for this effect.

Large delocalized electronic systems are good candidates for NLO materials because they contain many low-lying states that can temporarily be occupied by electrons, perhaps introducing charge-transfer character to the ground state. The nonlinear response of electrons to external fields is often described using such states as intermediates. Therefore, the computational requirements for describing the NLO processes are much more demanding than those for computing just the total energy of the system. Although *ab initio* correlated methods have been quite successful in providing chemical descriptions of molecules, they are not feasible at present for the evaluation of nonlinear response properties of large systems. Density functional theory (DFT) methods have been shown capable of reproducing and predicting a variety of chemical properties, such as atomization energies, bond lengths, and vibrational frequencies, while requiring much less computational effort than do rigorous *ab initio* correlated methods.² Despite their manifold successes in predicting a wide range of chemical properties, DFT has been found to give poor results for some properties, including weakly bound systems and

charge-transfer systems, as well as for the electronic response in large conjugated systems.³ The latter aspect is the subject that we discuss in this chapter, demonstrating how these problems can be overcome to yield effective and practical computational methods for the NLO properties of materials. Traditionally, DFT catastrophically overestimates the rate of increase in the polarizability of a long molecule as its length increases.^{4,5}

The well-known deficiency in evaluating polarizabilities comes from inadequacies in the conventional exchange functional used in DFT. Conventional exchange correlation functionals are local and cannot represent correctly the response of the electrons at long distance. The effects are modest in small molecular systems but become nonnegligible in large molecules. Conventional exchange functionals thus fail to evaluate correctly such properties as the polarizability and hyperpolarizability of large molecules. The gradient correction for nonlocality that is commonly applied through the generalized-gradient approximation (GGA) is ineffective in relieving the problem, which instead needs to be solved as a many-body interaction involving different energy levels. Conventional *hybrid* methods such as B3LYP⁶ do not improve the situation either, making the search for new functionals a key focus. A variety of approaches have been developed.

The optimized effective potential (OEP) method has been advanced as a solution that seems to provide useful results,⁷ at least when it is implemented appropriately.⁸ Unfortunately, the OEP method is rather complicated in that an extra equation must be solved to obtain the optimized potential,⁹ and this equation is also technically difficult to solve. Care must be taken in the choice of appropriate auxiliary basis functions to properly represent the extra equation with in particular the use of large basis sets leading to a *deterioration* of the solution. Other methods include the Krieger–Li–Iafrate (KLI) approximation¹⁰ and the common-energy-denominator approximation (CEDA)¹¹ for large-molecule applications. Unfortunately, these approximations adversely influence calculated response properties even when the ground-state energy is well represented.¹² The current density functional theory (CDFT)^{13,14} provides another alternative for the evaluation of NLO response properties. It predicts reasonable polarizabilities and hyperpolarizabilities^{15,16} for long molecules (except for hydrogen chains). There has also been a study on the optical properties of molecules using a many-body f_{xc} kernel that yielded good polarizabilities and optical spectra.¹⁷ Although such approaches provide deep insights into the origin and evaluation of the NLO properties, their implementation is also rather complicated. Heavier computational demand also makes these methods less accessible for the large molecules that appear in nano or bio systems.

Recently, we introduced a simple hybrid method with long-range correction (LC) using an Ewald partitioning technique on the electron repulsion operator to account for the nonlocal effect of long-distance interactions.^{18,19} The use of this method to evaluate the hyperpolarizabilities of long conjugated systems has been successful.^{20–24} In this chapter we explain briefly the basic theory for the evaluation of molecular hyperpolarizabilities and describe the LC-DFT method. We also discuss the classic π -electron-only hypothesis and its validity for

conjugated systems in the context of NLO property evaluation. Finally, we discuss the effects of different types of conjugations.

14.2 NONLINEAR OPTICAL RESPONSE THEORY

The response of the electrons in matter to applied optical fields can be measured in terms of the energy $W(E)$ of the electronic systems as a function of the electric field E caused by the incident light:

$$W(E) = W_0 + W^{(1)}E + W^{(2)}E^2 + W^{(3)}E^3 + \dots \quad (14.1)$$

Here, $W^{(n)}$ is the n th-order energy in the expansion with respect to the applied field. $W^{(1)}$ is related to permanent dipole moment, and $W^{(2)}$, $W^{(3)}$, $\dots$ are related to the linear and nonlinear polarizabilities, respectively. The total energy of the electronic state in equilibrium with the optical field is well defined and can be computed by solving the time-independent Schrödinger equation. In most approximation schemes employed in quantum chemistry, the solutions are upper bounded and well behaved as the level of theory and basis set are improved. An alternative finite-field expression based on the analogous expansion of the induced dipole,

$$\mu(E) = \mu_0 + \alpha E + \beta \frac{E^2}{2} + \gamma \frac{E^3}{2 \cdot 3} + \dots \quad (14.2)$$

is also in common use. Here, the zeroth order term, μ_0 , is the permanent dipole, while α is the linear polarizability and β , γ , $\dots$ are the hyperpolarizabilities. The advantage of the latter approach is that the dipole moment is a physical observable that can be compared directly with experimental results. However, computation of the induced dipole generally involves more elaborate computations. The direction of the induced moment does not necessarily coincide with that of the applied field, and therefore the expansion coefficients (α , β , γ , $\dots$) are, in fact, tensors. The key observables, the macroscopic polarization projected against the molecular orientation vectors, are obtained from the ensemble average of the microscopic polarization tensors over the time scale of resolution for the experiments. The shape of the mobile electron cloud is intimately related to the polarization tensor. While all the tensor components are needed, in principle, to evaluate the macroscopic polarization, many NLO materials consist of molecules whose dimension is enlarged in one direction, and thus the corresponding components of the tensors dominant. Since we focus on such a case, that of linearly prolonged conjugated systems, we are concerned primarily with the absolute values of the longitudinal component of α , β , γ , $\dots$ in the expansion above.

To achieve intense electric fields, optical laser beams of specific frequency ω are used. This is modeled using the frequency-dependent Hamiltonian

$$H_{\text{int}}(\omega) = \mu \cdot \frac{1}{2}(e^{+i\omega t} + e^{-i\omega t})E \quad (14.3)$$

The induced moment is observed at the frequency of the corresponding NLO process. For example, the induced moment from second-harmonic generation (SHG) is observed at the doubled frequency 2ω , that from third harmonic generation (THG) is observed at tripled frequency 3ω , and so on. Therefore, the expressions with only static electric field E , such as in Eqs. (14.1) and (14.2), are inappropriate for specific NLO process and need to be enhanced as

$$\begin{aligned}\mu(E) = & \mu_0 + \alpha_0 E_0 + \alpha(-\omega; \omega) E_\omega e^{i\omega t} \\ & + \beta_0 \frac{E_0^2}{2} + \beta(-2\omega; \omega, \omega) \frac{E_\omega^2}{2} e^{2i\omega t} + \beta(-\omega; \omega, 0) E_0 E_\omega e^{i\omega t} + \dots \\ & + \gamma_0 \frac{E_0^3}{2 \cdot 3} + \gamma(-3\omega; \omega, \omega, \omega) \frac{E_\omega^3}{2 \cdot 3} e^{3i\omega t} + \gamma(-2\omega; \omega, \omega, 0) \frac{E_\omega^2}{2} E_0 e^{2i\omega t} \\ & + \gamma(-\omega; \omega, \omega, -\omega) \frac{E_\omega^2}{2} E_{-\omega} e^{i\omega t} + \dots\end{aligned}\quad (14.4)$$

Typically, the frequency-dependent expansion coefficients, $\alpha(-\omega; \omega)$, $\beta(-2\omega; \omega, \omega)$, $\beta(-\omega; \omega, 0)$, $\gamma(-3\omega; \omega, \omega, \omega)$, $\gamma(-2\omega; \omega, \omega, 0)$, $\gamma(-\omega; \omega, \omega, -\omega)$, $\dots$ are formulated in the sum-over-states (SOS) representation using time-dependent perturbation theory as

$$\alpha(-\omega; \omega) = 2P_{-\omega, \omega} \sum_k' \frac{\langle n | \mu | k \rangle \langle k | H_{\text{int}}(\omega) | n \rangle}{\Omega_{kn} - \omega} \quad (14.5a)$$

$$\beta(-\omega_\sigma; \omega_1, \omega_2) = 3K(-\omega_\sigma; \omega_1, \omega_2) P_{-\sigma, 1, 2} \cdot \sum_{k, l}' \frac{\langle n | \mu | l \rangle \langle l | \bar{H}_{\text{int}}(\omega_2) | k \rangle \langle k | H_{\text{int}}(\omega_1) | n \rangle}{(\Omega_{ln} - \omega_\sigma)(\Omega_{kn} - \omega_1)} \quad (14.5b)$$

$$\begin{aligned}\gamma(-\omega_\sigma; \omega_1, \omega_2, \omega_3) = & 4K(-\omega_\sigma; \omega_1, \omega_2, \omega_3) P_{-\sigma, 1, 2, 3} \\ & \cdot \left[\sum_{k, l, m}' \frac{\langle n | \mu | m \rangle \langle m | \bar{H}_{\text{int}}(\omega_3) | l \rangle \langle l | \bar{H}_{\text{int}}(\omega_2) | k \rangle \langle k | H_{\text{int}}(\omega_1) | n \rangle}{(\Omega_{mn} - \omega_\sigma)(\Omega_{ln} - \omega_1 - \omega_2)(\Omega_{kn} - \omega_1)} \right. \\ & \left. - \sum_{k, l}' \frac{\langle n | \mu | l \rangle \langle l | H_{\text{int}}(\omega_3) | n \rangle \langle n | H_{\text{int}}(\omega_2) | k \rangle \langle k | H_{\text{int}}(\omega_1) | n \rangle}{(\Omega_{ln} - \omega_\sigma)(\Omega_{ln} - \omega_1)(\Omega_{kn} + \omega_2)} \right] \quad (14.5c)\end{aligned}$$

Here, $P_{-\sigma, 1, 2, 3, \dots}$ denotes the average of all terms generated by simultaneous permutations at frequencies $\omega_\sigma, \omega_1, \omega_2, \omega_3, \dots$, while corresponding operators, $\mu, H(\omega_1), H(\omega_2), H(\omega_3), \dots$ and the notation $\sum'$ means a summation of all states except the initial state n . Here, $\Omega_{kl} = \omega_k - \omega_l - \frac{1}{2}i\Gamma_{kl}$ is defined by the energy difference of states k and l corrected by a radiative damping factor, a complex number that plays an important role in resonant situations. Also, $K(-\omega_\sigma; \omega_1, \omega_2)$, $K(-\omega_\sigma; \omega_1, \omega_2, \omega_3), \dots$ are the numerical prefactors that depend on the NLO process of interest. The prefactors typically are established so as to provide a consistent identical hyperpolarizability value at zero-frequency limit in the expressions corresponding to different NLO processes. However, care must be taken when the theoretical values thus evaluated are compared with experimental values, since ensemble averaging of microscopic tensor

components contributes to the observable differently for each experimental setting.

As is seen in the equations above, the properties are expressed as a summation of products of transition moments between ground and intermediate excited states, divided by energy denominators. The latter are the energy difference between the electronic states shifted by multiples of the applied frequency, ω , 2ω , 3ω , and so on, in the nonresonant situation. For the resonant-frequency region, the damping factors Γ_{kl} play an important role in evaluating the lifetime and lineshape, but in this chapter we are concerned primarily with NLO properties in the nonresonant region where no significant absorption occurs. However, it should be noted that the effects of dispersion on nonlinear process are enhanced compared to those in a linear process because of the multipliers in the denominators. When the energy difference between the electronic states (excitation energy) approaches the doubled or tripled frequency of the applied field, the dispersion becomes nontrivial.

While the SOS representation provides a wealth of information concerning the NLO process of interest, it involves the infinite number of intermediate states whose evaluation is impossible in practice. Unfortunately, truncation of the intermediate states is not a successful strategy because the expansion is poorly convergent.²⁵ Of course, it is possible to compute the dynamic properties by directly solving the perturbed equation of appropriate order and the corresponding NLO process at a given frequency, and the frequency-dependent NLO property has been evaluated by the time-dependent coupled Hartree–Fock (TDCHF) method.²⁶ LC-DFT implementation of such an algorithm for NLO property evaluation is in progress.²⁰ We here compute hyperpolarizabilities of long conjugated molecules at zero frequency in order to evaluate their dependence on the length of the molecule. In the zero-frequency limit, we can use finite-field techniques based on Eq. (14.1) and therefore almost all quantum chemical methods can be employed. While a property evaluated in the zero-frequency limit may be quite different from that observed at the specific frequency in a certain kind of experimental setting, this approach provides much information concerning NLO materials. We explain our hybrid DFT method developed recently, introducing a range-dependent partition of the Coulomb force known as the range separation hybrid (RSH) scheme.

14.3 LONG-RANGE-CORRECTED DENSITY FUNCTIONAL THEORY

As explained in the introduction, DFT is the most appropriate quantum chemical method for large-molecular systems such as long conjugated molecules but suffers from a few pertinent problems. To correct for the long-range deficiencies of traditional exchange functionals, a partitioning technique is introduced. Following to original idea of Savin,²⁷ we partition the Coulomb force into short- and long-range parts using the error function

$$\frac{1}{r_{ij}} = \frac{1 - \text{erf}(\bar{\mu}r_{12})}{r_{12}} + \frac{\text{erf}(\bar{\mu}r_{12})}{r_{12}} \quad (14.6)$$

where $\bar{\mu}$ is a parameter that determines the ratio of the partition. The short-range exchange energy E_x^{sr} is computed by modifying the usual exchange energy expression from

$$E_x = -\frac{1}{2} \sum_{\sigma} \int \rho_{\sigma}^{4/3} K_{\sigma} d^3\mathbf{R} \quad (14.7)$$

into

$$E_x^{\text{sr}} = -\frac{1}{2} \sum_{\sigma} \int \rho_{\sigma}^{4/3} K_{\sigma} \left\{ 1 - \frac{8}{3} a_{\sigma} \left[\sqrt{\pi} \text{erf} \left(\frac{1}{2a_{\sigma}} \right) + 2a_{\sigma}(b_{\sigma} - c_{\sigma}) \right] \right\} d\mathbf{r} \quad (14.8)$$

where a_{σ} , b_{σ} , and c_{σ} are

$$a_{\sigma} = \frac{\bar{\mu} K_{\sigma}^{1/2}}{6\sqrt{\pi} \rho_{\sigma}^{1/3}} \quad (14.9)$$

$$b_{\sigma} = \exp \left(-\frac{1}{4a_{\sigma}^2} \right) - 1 \quad (14.10)$$

$$c_{\sigma} = 2a_{\sigma}^2 b_{\sigma} + \frac{1}{2} \quad (14.11)$$

and K_{σ} is called the enhancement factor. The use of K_{σ} allows the modification of GGA functionals. The long-range part of the exchange energy E_x^{lr} is evaluated using Hartree–Fock (HF) exchange integrals as

$$E_x^{\text{lr}} = - \sum_i^{\text{occ}} \sum_j^{\text{occ}} (ij|ji)^{\text{lr}} \quad (14.12)$$

and

$$(pq|rs)^{\text{lr}} = \psi_p \psi_q \left| \frac{\text{erf}(\bar{\mu}r_{12})}{r_{12}} \right| \psi_r \psi_s \quad (14.13)$$

where $\psi_{i\sigma}$ is the i th molecular orbital (MO). In contrast to density partitioning schemes such as B3LYP, the proportion of the nonlocal HF contribution varies according to the range of the interaction in the present LC scheme. The ratio of the nonlocal HF part to the local DFT part becomes larger at greater distances, thus including the nonlocal effect more efficiently. In all the DFT calculations using the LC scheme, Becke's exchange and one-parameter functional (BOP) is used with a parameter of $\bar{\mu} = 0.47^{28}$ (except for one example discussed in Section 14.4.1), and all the calculations are performed using the development version of GAUSSIAN03.²⁹

14.4 EVALUATION OF HYPERPOLARIZABILITY FOR LONG CONJUGATED SYSTEMS

14.4.1 Examination of the Classical Hypothesis: The Role of π -Conjugation in Determining NLO Properties

Chemists have categorized conjugated molecules quite differently from other hydrocarbons because of their distinguished reactivities and spectroscopic properties; it is apparent that these molecules have enhanced responses to light irradiation. The reason for this sensitivity has been attributed to their mobile π -electrons, which can move freely through the conjugation pathway in the system. Since the early theoretical development of spectroscopic quantum chemistry, it has been recognized that it is a good approximation to ignore the effects of the many σ -electrons in a large molecule and focus purely on the contribution from the much fewer π -electrons, providing an enormous computational simplicity. With modern software and hardware, such ideas have seemed to become obsolete. However, as is evident from the SOS representation of NLO properties given above, π -electrons do play a role of paramount importance in the nonlinear optical process.

An important question is therefore the quantitative reliability of the π -electron approximation in practical NLO applications. We show in Table 14.1 the longitudinal polarizabilities of polyenes with different lengths evaluated using the π -electron approximation together with those obtained using all electrons. All properties are evaluated by a finite-field method using energies computed by the HF, BLYP, B3LYP, and LC-BLYP ($\bar{\mu} = 0.33$)¹⁹ methods as a function of the applied field, but for the π -electron approximation, the finite field is applied only on the π -space of the Hamiltonian. There is found a systematic difference in the evaluated absolute value of the polarizabilities with, in particular, the π -electron approximation significantly underestimating the property. This comes from the omission of the σ -electron response, with the error increasing as the size of the system increases. However, the neglected contribution does not increase

TABLE 14.1 Longitudinal Polarizabilities α (a.u.) of Polyenes Computed by the HF, BLYP, B3LYP, and LC-BLYP Methods Using the 6-31G Basis Set

	HF	BLYP	B3LYP	LC-BLYP
Ethylene				
Total	33.66	30.90	26.84	31.06
π only	21.94	17.31	18.27	17.96
Butadiene				
Total	80.91	78.09	70.32	75.08
π only	63.33	58.81	59.63	55.15
C ₂₀ H ₂₂				
Total	1328	2046	1609	1253
π only	1225	1995	1548	1147

with length as much as the contribution from the π -electron part. Consequently, for the longer polyenes, the relative error of the approximation becomes more acceptable. Indeed, for the longer molecules, the variation in the computed value with computational methods significantly exceeds the error introduced by the π -electron approximation. It is interesting to note that even the error caused by crude representation of the space using STO-3G (921 for total and 858 for π -only compared with 1253 and 1147 of 6-31G LC-BLYP) seems to be similar or even less than the one from a deficiency of conventional DFT functional (1633 for total and 1603 for π -only compared with 2046 and 1995 of 6-31G BLYP) for $C_{20}H_{22}$. In Table 14.2 we summarize the longitudinal hyperpolarizabilities of $C_{20}H_{22}$. For this molecule, the π -electron approximation results in an overestimation, indicating a more complicated mechanism for this NLO process than for the linear response process, even in the interplay between σ - and π -electrons responding to the applied field. The error in the π -electron approximation remains less than the variation with computational methods, however.

14.4.2 Double- and Triple-Bonded Systems

We calculate the hyperpolarizabilities (γ) of polyyne and polyene to examine the NLO properties of different conjugated systems using DFT, HF, and ab initio electron correlation methods such as Müller–Plesset MP2, MP3, MP4(SDQ) theory^{30,31} and coupled-cluster CCSD, and CCSD(T) theory.^{32,33} For the geometries of the polyyne $H-(C\equiv C)_n-H$, a single (C—C) bond length of 1.3650 Å and triple (C $\equiv$ C) bond length of 1.2050 Å are used, taken from the averaged experimental values obtained from x-ray diffraction data of the i -Pr₃Si—(C $\equiv$ C)_{*n*}—Si*i*-Pr₃ (*n* = 4, 5, 6, and 8) molecules.³⁴ For the polyenes $H-(HC=CH)_n-H$, we used the geometries obtained from B3LYP/6-311G geometry optimizations.⁴ In all calculations, the cc-pVDZ basis set³⁵ is used. Hyperpolarizabilities γ are computed by the finite-field (FF) method using Eq. (14.1) by numerical Romberg iteration.³⁶

Figure 14.1 and Table 14.3 show, respectively, the γ -values of polyyne obtained using DFT and several wavefunction methods. As reported by other researchers,³ the pure functional [BOP (B88x exchange³⁷ and the one-parameter progressive correlation functional³⁸)] and the hybrid functional, B3LYP,^{39,40} which do not have long-range correction, overestimate γ -values. The tendency becomes more enhanced as the chain length, *n*, increases. The LC-DFT (LC-BOP) functional provides γ -values reasonably close to those from the

TABLE 14.2 Longitudinal Second Hyperpolarizabilities γ (10^7 a.u.) of $C_{20}H_{22}$ Computed Using the 6-31G Basis Set^a

	HF	BLYP	B3LYP	LC-BLYP
Total	2.0 (2.0)	5.8 (5.6)	5.6 (5.5)	2.8 (3.1)
π only	2.3	6.6	6.4	3.2

^aThe values in parentheses were obtained using cc-pVDZ.

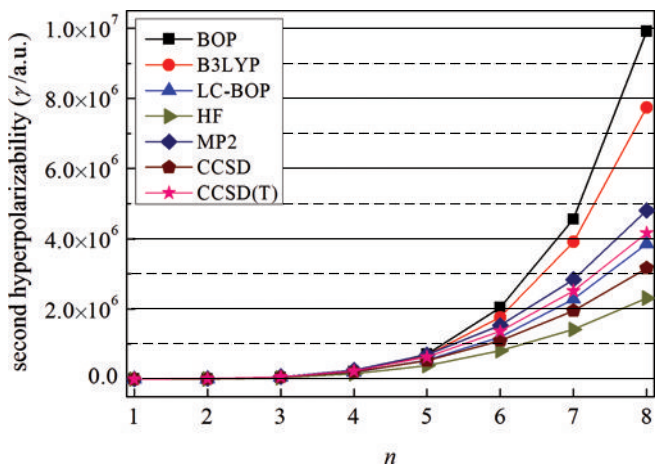


Fig. 14.1 (color online) Longitudinal second hyperpolarizabilities (γ) of the polyynes, $\text{H}-(\text{C}\equiv\text{C})_n-\text{H}$.

TABLE 14.3 Calculated Longitudinal Second Hyperpolarizabilities ($\gamma/\text{a.u.}$) of the Polyynes, $\text{H}-(\text{C}\equiv\text{C})_n-\text{H}^a$

γ	$1(\times 10^2)$	$2(\times 10^3)$	$3(\times 10^4)$	$4(\times 10^5)$	$5(\times 10^5)$	$6(\times 10^6)$	$7(\times 10^6)$	$8(\times 10^6)$
BOP	6.04	6.84	4.86	2.14	7.13	2.04	4.56	9.91
B3LYP	5.11	7.11	5.03	2.14	6.79	1.76	3.91	7.74
LC-BOP	4.96	8.32	5.42	1.96	5.40	1.20	2.28	3.86
HF	2.42	6.63	4.20	1.50	3.87	0.81	1.42	2.31
MP2	6.10	10.8	6.86	2.56	6.97	1.53	2.84	4.81
MP3	5.84	9.16	5.50	1.94	5.00	1.04	1.87	3.03
MP4	5.32	9.44	5.77	2.07	5.39	1.14	2.05	3.36
CCSD	5.74	9.91	5.82	2.03	5.27	1.10	1.95	3.16
CCSD(T)	5.11	10.2	6.45	2.32	6.29	1.36	2.51	4.17

^aThe numbers in the first row are the unit number n .

CCSD and CCSD(T). On the other hand, HF shows the lowest value and MP2 shows the highest value among the wavefunction methods.

Figure 14.2 and Table 14.4 show, respectively, the γ -values of polyenes obtained with the DFT and wavefunction methods. Although the complete set of γ -values as a function of chain length n is not presented, key features can be identified. Similar to the results obtained for the polyynes, MP2 predicts the highest and HF the lowest values among the wavefunction methods. The conventional functionals (BOP and B3LYP) also predict large values, while the LC-DFT (LC-BOP) functional again predicts γ -values surprisingly close to those from CCSD and CCSD(T). On the other hand, MP2 predicts the largest γ -values for the entire range of the polyynes and the polyenes in all the methods, except for conventional DFT methods which present gradual divergence of hyperpolarizabilities as the chain numbers are larger.

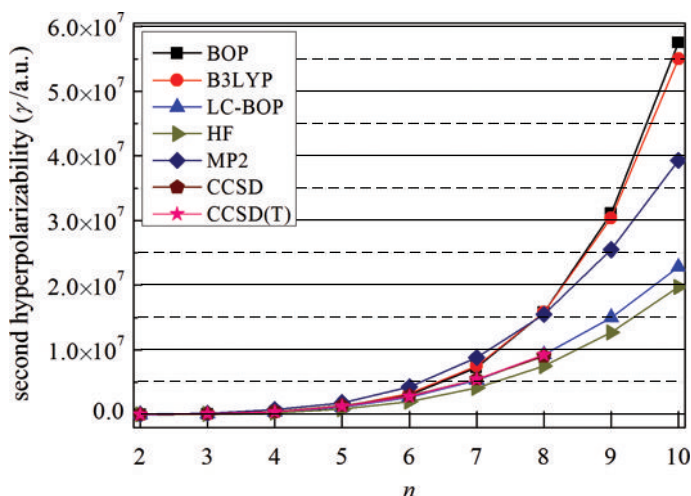


Fig. 14.2 (color online) Longitudinal second hyperpolarizabilities (γ) of the polyenes, $\text{H}-(\text{HC}=\text{CH})_n-\text{H}$.

TABLE 14.4 Longitudinal Second Hyperpolarizabilities ($\gamma/\text{a.u.}$) of the Polyenes, $\text{H}-(\text{HC}=\text{CH})_n-\text{H}^a$

γ	$2(\times 10^4)$	$3(\times 10^5)$	$4(\times 10^5)$	$5(\times 10^6)$	$6(\times 10^6)$	$7(\times 10^6)$	$8(\times 10^7)$	$9(\times 10^7)$	$10(\times 10^7)$
BOP	0.31	0.77	3.43	1.12	3.06	7.27	1.58	3.11	5.75
B3LYP	1.07	0.83	3.71	1.21	3.25	7.54	1.58	3.04	5.50
LC-BOP	1.09	1.05	4.01	1.16	2.67	5.34	0.93	1.50	2.28
HF	0.75	0.68	2.82	0.85	2.04	4.14	0.75	1.27	1.97
MP2	4.40	1.69	7.88	1.82	4.30	8.84	1.55	2.55	3.93
MP3	2.19	1.62	6.16	1.82	3.95	7.12	1.27		
MP4	1.81	1.49	5.74	1.72	3.73	6.86	1.21		
CCSD	1.52	1.25	4.54	1.38	2.88	5.47	0.90		
CCSD(T)	1.35	1.16	4.16	1.32	2.80	5.45	0.93		

^aThe numbers in the first row are unit number n .

On moving from CCSD to CCSD(T), the γ -values of the polyynes change significantly, suggesting that even the CCSD(T) hyperpolarizabilities are not converged with respect to the inclusion of correlation effects (see Fig. 14.3). The calculation of γ for polyynes appears to be a challenging case problem for conventional correlated methods.^{41–44} On the other hand, the differences between the hyperpolarizabilities calculated for the polyynes by CCSD and CCSD(T) are small, perhaps suggesting that the values for the polyynes are nearly converged.

Although direct comparison to the experimental values of the absolute values evaluated theoretically should be the final goal for theorists, it is well known that the absolute value of third-order hyperpolarizabilities in the condensed phase is strongly pronounced through intermolecular interactions.⁴³ Some of those effects can be taken conveniently in local field correction, which assumes a continuous medium, but the large deviation of absolute molecular hyperpolarizability values

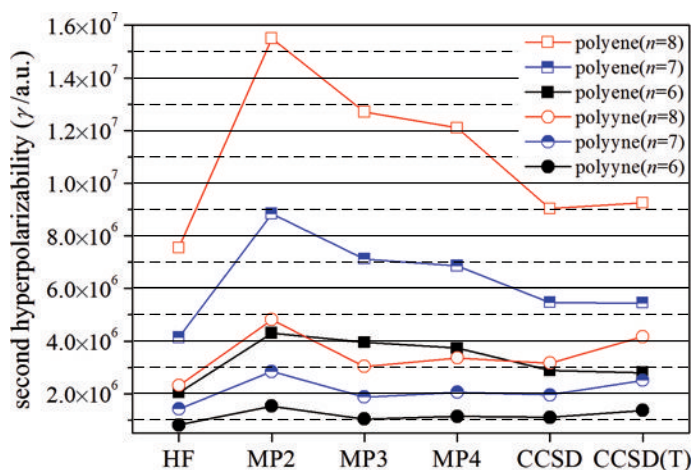


Fig. 14.3 (color online) Variation in the longitudinal second hyperpolarizabilities (γ) of the polyenes, $\text{H}-(\text{HC}=\text{CH})_n-\text{H}$ ($n = 6, 7$, and 8), and the polyynes, $\text{H}-(\text{C}\equiv\text{C})_n-\text{H}$ ($n = 6, 7$, and 8) with the calculation method used: HF, MP2, MP3, MP4, CCSD, and CCSD(T).

computed by rigorous ab initio methods from the third-order NLO coefficients observed for those systems suggests that the intermolecular interaction effect in those systems is paramount under the experimental settings, and it is clear that the much more sophisticated and/or computationally demanding methods must be used for taking the effects. We introduce here an argument using a power-law function to calibrate the length effect on the molecular property. To identify the length dependence of the molecular properties, simple power-law functions provide a useful tool. We fit the calculated γ -values for polyynes $n = 1$ to 8 and polyenes $n = 2$ to 10 to the power-law function, $\gamma = bn^c$, and the results are given in Table 14.5 [only $n = 1$ to 8 are used for MP2, MP3, MP4, CCSD, and CCSD(T)]. For both the polyynes and polyenes, the exponents c calculated by the pure and hybrid functionals exceed 5 and are very much larger than those obtained using wavefunction methods. This is consistent with the fact that the γ -values for large molecules calculated using conventional DFT are overestimated^{4,5,21}; hence, these methods cannot provide reliable information on the length dependence of NLO properties. On the other hand, the exponents evaluated using LC-DFT are rather close to those from CCSD(T) for the polyynes. It is notable that the HF exponent for the polyenes is larger than that from other wavefunction methods, whereas that for the polyynes is smaller.

The hyperpolarizability exponent c observed for the polyynes, 4.3 ,³⁴ is higher than that for the polyenes, 2.3 to 3.6 .^{45,46} Contrary to the experimental findings, all values computed for the polyenes exceed those for the polyynes.

It is well known that for a reliable comparison with experiment, vibrational NLO effects should be considered. To estimate these contributions for the polyenes and polyynes, we use RHF/6-31G calculated values⁴⁴ for the ratio of

TABLE 14.5 c Values of the γ Power Law ($\gamma = bn^c$) for the Polynes $[H-(C \equiv C)_n-H]$ and the Polyenes $[H-(HC=CH)_n-H]^a$

	Polyyne				Polyene			
	γ		γ_{vib}^b		γ		γ_{vib}	
	b	c	b	c	b	c	b	c
BOP	79 (± 11)	5.64 (± 0.066)	145 (± 20)	5.29 (± 0.071)	92 (± 3)	5.80 (± 0.013)	102 (± 12)	5.89 (± 0.052)
B3LYP	176	5.14	171	5.13	142	5.59	163	5.66
LC-BOP	(± 4)	(± 0.011)	(± 6)	(± 0.018)	(± 4)	(± 0.013)	(± 18)	(± 0.049)
	778	4.09	620	4.19	1812	4.10	2271	4.14
HF	(± 79)	(± 0.050)	(± 49)	(± 0.042)	(± 67)	(± 0.041)	(± 109)	(± 0.021)
	971	3.74	882	3.78	881	4.35	1129	4.38
MP2	(± 99)	(± 0.050)	(± 123)	(± 0.070)	(± 86)	(± 0.043)	(± 40)	(± 0.015)
	1086	4.04	937	4.09	2397	4.22	2994	4.25
MP3	(± 93)	(± 0.042)	(± 89)	(± 0.050)	(± 198)	(± 0.037)	(± 214)	(± 0.032)
	1196	3.77	968	3.85	2158 ^c	4.17 ^c	3711 ^c	4.06 ^c
MP4	(± 125)	(± 0.052)	(± 85)	(± 0.051)	(± 282)	(± 0.065)	(± 658)	(± 0.088)
	1186	3.82	980	3.90	2052 ^c	4.17 ^c	3655 ^c	4.04 ^c
CCSD	(± 117)	(± 0.049)	(± 93)	(± 0.051)	(± 201)	(± 0.048)	(± 468)	(± 0.063)
	1314	3.75	1132	3.84	2345 ^c	3.97 ^c	4081 ^c	3.85 ^c
CCSD(T)	(± 140)	(± 0.053)	(± 148)	(± 0.052)	(± 361)	(± 0.076)	(± 569)	(± 0.069)
	1142	3.95	917	4.04	1697 ^c	4.14 ^c	2950 ^c	4.02 ^c
	(± 119)	(± 0.052)	(± 82)	(± 0.047)	(± 225)	(± 0.066)	(± 382)	(± 0.064)

^aThe values in parentheses are estimates of the fitting error in each method. The cc-pVDZ basis set is used in all calculations.

^bFor polyyne, we included data only for $n = 1$ to 7 as Ref. 44 does not give values for $n = 8$.

^cFor polyene, MP3, MP4, CCSD, and CCSD(T) data are used only for $n = 1$ to 8.

vibrational γ -values (γ_{vib}) and the electronic γ -values. In Table 14.5 we also present the calculated exponents c and its prefactor b for the polyynes and polyenes calculated using this correction. While more sophisticated vibrational correction are typically required,^{1,41} their explicit determination remains impractical for large-molecular systems. The vibrational corrections used here look not to change the exponents so much, but it is noticeable that in the CCSD(T) and LC-DFT methods, which are thought to predict hyperpolarizability values more reliably than other methods do, the exponents for polyynes become slightly larger than those of polyenes. This shows us that the vibrational corrections can be a key to explaining the reason that the hyperpolarizability exponent observed for the polyynes is higher than that for the polyenes. We expect that more sophisticated vibrational correction will be able to address this problem.

Besides the vibrational effect, considering the geometries of the polyenes, we also notice that the molecules used in the hyperpolarizability measurements have varying conformations,^{46,47} whereas all-*trans* —C=C— conformations are used in the calculations.⁴⁸ Further, the geometries of the polyenes used in the calculations were optimized using B3LYP, a method that underestimates bond-length alternation and hence is expected to overestimate hyperpolarizabilities.^{6,21} For the polyynes, only one experimental configuration is possible, but various end-group cappings were used in the experiments.⁴⁵ Another possible explanation for the difference between the length dependences of the hyperpolarizabilities observed and those calculated is that the longitudinal second hyperpolarizability, γ_{zzzz} , is calculated, whereas the experimental values refer to the isotropic second hyperpolarizability γ .³³ Finally, we must keep in mind that the experimental γ -values are also affected by solvent effects that can significantly alter the energies of excited charge-transfer states, effects absent in the calculations.⁴⁹

14.5 CONCLUSIONS

We revisited the basic response theory for NLO property evaluation of materials using time-dependent perturbation theory to present a basic strategy for the theoretical investigation of NLO materials. Although the SOS representation is intuitive and may be useful for predicting the behavior of NLO properties in the vicinity of a resonance, it is not practical for the nonresonant situations important for the NLO materials of interest. Direct evaluation of dynamic NLO properties by solving the perturbed equation at the frequency of the applied field also involves considerable computational effort. Finite-field studies of the static hyperpolarizability can provide reliable information about the NLO materials far from resonance; they are limited, however, in that they cannot provide information relating to the specific NLO process with the frequency of the applied oscillating field. Because of the deficiencies in conventional DFT functionals, these methods are not applicable to NLO studies of large conjugated molecules. We introduce a practical method that incorporates long-range corrections into conventional DFT methods. It is based on the simple idea of range-dependent partitioning of the Coulomb interaction. We find that this method provides a

qualitatively correct description of the NLO properties of large molecules without requiring prohibitive computational effort. We investigate further the validity of the π -electron approximation. This approach is found inadequate for an evaluation of the response properties of small molecules, but for larger systems the dominant terms are properly included so that the error diminishes in relative magnitude. Indeed, the error from this approximation becomes much smaller than the variation in the results associated with the choice of the computational method. These results provide an optimistic perspective for the theoretical prediction of the properties of NLO materials, since this approximation considerably reduces the computational resources required.

We further investigated the influence of different types of π -conjugation on NLO properties by contrasting polyynes with polyenes. For both systems, LC-DFT gives γ -values close to those predicted by CCSD and CCSD(T), whereas conventional DFT methods such as BOP, as well as hybrid DFT methods such as B3LYP, considerably overestimate the response. MP2 predicts the highest and HF predicts the lowest γ -values among all the wavefunction methods tested. The CCSD and CCSD(T) methods predict similar hyperpolarizabilities for the polyenes but not for the polyynes, indicating that electron correlation may not be described properly in the dense π -electron polyynes. For the exponential scale factor c (from the fit $\gamma = bn^c$), LC-DFT also predicts results similar to those of CCSD(T). The theoretical prediction that hyperpolarizabilities increase much faster with increasing length for polyenes compared to polyynes is inconsistent with experimental observations, however. This could arise from the differences in the chemical structures considered, solvent effects, or the approximation that the diagonal hyperpolarizability component dominates the values observed. Even though the vibrational effect considered here shows a small influence on the γ -value and γ scaling factor, more sophisticated vibrational effects may correct the theoretical inconsistency with the experimental observations.

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REFERENCES

1. Torrent-Sucarrat, M.; Sola, M.; Duran, M.; Luis, M. J.; Kirtman, B. *J. Chem. Phys.* **2004**, *120*, 6346.
2. Koch W.; Holthausen, M. C. *A Chemist's Guide to Density Functional Theory*, Wiley-VCH, New York, 2000.
3. (a) Reimers, J. R.; Cai, Z.-L.; Bilić, A.; Hush, N. S. *Ann. N.Y. Acad. Sci.* **2003**, *1006*, 235. (b) Cai, Z.-L.; Sendt, K.; Reimers, J. R. *J. Chem. Phys.* **2002**, *117*, 5543.

4. Champagne, B.; Perpète, É. A.; Jacquemin, D.; van Gisbergen, S. J. A.; Baerends, E.-J.; Soubra-Ghaoui, C.; Robins, K. A.; Kirtman, B. *J. Phys. Chem. A* **2000**, *104*, 4755.
5. Champagne, B.; Perpète, É. A.; van Gisbergen, S. J. A.; Baerends, E.-J.; Snijders, J. G.; Soubra-Ghaoui, C.; Robins, K. A.; Kirtman, B. *J. Phys. Chem. A* **1998**, *109*, 10489.
6. Stevens, P. J.; Devlin, J. F.; Chabalowski, C. F.; Frisch, M. J. *J. Phys. Chem.* **1994**, *98*, 11623.
7. Sahní, V.; Gruenebaum, J.; Perdew, J. P. *Phys. Rev.* **1982**, B26, 4371.
8. Mori-Sánchez, P.; Wu, Q.; Yang, W. *J. Chem. Phys.* **2003**, *119*, 11001.
9. (a) Kummel, S.; Perdew, J. P. *Phys. Rev. B* **2003**, *68*, 035103. (b) Kummel, S.; Perdew, J. P. *Phys. Rev. Lett.* **2003**, *90*, 043004.
10. Krieger, J. B.; Li, Y.; Iafrate, G. J. *Phys. Rev.* **1992**, A46, 5453.
11. Gritsenko, O. V.; Baerends, E. J. *Phys. Rev.* **2001**, A64, 042506.
12. Kümmel, S.; Kronik, L.; Perdew, J. P. *Phys. Rev. Lett.* **2004**, *93*, 213002.
13. van Faassen, M.; de Boeij, P. L.; van Leeuwen, R.; Berger, J. A.; Snijders, J. G. *J. Chem. Phys.* **2003**, *118*, 1044.
14. van Faassen, M.; Jensen, L.; Berger, J. A.; de Boeij, P. L. *Chem. Phys. Lett.* **2004**, *395*, 274.
15. van Faassen, M.; de Boeij, P. L.; van Leeuwen, R.; Berger, J. A.; Snijders, J. G. *Phys. Rev. Lett.* **2002**, *88*, 186401.
16. van Faassen, M. *Int. J. Mod. Phys.* **2006**, B20, 3419.
17. Marini, A.; Del Sole, R.; Rubio, A. In *Time-Dependent Density Functional Theory*, Lecture Notes in Physics, Vol. 706, Marques, M. A. L., Ullrich, C. A., Nogueira, F., Rubio, A., Burke, K., and Gross, E. K. U., Eds., Springer-Verlag, Berlin, 2006, Chap. 20.
18. Iikura, H.; Tsuneda, T.; Yanai, T.; Hirao, K. *J. Chem. Phys.* **2001**, *115*, 3540.
19. Tawada, Y.; Tsuneda, T.; Yanagisawa, S.; Yanai, T.; Hirao, K. *J. Chem. Phys.* **2004**, *120*, 8425.
20. Kamiya, M.; Sekino, H.; Tsuneda, T.; Hirao, K. *J. Chem. Phys.* **2005**, *122*, 234111.
21. Sekino, H.; Maeda, Y.; Kamiya, M.; Hirao, K. *J. Chem. Phys.* **2007**, *126*, 014107.
22. Kirtman, B.; Bonness, S.; Ramirez-Solis, A.; Champagne, B.; Matsumoto, H.; Sekino, H. *J. Chem. Phys.* **2008**, *128*, 114108.
23. Song, J.-W.; Watson, M. A.; Sekino, H.; Hirao, K. *J. Chem. Phys.* **2008**, *129*, 024117.
24. Song, J.-W.; Watson, M. A.; Sekino, H.; Hirao, K. *Int. J. Quantum Chem.* **2009**, *109*, 2012.
25. Sekino, H.; Bartlett, R. J. *Theoretical and Computational Modeling of NLO and Electronic Materials*, Karna, S. P., and Yeates, A. T., Eds., ACS Symposium Series, 1994, pp. 79–101.
26. Sekino, H.; Bartlett, R. J. *J. Chem. Phys.* **1986**, *85*, 976.
27. Savin, A. In *Recent Developments and Applications of Modern Density Functional Theory*, Seminario, J. J., Ed., Elsevier, Amsterdam, 1996, Chap. 9.
28. Song, J.-W.; Hirosawa, T.; Tsuneda, T.; Hirao, K. *J. Chem. Phys.* **2007**, *126*, 154105.
29. Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; et al. *Gaussian 03, Revision D.02*, Gaussian Inc., Wallingford CT, 2004.

30. Sekino, H.; Maeda, Y.; Kamiya, M. *Mol. Phys. (Bartlett Special Issue)* **2005**, *103*, 2183.
31. Müller, C.; Plesset, M. S. *Phys. Rev.* **1934**, *46*, 0618.
32. Bartlett, R. J.; Purvis, G. D., III. *Int. J. Quantum Chem.* **1978**, *14*, 561.
33. Pople, J. A.; Krishnan, R.; Schlegel, H. B.; Binkley, J. S. *Int. J. Quantum Chem.* **1978**, *14*, 545.
34. Eisler, S.; Slepko, A. D.; Elliott, E.; Luu, T.; McDonald, R.; Hegmann, F. A.; Tykwinski, R. R. *J. Am. Chem. Soc.* **2005**, *127*, 2666.
35. Dunning, T. H., Jr. *J. Chem. Phys.* **1989**, *90*, 1007.
36. Jaquemin, D.; Champagne, B.; André, J.-M. *Int. J. Quantum Chem.* **1997**, *65*, 679.
37. Becke, A. D. *Phys. Rev. A* **1988**, *38*, 3098.
38. Tsuneda, T.; Suzumura, T.; Hirao, K. *J. Chem. Phys.* **1999**, *110*, 10664.
39. Lee, C.; Yang, W.; Parr, R. G. *Phys. Rev. B* **1988**, *37*, 785.
40. Becke, A. D. *J. Chem. Phys.* **1993**, *98*, 5648.
41. Torrent-Sucarrat, M.; Solá, M.; Duran, M.; Luis, J. M.; Kirtman, B. *J. Chem. Phys.* **2003**, *118*, 711.
42. Toto, J. L.; Toto, T. T.; de Melo, C. P. *Chem. Phys. Lett.* **1996**, *104*, 8586.
43. Bredas, J. L.; Adant, C.; Tackx, P.; Persoons, A.; Pierce, B. M. *Chem. Rev.* **1994**, *94*, 243.
44. Kirtman, B.; Champagne, B. *Int. Rev. Phys. Chem.* **1997**, *16*, 389.
45. Luu, T.; Elliott, E.; Slepko, A. D.; Eisler, S.; McDonald, R.; Hegmann, F. A.; Tykwinski, R. R. *Org. Lett.* **2005**, *7*, 51.
46. Samuel, I. D. W.; Ledoux, I.; Dhenaut, C.; Zyss, J.; Fox, H. H.; Schrock, R. R.; Silbey, R. J. *Science* **1994**, *265*, 1070.
47. Craig, G. S. W.; Cohen, R. E.; Schrock, R. R.; Silbey, R. J.; Puccetti, G.; Ledoux, I.; Zyss, J. *J. Am. Chem. Soc.* **1993**, *115*, 860.
48. Rossi, G.; Chance, R. R.; Silbey, R. *J. Chem. Phys.* **1989**, *90*, 7594.
49. Ray, P. C. *Chem. Phys. Lett.* **2004**, *395*, 269.

15 Calculating the Raman and HyperRaman Spectra of Large Molecules and Molecules Interacting with Nanoparticles

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This chapter describes calculations of the Raman and hyperRaman spectra of large molecules and molecules interacting with nanoparticles using time-dependent density functional theory with the Amsterdam density functional (ADF) program package. The ADF code uses Slater basis functions, which provides a very efficient basis set for optical property calculations using density functional theory (DFT). In addition, ADF has special capabilities for determining resonant Raman spectra, which is enabled by the inclusion of excited-state lifetimes in the calculations, and therefore polarizabilities and polarizability derivatives for wavelengths close to resonance can be determined. Specific details of the theory are described, and examples of applications to pyridine (for nonresonant properties) and uracil (for resonant properties) are provided.

15.1 INTRODUCTION

Raman spectroscopy is an inelastic linear light-scattering method that provides a vibrational fingerprint of a molecule. This fingerprint can be used to identify molecules, so there has been increasing interest in using Raman in analytical chemistry applications and medical diagnostics,^{1–6} particularly with the development of lasers and detectors which allow Raman measurements to be made over a wide range of wavelengths from the near-infrared to the ultraviolet (UV). HyperRaman spectroscopy is an analogous optical technique that involves inelastic light scattering relative to the second harmonic of the incident light, so this nonlinear optics technique also provides a vibrational fingerprint, but for an incident frequency which is only half of the frequency needed to produce the same scattered photon as in Raman scattering.^{7–10} In addition, the selection rules for hyperRaman are different from those for Raman, as the latter involves two photons (incident and scattered) while the former involves three photons (two incident plus one scattered). This means that vibrations that are silent in Raman become active in hyperRaman. Both techniques are intrinsically weak processes, however both can be amplified by placing the molecule next to a silver or gold nanoparticle, as plasmon excitation in the particle can produce enhanced electromagnetic fields near the particle surface, leading to surface-enhanced Raman and hyperRaman spectroscopies (SERS and SEHRS, respectively).^{11,12} In addition, enhancement can also arise if the molecule has a resonant electronic state at the excitation wavelength, leading to resonance Raman and resonance hyperRaman spectroscopy. Under favorable conditions it is possible to combine resonance and surface enhancement effects, leading to surface-enhanced resonance Raman spectroscopy (SERRS) and surface-enhanced resonance hyperRaman spectroscopy (SERHRS).^{13,14}

Raman intensities are proportional to the square of the derivative of the polarizability of the molecule with respect to vibrational normal coordinates,¹⁵ so the calculation of Raman intensities requires a determination of the frequency-dependent polarizabilities, usually by determining the first-order response of the molecule to the applied electromagnetic field. Many electronic structure codes have the ability to produce Raman spectra in the static limit (low frequency) through analytical determination of the polarizability derivative. This works well for small molecules that do not have important electronic transitions in the visible. However, for larger systems, especially for molecules with transitions at optical frequencies, or for molecules interacting with metal particles (as in SERS), this approximation is not appropriate. In this chapter we describe calculations of Raman intensities based on the Amsterdam density functional (ADF) code,^{16–18} a code specifically developed to determine response properties using time-dependent density functional theory (TDDFT). The basics of density functional theory (DFT) and TDDFT are described in detail in Chapter 1.

ADF and a recently developed local version of ADF have some unique features for calculating Raman, resonance Raman, and SERS intensities at finite frequencies.^{19–21} ADF can also determine hyperRaman intensities, but

in an automated fashion only in the static limit at this point. The capability of calculating dynamic hyperpolarizabilities is available^{22–24} and will soon be combined with near-resonance damping functionality. In either case, ADF provides an efficient approach to studying large-molecular systems due to the use of Slater orbital basis functions in the calculations. These functions mimic the slow fall-off of atomic orbitals, a property that is especially important for response properties, much better than do Gaussian orbitals. Hence, they provide a more efficient representation of the change in density that arises in response to an applied electromagnetic field. As such, ADF enables the determination of Raman intensities for a number of challenging problems,²⁵ including studies of the resonance Raman scattering for molecules with multiple excited states,²⁰ and the study of SERS intensities for molecules interacting with silver and gold metal clusters.^{26,27} In all these SERS calculations, the atoms in the molecule and in the metal cluster are described using basis sets of comparable quality and the same density functional [the same combination of exchange–correlation (XC) potential and XC response kernel]. This has the advantage of providing a completely balanced electronic structure description of the entire system, but a limitation with this approach is that the calculations are restricted to a total system size on the order of 100 to 200 atoms. To go beyond this requires methods that partition between components of the system that are described with quantum mechanics and components described using classical electrodynamics. The formal theory of such calculations was recently developed²⁸ but has not yet been implemented.

The Raman intensity calculation begins with a determination of the harmonic frequencies and normal coordinates of vibration for the molecule of interest by using density functional theory to calculate the Hessian matrix (second derivative of the energy with respect to the nuclear positions). Diagonalization of the mass-weighted Hessian determines the vibrational frequencies, and the eigenvectors define the normal coordinates. Subsequently, the polarizabilities (second derivative of energy with respect to applied finite field) are determined from TDDFT. For the Raman intensity, the polarizability calculations are performed for geometries that are displaced from equilibrium so that the derivatives of the polarizability with respect to each normal mode vibration can be calculated by finite differencing. Both normal Raman differential cross sections and relative surface-enhanced Raman intensities can be calculated from combinations of the polarizability derivatives. This approach can also be expanded to allow for the calculation of resonance Raman spectra. HyperRaman and surface-enhanced Raman spectra can also be calculated using ADF. While the use of finite differencing may seem to be inefficient relative to the analytical evaluation of the polarizability derivatives, for large molecules one often does not want or need derivatives with respect to all the modes. Indeed, for applications in SERS, where the system of interest is a molecule plus a large metal cluster, only a small fraction of the possible modes, those referring to vibrations of the molecule, is of interest, and in any case the finite-difference procedure is trivially parallelized.

In the following sections we describe the underlying theory of Raman/hyperRaman intensity calculations, and specific details of these calculations based on the ADF code.

15.2 DISPLACEMENT OF COORDINATES ALONG NORMAL MODES

To construct the Raman or hyperRaman spectrum of a computationally large molecule, it is first necessary to calculate the vibrational normal modes of an optimized local minimum structure. Details of these calculations, which involve diagonalization of the mass-weighted Hessian matrix (second derivative of energy with respect to atomic coordinates), are described in standard textbooks, so we omit the steps here. The Raman intensity is easily calculated by making the double harmonic approximation. This approximation is composed of two parts.²⁹ First, each vibration is assumed to be described by a harmonic potential (i.e., a linear expression for the intramolecular forces). Second, the dipole moment function $\mu(r)$ is assumed to vary linearly with the normal mode coordinate r in the region where r is close to the equilibrium structure denoted by r_e .

In ADF it is possible to calculate the energies (in wavenumbers, cm^{-1}) and Cartesian displacements (in bohr) of the vibrational normal modes of a molecule by using the `FREQUENCIES` keyword under the `GEOMETRY` block. To calculate the polarizability derivatives, the components of the polarizability tensor are calculated at two structures that have been displaced in different directions along a vibrational mode. Starting with the equilibrium geometry, the coordinates $R_{\text{eq},i}$ of each atom are changed by a small amount $\pm s_R \Delta R_{k,i}$ where $\Delta R_{k,i}$ is the Cartesian displacement of the i th coordinate in the k th vibrational normal mode and s_R is the step size. Ideally, s_R should be mode specific, such that a more shallow potential (low harmonic frequency) should be treated with a somewhat larger displacement.³⁰

The s_R should be chosen so that the norm (root of the sum of squares) of $s_R \Delta R_{k,i}$ for each k is on the order of a few hundredths of a bohr.³¹ If the s_R is too large, the double harmonic approximation breaks down, while if it is too small, there will not be an appreciable change in the polarizability tensor. Both cases will lead to errors in the polarizability derivatives and thus the calculated Raman intensities.

Once a suitable s_R has been chosen, the equilibrium coordinates are displaced to obtain two sets of coordinates. The set created by using $R_{\text{eq},i} - s_R \Delta R_{k,i}$ will be denoted as the *minus structure*, and those created by $R_{\text{eq},i} + s_R \Delta R_{k,i}$ will be denoted as the *plus structure*. Polarizability derivatives are then calculated by finite differencing.

15.3 CALCULATION OF POLARIZABILITIES USING TDDFT

Polarizabilities can be calculated using time-dependent DFT (TDDFT) response theory. In the ADF program, this functionality can be reached by specifying

the input “block” keyword RESPONSE or AORESPONSE [conveniently, also via the graphical user interface (GUI)]. ADF input files consist of a list of keywords (e.g., BASIS, ATOMS, GEOMETRY) which provide the program with specifics of the chemical system (e.g., charge, atomic positions), type of calculation desired (e.g., geometry optimization, Hessian matrix diagonalization), and specifics of the calculation (e.g., basis set, level of theory). Many keywords have specific options that can be enumerated on lines following the keyword forming a block which is ended with the line END. For more details on using ADF, refer to the documentation, including a user guide and input examples, available at <http://www.scm.com>. The RESPONSE keyword triggers the original implementation of TDDFT response theory by van Gisbergen et al.,^{16,22} which is capable of using symmetry. AORESPONSE triggers a more recently developed code^{32,33} that offers additional functionality, such as the near-resonance dynamic response capability,^{19,34} or enhanced analysis features,^{23,24} but lacks symmetry. Both blocks allow calculation of frequency-dependent polarizabilities, but the AORESPONSE block is needed to calculate the resonance Raman spectra. For the examples in this chapter the RESPONSE key was used to calculate hyperpolarizabilities from which hyperRaman spectra can be predicted. In our explanations of how to calculate Raman spectra, use of the AORESPONSE block will be assumed. Any specifics for calculating hyperRaman spectra will assume use of the RESPONSE block. In an upcoming version of the program the hyperRaman and resonance hyperRaman functionality will be combined with the AORESPONSE functionality. An example of an input (more example inputs can be found in the supporting information) to calculate the static polarizability tensor of a displaced structure of pyridine using the AORESPONSE block is as follows:

BASIS

```
C /share/apps/adf2007.01/atomicdata/ET/DIFFUSE/ET-QZ3P-polar/C
H /share/apps/adf2007.01/atomicdata/ET/DIFFUSE/ET-QZ3P-polar/H
N /share/apps/adf2007.01/atomicdata/ET/DIFFUSE/ET-QZ3P-polar/N
END
```

DEPENDENCY

```
XC
model SAOP
END
```

ATOMS

N	0.000000	0.000002	0.043787
C	0.000000	-0.000002	2.855245
C	0.000000	1.197479	2.143113
C	0.000000	-1.197480	2.143111
C	0.000000	1.141525	0.748759
C	0.000000	-1.141523	0.748756

H	0.000000	2.064563	0.165373
H	0.000000	-2.064560	0.165368
H	0.000000	2.159648	2.654667
H	0.000000	-2.159650	2.654661
H	0.000000	-0.000003	3.944999

END

SYMMETRY NOSYM

AORESPONSE

ALDA

END

END INPUT

The real parts of the nine Cartesian components of the frequency-dependent polarizability tensor of an input structure are calculated if the AORESPONSE block is included in the input.

When an external potential $v_{\text{ext},i}(\mathbf{r}, t) = E r_i \cos \omega t$ is applied to a molecule, the components of the polarizability, $\alpha_{ij}(\omega)$, can be determined from the change in the electron density using

$$\alpha_{ij}(\omega) = - \int d^3r \rho_i^{(1)}(\mathbf{r}, \omega) r_j \quad (15.1)$$

where i and j are the Cartesian directions and $\rho_i^{(1)}(\mathbf{r}, \omega)$ is the *linear* change in density (linear response) due to the external potential.³⁵

In TDDFT, the density change is found using the linear response function of the noninteracting Kohn–Sham system $\chi_s(\mathbf{r}, \mathbf{r}', \omega)$ and the linear change in the effective potential $v_{\text{eff}}^{(1)}(\mathbf{r}, \omega)$ with the relation³⁵

$$\rho^{(1)}(\mathbf{r}, \omega) = \int d\mathbf{r}' \chi_s(\mathbf{r}, \mathbf{r}', \omega) v_{\text{eff}}^{(1)}(\mathbf{r}', \omega) \quad (15.2)$$

where for the potential given above, the external field part of the linear perturbation operator ($v_{\text{ext}}^{(1)}$ below) is obtained through division by the field amplitude. In the absence of finite-lifetime or other damping terms, the expression for the Kohn–Sham response function, constructed from the occupied and virtual Kohn–Sham orbitals (ϕ), energies (ϵ), and occupation numbers (n), is³⁵

$$\begin{aligned} \chi_s(\mathbf{r}, \mathbf{r}', \omega) = & \sum_i^{\text{occ.}} \sum_m^{\text{virt.}} n_i \phi_i(\mathbf{r}) \phi_m(\mathbf{r}) \phi_m(\mathbf{r}') \phi_i(\mathbf{r}') \\ & \times \left[\frac{1}{(\epsilon_i - \epsilon_m) + \omega} + \frac{1}{(\epsilon_i - \epsilon_m) - \omega} \right] \end{aligned} \quad (15.3)$$

When adopting the finite lifetime damping technique, the frequencies are formally substituted for by $\omega \rightarrow \omega + i\gamma$, where γ is a common damping parameter, and thus the response function as well as the linear density response become complex. This allows calculation of both the real and imaginary parts of the polarizability. The change in effective potential is³⁵

$$v_{\text{eff}}^{(1)}(\mathbf{r}', \omega) = v_{\text{ext}}^{(1)}(\mathbf{r}', \omega) + \int d\mathbf{r}' \frac{\rho^{(1)}(\mathbf{r}', \omega)}{|\mathbf{r} - \mathbf{r}'|} + \int d\mathbf{r}' f_{\text{xc}}(\mathbf{r}, \mathbf{r}', \omega) \rho^{(1)}(\mathbf{r}', \omega) \quad (15.4)$$

and contains terms for the external field as noted above, the linear response of the Coulomb potential, and the linear response of the exchange-correlation potential; f_{xc} is called the exchange-correlation kernel. The change in the effective potential is constructed in such a way that it will result in the correct change in density for the fully interacting system even though the noninteracting response function is being used, assuming that one would know the exact expression for the XC kernel. Of course, in practice, this is the term that gets approximated. In most cases an adiabatic approximation is used (i.e., one uses a frequency-independent f_{xc} , which neglects all memory effects). With the adiabatic approximation, XC kernels can be obtained simply by taking functional derivatives of the XC potential used for the ground-state calculation, based on popular functionals such as VWN, LYP, BP86, B3LYP, and PBE0. It is particularly efficient to use an XC kernel based on a local-density approximation (LDA) such as the VWN or Xalpha functional (ALDA keyword in AORESPONSE block, default in RESPONSE). Used in the examples, the adiabatic LDA (ALDA) exchange correlation kernel f_{xc} is local in space and time.³⁵ With a hybrid functional the kernel contains some nonlocal Hartree-Fock exchange. An implementation based on ADF's Slater-type basis and density-fitting approach has been reported by Ye et al.²³

The last two terms in the expression for the change in effective potential are dependent on the change in the density. Calculation of the density change must therefore be done in a self-consistent manner. The initial density change is calculated using $v_{\text{eff}}^{(1)} = v_{\text{ext}}^{(1)}$. Then the new effective field is determined using the updated density change. A new density change is calculated using the new effective field and the cycle continues until the change in the density change is below a set threshold. As in other self-consistent field codes, the iterations incorporate procedures to accelerate and stabilize the solution such that convergence is virtually guaranteed.^{36,37} The number of iterations and the convergence threshold can be set in the SCF block. With the change in density converged, the polarizability components are calculated. Similar procedures are adopted for calculating electric hyperpolarizabilities; see articles by van Gisbergen et al.^{16,22} and Ye et al.^{23,24} for further details regarding implementations in the ADF package and benchmark data.

To calculate Raman spectra for nonresonant molecules where the frequency dependence of the polarizability derivatives is weak, it is often sufficient to calculate the static polarizabilities (polarizabilities at zero frequency: $\omega = 0$) and

include a nonzero frequency only when calculating cross sections.^{38,39} (There are prefactors in the cross sections that cause the cross sections to be zero at zero frequency, so a finite frequency estimate of the cross sections requires inputting a frequency other than zero.) If the frequency-dependent polarizabilities are desired, the FREQUENCY keyword can be put into AORESPONSE, followed by a list of frequencies and the units (EV, HARTREE, or ANGSTROM) used for the frequencies. The necessary components of the static hyperpolarizability are obtained by adding the ALLCOMPONENTS and HYPERPOL keywords to the RESPONSE block. To obtain frequency-dependent hyperpolarizabilities, the DYNHYP keyword must also be added to the "block" and the frequency in hartrees must be specified after the HYPERPOL keyword. Use of the DEPENDENCY keyword in the input as well as specifying SYMMETRY NOSYM is suggested for both types of calculations.

15.4 DERIVATIVES OF THE POLARIZABILITIES WITH RESPECT TO NORMAL MODES

With either the static or frequency-dependent polarizabilities at hand for both the plus and minus structures for a normal mode, the polarizability derivatives can be calculated using the quotient of the change in polarizability and twice the normal-mode step size. This step size, s_{Qk} , is different from the step size s_R used earlier to make the displaced structures, and must be calculated separately for each normal mode. Note that this is not contained in ADF. The two step sizes are related by the equation

$$s_{Qk} = s_R / \sqrt{\sum_i^{3N} \left(\frac{\Delta R_i}{\sqrt{\sum_i^{3N} (\Delta R_i \sqrt{m_i})^2}} \right)^2} = \frac{s_R}{\Delta R^{\text{norm}} / \Delta Q^{\text{norm}}} \quad (15.5)$$

where m_i is the mass of the atom being displaced by ΔR_i , and ΔQ^{norm} is the square root of the sum of the squares of the mass-weighted displacements.³¹

The coordinates were displaced both backward and forward along the vibration, so the change in the polarizabilities must be divided by twice the s_{Qk} step size. The polarizability derivatives, α'_{ij} are therefore given as

$$\alpha'_{ij} = \frac{\alpha_{ij}(\text{plus}) - \alpha_{ij}(\text{minus})}{2s_{Qk}} \quad (15.6)$$

Polarizabilities in ADF are reported as polarizability volumes in atomic units and so have units of cubic bohr. By calculating s_{Qk} using the displacements in bohr and the masses in atomic mass units, the polarizability derivatives will have units of square bohr per square root of amu. Hyperpolarizabilities are also given in atomic units (quintic bohr per electron charge), which can be converted to quintic angstroms per electrostatic unit and then to quartic angstroms per statvolt.⁴⁰ The

components of the polarizability are also given with respect to a molecule fixed coordinate frame. Results for the specific components will therefore vary if the molecule coordinates are transformed with respect to this frame. Although there are times when the molecular orientation is important, most manipulations to produce spectra are invariant to orientation as they involve orientation averaging.

15.5 ORIENTATION AVERAGING

Certain combinations of the polarizability derivatives will give values that accurately predict the relative Raman peak intensities. When trying to reproduce spectra of systems that sample over all orientations of the molecule, the intensity of Raman scattered light will be

$$I_{\text{Raman}} = \frac{\omega^4}{c^4} I_0 \sum_{ij} \langle \tilde{\alpha}_{ij}'^2(\omega, Q) \rangle \quad (15.7)$$

where ω is the frequency of the scattered light. The tilde denotes that the components of the polarizability derivatives are defined relative to a space fixed-coordinate system, and the brackets denote that the value within is orientation averaged. For hyperRaman scattering, the expression for the intensity is

$$I_{\text{hyperRaman}} = \frac{8\pi\omega^4}{c^4} I_0 \sum \langle \tilde{\beta}_{ijj}'^2(\omega, Q) \rangle \quad (15.8)$$

If a common experimental setup is assumed where the scattering observed is 90° relative to the direction of the incident light and the scattered beam polarization is not resolved, the expression for the Raman intensity for a normal mode k becomes⁴¹

$$I_k^{\text{Raman}} = \frac{\omega^4}{c^4} I_0 \left(a_k'^2 + \frac{7}{45} \gamma_k'^2 \right) \quad (15.9)$$

The value $45a_k'^2 + 7\gamma_k'^2$ is called the Raman scattering factor, S_k , and is dependent on the polarizability derivatives through a_k' , the trace, and γ_k' , the anisotropy, of the polarizability derivatives. The trace and anisotropy in terms of the polarizability derivatives in the molecule fixed-coordinate system are³¹

$$\begin{aligned} a_k' &= \frac{1}{3} [(\alpha'_{xx})_k + (\alpha'_{yy})_k + (\alpha'_{zz})_k] \\ \gamma_k'^2 &= \frac{1}{2} \{ [(\alpha'_{xx})_k - (\alpha'_{yy})_k]^2 + [(\alpha'_{yy})_k - (\alpha'_{zz})_k]^2 + [(\alpha'_{zz})_k - (\alpha'_{xx})_k]^2 \\ &\quad + 6[(\alpha'_{xy})_k^2 + (\alpha'_{yz})_k^2 + (\alpha'_{zx})_k^2] \} \end{aligned} \quad (15.10)$$

Raman scattering factors are generally reported in quartic angstroms per atomic mass unit.

Under the same experimental conditions as those outlined for Raman scattering, the hyperRaman intensity expression becomes

$$I_{\text{hyperRaman}} = \frac{8\pi\omega^4}{c^5} I_0^2 (\langle \tilde{\beta}_{iii}^{\prime 2} \rangle + \langle \tilde{\beta}_{ijj}^{\prime 2} \rangle) \quad (15.11)$$

In terms of the molecule fixed hyperpolarizability derivatives, $\langle \tilde{\beta}_{iii}^{\prime 2} \rangle$ and $\langle \tilde{\beta}_{ijj}^{\prime 2} \rangle$ are⁹

$$\begin{aligned} \langle \tilde{\beta}_{iii}^{\prime 2} \rangle = & \frac{1}{7} \sum_i \beta_{iii}^{\prime 2} + \frac{4}{35} \sum_{i \neq j} \beta_{iij}^{\prime 2} + \frac{2}{35} \sum_{i \neq j} \beta'_{iii} \beta'_{ijj} + \frac{4}{35} \sum_{i \neq j} \beta'_{jii} \beta'_{ijj} \\ & + \frac{4}{35} \sum_{i \neq j} \beta'_{iii} \beta'_{jji} + \frac{1}{35} \sum_{i \neq j} \beta_{jii}^{\prime 2} + \frac{4}{105} \sum_{i \neq j \neq k} \beta'_{ijj} \beta'_{jkk} \\ & + \frac{1}{105} \sum_{i \neq j \neq k} \beta'_{jii} \beta'_{jkk} + \frac{4}{105} \sum_{i \neq j \neq k} \beta'_{ijj} \beta'_{kkj} \\ & + \frac{2}{105} \sum_{i \neq j \neq k} \beta_{ijk}^{\prime 2} + \frac{4}{105} \sum_{i \neq j \neq k} \beta'_{ijk} \beta'_{jik} \end{aligned} \quad (15.12)$$

$$\begin{aligned} \langle \tilde{\beta}_{ijj}^{\prime 2} \rangle = & \frac{1}{35} \sum_i \beta_{iii}^{\prime 2} + \frac{4}{105} \sum_{i \neq j} \beta'_{iii} \beta'_{ijj} - \frac{4}{70} \sum_{i \neq j} \beta'_{iii} \beta'_{jji} + \frac{8}{105} \sum_{i \neq j} \beta_{ijj}^{\prime 2} \\ & + \frac{3}{35} \sum_{i \neq j} \beta_{ijj}^{\prime 2} - \frac{4}{70} \sum_{i \neq j} \beta'_{ijj} \beta'_{jii} + \frac{1}{35} \sum_{i \neq j \neq k} \beta'_{ijj} \beta'_{ikk} \\ & - \frac{4}{210} \sum_{i \neq j \neq k} \beta'_{iik} \beta'_{jjk} - \frac{4}{210} \sum_{i \neq j \neq k} \beta'_{ijj} \beta'_{jkk} \\ & + \frac{2}{35} \sum_{i \neq j \neq k} \beta_{ijk}^{\prime 2} - \frac{4}{210} \sum_{i \neq j \neq k} \beta'_{ijk} \beta'_{jik} \end{aligned} \quad (15.13)$$

15.6 DIFFERENTIAL CROSS SECTIONS

Although Raman scattering factors will give a good idea of relative intensities, it is the differential cross sections that are directly comparable to experimental measurements. The frequency of the incident light is part of the expression of the differential cross section which allows normal Raman spectra for a specific wavelength of incident light to be calculated even while using static polarizabilities.³⁹ This approach should give reasonable estimates for any off-resonance situation as long as the dispersion of the polarizability is relatively small. The computational effort to calculate the scattering factors using dynamic polarizabilities is higher, but is recommended for improved accuracy.

For the Q branch in an experiment where the scattering angle is 90° and the incident light is perpendicularly plane polarized with respect to the scattering plane, the differential cross section is^{31,39}

$$\frac{d\sigma}{d\Omega} = \frac{h}{8\epsilon_0^2 c \tilde{\nu}_k} (\tilde{\nu}_{\text{in}} - \tilde{\nu}_k)^4 \frac{S_k}{45} \frac{1}{1 - \exp(-hc\tilde{\nu}_k/k_B T)} \quad (15.14)$$

where $\tilde{\nu}_{\text{in}}$ is the frequency of the incident light and $\tilde{\nu}_k$ is the frequency of the k th normal mode, both in wavenumbers. If the Raman scattering factors in quartic angstroms per atomic mass unit are converted to $\text{C}^2 \cdot \text{m}^2/\text{V}^2 \cdot \text{kg}$ using a factor of $1/4\pi\epsilon_0$ along with the appropriate length and mass conversions, the differential cross section can be made to have units of cm^2/sr (sr is the abbreviation for *steradians*). These are the standard units for reporting Raman scattering differential cross sections.

Example 1: Raman Spectra of Pyridine and Pyridine on a Silver Cluster As an example of the results that can be expected using the method described above, simulated Raman spectra for pyridine and pyridine on the surface of a tetrahedral 20-silver-atom cluster will be shown. The orientationally averaged off-resonance spectra calculated are referred to as normal or bulk Raman spectra, and are comparable to those obtained in experiments performed on solutions of the species modeled. Geometry optimization and normal-mode frequency calculations were performed using the PW91 functional and a polarized triple-zeta Slater-type basis (TZP) for all atoms. Relativistic effects, which have been shown to be important in the modeling of optical properties of silver clusters,⁴² are included with the use of the zeroth-order regular approximation (ZORA)^{43,44} in its spin-free (scalar relativistic) version. An extension of AORESPONSE to include spin-orbit coupling has also been developed recently,⁴⁵ but for an Ag cluster, such effects can be considered negligible. The normal-mode frequencies calculated were compared to those from experiment to ensure decent agreement. Normal-mode frequencies and atomic coordinates for the optimized geometries are available in the supporting information. Polarizability calculations used an asymptotically correct XC potential, SAOP,⁴⁶ and the larger ET-QZ3P-polar basis set for the carbon, hydrogen, and nitrogen atoms (still using TZP for the silver atoms). Use of the SAOP model potential gives the correct long-distance behavior, which is important for obtaining accurate polarizabilities (although for the systems at hand, BP86 and TZP give similar results) and even more so for hyperpolarizabilities.⁴⁷

The normal Raman spectrum for pyridine, calculated from static polarizabilities and using an incident wavelength of 514.5 nm in the equation for the cross section, is shown in Fig. 15.1 (the differential cross section is given in units of $10^{-30} \text{ cm}^2 \text{ sr}^{-1}$ and wavenumbers are given in cm^{-1}). The stick spectrum (note: it has been scaled) obtained from calculation of intensities at each normal-mode frequency is overlaid by the spectrum where each peak has been convoluted with a Lorentzian with a width of 20 cm^{-1} . Peaks and intensities seen in the experimental spectrum^{48,49} are reproduced well by the calculations. The minor

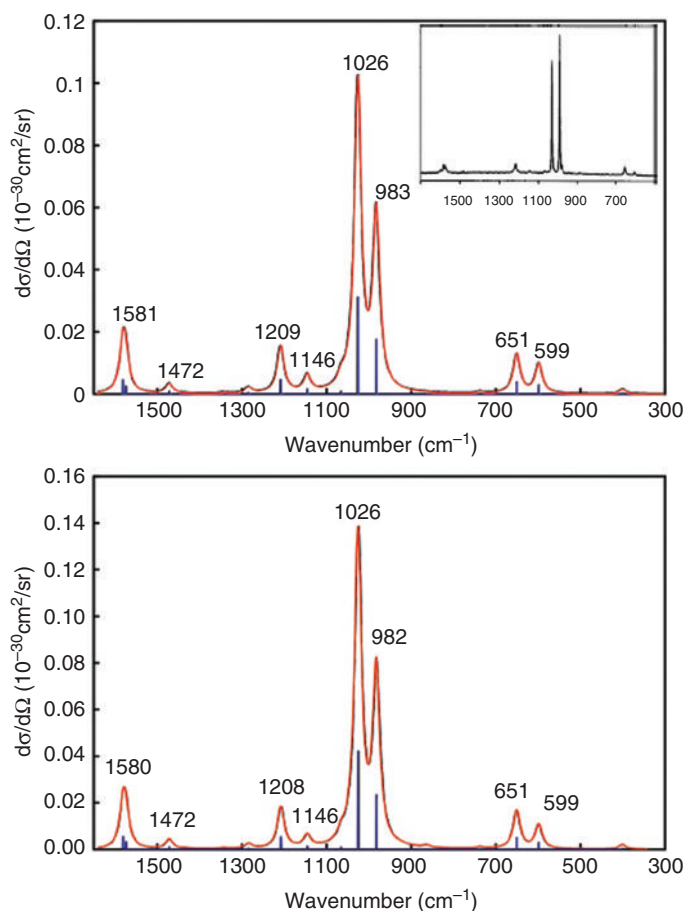


Fig. 15.1 (color online) Simulated normal Raman spectrum of pyridine at an incident wavelength of 514.5 nm using static (top) and frequency-dependent (bottom) polarizability derivatives. The spectrum is broadened by a Lorentzian with a full width at half-maximum of 20 cm⁻¹. The scale is for a broadened spectrum. Inset: Experimental spectrum from Golab et al.⁴⁸

peaks, however, are relatively more intense and the intensity ordering of the peaks at 983 and 1026 cm⁻¹ is the opposite of what is seen experimentally.

Adding a tetrahedral 20-silver-atom cluster allows the investigation of phenomena such as the chemical enhancements observed in SERS.³⁹ Though the pyridine-Ag₂₀ system has a large number of normal modes, only those in the range 300 to 1600 cm⁻¹, which correspond primarily to motions of the atoms in pyridine, are of interest. Figure 15.2 shows the optimized pyridine-Ag₂₀ complex geometry (where the pyridine is perpendicular to a face of the cluster and binds through the N atom to the Ag atom at the center of the face) and the calculated normal Raman spectrum for the structure with the cross section

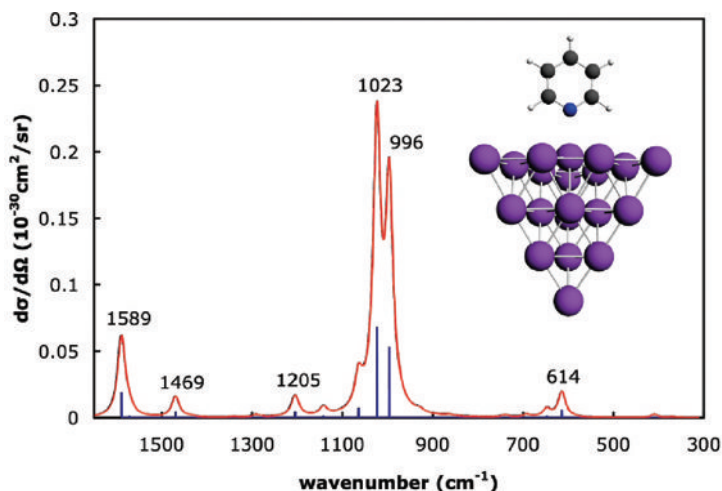


Fig. 15.2 (color online) Optimized geometry and simulated normal Raman spectrum of the surface pyridine- Ag_{20} complex at an incident wavelength of 514.5 nm using static polarizability derivatives. The spectrum is broadened by a Lorentzian with a full width at half-maximum of 20 cm^{-1} . The scale is for a broadened spectrum.

once again assuming an excitation wavelength of 514.5 nm. Comparing the intensities of the peaks in this spectrum to those in the pyridine spectrum, the chemical enhancement is approximately one order of magnitude. These results are comparable to results presented by Zhao et al.,³⁹ where it was also found that the corresponding spectra at wavelengths that are on-resonance for the Ag_{20} are enhanced by 10^5 or greater. This provides a model for understanding SERS.

Example 2: HyperRaman Spectrum of Pyridine Using the same geometry and frequencies for pyridine as in the normal Raman example, the hyperRaman spectra can also be simulated. The hyperpolarizability calculations at the displaced geometries were run with the SAOP model potential and an ET-QZ3P-polar basis set for all atoms. The orientationally averaged hyperRaman spectrum is shown in Fig. 15.3 [intensities are given in $\text{angstrom}^6/(\text{amu} \cdot \text{statvolt}^2)$].

The differential cross section is not calculated because the equation outlined is only applicable to Raman spectroscopy with a specific experimental setup.^{31,39} Although an effective excitation wavelength cannot be added into the spectrum, the relative intensities of the peaks should still be able to be compared to experimental spectra. In general, experimental hyperRaman spectra are rarely determined, due to the hyperRaman signal being even weaker than the already weak Raman signal. Luckily, for pyridine there are experimental measurements, which are matched rather well by the calculated spectrum.⁴⁹ Not all the peaks calculated can be verified due to noise in the experiment, but the relative intensities of those that are observed matches well.

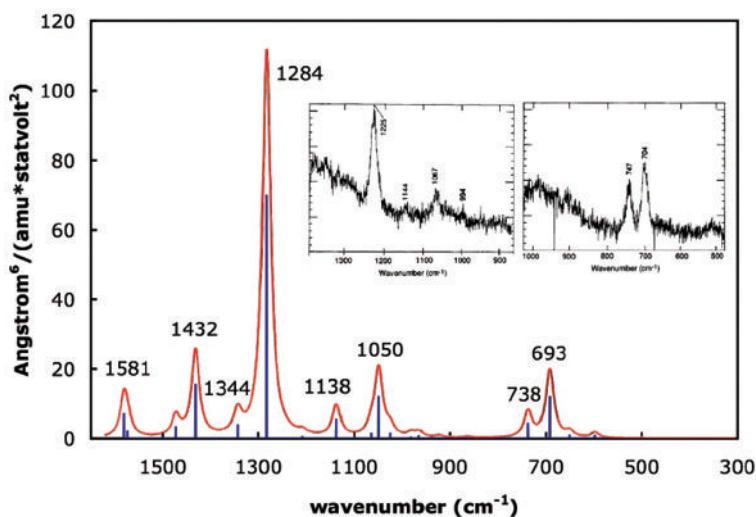


Fig. 15.3 (color online) Simulated normal hyperRaman spectrum of pyridine using static hyperpolarizability derivatives. The spectrum is broadened by a Lorentzian with a full width at half-maximum of 20 cm^{-1} . The scale is for a broadened spectrum. Inset: Experimental spectrum from Neddersen et al.⁴⁹

15.7 SURFACE-ENHANCED RAMAN AND HYPERRAMAN SPECTRA

The previous discussion is applicable to the Raman and hyperRaman spectra of any system where orientation averaging applies. The intensities are generally small, but they can be greatly enhanced by placing the molecules on a surface. Molecules adsorbed to a surface are generally restricted to a finite set of orientations relative to the surface, so the expressions based on orientation averaging no longer apply. If a specific orientation to the surface is assumed, the Raman intensities are proportional to the polarizability component perpendicular to the surface, as the plasmon-enhanced electromagnetic field near the surface is dominated by this component. For calculations which assume that the z -direction is normal to the surface, $\alpha_{zz}^2(\beta_{zzz}^2$ for hyperRaman intensities) will give the relative peak intensities.

As the interest is only in one of the components of the polarizability tensor of the molecule, the orientation of the molecule in the input becomes important. For example, to calculate the surface-enhanced Raman spectrum of a molecule standing straight up on a surface, the molecule should be appropriately oriented along the z -axis (as determined by its adsorption behavior) in all of the inputs. Also, the frequency calculation and polarizability calculations at the displaced coordinates would be performed as for a normal Raman calculation. The difference is that it is only necessary to calculate the polarizability derivative for the α_{zz} component.

15.8 APPLICATION OF TENSOR ROTATIONS TO RAMAN SPECTRA FOR SPECIFIC SURFACE ORIENTATIONS

In cases where the molecular orientation is uncertain, the comparison of simulated spectra with experiment can be used to infer the correct orientation. In this case the complete polarizability tensor needs to be determined for an arbitrary orientation, and then the polarizability is rotated to the desired orientation.

A second-order tensor (the polarizability tensor $[\alpha_{lm}]$) or third-order tensor (the hyperpolarizability tensor $[\beta_{ijk}]$) tensor can be rotated into a new coordinate frame by applying a rotation matrix $[R]$ and its inverse. The tensor in the new coordinate frame is given by

$$[\alpha_{ij}^*] = [R][\alpha_{lm}][R]^{-1} \quad (15.15)$$

Here R is an orthogonal matrix ($[R]^{-1} = [R]^T$) whose components r_{il} are the cosines of the angle between the i th axis of the original coordinate frame and the l th axis of the target coordinate frame:

$$r_{il} = \cos(i, l) \quad (15.16)$$

For surface-enhanced Raman, only the perpendicular component of the polarizability tensor is of interest. This can easily be calculated using the formula

$$\alpha_{ij}^* = \sum_{lm} r_{il} r_{jm} \alpha_{lm} \quad (15.17)$$

for polarizabilities, and

$$\beta_{ijk}^* = \sum_{lmn} r_{il} r_{jm} r_{kn} \beta_{lmn} \quad (15.18)$$

for hyperpolarizabilities. Of course, this work can be avoided completely if the molecular structure is defined in coordinates where one axis is along the surface normal.

Example 3: Surface-Enhanced Raman Spectrum of Pyridine If a normal Raman spectrum has already been calculated for the molecule of interest, it takes only minor modifications to obtain a surface-enhanced Raman spectrum. For the example molecule pyridine, the results of the polarizability calculations from the pyridine normal Raman example will be used. To model the surface-enhanced spectrum using only the polarizability derivatives of the molecule (so that plasmon enhancement effects are left out), an orientation relative to a fictional surface must be assumed. For pyridine, it will be assumed that the nitrogen atom binds to the surface and that the molecule stands straight up. This orientation places the C_2 -axis of pyridine along the surface normal.

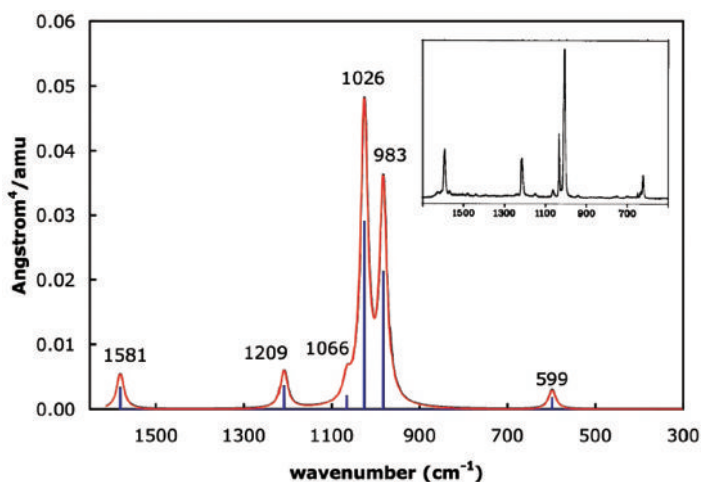


Fig. 15.4 (color online) SERS spectrum of pyridine standing straight up on a fictional surface. The spectrum is broadened by a Lorentzian with a full width at half-maximum of 20 cm^{-1} . The scale is for a broadened spectrum. Inset: Experimental spectrum from Golab et al.⁴⁸

The equilibrium structure of pyridine used for the polarizability calculations has its C_2 -axis along the z -axis. This means that the surface normal is along the z -axis in the calculations and that the squares of the derivatives of the α_{zz} components of the polarizabilities will be proportional to the experimental SERS intensities. The SERS spectrum for pyridine obtained in this manner is shown in Fig. 15.4 (intensities are given in $\text{angstrom}^4/\text{amu}$). Once again, the differential cross-section equation does not apply to what is being modeled. The surface is fictional, so only the relative intensities have any real significance.

The calculated spectrum compares well with experimental data,⁴⁸ except for the peak at 1026 cm^{-1} , which should be only slightly more intense than the peaks at 1581 and 1209 cm^{-1} . While the correct intensity ordering is not observed, the peak at 983 cm^{-1} does increase in intensity relative to the peak at 1026 cm^{-1} going from the nonresonant Raman spectrum to the SERS spectrum, which is seen experimentally.⁴⁸ It may be possible that the differences observed are occurring because the orientation of pyridine relative to the surface is not what has been assumed in the calculations. A rigorous study would consider other orientations and possibly average over a range of orientations to see if better agreement can be achieved.

15.9 RESONANCE RAMAN

Another phenomenon used to increase Raman intensities in experiments is the resonance Raman effect. Resonance Raman involves using incident light with an energy that matches the energy needed to put the molecule in an electronically

excited state.⁵⁰ In the expression for the Kohn–Sham response function, this would mean $\omega = \varepsilon_i - \varepsilon_m$, which leads to division by zero in the response function described above.¹⁹ The zero occurs because it was assumed that the excited state has an infinite lifetime. However, the excited states of molecules in a condensed phase always have a significant width, due to dephasing of the excited state through interaction with the environment.

The AORESPONSE functionality in ADF allows calculation of polarizabilities at resonant wavelengths by adding in an effective lifetime by way of a damping parameter in the response function. This is not a perfect fix, though, because it assumes that all excited states have the same lifetime, which is generally not true. Damping parameters are best obtained by fitting experimental absorption data for the molecule of interest.¹⁹ If there are no available data, it is possible to use the value for a similar molecule if the short-time approximation is valid. A value of 0.004 atomic unit (0.1 eV) has been found to be reasonable for many large organics, as well as pyridine interacting with silver clusters.³⁹ In the AORESPONSE block, the keyword LIFETIME followed by the lifetime in atomic units will tell the program to account for the excited-state lifetime provided. With a lifetime specified, ADF will be able to calculate both the real and imaginary parts of the polarizability. The imaginary polarizabilities should be treated like their real counterparts until the scattering factors are calculated. At that point, the real and imaginary scattering factors can be summed to give the total scattering factor.

15.10 DETERMINATION OF RESONANT WAVELENGTH

Using the AORESPONSE lifetime functionality, it is possible to calculate polarizabilities for the displaced structures at resonant wavelengths, but it is important to have an idea of where the resonance is located before doing the calculations. Experimental resonance Raman literature or absorption maximum data for the system provide a good place to start. Using the optimized geometry for the system, polarizability calculations should then be run for a range of incident light frequencies close to where the resonant frequency is believed to be. The polarizability calculations should also be using the finite lifetime that was found to be appropriate for the system. The absorption maximum for the system occurs where the imaginary polarizability has its maximum and is an appropriate frequency to choose for the resonance Raman calculations.³⁹

Of course, another way to determine the excitation energies of the system for a given combination of basis set, XC potential, and XC kernel would simply be to run a calculation of the excitation spectrum using TDDFT. This can be accomplished using the EXCITATIONS keyword in ADF. The equivalence of the two approaches, $\text{Im}[\alpha]$ versus TDDFT excitation spectra, was demonstrated explicitly by Jensen et al.,¹⁹ Devarajan et al.,⁴⁵ and Krykunov et al.⁵¹ for the closely related case of optical rotatory dispersion versus TDDFT circular dichroism spectra.

Example 4: Resonance Raman Spectrum of Uracil To detail the steps necessary to calculate a resonance Raman spectrum, the molecule uracil will be used

as an example. An excitation in uracil that can be used to study RRS corresponds to the lowest-energy $\pi \rightarrow \pi^*$ transition.²⁰ This excitation is found experimentally at 5.08 eV (244 nm) in the gas phase and 4.77 eV (260 nm) in the aqueous phase.⁵² To discern what excitation energy to use in the calculation of resonance polarizabilities, the real and imaginary polarizabilities of the equilibrium geometry were calculated at discrete points between incident light wavelengths of 240 and 280 nm. For all the calculations in this example, a value of $\Gamma = 0.004$ a.u. was chosen for the damping parameter, the BP86 functional was used, and all atoms were treated with a TZP basis set. The real and imaginary polarizabilities as a function of the wavelength of the incident light are shown in Fig. 15.5.

A maximum is seen in the imaginary polarizability of the system at 263 nm. For the polarizability derivative calculations, it is reasonable to use 263 nm for the incident light wavelength in the input to the displaced geometry calculations, or a nearby wavelength that was used in experiments. Using an incident light wavelength of 263 nm, the spectrum displayed in Fig. 15.6 can be obtained. The spectrum assumes an average over all molecular orientations, and the stick spectrum has been broadened by a Lorentzian as in the pyridine nonresonant Raman example.

Close agreement with experiment is seen except for the peak at 1737 cm^{-1} , which is much too intense in the calculations, and the peaks at 1448 and 1353 cm^{-1} which are seen as a single peak at 1401 cm^{-1} in experiments. The second issue appears to be due to solvent effects since adding two water molecules to the calculations shifts the two peaks together around 1400 cm^{-1} .²⁰ This does not, however, correct the peak at 1737 cm^{-1} . This error probably arises due to Fermi resonance (not included in the calculations) between the

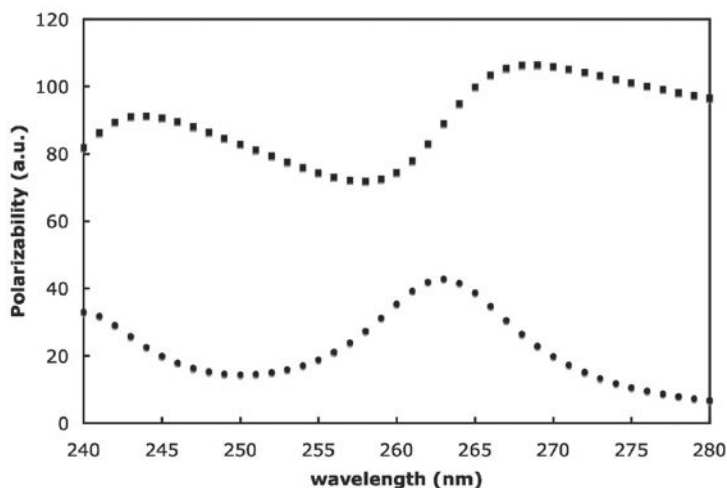


Fig. 15.5 Real (squares) and imaginary (circles) polarizabilities of uracil as a function of the wavelength of incident light between 240 and 280 nm.

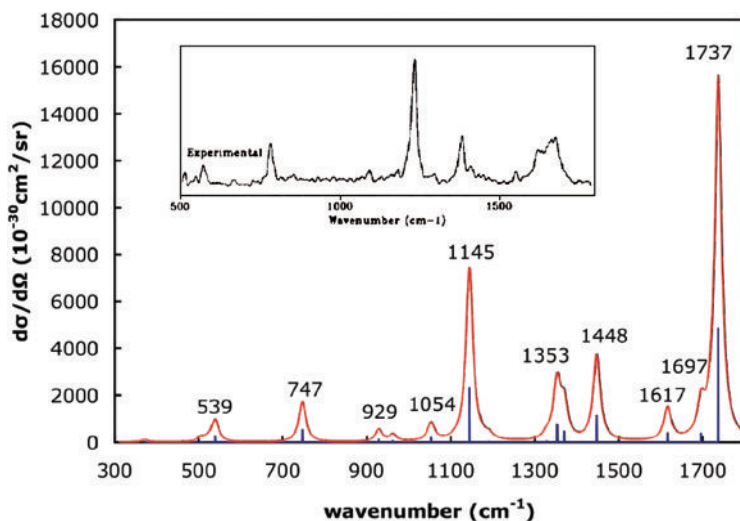


Fig. 15.6 (color online) Simulated resonance Raman spectrum of uracil at an incident wavelength of 263 nm. The spectrum is broadened by a Lorentzian with a full width at half-maximum of 20 cm^{-1} . The scale is for a broadened spectrum. Inset: Experimental spectrum from Jensen et al.²⁰

C—O and N—H bending modes.^{53,54} Fermi resonances and overtones are not accounted for in the harmonic approximation that has been made in the calculation of the vibrations.²⁰ Raman spectra calculated for molecules in which these processes play a visible role will not accurately reproduce all the peak intensities.

15.11 SUMMARY

In this chapter we have provided a detailed discussion of the calculation of Raman and hyperRaman spectra for large molecules and molecules interacting with metal clusters using the ADF computer program and time-dependent density functional theory. Both static- and frequency-dependent Raman spectra are considered, and the frequency-dependent spectra include the possibility of excitation on resonance through the input of an empirical width factor in the resonant optical response. In addition, we describe the calculation of spectra for specific molecular orientations and an average over orientations.

Specific examples are presented for pyridine in vacuum, for pyridine interacting with a silver cluster, and for pyridine oriented on a fictitious surface to mimic orientation effects that can occur in SERS. In addition, we examined the resonance polarizability and resonance Raman spectrum of uracil as an example of a resonance Raman calculation.

Although these examples reveal important capabilities that are now available using TDDFT, there remain important limitations in the use of this method for large systems. The current technology can handle up to 100 to 200 atoms but becomes impractical for much larger systems. Even for 100 to 200 atoms, it can be quite challenging to calculate spectra for a lot of normal modes. In addition, the excited-state widths are purely empirical factors in the current version of the code and are assumed not to depend on the nature of the excited state. Finally, we note that the models of SERS which replace the metal particles by silver clusters that have less than 100 atoms make important approximations whose validity is still uncertain. Plasmon resonances are size dependent for small clusters, so the resonance wavelengths do not match the observations, and the cluster-size dependence of the widths is unknown. In addition, the behavior of the electromagnetic fields around the cluster are unlikely to match the fields associated with large particles, so the field enhancements that lead to SERS are not likely to be described accurately.

Supporting Information

Supporting information including atomic coordinates and vibrational frequencies for all example species may be found on the book Web site.

Acknowledgments

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REFERENCES

1. Camden, J. P.; Dieringer, J. A.; Zhao, J.; Van Duyne, R. P. *Acc. Chem. Res.* **2008**, *41*, 1653.
2. LaFratta, C. N.; Walt, D. R. *Chem. Rev.* **2008**, *108*, 614.
3. Jain, P. K.; Huang, X.; El-Sayed, I. H.; El-Sayed, M. A. *Plasmonics* **2007**, *2*, 107.
4. Lal, S.; Link, S.; Halas, N. J. *Nat. Photon.* **2007**, *1*, 641.
5. Murphy, C. J.; Gole, A. M.; Hunyadi, S. E.; Stone, J. W.; Sisco, P. N.; Alkilany, A.; Kinard, B. E.; Hankins, P. *Chem. Commun.* **2008**, 544.
6. Willets, K. A.; Van Duyne, R. P. *Annu. Rev. Phys. Chem.* **2007**, *58*, 267.
7. Kneipp, J.; Kneipp, H.; Kneipp, K. *Chem. Soc. Rev.* **2008**, *37*, 1052.
8. Kelley, A. M. *J. Phys. Chem. A* **2008**, *112*, 11975.
9. Yang, W. H.; Schatz, G. C. *J. Chem. Phys.* **1992**, *97*, 3831.
10. Yang, W.-H.; Hultheen, J.; Schatz, G. C.; Van Duyne, R. P. *J. Chem. Phys.* **1996**, *104*, 4313.
11. Jeanmaire, D. L.; Van Duyne, R. P. *J. Electroanal. Chem.* **1977**, *84*, 1.

12. Hulteen, J. C.; Young, M. A.; Van Duyne, R. P. *Langmuir* **2006**, *22*, 10354.
13. Schatz, G. C. *Acc. Chem. Res.* **1984**, *17*, 370.
14. Moskovits, M. *Rev. Mod. Phys.* **1985**, *57*, 783.
15. Schatz, G. C.; Ratner, M. A. *Quantum Mechanics in Chemistry*, Dover, Mineola, NY, 2002.
16. van Gisbergen, S. J. A.; Snijders, J. G.; Baerends, E. J. *Comput. Phys. Commun.* **1999**, *118*, 119.
17. Velde, G. T.; Bickelhaupt, F. M.; Baerends, E. J.; Guerra, C. F.; Van Gisbergen, S. J. A.; Snijders, J. G.; Ziegler, T. *J. Comput. Chem.* **2001**, *22*, 931.
18. ADF2008.01; SCM: *Theoretical Chemistry*, Vrije Universiteit, Amsterdam, <http://www.scm.com>, click on "Theoretical Chemistry."
19. Jensen, L.; Autschbach, J.; Schatz, G. C. *J. Chem. Phys.* **2005**, *122*, 224115/1.
20. Jensen, L.; Zhao, L. L.; Autschbach, J.; Schatz, G. C. *J. Chem. Phys.* **2005**, *123*, 174110/1.
21. Mort, B. C.; Autschbach, J. *J. Phys. Chem. A* **2006**, *110*, 11381.
22. van Gisbergen, S.; Snijders, J. G.; Baerends, E. J. *J. Chem. Phys.* **1998**, *109*, 10644.
23. Ye, A.; Patchkovskii, S.; Autschbach, J. *J. Chem. Phys.* **2007**, *127*, 074104.
24. Ye, A.; Autschbach, J. *J. Chem. Phys.* **2006**, *125*, 234101.
25. Jensen, L.; Aikens, C. M.; Schatz, G. C. *Chem. Soc. Rev.* **2008**, *37*, 1061.
26. Jensen, L.; Zhao, L. L.; Schatz, G. C. *J. Phys. Chem. C* **2007**, *111*, 4756.
27. Aikens, C. M.; Schatz, G. C. *J. Phys. Chem. A* **2006**, *110*, 13317.
28. Masiello, D. J.; Schatz, G. C. *Phys. Rev. A* **2008**, *78*, 042505/1.
29. Bernath, P. F. *Spectra of Atoms and Molecules*, 2nd ed., Oxford University Press, New York, 2005.
30. Mort, B. C.; Autschbach, J. *J. Phys. Chem. A* **2005**, *109*, 8617.
31. Reiher, M.; Neugebauer, J.; Hess, B. A. *Z. Phys. Chem.* **2003**, *217*, 91.
32. Krykunov, M.; Autschbach, J. *J. Chem. Phys.* **2005**, *123*, 114103.
33. Krykunov, M.; Autschbach, J. *J. Chem. Phys.* **2007**, *126*, 024101.
34. Autschbach, J.; Jensen, L.; Schatz, G. C.; Tse, Y. C. E.; Krykunov, M. *J. Phys. Chem. A* **2006**, *110*, 2461.
35. van Gisbergen, S. J. A.; Snijders, J. G.; Baerends, E. J. *J. Chem. Phys.* **1995**, *103*, 9347.
36. Pulay, P. Analytical derivative techniques and the calculation of vibrational spectra. In *Modern Electronic Structure Theory*, Part II, Vol. 2, Yarkony, D. R., Ed., World Scientific, Singapore, 1995, p. 1191.
37. Pople, J. A.; Raghavachari, K.; Schlegel, H. B.; Binkley, J. S. *Int. J. Quantum Chem.* **1979**, *S13*, 225.
38. Neugebauer, J.; Reiher, M.; Kind, C.; Hess, B. A. *J. Comput. Chem.* **2002**, *23*, 895.
39. Zhao, L.; Jensen, L.; Schatz, G. C. *J. Am. Chem. Soc.* **2006**, *128*, 2911.
40. Kanis, D. R.; Ratner, M. A.; Marks, T. J. *Chem. Rev.* **1994**, *94*, 195.
41. Califano, S. *Vibrational States*, Wiley, New York, 1976.
42. Aikens, C. M.; Li, S. Z.; Schatz, G. C. *J. Phys. Chem. C* **2008**, *112*, 11272.
43. van Lenthe, E.; Baerends, E. J.; Snijders, J. G. *J. Chem. Phys.* **1993**, *99*, 4597.
44. van Lenthe, E.; Baerends, E. J.; Snijders, J. G. *J. Chem. Phys.* **1994**, *101*, 9783.

45. Devarajan, A.; Gaenko, A.; Autschbach, J. *J. Chem. Phys.* **2009**, *130*, 194102.
46. Gritsenko, O. V.; Schipper, P. R. T.; Baerends, E. J. *Chem. Phys. Lett.* **1999**, *302*, 199.
47. Schipper, P. R. T.; Gritsenko, O. V.; van Gisbergen, S. J. A.; Baerends, E. J. *J. Chem. Phys.* **2000**, *112*, 1344.
48. Golab, J. T.; Sprague, J. R.; Carron, K. T.; Schatz, G. C.; Van Duyne, R. P. *J. Chem. Phys.* **1988**, *88*, 7942.
49. Neddersen, J. P.; Mounter, S. A.; Bostick, J. M.; Johnson, C. K. *J. Chem. Phys.* **1989**, *90*, 4719.
50. Albrecht, A. C. *J. Chem. Phys.* **1961**, *34*, 1476.
51. Krykunov, M.; Kundrat, M. D.; Autschbach, J. *J. Chem. Phys.* **2006**, *125*, 194110.
52. Clark, L. B.; Peschel, G. G.; Tinoco, I. *J. Phys. Chem.* **1965**, *69*, 3615.
53. Peticolas, W. L.; Rush, T. *J. Comput. Chem.* **1995**, *16*, 1261.
54. Szczesniak, M.; Nowak, M. J.; Rostkowska, H.; Szczepaniak, K.; Person, W. B.; Shugar, D. *J. Am. Chem. Soc.* **1983**, *105*, 5969.

16 Metal Surfaces and Interfaces: Properties from Density Functional Theory

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In this chapter we describe comprehensive theoretical studies of metallic surfaces and interfaces using density functional theory (DFT) calculations. First, we provide a general introduction and background, then describe the methodology used and validation studies performed. Calculations performed on Fe(100), Fe(110), and Fe(111) surfaces to investigate their structure, energetics, electronic, magnetic, and adsorption properties are then discussed. Interfaces between these surfaces and, specifically, adhesion and the associated electronic and magnetic properties are then presented. Adhesion is studied between the surfaces in match (in registry) and mismatch (out of registry), ideal and relaxed, and clean and sulfur-contaminated states. Finally, we provide summaries, conclusions, and suggestions for future work.

16.1 BACKGROUND, GOALS, AND OUTLINE

Iron surfaces have been of interest to both pure and applied sciences since the Iron Age. Despite their crucial importance for many industries,^{1–3} from crude heavy industry to refined electronics, there is a gap in the fundamental understanding of many important properties of iron surfaces, such as magnetic properties and adhesion, which may slow their application in new and innovative technologies. This gap in understanding arises partly because of the inherent difficulty of studying the material both experimentally, due to its high susceptibility to corrosion,⁴ and theoretically due to its transition metal nature and hence complex electronic properties.

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Specifically, adhesion between metallic iron surfaces plays an important role in many industrial processes.^{5,6} For example, in the extraction of metallic iron (Fe) via the fluidized-bed iron ore reduction process of powdered ores, the process often suffers from the buildup of deposits, known as *accretions*, in various parts of the reactors, and component particles may strongly adhere, forming large clumps resulting in defluidization of the bed.⁷ As iron forms a major constituent of the accretions, a fundamental understanding of the mechanism by which the metal particles adhere, as well as identification of the species capable of preventing severe adhesion, is of vital importance.

Previous investigations of Fe surfaces and interfaces have looked at a number of their properties, including structural and magnetic features. However, generally speaking, previous studies have not provided a systematic fundamental description, particularly of the dynamic properties associated with thermal and impurity-induced transformations and the effects of the material which are crucial for the ability to design and manipulate its properties at the macro- and nanoscale.

Here we present an account of our theoretical work on Fe, which includes new results and those obtained previously. Specifically, after describing the methodology in Section 16.2, in Section 16.3.1 we review results on the computed relaxations and energies of the three low index surfaces—(100), (110), and (111)—of body-centered cubic (bcc) Fe and compare the computational results with experimental observations. In Section 16.3.2 we describe new results on the magnetic properties of the Fe(100), Fe(110), and Fe(111) surfaces, such as changes in the magnetic and electronic properties after relaxation and the layer-resolved magnetic moment values, as well as up- and down-spin-resolved density of states. In Sections 16.3.3 and 16.3.4 we present results on the adsorption of atomic S on the atop, bridge, and hollow sites of Fe(100) and Fe(110) surfaces at $1/2$ and $1/4$ monolayer (ML) coverages. The most stable site, the effects of S adsorption on surface reconstruction, and magnetic and electronic properties are considered. A summary of the effect of higher S coverages on these properties is also presented. In Sections 16.3.5 and 16.3.6 we discuss our calculations on the dynamic behavior of S and H₂S adsorbed on Fe(100) and Fe(110), including *ab initio* molecular dynamics (AIMD) simulations to examine the effect of elevated temperatures.

In Section 16.4.1 we review our studies on adhesion between clean, bulk-terminated bcc Fe(100), Fe(110), and Fe(111) matched and mismatched interfaces. The parameters obtained from this work allowed the behavior of the work of separation (W_{sep}) to be determined and examined. In Section 16.4.2 we examine newly obtained results on the relationship between magnetic and electronic properties and adhesion of the Fe(100), Fe(110), and Fe(111) surfaces in match and mismatch. In Section 16.4.3 we discuss the avalanche effect in adhesion between Fe(100) surfaces, in match and mismatch, where the role of model constraints has been focused on specifically. In Section 16.4.4 we give a brief summary of our study of the effect of adsorbed S on the adhesion of Fe(100) and Fe(110) surfaces in the atop, bridge, and hollow sites at $1/2$ and $1/4$ ML coverages

in match and mismatch interfaces. The effect of adsorbed S on the charge-density distribution and magnetic properties of the interface are also examined and related to the interfacial geometry. Also discussed is the effect of relaxation of the interfaces and different coverages of S at the interface. We conclude this chapter with a summary and outline of future work in Section 16.5.

16.2 METHODOLOGY

Density functional theory (see Chapter 1) is a technique that can provide fundamental understanding of the structural, electronic, magnetic, and adhesion properties of materials and their surfaces and interfaces at the electronic level.^{8–16} Theoretically, it is possible to construct a model interfacial system of any two surfaces with any degree of lattice match or mismatch and to make arbitrary alterations to the surfaces: for example, to introduce atomic and molecular impurities and then systematically investigate their effects on the system's properties. A range of atomic simulation methods, including DFT, have already been applied successfully to the investigation of various metallic and ceramic interfaces (e.g., MgO/Ag,¹⁷ Mo/MoSi₂,¹⁸ NiAl/Cr,¹⁹ and Fe^{20,21}) and on the effects of impurities (S, C, N, O, P, etc.) on adhesion between surfaces.^{18,22–29}

A fairly comprehensive review of applications of various theoretical simulation techniques to study material interfaces can be found in the literature by Finnis³⁰ and is beyond the scope of this publication.

We have developed a number of methods using classical empirical potentials based on the embedded-atom method (EAM) to study Fe surfaces and interfaces^{31–34}; the advantage of this approach is that it is significantly less computationally expensive and it is possible to estimate the system free energies for much larger models and hence simulate a wider variety of surface structures and defects. However, in this chapter we describe our investigations of the surface and interfaces using the DFT approach.

16.2.1 Choice and Validation of the Computational

Method: Bulk Iron Studies

All calculations were performed using the Vienna ab initio simulation package (VASP),^{35–37} which performs fully self-consistent DFT calculations to solve the Kohn–Sham equations³⁸ within the local spin density approximation (LSDA) using the functional of Perdew and Zunger³⁹ (PZ) or the generalized-gradient spin approximation (GGSA), using the functional of Perdew and Wang⁴⁰ (PW91). The electronic wavefunctions are expanded as linear combinations of plane waves (see Chapter 3), truncated to include only plane waves with kinetic energies below a prescribed cutoff energy, E_{cut} . Due to the delocalized nature of conduction electrons in metals, a delocalized plane-wave basis provides a good representation of metallic systems. The core electrons are replaced by ultrasoft pseudopotentials by Vanderbilt,⁴¹ and k -space sampling was performed using the scheme of Monkhorst and Pack.⁴²

TABLE 16.1 Calculated Structure and Properties of Bulk Fe, Using Both LSDA and GGSA Functionals^a

Property	LSDA	GGSA	Experimental
Lattice parameter, a_0 (Å)	2.767 (−3.5%)	2.869 (+0.11%)	2.866
Bulk modulus, B (GPa)	195 (+16%)	140 (−16%)	168
Magnetic moment/atom (μ_B)	1.98 (−11%)	2.37 (+6.8%)	2.22

Source: Ref. 20.

^aThe percent deviation from known experimental values⁴³ is shown in parentheses.

The bulk, surfaces, and interfaces of Fe are modeled using the supercell approach, where periodic boundary conditions are applied to the central supercell so that it is reproduced periodically throughout space. Tests were performed on the bulk bcc phase of Fe, using both LSDA and GGSA functionals as well as different E_{cut} and k -space sampling values, to ensure that the bulk properties were converged.²⁰ The optimized bulk structure was then used to create different surface and interface models.

The total energy, E_{tot} , and lattice parameter, a_0 , of bulk bcc Fe were calculated using different plane-wave cutoff energy values and k -point sampling sets to ensure the reliability of the calculations. It was found that an E_{cut} of 300 eV and k -point mesh of $12 \times 12 \times 12$ gave convergence of E_{tot} and a_0 to 10^{-4} eV/atom and 0.001 Å, respectively. The lattice parameter, bulk modulus, and magnetic moment values calculated using these converged parameters with both LSDA and GGSA functionals are presented in Table 16.1, along with the experimental values.⁴³

The values calculated using GGSA were found to give better agreement with the known experimental values than those calculated with LSDA. In particular, the LSDA functional was shown to predict the face-centered-cubic (fcc) Fe phase to be more energetically stable at 0 K than the bcc phase, while the GGSA functional correctly predicted the order of stability, consistent with previous findings (see, e.g., Jansen and Peng⁴⁴).

16.2.2 Surface and Interface Models

The relaxed-bulk bcc Fe cell (with the lattice parameter determined using the GGSA PW91 functional) was cut along the (100), (110), and (111) Miller planes to form the three low-index Fe surface models (see Fig. 16.1). These models also served as our interface models. Using the supercell approach, the interfacial separation distance was defined by the vacuum layer thickness between image cells adjacent to each other in the z -direction (Fig. 16.1).

Interfaces were modeled in two different orientations corresponding to a perfect lattice match between the two surfaces (i.e., epitaxial interfaces) and maximum lattice mismatch (i.e., where surface atoms of the two surfaces share the same coordinates in the x,y -plane within the supercell). An even number of

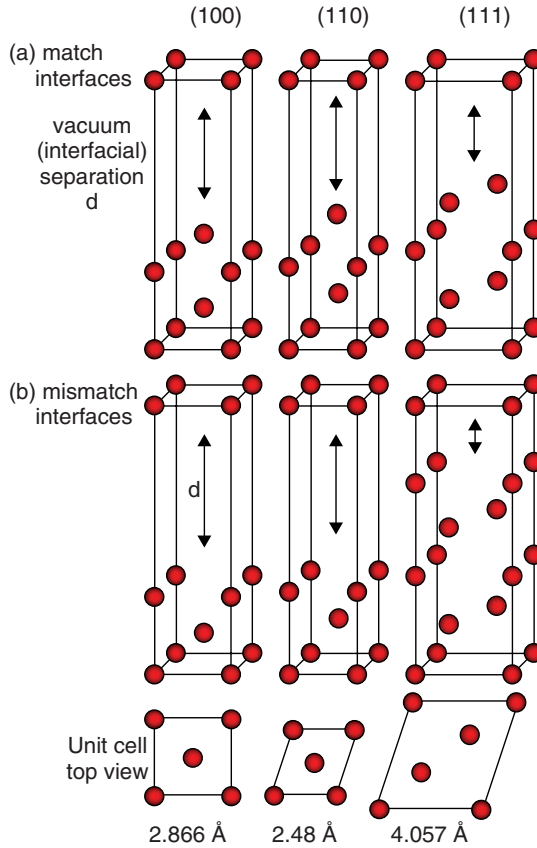


Fig. 16.1 (color online) Surface/interface models: (a) (100), (110), and (111) match interfaces; (b) (100), (110), and (111) mismatch interfaces. Profiles of the surface unit cells are displayed below each surface supercell model. The interfacial separation, d , is indicated.

atomic layers was used to model the match interfaces, while an odd number of layers was used to model the mismatch interfaces (Fig. 16.1).

To determine the number of layers in the surface model required for convergence, the surface energies of the unrelaxed surfaces (E_{surf}) were calculated as a function of slab thickness, using a vacuum layer separation of 10 Å. E_{surf} was calculated using the expression:

$$E_{\text{surf}} = \frac{E_{\text{tot}}(\text{slab}) - nE_{\text{tot}}(\text{bulk})}{2A} \quad (16.1)$$

where $E_{\text{tot}}(\text{slab})$ and $E_{\text{tot}}(\text{bulk})$ are the total energies of the slab and bulk, respectively; n is the number of bcc Fe unit cells present in the slab; and A is the cross-sectional surface area of the slab.

All surface and interface calculations also used the PW91 functional and GGSA approach. Further specific computational details for each case are given in relevant sections, as appropriate.

16.2.3 Interfacial Adhesion: Work of Separation and UBER

Most of the calculations we report here (except those described in Section 16.4.3) have been performed for interfaces between ideal Fe surfaces; namely, we calculate the *work of separation* (W_{sep}). The concept of the work of separation versus the work of adhesion has been introduced by Finnis³⁰ and was discussed by us previously in detail.²⁰

In terms of the surface and interfacial excess free energies of the materials, the ideal W_{sep} is given by the Dupre equation⁴⁵:

$$W_{\text{sep}} = \sigma'_1 + \sigma'_2 - \sigma_{12} \quad (16.2)$$

where σ'_1 and σ'_2 are ideal surface free energies of materials 1 and 2, and σ_{12} is the interfacial free energy. This quantity should be distinguished from the work of adhesion, which is defined as the energy required to separate two surfaces from the equilibrium separation to infinity, taking full account of all relaxation and diffusion processes.

W_{sep} can be calculated directly from the molecular simulation of isolated surfaces and of these surfaces when brought into close contact to form an interface.³⁰ By calculating the single-point energy at discrete separation distances, d , one can obtain an interaction energy curve $E_{\text{ad}}(d)$:

$$E_{\text{ad}}(d) = \frac{E(d) - E(\infty)}{A} \quad (16.3)$$

where $E(d)$ is the total computed energy at separation distance d , $E(\infty)$ is the total energy at infinite separation, and A is the cross-sectional area of interaction. The well depth of this curve, E_0 , is equivalent to the W_{sep} . The adhesion curves calculated can be fitted to the universal binding-energy relation (UBER),⁴⁶ which is given by a Rydberg-type function adapted for the case of interfacial adhesion and is considered to give a valid representation of binding in situations where bonding results mainly from overlap of the tails of wavefunctions⁴⁷:

$$E'_{\text{ad}}(d) = -E_0(1.0 + d^*)e^{d^*} \quad (16.4)$$

where $E'_{\text{ad}}(d)$ is the fitted adhesion interaction energy, $d^* = (d - d_0)/l$ (scaled distance), E_0 is the depth of the adhesion energy well at equilibrium interfacial separation (equivalent to the work of separation W_{sep}), d_0 is the interfacial separation at the adhesion energy minimum, and l is the scale factor, which for transition metals may be interpreted as the surface scaling length, and sets the approximate scale for the distance over which electronic forces can act. The value of E_0 represents the work of separation for a particular interface.

16.2.4 Calculation of Binding Energies for Surface Impurities

We have computed the binding energies of sulfur impurity adsorbed in various adsorption sites by the equation



The binding energy is the difference in total energy of the products minus the reactants:

$$\begin{aligned} \text{BE} &= E_{\text{tot}}(\text{products}) - E_{\text{tot}}(\text{reactants}) \\ &= E_{\text{tot}}(\text{S} \cdot \text{Fe}) - [E_{\text{tot}}(\text{S}) + E_{\text{tot}}(\text{Fe})] \end{aligned} \quad (16.6)$$

where $E_{\text{tot}}(\text{S})$ is the total energy of an isolated S atom and $E_{\text{tot}}(\text{S} \cdot \text{Fe})$ and $E_{\text{tot}}(\text{Fe})$ are the total energies of the relaxed clean Fe surface and S-adsorbed Fe(110) surface, respectively.

16.3 STRUCTURE AND PROPERTIES OF IRON SURFACES

16.3.1 Structural Relaxation and Stability of Fe(100), Fe(110), and Fe(111) Surfaces

16.3.1.1 Introduction and Previous Studies Relaxation of metal surfaces after cleavage from the bulk is a well-known phenomenon. The reduction in atomic interactions perpendicular to the surface can cause the topmost surface layers to contract toward the bulk or expand away from it. In addition, movements of the surface atoms within the plane of the surface can lead to surface reconstructions.

Some previous theoretical findings have differed from those obtained experimentally.⁴⁸ For the low-index surfaces of Fe in particular, there is also some conflict; however, it has been shown that the surfaces do not reconstruct, while they do relax.^{49–51}

We have already reviewed the findings of the experimental studies,⁵² which used low-energy electron diffraction (LEED)^{49–51,53} and medium-energy ion scattering (MEIS)^{54–56} to examine relaxation of the Fe(100), Fe(110), and Fe(111) surfaces and find that there is some conflict between the reported surface relaxations.

The relaxations that occur from cleavage of a bulk structure to yield a surface result from the drive to minimize the energy of the surface. The measurement of surface energy values experimentally, however, can be very difficult to perform for a number of reasons, one being the difficulty to control the presence or absence of contaminants. In calculations, the state of the surface and level of impurities can be examined systematically. Theoretical studies that have determined surface energy values of the low-index Fe surfaces have included mainly molecular mechanics (MM) techniques,^{34,57–68} with fewer studies using a quantum mechanical (QM) approach.^{20,69,70} In particular, the latter studies did not

take into account the effect of surface relaxation on the calculated surface energy values. Furthermore, there were conflicting trends obtained for the stability of the three low-index surfaces. Hence, we performed DFT calculations⁵² to model these properties and to try to clarify the situation.

16.3.1.2 Surface Models The Fe surfaces were modeled using the supercell approximation as described in Section 16.2.2. All models used a $[1 \times 1]$ crystal unit cell; however, a number of $[2 \times 2]$ unit cell slab calculations were performed as well in order to test for convergence. k -Space sampling was performed using the scheme of Monkhorst and Pack.⁴² A k -point mesh of $12 \times 12 \times 1$ for the $[1 \times 1]$ unit cells and a $6 \times 6 \times 1$ mesh for the $[2 \times 2]$ unit cell calculations were employed. A lattice constant value of 2.869 Å was used, as this was the optimized value obtained in our previous study²⁰ of bulk bcc Fe using the same computational parameters. Models with different numbers of layers (ranging from 7 to 17 layers) were constructed to determine the size of slab needed to converge the surface geometry and energy values. Either one middle layer (for an odd-number layered model) or two middle layers (for an even-number layered model) were fixed, to provide a reference point for comparing the relaxed Fe positions, while all other atoms were allowed to relax in the x -, y -, and z -directions. The models selected are described in Section 16.3.1.3.

16.3.1.3 Surface Relaxation Our calculations of the relaxed surface models⁵² indicated that only relaxations perpendicular to the surface (in the z -direction) occurred and that these surfaces do not reconstruct (showing no atomic displacements in the x, y -directions), in agreement with experimental studies.^{49–51} For each layer in our model we calculated the values of δz_n , which is a measure of the distance the n^{th} layer of the surface moves as a percentage of the interlayer spacing. A positive value indicates an expansion or upwards movement (towards the surface), whereas a negative value indicates a contraction or downwards displacement. The relaxation values for the (100), (110) and (111) surfaces were found to be converged by 0.01, 0.0005 and 0.005 Å for a 13, 7 and 12 layer model, respectively. The $[2 \times 2]$ surface models showed close agreement with the $[1 \times 1]$ slabs. These models are employed in our further work.

The relaxation values obtained (Table 16.2) showed good agreement with experiment, with the open surface relaxing more, in the order of (110) < (100) < (111). The magnitude of the relaxations was found to be smaller as the bulk layers were approached.

For all surfaces, the topmost layer contracted toward the bulk, with the (111) surface showing the largest relaxation, followed by the (100), then the (110) surface. The relaxation of the (110) surface layer was essentially zero, indicating that it is basically bulk cleaved. The second layer was found to relax outward for the (100) and (110) surfaces, while it expanded away from the bulk for the (111) surface. Again, the relaxations were largest for the (111) surface and smallest for the (110) surface.

TABLE 16.2 Calculated Relaxation Measurements, δz_n ($n = 1, 2, \dots$) as a Percentage of the Bulk Interlayer Spacing for the First Five Layers of Fe(100), Fe(110), and Fe(111)^a

	Surface Relaxation (%)					Surface Energy (J m^{-2})	
	δz_1	δz_2	δz_3	δz_4	δz_5	Relaxed	Unrelaxed
(100)	-1.89	+2.59	+0.21	-0.56	-0.14	2.29	2.32
(110)	-0.13	+0.197	-0.06	—	—	2.27	2.27
(111)	-13.3	-3.6	+13.3	-1.2	+0.35	2.52	2.62

Source: Ref. 52.

^aCalculated surface energy values.

For the (100) and (110) surfaces, the magnitude of our calculated surface relaxations agreed well with the experimentally determined values^{50,51,55} and fell within the error of these measurements. For the (111) surface, there was a discrepancy between the relaxation values measured experimentally using MEIS^{54,56} and LEED.^{49,53} The MEIS measurements^{54,56} indicated that the first layer contracted and the second expanded, whereas the LEED study⁵³ indicated that the first two layers contracted and the third expanded. Our calculations agreed with the LEED measurements. The magnitude of the surface relaxations can be related to the openness of the surface, with the more open (111) surface showing larger relaxation and the most close-packed (110) surface being almost bulk cleaved.

16.3.1.4 Surface Energy The calculated surface energy values (Table 16.2) for all three surfaces was found to be converged to at least 0.01 J m^{-2} by nine layers with the unrelaxed models having slightly higher or the same surface energy values.

Experimentally, the surface energy of Fe has been determined using liquid surface tension measurements by extrapolating the data to 0 K to give a numerical value for the solid of 2.41 J m^{-2} .⁷¹ As this value does not represent a particular surface of Fe, we cannot make a direct comparison; however, our values were generally in line with this value, especially if the average for all three surfaces was calculated.

It was also found that the results obtained from previous MM calculations are dependent on the quality of the potentials employed, while the QM calculations, including our work, all give values that are close to the experiment.

The surface energy values that were calculated showed the order of the surface stability to be (110) < (100) < (111), before and after relaxation. This relative order could be explained in terms of bond cutting arguments as well as the openness of the surface.⁵²

In summary, our models provide a good approximation of the surface energy values, with the extent of the decrease in surface energy after relaxation being related to the magnitude of the relaxation and are therefore used in subsequent studies. Our calculations described above provided the first fully converged study of the relaxation and surface energies of the three low-index Fe surfaces.

16.3.2 Electronic and Magnetic Properties of Fe(100), Fe(110), and Fe(111) Surfaces

16.3.2.1 Introduction and Previous Studies It is well known that the magnetic properties of metals at a surface are different from those in the bulk and the magnetic moments of Fe surfaces have been studied both theoretically^{69,72–80} and experimentally.⁸¹ Table 16.3 summarizes available computational results. It is well established that the magnetic moment (μ_B) at the surface is enhanced compared to the bulk, due to loss of coordination upon formation of the surface. However, only a few such studies that have investigated this effect theoretically consider surface relaxations,^{72,73,75} with most only examining bulk-terminated surfaces.^{69,73,74,76–80}

Despite the number of studies that have investigated magnetic properties of surfaces, we are unaware of any published computational studies of how the magnetic properties of Fe surfaces are related to Fe adhesion and interface formation. At an interface, the magnetic properties can differ from those of the surface or the bulk. Understanding this is particularly important for magnetic device technology.⁸²

TABLE 16.3 Computed Magnetic Moments (μ_B) of the Relaxed and (Unrelaxed) Fe(100), Fe(110), and Fe(111) Surfaces of Fe, Along with the Values Determined Previously^a

Surface	Year [Ref]	Magnetic Moment, μ_B					
		S	S-1	S-2	S-3	S-4	C
(100)	[this work]	3.03	2.47	2.59	2.48	2.45	2.42
		(3.06)	(2.50)	(2.55)	(2.47)	(2.46)	(2.43)
	1996 ⁷³	2.74	2.62	—	—	—	2.60
	1995 ⁷⁴	(3.01)	(2.36)	(2.42)	—	—	(2.32)
	1994 ⁶⁹	(2.97)					
	1992 ⁷⁶	(2.87)	(2.34)	(2.33)			
	1992 ⁷⁷	(2.97)	(2.30)	(2.37)	(2.25)	(2.24)	
	1987 ⁷⁸	(2.98)					
	1983 ⁷⁹	(2.98)	(2.35)	(2.39)	—	—	(2.25)
	1981 ⁸⁰	(3.01)	(1.68)	(2.13)	—	—	(1.84)
(110)	[this work]	2.75	2.53	2.40	2.43	2.41	2.39
		(2.75)	(2.53)	(2.48)	(2.44)	(2.41)	(2.39)
	2002 ⁷²	2.47	2.29	2.32	2.26	—	2.22
	1994 ⁶⁹ /1992 ⁷⁷	(2.57)	(2.35)	(2.25)	(2.24)	(2.24)	
	1987 ⁷⁸	(2.65)	(2.37)	(2.28)	(2.25)	—	(2.22)
(111)	[this work]	2.96	2.50	2.66	2.56	2.55	2.56
		(3.01)	(2.57)	(2.66)	(2.54)	(2.56)	(2.53)
	1993 ⁷⁵	2.62	2.25	2.34	2.15	2.17	2.11/2.00 ^b

^aS is the surface layer, S-*n* (*n* = 1 to 4) are the second to fifth layers, and C is the center of the slab.
^bThe calculation also included an S-5 value; hence, the values indicated are S-5/C.

16.3.2.2 Magnetic Moments and Density of States of Fe Surfaces To relate magnetic and electronic properties to adhesion we first examined the properties of the isolated relaxed and unrelaxed Fe(100), Fe(110), and Fe(111) surfaces. The layer-resolved magnetic moment values obtained from our calculations for the three low-index surfaces before and after relaxation are shown in Table 16.3 together with a summary of previously determined values for the same Fe surfaces. It can be seen that the magnetic moment values are enhanced at the surface, due to the loss in coordination at the surface resulting in localized surface states (see, e.g., Alden et al.⁷⁷ and Freeman and Fu⁷⁸). For the surfaces studied, the enhancement is 25%, 15%, and 16% for the (100), (110), and (111) surfaces, respectively, using the relaxed surface models. The difference in surface layer magnetic moment enhancement can be attributed to the coordination of Fe atoms at each surface, where the (110) surface atoms have a higher surface coordination number and hence the lowest surface enhancement. However, the difference between the (100) and (111) surfaces, which both have a surface Fe coordination number of 4, indicates that additional features of the surface atomic arrangement, such as packing, affect the magnetism of Fe, as seen previously.⁷⁵ The enhancement of the surface magnetic moment value observed for all Fe(100), Fe(110), and Fe(111) surfaces has been attributed by Wu and Freeman⁸³ to the difference in density of surface layer up- and down-spin states at the Fermi level (E_F) as compared to the bulk. They showed that for the bulk (or center layer) density of states (DOS), the Fermi level lies on an up-spin peak and in the valley of the down-spin DOS. At the surface layer, however, the DOS are significantly narrowed due to a loss in coordination. As a result, there is a decrease in up-spin states at E_F and an increase of down-spin states due to surface states and resonances. It is this increased number of down-spin states relative to the up-spin states that gives rise to the surface magnetic moment enhancement.

The total DOS resolved to up- and down-spin states of each of the unrelaxed (and relaxed) surfaces is shown in Fig. 16.2 (dashed line). The bulk DOS are shown in Fig. 16.3. As can be seen from Fig. 16.2, there is an increased density of down-spin states compared to up-spin states present at E_F for all three surfaces, leading to the enhanced surface magnetic moment. Comparison of the DOS for the three surfaces with those obtained previously shows good agreement. The atoms in the lower layers of our surface models show magnetic moment values that generally decrease and are identical within 1.2% for the S-4 and C layers, indicating that the surface models are large enough to achieve convergence. The magnetic moment values for the center layer of the (100) and (110) surfaces are similar, within 1.6%, and are converged to less than 1.25% compared to the bulk value, calculated to be $2.40 \mu_B$ using the same computational parameters. They are, however, up to 7% different when compared to the (111) surface value where the central layer μ_B is 5% larger than the bulk value, indicating that the surface model may not be large enough for convergence of this property. It is important to note, though, that other properties, including surface energy and relaxation, do converge for models of the same size.⁵² We therefore consider the models to be appropriate for this study and for comparison with previous work.

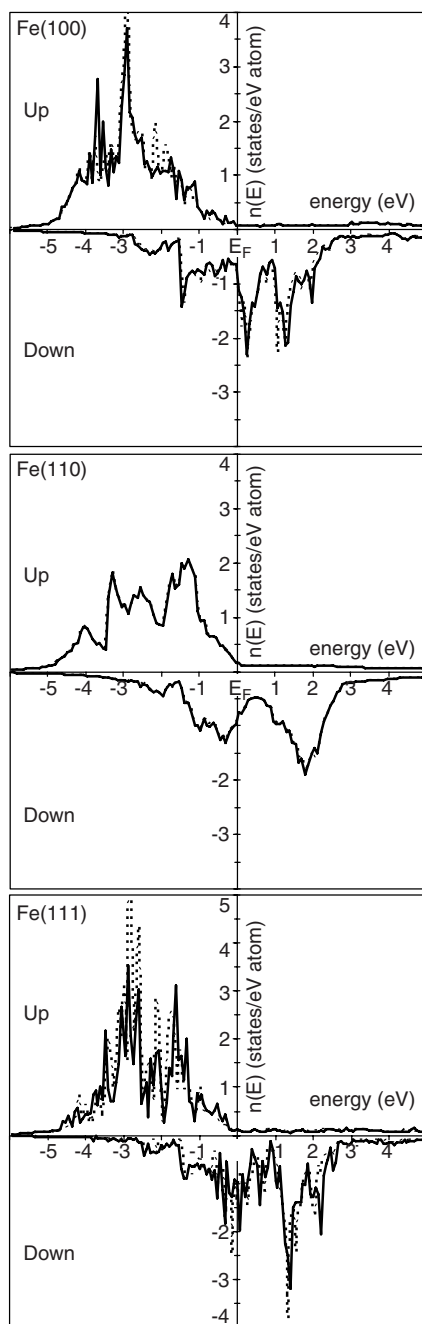


Fig. 16.2 Total density of states (DOS) resolved to up- and down-spin for the surface/top layer of the unrelaxed (dashed line) and relaxed (solid line) Fe (100), (110), and (111) surfaces. The DOS values have not been smoothed.

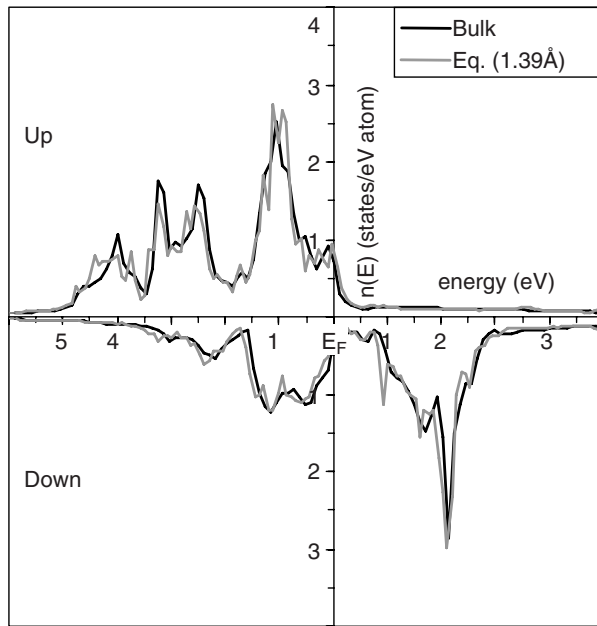


Fig. 16.3 Total density of states (resolved to up- and down-spins) for the surface layer of the (100) matching interface at equilibrium separation compared to the bulk. The DOS values have not been smoothed.

Even though our calculated layer-resolved magnetic moment values decrease toward the bulk (i.e., away from the surface), the (100) and (111) surfaces show some small oscillations. As can be seen from Table 16.3, most previous studies show an oscillation as well, which has been explained by rearrangements in the electron density (i.e., Friedel oscillations). The two exceptions are given by Kishi and Itoh,⁷³ whose surface model is not large enough to observe oscillations, and Eriksson et al.,⁷⁶ who incorporate spin-orbit coupling into their calculations. We do not observe such oscillations for the (110) surface, similar to previous calculations by Freeman and Fu,⁷⁸ Alden et al.,^{69,77} and Braun et al.⁷² We do see a 1.2% increase in the magnetic moment value at the S-3 layer, similar to the results of Braun et al.⁷²; however, this change is probably within computational uncertainty.

Comparison of the magnetic moment values after relaxation shows that the μ_B of the surface atom decreases for the (100) and (111) surfaces, while it remains the same for the (110) surface. This appears to be related directly to the magnitude of the surface relaxation of the outermost layer.⁵² The DOS of the outermost layer shows little change after relaxation (Fig. 16.2, dashed line). Thus, surface relaxation does not affect the surface magnetic moments or DOS to a significant extent, and therefore the “frozen surface” adhesion model we employ in Section 16.4.2 is justified.

16.3.3 Sulfur Adsorption on Fe(110)

16.3.3.1 Introduction and Previous Studies The presence of S on Fe surfaces has been shown to affect adhesion, corrosion, and catalysis and is thus of importance in industrial processes. Impurities, in general, can either increase or decrease the strength of adhesion, depending on conditions. Prior to studying the effect of impurities on adhesion we needed to examine the adsorption of these impurities on the clean Fe surfaces. The experimental S adsorption data on Fe(110)^{84–86} has concentrated primarily on the $\frac{1}{4}$ ML coverage with S adsorbed in a $p(2 \times 2)$ arrangement.

Below we summarize our findings on adsorption of S on Fe(110) in three different high-symmetry adsorption sites: atop, bridge, and four-fold hollow at $\frac{1}{4}$ ML coverage, followed by the effect of different S coverages on the foregoing properties of the Fe(110) surface^{87,88} (Section 16.3.3.8).

16.3.3.2 Adsorption Models and Computational Details The Fe surfaces were modeled using the supercell approach (Section 16.2.2). S adsorption at the experimentally observed coverage of $\frac{1}{4}$ ML and $p(2 \times 2)$ arrangement^{84,85} was modeled by placing an S atom on one side of the slab (see Fig. 16.4). S was adsorbed in atop, bridge, or four-fold hollow sites. The S atom and only the three top Fe layers were allowed to relax. A k -point mesh of $6 \times 6 \times 1$ was employed, as this gives a good description of FeS₂^{89,90} and clean Fe(110).⁵²

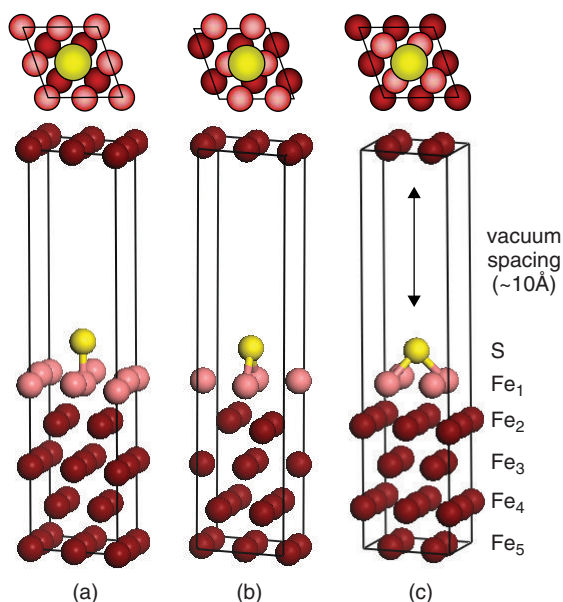


Fig. 16.4 (color online) Top and side views of the supercells used to model sulfur adsorbed in a $p(2 \times 2)$ arrangement ($\frac{1}{4}$ ML coverage) in (a) atop, (b) bridge, and (c) four-fold hollow sites.

To determine the workfunction (defined as the energy required to remove an electron from the Fermi level, E_F , to the vacuum) of the foregoing systems, a dipole correction was added in the direction perpendicular to the surface. As we have an asymmetric slab with the adsorbate placed on only one side of the slab, the electrostatic potential in the vacuum region will show a clear distinction between the each side of the slab, representing the adsorbed surface or the clean surface. The workfunction value, Φ , is represented as

$$\Phi = E_{\text{vac}} - E_F \quad (16.7)$$

where E_{vac} is the electrostatic potential in the vacuum region of the supercell on the adsorbate side of the supercell and E_F is the energy of the Fermi level. The change in workfunction value, $\Delta\Phi$, is calculated by subtracting the workfunction of the clean surface from that of the adsorbed surface.

16.3.3.3 Binding Energy and Workfunction Measurements The calculated binding energy values (Table 16.4) indicated that the hollow site is the most favored, and is in agreement with experimental data,⁸⁴ followed by the bridge and then the atop sites.

The calculated workfunction values and workfunction changes for S/Fe(110) in the three adsorption sites are shown in Table 16.4; our calculated values compared well to the experimental segregation energy value of 5.2 eV⁹¹ as does the calculated clean surface value with the experimental value of 5.12 ± 0.06 eV.⁹² The change in sign of the workfunction values after S adsorption was similar to other atomic adsorbates, such as oxygen, which also show a negative workfunction change,⁹³ indicating a negatively charged surface species. As the magnitude of the workfunction change was only very small, it suggested that there is little transfer of charge from the Fe to the S. The change in workfunction values after S adsorption were largest for the atop site, followed by the bridge and then four-fold hollow site.

16.3.3.4 Adsorption Geometry After adsorption of S, the calculations showed that both relaxation and surface reconstruction occurred.⁸⁷ Table 16.5 shows the

TABLE 16.4 Parameters Calculated for S Adsorbed on Fe(110) in the Atop, Bridge, and Four-fold Hollow Sites^a

Parameter	Adsorption Site		
	Atop	Bridge	Hollow
BE (eV)	4.52	5.32	5.82
Φ (eV)	5.08	4.999	4.98
$\Delta\Phi$ (eV)	0.24	0.15	0.14

Source: Ref. 87.

^aBE, binding energy; Φ , workfunction; $\Delta\Phi$, change in workfunction after S adsorption. The workfunction for the clean Fe(110) surface was calculated to be 4.84 eV, using a five-layer slab.

TABLE 16.5 Calculated Distances for S Adsorbed on Fe(110) in a $p(2 \times 2)$ Arrangement in Atop, Bridge, and Four-Fold Hollow Sites^a

Distance (Å)	Adsorption Site			
	Atop ⁸⁷	Bridge ⁸⁷	Four-fold Hollow ⁸⁷	Four-fold Hollow ⁸⁴
$d_{\perp}(\text{S}-\text{Fe}_S)$	2.06 (1.797)	1.70	1.49	1.43
$d(\text{S}-\text{Fe})$	2.06	2.15	2.19	2.17

Source: Ref. 87.

^aIncluded are the corresponding values determined from LEED measurements⁸⁴ for the four-fold hollow site: the perpendicular height of S above the highest atom in the topmost Fe layer, $d_{\perp}(\text{S}-\text{Fe}_S)$, and the shortest S–Fe distance, $d(\text{S}-\text{Fe})$.

calculated distances between the adsorbed S and closest Fe atom, the height of S above the surface and the experimental LEED⁸⁴ values for the 4-fold hollow site.

The perpendicular height of the adsorbed S above the top Fe layer (Table 16.5) increases going from the four-fold hollow to the bridge and atop sites, as the S lies closer to the surface for the more highly coordinated adsorption sites. The shortest S–Fe bond distances were again related to the coordination number of the adsorption site; the S–Fe bond distance is shorter for the atop site, where it is bonding directly to one atom but is longer for the bridge and four-fold hollow sites, where the bonding is distributed over more atoms.

Interestingly, some buckling of the surface layers was observed after S adsorption. For the four-fold hollow site all Fe atoms in the top layer relax upward slightly, opposite to the clean Fe(110) surface. In addition, the two Fe atoms lying farther from the S moved upward, while the two atoms closest to the S only moved upward, which resulted in the S–Fe distances to these four surface atoms being equalized, maximizing the S–Fe coordination. The second-layer Fe atoms were less buckled and the third-layer Fe atoms were bulklike, in good agreement with experimental data.⁸⁴ For the bridge site, there was also some buckling of the surface layer, similar to the four-fold hollow site; for the second layer there was some small buckling, while the third layer was bulklike. For the atop site, all surface layer Fe atoms relaxed upward slightly, except for the atom directly below the adsorbed S, which moved downward. The atoms next closest to the S in the top layer relaxed upward, with the farthest ones also relaxing upward, but only slightly.

The small displacement in the x - or y -direction indicated that the four-fold hollow site reconstructs the most and the atop site the least.⁵² For the four-fold hollow site, the second-layer Fe atoms showed no reconstruction, while those in the third layer reconstructed slightly but the movement was negligible (<0.003 Å).

16.3.3.5 Charge Density The charge-density distribution plots for S adsorbed in atop, bridge, and four-fold hollow sites on Fe(110)⁵² indicate that in all adsorption sites the S bonding is covalent in nature, as illustrated by an accumulation of charge in the S–Fe interatomic region. In addition, the differences between the

adsorption sites could be related to the binding energy values, showing that the S bonds more strongly in the four-fold hollow site, where the electron density is more evenly dispersed between four Fe atoms, and weakest in the atop site, where the electron density is concentrated on only one Fe atom.

16.3.3.6 Magnetic Properties The magnetic moment values for the S and Fe atoms in each adsorption system were previously calculated and reported by Spencer et al.⁸⁷ The magnetic moment of the middle Fe layers of each model represented a bulklike value. At the bulk-terminated side of the model, the magnetic moments were enhanced, in agreement with the value seen for the clean bulk-terminated (110) surface (see Section 16.3.2.2).

The magnetic moment of the S was found to decrease from the atop sites to the bridge sites and then to the four-fold hollow sites. This trend may be related to the length of the S–Fe bond that is formed, which is in turn related to the coordination of the site. For the atop site there was only one atom that interacted directly with the S, with the S $3p_z$ and Fe $3d_{z^2}$ adsorbate–substrate orbitals being the strongest likely interaction. For the other two sites the coordination number is higher and hence the S moment was quenched more. The strongest symmetry-allowed interactions for these sites would probably be between the S $3p_{x,y}$ and Fe $3d_{xz,yz}$ orbitals. The small difference between the values of the bridge and four-fold hollow sites may be related to differences in the geometry of the two sites.

16.3.3.7 Density of States Analysis of the DOS before and after S adsorption gave us an indication of the adsorbate–substrate bonding interactions that occur. As it was the Fe d orbitals that are most affected by the bonding to S, we examined the changes in these DOS resolved to the different substrate layers for S adsorbed in the favored four-fold hollow site.⁸⁷

The DOS of the two surface Fe atoms were found to show some significant changes after S adsorption, with the main change being the increase in intensity of the states between 4 and 5 eV below E_F , arising from the S–Fe interaction. This is consistent with ultraviolet photoelectron data that show an S-induced peak at ~ 5.5 eV below E_F .⁹⁴ A corresponding decrease in the clean DOS was also seen near E_F at ~ 0.1 eV. The S-induced changes in the surface Fe DOS depended on the S–Fe distances, showing that the S atom interacted more strongly with the closest Fe atom, in line with the changes in other properties discussed in earlier sections. The large number of states at ~ 12 eV below E_F was attributed to S 3s states. The DOS for the second-layer Fe atoms showed very little change after adsorption, indicating that the interaction was not as strong as with the surface Fe atoms.

16.3.3.8 Coverage Dependence of Adsorbed Sulfur As well as the $1/4$ ML coverage, S has been shown experimentally to adsorb at higher arrangements and coverages on this surface. At $1/3$ ML coverage, different overlayer arrangements have been observed, while more complicated arrangements and coverages can form^{85,86,95} at higher temperatures and pressures.

At $1/3$ ML coverage, two overlayer arrangements have been observed experimentally. Overlayer A is shown in Fig. 16.5, and Kelemen et al.⁸⁵ described this as $c(3 \times 1)$ after adsorption of H_2S on Fe(110) at 150°C . They also observed the same overlayer after segregation of S from the bulk via heating, but found that it did not afford the same degree of control over the S surface concentration as after H_2S adsorption. Experiments by Oudar⁸⁶ also showed this $c(3 \times 1)$ structure and suggested that S adsorbs in sites where the coordination of the substrate atoms is maximized. This arrangement was also observed by Berbil-Bautista et al.⁹⁶ after H_2S adsorption and dissociation on three-dimensional islands of Fe on W(110). By contrast, a different arrangement, referred to as overlayer B (Fig. 16.5), was found by Weissenrieder et al.,⁹⁵ who annealed an Fe(110) single crystal to 700°C to yield an S coverage corresponding to $1/3$ ML and referred to it as $p(3 \times 1)$. This arrangement was also observed by Taga et al.⁹⁷ after annealing an Fe(111) surface to 600°C . As the temperature was raised, (110) facets developed on the surface, and the S concentration increased, resulting in a surface reconstruction and facet structure described as Fe(110) $p(3 \times 1)$ -S.

We have examined S adsorption on Fe(110) at the higher coverages of $1/3$,⁹⁸ $1/2$, and 1 ML⁸⁸ [see Fig. 16.5 for the arrangements described above as well as a third arrangement (overlayer C)]. At $1/3$ ML coverage, S was most stable in the arrangement formed after H_2S dissociation on Fe(110) (overlayer A) and prefers to adsorb in four-fold hollow sites. A minor adsorbate-induced reconstruction is caused that leads to an increase in the coordination of the adsorbed S and quenching of the magnetic moment of the Fe atoms most loosely bound to the adsorbate. The bond to the surface is polar covalent in nature and results in a positive workfunction change. The bonding to the surface is rather delocalized, with the strongest interaction being with orbitals of x,y -character. For overlayer B (formed via the segregation of S from the bulk), no minimum-energy structures were found, indicating that the presence of subsurface S atoms may stabilize this

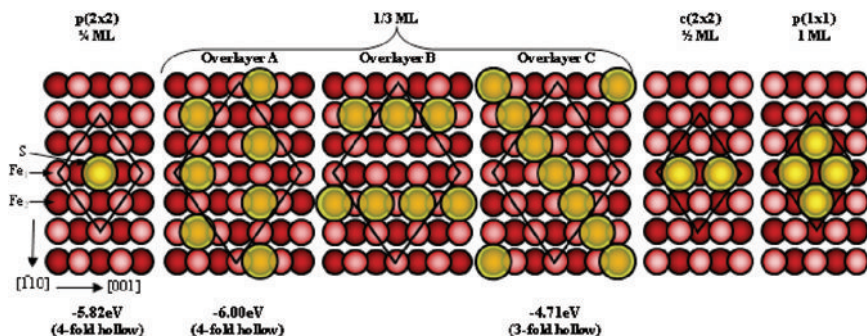


Fig. 16.5 (color online) Top view of the Fe(110) surface showing unit cells used to model the various S coverages and arrangements indicated. S was adsorbed in atop, bridge, and four-fold hollow sites (only the supercells for the atop site are shown). The binding energies of the most stable minimum-energy structures (and their adsorption sites) are also indicated.

overlayer. In the alternate overlayer arrangement (overlayer C), S was most stable in three-fold hollow sites, but was less stable than the other arrangements. The reduced stability of this overlayer was attributed to the interaction of adjacent S atoms located closest together in this arrangement, leading to a more localized bonding, with the interaction of S with the surface being stronger with orbitals of γ -character.

At coverages of $1/2$ and 1 ML, none of the high-symmetry sites were determined to be minima. As the coverage is increased, the S atoms sit higher on the surface and the workfunction change becomes negative. Overall, it was found that as the S coverage increases, the bonding goes from being S–Fe dominated at the low coverages to being S–S dominated at the higher coverages where the S atoms are located closer together on the surface and interact with each other. Full details were described previously by Spencer et al.⁸⁸

16.3.3.9 Conclusions Our calculations indicated that the most likely adsorption site of S on Fe(110) [in a $p(2 \times 2)$ coverage] is a four-fold hollow site, with the bridge and atop sites being less favored, and is in agreement with experimental data. The surface charge-density distribution for the three sites is related to the adsorption energetics and geometry. Adsorption in all three sites causes the substrate to reconstruct and relax, with the atop site showing very little reconstruction ($<9\%$), and the four-fold hollow site reconstructing the most, which agrees well with available experimental data. The adsorption of S quenches the magnetic moment of the surface Fe atoms and causes an increase in the d-orbital DOS. The data indicate that S interacts only very weakly with the second layer and not with subsequent layers.

16.3.4 Sulfur Adsorption on Fe(100): A Brief Summary

A number of experimental^{99–109} and theoretical^{110–118} studies have examined the S-contaminated Fe(100) surface, formed primarily from segregation of S from the bulk but also from dissociative molecular adsorption of H_2S in the gas phase. One of the first experimental studies¹⁰⁶ showed that S prefers to adsorb in hollow sites; however, none of the theoretical studies mentioned above examined the possibility of S adsorbing in an atop or bridge site and the effect on the related structural, energetic and magnetic properties of the surface.

We have examined the adsorption of S in the atop, bridge, and hollow sites of Fe(100) in a $c(2 \times 2)$ arrangement (i.e., 50% coverage), and the related energetics, surface geometry, charge-density distribution, magnetic moments, electronic density of states, and workfunction values of the contaminated surface.¹¹⁹ These show that S adsorbs preferentially in the hollow site on Fe(100), in agreement with experiment. The binding energy of S in the atop and bridge sites is smaller, with the atop site being least favored. The presence of S is found to affect the underlying structure, inducing an expansion of the second Fe layer, in contrast to the clean surface (see Section 16.3.3.1). Workfunction changes indicate that the adsorbed S behaves as an electronegative species and in all sites leads to a

quenching of the magnetic moments values of the Fe atoms bonded directly to the adsorbed S.

16.3.5 Dynamic Behavior of S Adsorbed on Fe(100) and Fe(110)

The sulfidation of Fe surfaces has been studied experimentally by a number of groups; however, the diffusion pathways of adsorbed S on Fe surfaces, which characterize the initial stage of S deposition, is not well understood. In previous computational studies the effect of temperature and mobility of S on different Fe surfaces has not been examined.

We have used density functional theory to calculate the vibrational frequencies of S/Fe(100) at $1/4$ ML coverage as well as of S/Fe(110) at $1/4$, $1/3$, $1/2$, and 1 ML coverages,^{120,121} in order to determine the possible diffusion pathway of S on these surfaces. The activation energies, prefactors, and rate constants for the transition of S between local minima on both surfaces at $1/4$ ML coverage are calculated and compared. Ab initio molecular dynamics simulations were then performed to monitor the mobility of S on the (100) and (110) surfaces at $1/4$ ML coverage at temperatures up to the melting point of Fe (1808 K). A time step of 2 fs was employed except for the 1808 K simulation, where a time step of 1 fs was used. The S was initially adsorbed in an atop site and each MD simulation was performed for 2 ps, which was sufficient to see the diffusion of adsorbed S from the atop site to the most stable hollow site. The Verlet¹²² algorithm was used to integrate the equations of motion, with the temperature being controlled by the algorithm of Nose.¹²³

16.3.5.1 Vibrational Frequencies Our calculated vibrational frequency values have been presented elsewhere,^{120,121} along with those calculated by Jiang and Carter.¹²⁴ At $1/4$ ML coverage, the hollow sites on both surfaces are minima, the atop sites are higher-order saddle points, and the bridge sites are transition states. The distortion in geometry of the imaginary frequencies indicate that S will diffuse from the atop site to the minimum (hollow site) via the transition state (bridge site). The stretching frequency increases as the atom sits higher on the surface and is less highly coordinated. The translational and stretching frequencies are much greater on the (110) surface than on the (100) surface, suggesting that the mobility of S is higher on the (110) surface.

16.3.5.2 Kinetic Rate Constants The difference in mobility of adsorbed atoms on the surfaces is governed by the activation energy barrier required for the adsorbate to move from one site on the surface to another, and transition-state theory (TST) can be used to describe hopping of an adatom from one site to a nearest-neighbor site via a transition state.^{125,126} The rate constants for an S atom hopping from one energy minimum (hollow site) to another via a transition state (bridge site) on the (100) and (110) surfaces at $1/4$ ML S coverage have been calculated (see Todorova et al.¹²¹ for details). These values are presented in Table 16.6, along with our calculated activation energy values, preexponential factors, and time constants.

TABLE 16.6 Activation Energies E_a , Preexponential Factors A , Reaction Rates k , and Time Constants τ

Surface	E_a (eV)	A (s^{-1})	$k(s^{-1})$	$\tau(s)$
Fe(100)	1.20	4.83×10^{12}	2.38×10^{-8}	4.21×10^1
Fe(110)	0.51	3.84×10^{12}	8.58×10^3	1.17×10^{-10}

Calculated (zero-point corrected) at $T = 298$ K for S site-to-site movement on Fe(100) and Fe(110) surfaces.

The rate constants and time constants calculated indicate that diffusion of S from one minimum-energy site to another minimum-energy site is faster on the (110) surface than on the (100) surface. At 298 K, such an event is probable on Fe(110) but highly unlikely on Fe(100). At 1808 K, the rates are much faster and indicate that such events are highly probable on both surfaces. However, these values, particularly those at 1808 K, are only an estimate of the mobility of S on the surface. As the calculations are taken from 0 K values, they do not account for surface reconstructions, which occur at elevated temperatures and which can be significant. It is in this instance where *ab initio* MD simulations can provide unique insight into the adsorption dynamics at elevated temperatures.

16.3.5.3 *Ab Initio* Molecular Dynamics The trajectories of the *ab initio* MD simulations of S adsorbed at $\frac{1}{4}$ ML coverage on both Fe(100) and Fe(110) during simulations at 298 and 1808 K are presented in Fig. 16.6. A full analysis has been made elsewhere¹²¹; however, the major outcomes are that S shows more localized lateral displacements on Fe(100) than Fe(110) but larger outward relaxations on Fe(100), indicating that the mobility of S on Fe(110) is greater than that on Fe(100), in agreement with the time constants calculated. At 1808 K, the MD simulations confirm surface melting and show S diffusing readily across the surface, consistent with the faster time constants. Diffusion is again faster on Fe(110) than on Fe(100). S does not desorb or diffuse into the bulk at any temperature examined within our simulation time frame. The use of *ab initio* MD provides an advantage over static 0 K calculations in that the mobility and geometric changes of the surface, including surface melting, can be taken into account as a function of temperature.

16.3.6 H₂S Dissociation on Fe(100) and Fe(110) with *Ab Initio* Molecular Dynamics

Poisoning of metal surfaces with both sulfur and hydrogen can be caused by the presence of H₂S in industrial processes. When S is present in Fe, it has been shown to form stable iron sulfides. Furthermore, the presence of both S and H can cause corrosion and embrittlement of the metal. The extent to which the surface is damaged is a result of the reaction conditions and in particular the temperatures involved. Hence, there is a need to better understand how these

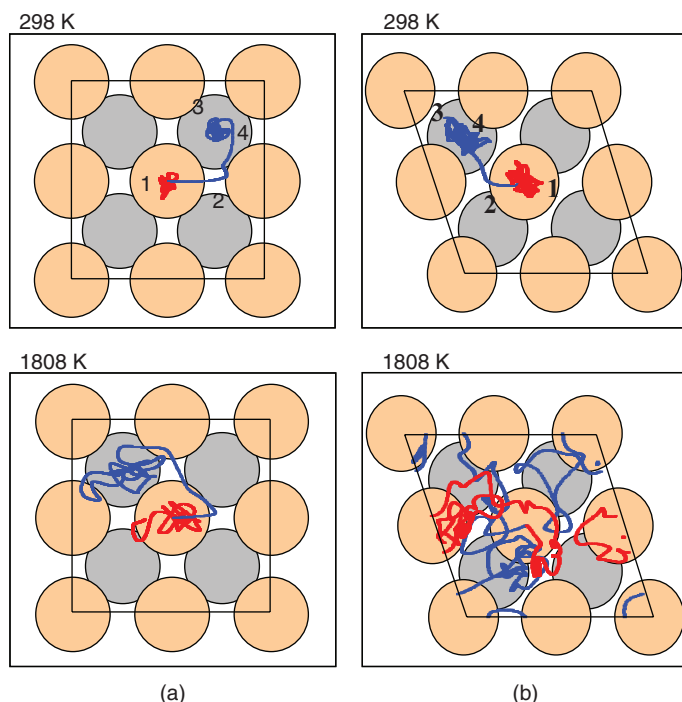


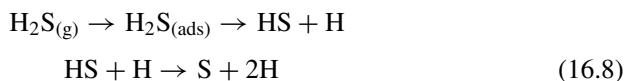
Fig. 16.6 (color online) Trajectories of the S atom (blue) and central surface Fe layer atom (red) during the MD calculations at the temperatures indicated. The different adsorption sites are labeled in the first cell only: (a) Fe(100) and (b) Fe(110) surfaces.

reactions are affected by the reaction conditions and how they are affected by different temperatures.

We have investigated the reaction of H_2S with both the Fe(100)¹²⁷ and Fe(110)¹²⁸ surfaces at different temperatures using ab initio MD simulations. Previously, Narayan et al.¹²⁹ used ESCA (electron spectroscopy for chemical analysis) to examine the reaction of H_2S with polycrystalline Fe from 100 to 773 K. They found that H_2S dissociated between 190 K and ambient temperature. Above 423 K, a nonstoichiometric iron sulfide (FeS) formed, which converted to stoichiometric FeS with increased H_2S pressure and exposure time. Adsorption kinetic experiments performed by Shanabarger¹³⁰ indicated that H_2S dissociates on polycrystalline Fe and Fe(111) surfaces. The findings were consistent with a precursor adsorption model. The adsorption kinetics on other low-index single-crystal surfaces of Fe were not performed. Density functional theory calculations by Jiang and Carter¹²⁴ examined the adsorption sites, diffusion barriers, and dissociation pathways of $1/4$ ML coverage of H_2S adsorbed on Fe(100) and Fe(110). The effect of temperature on these reactions, however, was not taken into consideration.

Our MD simulations were performed at 298, 800, 1000, and 1808 K, using a time step of 1 or 0.5 fs. A Verlet¹²² algorithm was used to integrate the equations of motion, with the temperature controlled by the algorithm of Nose.¹²³ At the beginning of the simulations, H₂S was adsorbed ~ 3 Å above a $[2 \times 2]$ surface slab (as employed previously for adsorbed S). The surface and all atoms, except the bottom two surface layers were allowed to relax during the simulations, which were performed for up to 5.25 ps.

16.3.6.1 Stages of the Adsorption Process On both surfaces, H₂S was found to dissociate via a two-step process:



A schematic diagram representing the different stages of adsorption found from the ab initio MD simulations is presented in Fig. 16.7. Stage 1 represents the initial configuration with the H₂S molecule sitting approximately 3 Å above the surface. At the beginning of the simulation, at 298 K, the molecule rotates about its axis before adsorbing on the surface (stage 2). It then dissociates to leave an adsorbed SH species and an adsorbed H atom, represented by stage 3. Further into the simulation, the SH species then dissociates to leave adsorbed S and two H atoms (stage 4). At higher temperatures, the same dissociation process occurs; however, after complete dissociation, one of the H atoms diffuses below the top surface layer (stage 5). At the highest temperature simulated, 1808 K, complete dissociation and H dissolution is again observed; however, after ~ 2.34 ps, the two H atoms return to the surface, combine, and desorb from the surface as H₂ (stage 6).

On the Fe(100) surface, complete dissociation also occurs at 298 K. At 1808 K, the same process occurs, with H dissolution again being observed; however, on this surface, the H atoms diffuse further into the surface layers. No H₂ desorption is observed after 1.5 ps on this surface; however, as subsurface diffusion is more facile on this surface than on Fe(110), longer simulation times may be required to simulate this process.

16.3.6.2 Dissociation Mechanism On both surfaces, the dissociation mechanism has been shown to change with temperature,^{127,128} highlighting the benefit

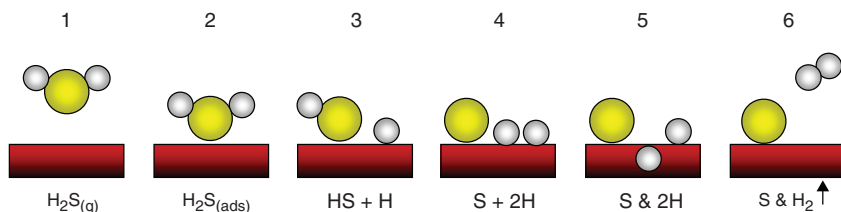


Fig. 16.7 (color online) Stages of the adsorption process of H₂S on Fe(100) and (110).

of studying these effects by MD. Specifically, large reconstructions of the Fe surface atoms are found to alter the structure of the transition state for H₂S dissociation to SH and H. At lower temperatures, where the substrate atoms only oscillate around their crystal lattice positions, the adsorbate structure is more distorted during dissociation. In contrast, at elevated temperatures, greater movement of the substrate atoms means that the adsorbate structure is little changed in order to overcome the dissociation barrier.

16.4 STRUCTURE AND PROPERTIES OF IRON INTERFACES

In this section we discuss the properties of clean Fe(100), Fe(110), and Fe(111) interfaces and how these properties are affected by S contamination at the interface. Our studies of adhesion between clean bulk-cleaved Fe(100), Fe(110),²⁰ and Fe(111)²¹ interfaces in match and mismatch are summarized in Section 16.4.1. The fitted UBER curves allowed the work of separation (W_{sep}), the equilibrium interfacial separation (d), the scale length (l), and peak interfacial strength (σ_{max}) to be studied. In Section 16.4.2 we present our new results on the relationship between adhesion and electronic and magnetic properties of the low-index Fe interfaces. In Section 16.4.3 we discuss the findings from our study on the effect of relaxation on the adhesion of Fe(100) match and mismatch interfaces.¹³¹ In Section 16.4.4 we present the findings of our studies on the effect of 1/4 ML coverage S impurity on Fe(110) adhesion¹³² and the effect of S contamination on the Fe(100) interfacial properties at the experimentally observed surface coverage of 1/2 ML of S.¹¹⁹

16.4.1 Adhesion Between Ideal Surfaces

The adhesion curves calculated for the Fe(100), Fe(110),²⁰ and Fe(111)²¹ interfaces are presented in Fig. 16.8, and the UBER parameters E_0 , d_0 , and l are given in Table 16.7. The matching (111) interface was shown to be the strongest interface having the highest W_{sep} , followed by the (100) and then (110) interfaces. For

TABLE 16.7 Fitted UBER Parameters for Fe(100), Fe(110)²⁰, and Fe(111)²¹ Matching and Mismatching Interfaces

Surface	Interface	Interface Properties					
		$E_0 = W_{\text{sep}}$ (J m ⁻²)	γ (J m ⁻²)	d_0 (Å)	l (Å)	σ_{max} (GPa)	R^2
(100)	Match	4.690	2.345	1.390	0.600	57.51	0.998
	Mismatch	1.422	N.A.	2.427	0.567	18.45	0.996
(110)	Match	4.494	2.247	1.991	0.590	56.04	0.996
	Mismatch	2.795	N.A.	2.427	0.588	34.97	0.992
(111)	Match	5.381	2.690	0.809	0.623	63.59	0.999
	Mismatch	0.733	N.A.	2.389	0.476	11.33	0.998

^aN.A., not available.

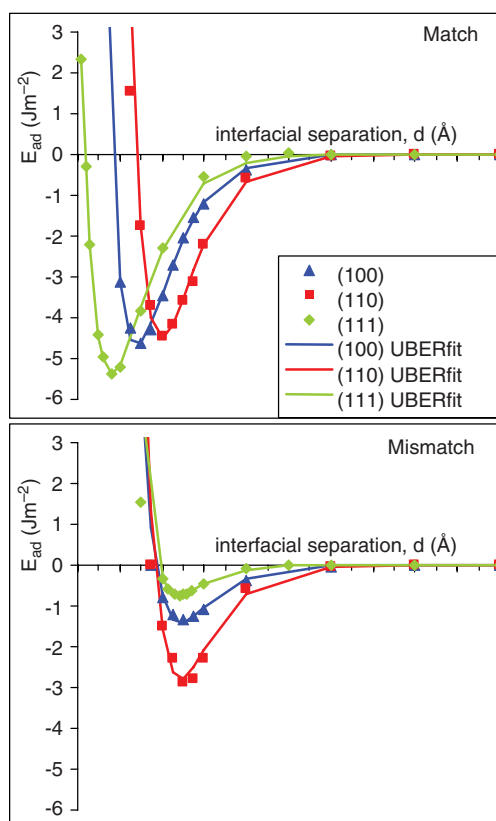


Fig. 16.8 (color online) Adhesion energy curves for the (100), (110), and (111) matching and mismatching interfaces and fitted UBER.

the mismatching interfaces, the order was opposite, showing the (100) to be the strongest interface, followed by the (110) and then (111) interfaces. For all the low-index surfaces, however, the matching interfaces were found to be stronger than the mismatching interfaces. This relative order is related to the number of dominant Fe–Fe interactions at the respective interfaces. For the matching interfaces, there are four nearest-neighbor (NN) interactions for the (100) and (111) interfaces and two for the (110) interface, acting over an area of 8.21, 14.25, and 5.8 Å², respectively, but for the mismatching interfaces there is only one NN interaction acting over the respective surface areas. Hence, the matching interfaces have “more bonding” over the same surface area than the corresponding mismatching interfaces, giving rise to stronger interfaces. The relative order of the W_{sep} values for the mismatching interfaces could also be related to the number of Fe–Fe interactions across the interface but for the matching interfaces, the relation is more complicated, as there are different numbers of NN interactions as well as bond separations.

The ideal surface energy values (Table 16.7) from the W_{sep} values for epitaxy (Table 16.7) showed the relative order of stability of the three surfaces to be (110)>(100)>(111). This order is the same as that calculated for the isolated surfaces in Section 16.3.1.4. As a result of the relative stability of the surfaces, despite the (111) matching interface having the largest W_{sep} of all the low-index interfaces, the lower stability of the surface indicates that it is less likely to exist as the clean bulk-terminated face.

The d_0 values calculated (Table 16.7) were found to be smaller for the matching interfaces than for the mismatching interfaces. In fact, the d_0 values for the matching interfaces indicate that the interface forms the bulk structure at the equilibrium separation. For the mismatching interfaces, the d_0 values were found to be approximately equal to the Fe–Fe bond distance of $2.482 \text{ \AA}^{133}$, as the topmost Fe atoms on each surface forming the interface directly face each other.

The l values (Table 16.7) calculated for the matching and mismatching interfaces were all close to each other and agreed with the empirically estimated average screening length value of $0.56 \text{ \AA}$ for several Fe surfaces,⁴⁶ except for the (111) mismatching interface, suggesting again that this interface is unlikely to form. The l values were slightly larger for the matching interfaces, indicating that the electronic interactions between the approaching surfaces forming the interface begin at a larger separation. The ideal peak interfacial stress values (Table 16.7), which give a measure of the maximum tensile stress that the interfaces can withstand without spontaneous cleavage, were shown to be in the same order as the W_{sep} values.

16.4.2 Relationship Between Adhesion and Electronic and Magnetic Properties

In this section we present new results investigating the relationship between adhesive energy and interfacial separation for the body-centered cubic (bcc) Fe(100), Fe(110), and Fe(111) interfaces. Both ideally matching and mismatching interfaces were considered in order to cover the endpoints of the range of adhesion of real surfaces.

16.4.2.1 Magnetic Properties and Adhesion of Fe Interfaces The computed layer-by-layer local atomic magnetic moments for the Fe(100), Fe(110), and Fe(111) interfaces in match and mismatch at three interfacial separation distances (d)^{20,21}: $10 \text{ \AA}$ separation, the interfaces at approximately infinite separation; $4 \text{ \AA}$, the approximate distance at which metallic interactions begin to dominate; and the equilibrium separation (Eq.) are shown in Fig. 16.9.

Figure 16.9 shows that for the (100) match interface, the top surface layer μ_B changes considerably as the surfaces approach, while the second and third layers change only slightly and the lower layers, hardly at all. At the equilibrium interfacial separation, the μ_B values differ very little from layer to layer, consistent with the fact that at this separation the system is essentially bulk Fe. For the mismatch interface, it is again the surface μ_B that is most changed upon

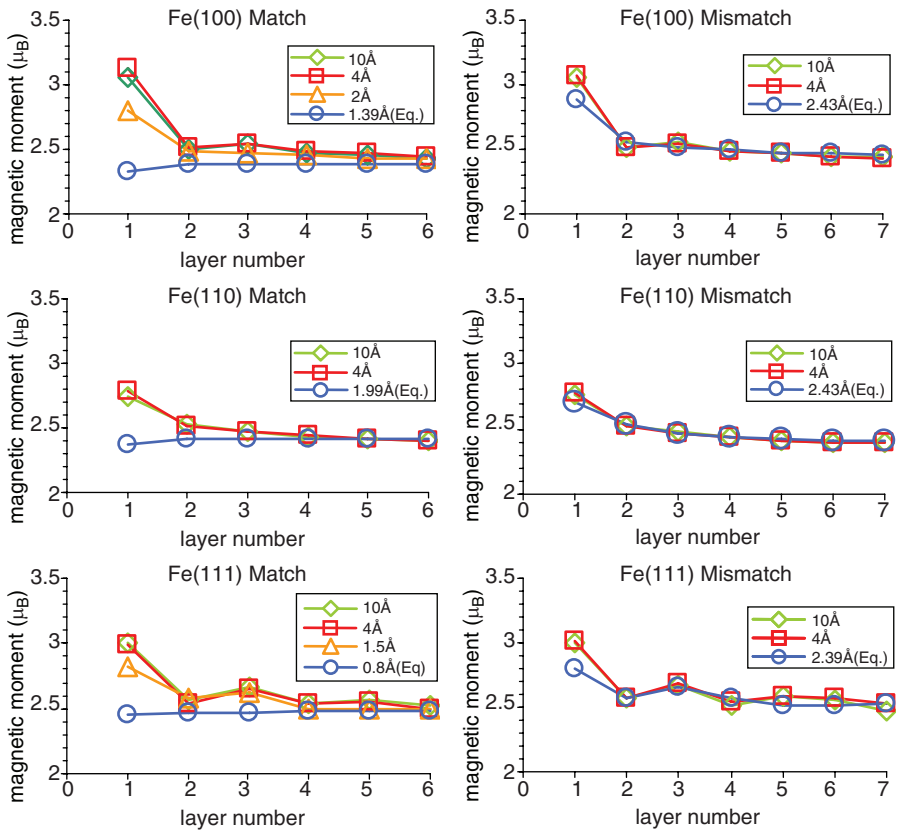


Fig. 16.9 (color online) Calculated layer-by-layer magnetic moment values (μ_B) for the match and mismatch Fe(100), (100), and (111) interfaces at the interfacial separations indicated; Eq. is the equilibrium separation.

formation of the interface, while the lower layers stay almost constant. At the equilibrium interfacial separation the surface μ_B is still enhanced, as the bulk crystal is not formed when the surfaces are out of epitaxy.

The (110) match and mismatch interfaces display similar trends to the (100) interfaces where the second- and third-layer μ_B values stay almost the same as those of the lower layers. The third layers of both (110) interfaces, however, appear to be less affected than they are on the (100) interface. This surface is more closely packed than the (100) surface, and hence it would be expected that the lower layers would be less affected by changes occurring at the surface layer.

The (111) match and mismatch interfaces also show a surface layer magnetic moment enhancement; however, in addition to the surface layer, the second- and third-layer μ_B values are clearly altered as the interfacial separation is decreased. For this less close-packed surface, the second and third atomic layers are more exposed. It can therefore be suggested that there are surface states localized on

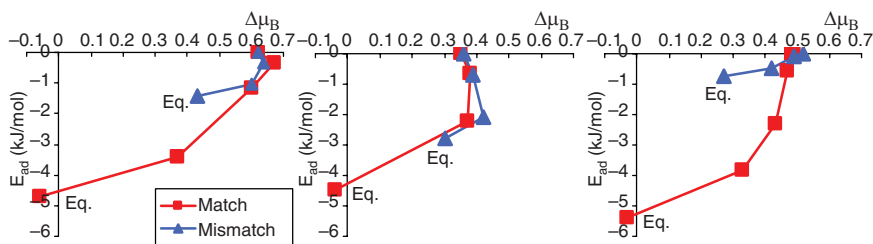


Fig. 16.10 (color online) Adhesion energy values, $E_{ad}^{20,21}$ plotted against surface layer magnetic moment enhancements, $\Delta\mu_B = \mu_{B\text{surface}} - \mu_{B\text{bulk}}$, corresponding to the same interfacial separations for the (100), (110), and (111) interfaces (from left to right) in match and mismatch (triangles).

these “lower-layer” atoms, and as the surfaces are brought together, the lower-layer surface states also begin to interact, resulting in changes in their computed magnetic moments. This is in contrast to the (100) and (110) surfaces, where atoms below the topmost layer are fully (i.e., bulk) coordinated; their magnetic moment values are therefore close to those computed for the bulk, and changes in interfacial separations have negligible influence. This observation is consistent with this surface being more open.

The relation between the surface μ_B changes and the adhesion energy can be seen from Fig. 16.10, where the values for the surface μ_B enhancement, $\Delta\mu_B$ (the difference between the surface atomic layer $\mu_{B\text{surface}}$ and the computed bulk $\mu_{B\text{bulk}}$), and the adhesion energy, E_{ad} , for the interfaces have been plotted. For all three matching interfaces the adhesion energy decreases with decreasing $\Delta\mu_B$ until the adhesion energy reaches a minimum when the interface is most stable (bulklike), and the enhancement is essentially zero. For the mismatching interfaces, the adhesion energy decreases as the $\Delta\mu_B$ decrease but $\Delta\mu_B$ does not reach zero at the minimum adhesion energy because the bulk crystal structure is not formed.

16.4.2.2 Density of States

DOS of Matching Interfaces The surface layer density of states (S-DOS), resolved to up- and down-spin states, for all interfaces were calculated at four interfacial separations: 10 Å, 4 Å, Eq., and a separation between 4 Å and Eq. As the difference in magnitude of the up- and down-spin states at the Fermi level affects the surface μ_B enhancement, we examine how these states change as a function of interfacial separation.

The S-DOS for the matching (100) interface are shown in Fig. 16.11. At 10-Å separation, the S-DOS are identical to those seen earlier for the unrelaxed surface (Fig. 16.2), as this separation represents the isolated surfaces.²⁰ The values calculated for the up- and down-spin DOS at E_F (Table 16.8) show the presence of more down-spin states at E_F , which gives rise to the surface μ_B enhancement.

TABLE 16.8 Number of Up- and Down-Spin States at the Fermi Energy (in States/eV Atom) for Match and Mismatch Interfaces at Interfacial Separations of 10 Å and Equilibrium (Eq.)

Interface	Interfacial Separation	Match		Mismatch	
		Up	Down	Up	Down
(100)	10 Å	0.09	1.02	0.10	0.81
	Eq.	0.79	0.23	0.13	1.24
(110)	10 Å	0.16	0.86	0.14	0.87
	Eq.	1.00	0.52	0.20	0.45
(111)	10 Å	0.08	1.45	0.07	1.33
	Eq.	0.45	0.33	0.15	2.03

That there is little chemical interactions of the surfaces for $d > 4$ Å is shown by the similarity in the S-DOS, consistent with the similarity of the adhesion energy curves²⁰ and the values of the surface μ_B enhancements.

At the equilibrium interfacial separation, the number of down-spin states at E_F has decreased significantly (see Table 16.8), the overall features of the S-DOS are those of bulk Fe (Fig. 16.3), and the up-spin S-DOS change significantly at E_F , with an increased number of states at E_F . As a result of these changes, there is a larger number of up-spin states at E_F , as compared to larger separation distances, leading to a significant decrease in the surface μ_B at this separation.

For the (110) matching interface, similar behavior is observed as the interfacial separation is decreased, but at the equilibrium separation there is a decrease in the down-spin states, whereas there is an increase in the up-spin states at E_F , and the DOS resemble those of the bulk crystal structure (Fig. 16.3).

The up- and down-spin S-DOS of the matching (111) interface (Fig. 16.11) show behavior similar to that of the other two interfaces, with the down-spin states dominating at larger interfacial separations. At the equilibrium separation the up-spin states dominate at E_F and the S-DOS resemble those of the bulk. This is consistent with the very small value computed for the μ_B enhancement.

DOS of Mismatching Interfaces The resolved surface layer DOS values for the three mismatching interfaces were calculated and the up- and down-spin states at E_F are shown in Table 16.8. For the (100) interface at 10 Å separation, the S-DOS represents the isolated noninteracting surface. As the interfacial separation is decreased, the down-spin states present near E_F vary slightly in number, but unlike the matching interface, they are still present at the equilibrium separation, still having an increased number of down-spin states, indicating an enhanced surface μ_B value. Similar behavior is seen for the DOS of the (110) and (111) mismatching interfaces.

16.4.2.3 Charge Density The charge-density distribution of the (100), (110), and (111) matching and mismatching interfaces was examined at two different interfacial separations: equilibrium separation and a separation greater than

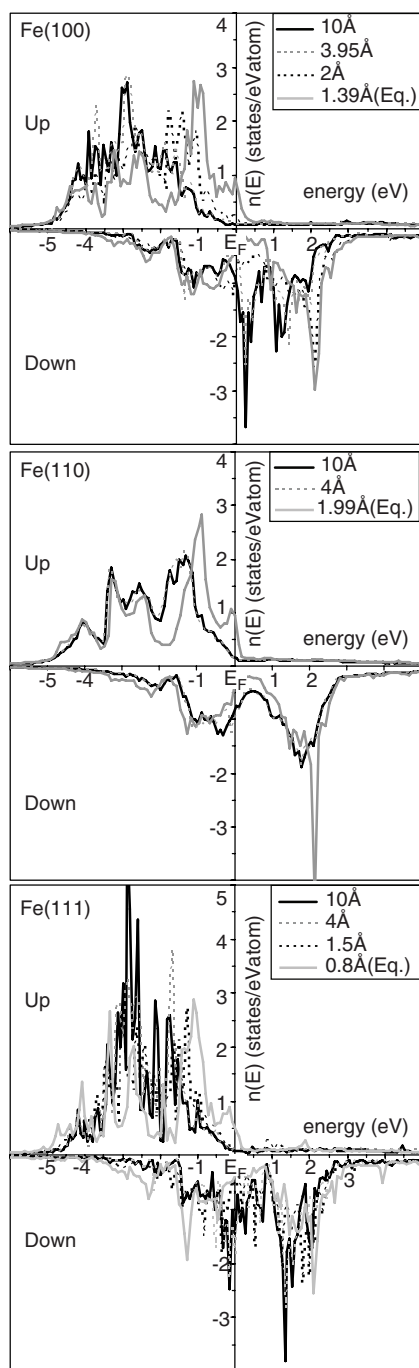


Fig. 16.11 Surface layer density of states (resolved to up- and down-spin states) for the (100), (110), and (111) matching interfaces at the interfacial separation indicated, including equilibrium (Eq.). The DOS values have not been smoothed.

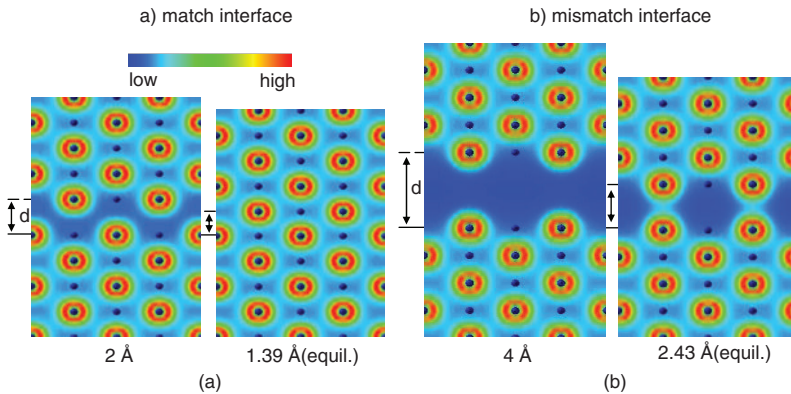


Fig. 16.12 (color online) Charge-density plots of (a) matching and (b) mismatching Fe(100) interfaces at the interfacial separation d indicated.

equilibrium. The plots shown in Fig. 16.12 correspond to a slice taken perpendicular to the (100) match and mismatch interfaces.

For the (100) matching interface (Fig. 16.12a) at a separation of 2 Å (greater than equilibrium), the plot shows a region of low charge density between the two surfaces forming the interface, indicating that negligible metallic bond formation occurs. At the equilibrium separation (1.39 Å) there is a uniform distribution of the charge density between the atoms at the interface and the bulk, signifying bond formation has occurred and the bulk material formed. The (110) and (111) matching interfaces (not shown) show identical behavior at the corresponding interfacial separations. Hence, irrespective of the crystal face forming the interface in epitaxy, the interface is most stable when the charge density is evenly distributed between the atoms at the interface and those within the bulk.

The charge-density plot for the corresponding (100) mismatching interface (Fig. 16.12b) shows that at an interfacial separation greater than equilibrium separation (4 Å), there is a region of very low charge density at the interface, similar to the matching interface. At the equilibrium separation (2.43 Å), an increase in the charge density between the closest surface atoms forming the interface indicates that some bonding occurs. However, there are large areas of low charge density between the directional bonds, which result in a much weaker interfacial energy than that in the epitaxial arrangement.^{20,21} The mismatching (110) and (111) interfaces show similar behavior.

16.4.2.4 Conclusions For all three surfaces studied, there is an enhanced magnetic moment at the surface due to an increased number of down-spin states as opposed to up-spin states at the Fermi level in the DOS, consistent with previous studies. The inclusion of surface relaxation in the calculations had little effect on the magnetic moment values and DOS.

The magnetic moments calculated for the interfaces at a number of special interfacial separation distances were found to be related and were consistent with

the adhesion properties obtained previously. The surface layer magnetic moment is most affected upon formation of the interface, with lower layers being less affected but most altered for more open surfaces.

For the matching interfaces the surface layer magnetic moment enhancement decreases as the interfacial separation is reduced, until it reaches zero at the equilibrium separation. In contrast, for mismatching interfaces an enhanced surface magnetic moment is still present at the equilibrium separation, as manifested by the increased number of down-spin states at E_F .

The charge-density plots for different interfacial separations show rearrangement of the electron density as the surfaces are brought into contact in and out of epitaxy. There is little interaction between the surfaces at large interfacial separations, in agreement with the DOS and magnetic moment enhancement values, but for shorter separations they indicate bond formation.

16.4.3 Effect of Relaxation on Adhesion of Fe(100) Surfaces: Avalanche

16.4.3.1 Introduction and Previous Studies Avalanche is a process whereby the mutual attraction between two surfaces, at a critical interfacial separation, causes the surface atoms to displace toward the opposing surface, resulting in a collapse of the two slabs to form a single slab. A number of studies have examined this effect using a range of computational methods.^{134–138} Good and Banerjee¹³⁹ performed Monte Carlo simulations at room temperature on bcc Fe and W^{140,141} and found that avalanche still occurred for Fe(110) interfaces that were out of registry; however, it was inhibited when the surfaces were far out of registry and when only a few layers near the surface were allowed to relax. Also, the energy released in the avalanche decreased as the loss of registry increased.

A study of the avalanche effect for silicon (111) surfaces¹⁴² showed covalent bond effects, indicating the importance of using quantum mechanical methods. None of these studies, however, employed quantum mechanical techniques to examine avalanche in adhesion between metallic surfaces. Furthermore, no lateral displacements were allowed during the simulations, preventing the study of avalanche formation, or avalanche of a mismatching interface into a matching one.

16.4.3.2 Interface Models The Fe interfaces were modeled using the supercell approximation, described in Section 16.2.2. Surfaces were cleaved from a crystal structure of bcc Fe, corresponding to the (100) Miller plane; the specific details of the individual models and their graphical representations have been explained by Spencer et al.¹³¹

In model I¹³¹ the sandwich approach was used to represent the match and mismatch interfaces, which means that only one vacuum spacer was positioned between the surfaces, comprising six layers each for the match interface and six and five layers for the mismatch interface. The three-dimensional periodic boundary conditions (PBCs) were then applied to the cell. For the match interface, the two middle-layer atom positions were fixed; for the mismatch interface the

middle layer of atoms was fixed. All other atoms were allowed to relax. We defined the initial and final interfacial separations as the distance between the boundary layers of the original and relaxed separated surfaces, respectively. The total energies were calculated for separations from approximately 1 to 10 Å.

Model II¹³¹ was identical to model I except that no surface layers were fixed and an additional vacuum spacer of >30 Å was added in the z -direction to allow the entire slab to move in the z -direction during relaxation. The initial interfacial separation was approximately 3 Å for both match and mismatch interfaces. The systems were then subject to the full geometry optimization, keeping the total volume of the supercell fixed. The energy at the final interfacial separation was calculated.

In model III,¹³¹ vacuum spacers of approximately 8 Å were introduced in the x -, y -, and z -directions, creating a periodic cluster-type model. The number of layers was similar to those of models I and II, but only a mismatch initial configuration was used for the geometry optimization. One surface (i.e., cluster) was fixed during the geometry optimization, while another one was free to move in all three directions. The initial interfacial separation was 4.8 Å, and the final geometry was examined.

16.4.3.3 Summary of Findings In model I, the relaxation resulted in increasing the interlayer spacing throughout the surfaces. For the relaxed system, the interlayer spacing was approximately 1.58 Å and for the unrelaxed surface it was 1.4345 Å, making the relaxed interlayer spacing approximately 10% larger than the unrelaxed spacing. Further detailed analyses¹³¹ indicated that in such a system setup, a proper avalanche effect cannot occur because of the additional constraint on the fixed layers of the slabs as well as the periodic boundary conditions in all three dimensions, which cause unrealistic stretching of the interlayer spacing and formation of a highly strained crystal region.

In model II, relaxation of the periodic boundary condition in one (z -) dimension resulted in the two surfaces jumping together. The equilibrium interfacial separation of 1.437 and 2.4996 Å was achieved for the match and mismatch interfaces, respectively. The match interface value was approximately equal to the bulk interlayer spacing (1.4345 Å), as was expected. Similarly, the mismatch interface was close to the bulk Fe–Fe distance of 2.47 Å. The overall geometry at the center of the interface formed upon avalanche was bulklike, as opposed to the strained model I.

The adhesion energy for the match interface after relaxation compared well with that obtained for the minimum-energy structure with the same interfacial separation using model I, but as the outer layers of model II were allowed to move, this resulted in surface relaxation and hence in slightly lower energy.

In our model III, the two clusters were found to approach each other, forming a nearly matching interface with some minor structural imperfections due to a limited simulation time. However, the calculation clearly illustrated that if no constraints are imposed on the system, it will undergo avalanche and relax toward perfect registry.

16.4.4 Effect of Sulfur Impurity on Fe(110) Adhesion

16.4.4.1 Introduction and Previous Studies In Section 16.3.3 we discussed the effects that S impurity can have on the properties of Fe surfaces. Experimentally, the presence of S contamination affects the adhesive strength of the interface compared to the clean surfaces^{143–145} but there are some conflicting findings. Also, the effect that S has on the structural, electronic, and magnetic properties has not been examined. Below we summarize our findings on the effect of the experimentally observed $1/4$ ML coverage of S adsorbed in atop, bridge, and four-fold hollow sites on the adhesion properties of Fe(110) surfaces¹³² and how they compared to the clean interfaces. We also provide a brief summary of the effect of different S coverages on the properties of Fe(110) in Section 16.4.4.3.

16.4.4.2 Interface Models and Computational Parameters Adhesion between a relaxed S/Fe(110) surface and an unrelaxed clean Fe(110) surface was investigated in order to make a comparison with our previous study of adhesion between unrelaxed clean Fe(110) surfaces.²⁰ Our S/Fe(110) surface models obtained previously⁸⁷ and described in Section 16.4.4.1 were used to model the S-contaminated interfaces. The relaxed five-layer model with a S atom adsorbed in either an atop, bridge, or four-fold hollow site on one side of the slab in a $p(2 \times 2)$ arrangement represented a mismatch interface, where insertion of the vacuum spacer in the z -direction resulted in formation of the interface. An additional layer was added to the relaxed five-layer model to form the match interfaces. The definitions of match and mismatch are described according to the geometry of the interface formed when the S is removed. By adjusting the size or thickness of the vacuum spacer, different interfacial separations were modeled. The two surfaces forming the interfaces were defined as surface A [the relaxed S/Fe(110) surface] and surface B [the unrelaxed clean Fe(110) surface]. The interfacial separation was defined as the distance between the topmost Fe atoms on each surface. A diagram of the models employed can be found elsewhere.¹³²

For all three matching interfaces the S atom lies between two different adsorption sites, one on surface A and the other on surface B. On surface A, the S atom lies above an atop, bridge, or four-fold hollow site, whereas on surface B, the S atom lies above a four-fold hollow, bridge, and atop site, respectively. For the bridge–site interface, the two Fe atoms forming the bridge site on surface B are oriented at right angles to those forming the bridge site on surface A. As the topmost Fe atoms and S atoms on surface A were relaxed, they showed some buckling (described previously by Spencer et al.⁸⁷). The Fe atoms on surface B represented a clean bulk-terminated surface which did not show any buckling. The interfaces were described as atop, bridge, or hollow, depending on the site to which the S atom was adsorbed on surface A.

As the work of separation, by definition, disregards the effect of plastic or diffusional processes, we performed further calculations to remove some of the constraints applied to our interface models and to examine the effect of relaxation of the interface at equilibrium. These calculations were performed on the

interfaces at the equilibrium separation and allowed all S and Fe atoms to relax while also allowing the cell volume to change.

16.4.4.3 Results

Adhesion Energetics The adhesion energy values calculated for each interface¹³² are presented in Fig. 16.13, along with the fitted UBER parameters in Table 16.9.¹³² In all adsorption sites and for both match and mismatch interfaces, the UBER provides a good description of the adhesion values. The S was found to decrease the adhesion energy compared to the clean interface²⁰ in all adsorption sites and alignments of match and mismatch. The strongest interface was with S adsorbed in atop sites in a matching orientation. For all interfaces, except the hollow interface, the match interfaces were stronger than the corresponding mismatching interfaces. Relaxation of the interfaces at the equilibrium separation led to an increase in the adhesion energy, but the interfaces were still weaker than the corresponding clean ones.

For all interfaces, the S was found to increase the equilibrium interfacial separation, with the S–Fe distances to different adsorption sites on the two surfaces being consistent with the distances on the same sites on the isolated surface. The shortest S–Fe distances to surfaces A and B were found to be smaller than on the isolated surface, due to the attraction between the Fe atoms across the interface, bringing the two surfaces closer together. The relaxation introduced surface buckling of the clean surface due to the presence of S, as it did on the isolated surface, but of larger magnitude. A comparison of the S–Fe distances at the interface with those found in naturally occurring iron sulfide minerals indicated the presence of chemical bonds across the interface.

Similar to the W_{sep} values, the screening length, l (Table 16.9), for each interface was reduced by the presence of S-contamination, showing that the attraction

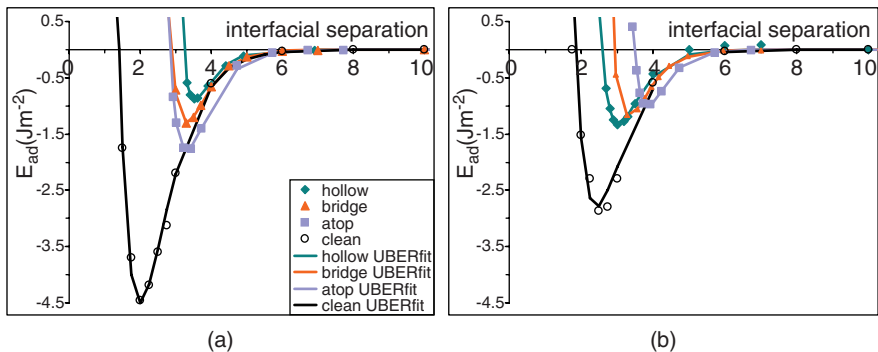


Fig. 16.13 (color online) Adhesion energy data calculated and fitted UBER curves for the $1/4$ -ML S-contaminated Fe(110) match (a) and mismatch (b) interfaces with S adsorbed in atop, bridge, and hollow sites. The clean Fe(110) interface data²⁰ are shown for comparison. (From Ref. 132.)

TABLE 16.9 UBER Parameters Calculated for the S-Contaminated Match and Mismatch Interfaces¹³² and Values for Clean Interfaces^a

Adsorption Site	Atop	Bridge	Hollow	Clean
<i>Match Interface</i>				
$E_0 = W_{\text{sep}}(E_{\text{ad}})$ (J m ⁻²)	1.79 (2.41)	1.30 (1.95)	0.88 (1.50)	4.494
d_0 (Å)	3.30 (2.30)	3.30 (2.25)	3.55 (2.29)	1.991
l (Å)	0.47	0.43	0.37	0.590
R^2	0.998	0.998	1.000	0.99
<i>Mismatch Interface</i>				
$E_0 = W_{\text{sep}}(E_{\text{ad}})$ (J m ⁻²)	1.02 (1.16)	1.19 (1.42)	1.32 (1.72)	2.795
d_0 (Å)	3.86 (3.10)	3.33 (2.78)	3.03 (2.60)	2.427
l (Å)	0.37	0.43	0.45	0.588
R^2	0.999	1.000	0.995	0.99

Source: Ref. 20.

^aThe adhesion energy, E_{ad} , and d_0 values calculated for the relaxed S-contaminated interfaces are shown in parentheses.

between the contaminated surfaces occurs over a shorter separation distance than with a clean interface. The relative order of the l values is correlated to the distance of the S atom from the underlying surface. In particular, from 6 to ~ 3.5 Å the attraction was greater than between the clean surface at the same separation, indicating that it is more likely to adhere.

Charge Density Charge-density plots taken along the directions that cut the shortest S–Fe bonds across the interface were examined and compared for each interface (see Ref. 132). For both match and mismatch interfaces at the equilibrium separation, they showed that the S bonds to both surfaces A and B, bonding to the same atoms as on the isolated surface as well as the closest Fe atoms on the other surface. They also further supported the chemical as opposed to physical nature of the bonds formed at the interface.

Bonding across the interface was in line with the interfacial geometry, being symmetrical for the mismatching interfaces. For each interface, however, there were regions of low charge density between adjacent S atoms which were not seen for the clean interfaces, as the S atom prevents the Fe atoms from getting close enough to interact as strongly across the interfacial boundary. After relaxation of these interfaces, these large regions of low charge density were reduced due to the structural changes that lead to a more even distribution of charge at the interface.

Magnetic Moments The magnetic moment enhancements, $\Delta\mu_B$, calculated for the Fe atoms most strongly bonded to the S atom on surfaces A and B were calculated as a function of interfacial separation. At an interfacial separation of 12 Å for both match and mismatch interfaces, the magnetic moment enhancements of Fe atoms on surfaces A and B were the same as seen on the isolated S-contaminated

surfaces⁸⁷ and clean surface (see Section 16.3.2.2), respectively, in line with the adhesion energy curves. Hence, for the clean surface B, the enhancements were positive, as seen on the clean isolated surface, whereas they were negative for the S-contaminated surface A, as S quenches the enhancement seen on the clean surface.

At smaller separations, the enhancements were found to stay the same until the separation where the surfaces began being attracted to each other. The values then generally decreased significantly by the equilibrium separation, with the values for surface A being largest for the hollow site, and smaller for the bridge and then atop sites. For surface B they were in the opposite order.

After relaxation, the enhancements for all interfaces were found to decrease, becoming more negative as a result of the stronger interaction between the surfaces, giving rise to more spin pairing. Also, the magnetic moment enhancements for S bonding to the same sites on the different surfaces became identical, in line with the changes in geometry and charge density.

Effect of Sulfur Coverage on Adhesion To determine how other coverages of S affect the interfacial properties of Fe, we performed density functional theory calculations of S adsorbed in three adsorption sites (atop, bridge, and four-fold hollow) at two different arrangements, $c(2 \times 2)$ and $p(1 \times 1)$, corresponding to coverages of $1/2$ and 1 ML, respectively. We examine the same parameters as calculated for the $1/4$ ML coverage for interfaces, both in and out of epitaxy.

Different experimental studies of the effect of different coverages of S impurity on the adhesion of different Fe^{143–145} surfaces led to some conflict as to whether it increases or decreases the Fe adhesion. Buckley¹⁴⁴ found that S appreciably decreased the adhesive strength of the Fe(110) interface formed through S segregation at $1/4$ ML coverage and $c(2 \times 4)$ arrangement. In contrast, later studies by Hartweck and Grabke,^{143,145} found that segregated S increased the strength of adhesion of polycrystalline surfaces at submonolayer coverages, showing a maximum in the adhesive force at an estimated S coverage of 0.6 ML. S reduced the strength of adhesion compared to that of the clean surfaces at coverages greater than 1 ML. The differences have been suggested to be due to grain boundary effects.

The adhesion energy curves and UBER parameters calculated from the fitted curve¹⁴⁶ indicate that S reduces the adhesive strength of Fe(110) surfaces in match and mismatch orientations at all coverages examined ($1/4$, $1/2$, and 1 ML). The largest work of separation was for the matching atop interface with $1/2$ ML S coverage. For the mismatching configuration, the bridge $1/2$ ML mismatching interface has the largest work of separation; however, it is still weaker than the strongest matching interface. The mismatching four-fold hollow 1 ML interface has such a low work of separation that it is unlikely to form. The charge-density slices of the matching and mismatching interfaces of the strongest match and mismatch interfaces examined are presented in Fig. 16.14. The magnetic moment enhancement values, μ_B , calculated for the Fe atoms closest to the S atoms on either side of the interface are also indicated.

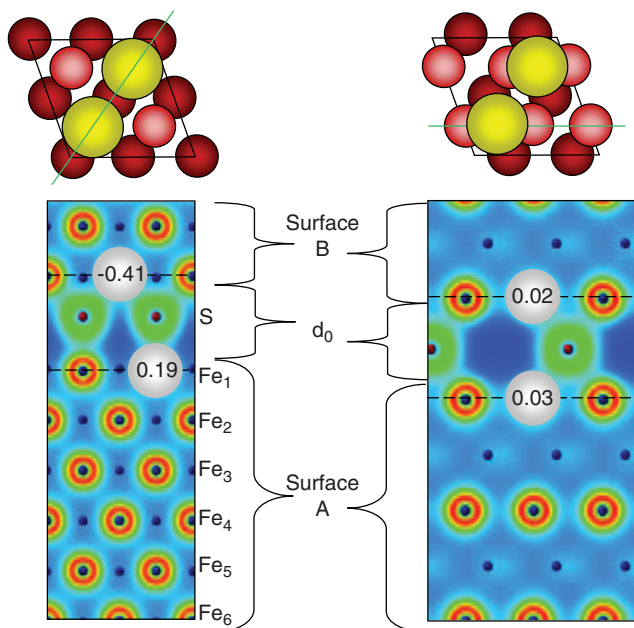


Fig. 16.14 (color online) Charge-density plots of the atop match and bridge mismatch interfaces with $1/2$ -ML S coverage. Slices are taken through the azimuths indicated. The calculated magnetic moment enhancement values, $\Delta\mu_B$, of the Fe atoms closest to the S atoms on either side of the interface are also indicated.

Overall, compared to the results for the clean interface, we found that the interfacial separation was increased by the presence of S. The distance of S from the two surfaces was also found to be related directly to the type of adsorption site in which S sits at the two surfaces.

16.4.5 Effect of Sulfur Impurity on Fe(100) Adhesion: A Brief Summary

We have performed a detailed study of the effects of S on the adhesion of the (100) surface of Fe using methodology similar to that employed for Fe(110), described in Section 16.4.4 and in the literature.¹¹⁹

Adhesion energy calculations show that at $1/2$ ML coverage, S decreases the adhesive energy between the Fe(100) surfaces in both match and mismatch orientations, as was also seen for the Fe(110) match and mismatch interfaces with $1/4$ ML coverage of adsorbed S. The strongest S-contaminated Fe(100) interface was found to be the atop match interface. The difference between the W_{sep} values calculated for the clean and S-contaminated atop and bridge mismatch interfaces, however, was only 6.5%, which is smaller than the difference for the corresponding Fe(110) interfaces. In particular, for these two interfaces (as well as for their matching counterparts), the adhesive attraction was found to be stronger at larger interfacial separations than it was for the corresponding clean interface. Hence,

this indicates that the S-contaminated interfaces can be more prone to adhesion. A complete report of the effects of $1/2$ ML coverage of S on the adhesion properties of Fe(100) surfaces has been published elsewhere.¹¹⁹

16.5 SUMMARY, CONCLUSIONS, AND FUTURE WORK

The results above show that the (100) and (110) surfaces have almost identical surface energies, with the (110) being slightly lower while the (111) surface has the highest energy. The surface relaxation results demonstrate that for the (100) surface a contraction of the outer layer is observed while the second and third layers expand perpendicular to the surface plane; for the (110) surface, little relaxation occurs, indicating that it is essentially bulk cleaved; and for the (111) surface, the first two layers contract while the third expands, with the magnitude of the relaxations being much larger than for the other surfaces.

The layer-resolved magnetic moment values, as well as up- and down-spin-resolved density of states, indicate the presence of an enhanced magnetic moment at the surface which is only slightly affected by relaxation, with the more open (111) surface showing larger changes and the most closely packed (110) surface showing little change.

The adsorption of atomic S on the Fe(100) and Fe(110) surfaces at different adsorbent surface densities at the atop, bridge, and hollow sites shows that for both the Fe(100) and Fe(110) surfaces, the hollow site is the most stable, followed by the bridge and atop sites. At all three sites, S adsorption results in minor surface reconstruction, the most significant being for the hollow site. All three adsorption configurations affect the underlying surface geometry, with S causing a buckling of the top Fe layer when adsorbed in an atop site. Comparisons between S-adsorbed and clean Fe surfaces revealed a reduction in the magnetic moments of surface layer Fe atoms in the vicinity of the S. At the hollow site, the presence of S causes an increase in the surface Fe d-orbital density of states but has no significant effect on the structure and magnetic properties of lower substrate layers.

We have also modeled adhesion energy as a function of surface separation between clean, bulk-terminated Fe(100), Fe(110), and Fe(111) matched and mismatched surfaces. The values of the adhesion parameters obtained suggested that the (110) interface was slightly more stable than the (100) interface. However, the order of stability is reversed if the effects of both matching and mismatching interfaces are taken into consideration, in agreement with experimental findings. The (111) interface in epitaxy is much stronger than the mismatch interface. Compared to the (100) and (110) interfaces, the (111) match interface is strongest, whereas the (111) mismatch interface is the weakest.

In addition, we have examined the relationship between magnetic and electronic properties and adhesion of the Fe(100), Fe(110), and Fe(111) surfaces and found that for matching interfaces, the surface layer magnetic moment is enhanced for larger interfacial separations and decreases to the bulk value as the surfaces are brought together. The enhancement approaches zero at the minimum

adhesion energy, where the bulk solid is formed. The lower layers show smaller enhancements with little or no enhancement at the centre of the slab. The mismatch interfaces show similar behavior, but the enhancement does not reach zero at the equilibrium separation, as the bulk structure is not formed.

To consider the dynamics of the interface formation, we have studied the avalanche effect between Fe(100) surfaces, in match and mismatch, and the role of model constraints on the results. When the central layers of the two surfaces are constrained, the surface layers are attracted toward each other, forming a strained crystal region at intermediate interfacial separations, but if the constraints in the *z*-direction are lifted, the surfaces avalanche together. When the surfaces are allowed to move sideways, an interface initially out of registry (mismatch) will tend to avalanche toward an interface that is in registry (match).

The effects of adsorbed S on the adhesion of Fe(100) and Fe(110) surfaces have been studied by introducing S impurity in atop, bridge, and hollow sites at a range of coverages in match and mismatch interfaces. The calculated minima of the adhesion energy curves show that the presence of S on the surface reduces the strength of the interface. However, the contaminated interfaces can be more prone to adhesion, as the increased adhesive energy values at larger separations show. The effect of adsorbed S on the charge-density distribution and magnetic properties of the interface have also been examined and related to the interfacial geometry. The effect of relaxation of the interfaces at equilibrium was also investigated and was shown to increase the strength of the interface while reducing the equilibrium interfacial separation.

Some recent studies have included modeling of the surface properties of the three low-index faces of Fe^{33,147–150}; experiments and modeling of various properties of Fe nanoparticles,^{151,152} nanowires,¹⁵³ and nanosized clusters¹⁵⁴; adhesion and other properties of high-toughness steels^{155,156}; and the behavior of segregated S at an Fe grain boundary.¹⁵⁷

Finally, it must be emphasized that having developed several approaches to model Fe substrate structures, we can now create various surface defects and impurities as well as controlled modified surface models, with modifications ranging from individual atoms, molecules, nanoclusters, and thin layers to study their effects on the surface and interface properties and the effects of temperature and pressure on the structure and properties of surfaces and interfaces. With the current focus on miniaturization, the ability to modify surfaces atomically for specific applications opens up enormous possibilities for theoretical experimentation with various conditions, surface modifications, and resultant properties, which has a great potential to aid laboratory synthesis and fabrication.

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REFERENCES

1. Baddoo, N. R. *J. Constr. Steel Res.* **2008**, *64*, 1199.
2. Kuziak, R.; Kawalla, R.; Waengler, S. *Arch. Civ. Mech. Eng.* **2008**, *8*, 103.
3. Camley, R. E.; Celinski, Z.; Fal, T.; Glushchenko, A. V.; Hutchison, A. J.; Khivintsev, Y.; Kuanr, B.; Harward, I. R.; Veerakumar, V.; Zagorodnii, V. V. *J. Magn. Mater.* **2009**, *321*, 2048.
4. Grabke, H. J. *Mater. Corros.* **2003**, *54*, 736.
5. Georg, D. *Eng. Aus.* **2000**, *72*, 30.
6. Castle, J. E. *J. Adhes.* **2008**, *84*, 368.
7. Hayashi, S.; Sawai, S.; Iguchi, Y. *ISIJ Int.* **1993**, *33*, 1078.
8. Payne, M. C.; Teter, M. P.; Allan, D. C.; Arias, T. A.; Joannopoulos, J. D. *Rev. Mod. Phys.* **1992**, *64*, 1045.
9. Greeley, J.; Norskov, J. K.; Mavrikakis, M. *Annu. Rev. Phys. Chem.* **2002**, *53*, 319.
10. Gross, A. *Surf. Sci.* **2002**, *500*, 347.
11. Segall, M. D.; Lindan, P. J. D.; Probert, M. J.; Pickard, C. J.; Hasnip, P. J.; Clark, S. J.; Payne, M. C. *J. Phys. Condes. Matter* **2002**, *14*, 2717.
12. Velde, G. T.; Bickelhaupt, F. M.; Baerends, E. J.; Guerra, C. F.; Van Gisbergen, S. J. A.; Snijders, J. G.; Ziegler, T. *J. Comput. Chem.* **2001**, *22*, 931.
13. Nagy, A. *Phys. Rep. Rev. Sec. Phys. Lett.* **1998**, *298*, 2.
14. Ordejon, P. *Phys. Status Solidi B* **2000**, *217*, 335.
15. Schwarz, K.; Blaha, P. *Comput. Mater. Sci.* **2003**, *28*, 259.
16. Pisani, C. *J. Mol. Struct. (Theochem)* **1999**, *463*, 125.
17. Hong, T.; Smith, J. R.; Srolovitz, D. J. *J. Adhes. Sci. Technol.* **1994**, *8*, 837.
18. Hong, T.; Smith, J. R.; Srolovitz, D. J. *Phys. Rev. B* **1993**, *47*, 13615.
19. Raynolds, J. E.; Smith, J. R.; Zhao, G.-L.; Srolovitz, D. J. *Phys. Rev. B* **1996**, *53*, 13883.
20. Hung, A.; Yarovsky, I.; Muscat, J.; Russo, S.; Snook, I.; Watts, R. O. *Surf. Sci.* **2002**, *501*, 261.
21. Spencer, M. J. S.; Hung, A.; Snook, I. K.; Yarovsky, I. *Surf. Sci.* **2002**, *515*, L464.
22. Hong, S. Y.; Anderson, A. B.; Smialek, J. L. *Surf. Sci.* **1990**, *230*, 175.
23. Hong, T.; Smith, J. R.; Srolovitz, D. J. *Phys. Rev. Lett.* **1993**, *70*, 615.
24. Hong, T.; Smith, J. R.; Srolovitz, D. J. *Acta Metall. Mater.* **1995**, *43*, 2721.
25. Raynolds, J. E.; Roddick, E. R.; Smith, J. R.; Srolovitz, D. J. *Acta Mater.* **1999**, *47*, 3281.
26. Smith, J. R.; Cianciolo, T. V. *Surf. Sci.* **1989**, *210*, L229.
27. Smith, J. R.; Hong, T.; Srolovitz, D. J. *Phys. Rev. Lett.* **1994**, *72*, 4021.
28. Smith, J. R.; Raynolds, J. E.; Roddick, E. R.; Srolovitz, D. J. *J. Comput. Aided Mater. Des.* **1996**, *3*, 169.

29. Smith, J. R.; Raynolds, J. E.; Roddick, E. R.; Srolovitz, D. J. *Processing and Design Issues in High Temperature Materials: Proceedings of the Engineering Foundation Conference*, 1997, p. 37.
30. Finnis, M. W. *J. Phys. Condens. Matter* **1996**, 8, 5811.
31. Grochola, G.; Russo, S. P.; Snook, I. K.; Yarovsky, I. *J. Chem. Phys.* **2002**, 117, 7685.
32. Grochola, G.; Russo, S. P.; Snook, I. K.; Yarovsky, I. *J. Chem. Phys.* **2002**, 117, 7676.
33. Grochola, G.; Russo, S. P.; Yarovsky, I.; Snook, I. K. *J. Chem. Phys.* **2004**, 120, 3425.
34. Grochola, G.; Russo, S. P.; Snook, I. K.; Yarovsky, I. *J. Chem. Phys.* **2002**, 116, 8547.
35. Kresse, G.; Furthmuller, J. *Phys. Rev. B* **1996**, 54, 11169.
36. Kresse, G.; Furthmuller, J. *Comput. Mater. Sci.* **1996**, 6, 15.
37. Kresse, G.; Hafner, J. *Phys. Rev. B* **1993**, 48, 13115.
38. Kohn, W.; Sham, L. J. *Phys. Rev.* **1965**, 140, 1133.
39. Perdew, J. P.; Zunger, A. *Phys. Rev. B* **1981**, 23, 5048.
40. Perdew, J. P.; Yue, W. *Phys. Rev. B* **1992**, 45, 13244.
41. Vanderbilt, D. *Phys. Rev. B* **1990**, 41, 7892.
42. Monkhorst, H. J.; Pack, J. D. *Phys. Rev. B* **1976**, 13, 5188.
43. Herper, H. C.; Hoffmann, E.; Entel, P. *Phys. Rev. B* **1999**, 60, 3839.
44. Jansen, H. J. F.; Peng, S. S. *Phys. Rev. B* **1988**, 37, 2689.
45. Dupre, A. *Theorie mechanique de la chaleur*, Gauthier-Villars, Paris, 1869.
46. Rose, J. H.; Smith, J. R.; Ferrante, J. *Phys. Rev. B* **1983**, 28, 1835.
47. Banerjee, A.; Smith, J. R. *Phys. Rev. B* **1988**, 37, 6632.
48. Feibelman, P. J. *Surf. Sci.* **1996**, 360, 297.
49. Shih, H. D.; Jona, F.; Jepsen, D. W.; Marcus, P. M. *Surf. Sci.* **1981**, 104, 39.
50. Shih, H. D.; Jona, F.; Bardi, U.; Marcus, P. M. *J. Phys. C* **1980**, 13, 3801.
51. Legg, K. O.; Jona, F.; Jepsen, D. W.; Marcus, P. M. *J. Phys. C* **1977**, 10, 937.
52. Spencer, M. J. S.; Hung, A.; Snook, I. K.; Yarovsky, I. *Surf. Sci.* **2002**, 513, 389.
53. Sokolov, J.; Jona, F.; Marcus, P. M. *Phys. Rev. B* **1986**, 33, 1397.
54. Xu, C.; O'Connor, D. J. *Nucl. Instrum. Methods Phys. Res.* **1990**, 51, 278.
55. Xu, C.; O'Connor, D. J. *Nucl. Instrum. Methods Phys. Res.* **1991**, 53, 315.
56. Yalisove, S. M.; Graham, W. R. *J. Vac. Sci. Technol. A* **1988**, 6, 588.
57. Rodriguez, A. M.; Bozzolo, G.; Ferrante, J. *Surf. Sci.* **1993**, 289, 100.
58. Johnson, R. A.; White, P. J. *Phys. Rev. B* **1976**, 13, 5293.
59. Kato, S. *Jpn. J. Appl. Phys.* **1974**, 13, 218.
60. Tyson, W. R. *J. Appl. Phys.* **1976**, 47, 459.
61. Tyson, W. R.; Ayres, R. A.; Stein, D. F. *Acta Metall.* **1973**, 21, 621.
62. Haftel, M. I.; Andreadis, T. D.; Lill, J. V.; Eridon, J. M. *Phys. Rev. B* **1990**, 42, 11540.
63. Linford, R. G.; Mitchell, L. A. *Surf. Sci.* **1971**, 27, 142.
64. Schweitz, J. A.; Vingsbo, O. *Mater. Sci. Eng.* **1971**, 8, 275.

65. Gvozdev, A. G.; Gvozdeva, L. I. *Fiz. Met. Metalloved.* **1971**, 31, 640.
66. Avraamov, Y. S.; Gvozdev, A. G. *Fiz. Met. Metalloved.* **1967**, 23, 405.
67. Gilman, J. J. Cleavage, ductility and tenacity in crystals. In *Fracture in Solids*, Averbach, B. L., Felbeck, D. K., Hahn, G. T., and Thomas, B. L., Eds., Wiley, New York, 1959, p. 193.
68. Nicholas, J. F. *Aust. J. Phys.* **1968**, 21, 21.
69. Alden, M.; Skriver, H. L.; Mirbt, S.; Johansson, B. *Surf. Sci.* **1994**, 315, 157.
70. Vitos, L.; Ruban, A. V.; Skriver, H. L.; Kollar, J. *Surf. Sci.* **1998**, 411, 186.
71. Tyson, W. R.; Miller, W. A. *Surf. Sci.* **1977**, 62, 267.
72. Braun, J.; Math, C.; Postnikov, A.; Donath, M. *Phys. Rev. B* **2002**, 65, 184412.
73. Kishi, T.; Itoh, S. *Surf. Sci.* **1996**, 358, 186.
74. Ostroukhov, A. A.; Floka, V. M.; Cherepin, V. T. *Surf. Sci.* **1995**, 333, 1388.
75. Wu, R. Q.; Freeman, A. J. *Phys. Rev. B* **1993**, 47, 3904.
76. Eriksson, O.; Boring, A. M.; Albers, R. C.; Fernando, G. W.; Cooper, B. R. *Phys. Rev. B* **1992**, 45, 2868.
77. Alden, M.; Mirbt, S.; Skriver, H. L.; Rosengaard, N. M.; Johansson, B. *Phys. Rev. B* **1992**, 46, 6303.
78. Freeman, A. J.; Fu, C. L. *J. Appl. Phys.* **1987**, 61, 3356.
79. Ohnishi, S.; Freeman, A. J. *Phys. Rev. B* **1983**, 28, 6741.
80. Wang, C. S.; Freeman, A. J. *Phys. Rev. B* **1981**, 24, 4364.
81. Danan, H.; Herr, A.; Meyer, A. J. *J. Appl. Phys.* **1968**, 39, 669.
82. Binns, C.; Baker, S. H.; Demangeat, C.; Parlebas, J. C. *Surf. Sci. Rep.* **1999**, 34, 107.
83. Wu, R. Q.; Freeman, A. J. *Phys. Rev. Lett.* **1992**, 69, 2867.
84. Shih, H. D.; Jona, F.; Jepsen, D. W.; Marcus, P. M. *Phys. Rev. Lett.* **1981**, 46, 731.
85. Kelemen, S. R.; Kaldor, A. *J. Chem. Phys.* **1981**, 75, 1530.
86. Oudar, J. *Bull. Soc. Fr. Mineral. Cristallogr.* **1971**, 94, 225.
87. Spencer, M. J. S.; Hung, A.; Snook, I.; Yarovsky, I. *Surf. Sci.* **2003**, 540, 420.
88. Spencer, M. J. S.; Snook, I. K.; Yarovsky, I. *J. Phys. Chem. B* **2005**, 109, 9604.
89. Hung, A.; Muscat, J.; Yarovsky, I.; Russo, S. P. *Surf. Sci.* **2002**, 513, 511.
90. Hung, A.; Muscat, J.; Yarovsky, I.; Russo, S. P. *Surf. Sci.* **2002**, 520, 111.
91. Broden, G.; Gafner, G.; Bonzel, H. P. *Appl. Phys.* **1977**, 13, 333.
92. Fischer, R.; Fischer, N.; Schuppler, S.; Fauster, T.; Himpsel, F. J. *Phys. Rev. B* **1992**, 46, 9691.
93. Delchar, T. A. *Surf. Sci.* **1971**, 27, 11.
94. Schonhense, G.; Getzlaff, M.; Westphal, C.; Heidemann, B.; Bansmann, J. *J. Phys.* **1988**, C8, 1643.
95. Weissenrieder, J.; Gothelid, M.; Le Lay, G.; Karlsson, U. O. *Surf. Sci.* **2002**, 515, 135.
96. Berbil-Bautista, L.; Krause, S.; Hanke, T.; Bode, M.; Wiesendanger, R. *Surf. Sci.* **2006**, 600, L20.
97. Taga, Y.; Isogai, A.; Nakajima, K. *Trans. Jpn. Inst. Met.* **1976**, 17, 201.
98. Spencer, M. J. S.; Snook, I.; Yarovsky, I. *J. Phys. Chem. B* **2006**, 110, 956.

99. Sinkovic, B.; Johnson, P. D.; Brookes, N. B.; Clarke, A.; Smith, N. V. *Phys. Rev. B* **1995**, *52*, R6955.
100. Sinkovic, B.; Johnson, P. D.; Brookes, N. B.; Clarke, A.; Smith, N. V. *Phys. Rev. Lett.* **1989**, *62*, 2740.
101. Johnson, P. D.; Clarke, A.; Brookes, N. B.; Hulbert, S. L.; Sinkovic, B.; Smith, N. V. *Phys. Rev. Lett.* **1988**, *61*, 2257.
102. Clarke, A.; Brookes, N. B.; Johnson, P. D.; Weinert, M.; Sinkovic, B.; Smith, N. V. *Phys. Rev. B* **1990**, *41*, 9659.
103. Fujita, D.; Ohgi, T.; Homma, T. *Appl. Surf. Sci.* **2002**, *200*, 55.
104. Zhang, X. S.; Terminello, L. J.; Kim, S.; Huang, Z. Q.; Vonwittenau, A. E. S.; Shirley, D. A. *J. Chem. Phys.* **1988**, *89*, 6538.
105. Didio, R. A.; Plummer, E. W.; Graham, W. R. *Phys. Rev. Lett.* **1984**, *52*, 683.
106. Legg, K. O.; Jona, F.; Jepsen, D. W.; Marcus, P. M. *Surf. Sci.* **1977**, *66*, 25.
107. Grabke, H. J.; Paulitschke, W.; Tauber, G.; Viehhaus, H. *Surf. Sci.* **1977**, *63*, 377.
108. Grabke, H. J.; Petersen, E. M.; Srinivasan, S. R. *Surf. Sci.* **1977**, *67*, 501.
109. Didio, R. A.; Plummer, E. W.; Graham, W. R. *J. Vac. Sci. Technol. A* **1984**, *2*, 983.
110. Fernando, G. W.; Wilkins, J. W. *Phys. Rev. B* **1986**, *33*, 3709.
111. Fernando, G. W.; Wilkins, J. W. *Phys. Rev. B* **1987**, *35*, 2995.
112. Kishi, T.; Itoh, S. *Surf. Sci.* **1996**, *363*, 100.
113. Huff, W. R. A.; Chen, Y.; Zhang, X. S.; Terminello, L. J.; Tao, F. M.; Pan, Y. K.; Kellar, S. A.; Moler, E. J.; Hussain, Z.; Wu, H.; Zheng, Y.; Zhou, X.; von Wittenau, A. E. S.; Kim, S.; Huang, Z. Q.; Yang, Z. Z.; Shirley, D. A. *Phys. Rev. B* **1997**, *55*, 10830.
114. Chubb, S. R.; Pickett, W. E. *J. Appl. Phys.* **1988**, *63*, 3493.
115. Chubb, S. R.; Pickett, W. E. *Phys. Rev. B* **1988**, *38*, 10227.
116. Chubb, S. R.; Pickett, W. E. *Phys. Rev. B* **1988**, *38*, 12700.
117. Anderson, A. B.; Hong, S. Y. *Surf. Sci.* **1988**, *204*, L708.
118. Hong, S. Y.; Anderson, A. B. *Phys. Rev. B* **1988**, *38*, 9417.
119. Nelson, S. G.; Spencer, M. J. S.; Snook, I.; Yarovsky, I. *Surf. Sci.* **2005**, *590*, 63.
120. Todorova, N.; Spencer, M. J. S.; Yarovsky, I. Dynamic properties of the sulfur-contaminated Fe(110) surface. In *Proceedings of the Australian Institute of Physics 16th Biennial Congress*, Canberra, Australia, 2005.
121. Todorova, N.; Spencer, M. J. S.; Yarovsky, I. *Surf. Sci.* **2007**, *601*, 665.
122. Verlet, L. *Phys. Rev.* **1967**, *159*, 98.
123. Nose, S. *Prog. Theor. Phys. Suppl.* **1991**, *1*.
124. Jiang, D. E.; Carter, E. A. *J. Phys. Chem. B* **2004**, *108*, 19140.
125. Kamakoti, P.; Sholl, D. S. *J. Membr. Sci.* **2003**, *225*, 145.
126. Haug, K.; Jenkins, T. *J. Phys. Chem. B* **2000**, *104*, 10017.
127. Spencer, M. J. S.; Todorova, N.; Yarovsky, I. *Surf. Sci.* **2008**, *602*, 1547.
128. Spencer, M. J. S.; Yarovsky, I. *J. Phys. Chem. C* **2007**, *111*, 16372.
129. Narayan, P. B. V.; Anderegg, J. W.; Chen, C. W. *J. Electron Spectrosc. Relat. Phenom.* **1982**, *27*, 233.
130. Shanabarger, M. R. A comparison of adsorption kinetics on iron of H₂ and H₂S. In *Hydrogen Effects in Metals*, Bernstein, J. M., and Thompson, A. W., Eds., The Metallurgical Society of AIME, Warrendale, PA, 1981, p. 135.

131. Spencer, M. J. S.; Hung, A.; Snook, I.; Yarovsky, I. *Surf. Rev. Lett.* **2003**, *10*, 169.
132. Spencer, M. J. S.; Snook, I. K.; Yarovsky, I. *J. Phys. Chem. B* **2004**, *108*, 10965.
133. *Handbook of Chemistry and Physics*, 70th ed., CRC Press, Metals Park, OH, 1989–1990.
134. Taylor, P. A.; Nelson, J. S.; Dodson, B. W. *Phys. Rev. B* **1991**, *44*, 5834.
135. Taylor, P. A. *Phys. Rev. B* **1991**, *44*, 13026.
136. Smith, J. R.; Bozzolo, G.; Banerjea, A.; Ferrante, J. *Phys. Rev. Lett.* **1989**, *63*, 1269.
137. Good, B. S.; Banerjea, A.; Smith, J. R.; Bozzolo, G.; Ferrante, J. *Mater. Res. Soc. Symp. Proc.* **1990**, *193*, 313.
138. Lynden-Bell, R. M. *Surf. Sci.* **1991**, *244*, 266.
139. Good, B. S.; Banerjea, A. *J. Phys. Condens. Matter* **1996**, *8*, 1325.
140. Banerjea, A.; Good, B. S. *Int. J. Mod. Phys. B* **1997**, *11*, 315.
141. Banerjea, A.; Good, B. S. *Indian J. Phys.* **1995**, *69A*, 105.
142. Nelson, J. S.; Dodson, B. W.; Taylor, P. A. *Phys. Rev. B* **1992**, *45*, 4439.
143. Hartweck, W.; Grabke, H. J. *Surf. Sci.* **1979**, *89*, 174.
144. Buckley, D. H. *Int. J. Nondestructive Test.* **1970**, *2*, 171.
145. Hartweck, W. G.; Grabke, H. J. *Acta Metall.* **1981**, *29*, 1237.
146. Spencer, M. J. S.; Snook, I. K.; Yarovsky, I. *J. Phys. Chem. B* **2005**, *109*, 10204.
147. Jiang, D. E.; Carter, E. A. *Surf. Sci.* **2003**, *547*, 85.
148. Zhang, J. M.; Ma, F.; Xu, K. W. *Surf. Interface Anal.* **2003**, *35*, 662.
149. Blonski, P.; Kiejna, A. *Vacuum* **2004**, *74*, 179.
150. Wang, X. C.; Jia, Y.; Qian, Y.; Wang, F.; Ma, J. X.; Hu, X. *Surf. Sci.* **2004**, *551*, 179.
151. Postnikov, A. V.; Entel, P.; Soler, J. M. *Eur. Phys. J. D* **2003**, *25*, 261.
152. Postnikov, A. V. Surface relaxation in solids and nanoparticles. In *Computational Materials Science*, Vol. 187, Catlow, R., and Kotomin, E., Eds., IOS Press, Amsterdam, 2003, p. 245.
153. Mohaddes-Ardabili, L.; Zheng, H.; Ogale, S. B.; Hannoyer, B.; Tian, W.; Wang, J.; Lofland, S. E.; Shinde, S. R.; Zhao, T.; Jia, Y.; Salamanca-Riba, L.; Schlom, D. G.; Wuttig, M.; Ramesh, R. *Nat. Mater.* **2004**, *3*, 533.
154. De Hosson, J. T. M.; Palasantzas, G.; Vystavel, T.; Koch, S. *JOM* **2004**, *56*, 40.
155. Hao, S.; Moran, B.; Liu, W. K.; Olson, G. B. *J. Comput. Aided Mater. Des.* **2003**, *10*, 99.
156. Hao, S.; Liu, W. K.; Moran, B.; Vernerey, F.; Olson, G. B. *Comput. Methods Appl. Mech. Eng.* **2004**, *193*, 1865.
157. Gesari, S. B.; Pronisato, M. E.; Juan, A. *J. Phys. Chem. Solids* **2004**, *65*, 1337.

17 Surface Chemistry and Catalysis from Ab Initio–Based Multiscale Approaches

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Chemical problems involving heterogeneous catalysis, diffusion, and related processes occur in systems that are too large to simulate using electronic structure methods directly, requiring either the use of prohibitively large samples and/or prohibitively long simulation times. However, methods such as density functional theory, augmented by statistical mechanics techniques such as kinetic Monte Carlo, can directly address the critical issues using multiscale techniques. As a result, phase diagrams for catalytic processes can be calculated and used to model real-time catalytic processes. Significant applications considered include CO catalytic conversion, hydrogen storage, and fuel cell operation.

17.1 INTRODUCTION

Theory, computation, and simulation have been identified repeatedly in international reports and technology road maps as key components of a successful strategy toward the implementation of new energy technologies.^{1,2} Indeed, they play a crucial role in the advancement and development of all new technologies that require knowledge and understanding on the atomic level as well as on the nanoscale. Materials by design and the growing, exciting role of computation/simulation are making impacts across multidisciplinary fields such as physics, chemistry, engineering, and biology.

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Advances in catalytic science laid the foundation for the rapid development of the petroleum and chemical industries in the twentieth century, which contributed directly to the substantial increase in the standard of living in industrialized countries. Traditionally, catalytic science has progressed through trial and error, requiring many thousands of experiments involving complex combinations of metals, metal compounds, promoters, and inhibitors.³ With increased awareness of the need for new and improved green energy technologies and processes for an environmentally clean and sustainable future, catalysis researchers are focusing on ways to improve existing applications and develop new ones. Control and understanding on the atomic level of surface and material properties is crucial for the development of cutting-edge technologies. Lack of such knowledge presently hinders further progress in already established applications and prevents real advances in promising ones which are still at the conceptual level. Modern imaging and spectroscopic techniques are being extended to operate under increasingly realistic conditions (e.g., high pressures, high temperatures),⁴ and can provide quantitative information at an unprecedented level. However, determination of important properties such as adsorption and reaction energetics, structure of surface species, and the nature of transient intermediates and transition states are still highly challenging.

Increasingly, accurate quantum mechanical calculations are being used to investigate such quantities and to predict new materials and structures that may lead to improved efficiencies and selectivities. Indeed, an ultimate goal of catalysis and materials research is to control chemical reactions and materials properties so that one can synthesize any desired molecule or material. Understanding the mechanisms and dynamics of such transformations has been identified as a grand challenge for catalysis and advanced materials research.⁵ Calculation methods derived from advanced theoretical models and implemented in efficient algorithms are crucial for fundamental understanding and ultimately for steps toward first-principles design. By combining density functional theory (DFT) calculations with statistical mechanical approaches, phenomena and properties occurring on macroscopic length and long time scales can be achieved, affording accurate predictions of surface structures, phase transitions, diffusion, and increasingly, heterogeneous catalysis.^{6–10}

The present chapter contains some recent applications of first-principles-based multiscale modeling approaches for describing and predicting surface structures, phase transitions, and catalysis. In particular, through specific applications, these approaches are highlighted: (1) *ab initio* atomistic thermodynamics, which predicts stable (and metastable) phases, from a pool of considered structures, in equilibrium with a gas-phase environment; (2) the *ab initio* lattice-gas Hamiltonian plus equilibrium Monte Carlo method, which can predict stable surface structures (without their explicit consideration), including order–disorder phase transitions; and (3) *ab initio* kinetic Monte Carlo simulations, which in addition to the above can describe the kinetics of a system (e.g., reaction rates).

17.2 PREDICTING SURFACE STRUCTURES AND PHASE TRANSITIONS

17.2.1 Oxygen on Pd(111): The Lattice-Gas Plus Monte Carlo Approach

The surface structures that form on adsorbing species on a solid surface are dictated by the lateral interactions between them. Such interactions can also significantly affect the stability of the adsorption phase and thus affect the surface function and properties. This has important consequences, for example, for heterogeneous catalysis, which involves surface processes such as adsorption, diffusion, desorption, and chemical reactions. In particular, the carbon monoxide oxidation reaction has long served as a prototypical “simple” chemical reaction for experimental study, with the aim of achieving a deeper understanding on the microscopic level.¹¹ This reaction is the basic reaction step in many industrial reactions and is also an important reaction in its own right, as illustrated, for example, by the fact that it is one of the main reactions that the three-way automotive catalytic converter catalyzes for pollution control and environmental protection. If atomic oxygen, adsorbed on transition metal surfaces, is exposed to CO gas, the metal catalyzes the formation of carbon dioxide through a Langmuir–Hinshelwood mechanism, in which both reactants are adsorbed on the surface prior to product formation, in this case CO₂.¹² The activation energy of this reaction depends on the coverage of adsorbates, indicating that the lateral interactions are significant.¹³

In particular, for the O/Pd(111) system, it was found that upon exposure to CO, the $p(2 \times 2)$ islands, which initially form on adsorption of oxygen, compress into $(\sqrt{3} \times \sqrt{3})R30^\circ$ (hereafter denoted by “ $\sqrt{3}$ ”) domains and finally into $p(2 \times 1)$ domains.¹⁴ These structural rearrangements have profound effects on the reactivity of CO₂ formation: While the $p(2 \times 2)$ phase is unreactive for temperatures in the range 190 to 320 K, the $\sqrt{3}$ phase displays half-order kinetics with respect to oxygen coverage, suggesting that the reaction site is at the periphery of the O islands. For the $p(2 \times 1)$ phase, the reaction is first order, implying that the reaction proceeds uniformly over the O islands. As an initial step toward a detailed understanding of the role played by lateral interactions in the CO oxidation reaction over Pd(111), it is appropriate to investigate the behavior of the system in the presence of just the oxygen adsorbate.

In the following, the lattice-gas Hamiltonian plus (LGH) Monte Carlo (MC) approach^{15,16} will be used to describe the O/Pd(111) system and to predict order–disorder phase transition temperatures for varying oxygen coverages.¹⁷ Such an approach affords identification of unanticipated geometries and stoichiometries and can be used to describe the coexistence of phases and disordered phases, as well as associated order–order and order–disorder phase transitions. The first step is to create a sufficiently accurate lattice-gas Hamiltonian (LGH),

which can be written as

$$H^{\text{LGH}} = V^1 \sum_i n_i + \sum_{m=1}^r V_m^2 \sum_{\langle ij \rangle_m} n_i n_j + \sum_{m=1}^q V_m^3 \sum_{\langle ijk \rangle_m} n_i n_j n_k + \cdots \quad (17.1)$$

where n_i indicates the occupation of site i , which is 0 if the site is empty or 1 if it is occupied; V^1 is the one-body term, which represents the adsorption energy of the isolated adsorbate; V_m^2 are the two-body, or pair, interactions (where r pair interactions are considered, with $m = 1$ corresponding to nearest-neighbor interactions, $m = 2$ second nearest-neighbor interactions, and so on); V_m^3 are the three-body, or trio, interactions (where q trio interactions are considered); and so on. The LGH [Eq. (17.1)] contains an infinite number of terms, but in practice it can be truncated, since higher-order interactions become negligible compared to the lower-order terms. The interactions considered to describe the O/Pd(111) system are illustrated in Fig. 17.1. The values of the interactions are determined from least-squares fits of energies for structures calculated using density functional theory, with oxygen coverages ranging from $\frac{1}{9}$ monolayer (ML) to 1 ML. To determine which interactions to include in the expansion, and to evaluate the accuracy of the LGH, we use the leave-one-out cross-validation (LOO-CV) scheme (see Refs. 18–21). It is found for this system that the set of interaction

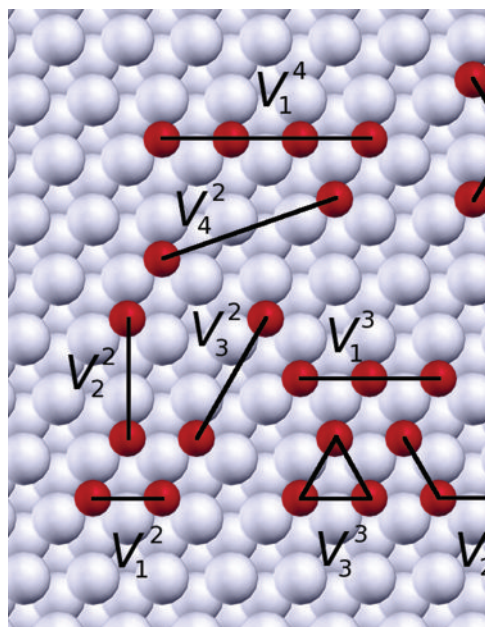


Fig. 17.1 (color online) Top view of the oxygen adsorbates on Pd(111), where the lateral interactions between O atoms considered in the lattice-gas Hamiltonian are shown. Light gray spheres represent Pd atoms, and small dark spheres, O atoms. (From Ref. 17.)

parameters which yield a high accuracy consist of six lateral interactions: three two-body interactions (V_1^2, V_2^2, V_3^2 , with respective values of 244, 39, and -6 meV; see Fig. 17.1), and three three-body interactions (V_1^3, V_2^3, V_3^3 , with values 31, 30–49 meV) interactions.¹⁷ It is interesting to see that the values of the two-body interactions are remarkably similar to what has been reported for O/Pt(111) (238, 39, -6 meV)¹⁸ and for the O/Ru(0001) system (265, 44, -25 meV).²²

Once the LGH has been constructed, its reliability can be tested by calculating the *ground-state line* (or *convex hull*), which identifies the lowest-energy surface structures for a given coverage. In particular, it can be observed whether it correctly reproduces that obtained directly from DFT. The formation energies (from DFT or the LGH) are calculated as

$$\Delta E_f = \Theta[E_b^{\text{O/Pd}} - E_b^{\text{O}(1 \times 1)/\text{Pd}}] \quad (17.2)$$

which shows the stability of a structure with respect to phase separation into a fraction Θ of the full monolayer $\text{O}(1 \times 1)/\text{Pd}$ and a fraction, $1 - \Theta$, of the clean slab. In Eq. (17.2), E_b represent the binding energy per oxygen atom of a given oxygen adsorption structure on the Pd(100) surface. For example, the binding energy of oxygen on a surface with 1 ML coverage is given by $E_b^{\text{O}(1 \times 1)/\text{Pd}} = E_{\text{tot}}^{\text{O}(1 \times 1)/\text{Pd}} - E_{\text{tot}}^{\text{Pd}} - 1/2 E_{\text{tot}}^{\text{O}_2}$, where $E_{\text{tot}}^{\text{O}(1 \times 1)/\text{Pd}}$, $E_{\text{tot}}^{\text{Pd}}$, and $E_{\text{tot}}^{\text{O}_2}$ are the total energies of the $\text{O}(1 \times 1)/\text{Pd}(100)$ structure, the clean Pd(100) surface, and an oxygen molecule, respectively. In Fig. 17.2, the formation energy as a function of oxygen coverage is shown. From it, the structures belonging to the convex hull (lowest-energy line) can be identified. All structures with a formation energy higher than that for the same coverage are unstable against phase

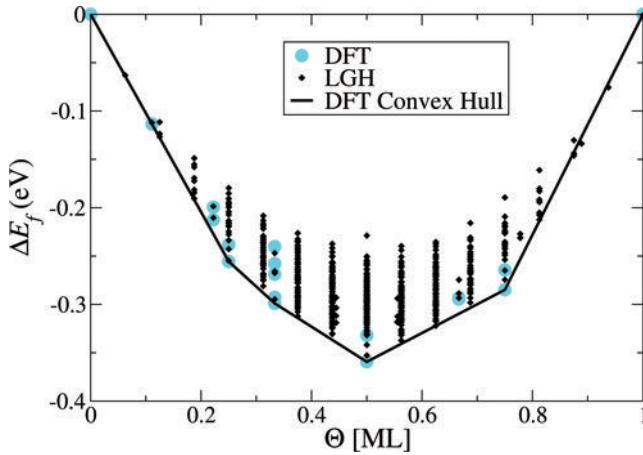


Fig. 17.2 (color online) Formation energy, ΔE_f , versus coverage, Θ , of the twenty-two structures calculated directly from density-functional theory (DFT) (large pale dots) and those obtained from the lattice-gas Hamiltonian (LGH). The continuous (lowest energy) line represents the convex hull. (From Ref. 17.)

separation into the two closest structures belonging to the convex hull. It can be seen that there is an excellent agreement between the DFT and the LGH formation energies, except for very high coverages, where there are large atomic relaxations which are difficult to capture in the LGH. The ground-state geometries lying on the convex hull are the $p(2 \times 2)$, $\sqrt{3}$, and $p(2 \times 1)$ structures. The former two agree with experimental results.²³ The $p(2 \times 1)$ structure is also observed experimentally, but only, for example, when the O/Pd(111) system is exposed to CO gas.^{14,23} Importantly, both DFT and the LGH calculations predict the same ground-state structures, indicating that the LGH is sufficiently accurate to describe the correct ordering of the adsorbates on the surface.

Having constructed the LGH, it can be used, for example, to predict temperature-driven phase transitions. Although there are no experimental results for the O/Pd(111) system published to date, it can be expected, for example, that configurational entropy will drive a phase transition to a disordered phase at elevated temperatures. Such phase transitions have been reported for O/Ru(0001),^{15,24} where it was shown that the transition temperature depends strongly on the oxygen coverage. For this latter system, two peaks occur, one at 0.25 ML ($\simeq 800$ K) and the other at 0.50 ML ($\simeq 600$ K), which correspond to the stable $p(2 \times 2)$ and $p(2 \times 1)$ phases. Qualitatively, the same behavior was found for the O/Pt(111) system through similar theoretical simulations.¹⁸ Also, the O/Ni(111) system forms a stable $p(2 \times 2)$ structure, which exhibits a pronounced peak in the order–disorder transition temperature versus coverage curve.²⁵

To investigate order–disorder phase transitions, Monte Carlo (MC) simulations can be carried out. In particular, we employ the Wang–Landau scheme, which affords an efficient evaluation of the configurational density of states, $g(E)$, (i.e., the number of system configurations with a certain energy, E).^{26–29} From this, all major thermodynamic functions can be directly calculated, including the free energy,

$$F(T) = -k_B T \ln \sum_E g(E) e^{-E/k_B T} = k_B T \ln(Z) \quad (17.3)$$

where Z is the partition function, k_B is the Boltzmann constant, and T is the temperature. The internal energy is given as

$$U(T) = \langle E \rangle_T = \frac{\sum_E E g(E) e^{-E/k_B T}}{Z} \quad (17.4)$$

the specific heat as

$$C_v(T) = \frac{\langle E^2 \rangle_T - \langle E \rangle_T^2}{T^2} \quad (17.5)$$

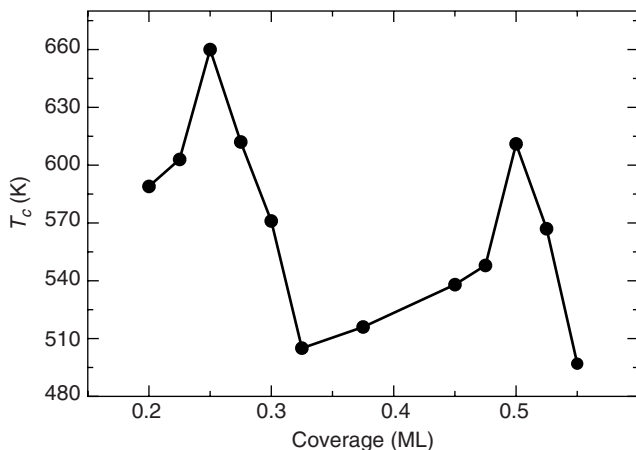


Fig. 17.3 Order–disorder transition temperature, T_c , as a function of the oxygen coverage. (From Ref. 17.)

and the entropy as

$$X = \frac{U - F}{T} \quad (17.6)$$

Using the Wang–Landau scheme for a given coverage, a single simulation yields $g(E)$ and hence the transition temperature, T_c , while in traditional MC studies based on the Metropolis algorithm, one needs to perform a series of simulations at various temperatures to check the variations of a properly defined order parameter. From the divergence of the specific heat at the order–disorder transition temperature, the dependence on coverage of the transition temperature is obtained as shown in Fig. 17.3. In this figure two pronounced peaks occur, corresponding to the $p(2 \times 2)$ and $p(2 \times 1)$ phases. As noted above, to date, no experimental results have been reported for order–disorder phase-transition temperatures as a function of coverage for this system; thus, the predictions in Fig. 17.3 await experimental confirmation.

A similar theoretical approach has been used to study the O/Pd(100) system.¹⁹ This study was limited to low oxygen coverages (i.e., 0 to 0.35 ML), but a similar peak of T_c at 0.25 ML was observed. Zhang et al.,¹⁹ through comparison with experiment and from investigation of different theoretical treatments found that the main source of uncertainty in the lateral interactions is the exchange-correlation functional employed, and other approximations, such as a finite number of lateral interactions, neglect of vibrational contributions, and neglect of population of other sites besides the most favorable one, have relatively negligible effects.

17.3 SURFACE PHASE DIAGRAMS FROM AB INITIO ATOMISTIC THERMODYNAMICS

17.3.1 Ag–Cu Alloy Surface and Chemical Reactions in an Oxygen and Ethylene Atmosphere

The ab initio atomistic thermodynamics approach describes systems in thermodynamic equilibrium, taking into account the effect of the atmosphere or “environment” (e.g., a gas phase of one or more species) through the chemical potential.^{30–35} This method uses results from first-principles electronic structure theory to calculate the Gibbs free energy. Various surface structures can be compared to determine which is the most stable for certain temperature and gas pressure conditions, which is correlated to the chemical potential. It is an indirect approach in that its reliability depends on the structures explicitly considered. These structures are restricted to being ordered, due to the periodic boundary conditions employed in the supercell approach which most modern density functional theory codes use. Despite these restrictions, it represents a very valuable first step in the study of surfaces under realistic conditions.

In the following, this approach is used for the study of ethylene epoxidation over an Ag–Cu alloy catalyst. On the basis of experiments and first-principles calculations, it has been proposed that if an Ag–Cu alloy is used instead of the traditional Ag catalyst, the selectivity toward ethylene oxide is improved. Experimentally, it was shown through ex situ x-ray photoelectron spectroscopy (XPS) measurements that the copper surface content is much higher than the overall content of the alloy, indicating copper segregation to the surface.³⁶ This led to the theoretical consideration of a model in which one out of four silver atoms is replaced by a copper atom (i.e., representing a two-dimensional surface alloy).^{37,38} At the temperatures and pressures used in the experiments (e.g., ~530 K, 0.1 atm), however, copper oxidizes to CuO, and at higher temperatures or lower pressures, to Cu₂O. Therefore, it is possible that more complex structures are present on the catalyst surface. Indeed, our recent studies show that a two-dimensional Ag–Cu surface alloy is not stable in an environment containing oxygen and ethylene at temperatures and pressures relevant for industrial applications, as explained below. Rather, the results show that thin surface copper oxide–like films form. These predictions are supported by recent XPS measurements and high-resolution transmission electron microscopy results.³⁹

As a first step into the theoretical study of this system, the Ag–Cu alloy surfaces are considered in contact with a pure oxygen environment. As a second step, the effect of the ethylene gas phase is investigated. The most stable surface structures are those that minimize the change in the Gibbs surface free energy,

$$\Delta G(\mu_{\text{O}}) = \frac{1}{A} (G^{\text{O/Cu/Ag}} - G^{\text{slab}} - \Delta N_{\text{Ag}} \mu_{\text{Ag}} - N_{\text{Cu}} \mu_{\text{Cu}} - N_{\text{O}} \mu_{\text{O}}) \quad (17.7)$$

where ΔN_{Ag} is the difference in the number of Ag atoms between the adsorption system and the clean Ag slab, and N_{Cu} is the number of Cu atoms. μ_{Cu} , μ_{Ag} ,

and μ_{O} are the copper, silver, and oxygen chemical potentials, respectively. The Ag and Cu chemical potentials are taken to be that of an Ag and Cu atom in the respective bulk material. This assumes that the system is in equilibrium with bulk Ag, which acts as the reservoir. $G^{\text{O/Cu/Ag}}$ and G^{slab} are the free energies of the adsorbate structure and the clean Ag slab, respectively. Normalization to the surface area, A , allows comparison of structures with different unit cells. The temperature and pressure dependence enters through the oxygen chemical potential,³¹

$$\mu_{\text{O}}(T, p) = \frac{1}{2} \left[E_{\text{O}_2}^{\text{total}}(T, p^0) + \tilde{\mu}_{\text{O}_2}(T, p^0) + k_{\text{B}} T \ln \frac{p_{\text{O}_2}}{p^0} \right] \quad (17.8)$$

Here p^0 is the standard pressure (1 atm) and $\tilde{\mu}_{\text{O}_2}(T, p^0)$ is the chemical potential at the standard pressure. This can be obtained either from thermochemical tables⁴⁰ (as done in this case) or calculated directly. Contributions to the free energy due to vibrations should be taken into account. For O/Ag³⁴ and O/Cu⁴¹ systems studied in the literature, such contributions have been shown to be sufficiently small (e.g., $< 10 \text{ meV}/\text{\AA}^2$) as not to play an important role. This was also found for two of the most favorable O/Cu/Ag structures, CuO(1L) and $p2$, described below).³⁰ Neglecting vibrational contributions as done for the O/Cu/Ag system results in the free energies being approximated by the total energies, $E^{\text{O/Cu/Ag}}$ and E^{slab} .

Defining the average oxygen binding energy of a surface structure as

$$E_b^{\text{O/Cu/Ag}} = -\frac{1}{N_{\text{O}}} [E^{\text{O/Cu/Ag}} - (E^{\text{slab}} + \Delta N_{\text{Ag}} \mu_{\text{Ag}} + N_{\text{Cu}} \mu_{\text{Cu}} + N_{\text{O}} E^{\text{O}})] \quad (17.9)$$

where E^{O} is the total energy of a free O atom, the change in Gibbs surface free energy can be written

$$\Delta G(\Delta \mu_{\text{O}}) = -\frac{1}{A} (N_{\text{O}} E_b^{\text{O/Cu/Ag}} + N_{\text{O}} \Delta \mu_{\text{O}}) \quad (17.10)$$

where the oxygen chemical potential is now referenced with respect to half the total energy of an isolated O_2 molecule: $\Delta \mu_{\text{O}} = \mu_{\text{O}} - \frac{1}{2} E_{\text{O}_2}^{\text{total}}$.

Through considering a host of O/Cu/Ag(111) surface structures, the change in Gibbs surface free energy can be plotted as a function of oxygen chemical potential (see Fig. 17.4). The lower the value of the free energy, the more favorable (stable) the structure is. The slope of the lines is proportional to the oxygen coverage; that is, the higher the oxygen content, the steeper the slope. The vertical dashed lines indicate the value of the chemical potential above which Cu oxidizes to Cu_2O and to CuO . In this figure, the pressures corresponding to the oxygen chemical potential are shown for three selected temperatures, 300, 600, and 900 K. From Fig. 17.4 it can therefore be predicted that the thermodynamically stable structures are: clean Ag(111) for $\Delta \mu_{\text{O}} < -1.43 \text{ eV}$, $p2$ and $p4\text{-OCu}_3$ structures (almost degenerate) between $-1.43 < \Delta \mu_{\text{O}} < -1.26 \text{ eV}$, bulk copper(I) oxide Cu_2O for $-1.26 < \Delta \mu_{\text{O}} < -1.23 \text{ eV}$, and bulk copper(II) oxide

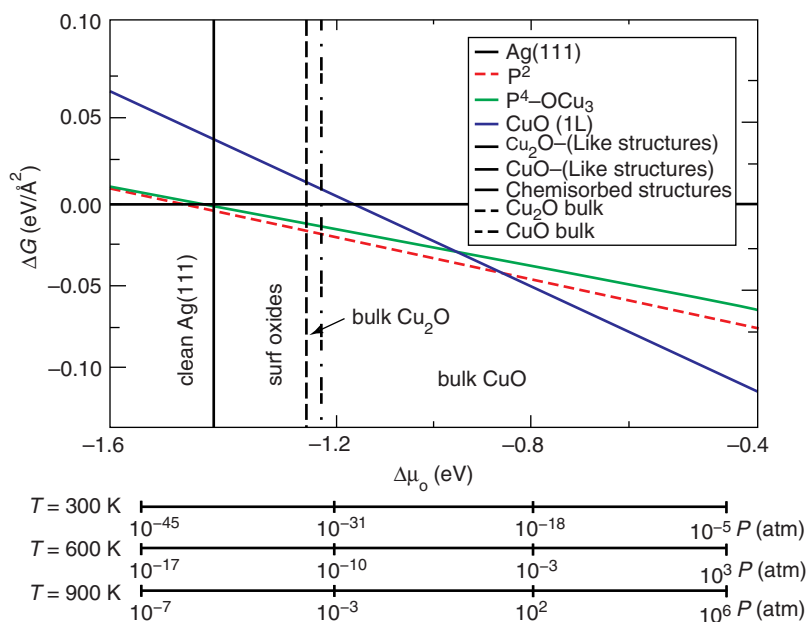


Fig. 17.4 (color online) Change in Gibbs surface free energy, ΔG , as a function of the change in oxygen chemical potential, $\Delta\mu_{\text{O}}$. The vertical dashed lines separate the regions of stability of the clean Ag(111) surface, the surface oxides, and bulk oxides Cu_2O and CuO. A detailed description of the surface atomic structures is given in Ref. 30 and some are depicted in Fig. 17.5. (From Ref. 30.)

CuO for $\Delta\mu_{\text{O}} > -1.23$ eV. Figures 17.5a and 17.5b show the atomic geometry of the $p2$ and $p4\text{-OCu}_3$ structures, as well as a CuO-like structure CuO(1L) (Fig. 17.5c), which is like a layer of bulk CuO forced to match the (2×2) lattice of the underlying Ag(111) surface. Also shown is a structure with 1 ML of Cu and 1 ML of O on top of the Cu layer, labeled O1ML (Fig. 17.5d). It is worth noting that in the absence of oxygen, Cu prefers to be located in the subsurface layer, that is, beneath the outermost Ag layer, but when there is oxygen in the atmosphere, the copper atoms segregate to the surface and form thin surface oxide-like structures. Moreover, a two-dimensional surface Ag–Cu alloy is not stable anywhere in the range of chemical potential considered. On the other hand, there is a narrow region in which two-dimensional O–Cu surface oxides are stable. This is indicated in Fig. 17.4 by the region labeled “surface oxides.” In this region thin O–Cu structures have the lowest Gibbs surface free energy.

The results presented in Fig. 17.4 correspond to the situation where there is no limit to the Cu concentration. For the Ag–Cu alloy catalysts, however, there is only $\approx 2.5\%$ Cu. At the surface, in an oxygen and reaction atmosphere, it is estimated from experiment that the surface has around 50 times more Cu atoms compared to the nominal bulk component. Moreover, from XPS studies, the Cu content on the surface is suggested to be in the range 0.1 to 0.75 ML.⁴²

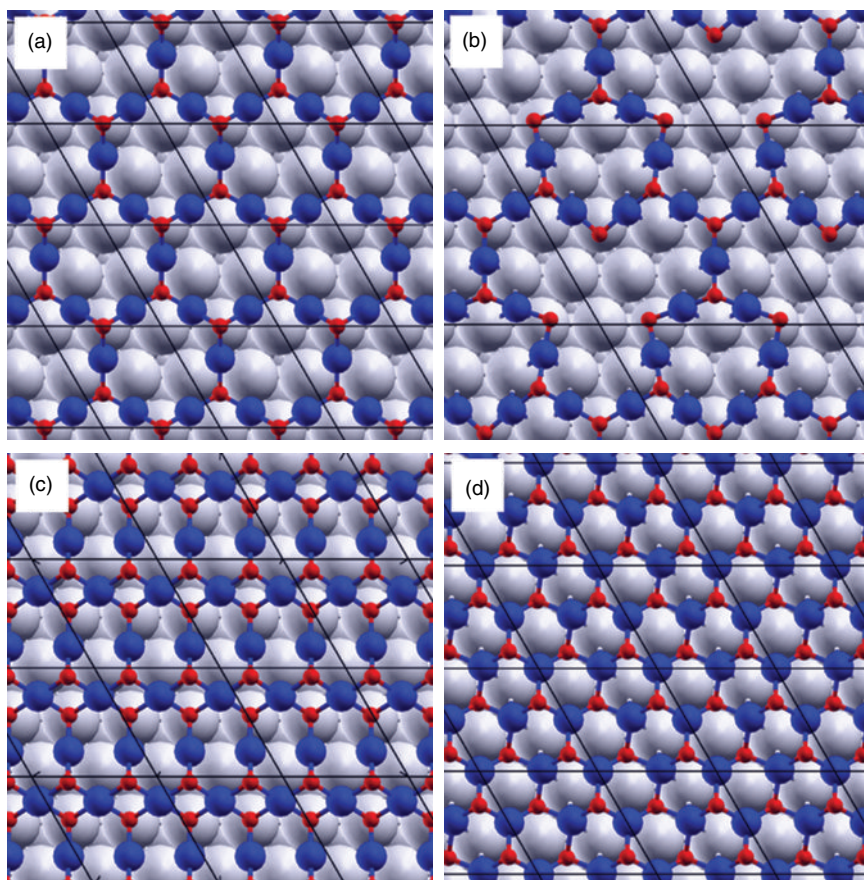


Fig. 17.5 (color online) Top view of four surface structures considered: (a) $p2$; (b) $p4\text{-OCu}_3$; (c) $\text{CuO}(1\text{L})$; (d) O1ML/Cu1ML . The gray spheres represent the underlying $\text{Ag}(111)$ substrate. Copper atoms are shown as large dark circles, and oxygen atoms are the small dark circles. The black lines represent the surface unit cells. (From Ref. 30.)

To consider explicit Cu concentrations in the theory, we can use the results of Fig. 17.4 to determine the structures that will be present on the surface as a function of copper content and the oxygen chemical potential. In doing this, published results for many O–Ag structures were also utilized for the system in the absence of copper.

To construct such a surface phase diagram, for a given value of the oxygen chemical potential, the surface free energy is plotted versus the copper content in the various considered structures. From this, the convex hull of the stable structures can be identified. By repeating this for the other values of the oxygen chemical potential in the range considered, the phase diagram as a function of the oxygen chemical potential and Cu content can be constructed. This is shown in Fig. 17.6. It can be seen that for a value of $\Delta\mu_{\text{O}} = -0.61\text{eV}$, which

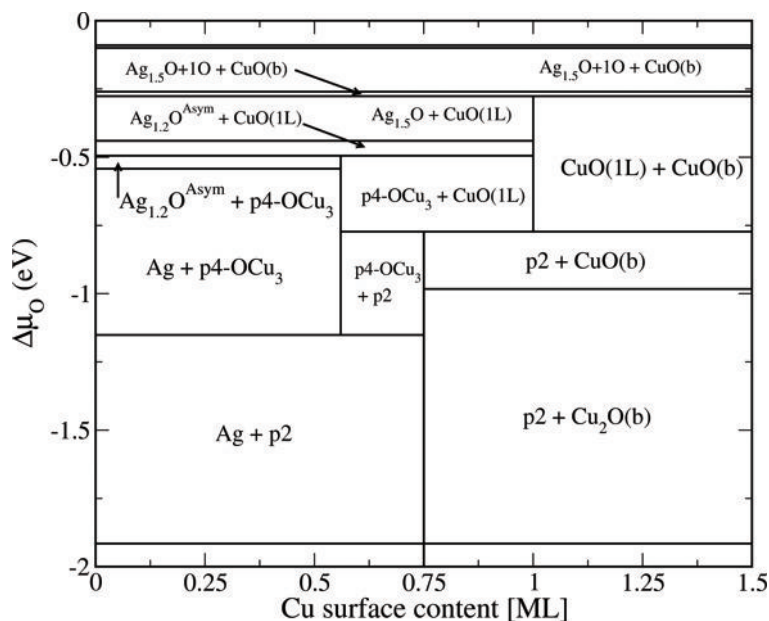


Fig. 17.6 Surface phase diagram showing structures belonging to the convex hull as a function of the Cu surface content and the change in oxygen chemical potential, $\Delta\mu_{\text{O}}$. (From Ref. 30.)

corresponds to conditions typical of industrial applications ($p = 1$ atm, $T = 600$ K) and for Cu content below 0.5 ML, the results predict that there will be patches of one-layer oxidic structures (i.e., $p4\text{-Cu}_3$) which coexist with the clean Ag surface. For higher values of $\Delta\mu_{\text{O}}$, O–Ag structures are predicted in coexistence with the $p4\text{-Cu}_3$ structure. For higher Cu contents, the $\text{CuO}(1\text{L})$ and $p2$ structures are predicted to be present above and below $\Delta\mu_{\text{O}} = -0.75\text{eV}$, respectively. For even higher Cu contents, bulk CuO is predicted to form on the surface. These predictions are consistent with recent experiments performed on the Ag–Cu system under catalytic conditions,⁴³ where through a combination of in situ XPS and near-edge x-ray absorption fine structure measurements, thin layers of CuO are found to be present on the surface. Areas of clean Ag are also present on the surface, in agreement with theory. Analogous calculations have been carried out for the other two low-index surfaces, (100) and (110).⁴⁴ A scenario similar to that of the (111) surface is found; that is, the presence of oxygen leads to copper segregation to the surface, and thin copper oxide-like layers are predicted on top of the silver surface, as well as copper-free structures.

Having studied Ag–Cu alloy surfaces in a pure oxygen environment, it is important to consider the effect of the (reducing) reactant ethylene. This is discussed below for the (111) surface. To do this, a “constrained thermodynamic equilibrium” approach is assumed, which considers the stability of the thin oxide-like layers toward the oxidation of ethylene to acetaldehyde

(thermodynamically favored reaction product). For a surface with stoichiometry $\text{Ag}_x\text{Cu}_y\text{O}_z$, the condition of stability is

$$\Delta\mu_{\text{C}_2\text{H}_4} - \Delta\mu_{\text{O}} \leq \frac{-2}{z} H_f(T = 0 \text{ K}) + \Delta E^{\text{mol}} \quad (17.11)$$

where $\mu_{\text{C}_2\text{H}_4}$ is the ethylene chemical potential with respect to its zero-temperature value. $H_f(T = 0 \text{ K})$ is the zero-temperature formation energy of the surface structure, and

$$\Delta E^{\text{mol}} = E_{\text{CH}_3\text{CHO}} - E_{\text{C}_2\text{H}_4} - \frac{1}{2} E_{\text{O}_2} \quad (17.12)$$

calculated to be -2.18 eV . Considering a Cu surface coverage of 0.5 ML, the surface phase diagram, as a function of oxygen and ethylene chemical potentials is shown in Fig. 17.7. The region corresponding to typical experimental conditions is indicated as that enclosed by the black dashed lines. It can be seen that

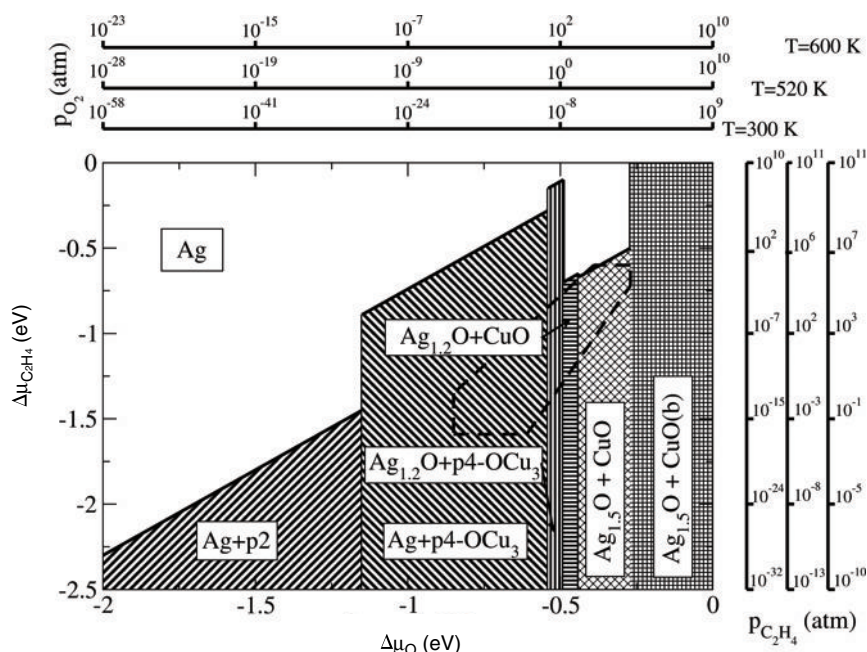


Fig. 17.7 Surface phase diagram for the (111) surface of the Ag–Cu alloy under constrained thermodynamic equilibrium with an atmosphere of oxygen and ethylene. The shaded areas represent the region of stability of a combination of two surface structures giving a Cu coverage of 0.5 ML. The white area corresponds to the clean Ag(111) surface, where Cu is assumed to be in a bulk reservoir, and ethylene is oxidized to acetaldehyde. The dashed polygon encloses the region that corresponds to typical values of temperature and pressure used in experiments ($T = 300$ to 600 K and $p_{\text{O}_2}, p_{\text{C}_2\text{H}_4} = 10^{-4} - 1 \text{ atm}$). (From Ref. 39.)

several structures can be present, all stable with respect to reduction by ethylene. Neglecting the effect of ethylene, therefore, the relative stability of the structures from all the low-index surfaces can be investigated as a function of the Cu surface content for a representative oxygen chemical potential ($\Delta\mu_{\text{O}} = -0.61$ eV). Here the chemical potential of Cu is used as a parameter to control the Cu content. The results are shown in Fig. 17.8, where for several values of $\Delta\mu_{\text{Cu}}$ the shapes predicted for the particles are shown, obtained by minimizing the surface free energy according to the Wulff construction.⁴⁵ For the value selected of $\Delta\mu_{\text{O}}$ selected, the value of $\Delta\mu_{\text{Cu}}$ above which Cu oxidizes to bulk copper oxide is -0.62 eV. The values of $\Delta\mu_{\text{Cu}}$ compatible with the experimentally indicated Cu coverages (0.1 to 0.75 ML) are those close to the formation of bulk copper oxide. Around this region, both the (100) and (110) surfaces are covered with

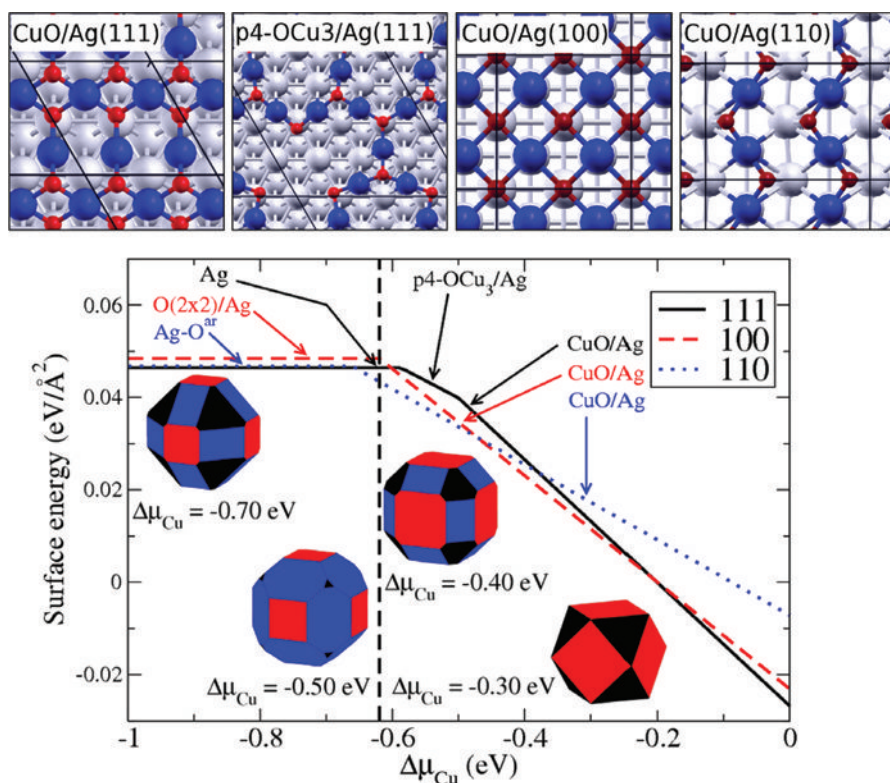


Fig. 17.8 (color online) (Top) Atomic geometry of four of the most stable oxidelike structures on the surface of Ag–Cu particles in an oxidizing atmosphere. Large light gray spheres represent Ag atoms, small spheres, O atoms; and dark spheres, Cu atoms. (Bottom) Surface energy versus the Cu chemical potential for $\Delta\mu_{\text{O}}$ of -0.61 eV (corresponding to $T = 600$ K and $p_{\text{O}_2} = 1$ atm). At selected values of $\Delta\mu_{\text{Cu}}$, the predicted particle shape, as obtained through the Wulff construction, is presented. (From Ref. 39.)

a one-layer oxidelike structure with a ratio of Cu to O of 1, denoted “CuO/Ag.” For values of $\Delta\mu_{\text{Cu}} < -0.65$ eV, all facets are covered with Cu-free structures.

Having predicted the equilibrium shape and surface structures of the Ag–Cu catalyst under conditions of practical interest, the adsorption of ethylene and the two competing chemical reactions leading to the formation of acetaldehyde (Ac) and ethylene oxide (EO) (see Fig. 17.9) can be investigated. For the (2×2) -O/Ag(111) and (2×2) -O/Ag(100) surfaces, both reactions are known to proceed through a common oxametallacycle (OMC)^{37,38,46,47} intermediate, where ethylene is bonded with one C atom to a surface metal atom and with the other C atom bonded to oxygen. The OMC is shown in Fig. 17.9 (leftmost panel). Similar findings have also been reported for Ag oxides.⁴⁸

From calculations of the reaction pathways for Ac and EO formation over the predicted stable surface structures, it is found that the behavior can be quite varied,⁴⁹ depending on the surface structure; in particular, for the (111) surface formation of EO does not involve the formation of any intermediate for the $p2/\text{Ag}(111)$, $p4\text{-OCu}_3/\text{Ag}(111)$, and CuO/Ag(111) structures. For formation of Ac over the CuO/Ag(111) surface, the reaction does, however, proceed by an OMC, but this is a metastable state. Ac formation over the $p2/\text{Ag}(111)$ surface involves the formation of a different stable intermediate in which ethylene is bound to one oxygen on each carbon. The OMC, on the other hand, is a common intermediate for both Ac and EO formation over the (2×2) -O/Ag(111), CuO/Ag(100), and CuO/Ag(110) surfaces. In Fig. 17.10 the transition states for Ac and EO formation over the (2×2) -O/Ag(111) and CuO/Ag(111) surfaces are shown as an example. The activation barrier for EO formation is lower than that of Ac for the CuO/Ag(111) structure, while the trend is the opposite for the (2×2) -O/Ag(111) surface. This is consistent with, and possibly partially explains, the greater selectivity reported experimentally for the Ag–Cu catalysts compared to pure silver. As mentioned above, the nature of the reaction pathways for the surface structures identified to be potentially catalytically relevant for

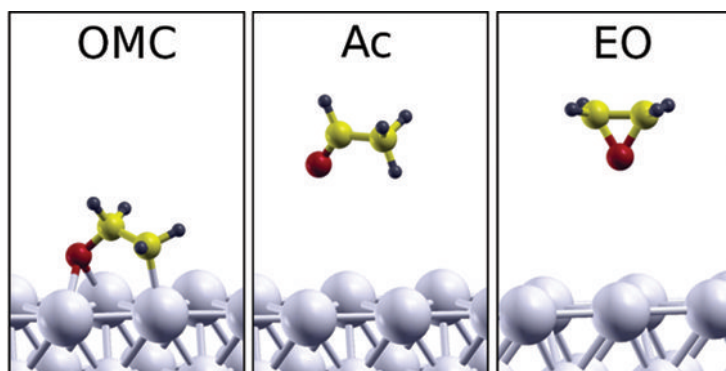


Fig. 17.9 (color online) Atomic geometry of the oxametallacycle (OMC) intermediate (left) and final states acetaldehyde (Ac) (center) and ethylene oxide (EO) (right) on (2×2) -O/Ag(111). (From Ref. 49.)

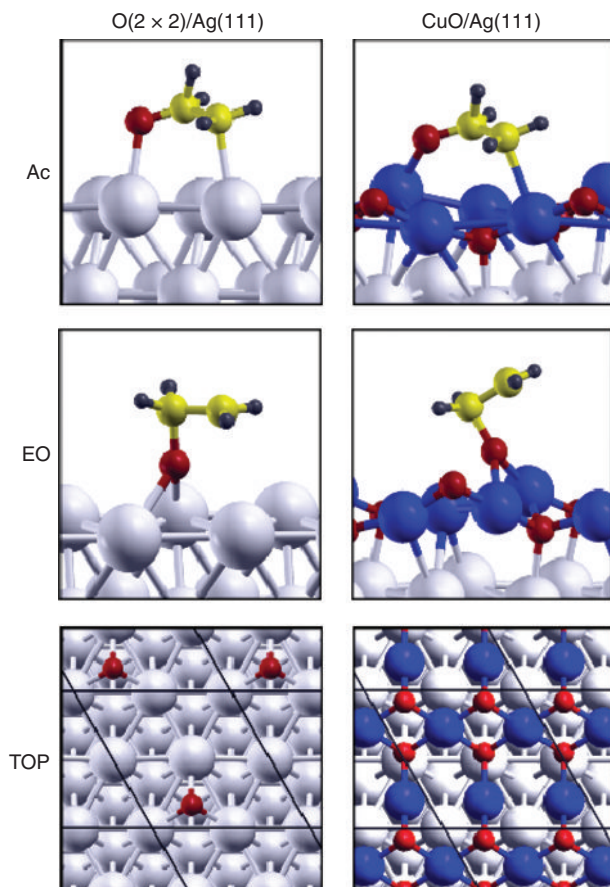


Fig. 17.10 (color online) Transition-state geometries for the formation of acetaldehyde (top panels) and ethylene oxide (central panels) and top view of the surface for the reaction over (2×2) -O/Ag(111) and for the CuO/Ag(111) structure (bottom panels). The large light gray spheres represent Ag atoms; the large dark ones, Cu; the medium dark ones, O; and the very small spheres, H atoms. (From Ref. 49.)

the low-index surfaces are quite varied, but the preliminary results point to the Cu-containing structures providing better selectivity toward EO formation, consistent with experimental measurements. For more details, see Ref. 49.

17.4 CATALYSIS AND DIFFUSION FROM AB INITIO KINETIC MONTE CARLO SIMULATIONS

17.4.1 CO Oxidation Reaction over Pd(100)

The importance of molecular-level mechanisms and their interplay for determining observable macroscopic (and microscopic) material phenomena is without

question. Often, as, for example, in the study of order–disorder phase transition temperatures discussed in Section 17.2, there is no direct link between the microscopic (electronic) theory and experimental measurables, and appropriate “hierarchical” approaches have to be developed that link the physics across all relevant length and time scales into one multiscale simulation.⁵⁰ A particularly successful approach is that of *ab initio* kinetic Monte (kMC). Considering, for example, the study of heterogeneous catalysis, for given gas-phase conditions, such calculations can determine the detailed surface composition and the occurrence of each individual elementary process at any time. From the latter, the catalytic activity (i.e., product formation) per surface area can also be obtained, either time-resolved (e.g., during induction, when the catalyst surface is being restructured to its active form) or time-averaged, during steady state. A recent comprehensive description of the kMC approach using microscopic parameters obtained from *ab initio* electronic structure total energy calculations for heterogeneous catalysis is given in Ref. 7.

First-principles-based kMC involves, first, a determination of the elementary steps involved in the particular process to be studied, and their calculation by electronic structure, total energy calculations (most typically using density functional theory). For catalysis, these would include adsorption and desorption of reactants and reaction intermediates, as well as surface diffusion and surface reactions. The second step concerns describing the statistical interplay of the elementary processes as achieved by kinetic Monte Carlo simulations.⁵¹

In kMC the relationship between “MC time” and “real time” is obtained by regarding the MC process as providing a numerical solution to the Markovian master equation describing the dynamic system evolution.^{52–56} A sequence of configurations is generated using random numbers. For each step (new configuration), all possible elementary processes and the rates with which they occur are calculated. These processes are weighted by the rates, and one of the processes is executed randomly to achieve the new system configuration. In this way the kMC algorithm effectively simulates stochastic processes, and a direct relationship between kMC time and real time is established. The flow diagram for the kMC process is shown in Fig. 17.11. Properly evaluating the time evolution requires simulation cells that are large enough to capture the effects of correlation and spatial distribution of the species at the surface. Most processes considered in kMC are highly activated and occur on time scales orders of magnitude longer than, for example, a typical vibration (10^{-12} s). Due to these “rare events,” the statistical interplay of the elementary processes need to be evaluated over time scales that can reach to seconds and more.

A recent application demonstrating the power of this approach is the study of the CO oxidation reaction over the Pd(100) surface. The motivation for this study is related to the increasing awareness that for oxidation catalysis (i.e., under atmospheric oxygen conditions) the surface of a transition metal (TM) catalyst may be oxidized, and instead of being the pure TM surface, which is often the subject of quantitative ultrahigh-vacuum (UHV) surface science studies, the oxidized material may be active for the catalysis. This has recently

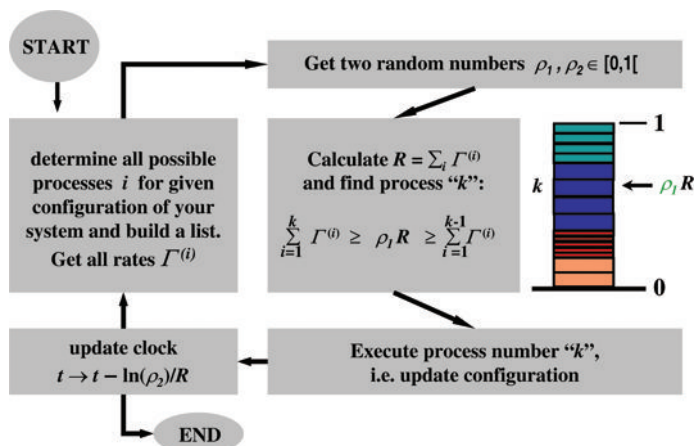


Fig. 17.11 (color online) Flow diagram showing the basic steps in a kinetic Monte Carlo simulation. First, loop over all the lattice sites and determine the elementary atomic processes that are possible for the current system configuration. Then generate two random numbers and advance the system configuration according to the process selected by the first random number. Then, increment the clock according to the rates and the second random number as prescribed by an ensemble of Poisson processes, and then start all over again or stop if the simulation time is sufficiently long. (From Ref. 6.)

been revealed for CO oxidation employing Ru catalysts. In this case, bulk oxide RuO_2 is, in fact, the stable phase under reactive conditions.^{57,58} For TMs farther to the right in the periodic table, the late TM and noble metals, which are also used in oxidation catalysis, the situation is different; thus, it is of great interest to consider the analogous reaction of CO oxidation over the more noble metal, Pd. Briefly, from the kMC simulations described below, it was found that oxide formation in the reactive environment also plays a significant role, but a difference is that this oxide is not a bulklike film that once it becomes stable, actuates the catalysis; rather, the study indicates the relevance of a subnanometer surface oxide structure which is probably formed continuously and reacted away in the sustained catalytic operation.

As a first step in this study, using the approach of ab initio atomistic thermodynamics described in Section 17.3, the surface structure and stability of the Pd(100) surface in an atmosphere containing oxygen and carbon monoxide, for a wide range of partial pressures and temperatures, is studied. The resulting phase diagram is shown in Fig. 17.12.^{59,60} Here, a constrained atomic thermodynamics approach was employed,^{61,62} as for the Ag–Cu alloy catalysts described in Section 17.3 for ethylene oxidation, in which it is assumed that the surface is in equilibrium with i separate reservoirs representing the i gas-phase species, each characterized by the chemical potential $\Delta\mu_i(T, p_i)$ with partial pressure p_i and temperature T . The character of the surface phase diagram can be described in terms of three regions: first, a region where bulklike thick oxide films are stable (crosshatched region); then a region consisting of adsorption

phases on a $(\sqrt{5} \times \sqrt{5})R27^\circ$ (hereafter denoted “ $\sqrt{5}$ ”) surface oxide (hatched area), which has recently been characterized and resembles a layer of PdO(101) on the surface⁶³; and finally, a region with different CO and O adsorption phases on Pd(100). Gas-phase conditions, representative of technological CO oxidation catalysis ($p_i \sim 1$ atm, $T \sim 300$ to 600 K), correspond to the phase boundary between the regions of adsorption on the $\sqrt{5}$ surface oxide and that of CO-covered Pd(100). Thus, unlike for Ru, the presence of bulk oxides in the reactive environment can be ruled out, while the stability region of the thin $\sqrt{5}$ surface oxide structure extends into such conditions.

To investigate the reactivity of the $\sqrt{5}$ phase, and to see if its stability region changes when the kinetic effects of catalytic reaction on the surface are taken into account, kinetic Monte Carlo calculations are carried out. In these simulations, hollow and bridge sites are considered and all nonconcerted adsorption, desorption, diffusion, and Langmuir–Hinshelwood reaction processes (where both reactants are adsorbed on the surface prior to reaction to the product) involving these sites: in all, 26 elementary processes. Also, nearest-neighbor lateral interactions are taken into account in the elementary process rates. The required (14) interaction parameters are determined from DFT calculations of 29 ordered configurations with O and/or CO in bridge and hollow sites of the $\sqrt{5}$ surface unit cell. The resulting adsorption energies are expressed in terms of the LGH expansion. The kMC simulations are performed on a lattice comprising (50×50) surface unit cells for fixed (T , p_{O_2} , p_{CO}) conditions, in particular for $p_{O_2} = 1$ atm and temperatures in the range 300 to 600 K. Initially, the CO partial pressure was chosen to be low, 10^{-5} atm, corresponding to the middle of the stability region of the $\sqrt{5}$ phase, and subsequently increased, moving closer and closer to the boundary of the stability region of the $\sqrt{5}$ phase. This is indicated by the vertical arrows in Fig. 17.12. When the surface reaction consumes surface oxygen faster than it is replenished, the $\sqrt{5}$ phase becomes destabilized. To determine the onset of the structural destabilization from the kMC simulations, the percentage occupation of O atoms in hollow sites is monitored as a function of CO partial pressure. Full occupation of these sites corresponds to the intact $\sqrt{5}$ phase. The results are shown in Fig. 17.13. Interpreting a reduction to 95% occupation as the onset of decomposition, the results predict critical CO pressures of 5×10^{-2} , 10^{-1} , and 10 atm at 300, 400, and 600 K, respectively. These results are rather similar to those obtained from the constrained atomistic thermodynamics approach, which are shown in Fig. 17.13 as the vertical lines. The critical pressures obtained (e.g., at 400 K $p_{O_2}/p_{CO} \approx 10 : 1$) are in good accord with reactor scanning tunneling microscopy (STM) experiments⁶⁴ performed under such gas-phase conditions. Importantly, the theoretical results show that for relevant p_{O_2}/p_{CO} ratios, the turnover frequencies (number of CO₂ molecules produced per site per second) for the intact $\sqrt{5}$ surface oxide alone are already of a similar order of magnitude to those reported experimentally⁶⁵ for the Pd(100) surface under comparable gas-phase conditions. This shows that this particular surface oxide is certainly not “inactive” with respect to the oxidation of CO, which is contrary to early prevalent general preconceptions.

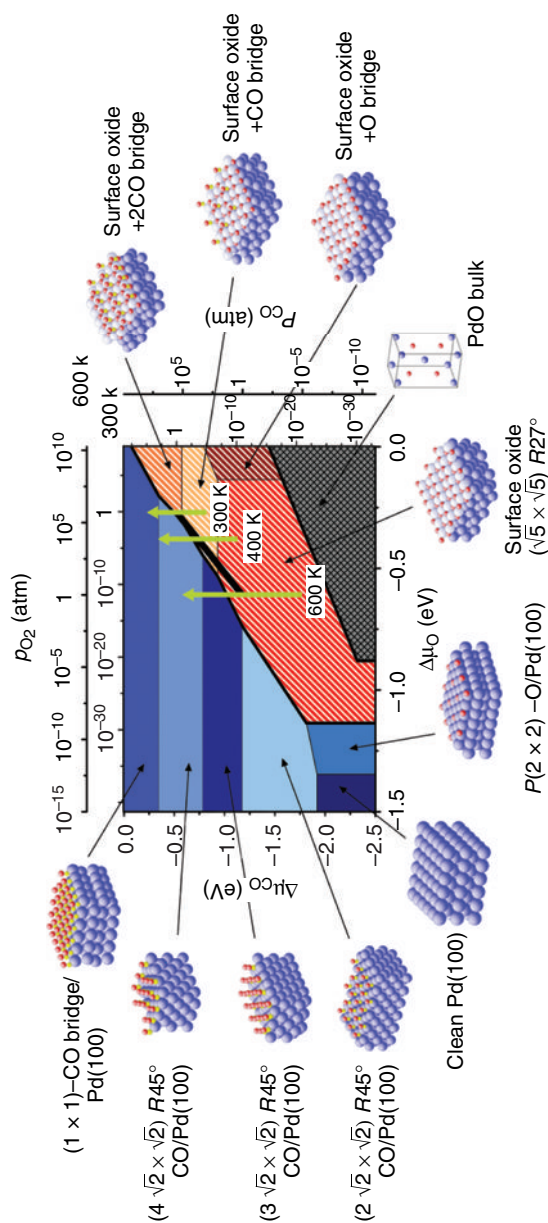


Fig. 17.12 (color online) Surface phase diagram for the Pd(100) surface in constrained thermodynamic equilibrium with an environment containing O_2 and CO. The various surface structures corresponding to the regions in the phase diagram are illustrated. The pressures corresponding to the O_2 and CO chemical potentials are shown for temperatures of 300 and 600 K. The thick black line marks gas-phase conditions representative of that employed for technological CO oxidation catalysis (i.e., partial pressures of 1 atm and temperatures between 300 and 600 K). The three vertical lines correspond to the gas-phase conditions employed in the kinetic Monte Carlo simulations shown in Fig. 17.13. (From Ref. 60.)

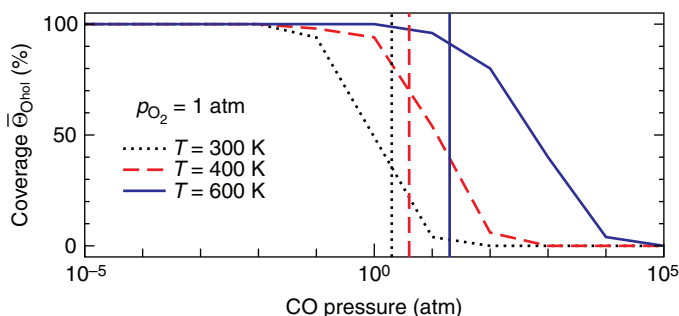


Fig. 17.13 (color online) Average coverage (occupation) of oxygen atoms in hollow sites of the $(\sqrt{5} \times \sqrt{5})R27^\circ(\sqrt{5})$ surface oxide-like structure as obtained from kinetic Monte Carlo simulations. 100% corresponds to the intact $\sqrt{5}$ structure. The reduction of the $\sqrt{5}$ surface oxide-like phase occurs at CO pressures close to those corresponding to the stability boundary (transition from the hatched to the plain areas in Fig. 17.12) and indicated by the vertical lines in Fig. 17.12. (From Ref. 59.)

17.4.2 Permeability of Hydrogen in Amorphous Materials

In a new application, first-principles kinetic Monte Carlo–based simulations have recently been used for the study of the permeability of hydrogen through crystalline and amorphous membranes.^{9,66,67} The use of metal membranes can potentially play an important role in the large-scale production of high-purity hydrogen, which is required for its use as a fuel in (polymer electrolyte) fuel cell technologies.⁶⁸ In these membranes, hydrogen permeates through the film by dissociation of molecular hydrogen, diffusion of atomic H through interstitial sites, and then recombination to H₂. Permeation of hydrogen occurs at much greater rates than other elements; thus, the membranes, can deliver high-purity H₂ from gas mixtures containing large concentrations of other species. There has been a recent focus on exploring the possibility that amorphous metals may represent a promising new class of membranes, which are to date relatively unexplored compared to crystalline metals and alloys.

Hao and Sholl⁹ have recently investigated hydrogen permeability through amorphous and crystalline Fe₃B metal films. The scheme involves kinetic Monte Carlo simulations and the goal is that this approach could be used to identify materials with high potential for improved performance through an efficient screening of candidate structures. The structure of crystalline Fe₃B is shown in Fig. 17.14b, while an amorphous structure obtained from molecular dynamics simulations is shown in Fig. 17.14a. Considering H₂ transport through a film, the rate is often limited by interstitial diffusion of H through the bulk material. In this case, the flux can be related to the operating conditions if the solubility and diffusion coefficient of interstitial H is known. The latter quantity can be accurately calculated for crystalline materials from first-principles-based approaches. For amorphous solids the situation is, however, more complex. In this case a detailed model for the atomic structure must first be generated. Once this is established, the sites can

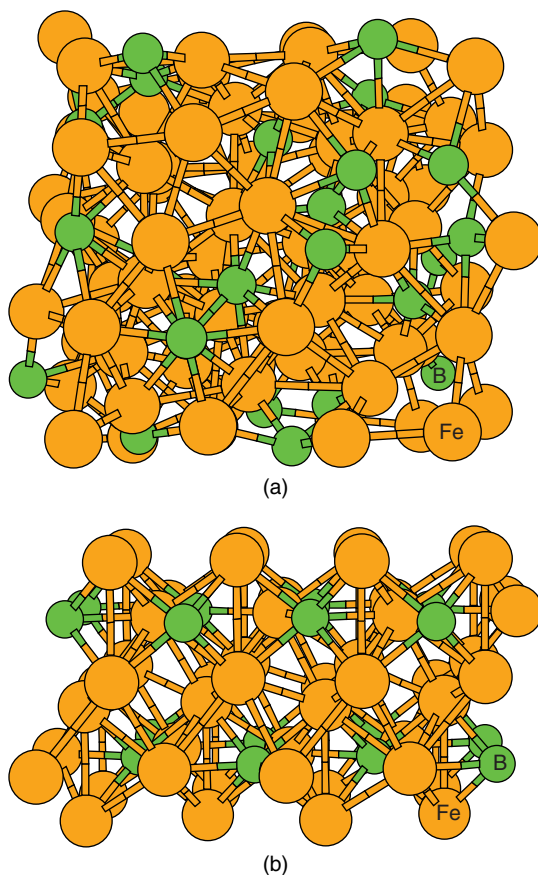


Fig. 17.14 (color online) Atomic structure of crystalline Fe₃B (b) and an example of an amorphous structure of Fe₃B (a) as generated from a molecular dynamics simulation. (From S. Hao, private communication.)

be occupied with interstitial hydrogen and the transition states for diffusion of H atoms between sites can be identified. For amorphous materials, the solubility is typically stronger than in the crystalline counterpart, due to the greater range of interstitial binding sites, some of which can bind H notably stronger. This results in the effects of H concentration being greater for amorphous systems, and this must be taken into account. To investigate this, Hao and Sholl⁹ carried out simulations for various concentrations for both crystalline *c*-Fe₃B and amorphous *a*-Fe₃B.

As the first step, the amorphous geometry was created through an *ab initio* molecular dynamics simulation of a representative liquidlike sample of 100 atoms, which was rapidly quenched and then an energy minimization carried out. Subsequently, the interstitial sites were identified. This was done using an automatic procedure for the amorphous structure, due to the great number of them.

The binding energies and the interactions between H atoms in the interstitial sites were then calculated using density functional theory. From the site energies and the H–H interaction energies, the solubility of H in α -Fe₃B and ϵ -Fe₃B was obtained using grand canonical Monte Carlo calculations.⁶⁹ The result is shown in Fig. 17.15, plotted as a function of temperature and H₂ pressure. An important finding (Fig. 17.15) is that the H solubility is far larger in the amorphous material than in the crystalline material (e.g., two to three orders of magnitude at 600 K). It can also be noticed that the qualitative dependence of the solubility on temperature is different for the amorphous and crystalline materials, which is attributed to the broad distribution of site energies in the amorphous material.⁹

Calculation of H diffusion requires the calculation of transition states between adjacent H sites. Initially, Hao and Sholl employed an approximation for the positions of the transition states before carrying out the more computationally expensive DFT calculations. For α -Fe₃B, this involved determining a huge number (462) of transition states, highlighting the complexity of treating the amorphous structure. Once determined, the rates and the H diffusion can be calculated using kinetic MC. On investigating the concentration dependence of the diffusion coefficient for amorphous Fe₃B, it was found that for increasing concentration, the diffusion coefficient increases (e.g., at 600 K by around three orders of magnitude for H concentration varying from 0 to 0.2H/M) and then begins to decrease again. This behavior was explained by the fact that at low concentrations the strongest binding sites are occupied, which have associated large diffusion barriers. For higher concentrations, these sites are occupied, and less favored sites become populated which have smaller barriers for diffusion. For even higher concentration, the diffusion coefficient decreases due to blocking effects by the interstitial H atoms.

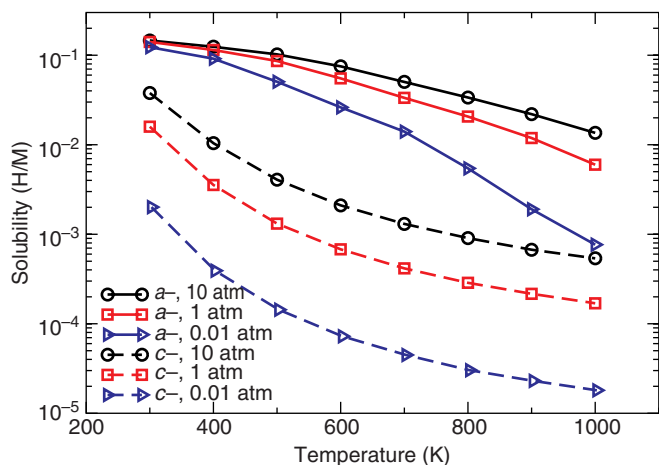


Fig. 17.15 (color online) Calculated H solubility in α -Fe₃B (solid curves) and ϵ -Fe₃B (dashed curves) as a function of temperature for several H₂ pressures. Lines are guides to the eye. (From Ref. 9.)

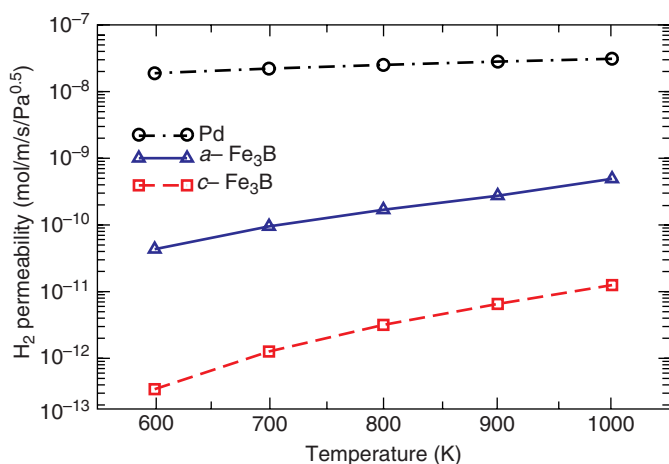


Fig. 17.16 (color online) Calculated permeability of H₂ in *a*-Fe₃B and *c*-Fe₃B at different temperatures. The “feed pressure” was 10 atm and the permeate pressure was 1 atm. The permeability of pure Pd is also shown for comparison. (From Ref. 9.)

To make contact with the experimental results, the more relevant quantity is H permeation through these materials, which involves calculation of the flux through the membrane. Here it was assumed that the net transport is dominated by diffusion through the bulk of the membrane. The results obtained are shown in Fig. 17.16 for particular pressures. It can be seen that the permeability of the amorphous material is about 1.5 to 2 orders of magnitude larger than the crystalline material, supporting the notion that amorphous structures can indeed have higher permeabilities. It is noted that the permeability of pure Pd is greater than that of both *a*-Fe₃B and *c*-Fe₃B, although the latter material was chosen not because it was thought it may yield greater permeabilities than Pd, but because it represented a system in which a detailed comparison of the behavior of a crystalline versus an amorphous system could be achieved.

17.5 SUMMARY

In this chapter, recent applications and results of first-principles-based approaches to describing and predicting surface properties, such as structures, stoichiometry, phase transitions, and heterogeneous catalysis, and also bulk properties, including solubility, diffusivity, and permeability, were discussed. Three particular calculation approaches were highlighted which are often described under the label “multiscale modeling.” First, using the lattice-gas Hamiltonian (LGH) in combination with equilibrium Monte Carlo (MC) simulations, order–disorder phase transitions for the O/Pd(111) system were presented. This approach is truly predictive in nature in that completely unanticipated structures can be found. It can, in principle, also describe the coexistence of phases and configurational

entropy. For the case of O/Pd(111) the recently introduced MC scheme of Wang and Landau was used. This algorithm enables direct evaluation of the density of (configurational) states, and thus straightforward determination of the main thermodynamic functions.

Using the *ab initio* atomistic thermodynamics approach, the alloy catalyst Ag–Cu was investigated regarding its surface structure and activity for the ethylene epoxidation reaction. In this approach the free energy for surface structures are calculated, from which the stability range of various identified low-energy phases are predicted. The main limitation of this method is that its predictive power is limited to the explicitly considered surface structures, and that due to the supercell approach used in most modern first-principles approaches, the structures investigated are restricted to be periodic. From investigation of the chemical reactions over the surface phases identified, the calculations showed that first under reaction conditions the catalyst surface is very different to a hitherto assumed AgCu surface alloy. In particular, the results point to a dynamical coexistence of thin CuO and AgO–CuO films on the Ag substrate. This is likely to have important consequences regarding the mechanism by which Cu enhances the catalyst selectivity since the active O species will be part of the oxide layer rather than adsorbed O atoms on a metal surface. Preliminary investigations indicate that some reaction pathways for ethylene oxidation over such Cu-oxide layers have a lower activation energy than that of the (undesired) competing reaction to acetaldehyde. These findings may also be of high relevance for understanding the activity of other dilute alloy catalysts.

The most complex approach discussed, kinetic MC, links an accurate description of the elementary processes, which have a clear microscopic meaning (obtained through use of first-principles calculations) with a proper evaluation of their statistical interplay. Important to the success of this approach is the identification of all relevant elementary processes, which can be nontrivial. Further, for increasingly complex systems, the number of elementary processes can virtually explode. In the literature there have been some attempts to generate the list of elementary reactions “on the fly” (see, e.g., Refs. 70 and 71, where this approach is discussed in more detail and distributed). Typically, *ab initio* kMC studies have been carried out with “home-grown” codes written around a particular application. In the present chapter, two recent examples were described: the first, the carbon monoxide oxidation reaction over Pd(100) in which the importance of the formation of a thin surface-oxide-like film was identified, and the second, the permeability of hydrogen through amorphous and crystalline films of Fe₃B. In the latter study, the calculations predicted a greater permeability for the amorphous membrane, pointing to amorphous structures possibly representing a new class of higher-efficiency membranes for hydrogen purification.

Over the years there has been a considerable increase in the atomic-level understanding of material systems, which has arisen primarily due to the synergy between experiment and first-principles-based studies. It is envisaged that this trend will continue, with the theoretical methods described here, as well as new

approaches that will be developed together with the seemingly ever-increasing computer power, proving very valuable for advancing the performance of technological applications right across the multidisciplinary fields of physics, chemistry, biology, engineering, and materials science, yielding many exciting discoveries along the way.

REFERENCES

1. Basic research needs for the hydrogen economy. Presented at the Workshop on Production, Storage and Use, U.S. Department of Energy, Office of Basic Energy Sciences, Washington, DC, 2003.
2. Basic research needs for solar energy utilization. Report of the Basic Energy Sciences Workshop on Solar Energy Utilization, 2005.
3. Satterfield, C. N. *Heterogeneous Catalysis in Industrial Practice*, McGraw-Hill, New York, 1991.
4. Lundgren, E.; Over, H. *J. Phys. Condens. Matter* **2008**, *20*, 180302, and references therein.
5. Basic research needs: catalysis for energy. Presented at the Workshop on Production, Storage and Use, U.S. Department of Energy, Office of Basic Energy Sciences, Washington, DC, 2007.
6. Reuter, K.; Stampfl, C.; Scheffler, M. Ab initio atomistic thermodynamics and statistical mechanics of surface properties and functions. In *Handbook of Materials Modeling*, Vol. 1., Yip, S., Ed., Springer-Verlag, Berlin, 2005, pp. 149–194.
7. Reuter, K. First-principles kinetic Monte Carlo simulations for heterogeneous catalysis: Concepts, status and frontiers. In *Modeling Heterogeneous Catalytic Reactions: From the Molecular Process to the Technical System*, Deutschmann, O., Ed., Wiley-VCH, Weinberg, Germany, 2009.
8. Stampfl, C. *Catal. Today* **2005**, *105*, 17.
9. Hao, S.; Sholl, D. S. *Energy Environ. Sci.* **2008**, *1*, 175.
10. Sholl, D. S.; Steckel, J. A. *Density Functional Theory: A Practical Introduction*, Wiley, New York, 2009.
11. Engel, T.; Ertl, G. *J. Chem. Phys.* **1978**, *69*, 1267; *Adv. Catal.* **1979**, *28*, 1; *The Chemical Physics of Solid Surfaces and Heterogeneous Catalysis*, Vol. 4, King, D. A. and Woodruff, D. P., Eds., Elsevier, Amsterdam, 1982.
12. Campbell, C. T.; Ertl, G.; Kuipers, H.; Segner, J. *J. Chem. Phys.* **1980**, *73*, 5862.
13. Zaera, F. *Prog. Surf. Sci.* **2002**, *69*, 1.
14. Nakai, I.; Kondoh, H.; Shimada, T.; Resta, A.; Andersen, J.; Ohta, T. *J. Chem. Phys.* **2006**, *124*, 224712.
15. McEwen, J.-S.; Payne, S. H.; Stampfl, C. *Chem. Phys. Lett.* **2002**, *361*, 317.
16. Borg, M.; Stampfl, C.; Mikkelsen, A.; Gustafson, J.; Lundgren, E.; Scheffler, M.; Andersen, J. N. *ChemPhysChem* **2005**, *6*, 1923.
17. Piccinin, S.; Stampfl, C. *Phys. Rev. B* **2010**, *81*, 155427.
18. Tang, H.; Van der Ven, A.; Trout, B. L. *Phys. Rev. B* **2004**, *70*, 045420.
19. Zhang, Y.; Blum, V.; Reuter, K. *Phys. Rev. B* **2007**, *75*, 235406.

20. Shao, J. *J. Am. Stat. Assoc.* **1993**, 88, 486.
21. Zhang, P. *Ann. Math. Stat.* **1993**, 21, 299.
22. Stampfl, C.; Kreuzer, H. J.; Payne, S. H.; Pfnür, H.; Scheffler, M. *Phys. Rev. Lett.* **1999**, 83, 2993.
23. Mendez, J.; Kim, S. H.; Cerdá, J.; Wintterlin, J.; Ertl, G. *Phys. Rev. B* **2005**, 71, 085409.
24. Piercy, P.; De' Bell, K.; Pfnür, H. *Phys. Rev. B* **1992**, 45, 1869.
25. Kortan, A. R.; Park, R. L. *Phys. Rev. B* **1981**, 23, 6340.
26. Wang, F.; Landau, D. P. *Phys. Rev. Lett.* **2001**, 86, 2050.
27. Wang, F.; Landau, D. P. *Phys. Rev. E* **2001**, 64, 056101.
28. Schulz, B. J.; Binder, K.; Müller, M.; Landau, D. P. *Phys. Rev. E* **2003**, 67, 067102.
29. Keil, F. J. *J. Univ. Chem. Technol. Metall.* **2008**, 43, 19.
30. Piccinin, S.; Stampfl, C.; Scheffler, M. *Phys. Rev. B* **2008**, 77, 075426.
31. Reuter, K.; Scheffler, M. *Phys. Rev. B* **2002**, 65, 035406.
32. Weinert, C.; Scheffler, M. *Mater. Sci. Forum* **1986**, 10–12, 25.
33. Scheffler, M.; Dabrowski, J. *Phil. Mag. A* **1988**, 58, 107.
34. Li, W.-X.; Stampfl, C.; Scheffler, M. *Phys. Rev. B* **2003**, 67, 045408.
35. Stampfl, C. *Catal. Today* **2005**, 105, 17.
36. Linic, S.; Jankowiak, J.; Barteau, M. A. *J. Catal.* **2004**, 224, 489.
37. Linic, S.; Barteau, M. A. *J. Am. Chem. Soc.* **2002**, 124, 310.
38. Linic, S.; Barteau, M. A. *J. Am. Chem. Soc.* **2004**, 125, 4034.
39. Piccinin, S.; Zafeiratos, S.; Stampfl, C.; Hansen, T.; Hävecker, M.; Teschner, D.; Knop-Gericke, A.; Schlögl, R.; Scheffler, M. *Phys. Rev. Lett.* **2010**, 104, 035503.
40. Stull, D. R.; Prophet, H. *JANAF Thermochemical Tables*, 2nd ed., U.S. National Bureau of Standards, Washington, DC, 1971.
41. Soon, A.; Todorova, M.; Delley, B.; Stampfl, C. *Phys. Rev. B* **2006**, 73, 165424.
42. Jankowiak, J. T.; Barteau, M. A. *J. Catal.* **2005**, 236, 366.
43. Zafeiratos, S.; Hävecker, M.; Teschner, D.; Vass, E.; Schnörch, P.; Girgsdies, F.; Hansen, T.; Knop-Gericke, A.; Schlögl, R.; Bukhiyarov, V. Unpublished.
44. Piccinin, S.; Stampfl, C.; Scheffler, M. *Surf. Sci.* **2009**, 603, 1467.
45. Wulff, G. *Z. Kristallogr.* **1901**, 34, 449.
46. Kokalj, A.; Gava, P.; de Gironcoli, S.; Baroni, S. *J. Catal.* **2008**, 254, 304.
47. Torres, D.; Lopes, N.; Illas, F.; Lambert, R. *J. Am. Chem. Soc.* **2005**, 127, 10774.
48. Bocquet, F.; Loffreda, D. *J. Am. Chem. Soc.* **2005**, 127, 17207.
49. Piccinin, S.; Nguyen, N. L.; Stampfl, C.; Scheffler, M. *J. Mater. Chem.* **2010**, 20, 10521.
50. Yip, S., Ed. *Handbook of Materials Modeling*, Springer-Verlag, Berlin, 2005.
51. Voter, A. F. Introduction to the kinetic Monte Carlo method. In *Radiation Effects in Solids*, Sickafus, K. E., Kotomin, E. A., and Uberuaga, B. P., Eds., Springer-Verlag, Berlin, 2007.
52. Bortz, A. B.; Kalos, M. H.; Lebowitz, J. L. *J. Comput. Phys.* **1975**, 17, 10.
53. Gillespie, D. T. *J. Comput. Phys.* **1976**, 22, 403.
54. Voter, A. F. *Phys. Rev. B* **1986**, 34, 6819.

55. Kang, H. C.; Weinberg, W. H. *J. Chem. Phys.* **1989**, *90*, 2824.
56. Fichthorn, K. A.; Weinberg, W. H. *J. Chem. Phys.* **1991**, *95*, 1090.
57. Reuter, K.; Scheffler, M. *Appl. Phys. A* **2004**, *78*, 793.
58. Over, H.; Mühler, M. *Prog. Surf. Sci.* **2003**, *72*, 3.
59. Rogal, J.; Reuter, K.; Scheffler, M. *Phys. Rev. B* **2008**, *77*, 155410.
60. Rogal, J.; Reuter, K.; Scheffler, M. *Phys. Rev. Lett.* **2007**, *98*, 046101.
61. Reuter, K.; Scheffler, M. *Phys. Rev. B* **2003**, *68*, 045407.
62. Reuter, K.; Scheffler, M. *Phys. Rev. Lett.* **2003**, *90*, 046103.
63. Todorova, M.; Lundgren, E.; Blum, V.; Mikkelsen, A.; Gray, S.; Gustafson, J.; Borg, M.; Rogal, J.; Reuter, K.; Andersen, J. N.; Scheffler, M. *Surf. Sci.* **2003**, *541*, 101.
64. Hendriksen, B. L. M.; Bobaru, S. C.; Frenken, J. W. M. *Surf. Sci.* **2004**, *552*, 229.
65. Szanyi, J.; Goodman, D. W. *J. Phys. Chem.* **1994**, *98*, 2972.
66. Semidey-Flecha, L.; Sholl, D. S. *J. Chem. Phys.* **2008**, *128*, 144701.
67. Hao, S.; Sholl, D. S. *J. Chem. Phys.* **2009**, *130*, 244705.
68. Schlappbach, L.; Züttel, A. *Nature* **2001**, *414*, 353.
69. Ling, C.; Sholl, D. S. *J. Membr. Sci.* **2007**, *303*, 162.
70. Henkelman, G.; Jónsson, H. *J. Chem. Phys.* **2001**, *115*, 9657.
71. Pedersen, A.; Jónsson, H. *Math. Comput. Simul.* **2010**, *10*, 1487.

18 Molecular Spintronics

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Molecular spintronics is a new rising field to share and maximize the common area between spintronics and molecular electronics. This chapter offers a pedagogical introduction to the theoretical work on molecular spintronics. Theoretical backgrounds for both spintronics and molecular electronics are overviewed and their numerical implementation issues are discussed in detail. In particular, we review molecular analogs of conventional spin valve devices and graphene nanoribbon-based super magnetoresistance.

18.1 INTRODUCTION

Spintronics is a promising research field where electronic devices exploit the spin of an electron as a transport carrier rather than its charge in conventional electronics. Manipulation of the spin using external magnetic fields enables us to store information with high density in an electronic device.¹ In addition, nonvolatility of the spin empowers the device to keep the information without electric power. This new idea triggered by the discovery of the giant magnetoresistance (GMR) effect in 1988 has led to the innovation of information storage techniques, with successful application of the GMR device to the read head sensor in hard disk drives.^{2,3} It eventually advanced an information-oriented era. As a result, in 2007, Nobel prizes were awarded to A. Fert and P. Grünberg for their discovery of the GMR effect. In the meantime, popularization of small and portable electronic devices has led to increased demand to develop not only nonvolatile but also low power consumption, high-speed access, and high-density memory devices. Emergence of tunneling magnetoresistance (TMR) has opened a new way to develop high-performance magnetoresistive random access memory (MRAM), which has attracted great attention as a next generation of information storage.⁴

On the other hand, molecular electronics is a rapidly growing field where a single or a few molecules are used as an individual electronic device.^{5–9} Such a bottom-up approach would provide an ideal means to construct nanoscale devices, complementing or even replacing conventional top-down approaches.^{8,9} In addition, organic molecules have essential advantages to be used in spintronics. There are two intrinsic sources to collapse long spin coherence in materials: spin-orbit coupling and hyperfine interactions. Organic molecules are composed of low-mass atoms, while the strength of the spin-orbit coupling increases with the atomic number Z (proportional to Z^4 in the case of atoms). Carbon-12 (^{12}C), the most abundant isotopes of carbon as well as the main component of organic molecules, has zero nuclear spin, so that it has no hyperfine interactions. Moreover, delocalized orbitals of conjugated molecules have small hyperfine interactions. These properties of molecules promise long spin-relaxation length, which is vital to fabricate high-performance spintronic devices.

In this regard, novel combination of both spintronics and molecular electronics would be the natural evolution toward molecular-scale spintronic devices. This new emerging field, molecular spintronics, has already shown the feasibility of real applications with successful measurements of spin-dependent electrical currents in molecule-based devices.^{10–15} The first experiment was carried out by exploiting a multiwall carbon nanotube (CNT) sandwiched between cobalt electrodes.¹¹ CNTs have attracted much interest because of their superior properties, such as high carrier mobility, ballistic electron transport, and mechanical robustness. Furthermore, they are composed of only carbon atoms, so that they have negligible spin-orbit coupling and hyperfine interactions. Indeed, CNTs have shown very long spin relaxation length reaching over micrometers.¹⁴ Subsequently, organic molecules and graphene (a single graphite layer) have been used in spintronic devices.^{12–15}

In addition, a new type of spintronic devices can be made when exploiting a magnetic molecule in spintronics.^{16–20} Particular molecules comprised of transition metals show internal spin ordering whose orientation can be controlled by an external magnetic field. Electron transport through such a magnetic molecule shows nontrivial spin-dependent effects due to the internal spin dynamics of the molecule. All this experimental evidence shows the bright future of molecular spintronics.

Alongside experimental works, theoretical studies have also been active.⁸ As quantum chemistry, including density functional theory (DFT), the Hartree–Fock (HF) method, and post-HF methods, has offered versatile tools to study electronic structures for a variety of materials, theoretical modeling should be a powerful means to investigate transport properties in molecular spintronic devices. However, it is not straightforward to use conventional quantum chemistry for this purpose, since we are dealing not only with nonequilibrium states driven by a bias voltage (for which the variational principle is not valid) but also open boundary systems made by a contact between two semi-infinite metallic electrodes and a finite molecule. A general way to study such a system is to utilize the nonequilibrium Green's function (NEGF) method.^{21,22} At present, several schemes based

on the NEGF method to describe quantum transport quantitatively as well as qualitatively are available^{23–33} (see also Chapters 1 and 19). Some of them are also used for spin-polarized transport.^{29–33} Especially, parameter-free methods enable us to design novel spintronic devices as well as to interpret experimental observations.

The goal of this chapter is to offer a pedagogical introduction of the exciting molecular spintronics based on theoretical works. In the following sections we discuss theoretical backgrounds on spintronics and molecular electronics, practical schemes for numerical implementation, and interesting example studies.

18.2 THEORETICAL BACKGROUND

18.2.1 Magnetoresistance

A representative spintronic device is the spin valve that is composed of two ferromagnetic (FM) electrodes connected by a spacer as shown in Figs. 18.1 and 18.2. The resistance in the spin-valve device depends on the relative spin orientation between the two FM electrodes. In general, the resistance is smaller for the parallel spin orientation than for the antiparallel spin orientation. Consequently, the resistance in a spin-valve device is tuned by an external magnetic

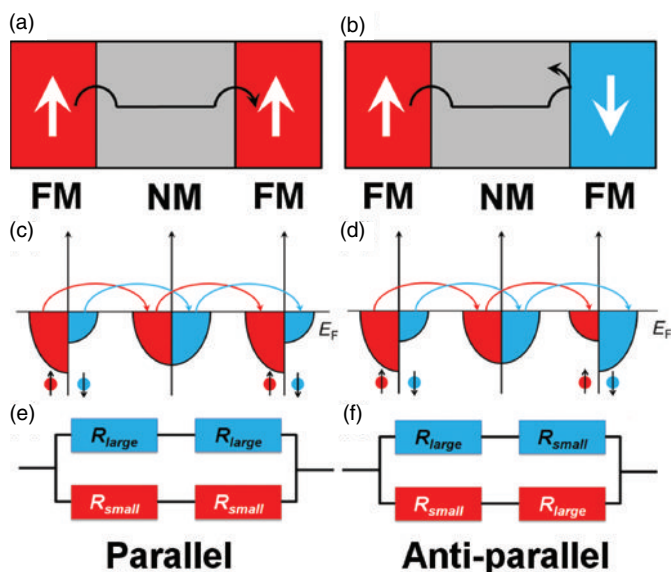


Fig. 18.1 (color online) (a,b) Schematic structure of a GMR device with parallel and antiparallel spin alignments; (c,d) corresponding density of states (with respect to energy) and spin-transfer paths (from the left to right electrode through a spacer); (e,f), schematic presentation of resistance for the spin-transfer paths.

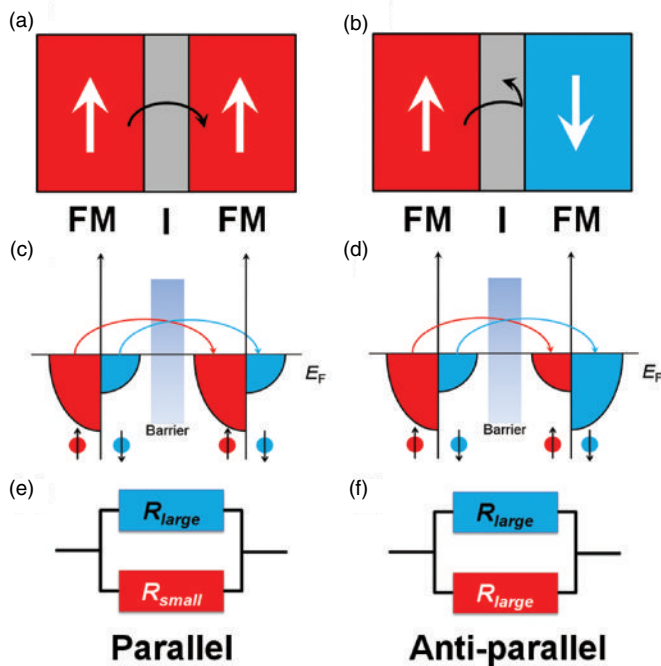


Fig. 18.2 (color online) Same as Fig. 18.1 for a TMR device.

field. Magnetoresistance (MR), the quantitative value measuring the effectiveness of a spin-valve device, is typically defined as follows:

$$\text{MR} = \frac{R_{\text{AP}} - R_{\text{P}}}{R_{\text{P}}} = \frac{G_{\text{P}} - G_{\text{AP}}}{G_{\text{AP}}} \quad (\text{optimistic}) \quad (18.1)$$

or

$$\text{MR} = \frac{R_{\text{AP}} - R_{\text{P}}}{R_{\text{AP}}} = \frac{G_{\text{P}} - G_{\text{AP}}}{G_{\text{P}}} \quad (\text{pessimistic}) \quad (18.2)$$

where R/G is resistance/conductance and P/AP is parallel/antiparallel. The optimistic version is most commonly used. However, the pessimistic MR is useful when a system has a vanishing G_{AP} , because in this case the pessimistic MR is bounded by 1, while the optimistic MR is unbounded. The type of MR is determined by a spacer material, since the mechanism of spin transport is different according to the spacer material. Figures 18.1 and 18.2 show schematic structures of two conventional spin-valve devices.

As shown in Fig. 18.1, a GMR device adopts a nonmagnetic metal (NM) as a spacer, so that spins injected from one of the FM electrodes travel through conducting channels of the NM spacer to the other FM electrode. Figure 18.1c and d show configurations of density of states (DOS) and spin-transfer paths from

FM to NM and from NM to FM for the parallel and antiparallel spin cases. Spins of the left FM electrode transfer to the nonmagnetic metal and then to the right FM electrode, which has the same spin DOS as that of the left FM electrode. In this process, spin-up and spin-down carriers have different resistance due to the asymmetric spin DOS at both electrodes as described in Fig. 18.1e and f. Resistances for the parallel and antiparallel spin configurations are as follows:

$$R_P = \frac{2(R_{\text{large}}R_{\text{small}})}{(R_{\text{large}} + R_{\text{small}})} \approx 2R_{\text{small}} \quad \text{and} \quad R_{\text{AP}} = \frac{R_{\text{large}} + R_{\text{small}}}{2} \approx \frac{R_{\text{large}}}{2}$$

Thus, the GMR device gives a substantial MR value.

When an insulator is used as a spacer, the spin transfer between two FM electrodes is achieved by quantum mechanical tunneling through the potential barrier due to the insulator, as shown in Fig. 18.2. The magnetoresistance through this mechanism is called TMR. As in the GMR device, both spin carriers have different resistance, as depicted in Fig. 18.2e and f. Resistance according to the relative spin configurations is given by

$$R_P = \frac{R_{\text{large}}R_{\text{small}}}{R_{\text{large}} + R_{\text{small}}} \approx R_{\text{small}} \quad \text{and} \quad R_{\text{AP}} = \frac{R_{\text{large}}}{2}$$

The spin flip during the tunneling process is negligible, so that the TMR can be directly expressed by spin polarization of the two FM contacts, as derived by Jullière³⁴:

$$\text{TMR} = \frac{R_{\text{AP}} - R_P}{R_P} = \frac{2P_1P_2}{1 - P_1P_2} \quad (18.3)$$

Here $P_{1(2)}$ is the polarization of the first (second) FM electrodes:

$$P_i = \frac{N_{i\uparrow}(E_F) - N_{i\downarrow}(E_F)}{N_{i\uparrow}(E_F) + N_{i\downarrow}(E_F)} \quad (18.4)$$

with the number of spin-up electrons $N_{i\uparrow}(E_F)$ and the number of spin-down electrons $N_{i\downarrow}(E_F)$ at the Fermi level (E_F). Typical TMR values ($\sim 100\%$) are larger than typical GMR values ($\sim 10\%$). Relatively low MR values in GMR devices may originate from spin flip occurred during the diffusion of the injected spins through a NM spacer.

18.2.2 Molecular Electronics

Figure 18.3 is a schematic of a two-terminal molecular electronic device. Under an applied bias voltage, electrical currents are driven through the molecule(center) from the source (left) to the drain (right) electrodes. For small molecules whose spatial extension is smaller than the mean free path of the system, electron transport shows the ballistic behavior if the device has continuum bands, while it

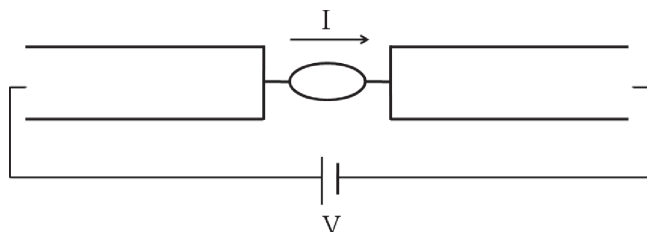


Fig. 18.3 Two-terminal molecular electronic device.

shows resonant or nonresonant tunneling behavior if the device has discrete energy levels.⁸ Molecular orbitals (MOs) of the device provide channels for electron transport. Therefore, an accurate description of molecular energy levels in the junction is vital to understanding transport properties.

As a molecule is bonded to metal electrodes, we need to take into account the following. First, there would be a significant charge transfer between electrodes and a molecule due to the dissimilarity of their electronic structures, resulting in the MO energy level shifts (Δ). Second, the molecular states are coupled to the continuum states of the electrodes, and this coupling results in a finite broadening (Γ) of molecular energy levels. Consequently, the MO energy levels are renormalized by the contact effects in the junction as depicted in Fig. 18.4. Here, we discuss how to calculate the renormalized molecular energy levels and electrical currents through them.

Before going into the detailed discussion, we describe how electrical currents are determined by alignment of the molecular energy levels with respect to the

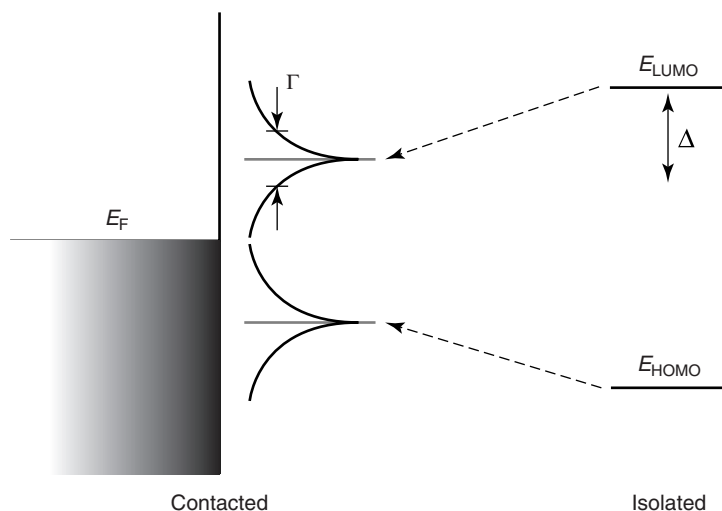


Fig. 18.4 Renormalization of the molecular energy levels in the metal–molecule contact. (From Ref. 8, with permission of RSC Publishing.)

energy bands of both leads. As an external bias voltage is applied, the chemical potential of both electrodes is split by the bias voltage, giving rise to two different Fermi functions at both electrodes. The two Fermi functions determine the energy range to allow transmission of electrons, which is called the *bias window*. The incoming electrons would transmit through the broadened energy levels as depicted in Fig. 18.5. Some of them transmit with high probability, especially at the resonance energy level, whereas others are reflected. In this way, the transmission probability as a function of energy $[T(\epsilon)]$ is determined by the renormalized molecular energy levels. Finally, we can calculate the current (I) by integrating this function over all energy ranges in the bias window restricted by the two Fermi functions $[f_L(\epsilon)$ and $f_R(\epsilon)]$ as follows:

$$I = \frac{2e}{h} \int_{-\infty}^{\infty} T(\epsilon) [f_L(\epsilon) - f_R(\epsilon)] d\epsilon \quad (18.5)$$

where h is the Planck constant and e is the electron charge. It should be emphasized that the energy-level shift and broadening are very important to determine the transmission probability and electrical currents.

Let us consider the simplest system having a single energy level. In this case, one can intuitively derive the explicit form of the transmission function. The energy broadening factor Γ is related to the electron hopping rate between the energy states of the molecule and one of the electrodes by the energy–time uncertainty principle:

$$\Delta E \Delta t = \Gamma \tau \sim h \quad (18.6)$$

where τ is the lifetime of an electron in the molecular state, and thus the hopping rate is given by $1/\tau (\sim \Gamma/h)$. Using the definition of the current, we obtain the

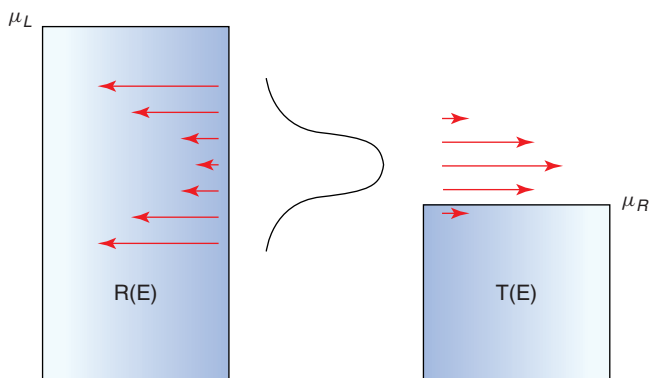


Fig. 18.5 (color online) Transmission probability in a molecular junction. $R/T(E)$ is a reflection/transmission probability as a function of energy. $\mu_{L/R}$ is the chemical potential of the left/right electrode. $T(E) + R(E) = 1$. $\mu_L - \mu_R = eV$, where V is the applied bias voltage.

following formula for the current (I_L) from the left electrode to the molecule:

$$I_L = \frac{e(N - N_L)}{\tau} = e \left(\frac{\Gamma_L}{h} \right) (N - N_L) \quad (18.7)$$

where Γ_L is the broadening factor due to the left contact, and N and $N_L [= 2f_L(\epsilon)]$ are the number of electrons in the molecule and the left electrode, respectively. In the same way, the current at the right contact is given by

$$I_R = \frac{e(N - N_R)}{\tau} = e \left(\frac{\Gamma_R}{h} \right) (N - N_R) \quad (18.8)$$

where $N_R = 2f_R(\epsilon)$. Assuming that $I = I_L = -I_R$, we calculate the number of electrons in the molecular energy level at the steady state. Then we have

$$N = \frac{\Gamma_L f_L(\epsilon) - \Gamma_R f_R(\epsilon)}{\Gamma_L + \Gamma_R} \quad (18.9)$$

and

$$I(\epsilon) = \frac{2e}{h} \frac{\Gamma_L \Gamma_R}{\Gamma_L + \Gamma_R} [f_L(\epsilon) - f_R(\epsilon)] \quad (18.10)$$

On the other hand, the molecular energy level is broadened with a factor $\Gamma (= \Gamma_L + \Gamma_R)$ due to the contact effect, as shown in Fig. 18.5. To take such an effect into account, the total current should be obtained by integrating the current as a function of energy in Eq. (18.10) over all the energy range with a weighting factor $[D(\epsilon)]$, which presents an energy-dependent distribution for the broadened molecular energy level:

$$I = \frac{2e}{h} \int_{-\infty}^{\infty} D(\epsilon) \frac{\Gamma_L \Gamma_R}{\Gamma_L + \Gamma_R} [f_L(\epsilon) - f_R(\epsilon)] d\epsilon \quad (18.11)$$

By comparing Eq. (18.11) with Eq. (18.5), we find that the transmission function for the single energy level is

$$T(\epsilon) = D(\epsilon) \frac{\Gamma_L \Gamma_R}{\Gamma_L + \Gamma_R} \quad (18.12)$$

To extend formula (18.12) for the realistic case comprised of multienergy levels, we need to deal with the Keldysh NEGF method.²²

18.2.3 Nonequilibrium Green's Function Method for Quantum Transport

A target system that we want to describe in terms of the NEGF method is composed of the device molecule and the left and right electrodes (Fig. 18.3). To establish the Hamiltonian for the system, we start from an uncoupled state where

each part is in its own equilibrium state independently, while the interaction terms between them are turned on later as a perturbative potential. By assuming that both electrodes are noninteracting systems, the Hamiltonian is

$$H_\alpha = \sum_k \varepsilon_{k\alpha} c_{k\alpha}^\dagger c_{k\alpha} \quad (18.13)$$

where $c_{k\alpha}^\dagger$ ($c_{k\alpha}$) is the creation (annihilation) operator of an electron with momentum k and kinetic energy $\varepsilon_{k\alpha}$ for the α ($=$ L,R) electrode region. For the device region, the form of the Hamiltonian depends on how to treat electron–electron or electron–phonon interactions. For the sake of simplicity, we concentrate on the noninteracting case. Then the Hamiltonian of the device part (H_{dev}) is

$$H_{\text{dev}} = \sum_n \varepsilon_n d_n^\dagger d_n \quad (18.14)$$

where $d_n^\dagger$ (d_n) is the creation (annihilation) operator of the electron in the state $|n\rangle$ with energy ε_n . We refer readers to the more specialized literature for generalization of the formalism in the case of interacting systems.^{22,35} In most practical calculations, the electron–electron interaction is effectively considered by the noninteracting Kohn–Sham potential using DFT.

The coupling effect is taken into account by turning on the interaction potential term $V_{\text{int},\alpha}$ between the device and electrode α :

$$V_{\text{int},\alpha} = \sum_{k,n} \tau_{k\alpha,n} c_{k\alpha}^\dagger d_n + \tau_{k\alpha,n}^* d_n^\dagger c_{k\alpha} \quad (18.15)$$

where $\tau_{k\alpha,n}$ denotes the hopping term from state $|n\rangle$ to state $|k\rangle$. Finally, the total Hamiltonian is given by

$$H = H_{\text{dev}} + H_L + H_R + V_{\text{int,L}} + V_{\text{int,R}} \quad (18.16)$$

By definition, electrical currents from the left electrode to the device part (I_L) can be calculated from Heisenberg's equation of motion^{22,35}:

$$I_L = \frac{d}{dt} \langle e N_L(t) \rangle = \frac{ie}{\hbar} \langle [H, N_L(t)] \rangle \quad (18.17)$$

where $N_L(t) [\equiv \sum_k c_{kL}^\dagger(t) c_{kL}(t)]$ is the number operator of electrons in the left electrode. Since $H_{L/R}$ and H_{dev} commute with the number operator, Eq. (18.17) is simplified as

$$I_L = \frac{ie}{\hbar} \langle [V_{\text{int,L}}, N_L(t)] \rangle = \frac{ie}{\hbar} \sum_{k,n} \tau_{kL,n} \langle c_{kL}^\dagger(t) d_n(t) \rangle - \tau_{kL,n}^* \langle d_n^\dagger(t) c_{kL}(t) \rangle \quad (18.18)$$

TABLE 18.1 Definition of Various Green's Functions

Definition of Various Green's Functions ^a	Name	Physical Meaning
$G_{ij}^r(t, t') = -i\theta(t - t')\langle\{c_i(t), c_j^\dagger(t')\}\rangle$	Retarded Green's function	
$G_{ij}^a(t, t') = \theta(t' - t)\langle\{c_i(t), c_j^\dagger(t')\}\rangle$	Advanced Green's function	
$G_{ij}^<(t, t') = i\langle c_j^\dagger(t')c_i(t)\rangle$	Lesser Green's function	Particle propagator
$G_{ij}^>(t, t') = -i\langle c_i(t)c_j^\dagger(t')\rangle$	Greater Green's function	Hole propagator
$G_{ij}^t(t, t') = -i\langle T[c_j^\dagger(t')c_i(t)]\rangle$	Time-ordered Green's function	
$G_{ij}^{\bar{t}}(t, t') = -i\langle \bar{T}[c_j^\dagger(t')c_i(t)]\rangle$	Anti-time-ordered Green's function	

Source: Ref. 22.

^a $c_i^\dagger$ (c_i) denotes the particle creation (annihilation) operator for state $|i\rangle$. T ($\bar{T}$) is the time-ordering (anti-time-ordering) operator. Symbol $\langle \hat{A} \rangle$ means the thermal average of the operator $\hat{A}$ over the grand canonical ensemble.

By introducing the lesser Green's function defined in Table 18.1, Eq. (18.18) becomes

$$I_L = \frac{e}{\hbar} \sum_{k,n} \tau_{kL,n} G_{kL,n}^<(t, t) + \tau_{kL,n}^* G_{n,kL}^<(t, t) \quad (18.19)$$

Equation (18.19) can be rewritten in the energy domain by using Fourier transform:

$$I_L = \frac{e}{\hbar} \sum_{k,n} \int_{-\infty}^{\infty} \frac{d\varepsilon}{2\pi} [\tau_{kL,n} G_{n,kL}^<(\varepsilon) + \tau_{kL,n}^* G_{kL,n}^<(\varepsilon)] \quad (18.20)$$

Equation (18.20) indicates that the current at the left contact equals the sum of all possible contributions of the particle (electron) propagations from the arbitrary state $|n\rangle$ in the device part to an arbitrary state $|k\rangle$ in the left electrode, or vice versa. According to the Keldysh nonequilibrium Green's function formalism, the lesser Green's function in Eq. (18.20) is decomposed into the propagation part in the electrodes and the propagation part in the device molecule with a corresponding hopping term between them²²:

$$G_{kL,n}^<(\varepsilon) = \sum_m \tau_{kL,m} [g_{kL,kL}^t(\varepsilon) G_{m,n}^<(\varepsilon) - g_{kL,kL}^<(\varepsilon) G_{m,n}^{\bar{t}}(\varepsilon)] \quad (18.21)$$

$$G_{n,kL}^<(\varepsilon) = \sum_m \tau_{kL,m}^* [g_{kL,kL}^<(\varepsilon) G_{n,m}^t(\varepsilon) - g_{kL,kL}^{\bar{t}}(\varepsilon) G_{n,m}^<(\varepsilon)] \quad (18.22)$$

Here we introduced time-ordered and anti-time-ordered Green's functions from Table 18.1. In Eqs. (18.21) and (18.22), $G_{n,m}(\epsilon)$ represents particle propagation between states $|n\rangle$ and $|m\rangle$ in the device part, and $g_{kL,kL}(\epsilon)$ denotes the Green's function for the noninteracting left electrode:

$$g_{kL,kL}^<(\epsilon) = 2\pi i f(\epsilon) \delta(\epsilon - \epsilon_k) \quad (18.23)$$

$$g_{kL,kL}^>(\epsilon) = -2\pi i [1 - f(\epsilon)] \delta(\epsilon - \epsilon_k) \quad (18.24)$$

By inserting Eqs. (18.21) and (18.22) into Eq. (18.20), one finally arrives at the following:

$$I_L = \frac{ie}{\hbar} \sum_{n,m} \int_{-\infty}^{\infty} d\epsilon \tau_{L,n} \tau_{L,m}^* \rho_L(\epsilon) \{G_{n,m}^<(\epsilon) + f_L(\epsilon) [G_{n,m}^r(\epsilon) - G_{n,m}^a(\epsilon)]\} \quad (18.25)$$

where $\rho_L(\epsilon)$ is the density of states for the left electrode and we use the following relations²²: $G^r(\epsilon) + G^{\bar{r}}(\epsilon) = G^>(\epsilon) + G^<(\epsilon)$ and $G^>(\epsilon) - G^<(\epsilon) = G^r(\epsilon) - G^a(\epsilon)$. In Eq. (18.25), $G^r(\epsilon)$ and $G^a(\epsilon)$ denote the retarded and advanced Green's functions for the device part, respectively, which can be obtained by Fourier transformation of the retarded and advanced Green's functions defined in Table 18.1 to the energy domain.

We can evaluate the current at the right contact I_R in the same way. For a steady state, which means that $I = I_L = -I_R$, the current in a matrix version is

$$I = \frac{ie}{2\hbar} \int_{-\infty}^{\infty} \text{Tr}\{[f_L(\epsilon)\Gamma_L(\epsilon) - f_R(\epsilon)\Gamma_R(\epsilon)][G^r(\epsilon) - G^a(\epsilon)]\} \\ + \text{Tr}\{[\Gamma_L(\epsilon) - \Gamma_R(\epsilon)]G^<(\epsilon)\} d\epsilon \quad (18.26)$$

where

$$\Gamma_{L/R}(\epsilon) = 2\tau_{L/R}^+ \rho_{L/R}(\epsilon) \tau_{L/R} = -2 \text{Im}[\tau_{L/R}^+ g_{L/R}^r(\epsilon) \tau_{L/R}] = -2 \text{Im}[\Sigma_{L/R}^r(\epsilon)] \quad (18.27)$$

The $\Gamma_{L/R}(\epsilon)$ is twice the imaginary part of the retarded self-energy for the left/right electrodes $[\Sigma_{L/R}(\epsilon)]$. The lesser Green's function in the device part for the noninteracting system is defined by³⁵

$$G^<(\epsilon) \equiv if_L(\epsilon)G^r(\epsilon)\Gamma_L(\epsilon)G^a(\epsilon) + if_L(\epsilon)G^r(\epsilon)\Gamma_R(\epsilon)G^a(\epsilon) \quad (18.28)$$

Finally, one obtains the electrical current:

$$I = \frac{e}{h} \int \text{Tr}[G^a(\epsilon)\Gamma_R(\epsilon)G^r(\epsilon)\Gamma_L(\epsilon)][f_L(\epsilon) - f_R(\epsilon)] d\epsilon \quad (18.29)$$

The final expression for the noninteracting system is exactly the same as Eq. (18.5) if Eq. (18.29) is multiplied by 2 to take into account the spin

degeneracy. Thus, the transmission in the noninteracting regime is given by

$$T(\epsilon) \equiv \text{Tr}[G^a(\epsilon)\Gamma_R(\epsilon)G^r(\epsilon)\Gamma_L(\epsilon)] \quad (18.30)$$

The next step is to calculate the retarded/advanced Green's function and the left/right coupling (i.e., self-energy) terms.

18.3 NUMERICAL IMPLEMENTATION

Theoretical description of quantum transport requires sophisticated calculations for a metal–molecule junction composed of a large number of atoms. Density functional theory (DFT), as reviewed in Chapters 1 to 3, enables us to perform accurate calculations of electronic structure for such a system at the first-principles level with computational efficiency. In addition, the NEGF method can easily be implemented in a usual DFT code, since an electron density, the main ingredient in DFT, can be obtained directly from the NEGF method for an open system. In this section we discuss the detailed numerical implementation issues of the NEGF method based on DFT.

18.3.1 Green's Function

Accurate description of the metal–molecule contact geometry is essential to take into account correctly the contact effects discussed above. As shown in Fig. 18.6, the entire device system has an infinite length without periodicity. However, each electrode has well-defined periodic conditions except near the surface region at the metal–molecule contact, since only electrons close to the surface region are redistributed to screen the potential induced from the metal–molecule junction. Indeed, typical metal electrodes such as gold are good conductors that can effectively screen the induced potential within a few atomic layers. Therefore, the remote part of electrodes can be regarded as remaining in their bulk state if a sufficient part of the electrode is included into the scattering part as a screening region.

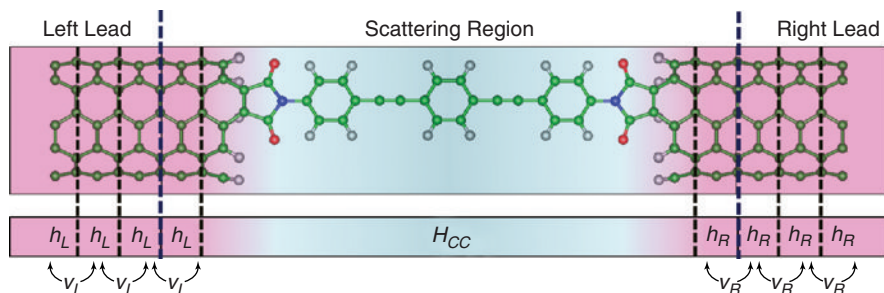


Fig. 18.6 (color online) Simulation box of an extended molecular system.

In this extended molecule model, most studies have used the electrode part with a few atoms or metal clusters, which may not be appropriate to take into account the screening effect. A few program codes have considered a realistic contact model by adopting sufficiently large atomic layers with a periodic boundary condition along the direction perpendicular to the current flow.^{28,30–33} Finally, as the bulk property stemming from the remote part of the electrodes is treated effectively by introducing self-energy, the effective Hamiltonian (H_{eff}) projected onto the extended molecular part is

$$H_{\text{eff}} = H_{\text{CC}} + \Sigma_{\text{L}} + \Sigma_{\text{R}} \quad (18.31)$$

Note that the self-energy terms should involve the contact effects; the real part of the self-energy gives the energy-level shift [$\Delta_{\text{L/R}} = 2 \text{Re}(\Sigma_{\text{L/R}})$], while the imaginary part of the self-energy results in the energy-level broadening [$\Gamma_{\text{L/R}} = -2 \text{Im}(\Sigma_{\text{L/R}})$], as depicted graphically in Fig. 18.4.

In a matrix notation, the Hamiltonian for an open system can be written as

$$H = \begin{bmatrix} \ddots & & & \vdots & & & \ddots \\ & h_{\text{L}} & v_{\text{L}} & 0 & 0 & 0 & \\ & v_{\text{L}}^{\dagger} & h_{\text{L}} & V_{\text{LC}} & 0 & 0 & \\ \cdots & 0 & V_{\text{CL}} & H_{\text{CC}} & V_{\text{CR}} & 0 & \cdots \\ & 0 & 0 & V_{\text{RC}} & h_{\text{R}} & v_{\text{R}} & \\ & 0 & 0 & 0 & v_{\text{R}}^{\dagger} & h_{\text{R}} & \\ \ddots & & & \vdots & & & \ddots \end{bmatrix} \quad (18.32)$$

To evaluate the retarded Green's function of the Hamiltonian, we have to invert the matrix of the infinite dimension as follows:

$$G^r(\epsilon) = [\epsilon S - H]^{-1} \quad (18.33)$$

where S denotes the overlap matrix. Since the remote part of each electrode retains its bulk state due to the screening effect, we only need to calculate the Green's function projected onto the extended molecular region:

$$G_{\text{MM}}^r(\epsilon) = [\epsilon S_{\text{MM}} - H_{\text{MM}} - \tau_{\text{ML}}(\epsilon) g_{\text{LL}}^r(\epsilon) \tau_{\text{LM}}(\epsilon) - \tau_{\text{MR}}(\epsilon) g_{\text{RR}}^r(\epsilon) \tau_{\text{RM}}(\epsilon)]_{\text{M} \times \text{M}}^{-1} \quad (18.34)$$

Here, the Hamiltonian of the extended molecular region is

$$H_{\text{MM}} = \begin{bmatrix} h_{\text{L}} & V_{\text{LC}} & 0 \\ V_{\text{CL}} & H_{\text{CC}} & V_{\text{CR}} \\ 0 & V_{\text{RC}} & h_{\text{R}} \end{bmatrix} \quad (18.35)$$

the surface Green's function of the lead α is

$$g_{\alpha\alpha}^r(\epsilon) = [\epsilon S_{\alpha\alpha} - H_{\alpha\alpha}]^{-1} \quad (18.36)$$

and the coupling term between the extended molecular region and the lead part α is

$$\tau_{M\alpha}(\epsilon) = \epsilon S_{M\alpha} - v_{M\alpha} \quad (18.37)$$

One can also calculate the Green's function for the extended molecular region using the effective Hamiltonian in Eq. (18.31):

$$G_{MM}^r(\epsilon) = [\epsilon S_{MM} - H_{\text{eff}}]^{-1} \quad (18.38)$$

Consequently, the explicit form of the self-energy terms is

$$\Sigma_{\alpha}^r(\epsilon) = \tau_{M\alpha}(\epsilon) g_{\alpha\alpha}^r(\epsilon) \tau_{\alpha M}(\epsilon) = \tau_{\alpha}^{\dagger}(\epsilon) g_{\alpha\alpha}^r(\epsilon) \tau_{\alpha}(\epsilon) \quad (18.39)$$

where we define $\tau_{\alpha}(\epsilon) = \tau_{\alpha M}(\epsilon)$ for the sake of simplicity. Note that the final expression of the self-energy in Eq. (18.39) is the same as that defined in Eq. (18.27). The surface Green's function in Eq. (18.39) can be obtained from separate calculations of the bulk system corresponding to the periodic part of electrodes.³²

18.3.2 Density Matrix

Once we obtain the self-energy matrix, the effective Hamiltonian will be given by Eq. (18.31). Then we can calculate the retarded Green's function matrix, which is related directly to the density matrix as follows:

$$\rho_{nm} \equiv \langle d_n^{\dagger}(t) d_m(t) \rangle = -i G_{nm}^{<}(t, t) = \int \frac{d\epsilon}{2\pi i} G_{nm}^{<}(\epsilon) \quad (18.40)$$

By inserting the relation in Eq. (18.28) into Eq. (18.40), the density matrix element becomes

$$\rho_{mn} = \int \frac{d\epsilon}{2\pi} [f_L(\epsilon) G^r(\epsilon) \Gamma_L(\epsilon) G^a(\epsilon) + f_L(\epsilon) G^r(\epsilon) \Gamma_R(\epsilon) G^a(\epsilon)]_{mn} \quad (18.41)$$

which gives an electron density of the molecular region in the non-equilibrium state. In equilibrium, $f_L = f_R = f$, so that

$$\begin{aligned} \rho_{nm} &= \int \frac{d\epsilon}{2\pi} [G^r(\epsilon) \Gamma(\epsilon) G^a(\epsilon)]_{nm} f(\epsilon) = \int \frac{d\epsilon}{2\pi} [G^r(\epsilon) - G^a(\epsilon)]_{nm} f(\epsilon) \\ &= -\frac{1}{\pi} \text{Im} \int d\epsilon [G^r(\epsilon)]_{nm} f(\epsilon) \end{aligned} \quad (18.42)$$

The electron density is

$$\rho(r) = \sum_{nm} \rho_{nm} \phi_n^*(r) \phi_m(r) \quad (18.43)$$

where $\phi_{n/m}(r)$ is a localized atomic basis orbital.

Direct numerical integration along the energy domain in Eq. (18.42) requires huge computational costs. According to the residue theorem, the equivalent result can be obtained by integration along a certain contour on the complex plane:

$$\int G^r(z) f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}[G(z) f(z)] \quad (18.44)$$

where $z_k = i(2k + 1)\pi k_B T$ and

$$\int_{-\infty}^{\infty} G(\varepsilon) f(\varepsilon) d\varepsilon = - \int_C G(z) f(z) dz - 2\pi i k_B T \sum_{k=1}^n G(z_k) \quad (18.45)$$

Here z_k is a singular point of the Fermi function on the complex plane, called the Matsubara frequency. Figure 18.7 shows an example of contour points to be used in numerical evaluation of the first term on the right-hand side in Eq. (18.45). In this way, the number of grid points can be drastically reduced to obtain a reasonable density matrix.³³

The retarded Green's function, which gives electron density for an open system, must be a functional of the electron density, since it is calculated from the Kohn–Sham Hamiltonian. Therefore, the electron density in the steady state

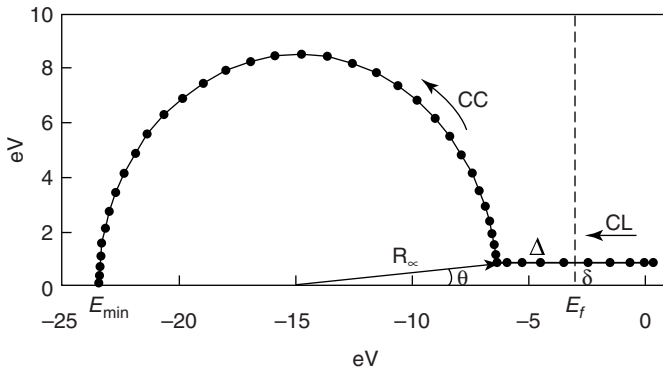


Fig. 18.7 Example of contour points on the complex plane obtained by the Gaussian quadrature method. CL and CC represent the direction of the contour integral. Δ , δ , R_{cc} , and θ are parameters to determine the shape of the contour. $E_{\min}$ is the minimum energy point on the contour. E_F is the Fermi energy. (From Ref. 33, with permission of John Wiley & Sons, Inc.)

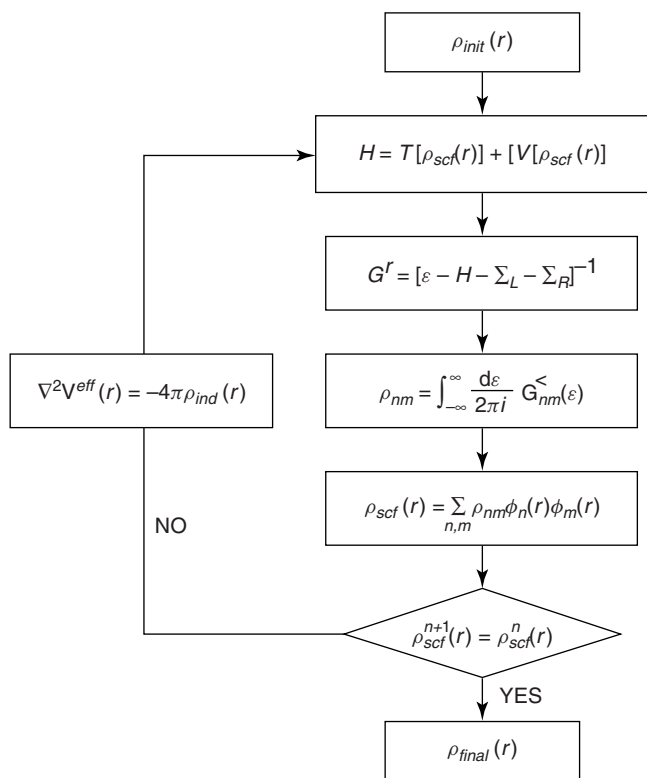


Fig. 18.8 Self-consistent loop of NEGF + DFT. (From Ref. 8, with permission of RSC Publishing.)

for an open system should be converged via a self-consistent loop as shown in Fig. 18.8. Based on the final converged electron density, one can calculate transmission values and in turn electrical currents at a given bias voltage using Eq. (18.29).

18.4 EXAMPLES

18.4.1 Molecular Analogs of Conventional Spin-Valve Devices

The most studied spintronic devices based on molecules are analogs to the prototype of conventional spin-valve devices shown in Figs. 18.1 and 18.2.^{36–39} Organic molecules simply replace spacer materials bridging two FM electrodes. A theoretical work in this direction was done by Emberly and Kirczenow with empirical parameters in 2002.³⁶ This study was carried out later at the first-principles level by Rocha et al.³⁹ They adopted two different molecules between Ni electrodes; one of them is nonconjugated, the other is conjugated. These two

molecules play a role as an insulator and nonmagnetic metal of the conventional TMR and GMR devices, respectively, since the nonconjugated molecule has a large energy gap between the highest-occupied molecular orbital (HOMO) and the lowest-unoccupied molecular orbital (LUMO), while the conjugated molecule has a small HOMO–LUMO energy gap.

Figure 18.9 shows a nonconjugated octane-based device (Ni/octane/Ni) and its transport properties. The transmission value of the device near the E_F in

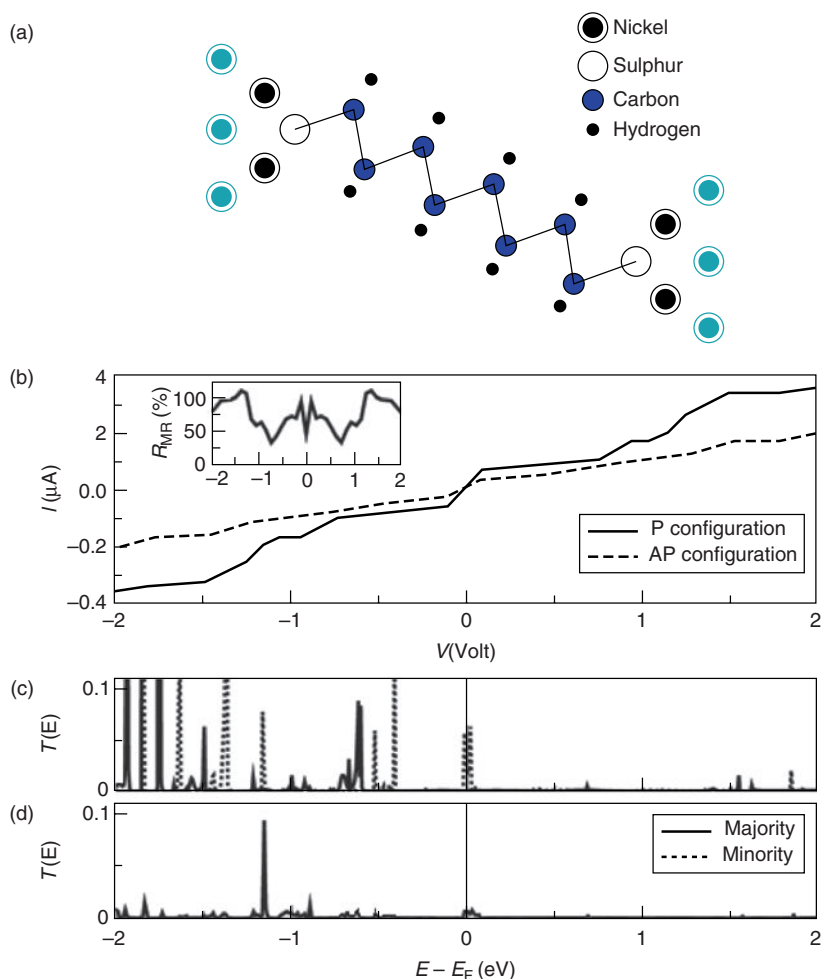


Fig. 18.9 (color online) Structure of the octane molecule attached to the (001) Ni surface (a), the corresponding current–voltage (I – V) characteristics (b), and the zero-bias transmission coefficients $[T(E)]$ for the parallel (P) (c) and antiparallel (AP) (d) configurations of an octane-based nickel spin valve. In the antiparallel case the transmission coefficients for both spin directions are identical. The inset in (b) shows the magnetoresistance ratio. E_F is the position of the Fermi level of the nickel leads. (From Ref. 39, with permission of Nature Publishing Group.)

Fig. 18.9c decays exponentially as the length of the alkane chain increases. This result, which is consistent with experimental observation,¹² demonstrates that the transport is due to tunneling, so that the device really behaves as a TMR device. The parallel spin configuration of the two Ni electrodes shows a significant resonance peak at the E_F for minority spins, while such a peak diminishes for the antiparallel configuration. As a result, the $I-V$ curves of the parallel and antiparallel configurations are quite different at a low bias region, as shown in Fig. 18.9b. The $I-V$ curve is nonlinear for the parallel configuration, whereas it is almost linear for the antiparallel configuration. The bias-dependent MR ratio in the optimistic version [Eq. (18.1)] shows the maximum value over 100% at a certain finite voltage as shown in the inset of Fig. 18.9b.

Figure 18.10 shows a conjugated tricene-based device (Ni/tricene/Ni) and its transport properties. In contrast to the Ni/octane/Ni device, it has finite

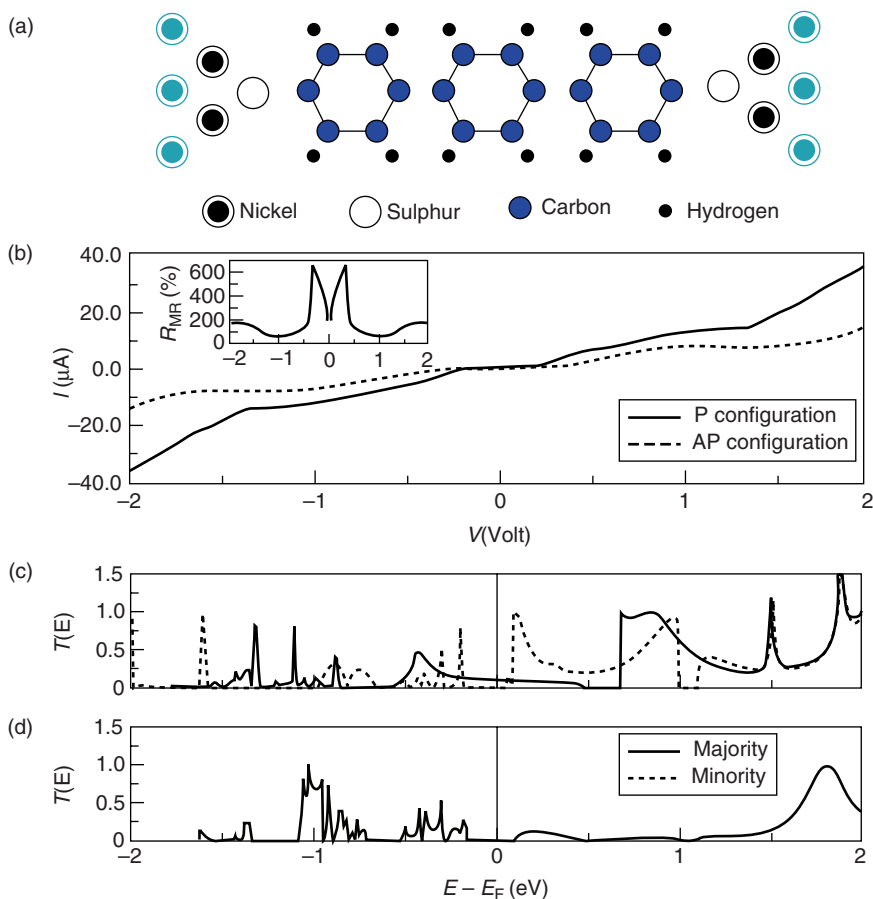


Fig. 18.10 (color online) Same as Fig. 18.9 for the tricene molecule. (From Ref. 39, with permission of Nature Publishing Group.)

transmission values over a broad range around E_F for the parallel spin configuration between both Ni electrodes, as shown in Fig. 18.10c. Moreover, theoretical results confirm that the transmission values do not show a strong dependence on the molecular length according to the increasing number of phenyl groups in the molecule. Thus, the conjugated molecular device is analogous to the GMR device. The parallel spin configuration shows finite current, whereas the antiparallel spin configuration shows considerably suppressed current at a low bias region, since the transmission values are very small around E_F . Consequently, the maximum MR ($\sim 600\%$) is obtained at a low bias, as shown in the inset of the Fig. 18.10b.

Molecular spintronic devices introduced in this section exhibit considerably larger MR values than those reported in experiments,¹² because the present theoretical method did not take into account the reduction of MR values due to spin-flip and electron–phonon coupling. Therefore, the calculated results should be considered as an upper limit for observable experimental values.

18.4.2 Single-Molecular Magnets

Up to now we have considered a prototype of spin-valve devices with a variation of spacer materials. This section introduces a different type of spintronic devices. A molecule incorporating transition metal ions may have a nonzero spin state due to the magnetic coupling between spins of the transition metals. Such a metal complex molecule would show spin-dependent transport phenomena, as it is attached to electrodes as depicted in Fig. 18.11. An interesting point in

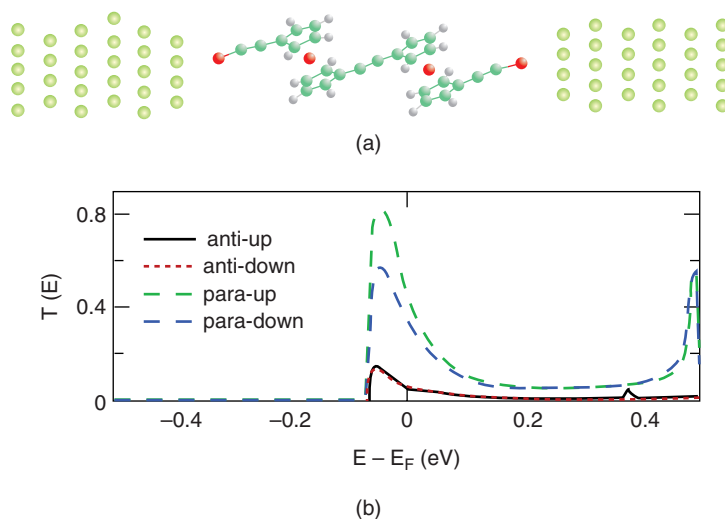


Fig. 18.11 (color online) (a) Structure of two molecules containing two cobaltocene moieties (di-Co) which are adsorbed at hollow sites on Au(001) leads (Au, yellow; H, white; C, blue; S, orange); (b) corresponding transmission functions. (From Ref. 40, with permission of The American Chemical Society.)

this type of device is that electrodes do not need to be a FM metal. The spin-dependent effects can be driven from control of the intrinsic magnetic property of the molecule, in contrast to conventional analogs, where the spin-dependent effects come from FM contacts. The magnetic properties in the molecule can be controlled not only by a magnetic field but also an electric field.⁴¹

Figure 18.11 shows a cobaltocene moiety attached to two gold electrodes. The dicobaltocene molecule has two cobalt ions whose spins favor the antiparallel configuration due to the superexchange interaction between them. One can control the spin configuration to be parallel or antiparallel by applying large magnetic fields or electric fields. Theoretical calculations based on the NEGF + DFT method show that the transmission values in the parallel spin state are much larger than that in the antiparallel spin state (Fig. 18.11b). In this way, the dicobaltocene device plays a role as a complete spin-valve device without FM contacts.

Another type of molecular spintronic device can be achieved by using a single molecular magnet which exhibits a remnant magnetization, since individual molecules can store information. A recent experiment has indeed demonstrated that a molecular magnet (Fe_4) on the gold surface exhibited magnetic hysteresis.¹⁹ However, such a magnetic state cannot be described by ground-state calculations, so the present method would not be appropriate for studying a single molecular magnet.

18.4.3 Super Magnetoresistance Based on a Graphene Nanoribbon

Here we discuss the fascinating phenomena of an extreme enhancement in magnetoresistances which originate entirely from the wave property of an electron. Graphene has great advantages to be used in spintronics such as CNTs: extreme flexibility, stability, and as high carrier mobility.⁴² This has been proven from the demonstration of spin injection into graphene.⁴³ The spin-relaxation length of the injected spins is about a few micrometers even at room temperature. In particular, a zigzag graphene nanoribbon (ZGNR) shows intriguing ferromagnetic spin ordering along their edges.^{44,45} The ZGNR can be utilized as a spacer material of a spin-valve device as depicted in Fig. 18.12. The behavior of this device is unique compared with the previous conventional analogs.^{41,46} As the magnetic configuration at both electrodes is parallel/antiparallel, the spin magnetization on the nanoribbon would follow the same parallel/antiparallel configuration. This behavior offers a new type of magnetoresistance.

Figure 18.13 exhibits two orbitals of the ZGNR associated with two different bands. The lower energy bands with respect to the E_F have C_2 symmetry regardless of the spin polarization, whereas the higher-energy bands have σ symmetry. To find out how the symmetry affects transmission values, we plot transmission curves together with band structures for the left and right electrodes only in the case of the α -spin (to avoid complexity) as shown in Fig. 18.14. The band structure of each electrode is calculated from the bulk ZGNR with the ferromagnetic spin configuration. For the parallel case, bands

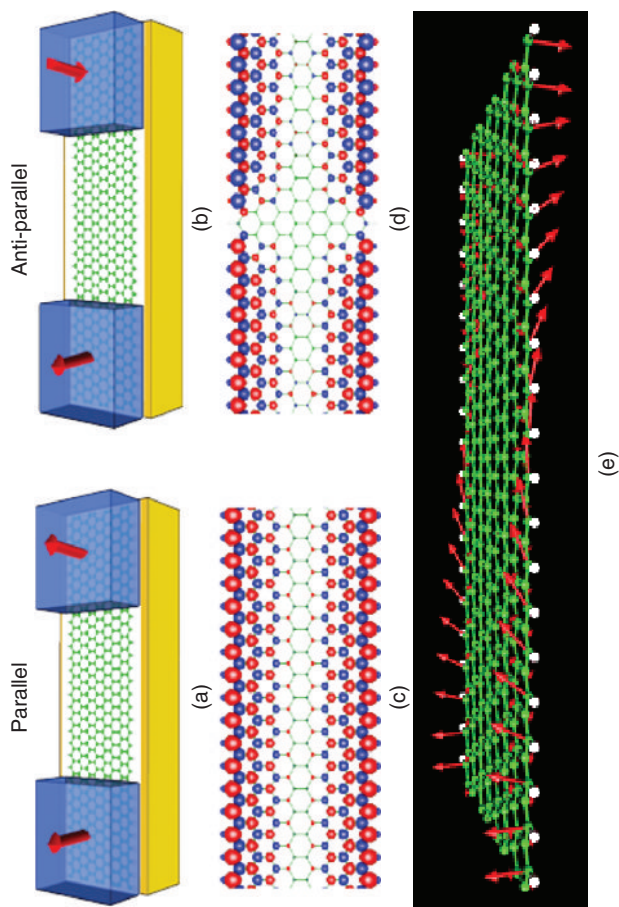


Fig. 18.12 (color online) Schematic ZGNR-based spin-valve device with parallel (a) and antiparallel (b) spin configurations, the corresponding spin-magnetization density isosurfaces (c, d), and the noncollinear spin orientations in the ZGNR with a domain wall for the anti-parallel case (e). (From Ref. 46, with permission of Nature Publishing Group.)

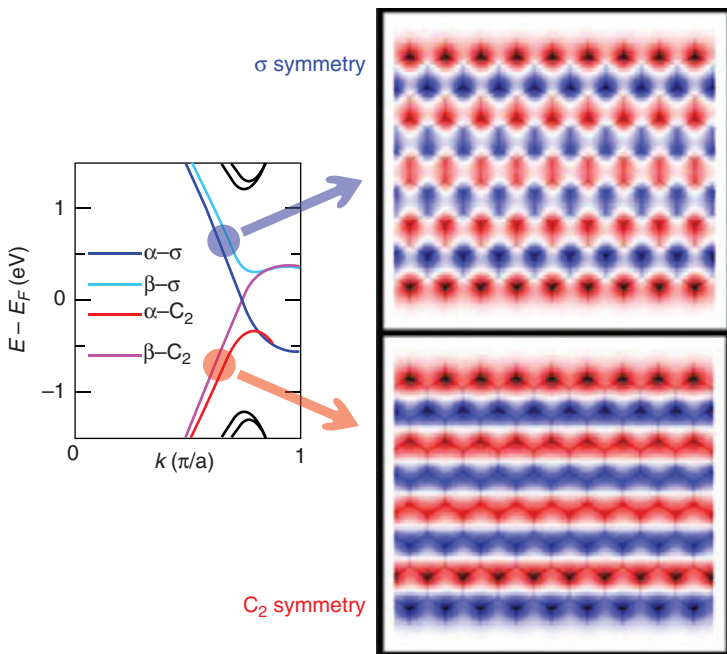


Fig. 18.13 (color online) Orbital symmetries of the band structure of the ZGNR. The upper and lower panels on the right exhibit the orbitals (wavefunctions) corresponding to the bands on the left panel, respectively. The upper panel shows σ symmetry with respect to the middle horizontal line, while the lower panel shows C_2 symmetry. (From Ref. 46, with permission of Nature Publishing Group.)

having the same symmetry are aligned for all energy ranges, yielding perfect transmission (upper panel in Fig. 18.14). In contrast, for the antiparallel case, the transmission curve has perfect reflection in a particular energy range around the E_F where the orbital symmetries are mismatched (lower panel in Fig. 18.14).

Spin-dependent currents are calculated using a spin-polarized version of Eq. (18.29). Figure 18.15 shows the calculated $I-V$ characteristics for the ZGNR spin-valve device. When the spin configuration is parallel, the $I-V$ curve is linear due to the constant transmission region around E_F . The slope of the $I-V$ curve (i.e., conductance) is quantized ($2e^2/h$), which means that each spin state contributes one complete transport channel. In contrast, there is no transport channel around E_F in the case of the antiparallel spin configuration. Therefore, the current is suppressed up to a certain threshold voltage. In this way, magnetoresistance in the graphene nanoribbon spin-valve device is modulated not only by spin symmetry matching but also orbital symmetry matching, which is contrasted to the fact that only the spin symmetry matching is relevant in conventional spin-valve devices.

The MR value calculated exceeds a million percent, which is ten thousands larger than the maximum value experimentally reported so far. In conclusion, the

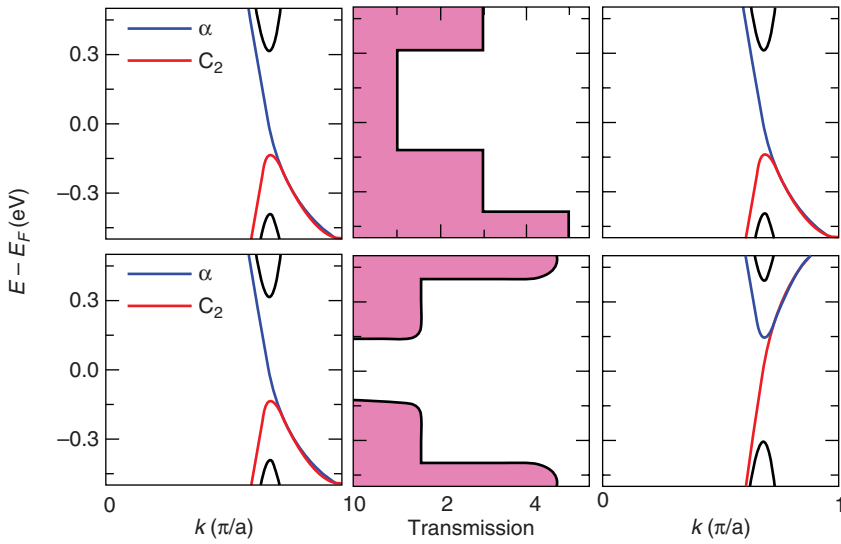


Fig. 18.14 (color online) Band structures for the left lead (left), the right lead (right), and the corresponding transmission curve (middle) for the α -spin in the parallel (upper panel) and antiparallel (lower panel) configurations of the ZGNR for the zero bias. (From Ref. 46, with permission of Nature Publishing Group.)

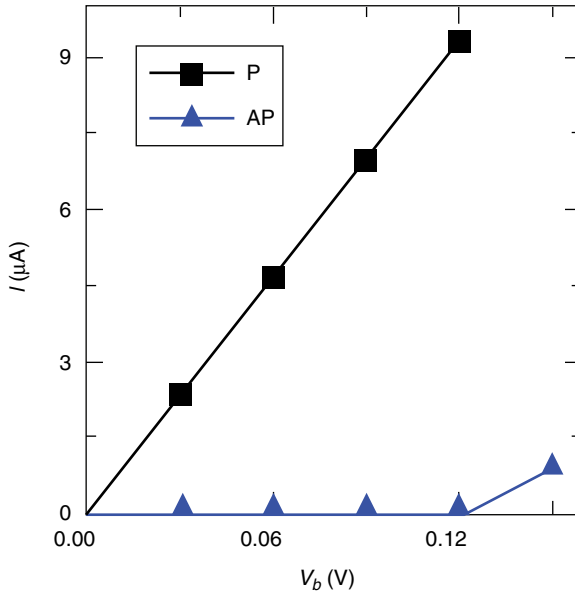


Fig. 18.15 (color online) I - V curves of the ZGNR spin-valve device for the parallel (P) configuration (black) and the antiparallel (AP) configuration (gray). (From Ref. 46, with permission of Nature Publishing Group.)

double spin-filtering effect offers a new type of magnetoresistance to achieve an ideal spin-valve device, called supermagnetoresistance.

18.5 CONCLUSIONS

The advance in information technology requires constantly smaller and faster devices. Molecular spintronics that has emerged with the ideal combination of both molecular electronics and spintronics has been proposed as an ultimate solution to catering for such a demand. Theoretical tools based on a first-principles method for studying quantum transport offer a great opportunity to investigate spin-dependent phenomena in a variety of molecular devices with an accurate description of molecular electronic structures. In particular, molecular analogs of the conventional magnetic tunnel junction have been widely studied. Magnetic control of the molecular orbital symmetry as well as spin symmetry in graphene nanoribbons has led to a new type of magnetoresistance, offering a key idea to make an ideal spin-valve device. Despite successful examples reviewed in this chapter, theoretical results based on the present method should be regarded as the upper limit to be observed in reality, because of the absence of spin-flip process in calculations. Spin flip can occur during the spin-injection or detection process at metal–molecule contacts. The electron–phonon interaction in molecules could be another source of the spin flip. These effects should be taken into account for more quantitative description of spin-dependent transport. From the practical point of view, it is desirable to design molecular spintronic devices comprised purely of organic materials, including ferromagnetic electrodes, to achieve efficient spin injection through a low potential barrier at the molecular junction due to the electronic similarity between the organic molecule and the electrodes.

REFERENCES

1. Wolf, S. A.; et al. *Science* **2001**, *294*, 1488–1495.
2. Fert, A. *Rev. Mod. Phys.* **2008**, *80*, 1517–1530.
3. Grünberg, P. A. *Rev. Mod. Phys.* **2008**, *80*, 1531–1540.
4. Akerman, J. *Science* **2005**, *308*, 508–510.
5. Joachim, C.; Gimzewski, J. K.; Aviram, A. *Nature* **2000**, *408*, 541–548.
6. Nitzan, A.; Ratner, M. A. *Science* **2003**, *300*, 1384–1389.
7. Tao, N. J. *Nat. Nanotech.* **2006**, *1*, 173–181.
8. Kim, W. Y.; Choi, Y. C.; Min, S. K.; Cho, Y.; Kim, K. S. *Chem. Soc. Rev.* **2009**, *38*, 2319–2333.
9. Kim, W. Y.; Choi, Y. C.; Kim, K. S. *J. Mater. Chem.* **2008**, *18*, 4510–4521.
10. Naber, W. J. M.; Faez, S.; Wiel, W. G. *J. Phys. D* **2007**, *40*, R205–R228.
11. Tsukagoshi, K.; Alphenaar, B. W.; Ago, H. *Nature* **1999**, *401*, 572–574.
12. Petta, J. R.; Slater, S. K.; Ralph, D. C. *Phys. Rev. Lett.* **2004**, *93*, 136601.

13. Xiong, Z. H.; Wu, D.; Vardeny, Z. V.; Shi, J. *Nature* **2004**, *427*, 821–824.
14. Hueso, L. E.; et al. *Nature* **2007**, *445*, 410–413.
15. Hill, E. W.; Geim, A. K.; Novoselov, K.; Schedin, F.; Blake, P. *IEEE Trans. Magn.* **2006**, *42*, 2694–2696.
16. Heersche, H. B.; et al. *Phys. Rev. Lett.* **2006**, *96*, 206801.
17. Jo, M.-H.; et al. *Nano Lett.* **2006**, *6*, 2014–2020.
18. Grose, J. E.; et al. *Nature Mater.* **2008**, *7*, 884–889.
19. Mannini, M.; et al. *Nature Mater.* **2009**, *8*, 194–197.
20. Bogani, L.; Wernsdorfer, W. *Nature Mater.* **2008**, *7*, 179–186.
21. Datta, S. *Electronic Transport in Mesoscopic Systems*, Cambridge University Press, Cambridge, UK, 1995.
22. Haug, H.; Jauho, A.-P. *Quantum Kinetics in Transport and Optics of Semiconductors*, Springer-Verlag, Berlin, 1996.
23. Datta, S.; et al. *Phys. Rev. Lett.* **1997**, *79*, 2530–2533.
24. Nardelli, M. B. *Phys. Rev. B* **1999**, *60*, 7828–7833.
25. Ventra, M. D.; Pantelides, S. T.; Lang, N. D. *Phys. Rev. Lett.* **2000**, *84*, 979–982.
26. Derosa, P. A.; Seminario, J. M. *J. Phys. Chem. B* **2001**, *105*, 471–481.
27. Taylor, J.; Guo, H.; Wang, J. *Phys. Rev. B* **2001**, *63*, 245407.
28. Kim, Y.-H.; Tahir-Kheli, J.; Schultz, P. A.; Goddard, W. A., III. *Phys. Rev. B* **2006**, *73*, 235419.
29. Palacios, J. J.; Perez-Jimenez, A. J.; Louis, E.; Verges, J. A. *Phys. Rev. B* **2001**, *64*, 115411.
30. Brandbyge, M.; Mozos, J.-L.; Ordejon, P.; Taylor, J.; Stokbro, K. *Phys. Rev. B* **2002**, *65*, 165401.
31. Ke, S.-H.; Baranger, H. U.; Wang, W. *Phys. Rev. B* **2004**, *70*, 085410.
32. Rocha, A. R.; García-Suárez, V. M.; Bailey, S. W.; Lambert, C. J.; Ferrer, J.; Sanvito, S. *Phys. Rev. B* **2006**, *73*, 085414.
33. Kim, W. Y.; Kim, K. S. *J. Comput. Chem.* **2008**, *29*, 1073–1083.
34. Jullière, M. *Phys. Lett. A* **1975**, *54*, 225–226.
35. Meir, Y.; Wingreen, N. S. *Phys. Rev. Lett.* **1992**, *68*, 2512–2515.
36. Emberly, E. G.; Kirczenow, G. *Chem. Phys.* **2002**, *281*, 311–324.
37. Pati, R.; Scnapati, L.; Ajayan, P. M.; Nayak, S. K. *Phys. Rev. B* **2003**, *68*, 100407(R).
38. Waldron, D.; Haney, P.; Larade, B.; MacDonald, A.; Guo, H. *Phys. Rev. Lett.* **2006**, *96*, 166804.
39. Rocha, A. R.; García-Suárez, V. M.; Bailey, S. W.; Lambert, C. J.; Ferrer, J.; Sanvito, S. *Nature Mater.* **2005**, *4*, 335–339.
40. Liu, R.; Ke, S.-H.; Baranger, H. U.; Yang, W. *Nano Lett.* **2005**, *5*, 1959–1962.
41. Kim, W. Y.; Kim, K. S. *Acc. Chem. Res.* **2010**, *43*, 111–120.
42. Geim, A. K.; Novoselov, K. S. *Nature Mater.* **2007**, *6*, 183–191.
43. Tombros, N.; Jozsa, C.; Popinciuc, M.; Jonkman, H. T.; Wees, B. J. V. *Nature* **2007**, *448*, 571–574.
44. Fujita, M.; Wakabayashi, K.; Nakada, K.; Kusakabe, K. *J. Phys. Soc. Jpn.* **1996**, *7*, 1920–1923.
45. Pisani, L.; Chan, J. A.; Montanari, B.; Harrison, N. M. *Phys. Rev. B* **2007**, *75*, 064418.
46. Kim, W. Y.; Kim, K. S. *Nature Nanotechnol.* **2008**, *3*, 408–412.

19 Calculating Molecular Conductance

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In this chapter, the theory of electron transport through single molecule junctions is reviewed and applications presented. The nonequilibrium Green's function theory commonly used to reduce the system, which involves semiinfinite leads, to a size amenable to high-level electronic structure calculations is introduced and illustrated with model system calculations. The significance of basic chemical properties such as the nature of the metal–organic interface is stressed, along with physical properties such as elastic and inelastic scattering, device heating and dissipation, and current-induced forces. Applications discussed include rectification, negative differential resistance, molecular switches, thermoelectric effects, photoactive switching, spintronics, logic gate design, and DNA sequencing.

19.1 INTRODUCTION

In the past decade, the world of molecular nanotechnology has opened up in almost unimaginable ways. The once visionary predictions that electrons could tunnel under bias through molecular monolayers¹ and that single molecules could function as electronic components² and be wired into large-scale devices³ have been realized. Experimental techniques have been developed to allow measurements of electron transport through single molecules bound to metallic electrodes⁴ with large numbers of measurements and statistical techniques used to determine single-molecule conductance reliably.^{5–8} Together these developments present a relatively new and unexplored domain for theoretical efforts: molecules bound in electrically conducting junctions.

Environmental effects on molecular properties are well known: for example, the many effects of solvent; however, binding molecules in conducting junctions introduced hitherto unseen environmental effects. In some sense, the junction

behaves as a heterogeneous solvent, shifting vibrational frequencies and molecular energy levels. When molecules are strongly bound to electrodes, however, the solvent analogy fails to capture the details of the system, as covalent bonds result in charge-transfer and structural changes beyond anything that could be introduced by intermolecular interactions. These systems have provided an enduring challenge as the details of the electronic structure depend on the precise structure of the system studied, sometimes down to the number of junction atoms included explicitly in the calculation. Whereas many system properties, such as vibrational spectra, may be insensitive to the details of the electronic structure, transport can be exquisitely sensitive. This sensitivity is both a blessing and a curse. On the one hand, it should, one day, provide extremely fine control to benchmark theory against experiment. Yet, on the other, until that day comes it can result in stark disparity between results from relatively similar theoretical methods.

In addition to the challenge of describing the equilibrium electronic structure of the junction, transport calculations require a method to describe the nonequilibrium behavior of the system when subjected to some external perturbation. The simplest molecular electronic devices have as the only perturbation an external electric field; however, more complex function and varied applications can result from the effects of light, temperature gradients, or chemical change in the junction. The strength with which a molecule is bound in the junction can influence the nature of the predominant transport processes. For example, weakly electronically bound systems are more likely to exhibit Coulomb blockaded transport with single electron charging events visible as the electric field in the junction is increased and the molecule moves through different redox states.⁹ Conversely, in junctions where the molecule is strongly bound to the electrodes, it is more likely that tunneling processes will dominate transport; this is the regime we focus on in this chapter. In this instance, a molecule in a junction effectively acts as a tunnel barrier to transport.

Understanding the challenges involved in providing a good description of the electronic structure for transport calculations is intimately linked to understanding what controls transport properties. For this reason, we proceed in five sections. In Section 19.2 we give a very brief outline of the nonequilibrium Green's function (NEGF) formalism; for a more thorough treatment of the theory and its applications, see Chapter 1 (basics) and Chapter 18 (spintronics applications). We also outline the connection between basic aspects of a system's structure and the resulting transport characteristics. Section 19.3 provides an overview of the various electronic structure methods used in transport calculations and the errors and approximations involved. The final three sections then shift focus to what can be understood from molecular transport calculations. More specifically, what can be understood about the nature of molecular electron transport itself in Section 19.4, chemical trends in molecular electron transport in Section 19.5, and the design of molecular electronic devices in Section 19.6.

As a final introductory remark, we note that there are many related areas that have been instructive and inspirational for researchers in molecular electron transport that will not be covered in this chapter. For example, electron transport

through quantum dots, atomic wires, DNA, nanotubes, C₆₀, nanowires, proteins, and intramolecular electron transfer have all been highly influential and have been studied in great detail. Many of the ideas highlighted in this chapter may be applicable to these systems; however, readers should look elsewhere for a complete treatment of the developments in these areas.

19.2 OUTLINE OF THE NEGF APPROACH

The choice of computational transport method, even when simply considering tunneling in strongly bound systems, is as varied as is the choice of electronic structure method. At the highest level there are a variety of formalisms, but the details involved in the implementation of a method mean that codes using the same formal approach may differ substantially. The NEGF approach outlined here features prominently in early work in the area^{10–12} and forms the basis of a number of widely used implementations.^{13–24} Alternative approaches have been employed, both in the early stages of development^{25–28} and in more recent implementations.^{29–31} Importantly, it can be shown that reasonable agreement between different theoretical approaches can be achieved.^{32,33}

In this section we will outline the NEGF formalism in its most simple form at the Landauer (coherent tunneling) level of transport^{34–36}; for a more extensive introduction, see Chapter 1. The power of the approach is not only in what it offers at this level of theory, but also the extent to which it can be extended to include many more complex processes, such as electron correlation,³⁷ inelastic transport,^{19,38–40} light induced by transport, and transport induced by light,^{41,42} to name a few.

19.2.1 Formal Details

Before we detail the formal aspects of the approach, it is important to understand the partitioning of the system in a physical sense. The junction is divided into three regions, illustrated in Fig. 19.1. It should be noted that all parts of the system are not treated equally in the transport formalism, so care is required to ensure that the partitioning is sensible. In Section 19.3.3 we explore further what constitutes sensible choices for the partitioning; at this stage we simply highlight that the system comprises three regions, two leads, and an extended molecule region, which may or may not include some number of lead atoms. In Fig. 19.1 the extended molecule includes an entire layer of the electrode, but this need not be the case, and the partitioning may simply include some number of atoms that form part of one or several layers.

The first step in the transport calculation is an electronic structure calculation for the full system. This may be performed with periodic boundary conditions or without, effectively modeling the leads as clusters and then the partitioning is invoked. The Hamiltonian (Kohn–Sham or Fock matrix) that is obtained from the electronic structure calculation is divided according to the partitioning; in

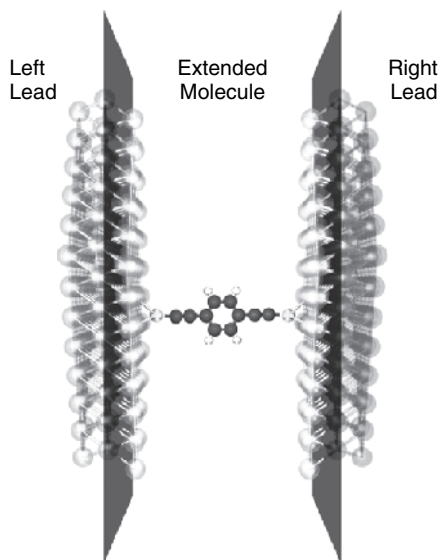


Fig. 19.1 Partitioning of the junction into three regions: two leads and an extended molecule.

the usual case that the calculation is performed with a nonorthogonal basis, the overlap matrix will be partitioned similarly.

$$H = \begin{bmatrix} H_L & V_{ML}^\dagger & 0 \\ V_{ML} & H_M & V_{MR} \\ 0 & V_{MR}^\dagger & H_R \end{bmatrix} \quad (19.1)$$

Here the subscripts L and R designate the left and right leads, and the subscript M denotes the extended molecule.

The NEGF approach leads to a computationally convenient setup by allowing the problem to be described with matrices that are only the size of the extended molecule. The effect of the, possibly very large, leads is subsumed into self-energies which enter into the extended molecule's Green's function. First we construct the unperturbed Green's functions for the leads:

$$\begin{aligned} g_L(E) &= (zS_L - H_L)^{-1} \\ g_R(E) &= (zS_R - H_R)^{-1} \end{aligned} \quad (19.2)$$

where $z = E + i\eta$ and η is a positive infinitesimal. These Green's functions are then used to construct the self-energies:

$$\begin{aligned} \Sigma_L(E) &= (zS_{ML} - V_{ML})g_L(E)(zS_{ML}^\dagger - V_{ML}^\dagger) \\ \Sigma_R(E) &= (zS_{MR} - V_{MR})g_R(E)(zS_{MR}^\dagger - V_{MR}^\dagger) \end{aligned} \quad (19.3)$$

These self-energies are complex and, in a general sense, the real part can be considered to shift the spectrum of the extended molecule energetically while the imaginary part will broaden it. At this point we should note that the real part of the self-energy is nonzero only when there is some energy dependence in the density of states of the leads. This distinction is important, as the density of states of a metal lead may be approximated as a constant, leading to the wide-band approximation, meaning that there is only an imaginary component to the self-energy.

The convenience of the method remains only as long as a computationally efficient method for calculating the Green's functions of the leads is obtained. For example, the finite spatial extent of the range of interaction means that block iterative schemes can be developed¹⁴; however, we do not discuss these approaches in detail.

With the self-energies for the two leads, the retarded Green's function for the extended molecule is constructed:

$$G_M^r(E) = [zS_M - H_M + \Sigma_L(E) + \Sigma_R(E)]^{-1} \quad (19.4)$$

The advanced Green's function is then simply obtained, $G_M^a(E) = G_M^{r\dagger}(E)$.

The final pieces that are required to calculate the transmission are obtained directly from the imaginary component of the self-energies:

$$\Gamma_L(E) = i[\Sigma_L(E) - \Sigma_L^\dagger(E)] \quad (19.5)$$

$$\Gamma_R(E) = i[\Sigma_R(E) - \Sigma_R^\dagger(E)] \quad (19.6)$$

Together the transmission is then obtained as the trace over a matrix product:

$$T(E, V) = \text{Tr}[\Gamma_L(E)G^r(E)\Gamma_R(E)G^a(E)] \quad (19.7)$$

The bias enters the formalism in a second way through the effective window of integration given by the difference in the Fermi functions of the two leads. The electronic structure of some molecular systems is relatively invariant to small changes in the bias, and in these cases the current–voltage characteristics of the junction may be reasonably approximated by integrating the zero-bias transmission over a varying window.

Throughout this chapter we also refer to the conductance of the junction, which is properly the differential conductance and is defined as

$$g(V) = \frac{dI(V)}{dV} \quad (19.8)$$

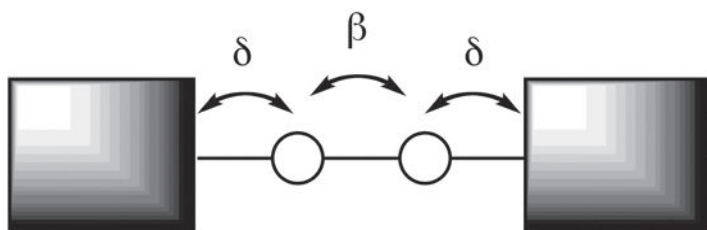


Fig. 19.2 Simple two-site model for a junction.

19.2.2 Model System

With this simple introduction to the transport formalism, we illustrate some of the general features with a model system. We consider a simple two-site model for the extended molecule with a single coupling element to each lead as shown in Fig 19.2.

We can write down a Hückel Hamiltonian (see Chapter 10) to model this system where each electrode is represented as a single site:

$$H = \begin{bmatrix} \alpha & \delta & 0 & 0 \\ \delta & \alpha & \beta & 0 \\ 0 & \beta & \alpha & \delta \\ 0 & 0 & \delta & \alpha \end{bmatrix} \quad (19.9)$$

We assume that the single site in each lead that couples to the molecule is part of some large semi-infinite lead with a constant density of states $\rho(E) = \rho_0$. This assumption yields self-energies that will broaden, but not shift, the features in the transmission:

$$\Sigma_L = \begin{bmatrix} \frac{i\delta^2\rho_0}{2} & 0 \\ 0 & 0 \end{bmatrix} \quad \Sigma_R = \begin{bmatrix} 0 & 0 \\ 0 & \frac{i\delta^2\rho_0}{2} \end{bmatrix} \quad (19.10)$$

This model can be used to explain some basic correlations between chemical features and transmission characteristics. In all cases we set both α and the electrode Fermi energy equal to zero and investigate the effect of varying β and δ away from their initial values of $\beta = -2.7$ eV and $\delta = \beta/3 = -0.9$ eV. First, by varying the magnitude of β , the effect of the strength of the intramolecular coupling can be investigated. This can simply be the electronic coupling matrix element between two atoms or, alternatively, two subunits of a molecule. Figure 19.3 shows the changes in transmission that result from a moderate increase or decrease in the β , as well as a substantial decrease, to just 5% of

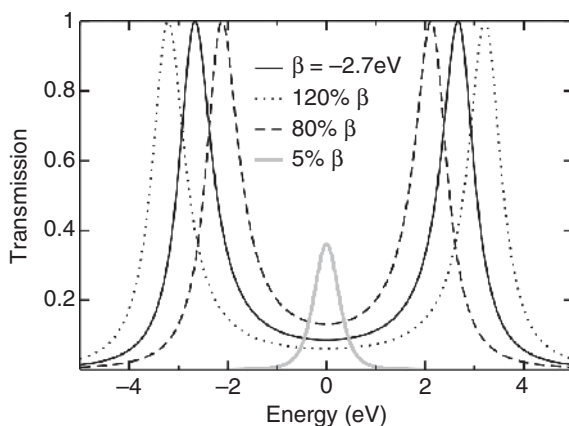


Fig. 19.3 Variation in the transmission as the value of β changes.

the original value to indicate the effect of electronic coupling matrix element approaching zero.

As β increases, the splitting between the bonding and antibonding orbitals of the isolated molecule increases and so does the position of the corresponding molecular resonances in the transmission. The width of the resonances, which we will show is controlled by δ , remains constant, so the transmission at $E = 0$ actually decreases, despite the stronger interaction between the components of the molecule. Conversely, decreasing β by 20% actually results in increased transmission at $E = 0$. As the coupling becomes very low, the molecular orbitals of the isolated system approach degeneracy and the transmission decreases substantially. This is a physically intuitive picture and could correspond to two electrodes, each with a single hydrogen atom adsorbed but sufficiently far apart that there was very little interaction between the atoms. This is not a situation where high levels of electronic transmission would be expected. This example illustrates an important aspect of transmission calculations: The location of molecular resonances is critical, in many cases, for determining the magnitude of the transmission near the Fermi energy and weakly bound systems may exhibit large transmission by this means.

There is one very important point to note at this stage regarding the interpretation of transmission features in terms of molecular orbitals. Sometimes it has been suggested that the form of a molecular orbital (e.g., the delocalized orbitals typical of conjugated systems) is indicative of high transmission; however, this is not the case. In this example, it is clear that the form of the molecular orbitals, the bonding and antibonding orbitals, are invariant to the change in the coupling. The eigenvalues shift as a response to the coupling strength, and it is the splitting of the orbitals that indicates the strength of the electronic coupling and therefore transport through the system. Conjugated molecules with delocalized electron density will also, generally, have an eigenvalue spectrum with split pairs of orbitals indicating strong coupling through the system, and this is the

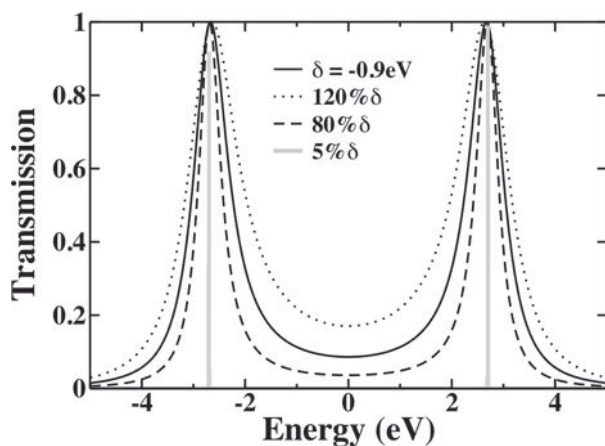


Fig. 19.4 Variation in the transmission as δ changes. The 5% δ transmission appears as a vertical line below each resonance.

critical factor in the magnitude of the transmission at any particular resonance. It is not the form of a molecular orbital but the fact that it is split from its symmetry-related pair that controls the magnitude of transmission, and as the orbitals coalesce, the transmission through both will decrease.

The second aspect we illustrate with this model is the way the transmission changes with increasing or decreasing strength of the coupling to the electrodes, shown in Fig. 19.4. The parameter δ directly controls the magnitude of the imaginary part of the self-energy, and when the transmission is examined it is clear why this component is described as broadening the features. As the coupling to the electrodes, δ , goes to zero, the transmission resonances become infinitely sharp, appearing in Fig. 19.4 as two vertical lines below the resonances. Integrating this curve to give a measure of the current, we find that the current goes to zero, exactly as we would expect for a system with no coupling between the various components.

In a real molecule, in contrast with our model system, changes in the coupling to the electrode can also be accompanied by changes in the charge transferred between the molecule and the leads. This can shift resonances energetically, potentially resulting in dramatic differences in the transmission and current through the junction.

In this example we have set both the site energies, α , and the electrode Fermi energy to zero, resulting in transmission resonances symmetrically positioned about the Fermi energy. There is no requirement, however, that this will be the case, and in many molecular junctions it will not be. When the Fermi energy falls closer to the resonances of either the occupied or virtual orbitals, it is common to discuss the transport as being predominantly either holes or electrons, respectively. It is also common for transport to be described as being “through the HOMO” (the highest-occupied molecular orbital) or “through the LUMO” (the

lowest-unoccupied molecular orbital). This is simply a verbal crutch and should not be taken literally; charge transport does not really occur in the mean-field molecular orbital space in which it is convenient to work. When charge transport is said to be “through” a particular molecular orbital, it indicates that the position of the Fermi energy results in a dominant contribution to the transport coming from the tail of resonances that are energetically proximate to either the HOMO or the LUMO of the isolated molecule. In both Figs. 19.3 and 19.4 it is clear that the proximity of the Fermi energy to the resonances will have a significant impact on the magnitude of the transmission, and therefore the current, through the system.

In subsequent sections, many of the physical and chemical changes to the junction can be understood simply in terms of the extent to which they either increase or decrease either the intramolecular or molecule–electrode electronic coupling matrix elements or they shift the position of resonances with respect to the Fermi energy. Considering the different systems in these terms can provide some insight into why the transmission features vary as they do.

19.3 ELECTRONIC STRUCTURE CHALLENGES

As an electronic structure calculation always underlies the transmission, any errors and assumptions inherent in the electronic structure may have considerable impact on the conductance characteristics predicted. The development of a good description of electron transport through junctions must therefore start with a strong electronic structure foundation.

19.3.1 Methods

A variety of methods have been used to provide the requisite electronic structure calculations for transport. There is an essential compromise involved in the choice of electronic structure method. On the one hand, it would be desirable to move toward higher-level methods to ensure a more accurate treatment of the electronic structure. On the other, however, a molecular junction is truly described only with a considerable number of lead atoms included. No matter how high the level of theory, a large electrode induces electronic structure changes on a chemisorbed molecule, which cannot be modeled well by a single atom.

Much of the early work in the area made use of Hückel models^{43,44} or semiempirical methods^{10,12,25,45} to treat the system. Indeed, these methods were very successful at capturing many aspects of the transport properties and are still in use today.^{16,46,47} These methods are extremely fast, making it possible to include large numbers of atoms explicitly and perform transport calculations with relative ease. More recently, SCC-DFTB (self-consistent-charge density functional tight binding) has been used to study a variety of transport properties in molecular junctions.^{14,15,40,48} SCC-DFTB is also an approximate method, effectively a tight-binding Hamiltonian parameterized using density functional theory, and also offers excellent computational efficiency.

By far the most widely used method in transport calculations is density functional theory (DFT).^{13,17,19,21,22,49–53} Over the years, increasing computer power and increasingly efficient DFT methods have extended the size of the system that can be treated explicitly. Today, very large systems with periodic or open boundary conditions are used, offering a very good description of molecular junctions within the inherent limitations of this level of theory (for more information, see, e.g., Chapters 2 and 3).

Molecular conductance calculations are extremely sensitive to the position of molecular energy levels relative to the lead Fermi level, called the *band lineup problem*.^{50,54} Higher-level theoretical methods have been implemented for transport applications in an effort to circumvent the known problems of the common approaches. Two different approaches using the GW^{55–57} and configuration interaction³⁰ techniques have been developed and applied to molecular junctions. Both of these methods show promise; however, as always there are compromises involved in moving to higher-level approaches.

There are a variety of electronic structure problems that have been shown to cause artifacts in transport calculations,^{47,51,58–60} and indeed it is likely that more will be discovered as the complexity of the systems studied increases. This is, perhaps, unsurprising. Many years of work have been involved in developing methods to obtain reliable electronic structures for molecules in the gas phase, and molecular transport junctions offer many additional complexities: in particular, the calculations must describe out-of-equilibrium situations, so the variational principle fails. Until a greater body of understanding is obtained, care should be taken and any transport results obtained should be interrogated to ensure that qualitative, if not quantitative, sense is maintained.

19.3.2 Basis Sets

As in computational chemistry generally, the effect of basis set changes cannot be underestimated. There have been a number of studies of basis set effects, using Gaussian-type basis sets,^{61–63} but understanding of basis set effects is far from complete. One question that remains is whether traditional atom-centered basis sets will provide a good description of transport through molecular monolayers where cooperative effects may be in play. An alternative approach has been to use plane-wave basis sets and Wannier functions to provide the localization required for partitioning.^{23,24} In the long term it remains to be seen which of these approaches may offer the best combination of efficiency and accuracy in describing transport in junctions.

19.3.3 Partitioning

As highlighted at the outset of Section 19.2, the partitioning of the system is of particular importance. A fundamental distinction between the part of the system that is treated as the extended molecule from the part treated as leads is that transmission resonances originating from the leads are not necessarily well treated. The electrode self-energies may shift, broaden, or in the case of

semiconductor electrode suppress regions of the transmission spectrum, but the features of the transmission spectrum will be treated most reliably when they come from the extended molecule. Calculations have looked at the effects of changing the size of the extended molecule^{64–67}; ideally, the extended molecule size will be increased until there is convergence in the transport properties.

The most dramatic changes are seen when the extended molecule size is changed from encompassing the molecule alone to including any number of lead atoms (typically, this number ranges from three to tens of lead atoms). This may change the symmetry of the system, as the leads will not generally have the symmetry of the molecule.⁶⁸ It will also introduce additional features between the resonances associated with the HOMO and LUMO, sometimes referred to as metal-induced gap states. The question that must be asked in any system design is what properties of the system will be interrogated. If a qualitative comparison between different molecules is desired, an extended molecule that does not include any lead atoms may be desirable, as it will accentuate the molecular features in the transmission. If quantitative agreement with experiment is desired, lead atoms clearly need to be included, as these may introduce the features that dominate the transmission spectrum at low bias.

19.3.4 Field Effects

The first point where nonequilibrium effects enter the transport calculation is with the application of a bias voltage, and this presents a challenge for theory. The effects of an electric field on molecular electronic structure are not necessarily insignificant and have been studied in transport junctions.^{69–73} Today, the most sophisticated approaches use a self-consistent method to calculate the potential drop across the junction in the presence of an applied bias.^{13,15,16,20,73,74} The challenge for these approaches is that the self-consistent calculation may be time consuming, and convergence is not always straightforward.

19.4 CHEMICAL TRENDS

The chemical trends in molecular electron transport are, unsurprisingly, one of the most studied aspects of the area. The work done has, in part, followed prior work on intramolecular electron transfer and elucidated the same relationships that were seen in that area. Nonetheless, the demonstration of the fundamental link between chemical properties and electron transmission, across different environments, is an important aspect of establishing that it is truly the nature of the molecule which controls a junction's transport characteristics.

Transport junctions with large leads also present additional structural aspects which need to be considered when examining trends. Many significant details are involved in the nature of the binding to the electrode, as well as the structure of the electrode itself, thermal flexibility in the junction, and intermolecular interactions in monolayers. Figure 19.5 illustrates some of the aspects considered in this section.

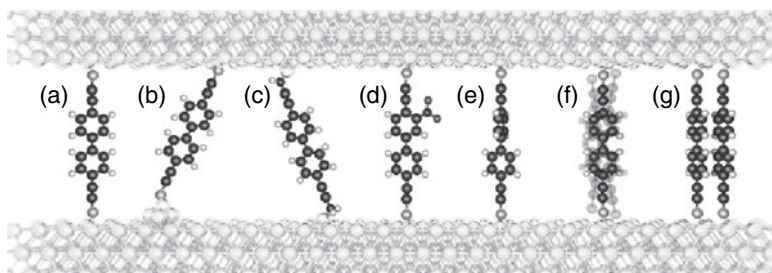


Fig. 19.5 Some of the details of a junction that may impact transport properties (a), the binding site (b), binding orientation and electrode structure (c), binding groups (d), substituent effects (e), conformational flexibility (f), thermal fluctuations, and intermolecular interactions (g).

19.4.1 Electrode Materials

Studies of chemical trends focus predominantly on chemical and physical changes in the molecule and how these affect transport properties. The effect of the electrode material and structure is, however, of equal if not greater importance. Transport calculations have largely used gold(111) electrodes, but other electrode materials have been studied,^{75–79} including semiconductor,^{80–83} ferromagnetic,^{84–91} and nanotube^{79,92} electrodes. Changing electrode materials introduces different electronic and spin densities of states and, consequently, can alter significantly the electronic and spin transport through the junction.

19.4.2 Binding Site

Starting with the simplest possible structure for a gold(111) electrode, a planar surface, there already is a variable in the calculation setup: the choice of binding site. Most commonly, researchers have investigated three sites, frequently with the molecule bound perpendicular to the surface: the atop site, where the binding group binds to a single surface atom; the face-centered cubic (fcc) and hexagonal close-packed (hcp) hollow sites, where the binding group sits at the midpoint of a triad of surface atoms; and the bridge site, where the binding group sits above a pair of surface atoms. The fcc and hcp hollow sites are distinguished by the structure of the atoms in the second layer of the lead, in the case of the fcc site there is a hollow site in the second layer while the hcp site sits above an atop site in the second layer.

Depending on the theoretical method, the variation predicted between binding sites can change considerably^{51,65,76,93–100}; in some cases the variation between sites is rather dramatic. Changing the binding site can change the electrode coupling and thereby charge transfer to the molecule, moving resonances closer or farther from the Fermi energy. Depending on the magnitude of this shift, it can have a significant influence on the transport properties.

19.4.3 Binding Orientation

In addition to the choice of binding site, the orientation of a molecule with respect to the electrode surface can also affect the transport properties,^{93,94,99–110} for similar reasons. In Fig. 19.5 we illustrated the effect of binding orientation with a simple tilt on the molecule and some additional surface structure on the electrode; however, the real system may be significantly more complicated.

In break-junction experiments the junction is being elongated with a changing electrode structure for each measurement. Calculations have shown that the effect of elongating a junction^{93,103,111,112} and changing the electrode structure with elongation^{113–116} can have a significant impact on the conductance.

19.4.4 Length Dependence

A very large number of theoretical studies have shown the well-known behavior that conductance decreases (usually, exponentially) with increasing molecular length.^{11,33,94,98,117–122} This property is an intuitive one, even from the basic understanding that can be derived from the conduction properties of macroscopic wires, although the reasons underlying the behavior differ substantially on these two dramatically different length scales. Two precautions should be appreciated here. First, the trend is observed only when the molecules in the series are truly similar. For example, two fully conjugated molecules are not necessarily similar if the nature of the conjugation is not in fact the same.¹²³ Second, even for a given series of molecules, small structural changes can result in different length-dependent decay.¹²⁰ For this reason, among others, care has to be taken when comparing the decay characteristics between methods, two methods may agree well for one series of molecules while they give significantly different results for another.³³

19.4.5 Binding Groups

The following two items deal with substituent effects; however, we separate out one type of substituent for particular attention: the binding group between the molecule and the electrodes. These groups are often called *alligator clips*,¹²⁴ in analogy with conventional macroscopic electronics. This description is apt; indeed, these groups are used to ensure strong binding with the surface; however, it captures only part of their role. As these groups control the chemical bond between the molecule and the electrode, they also control the charge transfer, thereby influencing the energetic relationship between the molecular energy levels and the Fermi energy, which we previously highlighted as being of particular importance.

Over the years many studies have compared the influence of different binding groups^{26,33,76,95,100,113,118,125–132}; here we simply discuss the most prevalent choices. The most commonly used binding group is a thiol termination on a metal (generally, gold) surface. It is understood that this group will chemisorb on the metal surface with the terminal hydrogen atoms removed. Other analogous groups

have been studied (O, Se, Te),^{26,76,125,127,128,131} although none have found such widespread use. Ligands with two sulfurs (dithiocarboxylates) have also been computed.¹³² In more recent years, —NH_2 binding groups have been studied increasingly.^{33,100,113,116,129} In these systems the molecule binds to the electrode through the nitrogen lone pair and the terminal hydrogen atoms remain. In a similar fashion, —PH_2 groups have also been studied.^{113,116} Finally, the effect of asymmetric combinations of binding groups^{101,118,133,134} has also been studied, including the extreme case where only one side chemisorbs.^{101,118,133} These systems have been used to create junctions with asymmetry in the coupling strength to the two electrodes, which is of particular interest for basic rectifier models.

19.4.6 Substituents

The second aspect of substituent effects is the role that functional groups, generally electron donating or withdrawing, can have on transmission. These effects have been studied in a variety of molecules,^{63,65,66,131,135–138} and they can have a variety of effects. Perhaps unsurprisingly, the influence of a substituent can even depend on its orientation,^{46,131,139} as the extent to which it influences the π -system transport in a conjugated system varies. Recent work has shown that in molecules where the transmission is dominated by destructive quantum interference effects, substituents can be used to induce particularly dramatic changes in transmission.¹³⁵ In molecules where destructive interference effects are not present, the substituents act primarily to shift the position of the molecular resonances. In systems with interference features, however, substituents can also shift the interference minima, which can result in particularly stark changes at low bias.

19.4.7 Interference Effects

The quantum nature of molecular electronic structure is one of the aspects that makes molecular junction transport so fascinating. Molecules are not simply small wires; interference effects can mean that seemingly insignificant changes to a system can result in dramatic changes in the transport properties. Destructive interference effects manifest as dips in the molecular transmission, and the width and depth of the dips can be tuned by changing the properties of the molecule.

One of the first areas where this effect was seen was in early Hückel model studies where the Hamiltonian was extended beyond its usual tridiagonal form to include non-nearest-neighbor interactions.^{140–142} In that case, it was shown that even small non-nearest-neighbor couplings could result in considerable changes in the transmission. These results highlight how the complex quantum nature of transport paths through molecules can yield interesting surprises.

There are common molecular systems where the electronic coupling and transport properties are known to be dominated by interference effects. The best known and studied is the variation between *ortho*-, *meta*-, or *para*-substituted phenyl rings,^{128,143–147} and similarly in other cyclic structures.^{145–150} Recent

work has also shown that dramatic interference effects can be seen in the transport properties of acyclic cross-conjugated molecules^{151,152} and other acyclic structures.^{153,154} Together these systems offer interesting prospects for fine-tuning molecular electronic transport over a large dynamic range. Importantly, from the perspective of the limited scope of this chapter, interference features can induce dramatic changes in the electronic transmission far from resonance, where the assumption of elastic tunneling is better founded.

19.4.8 Conformational Dependence

The dihedral between conjugated components in a molecule (e.g., phenyl rings) has long been known to control the electronic coupling through the system and thereby the transport.¹⁵⁵ This effect was clearly demonstrated with molecular conductance measurements⁸ and has been examined extensively theoretically.^{8,46,62,115,120,156–164} As the dihedral increases, the strength of the π transport through the system, which is the dominant component of the transmission, decreases. This relationship is so straightforward that it can even be modeled with simple Hückel models introducing a $\cos \theta$ dependence into the electronic coupling matrix element between the two phenyl rings.

More recent work had examined the less intuitive conformational dependence of σ -bonded systems. As the geometry of an alkane⁹⁷ or silane¹⁶⁵ is varied from the minimum energy all-*trans* structure to introduce a *gauche* defect, again with the variation of one dihedral, the transmission also decreases considerably. This result arises from a more complicated mechanism: the interaction in σ -systems between non-nearest neighbor atoms. Previous work has revealed this decreasing coupling through model system calculations using the ladder C or ladder H models.¹⁶⁶

19.4.9 Thermal Variation

Having already detailed the sensitivity to binding site, binding orientation, and molecular conformation, an issue clearly arises as to what extent thermal fluctuations will alter molecular conductance. Experimental measurements sample a distribution of molecular geometries, and theoretical studies have sought to replicate this by examining a range of geometries often generated from molecular dynamics simulations.^{114,152,167–171} The distribution of conductance values that thermal fluctuations produce might seem to suggest that it is impossible to distinguish similar molecules through measurement of molecular conductance alone. It has been shown, however, that multiple sampling can be used to distinguish overlapping distributions, even from very similar molecules.¹⁶⁹

19.4.10 Intermolecular Interactions

Experimental measurements generally start with a monolayer, or some large number, of molecules in the junction. In this situation it is easy to envisage that the

close proximity of other molecules may lead to variations in the conductance, as the presence of one molecule modifies the electronic structure of the another. This effect sets molecular junctions apart from analogous macroscopic circuits, where, for example, the behavior of one resistor is invariant to other resistors in parallel. The effects of intermolecular interactions have been studied theoretically at a large range of levels of theory and there are certainly regimes where the contributions of molecules in parallel do not simply sum.^{162,172–178}

19.5 FEATURES OF ELECTRONIC TRANSPORT

The transport calculations that have been performed on molecular junctions have revealed more than simply an estimate of the magnitude of the current. The nature of the transport process can be probed theoretically, yielding understanding that can be used to design systems with precise attributes.

19.5.1 Spatial Distribution

In the case of elastic tunneling through a molecular system, the question of what the spatial distribution of current is flowing through that system might seem to be spurious. One line of argument would suggest that the molecule is acting as a tunneling barrier, the electron never resides on the molecule, and consequently, the spatial distribution of the transmission channel is not something to be considered. Only a small step away from this system is required, however, to make the spatial distribution a very relevant question. When there is inelastic transport through the system, elastic channels coupled by interaction with vibrational modes of the molecule, the spatial distribution of the current appears to play a role in controlling in which regions of the molecule vibrational modes are excited.¹⁷⁹

Inelastic transport is not the only reason that some description of the spatial distribution of a channel might be desirable. Efforts to design spintronic devices where transport is perturbed by spin density on the molecule, or even simply the role that substituents may play in transport, would seem to be able to be enhanced by maximizing the extent to which these groups are involved in the conduction path. Two directions have been taken to quantitatively describe the spatial distribution of electronic transmission: eigenchannels^{180–186} and local currents.^{15,187,188}

The first point to note with regard to the spatial nature of transport is that (due to the electrodes) the matrices describing transmission do not necessarily retain the full symmetry properties of the underlying molecular geometry,⁶⁸ at least as commonly written in NEGF approaches. This is important, as it puts a limit on the symmetry of any spatial description of the transmission that will be obtained from the transport equations.

The concept of eigenchannels in transport goes back further than molecular conductance calculations¹⁸⁹; indeed, there was considerable success in characterizing the number of channels involved in transport junctions of metallic wires.

In molecular electronics, the number of conduction channels observed, through shot-noise measurements,¹⁹⁰ was used to determine that measurements of transport through a hydrogen molecule bound in a platinum junction were made with a single molecule bound lengthways in the junction^{49,191} rather than straddling the junction.¹⁹²

A number of methods have been put forward for obtaining transmission eigenchannels, the simplest of which is probably diagonalizing the transmission matrix [the matrix under the trace in Eq. (19.7)]. The problem with this approach is that the eigenvectors obtained by this method are localized at one end of the molecule, providing little insight into the spatial distribution of currents^{180,186} and certainly not corresponding to a real scattering state. Recent interest in transmission eigenchannels has been stimulated in part by efforts to describe the propensity rules for inelastic electron tunneling spectroscopy (IETS). In this case, success was achieved not by diagonalizing the transmission matrix but by transforming to a basis that diagonalized part of the transmission,^{193,194} with the two methods differing slightly in their approach. The significance of eigenchannels defined by more involved methods such as these is that the eigenvectors can be shown to correspond to scattering states.¹⁸³ The eigenchannels obtained have an energy-dependent form, taking a form close to that of the underlying molecular orbital at each transmission resonance. In the limit of zero coupling to the electrodes, the channel would take the form of each molecular orbital at resonance. Figure 19.6 shows the form of the dominant π conduction channel of chemisorbed 1,4-benzenedithiol at each of the resonances, calculated by one of the methods.¹⁹³

The weakness of what has been referred to as the *conduction channel approach* is that it has not been one approach but many. Effectively, the descriptions that have been used to date form little more than basis sets which may illuminate some aspect of the problem or provide a mathematically convenient description. Grounding eigenchannels in the base of scattering theory provides an argument for which approaches should be preferred and a promising way for this analysis to move forward.

An alternative approach is to describe the spatial distribution of current through a molecule in terms of local or “bond” currents; that is, the contributions to the current from pairs of atoms be they bonded or nonbonded. This description is not so open to the plethora of alternatives that have challenged channel analysis, and it has been shown to provide an intuitive picture of the flux through a system.¹⁵ The challenge that remains is whether local currents can be used to predict any useful features in a system.

19.5.2 Inelastic Effects: Heating and Dissipation

We now move away from the simple elastic tunneling picture that we have addressed so far and examine how inelastic tunneling affects transport in molecular junctions. Inelastic effects encompass a range of processes, from simply inelastic tunneling to polaron formation to heat conduction. The nature of the

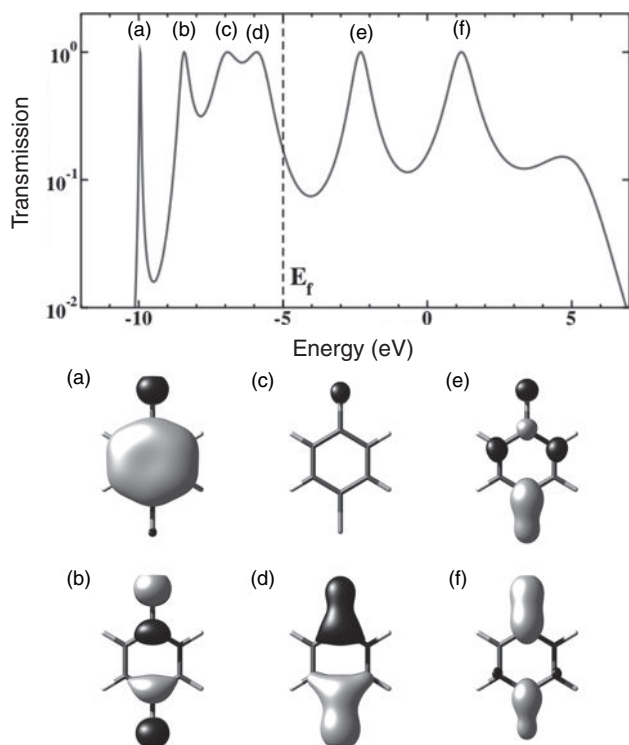


Fig. 19.6 Transmission through the dominant π channel¹⁹³ in 1,4-benzenedithiol between gold electrodes as calculated with gDFTB (top) and the form of the conduction channel at each of the transmission resonances (bottom).

processes that dominate is determined by the strength of the vibronic coupling and the time scale of the transfer process. The full range of inelastic processes have been discussed in reviews in the area^{39,195}; however, our focus in this section is more limited. Specifically, we look at the molecular vibrational excitation and dissipation processes that result from inelastic tunneling through the junction. These effects are important from two perspectives: first, the extent to which local heating, vibronic excitation, occurs due to the passage of current can have a direct bearing on the longevity of the junction, and second, understanding IETS necessarily requires an understanding of inelastic transport processes.

Heating^{40,48,196} and dissipation^{48,197,198} have been studied independent of studies on inelastic transport^{19,40,199} and IETS.^{38,101,179,200–205} The significance of IETS is that it provides clear experimental evidence for the inelastic processes that dominate in real junctions rather than the often-idealized systems studied theoretically. The vibrational spectrum obtained by IETS will show Raman-active modes and infrared active modes, but not necessarily all modes, which indicated that the selection rules governing this spectroscopy differed from other methods and warranted investigation.

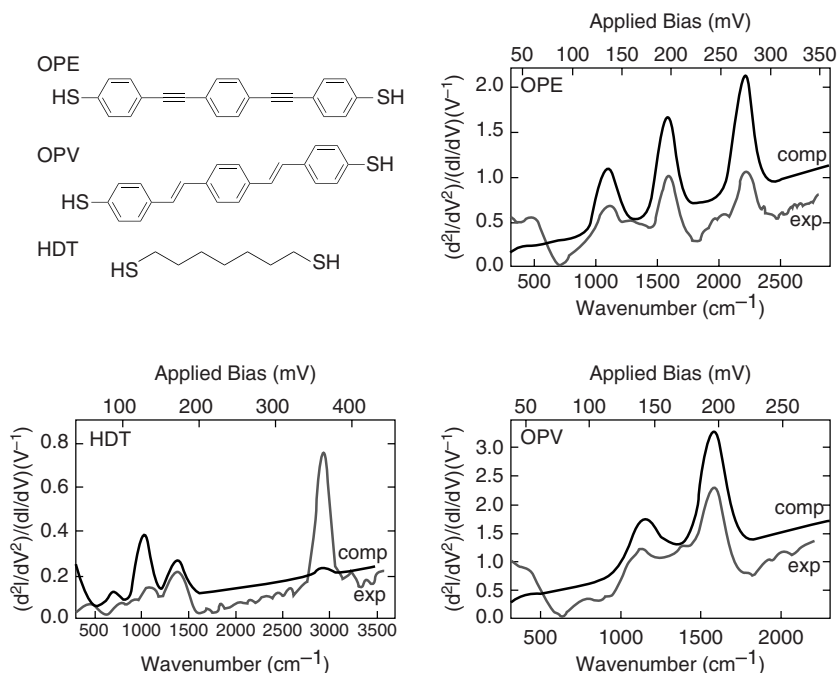


Fig. 19.7 Computed and experimental IETS results for a variety of molecules. (From Ref. 202, with permission. Copyright © 2009 by the American Physical Society.)

The excitation of vibrational modes by the passing current couples incoming and outgoing elastic channels according to symmetry selection rules. As there are contributions from incoming channels with a variety of symmetry properties rather than one clear symmetry ground state, the spectrum observed is governed by propensity rules.^{193,194,206,207} IETS calculations are really one of the great success stories of molecular conductance. Their dependence on vibrational frequencies, which are very well described by simple electronic structure methods rather than simply electronic properties has led to very good agreement between theory and experiment, as illustrated in Fig. 19.7.

19.5.3 Current-Induced Forces

Studies have also examined current-induced forces, the changes in geometry as a result of the changing charge distribution on the molecule.^{26,208–211} These changes will influence the vibrational properties, stability, and therefore longevity of the junction. Clear understanding and control of the extent to which such changes occur is obviously helpful for molecular device design.

19.5.4 Multiple States

Possibly the biggest assumption that underlies most theoretical work on molecular transport junctions is that it is simply the ground electronic state of the

system that dominates the properties. The nature of the transport process, even when dominated by elastic transport, is so reminiscent of charged and excited states that it is difficult to believe that these states will have no bearing on the properties of the junction. Even within the mean-field one-electron electronic structure picture that has been used to describe transport properties, the question is: Should different charged or excited states be used to describe the molecule? Equivalently, do we need a picture that allows transport to be described with contributions from different charged or electronic states of the molecule acting together? There has been some work examining the effects of multiple states on transport^{30,212–215}; however, this is an area that is most certainly going to be of interest for future work.

19.6 APPLICATIONS

Finally, we turn our attention to molecular electronic devices which have been calculated theoretically.

19.6.1 Rectifier

Since the initial proposal that single molecules could function as rectifiers,² these devices have captured researchers' attention. The essential element of a molecular rectifier is some symmetry-breaking property in the molecule that responds differently to forward or reverse bias. Symmetry, and symmetry-breaking interactions, arise naturally in chemistry; indeed, nature makes great use of these properties in biological systems. The question for researchers is how best to achieve a symmetry-broken response to applied bias from a molecule in a junction.

Essentially, molecular asymmetry is required for rectification, and by far the most thoroughly investigated approach is simply using structural asymmetry, frequently in the binding to the two electrodes, to ensure that different bias windows are accessed in the forward and reverse sweeps.^{72,133,134,216–220} This approach has yielded promising results; however, work has also shown that there are distinct limitations on the maximum rectification ratio that can be achieved by this approach.²²¹ Again utilizing molecular asymmetry, but this time using groups that are specifically electron donating or accepting, theoretical studies have examined systems drawing their inspiration directly from the Aviram–Ratner proposal.^{222,223}

The approaches to rectification outlined above relied on the bias window asymmetrically sampling the molecular transmission, with no requirement for the molecular transmission properties to change under bias. Two further approaches take the distinctly different approach of using the molecular response to the applied field to produce dramatically different electronic transmission depending on the direction of the applied bias. The first uses conformational change as a function of the field to produce rectification.^{224,225} The second used the extremely sensitive response to electric field of a molecule with multiple groups

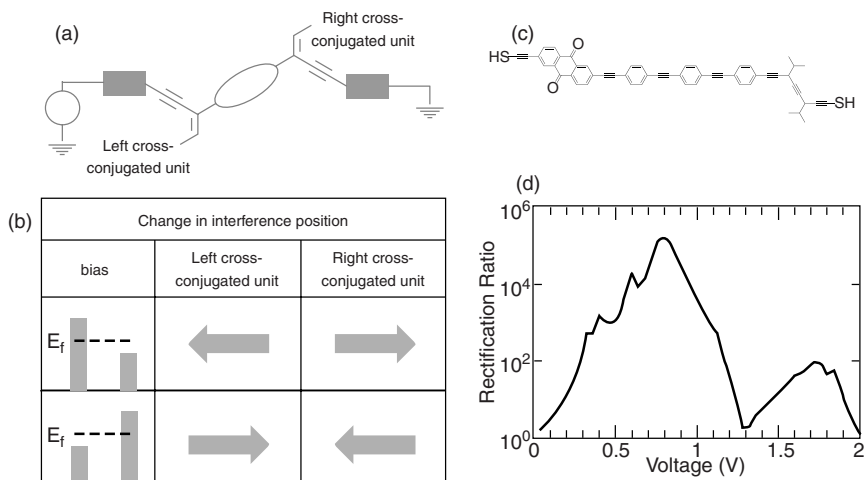


Fig. 19.8 Design of a molecular rectifier (a) and origin of the rectification (b) for a system using interference features. Sample molecule (c) and its rectification ratio as a function of voltage (d) calculated using Hückel-IV. (From Ref. 135, with permission. Copyright © 2008 by the American Chemical Society.)

inducing destructive interference features.¹³⁵ As noted in Section 19.4.7, destructive interference features manifest as dips in the transmission which can be tuned chemically and also shift with applied bias. The bias-dependent shifts in molecular transmission can then be utilized to design systems to act as rectifiers. Figure 19.8 shows schematically how a rectifier can be designed from a system with two functional groups inducing interference features in part (a) and the bias-dependent shifts on the interference features in part (b). In Fig. 19.8c a sample molecule is shown, and Fig. 19.8d shows the rectification ratio as a function of voltage, calculated using Hückel-IV.

19.6.2 Negative Differential Resistance

As the name suggests, negative differential resistance (NDR) is quintessentially nonohmic behavior: As the applied bias increases, the current measured through the system decreases. As a useful component for electronic devices, these effects have been studied theoretically in molecular junctions.^{81,83,135,226–233}

There are a variety of possible mechanisms by which NDR can occur in molecular systems: for example, charging of the system, conformational change, or some other less severely bias-dependent change in the molecule which lowers the underlying transmission. It remains to be seen which types of mechanisms will result in stable devices with desirable properties. Figure 19.9 shows how NDR can also result from the band structure of the electrodes. In this case, a semiconductor electrode suppresses transport at some bias voltages resulting in NDR.

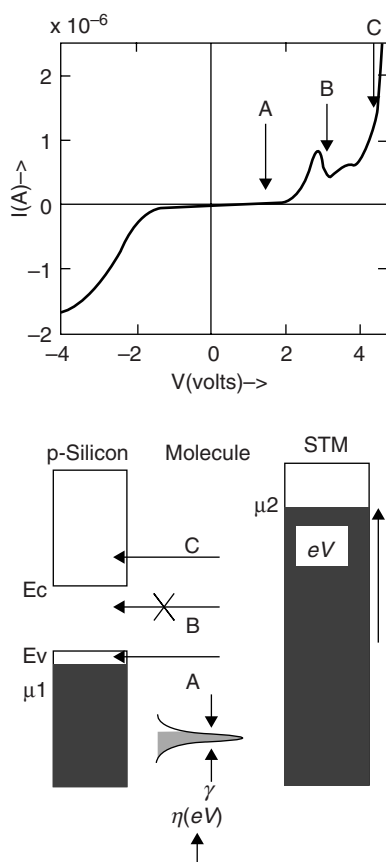


Fig. 19.9 With increasing bias voltage, a semiconductor electrode's band structure can result in NDR. (From Ref. 81, with permission. Copyright © 2009 by the American Physical Society.)

19.6.3 Switching

The ability of a molecular junction to switch from a low- to a high-conductance state opens up a wide range of possible device applications. Depending on the switching speed and reversibility of the process, the junction may function as a transistor, a memory device, or a sensor, to name but a few.

In order to have a switch fast enough for a transistor application, the “on” and “off” states of the system need to be accessible through electronic changes alone. This has driven considerable interest in studying the response of molecular junctions to the presence of a gate field or third terminal.^{14,16,99,234–241} Both experimentally and theoretically, quantifying the real effect of a gate electrode is a complicated exercise. The very small size of the molecule in the junction, and the presence of the large, often metallic electrodes means that there can be considerable screening of the gate field.

At the simplest level, the gate field can be approximated theoretically by a shift of all of the molecular eigenenergies, an approximation that can be improved somewhat if it is followed by a step to relax the electronic structure, allowing for polarization effects in the molecule. Strongly bound molecules with relatively featureless transmission spectra are unlikely to show any dramatic response to small applied gate voltages. If there is significant structure in the transmission spectrum, due to destructive interference features, for example,¹³⁵ there can be a dramatic response to the gate field.

Switching can also be induced by chemical changes which fundamentally alter how conductive the molecule in the junction can be. These effects have been examined using conformational change for the switching process^{104,242,243} and also through the effects of charging and polarization.^{231,232,244–246} These types of switching processes may not lead to devices that can be switched rapidly thousands of times, but they may yet prove their utility in memory devices or sensors that have no such requirement.

19.6.4 Thermoelectric

Efficient thermoelectric materials have the potential to revolutionize both power generation and industrial and domestic cooling. Nanoscale thermoelectrics have shown a lot of promise in recent years, and with some approximations the thermoelectric properties of molecular junctions can simply be related to properties of the electronic transmission.^{247,248} At the simplest level, the Seebeck coefficient is related to the derivative of the electronic transmission as a function of energy. This implies that the sorts of chemical and structural variations that can lead to large transmission gradients may also make molecular junctions useful thermoelectric materials. There have been a number of calculations of molecular thermoelectric properties,^{156,249–251} detailing interesting chemical trends. Further work is required to determine how reliable it is to relate the Seebeck coefficient to electronic transmission at the Landauer level and what types of molecules might lead to optimal thermoelectric properties.

19.6.5 Photoactive Switching

Both natural and synthetic molecular systems can exhibit very precise and controllable responses to light of particular frequencies. This property naturally leads to the idea of photo-switchable molecular devices. A variety of calculations have been performed in this area,^{252–257} most commonly focusing on reversible isomerization, driving molecules from high to low conductance states. Figure 19.10 shows one example of the types of systems studied. Upon irradiation the molecule isomerizes, resulting in a different transport path, with significantly different transmission.

19.6.6 Spintronics

The discovery of giant magnetoresistance brought an information storage revolution as compact memory became accessible. For future memory devices, as

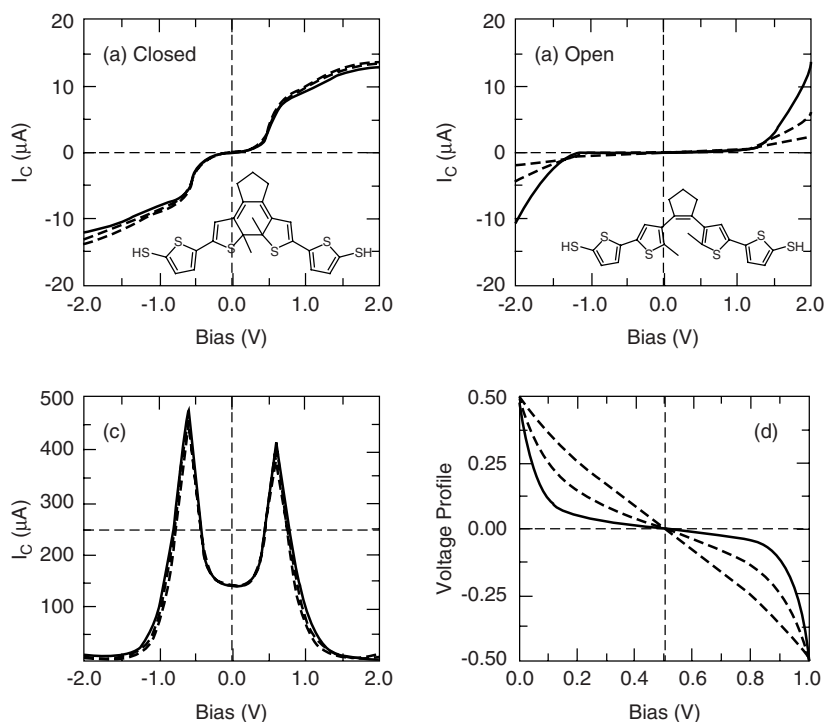


Fig. 19.10 The transmission through a photo-switchable molecule. (Reproduced with permission from Zhuang, M., Ernzerhof, M. *Phys. Rev. B* 2005, 72, 073104. Copyright © 2009 by the American Physical Society.)

well as a range of other applications, the spin transport properties of molecules have attracted considerable interest.^{84–87,89,90,182,258–261} This work is reviewed in detail in Chapter 18.

Spin transport properties may be controlled in the junction by spin transport properties of either the leads or the molecule, or a combination of the two. Ferromagnetic leads^{84–86,90,259,260} and transition metal complexes^{87,258,261} with unpaired spins have been studied, and organic radicals also offer intriguing prospects. A challenge for future work is how best to control the spin transport, and potentially spin selectivity, in molecular junctions. It is presently unknown what sorts of functionalities, energy levels, and spin densities in the structure will offer optimal device characteristics.

19.6.7 Logic

Using either two- or multiterminal junctions, it has been proposed that single molecules could function as logic components.^{150,262–267} Effectively, in these devices the computation is being performed by the chemical structure and coupling relationships inside the molecule. This mechanism differs substantially from

logic gates built from conventional components and starts to harness the unique properties that chemical complexity and quantum effects offer.

19.6.8 DNA Sequencing

One proposed application of molecular conductance junctions without an analog in conventional electronics is their use in DNA sequencing.^{169,268,269} Despite the similarity of the base pairs and their conductance variation with thermal motion, it was shown that repeated measurement would allow the bases to be distinguished,¹⁶⁹ and thereby DNA sequences, as it was passed through a nanopore.

19.7 CONCLUSIONS

Large systems present unique challenges for computational methods, and molecular transport junctions are no exception. Reliably modeling of the equilibrium properties of such junctions is challenge enough, yet nonequilibrium effects have to be addressed for reliable transport calculations. This area has attracted considerable theoretical attention and with it a great deal of understanding. Today, chemical and structural trends are increasingly well understood, details of the transport process are being elucidated, and the range of device applications proposed is ever-expanding. The information technology explosion of the twentieth century has brought incredible opportunities for computational methods, and we can only wait and see what magic the twenty-first century will bring.

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REFERENCES

1. Kuhn, H.; Möbius, D. *Angew. Chem. Int. Ed.* **1971**, *10*, 620–637.
2. Aviram, A.; Ratner, M. A. *Chem. Phys. Lett.* **1974**, *29*, 277–283.
3. Carter, F. L. *J. Vac. Sci. Technol. B* **1983**, *1*, 959–968.
4. Reed, M. A.; Zhou, C.; Muller, C. J.; Burgin, T. P.; Tour, J. M. *Science* **1997**, *278*, 252–254.
5. Reichert, J.; Ochs, R.; Beckmann, D.; Weber, H. B.; Mayor, M.; von Löhneysen, H. *Phys. Rev. Lett.* **2002**, *88*, 176804.
6. Smit, R. H. M.; Noat, Y.; Untiedt, C.; Lang, N. D.; van Hemert, M. C.; van Ruitenbeek, J. M. *Nature* **2002**, *419*, 906–909.
7. Xu, B.; Tao, N. J. *Science* **2003**, *301*, 1221–1223.

8. Venkataraman, L.; Klare, J. E.; Nuckolls, C.; Hybertsen, M. S.; Steigerwald, M. L. *Nature* **2006**, *442*, 904–907.
9. Kubatkin, S.; Danilov, A.; Hjort, M.; Cornil, J.; Bredas, J.-L.; Stuhr-Hansen, N.; Hedegard, P.; Bjornholm, T. *Nature* **2003**, *425*, 698–701.
10. Tian, W.; Datta, S.; Hong, S.; Reifenberger, R.; Henderson, J. I.; Kubiak, C. P. *J. Chem. Phys.* **1998**, *109*, 2874–2882.
11. Samanta, M. P.; Tian, W.; Datta, S.; Henderson, J. I.; Kubiak, C. P. *Phys. Rev. B* **1996**, *53*, R7626.
12. Hall, L. E.; Reimers, J. R.; Hush, N. S.; Silverbrook, K. *J. Chem. Phys.* **2000**, *112*, 1510–1521.
13. Ke, S.-H.; Baranger, H. U.; Yang, W. *Phys. Rev. B* **2004**, *70*, 085410.
14. Pecchia, A.; Penazzi, G.; Salvucci, L.; Di Carlo, A. *New J. Phys.* **2008**, *10*, 065022.
15. Pecchia, A.; Carlo, A. D. *Rep. Prog. Phys.* **2004**, *67*, 1497–1561.
16. Zahid, F.; Paulsson, M.; Polizzi, E.; Ghosh, A. W.; Siddiqui, L.; Datta, S. *J. Chem. Phys.* **2005**, *123*, 064707–064710.
17. Damle, P.; Ghosh, A. W.; Datta, S. *Chem. Phys.* **2002**, *281*, 171–187.
18. Taylor, J.; Guo, H.; Wang, J. *Phys. Rev. B* **2001**, *63*, 245407.
19. Frederiksen, T.; Paulsson, M.; Brandbyge, M.; Jauho, A.-P. *Phys. Rev. B* **2007**, *75*, 205413–205422.
20. Brandbyge, M.; Mozos, J.-L.; Ordeján, P.; Taylor, J.; Stokbro, K. *Phys. Rev. B* **2002**, *65*, 165401.
21. Palacios, J. J.; Pérez-Jiménez, A. J.; Louis, E.; SanFabián, E.; Vergés, J. A. *Phys. Rev. B* **2002**, *66*, 035322.
22. Rocha, A. R.; García-Suárez, V. M.; Bailey, S.; Lambert, C.; Ferrer, J.; Sanvito, S. *Phys. Rev. B* **2006**, *73*, 085414–085422.
23. Calzolari, A.; Marzari, N.; Souza, I.; Buongiorno Nardelli, M. *Phys. Rev. B* **2004**, *69*, 035108.
24. Thygesen, K. S.; Jacobsen, K. W. *Chem. Phys.* **2005**, *319*, 111–125.
25. Emberly, E. G.; Kirczenow, G. *Phys. Rev. B* **1998**, *58*, 10911.
26. Di Ventra, M.; Lang, N. D. *Phys. Rev. B* **2001**, *65*, 045402.
27. Lang, N. D.; Avouris, P. *Phys. Rev. B* **2001**, *64*, 125323.
28. Kergueris, C.; Bourgoin, J. P.; Palacin, S.; Esteve, D.; Urbina, C.; Magoga, M.; Joachim, C. *Phys. Rev. B* **1999**, *59*, 12505.
29. Ernzerhof, M.; Zhuang, M. *J. Chem. Phys.* **2003**, *119*, 4134–4140.
30. Delaney, P.; Greer, J. C. *Int. J. Quantum Chem.* **2004**, *100*, 1163–1169.
31. Goyer, F.; Ernzerhof, M.; Zhuang, M. *J. Chem. Phys.* **2007**, *126*, 144104–144108.
32. Strange, M.; Kristensen, I. S.; Thygesen, K. S.; Jacobsen, K. W. *J. Chem. Phys.* **2008**, *128*, 114714–114718.
33. McDermott, S.; George, C. B.; Fagas, G.; Greer, J. C.; Ratner, M. A. *J. Phys. Chem. C* **2009**, *113*, 744–750.
34. Landauer, R. *IBM J. Res. Dev.* **1957**, *1*, 223.
35. Landauer, R. *Phil. Mag.* **1970**, *21*, 863–867.
36. Datta, S. *Electronic Transport in Mesoscopic Systems*, Cambridge University Press, New York, 1997.

37. Meir, Y.; Wingreen, N. *Phys. Rev. Lett.* **1992**, *68*, 2512.
38. Galperin, M.; Ratner, M. A.; Nitzan, A. *J. Chem. Phys.* **2004**, *121*, 11965–11979.
39. Galperin, M.; Ratner, M. A.; Nitzan, A. *J. Phys. Condens. Matter* **2007**, *19*, 103201.
40. Pecchia, A.; Di Carlo, A.; Gagliardi, A.; Sanna, S.; Frauenheim, T.; Gutierrez, R. *Nano Lett.* **2004**, *4*, 2109–2114.
41. Galperin, M.; Nitzan, A. *J. Chem. Phys.* **2006**, *124*, 234709–234717.
42. Galperin, M.; Nitzan, A. *Phys. Rev. Lett.* **2005**, *95*, 206802.
43. Mujica, V.; Kemp, M.; Ratner, M. A. *J. Chem. Phys.* **1994**, *101*, 6849–6855.
44. Mujica, V.; Kemp, M.; Ratner, M. A. *J. Chem. Phys.* **1994**, *101*, 6856–6864.
45. Datta, S.; Tian, W.; Hong, S.; Reifenberger, R.; Henderson, J. I.; Kubiak, C. P.; *Phys. Rev. Lett.* **1997**, *79*, 2530.
46. Solomon, G. C.; Andrews, D. Q.; Duyne, R. P. V.; Ratner, M. A. *ChemPhysChem* **2009**, *10*, 257–264.
47. Solomon, G. C.; Reimers, J. R.; Hush, N. S. *J. Chem. Phys.* **2004**, *121*, 6615–6627.
48. Gagliardi, A.; Romano, G.; Pecchia, A.; Carlo, A. D.; Frauenheim, T.; Niehaus, T. A. *New J. Phys.* **2008**, *10*, 065020.
49. Cuevas, J. C.; Heurich, J.; Pauly, F.; Wenzel, W.; Schon, G. *Nanotechnology* **2003**, *14*, R29–R38.
50. Xue, Y.; Datta, S.; Ratner, M. A. *J. Chem. Phys.* **2001**, *115*, 4292–4299.
51. Evers, F.; Weigend, F.; Koentopp, M. *Phys. Rev. B* **2004**, *69*, 235411.
52. Stokbro, K.; Taylor, J.; Brandbyge, M.; Ordejón, P. *Ann. N.Y. Acad. Sci.* **2003**, *1006*, 212–226.
53. Stokbro, K.; Taylor, J.; Brandbyge, M.; Mozos, J. L.; Ordejón, P. *Comput. Mater. Sci.* **2003**, *27*, 151–160.
54. Ghosh, A. W.; Zahid, F.; Datta, S.; Birge, R. R. *Chem. Phys.* **2002**, *281*, 225–230.
55. Thygesen, K. S. *Phys. Rev. Lett.* **2008**, *100*, 166804–166804.
56. Thygesen, K. S.; Rubio, A. *Phys. Rev. B* **2008**, *77*, 115333–115322.
57. Thygesen, K. S.; Rubio, A. *J. Chem. Phys.* **2007**, *126*, 091101–091104.
58. Toher, C.; Filippetti, A.; Sanvito, S.; Burke, K. *Phys. Rev. Lett.* **2005**, *95*, 146402.
59. Sai, N.; Zwolak, M.; Vignale, G.; Di Ventra, M. *Phys. Rev. Lett.* **2005**, *94*, 186810.
60. Toher, C.; Sanvito, S. *Phys. Rev. B* **2008**, *77*, 155402.
61. Ke, S.-H.; Baranger, H. U.; Yang, W. *J. Chem. Phys.* **2007**, *127*, 144107.
62. Bauschlicher, C. W., Jr.; Ricca, A.; Xue, Y.; Ratner, M. A. *Chem. Phys. Lett.* **2004**, *390*, 246–249.
63. Bauschlicher, C. W., Jr.; Lawson, J. W.; Ricca, A.; Xue, Y.; Ratner, M. A. *Chem. Phys. Lett.* **2004**, *388*, 427–429.
64. Prociuk, A.; Van Kuiken, B.; Dunietz, B. D. *J. Chem. Phys.* **2006**, *125*, 204717.
65. Solomon, G. C.; Reimers, J. R.; Hush, N. S. *J. Chem. Phys.* **2005**, *122*, 224502.
66. Pantelides, S. T.; Di Ventra, M.; Lang, N. D.; Rashkeev, S. N. *IEEE Trans. Nanotechnol.* **2002**, *1*, 86–90.
67. Derosa, P. A.; Seminario, J. M. *J. Phys. Chem. B* **2001**, *105*, 471–481.
68. Solomon, G. C.; Gagliardi, A.; Pecchia, A.; Frauenheim, T.; Di Carlo, A.; Reimers, J. R.; Hush, N. S. *J. Chem. Phys.* **2006**, *125*, 184702–184705.
69. Basch, H.; Ratner, M. A. *J. Chem. Phys.* **2004**, *120*, 5761–5770.

70. Xue, Y.; Ratner, M. A. *Phys. Rev. B* **2003**, *68*, 115406.
71. Liang, G. C.; Ghosh, A. W.; Paulsson, M.; Datta, S. *Phys. Rev. B* **2004**, *69*, 115302.
72. Elbing, M.; Ochs, R.; Koentopp, M.; Fischer, M.; von Hänisch, C.; Weigend, F.; Evers, F.; Weber, H. B.; Mayor, M. *Proc. Natl. Acad. Sci. USA* **2005**, *102*, 8815–8820.
73. Galperin, M.; Nitzan, A. *Ann. N.Y. Acad. Sci.* **2003**, *1006*, 48–67.
74. Arnold, A.; Weigend, F.; Evers, F. *J. Chem. Phys.* **2007**, *126*, 174101–174114.
75. Basch, H.; Ratner, M. A. *J. Chem. Phys.* **2005**, *123*, 234704.
76. Yaliraki, S. N.; Kemp, M.; Ratner, M. A. *J. Am. Chem. Soc.* **1999**, *121*, 3428–3434.
77. Dalglish, H.; Kirczenow, G. *Phys. Rev. B* **2006**, *73*, 245431.
78. Rauba, J. M. C.; Strange, M.; Thygesen, K. S. *Phys. Rev. B* **2008**, *78*, 165116.
79. Cho, Y.; Kim, W. Y.; Kim, K. S. *J. Phys. Chem. A* **2009**, *113*, 4100–4104.
80. Mujica, V.; Ratner, M. A. *Chem. Phys.* **2006**, *326*, 197–203.
81. Rakshit, T.; Liang, G. C.; Ghosh, A. W.; Hersam, M. C.; Datta, S. *Phys. Rev. B* **2005**, *72*, 125305.
82. Rakshit, T.; Liang, G.-C.; Ghosh, A. W.; Datta, S. *Nano Lett.* **2004**, *4*, 1803–1807.
83. Bevan, K. H.; Kienle, D.; Guo, H.; Datta, S. *Phys. Rev. B* **2008**, *78*, 035303–035310.
84. Ning, Z.; Zhu, Y.; Wang, J.; Guo, H. *Phys. Rev. Lett.* **2008**, *100*, 056803–056804.
85. Waldron, D.; Liu, L.; H. Guo, *Nanotechnology* **2007**, *18*, 424026.
86. Waldron, D.; Haney, P.; Larade, B.; MacDonald, A.; Guo, H. *Phys. Rev. Lett.* **2006**, *96*, 166804.
87. Maslyuk, V. V.; Bagrets, A.; Meded, V.; Arnold, A.; Evers, F.; Brandbyge, M.; Bredow, T.; Mertig, I. *Phys. Rev. Lett.* **2006**, *97*, 097201.
88. Dalglish, H.; Kirczenow, G. *Phys. Rev. B* **2005**, *72*, 184407.
89. Dalglish, H.; Kirczenow, G. *Phys. Rev. B* **2005**, *72*, 155429.
90. Rocha, A. R.; García-Suárez, V. M.; Bailey, S. W.; Lambert, C. J.; Ferrer, J.; Sanvito, S. *Nature Mater.* **2005**, *4*, 335–339.
91. Pati, R.; Senapati, L.; Ajayan, P. M.; Nayak, S. K. *Phys. Rev. B* **2003**, *68*, 100407.
92. Ren, W.; Reimers, J. R.; Hush, N. S.; Zhu, Y.; Wang, J.; Guo, H. *J. Phys. Chem. C* **2007**, *111*, 3700–3704.
93. Andrews, D. Q.; Cohen, R.; Van Duyne, R. P.; Ratner, M. A. *J. Chem. Phys.* **2006**, *125*, 174718–174719.
94. Basch, H.; Cohen, R.; Ratner, M. A. *Nano Lett.* **2005**, *5*, 1668–1675.
95. Xue, Y.; Ratner, M. A. *Phys. Rev. B* **2003**, *68*, 115407.
96. Yaliraki, S. N.; Roitberg, A. E.; Gonzalez, C.; Mujica, V.; Ratner, M. A. *J. Chem. Phys.* **1999**, *111*, 6997–7002.
97. Li, C.; Pobelov, I.; Wandlowski, T.; Bagrets, A.; Arnold, A.; Evers, F. *J. Am. Chem. Soc.* **2008**, *130*, 318–326.
98. Lee, M. H.; Speyer, G.; Sankey, O. F. *Phys. Status Solidi (b)* **2006**, *243*, 2021–2029.
99. Bratkovsky, A. M.; Kornilovitch, P. E. *Phys. Rev. B* **2003**, *67*, 115307.
100. Li, Z.; Kosov, D. S. *Phys. Rev. B* **2007**, *76*, 035415–035417.
101. Troisi, A.; Ratner, M. A. *Phys. Chem. Chem. Phys.* **2007**, *9*, 2421–2427.
102. Bagrets, A.; Arnold, A.; Evers, F. *J. Am. Chem. Soc.* **2008**, *130*, 9013–9018.

103. Tanibayashi, S.; Tada, T.; Watanabe, S.; H. Sekino, *Chem. Phys. Lett.* **2006**, *428*, 367–370.
104. Emberly, E. G.; Kirczenow, G. *Phys. Rev. Lett.* **2003**, *91*, 188301.
105. Emberly, E. G.; Kirczenow, G. *Phys. Rev. B* **2001**, *64*, 235412.
106. Quek, S. Y.; Venkataraman, L.; Choi, H. J.; Louie, S. G.; Hybertsen, M. S.; Neaton, J. B. *Nano Lett.* **2007**, *7*, 3477–3482.
107. Yanov, I.; Kholod, Y.; Leszczynski, J.; Palacios, J. J. *Chem. Phys. Lett.* **2007**, *445*, 238–242.
108. Stojkovic, S.; Joachim, C.; Grill, L.; Moresco, F. *Chem. Phys. Lett.* **2005**, *408*, 134–138.
109. Kornilovitch, P. E.; Bratkovsky, A. M. *Phys. Rev. B* **2001**, *64*, 195413.
110. Yan, L.; Bautista, E. J.; Seminario, J. M. *Nanotechnology* **2007**, *18*, 485701.
111. Hoft, R. C.; Ford, M. J.; García-Suárez, V. M.; Lambert, C. J.; Cortie, M. B. *J. Phys. Condens. Matter* **2008**, *20*, 025207.
112. Hoft, R. C.; Ford, M. J.; Cortie, M. B. *Chem. Phys. Lett.* **2006**, *429*, 503–506.
113. Kamenetska, M.; Koentopp, M.; Whalley, A. C.; Park, Y. S.; Steigerwald, M. L.; Nuckolls, C.; Hybertsen, M. S.; Venkataraman, L. *Phys. Rev. Lett.* **2009**, *102*, 126803–126804.
114. Paulsson, M.; Krag, C.; Frederiksen, T.; Brandbyge, M. *Nano Lett.* **2009**, *9*, 117–121.
115. Strange, M.; Thygesen, K. S.; Jacobsen, K. W. *Phys. Rev. B* **2006**, *73*, 125424–125427.
116. Park, Y. S.; Whalley, A. C.; Kamenetska, M.; Steigerwald, M. L.; Hybertsen, M. S.; Nuckolls, C.; Venkataraman, L. *J. Am. Chem. Soc.* **2007**, *129*, 15768–15769.
117. Wohlthat, S.; Pauly, F.; Reimers, J. R. *Chem. Phys. Lett.* **2008**, *454*, 284–288.
118. Hong, S.; Reifengerger, R.; Tian, W.; Datta, S.; Henderson, J. I.; Kubiak, C. P. *Superlattices Microstruct.* **2000**, *28*, 289–303.
119. Kaun, C.-C.; Guo, H. *Nano Lett.* **2003**, *3*, 1521–1525.
120. Kondo, M.; Tada, T.; Yoshizawa, K. *J. Phys. Chem. A* **2004**, *108*, 9143–9149.
121. Piccinin, S.; Selloni, A.; Scandolo, S.; Car, R.; Scoles, G. *J. Chem. Phys.* **2003**, *119*, 6729–6735.
122. Seminario, J. M.; Yan, L. *Int. J. Quantum Chem.* **2005**, *102*, 711–723.
123. Solomon, G. C.; Andrews, D. Q.; Van Duyne, R. P.; Ratner, M. A. *J. Am. Chem. Soc.* **2008**, *130*, 7788–7789.
124. Tour, J. M. *Chem. Rev.* **1996**, *96*, 537–554.
125. Ke, S.-H.; Baranger, H. U.; Yang, W. *J. Am. Chem. Soc.* **2004**, *126*, 15897–15904.
126. Wohlthat, S.; Pauly, F.; Reimers, J. R. *J. Phys. Condens. Matter* **2008**, *20*, 295208.
127. Xue, Y.; Ratner, M. A. *Phys. Rev. B* **2004**, *69*, 085403.
128. Yaliraki, S. N.; Ratner, M. A. *Ann. N.Y. Acad. Sci.* **2002**, *960*, 153–162.
129. Kristensen, I. S.; Mowbray, D. J.; Thygesen, K. S.; Jacobsen, K. W. *J. Phys. Condens. Matter* **2008**, *20*, 374101.
130. Lang, N. D.; Kagan, C. R. *Nano Lett.* **2006**, *6*, 2955–2958.
131. Luo, Y.; Wang, C.-K.; Fu, Y. *J. Chem. Phys.* **2002**, *117*, 10283–10290.
132. Li, Z.; Kosov, D. S. *J. Phys. Chem. B* **2006**, *110*, 19116–19120.

133. Taylor, J.; Brandbyge, M.; Stokbro, K. *Phys. Rev. Lett.* **2002**, *89*, 138301.
134. Ford, M. J.; Hoft, R. C.; McDonagh, A. M.; Cortie, M. B. *J. Phys. Condens. Matter* **2008**, *20*, 374106.
135. Andrews, D. Q.; Solomon, G. C.; Van Duyne, R. P.; Ratner, M. A. *J. Am. Chem. Soc.* **2008**, *130*, 17309–17319.
136. Taylor, J.; Brandbyge, M.; Stokbro, K. *Phys. Rev. B* **2003**, *68*, 121101.
137. Mowbray, D. J.; Jones, G.; Thygesen, K. S. *J. Chem. Phys.* **2008**, *128*, 111103–111105.
138. Stadler, R.; Thygesen, K. S.; Jacobsen, K. W. *Nanotechnology* **2005**, *16*, S155–S160.
139. Di Ventra, M.; Kim, S. G.; Pantelides, S. T.; Lang, N. D. *Phys. Rev. Lett.* **2001**, *86*, 288.
140. Kemp, M.; Roitberg, A.; Mujica, V.; Wanta, T.; Ratner, M. A. *J. Phys. Chem.* **1996**, *100*, 8349–8355.
141. Kemp, M.; Mujica, V.; Ratner, M. A. *J. Chem. Phys.* **1994**, *101*, 5172–5178.
142. Cheong, A.; Roitberg, A. E.; Mujica, V.; Ratner, M. A. *J. Photochem. Photobiol. A* **1994**, *82*, 81–86.
143. Patoux, C.; Coudret, C.; Launay, J.-P.; Joachim, C.; Gourdon, A. *Inorg. Chem.* **1997**, *36*, 5037–5049.
144. Ke, S.-H.; Yang, W.; Baranger, H. U. *Nano Lett.* **2008**, *8*, 3257–3261.
145. Walter, D.; Neuhauser, D.; Baer, R. *Chem. Phys.* **2004**, *299*, 139–145.
146. Stafford, C. A.; Cardamone, D. M.; Mazumdar, S. *Nanotechnology* **2007**, *18*, 424014.
147. Cardamone, D. M.; Stafford, C. A.; Mazumdar, S. *Nano Lett.* **2006**, *6*, 2422–2426.
148. Tada, T.; Nozaki, D.; Kondo, M.; Hamayama, S.; Yoshizawa, K. *J. Am. Chem. Soc.* **2004**, *126*, 14182–14189.
149. Quinn, J. R.; Foss, F. W.; Venkataraman, L.; Hybertsen, M. S.; Breslow, R. *J. Am. Chem. Soc.* **2007**, *129*, 6714–6715.
150. Baer, R.; Neuhauser, D. *J. Am. Chem. Soc.* **2002**, *124*, 4200–4201.
151. Solomon, G. C.; Andrews, D. Q.; Goldsmith, R. H.; Hansen, T.; Wasielewski, M. R.; Van Duyne, R. P.; Ratner, M. A. *J. Am. Chem. Soc.* **2008**, *130*, 17301–17308.
152. Andrews, D. Q.; Solomon, G. C.; Goldsmith, R. H.; Hansen, T.; Wasielewski, M. R.; Duyne, R. P. V.; Ratner, M. A. *J. Phys. Chem. C* **2008**, *112*, 16991–16998.
153. Collepardo-Guevara, R.; Walter, D.; Neuhauser, D.; Baer, R. *Chem. Phys. Lett.* **2004**, *393*, 367–371.
154. Ernzerhof, M.; Zhuang, M.; Rocheleau, P. *J. Chem. Phys.* **2005**, *123*, 134704–134705.
155. Weitellier, S.; Launay, J. P.; Joachim, C. *Chem. Phys.* **1989**, *131*, 481–488.
156. Pauly, F.; Viljas, J. K.; Cuevas, J. C. *Phys. Rev. B* **2008**, *78*, 035315–035316.
157. Pauly, F.; Viljas, J. K.; Cuevas, J. C.; Schön, G. *Phys. Rev. B* **2008**, *77*, 155312–155319.
158. Cohen, R.; Stokbro, K.; Martin, J. M. L.; Ratner, M. A. *J. Phys. Chem. C* **2007**, *111*, 14893–14902.
159. Xue, Y.; Ratner, M. A. *Int. J. Quantum Chem.* **2005**, *102*, 911–924.
160. Xue, Y.; Ratner, M. A. *Phys. Rev. B* **2004**, *70*, 081404.

161. Delaney, P.; Nolan, M.; Greer, J. C. *J. Chem. Phys.* **2005**, *122*, 044710–044715.
162. Tomfohr, J.; Sankey, O. F. *J. Chem. Phys.* **2004**, *120*, 1542–1554.
163. Stadler, R.; Thygesen, K. S.; Jacobsen, K. W. *Phys. Rev. B* **2005**, *72*, 241401.
164. Seminario, J. M.; Derosa, P. A. *J. Am. Chem. Soc.* **2001**, *123*, 12418–12419.
165. George, C. B.; Ratner, M. A.; Lambert, J. B. *J. Phys. Chem. A* **2009**, *113*, 3876–3880.
166. Schepers, T.; Michl, J. *J. Phys. Org. Chem.* **2002**, *15*, 490–498.
167. Pecchia, A.; Gheorghe, M.; Di Carlo, A.; Lugli, P.; Niehaus, T. A.; Frauenheim, T.; Scholz, R. *Phys. Rev. B* **2003**, *68*, 235321.
168. Dreher, M.; Pauly, F.; Heurich, J.; Cuevas, J. C.; Scheer, E.; Nielaba, P. *Phys. Rev. B* **2005**, *72*, 075435.
169. Lagerqvist, J.; Zwolak, M.; Di Ventra, M. *Nano Lett.* **2006**, *6*, 779–782.
170. Andrews, D. Q.; Van Duyn, R. P.; Ratner, M. A. *Nano Lett.* **2008**, *8*, 1120–1126.
171. Hu, Y.; Zhu, Y.; Gao, H.; Guo, H. *Phys. Rev. Lett.* **2005**, *95*, 156803.
172. Liu, R.; Ke, S.-H.; Baranger, H. U.; Yang, W. *J. Chem. Phys.* **2005**, *122*, 044703–044704.
173. Lagerqvist, J.; Chen, Y.-C.; Ventra, M. D. *Nanotechnology* **2004**, *15*, S459–S464.
174. Yaliraki, S. N.; Ratner, M. A. *J. Chem. Phys.* **1998**, *109*, 5036–5043.
175. Magoga, M.; Joachim, C. *Phys. Rev. B* **1999**, *59*, 16011.
176. Lang, N. D.; Avouris, P. *Phys. Rev. B* **2000**, *62*, 7325.
177. Landau, A.; Kronik, L.; Nitzan, A. *J. Comput. Theor. Nanosci.* **2008**, *5*, 535–544.
178. Landau, A.; Nitzan, A.; Kronik, L. *J. Phys. Chem. A*, **2009**, *113*, 7451–7460.
179. Solomon, G. C.; Gagliardi, A.; Pecchia, A.; Frauenheim, T.; Di Carlo, A.; Reimers, J. R.; Hush, N. S. *J. Chem. Phys.* **2006**, *124*, 094704–094710.
180. Solomon, G. C.; Gagliardi, A.; Pecchia, A.; Frauenheim, T.; Di Carlo, A.; Reimers, J. R.; Hush, N. S. *Nano Lett.* **2006**, *6*, 2431–2437.
181. Heurich, J.; Cuevas, J. C.; Wenzel, W.; Schön, G. *Phys. Rev. Lett.* **2002**, *88*, 256803.
182. Wang, B.; Zhu, Y.; Ren, W.; Wang, J.; Guo, H. *Phys. Rev. B* **2007**, *75*, 235415–235417.
183. Paulsson, M.; Brandbyge, M. *Phys. Rev. B* **2007**, *76*, 115117.
184. Brandbyge, M.; Kobayashi, N.; Tsukada, M. *Phys. Rev. B* **1999**, *60*, 17064.
185. Brandbyge, M.; Sørensen, M. R.; Jacobsen, K. W. *Phys. Rev. B* **1997**, *56*, 14956.
186. Jacob, D.; Palacios, J. J. *Phys. Rev. B* **2006**, *73*, 075429–075424.
187. Sai, N.; Bushong, N.; Hatcher, R.; Di Ventra, M. *Phys. Rev. B* **2007**, *75*, 115410–115418.
188. Ernzerhof, M.; Bahmann, H.; Goyer, F.; Zhuang, M.; Rocheleau, P. *J. Chem. Theory Comput.* **2006**, *2*, 1291–1297.
189. Büttiker, M. *IBM J. Res. Dev.* **1988**, *32*, 63–75.
190. Djukic, D.; van Ruitenbeek, J. M. *Nano Lett.* **2006**, *6*, 789–793.
191. Thygesen, K. S.; Jacobsen, K. W. *Phys. Rev. Lett.* **2005**, *94*, 036807.
192. García, Y.; Palacios, J. J.; SanFabián, E.; Vergés, J. A.; Pérez-Jiménez, A. J.; Louis, E. *Phys. Rev. B* **2004**, *69*, 041402.
193. Gagliardi, A.; Solomon, G. C.; Pecchia, A.; Frauenheim, T.; Di Carlo, A.; Hush, N. S.; Reimers, J. R. *Phys. Rev. B* **2007**, *75*, 174306.

194. Paulsson, M.; Frederiksen, T.; Ueba, H.; Lorente, N.; Brandbyge, M. *Phys. Rev. Lett.* **2008**, *100*, 226604.
195. Galperin, M.; Ratner, M. A.; Nitzan, A.; Troisi, A. *Science* **2008**, *319*, 1056–1060.
196. D'Agosta, R.; Ventra, M. D. *J. Phys. Condens. Matter* **2008**, *20*, 374102.
197. Pecchia, A.; Romano, G.; Di Carlo, A. *Phys. Rev. B* **2007**, *75*, 035401–035410.
198. Romano, G.; Pecchia, A.; Carlo, A. D. *J. Phys. Condens. Matter* **2007**, *19*, 215207.
199. Sergueev, N.; Roubtsov, D.; Guo, H. *Phys. Rev. Lett.* **2005**, *95*, 146803.
200. Chen, Y.-C.; Zwolak, M.; Di Ventra, M. *Nano Lett.* **2005**, *5*, 621–624.
201. Chen, Y.-C.; Zwolak, M.; Di Ventra, M. *Nano Lett.* **2004**, *4*, 1709–1712.
202. Troisi, A.; Ratner, M. A. *Phys. Rev. B* **2005**, *72*, 033408.
203. Troisi, A.; Beebe, J. M.; Picraux, L. B.; van Zee, R. D.; Stewart, D. R.; Ratner, M. A.; Kushmerick, J. G. *Proc. Natl. Acad. Sci. USA* **2007**, *104*, 14255–14259.
204. Paulsson, M.; Frederiksen, T.; Brandbyge, M. *Nano Lett.* **2006**, *6*, 258–262.
205. Nakamura, H.; Yamashita, K.; Rocha, A. R.; Sanvito, S. *Phys. Rev. B* **2008**, *78*, 235420.
206. Troisi, A.; Ratner, M. A. *J. Chem. Phys.* **2006**, *125*, 214709–214711.
207. Troisi, A.; Ratner, M. A. *Nano Lett.* **2006**, *6*, 1784–1788.
208. Brandbyge, M.; Stokbro, K.; Taylor, J.; Mozos, J.-L.; Ordejón, P. *Phys. Rev. B* **2003**, *67*, 193104.
209. Girard, Y.; Yamamoto, T.; Watanabe, K. *J. Phys. Chem. C* **2007**, *111*, 12478–12482.
210. Di Ventra, M.; Pantelides, S. T.; Lang, N. D. *Phys. Rev. Lett.* **2002**, *88*, 046801.
211. Dundas, D.; McEniry, E. J.; Todorov, T. N. *Nature Nanotechnol.* **2009**, *4*, 99–102.
212. Muralidharan, B.; Ghosh, A. W.; Datta, S. *Phys. Rev. B* **2006**, *73*, 155410–155415.
213. Hettler, M. H.; Schoeller, H.; Wenzel, W. *Europhys. Lett.* **2002**, *57*, 571–577.
214. Galperin, M.; Nitzan, A.; Ratner, M. A. *Phys. Rev. B* **2008**, *78*, 125320–125329.
215. Yeganeh, S.; Ratner, M. A.; Galperin, M.; Nitzan, A. *Nano Lett.* **2009**, *9*, 1770–1774.
216. Zahid, F.; Ghosh, A. W.; Paulsson, M.; Polizzi, E.; Datta, S. *Phys. Rev. B* **2004**, *70*, 245317.
217. Liu, R.; Ke, S.-H.; Yang, W.; Baranger, H. U. *J. Chem. Phys.* **2006**, *124*, 024718.
218. Mujica, V.; Ratner, M. A.; Nitzan, A. *Chem. Phys.* **2002**, *281*, 147–150.
219. Gonzalez, C.; Mujica, V.; Ratner, M. A. *Ann. N.Y. Acad. Sci.* **2002**, *960*, 163–176.
220. Miller, O. D.; Muralidharan, B.; Kapur, N.; Ghosh, A. W. *Phys. Rev. B* **2008**, *77*, 125427.
221. Armstrong, N.; Hoft, R. C.; McDonagh, A.; Cortie, M. B.; Ford, M. J. *Nano Lett.* **2007**, *7*, 3018–3022.
222. Stokbro, K.; Taylor, J.; Brandbyge, M. *J. Am. Chem. Soc.* **2003**, *125*, 3674–3675.
223. Krzeminski, C.; Delerue, C.; Allan, G.; Vuillaume, D.; Metzger, R. M. *Phys. Rev. B* **2001**, *64*, 085405.
224. Troisi, A.; Ratner, M. A. *Nano Lett.* **2004**, *4*, 591–595.
225. Troisi, A.; Ratner, M. A. *J. Am. Chem. Soc.* **2002**, *124*, 14528–14529.
226. Liu, R.; Ke, S.-H.; Baranger, H. U.; Yang, W. *J. Am. Chem. Soc.* **2006**, *128*, 6274–6275.

227. Xue, Y.; Datta, S.; Hong, S.; Reifengerger, R.; Henderson, J. I.; Kubiak, C. P. *Phys. Rev. B* **1999**, *59*, R7852.
228. Hettler, M. H.; Wenzel, W.; Wegewijs, M. R.; Schoeller, H. *Phys. Rev. Lett.* **2003**, *90*, 076805.
229. Dalgleish, H.; Kirczenow, G. *Nano Lett.* **2006**, *6*, 1274–1278.
230. Lang, N. D.; *Phys. Rev. B* **1997**, *55*, 9364.
231. Yeganeh, S.; Galperin, M.; Ratner, M. A.; *J. Am. Chem. Soc.* **2007**, *129*, 13313–13320.
232. Galperin, M.; Ratner, M. A.; Nitzan, A. *Nano Lett.* **2005**, *5*, 125–130.
233. Kim, W. Y.; Kwon, S. K.; Kim, K. S. *Phys. Rev. B* **2007**, *76*, 033415.
234. Ke, S.-H.; Baranger, H. U.; Yang, W. *Phys. Rev. B* **2005**, *71*, 113401.
235. Ghosh, A. W.; Rakshit, T.; Datta, S. *Nano Lett.* **2004**, *4*, 565–568.
236. Damle, P.; Rakshit, T.; Paulsson, M.; Datta, S. *IEEE Trans. Nanotechnol.* **2002**, *1*, 145–153.
237. Emberly, E. G.; Kirczenow, G. *Phys. Rev. B* **2000**, *62*, 10451.
238. Emberly, E.; Kirczenow, G. *J. Appl. Phys.* **2000**, *88*, 5280–5282.
239. Lang, N. D.; Solomon, P. M. *Nano Lett.* **2005**, *5*, 921–924.
240. Yang, Z.; Lang, N. D.; Di Ventra, M. *Appl. Phys. Lett.* **2003**, *82*, 1938–1940.
241. Di Ventra, M.; Pantelides, S. T.; Lang, N. D. *Appl. Phys. Lett.* **2000**, *76*, 3448–3450.
242. Derosa, P. A.; Guda, S.; Seminario, J. M. *J. Am. Chem. Soc.* **2003**, *125*, 14240–14241.
243. Seminario, J. M.; Derosa, P. A.; Bastos, J. L. *J. Am. Chem. Soc.* **2002**, *124*, 10266–10267.
244. Seminario, J. M.; Zacarias, A. G.; Derosa, P. A. *J. Chem. Phys.* **2002**, *116*, 1671–1683.
245. Seminario, J. M.; Zacarias, A. G.; Derosa, P. A. *J. Phys. Chem. A* **2001**, *105*, 791–795.
246. Seminario, J. M.; Zacarias, A. G.; Tour, J. M. *J. Am. Chem. Soc.* **2000**, *122*, 3015–3020.
247. Paulsson, M.; Datta, S. *Phys. Rev. B* **2003**, *67*, 241403.
248. Galperin, M.; Nitzan, A.; Ratner, M. A. *Mol. Phys.* **2008**, *106*, 397–404.
249. Ke, S.-H.; Yang, W.; Curtarolo, S.; Baranger, H. U. *Nano Lett.* **2009**, *9*, 1011–1014.
250. Viljas, J. K.; Pauly, F.; Cuevas, J. C. *Phys. Rev. B* **2008**, *77*, 155119.
251. Dubi, Y.; M. Di Ventra, *Nano Lett.* **2009**, *9*, 97–101.
252. Zhang, C.; He, Y.; Cheng, H.-P.; Xue, Y.; Ratner, M. A.; Zhang, X. G.; Krstic, P. *Phys. Rev. B* **2006**, *73*, 125445.
253. Kondo, M.; Tada, T.; Yoshizawa, K. *Chem. Phys. Lett.* **2005**, *412*, 55–59.
254. Li, J.; Speyer, G.; Sankey, O. F. *Phys. Rev. Lett.* **2004**, *93*, 248302.
255. Zhuang, M.; Ernzerhof, M. *J. Chem. Phys.* **2009**, *130*, 114704–114708.
256. Zhuang, M.; Ernzerhof, M. *Phys. Rev. B* **2005**, *72*, 073104.
257. Zhang, C.; Du, M. H.; Cheng, H. P.; Zhang, X. G.; Roitberg, A. E.; Krause, J. L. *Phys. Rev. Lett.* **2004**, *92*, 158301.
258. Liu, R.; Ke, S.-H.; Yang, W.; Baranger, H. U. *J. Chem. Phys.* **2007**, *127*, 141104.
259. Dalgleish, H.; Kirczenow, G. *Phys. Rev. B* **2006**, *73*, 235436–235437.

- 260. Emberly, E. G.; Kirczenow, G. *Chem. Phys.* **2002**, *281*, 311–324.
- 261. Koleini, M.; Paulsson, M.; Brandbyge, M. *Phys. Rev. Lett.* **2007**, *98*, 197202–197204.
- 262. Jlidat, N.; Hliwa, M.; Joachim, C. *Chem. Phys. Lett.* **2009**, *470*, 275–278.
- 263. Duchemin, I.; Renaud, N.; Joachim, C. *Chem. Phys. Lett.* **2008**, *452*, 269–274.
- 264. Duchemin, I.; Joachim, C. *Chem. Phys. Lett.* **2005**, *406*, 167–172.
- 265. Stadler, R.; Ami, S.; Joachim, C. Forshaw, M. *Nanotechnology* **2004**, *15*, S115–S121.
- 266. Ami, S.; Hliwa, M. Joachim, C. *Chem. Phys. Lett.* **2003**, *367*, 662–668.
- 267. Baer, R.; Neuhauser, D. *Chem. Phys.* **2002**, *281*, 353–362.
- 268. Branton, D.; Deamer, D. W.; Marziali, A.; Bayley, H.; Benner, S. A.; Butler, T.; Di Ventra, M.; Garaj, S.; Hibbs, A.; Huang, X.; Jovanovich, S. B.; Krstic, P. S.; Lindsay, S.; Ling, X. S.; Mastrangelo, C. H.; Meller, A.; Oliver, J. S.; Pershin, Y. V.; Ramsey, J. M.; Riehn, R.; Soni, G. V.; Tabard-Cossa, V.; Wanunu, M.; Wiggin, M.; Schloss, J. A. *Nature Biotechnol.* **2008**, *26*, 1146–1153.
- 269. Zwolak, M.; Di Ventra, M. *Rev. Mod. Phys.* **2008**, *80*, 141.

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